

PROJECT CONTROLS INSTITUTE GLOBAL

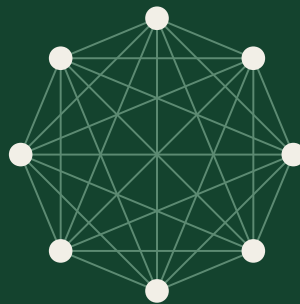
# PML-AI

## Body of Knowledge

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*The reference for the PCI Project Management Leader – AI*

Leadership · delivery systems · governance · the governed use of AI



$$n(n-1)/2 = 28$$

eight people · twenty-eight channels · every one a place to lose a decision

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*AI proposes; the professional verifies,  
decides and remains accountable.*

FIRST EDITION — DRAFT

For editorial and technical review · not for release or distribution

Finance intelligently. Control predictively. Deliver successfully.

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## Status of this draft

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## How to use this book

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This is a reference, not a course. It is written to be read in order once and consulted out of order thereafter, and the apparatus below exists to make the second use as easy as the first.

**How anything is addressed.** Content is numbered *Domain.KnowledgeArea.Topic* — so 7.3.2 is the second topic of the third Knowledge Area of Domain 7, and a cross-reference of the form *KA 10.2* is precise. Page numbers are never used for cross-references, because this volume is regenerated from source and its pagination is not stable across editions. Every reference in the text is therefore still valid in the next one.

**The worked examples are the spine of the book.** Each follows the same five steps — Setup, Formula, Substitution, Result, Interpretation — and the *Interpretation* is deliberately the longest and the most valuable. It is where the number is turned into a decision: what breaks the result, what the breakeven is, which assumption the conclusion is most sensitive to, and what a reviewer should check. If you read only one part of a worked example, read that one. The four steps before it exist so that you can reproduce the number and disagree with it.

**Every number can be checked, and should be.** Every figure printed as a result anywhere in this volume — in worked examples, in the text, in multiple-choice options, in exercise solutions, in case studies — is recomputed independently with decimal arithmetic by a verification suite that must pass in full before any chapter is accepted. Appendix C records how many checks stand behind each domain. If you find a number that does not reproduce from its stated method and inputs, that is a defect in this book and worth reporting.

**The AI in this KA sections are not an appendix to the subject.** Each states three things: where AI genuinely earns its place in that Knowledge Area, where it must not go, and how to verify its output concretely. They exist because the alternative — a single chapter about AI at the back — teaches that the question is separable from the work, and it is not. One principle governs all of them: *AI proposes; the professional verifies, decides and remains accountable.*

**What the sections at the end of each domain are for.** *Advanced topics* extend the domain past the level a candidate is assessed on, and each closes with a list of invariants a reviewer can test. *Industry variations* say what changes by sector, and are written so that a reader in one sector can see what a reader in another is dealing with. *Case studies* carry real arithmetic and are the place where several Knowledge Areas have to work together. *Executive perspective* is what a director cannot delegate. *Calculation exercises* each carry a full solution and a *Common error* note, and the common error is frequently the more useful half. The *Practitioner's toolkit* items are meant to be adopted and then left stable; adapt the headings to your organisation and stop changing them. *Exam preparation* lists what is assessed, the calculations to

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be able to do under time pressure, and the traps — each cross-referenced to where it is taught.

**Two conventions worth knowing before you meet them.** Where a result is established in one domain, the others *cite* it rather than re-derive it; if two domains appear to derive the same thing independently, that is a defect. And where one word carries two genuinely different concepts, the later use states the collision explicitly rather than leaving you to infer it. Appendix B lists both.

**If you are studying for the examination,** work the calculation exercises before reading their solutions, and treat the *Common error* notes as the syllabus they effectively are. If you are using this at work, start from the Practitioner's toolkit of the domain you need and follow its cross-references back into the treatment.

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## PART ONE

# Leading projects

Domains 1–4 — the profession, strategy and selection, governance and decision rights, and delivery architecture: what a project leader is for and answerable for.

DOMAIN

# 1

## The Project Leadership Profession

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## Why this domain exists

Every technique in this book (the networks of Domain 6, the earned value of Domain 7, the risk quantification of Domain 8) is instrumentation. This domain is about the person holding the instruments: what a project leader is *for*, what they are answerable for, and how they think. KA 1.1 establishes the delivery landscape and shows how leading temporary work differs from running a steady-state function. KA 1.2 defines accountability precisely, because the word is used loosely and the distinction decides who answers when things go wrong. KA 1.3 builds the two habits of mind that separate a leader from a coordinator — systems thinking, and the discipline of outputs versus outcomes versus benefits. Professional ethics and the governed use of AI are grounded in KA 1.4. A reader who finishes only this domain should already reason like the profession: *know what success actually means, know who owns it, and stay answerable for the judgment even when a machine produced the analysis.*

**Learning objectives.** After this domain a candidate can: distinguish projects, programmes and portfolios and explain what each optimises; **price the delivery verdict and the programme verdict on the same programme and state why the two are not directly commensurable**; contrast project leadership with operational leadership; explain why a temporary organisation makes trust and clarity urgent; distinguish accountability from responsibility and apply the distinction to a delegation; state the leader's obligations to sponsor, team, users and the public, and where they conflict; **compute the value of escalating a problem early and the number of weeks of earliness that funds the whole remedy**; explain the professional standard of care and show what exercising it looked like on a specific priced decision; analyse a project as a system with feedback and delay; **compute the transient trough that follows reinforcing a late team, derive its crossover horizon and state the condition under which no trough exists**; distinguish outputs, outcomes and benefits and demonstrate why output-based claims overstate value; **convert a benefit stream into a simple payback, show that payback scales as the reciprocal of adoption, and invert a payback rule into the adoption commitment it silently makes**; compute the value of time in benefit terms and the adoption level below which paying for speed destroys value; describe the ethical obligations of the role; apply the PCI responsible-AI principle to a realistic delivery decision; and **derive the breakeven error probability that makes a verification proportionate, and name the class of artefact for which that arithmetic must not be the deciding test.**

**The master case.** One programme runs through this domain and returns in Domains 2, 5, 15 and 16: **Meridian Care Records**, a fictional public-health programme rolling a shared clinical-records system out to **40 clinics** at an approved cost of **USD 2,400,000**. Its full-potential benefit is **USD 979,200** a year, its realistic benefit **USD 685,440** a year at 70 % adoption, and its cost of delay **USD 14,280 a week**: three figures derived in KA 1.3 and used, unchanged, by a dozen later

domains. This domain's job is to make them arithmetic rather than rhetorical, and to be explicit about what each rests on.

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## KNOWLEDGE AREA 1.1

# Projects, programmes and portfolios

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*Topics: 1.1.1 the delivery landscape · 1.1.2 project versus operational leadership · 1.1.3 the temporary organisation.*

## 1.1.1 The delivery landscape

**Definitions.** A **project** is a temporary endeavour undertaken to produce a defined result. A **programme** is a group of related projects and change activities managed together to achieve **outcomes** no single project could deliver alone. A **portfolio** is the set of projects and programmes an organisation chooses to fund, selected and balanced against strategy and capacity. The three are not sizes; they are different *objects of management*:

| LEVEL            | OPTIMISES                                           | PRIMARY SUCCESS TEST                      | THIS BOOK     |
|------------------|-----------------------------------------------------|-------------------------------------------|---------------|
| <b>Project</b>   | Delivery of a defined output to time, cost, quality | Was the thing delivered, fit for purpose? | Domains 4–13  |
| <b>Programme</b> | Coherent outcomes and benefits across components    | Did the change actually land?             | Domain 15     |
| <b>Portfolio</b> | Value and balance of the whole investment set       | Are we doing the right things at all?     | Domains 2, 15 |

The practical consequence is that **the same event can be a project success and a programme failure**: Meridian delivering all 40 clinic installations on time is a project triumph and worth nothing if clinicians do not use the system. The example below prices both verdicts, because a leader who can only assert the distinction will lose the argument to whoever has a number.

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## WORKED EXAMPLE

### Worked example 1.1.1 — the two verdicts on one programme, priced.

1. **Setup.** Meridian's completion report states the delivery verdict: **40 of 40** clinics installed, on time, **3 % under** the approved cost of **USD 2,400,000**. The programme verdict is taken separately, against benefit: the honest case assumed **70**

% adoption and **USD 685,440** a year (derived in 1.3.2); measured adoption is **40** %, worth **USD 391,680** a year. Both statements are true and they are reported to the same board in the same month.

2. **Formula.** Delivery verdict = approved cost × underspend rate (a one-off capital variance). Programme verdict = potential benefit × (planned adoption – actual adoption), an **annual flow**. The comparison ratio = annual benefit shortfall ÷ capital underspend.
3. **Substitution.** Underspend 2,400,000 × 0.03; shortfall 685,440 – 391,680, equivalently 979,200 × (0.70 – 0.40); ratio 293,760 / 72,000.
4. **Result.** Delivery verdict **+USD 72,000**, once. Programme verdict (**USD 293,760**) a year. The shortfall is **4.08 times** the underspend **in the first year alone**. Undiscounted over Domain 2's eight-year appraisal life the shortfall totals **USD 2,350,080**, or **97.92** % of what the whole programme cost to build; discounted at Domain 2's 7 % (annuity factor **5.971299**) it is **USD 1,754,128.79**, or **73.09** % of the approved cost.
5. **Interpretation.** Four readings, and the fourth is the one that keeps a leader honest.

**The verdicts are not commensurable, and saying so is part of reporting them.** One is a stock (a capital variance, banked once); the other is a flow (annual benefit, recurring for as long as the service runs). Dividing one by the other is legitimate only as a *scale* comparison, which is what the 4.08 is for; converting the flow into a single figure requires a horizon and a discount rate, and those belong to the business case (Domain 2, KA 2.2.2), not to the delivery report. A leader who quotes the undiscounted 2,350,080 without saying it is undiscounted has overstated by **USD 595,951.21** against the discounted figure.

**The delivery lever is small because it is bounded and the benefit lever is not.** To offset the first year of shortfall through cost alone, the programme would have had to underspend by **12.24** %: four times what it achieved, on a rollout whose costs are mostly contracted. The asymmetry is structural, not particular to Meridian: on benefit-generating work the accessible range of cost variance is a few per cent of a one-off, while the accessible range of adoption is tens of per cent of a perpetuity. That is the quantitative reason KA 1.3 spends its attention on the outcome link.

**The weekly form is the one that changes behaviour.** The 293,760 a year is **USD 6,120 a week**, which is exactly **12 clinics × 6 hours × USD 85**, the cost of delay of 1.3.3 applied to the twelve clinics that are not using the system. A board that hears "we are 293,760 light on the annual case" adjourns; a board that hears "the gap is costing 6,120 a week and has been for nine months" acts. Same number, different unit, different decision.

**The delivery verdict was provisional and was later reversed.** Domain 16's closing account records a final capital outturn of **USD 2,514,000**, against the

USD 2,328,000 implied by the 3 % underspend at installation, a swing of USD 186,000, most of it the recovery this domain's case study describes. The professional caution: an underspend reported before the outcome is measured is an *interim* figure, and a leader who banks it rhetorically ("we came in under") has made a claim the programme may still take back. A reviewer should ask, of any celebrated underspend, what remains to be spent to make the output work.

**How the discipline arrived at that distinction.** The vocabulary above is recent, and knowing why it exists prevents a candidate from treating it as bureaucratic layering. Four shifts, each of which added an obligation that had not previously been anyone's:

| SHIFT                                | WHAT WAS CODIFIED                                                                                                                                                                                                                                                                                                        | WHAT IT ADDED TO THE LEADER'S DUTY                                                         |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Technique</b>                     | Network scheduling, cost control and configuration control, developed on large capital, defence and aerospace programmes and generalised from there                                                                                                                                                                      | Produce a plan that can be checked by someone else                                         |
| <b>Process and profession</b>        | Bodies of knowledge, competence frameworks and certification; a common lifecycle vocabulary                                                                                                                                                                                                                              | Practise to a stated standard, not to personal habit                                       |
| <b>Governance and outcomes</b>       | Sponsorship, decision rights, gates, benefits management; internationally, the ISO 21500 family — context and concepts (ISO 21500), project management guidance (ISO 21502), programme (ISO 21503), portfolio (ISO 21504) and governance (ISO 21505), alongside ISO 31000 for risk and ISO 10006 for quality in projects | Answer for whether the change landed, not only for whether the thing was built             |
| <b>Adaptive and digital delivery</b> | Iterative and flow-based methods generalised out of software; data-centred delivery environments; and management-system standards for artificial intelligence itself, including ISO/IEC 42001 for AI management systems and ISO/IEC 23894 for AI risk                                                                    | Choose a delivery approach and defend the choice; govern the tools that now draft the work |

Those standards are named here, not reproduced: each is described in this book's own words, and a reader who needs the requirements themselves must obtain the standard. The relevant point for Domain 1 is directional. **The discipline has moved its test of success outward:** from "was the plan followed?" to "was the thing delivered?" to "did the change land, and can you show how you knew?" Every one of those moves increased what the leader must be able to evidence, and none of them removed an earlier obligation. That is why a modern project leader is answerable for a benefits chain (1.3.2) that a scheduler of the technique era was never asked about, and why the fourth shift puts a governed AI tool inside the same accountability frame rather than outside it (KA 1.4).

### 1.1.2 Project versus operational leadership

Operations optimise a repeating process; projects deliver a novel result once. The differences that change a leader's behaviour:

|                | OPERATIONAL<br>LEADERSHIP    | PROJECT LEADERSHIP                            |
|----------------|------------------------------|-----------------------------------------------|
| Work           | Repeating, refined over time | Novel, done once                              |
| Team           | Stable, known, yours         | Assembled, borrowed, temporary                |
| Authority      | Usually line authority       | Often influence without authority             |
| Time           | Continuous                   | Bounded, with an end that must be planned for |
| Learning curve | Amortised over many cycles   | Paid once, mid-delivery                       |
| Failure mode   | Drift, inefficiency          | Late discovery, then irreversibility          |

The last row is the one that shapes everything downstream. Operational problems are usually recoverable at similar cost whenever they are found; project problems get monotonically more expensive as commitments harden. That is why this book spends so much on early definition (Domain 5), honest schedules (Domain 6) and early warning (Domains 7–8): **in temporary work, lead time is the cheapest resource a leader has, and it is spent whether used or not.**

That claim is quantified elsewhere in this volume rather than asserted twice, and a candidate should know where: Domain 5 (KA 5.2.1) prices the rising cost of correcting a requirement defect by the stage at which it is found; Domain 9 (KA 9.2.2) computes the containment economics that make a later layer of checking an order of magnitude dearer than an earlier one; and Domain 3 (KA 3.3.1) values a gate by netting the cost of the delay it imposes against the failure cost it prevents. Domain 1's contribution is the leadership consequence, which is behavioural: because correction cost rises with commitment, **the profession's most valuable act is moving information earlier**, and KA 1.2 prices one instance of exactly that.

### 1.1.3 The temporary organisation

A project team is a temporary organisation: people arrive with divided loyalties, no shared history, and no accumulated trust to draw on. Three consequences a leader must design for. **Trust has no time to accrue naturally**, so it must be built deliberately and early: the team charter (Domain 12) exists because the informal norms that operations grow over years must here be stated in week one. **Clarity substitutes for familiarity**: roles, decision rights and done-definitions have to be explicit precisely because nobody can infer them (Domains 3–5). **Endings must be engin-**



**eered:** a temporary organisation dissolves, and the knowledge, the relationships and the operational readiness either transfer deliberately or evaporate (Domain 16).

### AI in this domain — first statement of the principle

AI now sits inside the delivery environment itself: drafting plans, summarising status, generating risk lists, answering questions the team used to bring to a senior colleague. The suite's governing principle, stated once here and applied in every later domain:

**AI proposes; the professional verifies, decides and remains accountable.**

Three things it means concretely for a leader, before any technique in this book is applied. **Verification is a duty, not a preference:** machine output entering a plan, report or decision is checked by a named human against evidence. **Accountability is not transferable:** "the model said so" is never a defence, because accountability cannot be delegated to software (KA 1.2 makes the reason precise). **Disclosure is part of honesty:** material AI use in a deliverable is stated, so colleagues can weigh it. Domain 14 builds the full governance architecture; this domain establishes that the obligation attaches to the *person*, not the tool.

#### KEY TERMS — KA 1.1

| TERM                            | MEANING                                                                                                                       |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Project</b>                  | Temporary endeavour producing a defined result.                                                                               |
| <b>Programme</b>                | Related projects managed together for outcomes no one project delivers.                                                       |
| <b>Portfolio</b>                | The funded, balanced set of projects and programmes.                                                                          |
| <b>Temporary organisation</b>   | A team with no accumulated trust or shared norms; both must be designed.                                                      |
| <b>Irreversibility</b>          | The rising cost of correction as commitments harden, projects' defining failure mode.                                         |
| <b>Delivery verdict</b>         | The project-level judgment: was the output delivered to time, cost and quality? A one-off variance.                           |
| <b>Programme verdict</b>        | The programme-level judgment: did the outcome and its benefit materialise? An annual flow.                                    |
| <b>Commensurability</b>         | Whether two figures may be compared directly; a capital variance and an annual benefit may not, without a horizon and a rate. |
| <b>Responsible-AI principle</b> | AI proposes; the professional verifies, decides and remains accountable.                                                      |

## SAMPLE MCQs — KA 1.1

**MCQ 1.1-A** [1.1.1 · Analysis] A programme's four projects each deliver on time and to budget, but the intended service change does not materialise. The most accurate statement is:

- A. the programme succeeded, since all its projects succeeded
- B. the projects succeeded on their own test while the programme failed on its own test — outcomes are not the sum of outputs CORRECT
- C. the projects must therefore have failed
- D. programme success cannot be assessed until every project closes

*Rationale:* Projects are judged on delivered outputs, programmes on outcomes; the two tests are different, which is exactly why the distinction exists. C rewrites history to preserve a single verdict, and D denies the measurement programmes exist to make.

**MCQ 1.1-B** [1.1.2 · Analysis] Why does project leadership place more weight on early definition than operational leadership does?

- A. project teams are less capable
- B. project problems grow monotonically more expensive as commitments harden, so lead time is the cheapest resource available CORRECT
- C. operations have no need for planning
- D. projects have larger budgets

*Rationale:* Irreversibility is the structural difference (1.1.2). A and D are unfounded; C misstates operational practice, which plans continuously, just with the ability to correct cheaply next cycle.

**MCQ 1.1-C** [1.1.3 · Recall] A team charter is written in week one primarily because:

- A. governance requires the document
- B. a temporary organisation has no accumulated norms, so what operations grow informally must here be made explicit CORRECT
- C. it replaces the need for a project plan
- D. it fixes the team's membership for the duration

*Rationale:* Explicit clarity substitutes for familiarity the team has had no time to build. A is a by-product, C confuses artefacts, D is false, membership usually changes.

**MCQ 1.1-D** [1.1.1 · Evaluation] A programme delivers 3 % under an approved cost of 2,400,000 while adoption reaches 40 % against a planned 70 % on a full-potential benefit of 979,200 a year. The soundest single sentence for the board is:

- A. the programme is 72,000 ahead, so the net position is positive
- B. the delivery result is +72,000 once, the benefit result is (293,760) a year — 4.08 times the underspend in year one alone — and the two are a stock and a flow, so no net figure should be quoted without a horizon and a rate
- C. the programme has lost 221,760 net in its first year
- D. no comparison is possible, since the two figures measure different things

CORRECT

*Rationale:* Both verdicts must be reported, scaled against each other, and flagged as non-commensurable (1.1.1). A nets a one-off against a perpetuity and keeps only the flattering half; C performs the same illegitimate subtraction ( $293,760 - 72,000$ ) and presents the remainder as a result; D uses non-commensurability as an excuse not to compare at all, when the ratio is precisely what makes the scale visible.

**MCQ 1.1-E** [1.1.1 · Comprehension] Which best describes what the governance-and-outcomes shift added to the project leader's duty?

- A. a requirement to follow a certified process
- B. an obligation to answer for whether the intended change landed, in addition to whether the output was built
- C. the replacement of scheduling technique by benefits management
- D. a transfer of delivery accountability to the sponsor

CORRECT

*Rationale:* Each shift added an obligation without removing an earlier one (1.1.1). A describes the preceding process-and-profession shift; C is the common misreading, technique was retained, not displaced; D inverts KA 1.2, where accountability for delivery stays with the leader.

#### SELF-CHECK — KA 1.1

1. *Can a project succeed while its programme fails?* — Yes: outputs delivered, outcomes not realised. The tests differ (1.1.1).
2. *Name the defining failure mode of temporary work.* — Late discovery followed by irreversibility; correction cost rises as commitments harden.
3. *State the responsible-AI principle.* — AI proposes; the professional verifies, decides and remains accountable.
4. *Why may a 72,000 underspend not simply be netted against a 293,760 annual benefit shortfall?* — One is a stock and the other a flow; netting them needs a horizon and a discount rate, which belong to the business case (1.1.1, Domain 2 KA 2.2.2).

- 
5. *Name the four shifts in the discipline and the obligation each added.* — Technique (a checkable plan), process and profession (practice to a stated standard), governance and outcomes (answer for the change landing), adaptive and digital delivery (defend the approach; govern the tools).
- 

## KNOWLEDGE AREA 1.2

# The project leader's accountability

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*Topics: 1.2.1 accountability and responsibility · 1.2.2 the obligation set · 1.2.2a interests, competence and confidentiality · 1.2.3 the professional standard of care.*

## 1.2.1 Accountability and responsibility

**Definitions. Responsibility** is the obligation to *do* the work. It can be shared and delegated. **Accountability** is the obligation to *answer* for the outcome. It sits with exactly one named person and **cannot be delegated**, only reassigned by whoever holds the authority to do so. A leader may hand a task to a specialist, an outsourcer or an AI tool; the answer for the result stays where it was.

Two working rules follow.

**Delegation transfers responsibility, never accountability.** The leader who delegates keeps the duty to specify, to verify at a proportionate depth, and to notice when the work is failing. "I gave it to the vendor" is a description of responsibility, not a defence of accountability.

**One accountable person per outcome.** Where two people are accountable for the same thing, nobody is: each defers, and the gap is discovered late. This is why Domain 3 builds explicit decision rights and why RACI-style matrices insist on a single "A" per row.

The mechanism deserves stating precisely, because it is not merely untidy. It has a cost and a signature. Two named holders of one outcome each face the same private calculation: acting costs effort and political capital now, while the consequence of not acting is shared, deferred and attributable to the other. Each therefore waits for evidence that the other will not act, and the evidence takes the form of the outcome deteriorating. **Dual accountability does not slow the response by a little; it converts the trigger from a measure into an incident.** Its signature in a report pack is an outcome measure that is reported by both parties, owned by neither, and never accompanied by a proposed action. On Meridian, the two holders were the delivery leader (installations) and the clinical directorate (practice change), the un-owned outcome was adoption, and the incident that finally triggered action was an external audit, twenty-six weeks after the measure first showed it. Worked example 1.2.2 prices those twenty-six weeks.

**Accountability is not the same as legal liability.** This domain uses accountability in the professional sense defined above. Legal liability (who must compensate whom, on what basis, and subject to what limitation or exclusion) is set by the contract and by the law of the jurisdiction, and it can sit with a party who is not professionally accountable, or be capped for a party who is. The two are routinely conflated, in both directions: a contractual cap is offered as a reason not to answer, or a professional duty is treated as if it created a payment obligation. Keep them separate in your own reasoning, and where a decision turns on which is engaged (an indemnity, a professional-indemnity notification, a duty to warn a third party), take the position from counsel for the applicable jurisdiction rather than from this book (Domain 10, KA 10.4 for the contractual machinery).

The AI corollary is now immediate rather than novel. A tool has no standing to be accountable. It cannot be asked to answer, cannot be sanctioned, cannot hold a duty. So *every* use of AI in delivery leaves accountability exactly where it was: with the professional who used the output. That is not a policy choice PCI made; it follows from what accountability is.

### 1.2.2 The obligation set

A project leader owes duties in five directions, and the skill is holding them together honestly rather than pretending they never conflict:

| To                            | OWES                                                                                                                             | TYPICAL CONFLICT                                                                        |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Sponsor / organisation</b> | Honest forecasts, stewardship of funds, no surprises                                                                             | Pressure to report reassurance rather than status                                       |
| <b>Team</b>                   | Clarity, safety, achievable asks, fair credit                                                                                    | Absorbing schedule pressure by quietly overloading people                               |
| <b>Users / customers</b>      | A result that is genuinely fit for purpose                                                                                       | Delivering on time by descoping what users needed most                                  |
| <b>Public / third parties</b> | Safety, legality, honest claims, environmental care                                                                              | Cost pressure meeting a safety or compliance margin                                     |
| <b>Profession / self</b>      | Work within competence, confidentiality of entrusted information, declaration of interests, honest representation of credentials | Accepting a role, or an opinion, the leader is not competent or not independent to give |

The fifth row is different in kind from the first four and is the one most often left out. The others are owed to a party who can notice the breach and complain about it; this one is owed to the standing of the discipline, and its breaches are invisible from outside: the panel member with an undisclosed stake, the assurance reviewer marking their own plan, the signature on an assessment the signatory was not qualified to give. Because nobody is placed to raise them, they have to be raised by

the person holding them, which is what makes the fifth row an *instrument* rather than a sentiment. 1.2.2a supplies the instrument.

**The escalation duty.** When these conflict beyond the leader's authority to resolve, the professional act is to escalate with options, not to absorb the conflict silently and hope. An unescalated conflict becomes a late surprise, and late surprises are the failure mode of §1.1.2.

**The honesty asymmetry.** Bad news travels badly: it costs the messenger and helps the project. A leader's most consequential cultural act is making early bad news *safe*, because a schedule or cost problem reported five weeks early is usually recoverable (Domain 6's case study turns on exactly this) and the same problem reported at the deadline is not. The asymmetry is worth making arithmetic, because "escalate early" is advice everyone agrees with and almost nobody prices.

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#### WORKED EXAMPLE

##### Worked example 1.2.2 — what the timing of one escalation was worth.

1. **Setup.** Meridian measures adoption monthly from go-live. At **week 13** the clinical lead's data shows **16** of 40 clinics in daily use: 40 % against the planned **70 %**, a shortfall of **12 clinics**, each worth  $6 \times 85 = \text{USD } 510$  a week (1.3.2). Nobody escalates: the measure is reported by two parties and owned by neither (1.2.1). The shortfall surfaces at **week 39**, in an external audit. The remedy, once authorised, is clinical champions in the **11** lowest-adoption clinics at **USD 12,000** each; it lifts adoption to **68 % (27.2 clinics)**, closing **11.2** of the 12-clinic gap, and it costs the same whenever it is authorised. (Domain 16's registered clinic-count measure reads **67.50 %** and reconciles the two definitions at KA 16.4.1; the 11.2 clinics used here is therefore the more generous of the two readings.) What did the 26-week silence cost?
2. **Formula.** Shortfall run rate = gap clinics  $\times$  weekly value per clinic. Recovered run rate = closed clinics  $\times$  weekly value. Timing value of the escalation = weeks of earliness  $\times$  recovered run rate. Weeks of earliness that fund the whole remedy = remedy cost  $\div$  recovered run rate.
3. **Substitution.** Run rate  $12 \times 510$ ; recovered  $11.2 \times 510$ ; timing value  $26 \times 5,712$ ; remedy  $11 \times 12,000$ ; funding earliness  $132,000 / 5,712$ .
4. **Result.** The shortfall ran at **USD 6,120 a week**; the remedy recovers **USD 5,712 a week**. Escalating at week 13 instead of week 39 was worth **USD 148,512**: against a remedy costing **USD 132,000**, so the *timing* of the escalation was worth **1.13 times** the entire remedy it would have paid for. **23.11 weeks** of earliness funds the remedy outright, so the 26 weeks available carried **2.89 weeks** of margin.
5. **Interpretation.** Five readings, and the last is the one a leader must not skip.

**The most valuable act in the sequence cost nothing.** The remedy required a business case, funding and eleven appointments; the escalation required one

person to send one message. On these figures the message was worth more than the programme it triggered. That is the honesty asymmetry in numbers, and it generalises: wherever a problem has a run rate, the value of early notice rises linearly with earliness while its cost stays fixed and small, so **the benefit-to-cost ratio of escalation is bounded only by how early the measure could have shown**, which is why the behaviour is worth engineering rather than exhorting.

**The recovered run rate, not the shortfall run rate, is the honest multiplier.** The gap was 12 clinics but the remedy closes 11.2, so valuing the earliness at 6,120 rather than 5,712 would overstate it by **USD 10,608** across the 26 weeks. The general error is pricing early warning at the size of the problem rather than at the size of the *fix that early warning brings forward*. Note the identity that falls out: 5,712 is exactly **40.00 %** of Meridian's USD 14,280 cost of delay, because 11.2 of 28 adopting clinics is 40 %. Early warning about part of a programme is worth the corresponding fraction of that programme's cost of delay, and no more.

**Everything rests on the remedy working.** Had clinical champions failed to move adoption, the 148,512 would be zero: earliness is worth exactly the value of the action it enables. The same dependency governs 1.3.3, and a leader who cannot name the action an escalation is meant to trigger does not yet have an escalation, only a complaint.

**The sensitivity runs the reassuring way.** The calculation assumes a constant 12-clinic gap. In practice an unaddressed adoption gap tends to widen as early enthusiasm decays and paper workarounds harden, so a constant-gap model **understates** the timing value and 148,512 should be read as a floor. If instead the gap had been closing unaided, the timing value would fall; and the test of which world you are in is whether the measure has a trend, which is the reason a single monthly number without a trend line is not yet a measure.

**The professional caution: this arithmetic must not be turned on the analyst.** The temptation on seeing 148,512 is to ask who failed to escalate. But the analyst reported the data on time; the failures were that no name owned the measure (1.2.1) and no route made raising it safe (1.2.2). The 148,512 is therefore the *leader's* number. Used to discipline a junior it destroys the next escalation and buys a second incident; used to fund an owned measure and a standing route to the board, it is the strongest business case available for the cultural work this KA describes.

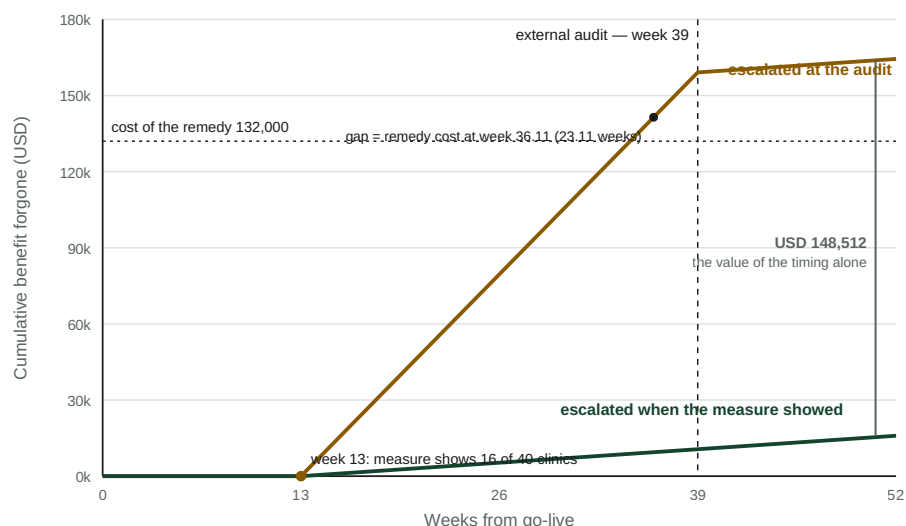


Fig 1.2.1 — The cost of a silent measure

### 1.2.2a Interests, competence and confidentiality

**Definition.** An **interest** is anything capable of affecting, or of appearing to affect, a person's judgement on a matter before them: a financial stake, an employment or commercial relationship, a personal or family connection, authorship of the thing being assured, or a future role that depends on the outcome. Where an interest bears on a matter the person is being asked to decide, advise on or assure, it is a **conflict of interest**: the term this profession and every neighbouring one uses, and the term a register, a code of conduct or an accreditation reviewer will look for. Nothing in this book treats a conflict of interest as a lapse of character: it is a structural fact about a role, and the professional act is to surface it, not to be free of it. The appearance is part of the definition, not a softening of it: 1.4.1's daylight test asks whether every party seeing the full picture would still regard the decision as impartial, and an interest that fails that test has already done its damage whether or not it moved a single judgement.

The duty is not to have no interests, which is impossible in a discipline where the same small population of people sponsors, delivers, assures and buys. It is a four-step sequence, and what makes it an instrument rather than an intention is that each step has somewhere to live.

- **Identify.** A standing question, asked at appointment to any role and again as each gate, selection panel, evaluation or approval is convened: *do I hold an interest in what is about to be decided?* Asking it once at induction produces a register that is accurate on the day it is written and wrong thereafter, because interests are acquired.
- **Declare.** In writing, to the chair, **before the item is taken**: not at the end of the discussion, and not after the decision. A declaration made afterwards is an explanation, not a declaration.



- **Record.** In the decision record itself, as a field of its own, with a **nil return** recorded when there is nothing to declare so that silence is evidence rather than an absence of evidence (Domain 3, KA 3.3.4 for the field, and Toolkit 3.T.3 for the monthly counts it makes possible).
- **Abstain.** The interested party takes no part in the discussion or the decision, and the minute records **who decided instead**. Abstention that is not minuted is indistinguishable from participation when the file is read later, which is when it will be read.

Two rules are hard, and each is hard because the soft version of it is the one that fails in practice.

**An interest that cannot be managed by abstention requires the role to change, not the record to be silent.** Abstention works for an item; it does not work for a standing position. Where the interest touches the substance of the role itself (the assurance lead who wrote the plan, the panel chair whose former employer is bidding, the reviewer whose next contract depends on the answer) the remedy is to move the person or move the work, and to say in the record that this was done. An organisation that meets a structural interest with a declaration and a decision taken anyway has recorded the conflict without managing it, which is a worse position than not knowing, because it now has evidence that it knew.

**A leader may not sign an opinion, an estimate or an assessment they are not competent to give.** The pressure runs the other way, because signing is quick and obtaining competence is slow, and because the request usually arrives as a courtesy. *You just need to sign this off.* The professional act is to **obtain the competence** (a qualified specialist, a named second signature, an independent reviewer of the right discipline), or to **decline**, in writing, saying what would have to be true for the signature to be available. A signature is a statement that the signatory checked something they were able to check; where that is not so, the file records diligence that did not happen, and 1.2.3's whole apparatus of care rests on a document that is not true.

**Confidentiality is a standing duty, not an AI failure mode.** Information entrusted for the work (a supplier's pricing, an employee's circumstances, a patient's record, an unannounced strategy) is held for the purpose it was given for and no other, and that duty survives the project, the employment and the tool. 1.4.2 lists confidentiality leakage among the failure modes to expect from AI use, and that entry is a *special case* of this duty rather than its origin: pasting project data into a tool is a disclosure for the same reason that discussing it in a lift is, and the obligation that both breach was in force before either was possible.

Everything downstream of this subsection is an application of it. Selection scoring depends on who scored (Domain 2, KA 2.2.3); steering membership depends on who may vote (Domain 3, KA 3.2.2); assurance depends on who may not review their own work (Domain 3, KA 3.3.2); procurement depends on declaration and exclusion

before the panel is constituted (Domain 10, KA 10.2.3); reserve draws depend on who may approve a release that covers their own overspend (Domain 8, KA 8.3.2); and the verification signature on an AI-assisted artefact depends on a verifier who is both competent and independent (1.4.3). None of those machineries states a rule about **who may not decide**. This one does, and they inherit it.

### 1.2.3 The professional standard of care

Professionals are judged not on outcomes alone but on whether they exercised the **care a competent practitioner would** in the circumstances. Practically, four things constitute care in this discipline: **method appropriate to the stakes** (a two-week internal task and a safety- critical programme do not warrant the same rigour); **evidence proportionate to the claim** (a forecast presented to a board carries its basis: Domain 7's estimate basis sheet); **records that let others check the work** (decision logs, assumption registers, audit trails); and **candour about uncertainty** (ranges, not false single numbers). A leader who did all four and still missed can defend the work. One who hit the date by luck and cannot show any of them has no defence available when the next project does not get lucky.

**What care looks like on one real decision.** Abstract lists of this kind are easy to agree with and hard to apply, so take the acceleration decision worked in 1.3.3 (spend USD 60,000 to arrive eight weeks sooner), and ask what each constituent required of the leader who took it.

| CONSTITUENT                                | WHAT IT REQUIRED HERE                                                                                                                                        | WHAT ITS ABSENCE WOULD HAVE LOOKED LIKE                                                                       |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Method proportionate to the stakes</b>  | A 60,000 decision does not warrant a simulation; it warrants a run rate, a breakeven and one sensitivity (the three lines of 1.3.3)                          | "The team thinks we can pull it in and it seems worth it"                                                     |
| <b>Evidence proportionate to the claim</b> | The USD 14,280 a week traced to 28 adopting clinics, 6 hours, USD 85: every factor with a source, and adoption identified as the one factor not yet observed | A weekly benefit figure quoted with no derivation, which cannot be challenged and therefore cannot be trusted |
| <b>Records others can check</b>            | The decision recorded with its inputs, its breakeven of 4.20 weeks, and the adoption assumption named as the thing that would invalidate it                  | A minute reading "acceleration approved", which tells a later reviewer nothing about what was known           |
| <b>Candour about uncertainty</b>           | The 50/70/90 % adoption range carried into the paper, and the statement that below 36.76 % adoption the spend destroys value                                 | A single-point benefit and a confident recommendation                                                         |

The leader who did those four things and then watched adoption stall at 40 % has a defensible file: the decision was right on the information available, the invalidating condition was named in advance, and the record shows it. That is the whole practical content of the standard of care. **It is not a promise about outcomes, it is a discipline about evidence, and it is built before the outcome is known, because it cannot be built afterwards.**

Two cautions. First, care is judged **in the circumstances**, which cuts both ways: the same one-page analysis that constitutes care on a 60,000 decision would be negligent on a safety-related one, and a leader who applies a uniform standard is wrong in one direction or the other (the same proportionality argument as KA 1.4.3). Second, where care is **codified** (regulated construction, pharmaceuticals, aviation, nuclear, financial services) the required records, competencies and sign-offs are set by that regime and by the relevant professional-body rules, and what counts as adequate there is a question for the regime and, where liability is engaged, for counsel in the applicable jurisdiction. This book states the professional discipline; it does not state anyone's legal duty.

#### KEY TERMS — KA 1.2

| TERM                                 | MEANING                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Responsibility</b>                | The obligation to do; shareable and delegable.                                                                                                                                                                                                                                                                                                                                 |
| <b>Accountability</b>                | The obligation to answer; single-holder, non-delegable.                                                                                                                                                                                                                                                                                                                        |
| <b>Escalation duty</b>               | The obligation to raise, with options, a conflict beyond one's authority.                                                                                                                                                                                                                                                                                                      |
| <b>Honesty asymmetry</b>             | Bad news costs the messenger and helps the project; leaders must make it safe.                                                                                                                                                                                                                                                                                                 |
| <b>Timing value of an escalation</b> | Weeks of earliness × the run rate the remedy recovers: what the <i>timing</i> alone is worth, separate from the remedy.                                                                                                                                                                                                                                                        |
| <b>Silent measure</b>                | An outcome measure reported by two parties, owned by neither, and never accompanied by a proposed action.                                                                                                                                                                                                                                                                      |
| <b>Legal liability</b>               | Who must compensate whom, set by contract and jurisdiction; distinct from professional accountability and capable of sitting elsewhere.                                                                                                                                                                                                                                        |
| <b>Declared interest</b>             | Anything capable of affecting, or of appearing to affect, judgement on a matter before a person (financial, employment, personal, authorship of what is being assured, or a future role dependent on the outcome) disclosed in writing to the chair before the item is taken. <i>Context flag: the stakeholder sense of "interest" (attention paid) is Domain 11, KA 11.1.</i> |
| <b>Conflict of interest</b>          | An interest that bears on a matter the person is being asked to decide, advise on or assure. Not a lapse of character but a structural fact about a role: the professional act is to declare and abstain, not to claim to have none. Where abstention cannot manage it, the role changes rather than the record staying silent.                                                |

| TERM                        | MEANING                                                                                                                                                                                                                       |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Interest duty</b>        | The four steps of 1.2.2a: identify at appointment and at each gate · declare in writing before the item is taken · record in the decision record, nil returns included · abstain, with the minute naming who decided instead. |
| <b>Abstention</b>           | The interested party taking no part in the discussion or the decision, minuted with the name of whoever decided in their place.                                                                                               |
| <b>Structural interest</b>  | An interest touching the substance of a role rather than a single item; managed by changing the role or the work, never by a declaration followed by the decision being taken anyway.                                         |
| <b>Competence limit</b>     | The boundary beyond which a leader may not sign an opinion, estimate or assessment; the professional act is to obtain the competence or to decline in writing.                                                                |
| <b>Confidentiality duty</b> | The standing obligation to hold entrusted information for the purpose it was given for and no other; it survives the project and the tool, and predates both.                                                                 |
| <b>Standard of care</b>     | The care a competent practitioner would exercise in the circumstances.                                                                                                                                                        |

## SAMPLE MCQs — KA 1.2

**MCQ 1.2-A [1.2.1 · Application]** A leader outsources the migration design to a specialist vendor under a fixed-price contract. Accountability for the migration's success now rests with:

- A. the vendor, under the contract
- B. the leader, who retains the duty to specify, verify proportionately and act on failure CORRECT
- C. jointly and equally with the vendor
- D. the sponsor, who approved the contract

*Rationale:* The contract transfers responsibility (and commercial risk); accountability is non-delegable. C describes the arrangement that guarantees nobody answers (1.2.1); D confuses approving a route with owning the delivery.

**MCQ 1.2-B [1.2.1 · Analysis]** An AI planning tool generates a schedule that the leader submits unchanged; it contains an infeasible dependency that causes a two-month slip. The accountability position is:

- A. shared with the tool's vendor, who supplied a defective product
- B. the leader's: a tool cannot answer for an outcome, and submitting unverified output is a failure of the verification duty CORRECT
- C. nobody's — the failure was technological
- D. the team's, for not catching it

*Rationale:* Accountability requires a party who can be asked to answer, so it cannot attach to software (1.2.1); the specific lapse is verification (1.1's principle). Vendor recourse (A) is a commercial question and does not relocate the professional answer; D inverts delegation.

**MCQ 1.2-C** [1.2.3 · Analysis] A leader's project overran despite a documented method, ranged forecasts, a decision log and escalated risks. Under the standard of care this is:

- A. professional negligence, because the project overran
- B. defensible practice — care is judged on the exercise of competent method, not on outcome alone CORRECT
- C. irrelevant, since standards of care apply only to regulated professions
- D. defensible only if the overrun was under 10 %

*Rationale:* The standard is conduct-based (1.2.3). A collapses care into outcome; C is false in substance. The expectation attaches to professional practice generally; D invents a threshold.

**MCQ 1.2-D** [1.2.2 · Application] An adoption shortfall of 12 clinics, each worth USD 510 a week, is visible at week 13 but escalated at week 39. The remedy costs USD 132,000 and recovers 11.2 of the 12 clinics. The value of the 26 weeks of earliness that were not taken is:

- A. USD 159,120
- B. USD 148,512 CORRECT
- C. USD 16,512
- D. USD 132,000

*Rationale:* Earliness is worth the run rate the remedy recovers:  $26 \times 11.2 \times 510 = 148,512$  (1.2.2). A values it at the full 12-clinic gap ( $26 \times 6,120$ ), crediting the remedy with a recovery it does not achieve; C deducts the remedy cost, which is incurred either way and so does not belong in a timing comparison; D is the remedy cost itself.

**MCQ 1.2-E** [1.2.1 · Analysis] A programme's adoption measure appears in both the delivery report and the clinical directorate's report, with no proposed action in either. The most accurate diagnosis is:

- A. duplicated reporting, which should be rationalised for efficiency
- B. two accountable names and therefore none — the measure has no owner, and its trigger has become an incident rather than a threshold CORRECT
- C. adequate governance, since two parties are monitoring the measure
- D. a data-quality problem in the reporting pipeline

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*Rationale:* Dual accountability converts the trigger from a measure into an incident, and the signature is exactly this: reported twice, owned once by nobody, never with an action (1.2.1). A treats the symptom as an administrative annoyance; C mistakes visibility for ownership; D relocates a governance defect to the tooling.

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SELF-CHECK — KA 1.2

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1. *What can be delegated and what cannot?* — Responsibility can; accountability cannot.
  2. *Why can accountability never attach to an AI tool?* — Accountability is the obligation to answer; a tool cannot be asked to answer, sanctioned, or hold a duty.
  3. *Name the four constituents of care in this discipline.* — Proportionate method, evidence matching the claim, checkable records, candour about uncertainty.
  4. *What is the signature of dual accountability in a report pack?* — An outcome measure reported by both parties, owned by neither, never accompanied by a proposed action (1.2.1).
  5. *At what run rate should the value of early warning be priced?* — The run rate the remedy **recovers**, not the size of the problem: 5,712 a week, not 6,120 (1.2.2).
  6. *Is professional accountability the same as legal liability?* — No: liability is set by contract and jurisdiction and may sit elsewhere or be capped; take that position from counsel (1.2.1).
  7. *Name the four steps of the interest duty.* — Identify at appointment and at each gate; declare in writing to the chair before the item is taken; record in the decision record, nil returns included; abstain, with the minute naming who decided instead (1.2.2a).
  8. *An interest cannot be managed by abstention. What changes?* — The role or the work, not the record: a declaration followed by the decision being taken anyway records the conflict without managing it (1.2.2a).
  9. *What is the professional response to a request to sign an assessment outside one's competence?* — Obtain the competence (a qualified specialist or a named second signature), or decline in writing, stating what would have to be true for the signature to be available (1.2.2a).
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KNOWLEDGE AREA 1.3

## Systems thinking and value

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*Topics:* 1.3.1 projects as systems · 1.3.2 outputs, outcomes, benefits, value · 1.3.3 leading under uncertainty.

### 1.3.1 Projects as systems

A project is a system: parts that interact, with **feedback** and **delay** between cause and visible effect. Three systems behaviours a leader must recognise, because each defeats common-sense management.

**Delay hides consequence.** Adding people to a late project consumes the productivity of those already on it (onboarding, communication paths) before adding any, so the intervention looks harmful before it looks helpful; and leaders often reverse it in the interval.

**Local optimisation degrades the whole.** A discipline lead maximising their own efficiency levels their team's workload smoothly and delivers the interface documents late, stalling three other teams. Every function can be efficient while the project is slow: the constraint logic of Domain 6 (6.A.1) is the schedule expression of this.

**Pressure moves, it does not vanish.** Squeeze the schedule and the pressure emerges as quality defects (Domain 9), scope disputes (Domain 5), team attrition (Domain 12) or supplier claims (Domain 10). A leader's question on receiving any "we absorbed it" report is *where did it go?*

The first of those three is the one that most often causes a leader to undo their own correct decision, and it is priced in KA 1.3.3 (Worked example 1.3.3b), where the cost of delay it needs has been derived: adding people to late work produces a measurable **transient** in which progress is worse than it would have been, and the arithmetic gives the week at which the intervention overtakes doing nothing. Domain 12 (KA 12.2.2) treats the *steady-state* counterpart (the coordination cost of team size, which has a hard capacity ceiling), and the two are different mechanisms with different remedies, so neither substitutes for the other: shortening a ramp does nothing about a mesh, and structuring a team does nothing about a ramp.

### 1.3.2 Outputs, outcomes, benefits and value

**Definitions.** An **output** is what the project delivers (a system installed, a road built). An **outcome** is the change in how things work as a result (clinicians actually using the records; journeys actually faster). A **benefit** is the measurable improvement that the outcome produces (hours released, cost avoided, harm reduced). **Value** is benefit weighed against cost and risk, for a stated stakeholder. The chain has a hard property: **each link can fail independently**, so an output claimed as a benefit is an overstatement by construction.

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#### WORKED EXAMPLE

**Worked example 1.3.2 — what Meridian actually delivers.**

1. **Setup.** Meridian Care Records installs the shared records system in **40 clinics** (the output). Realistic adoption at 12 months is **70 %** of clinics using it in daily practice. In an adopting clinic the system releases **6 clinician-hours per week**, valued at **USD 85 per hour**, over a **48-week** operating year.
2. **Formula.** Annual benefit = clinics × adoption × hours/week × rate × weeks. Contrast with the output-based claim, which counts all 40 clinics.
3. **Substitution.** Adopting clinics  $40 \times 0.70 = 28$ ; benefit  $28 \times 6 \times 85 \times 48$ . Output-based claim  $40 \times 6 \times 85 \times 48$ .
4. **Result.** Benefit **USD 685,440 per year** ( $\approx$  SAR 2.57 million indicatively). The output-based claim gives **USD 979,200**: an overstatement of **USD 293,760**, exactly **30.0 %**, which is the non-adoption rate reappearing as fictitious value.
5. **Interpretation.** The 30 % gap is the *outcome* link being skipped, not an estimating error. This is why a benefits register (Domain 16) tracks adoption as a measure in its own right, and why "40 clinics live" is a milestone, not a benefit. Five things follow.

**The overstatement is exactly the non-adoption rate, whatever the other numbers are.** Because both figures share the hours, the rate and the weeks, the ratio of the honest benefit to the output-based claim is the adoption term alone:  $685,440 / 979,200 = 0.70$ . So an output-based claim overstates by 1 – a (here 30.0 %) **irrespective of how many hours are released, what they are worth, or how long the operating year is.** That is a useful thing to know in a meeting: you do not need the model to say how wrong a claim is, only the adoption assumption it omitted. The corollary is a reviewer's shortcut; if a benefits case contains no adoption term, its overstatement is already known to be the non-adoption rate, and the discussion can move straight to what that rate is.

**Sensitivity is the leader's real lever.** At 50 % adoption the benefit is USD 489,600; at 90 %, USD 881,280. **A programme of this shape creates more value by moving adoption than by moving its installation date**, which should change where the leader spends personal attention (Domain 11's engagement work, not Domain 6's schedule compression). Per clinic the same arithmetic gives  $979,200 / 40 = \text{USD } 24,480$  of full potential and **USD 17,136** realistic (the per-clinic figures Domain 5 uses to price scope decisions at KA 5.1.3), so a single clinic won or lost is worth more than most of the change requests a rollout of this kind argues about.

**The definition of adoption is where the number is actually decided.** "Using it in daily practice" has to be operationalised (a proportion of encounters recorded in the system, a threshold of active clinicians, a fortnight without reversion to paper), and reasonable definitions differ. On these figures a **5-percentage-point** difference in where the threshold is drawn is worth  $0.05 \times 979,200 = \text{USD } 48,960$  a year. So the definition belongs in the benefits register

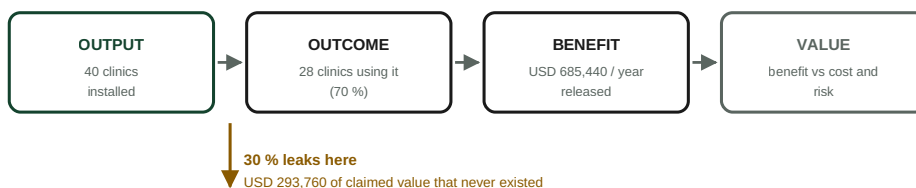


with the measure, agreed before the first report, or the programme will spend its second year arguing about the denominator (Domain 16, KA 16.4.1).

**Released hours are capacity, not cash, and the distinction is a professional obligation, not a quibble.** Valuing 6 clinician-hours at USD 85 states what those hours are *worth if they are used for something*. They become a cash saving only if headcount or agency spend falls, and a service improvement only if the hours go to patients; if neither happens they are absorbed and the benefit is real to nobody. A benefits case must therefore say which of the two it is claiming and who is accountable for making it happen: the same distinction Domain 2 draws between cost *removed* and cost *moved* (KA 2.1.2). A leader who lets a released-hours figure be read as cash has authorised a claim the finance function will later, correctly, refuse to recognise.

**What would falsify the whole chain.** Three things, in order of likelihood: adoption measured on a definition nobody agreed (above); hours released estimated rather than sampled, Domain 16's measurement found 5.4 hours, not 6.0, which alone removes **USD 66,096** a year at that domain's measured steady adoption of **67.50 %** ( $163,200 \times 0.675 \times 0.6$ ); and an attribution problem, where hours would have been released anyway by an unrelated change, which is why a comparison cohort matters (Domain 16, KA 16.4.3). None of these is a reason to omit the arithmetic. They are the reasons to publish it with its assumptions attached, so that the argument is about the assumptions rather than about the conclusion.

Each link can fail independently — an output claimed as a benefit overstates by construction



Adoption is the outcome measure that links an output to any benefit at all — so it needs its own owner (Toolkit 1.T.1).

Fig 1.3.1 — The value chain, and where it leaks

**The fourth link is the one leaders skip.** Benefit is not value: value is benefit weighed against cost and risk, for a stated stakeholder. A programme with a genuine benefit and a bad ratio of benefit to cost is a bad investment, and the leader who reports only the benefit has not yet said anything about value. The simplest instrument that closes the link is **simple payback** (the number of periods of benefit needed to repay the cost), and it repays study here because it is the instrument

boards actually use, it makes the adoption dependency arithmetic, and its limitations are instructive.

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#### WORKED EXAMPLE

**Worked example 1.3.2b — payback, and the adoption commitment a payback rule silently makes.**

1. **Setup.** Meridian's approved cost is **USD 2,400,000**. Full potential benefit is **USD 979,200** a year; the honest benefit at 70 % adoption is **USD 685,440**. The sponsoring organisation applies a **three-year simple payback** rule to discretionary investment. Compute the payback on both benefit figures, and find the adoption the rule actually requires.
2. **Formula.** Simple payback = cost ÷ annual benefit, where annual benefit = potential × adoption. Required adoption for a payback target  $T = \text{cost} \div (T \times \text{potential})$ .
3. **Substitution.** Output-based  $2,400,000 / 979,200$ ; honest  $2,400,000 / 685,440$ ; required annual benefit  $2,400,000 / 3 = 800,000$ ; required adoption  $800,000 / 979,200$ .
4. **Result.** Payback on the output-based claim **2.4510 years**; on the honest benefit **3.5014 years**. The ratio of the two is **1.428571**, which is exactly  $1 / 0.70$ . A three-year payback rule requires **USD 800,000** a year, and therefore **81.70 %** adoption — **11.70 percentage points**, or **4.68 clinics**, beyond what the case assumed.
5. **Interpretation.** Five readings, and the fourth is the one that changes how a leader reads a board's own rules.

**Payback scales as the reciprocal of adoption, exactly.**  $\text{payback} = \text{cost} / (\text{potential} \times a)$ , so halving adoption doubles payback and the 1.428571 ratio above is  $1/0.70$  with no residue. This is worth holding as an identity because it converts a governance question into an arithmetic one: any output-based payback is understated by the factor  $1/a$ , and a leader who knows the adoption assumption can correct a payback in their head.

**A payback rule is an adoption commitment in disguise.** Nobody in the approval meeting said "we undertake to reach 81.70 % adoption"; the three-year rule said it for them, and nobody noticed because the rule was expressed in years and the commitment is in percentage points. **Inverting a hurdle into the operational condition it implies is one of the highest-value habits in this book:** it is the same move Domain 2 makes when it reports breakeven adoption instead of NPV (KA 2.2.2), and Domain 3 makes when it converts a delegation threshold into an escalation volume (KA 3.2.3). The professional act on seeing a 3.5014-year payback against a three-year rule is not to argue about the rule but to say what the rule demands and ask who owns delivering it.

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**The marginal value of adoption is largest where adoption is worst, by a factor of 3.60.** Because payback is hyperbolic in adoption, moving from 40 % to 50 % cuts payback by **1.2255 years** while moving from 80 % to 90 % cuts it by **0.3404 years**; the first ten points are worth **3.60 times** the last ten. This contradicts the instinct to chase the final laggards for completeness, and it is the arithmetic behind the recovery in this domain's case study, which funded champions in the **eleven lowest-adoption** clinics rather than pushing the leaders higher. The same reasoning appears in Domain 11's engagement prioritisation: effort goes where the curve is steep.

**Simple payback is the wrong instrument for the decision and the right one for the conversation.** It ignores the time profile of benefits, the cost of capital, everything after the payback point, and (on Meridian) the operating cost the case omitted. Including Domain 16's measured run cost of **USD 108,000** a year, net benefit falls to **USD 577,440** and payback lengthens to **4.1563 years**, an addition of **0.6549 years** from a line that was simply absent. Domain 2 does the decision properly, discounted over eight years at 7 %, and reports a breakeven sustained adoption of **41.05 %**. Note what the two breakevens are for: **81.70 % is what a three-year payback rule demands; 41.05 % is what creating value demands.** The **40.65-point** gap between them is the price of using a payback rule as a proxy for value, and a programme killed at 70 % adoption for failing a payback rule would have been killed while creating value, which is a governance failure and not an arithmetic one.

**The caution.** Every figure here is undiscounted, in nominal money, and takes the approved cost as complete. Do not present a simple payback to a board without saying which of those three simplifications is doing the most work; on Meridian it was the third. Where the investment is material, the discounted appraisal is not an optional refinement: Domain 2 is where it belongs.

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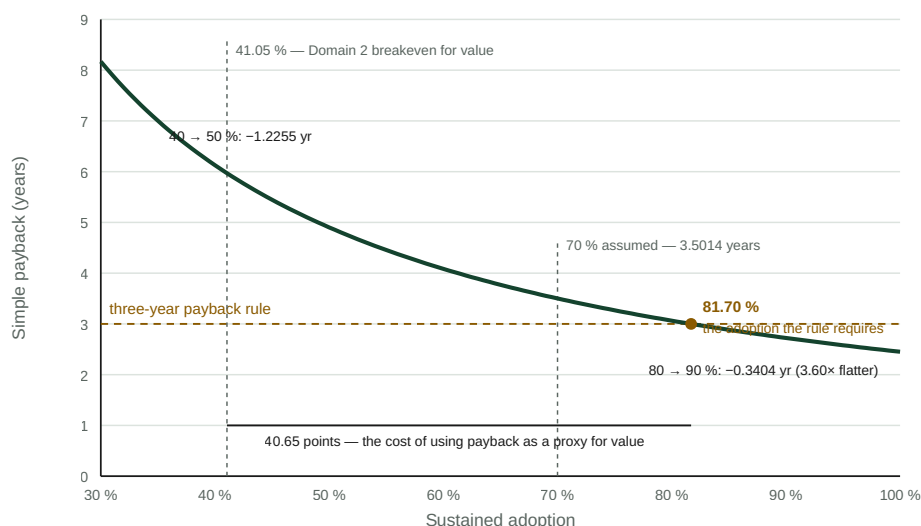


Fig 1.3.2 — Payback is hyperbolic in adoption, so the first points are worth the most

### 1.3.3 Leading under uncertainty

Uncertainty is the condition, not an exception to be eliminated before acting. Three disciplines constitute leading well inside it.

**Decide at the right moment.** Deciding early buys momentum and forecloses options; deciding late preserves options and costs time. The professional question is not "do we have enough information?" but *what is the last responsible moment, and what does waiting cost?*

**Price the time.** In benefit-generating work, delay has an arithmetic cost.

#### WORKED EXAMPLE

#### Worked example 1.3.3 — what a week of Meridian delay costs.

- Setup.** Meridian's benefit runs at the KA 1.3.2 rate once adopted. The team can compress the rollout by **8 weeks** at an incremental cost of **USD 60,000**. Should the leader buy it?
- Formula.** Benefit per week = adopting clinics × hours × rate. Net = (weeks × benefit per week) – cost. Breakeven weeks = cost ÷ benefit per week.
- Substitution.** Per week  $28 \times 6 \times 85 = 14,280$ ; eight weeks 114,240; net  $114,240 - 60,000$ .
- Result.** Benefit per week **USD 14,280**; eight weeks **USD 114,240**; net **+USD 54,240**. Breakeven at **4.20 weeks**, beyond which acceleration pays.
- Interpretation.** The decision is not "is 60,000 a lot?" but "does it buy more than it costs?", and here it clearly does, provided adoption materialises. Five further readings, and the second is the one that separates a priced decision from a lucky one.

**The breakeven is the sentence to take to the sponsor, not the net.** A net of +54,240 invites an argument about the inputs; a breakeven of 4.20 weeks invites a question that can be answered, *can we really pull in more than 4.20 weeks?* The identity is  $\text{breakeven weeks} = \text{cost} \div \text{cost of delay}$ , which is scale-free: it does not care about the size of the programme, only about the price of a week. The same identity governs Domain 3's gate durations and Domain 15's escalation latency, which is why the cost of delay is registered once and reused rather than re-derived.

**Every figure rests on the 70 % adoption of 1.3.2, and the sensitivity is severe.** Spending 60,000 to arrive sooner at an unadopted system buys nothing. Precisely: the acceleration pays for itself over eight weeks at any adoption above 36.76 %, because 60,000 over 8 weeks needs USD 7,500 a week, which is 14.7059 clinics at USD 510 each. That is comfortably below the planned 70 %, so the decision is robust *in direction*. But it is fragile in *magnitude*: at the adoption Meridian actually achieved, 40 %, the cost of delay is  $16 \times 510 = \text{USD } 8,160$  a week, the breakeven stretches to 7.3529 weeks of the 8 available, and the net collapses from USD 54,240 to USD 5,280: a 90.3 % reduction, from a variable nobody in the meeting was watching. **A decision can be correct and still have almost no margin, and the leader is the person who has to know which.**

**The maximum justified spend follows immediately and is worth stating in the paper.** Eight weeks at 14,280 is USD 114,240, so any compression package priced above that destroys value however attractive it looks; at the measured 40 % adoption the ceiling falls to  $8 \times 8,160 = \text{USD } 65,280$ . Quoting the ceiling rather than only the recommendation is what stops a 60,000 proposal becoming a 120,000 proposal during procurement without anyone re-testing it.

**Sequencing engagement before acceleration is what the numbers say, not a preference.** Adoption work raises both the benefit *and* the value of every subsequent timing decision, because the cost of delay is proportional to adoption. Acceleration bought before adoption is secured is a bet on an unverified assumption; adoption secured first makes the acceleration worth more.

**What breaks it.** The eight weeks must be genuinely available on the critical path: compression applied to non-critical work buys nothing at all, which Domain 6 (KA 6.4) computes properly, and a compression estimate produced by the same team that owns the date should be treated as an advocacy figure until tested. The 60,000 must be the *whole* incremental cost, including any overtime premium, additional supervision and the rework that rushed installation tends to generate (Domain 9, KA 9.2.3 prices the rework-capacity effect). And the benefit must actually begin when the output arrives: if go-live is gated by a training window or a licence date, arriving early buys nothing but idle readiness. A reviewer's single best question on any acceleration case is *what starts earning on the earlier date, and who has confirmed it can?*

**The other way to buy time is people, and it does not price the same way.** Acceleration bought with money answered cleanly in the example above. The same objective pursued by adding staff behaves differently, because capacity added mid-delivery arrives with a transient attached, which is 1.3.1's "delay hides consequence" property in its most expensive everyday form. The two decisions look alike to a sponsor and reach opposite verdicts, which is the reason to work both.

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#### WORKED EXAMPLE

#### Worked example 1.3.3b — the reinforcement trough, and when buying time with people is a mistake.

1. **Setup.** Eight clinics remain in Meridian's rollout. The crew of 6 deployment specialists completes 0.1 clinics each per week, so 0.6 a week in total. The programme is late, and the leader can add 3 more specialists at USD 4,200 per specialist-week. Newcomers take a 4-week ramp, during which each works at 25 % of full productivity and absorbs 50 % of an existing specialist's time in supervision. The cost of delay is the USD 14,280 a week derived above. Both the ramp length and the supervision load are **locally calibrated planning figures, not constants**: the same discipline Domain 12 applies to its coordination parameter.
2. **Formula.** Ramp-period rate = base rate – (newcomers × supervision × per-head rate) + (newcomers × ramp productivity × per-head rate). Post-ramp rate = base rate + newcomers × per-head rate. Duration = ramp weeks + (remaining work – ramp-period output) ÷ post-ramp rate. Crossover horizon = ramp weeks × (1 + deficit rate ÷ gain rate). Net = weeks saved × cost of delay – incremental specialist-weeks × rate.
3. **Substitution.** Ramp rate  $0.6 - (3 \times 0.5 \times 0.1) + (3 \times 0.25 \times 0.1) = 0.6 - 0.15 + 0.075$ ; post-ramp  $0.6 + 0.3$ ; baseline duration  $8 / 0.6$ ; new duration  $4 + (8 - 4 \times 0.525) / 0.9$ ; specialist-weeks  $9 \times 10.5556$  against  $6 \times 13.3333$ .
4. **Result.**

|                                 | CREW OF 6        | CREW OF 9          |
|---------------------------------|------------------|--------------------|
| Delivery rate, weeks 1–4        | 0.6 clinics/week | 0.525 clinics/week |
| Delivery rate thereafter        | 0.6              | 0.9                |
| Cumulative at week 4            | 2.4 clinics      | 2.1 clinics        |
| Duration for the last 8 clinics | 13.3333 weeks    | 10.5556 weeks      |
| Specialist-weeks consumed       | 80.0             | 95.0               |

Weeks saved 2.7778, worth USD 39,666.67 at the cost of delay. Incremental effort 15.0 specialist-weeks, costing USD 63,000. **Net (USD 23,333.33)**: the reinforcement loses money. The crossover horizon is  $4 \times (1 + 0.075/0.3) =$

**5.00 weeks**, and at week 4 the reinforced crew is **0.3 clinics: 12.5 %, behind** where the original crew would have been. 5. **Interpretation.** Four results, of which the third is the one that decides the case and the fourth is the one that keeps the model honest.

**The trough is real, measurable and misread.** For four weeks every progress metric says the decision was wrong, by 12.5 % at its worst. A leader who reverses in the trough pays the whole supervision cost, receives none of the added capacity, and ends later than either pure option: the single most expensive available move. The counter-measure is procedural, not stoical: **state the crossover week when the decision is taken**, and put it in the record, so the trough is a forecast being met rather than a surprise being explained.

**The crossover horizon is computable in advance, and it is the decision rule.**  $R \times (1 + d/g)$  needs only the ramp length, the net deficit rate during the ramp and the gain rate after it: three figures any team can state. Here it is 5.00 weeks: with more than five weeks of work left, reinforcement finishes earlier; with less, it finishes later, and no amount of commitment changes that. The identity also tells you which lever matters. Halving the ramp to two weeks moves the crossover to **2.50 weeks** — a gain of 2.50; halving the supervision load instead removes the deficit altogether ( $d = 0$ ) and moves the crossover to **4.00 weeks**, the ramp length itself — a gain of 1.00. **Shortening the ramp is worth two and a half times as much as easing supervision**, which is an argument for prepared onboarding material rather than for asking mentors to give newcomers less of their time.

**Being earlier is not the same as being better off.** The 2.7778 weeks saved are worth 39,666.67; the 15 extra specialist-weeks cost 63,000; the reinforcement therefore destroys **USD 23,333.33** of value while genuinely delivering sooner. It becomes worthwhile only above a cost of delay of **USD 22,680 a week**. And Meridian's maximum conceivable cost of delay, at 100 % adoption, is  $40 \times 6 \times 85 = \text{USD } 20,400$ , which is **11.18 % short** of the breakeven. **No adoption level makes this reinforcement pay on delay grounds.** So if the crew is added anyway, it must be for a reason that is stated and different (a statutory date, a contractual liquidated sum, the crew being needed for the next tranche regardless), and that reason belongs in the decision record (1.2.3). The alternative lever is the one that is cheap: the same 2.7778 weeks bought by shortening the ramp costs nothing like 63,000. The breakeven specialist-week rate is **USD 2,644.44**; at Meridian's 4,200 the arithmetic simply does not clear.

**What breaks the model.** It assumes the remaining work is divisible and rate-limited by specialist capacity, not by a fixed clinic-by-clinic sequence, a licence window or a single scarce approver, any of which makes added capacity worth nothing at all (Domain 6, KA 6.A.1 on the constraint). It assumes the newcomers are competent in the work and merely unfamiliar with this programme; recruiting into an unfamiliar *discipline* has a longer ramp and a larger supervision load, both of which push the crossover out. And it prices only duration and

effort: quality consequences of a rushed ramp sit in Domain 9's containment economics, and morale consequences in Domain 12. The invariant worth remembering, because it is parameter-free: **a trough exists precisely when the supervision load a newcomer imposes exceeds the newcomer's own ramp productivity**. At 50 % supervision against 25 % productivity there is a trough; at 25 % supervision against 50 % productivity the reinforced rate is 0.675 from day one and there is none. A leader who knows only that inequality already knows whether to expect a dip.

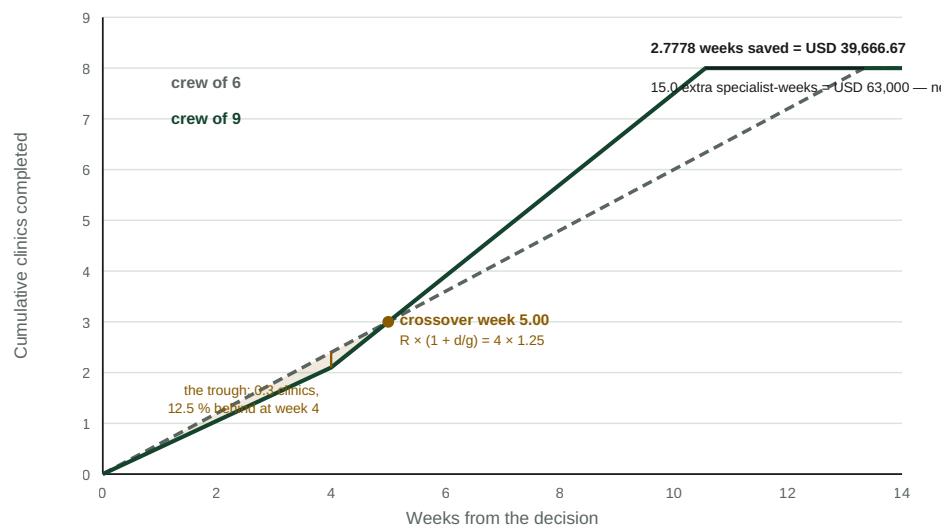


Fig 1.3.3 — The reinforcement trough and its crossover

**Act reversibly where you can.** Prefer decisions that can be unwound, pilots over commitments, and options kept open at low cost, then commit hard once uncertainty has genuinely reduced. This is the leadership behaviour behind rolling-wave planning (Domain 6, KA 6.3.3) and staged investment (Domain 2).

### AI in this KA

Systems and value reasoning is where AI assistance is *weakest* and most confidently wrong: a model will produce a fluent benefits case that silently treats outputs as benefits, because that is what most documents in its training data do. The governed use is narrow and real (stress-test a benefits chain by asking what would have to be true, generate adoption scenarios, surface second-order effects a team has not considered) followed by human judgment on which link is actually fragile. The check that catches the common failure is one question: *for each claimed benefit, which measured outcome produces it, and who owns that measure?* (Domain 16 turns that into the benefits register.)



## KEY TERMS — KA 1.3

| TERM                                      | MEANING                                                                                                                                                                                                                     |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Output / outcome / benefit / value</b> | Delivered thing · change in how things work · measurable improvement · benefit against cost and risk.                                                                                                                       |
| <b>Adoption</b>                           | The outcome measure linking an output to any benefit at all.                                                                                                                                                                |
| <b>Feedback and delay</b>                 | Systems properties that make interventions look wrong before they look right.                                                                                                                                               |
| <b>Local optimisation</b>                 | Function-level efficiency that degrades whole-system performance.                                                                                                                                                           |
| <b>Last responsible moment</b>            | The latest point a decision can be taken without foreclosing a needed option.                                                                                                                                               |
| <b>Cost of delay</b>                      | The benefit forgone per period of lateness.                                                                                                                                                                                 |
| <b>Simple payback</b>                     | Cost ÷ annual benefit, in periods; scales as the reciprocal of adoption.                                                                                                                                                    |
| <b>Reinforcement trough</b>               | The transient period after adding people in which progress is worse than it would have been; exists when supervision load per newcomer exceeds their ramp productivity.                                                     |
| <b>Crossover horizon</b>                  | $R \times (1 + d/g)$ : the remaining work below which reinforcement finishes later, not earlier.                                                                                                                            |
| <b>Released hours</b>                     | Capacity freed, not cash; a benefit only once redeployed to a stated use.                                                                                                                                                   |
| <b>Decision breakeven adoption</b>        | The adoption at which one priced decision stops paying, 36.76 % for Meridian's acceleration. Distinct from Domain 2's <b>breakeven adoption</b> , which is the level at which the whole investment's NPV is zero (41.05 %). |

## SAMPLE MCQS — KA 1.3

**MCQ 1.3-A** [1.3.2 · Application] 40 clinics installed; 70 % adoption; 6 hours/week released per adopting clinic at USD 85/hour over 48 weeks. The defensible annual benefit is:

- A. USD 979,200
- B. USD 685,440 CORRECT
- C. USD 293,760
- D. USD 14,280

*Rationale:*  $40 \times 0.70 \times 6 \times 85 \times 48 = 685,440$ . A is the output-based claim that skips adoption; C is the overstatement itself; D is the weekly benefit.

**MCQ 1.3-B** [1.3.2 · Analysis] A programme board is told "40 clinics live — benefits delivered". The soundest challenge is:

- A. ask for the installation evidence

- B. ask which measured outcome produces the benefit and who owns that measure, since installation is an output and adoption is unverified CORRECT
- C. accept it, since the output target was met
- D. ask for the cost variance instead

*Rationale:* The claim skips the outcome link, which is precisely where 30 % of the value leaked in 1.3.2. A verifies the wrong thing, C repeats the error, D changes the subject.

**MCQ 1.3-C** [1.3.3 · Application] Benefit accrues at USD 14,280 per week once adopted. Compressing the rollout by 8 weeks costs USD 60,000. The decision and its breakeven are:

- A. reject — 60,000 exceeds the weekly benefit
- B. accept — net +USD 54,240, breakeven at 4.20 weeks CORRECT
- C. accept — net +USD 114,240
- D. indifferent — the two are equal

*Rationale:*  $8 \times 14,280 = 114,240$ , less the 60,000 cost, nets **+54,240**; breakeven  $60,000/14,280 = 4.20$  weeks. A compares a total against a weekly rate; C forgets to deduct the cost; D asserts an equality the arithmetic denies.

**MCQ 1.3-D** [1.3.1 · Analysis] A discipline lead reports having "absorbed" a two-week schedule squeeze with no impact. The systems-literate response is:

- A. record the recovery and move on
- B. ask where the pressure went — into quality, scope, people or suppliers — because pressure moves rather than vanishing CORRECT
- C. squeeze the remaining teams equally
- D. treat it as evidence the original estimate was padded

*Rationale:* Pressure relocates (1.3.1); the leader's job is to find where. A accepts an unverified claim, C compounds it, D leaps to a conclusion the report does not support.

**MCQ 1.3-E** [1.3.3 · Evaluation] Reinforcing a crew saves 2.7778 weeks at a cost of delay of USD 14,280 a week and consumes 15.0 extra specialist-weeks at USD 4,200 each. The programme's cost of delay could not exceed USD 20,400 even at 100 % adoption. The strongest professional conclusion is:

- A. approve — the work finishes 2.7778 weeks earlier, which is the objective
- B. reject on delay grounds: the breakeven cost of delay is USD 22,680 a week, above the programme's maximum conceivable 20,400, so no adoption level makes it pay and any approval needs a different stated reason CORRECT
- C. approve — 39,666.67 of delay value exceeds the 23,333.33 net loss
- D. reject — adding people always makes a late project later

*Rationale:*  $63,000 / 2.7778 = 22,680$ , which exceeds the 20,400 ceiling (1.3.3b). A optimises earliness rather than value; C compares the gross benefit against the net result, double-counting the benefit; D over-applies the coordination result of Domain 12 as a law, here reinforcement genuinely does finish earlier, it simply costs more than it saves.

**MCQ 1.3-F** [1.3.2 · Application] A cost of USD 2,400,000 against a full-potential benefit of USD 979,200 a year is assessed under a three-year simple payback rule. The adoption the rule requires is:

- A. 70.00 %
- B. 81.70 % CORRECT
- C. 245.10 %
- D. 41.05 %

*Rationale:* The rule needs  $2,400,000/3 = 800,000$  a year, and  $800,000/979,200 = 81.70\%$  (1.3.2b). A is the case's assumption, which is the thing being tested; C divides the cost by the potential and reads the payback in years as a percentage; D is Domain 2's breakeven adoption for value creation, a different question, and 40.65 points lower.

**MCQ 1.3-G** [1.3.3 · Evaluation] Compressing the rollout by 8 weeks costs USD 60,000. Which statement best assesses the decision's robustness?

- A. it is robust: the net is +54,240, comfortably positive
- B. it pays at any adoption above 36.76 %, so the direction is robust, but at the 40 % adoption actually achieved the net falls to 5,280 — robust in direction, fragile in magnitude CORRECT
- C. it is not robust, because it depends on an adoption assumption
- D. robustness cannot be assessed without a discounted appraisal

*Rationale:*  $60,000/8 = 7,500$  a week is 14.7059 clinics, or 36.76 % adoption; at 40 % the weekly benefit is 8,160 and the net is  $8 \times 8,160 - 60,000 = 5,280$  (1.3.3). A quotes the point estimate as though it were the range; C treats any dependency as fatal and so never approves anything; D imports Domain 2's instrument into a decision whose horizon is eight weeks.

#### SELF-CHECK — KA 1.3

1. *Why is an output claimed as a benefit an overstatement by construction?* — Because the outcome link can fail independently; unadopted outputs produce no benefit (Meridian's 30 %).
2. *What is the leader's question on hearing "we absorbed it"?* — Where did the pressure go?

- 
3. *What must be true before paying to accelerate Meridian?* — That adoption materialises; arriving sooner at an unadopted system buys nothing.
  4. *By how much does an output-based benefit claim overstate, and what does the answer depend on?* — By exactly the non-adoption rate,  $1 - a$ ; it depends on nothing else, not the hours, the rate or the length of the operating year (1.3.2).
  5. *When does adding people to late work finish later rather than earlier?* — Below the crossover horizon  $R \times (1 + d/g)$ ; and a trough exists at all only when supervision load per newcomer exceeds their ramp productivity (1.3.3b).
  6. *What does a three-year payback rule commit an organisation to, on Meridian's figures?* — 81.70 % adoption, a commitment expressed in years and never discussed in percentage points (1.3.2b).
  7. *Are released clinician-hours a benefit?* — Not yet: they are capacity, and become a cash saving or a service improvement only when redeployed to a stated use with an owner (1.3.2).
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## KNOWLEDGE AREA 1.4

## Ethics and the responsible use of AI

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*Topics: 1.4.1 professional ethics in delivery · 1.4.2 the responsible-AI principle applied · 1.4.3 the leader's AI accountability.*

### 1.4.1 Professional ethics in delivery

Ethics in this discipline is rarely dramatic; it is a series of small reporting and allocation decisions taken under pressure. Four recurring tests:

- **Report what is, not what is wanted.** Presenting an optimistic case as the base case is a misrepresentation whatever the spreadsheet says (the same rule PFL-AI applies to forecasts).
- **Do not spend other people's margins silently.** Absorbing pressure into team overtime, a supplier's contingency or a safety margin is a decision someone is entitled to be told about.
- **Claim only what is true.** Milestone and benefit claims are statements of fact to people relying on them (1.3.2).
- **Compete and procure honestly.** Conflicts declared, evaluations run as published, suppliers not used as free consultancies (Domain 10's ethical sourcing).

**Conflicts of interest** are handled by disclosure and separation: declared before engagement, managed or declined, and tested by daylight, *would every party, seeing the full picture, still regard this as impartial?* Where the answer wavers, impartiality

has already failed. The operating instrument is 1.2.2a's four steps (identify, declare, record, abstain), and the daylight test is how the first of them is answered honestly.

### 1.4.2 The responsible-AI principle applied

The principle from KA 1.1 becomes operational through four obligations that recur in every later domain:

| OBLIGATION             | WHAT IT REQUIRES OF A LEADER                                                        |
|------------------------|-------------------------------------------------------------------------------------|
| <b>Appropriate use</b> | Match the tool to the task; prohibit it where the stakes or data forbid (Domain 14) |
| <b>Verification</b>    | Machine output checked against evidence by a named human before reliance            |
| <b>Transparency</b>    | Material AI use disclosed in deliverables and decisions                             |
| <b>Human decision</b>  | The judgment, and its record, belongs to a person                                   |

The failure modes to expect, stated honestly: **hallucination** (fluent, false specifics — the invented precedent, the plausible citation); **silent staleness** (confident answers from out-of-date data); **bias** inherited from training data, especially in anything touching people; **confidentiality leakage** (project data entering a tool is a disclosure); and **over-trust through fluency**: the deepest one, because polished output suppresses the scrutiny that rough output invites. None of these is a reason to refuse the technology; each is a reason the verification duty is not optional.

### 1.4.3 The leader's AI accountability

Three concrete leadership acts, which Domain 14 then systematises:

**Name the owner.** Every AI-assisted artefact that informs a decision has a human owner who verified it. Anonymous machine output has no place in a governed decision record (Domain 3, KA 3.3.4). The signature carries the duties of 1.2.2a with it: a verifier who is not competent to check the artefact has certified nothing, and one who is verifying work whose outcome they have an interest in has certified less than nothing, because the record now says the check was independent.

**Make verification proportionate and real.** A summary for internal reading needs a glance; a forecast in a board paper needs recomputation and a source trace; a safety-relevant analysis needs independent review. Uniform verification is either wasteful or negligent, usually both, in different places. "Proportionate" is only a defensible word if it can be derived, so the example below derives it.

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#### WORKED EXAMPLE

### Worked example 1.4.3 – how deep is proportionate, and what actually decides it.

- Setup.** Three AI-assisted artefacts on Meridian. A weekly internal **status summary**: a glance costs 0.5 hours at USD 85 (**USD 42.50**); a downstream control (the delivery meeting, where four people who know the work read it aloud) catches a material error with probability **0.90**; an uncaught error costs about **USD 5,000** in a week of misdirected attention. A **board benefits figure**: recomputation plus a source trace costs 6 hours at USD 85 (**USD 510**); once a paper is tabled the chance of anyone catching the error is **0.25**; an uncaught error carries the output-based overstatement of 1.3.2, **USD 293,760** a year. A **clinical-safety impact assessment** for the prescribing module: independent clinical review costs **USD 12,000**; the chance of another control catching the error is **0.05**; the modelled consequence is **USD 4,000,000**.
- Formula.** Verification is worth performing when its cost is below the expected loss it removes:  $C_v \leq P(\text{error}) \times (1 - q) \times L$ , where  $q$  is the probability a downstream control catches the error anyway and  $L$  the loss if it flows through. Rearranged, the **breakeven error probability** is  $P^* = C_v \div [(1 - q) \times L]$ : the error rate above which the check pays.
- Substitution.** Summary  $42.50 / (0.10 \times 5,000)$ ; board figure  $510 / (0.75 \times 293,760)$ ; safety assessment  $12,000 / (0.95 \times 4,000,000)$ .
- Result.**

| ARTEFACT                   | VERIFICATION |      | LOSS L    | EXPOSED        | BREAKEVEN<br>P* |
|----------------------------|--------------|------|-----------|----------------|-----------------|
|                            | C_v          | Q    |           | LOSS<br>(1-Q)L |                 |
| Weekly status summary      | 42.50        | 0.90 | 5,000     | 500.00         | <b>8.5000 %</b> |
| Board benefits figure      | 510          | 0.25 | 293,760   | 220,320        | <b>0.2315 %</b> |
| Clinical-safety assessment | 12,000       | 0.05 | 4,000,000 | 3,800,000      | <b>0.3158 %</b> |

- Interpretation.** Five readings. The third is the counter-intuitive one and the fifth is the limit of the method.

**The breakevens differ by a factor of 36.72, and that is what "proportionate" means.** A glance at the summary is justified only if such summaries are wrong more than **8.5 %** of the time, which, for AI-drafted narrative over live data, they plainly are, so a glance is warranted and a glance is all that is warranted. The board figure must be verified if the error rate exceeds **0.2315 %**, which is to say: always. Proportionality is not a sliding scale of

diligence applied by feel; it is the observation that the same duty produces very different depths once the exposure is written down.

**Uniform depth is arithmetically absurd, not merely wasteful.** Apply the board paper's six-hour recompute to the weekly summary and the breakeven becomes  $510 / 500 = 1.02$ : a required error probability **above 1**, meaning **no error rate whatever could justify it**. That is a sharper statement than "wasteful": it is a check that cannot be justified on any assumption. Meanwhile the same six hours applied to the safety assessment would not merely be too shallow, it would be the wrong *kind* of check: recomputation cannot find a clinical misjudgement. Depth and *kind* are separate choices, and conflating them is how organisations produce thick verification files that miss the thing that matters.

**The cost of checking barely matters; the exposed loss decides everything.** The safety review costs **23.53 times** the board recompute, yet its breakeven is only **1.36 times** higher. The two are effectively in the same band, while the trivial artefact sits two orders of magnitude away. The discriminating variable is  $(1 - q) \times L$ , not  $C_v$ . The practical consequence is that the familiar objection ("we cannot afford to check everything properly") is answering the wrong question: on anything with material downstream reliance, the check is cheap relative to the exposure, and the real constraint is attention, not money.

**q is where this arithmetic is most often abused.** A high  $q$  makes any check unnecessary, so a  $q$  is asserted rather than evidenced far more often than an  $L$  is. A reviewer's question is therefore not "what is your verification policy?" but *which named control would have caught this, and when has it caught anything?* On the summary,  $q = 0.90$  is credible only because the delivery meeting genuinely reads it; remove that meeting and the breakeven falls to  $42.50 / 5,000 = 0.8500\%$ , a tenfold increase in the case for checking, with no change to the artefact at all.

**The limit, stated plainly: row three should not be decided this way.** Where a duty arising under law or regulation, an operating licence or patient safety is engaged, the obligation to review is not contingent on an expected-value test, and computing  $P^*$  there has exactly one legitimate use: to show that the review is *also* cheap, which forecloses the cost argument. It must never be used to conclude that a safety review is not worth performing. What constitutes an adequate assessment, who is competent to sign it, and what must be retained are set by the applicable regulatory regime and the relevant professional body in the jurisdiction; take that from the regulator and from counsel, not from this table (Domain 14, KA 14.4 for the governance architecture, and Domain 9 for the assurance regime). Two further limits: expected-value reasoning is the wrong frame for a single catastrophic tail even where safety is not engaged, and this model prices one artefact at a time, whereas AI-assisted work arrives in volume — sizing a *sample* across many artefacts is Domain 14's problem (KA 14.3), and it starts where this example stops.

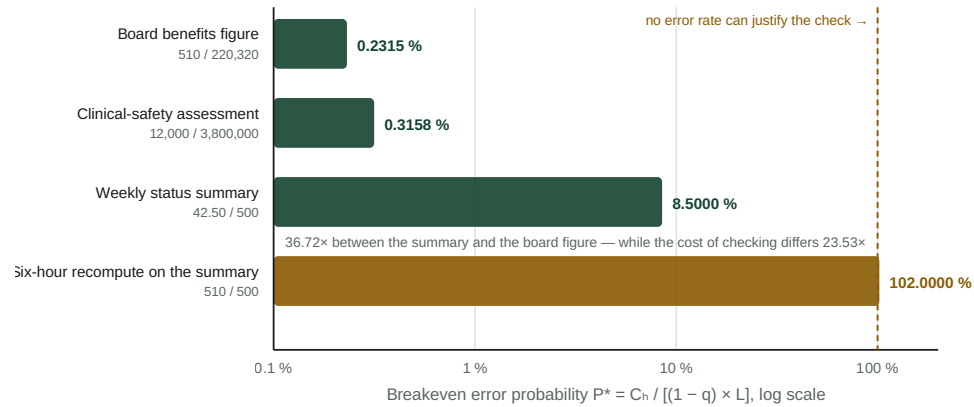


Fig 1.4.1 — Proportionate verification: the breakeven error probability

**Protect the team's judgment.** The subtler risk is not a wrong output but the atrophy of the skills needed to notice one. A leader who accepts AI-drafted plans no one on the team can critique has traded capability for speed, and will discover the price during the first recovery (Domain 6, KA 6.4).

#### KEY TERMS — KA 1.4

| TERM                                                  | MEANING                                                                                                    |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>Daylight test</b>                                  | Would every party, seeing the full picture, still regard this as impartial?                                |
| <b>Hallucination</b>                                  | Fluent, confidently-stated content that is false.                                                          |
| <b>Over-trust through fluency</b>                     | Polished output suppressing the scrutiny rough output would attract.                                       |
| <b>Verification proportionality</b>                   | Depth of checking matched to the stakes of reliance.                                                       |
| <b>Breakeven error probability (<math>P^*</math>)</b> | $C_v \div [(1 - q) \times L]$ : the error rate above which a verification pays for itself.                 |
| <b>Exposed loss</b>                                   | $(1 - q) \times L$ : the loss a verification is actually protecting against, after any downstream control. |
| <b>Depth and kind</b>                                 | Two separate verification choices: how hard to look, and what sort of looking finds this class of error.   |
| <b>Named owner</b>                                    | The human who verified an AI-assisted artefact and answers for it.                                         |



## SAMPLE MCQS — KA 1.4

**MCQ 1.4-A** [1.4.1 · Analysis] A sponsor asks that the optimistic scenario be presented as the base case "to keep the board confident". The professional response is:

- A. comply — scenario labels are presentational
- B. decline, and offer the honest base case with sensitivities and the evidence that supports confidence CORRECT
- C. comply while noting the true base case privately
- D. present both without indicating which is the base case

*Rationale:* Candour about status is a duty, not a style (1.4.1). C documents the misrepresentation without preventing it; D abdicates the professional judgment the board is relying on.

**MCQ 1.4-B** [1.4.2 · Application] Which AI failure mode most directly explains an authoritative-sounding schedule that encodes assumptions nobody made?

- A. confidentiality leakage
- B. over-trust through fluency, compounded by absent verification CORRECT
- C. model bias
- D. hardware limitation

*Rationale:* Polished output invites less scrutiny (1.4.2), and the missing control is verification. A concerns data egress, C concerns systematic skew (real, but not this symptom), D is irrelevant.

**MCQ 1.4-C** [1.4.3 · Analysis] A leader requires the same verification depth for every AI-assisted artefact, from internal summaries to safety analyses. This is:

- A. best practice — consistency is the point
- B. simultaneously wasteful and negligent: proportionality means depth matched to the stakes of reliance CORRECT
- C. acceptable only if the team is small
- D. unnecessary, since approved tools need no verification

*Rationale:* Uniform depth over-checks the trivial and under-checks the critical (1.4.3). A mistakes uniformity for rigour; D abandons the duty entirely.

**MCQ 1.4-D** [1.4.3 · Application] Recomputing an AI-drafted board benefits figure costs USD 510. An error surviving into the paper would carry an overstatement of USD 293,760, and the chance of anyone catching it after tabling is 0.25. The breakeven error probability above which the recompute pays is:

- A. 0.1736 %

B. 0.2315 % CORRECT

C. 0.6944 %

D. 0.1302 %

*Rationale:*  $510 / (0.75 \times 293,760) = 0.2315 \% (1.4.3)$ . A divides by the full loss and forgets that only 75 % of it is exposed; C multiplies by 0.25 instead of  $(1 - 0.25)$ , using the catch probability as if it were the escape probability; D applies the escape fraction to the verification cost  $(510 \times 0.75)$  rather than to the loss, which is the same slip in the opposite direction.

**MCQ 1.4-E** [1.4.3 · Evaluation] A safety review costing USD 12,000 has a breakeven error probability of 0.3158 %, while a USD 510 recompute has one of 0.2315 %. The most useful professional inference is:

A. the cheaper check is better value and should be preferred

B. the cost of checking is nearly irrelevant to whether to check — a 23.53-fold cost difference produces only a 1.36-fold difference in breakeven, because the exposed loss dominates CORRECT

C. both checks are unnecessary, since the breakevens are below 1 %

D. the safety review should be dropped, as its expected value is the weaker of the two

*Rationale:*  $P^* = C_v / [(1 - q)L]$ , and the exposed losses differ by more than the costs do (1.4.3). A compares the wrong quantity, the checks protect different exposures; C inverts the meaning of a breakeven, which is a threshold to exceed, not a hurdle to fail; D applies an expected-value test to a safety decision, which the example expressly rules out.

#### SELF-CHECK — KA 1.4

1. *State the daylight test.* — Would every party, seeing the full picture, still regard this as impartial?
2. *Which AI failure mode is hardest to counter, and why?* — Over-trust through fluency: polish suppresses the scrutiny that rough work attracts.
3. *What does "protect the team's judgment" mean in practice?* — Do not accept AI output the team cannot critique; capability traded for speed is repaid during recovery.
4. *State the breakeven error probability and say which of its inputs decides the answer.* —  $P^* = C_v / [(1 - q) \times L]$ ; the exposed loss  $(1 - q)L$  dominates, and the cost of checking barely moves it (1.4.3).
5. *What does a breakeven above 100 % tell you?* — That no error rate could justify that check on that artefact: it is unjustifiable rather than merely uneconomic (1.4.3).

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6. *Where must this arithmetic not be the deciding test?* — Where a duty arising under law or regulation, a licence or safety is engaged; there its only legitimate use is to show the review is also cheap (1.4.3).
- 

## Advanced topics — Domain 1

### 1.A.1 Authority, influence and the borrowed team

Most project leaders hold less authority than accountability, the defining asymmetry of the role. The sources of influence that actually work in temporary organisations: **competence** (visible command of the work), **reciprocity** (having helped others deliver), **information** (being the person with a true picture), **coalition** (sponsor and peer support secured before it is needed), and **fairness** (a record of allocating credit and blame accurately). Positional authority is the weakest of the six in borrowed teams, because the borrowed people's careers are decided elsewhere. The practical implication: influence is *built in advance*, and a leader who begins building it during a crisis is too late.

### 1.A.2 Success criteria and the multiple-verdict problem

Projects are judged simultaneously against delivery (time, cost, quality), outcome and benefit realisation, and stakeholder satisfaction, which is why the same project is often described as a success and a failure by different parties, honestly. The professional response is not to pick a flattering verdict but to **agree the criteria in advance, with owners and measures** (Domain 2's benefits mapping, Domain 5's acceptance criteria), and to report against all of them. Unagreed success criteria are settled retrospectively by whoever is most disappointed.

### 1.A.3 Dissent, compliance and the record that protects everyone

The obligation set of 1.2.2 has a case it does not cover: the leader escalates with options, the accountable authority chooses the option the leader advised against, and the leader must now deliver it. This is common, it is not misconduct by anyone, and handling it badly damages both the project and the leader.

Three positions are available and only one of them is professional. **Silent compliance** protects the relationship and destroys the record: when the risk materialises there is no evidence the question was asked, and the leader who warned privately is indistinguishable from the leader who did not notice. **Obstruction** (delivering the decision unenthusiastically, slowly, or with visible reservation to the team) is a breach of the duty to the sponsor and, because a team reads its leader accurately, converts a difficult decision into a failing one. **Dissent and comply** is the professional position: the disagreement is stated once, in writing, to the person who holds the authority; the reasoning and the condition that would change the answer are

recorded in the decision log (Domain 3, KA 3.3.4); the leader then implements the decision as if it were their own, including to the team; and the recorded condition is monitored, so that if it occurs the matter returns automatically rather than as an accusation.

What makes this work is that the record is **neutral in tone and specific in content**: the option chosen, the option advised, the reason, and the observable that would reopen it; for example, on Meridian, "proceed to full rollout without funded adoption support; the writer's advice was to fund champions first; reopen if adoption at week 13 is below 55 %." That sentence would have converted the 26-week silence of 1.2.2 into an automatic trigger, and it costs one line. A record written to apportion blame in advance ("for the avoidance of doubt, I do not accept responsibility for...") achieves the opposite: it invites the authority to stop asking for advice, and it is usually worth nothing anyway, because accountability is not reallocated by assertion (1.2.1).

Two boundaries. First, dissent and comply applies to decisions within the range of legitimate judgment. It does **not** apply where following the instruction would require the leader to act outside the law as they understand it, where safety or a regulatory obligation would be breached, or where the leader is being asked to make a statement they know to be untrue (1.4.1). Those are refusal-and-escalate situations, and they are the reason a profession has ethics rather than only a service ethic. Second, the protections available to someone who raises such a matter, the routes that count as a protected disclosure, and any obligation to report externally are entirely jurisdiction- and sector-specific, and can turn on details of how and to whom the disclosure was made. Before acting on that class of concern, take advice (from counsel, from the relevant professional body, or from the organisation's designated route) rather than from any general account, including this one.

#### 1.A.4 The reviewer's leadership eye

Invariants an experienced reviewer tests in the first hour on any project: exactly one accountable name per outcome; a benefits chain where each claimed benefit traces to a measured outcome with an owner; decision records that show what was known when; escalations that arrived early enough to matter; forecasts with ranges rather than single numbers; and AI-assisted artefacts carrying a named verifier. Every one of these is cheap to check and expensive to have missing: the same posture Domains 6 and 7 apply to schedules and cost.

Four questions convert that list into arithmetic in the same hour, and each has a one-line answer or a finding. *Does the business case contain an adoption term?*; if not, the overstatement is already known to be the non-adoption rate (1.3.2). *What operational condition does the approval hurdle imply?*: a payback rule restated as an adoption percentage often turns out to be a commitment nobody made (1.3.2b). *What is the cost of delay per week, and who computed it?*: a project that cannot state it cannot price a single timing decision, including its own escalations (1.3.3). *For the last AI-*

*assisted artefact that mattered, what was the exposed loss and which named control was relied on?* The answer distinguishes a verification regime from a verification policy (1.4.3). A reviewer who asks those four learns more in ten minutes than a document review yields in a day, because each of them is answerable only by someone who has actually done the arithmetic.

## Industry variations – Domain 1

- **Public programmes.** Accountability is doubled (professional and political), and the public obligation of 1.2.2 is enforceable through scrutiny bodies; benefits (not outputs) are the currency of accountability, and announced dates often precede plans (Domain 6's Case B).
- **Regulated industries (pharma, aviation, nuclear).** The standard of care is codified and auditable; the "care exercised" defence of 1.2.3 is a documented reality rather than an argument, and AI use faces explicit validation expectations.
- **Construction and engineering.** Authority is contractual as much as organisational; influence runs through commercial relationships, and the leader's obligation set includes site safety as a first-order duty.
- **Technology and product.** Outcome ownership frequently sits with a product role rather than the delivery leader, making the 1.3.2 chain a shared accountability that must be explicitly split: the commonest source of "who owns adoption?" disputes.
- **Professional services and consulting.** Duties to the client organisation and to the end-users can diverge, and independence (1.4.1) becomes a standing rather than occasional test.

## Case study – Domain 1: Meridian under scrutiny (public health)

**Situation.** Eighteen months in, Meridian Care Records reports 40 of 40 clinics installed, on time, 3 % under budget. The programme is publicly praised. Nine months later a health-service audit finds clinician hours released are running at roughly a third of the business case, and the programme is publicly described as a failure. Both accounts cite true facts.

**Analysis.** The delivery verdict was sound: the outputs landed. The benefits case had been written against **outputs** (all 40 clinics, no adoption term), so it claimed USD 979,200 per year where the honest figure at 70 % adoption was USD 685,440, and actual adoption reached only about 40 %, giving roughly USD 391,680. Two failures compounded: an arithmetic one (skipping the outcome link, 1.3.2) and an accountability one. **Nobody owned adoption.** The delivery leader owned installa-

tions, the clinical directorate owned practice change, and no single name owned the measure that connected them (1.2.1).

**The account in numbers.** Both public verdicts were arithmetically defensible, which is the whole difficulty.

| LINE                                                         | FIGURE                                                      | WHERE IT COMES FROM |
|--------------------------------------------------------------|-------------------------------------------------------------|---------------------|
| Approved cost                                                | USD 2,400,000                                               | The business case   |
| Delivery verdict (3 % underspend at installation)            | <b>+USD 72,000</b> , once                                   | WE 1.1.1            |
| Benefit claimed in the case (output basis, no adoption term) | USD 979,200 a year                                          | 1.3.2               |
| Honest benefit at the case's own 70 % adoption               | USD 685,440 a year                                          | 1.3.2               |
| Benefit at the 40 % adoption actually measured               | USD 391,680 a year                                          | WE 1.1.1            |
| Programme verdict (annual shortfall against the honest case) | <b>(USD 293,760) a year</b>                                 | WE 1.1.1            |
| Scale of the two verdicts                                    | shortfall = <b>4.08</b> × the underspend, in year one alone | WE 1.1.1            |
| Weekly form of the shortfall                                 | <b>USD 6,120 a week</b> = 12 clinics × 6 h × USD 85         | WE 1.2.2            |
| Simple payback, honest benefit against approved cost         | <b>3.5014 years</b>                                         | WE 1.3.2b           |
| Simple payback at the measured 40 % adoption                 | <b>6.1275 years</b>                                         | WE 1.3.2b           |

Two figures in that table explain the collapse of the public narrative. The programme was praised on a line worth 72,000 once and condemned on a line worth 293,760 a year, and **nobody had put the two in the same table**, which is a reporting failure, not a delivery one. And the payback the board believed it had approved, 2.4510 years on the output-based claim, became 6.1275 years on measured adoption: the same programme, the same installations, an investment case two and a half times longer.

**What was done.** The recovery was not technical. Adoption was made a named accountability with a monthly measure; the benefits case was restated with an adoption term and a sensitivity range (50/70/90 % → 489,600 / 685,440 / 881,280); clinical champions were funded in the eleven lowest-adoption clinics at USD 12,000 each, **USD 132,000** in total; and the programme board's report changed from "clinics live" to "clinics using, and hours released". Adoption reached 68 % within a year,

recovering **11.2** of the 12 missing clinics — **USD 5,712 a week**, or **USD 274,176** a year.

**What the timing cost.** The monthly measure had shown 16 of 40 clinics in daily use at **week 13**; the matter reached the board at **week 39**, through the audit. On WE 1.2.2's arithmetic those 26 weeks were worth **USD 148,512: 1.13 times** the entire remedy they would have funded, and more than twice the 72,000 underspend the programme had been congratulated for. The remedy needed only **23.11 weeks** of earliness to pay for itself outright.

**And the underspend did not survive.** Domain 16's closing account records a final capital outturn of **USD 2,514,000** against the **USD 2,328,000** implied by the 3 % underspend, a **USD 186,000** swing, most of it the recovery described above. The celebrated cost result was an interim figure that the programme later took back, which is the general lesson: **an underspend booked before the outcome is measured is a loan against the recovery, not a saving.**

**What the domain teaches here.** Every element of this domain appears: outputs are not benefits (1.3.2); one accountable name per outcome (1.2.1); success criteria agreed in advance or settled by the disappointed (1.A.2); and the leader's attention belongs where the value actually leaks, which the arithmetic, not instinct, identified.

## Case study B — Domain 1: the plan nobody could critique (financial services)

**Situation.** A platform-migration leader under time pressure used an AI planning assistant to generate the full delivery schedule and dependency network, reviewed it briefly, and issued it. It was detailed, internally consistent and professionally formatted. It also assumed the legacy data extract could run concurrently with the schema freeze: an assumption no one on the team had made, and which the two engineers who would have spotted it never saw.

**What happened.** The conflict surfaced eleven weeks later during rehearsal, costing a seven-week slip and a weekend of reputational damage with the regulator's supervisory team. The post-incident review found no fault in the tool: the schedule was a reasonable inference from an under-specified prompt. The failures were the leader's, no verification against the discipline leads (the 1.1 principle and 1.4.2 obligation), and no named verifier on the artefact (1.4.3). The compounding finding was cultural: the polish of the output had actively discouraged challenge, and two team members later said they had assumed the network "must have been checked".

**What it cost, and what the check would have cost.** This programme's cost of delay was **USD 38,000 a week** (dual-running licence and support on the legacy platform at **USD 12,000**, plus **USD 26,000** of deferred decommissioning saving), so the seven-week slip cost **USD 266,000** before any regulatory consequence. The verification that would have found the conflict was two discipline leads reading the

dependency network for four hours each:  $2 \times 4 \times 145 = \text{USD } 1,160$ . The loss was therefore **229.31 times** the check. Put through KA 1.4.3's test, the breakeven error probability was  $1,160 / 266,000 = 0.4361\%$ : the review paid for itself if an AI-generated dependency network carried one material error more than about four times in a thousand. It is not necessary to know the true rate to make that decision, only to observe that no honest estimate of it is that low.

Two details make this the general case rather than an unlucky one. First, the breakeven sits in the same band as the board-paper and safety-review rows of 1.4.3 (**0.2315 %** and **0.3158 %**), which is what one should expect: wherever an artefact carries material downstream reliance, the check is trivially cheap against the exposure, and the argument that verification is unaffordable does not survive being written down. Second, the eleven weeks between issue and discovery are 1.3.1's delay property doing its work; the seven-week loss was locked in from the day the plan was issued, and the rehearsal that finally revealed it was the only control in the chain, with the schedule freeze already committed by the time it fired.

**What the domain teaches here.** Fluency is not evidence. Accountability did not move to the vendor or the model: it stayed with the person who issued the plan (MCQ 1.2-B is this case in miniature). And the durable fix is structural, not exhortative: a named verifier per artefact, discipline leads walking their own links, a team kept capable of critique; and a rule that no artefact whose exposed loss exceeds a stated figure is issued without a recorded check, because at 229 to 1 the decision is not close enough to require judgment.

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## Executive perspective — Domain 1

What a project leader cannot delegate in this domain:

- **The single accountable name.** For every outcome, including the awkward ones like adoption. Where two names appear, the leader's job is to reduce it to one.
- **The benefits chain's honesty.** Each claimed benefit traced to a measured outcome with an owner. A leader who lets outputs be reported as benefits has authorised the failure of Meridian's first eighteen months.
- **The obligation set, held openly.** Sponsor, team, users, public (and the profession itself, which is the row that says who may not decide) with conflicts escalated with options rather than absorbed in silence.
- **The declaration that precedes the decision.** Interests identified at appointment and at every gate, declared in writing before the item is taken, recorded with nil returns, and answered by abstention with the minute naming who decided instead. A director's practical test is to open the last three gate papers and look for the field; where it is absent, the organisation has no way of knowing whether it has ever had a problem (1.2.2a).



- **Bad news made safe.** The single highest-leverage cultural act available, because it converts irreversible surprises into recoverable problems, and on Meridian it was worth **USD 148,512**, 1.13 times the remedy it would have funded, for the price of one message (1.2.2).
- **The inversion of every hurdle the organisation imposes.** A three-year pay-back rule is an **81.70 %** adoption commitment; a delegation threshold is an escalation volume; a gate is a priced delay. The director's contribution is insisting that hurdles be restated as the operational conditions they imply, because that is the form in which someone can own them (1.3.2b).
- **The AI accountability line.** Named verifiers, proportionate depth, and a team that remains able to critique what the tools produce. Proportionate is a derivable word: the exposed loss  $(1 - q) \times L$  decides the depth, and a check whose breakeven error probability exceeds 100 % is one the organisation should stop performing (1.4.3).

## Calculation exercises – Domain 1

**Exercise 1.1** A programme installs a system in 60 sites; adoption is 65 %; each adopting site releases 5 hours/week at USD 90/hour over 46 weeks. Compute the defensible annual benefit and the overstatement in an output-based claim. *Solution.* Adopting sites  $60 \times 0.65 = 39$ ; benefit  $39 \times 5 \times 90 \times 46 = \text{USD } 807,300$ . Output-based  $60 \times 5 \times 90 \times 46 = \text{USD } 1,242,000$ ; overstatement **USD 434,700 = 35.0 %**, exactly the non-adoption rate. Common error: applying adoption to the *rate* rather than the site count, the same product here, but it conceals which link failed and misdirects the fix.

**Exercise 1.2** Using Meridian's figures (28 adopting clinics, 6 h/week, USD 85/h), compute the cost of a 5-week delay and the maximum justified acceleration spend for 5 weeks. *Solution.* Per week  $28 \times 6 \times 85 = \text{USD } 14,280$ ; five weeks **USD 71,400**, which is also the maximum justified spend, since above it acceleration destroys value. Common error: comparing the spend against annual benefit rather than the benefit for the weeks actually saved.

**Exercise 1.3** Meridian's benefit at 50 %, 70 % and 90 % adoption (40 clinics, 6 h/week, USD 85/h, 48 weeks). What does the spread imply for where the leader spends attention? *Solution.* **USD 489,600 · 685,440 · 881,280**. The 40-point adoption swing moves annual benefit by **USD 391,680**: far more than plausible schedule compression could add (Exercise 1.2 values 5 weeks at 71,400). Attention belongs on adoption (Domain 11), not acceleration.

**Exercise 1.4** A programme is approved at USD 1,800,000 and delivers 2 % under budget. Its full-potential benefit is USD 540,000 a year; the case assumed 75 %

adoption and measurement finds 45 %. Report both verdicts and their scale, and state what may not be done with them. *Solution.* Delivery verdict  $1,800,000 \times 0.02 = \text{+USD } 36,000$ , once. Benefit at plan  $540,000 \times 0.75 = \text{USD } 405,000$ ; at outturn  $540,000 \times 0.45 = \text{USD } 243,000$ ; programme verdict (**USD 162,000 a year**, equivalently  $540,000 \times (0.75 - 0.45)$ ). The shortfall is **4.50 times** the underspend in the first year alone. What may not be done: netting them into a single figure. One is a stock, the other a flow, and a net requires a horizon and a discount rate from the business case. Common error: reporting "the programme is 126,000 down" ( $162,000 - 36,000$ ), which subtracts an annual flow from a one-off variance and produces a number that means nothing.

**Exercise 1.5** An investment of USD 3,150,000 has a full-potential benefit of USD 1,050,000 a year and assumes 60 % adoption. The sponsor applies a four-year simple payback rule. Compute the payback as assumed, and the adoption the rule requires. *Solution.* Benefit  $1,050,000 \times 0.60 = \text{USD } 630,000$ ; payback  $3,150,000 / 630,000 = \text{5.00 years}$ , which fails the rule. The rule needs  $3,150,000 / 4 = \text{USD } 787,500$  a year, so adoption of  $787,500 / 1,050,000 = \text{75.00 \%}$ : fifteen percentage points above the assumption, and the sentence the approval meeting should have heard. Common error: dividing cost by *potential* benefit ( $3,150,000 / 1,050,000 = 3.00$  years) and reporting a pass; that is the output-based payback, understated by the factor  $1/0.60 = 1.6667$ .

**Exercise 1.6** A crew of 5 completes 0.08 units a week each. Two people are added; they take a 3-week ramp at 20 % productivity and each absorbs 60 % of an existing member's time. Compute the crossover horizon and say whether a trough occurs. *Solution.* Supervision loss  $2 \times 0.6 \times 0.08 = 0.096$ ; newcomer output  $2 \times 0.2 \times 0.08 = 0.032$ ; deficit  $d = 0.064$  a week; post-ramp gain  $g = 2 \times 0.08 = 0.16$  a week. Crossover  $R \times (1 + d/g) = 3 \times (1 + 0.4) = \text{4.20 weeks}$ . A trough does occur, because supervision load per newcomer (60 %) exceeds ramp productivity (20 %); with fewer than 4.20 weeks of work remaining the reinforcement finishes later. Common error: comparing capacity before and after (0.40 against 0.56 a week) and concluding the addition helps whenever work remains, which never computes the crossover and so cannot distinguish the two cases.

**Exercise 1.7** Verifying an AI-drafted forecast costs USD 720. An error surviving into the decision would cost USD 250,000, and an existing control catches such errors with probability 0.40. Compute the breakeven error probability, and state what would change it most. *Solution.* Exposed loss  $(1 - 0.40) \times 250,000 = \text{USD } 150,000$ ; breakeven  $720 / 150,000 = \text{0.48 \%}$ . What changes it most is the control: removing it raises the exposed loss to 250,000 and lowers the breakeven to **0.288 %**, while halving the verification cost only moves it to 0.24 %, so evidence for  $q$  matters about as much as the price of checking. Common error: dividing by

the full loss and reporting 0.288 %, which is arithmetically the same as assuming no control exists; the error understates the breakeven and so overstates the case for checking, and it is the mirror of the commoner abuse in the other direction: asserting a high  $q$  for a control nobody has named.

## Practitioner's toolkit — Domain 1

*Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.*

### Toolkit 1.T.1 — Leader's accountability map (one page)

Per outcome: the outcome stated in outcome language (not output language) · **the single accountable name** · responsible parties · the measure that evidences it and its owner · the decision authority and threshold (Domain 3) · review cadence. Rule: if any row has two accountable names or an output in the outcome column, it is not finished.

### Toolkit 1.T.2 — Benefits-chain test (run on any benefits claim)

- ☐ Each claimed benefit names the measured **outcome** that produces it.
- ☐ Each outcome measure has an owner and a reporting cadence.
- ☐ Adoption (or the equivalent behavioural change) appears explicitly, with a value.
- ☐ The claim carries a sensitivity range, not a single number.
- ☐ Output milestones are labelled as milestones, never as benefits.
- ☐ Cost of delay per period is stated, so timing decisions can be priced (1.3.3).

### Toolkit 1.T.3 — AI-use and verification record

Per AI-assisted artefact: tool and approved environment · data classification cleared · prompt/intent summary · **verification performed and its depth** · **named verifier** · disclosure status in the deliverable · date. Standing agenda item, not an archive: the register's purpose is that the next artefact is verified, not that the last one was.

### Toolkit 1.T.4 — Verification-depth card (one per artefact class, not per artefact)

Six fields, filled once for a *class* of artefact and reused: the class · what relies on it · the **loss**  $L$  if a material error flows through · the **named downstream control** and its assumed catch rate  $q$ , with the evidence for that rate · the verification's **cost**  $C_v$  and **kind** (glance · recompute and source trace · independent review of a different discipline) · the resulting **breakeven**  $P^* = C_v / [(1-q)L]$  · the decision, and who verifies.

Three rules that make it usable rather than decorative. **A q with no named control is recorded as zero.** This is the single field that determines whether the card is honest. **A class whose  $P^*$  exceeds 100 % has its check removed, not reduced,** and the removal is recorded, because a check no error rate can justify is consuming attention that a real exposure needs. And **any class where a duty arising under law or regulation, a licence or safety is engaged is marked mandatory – not an expected-value decision,** with the  $P^*$  retained only as evidence that the review is cheap (1.4.3). Review the card when the tooling changes, not when the artefact changes.

## Exam preparation – Domain 1

**The traps.** Treating outputs as benefits (1.3.2: the domain's central trap, and Meridian's whole case study) · saying accountability is "shared" (1.2.1) · assigning accountability to a vendor or a tool (MCQ 1.2-A, 1.2-B) · comparing an acceleration cost against annual rather than saved-period benefit (Exercise 1.2) · reading a delivery success as a programme success (1.1.1) · accepting "we absorbed it" (1.3.1) · uniform AI verification depth (1.4.3) · judging the standard of care by outcome alone (1.2.3) · **treating a declared interest as a licence to decide the item**, when declaration is the second step of four and abstention is the fourth (1.2.2a) · **signing an opinion or assessment outside one's competence because the request arrived as a courtesy** (1.2.2a) · **netting a one-off capital variance against an annual benefit flow** (WE 1.1.1, Exercise 1.4) · **valuing early warning at the size of the problem rather than at the run rate the remedy recovers** (WE 1.2.2: 6,120 instead of 5,712) · **dividing cost by potential rather than realistic benefit when computing payback**, which understates it by exactly  $1/a$  (WE 1.3.2b, Exercise 1.5) · **approving a reinforcement because it finishes earlier, without testing it against the cost of delay** (WE 1.3.3b: earlier and worse off are compatible) · **omitting the downstream control from a breakeven error probability**, or asserting one that has no name (WE 1.4.3, Exercise 1.7).

**The calculations to be able to do under time pressure.** Annual benefit =  $\text{sites} \times \text{adoption} \times \text{hours} \times \text{rate} \times \text{weeks}$ , and the overstatement of an output-based claim as  $1 - a$ . Cost of delay =  $\text{adopting sites} \times \text{hours} \times \text{rate}$ , and breakeven weeks =  $\text{cost} \div \text{cost of delay}$ . Simple payback =  $\text{cost} \div (\text{potential} \times \text{adoption})$ , and the adoption a payback target  $T$  requires =  $\text{cost} \div (T \times \text{potential})$ . Crossover horizon =  $R \times (1 + d/g)$ . Breakeven error probability =  $C_v \div [(1 - q) \times L]$ . Five formulae, all of them one line, and between them they price every decision this domain contains.

### Reflection questions.

1. Take your current project: write the single accountable name for its principal outcome. If you hesitated, or wrote two, what does that predict?

2. For each benefit in your business case, which measured outcome produces it and who owns that measure? What percentage of your claimed value survives the question?
3. Which AI-assisted artefact in your last month of work had no named verifier, and what would have caught it if it had been wrong?

## Domain 1 summary

Project leadership is a distinct profession because temporary work fails distinctively: problems discovered late become irreversible, teams have no accumulated trust to draw on, and lead time is the cheapest resource available and is spent whether used or not. The discipline rests on a precise view of accountability (responsibility is delegable, the obligation to answer is not, one name per outcome) held across five directions at once (sponsor, team, users, public, and the profession itself) with conflicts escalated rather than absorbed, and judged by the care a competent practitioner would exercise rather than by outcome alone. The fifth direction is the one that says who may *not* decide: interests identified, declared in writing before the item is taken, recorded with nil returns and answered by abstention; a structural interest met by moving the role rather than by a silent record; and no signature given on an opinion, estimate or assessment the signatory is not competent to give. Two habits of mind make a leader rather than a coordinator: systems thinking, which expects feedback, delay, local optimisation and pressure that relocates instead of vanishing; and the outputs–outcomes–benefits–value chain, which Meridian turns into arithmetic: 40 clinics installed produce USD 685,440 of annual benefit at 70 % adoption, not the USD 979,200 an output-based claim asserts, and the 30 % gap is the outcome link being skipped, a proportion that is exactly the non-adoption rate whatever the hours or the rate. Against the approved cost of USD 2,400,000 that benefit repays in 3.5014 years, and a sponsor's three-year payback rule is therefore an undisclosed commitment to 81.70 % adoption: the habit of inverting a hurdle into the condition it implies being one of the most useful in the book. Timing decisions are priceable (USD 14,280 per week; acceleration breaking even at 4.20 weeks and paying at any adoption above 36.76 %), delay itself is priceable (a measure left silent for 26 weeks cost USD 148,512, more than the remedy it would have funded), and reinforcement is priceable in both directions: 2.7778 weeks earlier and USD 23,333.33 worse off, with a crossover horizon of 5.00 weeks computable before the decision. Attention belongs where value actually leaks. Around all of it sits professional ethics and the suite's governing principle (**AI proposes; the professional verifies, decides and remains accountable**) with named verifiers, a team kept capable of critique, and a depth of checking that is derived rather than felt: the breakeven error probability  $C_v / [(1-q)L]$  puts a weekly summary at 8.50 % and a board benefits figure at 0.2315 %, a difference of 36.72 times that is the whole content of the word "proportionate". Domain 2 turns strategy into selected work; Domain 3 gives the accountability of this domain its governance machinery.

DOMAIN

# 2

## Strategy, Selection and Business Alignment

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## Why this domain exists

Domain 1 established that a project is judged on outcomes and benefits, not delivered outputs, and proved it arithmetically. That leaves the prior question unanswered: **how does work get chosen at all, and how does anyone know it was worth choosing?** This domain answers it. It sets the strategic context and treats alignment as a test to be re-run rather than a box ticked at approval (KA 2.1), then builds the business case as a decision instrument: the options analysis and appraisal that make it one, and the selection models that rank competing candidates (KA 2.2). Benefits mapping, measurement baselines and the sustainability dimension follow, with the assumption and dependency management on which every forecast rests (KA 2.3). The domain closes on the hardest leadership act in the discipline, **stopping** (KA 2.4). The through-line is that a business case is a *promise about the future made to obtain money*, and the professional question is always whether anyone will be held to it.

**Learning objectives.** After this domain a candidate can: explain how strategy becomes a portfolio and why alignment decays; **compute a strategy-portfolio alignment index and the reallocation distance it implies, on both a mapped and a total-spend denominator, and say which denominator the finding belongs to**; convert an annual supersession hazard into a survival curve, an alignment half-life and an expected misaligned-spend figure, and price the re-test that removes it; describe the components of a decision-grade business case; construct and appraise a genuine options set including the do-nothing baseline; **quantify a counterfactual do-nothing cost and show how omitting it under-states a case, in the opposite direction to the flat-profile error**; build a benefits map from outputs through outcomes to benefits with owners and measures; **build a benefits profile that ramps rather than assuming steady state from day one, and quantify the overstatement a flat assumption produces**; distinguish flat-equivalent from ramped-basis breakeven adoption and convert between them; define measurement baselines, **separate an attributable improvement from a raw one with a comparison cohort, and state the invariant that fixes the over-claim share**; avoid double-counting; **distinguish sustainability and social value as a constraint, as a benefit and as a disclosed claim, and state the four provisions a disclosed claim carries and who approves it**; rank candidates with weighted scoring and under a binding constraint, and state each method's limits; **compute how far a criterion weight must move to reverse a ranking, and which criteria can never decide anything**; derive the marginal value of a unit of the binding constraint by enumeration and show why it is lumpy and non-monotone; manage assumptions and dependencies as live risks, price an assumption register against the NPV it supports and order the tests by exposure per unit of test cost; apply sunk-cost-free reasoning to a continuation decision; **distinguish the forward breakeven at a gate from the**

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**whole-investment breakeven and calibrate a kill criterion against the cost of its two errors; decompose a staged commitment into the price of staging and the value of the abandonment option, and state the probability of bad news at which staging begins to pay; and set kill criteria that make honest gates possible.**

**The master programme.** Meridian Care Records continues from Domain 1: the shared clinical-records rollout to **40 clinics**, whose verified benefit figures (full potential **USD 979,200** per year; **USD 685,440** at the realistic 70 % adoption) are now used to build the business case that should have been written for it.

**Reference points.** The portfolio material of KA 2.1 and the selection material of KA 2.2 have an international reference point worth naming: **ISO 21504**, which addresses guidance on portfolio management, sitting within the **ISO 21500** family Domain 1 introduced. Both are voluntary guidance describing practice (neither is legislation and neither is a certifiable requirement), and neither obliges anyone of itself unless an organisation, a contract or a regulator adopts it. They are named here and not reproduced: no clause, table or checklist from either appears in this volume, the treatment below is this book's own, and a reader who wants either document should obtain the current edition from its publisher. Naming them implies no endorsement in either direction.

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## KNOWLEDGE AREA 2.1

# Strategic context and alignment

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*Topics: 2.1.1 from strategy to portfolio · 2.1.2 drivers, constraints and environment · 2.1.3 alignment as a repeated test.*

## 2.1.1 From strategy to portfolio

Strategy states intent; a portfolio is what an organisation is actually doing about it. The gap between the two is where most strategic failure lives, and it is visible in three symptoms a leader should be able to name: **a portfolio that does not reflect the stated priorities** (the strategy says "digital first", the funded work is 80 % maintenance); **more work than capacity**, so everything is slow and nothing is finished (Domain 15's capacity management); and **no route for stopping**, so the portfolio only ever accretes.

The mechanism that closes the gap is a **portfolio process** with real teeth: candidate work described comparably, ranked against explicit criteria, funded to capacity rather than to appetite, and reviewed at intervals that allow re-decision. Where a project leader sits inside this, the obligation is specific: **describe your work hon-**



**estly enough to be compared**, including its weaknesses, because a portfolio process fed by advocacy rather than evidence selects the best storyteller.

The first symptom is the only one that can be measured directly from the ledger, and measuring it is what converts a complaint into a governance paper.

#### WORKED EXAMPLE

#### Worked example 2.1.1 – the alignment index, and the money it says must move.

1. **Setup.** Meridian's parent health authority publishes four strategic objectives with declared weights (**access and capacity 40 %**, **safety and quality 25 %**, **digital foundation 20 %**, **cost efficiency 15 %**), and a discretionary programme budget of **USD 30,000,000** a year. The funded portfolio, read off the finance system and mapped objective by objective, is: access **4,800,000**; safety **6,600,000**; digital **1,800,000**; cost efficiency **4,800,000**; and **12,000,000** of mandatory sustainment (estate compliance, licence renewals, end-of-life replacement) that maps to no strategic objective at all. Compute how aligned this portfolio is, and what the answer implies in money.
2. **Formula.** For each objective,  $\text{funded share} = \text{spend} \div \text{denominator}$ . The **alignment index** is the overlap between the declared and funded distributions,  $\text{index} = \sum \min(\text{declared weight}_i, \text{funded share}_i)$ . The **reallocation distance** is the money that must move to close the gap,  $\sum \text{over under-funded objectives of } (\text{declared} - \text{funded}) \times \text{denominator}$ , which is identically  $(1 - \text{index}) \times \text{denominator}$ .
3. **Substitution.** On the strategically mapped **18,000,000**: shares  $4,800,000/18,000,000 = 26.6667 \%$ ,  $6,600,000/18,000,000 = 36.6667 \%$ ,  $1,800,000/18,000,000 = 10.0000 \%$ ,  $4,800,000/18,000,000 = 26.6667 \%$ . Overlap  $\min(40, 26.6667) + \min(25, 36.6667) + \min(20, 10) + \min(15, 26.6667)$ . On the full **30,000,000**, shares are 16 %, 22 %, 6 %, 16 % and 40 % unmapped, and the declared weight on sustainment is zero.
4. **Result.**

| OBJECTIVE           | DECLARED WEIGHT | FUNDED SHARE OF MAPPED SPEND | GAP          | MONEY       |
|---------------------|-----------------|------------------------------|--------------|-------------|
| Access and capacity | 40.0000 %       | 26.6667 %                    | -13.3333 pts | (2,400,000) |
| Safety and quality  | 25.0000 %       | 36.6667 %                    | +11.6667 pts | +2,100,000  |
| Digital foundation  | 20.0000 %       | 10.0000 %                    | -10.0000 pts | (1,800,000) |

| OBJECTIVE       | DECLARED WEIGHT | FUNDED SHARE OF MAPPED SPEND | GAP          | MONEY      |
|-----------------|-----------------|------------------------------|--------------|------------|
| Cost efficiency | 15.0000 %       | 26.6667 %                    | +11.6667 pts | +2,100,000 |

**Alignment index on mapped spend 76.6667 %; on total discretionary spend 59.0000 %.** Reallocation distance **USD 4,200,000**; and the deficits (2,400,000 + 1,800,000) equal the surpluses (2,100,000 + 2,100,000) exactly, as they must. 5. **Interpretation.** Four readings, and the second is the one that decides whether this paper survives its first meeting.

**The number is not the finding; the denominator is.** The two indices differ by **17.6667 percentage points**, and the entire difference is the **40.0000 %** of discretionary money that serves no stated objective. Neither figure is wrong and neither is sufficient. Reported on mapped spend alone, the portfolio looks broadly aligned and the sustainment question never gets asked. Reported on total spend alone, the authority appears to be 41 % misaligned, which is unfair — sustainment is unstated strategy, not misalignment. The professional report gives both, and its recommendation is about the third figure: **either fund sustainment as a declared objective with a weight, or stop calling the other 60 % "the portfolio"**. An organisation that will not put a weight on keeping the lights on has no portfolio process; it has a discretionary fund and a large unexamined subsidy.

**The reallocation distance is the sentence that makes this actionable, and it has a scale a board already understands.** USD 4,200,000 is **1.75 times** Meridian's entire approved cost of 2,400,000; and the access deficit alone, 2,400,000, is *exactly* one Meridian. That is the comparison to put in the paper, because "the portfolio is 76.6667 % aligned" invites a debate about the measure while "the access objective is short by one programme the size of Meridian" invites a decision. The digital objective is funded at **50.00 %** of its declared weight, which is the arithmetic behind the classic symptom named above: the strategy says digital first and the money says otherwise, by a factor of two.

**The identity is worth holding.** Because the declared weights and the funded shares both sum to one, the positive gaps and the negative gaps are equal in magnitude, so  $\text{reallocation distance} = (1 - \text{index}) \times \text{denominator}$  with no residue, here  $0.233333 \times 18,000,000 = 4,200,000$ . One consequence: an index cannot be improved by adding money to a favoured objective without taking it from another, which is exactly the argument a portfolio board should be having and exactly the argument that "we will fund digital next year as well" avoids.

**What the index cannot see, and where it is dangerous.** It is a measure of *money against declared intent*, not of value, deliverability or benefit. A portfolio can score 100 % and consist entirely of failing projects; it can score 60 % and be the best available set because the under-weighted objective has no fundable

candidates, which is a **strategic coverage** finding (2.A.2), not an allocation one, and the two are routinely confused. The declared weights are themselves a governance artefact, often written for a strategy document rather than as spending targets, so the first question on seeing a low index is whether the weights were ever intended to bind. And the mapping of programmes to objectives is a judgement made by someone: a programme claiming three objectives can be counted once, split, or double-counted, and the index moves materially with that choice. State the mapping rule with the index, or the number is unauditable. Finally, the index is a **snapshot of commitments already made**; it changes slowly, because most of the portfolio is mid-flight. The lever that actually moves it is intake and stopping (KA 2.4), not exhortation.

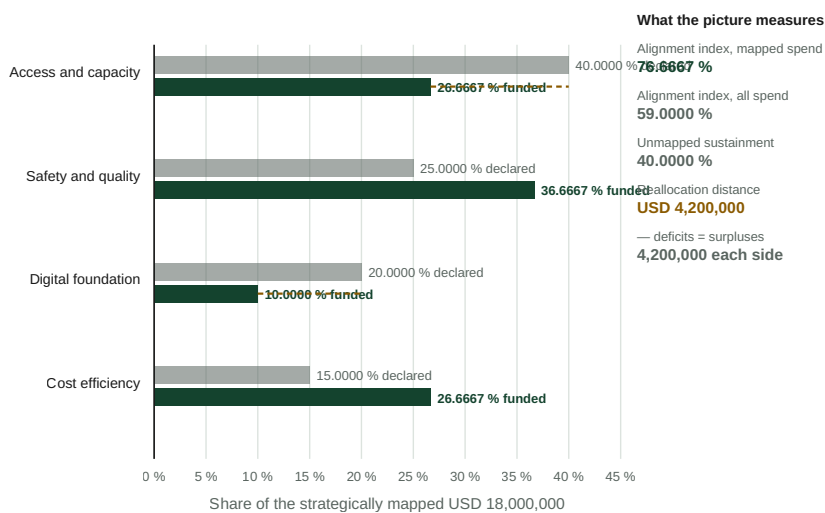


Fig 2.1.1 — The strategy-portfolio alignment index

**Intake cadence is the other half of the process, and it has a price.** A portfolio board that meets quarterly with a three-week paper deadline makes a candidate wait, on Domain 3's latency formula  $E[\text{wait}] = M/2 + L$  (KA 3.2.3), an expected  $13/2 + 3 = 9.5$  weeks before it can be decided at all; a monthly board with the same deadline makes it wait 5.0 weeks. For a candidate of Meridian's shape, whose cost of delay Domain 1 fixes at USD 14,280 a week, that is a difference of  $4.5 \times 14,280 = \text{USD } 64,260$  per candidate — paid before any work starts, and invisible in every business case because no case has a line for the time it spent in a queue. The counter-argument is real and must be stated: a board meeting more often decides on thinner evidence, and a bad selection costs far more than 64,260. The resolution is not a cadence but a **threshold** (small candidates decided continuously against pre-agreed criteria, large ones held for the full board), which is Domain 3's delegation design applied to intake rather than to escalation.

### 2.1.2 Drivers, constraints and environment

A business case that does not name its **driver** cannot be tested. The families, and what each implies:

| DRIVER                               | SUCCESS LOOKS LIKE                | FAILURE MODE IF MISIDENTIFIED                   |
|--------------------------------------|-----------------------------------|-------------------------------------------------|
| <b>Compliance / regulatory</b>       | The obligation is met by the date | Gold-plating a mandatory minimum                |
| <b>Cost reduction</b>                | Measured, sustained cost decrease | Cost moved rather than removed                  |
| <b>Revenue / growth</b>              | New or protected income           | Cannibalised existing revenue counted as new    |
| <b>Capability / enabling</b>         | A later benefit becomes possible  | Benefits claimed twice, here and downstream     |
| <b>Risk reduction</b>                | Exposure measurably lower         | Unquantified "insurance" spending               |
| <b>Service / outcome improvement</b> | Users measurably better served    | Output delivered, outcome unmeasured (Domain 1) |

**Constraints** (funding envelope, regulatory deadline, capacity, technical dependency) are not the same as objectives, and conflating them corrupts option generation: a "constraint" that is actually a preference silently eliminates the option that would have won. The professional habit is to write constraints down and mark each **hard** (physics, law, contract) or **soft** (preference, convention), because only soft ones can be traded.

### 2.1.3 Alignment as a repeated test

Alignment is granted at approval and **decays** thereafter: strategy moves, markets move, and the project's own understanding of what it can deliver moves. A project can be perfectly executed and strategically irrelevant by the time it lands, the most expensive form of success available.

Three practices keep alignment live. **Re-test at gates.** Each stage gate (Domain 3, KA 3.3.1) asks not only "is delivery on track?" but "is this still the right thing?". **Track the assumption set** rather than the conclusion: business cases fail through their assumptions (KA 2.3.4), and an assumption that has been falsified invalidates the case whether or not delivery is green. **Name a benefits owner outside the project** (usually the operational leader who will run the changed service), because a project that owns its own benefits case will never be the one to report that it has stopped making sense (Domain 1's accountability rule; Domain 16's realisation).

Decay is usually discussed as a mood. It is a hazard, and a hazard has a half-life.

## WORKED EXAMPLE

**Worked example 2.1.3 – the alignment half-life, and what a re-test is worth.**

1. **Setup.** The health authority's planning team, reviewing eight years of its own board minutes, assesses that a stated strategic priority is superseded (replaced, merged or de-prioritised) at a rate of about **15 % a year**. This is a **locally calibrated planning figure, not a constant**; in a regulated utility on a five-year price control it would be far lower, and in a consumer-facing technology business far higher. Meridian is a **three-year** delivery spending **USD 800,000** a year. Compute how long alignment survives, what spend is incurred after the driver has gone, and what an annual alignment re-test at each gate is worth if it detects a superseded driver **80 %** of the time.
2. **Formula.** Survival of alignment  $S(t) = (1 - h)^t$ . **Alignment half-life**  $t_{1/2} = \ln(0.5) \div \ln(1 - h)$ . Probability the driver is superseded during year  $k$  is  $(1 - h)^{(k-1)} \times h$ . Expected misaligned spend is the probability-weighted spend incurred after supersession and before detection; with detection probability  $d$  at each subsequent gate, a supersession missed at  $j$  consecutive gates costs the spend of those  $j$  years.
3. **Substitution.**  $S(1) = 0.85$ ,  $S(2) = 0.85^2 = 0.7225$ ,  $S(3) = 0.614125$ .  $t_{1/2} = \ln(0.5)/\ln(0.85)$ . Without a re-test:  $0.15 \times 1,600,000 + 0.1275 \times 800,000$ . With a re-test: year-1 supersession costs  $0.8 \times 0 + 0.2 \times 0.8 \times 800,000 + 0.2^2 \times 1,600,000 = 192,000$  in expectation; year-2 supersession costs  $0.2 \times 800,000 = 160,000$ .
4. **Result.**

|                                         | YEAR<br>1 | YEAR<br>2 | YEAR<br>3      | YEAR<br>4 | YEAR<br>5 |
|-----------------------------------------|-----------|-----------|----------------|-----------|-----------|
| Probability the driver is still current | 85.00 %   | 72.25 %   | <b>61.41 %</b> | 52.20 %   | 44.37 %   |

**Alignment half-life 4.2650 years.** Expected misaligned spend with **no** re-test  $240,000 + 102,000 = \text{USD } 342,000$ : **14.25 %** of the programme's cost. With an annual alignment re-test at 80 % detection,  $0.15 \times 192,000 + 0.1275 \times 160,000 = \text{USD } 49,200$ . The re-test is therefore worth **USD 292,800**, and its **breakeven cost is USD 97,600 per gate** across the three gates. 5. **Interpretation.** Four readings, and the first is the one to remember because it is parameter-free once the hazard is stated.

**The half-life is the sentence that changes how long programmes are allowed to be.** At a 15 % annual hazard, alignment is more likely than not to have survived a **four-year** delivery (52.20 %) and more likely than not to have failed a **five-year** one (44.37 %), with the crossing at **4.2650 years**. That is a design constraint on programme architecture, not a commentary on it: a delivery longer than its own alignment half-life is being asked to outlive the

reason it exists, and the professional response is to tranche it so that something of value lands and is *judged* inside the half-life (Domain 15, KA 15.1.2 builds the tranche architecture; Domain 13's incremental delivery is the same instinct at iteration scale). Note what the half-life is *not*: a forecast that the strategy will change at year 4.27. It is the point at which the odds turn, which is the only thing a design decision needs.

**The value of the re-test lives almost entirely in the early years.** A driver superseded during year 1 exposes two further years of spend; one superseded during year 3 exposes none. So of the 342,000 of expected misaligned spend, **240,000 – 70.18 % – comes from the first year alone**, and a re-test that only starts at the second gate captures a fraction of the value. This inverts the usual gate emphasis, which tightens as the money grows: on *alignment*, the early gate is worth the most, because it is the only one with anything left to protect.

**Detection quality is worth less than it looks, and this is the honest part.** Raising detection from 80 % to a perfect 100 % moves the expected misaligned spend from 49,200 to zero (a further **USD 49,200**), while introducing the re-test at all was worth 292,800, **5.95 times** as much. The large gain is in *asking the question*; the small gain is in asking it well. A leader who cannot get a rigorous strategic re-test into the gate pack should still get a crude one in: at  $d = 0.5$  the re-test is still worth **USD 201,000**, which is **58.77 %** of the 342,000 a perfect re-test would save, from a process amounting to one honest agenda item. The corresponding caution: the 80 % is a judgement about whether a committee reading a gate paper would notice that its own strategy had moved, and every organisation should suspect its own number is lower than it thinks.

**The machinery is Domain 3's, the loss is not, and the difference matters.** Domain 3, KA 3.3.1 prices a gate by its ability to detect a **delivery defect** and computes the remediation saved; this example prices the same gate by its ability to detect a **strategic** change and computes the irrelevant spend avoided. The two are additive (they detect different things and are not substitutes), which is why a gate paper that reports only delivery status is leaving the second value stream entirely uncollected. What the model omits is also worth naming: it assumes supersession is observable at a gate if looked for, that stopping on discovery is possible (Domain 3's decision rights, and the reason 2.4.1's "no route for stopping" symptom converts this value to zero), and that misaligned spend is a total loss, which overstates the case wherever the output has residual use. Where the residual value is material, replace the spend figure with the spend-less-salvage and the arithmetic runs unchanged.

## AI in this KA

Portfolio-level analysis is a legitimate AI application: normalising heterogeneous candidate descriptions into comparable form, surfacing dependency clashes across dozens of cases, flagging where a stated driver and the proposed benefits do not

match. The governed limits are two. **Strategic weighting is a governance judgment**, not an optimisation output, a model asked to "optimise the portfolio" will faithfully optimise whatever proxy it was given, and the proxy is the decision. And **advocacy is invisible to a model**: an AI reading business cases cannot tell an evidenced benefit from a confident one, so it amplifies whatever the strongest writer submitted. AI proposes; the professional verifies, decides and remains accountable.

**Verification, concretely.** The alignment index is where a model earns most of its keep and where its output must be checked hardest, because the arithmetic is trivial and the **mapping** is not: a model assigning programmes to objectives from their titles and summaries will produce a defensible-looking index built on classifications nobody agreed. So publish the mapping rule with the index, sample-check a stated proportion of the assignments by hand, and re-run the index under the alternative treatment of any multi-objective programme; if the index moves materially, the finding is about the mapping and not about the portfolio. The index itself is one addition of minima and should be reproduced in a spreadsheet whose formulae are visible. For the decay arithmetic, the supersession hazard must come from an organisation's own decision history, counted, and never from a model's sense of how fast strategies usually change: a plausible-sounding hazard drives a half-life that then drives programme architecture, which is a long way for an invented number to travel.

## KEY TERMS — KA 2.1

| TERM                          | MEANING                                                                                                                |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Portfolio process</b>      | Comparable description, explicit criteria, funding to capacity, periodic re-decision.                                  |
| <b>Driver</b>                 | The reason the work exists; determines what success means and how it is measured.                                      |
| <b>Hard / soft constraint</b> | Physics, law, contract vs preference and convention; only soft ones are tradeable.                                     |
| <b>Alignment decay</b>        | The erosion of strategic fit after approval, as strategy and understanding move.                                       |
| <b>Alignment index</b>        | $\sum \min(\text{declared weight}, \text{funded share})$ , the overlap between stated priorities and funded spend.     |
| <b>Reallocation distance</b>  | The money that must move to close the gaps: $(1 - \text{index}) \times \text{denominator}$ ; deficits equal surpluses. |
| <b>Supersession hazard</b>    | The annual probability that a stated strategic priority is replaced, merged or de-prioritised.                         |
| <b>Alignment half-life</b>    | $\ln(0.5) \div \ln(1 - h)$ : the delivery duration beyond which alignment is more likely lost than kept.               |
| <b>Benefits owner</b>         | The accountable person outside the project who will realise the benefit.                                               |

## SAMPLE MCQs — KA 2.1

**MCQ 2.1-A** [2.1.3 · Analysis] A project is delivering to plan, but the strategic priority it served has been superseded. The governance-sound response is:

- A. continue — the project is on track and the case was approved
- B. re-test the case at the next gate and decide on current strategy and remaining cost and benefit CORRECT
- C. continue but reduce scope proportionately
- D. transfer the project to another portfolio

*Rationale:* Alignment decays and gates exist to re-decide (2.1.3), on remaining cost and benefit (KA 2.4.2). A treats approval as permanent; C is an arbitrary compromise that decides nothing; D relocates the question without answering it.

**MCQ 2.1-B** [2.1.2 · Application] A case states its constraint as "must use the existing platform", which reflects an architectural preference rather than a contractual or legal requirement. The correct treatment is:

- A. accept it as a constraint, since the architects have stated it
- B. record it as a **soft** constraint, so it can be traded in option generation and its cost made visible CORRECT
- C. remove it from the case entirely
- D. re-classify it as an objective

*Rationale:* Only hard constraints (physics, law, contract) are untradeable; a soft one recorded as hard silently eliminates options (2.1.2). C loses real information; D confuses a limit with a goal.

**MCQ 2.1-C** [2.1.1 · Application] Declared weights are 40 / 25 / 20 / 15 %; the strategically mapped spend of USD 18,000,000 is funded 26.6667 / 36.6667 / 10.0000 / 26.6667 %. The reallocation distance is:

- A. USD 8,400,000
- B. USD 4,200,000 CORRECT
- C. USD 2,400,000
- D. USD 12,000,000

*Rationale:* The positive deficits are  $0.133333 \times 18,000,000 = 2,400,000$  and  $0.10 \times 18,000,000 = 1,800,000$ , totalling **4,200,000**, which equals  $(1 - 0.766667) \times 18,000,000$  (2.1.1). A adds the deficits to the surpluses, counting the same movement twice; C is the access deficit alone; D is the unmapped sustainment line, which is a separate finding about the denominator.



**MCQ 2.1-D** [2.1.3 · Application] A strategic priority is superseded at an annual rate of 15 %. The alignment half-life is:

- A. 6.6667 years
- B. 4.2650 years CORRECT
- C. 3.3333 years
- D. 4.6210 years

*Rationale:*  $\ln(0.5)/\ln(0.85) = 4.2650$  (2.1.3). A is  $1/h$ , the mean waiting time under a different model, not the median; C halves linearly ( $0.5/0.15$ ), ignoring compounding; D uses  $\ln(0.5)/h$ , treating the annual hazard as a continuous rate: a near miss that is systematically too long.

**MCQ 2.1-E** [2.1.1 · Evaluation] An alignment index of 76.6667 % is reported on mapped spend while the index on all discretionary spend is 59.0000 %, the difference being 40 % of spend that maps to no objective. The soundest treatment is:

- A. report the higher figure, since sustainment is not discretionary in practice
- B. report the lower figure, since 40 % of the money is misaligned
- C. report both, and make the recommendation about the unmapped 40 % – either give sustainment a declared weight or stop describing the other 60 % as the portfolio CORRECT
- D. exclude sustainment and re-baseline the declared weights to the mapped spend

*Rationale:* The denominator is the finding (2.1.1). A hides the largest single item; B misnames unstated strategy as misalignment; D quietly ratifies the omission by rebasing the weights to the spend they were meant to test.

**MCQ 2.1-F** [2.1.1 · Comprehension] A portfolio scores 100 % on the alignment index. It is therefore:

- A. well balanced, since every objective is funded to its declared weight
- B. correctly allocated against declared intent, which says nothing about risk profile, time to benefit, capability concentration or deliverability CORRECT
- C. optimal, since no money needs to move
- D. mis-measured, because a perfect score is not attainable in practice

*Rationale:* The index measures money against stated intent and nothing else (2.1.1, 2.A.2). A and C read an allocation measure as a balance or optimality measure, a portfolio funded exactly to weight can still be entirely high-risk, entirely long-dated and entirely dependent on one scarce team. D is false: a perfect score simply means the funded shares match the declared weights.

## SELF-CHECK — KA 2.1

1. *Name the three symptoms of a strategy-portfolio gap.* — Funded work not reflecting priorities; more work than capacity; no route for stopping.
2. *Why must a benefits owner sit outside the project?* — A project owning its own benefits case will not be the one to report that the case has stopped making sense.
3. *What distinguishes a hard from a soft constraint?* — Physics, law or contract versus preference; only soft constraints can be traded.
4. *State the alignment-index identity and why it constrains the argument.* — Reallocation distance =  $(1 - \text{index}) \times \text{denominator}$ , and deficits equal surpluses, so an objective can only be better funded by taking money from another.
5. *At a 15 % annual supersession hazard, what is the longest delivery whose alignment is more likely kept than lost?* — 4.2650 years; a five-year delivery has a 44.37 % chance of landing under the strategy that authorised it.
6. *Where does most of the value of an alignment re-test sit, and why?* — In the earliest gate: 70.18 % of Meridian's expected misaligned spend arises from a year-1 supersession, because that is the only case with two further years of spend still to protect.

## KNOWLEDGE AREA 2.2

## The business case and selection

*Topics: 2.2.1 the business case as a decision instrument · 2.2.2 options and appraisal · 2.2.3 selection and prioritisation models.*

### 2.2.1 The business case as a decision instrument

A decision-grade business case answers six questions and nothing else: **why** (the driver and the problem, with evidence); **what options** exist, including doing nothing; **what it costs**, with a range and an accuracy class (Domain 7, KA 7.1.1); **what it delivers**, as outcomes and benefits with owners and measures (KA 2.3); **what could go wrong**, with the exposure quantified (Domain 8); and **how it will be judged**, with success criteria agreed in advance (Domain 1, KA 1.A.2).

Its two failure modes are worth naming precisely. A **advocacy document** answers only "why" and "what it delivers", omitting ranges, options and risk: designed to obtain approval rather than support a decision. A **compliance artefact** answers all six in a template nobody reads, produced after the decision has been taken informally. Both are common; the test that separates them from the real thing is whether the case **could have concluded "no"**.

**The living-case principle.** The case is not filed at approval; it is the instrument the gates re-test (2.1.3) and the source of the benefits register Domain 16 tracks. A business case whose figures no longer match the project is an unmanaged forecast, not a historical document.

### 2.2.2 Options and appraisal

A **genuine options set** includes the **do-nothing / do-minimum baseline** (against which all value is measured: without it, benefits are unmeasurable), at least one option that is materially cheaper than the preferred one, and at least one that is materially more ambitious. An options set constructed to make the preferred option look inevitable (the "straw options" pattern) is the advocacy failure mode in structural form, and reviewers spot it by checking whether any option could plausibly have won.

**Appraisal** applies the discounted machinery PFL-AI Domain 4 builds in full: NPV as the primary value measure, with the range and sensitivity that make it honest. This book uses the results; the finance book derives them. What belongs to a delivery leader is the **benefits profile**, because that is where most business cases are wrong, and the error is systematic.

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#### WORKED EXAMPLE

##### Worked example 2.2.2 – the Meridian business case, twice.

1. **Setup.** Meridian costs **USD 2,400,000** to deliver. Domain 1 established the arithmetic: full potential at 100 % adoption is **USD 979,200** per year; realistic steady-state adoption is **70 %**, worth **USD 685,440** per year. The benefits are appraised over **8 years** at a **7 %** discount rate. Compare the case as commonly written (full potential from year one) with an honest **ramped** profile: 40 % adoption in year 1, 60 % in year 2, 70 % thereafter.
2. **Formula.** Flat case:  $PV = \text{annual benefit} \times AF(r, n)$ . Ramped case:  $PV = \sum (\text{potential} \times \text{adoption}_t) / (1 + r)^t$ .  $NPV = PV - \text{cost}$ .
3. **Substitution.**  $AF(0.07, 8) = 5.971299$ . Flat:  $979,200 \times 5.971299$ . Ramped:  $391,680/1.07 + 587,520/1.07^2 + 685,440 \times (\text{years 3–8 discounted})$ .
4. **Result.**

|                                  | YEAR<br>1 | YEAR<br>2 | YEARS<br>3–8 | PV OF<br>BENEFITS | NPV               |
|----------------------------------|-----------|-----------|--------------|-------------------|-------------------|
| <b>Flat, full potential</b>      | 979,200   | 979,200   | 979,200      | 5,847,096         | <b>+3,447,096</b> |
| <b>Ramped, 70 % steady state</b> | 391,680   | 587,520   | 685,440      | 3,732,898         | <b>+1,332,898</b> |

The flat case overstates NPV by USD 2,114,198: 158.6 % of the honest figure.

5. **Interpretation.** Both cases approve the programme, which is precisely why the error survives: the decision is unchanged, so nobody checks, and the *promise* is what gets remembered. Meridian was later judged a failure against benefits it never could have produced (Domain 1's case study), and this table is where that judgement was actually created: at approval, two years before anyone noticed. Five further readings follow, and the third is the one most business cases get subtly wrong even after they have fixed the profile.

**The ramp is not pessimism.** It is the adoption curve of the *same* 70 % figure, merely arriving when it actually arrives. Nothing in the honest column is a more cautious assumption than the flat column; the two differ only in *when* the identical steady state is reached. That is why the flat profile is not a judgement call a reasonable person might defend. It is an arithmetic claim that adoption is instantaneous, which nobody would sign if it were written in words.

**The ramp costs a fixed fraction of value, and the fraction is worth knowing.** The ramped PV of 3,732,898 is 91.2027 % of the flat PV at the same 70 % steady state ( $0.70 \times 5,847,096 = 4,092,967$ ), so the two ramp years give up 8.7973 % of the discounted value of the whole benefit stream. Because the profile 40 / 60 / 70 is the steady state scaled by 4/7, 6/7, 1, that fraction is **independent of the steady-state level**: halve the adoption and both figures halve. A leader can therefore correct any flat case for a two-year ramp of this shape by multiplying by 0.912027, in their head, in the meeting.

**Breakeven adoption comes in two flavours, and the difference is 3.9592 points.** The **flat-equivalent breakeven** is 41.05 %:  $2,400,000 / 5,847,096$ : the level a benefit stream *flat from year one* would have to reach. The **ramped-basis breakeven** is 45.01 % ( $2,400,000 / 5,332,711$ , where 5,332,711 is the PV of the ramped profile per unit of sustained adoption) the *steady state* a proportionally ramped profile must reach. Both are correct answers to different questions, and reporting the first while meaning the second understates the requirement by nearly four points. This book uses the flat-equivalent figure as its headline, because it is comparable across profiles, and states the basis every time. **A breakeven without its basis is not a number.**

**The breakeven, not the NPV, is the board sentence.** "This programme creates value at any sustained flat-equivalent adoption above 41.05 %" states the condition value depends on, in a unit the board can monitor monthly, and it survives every argument about the discount rate that an NPV invites. It also gives the sponsor a *falsifiable* commitment, which an NPV never does: an NPV can only be wrong in retrospect, while a breakeven can be tested every month from go-live.

**What breaks the whole example.** Three things. The 8-year horizon is a choice, and a benefit stream that in fact decays (as clinical workflows change and the released hours are re-absorbed) is not captured by a flat tail; where decay is plausible, profile it and let the horizon fall out. The 7 % rate is an input

the leader usually does not own, and the industry variations below compute what a point of it is worth. And the cost side is treated as a point estimate, which Domain 7, KA 7.1.1 forbids: the same table drawn with the cost at the top of its accuracy range moves every breakeven, and a case that ranges its benefits but not its costs has ranged the wrong side.

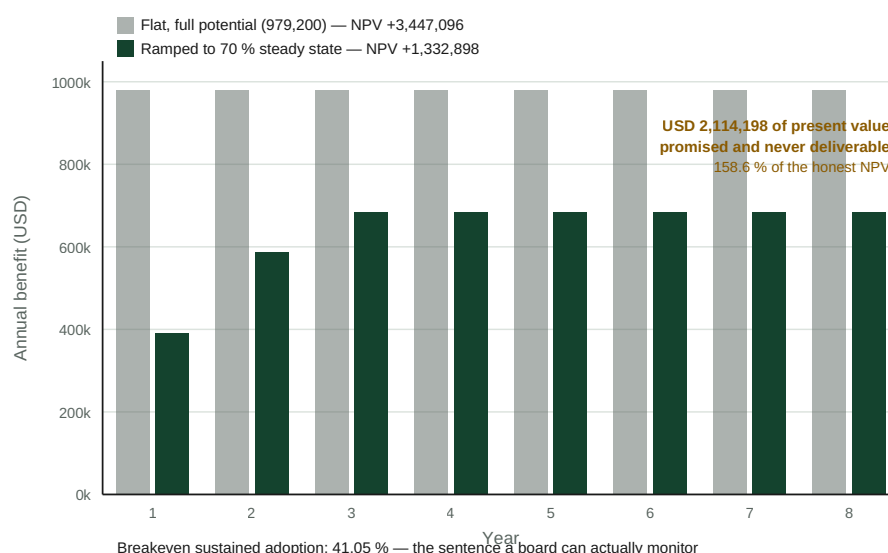


Fig 2.2.1 — Two Meridian business cases from identical facts

**The other half of the appraisal is the thing the project is being compared with,** and it is almost never zero. "Do nothing" is a course of action with its own cash flows: contracts that expire, equipment that fails, workarounds whose cost grows, obligations that fall due anyway. Where those costs are omitted, the case understates its own value, which makes this the mirror image of the flat-profile error, and the reason a reviewer must look for errors in *both* directions rather than assuming every case is inflated.

#### WORKED EXAMPLE

##### Worked example 2.2.2b — the do-nothing baseline that was not zero.

1. **Setup.** Meridian's approved case compared the programme against a do-nothing option valued at **zero**. Two facts were available at the time and omitted. First, the legacy records system's vendor support ends at the end of **year 3**; continuing without Meridian requires a mandatory re-platforming of the legacy estate costing **USD 600,000**, falling in **year 4**. Second, the manual reconciliation the legacy system requires between clinics grows as clinic numbers rise, adding **USD 40,000 a year in years 5 to 8**. Both are avoided if Meridian proceeds. Same horizon, same 7 % rate. Restate the case.

2. **Formula.** Incremental NPV = PV(project benefits) – project cost + PV(costs avoided in the do-nothing case). Equivalently, the do-nothing option is appraised in its own right and the project's value is the difference: NPV(project) – NPV(do nothing). The flat-equivalent breakeven adoption becomes (cost – PV of avoided cost) ÷ PV of full-potential benefits.

3. **Substitution.**  $600,000 / 1.07^4 = 600,000 \times 0.762895$ .  $40,000 \times (AF(0.07, 8) - AF(0.07, 4)) = 40,000 \times (5.971299 - 3.387211) = 40,000 \times 2.584087$ . Breakeven  $(2,400,000 - 561,101) / 5,847,096$ .

4. **Result.**

| COMPONENT                                                          | UNDISCOUNTED   | PRESENT VALUE  |
|--------------------------------------------------------------------|----------------|----------------|
| Legacy re-platforming avoided (year 4)                             | 600,000        | 457,737        |
| Manual reconciliation avoided (years 5–8)                          | 160,000        | 103,363        |
| <b>Do-nothing cost, and therefore value created by avoiding it</b> | <b>760,000</b> | <b>561,101</b> |

Incremental NPV rises from **+1,332,898** to **+1,893,998**: an uplift of **USD 561,101**, or **42.10 %** of the stated NPV. Flat-equivalent breakeven adoption falls from **41.0460 %** to **31.4498 %**, an improvement of **9.5962 percentage points**. 5. **Interpretation.** Five readings, and the third is the most consequential sentence in this domain.

**A zero do-nothing baseline is a claim, and usually a false one.** It asserts that the current arrangement can continue indefinitely at its current cost. That is true of very little: support contracts expire, regulation tightens, volumes grow, and the workaround that costs nothing today costs a post holder tomorrow. The professional discipline is to appraise the do-nothing option with the same rigour as the preferred one (its own cost profile, its own risks, its own owner), and to record that appraisal, because that is what makes the comparison a comparison rather than an assertion.

**The error runs in the opposite direction to the flat profile, which is why both must be looked for.** The flat profile flattered Meridian by 2,114,198 of present value; the missing counterfactual penalised it by 561,101. A reviewer who only knows that cases are optimistic will find the first and not the second, and will then be wrong in a way that is harder to correct; because arguing that a case is *understated* costs credibility that arguing the reverse does not. Net across the two corrections, Meridian's honest NPV is  $3,732,898 - 2,400,000 + 561,101 = +1,893,998$ , which is **54.94 %** of the 3,447,096 the approved case advertised. The case was wrong twice and still positive.

**Meridian's actual outcome sits between the two breakevens, and that is the whole lesson.** At its achieved flat-equivalent adoption of **40 %**, the pro-

gramme's NPV on the case as written is  $0.40 \times 5,847,096 - 2,400,000 =$  (USD 61,162): a small loss, and the arithmetic that made a public failure verdict defensible. On the honest baseline it is  $-61,162 + 561,101 = +\text{USD } 499,939$ : a clear success. **The same programme, the same 40 % adoption, the same cost: value-destroying or value-creating according to whether anyone appraised the do-nothing option.** The judgement passed on Meridian was not a measurement of the programme. It was a measurement of its business case.

**Do-minimum is usually the honest baseline, not do-nothing, and the difference is not presentational.** Pure inaction is often not permissible (the support contract cannot simply be allowed to lapse on a clinical system), so the correct comparator is the **cheapest compliant alternative**, here the 600,000 re-platforming on its own. Appraised that way, the 600,000 moves from "avoided cost" to "the do-minimum option's cost", and the incremental NPV is identical; what changes is that the options set now contains a genuine third option that could have won on a low adoption forecast. **Framing the counterfactual as an option rather than as an adjustment is what makes the options set genuine** (2.2.2's straw-options test), and it is the better habit for exactly that reason.

**What breaks it, and where to be careful.** Avoided costs must be *genuinely* avoided and not merely deferred: if the legacy re-platforming would still be needed for another system on the same estate, nothing has been avoided and the 457,737 is fictitious, the same single-claimant discipline KA 2.3.2 applies to benefits, applied to costs. The avoided cost must not also be counted as a benefit elsewhere in the case, which is the commonest form of the double count here. The timing carries real weight: the 600,000 is worth 457,737 at year 4 and would be worth **560,748** at year 1, so a vaguely dated obligation is a materially different number, and the year is a fact to be evidenced from the contract rather than assumed. And an avoided cost is not cash released unless the budget holding it is actually removed: 2.3.2's cash-releasing test applies with full force, because a finance function that never held a provision for the 600,000 will not recognise its avoidance. **This is an area where the accounting and budgetary treatment differs materially between organisations and jurisdictions; where the case turns on it, agree the treatment with the finance function in writing before the paper is written.**

### 2.2.3 Selection and prioritisation models

**Weighted scoring** ranks candidates against criteria carrying explicit weights. Its value is that it forces the criteria and weights into the open, where they can be argued; its limits are that scores are **ordinal judgments** (the arithmetic caution of Domain 8, KA 8.2.1 applies) and that weights are chosen by whoever runs the model. So so the model can be steered by anyone who understands it.

Which makes **who scores** a governance question rather than an administrative one. A scoring panel is a decision body, so the declaration duty of Domain 1, KA 1.2.2a

applies to it in full: each scorer identifies any interest in the candidates before the criteria are issued, declares it in writing to the chair, and takes no part in scoring the candidate concerned, with the record naming who scored in their place. The interest to watch for here is rarely financial. It is usually a sponsoring or delivering role in one of the candidates, which is precisely the person whose subject knowledge makes them worth having on the panel. Note the sequence, because it is the part that gets reversed under time pressure: interests are declared and the weights are fixed **before** any candidate is scored, since a weight set after the scores are visible is a way of choosing the winner while appearing to apply a model (Domain 10, KA 10.2.3 makes the same point for bids, where the stakes are contractual).

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#### WORKED EXAMPLE

#### Worked example 2.2.3 — four candidates, two rankings.

1. **Setup.** Four candidate programmes scored 1–5 against weighted criteria: strategic fit (0.35), benefit value (0.30), deliverability (0.20), risk — inverse, so higher is safer (0.15).
2. **Formula.** Weighted score =  $\Sigma(\text{weight} \times \text{score})$ . Then, under a binding constraint, rank by **NPV per unit of the scarce resource** instead.
3. **Substitution and result.**

| CANDIDATE | FIT | BENEFIT | DELIVER. | RISK | WEIGHTED SCORE |
|-----------|-----|---------|----------|------|----------------|
| Meridian  | 4   | 5       | 3        | 3    | 3.95           |
| Beta      | 5   | 3       | 4        | 4    | 4.05           |
| Gamma     | 3   | 4       | 5        | 4    | 3.85           |
| Delta     | 2   | 2       | 5        | 5    | 3.05           |

Now suppose the binding constraint is **integration-team capacity**, of which only 3 units exist: Meridian needs 3 units for an NPV of 1,693,072; Beta 2 units for 1,200,000; Gamma 1 unit for 900,000.

| CANDIDATE | NPV       | CAPACITY | NPV PER UNIT |
|-----------|-----------|----------|--------------|
| Meridian  | 1,693,072 | 3        | 564,357      |
| Beta      | 1,200,000 | 2        | 600,000      |
| Gamma     | 900,000   | 1        | 900,000      |

4. **Interpretation.** Two defensible methods give two different answers, and a leader who can say why commands the selection meeting. Four readings.



**The constraint changes the answer, not merely the ordering.** The scoring model ranks **Beta first** (4.05 against Meridian's 3.95) on strategic fit and deliverability. Under the capacity constraint, **Beta + Gamma** together consume the 3 units for a combined NPV of **2,100,000**, beating Meridian's 1,693,072 alone, so the constrained answer is to run the two smaller programmes, which neither the raw NPV ranking nor the scoring model would have selected. This is the delivery-side twin of PFL-AI's capital rationing (its Domain 4, KA 4.3.2). The leadership content is that **the binding constraint decides the method**, and naming that constraint honestly (usually a scarce team, not money) is the whole game.

**Enumeration is cheap and greedy ranking is not safe.** The feasible sets under 3 units are exactly {}, {Gamma} = 900,000, {Beta} = 1,200,000, {Meridian} = 1,693,072 and {Beta, Gamma} = **2,100,000**; {Meridian, Beta} needs 5 units and {Meridian, Gamma} needs 4, so both are infeasible. Here greedy ranking by NPV per unit (Gamma at 900,000, Beta at 600,000, Meridian at 564,357) happens to reach the optimum, and that is luck rather than method. Had a fifth candidate **Epsilon** needed all 3 units for an NPV of **2,200,000** (733,333 per unit), greedy would still have taken Gamma first and finished with Beta + Gamma at 2,100,000, missing the best set by **USD 100,000**. With  $n$  candidates there are  $2^n$  subsets, which for the ten or twenty candidates a real portfolio board considers is a spreadsheet exercise, not a research problem. **Enumerate; rank only to explain the answer afterwards.**

**Divisibility is the assumption that decides whether any of this is legitimate.** Ranking by value per unit is the correct optimum when candidates are *divisible*, when half of Beta can be bought for half the money and half the value. Projects are almost never divisible in that sense, which is what makes them lumpy and what makes greedy unsafe. Where a candidate genuinely can be **staged**, it becomes partly divisible and the arithmetic improves; that is one of the underrated benefits of the tranching argued for in 2.1.3 and priced in 2.A.1, and it belongs in the selection paper rather than being discovered later.

**What no model does.** Scoring and ratios cannot see option value, sequencing dependencies, or the strategic cost of *not* doing something. They order candidates; they do not decide. A portfolio board that treats a scoring output as the decision has automated its own accountability (Domain 1, KA 1.2.1). Two further blindnesses are specific to the constrained form: it takes the constraint as given, when the most valuable move is often to *relax* it (priced immediately below), and it assumes the candidates are independent, when Gamma may be the enabler on which Meridian's benefits depend, in which case the sets are not free to be chosen (Domain 15, KA 15.1.3 handles dependent candidates).

**Worked example 2.2.3b — what one unit of the binding constraint is worth.**

1. **Setup.** The three candidates above compete for one scarce integration team measured in **units**: Meridian needs 3 units for an NPV of 1,693,072; Beta needs 2 for 1,200,000; Gamma needs 1 for 900,000. The portfolio director can grow or shrink the team, and a unit of durable integration capacity (a qualified engineer with the platform knowledge, recruited and retained) costs **USD 400,000** a year all-in. How much capacity should the organisation hold?
2. **Formula.** For each capacity level  $c$ , solve  $V(c) = \max \sum NPV_i$  subject to  $\sum units_i \leq c$  by enumeration. The **marginal value of the  $n$ th unit** is  $V(n) - V(n-1)$ . Buy  $k$  further units when  $V(c + k) - V(c) > k \times \text{unit cost}$ .
3. **Substitution.** At  $c = 3$ , the feasible maximum is {Beta, Gamma} = 2,100,000. At  $c = 4$ , the admissible sets include {Meridian, Gamma} = 1,693,072 + 900,000. At  $c = 5$ , {Meridian, Beta} = 1,693,072 + 1,200,000. At  $c = 6$ , all three fit.
4. **Result.**

| CAPACITY | OPTIMAL SET             | PORTFOLIO NPV | MARGINAL VALUE OF THAT UNIT |
|----------|-------------------------|---------------|-----------------------------|
| 1        | Gamma                   | 900,000       | <b>900,000</b>              |
| 2        | Beta                    | 1,200,000     | <b>300,000</b>              |
| 3        | Beta + Gamma            | 2,100,000     | <b>900,000</b>              |
| 4        | Meridian + Gamma        | 2,593,072     | <b>493,072</b>              |
| 5        | Meridian + Beta         | 2,893,072     | <b>300,000</b>              |
| 6        | Meridian + Beta + Gamma | 3,793,072     | <b>900,000</b>              |

From a base of 3 units, at 400,000 a unit: **+1 unit** gains 493,072 for 400,000 (net **+93,072**; **+2 units** gains 793,072 for 800,000) net **(6,928)**; **+3 units** gains 1,693,072 for 1,200,000 — net **+493,072**. 5. **Interpretation.** Four readings, and the second contradicts the way capacity decisions are almost always taken.

**There is no such thing as "the value of a unit of integration capacity".** The marginal value runs 900,000 · 300,000 · 900,000 · 493,072 · 300,000 · 900,000. It is **lumpy and non-monotone**, and it is a property of the *candidate set*, not of the team. Anyone who quotes a single figure for what a scarce team is worth has averaged away the only information in the answer. The same applies to the shadow prices a solver reports: they are valid for the set that was solved and change when a candidate is added, withdrawn or re-estimated.

**The right answer here is to add one unit or three, and never two — which no incremental process would find.** Adding units one at a time, each judged on its own, accepts the first (it gains 493,072 for 400,000, a net **+93,072**) and then

rejects the second, which gains only 300,000 for 400,000 — a net **(100,000)** on its own, and (6,928) taken as a block of two. So the process stops there and never reaches the block of three, whose net is **+493,072: 5.30 times** the gain the incremental process settles for. Capacity decisions are non-convex because the candidates are lumpy, so they must be taken as a **block decision against enumerated plans**, which is precisely what an annual headcount round asking each manager to justify the next post cannot do. This is the arithmetic behind Domain 15's treatment of enterprise capacity as a portfolio-level decision (KA 15.3) rather than a departmental one.

**The comparison the portfolio board should actually be shown.** Not "should we hire?" but a table of feasible plans with their NPVs and their capacity costs, because on these figures the question "what is the best use of 1,200,000?" has the answer "three integration engineers", which no capital proposal would ever have contained. Note also the direction that is usually forgotten: at  $c = 2$  the marginal unit is worth only 300,000, **below its 400,000 cost**, so an organisation sitting at 2 units with these candidates should shrink rather than hold: a conclusion the same table produces and no advocacy paper ever reaches.

**What breaks it.** The unit cost must be the **durable** cost of holding capacity, not a contractor day rate for a peak, and if the capacity can be rented for a season the whole question changes shape (Domain 7, KA 7.4 prices resource acquisition modes; Domain 10 prices the make-or-buy). Recruitment lag matters and is absent here: three engineers who arrive in nine months do not enable a decision taken today, and Domain 1's reinforcement-trough arithmetic prices the transient they bring with them. The NPVs are point estimates on candidates whose own ranges overlap, so a marginal value of 300,000 is not reliably different from one of 493,072. And capacity has value beyond this candidate set (optionality against next year's candidates, and resilience), which this model prices at zero and Domain 15, KA 15.3 prices properly as protective capacity.

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#### WORKED EXAMPLE

##### Worked example 2.2.3c — how far must a weight move to reverse the ranking?

1. **Setup.** The same four candidates and the same weighted-scoring model: strategic fit 0.35, benefit value 0.30, deliverability 0.20, risk-inverse 0.15, scores on a 1–5 scale. Beta leads Meridian by 4.05 to 3.95. Two people who both accept the scores disagree about the weights: one believes strategic fit should carry less and benefit value more. **How much less?**

2. **Formula.** Shift  $\delta$  of weight from criterion a to criterion b, holding the rest fixed. A candidate's total changes by  $\delta \times (\text{score\_b} - \text{score\_a})$ . The ranking flips where  $\delta \times [(\text{score}_{\beta\_b} - \text{score}_{\beta\_a}) - (\text{score\_M\_b} - \text{score\_M\_a})] =$  the current margin. Separately, the most any criterion can move a total (its **criterion influence**) is  $(\text{score range}) \times \text{weight}$ .
3. **Substitution.** Meridian gains  $\delta(5 - 4) = +\delta$ ; Beta gains  $\delta(3 - 5) = -2\delta$ . The margin is 0.10, so  $\delta + 2\delta = 0.10$ .
4. **Result.**  $\delta = 0.033333$ , i.e. **3.3333 percentage points** of weight. At fit 0.316667 and benefit 0.333333 both candidates score exactly **3.983333**. Criterion influence, on a 1–5 scale (range 4): fit **1.40**, benefit **1.20**, deliverability **0.80**, risk **0.60**, summing to the full 4.00 range of possible totals. The four candidates' actual totals span only **1.00**, which is **25.00 %** of that maximum swing.
5. **Interpretation.** Five readings, and the last is the governance conclusion the arithmetic forces.

**A 3.33-point weight shift reverses the portfolio's top candidate, and 3.33 points is inside the noise of any weighting workshop.** Nobody can defend 0.35 against 0.3167 for strategic fit on any evidence; the two numbers are indistinguishable as statements of priority. So the model's answer here is not robust, and reporting "Beta ranks first" without reporting that it takes 3.33 points to change that is a material omission. The reviewer's question is exact: **what is the smallest defensible change of weight that reverses this ranking?**

**The score side is worse.** Meridian needs only **+0.50** on deliverability (3 to 3.5) to draw level, because deliverability carries a weight of 0.20 and  $0.10/0.20 = 0.50$ . On a 1–5 ordinal scale nobody can distinguish 3 from 3.5, which is Domain 8, KA 8.2.1's arithmetic caution about ordinal scales in its sharpest form: **the model is being asked to resolve a difference finer than its own inputs can express.** On the risk criterion, weighted 0.15, the required move is **+0.6667**, and the general rule falls out: the score movement needed to close a gap is  $\text{margin} \div \text{weight}$ , so the lightly weighted criteria are the ones where scoring error does least damage.

**Some weight shifts can never flip a ranking, and knowing which is free information.** Moving weight between strategic fit and *deliverability* changes Meridian's total by  $\delta(3 - 4) = -\delta$  and Beta's by  $\delta(4 - 5) = -\delta$ , identically, so the 0.10 margin is invariant to that trade no matter how large it is. A weighting argument between those two criteria is therefore not an argument about this decision at all, and a chair who knows it can stop the discussion. The pairs that matter are the ones where the two candidates' score *differences* differ.

**Criterion influence tells you what the model cannot decide.** With a 1–5 scale, the risk criterion at 0.15 can move a total by at most **0.60** across its whole range. If the real disagreement in the room is about risk, and the candidates' risk scores differ by 1 point (worth 0.15), the model literally cannot express it: **0.15**

against a spread of totals of 1.00. The remedy is not a bigger weight but a different instrument, quantified exposure per Domain 8, KA 8.2.2, reported beside the score rather than compressed into it.

**Scoring models compress, and compression is what makes them steerable.** The four candidates occupy 25.00 % of the available range, because a good candidate scores well on some criteria and badly on others and weighting averages the differences away. Compression is not a defect to be engineered out (it is the honest consequence of averaging), but it has a governance implication: **when the outputs are close, the weights decide, and the weights are chosen by whoever runs the model.** The countermeasures are procedural and cheap: agree and minute the weights *before* the candidates are scored, publish the flip point with the ranking, and where two candidates lie within the flip point of each other, report them as **not separated by this method** and decide on something else: constraint economics (2.2.3b), option value (2.A.1) or portfolio balance (2.A.2). A model that cannot separate two candidates has still done useful work by saying so.

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### AI in this KA

Business-case drafting is now routinely AI-assisted, and this KA identifies the specific hazard: a model will produce a **fluent, complete-looking case with a flat benefits profile**, because that is what most cases in its training data look like. It will also generate plausible options sets that are structurally straw. The verification duty is therefore concrete: check the benefits profile ramps, check at least one option could have won, check every benefit traces to a measured outcome with an owner (Domain 1's benefits-chain test), and check the range and class on the cost. An AI-drafted case that passes those four checks is genuinely useful work; one that has not been checked against them is the advocacy document of 2.2.1 with better formatting.

Four further checks belong on that list once this KA's arithmetic is available, and each catches a defect the four above do not. **A do-nothing option valued at zero** is as characteristic of generated cases as a flat profile, and for the same reason (most cases in the corpus do it), so ask the model explicitly what doing nothing costs in each year of the horizon, and then verify each item against a contract, licence or asset register rather than accepting the list. And **a breakeven quoted without its basis** is a near-certainty in generated text, because the distinction between flat-equivalent and ramped-basis is finer than the language usually carries; recompute it and label it. Where a model has been asked to run a scoring or constrained-selection model, the specific verification is the **flip point** and the **enumeration**: a model will produce a ranking with total confidence and will not volunteer that a 3.33-point weight shift reverses it, and a model asked to "optimise" a constrained selection will frequently return the greedy per-unit answer, which is not the optimum. Both are checkable in a spreadsheet in minutes, and both change recommendations.

## KEY TERMS — KA 2.2

| TERM                                    | MEANING                                                                                                                            |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Do-nothing baseline</b>              | The comparison against which all benefits are measured; without it value is unmeasurable.                                          |
| <b>Do-minimum option</b>                | The cheapest compliant alternative where inaction is not permissible; the honest comparator, and an option in its own right.       |
| <b>Counterfactual cost</b>              | Cost incurred <i>only</i> in the do-nothing case, and therefore value created by avoiding it.                                      |
| <b>Straw options</b>                    | An options set built so the preferred option cannot lose.                                                                          |
| <b>Benefits profile</b>                 | The time-phased benefit stream; ramps in reality, flat in most business cases.                                                     |
| <b>Breakeven adoption</b>               | The adoption level at which NPV is zero, a more useful board sentence than NPV.                                                    |
| <b>Flat-equivalent basis</b>            | Breakeven stated as the level a profile flat from year one would need (Meridian 41.05 %).                                          |
| <b>Ramped basis</b>                     | Breakeven stated as the <i>steady state</i> a proportionally ramped profile must reach (Meridian 45.01 %).                         |
| <b>Weighted scoring</b>                 | Ranking by $\Sigma(\text{weight} \times \text{ordinal score})$ ; forces criteria into the open, steerable by whoever sets weights. |
| <b>Value per unit of constraint</b>     | Ranking basis when a scarce resource binds; a heuristic, not an optimum.                                                           |
| <b>Marginal value of the constraint</b> | $V(n) - V(n-1)$ from enumeration; lumpy, non-monotone, and a property of the candidate set.                                        |
| <b>Flip point</b>                       | The smallest weight shift that reverses a ranking; report it with the ranking or the ranking is unqualified.                       |
| <b>Criterion influence</b>              | $(\text{score range}) \times \text{weight}$ : the most a criterion can move a total, and therefore what the model cannot decide.   |

## SAMPLE MCQs — KA 2.2

**MCQ 2.2-A** [2.2.2 · Application] Full benefit potential is 979,200 per year; steady-state adoption is 70 %; the profile ramps 40 %/60 %/70 % and is appraised over 8 years at 7 % (AF = 5.971299). Against a flat full-potential case, the ramped NPV is lower by:

- A. USD 979,200
- B. USD 2,114,198 CORRECT
- C. USD 1,332,898
- D. USD 293,760

*Rationale:* Flat PV 5,847,096 less ramped PV 3,732,898 = **2,114,198** of overstated present value. C is the honest NPV itself; A is one year's potential; D is Domain 1's single-year output-based overstatement, a different figure.

**MCQ 2.2-B** [2.2.2 · Analysis] Both the flat and ramped Meridian cases produce a positive NPV and the same approval decision. Why does the flat case still matter?

- A. it does not — the decision is unchanged, so the error is harmless
- B. because the case becomes the promise the programme is later judged against, and benefits that were never deliverable are recorded as failure CORRECT
- C. because the discount rate must be recalculated
- D. because the flat case understates cost

*Rationale:* The approval is identical; the *commitment* is not (2.2.2, and Domain 1's case study where Meridian was publicly called a failure). A is the reasoning that let the error survive.

**MCQ 2.2-C** [2.2.3 · Analysis] Integration capacity of 3 units is the binding constraint. Meridian needs 3 units for NPV 1,693,072; Beta 2 for 1,200,000; Gamma 1 for 900,000. The value-maximising selection is:

- A. Meridian alone — the highest single NPV
- B. Beta + Gamma — combined NPV 2,100,000 within the 3-unit constraint CORRECT
- C. Meridian + Gamma — the two highest NPVs
- D. all three, phased across two years

*Rationale:* Beta and Gamma together fit the constraint and beat Meridian's 1,693,072. C needs 4 units; D changes the premise rather than answering under it; A ignores the constraint's implications.

**MCQ 2.2-D** [2.2.1 · Recall] The clearest test that a business case is a decision instrument rather than an advocacy document is:

- A. whether it follows the corporate template
- B. whether it could have concluded "no" CORRECT
- C. whether it was approved
- D. whether its NPV is positive

*Rationale:* An options set and evidence that permit a negative conclusion are what make it a decision (2.2.1). A is the compliance failure mode; C and D are outcomes, not tests.

**MCQ 2.2-E** [2.2.2 · Application] A case shows NPV +1,332,898 against a do-nothing option valued at zero. In fact doing nothing forces a mandatory 600,000

spend in year 4 and 40,000 a year in years 5–8; the rate is 7 % ( $1/1.07^4 = 0.762895$ ;  $AF(0.07, 8) - AF(0.07, 4) = 2.584087$ ). The incremental NPV is:

- A. USD 1,332,898
- B. USD 1,893,998 CORRECT
- C. USD 2,092,898
- D. USD 771,797

*Rationale:* Avoided cost PV is  $457,737 + 103,363 = 561,101$ , added to the project's NPV (2.2.2b). A omits the counterfactual entirely; C adds the **undiscounted** 760,000; D subtracts the avoided cost instead of adding it, treating a cost the project removes as a cost it incurs.

**MCQ 2.2-F** [2.2.3 · Analysis] Beta leads Meridian 4.05 to 3.95. Scores on strategic fit are Beta 5, Meridian 4; on benefit value Beta 3, Meridian 5. Shifting weight from strategic fit to benefit value flips the ranking at a shift of:

- A. 10.00 percentage points
- B. 3.33 percentage points CORRECT
- C. 2.50 percentage points
- D. no shift between these two criteria can flip it

*Rationale:* Meridian gains 6 and Beta loses 26, so  $36 = 0.10$  and  $6 = 0.033333$  (2.2.3c). A removes weight from fit without adding it to benefit, so the weights no longer sum to one; C divides the 0.10 margin by the 4-point score range; D is true of a fit-to-deliverability shift, where both candidates move identically, but not of this pair.

**MCQ 2.2-G** [2.2.3 · Evaluation] With 3 units of integration capacity the optimal set is Beta + Gamma at 2,100,000; with 4 units it is Meridian + Gamma at 2,593,072. The marginal value of the fourth unit is:

- A. USD 900,000
- B. USD 1,693,072
- C. USD 493,072 CORRECT
- D. USD 564,357

*Rationale:*  $2,593,072 - 2,100,000 = 493,072$  (2.2.3b). A applies Gamma's best-in-set per-unit ratio to the marginal unit; B counts the whole NPV of the candidate newly admitted while ignoring that Beta is displaced; D is Meridian's own NPV per unit, which is an average and not a margin.



## SELF-CHECK — KA 2.2

1. *What must an options set contain to be genuine?* — A do-nothing baseline, a materially cheaper option, a materially more ambitious one; and at least one that could plausibly have won.
2. *Why is breakeven adoption a better board sentence than NPV?* — It states the condition the value depends on, which is what the board can actually influence and monitor.
3. *When does ranking by value per unit of constraint fail?* — With lumpy candidates: the feasible sets must be enumerated, as Beta + Gamma shows.
4. *State Meridian's breakeven adoption on both bases and the conversion between them.* — 41.0460 % flat-equivalent, 45.0053 % ramped-basis; the ramped PV is 91.2027 % of the flat PV at the same steady state, and  $41.0460 = 0.912027 \times 45.0053$ .
5. *Which two errors in a business case run in opposite directions?* — A flat benefits profile flatters (Meridian by 2,114,198 of PV); a zero do-nothing baseline penalises (by 561,101).
6. *Why should a capacity decision be taken as a block?* — The marginal value of the constraint is non-monotone, so judging one unit at a time stops at +93,072 and never reaches the +493,072 available from three.
7. *What should a scoring model report alongside its ranking?* — The flip point, and where two candidates lie within it, the statement that this method does not separate them.

## KNOWLEDGE AREA 2.3

## Benefits, value and sustainability

Topics: 2.3.1 benefits mapping · 2.3.2 measures and baselines · 2.3.3 sustainability and ESG value · 2.3.4 assumptions and dependencies.

**2.3.1 Benefits mapping**

A **benefits map** is a chain, drawn explicitly, from what the project delivers to why anyone wanted it: **output** → **enabling change** → **outcome** → **benefit** → **strategic objective**. Domain 1 established the links; this KA draws them and assigns them.

The **enabling change** is the step most maps omit and most programmes fail on. Meridian's outputs (installed clinics) produce no benefit without training, workflow redesign, clinical-champion support and data migration — none of which are software, and several of which sit outside the project's authority. A benefits map that

leaps from output to benefit has hidden exactly the work that determines whether anything is realised, which is why Meridian's adoption stalled at 40 % while installation ran to plan.

Every element on the map carries an **owner** and, for outcomes and benefits, a **measure**. The rule from Domain 1 (KA 1.2.1) applies unchanged: one accountable name per outcome, and the benefits owner sits outside the project (2.1.3).

Each link can fail independently — and the highlighted column is the one most maps omit

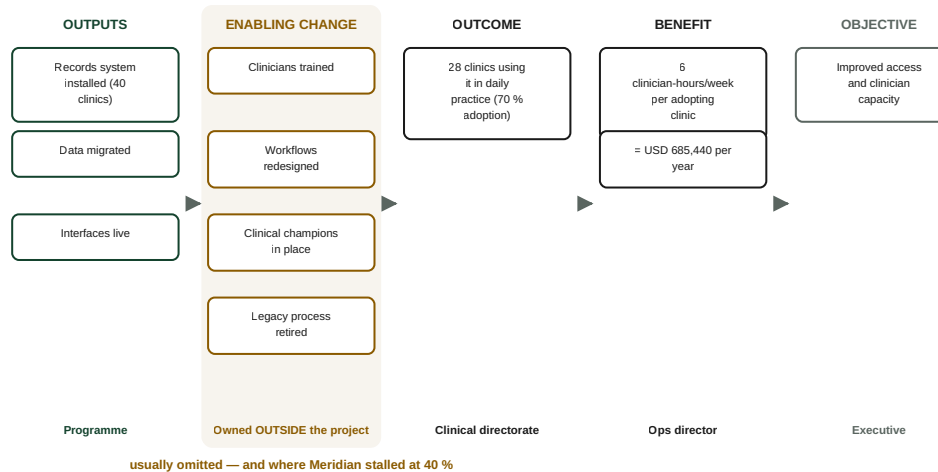


Fig 2.3.1 — Meridian's benefits map, with the enabling change restored

### 2.3.2 Measures and baselines

A benefit is only claimable against a **baseline**: the measured position before the change. Three disciplines make measurement honest:

- **Baseline before, not after.** Measure the current state before the change lands; a baseline reconstructed afterwards is an estimate shaped by the result it is used to justify.
- **Attribution.** Other things change too. Where a benefit could have other causes, say so and quantify what portion is plausibly attributable, over-claiming is the fastest way to lose the credibility that future cases depend on.
- **No double-counting.** The single commonest audit finding in benefits work: an enabling programme claims the benefit its dependent projects also claim, and the portfolio's total exceeds what the organisation could possibly realise. The portfolio benefits register (Domain 15) exists to catch this, and it only works if each benefit has exactly one claimant.

**Cash-releasing versus non-cash-releasing** benefits must be distinguished plainly. Meridian releases *clinician hours*; that becomes cash only if the organisation reduces cost or converts the hours into additional activity that has value. Reporting released hours as savings, when the headcount is unchanged and the hours are absorbed, is

the error that makes finance directors distrust benefits cases. And it is avoided by stating the conversion explicitly, or by claiming the benefit in its true unit (capacity) rather than in money it never became.

Attribution is the discipline of the three that is most often waved at and least often done, because doing it requires a **comparison cohort** (a group that did not receive the change), and choosing to keep one feels like withholding a benefit. It is worth the discomfort, and it is worth arithmetic.

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#### WORKED EXAMPLE

#### Worked example 2.3.2 – the measured benefit and the attributable benefit.

1. **Setup.** Twelve months after go-live, Meridian's benefits owner measures records-administration time per clinic per week. The pre-change baseline, measured before deployment, was **22.0 hours**. In the **28** adopting clinics it is now **15.6 hours**. In the **12** clinics that have not adopted (the comparison cohort) the same measure has fallen from 22.0 to **20.9 hours**, because an unrelated triage redesign was rolled out across the whole authority in the same period. The valuation basis is Domain 1's: **USD 85** an hour over a **48-week** operating year. The programme's business case assumed **6.0 hours** released per clinic per week. What may be claimed?
2. **Formula.** Raw improvement = baseline – post-change in the adopting group. Counterfactual improvement = baseline – post-change in the comparison group. Attributable improvement = raw – counterfactual (the difference of the two differences). Annual benefit = adopting clinics × attributable hours × rate × weeks.
3. **Substitution.** Raw  $22.0 - 15.6 = 6.4$ ; counterfactual  $22.0 - 20.9 = 1.1$ ; attributable  $6.4 - 1.1 = 5.3$ . Claimable  $28 \times 5.3 \times 85 \times 48$ ; the raw claim would be  $28 \times 6.4 \times 85 \times 48$ .
4. **Result.**

| BASIS                           | HOURS PER CLINIC PER WEEK | ANNUAL BENEFIT, 28 CLINICS |
|---------------------------------|---------------------------|----------------------------|
| Raw measured improvement        | 6.4                       | 731,136                    |
| Business-case assumption        | 6.0                       | 685,440                    |
| <b>Attributable improvement</b> | <b>5.3</b>                | <b>605,472</b>             |

The raw claim overstates the attributable benefit by **USD 125,664**, which is **17.1875 %** of the raw claim. 5. **Interpretation.** Five readings, and the first is the one that catches an unwary benefits owner in a good year.

**A measurement that beats the case can still fail to support it.** The raw figure, 6.4 hours, exceeds the case's 6.0, so the natural report is "benefits exceeding forecast". The attributable figure, 5.3 hours, is **11.67 % below** the case, and the case's 6.0 is **13.21 % higher** than what can honestly be claimed. Raw and attributable are different quantities, and it is the raw one that programmes report because it is the one the programme's own data produces. **The comparison cohort is what converts a measurement into evidence**, and a programme without one cannot distinguish its own effect from everything else that happened that year, in either direction.

**The over-claim share is an invariant, and it is worth knowing in a meeting.** Because the raw and attributable figures share the clinic count, the rate and the weeks, the over-claim as a fraction of the raw claim is  $\text{counterfactual} \div \text{raw} = 1.1/6.4 = 17.1875\%$ : irrespective of the valuation rate, the operating year or how many clinics adopted. It is the exact structural twin of Domain 1's adoption identity, where an output-based claim overstates by  $1 - a$  whatever the other numbers are (KA 1.3.2). Both identities exist for the same reason and are used the same way: **you do not need the model to say how wrong a claim is, only the term it omitted.**

**The correction moves the whole case, not just this year's report.** At 5.3 attributable hours, full potential falls from 979,200 to  $40 \times 5.3 \times 85 \times 48 = \text{USD } 864,960$ , which is **88.3333 %** of the original, exactly  $5.3/6.0$ , since nothing else changed. The flat-equivalent breakeven adoption therefore *rises* from 41.0460 % to  $2,400,000 / (864,960 \times 5.971299) = 46.4672\%$ , a deterioration of **5.4212 percentage points**. Taken together with the do-nothing correction of 2.2.2b, which improved the breakeven by 9.5962 points, the fully corrected flat-equivalent breakeven is  $(2,400,000 - 561,101) / (864,960 \times 5.971299) = 35.6035\%$ ; and Meridian's achieved 40 % adoption clears it with **4.3965 points** to spare, for an NPV of **+USD 227,074**. **Two errors of opposite sign, and you must fix both: fixing only the flattering one produces a verdict as wrong as fixing neither.**

**Choosing a comparison cohort is a real decision with real costs, and it should be made deliberately.** The 12 non-adopting clinics were not held back for measurement (they simply had not adopted), which makes them a **convenience cohort**, and convenience cohorts differ systematically from the adopters: the clinics that adopt first are usually the better-led, better-staffed ones, so part of the 5.3 may still be selection rather than software. A deliberately staged rollout gives a defensible cohort as a free by-product of the sequencing decision, which is another benefit of the tranching argued in 2.1.3, and it is one of the strongest arguments a benefits owner has against a simultaneous switch-on. Where withholding a change is not acceptable (and in clinical or safety settings it frequently is not) the honest alternatives are a pre-change trend line extended forwards, or a stated range with the attribution assumption on the face of the

report. **What is not acceptable is claiming the raw figure and calling the question unanswerable.**

**What breaks it.** The two groups must be measured the **same way at the same time**; a post-change measurement method that is more thorough in adopting clinics manufactures an effect. A counterfactual improvement can also be *negative* (the comparison group getting worse) in which case the attributable figure exceeds the raw one and the honest report is a larger claim, which is the case a leader should be most careful to actually make. Attribution and adoption interact and must not be double-corrected: the 5.3 hours is a per-adopting-clinic figure, and multiplying by 28 already applies the adoption term, so applying 70 % again is a common and severe error. And none of this converts the benefit to cash: 605,472 of attributable released time is **capacity**, and the cash-releasing test above still has to be passed separately.

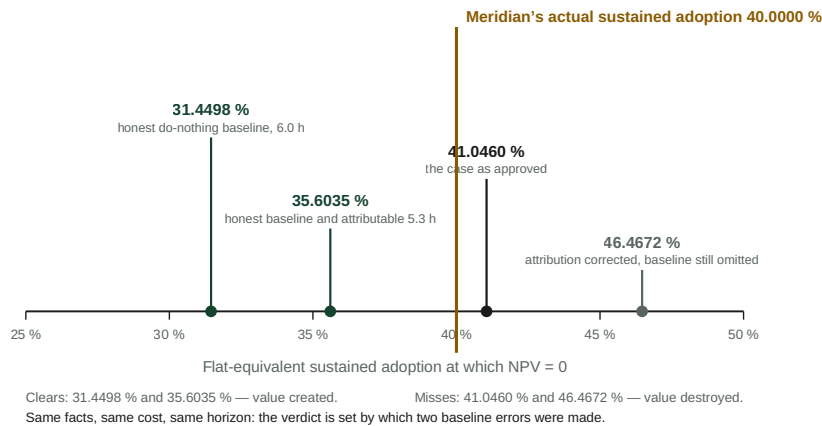


Fig 2.3.2 — Four breakevens from one programme

### 2.3.3 Sustainability and ESG value

Environmental, social and governance considerations enter this domain in two distinct ways, and conflating them causes most of the confusion.

**As constraints and obligations:** emissions limits, environmental permits, accessibility law, labour standards in the supply chain (Domain 10's ethical sourcing). These are hard constraints (2.1.2) and belong in the case as requirements, not benefits.

**As value:** whole-life carbon reduction, energy cost avoided, social value created, resilience to climate risk. These belong in the benefits map with measures and owners like any other benefit; and with the same honesty about attribution and monetisation. Where a carbon reduction has a price (a compliance cost avoided, a levy, an internal carbon price), it can be valued directly; where it does not, it should be reported in its physical unit alongside the financial case rather than converted with an invented price. **Whole-life thinking** is the practical contribution: a cheaper capital solution with higher operating energy and shorter life frequently loses on a

whole-life basis, and the appraisal horizon of KA 2.2.2 is what makes that visible (PFL-AI Domain 4's equivalent annual value handles unequal lives).

**As a disclosed claim:** the third way, and the one most often missed, because it is not an appraisal question at all. A benefit that stays inside the appraisal is an estimate a board may accept or challenge on the evidence. The moment the same benefit is reported *outside* the organisation (in a sustainability or annual report, in a bid or prequalification response, in a submission to a regulator, or in a financing or investor document) it stops being an estimate and becomes a **claim**, and a claim is held to a standard a forecast is not. Four provisions follow, and they are stated as professional obligations rather than as anyone's legal position:

- **A stated boundary and method.** What is counted, over what period, against which baseline, on whose emission or valuation factors, and what is deliberately excluded. A carbon or social-value number without its boundary is not a measure: two organisations applying different boundaries produce different figures from identical facts, and neither figure is checkable by the reader.
- **A named owner.** One person accountable for the figure: the rule KA 2.3.2 applies to every other benefit measure, and it does not weaken because the unit is tonnes or apprenticeship weeks rather than currency.
- **Retained evidence sufficient for someone else to test it.** External assurance, where it is obtained, tests the evidence rather than the assertion, so a figure whose working papers were a spreadsheet on one laptop cannot be assured whatever its accuracy. Domain 16, KA 16.4.4's retention economics apply directly, and the record class, custodian and retention period are set *before* the claim is published rather than after it is questioned. The benefits measurement plan of Domain 16, KA 16.4.1 is where a claimed benefit acquires the measured evidence that later supports it.
- **Approval by whoever signs the disclosure.** The project does not approve an external claim. That approval belongs to the function that owns the report, the bid or the submission; the project's obligation is to supply the measure, the boundary, the method and the evidence in a form that function can sign, and to say plainly where the evidence stops.

**The professional prohibition, in one line.** *A benefit that cannot be evidenced to the standard its intended audience requires is not reported as achieved.* Report the measured part as measured, the estimated part as estimated with its method beside it, and the unevidenced part not at all. Presenting a forecast as a result, counting a benefit the organisation did not cause, or moving the boundary until the number improves are the attribution failures of KA 2.3.2 committed in front of an audience that cannot see the working, which is why the profession treats the disclosed version as the graver of the two, and why the leader who supplies the number needs a written record of what was supplied and to whom.

**The standing caveat, in this volume's usual form.** Which external disclosure and assurance obligations apply, to which entities, on what timetable, in what form, and with what consequence for an unsupported claim, is jurisdiction- and entity-specific, changes, and is taken from the organisation's reporting function and from qualified counsel, never from this book. Nothing here states the position in any jurisdiction, states what any regime requires, or characterises any organisation's disclosure as compliant or otherwise. One distinction is worth holding because it is routinely blurred: a **voluntary reporting framework** is something an organisation chooses to adopt, and a **disclosure regime** is something that applies on its own terms to those it reaches; adopting the first tells you nothing about the second, and the two questions go to the same advisers separately. Where a quantified greenhouse-gas figure is being prepared, **ISO 14064** is the document usually named as addressing the quantification and reporting of greenhouse-gas emissions and removals; it is voluntary guidance, it is named here and not reproduced, and it is not itself a disclosure obligation.

#### 2.3.4 Assumptions and dependencies

**Business cases fail through their assumptions**, and the assumptions are usually stated once and never revisited. Meridian's case rested on an adoption assumption that was never tracked as a measure: the direct cause of everything Domain 1's case study describes.

An **assumption register** carries, per assumption: the statement, why it is believed, the impact if false, the owner, the date it will be tested, and the **trigger** that would falsify it. Its relationship to the risk register is exact: **every assumption is a risk in disguise** (Domain 8, KA 8.1.3), so material assumptions get register entries with **EMV** where quantifiable.

Quantifying them is the step that turns the register from a list into an instrument, because it permits two questions no list can answer: *is the case's promised value large relative to the exposure carried by its own assumptions?* and *which assumption should be tested first?*

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#### WORKED EXAMPLE

##### Worked example 2.3.4 – pricing Meridian's assumption register.

1. **Setup.** Meridian's honest case (2.2.2) gives a ramped PV of benefits of **3,732,898** against a cost of 2,400,000 (an NPV of **+1,332,898**), and, per unit of sustained adoption, a PV of **5,332,711**. Five material assumptions underpin it. Each has an assessed impact if false, in present-value terms, and a probability of being false, both **assessed by the programme team and recorded as judgements, not measurements**. The cost of testing each is also known.

2. **Formula.**  $EMV = \text{probability} \times \text{impact}$ . Assumption exposure ratio  $= \Sigma EMV \div NPV$ . Test priority ranks by  $EMV \div \text{cost to test}$  (infinite where the test is free, so those go first).

3. **Substitution.** A1: adoption falling from 70 % to 45 % costs  $0.25 \times 5,332,711 = 1,333,178$ . A2: attributable hours  $6.0 \rightarrow 5.3$  costs  $3,732,898 \times (1 - 5.3/6.0) = 435,505$ . A3: the legacy remediation is not in fact avoided,  $600,000/1.07^4 = 457,737$ . A4: the enabling change must be programme-funded, 320,000. A5: finance values clinician time at 70 rather than 85,  $3,732,898 \times (1 - 70/85) = 658,747$ .

#### 4. Result.

|    | ASSUMPTION                                   | IMPACT<br>IF FALSE<br>(PV) | P(FALSE) | EMV              | COST<br>TO<br>TEST | EMV<br>PER<br>UNIT<br>OF<br>TEST<br>COST |
|----|----------------------------------------------|----------------------------|----------|------------------|--------------------|------------------------------------------|
| A1 | Sustained adoption reaches 70 %              | 1,333,178                  | 0.35     | <b>466,612</b>   | 40,000             | 11.67                                    |
| A2 | 6.0 attributable hours per clinic-week       | 435,505                    | 0.50     | <b>217,752</b>   | 15,000             | 14.52                                    |
| A3 | Legacy remediation genuinely avoided         | 457,737                    | 0.25     | <b>114,434</b>   | 0                  | —<br>(free)                              |
| A4 | Enabling change funded outside the programme | 320,000                    | 0.45     | <b>144,000</b>   | 0                  | —<br>(free)                              |
| A5 | Finance accepts USD 85 an hour               | 658,747                    | 0.20     | <b>131,749</b>   | 0                  | —<br>(free)                              |
|    | <b>Total</b>                                 | 3,205,166                  |          | <b>1,074,548</b> | 55,000             |                                          |

Total EMV is **USD 1,074,548**, which is **80.62 %** of the NPV the case promises. A1 alone is **35.01 %** of it. The three free tests together carry **USD 390,184: 36.31 %** of the register's exposure. 5. **Interpretation.** Five readings, and the first is the single most useful sanity test on any business case.

**The exposure ratio tells a board how thin its own case is.** A positive NPV of 1,332,898 sitting above an assumption exposure of 1,074,548 is a ratio of **0.8062**: the case is right-side-up, but only just, and a board told "NPV +1.33m" hears something quite different from a board told "NPV +1.33m against 1.07m of expected exposure in the assumptions it rests on". The ratio also gives a usable



rule of thumb for challenge: **where  $\Sigma$  EMV approaches or exceeds the NPV, the case is not a value proposition but a bet**, and it should be structured as one — staged, with kill criteria (2.4.3) and a real abandonment option (2.A.1) — rather than approved as a commitment.

**The test-order rule is where the leader gets something for nothing.** Three of the five assumptions cost nothing to test: A3 is a letter to the vendor asking whether support will be extended; A4 is a written funding confirmation from the clinical directorate; A5 is a conversation with the finance business partner about the valuation rate. Together they carry **390,184** of exposure and can be resolved in a week, before approval. Ranking by **EMV ÷ cost to test** rather than by EMV alone is what surfaces that, and it inverts the usual instinct, which is to spend the analysis budget on the biggest number: A1; and to leave the cheap ones as "assumptions". **The professional habit: sort the register by exposure per unit of test cost, and clear the free rows before the paper is written.**

**The one that must be structural, not analytical, is adoption.** A1 cannot be resolved by analysis, only by evidence from operation, which is why its test costs 40,000 (a measured pilot cohort) and why it remains the largest residual after the free rows are cleared: 466,612 of a remaining 684,365. An assumption that can only be tested by doing part of the work is the definition of a case for staging, and 2.A.1 prices exactly that trade for Meridian. Note the symmetry with the alignment re-test of 2.1.3: both convert an unmanaged belief into a scheduled observation, and both are worth far more than the sophistication of the analysis around them.

**EMVs are not additive as a portfolio loss, and saying they are is a real error.** A1 (adoption) and A2 (attributable hours) are both statements about the same clinical behaviour and will fail together more often than independently; A5 (the valuation rate) is largely independent of both. So the 1,074,548 is the correct *expected* total (expectations add regardless of correlation), but it is **not** the right basis for a downside statement, because the correlated pair makes the tail worse than an independence assumption implies. Domain 8, KA 8.A.1 handles the correlation properly and KA 8.3 sizes contingency from it; the register's job here is to state the expected exposure and flag which entries move together.

**What breaks it, and the honest limits.** Every probability in the table is a judgement, and a register whose probabilities were set by the case's author will be systematically low: the same advocacy problem 2.2.1 identifies, appearing here as calibration rather than as omission. Impacts assessed as single points hide their own ranges. EMV also treats all failures as equivalent losses, when some are recoverable and some are not: A4 failing costs 320,000 and is survivable, while A1 failing changes what the programme *is*, and an expected-value table cannot express that difference, Domain 8's irreversibility treatment must be read alongside it. And a register is only alive if the test dates are in someone's calendar with a named owner; an assumption register reviewed at the same cadence as the risk register, by the same forum, with the same escalation,

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is the version that works, and a separate document nobody opens is the version that exists in most organisations.

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**Dependencies** (on other projects, on suppliers, on decisions not yet taken) are managed by naming them, naming the owner on the *other* side, and stating the date by which the dependency must be satisfied. Two failure modes recur: **assumed dependencies** (nobody on the other side knows they owe you anything) and **unmanaged reciprocals** (they are depending on you too, on a different date). Domain 15's programme dependency management handles the multi-project case; at project level the discipline is simply that a dependency without a named counterpart is a hope.

### AI in this KA

Benefits mapping is judgment work about causation, which is where AI is weakest: a model will happily assert that an output produces a benefit because most documents assert exactly that (Domain 1's KA 1.3 warning). Two legitimate uses. **Challenge:** ask a model to argue that the benefit will *not* materialise, and the enabling changes it names are often the ones the map omitted. **Cross-checking at portfolio scale:** detecting the same benefit claimed by two cases is pattern-matching over documents, which machines do well and humans do badly across fifty cases. The conclusions stay human, with named benefit owners.

**Verification, concretely.** Two specific hazards attach to the arithmetic of this KA. A model asked to compute an attributable benefit will happily do the subtraction and will **not** ask whether the comparison group is a fair counterfactual, so the professional check is on the cohort, not on the subtraction: were both groups measured the same way at the same time, and is the non-adopting group systematically different from the adopting one? That is a judgement about selection, and no amount of arithmetic substitutes for it. And a model asked to populate an assumption register will supply **probabilities**, fluently and without provenance, which then drive an exposure ratio that a board reads as analysis. Where the organisation has no history to calibrate from, the correct output is that it has none, plus the range of probability over which the recommendation is unchanged. What is safe to delegate is the reconciliation work: checking that every benefit in the register has exactly one claimant across the portfolio, that no avoided cost appears also as a benefit, and that the adoption term has been applied once rather than twice: all three are mechanical, all three are commonly wrong, and all three are reported with a location so a human can confirm them.

## KEY TERMS — KA 2.3

| TERM                                            | MEANING                                                                                                                                                                                         |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Benefits map</b>                             | Output → enabling change → outcome → benefit → objective, with owners and measures.                                                                                                             |
| <b>Enabling change</b>                          | The non-project work (training, workflow, adoption support) without which no benefit occurs.                                                                                                    |
| <b>Baseline</b>                                 | The measured pre-change position; must be measured before, not reconstructed after.                                                                                                             |
| <b>Attribution</b>                              | The portion of a measured improvement plausibly caused by this project.                                                                                                                         |
| <b>Comparison cohort</b>                        | A group that did not receive the change, used to measure the counterfactual improvement.                                                                                                        |
| <b>Attributable improvement</b>                 | Raw improvement less the counterfactual improvement; the difference of the two differences.                                                                                                     |
| <b>Over-claim share</b>                         | $\text{counterfactual} \div \text{raw}$ , invariant to the valuation rate, the operating year and the number of adopters.                                                                       |
| <b>Double-counting</b>                          | The same benefit claimed by more than one case; one claimant per benefit.                                                                                                                       |
| <b>Cash-releasing / non-cash-releasing</b>      | Benefits that reduce spend versus those that release capacity.                                                                                                                                  |
| <b>Assumption register</b>                      | Statement, basis, impact-if-false, owner, test date, falsifying trigger.                                                                                                                        |
| <b>Assumption exposure ratio</b>                | $\Sigma \text{EMV} \div \text{NPV}$ ; as it approaches one, the case is a bet rather than a value proposition.                                                                                  |
| <b>Test-order rule</b>                          | Rank assumptions by $\text{EMV} \div \text{cost to test}$ , so the free tests are cleared before the paper is written.                                                                          |
| <b>Disclosed claim</b>                          | A benefit reported outside the organisation; needs a stated boundary and method, a named owner, retained evidence and the signing function's approval, and is never approved by the project.    |
| <b>Claim boundary</b>                           | What a sustainability or social-value measure counts and excludes, over what period and against which baseline; without it the number is not checkable.                                         |
| <b>Voluntary framework vs disclosure regime</b> | A framework is adopted by choice; a regime applies on its own terms. Adopting the first says nothing about the second, and both are questions for the reporting function and qualified counsel. |

## SAMPLE MCQS — KA 2.3

**MCQ 2.3-A [2.3.1 · Analysis]** A benefits map runs directly from "system installed" to "USD 685,440 released per year". Its principal defect is:

- A. the benefit figure is too precise

B. the **enabling change** is missing — training, workflow redesign and adoption support, largely owned outside the project, are what convert the output into the outcome CORRECT

C. it should be expressed in hours rather than money

D. the map should start with the strategic objective

*Rationale:* Omitting enabling change hides the work that determines realisation (2.3.1), exactly how Meridian stalled at 40 % adoption with installation on plan. C is a separate (also real) question of unit; D is presentational.

**MCQ 2.3-B** [2.3.2 · Application] A programme releases 6 clinician-hours per week per clinic; headcount is unchanged and the hours are absorbed by existing demand. Reporting this as a cash saving is:

A. correct — released time has a value

B. incorrect: it is a non-cash-releasing (capacity) benefit unless cost is reduced or the capacity is converted to valued activity CORRECT

C. correct if the hourly rate is documented

D. acceptable provided it is discounted

*Rationale:* Value released is not cash released (2.3.2); claiming otherwise is the error that discredits benefits cases. A and C mistake a valuation rate for a cash effect; D discounts a figure that was never cash.

**MCQ 2.3-C** [2.3.4 · Analysis] An enabling platform programme and three dependent projects each claim the same USD 4m of downstream benefit. The portfolio total is:

A. USD 16m, since each case is individually valid

B. overstated by double-counting — one benefit, one claimant, with the enabler credited through the dependents or vice versa but not both CORRECT

C. USD 4m, and the dependent projects have no benefits

D. indeterminate until delivery completes

*Rationale:* Double-counting is the standing audit finding (2.3.2); the fix is a single claimant per benefit in the portfolio register. C overcorrects by denying the dependents any case; D defers a question answerable now.

**MCQ 2.3-D** [2.3.3 · Application] A carbon reduction has no priced compliance consequence for the organisation. The soundest treatment in the case is:

A. omit it, since it cannot be valued

B. report it in its physical unit alongside the financial case, rather than monetising it with an invented price CORRECT

C. monetise it using a rate found in a published study

D. record it as a constraint

*Rationale:* Unpriced benefits are reported honestly in their own unit (2.3.3). A discards real value; C imports a price the organisation does not face; D confuses value with obligation.

**MCQ 2.3-E** [2.3.2 · Application] A pre-change baseline of 22.0 hours a week falls to 15.6 in the 28 adopting clinics and to 20.9 in the non-adopting comparison clinics. At USD 85 an hour over 48 weeks, the attributable annual benefit across the adopters is:

- A. USD 731,136
- B. USD 605,472 CORRECT
- C. USD 685,440
- D. USD 125,664

*Rationale:* Attributable release is  $6.4 - 1.1 = 5.3$  hours, so  $28 \times 5.3 \times 85 \times 48 = 605,472$  (2.3.2). A is the raw claim with no counterfactual deducted; C is the business case's 6.0-hour assumption, which is a forecast and not a measurement; D is the over-claim itself, not the benefit.

**MCQ 2.3-F** [2.3.2 · Analysis] A programme measures a raw improvement above its business-case assumption and reports benefits exceeding forecast. The comparison cohort improved by 1.1 of the 6.4 hours. The strongest professional statement is that:

- A. benefits exceed forecast, since the measured improvement is larger than assumed
- B. the attributable improvement is 5.3 hours — below the assumed 6.0 — so the case is not supported, and the over-claim is 17.1875 % of the raw figure whatever the valuation rate CORRECT
- C. the result is unusable because no randomised comparison exists
- D. the counterfactual should be added to the claim, since both improvements are real

*Rationale:* Raw and attributable are different quantities, and the over-claim share is  $\text{counterfactual} \div \text{raw}$ , invariant to the rate, weeks and clinic count (2.3.2). A is the error the raw figure invites in a good year; C over-reaches: a convenience cohort is weak evidence, not no evidence; D claims an improvement the programme did not cause.

**MCQ 2.3-G** [2.3.4 · Evaluation] An assumption register carries EMVs of 466,612, 217,752, 114,434, 144,000 and 131,749 against a case NPV of 1,332,898; the last three cost nothing to test. The soundest first action is:

- A. commission the 40,000 adoption study, since 466,612 is the largest exposure
- B. clear the three free tests, resolving USD 390,184 — 36.31 % of the register's exposure — before the paper is written CORRECT

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C. reduce the case's NPV by the total EMV of 1,074,548 and re-submit

D. no action: an exposure ratio of 0.8062 is below 1, so the case stands

*Rationale:* Test priority ranks by EMV per unit of test cost, and a free test has no competitor (2.3.4). A is right on magnitude and wrong on order; C double-counts, since the EMVs are exposures around the NPV, not deductions from it; D treats a sanity ratio as an approval threshold.

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SELF-CHECK — KA 2.3

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1. *Which element do most benefits maps omit, and why does it matter most?* — The enabling change; it is the work that converts outputs into outcomes and it usually sits outside the project.
  2. *What makes a baseline trustworthy?* — Measurement before the change, not reconstruction after.
  3. *State the relationship between assumptions and risks.* — Every assumption is a risk in disguise; material ones get register entries with quantified exposure.
  4. *How is an attributable improvement computed, and what is the invariant?* — Raw improvement less the counterfactual improvement; the over-claim share is  $\text{counterfactual} \div \text{raw}$ , 17.1875 % on Meridian, whatever the rate, weeks or number of adopters.
  5. *Why must both of Meridian's baseline errors be corrected, not one?* — The flat profile flatters and the missing counterfactual penalises; correcting only one gives a breakeven of 46.4672 % or 31.4498 %, where the honest figure is **35.6035 %**.
  6. *What does an assumption exposure ratio approaching one tell a board?* — That the case is a bet rather than a value proposition, and should be staged with kill criteria rather than committed.
  7. *What changes when a sustainability or social-value benefit is reported outside the organisation?* — It becomes a claim rather than an estimate: it needs a stated boundary and method, a named owner, retained evidence someone else can test, and approval from the function signing the disclosure. The project supplies it; the project does not approve it.
  8. *State the professional prohibition on disclosed benefits.* — A benefit that cannot be evidenced to the standard its intended audience requires is not reported as achieved. Which external obligations apply, and to whom, is for the reporting function and qualified counsel, not for this book.
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## KNOWLEDGE AREA 2.4

## Strategic termination

*Topics: 2.4.1 stopping as a strategic decision · 2.4.2 sunk cost and escalation of commitment · 2.4.3 kill criteria and honest gates.*

### 2.4.1 Stopping as a strategic decision

Organisations that cannot stop projects cannot really select them either: a portfolio that only accretes has no capacity for anything new, and its selection process becomes a queue. Stopping is therefore a **portfolio capability**, not an admission of failure; and the leaders who can do it create the capacity that funds everything else.

The barriers are cultural and structural, and both are addressable. **Culturally**, cancellation is read as failure and punished, so nobody proposes it; the counter is to celebrate *early* stopping explicitly and to distinguish "we learned this will not work" from "we managed this badly". **Structurally**, no gate has authority to stop, or the decision is scheduled after the money is committed; the counter is Domain 3's decision rights with a real stop option and gates placed before irreversible commitments (Domain 1's irreversibility point).

### 2.4.2 Sunk cost and escalation of commitment

**The principle.** A continuation decision depends only on **remaining cost and remaining benefit**. Money already spent is irrelevant to it; it cannot be recovered by spending more, and it carries no information about the future beyond what is already in the forecast.

## WORKED EXAMPLE

#### Worked example 2.4.2 – Meridian at the reset point.

1. **Setup.** A hypothetical reset: USD 1,800,000 of the 2,400,000 has been spent. Completion requires a further **USD 900,000** (the original 600,000 plus 300,000 of newly discovered integration work). The benefits case has been re-based on measured adoption, giving a remaining benefit present value of **USD 780,000**. Continue or stop?
2. **Formula.** Forward-looking NPV = remaining benefit PV – remaining cost. Sunk cost excluded.
3. **Substitution.** 780,000 – 900,000.
4. **Result. Forward NPV = (USD 120,000).** The correct decision is to **stop** (or to descope to a variant whose remaining benefit exceeds its remaining cost).
5. **Interpretation.** Five readings, and the fourth is the one that keeps a leader from over-applying the rule.

**The sunk-cost frame does not merely mislead; it multiplies the loss by 7.5.** The pull to continue is powerful and entirely arithmetical in appearance: "we have spent 1,800,000 — stopping wastes it." That reasoning adds a sunk 1,800,000 back into a decision it cannot affect, and it converts a **120,000** loss into a **900,000** one. The 1,800,000 is already gone under *either* choice; the only question is whether the *next* 900,000 buys 780,000 of value. It does not.

**State the decision as a threshold, because a threshold can be checked.** The continuation is worth taking at any remaining cost below **780,000**, so the 900,000 estimate exceeds the justified ceiling by **USD 120,000**, or **15.38 %** of it. Equivalently, continuation needs the remaining benefit to be at least 900,000, which is **15.38 %** more than the re-based 780,000. That is a far better gate sentence than a negative NPV, because it tells the meeting exactly what would have to be true. And it is testable: a descope that removes 120,000 of remaining cost without removing benefit turns the decision round, which is why "stop" and "descope" belong in the same paper (Case study B does exactly this at scale).

**Stopping releases the money into the constrained selection problem, which is where its real value appears.** The 900,000 is redeployable to candidates the portfolio could not previously fund (KA 2.2.3's constraint arithmetic), and if the released capacity also includes the scarce team, 2.2.3b prices what that is worth, which may exceed the money. **This is why stopping is a strategic act and not merely a loss**, and it is also why the paper must say what the released resources will do next: a board asked to accept a write-off with no redeployment proposal is being asked to take the whole loss and none of the gain.

**The rule is about unrecoverable spend, and three things are commonly mis-sorted into it.** Sunk cost is spend that cannot be recovered *and* carries no information about the future beyond what the forecast already holds. Recoverable amounts are not sunk and belong in the forward arithmetic: resale or redeployment value of equipment and licences, recoverable prepayments, and reusable assets such as cleaned data or a qualified team. Contractual exit costs are not sunk either: they are a **forward** cost of stopping, and if terminating the supplier costs 150,000 the comparison is  $780,000 - 900,000 = (120,000)$  for continuing against  $(150,000)$  for stopping, and the correct decision reverses. And a cost already incurred *does* carry information where it revises the forecast: an overspend that reveals the work is harder than believed should change the estimate of the remaining 900,000, which is a legitimate use of history and not the sunk-cost fallacy. **The discipline is to exclude sunk cost from the comparison while using the evidence it generated.**

**What breaks it.** The remaining-benefit figure must be re-based on measured evidence, not inherited from the original case: using the approval-time benefit here would give  $3,732,898 - 900,000$  and a spurious continuation. Both sides must be on the same present-value basis and the same horizon. And the



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decision must be available: where an obligation is contractual or statutory, there is no stop option to exercise and the arithmetic is describing a choice nobody holds, in which case the honest output is the *size of the loss* and a claim or renegotiation strategy (Domain 10), not a recommendation to stop. **What a termination actually costs, and whether the exit right exists at all, are questions of the particular contract and the governing law: take legal advice on the exit position before the 150,000 goes into a gate paper as a number.**

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**Escalation of commitment** is the behavioural pattern that makes this hard: personal identification with the project, reputational exposure for the approver, and the asymmetry that continuing defers the reckoning while stopping realises it now. Its countermeasures are structural, not exhortative: pre-set kill criteria (2.4.3), decision-makers who did not author the case, and forward-only framing in every gate paper (Domain 8's bias table).

### 2.4.3 Kill criteria and honest gates

**Kill criteria** are conditions, agreed **in advance**, under which the project will stop: "if adoption in the pilot cohort is below 50 % at month six", "if the regulatory approval is not granted by Q3", "if remaining cost exceeds remaining benefit at any gate". Their power is entirely in the *advance* agreement: the same evidence assessed against a pre-agreed threshold is a decision, while assessed against no threshold it is a negotiation, and negotiations are won by whoever has most invested.

That establishes *why* a criterion works. It says nothing about **where the number goes**, and a criterion set at the wrong level is worse than none, because it stops the wrong projects with full procedural authority. Setting it is an arithmetic problem with two error costs.

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#### WORKED EXAMPLE

**Worked example 2.4.3 – calibrating the kill criterion, and the two errors it trades.**

1. **Setup.** Meridian's successor programme carries a kill criterion at the **month-6** gate, by which point **USD 900,000** of the 2,400,000 has been committed and **USD 1,500,000** remains. The month-6 signal is early adoption in the first cohort, which classifies a programme as **Weak** or **Strong**. From the authority's own history of comparable rollouts, eventual sustained adoption falls into three states — **Low 25 %** (probability 0.25), **Mid 45 %** (0.45), **High 70 %** (0.30) — and the signal is imperfect: it reads Weak for **80 %** of Low programmes, **35 %** of Mid programmes and **10 %** of High programmes. All figures are **locally calibrated judgements from a small sample and must be labelled as such**. Should the criterion stop programmes flagged Weak?

2. **Formula.** Forward NPV at the gate = PV of remaining benefit – remaining cost, where the PV of the remaining benefit at sustained adoption  $s$  is  $5,332,711 \times s$  (the ramped PV per unit of sustained adoption, 2.2.2). Forward breakeven adoption = remaining cost  $\div$  5,332,711. Then compare two policies: **A** continue everything,  $E[NPV] = \sum P(s) \times NPV(s)$ ; **B** stop on Weak,  $E[NPV] = \sum P(s) \times P(\text{Strong}|s) \times NPV(s)$ , taking a stopped programme's forward value as zero.
3. **Substitution.** Forward breakeven  $1,500,000 / 5,332,711$ .  $NPV(0.25) = 1,333,178 - 1,500,000$ ;  $NPV(0.45) = 2,399,720 - 1,500,000$ ;  $NPV(0.70) = 3,732,898 - 1,500,000$ . Policy B:  $0.25 \times 0.20 \times (-166,822) + 0.45 \times 0.65 \times 899,720 + 0.30 \times 0.90 \times 2,232,898$ .
4. **Result.** Forward breakeven adoption at this gate is **28.1283 %**, against a whole-investment breakeven on the same ramped basis of **45.0053 %**.

| EVENTUAL STATE | PRIOR | FORWARD NPV | P(WEAK) | EFFECT OF STOPPING ON WEAK |
|----------------|-------|-------------|---------|----------------------------|
| Low 25 %       | 0.25  | (166,822)   | 0.80    | loss avoided +33,364       |
| Mid 45 %       | 0.45  | 899,720     | 0.35    | value destroyed (141,706)  |
| High 70 %      | 0.30  | 2,232,898   | 0.10    | value destroyed (66,987)   |

Policy A (continue everything)  $E[NPV] = \text{USD } 1,033,038$ . Policy B (stop on Weak)  $E[NPV] = \text{USD } 857,709$ . The criterion **destroys USD 175,328** of expected value: it saves 33,364 and costs 208,693. 5. **Interpretation.** Six readings. The first two are the calibration errors that produce most bad criteria; the last reconciles this result with everything 2.4.3 says in favour of kill criteria.

**A criterion set at the investment breakeven kills programmes that are still worth completing.** The intuitive threshold ("stop unless it will clear the adoption the business case needed") is **45.0053 %** on this basis. The threshold that maximises value at *this* gate is **28.1283 %**, because 900,000 is already sunk and the decision is about the next 1,500,000 only. A criterion pitched at the investment breakeven condemns every programme whose eventual adoption lands between 28.1283 % and 45.0053 % (a **16.8770-point** band containing the entire Mid state) even though completing them returns 899,720 each. **The forward breakeven is the only defensible level for a criterion applied at a gate**, and the fact that it moves later in delivery is a feature: a criterion should get harder to trigger as the remaining cost falls, which is the exact opposite of how most escalating governance behaves.

**The signal, not the threshold, is what makes this criterion fail.** Even at the right threshold, the criterion is applied to a proxy. Of the programmes it flags Weak, most are not Low: the flagged population is  $0.25 \times 0.80 = 0.20$  Low,  $0.45 \times 0.35 = 0.1575$  Mid and  $0.30 \times 0.10 = 0.03$  High, so **48.39 %** of everything it stops would have returned positive value. The arithmetic

requirement is severe and worth stating exactly: even a criterion that detected **every** Low programme and never flagged a High one could tolerate a false-flag rate on Mid programmes of no more than  $41,706 / 404,874 = 10.3009\%$ . A proxy signal that good does not exist at month six on a two-year rollout, which is the honest reason early adoption thresholds so often do damage.

**The asymmetry is structural, not a quirk of these numbers.** The loss avoided by stopping a bad programme is bounded by how negative its forward NPV is (here 166,822), while the value destroyed by stopping a good one is its whole remaining upside, 899,720 or 2,232,898. When the upside is an order of magnitude larger than the downside, a criterion has to be far more accurate than intuition suggests. The corollary, and it is important, is that **the asymmetry reverses where the downside is large**: a programme whose failure carries a regulatory penalty, a safety consequence or a contractual liability has a much more negative forward NPV, and the same signal quality then makes the criterion clearly worth having. **Kill criteria pay where failure is expensive, not where success is uncertain.**

**The fix is not to abandon the criterion but to change what it does.** Three changes each make it value-positive. Set the threshold on the **decision-relevant quantity** (forward NPV computed from the re-based benefit case) rather than on a proxy, which is what the third of 2.4.3's example criteria ("if remaining cost exceeds remaining benefit at any gate") already does and is why it is the strongest of the three. Where a proxy is unavoidable, make the criterion trigger a **mandatory re-appraisal with a real stop option** rather than an automatic stop: the re-appraisal costs a few tens of thousands and removes the false-stop loss almost entirely, which on these figures is a trade of 208,693 against a fraction of it. And **improve the signal by buying information before the money is committed**, which is precisely the staged commitment 2.A.1 prices, and the reason real options and kill criteria are two halves of one instrument rather than two topics.

**What this does not license.** Nothing here argues against pre-agreed criteria, and the reasoning must not be borrowed to defend a portfolio that never stops anything. The result is narrower and sharper: **a criterion is an instrument with a computable expected value, and it must be calibrated rather than asserted.** The gate that has never stopped a project (MCQ 2.4-B) is failing a different test: it has no stop option at all, and this arithmetic assumes throughout that the option exists and can be exercised. A leader who cannot compute the two error costs should still insist on criteria; a leader who can should insist on the right ones.

**The limits of the model.** It assumes a stopped programme has zero forward value, which understates Policy B wherever partial delivery is usable or the released capacity has an alternative use worth more than nothing; reinstating either shrinks the 175,328 and could reverse it, and the honest version of this table has a redeployment value in it. The three states and the signal rates come

from a small internal sample, so the conclusion should be tested across a range rather than taken at a point; the direction of the result is robust to plausible variation, its magnitude is not. And it prices only expected value, so it says nothing about a board's willingness to carry the variance, which is a legitimate risk-appetite judgement belonging to the sponsor (Domain 8, KA 8.4).

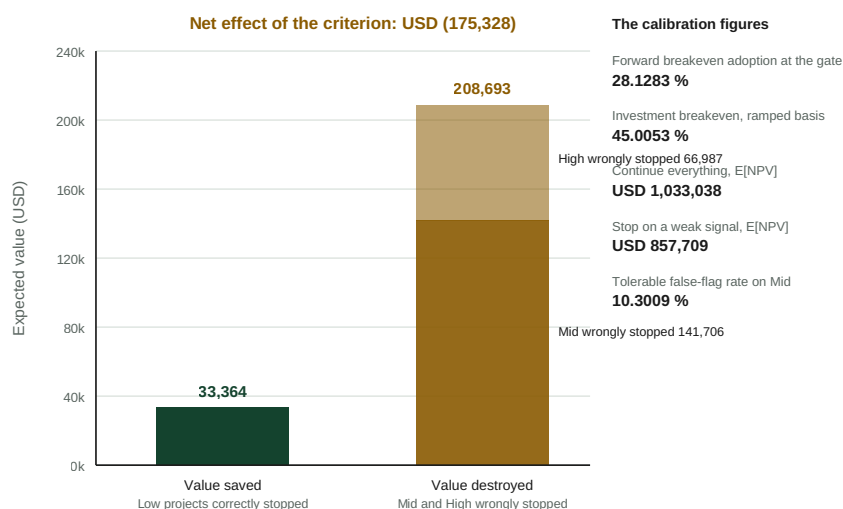


Fig 2.4.1 — Why a kill criterion on a weak signal destroys value

An **honest gate** has four properties: a **real stop option** with someone authorised to exercise it; **evidence prepared to a standard set beforehand**, not assembled to support a conclusion; **decision-makers who did not write the case**; and a **recorded decision with reasons** (Domain 3's auditability). A gate that has never stopped anything is a milestone with a meeting attached, not a control.

**Stopping well** is its own competence: capture what was learned and make it findable (Domain 9's lessons); harvest anything reusable (code, designs, cleaned data, supplier relationships, qualified people); close contracts deliberately rather than by abandonment (Domain 10); redeploy the team with their reputations intact; because how a cancellation is handled determines whether anyone proposes the next one; and tell stakeholders honestly, early (Domain 11).

### AI in this KA

The relevant AI use here is monitoring: tracking assumption triggers and kill-criteria thresholds across a portfolio, and flagging when a project's own reported data has crossed one. That is genuinely valuable because the crossing is usually visible in data nobody is reading. What must remain human is the decision and the *framing*: a model asked whether to continue will optimise whatever objective it was handed, and the escalation-of-commitment bias lives in the humans around it, not in the

arithmetic. The leader's use of AI here is to make the trigger impossible to miss, not to outsource the courage.

**Verification, concretely.** Monitoring is only as good as the threshold it monitors, so the first check is on the criterion itself: is it set at the **forward** breakeven, recomputed at this gate, or at the investment breakeven inherited from approval? A monitoring system faithfully alerting against a mis-calibrated threshold industrialises the error of 2.4.3. Where a model is used to score the two error costs of a candidate criterion, its signal-quality inputs (the false-flag rates) must come from counted outcomes on comparable past projects, and where those do not exist the honest output is the **breakeven signal quality** (10.3009 % on Meridian's successor) rather than a recommendation: it tells a sponsor how good the signal would have to be, which is a decidable question. Reproduce the two policy expectations by hand; each is three multiplications and a sum. And no model output should ever be the trigger of an automatic stop: a stop is an attributable decision by a named authority (Domain 3, KA 3.A.2), and the value of automation here is entirely in making the crossing visible on the day it happens rather than in the quarter it is noticed.

#### KEY TERMS — KA 2.4

| TERM                               | MEANING                                                                                                                          |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Forward-looking NPV</b>         | Remaining benefit PV less remaining cost; the only basis for continuation.                                                       |
| <b>Sunk cost</b>                   | Unrecoverable spend carrying no information beyond the forecast; irrelevant to a continuation decision.                          |
| <b>Forward exit cost</b>           | The cost of stopping; termination, demobilisation, disposal; a forward cost, never sunk.                                         |
| <b>Forward breakeven</b>           | The adoption or benefit level at which the <i>remaining</i> cost is just repaid; the only defensible level for a gate criterion. |
| <b>Escalation of commitment</b>    | Continuing because of what is invested; countered structurally.                                                                  |
| <b>Kill criteria</b>               | Pre-agreed stop conditions; powerful only because agreed in advance, and only useful if calibrated.                              |
| <b>False stop / false continue</b> | Stopping a project that would have created value; continuing one that will not. The two errors a criterion trades.               |
| <b>Honest gate</b>                 | Real stop option, pre-set evidence standard, independent deciders, recorded reasons.                                             |

## SAMPLE MCQs — KA 2.4

**MCQ 2.4-A** [2.4.2 · Application] 1,800,000 is spent; completion needs a further 900,000; remaining benefit PV is 780,000. The correct decision and its basis are:

- A. continue — stopping wastes the 1,800,000 already spent
- B. stop — forward NPV is (120,000); the sunk 1,800,000 is irrelevant to the remaining choice CORRECT
- C. continue — total spend 2,700,000 against total benefits exceeds the original case
- D. continue at reduced pace to spread the cost

*Rationale:* Only remaining cost and benefit bear on the decision (2.4.2). A is the sunk-cost fallacy stated plainly; C reintroduces sunk cost as "total"; D changes the schedule without changing the negative economics.

**MCQ 2.4-B** [2.4.3 · Analysis] A portfolio's gates have never stopped a project in four years. The soundest inference is:

- A. selection is excellent, so no project has needed stopping
- B. the gates are not functioning as controls — no real stop option, evidence assembled to support continuation, or deciders too close to the cases CORRECT
- C. the gate criteria are too lenient and should be tightened numerically
- D. stopping is unnecessary if delivery is well managed

*Rationale:* A control that never fires is not demonstrably a control (2.4.3). A is implausible across a whole portfolio; C addresses thresholds when the defect is process and authority; D mistakes delivery quality for strategic validity.

**MCQ 2.4-C** [2.4.3 · Recall] Kill criteria derive their power from:

- A. the severity of the thresholds
- B. being agreed in advance, so the same evidence produces a decision rather than a negotiation CORRECT
- C. being set by the project manager
- D. being confidential until invoked

*Rationale:* Advance agreement converts assessment into decision (2.4.3); without it, whoever has most invested wins the argument. C removes independence; D prevents the behavioural effect entirely.

**MCQ 2.4-D** [2.4.3 · Application] At a gate, USD 900,000 is spent and USD 1,500,000 remains; the PV of remaining benefit is 5,332,711 × s for sustained adoption s. The adoption at which the remaining spend is just repaid is:

- A. 45.0053 %
- B. 28.1283 % CORRECT
- C. 41.0460 %
- D. 62.5000 %

*Rationale:*  $1,500,000 / 5,332,711 = 28.1283\%$  (2.4.3). A is the *whole-investment* breakeven on the same ramped basis, which reintroduces the sunk 900,000 into a forward decision; C is the flat-equivalent whole-investment breakeven, wrong on both counts; D divides remaining cost by total cost, which is a completion percentage and not a breakeven at all.

**MCQ 2.4-E** [2.4.3 · Evaluation] A criterion flags 80 % of Low-adoption programmes, 35 % of Mid and 10 % of High. Priors are 0.25 / 0.45 / 0.30 and forward NPVs (166,822) / 899,720 / 2,232,898. Stopping on a flag:

- A. saves USD 33,364 of expected loss, so it is worth applying
- B. destroys USD 175,328 of expected value, because the value of the good programmes it stops exceeds the loss of the bad ones it prevents CORRECT
- C. is value-neutral, since the criterion is applied consistently to every programme
- D. cannot be assessed without knowing each programme's actual outcome

*Rationale:*  $33,364 - (141,706 + 66,987) = (175,328)$  (2.4.3). A counts only the benefit side of the trade, the standard omission; C confuses procedural consistency with expected value; D is the argument that prevents any criterion from ever being calibrated, since actual outcomes are by definition unavailable at the gate.

**MCQ 2.4-F** [2.4.2 · Analysis] Remaining cost is 900,000 and remaining benefit PV is 780,000, so forward NPV is (120,000). Terminating the principal supplier would cost 150,000. The correct decision is:

- A. stop, since forward NPV is negative
- B. continue, because the 150,000 exit cost is a forward cost of stopping and exceeds the 120,000 cost of completing CORRECT
- C. stop, and treat the 150,000 as a sunk cost of the original decision
- D. continue, because the 1,800,000 already spent would otherwise be wasted

*Rationale:* Exit costs are forward costs, so the comparison is (120,000) against (150,000) (2.4.2). A applies the rule without completing the comparison; C misclassifies a future payment as sunk; D is the sunk-cost fallacy, which reaches the same answer here for entirely the wrong reason, and would reach the wrong answer if the exit cost were 50,000.

## SELF-CHECK — KA 2.4

1. *What is the only correct basis for a continuation decision?* — Remaining benefit against remaining cost; sunk cost excluded, forward exit costs included.
2. *Why is stopping a strategic act rather than a loss?* — It releases capacity and funding for candidates the portfolio could not otherwise support.
3. *Name two properties of an honest gate.* — A real stop option with authority; deciders who did not write the case (also: pre-set evidence standards; recorded reasons).
4. *Which three things are commonly misclassified as sunk cost?* — Recoverable value (resale, redeployment, reusable assets); forward exit costs; and the *information* an overspend generated, which should revise the remaining estimate.
5. *At what level should a gate criterion be set, and why does it move?* — At the forward breakeven (28.1283 % on Meridian's successor at month six against a 45.0053 % investment breakeven), and it rises as remaining cost falls, so a criterion should get harder to trigger over time.
6. *When do kill criteria clearly pay?* — Where failure is expensive rather than where success is uncertain: the false-stop loss scales with the upside forgone, so the trade improves as the downside grows.

## Advanced topics — Domain 2

### 2.A.1 Real options thinking in selection

Static NPV values a committed plan; much project value lies in **flexibility**: the option to scale if a pilot succeeds, to defer while uncertainty resolves, to abandon cheaply, to switch technology. Treating these as options changes selection behaviour in a specific way: a small, staged first phase with a negative standalone NPV can be the highest-value choice if it buys the right to a large second phase at a decision point where far more is known. The practical rule for a delivery leader is **structure the work so that decision points exist**: phases sized to reach a genuine learning milestone, contracts that permit exit (Domain 10), architectures that permit switching. Formal option valuation is specialist; the *thinking* is not, and its absence is why organisations routinely commit fully to things they could have tested for a fraction.

The thinking is also arithmetic, and doing the arithmetic once cures the two opposite errors it attracts: treating a pilot as automatically prudent, and treating it as automatically a delay.

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#### WORKED EXAMPLE



**Worked example 2.A.1 – staging Meridian, and what the option to abandon is actually worth.**

1. **Setup.** Two ways to acquire Meridian's 40 clinics. Sustained adoption will turn out to be **High, 70 %** (probability 0.60) or **Low, 35 %** (0.40), an honest statement of the uncertainty the case carried. The PV of the ramped benefit stream per unit of sustained adoption is **5,332,711** (2.2.2), so High is worth 3,732,898 and Low 1,866,449 of benefit PV. **Option F – full commitment:** 2,400,000 now for all 40 clinics, at 60,000 a clinic. **Option S – staged:** a first phase of **12 clinics** now for **840,000** (70,000 a clinic, a 16.6667 % premium, because a small deployment loses the scale of a rollout), which reveals the adoption state at the end of year 1; then either commit the remaining 28 clinics for **1,680,000** at the original unit rate, or abandon. Staging delays the whole estate's benefit stream by **one year**; if abandoned, the 12 installed clinics stay in service and earn their 30 % share at the Low adoption. Rate 7 % throughout.
2. **Formula.** Value each option as a probability-weighted NPV at  $t = 0$ , discounting the year-1 commitment and the shifted benefit stream by  $1/1.07$ . Then decompose: **price of staging** =  $E[\text{NPV of F}] - E[\text{NPV of staging with no abandonment option}]$ , and **value of the option to abandon** =  $E[\text{NPV of staging}] - E[\text{NPV of staging committed to complete}]$ . The verdict is the option's value less its price, which is identically  $E[\text{NPV of S}] - E[\text{NPV of F}]$ .
3. **Substitution.** F:  $0.60 \times (3,732,898 - 2,400,000) + 0.40 \times (1,866,449 - 2,400,000)$ . S, High branch:  $-840,000 - 1,680,000/1.07 + 3,732,898/1.07$ . S, Low branch (abandon):  $-840,000 + 0.30 \times 1,866,449/1.07$ . S, Low branch if committed anyway:  $-840,000 - 1,680,000/1.07 + 1,866,449/1.07$ .
4. **Result.**

|                                   | HIGH (0.60) | Low (0.40) | EXPECTED NPV    |
|-----------------------------------|-------------|------------|-----------------|
| <b>F (full commitment)</b>        | +1,332,898  | (533,551)  | <b>+586,318</b> |
| <b>S (staged, abandon on Low)</b> | +1,078,596  | (316,697)  | <b>+520,479</b> |
| S (staged, committed to complete) | +1,078,596  | (665,749)  | +380,858        |

**Price of staging USD 205,460** (delay plus the 120,000 scale premium). **Value of the option to abandon USD 139,621.** Net (**USD 65,839**): staging loses money on these figures. Staging becomes the better choice once the probability of the Low state exceeds **53.9740 %**. Priced instead at the programme's own unit rate (720,000 for 12 clinics, no premium), staging is worth **+640,479** and beats full commitment by **54,161**. So so the **maximum justifiable first-phase cost is USD 774,161**, a premium ceiling of **7.5223 %** against the 16.6667 % assumed. 5. **Interpretation.** Six

readings. The decomposition in the second is the transferable technique; the fourth is the actionable finding.

**\*\*The option to abandon is real, computable and worth USD 139,621; and that is not the answer.\*\***  
 Everything the received wisdom says about piloting is true: the option has positive value, it is worth more the more uncertain the adoption, and abandoning after 840,000 is enormously better than abandoning after 2,400,000 (the Low branch improves from (665,749) to (316,697), a gain of **\*\*349,052\*\***). None of that settles the decision, because the option has a **\*\*price\*\***, and here the price is larger.

**\*\*Always separate the price of staging from the value of the option, because they have different owners and different fixes.\*\*** The price, 205,460, is delay plus scale premium – both consequences of **\*how\*** the staging is designed, and both reducible by design. The value, 139,621, comes from the uncertainty and the abandonment right – properties of the **\*situation\***, largely not in the leader's gift. Reporting one net figure hides the only lever that exists. **\*\*The professional move is not "should we pilot?" but "what would the pilot have to cost and how long could it take?"\*\***

**\*\*The whole result turns on 120,000 of scale premium, which is the actionable finding.\*\*** Remove the premium and staging wins by 54,161; keep it and staging loses by 65,839. The ceiling is **\*\*7.5223 %\*\***, so any pilot priced above about a 7.5 % premium over the programme's unit economics destroys value on these probabilities. That converts an abstract exhortation, "structure the work so decision points exist", into a procurement instruction: **\*\*the pilot must be bought at rollout unit rates, or it is not a pilot but a small expensive project.\*\*** In practice that means negotiating the whole-estate rate up front with a phased commitment, which is Domain 10's framework-with-call-off pattern (KA 10.3), and it is the single highest-value thing a leader can do to make staging viable.

**\*\*Delay is the other half of the price, and it is priced by the cost of delay, not by sentiment.\*\*** Shifting the whole benefit stream by a year costs  $3,732,898 \times (1 - 1/1.07) =$  **\*\*244,208\*\*** in the High state. That is why staging fails most often on benefit-generating work with a high cost of delay (Meridian's USD 14,280 a week), and succeeds most often where benefits are far off or the downside is catastrophic. **\*\*A pilot that does not shorten its own decision**

point is paying full price for the option and taking delivery late.\*\*

\*\*The breakeven probability is the sentence for the board.\*\*  
 "Staging pays if we think there is more than a 53.9740 % chance this does not achieve the adoption the case needs" is answerable in a room; "the real option is worth 139,621" is not. And it exposes something uncomfortable and useful: an organisation that wants to pilot everything is asserting that it usually expects failure, which may be true, and if it is, the correct response is to fix the selection process rather than to pay a premium on every programme to hedge it.

\*\*What this is not, and the honest limits.\*\* This is an \*\*expected-value decision tree\*\* (Domain 8, KA 8.2.2), not an option valuation: it discounts both branches at the same 7 % and takes the probabilities as given, where formal option pricing would treat the risk differently, PFL-AI Domain 4 sets out the appraisal machinery and its assumptions, and a genuinely large staged commitment warrants specialist valuation. It also assumes the pilot \*\*resolves\*\* the uncertainty, which is generous: a 12-clinic pilot with early adopters may say very little about clinic 37, and a pilot that only partly resolves the question is worth proportionately less, the same signal-quality problem 2.4.3 computes for kill criteria, and the reason the two topics belong together. It ignores the learning that makes phase 2 cheaper or better, which biases \*against\* staging, and it ignores the organisational cost of restarting a stalled programme, which biases \*for\* it. Finally, the two states are a simplification of a continuum; the sensitivity that matters (the premium ceiling) should be recomputed across the plausible probability range before it is quoted.

## 2.A.2 Portfolio balance, not just portfolio ranking

Ranking selects the best individual candidates; **balance** asks whether the resulting set is survivable and coherent. Four balance questions: **risk profile** (is everything high-risk, or everything safe and incremental?), **time-to-benefit** (does anything land this year, or is all value in years 3–5?), **capability spread** (does the whole set depend on one scarce team: the correlation problem of Domain 8, KA 8.A.1, at portfolio level?), and **strategic coverage** (are some stated objectives served by nothing at all?). A portfolio of individually optimal projects can be collectively

unbalanced, and no scoring model detects it, which is why Domain 15 treats portfolio balancing as its own discipline.

The alignment index of 2.1.1 sits on the *allocation* side of this line and answers none of the four questions: a portfolio funded exactly to weight can still be entirely high-risk, entirely long-dated, and entirely queued behind the one integration team whose marginal value 2.2.3b computes. **Read the index for allocation, then ask the four balance questions separately**, because each has a different remedy and the index cannot see any of them.

### 2.A.3 The reviewer's business-case eye

Invariants testable in an hour: a do-nothing baseline exists and is quantified; at least one option could plausibly have won; costs carry a range and an accuracy class; **the benefits profile ramps**; every benefit traces to a measured outcome with an owner outside the project; enabling changes are named with their owners; no benefit is claimed by another case in the portfolio; assumptions have test dates and falsifying triggers; risk exposure is quantified and reconciles to the register; kill criteria exist and are numeric; and success criteria were agreed in advance. The single highest-yield check is the benefits profile. It is wrong in most cases, and it is wrong in the direction that flatters.

Six further checks, each of which the domain's arithmetic makes into a one-line test, and each of which a case can fail while passing all of the above.

- **The breakeven carries its basis.** A breakeven adoption stated without "flat-equivalent" or "ramped-basis" is ambiguous by 3.9592 points on Meridian's figures (2.2.2). Ask which.
- **The counterfactual is priced, not asserted as zero.** Ask what doing nothing costs in years 3 to 8, and whether the avoided costs are genuinely avoided rather than deferred or claimed elsewhere (2.2.2b). Meridian's omission was worth 561,101, in the case's favour.
- **The benefit measurement has a comparison cohort, or says it has not.** Raw and attributable are different quantities, and the over-claim share is `counterfactual ÷ raw` (2.3.2).
- **Σ EMV of the assumption register against the NPV.** Meridian's ratio is 0.8062; anything near or above 1 means the case should be staged rather than committed (2.3.4).
- **The ranking's flip point is reported.** If a 3.33-point weight shift reverses the recommendation, the recommendation is a preference (2.2.3c).
- **Gate criteria are set at the forward breakeven, not the investment breakeven.** A criterion pitched at the latter condemns everything in a 16.8770-point band that is still worth completing (2.4.3).

## Industry variations — Domain 2

- **Public sector.** Appraisal is often mandated (published guidance, prescribed discount rates, optimism-bias uplifts), social value and distributional effects sit beside financial NPV, and political commitment can precede the case, making 2.4's stopping problem acutely hard and 2.1.3's re-testing acutely necessary. The specific arithmetic consequence is that a mandated uplift is a **breakeven** change, not a presentational one: a 20 % optimism-bias uplift on Meridian's cost takes it to 2,880,000 and the flat-equivalent breakeven adoption from 41.0460 % to **49.2552 %**, a rise of **8.2092 points** that puts the achieved 40 % decisively out of reach. A leader working under a mandated uplift should therefore compute the breakeven *after* it and manage adoption to that figure from day one, rather than discovering at evaluation that the standard was moved before approval.
- **Regulated utilities.** Investment cases are made to a regulator on a periodic cycle; benefits are defined by the regulatory framework, and the "customer" for the case is external, so the discount rate is set by the regulatory determination rather than chosen. That makes rate sensitivity a live governance figure: on Meridian's shape, one percentage point of discount rate (7 % to 8 %, [AF 5.971299](#) to 5.746639) moves the flat-equivalent breakeven from 41.0460 % to **42.6507 %**, a rise of **1.6047 points**. It also means the supersession hazard of 2.1.3 is low within a price control and spikes at its boundary, which is where the alignment half-life should be recomputed rather than assumed.
- **Technology and product.** Real options thinking (2.A.1) is the native mode (staged funding, pilots, kill criteria on usage metrics), and benefits are frequently non-cash-releasing capacity or optionality, so 2.3.2's honesty about units matters most here. The premium ceiling is the figure to carry across: staging pays only while the first phase can be bought near rollout unit economics (7.5223 % on Meridian's numbers), which in a product context usually holds, because a smaller first release is genuinely cheaper rather than merely smaller. And that structural difference, not cultural preference, is why staging is the default here and not in construction.
- **Energy and infrastructure.** Whole-life and carbon considerations are first-order value (2.3.3), horizons are decades, and PFL-AI's bankability tests (its Domain 5) run alongside the internal case because external capital must also be convinced. Two of this domain's mechanisms invert here. The alignment half-life is short relative to everything else in the appraisal (a 30-year asset outlives every strategy that will ever be written about it), so alignment is managed by designing for optionality rather than by re-testing fit. And the scale premium on staging is usually prohibitive, because civil works do not divide, which is why the flexibility is bought in the *design* (spare capacity, provision for a second circuit) rather than in the commitment.
- **Health and social programmes.** Benefits are largely non-financial and attribution is genuinely hard; the professional response is to measure outcomes in their own units with pre-change baselines rather than to force monetisation.

Meridian's own attribution correction is the worked illustration (a comparison cohort removed 17.1875 % of a claim that had looked like an over-performance), and it is representative rather than exceptional: where a whole system is changing at once, the counterfactual is frequently a large fraction of the measured effect. Where withholding the change is not acceptable, a staged rollout gives the cohort as a by-product of the sequencing, which is usually the only defensible route available.

## Case study – Domain 2: the Meridian case that should have been written (public health)

**Situation.** Meridian's approved business case claimed **USD 979,200** of annual benefit from year one (the full-potential figure), supporting an NPV of **+3,447,096** over eight years at 7 % against a delivery cost of 2,400,000. Every number in it was arithmetically correct. No adoption term appeared anywhere, and no benefit had an owner outside the programme.

**What the honest case looks like.** Same cost, same horizon, same rate, same 70 % steady-state adoption Domain 1 establishes; but profiled as it would actually arrive (40 % / 60 % / 70 %): PV of benefits **3,732,898**, NPV **+1,332,898**. The approved case overstated present value by **USD 2,114,198**, or **158.6 %** of the honest NPV. It also omitted the sentence that would have mattered most: **breakeven sustained adoption is 41.05 %**, so the programme creates value at any adoption above roughly 41 %, a threshold a board can monitor, unlike an NPV.

**What followed, and why it was predictable.** Both cases approve the programme, so the flat profile changed no decision, and was therefore never challenged. Two years later, delivery complete and adoption at 40 %, the programme was measured against a promise of 979,200 and publicly judged a failure (Domain 1's case study). Actual annual benefit at 40 % adoption was **391,680**, and against the case as approved the programme's NPV at that adoption is  $0.40 \times 5,847,096 - 2,400,000 = \text{(USD 61,162)}$ : a small negative, and the arithmetic that made the failure verdict defensible.

**The verdict does not survive an honest baseline.** Two corrections were available at approval and neither was made, and they run in opposite directions.

| TREATMENT OF THE SAME FACTS                                 | FLAT-EQUIVALENT<br>BREAKEVEN ADOPTION | NPV AT THE<br>ACHIEVED 40 % |
|-------------------------------------------------------------|---------------------------------------|-----------------------------|
| The case as approved (zero counterfactual, 6.0 h assumed)   | 41.0460 %                             | <b>(61,162)</b>             |
| Attribution corrected only (5.3 attributable hours)         | 46.4672 %                             | (334,026)                   |
| Counterfactual priced only (do-nothing costs 561,101 of PV) | 31.4498 %                             | +499,939                    |

| TREATMENT OF THE SAME FACTS              | FLAT-EQUIVALENT<br>BREAKEVEN ADOPTION | NPV AT THE<br>ACHIEVED 40 % |
|------------------------------------------|---------------------------------------|-----------------------------|
| <b>Both corrections: the honest case</b> | <b>35.6035 %</b>                      | <b>+227,074</b>             |

The programme was **1.05 percentage points short** of the breakeven it was judged against and **4.3965 points clear** of the breakeven that was true. Meridian at 40 % adoption created **USD 227,074** of value and was publicly recorded as a failure. **The verdict measured the business case, not the programme.** And the two errors that produced it are of opposite sign, which is why a reviewer who only knows that cases are optimistic would have found one of them and made the finding worse.

**A third figure was available and would have changed the monthly conversation.** At the assumed 70 % adoption the honest case's assumption register carried **USD 1,074,548** of expected exposure against an NPV of 1,332,898 (a ratio of **0.8062**) of which **USD 390,184** could have been resolved for nothing before approval by a letter to the legacy vendor, a written funding confirmation for the enabling change and one conversation with the finance business partner about the valuation rate (KA 2.3.4). None of the three was done.

**What was done differently on the successor programme.** The case carried a ramped profile with adoption as a tracked measure; a named benefits owner in the clinical directorate; the enabling changes (training, workflow redesign, champions) itemised with their own owners and costs; the breakeven adoption stated on the front page; and a kill criterion at 50 % adoption by month six. The board's monthly question changed from "are clinics live?" to "what is adoption, against 41 %?".

**What the domain teaches here.** A business case is a promise, and the profile is the promise's shape. An overstatement that changes no decision still changes the standard the programme will be held to, which makes the flat-profile error uniquely dangerous, because nothing at approval time creates any pressure to catch it.

## Case study B — Domain 2: the platform that could not be stopped (financial services)

**Situation.** A core-platform replacement, approved at USD 40m over three years, reached year four having spent USD 62m with roughly 55 % of scope delivered. A re-based assessment put remaining cost at USD 21m and remaining benefit present value at USD 14m, a forward NPV of **(USD 7m)**.

**What happened at the gate.** The paper presented total investment (62m spent, 21m to go) against total lifetime benefits, showing a positive "programme return", and recommended continuation. Three structural features made that recommendation almost inevitable: the gate had **no stop option** defined; the paper was written by the programme's own director; and there were **no kill criteria** from the original



approval to test against. The sponsor who had championed the original case chaired the gate.

**How it was resolved.** An independent review, commissioned only after a regulator asked about delivery timescales, reframed the decision as forward-only: 21m to buy 14m. The programme was descoped to a variant whose remaining benefit (9m) exceeded its remaining cost (6m), the balance of capacity was redeployed, and the write-off was taken and disclosed. The descoped variant delivered in eleven months.

**The arithmetic of the reframing, because it is the transferable part.** Continuing as planned had a forward NPV of  $14 - 21 = (7\text{m})$ . The descoped variant has  $9 - 6 = +3\text{m}$ . The swing between the two decisions available at that gate is therefore **10m**, which is **25 %** of the programme's original 40m approval and **16.13 %** of the 62m spent, a decision larger than most of the change requests the programme had spent four years processing. Two secondary figures made the descope defensible rather than merely attractive. The **discarded 5m of remaining benefit** costs 15m of remaining spend, a ratio of **0.3333**: value per unit of spend far below the 1.5 the retained scope achieves, which is the same value-per-unit-of-constraint reasoning as KA 2.2.3 applied inside a single programme rather than across candidates. And the **released capacity had a named use**: the 15m and the integration team went to two funded candidates, so the board was not asked to accept a write-off with nothing on the other side of it.

**The gate criterion that would have caught it years earlier.** A criterion set at the forward breakeven (*remaining benefit must exceed remaining cost at every gate*) is the one criterion in 2.4.3's list that needs no proxy signal and no calibration, because it is stated in the decision's own units. On this programme it would have been breached well before year four, and it is available to any gate willing to compute two numbers.

**What the domain teaches here.** Escalation of commitment is structural before it is personal: absent kill criteria, an independent decider and a real stop option, a rational actor presenting a "total investment" frame will continue. Note that the resolution was not "stop everything": descoping to the part where remaining benefit exceeds remaining cost is the same arithmetic applied with more imagination.

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## Executive perspective — Domain 2

What a project leader cannot delegate in this domain:

- **The benefits profile.** Not the NPV, the *shape*. The leader who lets a flat full-potential profile through has set the standard their programme will be judged against, and Meridian is what that costs.
- **The breakeven sentence, with its basis.** Stating the condition on which value depends (41.0460 % flat-equivalent, 45.0053 % ramped-basis) in a form the board can monitor monthly, and never quoting one while meaning the other.

- **The counterfactual.** Insisting that doing nothing is appraised rather than assumed to cost zero. It is the one error that runs in the leader's favour, which is exactly why nobody else will find it, and on Meridian it was the difference between a recorded failure and a 227,074 success.
- **The enabling change and who owns it.** Naming the non-project work that converts outputs into outcomes, and securing owners for it outside the project, before approval rather than after.
- **The honest options set.** Ensuring at least one option could have won, including doing nothing.
- **The priced assumption register.** Two acts nobody else will perform: clearing the zero-cost rows before the paper is written (390,184 of Meridian's 1,074,548, because after approval they stop being questions and become risks), and then saying the ratio out loud, since "NPV +1.33m against 1.07m of expected exposure in the assumptions it rests on" is a different sentence from "NPV +1.33m".
- **Forward-only framing.** Refusing every "total investment" frame at every gate, and personally insisting on remaining cost against remaining benefit, including the forward exit cost, which is not sunk.
- **Kill criteria at approval, calibrated rather than asserted.** The leader is the only person who can get them agreed while goodwill is high; after the first slip, nobody will. And the level matters as much as the existence: set at the forward breakeven, not at the investment breakeven, or the criterion will stop programmes that are still worth completing.
- **The price of staging, separated from the value of the option.** Piloting is not automatically prudent. The leader owns the two numbers that decide it: what the first phase costs above rollout unit rates, and how long it delays everything.

## Calculation exercises — Domain 2

**Exercise 2.1** A programme's full benefit potential is USD 600,000 per year, steady-state adoption 75 %, profiled 30 % / 60 % / 75 % thereafter, appraised over 6 years at 8 %. Delivery costs USD 1,300,000. Compute the ramped NPV and the flat full-potential NPV, and the overstatement. *Solution.*  $AF(0.08, 6) = 4.622880$ . Flat:  $600,000 \times 4.622880 = 2,773,728$ ; NPV +1,473,728. Ramped:  $180,000/1.08 + 360,000/1.08^2 + 450,000 \times (\text{years 3–6 factors } 0.793832 + 0.735030 + 0.680583 + 0.630170 = 2.839615) = 166,667 + 308,642 + 1,277,827 = 1,753,136$ ; NPV +453,136. Overstatement USD 1,020,592. Common error: applying the steady-state percentage to all years including the ramp period.

**Exercise 2.2** Using Exercise 2.1's flat form, find the sustained adoption at which NPV is zero. *Solution.* Required annual benefit =  $1,300,000 / 4.622880 = 281,210$ ; as a share of the 600,000 potential, 46.87 %. Common error: dividing

cost by total undiscounted benefits ( $1,300,000 / 3,600,000 = 36.1\%$ ), which ignores discounting and understates the threshold.

**Exercise 2.3** Candidates scored 1–5 on strategic fit (0.40), benefit (0.30), deliverability (0.20), risk-inverse (0.10):  $P = 4/5/2/2$ ;  $Q = 5/3/4/3$ . Rank them. *Solution.*  $P: 1.60 + 1.50 + 0.40$

$\bullet 0.20 = **3.70**$ .  $Q: 2.00 + 0.90 + 0.80 + 0.30 = 4.00$ . **Q ranks first.** Common error: unweighted summation ( $P = 13$ ,  $Q = 15$ , same order here, but it is coincidence, and the method loses the weights that were argued over).

**Exercise 2.4** Spent USD 4,200,000; remaining cost USD 1,600,000; remaining benefit PV USD 1,950,000. Decide, and state what the sunk-cost frame would wrongly add. *Solution.* Forward NPV  $1,950,000 - 1,600,000 = +\text{USD } 350,000 \rightarrow \text{continue}$ . The sunk-cost frame would add the irrelevant 4,200,000 to the comparison (as "total spend of 5,800,000 against benefits"), which here happens to reach the same decision: the point being that it reaches it for the wrong reason and would reach the wrong one whenever forward NPV is negative.

**Exercise 2.5** A directorate declares strategic weights of 45 % / 30 % / 25 % across three objectives. Its strategically mapped spend is USD 12,000,000, funded 3,600,000 / 5,400,000 / 3,000,000. Compute the alignment index and the reallocation distance. *Solution.* Funded shares **30.0000 % / 45.0000 % / 25.0000 %**. Index  $\min(45, 30) + \min(30, 45) + \min(25, 25) = 30 + 30 + 25 = 85.0000\%$ . Reallocation distance  $(1 - 0.85) \times 12,000,000 = \text{USD } 1,800,000$ , which is the single deficit on objective one ( $0.15 \times 12,000,000$ ) and equals the single surplus on objective two. Common error: adding the deficit and the surplus to report 3,600,000, which counts one movement of money twice.

**Exercise 2.6** A programme costs USD 1,500,000 and has a full benefit potential of USD 540,000 a year, appraised over 6 %, 8 years. Doing nothing forces a mandatory USD 450,000 replacement in year 3 and USD 25,000 a year of work-around cost in years 4 to 8. Compute the flat-equivalent breakeven adoption with and without the counterfactual. *Solution.*  $AF(0.06, 8) = 6.209794$ , so full-potential PV is  $540,000 \times 6.209794 = 3,353,289$ . Without the counterfactual, breakeven  $1,500,000 / 3,353,289 = 44.7322\%$ . Avoided cost PV  $450,000/1.06^3 = 377,828.68$  plus  $25,000 \times 3.536782 = 88,419.55$  (the years 4–8 discount factors), total **466,248**. With it, breakeven  $(1,500,000 - 466,248) / 3,353,289 = 30.8280\%$ : an improvement of **13.9042 percentage points**. Common error: treating the avoided costs as additional *benefits* and adding them to the numerator of the adoption term, which mixes an adoption-

dependent stream with one that is not adoption-dependent and understates the breakeven further.

**Exercise 2.7** Candidates P (fit 4, benefit 5, deliverability 2, risk-inverse 2) and Q (5, 3, 4,

3) are scored under weights 0.40 / 0.30 / 0.20 / 0.10. Q leads. Find the shift of weight from strategic fit to benefit value that flips the ranking, and state which criterion can never decide the outcome. *Solution.* Totals P **3.70**, Q **4.00**: a margin of **0.30**. Shifting  $\delta$  from fit to benefit, P gains  $\delta(5 - 4) = +\delta$  and Q gains  $\delta(3 - 5) = -2\delta$ , so  $3\delta = 0.30$  and  $\delta = 0.100000 - 10.0000$  **percentage points** (weights become fit 0.30, benefit 0.40; both score **3.80**). Criterion influence on a 1–5 scale is  $4 \times \text{weight}$ : **1.60 / 1.20 / 0.80 / 0.40**, so the risk criterion can move a total by at most **0.40** across its whole range, and closing the 0.30 margin through it would need a **3.00-point** swing in the risk scores (0.30 / 0.10): three-quarters of the entire scale, which no assessor can defend. Common error: reporting only the ranking. A 10-point flip point is defensible where a 3-point one is not, and the reviewer cannot tell which case they are looking at unless it is stated.

**Exercise 2.8** A case with an NPV of USD 1,100,000 rests on four assumptions whose impacts if false are 900,000 (P 0.30), 400,000 (0.45), 250,000 (0.20) and 150,000 (0.60). Compute the assumption exposure ratio and state what it implies. *Solution.* EMVs **270,000 / 180,000 / 50,000 / 90,000**, total **USD 590,000**. Exposure ratio  $590,000 / 1,100,000 = 53.64\%$ . The case is comfortably right-side-up (about two-thirds of Meridian's 0.8062), so it can be committed rather than staged, though the 270,000 entry still warrants a test date and a falsifying trigger. Common error: deducting the 590,000 from the NPV to report a "risk-adjusted NPV" of 510,000. The EMVs describe exposure *around* a forecast that already embodies its own expectations; deducting them double-counts, and it also hides the ratio, which is the useful output.

**Exercise 2.9** Four candidates compete for a scarce integration team: R needs 3 units for an NPV of 2,100,000; S needs 2 for 1,500,000; T needs 2 for 1,350,000; U needs 1 for 800,000. Capacity is 4 units. Find the optimal set, the shortfall a greedy per-unit ranking incurs, and the marginal value of a fifth unit. *Solution.* Per-unit values: U **800,000**, S **750,000**, R **700,000**, T **675,000**. Greedy takes U (1 unit), then S (2 units), then cannot fit R or T — total **2,300,000** with one unit idle. Enumerating the feasible sets at 4 units gives  $\{R, U\} = 2,900,000$ ,  $\{S, T\} = 2,850,000$ ,  $\{S, U\} = 2,300,000$ ,  $\{T, U\} = 2,150,000$ ,  $\{R\} = 2,100,000$ . The optimum is  $\{R, U\} = 2,900,000$ , so greedy gives up **USD 600,000**, **20.69%** of the available value. At 5 units the best set is  $\{S, T, U\} = 3,650,000$ , so the marginal value of the fifth unit is **USD 750,000**. Common error: stopping at the per-unit ranking because it is the

method the constrained example in KA 2.2.3 introduces. Ranking explains an answer; enumeration finds it.

## Practitioner's toolkit — Domain 2

*Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.*

### Toolkit 2.T.1 — Business-case skeleton (the six questions)

**Why** (driver named from 2.1.2, evidence, do-nothing consequence) · **Options** (including do-nothing and one materially cheaper; why each was rejected) · **Cost** (range, accuracy class, whole-life not just capital) · **Benefits** (map per 2.T.2; **profiled, not flat**; owners outside the project; measures and baselines; cash-releasing or not, stated) · **Risk** (quantified exposure reconciling to the register) · **Judgment** (success criteria agreed now, kill criteria numeric, gate schedule). Front page carries the **breakeven sentence**, not only the NPV.

### Toolkit 2.T.2 — Benefits map and register row

Per benefit: description · **the outcome measure that produces it** and its owner (outside the project) · **enabling changes required**, each with an owner and whether funded · baseline value, measurement method and date measured · profile by year (**ramp explicit**) · cash-releasing or capacity · attribution note · single-claimant confirmation against the portfolio register · review cadence. Three further fields wherever the benefit may leave the organisation: **whether it is externally disclosed** and in what (report, bid, regulatory submission, financing document) · the **stated boundary and method** on which the figure rests, with what is excluded · and the **approving function and record class**, since the project supplies a disclosed claim and never approves one (2.3.3; Domain 16, KA 16.4.4 for the retention).

### Toolkit 2.T.3 — Gate decision pack rules

One page of forward-only economics (**remaining cost, remaining benefit, forward NPV**, and the **forward exit cost** of stopping: total spend may be reported for information but never as the decision basis) · the **forward breakeven** at this gate, recomputed, with the current measure against it · kill criteria with current values against thresholds · assumption register extract showing which assumptions have been tested and which falsified · alignment re-test against current strategy · options at this gate, including stop and descope, each with its own forward NPV · decider names (none of whom authored the case) · recorded decision and reasons.

### Toolkit 2.T.4 — Assumption register row, and the test-order rule

Per assumption: the statement, in one falsifiable sentence · why it is believed, with the evidence named · **impact if false, in present-value terms** · probability, with who assessed it (never the case's author alone) · **EMV** · **cost to test**, including "nil"

where the test is a letter or a conversation · **EMV per unit of test cost**, which sets the order · owner · test date, in a named person's calendar · the **falsifying trigger**: the observation that would settle it · whether the entry is correlated with another (mark the pairs; expectations add, tails do not) · and whether failure is recoverable or changes what the project is.

Two rules make the register an instrument rather than a list. **Sort by EMV per unit of test cost and clear the free rows before the paper is written**: on Meridian that was 390,184 of exposure resolvable in a week. And **report  $\Sigma \text{ EMV} \div \text{NPV}$  on the case's front page**: as it approaches one, the recommendation should change from "approve" to "stage" (2.3.4, 2.A.1).

## Exam preparation — Domain 2

**The traps.** A flat full-potential benefits profile (2.2.2 — the domain's central error) · applying steady-state adoption to the ramp years (Exercise 2.1) · computing breakeven on undiscounted benefits (Exercise 2.2) · **quoting a flat-equivalent breakeven while meaning a ramped-basis one, or the reverse** (2.2.2 — worth 3.9592 points on Meridian) · **valuing the do-nothing option at zero** (2.2.2b) · **using undiscounted avoided costs** (MCQ 2.2-E) · **subtracting an avoided cost instead of adding it** (MCQ 2.2-E) · reading sunk cost into a continuation decision (2.4.2, Exercise 2.4) · **misclassifying a forward exit cost as sunk** (MCQ 2.4-F) · reporting non-cash-releasing benefits as savings (2.3.2) · **claiming a raw improvement where a comparison cohort shows part of it was happening anyway** (2.3.2) · **applying the adoption term twice, once through the clinic count and once as a percentage** (2.3.2) · double-counting a benefit across an enabler and its dependents (MCQ 2.3-C) · **deducting assumption EMVs from an NPV to produce a "risk-adjusted" figure** (Exercise 2.8) · treating a soft constraint as hard (2.1.2) · ranking by raw NPV when a resource constraint binds (MCQ 2.2-C) · **taking a greedy per-unit ranking for an optimum** (2.2.3, Exercise 2.9) · **judging capacity one unit at a time when the marginal value is non-monotone** (2.2.3b) · adding deficits to surpluses when reporting a reallocation distance (Exercise 2.5) · **treating an annual hazard as a continuous rate when computing a half-life** (MCQ 2.1-D) · **setting a gate criterion at the investment breakeven rather than the forward breakeven** (2.4.3) · mistaking a gate that has never stopped anything for a functioning control (2.4.3).

### Reflection questions.

1. Take your current business case: is the benefits profile flat? What is the breakeven condition, and could you state it in one sentence to a board, with its basis?
2. What does doing nothing actually cost your organisation over the appraisal horizon? If the answer is "nothing", name the contract, licence or asset that will not need attention in the next five years.

3. Which enabling changes does your case depend on, and does each have a funded owner outside your project? What happens to the benefits if they do not?
4. List your case's five material assumptions with an impact and a probability. What is  $\Sigma \text{ EMV} \div \text{NPV}$ , and which of the five could you resolve this week for nothing?
5. What are your project's kill criteria, and if there are none, who would have to agree them, and would they agree today? If there are, are they set at the forward breakeven or at the investment breakeven, and what would each of the two errors cost?
6. If your programme were staged instead of committed, what premium would the first phase carry over your rollout unit rates, and how long would it delay the rest? Which of those two numbers could you negotiate away?

## Domain 2 summary

Work gets chosen through a portfolio process, and the quality of that choice depends on candidates being described honestly enough to be compared, which makes a business case a decision instrument rather than an advocacy document only when it could have concluded "no". How far the funded work matches the stated strategy is measurable: an **alignment index** of  $\Sigma \min(\text{declared}, \text{funded})$  gave 76.6667 % on the mapped spend of Meridian's parent authority, 59.0000 % on all of it, and a **reallocation distance** of **USD 4,200,000**, 1.75 Meridians, with the whole difference between the two indices being the 40 % of money that serves no declared objective. Alignment granted at approval then **decays**: at a 15 % annual supersession hazard the **alignment half-life is 4.2650 years**, so gates re-test the case and not merely delivery, and an annual re-test was worth **USD 292,800** against 342,000 of expected misaligned spend. The domain's central arithmetic is the benefits profile: Meridian's approved case claimed full potential from year one for an NPV of +3,447,096, where the same facts profiled as adoption actually arrives (40 % / 60 % / 70 % of a 979,200 potential) give **+1,332,898**: an overstatement of **USD 2,114,198**, 158.6 % of the honest figure, which changed no approval decision and set the standard the programme was later judged to have failed. The more useful board sentence is the breakeven, stated with its basis: **41.0460 % flat-equivalent**, or 45.0053 % on the ramped basis. **The opposite error matters equally**: valuing the do-nothing option at zero cost Meridian **USD 561,101** of present value and 9.5962 points of breakeven headroom, while an honest attribution correction (a comparison cohort showing 1.1 of a 6.4-hour improvement was happening anyway, an over-claim of **17.1875 %**) moved it 5.4212 points the other way. Corrected both ways the breakeven is **35.6035 %**, and Meridian at its achieved 40 % adoption created **USD 227,074** of value while being publicly recorded as a failure. Selection methods each have limits: weighted scoring exposes criteria and can be steered, and a **3.3333-point** weight shift reverses this domain's ranking; ranking by value per unit of a binding constraint beats raw NPV (Beta + Gamma's 2,100,000 over Meridian's 1,693,072) but remains a heuristic for



lumpy candidates, and the constraint's own marginal value is lumpy too —  $900,000 \cdot 300,000 \cdot 900,000 \cdot 493,072 \cdot 300,000 \cdot 900,000$ , so capacity is a block decision, add one unit or three and never two. Benefits require maps that include the **enabling change** most omit, baselines measured before the change, comparison cohorts, single claimants, honesty about cash-releasing versus capacity, and sustainability treated as constraint, as value or as a disclosed claim but never confused between them, a disclosed claim carrying a stated boundary and method, a named owner, retained evidence and the signing function's approval, and never being reported as achieved on evidence its audience would not accept; and an assumption register is priced, not listed: meridian's carried **USD 1,074,548** of expected exposure against a 1,332,898 NPV, a ratio of **0.8062**, of which 390,184 was resolvable for nothing. And the discipline closes with stopping: continuation depends only on remaining cost against remaining benefit, plus the forward exit cost (780,000 of value for 900,000 of spend is a stop, whatever the 1,800,000 already gone suggests) made possible by kill criteria agreed in advance, set at the **forward** breakeven (28.1283 % at Meridian's successor's month-6 gate, not the investment breakeven of 45.0053 %) and calibrated against their two errors, since a criterion on a weak proxy signal destroyed **USD 175,328** of expected value by stopping good programmes to prevent bad ones. Where the uncertainty is genuine, stage instead: Meridian's abandonment option was worth **USD 139,621** and its staging cost **USD 205,460**, so staging paid only above a 53.9740 % chance of failure, or at a first-phase premium below **7.5223** %. Domain 3 supplies the governance and decision rights this domain assumes; Domain 15 scales the selection arithmetic to a portfolio; Domain 16 measures what it promised.



DOMAIN

# 3

## Governance, Organisation and Decision Rights

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## Why this domain exists

Domain 1 established what a project leader is answerable for; Domain 2 established how work gets chosen and whether the promise was honest. Both assumed something this domain has to supply: **that someone, somewhere, can actually decide.**

Governance is where most delivery failures are finally located, and almost never in the terms practitioners use for it. Post-project reviews record "poor governance" as though it were a culture problem. It is usually a *design* problem, and a measurable one: authority set at the wrong level, committees meeting on a cadence that cannot keep up with the work, escalation paths whose total latency nobody has ever added up, and decisions that are made but not recorded, so they are made again. Each of those has an arithmetic consequence, and this domain computes it. That is the domain's central claim: **governance is a delivery variable with a price, not a compliance overhead**, and a leader who cannot price it will be asked to accept a structure that guarantees delay.

The domain proceeds from purpose to mechanism. KA 3.1 asks what governance is *for*, and separates it from management, assurance and administration (the three things it is most often confused with) then works structures across organisational forms and the specific problem of governing iterative delivery. KA 3.2 covers the two roles that do the deciding: the sponsor, whose obligations are almost always understated, and the steering body, whose failure modes are predictable; it closes with decision authorities and thresholds, and prices the cost of setting them too low. KA 3.3 builds the assurance and gate machinery, designs escalation as a *timed* pathway rather than an organisation chart, and insists on the decision record; because an unrecorded decision is a memory, not a decision.

**Learning objectives.** After this domain a candidate can: state what governance is for and distinguish it from management and assurance; describe governance structures across functional, matrix, projectised and multi-party organisational forms, and identify the accountability weakness of each; **price governance by unanimity against a defined majority and show why its cost grows exponentially in the number of parties**; govern iterative and hybrid delivery without either abandoning control or destroying cadence, and **compute what share of an iterative stream's decisions a periodic committee can serve at all**; specify the sponsor role as a set of testable obligations and **size it in diary hours and decision turnaround**; diagnose the standard steering-committee failure modes and compute a committee's average, peak and off-peak utilisation; **compute the expected latency of a committee decision from its meeting interval and paper lead time, and use it to price a governance design**; set delegation thresholds on value, reversibility and externality, and demonstrate arithmetically when a threshold is too low; design stage gates, compute whether a gate is worth its elapsed time, state its

breakeven detection rate, and **show how an untracked conditional pass converts a gate's detection rate into an effective one and can invert its value**; distinguish the three lines of assurance, **compute residual exposure across an assurance map and reallocate effort at constant cost**, and stop the lines duplicating; design an escalation path with a stated total latency, an out-of-cycle mechanism and **a decision action window tested against the remaining duration**; set objective escalation triggers on named forecast methods; maintain an auditable decision record, **price its re-decision tax**, and **design a cumulative test that catches clusters without re-centralising the delegated band**; **specify custody, version integrity, retention and closure handover for each class of governance record**, and **say who sets the period**; and govern AI-assisted governance analysis without delegating any decision to it.

**The master programme, and the master project.** Meridian Care Records continues from Domains 1 and 2: the clinical-records rollout to **40 clinics**, approved cost **USD 2,400,000**, benefits **USD 685,440** per year at the realistic 70 % adoption, and the **cost of delay of USD 14,280 per week** derived in Domain 1. That last figure is what makes this domain quantitative: every week a decision waits has a price, and Meridian's governance design spends it without ever appearing in a budget line. Where governance has to act on a *forecast* rather than on a *proposal*, the domain uses the volume's single-project thread, **Project Auriga** (25 weeks, BAC **USD 4,000,000**, and at week 13 the position PV 2,080,000 / EV 1,920,000 / AC 2,120,000 that Domains 6 and 7 develop), because an escalation trigger has to be written against a forecast method, and 3.3.3 shows that the choice of method decides whether the trigger fires.

**Reference points.** The international document most often named in connection with this subject is **ISO 21505**, which addresses guidance on the governance of projects, programmes and portfolios. It is voluntary guidance describing practice (not legislation, and not a certifiable requirement), and it obliges nobody of itself unless an organisation, a contract or a regulator adopts it. It is named here and not reproduced: no clause, table or checklist from it appears in this volume, everything below is this book's own, and a reader who wants the document should obtain the current edition from its publisher. Naming it implies no endorsement in either direction. KA 3.3.2 names a second reference point, an assurance architecture published by a professional body, on the same terms.

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## KNOWLEDGE AREA 3.1

### Governance models

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*Topics: 3.1.1 what governance is for · 3.1.2 structures across organisational forms · 3.1.3 governance in agile and hybrid environments.*

### 3.1.1 What governance is for

**Definition.** Project governance is the set of **decision rights, accountabilities and information flows** through which an organisation directs and controls a project: who may decide what, on whose authority, on what information, by when, and how the decision is recorded.

Every word of that carries weight, and the definition earns its keep mainly by exclusion. Governance is **not management**: management runs the work inside the authority granted; governance grants, bounds and withdraws that authority. It is **not assurance**: assurance forms an independent opinion on whether the work is likely to succeed; governance decides what to do about the opinion. And it is **not administration**: a portal, a report template and a monthly slide pack are governance *artefacts*, and an organisation can have all of them and no governance at all, because no one in the chain can actually decide anything. The confusion is not academic: the commonest remedy applied to a struggling project is more reporting, which adds administration, consumes the scarce attention of the people who could decide, and leaves the decision rights exactly where they were.

**What governance exists to produce.** Four things, and a governance design should be testable against each:

1. **Decidability.** Every decision the project will face has exactly one accountable decision-maker with sufficient authority. A decision with two owners is undecidable; one with none is unowned. This is a countable property, and KA 3.2 counts it.
2. **Timeliness.** The decision arrives while it can still change the outcome. This is the property most governance designs never examine, and KA 3.2 makes it computable.
3. **Legitimacy.** The decision is made by someone whose authority the organisation recognises, on information it can rely on, through a process it accepts, so the decision *sticks* rather than being relitigated by whoever dislikes it.
4. **Traceability.** The decision, its basis and its date can be reconstructed afterwards. This is the property that converts a decision from a memory into an institutional fact (3.3.4).

**The two failure directions.** Governance fails by being too heavy or too light, and both are common in the same organisation at once: heavy on the small decisions that are easy to control and light on the large ones that are uncomfortable. Heavy governance shows as latency: escalation for decisions that could have been delegated, agendas too full to reach the difficult item, and a reporting burden that consumes the delivery capacity it was meant to protect. Light governance shows as drift: decisions made by whoever is present, scope and budget changing without a decision-maker, and a project that no one can stop because no one is entitled to.

The professional discipline is therefore **proportionality, designed deliberately**. A governance structure is a design artefact with inputs (the project's value, novelty,

risk, reversibility and external exposure) and outputs (tiers, thresholds, cadences, gates). Copying the structure used for the last programme is inheritance, not proportionality.

**Governance and the leader's accountability.** Domain 1's distinction returns here with force. Governance can allocate *responsibility* (the doing) freely, and should. It cannot dissolve the project leader's **obligation to answer**. A leader who says "the steering committee decided" has described a fact, not a defence: the obligations to have framed the decision honestly, to have surfaced the material information, to have stated the recommendation and the risk of the alternative, and to have escalated in time all remain the leader's, and every one of them is assessable afterwards from the decision record.

### 3.1.2 Structures across organisational forms

The same governance intent produces different structures depending on where authority sits in the host organisation, and each form has a characteristic weakness a leader should assume is present.

**Functional.** Resources and authority stay with functional line managers; the project leader coordinates. *Characteristic weakness:* the leader has accountability without authority over the people doing the work, so every resource conflict is an escalation. The governance countermeasure is explicit, written resource commitments with named individuals and dated availability, agreed by the functional manager and treated as breached when unilaterally changed, not a request, a commitment.

**Matrix (weak, balanced, strong).** Authority is shared on a spectrum. *Characteristic weakness:* **dual reporting without a stated precedence rule**, which does not split authority evenly but transfers it to whichever manager escalates harder. The countermeasure is a precedence rule written in advance for the specific conflicts that will arise (who wins on priority of work, on technical approach, on performance assessment), because a matrix without one is a functional organisation that holds project meetings.

**Projectised.** The project owns its resources and the leader holds full authority. *Characteristic weakness:* isolation from the enterprise, locally optimal decisions, divergent standards, and a benefits chain that ends where the project ends, precisely the failure Domain 2's benefits map addresses. The countermeasure is enterprise representation in the governance body with a real veto on the standards that matter, and a receiving-organisation owner for each benefit.

**Multi-party (joint ventures, consortia, public-private).** Two or more organisations with distinct interests, and often distinct legal duties. *Characteristic weakness:* **governance by unanimity**, which is indistinguishable from an inability to decide, particularly under stress when interests diverge. The countermeasures are structural and belong in the agreement, not in a terms of reference: a defined majority for defined classes of decision, a reserved-matters list that genuinely requires unanimity

and is kept short, a deadlock-breaking mechanism with a deadline, and a single integrating authority for day-to-day direction. PFL-AI Domain 11 treats the risk-allocation side of the same problem; the governance side is that a structure where any party can stop everything will eventually be used that way.

**Unanimity has a price, and it is not linear.** Suppose a multi-party board meets on interval  $M$  with paper lead time  $L$ , and that at any given meeting each of  $n$  parties independently arrives able to approve with probability  $p$ : able, that is, in the practical sense of having its own internal clearance, its own legal sign-off and no unresolved question. Under unanimity the decision passes in a cycle with probability  $p^n$ ; a failed cycle costs a whole further meeting interval. So

$$E[\text{cycles}] = 1 / p^n \quad E[\text{wait}] = M/2 + L + (E[\text{cycles}] - 1) \times M$$

The first expression is the whole argument. Approval probability enters as a **power of the party count**, so the expected wait rises geometrically as parties are added, and it rises much faster still as  $p$  falls, which is exactly what happens under stress, when interests diverge and internal clearances become slower and more conditional. A defined majority replaces  $p^n$  with the probability that *enough* parties are ready, which is far less sensitive to both.

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#### WORKED EXAMPLE

##### Worked example 3.1.2 — pricing Meridian's consortium board: unanimity against a defined majority.

1. **Setup.** Meridian's rollout is governed by a three-party board: the health provider, the clinical systems supplier and the regional commissioner. It meets every **4 weeks** with a **2-week** paper lead time, so a single-authority decision would wait  $4/2 + 2 = 4.0$  weeks (3.2.3). Reviewed history of the parties' behaviour on comparable items puts each party's per-cycle readiness at  $p = 0.90$  in normal conditions and **0.70** once an item is contested. Cost of delay **USD 14,280** per week. Compare unanimity with a **two-of-three** defined majority.
2. **Formula.** Unanimity pass per cycle =  $p^n$ . Two-of-three pass per cycle =  $3p^2(1 - p) + p^3$ . Then  $E[\text{wait}] = M/2 + L + (1/\text{pass} - 1) \times M$ , costed at the cost of delay.
3. **Substitution.** Unanimity at  $p = 0.90$ :  $0.90^3 = 0.729$ ,  $1/0.729 = 1.3717$  cycles,  $4.0 + 0.3717 \times 4$ . Majority at  $0.90$ :  $3 \times 0.81 \times 0.10 + 0.729 = 0.243 + 0.729 = 0.972$ ,  $1/0.972 = 1.0288$  cycles,  $4.0 + 0.0288 \times 4$ . Contested,  $p = 0.70$ : unanimity  $0.343$ ,  $2.9155$  cycles; majority  $3 \times 0.49 \times 0.30 + 0.343 = 0.784$ ,  $1.2755$  cycles.
4. **Result.**

| RULE                                 | PASS PER<br>CYCLE | EXPECTED<br>CYCLES | $E[\text{WAIT}]$     | COST                      |
|--------------------------------------|-------------------|--------------------|----------------------|---------------------------|
| Single integrating<br>authority      | 1.000             | 1.0000             | <b>4.0000 w</b>      | <b>USD<br/>57,120.00</b>  |
| Two-of-three<br>majority, $p = 0.90$ | 0.9720            | 1.0288             | <b>4.1152 w</b>      | <b>USD<br/>58,765.43</b>  |
| Unanimity, $p = 0.90$                | 0.7290            | 1.3717             | <b>5.4870 w</b>      | <b>USD<br/>78,353.91</b>  |
| Two-of-three<br>majority, $p = 0.70$ | 0.7840            | 1.2755             | <b>5.1020 w</b>      | <b>USD<br/>72,857.14</b>  |
| Unanimity, $p = 0.70$                | 0.3430            | 2.9155             | <b>11.6618<br/>w</b> | <b>USD<br/>166,530.61</b> |

Unanimity costs **USD 19,588.48** more than a two-of-three majority per decision in normal conditions, and **USD 93,673.47** more once the item is contested. 5. **Interpretation.** Read the two rows that matter together. In normal conditions unanimity looks affordable: a **1.37-week** penalty, which is the sort of number a governance review waves through as the price of consensus. Under contest it becomes an **11.66-week** wait, a **2.13-fold** increase from a change in the parties' readiness of only 0.20, while the majority rule moves from 4.12 to 5.10 weeks, a rise of under a week. **The two rules are close in good conditions and divergent in bad ones, and governance is only load-bearing in bad ones.** That asymmetry is the reason the countermeasure is structural rather than behavioural: nothing about a consortium's good intentions in month one predicts its  $p$  in month fourteen.

The identity to carry forward is that unanimity's cost is driven by  $p^n$ , so **each additional party multiplies the failure probability rather than adding to it.** At  $p = 0.90$  the expected wait runs 4.94 weeks at two parties, 5.49 at three, 6.77 at five and **8.36 at seven**; and at  $p = 0.70$  the same series runs 8.16, 11.66, 23.80 and **48.57 weeks**, at which point the structure has not slowed the decision, it has removed it. This is why the professional prescription is not "avoid unanimity" but **keep the reserved-matters list short**: unanimity is the correct rule for a handful of decisions that genuinely should not be taken over a party's objection (a change of purpose, a change of contribution, admission of a new member), and applying it to the ordinary traffic of a programme imposes  $p^n$  on decisions that never needed it.

Three cautions, each of which a reviewer should raise. **Independence is the model's weak assumption.** Parties' positions are correlated: two commissioners aligned against a supplier fail together, so the realised pass probability under contest is worse than  $p^n$  in the direction of a bloc, and better than  $p^n$  where a single party's readiness genuinely drives the others'. State the assumption and treat the figures as a lower bound on the penalty, not a point forecast.  $p$  is a

**measured quantity or it is nothing.** It is countable directly from a board's own minutes as the share of items approved at first presentation, and a consortium that has never counted it is guessing at the cost of its own constitution. And **the deadlock-breaking mechanism is what caps the tail:** with an expert-determination or escalation-to-principals route carrying a stated deadline, the series above is truncated rather than geometric, which is worth more than any improvement in  $p$ . Deadlock provisions and reserved-matters lists are enforceability-sensitive and sit in the shareholders' or consortium agreement rather than in a terms of reference; take local counsel on both, since what a deadlock clause can compel varies by jurisdiction.

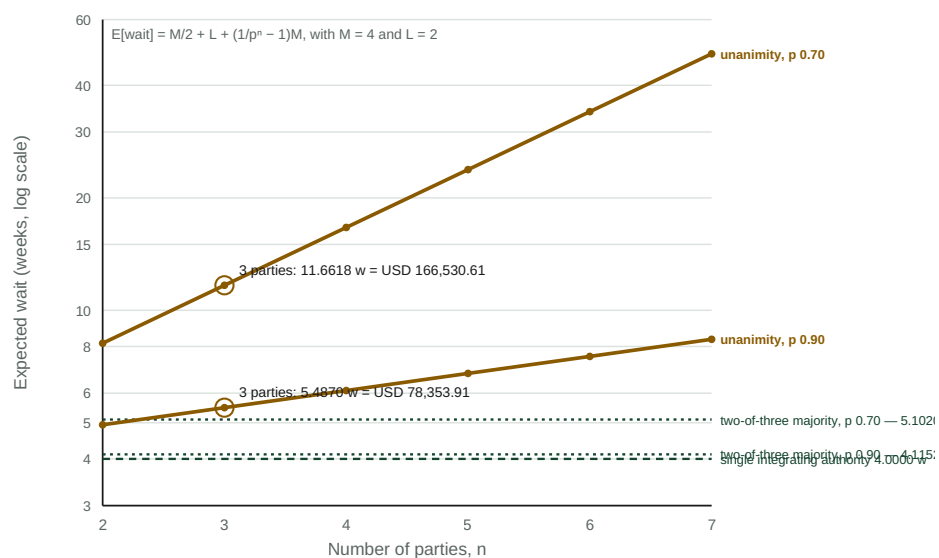


Fig 3.1.1 — Why unanimity is a power law

**Programme and portfolio layers.** Where a project sits inside a programme inside a portfolio, each layer adds a tier of potential escalation, and, unmanaged, adds its latency to every decision that travels up. KA 3.3 computes what that costs and why the design rule is **the fewest tiers a decision must legitimately pass through**, with the others informed rather than consulted.

**A note on outsourced delivery.** Where a supplier delivers, governance has to span the contract boundary, and the boundary is where decision rights are most often left undefined. The specific questions to settle before mobilisation: who may approve a change (and at what value on each side. The two thresholds are rarely equal and the mismatch is where change control fails); whose governance body is authoritative when they disagree; what the supplier is contractually obliged to escalate and within what period; and what information the client is entitled to, in what form, on what cadence. Domain 7 handles the commercial mechanics; the governance point is that a contract that does not allocate decision rights has not allocated the work.



### 3.1.3 Governance in agile and hybrid environments

Iterative delivery does not remove the need for governance; it changes what governance is asked to decide, and a structure designed for sequential delivery applied unchanged to iterative delivery fails in a specific, predictable way.

**What changes.** In sequential delivery, governance approves a plan and then controls variance against it. In iterative delivery the increment is the unit of decision, the backlog is the plan, and the governance question is not "is the project on plan?" but "**is the value being produced worth the next increment's cost, and is the direction still right?**" That converts governance from variance control to **continuation decisions at a cadence**, which is a more honest posture, and also a more demanding one, because it requires a decision-maker who can say "stop" repeatedly rather than once at a gate.

**The characteristic failure.** A monthly steering committee governing two-week sprints is *always behind*, and worse, its latency (computed in KA 3.2) exceeds the cycle it is governing, so the team either waits, destroying flow, or proceeds and seeks retrospective approval, destroying the governance. Both outcomes are attributed to the team. Neither is the team's doing.

**The design response**, in four moves:

- **Delegate inside a bounded envelope.** The team decides freely within stated bounds (scope within the agreed increment, technical approach within the architecture, spend within a period budget), and the governance body sets and reviews the envelope rather than the decisions inside it. This is the single highest-value governance decision in iterative delivery, and the worked example below shows that it is not an efficiency measure but a **feasibility requirement**.
- **Match cadence to the work.** Governance interaction at the increment's rhythm, increment reviews as the decision point, with the committee attending rather than being reported to.
- **Govern outcomes, not activity.** Working software or a released service demonstrated, with the benefit measure attached (Domain 2's map), rather than percentage-complete against a plan that the method deliberately does not have.
- **Keep the gates that carry real optionality.** Funding tranches, go-live authorisation and regulatory approvals remain genuine gates because they are genuinely irreversible. What should be removed are the gates that merely re-approve a decision already taken.

**The test that settles the argument.** Envelope width is usually debated as a matter of trust, and it can be settled as a matter of arithmetic. Each decision an iterative team raises has a **need-by time**: the point beyond which the answer no longer changes what the team does, because the work has either been done a different way or abandoned. A periodic body can serve a decision only if its expected latency is shorter than that decision's need-by time. So the governance question is not "how

much should we delegate?" but "**what share of the stream's decisions can this body serve at all?**", and the answer is countable from a single sprint's decision log.

#### WORKED EXAMPLE

#### Worked example 3.1.3 — how much of Meridian's iterative stream can a monthly committee serve?

- Setup.** Meridian's records application is built in **2-week** increments over a **26-week** build, so **13** increments. The team's decision log records **5** decisions per increment that exceed its current authority, **65** in total. Each was tagged at the time with its need-by time: **41** needed within the current increment ( $\leq 2$  weeks), **9** at 2–3 weeks, **7** at 3–4 weeks and **8** beyond 4 weeks. The steering committee's expected latency is **4.0 weeks** ( $M = 4$ ,  $L = 2$ ). Cost of delay **USD 14,280** per week; **25 %** of escalated decisions sit on the rollout critical path.
- Formula.** Decisions servable in time = count whose need-by time exceeds  $E[\text{wait}]$ . Servable share = that count  $\div$  total. Minimum envelope coverage = the share whose need-by time is inside the increment. Upper-bound wait cost = unservable count  $\times$  critical-path share  $\times E[\text{wait}] \times$  cost of delay.
- Substitution.** At  $E[\text{wait}] = 4.0$  only the  $> 4$  bucket is servable:  $8/65$ . At  $3.0$ , the 3–4 bucket joins:  $15/65$ . At  $2.0$  (a fortnightly committee with a one-week deadline), the 2–3 bucket joins:  $24/65$ . Minimum envelope  $41/65$ . Wait cost  $57 \times 0.25 \times 4 \times 14,280$ .
- Result.**

| GOVERNANCE DESIGN                    | $E[\text{wait}]$ | DECISIONS SERVABLE IN |                |
|--------------------------------------|------------------|-----------------------|----------------|
|                                      |                  | TIME                  | SHARE          |
| Monthly, 2-week papers (as designed) | 4.0 w            | 8 of 65               | <b>12.31 %</b> |
| Monthly, 1-week papers               | 3.0 w            | 15 of 65              | <b>23.08 %</b> |
| Fortnightly, 1-week papers           | 2.0 w            | 24 of 65              | <b>36.92 %</b> |

**87.69 %** of the stream's decisions cannot be served in time by the committee as designed. If the team waited for all 57 of them, the upper bound on delay is  $57 \times 0.25 = 14.25$  decisions delaying 4.0 weeks each: **57.00 delay-weeks**, or **USD 813,960**. The envelope must therefore cover at least the **63.08 %** of decisions needed inside the current increment before the team can keep cadence at all. 5. **Interpretation.** The decisive number is the third column of the table, and specifically what happens to it as the cadence is tuned. **Doubling the committee's meeting frequency and halving its paper deadline together take servable coverage from 12.31 % to 36.92 %, still under two decisions in**

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**five.** Cadence tuning cannot solve this problem, because the binding constraint is not the committee's speed but the *ratio* of its latency to the increment: at  $E[\text{wait}] = 4.0$  weeks against a 2-week increment the ratio is **2.0**, meaning the answer arrives two increments after the question, by which time the increment that raised it has shipped. Only delegation changes the ratio, which is why the bounded envelope is a feasibility requirement rather than a courtesy.

The **USD 813,960** is an upper bound and should be presented as one, because no team actually pays it. What a team does instead is the second failure of 3.1.3: it proceeds and seeks retrospective approval. So the honest reading is that the figure is the size of the pressure the design creates. And the *governance* consequence of that pressure is not delay but **the quiet transfer of 57 decisions to people who were never given authority over them**, unrecorded, and therefore invisible until an audit reconstructs them (Case study B is exactly this mechanism at a different scale). A design that cannot serve 87.69 % of its stream's decisions has not delegated them; it has lost track of them.

Three cautions. The upper bound **adds delays that in reality overlap** (two decisions waiting in the same four weeks do not cost eight weeks of programme), so the figure bounds the exposure and does not forecast it; where a defensible point estimate is needed, model the critical-path decisions serially and the rest as concurrent (Domain 6 supplies the machinery). The need-by tags are **the team's own judgement recorded at the time**, which is the right source and a biased one: a team under pressure to keep flow will tag optimistically, so the tag should be captured before the wait, never reconstructed after it. And the residual **8 decisions that the committee can serve are the ones it should keep**: they are, by construction, the decisions with time in them, and they will tend to be the irreversible and externally visible ones. The envelope's boundary is therefore not a value line but a **time-and-reversibility line**, which is the same conclusion 3.2.3 reaches from the other direction.

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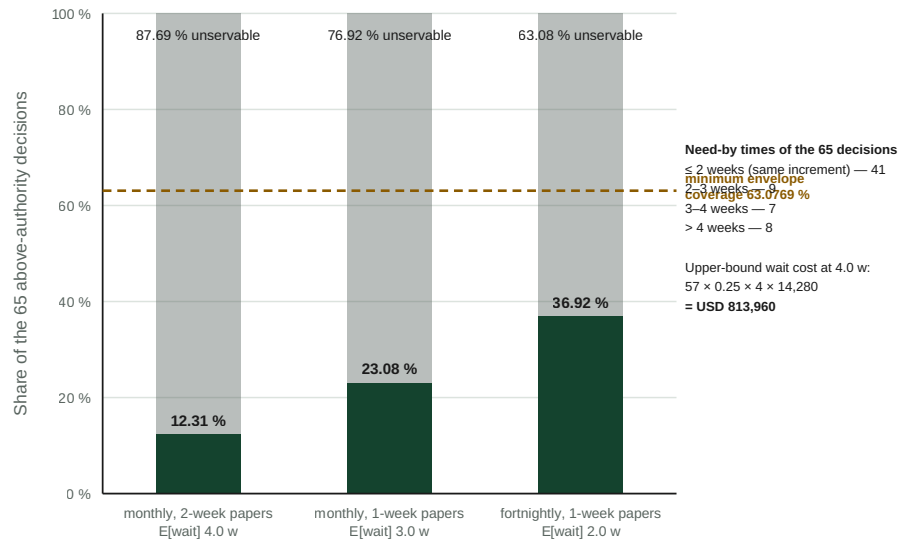


Fig 3.1.2 — What share of an iterative stream a periodic committee can serve

**Hybrid honestly.** Most real programmes are hybrid (an iterative build inside a sequential infrastructure or regulatory frame), and the governance failure is to apply one model to both parts. Meridian is exactly this shape: an iteratively developed records application, a sequential clinic rollout with immovable estate and training dependencies, and a regulatory approval that is a genuine gate. The workable design governs each part on its own terms and holds a **single integrated view of the whole** for the decisions that span them, which is the topic of Domain 4.

### AI in this KA

AI is useful in governance analysis and dangerous in governance decisions, and the line is bright.

**Where it earns its place.** Reading a set of terms of reference, delegation schedules and contract schedules against each other and listing the decisions with no named owner, two owners, or conflicting thresholds: a document-comparison task at which it is fast and thorough, and one humans do badly because it is tedious. Extracting the decision log from meeting minutes into a structured register, flagging items recorded without an owner or a date. Summarising a decision's history for a board paper. Modelling the latency of a proposed governance design (KA 3.2's arithmetic) across alternative cadences, decision rules and party counts, which is deterministic and verifiable. Tagging a sprint's decision log with need-by times drawn from the decision text and computing the servable share of 3.1.3, which is a counting task nobody performs because it is dull rather than difficult.

**Where it must not go.** No decision, and no *recommendation presented as a decision*. Governance authority is conferred on accountable people; it cannot be exercised by a system that cannot be answerable, and an organisation that lets a model's output

stand unchallenged as a governance conclusion has created accountability without a holder, the exact defect Domain 1 identified.

**The governing principle, applied.** AI proposes; the professional verifies, decides and remains accountable. Concretely: every AI-produced governance analysis is reviewed by the named accountable person before it reaches a decision body, its inputs are stated, its conclusions are independently checked on the material items, and the decision record shows a human decision-maker (never a tool) as the author of the decision.

#### KEY TERMS — KA 3.1

| TERM                          | MEANING                                                                                                                          |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Governance</b>             | The decision rights, accountabilities and information flows through which an organisation directs and controls a project.        |
| <b>Decision right</b>         | The authority to make a specified class of decision, bounded by value, scope and reversibility.                                  |
| <b>Decidability</b>           | The property that every decision has exactly one accountable decision-maker with sufficient authority.                           |
| <b>Reserved matter</b>        | A decision class that requires the highest (often unanimous) authority; kept deliberately short.                                 |
| <b>Bounded envelope</b>       | Stated limits within which a delivery team decides freely, reviewed rather than approved decision by decision.                   |
| <b>Precedence rule</b>        | The pre-agreed rule for which authority prevails in a matrix conflict.                                                           |
| <b>Governance artefact</b>    | A report, portal or template, evidence of governance, never a substitute for decision rights.                                    |
| <b>Party readiness (p)</b>    | The probability that a party arrives at a cycle able to approve; countable as the share of items approved at first presentation. |
| <b>Need-by time</b>           | The point beyond which a decision no longer changes what the team does; the test of whether a body can serve it.                 |
| <b>Envelope coverage</b>      | The share of a stream's above-authority decisions the delegated envelope contains.                                               |
| <b>Latency-to-cycle ratio</b> | Governance latency ÷ the delivery cadence it governs; above 1.0 the answer arrives after the increment that asked.               |

#### SAMPLE MCQS — KA 3.1

**MCQ 3.1-A** [3.1.1 · Comprehension] Which statement best distinguishes governance from management?

A. governance is performed by senior people and management by junior people

- B. governance grants, bounds and withdraws the authority within which management runs the work CORRECT
- C. governance is concerned with reporting and management with delivery
- D. governance applies to programmes and management to projects

*Rationale:* The distinction is about authority, not seniority, reporting or scale (3.1.1). C describes governance artefacts, which an organisation can have in full while having no governance.

**MCQ 3.1-B** [3.1.2 · Analysis] A balanced matrix has no written precedence rule for conflicts between project and functional priorities. The most likely consequence is that authority:

- A. is shared evenly, as intended
- B. transfers to whichever manager escalates harder CORRECT
- C. defaults to the project leader
- D. defaults to the steering committee automatically

*Rationale:* Undefined precedence does not split authority; it awards it to escalation behaviour (3.1.2). The countermeasure is a precedence rule written before the conflict arises.

**MCQ 3.1-C** [3.1.3 · Application] A monthly steering committee governs two-week sprints. The predictable failure is that:

- A. the team will produce lower-quality increments
- B. governance latency exceeds the cycle being governed, so the team either waits or proceeds and seeks retrospective approval CORRECT
- C. the committee will meet too often
- D. the sprints must be lengthened to a month

*Rationale:* The mismatch is one of latency against cycle time (3.1.3); the design response is a bounded envelope and cadence matching, not slower delivery.

**MCQ 3.1-D** [3.1.2 · Analysis] In a three-party consortium, the governance provision most likely to produce paralysis under stress is:

- A. a defined majority for defined decision classes
- B. a short reserved-matters list
- C. unanimity for all substantive decisions CORRECT
- D. a deadlock-breaking mechanism with a deadline

*Rationale:* Unanimity is indistinguishable from an inability to decide once interests diverge (3.1.2); the other three are the countermeasures.

**MCQ 3.1-E** [3.1.2 · Application] A three-party board meets every 4 weeks with a 2-week paper lead time. Each party is independently ready to approve in a

given cycle with probability 0.70. Under unanimity, the expected wait for one decision is closest to:

- A. 4.0 weeks
- B. 5.7 weeks
- C. 11.7 weeks CORRECT
- D. 12.0 weeks

*Rationale:*  $p^3 = 0.343$ , so  $E[\text{cycles}] = 1/0.343 = 2.9155$  and  $E[\text{wait}] = 4/2 + 2 + 1.9155 \times 4 = 11.6618$  weeks (3.1.2). A is the single-authority wait, ignoring the failure probability altogether. B applies the readiness probability once rather than to the power of three ( $1/0.70 = 1.4286$  cycles  $\rightarrow$  5.71 weeks): the commonest error, because it treats the board as one actor. D rounds the 2.9155 expected cycles up to 3 whole cycles and so adds two whole meeting intervals ( $4 + 2 \times 4 = 12.0$ ) instead of the expected 1.9155 of them.

**MCQ 3.1-F** [3.1.3 · Evaluation] A monthly committee with a 2-week paper deadline governs 2-week increments. Of 65 above-authority decisions, 41 are needed within the current increment, 9 at 2–3 weeks, 7 at 3–4 weeks and 8 beyond 4 weeks. Management proposes meeting fortnightly with a one-week deadline. The strongest evaluation of that proposal is that it:

- A. solves the problem, because the committee will then be faster than the increment
- B. raises servable coverage from 12.31 % to 36.92 % and therefore still leaves the majority of decisions unservable — only delegation changes the outcome CORRECT
- C. fails, because committee latency is irrelevant to iterative delivery
- D. raises servable coverage to 63.08 %, which is the minimum required

*Rationale:* At  $E[\text{wait}] = 2.0$  weeks the servable buckets are the 8, 7 and 9, i.e. 24 of 65 = 36.92 %, against 8 of 65 = 12.31 % as designed (3.1.3). A mistakes latency *equal* to the increment for latency shorter than it: at  $E[\text{wait}] = 2.0$  against a 2-week increment the latency-to-cycle ratio is 1.0, and the 41 same-increment decisions need their answer *inside* the increment, not at the end of it. C overcorrects: latency matters, it is simply not sufficient. D misreads 63.08 %; that is the minimum *envelope coverage* the same-increment decisions require, not a coverage the committee achieves.

SELF-CHECK — KA 3.1

1. Name the four things a governance design must produce. — Decidability, timeliness, legitimacy, traceability.

2. *Why is "more reporting" the wrong remedy for a struggling project's governance?* — It adds administration and consumes decision-makers' attention while leaving decision rights unchanged.
3. *What does a project leader still owe after a steering committee decides badly?* — The obligations to have framed the decision honestly, surfaced the material information, stated the recommendation, and escalated in time: all assessable from the record.
4. *Why does unanimity fail suddenly rather than gradually?* — Because pass probability is  $p^n$ : each party multiplies the failure probability, so a modest fall in readiness produces a large rise in expected wait (3.1.2, 5.49 weeks at  $p = 0.90$ , 11.66 at  $p = 0.70$ , three parties).
5. *What makes a bounded envelope a feasibility requirement rather than an efficiency measure?* — Governance latency of 4.0 weeks against a 2-week increment gives a latency-to-cycle ratio of 2.0, so 87.69 % of Meridian's stream decisions cannot be served in time at any realistic cadence (3.1.3).

## KNOWLEDGE AREA 3.2

# Sponsorship and steering

*Topics: 3.2.1 the sponsor role · 3.2.2 steering committees that work · 3.2.3 decision authorities and thresholds.*

## 3.2.1 The sponsor role

**Definition.** The sponsor is the individual accountable for the project's **business outcome**: for its being worth doing, remaining worth doing, and being adopted by the organisation that must use it. The project leader is accountable for delivery within the mandate; the sponsor is accountable for the mandate.

That is a demanding role and it is routinely treated as an honorific. The remedy is to state it as **testable obligations**, each of which can be evidenced or found absent:

| OBLIGATION                        | EVIDENCE IT IS BEING MET                                                                                            |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Own the business case             | The sponsor can state the benefit, its measure and its owner without notes (Domain 2).                              |
| Secure and defend funding         | Funding decisions carry the sponsor's name; funding challenges are answered by the sponsor, not the project leader. |
| Decide within authority, promptly | Decisions escalated to the sponsor have recorded dates in and out; the interval is monitored.                       |



| OBLIGATION                                           | EVIDENCE IT IS BEING MET                                                                           |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Own the receiving organisation's readiness           | The enabling change is resourced and progressing (Domain 2's omitted column).                      |
| Resolve conflicts between functions                  | Escalations that are genuinely inter-functional stop at the sponsor rather than travelling upward. |
| Hold the project leader to account, and support them | Recorded, dated performance conversations; and visible backing when the project is under pressure. |
| Be willing to stop it                                | Kill criteria agreed in advance and revisited at gates (Domain 2, KA 2.4).                         |

**The sponsor failure modes**, in descending order of frequency and each with its detection test. *The absent sponsor*, a name on a chart, no diary time; test: the count of sponsor decisions in the last quarter, and their turnaround. *The delegating sponsor*, a deputy attends everything, so authority is in the room but accountability is not; test: whether the last three material decisions were made by the sponsor or reported to them afterwards. *The advocate sponsor*, committed to the project rather than the outcome, and therefore unable to stop it; test: whether they can state a condition under which they would recommend stopping (Domain 2's escalation of commitment, at the governance level). *The operational sponsor*, drawn into managing delivery, which vacates the role that only they can hold; test: whether the sponsor's contributions in the last three meetings were about outcome and mandate, or about task sequence.

**The absent sponsor is a capacity problem before it is a commitment problem.** "No time" is the universal explanation for the first failure mode and it is almost never tested, which is unfortunate, because the role's time demand is a small sum that anyone can do; and doing it changes the conversation from an accusation about priorities into a question about portfolio load.

#### WORKED EXAMPLE

**Worked example 3.2.1a — what Meridian's sponsor role actually costs in diary time.**

- Setup.** The seven obligations above imply a specific set of commitments for Meridian: **13** steering meetings a year at **2 hours** (the 4-weekly cadence of 3.2.2); **22** sponsor-level decisions a year at **1.5 hours** each of reading, consultation and decision; **4** gate reviews at **4 hours**; **12** monthly one-to-one sessions with the project leader at **1 hour**; and enabling- change oversight with the receiving organisation at **2 hours** a month. Assume **46** working weeks and a nominal **40-hour** week.
- Formula.** Annual hours =  $\Sigma (\text{frequency} \times \text{duration})$ . Weekly load = annual hours ÷ working weeks. Share of a week = weekly load ÷ 40.

- 
3. **Substitution.**  $13 \times 2 + 22 \times 1.5 + 4 \times 4 + 12 \times 1 + 12 \times 2 = 26 + 33 + 16 + 12 + 24$ . Then  $111/46$ , then  $2.4130/40$ .
  4. **Result.** **111.0 hours** a year, **2.4130 hours** a week, or **6.03 %** of a nominal working week. Held across **five** such sponsorships the same executive carries **555.0 hours**, **12.0652 hours** a week, **30.16 %** of the week.
  5. **Interpretation.** Six per cent is the number that ends the argument. A role costing under two and a half hours a week cannot honestly be declined for want of time, and a sponsor who is absent at that price is making a **priority** choice, which is a legitimate thing to surface to whoever allocates sponsors. But the second half of the result is the more useful half, and it points away from the individual: at 2.4130 hours a week, an executive prepared to commit **20 %** of their week to sponsorship can carry  $8/2.4130 = 3.32$ , that is **three** programmes of Meridian's size, and organisations routinely name the same four or five names across a dozen. **Sponsor capacity is a portfolio constraint, not a personal virtue**, and the portfolio body that allocates sponsors without a load model is manufacturing absent sponsors and then complaining about them (Domain 15 treats the allocation; the arithmetic here is what a project leader brings to it).

Two cautions on using this. The estimate is **load, not latency**: 111 hours spread evenly is a different object from 111 hours that arrive in three clusters around the gates, and it is the clustering that produces the missed decision, which is why 3.2.1b measures turnaround separately rather than inferring it from availability. And the model **excludes the unschedulable half of the role**: defending the funding when it is challenged, standing behind the leader when the programme is under pressure, having the conversation about stopping. Those cost little time and most of the role's difficulty, so a sponsor who meets the 111 hours and none of them has met the diary and not the obligation.

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**Turnaround is the sponsor's own latency, and it has a tail.** A sponsor is not a committee, so  $M/2 + L$  does not apply; what applies is an empirical turnaround distribution, and the obligation "decide within authority, promptly" is only testable once that distribution is counted. It is worth counting because it behaves in a way the mean conceals.

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#### WORKED EXAMPLE

### Worked example 3.2.1b – Meridian's sponsor decision turnaround, and where the mean comes from.

1. **Setup.** The decision log records **22** decisions escalated to Meridian's sponsor over twelve months, with dates in and out: **9** returned in **1.0 week**, **8** in **2.5 weeks**, and **5** in **9.0 weeks**. Cost of delay **USD 14,280** per week; **25 %** of escalated decisions sit on the critical path. The steering committee's expected latency, for comparison, is **4.0 weeks** (3.2.3).
2. **Formula.** Mean turnaround =  $\Sigma (\text{count} \times \text{weeks}) \div \text{count}$ . Expected delay-weeks =  $\text{critical-path share} \times \Sigma (\text{count} \times \text{weeks})$ . Cost = delay-weeks  $\times$  cost of delay.
3. **Substitution.**  $9 \times 1.0 + 8 \times 2.5 + 5 \times 9.0 = 9 + 20 + 45 = 74$  decision-weeks; mean  $74/22$ . Delay-weeks  $0.25 \times 74$ . Committee comparison  $0.25 \times 22 \times 4.0$ .
4. **Result.** Mean turnaround **3.3636 weeks**, median **2.5 weeks**. Expected delay **18.500 weeks**, costing **USD 264,180.00**. The same 22 decisions taken by the committee at 4.0 weeks would cost  $0.25 \times 88 \times 14,280 = \text{USD } 314,160.00$ , so the sponsor route is **USD 49,980.00** cheaper. The **5** slow decisions (**22.73 %** of the count) carry **60.81 %** of the total wait. Bringing those five to the middle group's 2.5 weeks takes the total to **41.5** decision-weeks, the mean to **1.8864 weeks** and the cost to **USD 148,155.00**: a saving of **USD 116,025.00**, a **43.92 %** reduction, from changing the handling of five decisions.
5. **Interpretation.** Three results, in ascending order of usefulness. First, the sponsor route is *faster than the committee*, 3.36 weeks against 4.0, which is the arithmetic case for routing single-owner decisions to a single owner rather than to a body, and it is worth stating because organisations under stress do the reverse and add the decision to a committee agenda "for visibility". Second, **the mean is a tail phenomenon**: the median sponsor decision returns in 2.5 weeks and the mean is 3.36, and the whole of that gap is five items. Reporting a mean turnaround therefore describes a distribution the sponsor does not experience and the project does not suffer from. The right metric is the **count and cause of decisions beyond a stated service level**, not the average.

Third, and this is the finding that changes behaviour: **the fix is worth 43.92 % of the cost and touches 22.73 % of the decisions**, so a project leader arguing for a change to sponsor handling should argue about five items, not about diligence in general. And the five will have a nameable common cause; in practice one of three: the decision was framed without a recommendation, so the sponsor had to do the analysis; it required a peer's agreement the sponsor had to go and get, which is a decision-rights defect rather than a sponsor defect (3.1.2's precedence rule); or it was one the sponsor did not want to make, which is 3.2.1's advocate failure mode arriving as slowness rather than as refusal. Each has a different remedy, and none of them is "remind the sponsor".

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Two cautions. The delay figure **prices the wait as if each decision's weeks were additive**, which bounds rather than forecasts the programme effect, exactly as in 3.1.3. The defensible presentation is delay-weeks and the assumption alongside the currency figure. And a fast turnaround is not by itself evidence of a good sponsor: a sponsor who returns everything in a day may be approving without reading, which the record will show as decisions carrying no options considered and no versioned information reference (3.3.4). **Turnaround and record quality must be read together**, or the metric selects for the wrong behaviour.

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**What a project leader does with a weak sponsor.** This is a real and common situation, and the professional response is neither to complain nor to absorb the gap silently: absorbing it is attractive, because it feels like competence, and it is how a leader ends up accountable for a mandate they were never given. The workable response is procedural: write the decisions needed and their dates; send them with a recommendation and a stated consequence of non-decision; record the non-decision when it occurs; escalate the *pattern* rather than the individual instance once it recurs; and keep the record. That is not politics. It is the mechanism by which an organisation's governance failure becomes visible to the organisation while there is still time to fix it, and, incidentally, the only defensible position for the leader afterwards.

### 3.2.2 Steering committees that work

A steering committee exists to make the decisions that exceed the project leader's authority, in time for them to matter. Almost every observed dysfunction traces to one of five design faults.

**Fault 1: membership without authority.** Attendees who must consult before agreeing. The committee's effective authority is the *minimum* of its members' authority on the decision in hand, not the maximum, so one under-empowered member on a decision class blocks it. Test: for each decision class in the delegation schedule, name the member who can commit.

**Fault 2, membership too large.** Beyond roughly six to eight decision-makers the body becomes a briefing audience: contributions become positional, dissent becomes private, and the difficult item gets deferred. Attendance is not membership, the separation of *members* (who decide) from *attendees* (who inform) is the cheapest available fix.

**Membership carries a standing declaration.** Both faults above are about whether a member *can* decide; the companion question is whether a member *may* decide the item in front of them. Membership of a steering committee is an appointment, so it triggers the identification step of Domain 1, KA 1.2.2a at the point of joining and again as each item is taken: an interest is declared to the chair before the item, the interested member takes no part in it, and the minute records who decided instead.

The chair's practical instrument is a standing agenda item ahead of the first substantive one, which costs a minute and makes the nil return routine rather than pointed, and a body whose minutes never record a declaration is far more likely to be one that never asks than one that never has any.

**Fault 3: the agenda consumed by reporting.** Status occupies the meeting and the decisions arrive in the last ten minutes, which is why the reliable diagnostic of committee health is not attendance but the **share of agenda time spent on decisions**. Status is read in advance; the meeting decides.

**Fault 4 (cadence mismatched to the work.** Treated quantitatively in 3.2.3), and it is the fault with the largest measurable price.

**Fault 5, no capacity model.** A committee has finite throughput, and a design that routes more decisions to it than it can hear does not slow decisions down evenly: the items that get deferred are the contentious ones, which are the ones that needed the decision. The capacity arithmetic is elementary and almost never done.

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#### WORKED EXAMPLE

#### Worked example 3.2.2 — does Meridian's steering committee have the capacity to govern?

1. **Setup.** Meridian's steering committee meets every **4 weeks** (**13** meetings a year) and can handle **8** substantive agenda items per meeting. Annual demand: **36** escalated change requests (the count above the current delegation threshold, from 3.2.3), **26** standing reports (two per meeting), and **15** gate and assurance decisions.
2. **Formula.** Capacity = meetings × items per meeting. Utilisation = demand ÷ capacity.
3. **Substitution.** Capacity  $13 \times 8 = 104$ . Demand  $36 + 26 + 15 = 77$ .  $77/104$ .
4. **Result.** Capacity **104** item-slots; demand **77**; utilisation **74.0 %**.
5. **Interpretation.** Seventy-four per cent looks comfortable and is not, and the reasons are computable rather than impressionistic.

**First, demand is not uniform, and the peak is what a committee experiences.** The **51** decision items (36 changes plus 15 gate and assurance decisions), do not arrive evenly: they cluster at phase boundaries, and Meridian's history puts **40 %** of them in the **3** meetings that straddle the design-completion, first-clinic and full-rollout boundaries. That is  $51 \times 0.40 = 20.40$  items into 3 meetings, or **6.80** decision items per peak meeting, plus the 2 standing reports — **8.80 items against a capacity of 8**, a peak utilisation of **110.00 %**. The remaining **30.60** items spread over 10 meetings give  $3.06 + 2 = 5.06$ , an off-peak utilisation of **63.25 %**. So the true profile is not 74 % but **110 % at the boundaries and 63 % between them**, and a committee at 110 % does not defer items at

random: it defers the ones that will take longest to discuss, which are the contentious ones, which are the ones that needed the meeting. **An average utilisation below capacity is not evidence of sufficient capacity; it is evidence that nobody has looked at the profile.**

**Second, a quarter of the scarcest resource in the programme produces no decisions.** The 26 standing reports consume  $26/104 = 25.0$  % of total capacity (Fault 3). Moving them to pre-reading releases 26 slots, drops average utilisation to **49.04** % and (the number that matters) takes **peak** utilisation from 110.00 % to **85.00** %, which is the first design in this sequence that can actually hear a contentious item at a phase boundary. Raising the delegation threshold as 3.2.3 recommends removes a further 12 items, taking demand to **65** and utilisation to **62.50** % on the original agenda design, or to **37.50** % if both changes are made.

**Third, the demand figure above is incomplete, and knowing why is the point of 3.3.4.** It counts the decisions the committee is *asked* to make and omits the ones it makes twice; Worked example 3.3.4b counts those at **25** further slots a year and takes real utilisation to **98.08** %, which is the number that matches what the members report. A capacity model built only from the forward demand estimate will always understate, because re-decisions are invisible to everyone except the decision log.

Two cautions on the model. Capacity in **item-slots assumes items are comparable**, and they are not: a contested threshold change and a routine assurance acceptance are not one slot each, so a committee near its limit should weight items by expected discussion time and re-run the sum. And **raising capacity is the weakest of the three available levers**: adding meetings or lengthening them increases the demand on exactly the people whose attention is scarcest, whereas pre-reading and delegation reduce demand at no cost in scrutiny. The professional point is that committee capacity is a *designed* quantity with an average, a peak and a hidden component, and a governance structure proposed without this arithmetic has not been designed.

### 3.2.3 Decision authorities and thresholds

**The structure.** A delegation schedule states, for each decision class, the level of authority required, typically as a monetary threshold, but the better schedules use three dimensions: **value** (how much), **reversibility** (how hard to undo), and **externality** (who outside the project is affected). A cheap, irreversible, externally visible decision may deserve more authority than an expensive, reversible, internal one; a schedule that reads only on value cannot express that.

**Making reversibility a number, so the schedule can read on it.** Reversibility is usually left as an adjective, which is why it drops out of practice; it becomes operable as a ratio. Define the **reversal-cost ratio**  $\rho = (\text{cost to undo the decision}) \div (\text{value of the decision})$ , and read the required authority on  $\max(\text{value}, \text{cost to undo})$  rather than on value. The rule is one line and it catches the

decisions a value-only schedule is structurally blind to. On Meridian, a **USD 15,000** change to the clinical template set that has already been deployed to the first cohort of clinics carries a reversal cost of **USD 180,000** (re-configuration, re-validation and re-training across those clinics), so  $\rho = 12$ , and the authority appropriate to the decision is that of the **100,001–500,000** band, not the delegated band its price tag puts it in. Externality takes the same treatment as a three-point rating (internal / cross-functional / external or public) with a stated escalation of one authority level per step, because it is the dimension least amenable to a ratio: what an external decision costs is the cost of being wrong in public, which is not estimable in advance and is precisely why it is handled by rule rather than by calculation. Both dimensions, with the cumulative test of Worked example 3.3.4c, are the columns of Toolkit 3.T.2.

**Governance latency: the formula this domain contributes.** Committees meet periodically and require papers in advance, and those two facts alone determine how long a decision waits. Let

M = the meeting interval (e.g. 4 weeks for a monthly committee)  
 L = the paper lead time – how far before the meeting submissions close

For a decision that arises at a uniformly random point in the cycle:

$$E[\text{wait}] = M/2 + L$$

The derivation is elementary. A decision arising more than  $L$  before the next meeting makes that meeting, and waits on average  $M/2$  of the remaining interval; one arising inside the closed window misses it and waits for the following meeting. Averaging over arrival times gives half the meeting interval plus the *whole* paper lead time.

**Why the formula matters more than it looks.** It settles a question organisations habitually get wrong. The two available levers do **not** have equal effect: cutting the paper lead time by one week saves **one full week** of expected wait, while shortening the meeting interval by one week saves only **half a week**. Administrative deadlines, which feel unchangeable and cost nothing to change, are twice as powerful as meeting frequency, which is expensive and resisted. And a governance design's latency can be computed at design time, from two numbers, before anyone has waited.

Meridian's steering committee:  $M = 4$ ,  $L = 2$ , so  $E[\text{wait}] = 4/2 + 2 = 4$  **weeks**. Note what that says, a *monthly* committee with a two-week paper deadline imposes an expected wait of a full month, not the fortnight most people assume. At Domain 1's cost of delay of 14,280 per week, one escalated decision on the critical path costs **USD 57,120** in delay alone, before anyone in the room disagrees about anything.



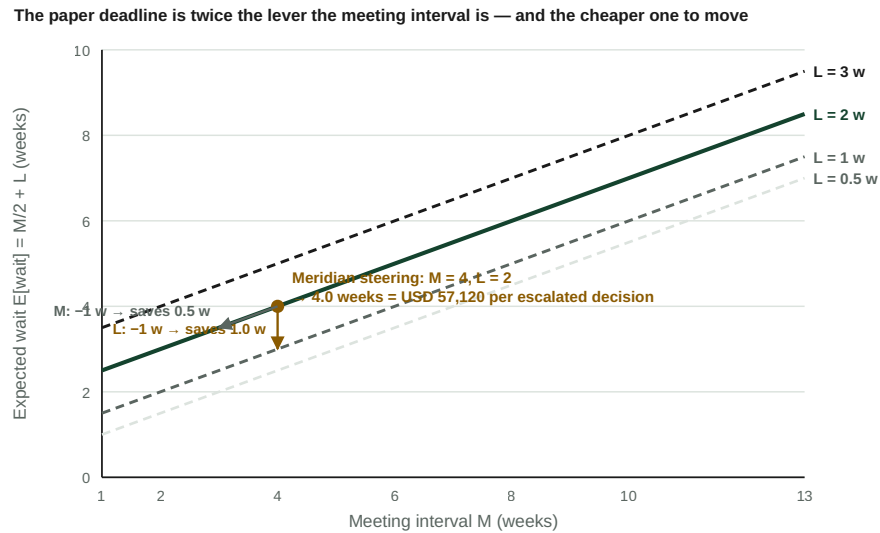


Fig 3.2.1 — Governance latency and its two levers

**Setting the threshold.** The threshold question is an economic one and can be answered as one. Setting a threshold too high risks decisions taken with insufficient authority or scrutiny; setting it too low guarantees latency on decisions that did not need it. The second cost is certain and computable; the first is probabilistic. Comparing them is the whole of the analysis.

#### WORKED EXAMPLE

##### Worked example 3.2.3 — Meridian's delegation threshold, priced.

- Setup.** Meridian's project leader may approve changes up to **USD 10,000**; everything above goes to the steering committee ( $E[\text{wait}] = 4$  weeks, cost of delay **14,280** per week). Year-one change requests, by value:  $\leq 10,000 \rightarrow 24$ ; **10,001–25,000**  $\rightarrow 12$  (average value **17,000**); **25,001–100,000**  $\rightarrow 15$ ; **100,001–500,000**  $\rightarrow 7$ ;  $> 500,000 \rightarrow 2$ . Total **60**. Reviewed history shows **25 %** of escalated changes sit on the critical path and therefore convert their wait into programme delay; the rest do not. Should the threshold move to 25,000?
- Formula.** Escalated count  $\times$  critical-path share  $\times E[\text{wait}] \times$  cost of delay, evaluated at each threshold. Then the worst-case cost of delegating the affected band.
- Substitution.** At 10,000: escalated  $60 - 24 = 36$ , delaying  $36 \times 0.25 = 9$ , each costing  $4 \times 14,280 = 57,120$ . At 25,000: escalated  $36 - 12 = 24$ , delaying  $24 \times 0.25 = 6$ .
- Result.** Annual governance delay cost **USD 514,080** at a 10,000 threshold and **USD 342,720** at 25,000, a saving of **USD 171,360** a year. The band being delegated is 12 changes worth **204,000** in total. Even in the **worst imaginable case** (the delegate decides *every one of the twelve* wrongly and destroys **40 %** of the



value each time) the loss is  $12 \times 17,000 \times 0.40 = \text{USD } 81,600$ , leaving the delegation ahead by **USD 89,760**. The threshold at which the two are equal is a value destruction of  $171,360/204,000 = 84\%$  of every delegated decision.

5. **Interpretation.** The escalation cannot be justified at any plausible error rate. The delegate would have to destroy 84 % of the value of *every* delegated decision for the committee's involvement to break even, and a delegate who did that would be removed for reasons unrelated to thresholds. That is the general shape of the result, and it is why over-centralised delegation schedules are so common and so expensive: **the cost of escalation is certain, recurring and invisible, while the cost of a delegated error is uncertain, occasional and highly visible.** Organisations optimise against the visible cost. Three professional cautions, however. The calculation prices *delay*, not scrutiny; where a decision class carries irreversibility or external exposure the value test is the wrong test, which is why the schedule reads on three dimensions and not one. The 25 % critical-path share is an *assumption from history* and must be stated as one, since the conclusion is proportional to it; though note that it would have to fall below  $81,600/(12 \times 57,120) = 11.9\%$  before the worst case even competed. And a raised threshold requires the delegate to have the *information* to decide, which is a real prerequisite: delegation without information is abdication.

## AI in this KA

**Where it earns its place.** Modelling latency across candidate governance designs: the arithmetic above, over dozens of cadence and threshold combinations, is deterministic and a natural fit. Classifying a change-request history into value bands and testing whether a proposed threshold would have altered any actual outcome, a genuinely useful retrospective test, and one nobody has time to do by hand. Drafting a delegation schedule from an existing one plus a set of stated changes, for human review. Detecting the decidability defects of 3.1.1 across a document set: decisions with no owner, two owners, or thresholds that conflict between the contract and the terms of reference.

**Where it must not go.** Setting a threshold, which is a risk-appetite decision belonging to the sponsor and the accountable body. Estimating the critical-path share or the error rate from plausibility rather than data: a model asked for these will produce confident numbers with no provenance, and they will then drive a real decision. Nor should its latency model be trusted unverified: the formula is two operations, and a leader who cannot reproduce  $M/2 + L$  on paper should not be quoting it.

**Verification, concretely.** Reproduce every number by hand, state the assumptions with their sources, and put the sensitivity in the paper: the breakeven error rate and the breakeven critical-path share, not merely the point estimate, because those are what tell a board whether the recommendation is robust or fragile.

## KEY TERMS — KA 3.2

| TERM                                           | MEANING                                                                                                                 |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Sponsor</b>                                 | The individual accountable for the project's business outcome and mandate.                                              |
| <b>Steering committee</b>                      | The body that makes decisions exceeding the project leader's authority, in time for them to matter.                     |
| <b>Delegation schedule</b>                     | The statement of authority by decision class, on value, reversibility and externality.                                  |
| <b>Governance latency</b>                      | The expected wait for a committee decision: $E[\text{wait}] = M/2 + L$ .                                                |
| <b>Meeting interval (M)</b>                    | The period between meetings of a decision body.                                                                         |
| <b>Paper lead time (L)</b>                     | How far before a meeting submissions close.                                                                             |
| <b>Committee capacity</b>                      | Meetings per year $\times$ substantive items per meeting; compared against decision demand, average, peak and off-peak. |
| <b>Member vs attendee</b>                      | Members decide; attendees inform. Conflating them enlarges the body and reduces its authority.                          |
| <b>Sponsor turnaround</b>                      | The measured interval from a decision reaching the sponsor to its return; read as a distribution, not a mean.           |
| <b>Reversal-cost ratio (<math>\rho</math>)</b> | Cost to undo a decision $\div$ its value; authority reads on $\max(\text{value}, \text{cost to undo})$ .                |
| <b>Sponsor load</b>                            | The annual diary hours the sponsor obligations imply; a portfolio allocation constraint, not a personal one.            |

## SAMPLE MCQs — KA 3.2

**MCQ 3.2-A** [3.2.3 · Application] A committee meets every 6 weeks and closes papers 2 weeks before each meeting. The expected wait for a decision arising at a random point in the cycle is:

- A. 3 weeks
- B. 5 weeks CORRECT
- C. 6 weeks
- D. 8 weeks

*Rationale:*  $E[\text{wait}] = M/2 + L = 6/2 + 2 = 5$  weeks (3.2.3). A counts only half the interval and omits the paper lead time; C is the interval itself; D adds the whole interval to the lead time.

**MCQ 3.2-B** [3.2.3 · Analysis] A committee meets every 4 weeks with a 2-week paper lead time. Two options are on the table: cut the paper lead time by one week, or cut the meeting interval by one week. Which reduces expected latency more, and by how much?

- A. cutting the meeting interval, by 1.0 week
- B. cutting the paper lead time, by 1.0 week — twice the 0.5-week saving from the interval CORRECT
- C. both, equally, by 1.0 week
- D. both, equally, by 0.5 weeks

*Rationale:*  $E[\text{wait}] = M/2 + L$  is 4.0 weeks initially; cutting  $L$  to 1 gives 3.0 (saves 1.0), cutting  $M$  to 3 gives 3.5 (saves 0.5), a one-week cut in the paper lead time always saves a full week, a one-week cut in the meeting interval only half of one (3.2.3). The administrative deadline is also the cheaper lever to move.

**MCQ 3.2-C** [3.2.3 · Evaluation] Raising Meridian's threshold from 10,000 to 25,000 saves 171,360 a year in delay. The band delegated comprises 12 changes worth 204,000. The strongest argument for the change is that:

- A. the project leader is experienced
- B. the delegate would have to destroy 84 % of the value of every delegated decision for the escalation to break even CORRECT
- C. the committee is too busy
- D. 12 changes is a small number

*Rationale:* The decisive argument is the breakeven value-destruction rate, which makes the comparison explicit and quantitative (3.2.3). A and D are assertions; C is a capacity argument, which is real but secondary.

**MCQ 3.2-D** [3.2.2 · Analysis] A steering committee's effective authority on a given decision class is best described as:

- A. the authority of its chair
- B. the highest authority among its members
- C. the lowest authority among the members whose agreement is required CORRECT
- D. the authority delegated to the project leader

*Rationale:* One member who must consult before agreeing blocks the class, so the binding constraint is the minimum, not the maximum (3.2.2, Fault 1).

**MCQ 3.2-E** [3.2.1 · Analysis] The most reliable test of whether a sponsor is an *advocate* sponsor rather than an effective one is whether they:

- A. attend every steering committee

- B. can state a condition under which they would recommend stopping the project CORRECT
- C. defend the project's funding
- D. know the project's schedule in detail

*Rationale:* An advocate sponsor is committed to the project rather than the outcome and therefore cannot stop it (3.2.1). C is an obligation of the role; D is closer to the operational failure mode.

**MCQ 3.2-F** [3.2.1 · Analysis] A sponsor returned 22 decisions in a year: 9 in 1.0 week, 8 in 2.5 weeks and 5 in 9.0 weeks. Which statement best supports a proposal to change how sponsor decisions are handled?

- A. the mean turnaround is 3.3636 weeks, which is too slow
- B. the median is 2.5 weeks, so performance is acceptable
- C. 22.73 % of the decisions carry 60.81 % of the total wait, so handling five items differently removes 43.92 % of the cost CORRECT
- D. the sponsor is slower than a 4-week committee and the decisions should go to the committee

*Rationale:* The mean (A) describes a distribution nobody experiences, and the median (B) hides the tail; the decisive presentation is the tail's share of the wait and the saving available from it (3.2.1). D is arithmetically wrong: 3.3636 weeks is *faster* than the committee's 4.0 weeks. Common error: arguing from the average, which invites a debate about diligence instead of about five items.

**MCQ 3.2-G** [3.2.3 · Application] A delegated change is worth USD 15,000 but would cost USD 180,000 to undo once deployed. Under a schedule that reads on value, reversibility and externality, the authority required is that appropriate to:

- A. USD 15,000, since that is the change's value
- B. USD 165,000, the net exposure
- C. USD 180,000, because authority reads on `max(value, cost to undo)` CORRECT
- D. USD 195,000, the sum of value and reversal cost

*Rationale:* The reversal-cost ratio is  $180,000/15,000 = 12$ , and the rule reads authority on the larger of the two figures (3.2.3). A is the value-only failure of Case study B. B nets the change's value off the reversal cost as though the value earned offset the cost of undoing it; there is no such offset. The reversal cost is what it costs to return to the prior state. D adds two quantities that are not additive: the reversal cost already includes undoing the 15,000 of work.

## SELF-CHECK — KA 3.2

1. *State the governance latency formula and both of its levers.* —  $E[\text{wait}] = M/2 + L$ ; shorten the paper lead time (a full week per week cut) or the meeting interval (half a week per week cut).
2. *Why do organisations systematically set delegation thresholds too low?* — The cost of escalation is certain, recurring and invisible; the cost of a delegated error is uncertain, occasional and highly visible, so optimisation runs against the visible cost.
3. *What must accompany a raised threshold for it to be delegation rather than abdication?* — The information, criteria and support the delegate needs to decide, plus a record of what they decided.
4. *Why is average committee utilisation a misleading measure?* — Demand clusters at phase boundaries; Meridian's 74.0 % average conceals **110.00 %** at the boundaries and **63.25 %** between them, and it is at the peak that contentious items are deferred (3.2.2).
5. *How does a delegation schedule express reversibility?* — Through the reversal-cost ratio  $\rho = \text{cost to undo} \div \text{value}$ , with authority read on  $\max(\text{value}, \text{cost to undo})$ , which put a USD 15,000 Meridian template change into the 100,001–500,000 band at  $\rho = 12$  (3.2.3).
6. *Roughly how much diary time does a Meridian-sized sponsorship require, and why does the answer matter?* — **111.0 hours a year**, about **6.03 %** of a working week; small enough that absence is a priority choice, and large enough that no executive carries more than about three (3.2.1).

## KNOWLEDGE AREA 3.3

## Assurance, gates and escalation

*Topics: 3.3.1 stage gates · 3.3.2 assurance lines · 3.3.3 escalation design · 3.3.4 auditability and the decision record.*

**3.3.1 Stage gates**

**Definition.** A stage gate is a point at which continuation requires an explicit decision by a named authority against stated criteria, with authority to stop, hold or redirect as well as proceed. A checkpoint without the authority to stop is a milestone with a meeting attached.

**What a gate is for.** Gates exist to convert **irreversibility into optionality**: before committing the next tranche of cost, an organisation buys the chance to stop or change direction while stopping is still cheap. It follows that gates belong where irreversibility genuinely steps up (before major commitment, before build, before go-

live, before regulatory submission), and not at regular calendar intervals, which is how they proliferate into re-approval ceremonies.

**Gate criteria.** Criteria must be set **in advance**, be **objectively assessable**, and cover both the work and the decision: is the deliverable adequate; is the business case still valid (Domain 2's benefits and assumptions, re-tested rather than restated); are the risks acceptable and owned; is the receiving organisation ready; and is the plan for the next stage credible? Criteria written after the evidence is available are not criteria.

**The honest gate.** A gate that has never held or stopped anything is not evidence of good delivery, it is evidence of a gate that does not function. And its cost is being paid for nothing. The pattern to watch for is the **conditional pass**: proceed subject to conditions, which are then neither tracked nor enforced. A conditional pass is a genuine and useful instrument, and it is only that if the conditions have owners, dates and a stated consequence of non-completion, verified at the next gate.

**Gates cost time, and time has a price.** This is the part of gate design that is almost never computed, and it decides whether the gate is worth having.

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#### WORKED EXAMPLE

##### Worked example 3.3.1 — is Meridian's design gate worth its elapsed time?

1. **Setup.** Meridian's design-completion gate requires **USD 45,000** of review effort and **6 weeks** of elapsed time (cost of delay **14,280** per week). Historical evidence on comparable work: a material design defect is present with probability **0.30**; the gate detects it with probability **0.80**; a defect corrected at design costs **120,000**; the same defect found in build costs **900,000**.
2. **Formula.** Expected total cost with the gate = review + delay +  $P(\text{defect}) \times [P(\text{detect}) \times \text{design-fix} + P(\text{miss}) \times \text{build-fix}]$ . Without the gate =  $P(\text{defect}) \times \text{build-fix}$ .
3. **Substitution.** Without:  $0.30 \times 900,000$ . With:  $45,000 + 6 \times 14,280 + 0.30 \times (0.80 \times 120,000 + 0.20 \times 900,000)$ .
4.  $0.20 \times 900,000$ .
5. **Result.** Without the gate **USD 270,000**. With the gate  $45,000 + 85,680 + 82,800 = \text{USD } 213,480$ . The gate is worth **USD 56,520**.
6. **Interpretation.** The gate pays, and the useful output is not the 56,520 but the two **breakeven points** it implies, because those are what a leader negotiates with. Holding everything else, the gate stops paying once its elapsed time reaches  $(270,000 - 45,000 - 82,800) / 14,280 = \mathbf{9.96 \text{ weeks}}$ : a gate that takes ten weeks destroys the value it exists to protect, which is the arithmetic behind the familiar and usually unquantified complaint that assurance has become an obstacle. And at the actual 6 weeks, the gate needs a detection probability above **55.85 %** to be worth holding, so a review staffed by people

who cannot competently detect the defect is worse than no review, because it costs the money and the delay and returns nothing. Both results generalise: **gate value is destroyed by elapsed time and by weak detection**, and a leader whose gate is under challenge should compute which of the two is the actual problem before defending or abandoning it.

**The conditional pass, priced, and the identity that makes it dangerous.** The instrument was described above as real only if its conditions have owners, dates and consequences. That is not a counsel of tidiness; it follows from the gate arithmetic. A gate detects a defect and then *directs a remedy*, and a directed remedy that is not completed leaves the defect in the work. So if a gate's detection rate is  $q$  and the share of conditions actually closed is  $r$ , the gate's **effective detection rate** is

$$q_{\text{eff}} = q \times r$$

— an identity worth stating because it converts an administrative failure into the same variable the gate's value already depends on. A gate that detects perfectly and closes half its conditions is a gate that detects half.

#### WORKED EXAMPLE

##### Worked example 3.3.1b — Meridian's conditional passes, and the gate that inverted.

1. **Setup.** Meridian's four gates in the last twelve months issued **23** conditions between them. At the following gate: **9** were verified closed, **8** were open past their stated date, and **6** had no owner or date recorded and were never tracked at all. Everything else is Worked example 3.3.1's design gate: review effort **USD 45,000**, elapsed **6 weeks** at **14,280** per week,  $P(\text{defect})$  **0.30**, nominal detection  $q$  **0.80**, design fix **120,000**, build fix **900,000**; the gate's nominal value was **USD 56,520**.
2. **Formula.** Closure rate  $r = \text{closed} \div \text{issued}$ .  $q_{\text{eff}} = q \times r$ . Then re-run 3.3.1's comparison with  $q_{\text{eff}}$  in place of  $q$ , and solve  $q \times r \geq d^*$  for the closure rate the gate needs, where  $d^*$  is 3.3.1's breakeven detection probability.
3. **Substitution.**  $r = 9/23$ .  $q_{\text{eff}} = 0.80 \times 0.3913$ . Expected remediation  $0.30 \times (0.3130 \times 120,000 + 0.6870 \times 900,000)$ . Total with gate  $45,000 + 6 \times 14,280 + \text{that}$ . Breakeven  $d^* = 130,680/234,000$ ;  $r^* = d^*/0.80$ .
4. **Result.** Closure rate **39.13 %**; effective detection **31.30 %**, against a nominal 80 %. Expected remediation rises from 82,800 to **USD 196,747.83**, so the total with the gate is  $45,000 + 85,680 + 196,747.83 = \text{USD } 327,427.83$  against **USD 270,000** without it. **The gate now destroys USD 57,427.83**: a swing of **USD 113,947.83** from its nominal value, caused by no change whatever to the gate

itself. The closure rate the gate needs to break even is  $d^*/q = 0.5585/0.80 = 69.81\%$ , i.e. 17 of the 23 conditions rather than 9; at 16 of 23 (69.57 %) it is still USD 453.91 underwater, and at 17 of 23 (73.91 %) it returns USD 7,685.22.

5. **Interpretation.** The headline is the swing. **A gate worth +56,520 became a gate worth -57,428 without anyone changing its criteria, its staffing or its duration:** the whole of the movement is in a tracking failure that no governance report would have shown, because governance reports count gates held and conditions issued, not conditions closed. That is the practical case for making condition closure a *counted* field rather than an exhortation, and it is why Toolkit 3.T.3 makes "conditions past their date" one of five monthly integers.

The identity  $q_{\text{eff}} = q \times r$  also relocates the remedy. When a gate is challenged as an obstacle, the instinct is to argue about its criteria or its duration; here neither is the problem. At the observed closure rate the gate's **breakeven elapsed time collapses from 9.96 weeks to 1.9784 weeks**, so a leader defending a six-week gate on this evidence is defending the indefensible, and a leader *attacking* it would be attacking the wrong object, since the gate at full closure is worth 56,520. **The question to compute first is which of  $q$ ,  $r$  and elapsed time is the binding constraint**, and only one of the three is cheap to fix.

Note also the composition of the 14 unclosed conditions, because the two halves are different defects. The **8 open past their date** are a follow-through failure: the owner and date existed, so the register can chase them, and a monthly count will close most of them. The **6 with no owner or date** were never conditions at all. They were reservations minuted as conditions, and they are the more serious finding, because a gate that issues them has recorded a decision it did not make. Closing the 8 alone takes  $r$  to  $17/23 = 73.91\%$  and the gate back into value, which is the cheapest available governance intervention in this domain.

Three cautions. The identity assumes a condition's remedy, if completed, restores the detection the gate credited itself with — reasonable for a defined corrective action, wrong for a condition that merely defers the question ("proceed subject to confirming the interface design"), which detects nothing and should never have been a conditional pass. **A conditional pass whose condition is "find out" is a hold recorded as a pass.** The model also treats closure as binary, and partial closure is common; where it matters, weight  $r$  by the share of each condition's remedy completed rather than by count. And  $r$  should be computed **per gate and per condition type**, not as one programme number, because an aggregate of 39.13 % can conceal one gate closing everything and another closing nothing; and it is the second that will produce the escape.



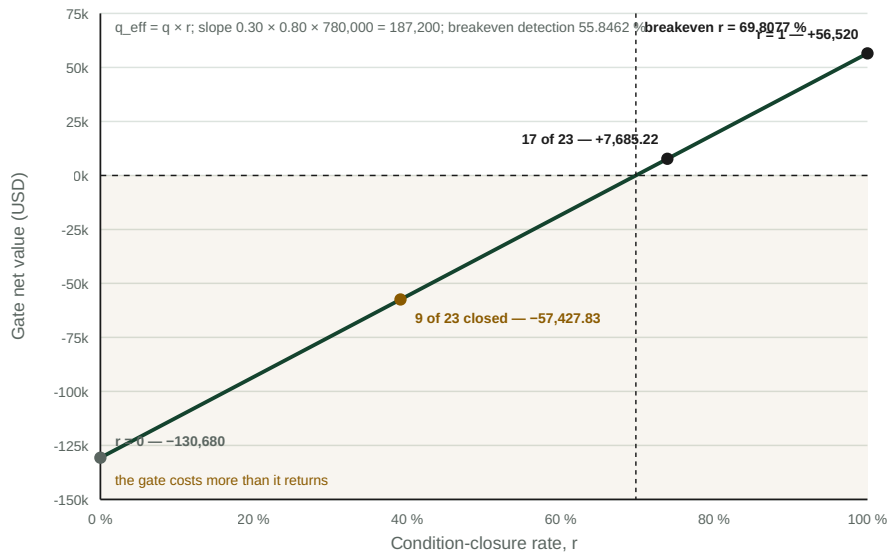


Fig 3.3.2 — The conditional pass: how closure rate decides a gate's value

### 3.3.2 Assurance lines

**Where the vocabulary comes from, before it is used.** The "three lines" language used below is not generic professional vocabulary; it belongs to an assurance architecture published by the **Institute of Internal Auditors**. It is voluntary guidance owned by a named body rather than a standard, a regulation or a requirement, and its owner has revised it, so the current formulation is better read as a model of *roles and their relationships* than as sequential lines of defence: the older "three lines of defence" framing, with its implication that assurance is a series of barriers an error passes through, is the reading its own publisher moved away from. What follows is described in this book's own words; nothing from the model is reproduced, and a reader who wants it should obtain the current version from its publisher, who has not reviewed or endorsed this volume.

It is also **one architecture among several**. The assurance structure that actually applies to a project is set by the organisation's own governance and, in regulated sectors, by what the regulator expects of that organisation, so three-line vocabulary is a *lens for finding gaps and duplication*, not an obligation to arrange anything in threes. Where an organisation's structure does not map onto it, the useful questions survive the translation: who prevents, who checks independently, who forms an opinion on the whole, and who can be told no.

#### The three lines, and what each is for.

- **First line, management.** The project's own controls: reviews, testing, quality checks, reporting. Owned by the project leader, and the only line that can prevent a defect rather than detect it.
- **Second line, oversight function.** A PMO, risk or quality function providing independent challenge while **typically** remaining inside management's chain.

Its value is comparability across the portfolio and pattern detection no single project can see.

- **Third line: internal audit (and external assurance).** Independent of management, **typically** reporting to an audit committee or equivalent body, forming an opinion on whether the whole control system works.

Both "typically" qualifications are load-bearing. Reporting lines vary with organisational form: the functional, matrix, projectised and multi-party structures of KA 3.1.2 place the second and third lines differently, and in a consortium or joint venture there may be two of each, answering to different parents with different appetites. Establish where each line actually reports **before** relying on its independence, because an assurance product's value is set by who can overrule it.

**The failure modes.** *Duplication*: three lines asking the same questions, which multiplies cost and destroys the project team's willingness to engage. *Gap*: everyone assumes another line covers a risk. *Capture*: the second line drafts the plan it later assures, and so cannot challenge it; this is the most damaging and the least visible, because the assurance product still looks independent. *Assurance as accountability transfer*: a leader treats a favourable assurance opinion as a discharge of their own obligation; Domain 1's principle applies unchanged, an opinion is information and not a transfer of accountability.

**Capture has an instrument, and it is not a reorganisation.** Because capture is a defect of *independence* rather than of structure, it is reached by the declaration duty of Domain 1, KA 1.2.2a: the reviewer who drafted, advised on or is otherwise interested in the thing being assured declares it to the commissioning authority before the review starts, and either the review passes to someone who did not, or, where no independent alternative exists, the opinion states the authorship on its face, so that a reader can discount it knowingly. Prior authorship of the assured artefact is a **structural** interest, not an item-by-item one, so the honest remedy is usually to move the work rather than to caveat the opinion. The countable check is on the assurance product itself: every opinion carries the reviewer's declaration, nil returns included, and the count of opinions where the declaration is missing is the measure of how much of the assurance programme is of unknown independence.

**The countermeasure** is an **assurance map**: risks and controls down the side, lines across the top, and each cell marked as covered, not covered or duplicated, reviewed by the accountable authority. It is a half-day artefact that surfaces both gaps and duplication, and its absence is the usual reason a project is simultaneously over-assured and unassured.

**The map becomes a decision tool when the cells carry numbers.** A map marked "covered / not covered" identifies gaps and cannot rank them, which is why maps are drawn, admired and not acted on. Give each material risk  $i$  a probability  $p_i$  that the control failure it describes is present, a consequence  $u_i$  if it escapes to

where it does damage, and each line  $j$  covering it a detection rate  $q_{ij}$ , and the map computes the only quantity that ranks its rows:

$$\text{residual exposure} = \sum_i p_i \times \prod_j (1 - q_{ij}) \times u_i$$

Two properties follow immediately and both are counter-intuitive. Because the miss probabilities **multiply**, the marginal contribution of an additional line to a risk already covered twice is small: the third line on a risk with two 0.50-plus detectors is buying the last slice of a small number. And because uncovered risks contribute  $p_i u_i$  in full, **a single gap almost always outranks every duplication in the map**. The reallocation that follows is therefore available at *constant cost*, which makes it the rare governance improvement that needs no budget decision.

#### WORKED EXAMPLE

##### Worked example 3.3.2 – Meridian's assurance map, and the reallocation that costs nothing.

1. **Setup.** Meridian's five material control risks are assured as follows, at a blended assurance day rate of **USD 950**. First line **90 days** (USD 85,500), second line **40 days** (USD 38,000), third line **30 days** (USD 28,500) – **160 days, USD 152,000** in total.

| Risk                                                | P    | U<br>(USD) | 1ST<br>LINE Q | 2ND<br>LINE Q | 3RD<br>LINE Q |
|-----------------------------------------------------|------|------------|---------------|---------------|---------------|
| R1 clinical data migration integrity                | 0.30 | 900,000    | 0.60          | 0.50          | 0.40          |
| R2 access control and confidentiality               | 0.20 | 1,200,000  | 0.50          | —             | —             |
| R3 benefits ownership in the receiving organisation | 0.35 | 685,440    | —             | —             | —             |
| R4 clinic readiness and training                    | 0.40 | 85,680     | 0.70          | 0.40          | —             |
| R5 third-party interface conformance                | 0.25 | 214,200    | 0.55          | 0.45          | —             |

$u$  for R3 is one year of the programme's realistic benefit (USD 685,440); for R4 and R5, six and fifteen weeks of delay at USD 14,280 per week. 2. **Formula.** Residual exposure =  $\sum p_i \prod (1 - q_{ij}) u_i$ . Marginal value of a line on a risk =  $p_i \times \Delta(\text{miss}) \times u_i$ . Breakeven consequence for a line on a risk =  $\text{line cost} \div (p_i \times \Delta q)$ . 3. **Substitution.** R1  $0.30 \times (0.40 \times 0.50 \times 0.60) \times 900,000$ ; R2  $0.20 \times 0.50 \times 1,200,000$ ; R3  $0.35 \times 1.00 \times 685,440$ ; R4  $0.40 \times (0.30 \times 0.60) \times 85,680$ ; R5  $0.25 \times (0.45 \times 0.55) \times 214,200$ . 4. **Result (as assured.)**

| Risk                 | COMBINED<br>MISS | RESIDUAL EXPOSURE<br>(USD) | SHARE OF<br>RESIDUAL |
|----------------------|------------------|----------------------------|----------------------|
| R1 (all three lines) | 0.1200           | 32,400.00                  | 7.87 %               |
| R2 (first line only) | 0.5000           | 120,000.00                 | 29.15 %              |
| R3 (no line)         | 1.0000           | <b>239,904.00</b>          | <b>58.27 %</b>       |
| R4                   | 0.1800           | 6,168.96                   | 1.50 %               |
| R5                   | 0.2475           | 13,253.63                  | 3.22 %               |
| <b>Total</b>         |                  | <b>411,726.59</b>          | 100 %                |

Now move the third line's **30 days** off R1 (where all three lines sit) onto R3, at the same detection rate of 0.40 and the same cost. R1's residual rises to **USD 54,000.00**; R3's falls to **USD 143,942.40**. Total residual **USD 337,364.99**: a reduction of **USD 74,361.60** (**18.06 %** of residual, **13.19 %** of residual plus assurance cost) for **no additional spend**. 5. **Interpretation.** The distribution in the fourth column is the finding, and it is the finding in almost every assurance map drawn for the first time. **All three lines sit on the risk carrying 7.87 % of residual exposure, while 87.41 % of it sits in the two risks with the least coverage**, and the largest single row, at **58.27 %**, has no assurance at all. This is the predictable result of assurance following *auditability*, not incompetence. Migration integrity is testable, sampleable and comfortable to review, so three functions review it. Benefits ownership in the receiving organisation is diffuse, politically awkward and owned outside the project, so nobody does, which is precisely Domain 2's omitted-column failure reappearing as an assurance gap. **Assurance concentrates where it is easy, not where residual exposure is, and only the arithmetic makes that visible.**

The reallocation arithmetic explains itself once the marginal values are written down. The third line on R1 was buying  $0.30 \times (0.20 - 0.12) \times 900,000 = \text{USD } 21,600$  of exposure reduction; on R3 the same 30 days buys  $0.35 \times 0.40 \times 685,440 = \text{USD } 95,961.60$ : **4.44 times** as much. Against its **USD 28,500** cost, the third line on R1 was running at a net **loss of USD 6,900**, and on R3 it returns a net **USD 67,461.60**. The breakevens make the point transferable: for the third line to pay on R1, either the consequence would have to exceed  $28,500 / (0.30 \times 0.08) = \text{USD } 1,187,500$  (against R1's 900,000) or the failure probability would have to exceed  $28,500 / (0.08 \times 900,000) = 39.58 \%$  (against R1's 0.30). **Neither is close**, and both are the sort of test a reviewer can run on a single row in a minute.

**The professional caution that must accompany this, because the arithmetic over-applies.** Nothing above justifies deleting the third line. Third-line assurance exists to form an **independent opinion on whether the**

**control system works:** a portfolio-level and constitutional product, reporting outside management, whose value is not the marginal detection it contributes to any one project's risk register. Charging its cost to a single project and testing it on detection economics misprices it by construction, and the honest conclusion from the numbers above is narrower and more useful: **the third line's effort was pointed at the wrong risk on this project.** Where an organisation genuinely wants the detection contribution of a third pass on a single risk, the marginal arithmetic says it will rarely pay, which is an argument about *where* independent assurance looks, never about whether it should exist.

Three further cautions. The multiplication of miss probabilities assumes the lines are **independent detectors**, and Domain 9, KA 9.2.2 shows why they usually are not: three functions reading the same design document with the same blind spot miss the same defect, so realised coverage on R1 is worse than 88.00 % and closer to the 80.00 % that the first two lines achieve alone. Make consecutive lines *methodologically different* or do not count them separately. The detection rates are **estimates**, and where no history exists the defensible presentation is a range with the reallocation decision computed across it: note that the R1-to-R3 conclusion survives any  $q$  for the third line above about 0.09 on R3, so it is robust in a way the point estimate does not show. And **capture invalidates a cell rather than reducing it:** a line assuring work it helped produce should be entered as  $q = 0$ , not as a lower  $q$ , because its opinion is not weak evidence, it is not evidence.

**Proportionality, and how to size it.** Assurance effort should scale with novelty, value, irreversibility and external exposure, not uniformly, and not with organisational anxiety. A high-value, low-novelty, reversible project may need less assurance than a small, novel, irreversible, publicly visible one, and a regime that cannot express that will over-assure the first and under-assure the second. The residual-exposure column above is what makes proportionality operable *within* a project: allocate the next assurance day to the row with the largest  $p_i \prod (1 - q_{ij}) u_i$ , and stop when the marginal reduction falls below the day rate. Across projects, the same logic is a portfolio judgement and belongs to the second line, whose comparability across the portfolio is the whole of its value.

### 3.3.3 Escalation design

**The principle.** Escalation is a **designed pathway with stated latency**, not an instruction to tell someone senior. A usable design states, for each escalation class: the trigger (objective, not "if concerned"), the destination, the decision required, the **time within which the decision will be made**, and the out-of-cycle mechanism if the ordinary cadence is too slow.

That fourth element is the one usually missing, and its absence has a computable cost.

## WORKED EXAMPLE

**Worked example 3.3.3 – the total latency of Meridian's escalation path.**

1. **Setup.** A decision requiring executive authority passes three tiers: the **project board** (meets every 2 weeks, papers close 1 week ahead), the **programme board** (every 4 weeks, papers 2 weeks ahead), and the **executive committee** (quarterly (every 13 weeks) papers 3 weeks ahead). Cost of delay **14,280** per week.
2. **Formula.** Total expected latency =  $\Sigma$  over tiers of  $M/2 + L$ . Cost = latency  $\times$  cost of delay.
3. **Substitution.** Project board  $2/2 + 1 = 2.0$ ; programme board  $4/2 + 2 = 4.0$ ; executive committee  $13/2 + 3 = 9.5$ . Total  $2.0 + 4.0 + 9.5$ .
4. **Result.** **15.5 weeks** of expected latency for a single decision, costing **USD 221,340** in delay; before anyone in any of the three rooms disagrees with the recommendation.
5. **Interpretation.** Two hundred and twenty-one thousand dollars is the price of an organisation chart, and it is invisible: it appears in no budget, is attributed to no decision, and is generally described afterwards as the project having been slow. Notice the distribution, the quarterly committee alone accounts for **9.5 of the 15.5 weeks (61 %)** and **USD 135,660**, which is what makes "add the executive committee to the approval path" such an expensive sentence when spoken casually in a governance review. Two redesigns follow directly. **Reduce the tiers a decision must legitimately pass:** if the programme board can decide with the executive committee *informed*, latency falls to **4.0 weeks** and **USD 57,120**, a saving of **USD 164,220**, or **74.2 %**, from a change that removes no scrutiny that anyone can name. **Add an out-of-cycle mechanism:** a written-resolution procedure with a five-working-day turnaround makes the single-tier path **1.0 week** and **USD 14,280**, a **93.5 %** reduction against the original. The general rule this supports: **count the tiers, add the latency, price it, and then justify each tier against its cost**, which is a conversation governance reviews almost never have, because until the latency is added up there is nothing to weigh the tier against.

Count the tiers, add the latency, price it — then justify each tier against its cost

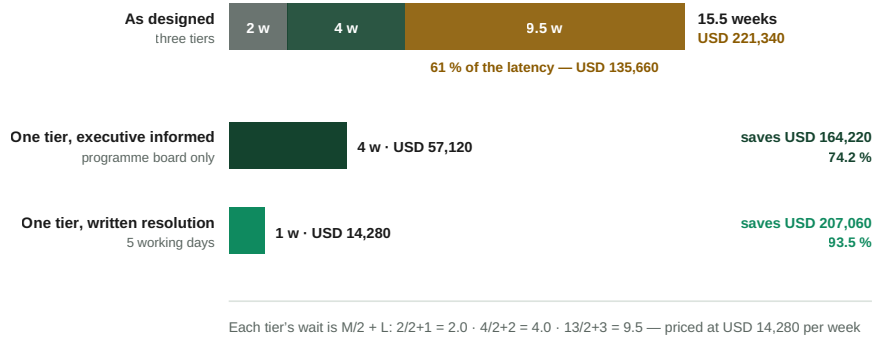


Fig 3.3.1 — The price of an escalation path

**Designing the trigger.** Objective triggers only: a forecast variance beyond a stated tolerance, a risk exposure above a stated threshold, a decision required above the delegated authority, a dependency breached. "Escalate if concerned" produces two failure modes at once (late escalation by those who fear it and constant escalation by those who fear the alternative), and both are then treated as individual judgement problems rather than as the design defect they are.

**But an objective trigger is not yet a well-specified one.** "Escalate if the forecast overrun exceeds 10 % of budget" reads as objective and is not, because *forecast* is not a single number: the earned value family produces several legitimate estimates at completion from the same data (Domain 7, KA 7.3), and they can sit on opposite sides of a tolerance. A trigger must therefore name three things the usual wording omits (**the forecast method, the measurement point and the confirmation rule**), and the cost of omitting them is not ambiguity in the abstract but a real escalation that either fires on noise or fails to fire at all. The volume's single-project thread makes this concrete.

#### WORKED EXAMPLE

##### Worked example 3.3.3b — writing Auriga's escalation trigger, and testing whether it can act.

- Setup.** Project Auriga is a **25-week** project with **BAC USD 4,000,000**. At the **week 13** data date: **PV 2,080,000**, **EV 1,920,000**, **AC 2,120,000**, giving **CPI 0.91** and **SPI 0.92** to two places. Domain 7's three estimates at completion are **4,200,000** (remaining work at plan), **4,416,667** ( $BAC/CPI$ ) and **4,608,056** ( $AC + (BAC - EV)/(CPI \times SPI)$ ). The governance framework says: *escalate to the programme board where the forecast overrun exceeds 10 % of BAC*. The escalation path available is 3.3.3's three-tier route at **15.5 weeks**, with a single-tier alternative at **4.0 weeks** and a written-resolution route at **1.0 week**. Planned average burn is  $4,000,000/25 = \text{USD } 160,000$  a week.

2. **Formula.** Variance at completion  $VAC = EAC - BAC$ , expressed as a share of BAC and tested against the tolerance. Then the **decision action window** =  $(\text{remaining duration} - \text{escalation latency}) \div \text{remaining duration}$ , and the last week at which a trigger can still produce an actionable decision =  $\text{planned finish} - \text{escalation latency}$ .
3. **Substitution.**  $4,200,000 - 4,000,000$ ;  $4,416,666.67 - 4,000,000$ ;  $4,608,055.56 - 4,000,000$ ; each over 4,000,000. Tolerance  $0.10 \times 4,000,000 = 400,000$ . Remaining duration  $25 - 13 = 12$ . Windows  $(12 - 15.5)/12$ ,  $(12 - 4.0)/12$ ,  $(12 - 1.0)/12$ .
4. **Result.**

| FORECAST METHOD                      | EAC<br>(USD) | VAC<br>(USD) | VAC AS %<br>OF BAC | TRIGGER AT<br>10 %? |
|--------------------------------------|--------------|--------------|--------------------|---------------------|
| Remaining work at plan               | 4,200,000.00 | 200,000.00   | 5.00 %             | No                  |
| BAC/CPI                              | 4,416,666.67 | 416,666.67   | 10.42 %            | Yes                 |
| $AC + (BAC - EV) / (CPI \times SPI)$ | 4,608,055.56 | 608,055.56   | 15.20 %            | Yes                 |

The three methods span **USD 408,055.56** on identical data, and the tolerance line at 400,000 falls inside that span. And the action windows:

| ESCALATION ROUTE           | LATENCY | WEEKS LEFT AFTER THE<br>DECISION | ACTION<br>WINDOW |
|----------------------------|---------|----------------------------------|------------------|
| Three tiers as designed    | 15.5 w  | -3.5                             | -29.17 %         |
| Single tier, board decides | 4.0 w   | 8.0                              | 66.67 %          |
| Written resolution         | 1.0 w   | 11.0                             | 91.67 %          |

The as-designed path returns its decision **3.5 weeks after Auriga's planned completion**. Working backwards, the latest week at which that path can still deliver an actionable answer is  $\text{week } 25 - 15.5 = 9.5$ : the first **38.00 %** of the project. 5. **Interpretation.** Two results, and the second is the more important.

**The trigger's answer depends entirely on which forecast method it names, and nobody has named one.** *BAC/CPI* clears the tolerance by USD 16,666.67 and the plan-based estimate misses it by USD 200,000; the composite clears it by USD 208,055.56. A framework that says "the forecast" has therefore delegated the escalation decision to whoever prepares the report, which is the opposite of what an objective trigger is for, and it does so invisibly, because each of the three numbers is correctly calculated. Worse, the *BAC/CPI* trigger is **mar-**



**ginal to the point of being noise.** The cost performance index at which  $BAC/CPI$  exactly equals  $1.10 \times BAC$  is  $1/1.10 = 0.9091$ , and Auriga's CPI is **0.9057**: below it by **0.0034**. In earned-value terms,  $EV$  would need to be **USD 1,927,272.73** rather than 1,920,000 for the trigger not to fire: **USD 7,272.73**, or **0.18 %** of BAC, one accrual timing difference. **A trigger stated on a single ratio at a single data date fires on measurement noise**, which is how organisations acquire a reputation for escalating trivia and then stop escalating at all. The corrections are cheap and specific: name the method ( $BAC/CPI$  is the defensible default because it is the method whose assumption, that present cost performance persists, is the one a governance body should be asked to accept or reject); state the measurement point (a defined data date, not "when noticed"); and add a **confirmation rule** (two consecutive data dates beyond tolerance, or one date beyond a wider secondary tolerance), so that a single period's noise cannot fire it and a sustained trend cannot fail to.

**The second result is that on this project the trigger cannot help, whatever it says.** An escalation latency of 15.5 weeks against 12 weeks of remaining duration gives a **negative action window**: the decision is delivered after the work it concerns has finished, so the escalation is **ceremonial**, not slow. This is the test that belongs beside every escalation path and is almost never applied (*is the latency shorter than the remaining duration of the thing being decided?*), and its diagnostic form is sharper still: the as-designed path is actionable only during the first **38.00 %** of a 25-week project, so from week 9.5 onward Auriga has an escalation route that can produce decisions and cannot produce *useful* ones. Note what this does to the two redesigns of Worked example 3.3.3. There they saved **USD 164,220** and **USD 207,060**; here they do something that cannot be expressed in money at all. They convert a decision from impossible to possible, moving the action window from  $-29.17 \%$  to **66.67 %** and **91.67 %**. **Latency priced in currency understates the case for a shorter path; latency compared with remaining duration is what shows when the path is not a cost but a nullity.**

Three cautions. The action window uses **planned** remaining duration, and at SPI 0.92 Auriga's remaining duration is itself likely longer than 12 weeks, which flatters the as-designed path. The defensible practice is to compute the window on the current schedule forecast (Domain 6) and to state which duration was used. The window is a **necessary and not a sufficient** test: a decision returned inside the window still has to arrive early enough for the *response* to be implemented, so a fuller form subtracts the implementation lead time as well, and for an irreversible response that lead time can exceed the decision latency. And a negative window is **not an argument for skipping the escalation**: the decision may be needed for reasons that outlive the project, and the honest response is to escalate *and* record that the ordinary path cannot answer in time, which is the evidence that gets the out-of-cycle mechanism built before the next project needs it.

**The escalation culture problem.** Escalation is only a functioning mechanism if using it is safe. Where escalation is read as failure, it happens late, which is precisely when the decision can no longer help. The countermeasures are governance ones and belong to the sponsor and the committee: respond to early escalation visibly and well; separate the escalation of an *issue* from any judgement about the person raising it; and track the **lead time** of escalations (how far before the impact they arrive), because a shortening lead time is the earliest available signal that the mechanism is decaying, and it appears in no standard report.

### 3.3.4 Auditability and the decision record

**The principle.** A decision that is not recorded has not been made. It has been remembered, and memories diverge exactly when the stakes rise. The decision record is the mechanism that converts a meeting into an institutional fact.

**What a decision record must contain**, per decision, and it is short enough that its absence is never a resourcing problem: a unique reference; the date; the decision-maker by name and role (never a committee alone, a body records the decision, a person is accountable for it); the decision, in words that permit only one reading; the options considered and why the chosen one was chosen; the information relied on, referenced to its version; **the interests declared in relation to the decision, naming who declared what and what was done about it, with a nil return recorded where there were none**; the conditions attached, with owners and dates; and the review date if the decision is provisional.

**The declared-interest field is the cheapest of those and the only one that is worthless when it is optional.** A field completed only where there was something to declare cannot be distinguished, a year later, from a field nobody filled in, so the nil return is the part that carries the evidence. Where an interest was declared, the entry states the management action (most often that the interested party took no part and who decided instead), because a declaration recorded without an action is a record of a conflict rather than of its management. The duty itself, its four steps and the case where abstention is not enough are Domain 1, KA 1.2.2a; this is where it lands in the governance machinery.

**Why the "information relied on, referenced to its version" line matters more than it looks.** When a decision is examined afterwards (in a lessons review, a dispute, an audit or an inquiry) the question is almost never "was the decision right?" but "**was it reasonable on what was known at the time?**" Only a versioned reference can answer that, and its absence converts a defensible decision into an indefensible one at exactly the moment the defence is needed.

**Two governance defects the record exposes**, and this is the practical case for keeping it well. The **re-decided decision**: the same question arriving at the same body a third time is a symptom of either an unrecorded decision or an illegitimate one, and the log makes the recurrence visible where minutes do not. And the **decision nobody made**: a change that took effect without an entry, which is how

scope and budget move without anyone having decided, the most common finding of a governance audit and the hardest to see from inside.

**RACI and its integrity check.** Responsibility assignment matrices (Responsible, Accountable, Consulted, Informed) are the standard artefact for decision rights, and their standard failure is being drafted, circulated and never checked. The checks are countable, and a matrix that fails any of them is not a matrix but a document: **exactly one A** per decision (two make it undecidable, none make it unowned); the A holds sufficient authority under the delegation schedule; C and R are distinguished (consultation is not agreement); and the count of C's per decision is bounded, since consultation is where latency accumulates invisibly.

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#### WORKED EXAMPLE

##### Worked example 3.3.4 – auditing Meridian's decision-rights matrix.

1. **Setup.** Meridian's matrix covers **12** decision classes across 7 roles. On review: **9** classes carry exactly one Accountable, **2** carry two, and **1** carries none.
2. **Formula.** Defect rate = classes failing the single-A test ÷ total classes.
3. **Substitution.**  $(2 + 1)/12$ .
4. **Result.** 3 of 12 classes are defective: a **25.0 %** defect rate.
5. **Interpretation.** A quarter of the programme's decision classes cannot be decided as documented, and the two failure types behave differently under stress, which is why they are worth distinguishing rather than totalling; and why the 25.0 % headline is the least useful number in the example.

**Price them separately.** The **two-A** classes will be decided, badly: by whichever holder acts first, or after a delay while the two reconcile, and reconciliation in practice means the item returns to the next meeting. Those two classes generated **11** decisions over the year; each incurring one extra committee cycle at 3.2.3's rates, with the standing **25 %** critical-path share, costs  $11 \times 0.25 \times 4 \times 14,280 = \text{USD } 157,080$ : **USD 78,540** per defective class. The **zero-A** class behaves differently and worse: nothing forces it to a decision at all, so it drifts until the consequence forces an escalation, and an escalation that arrives without an owner arrives at the top. Its **4** decisions, drifting to the full three-tier path of Worked example 3.3.3 at **15.5 weeks**, cost  $4 \times 0.25 \times 15.5 \times 14,280 = \text{USD } 221,340$ : coincidentally exactly one expected three-tier escalation, and **2.82 times** the cost of a two-A class. Total documented cost of a 25.0 % defect rate: **USD 378,420**.

That ordering is the professional content of the example. **A decision class with two accountable roles is expensive; a class with none is nearly three times as expensive**, because the two-A defect delays a decision and the zero-A defect *relocates* it: upward, late, and to people with no preparation. Yet a reviewer scanning a matrix finds the two-A cases first, since they are visible as duplicated letters, while the zero-A case is visible only as a blank in a row

nobody thought to write. The practical instruction is to audit the matrix **by decision class, not by role**: list every decision the project will face, then look for its A, which finds the missing rows that a column-by-column read never will. Two cautions. The costing assumes each defective class's decisions are **independent occasions of delay**, which bounds rather than forecasts as elsewhere in this domain; present the counts and the delay-weeks alongside the currency. And a single A is **necessary and not sufficient**: the A must also hold authority sufficient for the class under the delegation schedule, which is a second and separate check, a matrix can pass the single-A test in full while every A sits below the threshold its class requires, at which point the matrix is decorative, not wrong. The test costs an hour and it is the highest-yield governance check available before mobilisation.

**The re-decided decision has a price, and it is the largest hidden number in this domain.** 3.3.4 named the defect; counting it is what makes it actionable, and the count comes from the decision log because it comes from nowhere else: minutes record what a meeting decided, never that the meeting had decided it before.

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#### WORKED EXAMPLE

##### Worked example 3.3.4b — Meridian's re-decision tax.

1. **Setup.** Meridian's decision log holds **148** decisions over twelve months. Clustering the decision text shows **19** questions that arrived at the same body more than once, of which **6** arrived three or more times. Of the 19, **13** carried no versioned reference to the information relied on and **6** carried no named decision-maker. The committee's capacity is **104** item-slots a year against the **77** items of forward demand computed in 3.2.2;  $E[\text{wait}]$  is **4.0 weeks**, cost of delay **14,280** per week, critical-path share **25 %**.
2. **Formula.** Repeat slots consumed = second appearances + third appearances. Corrected demand = forward demand + repeat slots. Delay cost = repeat slots  $\times$  critical-path share  $\times E[\text{wait}] \times$  cost of delay.
3. **Substitution.** Repeat slots  $19 + 6 = 25$ . Corrected utilisation  $(77 + 25) / 104$ . Delay cost  $25 \times 0.25 \times 4 \times 14,280$ .
4. **Result.** Re-decision rate **12.84 %** of all logged decisions. Repeat slots **25**, which is **24.04 %** of the committee's annual capacity. Corrected demand **102** against capacity 104, utilisation **98.08 %**, not the 74.0 % of 3.2.2. Delay cost **USD 357,000.00**, which is **2.08 times** the entire annual saving from raising the delegation threshold (USD 171,360).
5. **Interpretation.** Start with the reconciliation, because it resolves a contradiction the domain has been carrying. Worked example 3.2.2 computed 74.0 % utilisation and the members reported a committee that was full; **98.08 %** is why. The forward demand model counted the decisions the committee was asked to make

and could not count the ones it made twice, and a capacity model that omits re-decisions will always read low by roughly the re-decision rate. **The single most useful correction to a committee capacity estimate is to add last year's repeat count**, and it takes an afternoon with the log.

Then the size of it. **USD 357,000 a year is spent deciding things that had already been decided**: more than twice the celebrated saving from the threshold change, and unlike that saving it requires no delegation, no risk appetite conversation and no approval from anybody. It is also, uniquely in this domain, a defect whose remedy costs nothing: the causal split says **68.42 %** of the re-decisions lacked a versioned information reference and **31.58 %** lacked a named decision-maker, and both are **fields on a template**. A decision re-arrives because its basis can be disputed ("that was decided on the old numbers") or its authority can be disputed ("the committee noted it, nobody decided it"), and the two fields close both routes. This is the concrete answer to the question 3.3.4 raises and most organisations treat as rhetorical: *what is the decision record actually worth?* Here, **USD 357,000 a year and 25 committee slots**, from two fields.

Three cautions. Not every repeat is a defect: a **provisional** decision with a stated review date is supposed to return, and a decision returning because the world changed is governance working, not failing, so the count must exclude scheduled reviews and material-change re-openings, and the defensible metric is *unplanned* re-decisions. The clustering that produces the count is a judgement about whether two differently worded questions are the same question, which is exactly the task AI is good at and exactly the task where a false positive is embarrassing, so every flagged pair is confirmed by a human before it is reported (see the AI treatment below). And the delay cost, as always here, **adds waits that may overlap**; the slot count (25 of 104) is the harder number and the one to lead with, because nobody can argue with it.

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**The cumulative test, designed rather than asserted.** Toolkit 3.T.2 and Case study B both call for a rule that aggregates related decisions, and a rule of that shape has two parameters: the aggregate threshold **X** and the **relatedness class** over which decisions are summed within a period **P**. Neither can be set without the other, and setting them badly does more damage than having no rule, because a cumulative test that trips constantly re-centralises the whole delegated band while appearing to be a control improvement.

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#### WORKED EXAMPLE

### Worked example 3.3.4c — sizing Meridian's cumulative test without re-centralising the delegation.

- Setup.** Meridian has raised its delegation threshold to **USD 25,000** (3.2.3), delegating **36** changes a year: the 24 at or below 10,000 averaging **5,000**, and the 12 in the 10,001–25,000 band averaging **17,000** — **USD 324,000** of delegated value a year. Separately, one genuine cluster occurred: 7 related changes to the clinical template set, in sequence **8,400 · 6,200 · 11,500 · 9,800 · 4,600 · 13,200 · 7,300**, none individually above the threshold. The programme has **3** workstreams and **14** deliverable-level change classes. Period **P** = one quarter. Escalating one change costs **4.0 weeks** at 25 % critical-path share, i.e. an expected  $0.25 \times 57,120 = \text{USD } 14,280$ .
- Formula.** Cluster aggregate =  $\Sigma$  of the related changes; the rule trips at the first change whose running total exceeds **X**. Base-rate aggregate per relatedness class per period = delegated value  $\div$  (classes  $\times$  periods). Annual cost of the rule = expected trips  $\times$  expected escalation cost.
- Substitution.** Running totals **8,400** → **14,600** → **26,100**. Broad relatedness (workstream):  $324,000 / (3 \times 4)$ . Narrow relatedness (deliverable class):  $324,000 / (14 \times 4)$ .
- Result.** The cluster totals **USD 61,000** and trips at the **third** change (running total **26,100** against **X** = 25,000), leaving **USD 34,900** of the cluster to be authorised at the aggregate's level.

| RELATEDNESS CLASS             | CLASSES<br>$\times$<br>PERIODS | BASE-RATE<br>AGGREGATE<br>PER CLASS-<br>PERIOD | X =<br>25,000<br>TRIPS?              | TRIPS<br>A<br>YEAR | ANNUAL<br>COST     |
|-------------------------------|--------------------------------|------------------------------------------------|--------------------------------------|--------------------|--------------------|
| Workstream<br>(broad)         | $3 \times 4 = 12$              | <b>USD 27,000</b>                              | Yes, on<br>the base<br>rate<br>alone | <b>12</b>          | <b>USD 171,360</b> |
| Deliverable<br>class (narrow) | $14 \times 4 = 56$             | <b>USD 5,785.71</b>                            | No                                   | <b>1</b>           | <b>USD 14,280</b>  |

The broad rule costs **USD 171,360** a year (**exactly** the saving the threshold increase produced), while the narrow rule costs **USD 14,280**, a factor of **12** less, and catches the same cluster. 5. **Interpretation.** The equality in the first row is not a coincidence and is worth naming, because it generalises. Raising the threshold delegated 12 changes and saved  $12 \times 0.25 \times 57,120$ ; a cumulative rule that trips 12 times a year escalates 12 changes and costs  $12 \times 0.25 \times 57,120$ . **A cumulative test cancels a delegation exactly when its annual trip count equals the number of decisions the delegation released.** So the trip count, not the threshold, is the quantity to design against; and the trip count

is driven overwhelmingly by the *relatedness class*, because that is what determines how much unrelated traffic is being summed.

The design rule that falls out is a **window**, and it is the transferable result:  $x$  must sit comfortably **above** the base-rate aggregate of its relatedness class per period and comfortably **below** the aggregates of the clusters it must catch. At deliverable-class relatedness,  $x = 25,000$  sits **4.32 times** above the base rate of 5,785.71 and **2.44 times** below the 61,000 cluster, a working window. Broadening relatedness to the workstream multiplies the base rate by **4.67**, to 27,000, which closes the window from below: every workstream now trips every quarter on ordinary traffic, and the rule has become a threshold reduction wearing a control's clothing. **Widening the relatedness class is the error, and it is the natural error**, because a broad class feels safer and costs nothing to write.

Two further observations belong in the record. The rule catches the cluster **after three of seven changes**: the first 26,100 was already decided under delegated authority, and the rule cannot undo it. So the cumulative test is a **containment** control, not a prevention control, and it must be paired with the provision Case study B actually needed: every change generates a **register entry regardless of value**, which costs no latency at all and is what makes the aggregate visible before it trips. A cumulative test without universal registration is a rule that cannot see its own inputs. And relatedness should be defined by **what the changes touch** (the same deliverable, the same assured control, the same interface) not by who requested them or which budget they hit, because the exposure the rule exists to catch is a coherent change to one thing, arriving in instalments.

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### **Custody and retention: the part that decides whether any of this survives.**

Everything above assumes the record still exists when it is wanted. The questions that make a decision record valuable (*was it reasonable on what was known at the time?*) are asked in a lessons review, a dispute, an audit or an inquiry, all of which happen years after the decision and long after the temporary organisation that made it has been dissolved. A record with no custodian and no retention period is the artefact most likely to be missing at the moment it is needed, and its absence converts a defensible decision into an indefensible one exactly as reliably as a missing version reference does.

Four provisions close that gap, and they are decided in advance rather than at closure.

**A named custodian, by class.** The decision log, the change log (Domain 4, KA 4.4), gate packs, the baseline archive (Domain 4, KA 4.3.3), the assurance opinions of 3.3.2 and acceptance evidence (Domain 5, KA 5.4.3) each have one named role accountable for their existence, completeness and retrievability. A role, not a person, because people move; and one role per class, because a record class custodied by "the project" is custodied by nobody once the project ends.



**A system that preserves version and prevents silent amendment.** The record is held where an earlier version cannot be quietly replaced by a later one. Where a record is amended, the amendment travels with it (**what changed, who changed it, when, and why**), because the difference between a corrected record and a rewritten one is the whole of its evidential value. An "amended" record that looks identical to an original is worse than no record, since it invites reliance it cannot support.

**A retention period set as a governance decision, taken in advance.** The period for each class is set at the longest of: the limitation period under the contract, any retention requirement the organisation is subject to, the benefits-realisation horizon the record must outlive (Domain 16, KA 16.4.1's measurement plan can run for years after closure), and the organisation's own records policy. It is stated as a schedule by class, never as a single rule for everything: Domain 16, KA 16.4.4 works the economics, and finds that retaining contractual and technical evidence pays as insurance at a very low probability of ever needing it while retaining personal data is the opposite trade. **The standing caveat:** retention requirements, limitation periods and disposal obligations are jurisdiction-, sector- and record-class-specific, they differ substantially, and they are taken from the organisation's records and legal functions and from qualified counsel. Nothing here states a legal minimum or maximum, and nothing here should be relied on as stating one.

**An explicit handover of custody at closure.** The project ends; the records do not. Closure transfers each class to a named continuing custodian in a permanent organisation, with the transfer itself recorded: what was transferred, in what form, to whom, and until when. This is Domain 16's responsible-archive obligation seen from the governance end, and it is the provision most often discovered to be missing at the point where the only people who knew where anything was kept have left.

**Who may see it, and who may not.** Access to the decision record is a governance decision too. Declared-interest entries, escalation notes and dissents name identified individuals and record positions attributed to them, so the log is not a general-circulation document: it carries a stated access list, and any entry about an identified person attracts the data-protection considerations Domain 11, KA 11.1.2 sets out for the stakeholder register and Domain 16, KA 16.4.4 sets out for the archive. Settle the wording, the holding and the access with whoever holds data-protection accountability in the organisation **before** the log is built rather than after it is requested; keeping the record itself remains non-optional.

## AI in this KA

**Where it earns its place.** Extracting a structured decision register from meeting minutes and flagging entries missing an owner, a date or a versioned information reference, a genuinely tedious task with a clear right answer. Running the RACI integrity checks above across a large matrix. Detecting re-decided decisions by clustering decision text across a long log, which is exactly the pattern humans miss because the recurrences are months apart. Testing gate criteria for assessability and



flagging the unmeasurable ones. Modelling gate and escalation economics across alternative designs, as computed above.

**Where it must not go.** No gate decision, no assurance opinion, and no authorship of a decision record entry. The record must show the accountable human, and a record generated wholesale by a tool cannot evidence that a person applied judgement. Nor should a model's summary of a decision's history be relied on for a dispute or an audit without verification against the source documents: the summary is a convenience, the versioned source is the evidence.

**Verification, concretely.** Any AI-produced register or audit is checked against the source minutes on a sampled basis with the sample size stated; every flagged defect is confirmed by a human before it is reported as a finding; and the gate and latency arithmetic is reproduced by hand, because all of it is a handful of operations and none of it should ever be taken on trust.

#### KEY TERMS — KA 3.3

| TERM                                              | MEANING                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Stage gate</b>                                 | A continuation decision by a named authority against pre-set criteria, with power to stop, hold or redirect.                                                                                                                                                                                             |
| <b>Conditional pass</b>                           | Proceeding subject to conditions: a real instrument only if the conditions have owners, dates and consequences.                                                                                                                                                                                          |
| <b>Three lines of assurance</b>                   | Management controls; independent oversight typically inside management; independent audit typically reporting outside it. A model of roles published by a professional body, revised by its owner, voluntary rather than required, a lens for finding gaps and duplication, not an obligatory structure. |
| <b>Assurance map</b>                              | Risks and controls against assurance lines, marking coverage, gaps and duplication.                                                                                                                                                                                                                      |
| <b>Assurance capture</b>                          | An assurance function assuring work it helped produce, and therefore unable to challenge it.                                                                                                                                                                                                             |
| <b>Escalation class</b>                           | A defined trigger, destination, decision, latency and out-of-cycle route.                                                                                                                                                                                                                                |
| <b>Out-of-cycle mechanism</b>                     | A written-resolution or delegated-authority route used when the ordinary cadence is too slow.                                                                                                                                                                                                            |
| <b>Escalation lead time</b>                       | How far before impact an escalation arrives; a shortening trend signals a decaying mechanism.                                                                                                                                                                                                            |
| <b>Decision record</b>                            | The versioned, attributable log that converts a decision from a memory into an institutional fact.                                                                                                                                                                                                       |
| <b>Single-A test</b>                              | The check that each decision class has exactly one Accountable role.                                                                                                                                                                                                                                     |
| <b>Effective detection (<math>q_{eff}</math>)</b> | A gate's detection rate multiplied by its condition-closure rate: $q_{eff} = q \times r$ .                                                                                                                                                                                                               |

| TERM                                           | MEANING                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Condition-closure rate (<math>r</math>)</b> | The share of a gate's conditions verified closed by the following gate.                                                                                                                                                                                                                                                                                                           |
| <b>Residual exposure</b>                       | $\sum p_i \prod (1 - q_{i,j}) u_i$ across an assurance map; the only quantity that ranks its rows.                                                                                                                                                                                                                                                                                |
| <b>Decision action window</b>                  | $(\text{remaining duration} - \text{escalation latency}) \div \text{remaining duration}$ ; negative means the escalation is ceremonial.                                                                                                                                                                                                                                           |
| <b>Confirmation rule</b>                       | The provision that stops a tolerance trigger firing on one period's measurement noise.                                                                                                                                                                                                                                                                                            |
| <b>Re-decision tax</b>                         | The committee slots and delay cost consumed by questions that had already been decided.                                                                                                                                                                                                                                                                                           |
| <b>Cumulative test</b>                         | The delegation-schedule provision that related decisions aggregating above $X$ within period $P$ require the authority appropriate to the aggregate. Setting $X$ and $P$ is arithmetic, not judgement, and is derived in Domain 4, KA 4.3.3, a round number chosen without reference to the observed decision rate provides the appearance of a control and none of the function. |
| <b>Relatedness class</b>                       | The set over which a cumulative test sums; widening it multiplies false trips and re-centralises the delegation.                                                                                                                                                                                                                                                                  |
| <b>Record custodian</b>                        | The named role accountable for a record class existing, being complete and being retrievable: one role per class, so that no class is custodied by "the project".                                                                                                                                                                                                                 |
| <b>Retention period</b>                        | How long a record class is held, set in advance at the longest of contractual limitation, applicable retention requirement, benefits-realisation horizon and records policy; a schedule by class, never one rule, and never a legal position stated by this book.                                                                                                                 |
| <b>Custody handover</b>                        | The recorded transfer of each record class to a continuing custodian at closure, so records outlive the temporary organisation.                                                                                                                                                                                                                                                   |

## SAMPLE MCQs — KA 3.3

**MCQ 3.3-A** [3.3.1 · Application] A gate costs 45,000 in review effort and 6 weeks of elapsed time at a delay cost of 14,280 per week. Expected remediation cost with the gate is 82,800; without the gate it is 270,000. The gate's net value is:

- A. USD 141,480
- B. USD 56,520 CORRECT
- C. USD 85,680
- D. USD 187,200

*Rationale:*  $270,000 - (45,000 + 85,680 + 82,800) = 56,520$  (3.3.1). A omits the delay cost; C is the delay cost alone; D omits the remediation cost.

**MCQ 3.3-B** [3.3.1 · Evaluation] For the gate above, the elapsed time at which it stops adding value is closest to:

- A. 6 weeks
- B. 8 weeks
- C. 10 weeks CORRECT
- D. 19 weeks

*Rationale:*  $(270,000 - 45,000 - 82,800) / 14,280 = 9.96$  weeks (3.3.1), the arithmetic behind the complaint that assurance has become an obstacle.

**MCQ 3.3-C** [3.3.3 · Application] Three tiers must approve a decision: 2-weekly with 1-week papers, 4-weekly with 2-week papers, and 13-weekly with 3-week papers. Total expected latency is:

- A. 9.5 weeks
- B. 15.5 weeks CORRECT
- C. 19.0 weeks
- D. 6.0 weeks

*Rationale:*  $(2/2+1) + (4/2+2) + (13/2+3) = 2.0 + 4.0 + 9.5 = 15.5$  (3.3.3). A is the top tier alone; D counts only half the intervals and omits the paper lead times.

**MCQ 3.3-D** [3.3.2 · Analysis] A PMO drafts the delivery plan and later provides second-line assurance on it. The defect is:

- A. duplication of first-line controls
- B. assurance capture — the function cannot challenge what it produced CORRECT
- C. a proportionality failure
- D. an assurance gap

*Rationale:* Capture is the most damaging and least visible line failure, because the product still looks independent (3.3.2).

**MCQ 3.3-E** [3.3.4 · Comprehension] The decision-record field most often missing and most consequential when a decision is later examined is:

- A. the date
- B. the versioned reference to the information relied on CORRECT
- C. the decision-maker's role
- D. the decision reference number

*Rationale:* The retrospective question is whether the decision was reasonable on what was known at the time, which only a versioned reference can answer (3.3.4).

**MCQ 3.3-F** [3.3.4 · Analysis] In a 12-class decision-rights matrix, 2 classes carry two Accountable roles and 1 carries none. Which statement is most accurate?

- A. the defect rate is 8.3 % and only the zero-A class matters
- B. the defect rate is 25.0 %; the two-A classes will be decided late or twice, and the zero-A class will drift until it becomes an escalation CORRECT
- C. the matrix is acceptable because 9 of 12 classes are correct
- D. the defect rate is 25.0 % and all three classes fail identically

*Rationale:*  $3/12 = 25.0\%$ , and the two failure types behave differently under stress, which is why they are distinguished rather than totalled (3.3.4).

**MCQ 3.3-G** [3.3.1 · Evaluation] A gate with a nominal detection rate of 0.80 has a net value of USD 56,520 and a breakeven detection probability of 55.85 %. Of the 23 conditions it issued, 9 were verified closed. The best evaluation of the gate is that it:

- A. remains worthwhile, since its detection rate is well above the breakeven
- B. has an effective detection rate of 31.30 %, is therefore below breakeven, and destroys USD 57,427.83 CORRECT
- C. is worthwhile but should be shortened from 6 weeks to 4
- D. has an effective detection rate of 39.13 % and is therefore below breakeven

*Rationale:*  $q_{eff} = q \times r = 0.80 \times 9/23 = 0.3130$ , which is below the 0.5585 breakeven, inverting the gate's value (3.3.1b). A ignores closure altogether, the commonest error, because governance reports count conditions issued, not closed. C treats elapsed time as the binding constraint when the binding constraint is  $r$ . D mistakes the closure rate itself for the effective detection rate, omitting the multiplication by  $q$ . It reaches the right conclusion from the wrong quantity, and will not survive a gate where  $q \times r$  and  $r$  fall on opposite sides of the breakeven.

**MCQ 3.3-H** [3.3.2 · Evaluation] An assurance map shows all three lines on a risk carrying 7.87 % of residual exposure and no line on the risk carrying 58.27 % of it. The soundest conclusion is that:

- A. the third line should be discontinued, since its marginal detection value is negative
- B. the third line's effort is pointed at the wrong risk; reallocating it reduces residual exposure at constant cost, but third-line existence is not justified on project-level detection economics CORRECT
- C. the coverage is acceptable because the highest-consequence risk is the one with three lines
- D. the uncovered risk should be accepted, since no line has the competence to assure it

*Rationale:* Reallocating the third line's 30 days from R1 to R3 cut residual exposure by USD 74,361.60 for no extra spend, but third-line assurance exists to opine on the control system across a portfolio and is mispriced by a single project's detection arithmetic (3.3.2). A over-applies the arithmetic, the named error. C confuses consequence with residual exposure, which is consequence *after* detection. D treats a coverage gap as a capability fact.

**MCQ 3.3-I** [3.3.3 · Application] A 25-week project reaches week 13 and a trigger fires. The available escalation path has a total expected latency of 15.5 weeks. The decision action window is:

- A. 66.67 %
- B. 0 %
- C. -29.17 %, so the decision arrives 3.5 weeks after planned completion
- CORRECT
- D. 129.17 %

*Rationale:*  $(12 - 15.5)/12 = -0.2917$  (3.3.3b). A is the window for the 4.0-week single-tier path. B assumes the window floors at zero, which hides the finding. D computes latency as a share of remaining duration (15.5/12) rather than the window itself, a real ratio, but not this one.

**MCQ 3.3-J** [3.3.4 · Analysis] A programme delegates 12 changes a year by raising its threshold, saving USD 171,360 in delay. It then adds a cumulative test at the same threshold, summed over workstreams, which trips 12 times a year. The consequence is that:

- A. the control improves at negligible cost
- B. the trip count equals the number of decisions the delegation released, so the rule cancels the saving exactly CORRECT
- C. the saving falls by about a quarter
- D. the rule is sound but the threshold should be lowered again

*Rationale:*  $12 \times 0.25 \times 57,120 = 171,360$ , identical to the saving (3.3.4c). The trip count, driven by the width of the relatedness class, is the quantity to design against. A is the assumption a governance review makes when it does not count trips; C invents a partial figure; D compounds the error.

#### SELF-CHECK — KA 3.3

1. *What does a gate buy, and what does that imply about where gates belong?* — Optionality against irreversibility; so gates belong where irreversibility steps up, not at calendar intervals.
2. *Name the assurance failure mode that is hardest to see, and why.* — Capture: the assurance product still looks independent while the function cannot challenge what it helped produce.

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3. *Which element of an escalation class is most often missing?* — The stated time within which the decision will be made, and with it the out-of-cycle mechanism.
  4. *State the identity that connects conditional passes to gate value.* —  $q_{\text{eff}} = q \times r$ : an untracked condition leaves the defect in the work, so closure rate multiplies detection rate (3.3.1b).
  5. *Which row of an assurance map should receive the next assurance day?* — The one with the largest  $p \prod (1 - q) u$ , which is almost always an uncovered risk rather than a duplicated one (3.3.2).
  6. *Why must an escalation trigger name a forecast method?* — Because the earned-value family produces several legitimate estimates from the same data, and they can straddle the tolerance: auriga's three sit at 5.00 %, 10.42 % and 15.20 % of BAC against a 10 % trigger (3.3.3b).
  7. *What are the two parameters of a cumulative test, and which one usually goes wrong?* — The aggregate threshold  $x$  and the relatedness class; widening the class is the natural error and it re-centralises the delegated band (3.3.4c).
  8. *Whose is the "three lines" vocabulary, and what does that change?* — It is an assurance architecture published by a professional body, voluntary and since revised by its owner into a model of roles rather than sequential defences. It is a lens for finding gaps and duplication; the applicable structure is set by the organisation's own governance and, in regulated sectors, by what the regulator expects (3.3.2).
  9. *Name the four provisions that keep a decision record usable years later.* — A named custodian by record class; a system that preserves version and carries any amendment with its amender and reason; a retention period set in advance at the longest of the applicable drivers, with its source named; and a recorded handover of custody at closure (3.3.4). Retention requirements themselves come from the records and legal functions, not from this book.
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## Advanced topics — Domain 3

### 3.A.1 Governance under stress, and the recovery structure

Governance designed for steady state is tested in crisis, and the failures are consistent: the cadence is too slow for the decision rate, authority fragments as senior figures intervene individually, information becomes contested at the moment it must be relied on, and the decision record (never robust) collapses first, so that a fortnight later nobody can reconstruct why a choice was made.

A **recovery governance** design is a legitimate and different structure, and it should be prepared in advance rather than improvised: a single accountable decision-maker with materially raised authority; a short daily or twice-weekly decision cadence with a standing agenda of decisions rather than reports; a *narrowed* membership, since crisis governance fails from too many participants before it fails from too few; a

single authoritative information source, with disputes about the data resolved before decisions rather than during them; and a decision log kept in the room, in real time, because reconstruction afterwards is precisely what will not be possible.

Two invariants are worth stating because they are what distinguishes recovery governance from an abandonment of governance. Raised authority must remain **bounded and time-limited**, with a stated review date, or it does not revert. And the escalation of *bad news* must be made safer, not harder, because the mechanism's failure mode under stress is silence; and silence is the one condition under which no governance design of any shape can work.

### 3.A.2 Governance of AI-assisted delivery

Where a project uses AI in its own delivery (estimating, scheduling, code generation, risk analysis, document review) governance acquires obligations that most existing structures do not express, and the honest position is that they must be added deliberately.

Four are minimal. **A register of AI uses**, stating for each what the tool does, whose decision it informs, and who is accountable for the output: the same decidability test as 3.1.1, applied to tools. **A verification standard proportional to consequence**: an AI-drafted internal summary and an AI-produced estimate that will set a baseline are not the same object and must not carry the same review. **A prohibition on unattributable authorship** in anything that will be relied on (a decision record, an assurance opinion, a board paper conclusion), because Domain 1's accountability principle requires a person to answer for it, and a tool cannot be that person. And **a data and confidentiality boundary** stated before use, not after an incident.

The governance body's own duty here is the one most often skipped: to ask, of any AI-informed recommendation it receives, *what was the model's input, who verified its output, and what would change the conclusion?* A body that cannot get those three answers is not governing the recommendation; it is ratifying it.

### 3.A.3 The reviewer's governance eye

Invariants to test on any governance design, each cheap and each diagnostic:

Every decision class has **exactly one** accountable role, and that role's authority is sufficient under the delegation schedule. The delegation schedule reads on **value, reversibility and externality**, not value alone. The **latency of every escalation path is stated**, computed as  $\sum (M/2 + L)$ , and each tier can be justified against its cost. Every decision body has a **capacity number** and a demand estimate. Committee agendas show the **share of time on decisions** rather than reports. Every gate has criteria set in advance, the authority to stop, and a **computed value** against its elapsed time and detection rate. Conditional passes have owners, dates and consequences, verified at the following gate. The assurance map shows **no gaps and no duplication**. Every escalation class has a stated decision deadline and an out-of-

cycle route. The decision log's entries carry a named decision-maker and a **versioned information reference**. And (the one test that subsumes many of the others) the **re-decision count** is tracked, because a question arriving at the same body a third time is proof that something upstream of it is broken.

Six further invariants, each answerable from documents that already exist:

Every decision body's utilisation is stated at **average, peak and off-peak**, and the peak is computed from the actual clustering of change and gate decisions rather than assumed uniform (3.2.2). Every escalation path carries a **decision action window** as well as a latency, tested against the remaining duration of the thing being decided; a negative window is a finding, not a nuance (3.3.3b). Every tolerance trigger names its **forecast method, measurement point and confirmation rule**, and a trigger stated on a single ratio at a single data date is treated as unspecified (3.3.3b). Every gate's **condition-closure rate** is counted and multiplied through its detection rate, because a gate reported as held is not a gate that worked (3.3.1b). The assurance map carries **residual exposure per row**, so that reallocation is arguable at constant cost rather than being a matter of taste (3.3.2). And any **cumulative test** is presented with its expected annual trip count and the base-rate aggregate of its relatedness class, since a rule that trips on ordinary traffic has reversed the delegation it was added to protect (3.3.4c).

Where the structure is multi-party, two more: the board's **first-presentation approval rate** is counted from its own minutes, because that is the *p* on which unanimity's cost depends; and the **reserved-matters list** is read against it, since every item on that list is priced at  $p^n$  (3.1.2).

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## Industry variations — Domain 3

- **Public sector and government.** Statutory decision rights, published assurance regimes and mandatory gates that are genuinely non-negotiable; latency is high by design and the leader's lever is almost entirely the paper lead time and the out-of-cycle route, not the cadence. Where a statutory path's latency exceeds the remaining duration of what it decides, the **action window is negative** and the honest report says so; because the alternative narrative, that the project was slow, is the one that will otherwise be recorded (3.3.3b).
- **Regulated industries (pharmaceutical, nuclear, aviation, financial services).** Some gates are external and unmovable, and the governance design must place the internal decision *before* the external submission with enough margin that a failed internal test does not force a submission. That margin is computable rather than a matter of comfort: it is the internal escalation path's action window against the time remaining to the submission date, and a negative window means the internal test cannot change the submission it was built to protect.



- **Construction and infrastructure.** Multi-party governance is the norm; decision rights follow the contract structure, so a governance design inconsistent with the contract loses to the contract every time (PFL-AI Domain 11). The **decision rule in the joint-venture agreement is usually the largest single latency term** in the whole structure: at four parties each 85 % ready, unanimity costs **10.4941 weeks** per decision against **5.7379** under a three-of-four majority (Exercise 3.5), and no schedule shows the difference.
- **Technology and product organisations.** Bounded envelopes and continuous funding decisions dominate; the characteristic risk is light governance on genuinely irreversible choices (architecture, data model, third-party dependency), because they do not present as gates. The **reversal-cost ratio** is the instrument that catches them, since it reads authority on  $\max(\text{value}, \text{cost to undo})$  and those decisions are cheap to take and expensive to unmake (3.2.3).
- **Healthcare.** Clinical governance runs in parallel with project governance and clinical authority is not delegable to a project body; Meridian's design must place clinical sign-off with clinical authority and integrate rather than absorb it. On the assurance map, clinical sign-off is a **line in its own right with its own detection rate**, not a component of the project's first line; and where the project helped produce what clinicians are asked to sign, the cell is entered as  $q = 0$  (3.3.2).
- **Energy and resources.** Stage-gated capital processes with substantial assurance at each gate, and the standard pathology is gate proliferation, re-approval ceremonies whose elapsed time exceeds 3.3.1's breakeven. The second pathology, less discussed and equally expensive, is the **unclosed condition**: it reduces a gate's effective detection to  $q \times r$  while reducing its elapsed time by nothing at all, so a stage-gated estate that does not count condition closure is paying full price for partial gates (3.3.1b).

### Case study — Domain 3: the four-week month (health, Meridian)

**Situation.** Twelve weeks into the clinic rollout, Meridian's steering committee was widely described as "the bottleneck". The committee met monthly, was well attended, and had never refused a request. The programme was six weeks behind, and the delay was attributed in the quarterly report to "slower than expected clinic readiness".

**What the arithmetic showed.** The project leader computed three numbers before the next governance review. Governance latency:  $M/2 + L = 4/2 + 2 = 4$  **weeks**, the committee that everyone called monthly imposed a full month's expected wait, and no one in the organisation had believed it was more than a fortnight. Escalation

volume: with a **10,000** delegation threshold, **36** of the year's **60** change requests required the committee, of which about a quarter sat on the critical path, costing  $9 \times 4 \times 14,280 = \text{USD } 514,080$  a year in delay. Committee capacity: **104** item-slots against **77** items of demand, of which **26** were standing reports producing no decision, so a quarter of the scarcest resource in the programme was being spent on status.

**What changed.** Three changes, none of which removed any scrutiny anyone could name. The paper lead time went from 2 weeks to 1, taking expected latency to **3.0 weeks**, a full week saved from a change to an administrative deadline. The delegation threshold went from 10,000 to **25,000**, worth **USD 171,360** a year against a worst case of 81,600 even if every delegated decision were decided wrongly and destroyed 40 % of its value. And the standing reports moved to pre-reading, releasing 26 slots and taking utilisation from 74.0 % to **49.0 %**, which is what created room for the contentious item to be reached.

**The two findings that were not in the original three.** The committee's own members had said the body was full, and the 74.0 % utilisation figure said it was not. Two further counts reconciled them. First, demand was not uniform: **40 %** of the 51 decision items fell in the **3** meetings straddling phase boundaries, giving  $51 \times 0.40/3 + 2 = 8.80$  items against a capacity of 8, a peak utilisation of **110.00 %** against **63.25 %** between the boundaries. Second, a clustering of the decision log's **148** entries found **19** questions that had arrived at the same body more than once, **6** of them three or more times: **25** repeat slots, **24.04 %** of the committee's annual capacity, which took real utilisation to **98.08 %** and cost **USD 357,000** a year in delay on the same 25 % critical-path assumption. **68.42 %** of those repeats carried no versioned reference to the information relied on. A fourth change followed, and it was the cheapest of the four: two mandatory fields on the decision-record template.

**The outcome, and the part that mattered.** Escalated decisions began arriving in about three weeks rather than four, and the peak-month deferrals stopped. But the durable change was to the *conversation*: "the committee is a bottleneck" is an accusation, while "our governance design costs 514,080 a year in delay, plus 357,000 a year deciding things twice, and here are four changes worth most of it" is a proposal. The second was approved in one meeting. The first had been raised, in various forms, for two quarters. Note the ranking the arithmetic produced, which was not the ranking anyone expected: the largest single item was not the threshold, the cadence or the agenda; it was the **re-decision tax**, at **2.08 times** the entire saving from the threshold change, and it was the only one of the four that required no risk-appetite decision from anybody.

**What the domain teaches here.** Governance is a delivery variable with a computable price. Until the price is computed, governance complaints are cultural and go nowhere; once computed, they are engineering, and engineering gets approved. And note which lever was largest per unit of organisational pain: the paper deadline, free, unglamorous, and twice as effective per week as meeting more often.

## Case study B — Domain 3: the decision nobody made (financial services)

**Situation.** A payments-platform programme completed build 5 % over budget with no approved change above 50,000, a clean change-control record. A routine internal audit then found that the delivered scope differed materially from the approved baseline in four respects, none of which appeared in any decision record.

**What had happened.** Each of the four differences had been assembled from a handful of individual changes: **20** in all, five per respect, averaging **35,000** and so **175,000** per respect and **700,000** in total. Each had been discussed in a working group, agreed in substance, and implemented. None had been raised as a change, because each had been individually below the project leader's **50,000** authority and had therefore, in the team's understanding, required no decision at all. One of the four altered a control the second line had assured on the basis of the original design. The programme's governance was not weak in the ordinary sense: the committee functioned, the gates were held, the delegation schedule existed. The defect was narrower and more instructive, **the schedule read on individual value only**, and the decision log recorded decisions *taken by bodies* rather than decisions *taken*.

**How it resolved.** Three corrections, each mapping to a specific defect. The delegation schedule acquired **reversibility and externality** dimensions, so a change touching an assured control required second-line agreement regardless of value. It acquired a **cumulative test**: related changes aggregating above a threshold within a period required the authority appropriate to the aggregate, which is the provision that would have caught all four. And the decision log was made the **register of record for every change to the baseline**, whoever decided it and at whatever value, so that a decision below a threshold still generated an entry. The audit finding was closed; the platform went live four weeks late.

**The parameter the first draft of the fix got wrong.** The cumulative rule was initially written at  $X = 50,000$ , the same value as the individual threshold, and summed **over each of the programme's four workstreams** each quarter, which sounded conservative and was the opposite. The programme logged **84** sub-threshold changes a year averaging **18,000**, so **USD 1,512,000** of delegated value, and a workstream-quarter therefore carried a base-rate aggregate of  $1,512,000/16 = \text{USD } 94,500$ : nearly twice the trip threshold on ordinary unrelated traffic alone. The rule would have tripped in **all 16** workstream-quarters, re-escalating routine change and leaving the delegation nominal. Redrawing the relatedness class at **deliverable level (22 classes rather than 4)** dropped the base-rate aggregate to **USD 17,181.82** per class-quarter, put  $X$  **2.91 times** above it and **3.5 times** below each respect's **175,000** aggregate, and cut the expected trip count from 16 a year to about **4**: one per respect, a **four-fold** reduction that caught every cluster the rule existed to catch. On that sizing each respect trips at its **second** change (running total 70,000 against 50,000), leaving **105,000** of it to be authorised at the aggregate's level. The lesson the audit report recorded was that **the relatedness class, not the threshold, is the para-**

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**meter that decides whether a cumulative test is a control or a re-centralisation.**

**What the domain teaches here.** A decision below a threshold is still a decision and still needs a record. A delegation schedule that reads only on value cannot see irreversibility, externality or accumulation (the three ways a small decision becomes a large one), and a decision log that records only what committees decided will systematically miss the changes that matter most, because those are exactly the ones that never reached a committee. And a control added to catch accumulation must be sized against the base rate of the class it sums over, or it will trip on the ordinary traffic it was never aimed at.

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## Executive perspective — Domain 3

What a programme director cannot delegate in this domain:

- **The latency of your own governance.** Compute  $\Sigma (M/2 + L)$  for every escalation path you own, price it at your cost of delay, and be able to state it. An unquantified governance design is one you are not managing (3.2.3, 3.3.3).
- **The delegation schedule, on three dimensions.** Value, reversibility, externality, and a cumulative test. Read on value alone, it will miss exactly the decisions that matter (Case study B).
- **The sponsor's obligations, in writing.** Not the title. The seven testable obligations, agreed and evidenced, with turnaround on sponsor decisions monitored like any other lead time (3.2.1).
- **Whether your gates function.** A gate that has never held anything is not working, and its elapsed time may already exceed the point at which it destroys value (3.3.1).
- **The decision record's integrity.** Named decision-maker, versioned information reference, conditions with owners. This is what makes a defensible decision defensible a year later (3.3.4).
- **Escalation lead time as a leading indicator.** Track how far before impact escalations arrive. A shortening trend is the earliest signal that people no longer believe escalation is safe, and it appears in no standard report (3.3.3).
- **The action window, not just the latency.** For every escalation path you own, compare its latency with the remaining duration of what it decides. Auriga's 15.5-week path on a 25-week project is actionable for the first **38.00 %** of it and ceremonial thereafter, and no currency figure shows that (3.3.3b).
- **Sponsor allocation as a portfolio decision.** A Meridian-sized sponsorship costs **111.0 hours** a year, about **6.03 %** of a working week, so no executive carries more than about **three**. Naming the same four people across a dozen pro-

grammes manufactures the absent sponsor you will later diagnose as a behaviour (3.2.1).

- **What your gates' conditions did next.** Count condition closure, not conditions issued.  $q_{\text{eff}} = q \times r$  turned a gate worth +56,520 into one destroying 57,428 on nothing but a tracking failure, and the swing appears in no governance report (3.3.1b).
- **Where your assurance is pointed.** Require residual exposure per row on the assurance map. All three of Meridian's lines sat on 7.87 % of it while 58.27 % sat uncovered; the correction cost nothing (3.3.2).
- **The cost of deciding things twice.** Track unplanned re-decisions. Meridian's ran at 12.84 % of logged decisions, 24.04 % of committee capacity and USD 357,000 a year — more than twice the saving from its threshold change, and remediable with two fields on a template (3.3.4b).

## Calculation exercises — Domain 3

**Exercise 3.1** A committee meets every 8 weeks and closes papers 2 weeks before each meeting; cost of delay is 14,280 per week. Compute the expected latency and its cost, then compute the saving from (a) halving the meeting interval to 4 weeks and (b) halving the paper lead time to 1 week. *Solution.*  $E[\text{wait}] = 8/2 + 2 = 6.0$  weeks, costing  $6 \times 14,280 = \text{USD } 85,680$ . (a)  $4/2 + 2 = 4.0$  weeks — saves 2.0 weeks, USD 28,560. (b)  $8/2 + 1 = 5.0$  weeks — saves 1.0 week, USD 14,280. Note the general rule the two results illustrate: a cut of  $x$  in  $L$  always saves  $x$ ; a cut of  $x$  in  $M$  saves  $x/2$ . Halving looks like the bigger intervention here only because  $M$  is four times  $L$ . Common error: assuming the expected wait is half the meeting interval, which omits the paper lead time entirely.

**Exercise 3.2** An organisation logs 80 change requests a year:  $\leq 5,000 \rightarrow 30$ ;  $5,001-20,000 \rightarrow 20$  (average value 11,000);  $20,001-75,000 \rightarrow 22$ ;  $> 75,000 \rightarrow 8$ . The committee meets every 4 weeks with a 1-week paper lead time; cost of delay is 9,500 per week; 30 % of escalated changes sit on the critical path. Compute the annual governance delay cost at a 5,000 threshold and at a 20,000 threshold, and the breakeven value-destruction rate on the delegated band. *Solution.*  $E[\text{wait}] = 4/2 + 1 = 3.0$  weeks, so each delaying change costs  $3 \times 9,500 = 28,500$ . At 5,000: escalated  $80 - 30 = 50$ , delaying  $50 \times 0.30 = 15$ , cost  $15 \times 28,500 = \text{USD } 427,500$ . At 20,000: escalated  $50 - 20 = 30$ , delaying 9, cost USD 256,500. Saving USD 171,000. The delegated band is worth  $20 \times 11,000 = 220,000$ , so the escalation breaks even only if the delegate destroys  $171,000/220,000 = 77.7$  % of the value of every delegated decision. Common error: comparing the saving with the *total value* of the delegated band rather than

with a plausible loss on it, which understates the case for delegating by an order of magnitude.

**Exercise 3.3** A gate costs 30,000 in review effort and 4 weeks of elapsed time at a delay cost of 9,500 per week. A material defect is present with probability 0.25; the gate detects it with probability 0.75; correction at design costs 80,000 and in build 600,000. Compute the gate's net value, its breakeven elapsed time and its breakeven detection probability. *Solution.* Without the gate:  $0.25 \times 600,000 = 150,000$ . With: review 30,000 + delay  $4 \times 9,500 = 38,000$  + expected remediation  $0.25 \times (0.75 \times 80,000 + 0.25 \times 600,000) = 0.25 \times 210,000 = 52,500$ , total **120,500**. Net value **USD 29,500**. Breakeven elapsed time  $(150,000 - 30,000 - 52,500)/9,500 = 7.105$  **weeks**. Breakeven detection probability: solve  $30,000 + 38,000 + 0.25 \times (600,000 - 520,000d) = 150,000 \rightarrow d = 52.31\%$ . Common error: omitting the elapsed-time cost, which makes every gate look worthwhile and is the reason gates proliferate.

**Exercise 3.4** A decision must pass a delivery board (every 2 weeks, 1-week papers), a portfolio board (every 6 weeks, 2-week papers) and an investment committee (every 12 weeks, 4-week papers). Cost of delay is 9,500 per week. Compute the total expected latency and cost, and the saving from a single-tier path at the portfolio board with a written-resolution procedure of one week. *Solution.*  $2/2 + 1 = 2.0$ ;  $6/2 + 2 = 5.0$ ;  $12/2 + 4 = 10.0$ . Total **17.0 weeks**, costing **USD 161,500**: of which the investment committee alone is 10.0 weeks and **95,000**. A one-week written-resolution path costs **USD 9,500**, a saving of **USD 152,000 (94.1%)**. For comparison, retaining the portfolio board on its ordinary cadence as the sole tier gives 5.0 weeks and **47,500**, saving 114,000. Common error: adding only the meeting intervals and ignoring the paper lead times, which understates latency here by 7 of the 17 weeks.

**Exercise 3.5** A four-party consortium board meets every 6 weeks with a 2-week paper lead time. Each party independently arrives able to approve in a given cycle with probability 0.85. Cost of delay is 9,500 per week. Compute the expected wait and cost for one decision under (a) unanimity and (b) a three-of-four majority, and compare both with a single integrating authority. *Solution.* Base wait  $M/2 + L = 6/2 + 2 = 5.0$  **weeks**, the single-authority case, costing **USD 47,500**. (a) Unanimity passes with  $0.85^4 = 0.5220$ , so  $E[\text{cycles}] = 1/0.5220 = 1.9157$  and  $E[\text{wait}] = 5.0 + 0.9157 \times 6 = 10.4941$  **weeks**, costing **USD 99,694.09**. (b) Three-of-four passes with  $4 \times 0.85^3 \times 0.15 + 0.85^4 = 0.3685 + 0.5220 = 0.8905$ , so  $E[\text{cycles}] = 1.1230$  and  $E[\text{wait}] = 5.0$

$\bullet 0.1230 \times 6 = 5.7379$  **weeks**, costing **USD 54,510.33**. Unanimity costs **4.7562 weeks** and **USD 45,183.76** more per decision than the majority rule. Common error: applying the readiness probability once instead of raising it to the power

of the party count. That gives  $1/0.85 = 1.1765$  cycles and 6.06 weeks, understating the unanimity penalty by more than four weeks, and it is the error that makes multi-party boards look survivable on paper.

**Exercise 3.6** The gate of Exercise 3.3 (review 30,000; 4 weeks elapsed at 9,500 per week;  $P(\text{defect})$  0.25; detection 0.75; design fix 80,000; build fix 600,000; net value 29,500; breakeven detection 52.31 %) issued 18 conditions on a conditional pass, of which 7 were verified closed by the following gate. Compute the effective detection rate, the gate's net value at that closure rate, and the closure rate the gate needs to break even. *Solution.*  $r = 7/18 = 38.89\%$ , so  $q_{\text{eff}} = 0.75 \times 0.3889 = 29.17\%$ . Expected remediation  $0.25 \times (0.2917 \times 80,000 + 0.7083 \times 600,000) = 112,083.33$ , so the total with the gate is  $30,000 + 38,000 + 112,083.33 = 180,083.33$  against 150,000 without it: the gate now **destroys USD 30,083.33**, a swing of **USD 59,583.33** from its nominal +29,500. Breakeven closure rate  $r^* = d^*/q = 0.5231/0.75 = 69.74\%$ , i.e. **13** of the 18 conditions: at 12 of 18 the gate is still 3,000 underwater, at 13 it returns **USD 2,416.67**. Common error: treating the closure rate as the effective detection rate and reporting 38.89 %, which omits the multiplication by  $q$  and makes the gate look closer to breakeven than it is. Note that  $r^*$  here (69.74 %) is close to Meridian's 69.81 % by coincidence, not by rule:  $r^* = d^*/q$  and both terms differ between the two gates.

**Exercise 3.7** Four material risks are assured by two lines. R1:  $p$  0.25,  $u$  600,000, first line 0.60, second line 0.50. R2:  $p$  0.30,  $u$  450,000, first line 0.50 only. R3:  $p$  0.40,  $u$  250,000, no line. R4:  $p$  0.20,  $u$  900,000, first line 0.40 only. The second line's contribution is 25 days at USD 800. Compute total residual exposure, then the reduction available from moving the second line's 25 days off R1 to (a) R4 and (b) R3, at the same detection rate of 0.50 and the same cost. *Solution.* Residual =  $\sum p \Pi(1 - q)$   $u$ : R1  $0.25 \times 0.20 \times 600,000 = 30,000$ ; R2  $0.30 \times 0.50 \times 450,000 = 67,500$ ; R3  $0.40 \times 1.00 \times 250,000 = 100,000$ ; R4  $0.20 \times 0.60 \times 900,000 = 108,000$ ; total **USD 305,500**. (a) To R4: R1 rises to 60,000, R4 falls to  $0.20 \times 0.30 \times 900,000 = 54,000$ ; total **USD 281,500**, a reduction of **USD 24,000**. (b) To R3: R3 falls to **50,000**, R4 stays at 108,000; total **USD 285,500**, a reduction of **USD 20,000**. Both reallocations cost nothing extra, and **(a) is better than (b) by USD 4,000, 16.67 % of the available gain**. Common error: reallocating to the *uncovered* risk by reflex. The rule is the largest **residual exposure**, not the largest gap: R4 at 108,000 outranks the wholly uncovered R3 at 100,000, because a high consequence with one weak detector can leave more exposure than a low consequence with none. (Marginal values: 30,000 on R1, 54,000 on R4, 50,000 on R3, against the line's 20,000 cost.)

**Exercise 3.8** A 40-week project has BAC 6,000,000. At week 22: PV 3,300,000, EV 3,000,000, AC 3,300,000. The governance framework escalates where "the forecast



overrun exceeds 10 % of BAC". The escalation path runs through three tiers: 3-weekly with 1-week papers, 5-weekly with 2-week papers, and 13-weekly with 4-week papers. Compute the three standard estimates at completion and test each against the tolerance; then compute the escalation path's total latency, the decision action window and the last week at which the path can still deliver an actionable decision. *Solution.*  $CPI = 3,000,000/3,300,000 = 0.9091$ ;  $SPI = 3,000,000/3,300,000 = 0.9091$ .  $EAC = AC + (BAC - EV) = 6,300,000$ ,  $VAC_{300,000} = 5.00\% \rightarrow$  no trigger.  $EAC = BAC/CPI = 6,600,000$ ,  $VAC_{600,000} = 10.00\% \rightarrow$  exactly on the tolerance of 600,000.  $EAC = AC + (BAC - EV)/(CPI \times SPI) = 6,930,000$ ,  $VAC_{930,000} = 15.50\% \rightarrow$  trigger. Latency  $(3/2 + 1) + (5/2 + 2) + (13/2 + 4) = 2.5 + 4.5 + 10.5 = 17.5$  weeks against a remaining duration of  $40 - 22 = 18$  weeks, so the action window is  $(18 - 17.5)/18 = 2.78\%$  (half a week), and the last week at which the path is actionable at all is  $40 - 17.5 =$  **week 22.5**, or **56.25 %** of the duration. Common errors: two of them. First, reporting "the forecast" without naming a method, when the three methods here straddle the tolerance and one sits exactly on it, a framework with no named method and no tie-break rule has not specified its trigger. Second, stopping at the latency and calling 17.5 weeks "slow"; it is not slow, it is **0.5 weeks from being impossible**, and only the action window shows that.

## Practitioner's toolkit — Domain 3

*Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.*

### Toolkit 3.T.1 — Governance design sheet

One page, completed before mobilisation and reviewed at every gate. Rows: each decision body, with its purpose, members (distinguished from attendees), decision classes it owns, meeting interval  $M$ , paper lead time  $L$ , computed  $E[\text{wait}]$ , capacity (meetings  $\times$  items) and estimated demand. Below it, each escalation path with its tiers, the summed latency  $\Sigma (M/2 + L)$ , that latency priced at the project's cost of delay, the **decision action window** against the remaining duration of what the path decides, and the out-of-cycle mechanism. Two columns that are usually omitted and are worth more than most of the others: **peak** utilisation as well as average, computed from the actual clustering of change and gate decisions; and, for a multi-party body, the **first-presentation approval rate** and the decision rule (unanimity / defined majority / single integrating authority), since  $p^n$  is what prices the constitution. The sheet's purpose is to make latency and capacity **visible at design time**; a governance design that cannot fill it in has not been designed.



### Toolkit 3.T.2 — Delegation schedule with a cumulative test

Columns: decision class · value threshold · **estimated cost to undo** and the reversal-cost ratio  $\rho$ , with authority read on  $\max(\text{value}, \text{cost to undo})$  · externality rating (internal / cross-functional / external or public), escalating one authority level per step · authority required · **cumulative rule** (related decisions aggregating above  $X$  within period  $P$  require the authority appropriate to the aggregate) · **relatedness class** for that rule, with its base-rate aggregate per period and the expected annual trip count · information the delegate must hold to decide. The three dimensions prevent the value-only failure of Case study B; the cumulative rule prevents its specific mechanism; the base-rate and trip-count columns are what stop the rule re-centralising the band it was added to protect (Worked example 3.3.4c); and the final column is what distinguishes delegation from abdication. Sizing test to apply before the rule goes live:  $X$  should sit comfortably above the relatedness class's base-rate aggregate per period and comfortably below the aggregates of the clusters it must catch, if it cannot, narrow the class rather than raising  $X$ .

### Toolkit 3.T.3 — Decision record entry and its integrity checks

Per-decision fields: reference · date · decision-maker (**named person and role**) · decision (one reading only) · options considered and why this one · information relied on **with versions · declared interests, with the management action taken and a nil return where there were none** (Domain 1, KA 1.2.2a) · conditions with owners and dates · review date if provisional. Monthly integrity checks, each a count: entries missing a named decision-maker; entries missing a versioned reference; **entries with the declared-interest field blank, neither a declaration nor a nil return; decisions in which an interested party did not abstain, or abstained without the minute naming who decided instead**; conditions past their date; decision classes with zero or multiple Accountable roles in the RACI (the single-A test); and **re-decisions**: the same question arriving at the same body more than twice, counted excluding scheduled reviews of provisional decisions and re-openings forced by a material change, so that the number measures the defect and not governance working properly. Every one of these is a number, and a governance function that reports them monthly will find its defects before an auditor does. Two derived figures belong on the same monthly page because they convert the counts into an argument: the **re-decision tax** (repeat slots × critical-path share ×  $E[\text{wait}]$  × cost of delay) and the repeat slots as a **share of committee capacity**, which is what corrects a capacity model that would otherwise read low by the re-decision rate.

**The custody block, completed once per record class and reviewed at every gate** (3.3.4): record class (decision log · change log · gate packs · baseline archive · assurance opinions · acceptance evidence) · **named custodian role** · where it is held, and whether that system preserves version and records amendments with the amender and the reason · **retention period**, with the source of the period named, contractual limitation, an applicable retention requirement, the benefits-realisation

horizon or the organisation's records policy, whichever is longest · disposal or de-identification action at the end of it · **access list**, since entries naming identified individuals are not general-circulation material · and the **closure handover**: the continuing custodian in the permanent organisation, the date custody transfers, and the record of the transfer itself. Retention requirements and limitation periods are jurisdiction- and sector-specific and come from the records and legal functions, never from this template. Two monthly counts: record classes with no named custodian, and classes whose retention period has no stated source.

### Toolkit 3.T.4 — Gate and conditional-pass checklist

Two halves, and the second is the one that is usually missing. **Before the gate**: criteria written and issued in advance, each objectively assessable, covering the deliverable, the business case re-tested (not restated), the risks with named owners, the receiving organisation's readiness and the next stage's plan · the named authority, with the power to stop, hold or redirect stated in writing · the gate's elapsed time and review cost estimated · and the gate's **computed value** against its breakeven elapsed time and breakeven detection probability, so that the decision to hold the gate is itself an evidenced decision.

**After the gate, per condition on any conditional pass**: the condition in words that permit only one reading (a condition that says "confirm" or "find out" is a hold, and should be recorded as one) · the named owner · the date · the consequence of non-completion · the verification method · and, at the following gate, the **closure state**. The three monthly integers this yields are conditions issued, conditions verified closed and conditions past their date, from which the **closure rate**  $r$  and the gate's **effective detection rate**  $q_{eff} = q \times r$  follow: computed per gate and per condition type, never as one programme average, because an aggregate closure rate hides the one gate that closed nothing.

## Exam preparation — Domain 3

**What is assessed.** Governance purpose and its separation from management and assurance; structural weaknesses by organisational form and **the price of a decision rule** (unanimity, defined majority, single integrating authority); governing iterative and hybrid delivery, and **what share of an iterative stream a periodic body can serve**; the sponsor's obligations, failure modes, **diary load and turn-around distribution**; steering-committee design faults and average, peak and hidden utilisation; **the governance latency formula and its application**; delegation thresholds and their economics on **three dimensions**; gate purpose, criteria, value and **the conditional pass through**  $q_{eff} = q \times r$ ; the three assurance lines, their failure modes and **residual exposure across an assurance map**; escalation design with stated latency, **objective triggers that name a forecast method**,

and **the decision action window**; and the decision record with its integrity checks, its **re-decision tax** and its **cumulative test**.

**The calculations to be able to do under time pressure.**  $E[\text{wait}] = M/2 + L$  for a single body and  $\Sigma (M/2 + L)$  for a multi-tier path, priced at a cost of delay. Multi-party latency  $M/2 + L + (1/p^n - 1)M$  under unanimity, and the majority-rule comparison. Servable share of a decision stream against need-by times. Sponsor load in annual hours and share of a week; turnaround mean, median and tail share. Escalation volume and delay cost at alternative thresholds, and the breakeven value-destruction rate on a delegated band. Reversal-cost ratio and authority on  $\max(\text{value}, \text{cost to undo})$ . Gate net value, breakeven elapsed time, breakeven detection probability, and effective detection  $q \times r$  with the breakeven closure rate  $d^*/q$ . Residual exposure  $\Sigma p \Pi(1 - q) u$ , the marginal value of a line on a risk, and the reallocation gain at constant cost. VAC as a share of BAC across the EAC family and the tolerance test. Decision action window  $(\text{remaining duration} - \text{latency})/\text{remaining duration}$ . Committee capacity, utilisation at average and peak, and repeat slots. RACI single-A defect rate and the cost of each defect type. Cumulative-test base-rate aggregate and expected trip count.

**The traps.** Taking expected wait as half the meeting interval and forgetting the paper lead time (Exercise 3.1) · adding meeting intervals without paper lead times in a multi-tier path (Exercise 3.4) · applying a multi-party readiness probability once instead of as  $p^n$ , which makes unanimity look survivable (Exercise 3.5) · treating governance artefacts as governance (3.1.1) · omitting elapsed-time cost from gate value, which makes every gate look worthwhile (Exercise 3.3) · mistaking a condition-closure rate for an effective detection rate, omitting the multiplication by  $q$  (Exercise 3.6) · comparing a delegation saving with the total value of the delegated band rather than a plausible loss on it (Exercise 3.2) · reading average committee utilisation as evidence of capacity when demand clusters (3.2.2) · reallocating assurance to the *uncovered* risk rather than the largest **residual** one (Exercise 3.7) · quoting "the forecast" against a tolerance without naming an EAC method, when the family can straddle the line (Exercise 3.8) · stopping at an escalation path's latency without testing it against the remaining duration (3.3.3b) · assuming a committee's authority is its highest member's rather than its binding minimum (3.2.2) · treating a favourable assurance opinion as a transfer of accountability (3.3.2) · recording only committee decisions and missing the ones taken below a threshold (Case study B) · sizing a cumulative test on its threshold and ignoring the trip count its relatedness class generates (3.3.4c) · reading a delegation schedule on value alone.

**How the domain connects.** Domain 1 supplies the accountability principle governance must respect and the cost of delay every calculation here is priced at. Domain 2 supplies the business case and kill criteria that gates re-test, and its omitted-column failure is what reappears here as the largest uncovered row on Meridian's assurance map. Domain 4 integrates the governance of parts into a whole. Domain 6 consumes governance latency as schedule input (a decision path of 15.5 weeks is a 15.5-week predecessor, whatever the plan says), and supplies the schedule

forecast on which the decision action window should properly be computed. Domain 7's change control is the delegation schedule in operation, and its earned-value forecasting supplies the EAC family that an escalation trigger must name. Domain 8's risk escalation uses the escalation classes designed here. Domain 9's independence rule governs whether two assurance lines may be counted as two detectors. Domain 14, KA 14.3.3 derives the verification standard that 3.A.2 requires here. Domain 15 owns the portfolio allocation of the scarce sponsor capacity that 3.2.1 sizes. And PFL-AI Domain 11 handles the risk-allocation face of the multi-party governance problem whose decision-rule cost 3.1.2 prices.

### Domain 3 summary

Governance is the decision rights, accountabilities and information flows through which an organisation directs a project, and it is not management, not assurance, and emphatically not the reporting apparatus that is usually offered in its place. A governance design must produce decidability, timeliness, legitimacy and traceability, and each of those is testable.

The domain's contribution is to make governance **computable**. The expected wait for a committee decision is  $M/2 + L$ : half the meeting interval plus the whole paper lead time. That single formula prices a governance design at design time, shows that the administrative deadline is twice the lever meeting frequency is, and adds across tiers to reveal what an escalation path actually costs: 15.5 weeks and **USD 221,340** for Meridian's three-tier path, of which the quarterly committee alone is 61 %. Extended by the failure probability of a decision *rule*, it prices a constitution:  $M/2 + L + (1/p^n - 1)M$  puts Meridian's three-party board at **5.4870 weeks** under unanimity against **4.1152** under a two-of-three majority when the parties are 90 % ready, and at **11.6618** against **5.1020** when they are 70 % ready. The two rules are close in good conditions and divergent in bad ones, which is when governance is load-bearing. The same arithmetic prices a delegation threshold: Meridian's 10,000 threshold cost **USD 514,080** a year, and raising it to 25,000 saves **USD 171,360** against a worst case of 81,600 even if every delegated decision were decided wrongly, a breakeven value destruction of **84 %** per decision, which no plausible delegate approaches. And it prices a gate: Meridian's design gate is worth **USD 56,520**, stops paying beyond **9.96 weeks** of elapsed time, and requires a detection probability above **55.85 %** to be worth holding at all; but only if its conditions are closed, because  $q_{eff} = q \times r$  turned that same gate into one destroying **USD 57,427.83** at an observed closure rate of **39.13 %**, a swing of **USD 113,947.83** with nothing changed but the tracking.

Three further quantities decide whether a governance design can function at all. **Servable coverage:** at 4.0 weeks of latency against 2-week increments, Meridian's committee can serve **12.31 %** of its iterative stream's decisions in time, and doubling its frequency while halving its paper deadline reaches only **36.92 %**, which is why a

bounded envelope is a feasibility requirement and not a courtesy. **Residual exposure:**  $\sum p \prod (1 - q) u$  showed all three of Meridian's assurance lines sitting on the risk carrying 7.87 % of residual exposure while 58.27 % sat wholly uncovered, and reallocating 30 days cut exposure by **USD 74,361.60** at no additional cost. And the **decision action window:** Project Auriga's 15.5-week escalation path against 12 weeks of remaining duration is -29.17 %. The decision arrives 3.5 weeks after planned completion, so the path is not slow but ceremonial, and it was actionable only during the first 38.00 % of the project.

The sponsor's role is a set of testable obligations, not a title: **111.0 hours** a year, about 6.03 % of a working week, which is small enough that absence is a priority choice and large enough that no executive carries more than about three; and the sponsor's own latency is a distribution with a tail, where 22.73 % of Meridian's decisions carried 60.81 % of the wait and handling five of them differently removed 43.92 % of the cost. Steering committees fail in five predictable ways, four of them designed in, and their true load is not the average: Meridian's 74.0 % concealed 110.00 % at phase boundaries and, once re-decisions were counted, 98.08 % overall. Gates buy optionality against irreversibility and belong where irreversibility steps up. Assurance is usefully read through the three-line model (a voluntary architecture published by a professional body and since revised by it, a lens rather than an obligation), whose worst failure is capture, and whose effort concentrates where review is comfortable rather than where exposure is. Escalation is a timed pathway with an out-of-cycle route and an objective trigger that **names its forecast method:** auriga's three estimates at completion sit at 5.00 %, 10.42 % and 15.20 % of BAC against a 10 % tolerance, so an unnamed method delegates the escalation to whoever prepares the report. And a decision that is not recorded, with a named decision-maker and a versioned information reference, has not been made. It has been remembered, which cost Meridian **USD 357,000** a year and 24.04 % of its committee's capacity in questions decided twice, remediable with two fields on a template. Case study B's four differences were assembled from 20 individually authorised changes totalling 700,000, which is the whole argument for a delegation schedule that reads on reversibility and externality and aggregates; and its first cumulative rule, summed over four workstreams, would have tripped in all 16 workstream-quarters on ordinary traffic, which is why such a rule must be sized against the base rate of the class it sums over. And a record only answers a question years later if somebody was made responsible for its still being there: every class of governance record carries a named custodian, a system that preserves version and carries any amendment with its reason, a retention period set in advance at the longest of the applicable drivers with its source named, an access list because these records name identified people, and a recorded handover of custody at closure: the retention requirements themselves coming from the records and legal functions and from qualified counsel, never from this book.

The through-line: **governance has a price, the price is computable, and until it is computed governance complaints are cultural and go nowhere.** Compute it, and they become engineering.

DOMAIN

# 4

## Integration and Delivery Architecture

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## Why this domain exists

Domain 1 established what a project leader is answerable for. Domain 2 established how work is chosen and whether the promise was honest. Domain 3 established who may decide, how quickly, and at what price. Each produced something real, and none of them produced a **project**.

That is what this domain does, and the word that names the gap is **integration**. A project is not a collection of well-run parts; it is a set of parts that must fit, arrive in an order, share interfaces, and change together. Every one of those relationships is a place where a project fails without any single part having failed, which is why integration failure is the hardest kind to attribute and the most expensive to repair. The specific and recurring form it takes: **the parts are managed and the joins are not**. Scope is controlled, the schedule is controlled, cost is controlled, and nobody owns the fact that a change to one moves all three.

The domain's second claim is that integration is **architectural**: a design choice with a price, made once and paid for continuously. Interfaces do not multiply linearly with components; they multiply combinatorially, and KA 4.2 computes the difference and what buying an integration layer saves. Baselines are not three documents but one three-dimensional statement, and KA 4.3 shows what the arithmetic of the hundred-per-cent rule catches that reading the documents does not. Changes do not cost what a change form says they cost, and KA 4.4 shows that the quoted figure is typically the smallest component of the true one, which is why a delegation threshold applied to a quoted cost is not a control at all.

The through-line: **integration is where the parts become a project, and the joins are where the money is**.

**Learning objectives.** After this domain a candidate can: state what a charter must contain to confer authority, and distinguish a charter from a plan; assemble a coherent plan of plans, **audit a plan set against itself and price the exposure the audit finds**; derive a rolling-wave horizon from the longest commitment lead time in the plan rather than asserting it; tailor defensibly; build a WBS that satisfies the hundred-per-cent rule and detect an unallocated or omitted element arithmetically; distinguish work from product breakdown, **cross-match the two and separate a genuine orphan from a matrix artefact**; **compute the interface count of a delivery architecture and price what an integration layer buys, including its breakeven cost and its marginal advantage as the programme grows**; integrate scope, schedule and cost into one baseline and **compute what a stale time-phased baseline does to each earned-value index and which indices it leaves exactly correct**; apply configuration management to a baseline, audit its integrity, and **derive the item-level audit tolerance from an interface-level target through the pairwise exponent**; maintain a baseline through change without losing traceability to the original, and **perform the original-to-current reconciliation**



**and decompose its residual gross rather than net;** design an integrated change-control flow and **price a screening misclassification in both of its directions;** **assess a change's true cost including schedule, rework, interface re-verification and regression, and explain why a threshold set on quoted direct cost fails;** detect and quantify baseline drift by accumulation and set a cumulative test that would actually have caught it; **price an untraced rejection;** and govern AI-assisted impact assessment without letting it decide.

**The master programme, and the handover.** Meridian Care Records continues from Domains 1–3: the clinical-records rollout to **40 clinics**, approved cost **USD 2,400,000** (Domain 2's business case), benefits **USD 685,440** a year at 70 % adoption, cost of delay **USD 14,280 per week** (Domain 1), and the governance design priced in Domain 3. This domain architects it. Part Two then picks up **Project Auriga** (the 25-week control-systems upgrade) to work planning, cost and risk at the scale of a single project, because the two scales need different illustrations and a book that uses only one has taught only half the discipline.

**Reference points.** The international document most often named in connection with integration and the management of a project as a whole is **ISO 21502**, which addresses guidance on project management. It is voluntary guidance describing practice (not legislation, and not a certifiable requirement), and it obliges nobody of itself unless an organisation, a contract or a regulator adopts it. It is named here and not reproduced: no clause, table or checklist from it appears in this volume, the treatment below is this book's own, and a reader who wants the document should obtain the current edition from its publisher. Naming it implies no endorsement in either direction.

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## KNOWLEDGE AREA 4.1

# Charter and management plans

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*Topics: 4.1.1 the charter · 4.1.2 the plan of plans · 4.1.3 tailoring.*

## 4.1.1 The charter

**Definition.** The charter is the document by which the organisation **authorises the project and confers authority on its leader**. It is not a plan, not a business case, and not a summary of either: the business case argues that the work is worth doing (Domain 2), the plan states how it will be done, and the charter states that it is to be done, by whom, within what bounds.

**What a charter must contain to do its job.** Each item earns its place by being something a leader cannot proceed without, and the test of a charter is whether its absence would be noticed:

| ELEMENT                                  | WHY IT MUST BE THERE                                                                                                            |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Purpose and objectives                   | The measurable statement of what success is (Domain 2's benefits, not activity).                                                |
| Named sponsor                            | The accountable owner of the outcome (Domain 3, KA 3.2.1), a role, and a person.                                                |
| Named project leader and their authority | The <b>bounds</b> within which they decide without asking: spend, scope, resource, commitment (Domain 3's delegation schedule). |
| Scope boundary, including exclusions     | Exclusions are the half most often omitted, and the half that prevents the argument.                                            |
| Key deliverables and success criteria    | What will be handed over, and how acceptance is judged.                                                                         |
| Milestones and constraints               | The dates that are genuinely fixed, distinguished from those that are planned.                                                  |
| Budget authority                         | The approved sum and the authority to commit against it.                                                                        |
| High-level risks and assumptions         | The material ones, carried forward into the risk register (Domain 8).                                                           |
| Governance and decision rights           | Bodies, thresholds and escalation, by reference to the design of Domain 3.                                                      |
| Approval                                 | Signature of the authority entitled to confer it.                                                                               |

**The two charter failures.** *The charter that authorises nothing:* a document of aspiration with no stated authority, so every decision is an ask, and the leader discovers their authority incident by incident, which is the functional-organisation weakness of Domain 3, KA 3.1.2, arriving through the front door. *The charter that is a plan:* dozens of pages of approach and schedule, signed once, immediately stale, and thereafter neither read nor updated; its length conceals the absence of the one paragraph that mattered.

**The test.** A usable charter answers, on one or two pages: *what is this for, who owns the outcome, who leads it, what may they decide alone, what is in and out, and what is fixed?* A leader who cannot answer the fourth of those from the charter does not have a charter; they have an announcement.

#### 4.1.2 The plan of plans

**The concept.** The project management plan is not a single narrative but an **integrated set** of subsidiary plans (scope, schedule, cost, quality, resource, communication, risk, procurement, stakeholder, change and configuration) plus the baselines they produce. Its integration is the whole point: the subsidiary plans are individually easy and collectively contradictory unless someone owns the consistency.

**The consistency checks that make it a plan rather than a folder**, and each is a specific, answerable question:

- Does the **schedule** contain the activities the **scope** requires: every deliverable traceable to work, and every work package to a deliverable?
- Does the **cost** baseline sum to the resourced schedule, at the resource rates the **resource** plan assumes (Domain 7's rate arithmetic)?
- Do the **quality** plan's verification activities appear in the schedule with duration and resource, or are they assumed to happen in gaps?
- Do the **risk** responses have owners, dates, and budget in the cost baseline (Domain 8's contingency)?
- Do the **procurement** lead times appear as schedule activities rather than as assumptions?
- Does the **change** plan's authority match the governance design's delegation schedule, on the same thresholds (Domain 3), and does it read on assessed impact rather than quoted cost (KA 4.4)?
- Does the **communication** plan's reporting cadence fit the governance bodies' paper lead times (Domain 3's L), so that reports exist when papers close?

That last check is the kind this domain exists to surface: it is invisible in any single plan and obvious the moment the plans are read against each other. A programme whose monthly report is produced three days *after* the steering committee's papers close will report last month's position every month, for its entire life, and will describe the problem as a reporting delay.

Those seven questions are usually treated as a review formality. They are an arithmetic exercise with a price attached.

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#### WORKED EXAMPLE

##### **Worked example 4.1.2 — auditing Meridian's plan set against itself, and pricing what it finds.**

1. **Setup.** Before baseline approval, Meridian's **eleven** subsidiary plans are read against one another using the checks above: two reviewers over three days, **6 person-days** at a blended **USD 880** per person-day (the USD 110 an hour of Domain 14, KA 14.2.1, over an eight-hour day). The plan set covers **172** work packages, **24** quality-verification activities, **19** risk responses with named owners and **7** procured items. Cost of delay **USD 14,280 per week** (Domain 1).
2. **Formula.** Run each check; convert each finding into money: unfunded work at the person-day rate or at its own stated cost, unscheduled elapsed time at the cost of delay, counting only what lands on the critical path. Total, then compare with the cost of the audit.

3. **Substitution.** Check by check. *Scope against schedule:* 4 deliverables with no scheduled activity, flagged, deliberately **not** priced. *Cost against the resourced schedule:* the cost baseline sums to the resourced schedule at the resource plan's rates, **this check passes**. *Quality:* of 24 verification activities, 9 appear nowhere in the schedule, carrying **46 person-days** of resource  $\rightarrow 46 \times 880$ ; 4 of those nine sit on the critical path with **12** working days between them  $\rightarrow (12/5) \times 14,280$ . *Risk:* of 19 responses, 6 have no line in the cost baseline, together planned at **118,000**. *Procurement:* 3 items have lead times stated only in the procurement plan; the longest is **14 weeks** against 6 allowed in the schedule  $\rightarrow (14 - 6) \times 14,280$ . *Change:* the change plan's threshold reads on quoted direct cost, the defect KA 4.4.2 prices. *Communication:* the report is produced after papers close; the staleness arithmetic belongs to Domain 14, KA 14.2.1 and is not repeated here.
4. **Result.** Unfunded verification labour **USD 40,480**; unfunded risk responses **USD 118,000**; procurement lead-time exposure 8 weeks = **USD 114,240**; critical-path verification exposure 2.4 weeks = **USD 34,272**. Quantified total **USD 306,992** (12.79 % of the approved 2,400,000) found by an audit costing **USD 5,280**, a ratio of **58.14** to one. Three classes of finding are recorded unpriced (the four unscheduled deliverables, the threshold basis and the reporting cadence), and one check passes clean.
5. **Interpretation.** The first thing to be honest about is what the 306,992 is not. It is **not a saving**. Most of it is money that will be spent whatever anyone decides. The verification activities are needed, the risk responses were chosen deliberately, the lead times are what the market quotes. What the audit changes is whether that money is spent inside an authorised baseline or outside it as an unexplained overrun, and an assurance function that reports the figure as a saving will be correctly disbelieved and will lose the argument it was right about. The defensible sentence is: *306,992 of committed but unauthorised cost and schedule moved from invisible to visible, for 5,280.*

Set that against WE 4.3.3. Meridian's baseline drifts **291,176** over a year, through 34 individually authorised approvals, and everybody involved works hard for it. The plan-set audit finds **more** (306,992, a ratio of **1.0543**) in three days, before any of the work has started, without a single change request being raised. **The cheapest change control is the one applied to the plan set before it is baselined**, because at that point a correction is a re-estimate rather than a change: no board, no assessment cost, no traceability obligation, no supplier conversation.

Now notice *which* check passes. Cost against the resourced schedule is the one organisations habitually test, and it is the one where both sides are owned by the same function and read from the same tool. Every failing check spans two owners: quality and planning, risk and cost, procurement and planning, change and governance. That is the general rule this KA exists to state — **plan-set defects live on ownership boundaries**, and a check that either party could

have run alone has almost certainly already been run. It follows that the audit should be sequenced by boundary rather than by plan, and that the reviewer must have standing with both owners, which is why this work fails when it is delegated to whoever has capacity.

The sensitivity is concentrated in one component. The 114,240 of procurement exposure assumes the eight-week lead-time gap lands on work that gates the rollout. If the affected item has eight weeks of float the component is **zero** and the total falls to **USD 192,752**: still **8.03 %** of the baseline, so the conclusion survives, but the headline drops by **37.21 %**. Every schedule-priced finding must therefore carry its float position (KA 4.4.2's third assessment rule), and an audit that prices delay without stating float has quoted an upper bound as a point estimate.

Finally, a caution about the four unpriced deliverables, because the instinct is to put a placeholder against them so that the total looks complete. Resist it. Those four are the findings a reviewer should care about most: an unpriced finding is one whose magnitude is unknown, and unknown magnitudes are where residual risk actually sits. A total that silently includes invented numbers for them is worse than a total that stops short and says why.

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**Progressive elaboration and rolling wave.** Detail is added as it becomes knowable rather than invented early: the near horizon is planned to work-package level, the far horizon to a coarser level, and the boundary moves forward at a stated cadence. Two disciplines make this honest rather than an excuse. The **horizon must be stated** ("detailed to work-package level for 12 weeks, planning-package level beyond"), so that nobody mistakes a coarse plan for an absent one. And the **far-horizon estimate must carry its uncertainty explicitly**, because a rolling-wave plan whose outer years are stated as point numbers has hidden its uncertainty in exactly the place it is largest (Domain 8's ranges).

**Setting the horizon rather than asserting it.** The horizon itself is usually chosen for administrative tidiness (a quarter, a phase, twelve weeks), and it has a derivable lower bound: **the horizon must be at least as long as the longest commitment lead time falling inside it**, because a plan that does not yet exist at work-package level cannot support a commitment that must be placed before it does. Meridian illustrates the failure exactly. Its stated horizon is **12 weeks**; the longest procurement lead time WE 4.1.2 found is **14 weeks**. The horizon is **two weeks shorter than its own supply chain**, and no amount of diligence inside the wave can close that gap, because the order must be placed before the work package that specifies it has been written. Two responses are legitimate: lengthen the horizon to at least 14 weeks on the affected path, or plan that one procurement to work-package level ahead of the wave and label it a **partial horizon**. What is not legitimate is to leave a 12-week horizon and a 14-week lead time in the same plan set and describe the consequence as a supplier problem. The test generalises to every commitment class

in the plan (procurement, recruitment, regulatory submission, facility access, decommissioning permission) compare each lead time with the horizon, and set the horizon requirement from the longest.

#### 4.1.3 Tailoring

**The principle.** Every method requires tailoring, and tailoring is a **decision with a rationale and an owner**, not the quiet omission of inconvenient parts. The distinction is procedural: a tailoring decision is recorded, states what was removed or added and why, names who approved it, and is reviewable. An omission is discovered later by someone else.

**What drives it.** Project size and value; novelty and technical uncertainty; regulatory and contractual obligation; delivery approach (sequential, iterative, hybrid); organisational maturity; and the receiving organisation's capacity to absorb process. A small, novel, publicly visible project may need *more* assurance and *less* documentation than a large, familiar, internal one, and a tailoring regime that cannot express that combination is not tailoring but scaling.

**The limits.** Three things are not tailorable and should be stated as such, because they are what tailoring is most often used to remove: the **decision record** (Domain 3, KA 3.3.4) — nothing about project size makes an unrecorded decision acceptable; **the traceability of a baseline change to an authority** (KA 4.3.3); and any **legal, regulatory or contractual** obligation, which does not scale with convenience. Everything else is a judgement, and every judgement is recorded.

#### AI in this KA

**Where it earns its place.** Reading a plan set against itself and listing the inconsistencies above: deliverables with no work packages, quality activities absent from the schedule, risk responses without budget, reporting cadences that miss governance paper deadlines. This is precisely the work integration requires, humans do it badly because it spans documents, and the answers are checkable. Drafting a charter or a subsidiary plan from an approved template plus stated inputs, for human completion. Proposing a tailoring set from project characteristics, as a starting position to be argued with.

**Where it must not go.** Approving a tailoring decision, which requires an accountable authority. Producing a charter's authority statement, which is a conferral of power and must be authored by the conferring body. And no AI-produced plan should enter a baseline unreviewed: a plan is a commitment, and Domain 1's principle requires a person to stand behind it.

**Verification, concretely.** Every flagged inconsistency is confirmed against the source documents before it is reported; every AI-drafted plan section is reviewed by the accountable owner of that plan; and the tailoring record states that the proposal was AI-assisted, which is an honesty obligation rather than a disclaimer.

## KEY TERMS — KA 4.1

| TERM                              | MEANING                                                                                                         |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Charter</b>                    | The document authorising the project and conferring bounded authority on its leader.                            |
| <b>Authority bounds</b>           | The spend, scope, resource and commitment limits within which the leader decides alone.                         |
| <b>Plan of plans</b>              | The integrated set of subsidiary plans and the baselines they produce.                                          |
| <b>Consistency check</b>          | A specific question testing whether two subsidiary plans agree.                                                 |
| <b>Plan-set consistency audit</b> | Reading the subsidiary plans against one another and pricing the findings, before baseline approval (WE 4.1.2). |
| <b>Progressive elaboration</b>    | Adding detail as it becomes knowable, at a stated horizon and cadence.                                          |
| <b>Planning package</b>           | A far-horizon element planned above work-package level, with its uncertainty stated.                            |
| <b>Commitment lead-time rule</b>  | The rolling-wave horizon must be at least as long as the longest commitment lead time inside it.                |
| <b>Partial horizon</b>            | One element planned to work-package level ahead of the wave because its lead time demands it, labelled as such. |
| <b>Tailoring</b>                  | A recorded, owned decision to adapt method, never a silent omission.                                            |

## SAMPLE MCQS — KA 4.1

**MCQ 4.1-A** [4.1.1 · Comprehension] The element of a charter whose absence most directly disables the project leader is:

- A. the list of high-level risks
- B. the statement of the leader's authority bounds CORRECT
- C. the milestone schedule
- D. the communication approach

*Rationale:* Without stated bounds every decision becomes an ask and authority is discovered incident by incident (4.1.1). The others are important and none of them confer power.

**MCQ 4.1-B** [4.1.2 · Analysis] A programme's monthly report is produced three days after its steering committee's papers close. The consequence is that:

- A. the report is slightly late
- B. the committee will systematically consider last month's position, every month CORRECT
- C. the reporting cadence must be made weekly

D. the paper lead time must be abolished

*Rationale:* The defect is a plan-consistency failure between the communication plan and the governance design's paper lead time (4.1.2), and it recurs for the project's whole life until the cadences are aligned.

**MCQ 4.1-C** [4.1.3 · Evaluation] Which is *not* legitimately tailorable?

- A. the number of subsidiary plans maintained
- B. the depth of the schedule beyond the planning horizon
- C. the traceability of a baseline change to an approving authority CORRECT
- D. the frequency of progress reporting

*Rationale:* Traceability of baseline change to authority, the decision record, and legal or contractual obligations are outside tailoring (4.1.3).

**MCQ 4.1-D** [4.1.2 · Application] In WE 4.1.2 the plan-set audit finds unfunded verification labour of 40,480, unfunded risk responses of 118,000, eight weeks of procurement lead-time exposure and 2.4 weeks of critical-path verification exposure, at a cost of delay of 14,280 per week. The quantified total is:

- A. USD 158,480
- B. USD 272,720
- C. USD 306,992 CORRECT
- D. USD 392,672

*Rationale:*  $40,480 + 118,000 + (8 \times 14,280) + (2.4 \times 14,280) = 306,992$  (4.1.2). A totals only the two money classes; B omits the critical-path verification exposure; D prices the whole 14-week procurement lead time rather than the 8-week gap against the schedule.

**MCQ 4.1-E** [4.1.2 · Evaluation] A rolling-wave plan states a 12-week detailed horizon; the longest procurement lead time in the plan set is 14 weeks. The correct conclusion is that:

- A. the procurement should be expedited to fit the horizon
- B. the horizon cannot support its own commitments and must be lengthened on that path, or the item planned ahead of the wave as a stated partial horizon CORRECT
- C. the horizon is a planning convention and the lead time is a supplier matter
- D. the plan should carry a risk for late procurement

*Rationale:* The order has to be placed before the work package specifying it exists, so no diligence inside the wave closes the gap (4.1.2). D converts a structural defect into a risk, which is the commonest way this finding is disposed of without being fixed.



## SELF-CHECK — KA 4.1

1. *What distinguishes a charter from a plan?* — The charter authorises and bounds; the plan states how. The charter's irreplaceable content is the authority statement.
2. *Name two plan-consistency checks that only appear when plans are read against each other.* — Quality verification activities missing from the schedule; reporting cadence missing the governance paper deadline.
3. *Why do plan-set defects cluster on ownership boundaries?* — Because a check both parties could run alone has already been run; the failing checks are the ones spanning two owners (WE 4.1.2).
4. *What sets the lower bound on a rolling-wave horizon?* — The longest commitment lead time inside it, since a commitment cannot be placed from a plan that does not yet exist at work-package level.
5. *What makes a tailoring decision legitimate?* — It is recorded, states what changed and why, names its approver, and is reviewable.

## KNOWLEDGE AREA 4.2

## Breakdown structures

*Topics: 4.2.1 WBS · 4.2.2 product and deliverable breakdown · 4.2.3 interfaces.*

### 4.2.1 The work breakdown structure

**Definition.** A WBS is a **deliverable-oriented hierarchical decomposition** of the total scope of work. Two words carry the weight. *Deliverable-oriented*: it decomposes what will be produced, not who will produce it or when, an organisational chart and a schedule are different artefacts, and a WBS built by team or by phase loses the property that makes it useful. *Total*: the decomposition covers the whole scope and nothing outside it.

**The hundred-per-cent rule.** The children of any element sum to exactly that element: no more, no less. More means duplication or scope not authorised; less means work that will be done and is not budgeted. The rule is not a style guideline; it is an arithmetic invariant, and because it is arithmetic it can be **checked**, which is the point of the next worked example.

**Work packages.** The lowest level, and the unit that is estimated, scheduled, assigned and measured. A work package should have a single accountable owner, a deliverable or verifiable outcome, an estimate the owner endorses, and a duration that permits progress to be assessed meaningfully: sized so that the answer to "is it done?" is not routinely "about half". Above the work packages, **control accounts**

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are the level at which performance is measured and reported, which is where Domain 7's earned value attaches.

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#### WORKED EXAMPLE

#### Worked example 4.2.1 — auditing Meridian's WBS against the hundred-per-cent rule.

1. **Setup.** Meridian's approved cost baseline is **USD 2,400,000** (Domain 2). Its five level-2 WBS elements are estimated at: records application build **780,000**; integration layer and interfaces **536,000** (the figure KA 4.2.3 derives); data migration **310,000**; clinic rollout across 40 clinics **520,000**; programme management and assurance **186,000**.
  2. **Formula.** Hundred-per-cent test:  $\Sigma$  children – parent. Then re-test with any omitted element restored.
  3. **Substitution.**  $780,000 + 536,000 + 310,000 + 520,000 + 186,000 = 2,332,000$ ; against 2,400,000.
  4. **Result.** The children sum to **2,332,000**, leaving **68,000 (2.83 %)** of the parent unallocated. The review then identifies the omitted element: **clinician training and enabling change**, estimated at **214,000**, the same column Domain 2's benefits map omitted. Restoring it gives an honest baseline of **2,546,000**, which is **146,000: 6.1 %: above** the approved figure.
  5. **Interpretation.** Both halves of that result matter and they are usually confused. The 68,000 of apparently spare budget is not spare; it is the residue of an incomplete decomposition, and in practice it will be consumed early by whatever arrives first, after which the omission becomes visible with no budget left to absorb it. And the omission is not random: it is the **enabling change** (the work that converts an output into an outcome), which Domain 2 identified as the column most benefits maps leave out, and which reappears here as the element most WBSs leave out, for the same reason. It belongs to somebody else, so nobody decomposes it. Note finally what the arithmetic does *not* say: at 2,546,000 the programme's NPV falls from Domain 2's **+1,332,898** to **+1,186,898**, a reduction of **11.0 %**, still comfortably positive. The omission bought nothing. It was not a decision to descope training in order to make the case work; it was a failure to look, and the honest baseline would have been approved.
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#### 4.2.2 Product and deliverable breakdown

**The distinction.** A **product breakdown structure** decomposes the *thing being delivered* into its constituent products and components; a WBS decomposes the *work*. They answer different questions and both are useful, but only one of them is a good starting point.

**Why to start with the product.** Building the product breakdown first and deriving the work from it prevents the commonest scope defect: **work that produces nothing**, and its mirror, **a product with no work**. If every product traces to work and every work package to a product, the two structures verify each other, and the verification is a matter of matching lists rather than judgement.

**Where each belongs.** The product breakdown drives configuration management (KA 4.3.2) — because configuration items are products, not activities — and acceptance, since acceptance is of products. The WBS drives estimating, scheduling and performance measurement. Confusing them produces two recognisable pathologies: configuration control applied to activities, which cannot work because activities do not have versions; and estimating against products, which omits the work that has no product (integration, testing, migration, training) and is therefore how the elements of the previous worked example go missing.

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#### WORKED EXAMPLE

##### Worked example 4.2.2 — cross-matching Meridian's product breakdown against its WBS.

1. **Setup.** Meridian's product breakdown identifies **148** products; its WBS contains **172** work packages. A traceability matrix relates the two,  $148 \times 172 = 25,456$  cells. The test is that every product traces to at least one work package and every work package to at least one product.
2. **Formula.** Product coverage = products with at least one work package ÷ products. Work-package coverage = work packages with at least one product ÷ work packages. Then **classify every exception**, because the raw counts are not the finding.
3. **Substitution.** Raw exceptions: **12** products with no work package and **9** work packages with no product, **21** in all. Classification, item by item, by the owner of each item. Of the 12: **5** are the clinician-training and enabling-change product set (WE 4.2.1's omitted **214,000**), and **7** are produced inside work packages named for their parent product, so they are matrix granularity rather than omission. Of the 9: **7** are legitimately product-free (integration testing, data migration, training delivery, commissioning, interface verification) and **2** are genuine orphans, budgeted at **26,000**, producing nothing anyone has asked for.
4. **Result.** Product coverage  $136/148 = 91.89\%$ ; work-package coverage  $163/172 = 94.77\%$ . Of 21 raw exceptions, **14** are legitimate and **7** are real findings, a false-positive rate of **66.67%**. The real findings are worth **214,000** of omitted work and **26,000** of unnecessary work: **USD 240,000**, exactly **10.00%** of the approved baseline.
5. **Interpretation.** The classification, not the matrix, is the professional work. Two thirds of the raw exceptions are artefacts, so a leader who takes raw counts to a steering committee will be corrected in the room on the seven granularity artefacts and will lose the five that mattered along with them. State the rule that

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way round: **a cross-match produces a candidate list, and the finding is the residue after each exception has been ruled on by the owner of the item.** Twenty-one rulings is an afternoon; 25,456 cells is not human work at all, which is exactly the division of labour the AI note below describes.

The two real classes are not symmetric, and only one of them is visible later. **214,000 of products with no work is an omission:** it arrives as an overrun, at the end, in the enabling-change column that Domain 2's benefits map dropped and that WE 4.2.1's WBS dropped again, the third time the same column has gone missing in this programme, which is what a systematic defect looks like as opposed to an accident. **26,000 of work with no product is waste:** it will be executed diligently, reported as complete, and never questioned, because a work package that produces nothing still reports 100 %. Earned value will confirm it. That is the uncomfortable corollary of Domain 7's machinery: performance measurement measures conformance to the plan, so work that should not be in the plan is measured as success.

Note what removing the orphans does to WE 4.2.1's arithmetic, because the direction surprises people. Allocated cost falls from 2,332,000 to **2,306,000**, so the unallocated residue **rises** from 68,000 to **94,000: 3.92 %** of the parent. Cleaning a WBS usually widens the hundred-per-cent gap rather than closing it, because part of the gap was being filled by work that should not have been there. A leader who expects the 214,000 omission and the 26,000 of waste to offset each other has misread both: one is outside the baseline and one is inside it, and they land on different sides of the sum.

A caution on the legitimate no-product class, which is where this check is most often misunderstood. It is the same class the paragraph above identified as the one a product-derived WBS omits by construction, so it has to be *added* rather than found, and the honest sequence is therefore: product breakdown first, for completeness of *what is delivered*; then the deliberate addition of the product-free work, as a decision; then the cross-match. On that sequence the seven legitimate exceptions are the **register of that addition**, not a defect list, and if that class comes back empty, the WBS is a product breakdown with different headings and the integration work has not been planned at all. An empty legitimate class is a worse finding than a large one.

What a reviewer should test is not the coverage percentages but whether each of the 21 exceptions carries a recorded ruling with a named ruler. An unrulled exception list is the commonest outcome of this check and is indistinguishable, six months later, from never having run it.

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#### 4.2.3 Interfaces — and why they are the expensive part

**The definition and the problem.** An interface is a defined relationship across a boundary (between components, teams, contracts, organisations or phases) at which something must be agreed and verified. Interfaces are where integration effort

actually lives, and they are systematically under-planned for one arithmetic reason: **components grow linearly and interfaces grow combinatorially**. A programme that has doubled its component count has quadrupled its interface count, and the plan that was written for the first count is still in use.

For  $n$  components each of which may connect to any other, the number of possible pairwise interfaces is:

|                                    |                |                   |
|------------------------------------|----------------|-------------------|
| Mesh interfaces                    | $= n(n - 1)/2$ | – every component |
| to every other                     |                |                   |
| Layered interfaces                 | $= n$          | – every           |
| component to one integration layer |                |                   |

#### WORKED EXAMPLE

##### Worked example 4.2.3 – what an integration layer buys Meridian.

1. **Setup.** Meridian must integrate **12** components: records core, patient master index, appointments, prescribing, laboratory results, imaging, billing, the national reporting gateway, identity and access management, the clinical document store, analytics, and the legacy records system during migration. Specifying, building, testing and documenting one interface costs **USD 18,000**. An integration layer would cost **USD 320,000** to build.
2. **Formula.** Mesh count  $n(n-1)/2$ ; layered count  $n$ . Cost = count  $\times$  unit cost (+ layer build).
3. **Substitution.** Mesh  $12 \times 11/2 = 66$ ; layered 12. Mesh cost  $66 \times 18,000$ ; layered  $12 \times 18,000 + 320,000$ .
4. **Result.** 66 point-to-point interfaces costing **USD 1,188,000**, against **12** interfaces plus the layer costing **USD 536,000**: the layer saves **USD 652,000**, or **54.9** %. It remains worth building while its cost is below  $1,188,000 - 216,000 =$  **USD 972,000**.
5. **Interpretation.** The saving is large and it is not the most important part of the result. The decisive number is **marginal**: adding a thirteenth component costs  $12 \times 18,000 =$  **USD 216,000** on a mesh architecture and **USD 18,000** on a layered one, a factor of **12**, which grows with every component added. Since programmes acquire components (a new department, an acquired system, a regulatory feed), the architecture is a bet on the *future* component count, and a mesh is a bet that the count will not grow. Three professional cautions, however, because this arithmetic is easy to over-apply. Real architectures are **partial meshes**: not every pair genuinely needs to connect, and the honest count is of *required* interfaces, which is why the interface register (Toolkit 4.T.2) is built from need rather than from the formula. The layer introduces a **single point of failure and a throughput constraint** that the point-to-point design does not have, and those are real costs not captured here. And the 18,000 unit cost is an

average over interfaces of very different difficulty: an internal API and a national reporting gateway are not the same object, and a leader quoting one figure for both should say so.

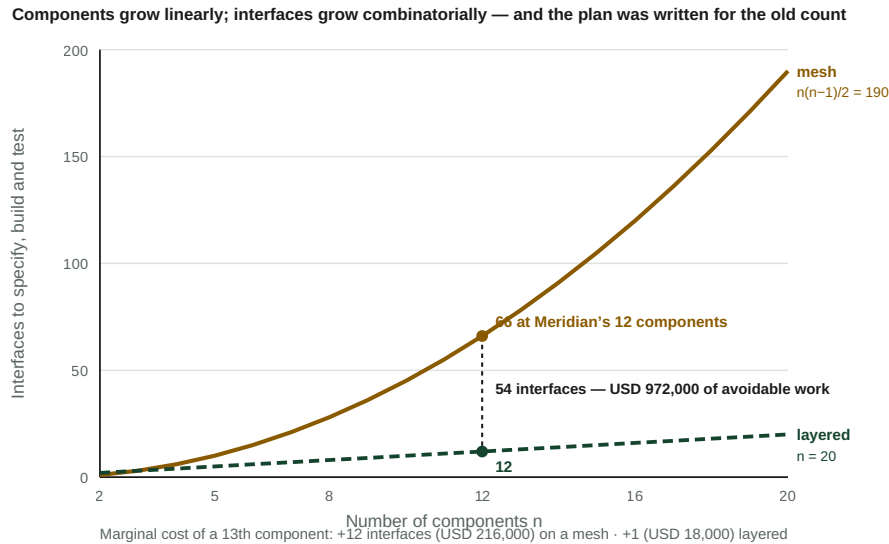


Fig 4.2.1 — Interfaces grow combinatorially; components do not

**Managing interfaces once counted.** The count is the beginning. Each required interface needs an **interface agreement**: the two parties, the thing exchanged, its format and content, the direction, the timing, the error and exception behaviour, who verifies it and when, and the version it is agreed at. Two disciplines matter more than the document. An interface has **exactly one owner on each side**, named: Domain 3's decidability test applied to a boundary, and the reason unowned interfaces are resolved late and by whoever notices. And interface verification is a **scheduled activity with duration and resource** (KA 4.1.2's consistency check), because the alternative is that verification happens at integration testing, which is where interface defects are most expensive to find and where they consume the schedule float that Domain 6 shows was never there.

### AI in this KA

**Where it earns its place.** Checking the hundred-per-cent rule across a large WBS and reporting every element whose children do not sum: mechanical, exhaustive, and exactly the check humans skip. Cross-matching a product breakdown against a WBS and listing products with no work and work with no product: 25,456 matrix cells in Meridian's case, which is not human work at any level of diligence. Extracting an interface register from architecture and design documents, which is tedious and error-prone by hand. Flagging interfaces with no named owner on one or both sides, or no verification activity in the schedule.

**Where it must not go.** Ruling on a cross-match exception. WE 4.2.2's arithmetic exists precisely because two thirds of the raw exceptions were legitimate, and telling a granularity artefact from an omission requires knowing what the work actually produces. The ruling belongs to the owner of the item and must be recorded with their name. Deciding the architecture. The mesh-versus-layer choice depends on availability requirements, throughput, organisational boundaries, contractual structure and the expected growth in components: several of which are judgements about the future that no model has grounds for, and all of which belong to an accountable architect and the governance body.

**Verification, concretely.** Reproduce the interface arithmetic by hand (it is one formula) and state the assumed unit cost with its basis. Confirm each AI-derived interface against the source design before it enters the register. And when a model reports a WBS as compliant, spot-check at least the largest elements, because a hundred-per-cent check is only as good as the estimate values it was given.

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#### KEY TERMS — KA 4.2

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| TERM                                  | MEANING                                                                                                                                                  |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>WBS</b>                            | Deliverable-oriented hierarchical decomposition of the total scope of work.                                                                              |
| <b>Hundred-per-cent rule</b>          | The children of any element sum to exactly that element: an arithmetic invariant, and checkable.                                                         |
| <b>Work package</b>                   | The lowest WBS level: one owner, a verifiable outcome, an endorsed estimate, an assessable duration.                                                     |
| <b>Control account</b>                | The level at which performance is measured and reported (Domain 7's earned value attaches here).                                                         |
| <b>Product breakdown structure</b>    | Decomposition of the thing delivered; drives configuration management and acceptance.                                                                    |
| <b>Product-to-work cross-match</b>    | The traceability test that every product has work and every work package a product: a candidate list, not a finding.                                     |
| <b>Orphan work package</b>            | Budgeted work producing no product anyone asked for: waste that still reports 100 % complete.                                                            |
| <b>Legitimately product-free work</b> | Integration, testing, migration, training delivery, commissioning: real work with no product, and the class a product-derived WBS omits by construction. |
| <b>Interface</b>                      | A defined relationship across a boundary at which something must be agreed and verified.                                                                 |
| <b>Mesh vs layered</b>                | $n(n-1)/2$ possible interfaces against $n$ , the architecture choice priced in WE 4.2.3.                                                                 |
| <b>Interface agreement</b>            | The parties, content, format, direction, timing, exceptions, verifier and version of one interface.                                                      |

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## SAMPLE MCQs — KA 4.2

**MCQ 4.2-A** [4.2.3 · Application] A programme integrates 12 components. The number of possible point-to-point interfaces is:

- A. 12
- B. 66
- C. 132
- D. 144

CORRECT

*Rationale:*  $n(n-1)/2 = 12 \times 11/2 = 66$  (4.2.3). A is the layered count; C counts each pair twice; D is  $n^2$ .

**MCQ 4.2-B** [4.2.3 · Evaluation] In WE 4.2.3, the strongest argument for the integration layer is:

- A. it saves 652,000 against the point-to-point design
- B. adding one further component costs 216,000 on a mesh and 18,000 layered — and programmes acquire components
- C. it reduces the interface count from 66 to 12
- D. point-to-point interfaces are technically inferior

CORRECT

*Rationale:* The marginal cost of growth is decisive because it compounds, whereas the one-off saving is a single number on today's component count (4.2.3). D is not established and would be an assertion.

**MCQ 4.2-C** [4.2.1 · Analysis] A WBS's five level-2 elements sum to 2,332,000 against an approved 2,400,000. The correct reading is that:

- A. the project has 68,000 of contingency
- B. the decomposition is incomplete, and the 68,000 will be consumed by whatever arrives first
- C. the baseline should be reduced to 2,332,000
- D. the hundred-per-cent rule permits a 3 % tolerance

CORRECT

*Rationale:* Unallocated budget is a symptom of incomplete decomposition, not contingency (4.2.1); the rule admits no tolerance, and reducing the baseline would lock in the omission.

**MCQ 4.2-D** [4.2.2 · Comprehension] Configuration management applies naturally to a product breakdown rather than a WBS because:

- A. products are more important than work
- B. configuration items are products, and activities do not have versions
- C. a WBS changes more often
- D. product breakdowns are more detailed

CORRECT



*Rationale:* Versioning is a property of things, not of activities (4.2.2); applying configuration control to activities is one of the two standard confusions.

**MCQ 4.2-E** [4.2.2 · Analysis] A product-to-work cross-match returns 21 raw exceptions, of which classification shows 14 to be granularity artefacts or legitimately product-free work. The correct reading is that:

- A. the matrix has 21 defects requiring correction
- B. the finding is the 7-item residue after each exception has been ruled on by the owner of the item CORRECT
- C. a 66.67 % false-positive rate makes the check unreliable and it should be dropped
- D. the false positives indicate the WBS is wrong at 14 places

*Rationale:* The cross-match produces a candidate list; the ruling produces the finding (4.2.2). A mistakes the candidate list for a defect report; C discards a check for doing exactly what a candidate-generation step is meant to do; D miscounts artefacts of matrix granularity as WBS defects.

**MCQ 4.2-F** [4.2.2 · Application] In WE 4.2.2 the classified findings are 214,000 of products with no work and 26,000 of work with no product. Removing the orphan work packages from the WBS:

- A. reduces the unallocated residue from 68,000 to 42,000
- B. raises the unallocated residue from 68,000 to 94,000 CORRECT
- C. leaves the residue unchanged, since the two findings offset
- D. reduces the approved baseline by 26,000

*Rationale:* Allocated cost falls to 2,306,000, so the gap against the approved 2,400,000 widens to 94,000 (4.2.2). A subtracts on the wrong side; C assumes an offset between a finding outside the baseline and one inside it; D confuses the estimate with the authorisation.

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SELF-CHECK — KA 4.2

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1. *State the hundred-per-cent rule and why it is checkable.* — Children sum exactly to their parent; it is arithmetic, so it can be audited rather than reviewed.
  2. *Why are interfaces systematically under-planned?* — Components grow linearly and interfaces combinatorially, so a plan written for an earlier component count is always an underestimate.
  3. *What two disciplines matter more than the interface document?* — Exactly one named owner on each side, and verification as a scheduled activity with duration and resource.
-

## KNOWLEDGE AREA 4.3

## Integrated baselines

*Topics: 4.3.1 scope–schedule–cost integration · 4.3.2 configuration management · 4.3.3 baseline maintenance.*

### 4.3.1 Scope, schedule and cost as one baseline

**The principle.** The performance measurement baseline is a **single three-dimensional statement**: the authorised scope, arranged in time, with cost attached. It is recorded in three documents for convenience, and treating those documents as three baselines is the origin of most integration failure.

**The integration invariants.** Each testable, and each a real defect when it fails:

- Every scope element appears in the schedule; every schedule activity traces to a scope element.
- The cost baseline sums to the resourced schedule at the resource plan's rates (Domain 7).
- The **time-phased** cost baseline (**PV**, Domain 7's planned value) follows the schedule's dates, so a schedule change is a **PV** change, always, and a baseline whose **PV** curve did not move when the schedule did is not integrated.
- Contingency and management reserve are held explicitly, at a stated level, and are not distributed invisibly into estimates (Domain 8).
- Every deliverable has acceptance criteria; every acceptance activity has duration and resource.

**Why the third invariant is the one that fails.** It requires the cost and schedule tools to be connected, or a person to maintain the connection, and it is the first thing abandoned under pressure. The consequence is precise and severe: **earned value becomes meaningless**. Domain 7's **SPI** and **CPI** compare achievement against a baseline; if the baseline's time-phasing no longer matches the schedule the project is executing, the indices measure the gap between two documents rather than the state of the work, and they will keep producing confident numbers while doing it.

By how much, and in which direction, is computable from the same three figures the report already prints.

## WORKED EXAMPLE

**Worked example 4.3.1 — what a stale time-phased baseline does to Auriga's indices, and what it leaves exactly alone.**

1. **Setup.** Project Auriga, Part Two's single-project thread, carries **BAC USD 4,000,000** over 25 weeks. At week 13 the reported position is **PV 2,080,000**, **EV 1,920,000**, **AC 2,120,000**, giving **CPI 0.91** and **SPI 0.92** (Domain 7, KA 7.3,

whose definitions are used unchanged). At week 8 a change was approved that re-sequenced downstream work; its cost effect was posted but the **PV** curve was never re-phased, so the reported PV is still 2,080,000. Two variants are examined, each moving **USD 240,000** of planned value across the week-13 status date: **(a)** the approved change **deferred** that value beyond week 13; **(b)** it **pulled it forward** inside week 13.

2. **Formula.**  $SPI = EV/PV$ ;  $SV = EV - PV$ ;  $CPI = EV/AC$ ; the composite forecast  $EAC = AC + (BAC - EV)/(CPI \times SPI)$ , alongside the two CPI-only forms  $BAC + AC - EV$  and  $BAC/CPI$ .

3. **Substitution.** Reported 1,920,000/2,080,000. Variant (a) 1,920,000/1,840,000; variant (b) 1,920,000/2,320,000.  $CPI = 1,920,000/2,120,000$  in all three columns.

#### 4. Result.

| MEASURE               | REPORTED<br>(STALE PV) | (A) DEFERRAL (TRUE<br>PV 1,840,000) | (B) ACCELERATION (TRUE<br>PV 2,320,000) |
|-----------------------|------------------------|-------------------------------------|-----------------------------------------|
| SPI                   | 0.9231                 | <b>1.0435</b>                       | <b>0.8276</b>                           |
| SV (USD)              | (160,000)              | <b>80,000</b>                       | <b>(400,000)</b>                        |
| CPI                   | 0.9057                 | 0.9057                              | 0.9057                                  |
| $EAC = BAC + AC - EV$ | 4,200,000              | 4,200,000                           | 4,200,000                               |
| $EAC = BAC/CPI$       | 4,416,667              | 4,416,667                           | 4,416,667                               |
| composite EAC         | 4,608,056              | <b>4,320,972</b>                    | <b>4,895,139</b>                        |

The same EV, the same AC and the same authorised scope produce an **SPI** anywhere between **0.8276** and **1.0435** (a spread of **0.2159**), according only to whether the PV curve was maintained. **CPI** and both CPI-only forecasts are **identical in all three columns**. 5. **Interpretation.** The identity that falls out of the table is the reason this defect is so durable: **a stale PV curve corrupts every schedule measure and leaves every cost-only measure exactly correct**. So the cost report reconciles to the ledger, the CPI-based forecast is defensible line by line, the finance function signs it; and the schedule indices printed on the facing page are wrong by up to a fifth of an index point. A reviewer who tests only the arithmetic *inside* the report will find nothing wrong with it, because nothing inside it is wrong. The defect is in the relationship between the report and the schedule, which is the one thing the report cannot show.

The composite 'EAC' is where the two halves meet, and it behaves in a way worth remembering because it makes the exposure quantifiable. Written as  $\text{'AC} + (\text{BAC} - \text{EV}) \times \text{PV} / (\text{CPI} \times \text{EV})$ , the composite is **linear in PV**, so its sensitivity is constant:  $(\text{BAC} - \text{EV}) / (\text{CPI} \times \text{EV}) =$  **1.1962** here, meaning **each USD 1 of mis-phased planned value moves the forecast by USD 1.20**. The 240,000 of mis-phasing therefore moves the composite forecast by **USD 287,083** (**7.18 %** of BAC), and, because the relationship is linear, by **exactly** that amount in either direction. The error cannot be defended as conservative, which is the usual defence offered.

The two directions have opposite management consequences, and the flattering one is the more common because deferral is the more common change. Variant (a) reports 0.92 on a project actually running at **1.0435** against the authorised schedule: a recovery plan will be launched, overtime authorised and float consumed to fix a problem the project does not have, real cost incurred to correct a reporting defect, and the composite forecast overstated by 287,083 into the bargain. Variant (b) reports the same 0.92 on a project running at **0.8276**: true schedule variance is (400,000) rather than (160,000), the composite forecast is understated by 287,083, and both facts surface at the next milestone rather than in the report designed to surface them. Neither variant contains a false statement and neither is detectable from the indices themselves.

What makes it detectable is one subtraction, and it is the check KA 4.3.3 formalises: does the sum of the approved changes' schedule effects equal the movement in the PV curve? Here it plainly does not, the curve did not move at all, and the finding is available in minutes to anyone who asks the question. That is why the invariant is better stated as a **sequence obligation** than as good practice: **the PV curve is re-phased in the same transaction that approves the change, or it will not be re-phased at all.** Anything deferred to a later housekeeping pass competes with delivery and loses, for the same structural reason KA 4.4.2 gives for impact assessment.

One caution against over-reading the 0.2159 spread. It is the consequence of a **single** 240,000 re-sequencing on a project whose phasing is otherwise sound. A programme with a year of

unmaintained phasing does not have a slightly wrong `SPI`; it has an `SPI` whose relationship to the work is unknown, and no sensitivity calculation recovers it. The honest treatment in that case is to **\*\*suspend the index and say so in the report\*\***, restating it only when the baseline has been re-phased against the current schedule. Continuing to publish a schedule index with a footnote about baseline quality is the specific failure to avoid: readers take the number and leave the footnote.

**The two-dimensional trap.** The commonest expression of a non-integrated baseline is a change approved on cost alone. A change is assessed for its cost, approved, and the schedule is not re-baselined. So so the project is delivered late against a baseline that never acknowledged the time, and the lateness is attributed to execution. KA 4.4 prices the general case.

#### 4.3.2 Configuration management

**Definition.** Configuration management is the discipline of **knowing, at any moment, exactly which version of every controlled item is the approved one**, and how it came to be. Its four functions: *identification* (what is controlled, and its identifier), *control* (how a version changes and on whose authority), *status accounting* (what the current version of everything is), and *audit* (verifying that the actual state matches the recorded state).

**Why it belongs in this domain rather than in quality.** Integration depends on it absolutely. When two components are integrated, the question is not whether each works but whether **these versions** work together, and an interface verified against version 3 of a specification tells you nothing about version 5. Every interface agreement is therefore version-bound (KA 4.2.3), and configuration management is what makes the binding meaningful.

---

#### WORKED EXAMPLE

##### Worked example 4.3.2 — auditing Meridian's configuration register.

1. **Setup.** A configuration audit covers Meridian's **340** controlled items. It finds **296** with a single identified current version and a complete change history; **28** with no version reference at all; **11** with two items marked current; and **5** whose recorded current version does not match the version actually deployed.
2. **Formula.** Defect rate = non-conforming items ÷ total. Classify by consequence.
3. **Substitution.**  $(28 + 11 + 5)/340$ .
4. **Result.** **44** non-conforming items (a **12.94** % defect rate) of which 28 are unidentified, 11 ambiguous and **5 actively wrong**.

- 
5. **Interpretation.** The three classes are not equally serious, and totalling them is the mistake a status report usually makes. The **28 unidentified** items cannot be integrated with confidence because no interface agreement can bind to them; the cost is rework at integration testing. The **11 ambiguous** items will each cause a decision by whoever picks first, and the wrong pick is discovered downstream. The **5 wrong** items are the serious finding: the register says something untrue, so verification performed against the register has verified nothing, and this is the class that produces the failure in which every component passed its own tests and the system did not work. A 12.94 % headline rate understates the position, because those five items alone can fail an integration.
- 

**The pairwise consequence, which is where the register is actually used.** An item-level defect rate answers a question nobody asks. What integration needs to know is whether **an interface** can be verified, and an interface is verifiable only if *both* of its sides are on their recorded version. That turns an item-level rate into a relationship-level rate through an exponent, and the exponent is where the surprises are.

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#### WORKED EXAMPLE

**Worked example 4.3.2b – why an 87 % register gives a 76 % integration, and what the audit tolerance therefore has to be.**

1. **Setup.** WE 4.3.2's audit found **296** of **340** controlled items conforming, **87.06 %**. Meridian has **31** genuinely required interfaces (Case study A's count), each side of each interface bound to a controlled item at a stated version. Treat all 31 as two-sided, which is the optimistic case; the three-party form (programme, national reporting gateway and identity provider) is computed alongside for comparison.
2. **Formula.** For a relationship spanning  $k$  controlled items whose individual conformance is  $p$ , and whose defects are independent, the probability that the whole relationship is verifiable is  $p^k$ . Expected at-risk relationships =  $\text{count} \times (1 - p^k)$ . Inverting for design: the item conformance required to achieve a relationship-conformance target  $q$  across  $k$  items is  $p = q^{(1/k)}$ .
3. **Substitution.**  $p = 296/340$ . Two-sided  $p^2 = 0.870588^2$ ; three-party  $p^3$ . At-risk  $31 \times (1 - p^2)$ . For an expected at-risk count of one interface in 31:  $p = (30/31)^{(1/2)}$ .
4. **Result.** Item conformance **87.06 %** yields two-sided interface conformance **75.79 %** and three-party conformance **65.98 %**. Of 31 required interfaces treated as two-sided, **7.50** are expected to be at risk: more where any of them is three-party, since  $1 - p^3 = 34.02 \%$  against  $1 - p^2 = 24.21 \%$ . Interface-level failure

(**24.21 %**) is **1.87 times** item-level failure (12.94 %). To hold the expected at-risk count at **one** interface, item conformance must reach **98.37 %**: at most **5** non-conforming items out of 340. The audit found **44**.

5. **Interpretation.** The exponent is the whole lesson and it is routinely missed: **integration quality is item quality raised to the number of parties**, so a register that looks tolerable where it is measured is materially worse where it is used. 87 % is the kind of number that earns a green status. 75.79 % is not, and 65.98 % on a three-party interface is a different conversation with a different audience. The direction never reverses (the assembled relationship is always worse than its parts), which is why an item-level dashboard over-reports integration readiness systematically rather than occasionally.

The inversion is the practically useful half, because it converts an integration target into an audit tolerance instead of leaving the tolerance to be chosen by whoever writes the procedure. Meridian's audit had **no stated tolerance at all**; had it wanted 95 % of interfaces verifiable it needed  $0.95^{(1/2)} = 97.47 \%$  item conformance, which is at most **8** non-conforming items. State the rule in that direction: **the tolerance on the register is derived from the target on the interfaces, and the derivation runs backwards through the exponent**. A configuration tolerance set directly at item level ("under 10 % non-conforming is acceptable") is a number with no argument behind it, and on a two-sided architecture a 10 % item tolerance silently permits a **19 %** interface failure rate that nobody has agreed to.

The audit *interval* falls out of the same arithmetic, and the answer is instructive because it is impossible. WE 4.3.2's 44 non-conforming items accumulated over the **6 months** since the previous audit, **7.33** items a month. To stay inside even the loose 5-item tolerance the audit would have to run every  $5/7.33 = 0.68$  **months**, about **3 weeks**, which no manual configuration audit sustains. That is not an argument for a heroic audit schedule; it is the diagnostic. **When the derived audit interval is shorter than the audit can feasibly run, the finding is about identification discipline, not about audit frequency**. The register is being written badly and has to be fixed at the point of writing, which is the automation case in the AI note below, and the reason software delivery largely solved this problem through version control while manual registers largely have not (see Industry variations).

Two cautions on the model, because it is a bound rather than a measurement. **Independence is a generous assumption**: register defects cluster by team, by system and by the week a migration happened, so where both sides of an interface are maintained by the same people with the same habit the real pairwise conformance is *worse* than  $p^k$ . Use  $p^k$  as an optimistic bound and label it as one. And  $p$  must be the conformance of the items **actually bound to interface agreements**, not of the whole register: the 340 items include many that no agreement references, and if the non-conforming 44 are concentrated among the referenced ones the interface exposure is worse again. A reviewer's

first question is therefore not the headline defect rate but *which items the interface register binds to, and what their conformance is measured separately.*

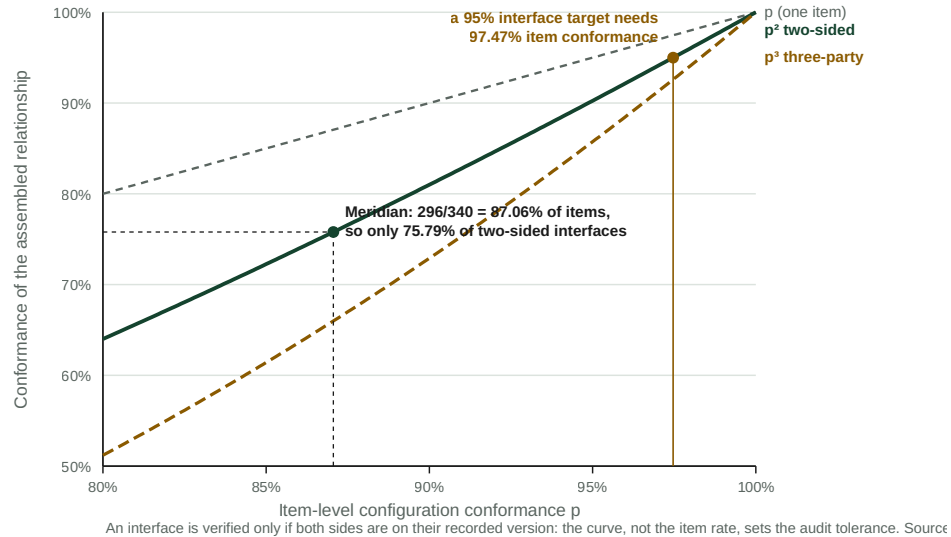


Fig 4.3.1 — Pairwise integrity is the power of item integrity

#### 4.3.3 Baseline maintenance

**The obligation.** A baseline is not a historical record; it is the current authorised statement of the project, and maintaining it means that every change is applied **with traceability to the authority that approved it**, while the original remains reconstructable.

**What must be preserved.** The original baseline, unaltered; every approved change with its reference, date, authority and effect on scope, schedule and cost; the current baseline as the sum of the original and the approved changes; and the ability to reproduce any intermediate state. That last requirement is what makes variance analysis honest a year later. And it is the reason a baseline is maintained by accumulation rather than by replacement.

**Re-baselining.** Occasionally a baseline becomes so distant from reality that variance against it conveys nothing, and re-baselining is legitimate. It is also the most abused instrument in project control, because it makes an adverse variance disappear without anything having improved. Three disciplines keep it honest: re-baselining requires **the authority that approved the original**, not the project's own; the **reason is recorded** and is a substantive change in circumstance rather than accumulated variance; and the **original remains visible in reporting**, so that performance against the original commitment can still be stated. A project on its third baseline with no visible original has lost the ability to answer the only question that matters to the organisation that funded it.



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**Drift: the failure mode with no decision in it.**


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**WORKED EXAMPLE**
**Worked example 4.3.3 — how Meridian's baseline moved 12.1 % without a decision.**

1. **Setup.** Over year one, Meridian approved **34** changes, each below the project leader's then **10,000** authority, at an average direct cost of **6,800**. Of those, **14** carried critical-path impact averaging **0.3 weeks** each; cost of delay **14,280** per week; baseline **2,400,000**.
  2. **Formula.** Direct drift = count × average cost. Schedule drift = affected count × average weeks × cost of delay. Express as a share of baseline, and per change.
  3. **Substitution.**  $34 \times 6,800$ ;  $14 \times 0.3 \times 14,280$ .
  4. **Result.** Direct **USD 231,200**; schedule impact **4.2 weeks** = **USD 59,976**; total **USD 291,176**: **12.1 %** of the baseline. Each individual change was **0.28 %** of it.
  5. **Interpretation.** There was no point at which anyone decided to spend 291,176 or to accept 4.2 weeks of delay, and no individual approval was wrong on its own terms. This is Domain 3's Case study B mechanism with a number attached, and it is why a delegation schedule needs a **cumulative test**. But the test's parameters have to be derived rather than chosen, and here the arithmetic is unforgiving: a rule of "related changes aggregating above 100,000 in a rolling 90 days requires steering authority" would **not** have caught this, because 34 changes a year is about  $34/4 = 8.5$  per quarter, or **57,800**: comfortably under 100,000. Catching it needs either a threshold below **57,800** on a 90-day window or the same 100,000 threshold on a **180-day** window ( $17 \times 6,800 = 115,600$ , which trips it). The professional point generalises past this example: **a cumulative test set at a round number without reference to the observed change rate provides the appearance of a control and none of the function**, and the observed change rate is available from the change log of any project that has run for a quarter.
- 

**The check that finds what the cumulative test cannot.** Drift of the kind above is entirely inside the change log, so summing the log detects it. The complementary failure, baseline movement that never reached the log, is invisible to any cumulative test, because cumulative tests read the log. It is caught instead by the reconciliation that maintenance by accumulation makes possible: *original plus the sum of approved changes equals the current baseline, exactly*. It is one addition and one subtraction, and it is habitually performed wrongly in the same way.

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**WORKED EXAMPLE**

### Worked example 4.3.3b — Meridian's reconciliation, and the residual that netting hid.

1. **Setup.** Meridian's original approved baseline is **USD 2,400,000**. The year-one change log holds the **34** small changes above (**231,200** in total) and **3** steering-approved changes at **86,000**, **142,000** and **57,500**. The cost tool reports a current baseline of **USD 2,948,900**.
2. **Formula.** Original +  $\Sigma$  approved changes = current baseline. Residual = current – expected. Then **decompose the residual by class before netting**: approved changes with no recorded baseline effect (which make the tool *lower* than the log implies) against baseline movements with no change reference (which make it *higher*).
3. **Substitution.** Approved total 231,200 + 86,000 + 142,000 + 57,500.  
Expected current 2,400,000 + 516,700. Residual 2,948,900 – 2,916,700.
4. **Result.** Approved changes total **USD 516,700**; expected current baseline **USD 2,916,700**; residual **USD 32,200**, or **1.34 %** of the original. Investigation decomposes it: **2** approved changes worth **44,900** (18,400 and 26,500) carry no baseline effect, and baseline movements totalling **77,100** carry no change reference —  $77,100 - 44,900 = 32,200$ , which closes. The **gross** error is **USD 122,000**: **23.61 %** of the approved changes and **5.08 %** of the original baseline. Netting concealed **USD 89,800** of it.
5. **Interpretation.** The arithmetic that matters is that **the two classes have opposite signs, so netting them destroys the finding**. A residual of 32,200 on a 2.4 m baseline is 1.34 % and will be written off as timing or rounding by any reviewer working at that resolution, while the actual defect is **3.79 times** larger. Hence the single most useful sentence in this KA: **reconcile gross, never net**. A reconciliation that balances to zero is not evidence of integrity; it is evidence that the errors happened to be of similar size. The concealment is not random either; it is exactly twice the smaller class,  $2 \times 44,900 = 89,800$ , which is a useful thing to be able to say out loud when someone proposes to report the net.  
  
The two classes are not equally serious and they have different remedies. **44,900 of approved-but-unrecorded changes** means work has authority and no budget line: it will be delivered and charged, and the variance will surface as an overrun against a baseline that never received the change. The remedy is bookkeeping, post the effects; the authority is already on file. **77,100 of movement with no change reference** is the governance finding: budget exists that no recorded decision created. Its remedy is not a posting but an investigation, because there are two possible answers and they are very different. Either a decision was taken and not recorded (a Domain 3, KA 3.3.4 failure, and one of the three things KA 4.1.3 places outside tailoring), or no decision was taken at all, and someone has moved a baseline without authority. A reconciliation report

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that states the residual and not its classes has withheld precisely the information that distinguishes those two.

Note the relationship to drift, because the two controls are complementary rather than overlapping. The 291,176 of drift above was **entirely inside the change log**: every dollar authorised, recorded and traceable, invisible only because nobody summed the column. The 77,100 here is **outside the log altogether**, so no cumulative test at any threshold or window would ever see it. **Drift controls and reconciliation controls catch disjoint failures**, and a programme with an excellently derived cumulative test and no reconciliation has covered one of the two and will believe it has covered both.

Cost and cadence are worth stating because they are the usual objection. The reconciliation is one addition and one subtraction over three figures a programme already holds: call it an hour a month, **USD 1,320** a year at the blended rate, against 122,000 of error found in year one. Run it monthly rather than at gates, and for an arithmetic reason rather than a diligence one: the *check* does not get harder with delay, but the *decomposition* does. Finding which movements are unreferenced is a search over one month's transactions if run monthly and over a year's if run annually, and it is the decomposition that carries all the value.

What a reviewer should test, in order: that the three figures are quoted from three systems of record rather than from one report that derives two of them; that the residual is decomposed before it is characterised; and that a **zero** residual has been tested rather than assumed, because the commonest way to produce one is to derive the current baseline *from* the change log, which makes the identity true by construction and the control worthless. A reconciliation that has never produced a non-zero residual is an unexamined formula, not a clean programme.

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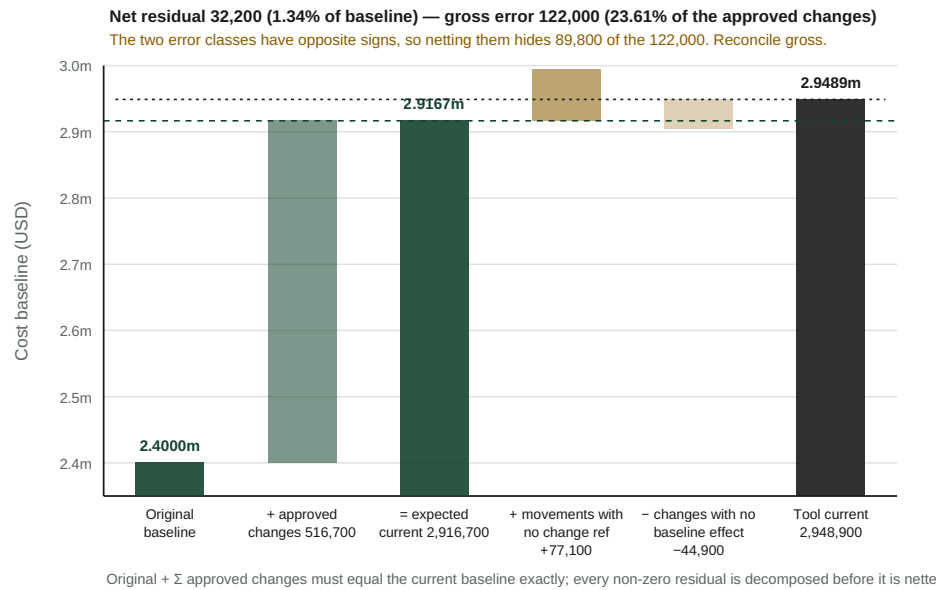


Fig 4.3.2 — Reconcile gross, never net

### AI in this KA

**Where it earns its place.** Continuous baseline-integrity checking: does the time-phased cost baseline still match the schedule's dates; do the change log's entries sum to the difference between the original and current baselines; are there approved changes with no baseline effect recorded, or baseline movements with no change reference? All four are reconciliations with definite answers, and the fourth in particular is how drift is detected early, and the second is the reconciliation of WE 4.3.3b, which must be decomposed by class rather than netted. Configuration status accounting, comparing the register against a deployed inventory and reporting the discrepancy classes of WE 4.3.2. Joining the configuration register to the interface register to compute the pairwise exposure of WE 4.3.2b, which is the only way to know which *interfaces* the item defects actually threaten. Trend analysis on the change log to derive the observed change rate a cumulative test should be set from, and the register's defect accumulation rate an audit interval should be set from.

**Where it must not go.** Authorising a baseline change or a re-baseline. Reconciling a discrepancy by adjusting the register to match reality, which destroys the evidence that the discrepancy existed. The correction must be a recorded, authorised change, and this is a real and tempting automation failure rather than a hypothetical one.

**Verification, concretely.** Every reported discrepancy is confirmed against both sources before it is treated as a finding; the change-log-sums-to-baseline-delta reconciliation is reproduced by hand at each reporting period, because it is a subtraction and it is the single most informative check in this KA; and no register is amended by a tool without an authorised change reference.

## KEY TERMS — KA 4.3

| TERM                                      | MEANING                                                                                                                 |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Performance measurement baseline</b>   | The authorised scope, arranged in time, with cost attached, one statement in three documents.                           |
| <b>Time-phased cost baseline</b>          | The PV curve; it must move whenever the schedule does.                                                                  |
| <b>Configuration management</b>           | Knowing which version of every controlled item is approved, and how it came to be.                                      |
| <b>Status accounting</b>                  | The record of the current version of every controlled item.                                                             |
| <b>Configuration audit</b>                | Verification that the actual state matches the recorded state.                                                          |
| <b>Pairwise configuration integrity</b>   | $p^k$ , the conformance of a relationship spanning $k$ controlled items, always worse than the item rate $p$ .          |
| <b>Derived audit tolerance</b>            | The item-level tolerance obtained by inverting $p^k$ from an interface-level target: $p = q^{(1/k)}$ .                  |
| <b>Original-to-current reconciliation</b> | Original baseline + $\Sigma$ approved changes = current baseline, exactly, every reporting period.                      |
| <b>Gross residual</b>                     | The sum of the reconciliation's error classes before netting. The figure that carries the finding.                      |
| <b>Baseline maintenance</b>               | Applying changes by accumulation, with traceability to authority, preserving the original.                              |
| <b>Re-baselining</b>                      | Replacing a baseline that no longer conveys information, legitimate, and the most abused instrument in project control. |
| <b>Baseline drift</b>                     | Cumulative movement through individually authorised small changes, with no decision on the total.                       |
| <b>Cumulative test</b>                    | A threshold on aggregated related changes over a stated period, set from the observed change rate.                      |

## SAMPLE MCQS — KA 4.3

**MCQ 4.3-A** [4.3.1 · Analysis] A schedule change is approved and the time-phased cost baseline is not updated. The most serious consequence is that:

- A. the cost baseline is slightly inaccurate
- B. earned value indices now measure the gap between two documents rather than the state of the work CORRECT
- C. the schedule must be re-baselined
- D. contingency is understated

*Rationale:* SPI and CPI compare achievement against the time-phased baseline; if it no longer matches the executing schedule they produce confident numbers about nothing (4.3.1).

**MCQ 4.3-B** [4.3.2 · Evaluation] In a 340-item configuration audit, 28 items have no version reference, 11 have two current versions and 5 have a recorded version that differs from what is deployed. Which is the most serious class, and why?

- A. the 28, because they are the most numerous
- B. the 11, because ambiguity blocks decisions
- C. the 5, because verification against the register has verified nothing

CORRECT

- D. all three are equally serious at a 12.94 % defect rate

*Rationale:* A register that states something untrue invalidates any verification performed against it, which is the class that produces "every component passed and the system failed" (4.3.2).

**MCQ 4.3-C** [4.3.3 · Application] 34 changes averaging 6,800 direct cost were approved over a year, 14 of them carrying 0.3 weeks of critical-path impact at a delay cost of 14,280 per week. Total baseline drift is closest to:

- A. USD 231,200
- B. USD 291,176
- C. USD 59,976
- D. USD 376,856

CORRECT

*Rationale:*  $34 \times 6,800 = 231,200$  direct plus  $14 \times 0.3 \times 14,280 = 59,976$  schedule impact (4.3.3). A omits the schedule impact; C is the schedule impact alone; D applies the schedule impact to all 34 changes rather than the 14 that carried it.

**MCQ 4.3-D** [4.3.3 · Evaluation] For the drift above, a cumulative test of "related changes above 100,000 in a rolling 90 days" would:

- A. have caught it, since total drift exceeds 100,000
- B. not have caught it, since a quarter's changes aggregate to about 57,800

CORRECT

- C. have caught it only if the threshold were raised
- D. be inapplicable to changes below the delegation threshold

*Rationale:* The 90-day aggregate is  $(34/4) \times 6,800 \approx 57,800$ , below the threshold (4.3.3), a cumulative test set at a round number without reference to the observed change rate has the appearance of a control and none of the function.

**MCQ 4.3-E** [4.3.3 · Comprehension] The discipline that most protects re-baselining from abuse is:

- A. re-baselining no more than once a year
- B. keeping the original baseline visible in reporting alongside the current one
- C. having the project leader approve it
- D. recalculating contingency at each re-baseline

CORRECT

*Rationale:* Visibility of the original preserves the ability to state performance against the original commitment (4.3.3). C is precisely the wrong authority.

**MCQ 4.3-F** [4.3.2 · Application] A configuration register holds 340 items of which 296 conform. Assuming independent defects, the expected share of **two-sided interfaces** whose verification cannot be relied on is closest to:

- A. 12.94 %
- B. 24.21 %
- C. 25.88 %
- D. 75.79 %

CORRECT

*Rationale:*  $1 - (296/340)^2 = 1 - 0.7579 = 0.2421$  (4.3.2b). A applies the item failure rate directly to interfaces; C doubles the item rate, which is the linear approximation and overstates slightly; D quotes the conformance rather than the failure.

**MCQ 4.3-G** [4.3.3 · Analysis] A reconciliation shows a residual of 32,200, which decomposes into 77,100 of baseline movement with no change reference and 44,900 of approved changes with no baseline effect. The correct reading is that:

- A. the residual is 1.34 % of the baseline and immaterial
- B. the gross error is 122,000 and must be reported as such, because the two classes have opposite signs
- C. the cost tool is correct and the change log needs updating
- D. netting is appropriate because the classes offset within the same baseline

CORRECT

*Rationale:* Opposite-signed classes make the net understate the gross by twice the smaller class (4.3.3b), so 89,800 of the 122,000 is concealed. A and D both read the net as the finding: A by applying a materiality test to it, D by asserting the offset is legitimate. C prejudges which record is wrong when the two classes have opposite remedies.

**MCQ 4.3-H** [4.3.1 · Analysis] A schedule change is approved and the PV curve is never re-phased. Which reported measure remains exactly correct?

- A. SPI
- B. SV

C. CPI CORRECTD. the composite  $EAC = AC + (BAC - EV) / (CPI \times SPI)$ 

*Rationale:*  $CPI = EV/AC$  contains no PV term, so it and the CPI-only forecasts are unaffected, which is why the defect passes a cost review (4.3.1). A and B are computed against PV directly; D is linear in PV and moves by USD 1.20 per USD 1 mis-phased on Auriga's figures.

## SELF-CHECK — KA 4.3

1. *Why is a schedule change always a cost-baseline change?* — Because the cost baseline is time-phased; if its curve does not move, earned value stops measuring the work.
2. *Which earned-value measures does a stale PV curve leave correct, and why does that matter?* — CPI and both CPI-only forecasts, because they contain no PV term; it matters because the cost review therefore passes and the defect survives.
3. *Which configuration-audit finding class is most serious, and why?* — Recorded versions that differ from deployed ones, because verification against the register then proves nothing.
4. *How is an item-level configuration tolerance derived?* — By inverting  $p^k$  from the interface-level target:  $p = q^{(1/k)}$ . Chosen directly at item level it has no argument behind it.
5. *Why must a reconciliation residual be reported gross?* — Because opposite-signed error classes make the net understate the gross by twice the smaller class, and a zero residual is not evidence of integrity.
6. *How should a cumulative-change threshold be set?* — From the observed change rate in the change log, not from a round number.

## KNOWLEDGE AREA 4.4

## Integrated change control

*Topics:* 4.4.1 change flow · 4.4.2 impact assessment · 4.4.3 the change board and the decision log.



#### 4.4.1 The change flow

**Why "integrated".** The word is the whole point: a change to scope is a change to schedule and to cost, and a control process that assesses one dimension has not controlled the change. The flow that does:

1. **Raise:** anyone may raise; the request states the change and its reason, and is logged on receipt with a reference, so that a rejected or withdrawn request still leaves a trace.
2. **Screen:** is it a change at all? Three outcomes are commonly confused and should be separated explicitly: a **change** (the baseline moves), a **clarification** (the baseline already covered it and someone misread it), and a **defect** (the work does not meet the baseline and must be corrected at no baseline movement). Misclassifying a defect as a change is how a supplier is paid twice; misclassifying a change as a clarification is how scope grows silently.
3. **Assess impact:** across all dimensions, per KA 4.4.2. This is the step that is under-resourced, and the reason is that assessment costs money before any decision has been taken to spend money.
4. **Decide:** at the authority the *assessed* impact requires, not the quoted cost (4.4.2's central point), with the decision recorded per Domain 3, KA 3.3.4.
5. **Implement and baseline:** update scope, schedule and cost together, with the change reference recorded against each, so that KA 4.3's reconciliation works.
6. **Verify and close:** confirm the change was implemented as approved, including its interface and documentation consequences.

Of the six steps, screening is the only one whose output is a **classification** rather than a number or a decision, and classifications are the cheapest thing in the flow to get right and the most expensive to reverse, because by the time a misclassification is discovered money has moved: in one direction or the other, and the two directions do not cost the same.

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#### WORKED EXAMPLE

**Worked example 4.4.1 — what Meridian's screening errors cost, in both directions.**

1. **Setup.** Meridian logged 76 requests in year one. Screening ruled 51 changes, 16 clarifications and 9 defects; of the 51 changes, 37 were subsequently approved (the 34 below the leader's threshold plus the 3 taken to steering), and 14 were rejected or withdrawn. An assurance re-screen against the baseline found 8 misclassifications: 5 of the approved "changes" were **defects** (work already owed under the baseline, at the change log's average approved value of 6,800), and 3 of the 16 "clarifications" were **changes**, carrying direct work of 9,200, 14,600 and 5,400, one of them with 0.5 weeks of critical-path impact (cost of delay 14,280 per week).

2. **Formula.** Error rate = misclassifications ÷ requests. Defect-as-change cost = count × average approved value (the sum paid a second time for an existing obligation). Change-as-clarification cost =  $\Sigma$  direct + schedule impact (unbudgeted, unlogged, and outside the baseline). Then compare the **net** effect on the recorded change total with the **gross** error.
3. **Substitution.**  $8/76.5 \times 6,800.9,200 + 14,600 + 5,400 + 0.5 \times 14,280.$
4. **Result.** Screening error rate **10.53** %. Defects paid as changes **USD 34,000**, which is **14.71** % of the 231,200 of direct drift computed in WE 4.3.3, so that figure includes that much work which was never a change at all. Changes taken as clarifications **USD 36,340**, none of it in any baseline or log. Total misclassification cost **USD 70,340**, or **USD 8,792.50** per error. The **net** effect on the recorded change total is  $36,340 - 34,000 = \text{USD } 2,340$  (**0.10** % of the baseline), giving a corrected drift of **293,516**, or **12.23** %, against WE 4.3.3's 291,176.
5. **Interpretation.** The two errors point in opposite directions and (for the second time in this domain, from wholly independent data), **netting them hides both**. A corrected drift figure that moves by 2,340 is inside anyone's materiality threshold while concealing 70,340 of gross misclassification, and the concealment is again exactly twice the smaller class:  $2 \times 34,000 = 68,000$  of the 70,340, where WE 4.3.3b concealed  $2 \times 44,900 = 89,800$  of 122,000. That recurrence is a property rather than a coincidence, and it is worth carrying out of the domain as one: **wherever a reconciliation has two error classes of opposite sign, the net understates the gross by twice the smaller class**. Report both figures or report neither.

The two errors also differ in **how they are discovered**, and that asymmetry decides where the control belongs. Paying twice for an existing obligation shows up in the cost report: money leaves, a variance appears, someone asks. Silent scope growth shows up nowhere, no change record, no baseline movement, no variance until the work is executed, at which point it is indistinguishable from ordinary productivity. So the direction that costs slightly *more* on this year's arithmetic (36,340 against 34,000) is also the direction with no downstream detector, and it is the one that needs the tighter control. A screening process designed only to stop overpayment has been designed against the visible half.

Whether the 34,000 can actually be recovered is a contractual question, not a project one. It turns on the terms, on any notice or claim period, on whether the work was accepted and paid without reservation, and on the governing jurisdiction. **Take contractual advice before treating a misclassification finding as recoverable income:** the arithmetic establishes that value was transferred, not that it can be reclaimed, and a leader who reports the second when they have only established the first will be corrected expensively.

The remedy is cheap and specific, and it is a change of form rather than of effort. Screening becomes a **recorded ruling** instead of an administrative sort: the

baseline clause or requirement relied on is cited, a named technical authority signs, and the three outcomes are separate fields rather than a checkbox. Costed at half an hour of a technical authority and half an hour of the change manager per request (**USD 110** at the blended rate, **USD 8,360** across 76 requests) against 70,340 of error, a ratio of **8.41**. The ratio is not the point and will vary; the point is that the ruling is the only artefact in the flow that makes a classification **contestable later**, and an uncontestable classification is one nobody will revisit.

One caution about the base rate, because a 10.53 % error rate invites the wrong conclusion. It is not in itself evidence of poor screening: some proportion of requests genuinely sits on the change/defect boundary and will be ruled differently by two competent reviewers, and the honest comparison is against that irreducible rate rather than against zero. What a reviewer should test is whether the disputed rulings were disputed **on the record**. A screening process that has reversed no ruling in a year is unexamined, not accurate.

**The discipline that holds it together.** No work begins on an unapproved change. That is easy to write and hard to hold, because the pressure is always to start, and the specific damage is not the work itself but that it removes the option to say no, converting a decision into a ratification. Where genuine urgency exists, the answer is an **emergency change route** with a named authority, a short deadline and mandatory retrospective ratification, which is the same instrument as Domain 3's out-of-cycle mechanism. An emergency route that exists is used and recorded; one that does not exist is used and not recorded.

**And the route's usage rate is a diagnostic, not a discipline problem.** Meridian's 76 requests included 5 routed as emergencies, **6.58 %**. A rising share is almost never indiscipline; it is the normal route's latency being priced by the people who have to live with it, and the correct response is to reduce that latency (the paper deadline before the meeting interval, per Domain 3's lever asymmetry) rather than to tighten the emergency criteria, which merely moves the same work into unrecorded starts and removes the only trace it was leaving. So set a **review trigger** on the share rather than a target for it: when the share rises, examine the normal route, not the requesters.

#### 4.4.2 Impact assessment — what a change actually costs

**The central claim of this KA.** The figure on a change request is the *direct* cost of doing the new work, and it is usually the smallest component of the change's true cost. Everything else (schedule consequence, rework of completed work, interface re-verification, regression testing, documentation and training) is real, incurred, and absent from the form.

A complete assessment covers: **direct cost** of the new work; **schedule** impact, on the critical path or on float, priced at the cost of delay; **rework** of work already com-

pleted to the old specification; **interface** re-verification for every affected interface (KA 4.2.3); **regression testing** of what was already proven; **documentation, training and communication** updates; **risk** profile change (Domain 8); and **benefit** change, since a change that reduces benefit has a cost that appears nowhere in a cost assessment (Domain 2).

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#### WORKED EXAMPLE

##### Worked example 4.4.2 — the change that cost 3.29 times its quoted price.

1. **Setup.** Meridian receives a change request to add a national-reporting field set. The quoted direct build cost is **USD 40,000**. Assessment establishes: **2 weeks** of critical-path impact (cost of delay **14,280** per week); **22,000** of rework to records-application screens already completed; **3** affected interfaces requiring re-verification at **6,000** each; **14,000** of regression testing; and **9,000** of documentation and clinician-training updates.
  2. **Formula.** True cost = direct + (schedule weeks × cost of delay) + rework + (interfaces × re-verification) + regression + documentation.
  3. **Substitution.**  $40,000 + 2 \times 14,280 + 22,000 + 3 \times 6,000 + 14,000 + 9,000$ .
  4. **Result.**  $40,000 + 28,560 + 22,000 + 18,000 + 14,000 + 9,000 = \text{USD } 131,560$ : **3.29 times** the quoted figure. The quoted cost is **30.4 %** of the true cost.
  5. **Interpretation.** Two consequences follow, and the second is the more important because it is structural. First, a change board deciding on the quoted 40,000 is deciding on 30 % of the information, and it will approve changes it would have refused, not through any failure of judgement but because the number in front of it was the wrong number. Second, and this connects directly to Domain 3: **a delegation threshold applied to quoted direct cost is not a control**. Meridian's threshold is 25,000, so this change escalates. But a change quoted at **22,000** with the same two weeks of critical-path impact carries a true cost of at least  $22,000 + 28,560 = \text{USD } 50,560$ , twice the threshold, and is decided by the project leader alone, because the form asked for the direct cost. The remedy is a single sentence in the delegation schedule (*the threshold reads on assessed total impact, not on quoted direct cost*), and it is worth more than most process improvements a programme will make. Note, finally, what this arithmetic is not: it is not an argument against change. Changes are frequently worth 3.29 times their quoted cost, and Domain 2's business-case logic still decides. It is an argument for **assessing before deciding**.
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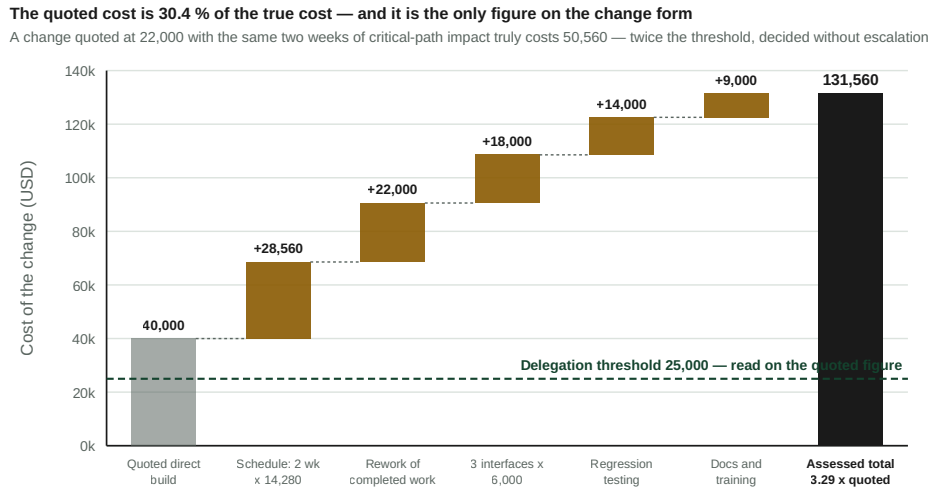


Fig 4.4.1 — What a change actually costs

**Assessment discipline.** Three rules make assessment reliable rather than performative. The assessment is done by the people who will do the work, not by the change board estimating on their behalf. It is **timeboxed and resourced**: an assessment budget exists, because otherwise assessment competes with delivery and loses, which is the actual reason changes are decided on quoted costs. And its **assumptions are stated**, particularly the schedule assumption, since whether two weeks lands on the critical path or on float is the difference between 28,560 and nothing, and it is a fact about the schedule rather than a matter of opinion (Domain 6's float).

#### 4.4.3 The change board and the decision log

**The change board.** A body with authority to approve baseline changes within stated limits and to escalate beyond them. Its membership must include the authority to commit cost, the authority to commit schedule, and the technical authority to judge feasibility. And the last of those is the one usually missing, which is how changes are approved that cannot be implemented as described.

**Cadence and latency.** Domain 3's arithmetic applies unchanged: a change board meeting fortnightly with a one-week paper deadline imposes  $E[\text{wait}] = 2/2 + 1 = 2$  weeks on every escalated change. Where change volume is high, that latency is a delivery cost and is reduced the same way: a shorter paper deadline first, then a written-resolution route for changes below a stated impact.

**The decision log, and the entry that must exist.** Every change decision (approved, rejected, deferred or withdrawn) generates an entry: reference, date, decision-maker by name, the decision, the assessed impact across all dimensions, the basis, and the baseline effect. **Rejections matter as much as approvals**, and are the entries most often missing: a rejected change that leaves no trace returns in three months as a new request, is assessed again at full cost, and may be approved by a different body in ignorance of the first decision. The log is what prevents that, and it

is the same log as Domain 3's, not a second one: one register of record for every decision that moves the baseline, whoever took it and at whatever value, which was Case study B's correction.

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#### WORKED EXAMPLE

#### Worked example 4.4.3 — pricing the rejection entries Meridian did not write.

1. **Setup.** Of the 51 requests WE 4.4.1 ruled as changes, 14 were rejected or withdrawn. The decision log records 5 of them with a rationale; 9 left no trace. Within the same year, 4 of those 9 returned as fresh requests. A full impact assessment consumes 5 person-days at the blended USD 880: USD 4,400. One of the four returning requests was approved by the programme board in ignorance of the earlier refusal, at an assessed impact of USD 63,000. Writing a rejection entry (reference, date, decision-maker, reason, and the assessment already performed) takes about a quarter of an hour, USD 27.50 at the same rate.
2. **Formula.** Duplicated assessment = recurrences × assessment cost. Total loss = duplicated assessment + the assessed impact of any decision reversed in ignorance. Compare with the cost of recording every rejection, and compare the mean loss per recurrence with the modal one.
3. **Substitution.**  $4 \times 4,400; + 63,000$ ; against  $14 \times 27.50$ .
4. **Result.** Rejections recorded 35.71 %; recurrence among the untraced 44.44 %. Duplicated assessment USD 17,600; one decision reversed in ignorance USD 63,000; total USD 80,600, or USD 20,150 per recurrence. Recording all 14 rejections would have cost USD 385.
5. **Interpretation.** The distribution matters more than the total, and a leader should present both figures or neither. Three of the four recurrences cost only the wasted re-assessment: USD 4,400 each, irritating and survivable. The fourth cost 63,000, so the mean of 20,150 is 4.58 times the modal case. Strip the reversed decision and the year's loss is 17,600 against 385 of recording cost, a ratio of 45.71; include it and the ratio is 209.35. Quote only the second and a competent reviewer will discount the whole argument as anecdote; quote only the first and you have understated the exposure by the one case that mattered. **The honest statement is that the modal cost of an untraced rejection is a wasted assessment and its tail cost is a reversed decision, and the control is bought for the tail.**

The tail case is also the only part the log uniquely prevents, and the mechanism is worth being precise about. Re-assessment waste can be attacked other ways: duplicate detection on request text, the third item in the AI note below, catches much of it. A **decision reversed in ignorance** cannot be caught that way: the second body was not careless, it simply had no record that the question had been asked and answered. That makes the rejection entry not an administrative courtesy but the instrument that makes a "no" **durable**, and a governance struc-

ture in which "no" does not persist is one where any sufficiently persistent requester eventually obtains a yes, from whichever authority happens to be sitting, at no point through anyone's misconduct.

Note the interaction with KA 4.4.2's assessment budget, because it converts this into an argument the budget holder cares about. Assessment is chronically under-resourced precisely because it precedes the decision to spend, and **17,600 of the year's assessment budget was spent twice on questions already answered**: four assessments of capacity that the same money could have applied to new requests. The rejection entry is therefore the cheapest available increase in assessment capacity, and it is worth putting to the budget holder in exactly those terms rather than as a compliance point.

The caution is about what the entry must contain. A rejection entry must carry the **assessment**, not merely the outcome: an entry reading "rejected — not affordable" saves nothing, because a returning request still has to be assessed from scratch to establish what it now costs. The entry that saves the 4,400 records the assessed impact, its basis and its date, so that a recurrence becomes a question about *what has changed since*: usually answerable in an hour, and occasionally answerable with a legitimate "the circumstances have changed and the answer is now yes", which is the outcome the log exists to make possible rather than to prevent.

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## AI in this KA

**Where it earns its place.** Assembling the *components* of an impact assessment: identifying which interfaces an affected component touches (from the interface register), which completed work packages are implicated, which test suites regress, which documents and training materials cite the changed behaviour. This is traversal of structured relationships, it is where human assessment is systematically incomplete, and it is checkable. Screening a request against the baseline to propose whether it is a change, a clarification or a defect, a proposal, for human ruling. Detecting duplicate or previously rejected requests in the log, which is exactly the failure the previous paragraph describes. Computing the assessed total against the delegation schedule and flagging where quoted and assessed costs fall on opposite sides of a threshold.

**Where it must not go.** Approving or rejecting a change. Estimating the direct cost or the schedule impact, which belongs to the people who will do the work and to the schedule respectively: a model asked for a schedule impact will produce a plausible number with no critical-path basis, and that number is worth 28,560 in the example above. And no classification of a defect as a change without human ruling, since that determination has commercial consequences.

**Verification, concretely.** Every AI-identified affected interface, work package and test suite is confirmed by the accountable owner before it enters the assessment; the

schedule impact is taken from the schedule, by the planner, with the float position stated; the arithmetic of the total is reproduced by hand; and the decision record names the human decision-maker, with the AI's contribution recorded as assessment input rather than as authorship.

## KEY TERMS — KA 4.4

| TERM                                     | MEANING                                                                                                                                                                    |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Integrated change control</b>         | Assessing and authorising a change across scope, schedule and cost together.                                                                                               |
| <b>Change vs clarification vs defect</b> | Baseline moves · baseline already covered it · work fails to meet the baseline.                                                                                            |
| <b>Screening ruling</b>                  | The recorded classification: the baseline clause relied on, the outcome, and a named technical authority; the only artefact that makes a classification contestable later. |
| <b>Emergency-route share</b>             | Emergency changes ÷ total requests, a latency diagnostic, so set a review trigger on it rather than a target.                                                              |
| <b>Assessed total impact</b>             | Direct + schedule + rework + interface re-verification + regression + documentation + risk + benefit.                                                                      |
| <b>Quoted direct cost</b>                | The figure on the form: typically a minority of the true cost, and the wrong basis for a threshold.                                                                        |
| <b>Emergency change route</b>            | A named authority, a short deadline and mandatory retrospective ratification.                                                                                              |
| <b>Change board</b>                      | The body authorised to approve baseline changes within limits, including a technical authority.                                                                            |
| <b>Rejection entry</b>                   | The log record of a change not approved: the entry most often missing and the reason requests recur.                                                                       |

## SAMPLE MCQs — KA 4.4

**MCQ 4.4-A [4.4.2 · Application]** A change is quoted at 40,000 direct. Assessment adds 2 weeks of critical-path impact at 14,280 per week, 22,000 of rework, 3 interfaces at 6,000 each, 14,000 of regression testing and 9,000 of documentation. The assessed total is:

- A. USD 40,000
- B. USD 103,000
- C. USD 131,560 CORRECT
- D. USD 109,560



*Rationale:*  $40,000 + 28,560 + 22,000 + 18,000 + 14,000 + 9,000 = 131,560$  (4.4.2). B omits the schedule impact; D omits the rework.

**MCQ 4.4-B** [4.4.2 · Evaluation] A programme's delegation threshold is 25,000 on quoted direct cost. A change quoted at 22,000 carries 2 weeks of critical-path impact at 14,280 per week. The structural defect is that:

- A. the threshold is too low
- B. a change with an assessed impact of at least 50,560 is decided without escalation, because the threshold reads on the quoted figure CORRECT
- C. the change should be split into smaller changes
- D. the cost of delay should be excluded from change assessment

*Rationale:* The threshold's *basis* is the defect, not its level (4.4.2); the remedy is that it reads on assessed total impact.

**MCQ 4.4-C** [4.4.1 · Analysis] Classifying a defect as a change results in:

- A. the baseline correctly reflecting the work
- B. the supplier being paid twice for the same obligation CORRECT
- C. faster resolution at no cost
- D. an unnecessary escalation

*Rationale:* A defect is work already owed under the baseline; treating it as a change adds budget for it (4.4.1). The mirror error (a change classified as a clarification) grows scope silently.

**MCQ 4.4-D** [4.4.3 · Comprehension] The change-log entry type most often missing and most consequential is:

- A. the approval entry
- B. the rejection entry CORRECT
- C. the implementation confirmation
- D. the assessed cost

*Rationale:* An untraced rejection returns as a new request, is re-assessed at full cost, and may be approved by a different body in ignorance of the first decision (4.4.3).

**MCQ 4.4-E** [4.4.2 · Analysis] Why is impact assessment systematically under-resourced?

- A. assessors lack the skill
- B. it costs money before any decision has been taken to spend money, so it competes with delivery and loses CORRECT
- C. change boards prefer quoted figures
- D. assessment adds no value to rejected changes

*Rationale:* The structural cause is that assessment is unfunded work preceding a decision (4.4.1, 4.4.2); the remedy is an explicit assessment budget.

**MCQ 4.4-F** [4.4.1 · Application] A re-screen finds 5 defects paid as changes at an average 6,800, and 3 changes ruled as clarifications carrying 29,200 of direct work plus 0.5 weeks of critical-path impact at 14,280 per week. The gross misclassification cost is:

- A. USD 34,000
- B. USD 36,340
- C. USD 70,340 CORRECT
- D. USD 2,340

*Rationale:*  $34,000 + 36,340 = 70,340$  (4.4.1). A and B are the two classes taken alone; D is the net, which understates the gross by twice the smaller class and is the specific error the example exists to name.

**MCQ 4.4-G** [4.4.3 · Evaluation] Four untraced rejections recurred: three cost only a duplicated 4,400 assessment, and one was approved by another body at an assessed impact of 63,000. The defensible way to present the case for recording rejections is to:

- A. quote the 209.35 ratio of total loss to recording cost
- B. quote the 45.71 ratio excluding the reversed decision
- C. quote both, stating that the modal cost is a wasted assessment and the tail cost is a reversed decision CORRECT
- D. quote the mean loss of 20,150 per recurrence as typical

*Rationale:* The mean is 4.58 times the modal case, so either ratio alone misleads, one as anecdote, the other by omitting the case that mattered (4.4.3). D presents a tail-dominated mean as typical.

#### SELF-CHECK — KA 4.4

1. *Name the three screening outcomes and the cost of confusing two of them.* — Change, clarification, defect; a defect treated as a change pays twice, a change treated as a clarification grows scope silently.
2. *Which screening error needs the tighter control, and why?* — A change ruled as a clarification, because overpayment surfaces in the cost report while silent scope growth has no downstream detector.
3. *Why is a threshold on quoted direct cost not a control?* — Because the quoted figure is typically a minority of the assessed impact, so changes above the real threshold are decided below it.

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4. *What must a rejection entry contain to be worth writing?* — The assessed impact, its basis and its date, so that a recurrence is a question about what has changed since.
  5. *What makes a schedule impact assessment a fact rather than an opinion?* — Whether the impact lands on the critical path or on float is a property of the schedule, and the float position must be stated.
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## Advanced topics — Domain 4

### 4.A.1 Integration across organisational boundaries

Where delivery spans organisations (client and supplier, joint venture partners, a programme and its constituent projects) integration acquires a failure mode that no internal design has: **each party integrates to its own baseline**. Version 4 of an interface specification on one side and version 5 on the other is not an argument about competence; it is the predictable consequence of two configuration registers with no reconciliation, and it is discovered at integration testing where it is most expensive.

Four provisions address it, and they belong in the contract rather than a plan (Domain 7; PFL-AI Domain 11): a **single interface register** identified as the register of record, with named owners on both sides of every interface; a **joint change process** for anything crossing the boundary, with one decision log rather than two reconciled ones; **matched configuration identification**, so that both parties refer to the same item by the same identifier at the same version; and **joint verification** of interfaces at a scheduled point, with resource committed on both sides. The recurring commercial mistake is to treat integration as each party's own responsibility up to its own boundary, which leaves the join itself unowned, and the join is where the failure happens.

### 4.A.2 Architectural decisions as governance decisions

Some integration choices are irreversible in practice long before they are irreversible in principle: the integration pattern of KA 4.2.3, the data model, the identity provider, a platform dependency. They share three properties that make them governance matters rather than technical ones. The cost of reversal rises steeply with time, the consequences fall outside the project's boundary and often outside its life, and they are typically taken by people whose delegated authority is defined in money, of which these decisions frequently involve very little.

That last point is the whole issue, and it is Domain 3's delegation schedule read on the wrong dimension: a decision costing 40,000 to take and 2,000,000 to reverse is not a 40,000 decision, and a schedule reading only on value cannot see it. The countermeasures are specific: an **architectural decision record** for each such

choice, with its alternatives, rationale and reversal cost stated; those records reviewed by the governance body on the **reversibility** dimension rather than the value one; and a standing question in gate criteria, *which decisions taken since the last gate are expensive to reverse, and who took them?* Case study B of Domain 3 is exactly this failure with a control attached rather than an architecture attached.

#### 4.A.3 The reviewer's integration eye

Invariants to test on any delivery architecture, each cheap and each diagnostic:

The plan set has been **read against itself before baseline approval**, with the findings priced, and the checks that failed are the ones spanning two owners. The **rolling-wave horizon is at least as long as the longest commitment lead time** inside it, or the exception is labelled a partial horizon. The WBS satisfies the **hundred-per-cent rule at every level**, and the value most likely to be missing is enabling change. Every product traces to work and every work package to a product. And and each cross-match exception carries a **recorded ruling with a named ruler**, since two thirds of them will be legitimate and an unrul'd exception list is indistinguishable from an unrun check. The **interface register is built from need**, has a named owner on each side of every entry, and every entry has a **scheduled** verification activity with resource. The **time-phased cost baseline matches the schedule's current dates**: the single most informative check in the domain, the first thing abandoned under pressure, and the one whose failure leaves every cost measure correct. The change log's approved changes **sum to the difference** between the original and current baselines, and any residual is **decomposed before it is netted**; there are no baseline movements without a change reference and no approved changes without a baseline effect. Screening outcomes are **recorded rulings** citing the baseline provision relied on. Rejections appear in the log, carrying their assessment rather than only their outcome. The delegation threshold reads on **assessed total impact**. A **cumulative test** exists and its threshold and period are derived from the observed change rate, and it is present *as well as* the reconciliation above, the two being disjoint controls (4.3.3b). Configuration status accounting has been audited against deployment within a stated period, and the item-level tolerance is **derived from an interface-level target** rather than chosen. Re-baselines carry the original's authority, a substantive reason, and the original still visible in reporting. And every architectural decision with a reversal cost materially above its decision cost has a record and a named decision-maker.

### Industry variations — Domain 4

- **Construction and infrastructure.** Interfaces are physical and contractual simultaneously, and interface management is a named discipline with its own register and manager; configuration management extends to as-built records, whose divergence from design is the standard handover dispute (Domain 16).

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- **Software and digital.** Configuration management is largely automated through version control and build pipelines, which solves identification and control while leaving the *interface agreement* and its ownership as human work, and integration failure migrates there accordingly. The pairwise arithmetic of WE 4.3.2b explains why the automation is worth so much: it lifts item conformance to where the exponent stops biting, so the interface-level rate follows without anyone having to compute it.
  - **Regulated manufacturing (pharmaceutical, medical devices, aerospace).** Configuration and change control are regulatory obligations with validated processes; change assessment must include re-validation cost, which frequently exceeds every other component and reverses the ranking of WE 4.4.2.
  - **Public-sector programmes.** Baselines are published commitments, so re-baselining is externally visible and politically costly, which makes drift the preferred failure mode; and therefore makes the cumulative test of KA 4.3.3 the highest-value control available.
  - **Multi-party ventures.** The reconciliation provisions of 4.A.1 are the domain's principal content; a single interface register named in the agreement is worth more than any internal process either party runs. And the exponent  $k$  in  $p^k$  is the number of parties, so a three-party interface is materially harder to keep verifiable than a two-party one at the same register quality (65.98 % against 75.79 % on Meridian's figures), which is a reason to reduce the number of parties to a join before it is a reason to audit harder.
  - **Healthcare.** Clinical safety cases are configuration-controlled artefacts bound to specific system versions, so a version change is a safety-case change, which makes Meridian's 5 mis-recorded configuration items a clinical-governance finding, not merely an administrative one.
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## Case study — Domain 4: the interface nobody owned (health, Meridian)

**Situation.** Meridian's integration testing began four weeks late and took eleven weeks against a planned five. The programme reported "integration complexity" and requested a schedule change. The assurance review that followed found something more specific.

**What the review found.** Of the 66 possible interfaces among the 12 components, 31 were genuinely required and had been specified. Of those 31, 9 had no named owner on one side, in every case the side outside the programme's own team: the national reporting gateway, the identity provider, the legacy records supplier. Those nine accounted for **all** of the four-week start delay and **four of the six weeks** of overrun, because each had to be renegotiated at the point of verification with someone who had never agreed to it. Separately, 5 configuration items were verified

against register entries that did not match what was deployed (WE 4.3.2's most serious class), which produced two defects that each passed component testing.

**What the arithmetic had already said.** The programme had planned interface work at 12 interfaces, the layered count, while operating a partial mesh of 31. The plan was not optimistic; it was counting the wrong thing. At 18,000 an interface, 31 interfaces is **558,000** against a planned **216,000**, and the 342,000 difference had been absorbed as unplanned effort and schedule.

**What the unowned interfaces cost, per interface.** Eight of the elapsed weeks lost (the four-week late start and four of the six weeks of overrun) were attributed to the nine unowned interfaces, which were **29.03 %** of the 31 required. At the programme's cost of delay that is  $8 \times 14,280 = \text{USD } 114,240$ , or **USD 12,693** per unowned interface, and none of it bought anything; the nine agreements had to be negotiated either way, and their negotiation effort already sat inside the 18,000 unit cost. What the programme paid for was negotiating them at the moment of *verification* rather than at the moment of *specification*. The register field that would have prevented it (*owner on side B, named*) costs nothing to add and nothing to maintain, which makes this the cheapest control in the domain and the one most often absent. The five mis-recorded configuration items compound it in exactly the way WE 4.3.2b predicts: with item conformance at 87.06 %, roughly **7.50** of the 31 interfaces were expected to be unverifiable on register grounds alone, and the two defects that passed component testing came out of that population rather than out of bad luck.

**What changed.** An interface register built from **need** rather than from either formula, with a named owner on **each** side of every entry and a scheduled verification activity for each, which made the nine unowned interfaces visible as nine missing agreements rather than as future complexity. Verification activities entered the schedule with duration and resource, which is what Domain 6 needed in order to show the float they consumed. And the configuration register was audited against deployment before integration rather than after.

**What the domain teaches here.** An unowned interface is a scheduled failure with an unknown date, not a risk. And the interface arithmetic is only useful applied to the *required* count: 12 was the architecture's promise, 66 was the theoretical maximum, and 31 was the number that had to be managed. Planning to the promise is the error.

## Case study B — Domain 4: the baseline that could no longer answer the question (financial services)

**Situation.** A payments programme (the same organisation as Domain 3's Case study B, two years later) reached its third baseline. Reported performance was **CPI 0.99** and **SPI 1.00**, on baseline three. An audit asked a simpler question: what is performance against the **original approved commitment**? Nobody could produce the answer within a fortnight.

**Why not.** Each re-baseline had been implemented by **replacement** rather than accumulation. The new baseline was loaded, the old one archived as a file, and the change log referenced changes to "the baseline" without stating which. Intermediate states were not reconstructable, so the original-to-current reconciliation (approved changes summing to the difference between original and current) could not be performed at all. Two of the three re-baselines had also been approved by the programme board rather than by the investment committee that approved the original, and neither recorded a substantive change in circumstance; both cited accumulated variance, which is the reason that does not qualify.

**How it resolved.** The reconstruction took six weeks of a two-person effort against the change log, the archived files and the finance ledger, and established performance against the original as **CPI 0.87**, a **14.9 %** cost overrun that three successive baselines had made invisible without ever stating an untruth. Corrections: baselines thereafter maintained by **accumulation** with every change referenced to its authority; the original baseline reported alongside the current one permanently; re-baselining reserved to the authority that approved the original, with a recorded substantive reason; and the original-to-current reconciliation performed and published every reporting period.

**What the domain teaches here.** A baseline exists to answer one question (how is this project performing against what was approved?), and a baseline maintained by replacement cannot answer it. Neither **CPI 0.99** nor **CPI 0.87** was a false statement; the first was true of a document and the second of the commitment, and only one of them was what the organisation had asked.

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## Executive perspective — Domain 4

What a programme director cannot delegate in this domain:

- **The authority statement in your own charter.** If you cannot state what you may decide alone without reading the document, you do not have it (4.1.1).
- **The interface count and its ownership.** Not the theoretical count and not the architecture's promise: the *required* count, with a named owner on each side and a scheduled verification for every entry. Unowned interfaces are scheduled failures with unknown dates (4.2.3, Case study A).
- **That the plan set has been read against itself before you sign the baseline.** Meridian's audit found 306,992 of unauthorised cost and schedule in three days for 5,280: more than a year of authorised drift, before any work had started (4.1.2).
- **That the time-phased cost baseline still matches the schedule.** This is one check, it is the first thing abandoned under pressure, and without it earned value measures nothing (4.3.1). Be aware that its failure leaves **CPI** and the CPI-only



forecasts exactly correct, so a clean cost review is not evidence that it has been done.

- **Reconciling gross, never net.** A residual of 32,200 concealed a gross error of 122,000, and the two classes had opposite remedies: one a posting, one an investigation (4.3.3b).
- **The tolerance on your configuration register, derived rather than chosen.** 87 % of items conforming is 76 % of two-sided interfaces verifiable; a 95 % interface target requires 97.47 % of items (4.3.2b).
- **The basis of your change threshold.** Assessed total impact, never quoted direct cost, a one-sentence change worth more than most process improvements (4.4.2).
- **A cumulative test derived from your own change log.** Meridian's baseline moved 12.1 % with no decision, and a 100,000/90-day rule would not have caught it (4.3.3).
- **Reversal cost as a governance dimension.** A decision costing 40,000 to take and 2,000,000 to reverse is not a 40,000 decision, and a value-only delegation schedule cannot see it (4.A.2).
- **Performance against the *original* baseline.** Keep it visible and reconcilable. It is the only question the funding organisation actually asked (4.3.3, Case study B).

## Calculation exercises — Domain 4

**Exercise 4.1** A programme integrates 9 components. One interface costs 24,000 to specify, build, test and document; an integration layer would cost 200,000. Compute the mesh and layered interface counts and costs, the saving, the breakeven layer cost, and the marginal cost of a tenth component under each architecture. *Solution.* Mesh  $9 \times 8/2 = 36$  interfaces at 24,000 = **USD 864,000**. Layered 9 interfaces = 216,000 plus 200,000 = **USD 416,000**. Saving **USD 448,000**. The layer is worth building while it costs below  $864,000 - 216,000 = \text{USD } 648,000$ . A tenth component adds  $9 \times 24,000 = 216,000$  on a mesh and **24,000** layered. Common error: counting  $n^2$  or  $n(n-1)$ , which double-counts each pair.

**Exercise 4.2** A parent WBS element is approved at 6,500,000. Its five children are estimated at 1,900,000, 1,250,000, 880,000, 1,640,000 and 410,000. A review then identifies an omitted element (commissioning and handover) estimated at 690,000. Compute the unallocated amount and its share of the parent, then the shortfall once the omission is restored. *Solution.* Children sum to **6,080,000**, leaving **420,000** unallocated, **6.46 %** of the parent. Restoring the omitted 690,000 gives an honest total of **6,770,000**, which exceeds the approved figure by **270,000**, or **4.15**



% Common error: reading the 420,000 as contingency, which locks in the omission and leaves nothing to absorb it.

**Exercise 4.3** A change is quoted at 75,000 direct. Assessment finds 3 weeks of critical-path impact at a cost of delay of 9,500 per week, 34,000 of rework, 4 affected interfaces at 7,500 each, 19,000 of regression testing and 11,000 of documentation. Compute the assessed total, the ratio to the quoted figure, and the quoted figure as a share of the true cost. *Solution.*  $75,000 + 28,500 + 34,000 + 30,000 + 19,000 + 11,000 = \text{USD } 197,500$ : **2.633 times** the quoted figure, of which the quote is **38.0 %**. Common error: omitting the schedule impact, which requires the float position from the schedule and is therefore the component most often left out on the grounds that it is "not yet known".

**Exercise 4.4** Over a year a project approved 48 changes averaging 5,200 direct cost, 18 of which carried 0.25 weeks of critical-path impact each. Cost of delay is 9,500 per week; the baseline is 3,200,000. Compute total drift, its share of the baseline, each change's individual share, and the 90-day cumulative threshold that would have caught it. *Solution.* Direct  $48 \times 5,200 = 249,600$ ; schedule  $18 \times 0.25 = 4.5$  weeks at 9,500 = **42,750**; total **USD 292,350**: **9.14 %** of the baseline, while each change was **0.16 %** of it. A 90-day window holds about  $48/4 = 12$  changes, aggregating to **62,400**, so a cumulative threshold at or below **62,400** on a 90-day window would trip; a 100,000 threshold would not. Common error: setting the cumulative threshold at a round number without reference to the observed change rate, which produces a control with the appearance of function and none of it.

**Exercise 4.5** A configuration register holds 500 controlled items, of which 468 conform. The programme has 44 required interfaces, 6 of them three-party and the remainder two-sided. Assuming independent defects, compute item conformance, two-party and three-party relationship conformance, the expected number of at-risk interfaces, and the item conformance that a 98 % two-party target would require. *Solution.* Item conformance  $468/500 = 93.60 \%$ . Two-party  $0.936^2 = 87.6096 \%$ ; three-party  $0.936^3 = 82.0026 \%$ . Expected at-risk: two-party  $38 \times (1 - 0.876096) = 4.7084$ , three-party  $6 \times (1 - 0.820026) = 1.0798$ , total **5.7882** interfaces. For a 98 % two-party target,  $p = 0.98^{(1/2)} = 98.99 \%$ , i.e. at most **5** non-conforming items against the 32 currently non-conforming. Common error: applying the item failure rate straight to interfaces,  $44 \times 0.064 = 2.8160$ , which understates the exposure by **51.35 %** because it ignores the exponent.

**Exercise 4.6** A programme's original approved baseline is 8,600,000. Its change log holds 27 approved changes totalling 742,300. The cost tool reports a current baseline of 9,318,600. Investigation finds 61,400 of approved changes with no

recorded baseline effect and 37,700 of baseline movement with no change reference. Compute the expected current baseline, the net residual, the gross error, and the ratio between them. *Solution.* Expected current  $8,600,000 + 742,300 = 9,342,300$ . Net residual  $9,318,600 - 9,342,300 = (23,700)$ : only **0.2756 %** of the original. Check:  $37,700 - 61,400 = (23,700)$ . Gross error  $61,400 + 37,700 = 99,100$ , **13.35 %** of the approved changes and **4.18 times** the net. Common error: treating a residual of a quarter of one per cent as rounding and closing the reconciliation, which discards a finding four times its size; note also that the residual here is *negative*, which does not make it benign. It means the log is ahead of the tool rather than behind it, and the two classes still require different remedies.

**Exercise 4.7** Of 84 change requests, screening ruled 46 changes, 24 clarifications and 14 defects. A re-screen finds that 7 of the "changes" were defects, at an average approved value of 9,400, and that 4 of the "clarifications" were changes, with direct work of 12,000, 8,500, 6,200 and 15,300, one carrying 0.75 weeks of critical-path impact at a cost of delay of 9,500 per week. Compute the error rate, both error classes, the gross total, the cost per error, and the net effect on the recorded change total. *Solution.* Errors **11** of 84, **13.10 %**. Defects paid as changes  $7 \times 9,400 = 65,800$ . Changes taken as clarifications  $42,000 + 0.75 \times 9,500 = 49,125$ . Gross total **USD 114,925**, or **USD 10,447.73** per error. Net effect on the recorded change total  $49,125 - 65,800 = (16,675)$ : **6.8921 times** smaller than the gross, and understating it by twice the smaller class ( $2 \times 49,125 = 98,250$ ). Common error: reporting the net, which here even has the wrong sign for the narrative. The log overstates the change total while scope has silently grown.

**Exercise 4.8** At a status date a project reports PV 6,300,000, EV 5,900,000 and AC 6,150,000 against a BAC of 12,000,000. A change approved two months earlier deferred 450,000 of planned value beyond the status date and the PV curve was never re-phased. Compute reported and true SPI and SV, CPI, and the composite EAC on each basis. *Solution.* Reported  $SPI = 5,900,000/6,300,000 = 0.9365$ ,  $SV = (400,000)$ . True PV  $6,300,000 - 450,000 = 5,850,000$ , so true  $SPI = 1.0085$  and  $SV = 50,000$ . The project is marginally *ahead* of the authorised schedule while reporting 0.94.  $CPI = 5,900,000/6,150,000 = 0.9593$  on both bases, and the CPI-only forecasts are unchanged at  $BAC + AC - EV = 12,250,000$  and  $BAC/CPI = 12,508,475$ . Composite  $EAC = AC + (BAC - EV)/(CPI \times SPI)$ : reported **USD 12,939,558**, true **USD 12,454,589**, an overstatement of **USD 484,968**, which is exactly the sensitivity  $(BAC - EV)/(CPI \times EV) = 1.0777$  multiplied by the 450,000 of mis-phasing. Common error: concluding that a stale baseline always flatters. Deferral makes performance look *worse* than it is, and the response (a recovery plan on a project that does not need one) costs real money to correct a reporting defect.

## Practitioner's toolkit — Domain 4

*Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.*

### Toolkit 4.T.1 — Charter on one page

Fields, in this order, and nothing else: purpose and measurable objectives · named sponsor · named project leader · **authority bounds** (spend, scope, resource, commitment — each a number or a sentence, not a paragraph) · scope boundary **including exclusions** · key deliverables and acceptance basis · fixed constraints, distinguished from planned dates · budget authority · material risks and assumptions · governance bodies and thresholds by reference · approval signature and date. If it exceeds two pages, the plan has begun; move the excess.

### Toolkit 4.T.2 — Interface register

One row per **required** interface (built from need, not from  $n(n-1)/2$ ): reference · the two components or parties · **owner on side A** and **owner on side B**, both named · content exchanged · format and protocol · direction · timing and frequency · error and exception behaviour · agreed specification **version** · verification activity, with its schedule reference, duration and resource · verification status and date. The two owner columns and the schedule reference are the fields that carry the value; an interface register without them is an inventory. Report monthly: entries with a missing owner, entries with no scheduled verification, and entries verified against a superseded version. Add one derived line to the same report — **expected at-risk entries**,  $\text{count} \times (1 - p^k)$  from the configuration register's current item conformance  $p$  and each entry's party count  $k$  — which converts the configuration audit into a number the integration plan can act on (WE 4.3.2b).

### Toolkit 4.T.3 — Integrated change request and baseline reconciliation

*Request side.* Reference · date raised · raiser · description and reason · **screening outcome** (change / clarification / defect, with the ruling recorded) · then the assessment, one line each: direct cost · schedule impact **with the float position stated** · rework · interfaces affected and re-verification cost · regression testing · documentation, training and communication · risk profile change · benefit change. Then: **assessed total impact** · authority required **on the assessed total** · decision, decision-maker by name and date · baseline effect on scope, schedule and cost · implementation and verification confirmation.

*Reconciliation side*, run every reporting period and published: original baseline · plus the sum of approved changes · equals current baseline (**the three must reconcile exactly**, each figure quoted from its own system of record rather than derived from another) · approved changes with no recorded baseline effect · baseline movements with no change reference · **the gross residual, stated before any netting, with**

**each class shown separately** · rolling-window cumulative total against the derived threshold · movement in the time-phased **PV** curve reconciled to the approved changes' schedule effects · and current performance reported against **both** the current and the original baseline.

## Exam preparation — Domain 4

**What is assessed.** Charter content and the primacy of the authority statement; the plan of plans and its consistency checks; tailoring as a recorded decision and its limits; WBS construction and the hundred-per-cent rule; work packages and control accounts; product versus work breakdown and where each belongs; **interface counting and integration architecture economics**; interface agreements and ownership; the integrated three-dimensional baseline and its invariants; configuration management's four functions and the audit finding classes; baseline maintenance by accumulation, and re-baselining discipline; **baseline drift and cumulative tests**; the change flow and the change/clarification/ defect distinction; **assessed impact versus quoted cost**; the change board and the decision log including rejections.

**The calculations to be able to do under time pressure.** Mesh and layered interface counts ( $n(n-1)/2$  and  $n$ ), their costs, the breakeven layer cost, and the marginal cost of one more component under each. The hundred-per-cent test and the shortfall once an omission is restored. Product and work-package coverage from a cross-match, and the classified residue after ruling. Plan-set audit exposure, converting unfunded work at a person-day rate and unscheduled elapsed time at the cost of delay, counting only the critical-path share. **SPI**, **SV**, **CPI** and the three **EAC** forms on both a stale and a re-phased **PV**, plus the composite forecast's sensitivity  $(BAC - EV)/(CPI \times EV)$  per unit of mis-phased planned value. Pairwise configuration integrity  $p^k$ , expected at-risk relationships, and the inversion  $p = q^{(1/k)}$  that turns an interface target into an item tolerance. The original-to-current reconciliation, its residual, and the decomposition of that residual into its two opposite-signed classes. Assessed total impact from its components, the ratio to the quoted figure, and the quoted share. Baseline drift from a change count, average cost, affected count, weeks and cost of delay; and the cumulative-window threshold that would catch it. Configuration defect rates by class.

**The traps.** Counting  $n^2$  or  $n(n-1)$  for interfaces · planning interface work at the architecture's promised count rather than the required count (Case study A) · reading unallocated budget as contingency (Exercise 4.2) · acting on raw cross-match exception counts without ruling on each one (WE 4.2.2) · assuming a stale **PV** curve always flatters, when deferral makes performance look worse (Exercise 4.8) · applying an item-level configuration rate to interfaces instead of  $p^k$  (Exercise 4.5) · choosing a configuration tolerance at item level rather than deriving it from an interface target · **netting a reconciliation residual instead of decomposing it**

(Exercises 4.6, 4.7, WE 4.3.3b, WE 4.4.1) · omitting schedule impact from a change assessment because the float position "is not yet known" (Exercise 4.3) · applying a delegation threshold to quoted direct cost (4.4.2) · setting a cumulative threshold at a round number rather than from the observed change rate (Exercise 4.4) · treating a defect as a change, or a change as a clarification · treating a rising emergency-route share as indiscipline rather than as a latency symptom · leaving rejections out of the decision log, or logging an outcome without the assessment · re-baselining by replacement (Case study B) · applying configuration control to activities · setting a rolling-wave horizon shorter than the longest commitment lead time inside it.

**How the domain connects.** Domain 1 supplies the accountability principle and the cost of delay every calculation here is priced at. Domain 2 supplies the business case whose benefits the baseline must still deliver, and the enabling-change column that WE 4.2.1 finds missing from the WBS. Domain 3 supplies the delegation schedule this domain corrects the basis of, the latency arithmetic the change board inherits, and the decision log the change log is part of. Part Two then works the baseline: Domain 6 schedules the WBS and supplies the float that change assessment needs, Domain 7 measures against the time-phased baseline this domain insists on maintaining, Domain 8 attaches uncertainty to it, and Domain 16 hands over the configuration record. PFL-AI Domain 11 handles the contractual face of cross-boundary integration.

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## Domain 4 summary

Domains 1 to 3 produced accountability, a justified choice and a working decision structure, and none of them produced a project. Integration does; and integration fails in a characteristic way: the parts are managed and the joins are not.

The charter authorises and bounds; its irreplaceable content is the statement of what the leader may decide alone, and a charter that has become a plan has lost it. The plan of plans is integrated by a short list of consistency checks that only appear when the plans are read against one another, including the one that makes a programme report last month's position every month for its whole life. Those checks are arithmetic, and running them on Meridian's eleven subsidiary plans found **USD 306,992** of unfunded cost and unscheduled time (**12.79 %** of the baseline, more than a full year of authorised drift) for an audit costing **USD 5,280**, three days before any work began. The checks that failed all spanned two owners; the one that passed had a single owner on both sides, which is the general shape of the finding. And the same audit exposed a horizon two weeks shorter than the programme's longest procurement lead time: a plan that could not support its own commitments, which no diligence inside the wave could have fixed.

The WBS obeys the hundred-per-cent rule, which is arithmetic and therefore auditable: Meridian's five level-2 elements sum to **2,332,000** against an approved

**2,400,000**, and the missing element is **clinician training and enabling change** at **214,000**: the same column Domain 2's benefits map omitted, leaving an honest baseline **6.1 %** above the approved figure and an NPV of **+1,186,898** that would still have been approved. The omission bought nothing.

Interfaces are where integration effort lives, and they grow combinatorially while components grow linearly: Meridian's 12 components admit **66** point-to-point interfaces costing **USD 1,188,000**, against **12** plus a layer at **USD 536,000**, a **54.9 %** saving, worth building while the layer costs below **972,000**, and decisive on the margin, where a thirteenth component costs **216,000** meshed and **18,000** layered. Case study A supplies the professional correction: the number to plan against is neither 66 nor 12 but the **31** genuinely required, of which **9** had no owner on the far side and accounted for the entire integration overrun.

Products and work verify each other, and the verification is arithmetic too, but only after each exception has been ruled on. Meridian's 148 products against 172 work packages produced **21** raw exceptions of which **14** were legitimate, a **66.67 %** false-positive rate, leaving **214,000** of omitted enabling change and **26,000** of work producing nothing: **USD 240,000**, exactly **10.00 %** of the baseline. Removing the orphans widened the unallocated gap from 68,000 to **94,000**, which is the counter-intuitive and correct direction.

The baseline is one three-dimensional statement, and its most informative invariant (that the time-phased cost baseline still matches the schedule) is the first abandoned under pressure, after which earned value measures the distance between two documents. How wrong is computable, and the shape of the answer explains the defect's durability: on Auriga's week-13 position, 240,000 of unre-phased planned value puts **SPI** anywhere between **0.8276** and **1.0435** against a reported **0.9231**, moves the composite **EAC** by **USD 287,083** in either direction at **USD 1.20** per USD of mis-phasing; and leaves **CPI** at **0.9057** and both CPI-only forecasts exactly correct. **A stale PV curve corrupts every schedule measure and leaves every cost-only measure right**, so the cost review passes and the defect survives.

Configuration management makes version-bound interface verification meaningful, and its audit classes are not equally serious: five items whose recorded version differs from what is deployed will fail an integration that a 12.94 % headline rate does not predict. Nor is the item rate the rate that matters. Integration quality is item quality raised to the number of parties, so Meridian's **87.06 %** of items gives **75.79 %** of two-sided interfaces and **65.98 %** of three-party ones, **7.50** of 31 interfaces expected to be unverifiable, and a 95 % interface target therefore requires **97.47 %** item conformance, at most **8** non-conforming items rather than the 44 found. **The tolerance on the register is derived from the target on the interfaces, backwards through the exponent**, and where the derived audit interval comes out at three weeks the finding is about how the register is written, not about how often it is audited.

Baselines are maintained by accumulation, never replacement, because the only question the funding organisation asked is performance against the original: the question Case study B's third baseline could not answer, and to which the answer turned out to be **CPI 0.87** rather than 0.99. Accumulation is what makes the domain's cheapest control possible: original plus approved changes equals current, exactly. Meridian's reconciliation showed a residual of **32,200** (1.34 %, dismissible), concealing a gross error of **USD 122,000**, since **77,100** of unreferenced movement and **44,900** of unposted approvals have opposite signs and hid **89,800** between them. **Reconcile gross, never net.** And the two controls are disjoint: summing the log catches drift inside it, the reconciliation catches what never entered it.

And change is integrated or it is not controlled. It begins with a classification, and Meridian got **10.53 %** of them wrong in both directions at once: **USD 34,000** of defects paid as changes (14.71 % of the year's direct drift, work that was never a change at all), and **USD 36,340** of changes ruled as clarifications, unbudgeted and unlogged. Gross **USD 70,340**; net **USD 2,340**. The same identity as the reconciliation, from unrelated data: **the net understates the gross by twice the smaller class.** The two errors also differ in detectability, and the one with no downstream detector is silent scope growth, which is where the tighter control belongs. Rejections are the other cheap entry nobody writes: **9** of Meridian's **14** left no trace, **4** returned, three costing a duplicated **USD 4,400** assessment and one costing **USD 63,000** because a different body approved what had already been refused, a total of **USD 80,600** against **USD 385** of writing, with a mean **4.58 times** the modal case, which is why the control is bought for the tail and both figures must be quoted.

Meridian's baseline moved **12.1 %** through 34 individually authorised changes averaging **0.28 %** each, with no decision anywhere on the total, and a 100,000-in-90-days cumulative rule would not have caught it because a quarter's changes aggregate to **57,800**, so a cumulative test must be derived from the observed change rate, not chosen for roundness. A change quoted at **40,000** truly cost **131,560**, **3.29 times** the quoted figure, of which the quote was **30.4 %**; and therefore the domain's single most valuable sentence, worth more than most process improvements a programme will make: **the delegation threshold reads on assessed total impact, not on quoted direct cost.**

PART TWO

# Delivering the work

Domains 5–10 — scope, planning and flow, cost and commercial, risk and resilience, quality and assurance, and procurement: the machinery of delivery.



DOMAIN

# 5

## Scope, Requirements and Value Definition

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## Why this domain exists

Part One produced a project. Domain 1 established what the leader is answerable for, Domain 2 whether the work was worth choosing and whether the promise was honest, Domain 3 who may decide and how quickly, and Domain 4 how the parts fit and how a change moves all three baseline dimensions. All four assumed something they never established: that the **content** of the scope was known.

It usually is not, and the gap is not the one practitioners expect. The familiar complaint is that requirements were incomplete. The more expensive and far commoner condition is that they were **complete, agreed, delivered, verified; and did not produce the value the business case promised**. Domain 4's WBS audit found a missing element worth USD 214,000, and what it found missing was clinician training and enabling change: not a forgotten cost line but a **forgotten requirement**, that the receiving organisation be able to use what it was given. The omission did not originate in a spreadsheet. It originated in a room where nobody asked the clinicians anything.

The domain's central claim follows: **scope is not a list of things to build; it is a chain of justified links from a benefit to an accepted deliverable, and the chain is only as good as its weakest link, which is almost never the build**. Four consequences organise the chapter. An unwritten boundary is decided later by whoever is most insistent: KA 5.1 prices one undrawn boundary at USD 143,000, then shows the correct answer was to *include* the disputed sites, a decision the boundary statement would have surfaced and its absence prevented. Requirement defects are not equally expensive; they cost four times more at each stage they survive, and KA 5.2 shows a USD 84,000 elicitation programme returning USD 610,800, with **a single avoided live-service defect paying for the whole of it**. Requirements do not carry equal value per unit of effort, and where capacity binds KA 5.3 shows the intuitive ranking method losing USD 44,000 a year (USD 262,737 of present value) to simple enumeration. And scope grows by accumulation with nobody approving anything, a different failure from Domain 4's authorised drift needing a different detector: KA 5.4 finds USD 220,920 of movement that **left no trace in the change log at all**, and the only control that catches it is a count.

The through-line: **the cheapest place to be wrong about scope is at the start, and the arithmetic of how much cheaper is the most under-used business case in project management**.

**Learning objectives.** After this domain a candidate can: write a scope statement that is decidable rather than descriptive, and distinguish deliverables from outcomes and from activity; assemble a scope baseline and state which artefacts constitute it; write exclusions and assumptions that settle boundaries in advance, and **price an undrawn boundary on both its cost and its forgone benefit, including the payback period**; select elicitation techniques by the kind of requirement being

sought and identify the stakeholder groups whose absence causes systematic requirement loss; **compute the correction cost of a requirement defect by the stage at which it is found, the saving from moving detection earlier, the return on an elicitation investment, and the schedule delay at which that investment stops paying**; specify a requirement so that it is unambiguous, testable, singular and traceable; **audit a traceability matrix forward and backward, report the defect rate by class rather than as a total, and say which class invalidates the record**; attribute benefit to requirement bundles and re-test a "must have" classification; **rank requirements by value per unit of constrained effort, and demonstrate where greedy ranking loses to enumeration and by how much**; distinguish controlled change from uncontrolled creep, **quantify creep by count reconciliation and price it at the cost of delay**; write acceptance criteria that are testable and **compute the breakeven dispute probability that justifies writing them**; distinguish verification from validation and price a conditional acceptance against the correction ladder; and govern AI-assisted requirements work without letting a model author, prioritise or accept scope.

**The master programme.** Meridian Care Records continues from Part One: the clinical-records rollout to **40 clinics**, approved cost **USD 2,400,000**, full-potential benefit **USD 979,200** a year, realistic benefit **USD 685,440** a year at 70 % adoption, cost of delay **USD 14,280 per week** (Domain 1), the ramped business case with NPV **+USD 1,332,898** (Domain 2, WE 2.2.2), steering latency  $E[\text{wait}] = M/2 + L = 4.0$  **weeks** (Domain 3), and the integrated baseline, interface architecture and authorised drift of **USD 291,176** (Domain 4). This domain works its requirements baseline of **480** approved requirements, its **425**-element design register, its Release 1 capacity of **20 development-weeks**, and the **USD 520,000** clinic-rollout element from Domain 4's WBS, whose per-clinic figure of **USD 13,000** prices the boundary question KA 5.1 opens with. From Domain 6, **Project Auriga** (the 25-week control-systems upgrade, BAC **USD 4,000,000**) takes over as the single-project illustration.

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## KNOWLEDGE AREA 5.1

### Scope definition and the scope baseline

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*Topics: 5.1.1 the scope statement · 5.1.2 the scope baseline · 5.1.3 exclusions, assumptions and the boundary.*

#### 5.1.1 The scope statement

**Definition.** The scope statement is the authorised description of **what the project will deliver and what it will not**, expressed in terms specific enough that a competent third party could judge whether a given item falls inside it. That last clause is the whole test, and it is the one most scope statements fail.

**Three registers, and only one of them is scope.** Practitioners habitually conflate them, and the conflation is where value definition goes wrong:

| REGISTER           | WHAT IT STATES                                         | EXAMPLE (MERIDIAN)                                                                      |
|--------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Outcome</b>     | The change in the world the organisation is paying for | Clinicians retrieve a complete patient history in under 20 seconds at any of 40 clinics |
| <b>Deliverable</b> | The thing that will be handed over and accepted        | A records application, 31 verified interfaces, migrated data, a trained user population |
| <b>Activity</b>    | The work that produces the deliverable                 | Build, configure, test, migrate, train, cut over                                        |

Scope is the **deliverable** register, bounded by the outcome register and decomposed into the activity register (Domain 4's WBS). A scope statement written in the activity register, "implement a records system", cannot be tested for completeness, because activities have no boundary of their own. A scope statement written purely in the outcome register cannot be accepted, because an outcome depends on behaviour the project does not control. The discipline is to write the deliverables, state the outcome each serves, and keep the two columns visibly distinct.

**What makes a scope statement decidable.** Four properties, each of which converts an argument into a lookup: **enumerated** deliverables (a countable list, not a category); **quantified** extent (40 clinics, not "the region"); **named** interfaces and dependencies, by reference to Domain 4's register; and **explicit exclusions**, which are the subject of 5.1.3 and the half most often omitted. A scope statement with all four is short. Length in a scope statement is almost always a symptom of the activity register having crept in.

### 5.1.2 The scope baseline

**What it is.** The scope baseline is the **approved** scope statement together with the WBS and the WBS dictionary: three artefacts, one authorised statement, and one of the three dimensions of the performance measurement baseline Domain 4, KA 4.3.1 insists is single. It changes only through integrated change control (Domain 4, KA 4.4), and every change to it is a change to schedule and cost by construction.

**The WBS dictionary is where scope actually lives.** The scope statement gives the boundary and the WBS gives the structure; the dictionary gives, per work package, the description of work, the deliverable, the acceptance basis, the responsible owner, the estimate and the dependencies. It is the document that answers "is this in scope?" at the level at which the question is actually asked, and it is the document most often left as a set of names. A WBS whose dictionary is a list of element titles has a structure and no content, and the deficit becomes visible at acceptance, when the acceptance basis has to be invented by whoever is in the room.

**The four scope-baseline defects**, in increasing order of cost to repair: *unbounded* – a category rather than a count, so extent is settled later by negotiation (5.1.3 prices one); *undocumented at dictionary level*, so acceptance criteria are improvised (KA 5.4.2 prices it); *untraceable to benefit* – deliverables no benefit requires, benefits no deliverable serves (KA 5.3.1); and *unversioned*, so nobody can say what was agreed, which is Domain 4's configuration failure arriving through the scope door.

### 5.1.3 Exclusions, assumptions and the boundary

**Why exclusions are the highest-return sentences in the document.** An exclusion costs a line to write and settles a dispute that would otherwise be resolved under time pressure, by whoever is most insistent, in month fourteen. It is Domain 3's decidability test applied to content rather than to authority: *if this question arises, who answers it, and from what?* Where the answer is "from the scope statement", the boundary holds. Where it is "from a conversation", the boundary is wherever the conversation ends.

**Assumptions are exclusions with a deadline.** An assumption states something the project has taken as true without controlling it: that the identity provider will be available, that clinical data quality in the legacy system meets a stated standard, that a named team is released on a stated date. Each carries an owner and a date by which it becomes a fact or a risk (Domain 8), because an assumption with neither is a hope in a register.

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#### WORKED EXAMPLE

**Worked example 5.1.3 – the boundary nobody drew, and the answer that was not "no".**

1. **Setup.** Meridian's approved scope statement says the records application will be rolled out to "all clinics in the region". The approved count in the business case is **40** clinics. The regional service register, however, also lists **6** branch surgeries, **3** out-of-hours hubs and **2** prison health units – **11** further sites, each of which has been told by its own management that it is included. From Domain 4's WBS the clinic-rollout element is **USD 520,000** across the 40 clinics. Full-potential benefit is **USD 979,200** a year across the 40 clinics; realistic adoption is **70 %** (Domain 1).
2. **Formula.** Per-site rollout cost = rollout element ÷ approved site count. Boundary exposure = additional sites × per-site cost. Per-site benefit at realistic adoption = (full potential ÷ site count) × adoption. Payback period = exposure ÷ additional annual benefit.
3. **Substitution.**  $520,000 / 40; 11 \times 13,000; (979,200 / 40) \times 0.70; 11 \times 17,136; 143,000 / 188,496.$

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4. **Result.** Per-site rollout cost **USD 13,000**. The undrawn boundary carries an exposure of **USD 143,000: 5.96 %** of the approved baseline. The same 11 sites would generate  $11 \times 17,136 = \text{USD } 188,496$  a year at realistic adoption, against a full-potential per-site figure of **USD 24,480**. Payback on including them is **0.76 years, about 9.1 months**.
5. **Interpretation.** The instinct on seeing a 143,000 boundary exposure is to write the exclusion, and that instinct is wrong here, which is exactly why the example is worth working. Eleven sites costing 143,000 once and returning 188,496 a year pay for themselves in **nine months**; excluding them destroys value, and including them by accident in month fourteen destroys almost as much, because by then the rollout waves are scheduled, the training capacity is committed and the 11 sites arrive as a disruption rather than as a plan. The value of the boundary statement is therefore not that it says no. **It is that it forces the question to be asked while the answer is still cheap to act on**, which is the same argument Domain 2 made about options and the same one Domain 4 made about assessing before deciding. Three cautions before this arithmetic is reused. The 13,000 per-site figure is the *rollout element only*: it excludes licence cost, clinical safety assessment and the training element Domain 4 found missing at USD 214,000, so the true marginal cost per site is higher and the payback longer: a leader quoting nine months must say which cost basis it rests on. The 17,136 per-site benefit assumes the new sites adopt at the same 70 % as the original 40, which is a genuine assumption and not a fact: prison health units in particular have a different clinical workflow, and a lower adoption assumption for them would be defensible. And the decision is a **change** under Domain 4, KA 4.4 whichever way it goes: an 11-site extension is not a clarification, and recording it as one is precisely the misclassification KA 5.4.1 quantifies.
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## AI in this KA

**Where it earns its place.** Reporting every unbounded extent in a scope statement: every category noun without a count, every "all", "as required" and "including but not limited to". It is mechanical, exhaustive, and exactly the reading humans do not perform on a document they have already approved. Cross-checking the deliverable list against an organisational asset or site register to surface candidate boundary questions, which is how Meridian's 11 sites would have appeared in week three rather than month fourteen. Drafting WBS dictionary entries from an approved structure and existing estimates, for owner completion. Checking that every assumption carries an owner and a resolution date.

**Where it must not go.** Deciding a boundary. The 11-site question turns on clinical workflow, regional politics, adoption expectations and a benefit judgement, and it is a change decision belonging to a named authority. Nor should a model author the exclusion list, because an exclusion is a commitment about what will *not* be delivered and someone must be accountable for having made it.

**Verification, concretely.** Every flagged extent is confirmed against the source document, because the flag rate is high and the false-positive rate is not zero. Every candidate boundary question is put to the sponsor as a question, never resolved as an inference. The per-site arithmetic is reproduced by hand (one division) with its cost basis stated. And any AI-drafted dictionary entry is endorsed by the work-package owner before it enters the baseline, since the dictionary is the acceptance basis.

## KEY TERMS — KA 5.1

| TERM                                              | MEANING                                                                                                                      |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope statement</b>                            | The authorised description of what will and will not be delivered, specific enough to be tested against.                     |
| <b>Outcome / deliverable / activity registers</b> | The three levels routinely conflated; scope is the deliverable register, bounded by outcomes and decomposed into activities. |
| <b>Decidable scope</b>                            | Enumerated deliverables, quantified extent, named interfaces, explicit exclusions.                                           |
| <b>Scope baseline</b>                             | The approved scope statement + WBS + WBS dictionary, one of Domain 4's three baseline dimensions.                            |
| <b>WBS dictionary</b>                             | Per work package: work, deliverable, acceptance basis, owner, estimate, dependencies, where scope actually lives.            |
| <b>Exclusion</b>                                  | A written statement of what is outside the boundary; the highest-return line in the document.                                |
| <b>Assumption</b>                                 | Something taken as true without being controlled, carrying an owner and a resolution date.                                   |
| <b>Boundary exposure</b>                          | The priced consequence of an unstated extent: additional units × unit cost.                                                  |

## SAMPLE MCQS — KA 5.1

**MCQ 5.1-A** [5.1.1 · Comprehension] A scope statement reads "implement a regional records system". Its principal defect is that:

- A. it is too short
- B. it is written in the activity register, so it has no boundary that can be tested

CORRECT

- C. it does not name the sponsor
- D. it omits the schedule

*Rationale:* Activities have no boundary of their own, so completeness cannot be assessed (5.1.1). Length is not the defect, decidability is; C and D belong to the charter and the schedule.

**MCQ 5.1-B** [5.1.3 · Application] A rollout element of USD 520,000 covers 40 approved sites. Eleven further sites believe they are included. The boundary exposure is:

- A. USD 13,000
- B. USD 143,000 CORRECT
- C. USD 60,000
- D. USD 188,496

*Rationale:*  $520,000/40 = 13,000$  per site,  $\times 11 = 143,000$  (5.1.3). A is the cost of one site, not of eleven; C divides the whole 2,400,000 baseline by 40 rather than the rollout element, giving a per-site 60,000: the wrong cost base, and a per-site figure rather than an exposure; D is the annual *benefit* of the 11 sites, not their cost.

**MCQ 5.1-C** [5.1.3 · Evaluation] The 11 sites would cost USD 143,000 to include and generate USD 188,496 a year at realistic adoption. The correct professional conclusion is that:

- A. they should be excluded, because the exposure exceeds the tolerance
- B. the boundary statement's value is that it forces the decision while it is still cheap to act on — and here the decision is probably to include them CORRECT
- C. they are in scope already, since the statement says "all clinics in the region"
- D. the exposure should be added to contingency

*Rationale:* A nine-month payback makes exclusion value-destroying (5.1.3); the boundary statement's function is to surface the question in time, not to say no. C is the reading that causes the dispute; D funds an unmade decision.

**MCQ 5.1-D** [5.1.2 · Analysis] A WBS dictionary consists only of element titles. The defect becomes visible:

- A. immediately, at baseline approval
- B. at acceptance, when the acceptance basis has to be invented by whoever is present CORRECT
- C. during estimating
- D. only if the scope changes

*Rationale:* The dictionary carries the acceptance basis; its absence is invisible until acceptance is attempted (5.1.2). That is what makes it an expensive defect rather than a documentation nicety.

SELF-CHECK — KA 5.1

1. Which register is scope, and what happens when the other two are used instead? — The deliverable register; the activity register removes any testable boundary, and



the outcome register cannot be accepted because it depends on behaviour the project does not control.

2. *Why is an exclusion the highest-return line in a scope statement?* — It costs a line and settles a dispute that would otherwise be resolved late, under pressure, by whoever is most insistent.
3. *What is the point of pricing a boundary if the answer turns out to be "include them"?* — The point is that the question gets asked while the answer is still cheap to act on; a nine-month payback accepted in week three is a plan, and the same decision in month fourteen is a disruption.

## KNOWLEDGE AREA 5.2

# Requirements elicitation, analysis and traceability

*Topics: 5.2.1 elicitation and its economics · 5.2.2 analysis and specification quality · 5.2.3 traceability.*

## 5.2.1 Elicitation and its economics

**Definition.** Elicitation is the deliberate discovery of what stakeholders need, as distinct from collection of what they ask for. The distinction is the discipline: a stated request is a proposed solution to an unstated need, and a requirements process that records requests has captured other people's designs without capturing the problem.

**Techniques, matched to what is being sought.** Each technique is good at one thing and blind to another, which is why a single-technique programme fails predictably:

| TECHNIQUE                   | GOOD AT                                           | BLIND TO                                                      |
|-----------------------------|---------------------------------------------------|---------------------------------------------------------------|
| Structured interview        | Depth, rationale, the <i>why</i> behind a request | Group disagreement; what the individual does not know they do |
| Facilitated workshop        | Conflict surfacing, shared prioritisation         | The quiet stakeholder and the absent one                      |
| Observation of work as done | The gap between documented and actual process     | Rare cases, exceptions, and the volume-weighted picture       |
| Document and data analysis  | Volumes, exception rates, statutory obligations   | Intent, and anything not already recorded                     |
| Prototype or walkthrough    | Tacit needs made visible by reaction              | Non-functional requirements (load, availability, security)    |

| TECHNIQUE                               | GOOD AT                   | BLIND TO                      |
|-----------------------------------------|---------------------------|-------------------------------|
| Existing-system defect and support logs | What actually fails today | What has never been attempted |

**The stakeholder groups whose absence causes systematic loss.** Requirement loss is not random; it clusters on groups that are structurally easy to omit; the people who will *operate* the service after handover, the people who will *support* it, the *external* users who never attend internal meetings, and whoever owns the **enabling change** that converts an output into an outcome. Domain 4 found the last of these missing as a USD 214,000 WBS element; Case study B of this domain finds the third of them missing at a cost of most of a programme's benefit.

**The economics, which is the part usually asserted rather than computed.** It is a commonplace that finding a requirement defect early is cheaper. The professional question is *how much cheaper, and therefore how much elicitation is worth buying*, and that is arithmetic.

#### WORKED EXAMPLE

##### Worked example 5.2.1 — the correction ladder, and what elicitation is worth.

- Setup.** Meridian's own change and defect records give the average cost of correcting one requirement defect by the stage at which it is found: **USD 400** at definition, **1,600** at design, **6,400** at build, **25,600** at test and integration, **102,400** in live service, a ladder rising by a factor of **four** at each stage. Projects of this type in the organisation surface about **96** requirement defects across a 480-requirement baseline (**0.2** per requirement), on this profile: **24** at definition, **22** at design, **26** at build, **18** at test, **6** in live service. A structured elicitation programme (clinician workshops across the 40 clinics, observation of records use as actually performed, prototype walkthroughs) costs **USD 84,000** (**USD 175** per requirement) and is expected to move **13** build-stage, **9** test-stage and **3** live-service defects into the definition stage.
- Formula.** Expected correction cost =  $\Sigma$  (defects at stage  $\times$  cost at stage). Saving from earlier detection =  $\Sigma$  (defects moved  $\times$  [cost at old stage – cost at definition]): the **difference**, not the old cost, because the defect still costs 400 to correct at definition. Return = saving  $\div$  investment. Breakeven delay = (saving – investment)  $\div$  cost of delay.
- Substitution.**  $24 \times 400 + 22 \times 1,600 + 26 \times 6,400 + 18 \times 25,600 + 6 \times 102,400$ ; then  $13 \times 6,000 + 9 \times 25,200 + 3 \times 102,000$ ; then  $610,800 / 84,000$ ; then  $(610,800 - 84,000) / 14,280$ .
- Result.** The unimproved expected correction cost is **USD 1,286,400**: an average of **USD 13,400** per defect. The elicitation programme saves  $78,000 + 226,800 + 306,000 = \text{USD } 610,800$ , for a net of **USD 526,800** and a return of **7.27**

**times** the investment. The expected correction cost falls to **USD 675,600**, or **USD 7,037.50** per defect. And the investment breaks even at a definition-phase extension of **36.89 weeks**.

5. **Interpretation.** Four things here matter more than the headline return. First, the single most useful sentence a leader can take from it: **moving one defect out of live service saves USD 102,000, which by itself pays for the entire USD 84,000 programme**. Nothing about a 7.27-times return is needed to authorise the spend; one defect is. Second, the *breakeven delay* answers the objection this investment always attracts. That elicitation delays the start. Elicitation would have to extend the definition phase by **more than 36.89 weeks** before it destroyed value, and a realistic four-week extension costs  $4 \times 14,280 = \text{USD } 57,120$ , leaving **USD 469,680** net. The objection is real, and the breakeven is **more than nine times** the extension anyone is proposing, an order of magnitude short of decisive; a leader who cannot say that has to argue about it instead. Third, the saving is computed on the **difference** between stages, never on the later cost alone: the gross figures give  $13 \times 6,400 + 9 \times 25,600 + 3 \times 102,400 = \text{620,800}$ , overstating the saving by 10,000 by forgetting that the defect still has to be fixed. Fourth, two real limits rather than hedges. The ladder is **this organisation's observed ladder**, not a law of nature: technology, regulatory regime and release cadence all change the ratio, and a programme that has not measured its own should say it is using an assumed one. And the arithmetic assumes the moved defects were **detectable at definition**; emergent integration behaviour, load characteristics that appear only at volume and clinical workflow effects that no prototype predicts are genuinely not, which is why the example moves half of the build and test defects rather than all of them. The honest claim is a large return on a *subset*, not the elimination of late defects.

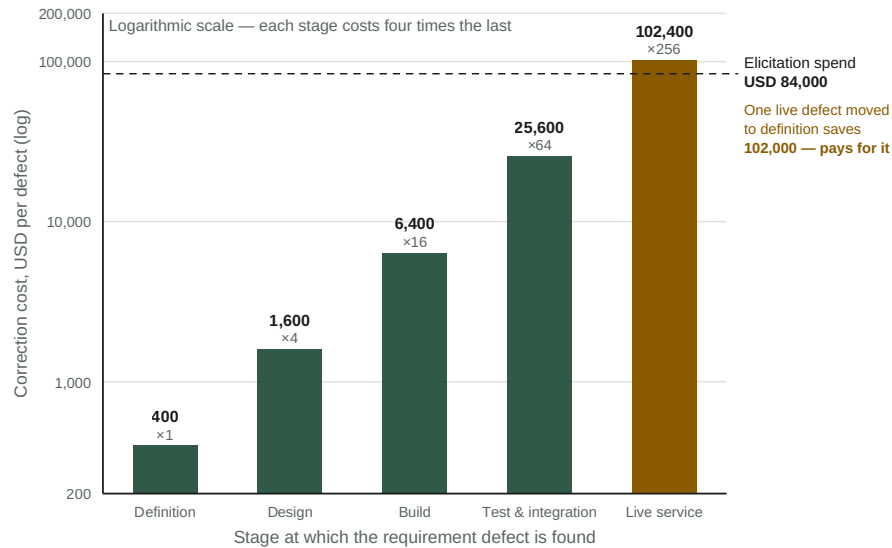


Fig 5.2.1 — What a requirement defect costs by the stage it is found

### 5.2.2 Analysis and specification quality

**From elicited need to specified requirement.** Analysis is the work of turning what stakeholders said into statements that can be designed against, built, tested and accepted. Six properties define a usable requirement, and each has a specific failure:

- **Necessary:** it traces to a benefit or an obligation. A requirement traceable to neither is someone's preference, and it will be built.
- **Unambiguous:** one reading only. The standard offenders are comparatives with no referent ("faster", "user-friendly"), passive constructions that hide the actor, and the conjunction "and/or".
- **Singular:** one requirement per statement. A compound requirement is accepted in part and disputed in part, and its status is permanently "in progress".
- **Testable:** a stated condition and a stated observable result. Untestable requirements are the subject of KA 5.4.2's arithmetic.
- **Feasible:** achievable within the architecture, the budget and the law, which requires a technical reader in the analysis loop.
- **Traceable:** identified, and linked forwards and backwards (5.2.3).

**Requirement classes, and why they must be separated.** *Functional* (what the system does), *non-functional* (how well — performance, availability, security, accessibility, usability), *constraint* (an imposed platform, standard or deadline), *data* (content, quality, retention), *transition* (migration, cutover, training, decommissioning) and *regulatory* (not negotiable and not prioritisable). They fail in different ways, which is why 5.2.3 reports them separately; and the two that go missing most reliably are **non-functional** requirements (because no user asks for availability),

and **transition** requirements, which are Domain 4's missing 214,000 in requirement form.

**The two analysis pathologies.** *Solution capture*: recording the stakeholder's proposed solution as the requirement, which locks the design before the problem is understood and forfeits every cheaper option; the test is whether the statement expresses a need several designs could satisfy. *Requirement inflation*: elaborating each need into derived statements until the count impresses and the meaning disperses. A 480-requirement baseline for a records rollout is traceable; the same content expressed as 3,000 is traceable by nobody, and the count itself then obstructs prioritisation (KA 5.3).

### 5.2.3 Traceability

**Definition.** Traceability is the maintained, bidirectional linkage from **benefit** → **requirement** → **design element** → **test case** → **accepted deliverable**, such that any link can be followed in either direction. Its purpose is not documentation. It is to make four questions answerable by lookup rather than by opinion: what will happen to the benefit if this requirement is dropped; what must be re-tested if this design element changes; is there anything being built that nobody asked for; and can this deliverable be accepted.

**Forward and backward are different tests.** *Forward* traceability asks whether every requirement reaches a design element, a test case and an accepted deliverable: it detects **unmet scope**. *Backward* traceability asks whether every design element and every test case traces to an approved requirement: it detects **unrequested scope**, which is the mechanism of gold-plating and one of the routes by which creep enters (KA 5.4.1). Programmes overwhelmingly run the forward test only, and the reverse test is the cheaper of the two.

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#### WORKED EXAMPLE

#### Worked example 5.2.3 – auditing Meridian's traceability matrix.

1. **Setup.** At entry to integration testing, Meridian's traceability audit covers the **480** approved requirements and the **425**-element design register. Forward: **416** requirements trace to a design element, a test case and a deliverable; **34** have no design element at all; **21** have a design element but no test case; **9** are recorded as accepted with no evidence of any verification having been performed. Backward: of the 425 design elements, **17** trace to no approved requirement.
2. **Formula.** Forward defect rate = non-conforming requirements ÷ approved requirements, reported **by class**. Reverse defect rate = orphaned design elements ÷ design register: a **different denominator**, and therefore not combinable with the forward rate.
3. **Substitution.**  $(34 + 21 + 9) / 480$ ; and separately  $34/480$ ,  $21/480$ ,  $9/480$ ; then  $17/425$ .

- 
4. **Result. 64** forward non-conformances, a **13.33 %** forward defect rate, comprising **7.08 %** with no design, **4.38 %** with no test and **1.88 %** accepted without verification. **86.67 %** of requirements are fully traced. Separately, **17** orphaned design elements: a **4.00 %** reverse defect rate.
  5. **Interpretation.** The classes are not totalled, for the reason Domain 4, KA 4.3.2 refused to total its configuration-audit classes: **they fail differently, and the total is the least informative number available.** The **34 with no design** are unmet scope that is still cheap, they surface at design review at 1,600 each on the ladder of 5.2.1, and the honest reading is that baseline and design are 34 requirements out of step. The **21 with no test case** will be delivered, accepted on assertion and found defective in live service at **102,400** each. The **9 accepted without verification evidence** are the serious finding, for a reason that has nothing to do with their number: the acceptance record states something untrue, so **every decision that relied on it (including the release decision) rested on nothing**, the same structural failure as Domain 4's five mis-recorded configuration items. A programme reporting "13.33 % traceability defects" has buried that. Two further observations. The rounded class rates sum to 13.34 % against a combined 13.33 %, display rounding, and one more reason not to present a total. And the **17 orphans** are design and build effort spent on work nobody approved, worth  $17 \times 6,400 = \text{USD } 108,800$  at build-stage correction cost if they must be removed: invisible to the forward test most programmes run alone, and the reverse test costs one query.
- 

### AI in this KA

**Where it earns its place.** Traceability auditing is close to ideal AI work: exhaustive matrix traversal with definite answers, done badly and partially by humans, every finding checkable against a source — both directions, by class, with the orphan list produced as a matter of course. Ambiguity detection in requirement text (comparatives without referents, passive constructions hiding the actor, statements joined by "and", "and/or" and "as appropriate") is linguistic pattern matching with a high true-positive rate, and is the highest-value AI application in the domain. Clustering interview notes and support-log entries into candidate themes, and flagging near-duplicates in a large baseline. Proposing candidate test conditions for analyst and tester ruling. And checking that every requirement carries a class and that the non-functional and transition classes exist at all: the absence of a whole class is a finding a model notices and a human does not.

**Where it must not go.** Authoring requirements. A requirement is a commitment made on behalf of stakeholders who exist, and a model generating plausible requirements from a project description produces exactly the failure this KA exists to prevent: a baseline that reads well and represents nobody. Ruling that a requirement is unnecessary or duplicated. Closing a traceability gap by creating the missing link, which converts a finding into a fabrication and is the most tempting automation

failure here. And no AI output may serve as verification evidence for the nine requirements of WE 5.2.3; that is the class the audit exists to catch.

**Verification, concretely.** Every reported gap is confirmed against both artefacts before it becomes a finding, and the orphan list is confirmed with the design owner, who can usually name the conversation the element came from. Ambiguity flags are triaged by the analyst, since some flagged constructions are correct in context. Clustered themes go back to the stakeholders who produced them, never straight into the baseline. The ladder arithmetic is reproduced by hand and its provenance stated as observed or assumed. And every AI contribution is recorded as an input, with the human author named.

#### KEY TERMS — KA 5.2

| TERM                               | MEANING                                                                                                                            |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Elicitation</b>                 | Deliberate discovery of need, as distinct from collection of requests.                                                             |
| <b>Solution capture</b>            | Recording a stakeholder's proposed solution as the requirement, locking the design before the problem is understood.               |
| <b>Correction ladder</b>           | The cost of correcting one requirement defect by the stage found; Meridian's observed ladder rises $\times 4$ per stage.           |
| <b>Breakeven elicitation spend</b> | The investment at which the saving from earlier detection is exactly recovered; expressed either as a sum or as a tolerable delay. |
| <b>Requirement classes</b>         | Functional · non-functional · constraint · data · transition · regulatory: reported separately because they fail differently.      |
| <b>Requirement inflation</b>       | Elaborating need into a count that impresses and disperses meaning, defeating traceability and prioritisation.                     |
| <b>Forward traceability</b>        | Requirement $\rightarrow$ design $\rightarrow$ test $\rightarrow$ accepted deliverable; detects unmet scope.                       |
| <b>Backward traceability</b>       | Design or test $\rightarrow$ approved requirement; detects unrequested scope, and costs one query.                                 |
| <b>Orphan</b>                      | A design element or test case tracing to no approved requirement.                                                                  |

#### SAMPLE MCQS — KA 5.2

**MCQ 5.2-A** [5.2.1 · Application] On a ladder of 400 / 1,600 / 6,400 / 25,600 / 102,400 per defect by stage, an elicitation programme moves 13 build-stage, 9 test-stage and 3 live-service defects to the definition stage. The saving is:

- A. USD 620,800
- B. USD 610,800 CORRECT
- C. USD 526,800

D. USD 1,286,400

*Rationale:* The saving is the **difference** at each stage:  $13 \times 6,000 + 9 \times 25,200 + 3 \times 102,000 = 610,800$  (5.2.1). A uses the gross later-stage costs and forgets the defect still costs 400 to fix at definition; C is the saving net of the 84,000 investment; D is the unimproved total correction cost.

**MCQ 5.2-B** [5.2.1 · Evaluation] The strongest single argument for the USD 84,000 elicitation programme is that:

- A. it returns 7.27 times its cost
- B. one defect moved out of live service saves USD 102,000 and pays for the whole programme CORRECT
- C. average correction cost falls from 13,400 to 7,037.50 per defect
- D. early detection is recognised good practice

*Rationale:* The single-defect argument is decisive because it needs no assumption about how many defects move (5.2.1); A and C are true but depend on the whole estimated shift, and D is an appeal to practice rather than a computation.

**MCQ 5.2-C** [5.2.3 · Analysis] In a 480-requirement audit, 34 have no design element, 21 no test case, and 9 are recorded as accepted with no verification evidence. Which finding most undermines the **release decision itself**, and why?

- A. the 34, because unmet scope is the largest group
- B. the 21, because untested requirements reach live service
- C. the 9, because the acceptance record states something untrue, so every decision that relied on it rested on nothing CORRECT
- D. all three equally, at a combined 13.33 % defect rate

*Rationale:* A false record invalidates the decisions taken on it, including the release decision (5.2.3) – the same structural failure as Domain 4's mis-recorded configuration items. A ranks by group size and B by future correction cost; both are real findings and both leave the release record truthful, which is why neither is the answer to *this* question. D is the reading the class breakdown exists to prevent.

**MCQ 5.2-D** [5.2.3 · Comprehension] Seventeen design elements trace to no approved requirement out of a 425-element register. This is:

- A. a forward traceability defect of 3.54 %
- B. a reverse traceability defect of 4.00 %, on a different denominator from the forward rate CORRECT
- C. immaterial, since the requirements are all still met
- D. evidence that 17 requirements were lost

*Rationale:*  $17/425 = 4.00$  %, and the design register is not the requirements baseline, so the two rates are not combinable (5.2.3). A puts the 17 over the 480-



requirement baseline ( $17/480 = 3.54\%$ ) and then mislabels a reverse finding as a forward one, the wrong denominator and the wrong test. C ignores unrequested effort worth 108,800 at build-stage cost. D inverts the finding: an orphan is work nobody asked for, not a requirement that went missing.

**MCQ 5.2-E** [5.2.2 · Evaluation] A requirement reads "the system shall provide faster record retrieval and improved reporting". Its defects are:

- A. it is untestable only
- B. it is ambiguous ("faster", "improved" — no referent) and compound (two requirements in one) CORRECT
- C. it is infeasible
- D. it is untraceable

*Rationale:* Comparatives without a referent make it ambiguous and therefore untestable, and the conjunction makes it two requirements that will be accepted in part and disputed in part (5.2.2).

#### SELF-CHECK — KA 5.2

1. *Why must the elicitation saving be computed on the difference between stages?* — Because the defect still costs 400 to correct at definition; using gross later-stage costs overstates the saving, here by 10,000.
2. *Which two requirement classes go missing most reliably, and why?* — Non-functional, because no user asks for availability; and transition, because migration, cutover and training belong to somebody else.
3. *What does backward traceability detect that forward traceability cannot?* — Unrequested scope: 17 design elements nobody approved, worth 108,800 at build-stage correction cost, invisible to the forward test most programmes run.

#### KNOWLEDGE AREA 5.3

## Value definition and prioritisation

*Topics:* 5.3.1 *attributing value to requirements* · 5.3.2 *prioritisation methods and their failure modes* · 5.3.3 *capacity-constrained selection*.

### 5.3.1 Attributing value to requirements

**The link that closes the chain.** Domain 2 built the benefits map from strategic driver to benefit to measure and owner. This KA connects the other end: from benefit to the **requirement bundle** that delivers it. Without that link, prioritisation has nothing to prioritise on and the benefits register has nothing to hold anyone to.

**Bundles, not individual requirements.** Value is rarely attributable to a single requirement, because benefits arise from usable capability and a capability needs several requirements to exist at all. The working unit is therefore the **requirement bundle**: the smallest set of requirements that together delivers a measurable benefit. Attributing value at the individual requirement level produces spuriously precise numbers and, worse, invites the delivery of half a capability, which returns nothing, at full cost.

**Two tests that make attribution honest.** The *sum test*: attributed benefit across all bundles must not exceed the business case's benefit, and where it falls short the balance must be explained rather than distributed. Meridian's five Release 1 candidate bundles carry **USD 957,000** of the **USD 979,200** full-potential annual benefit, and the remaining **USD 22,200** sits with minor requirements that support no bundle on their own, a statement, not a rounding. The *counterfactual test*: for each bundle, what happens to the benefit if it is not delivered? A bundle whose absence costs nothing has no value attributed to it, however popular it is, and a bundle whose absence collapses several other bundles' benefits is a dependency rather than a candidate.

**Where attribution is legitimately impossible.** Regulatory requirements, safety requirements and requirements imposed by contract are not prioritisable and should be removed from the value ranking entirely, held instead as constraints on capacity. Attempting to score them produces the absurdity of a statutory obligation ranking eleventh, and the process then has to override itself, which discredits the whole ranking.

### 5.3.2 Prioritisation methods and their failure modes

**The methods, and what each is actually for.** *Ordered ranking* (a strict sequence) is honest and does not scale past a few dozen items. *Category schemes* (must / should / could / will not, or an equivalent) scale and are the standard instrument in adaptive delivery (Domain 13), but they degrade in a specific way described below. *Weighted scoring* against explicit criteria is the Domain 2, KA 2.2.3 instrument applied at requirement level: it forces criteria into the open and remains an ordinal judgement, so its arithmetic must not be over-read. *Value per unit of effort* is the method that actually engages the constraint, and 5.3.3 shows both its power and its limit.

**Must-have inflation, quantified.** Category schemes fail because the categorisation is done by the requirement's originator, and every originator's requirement is a must-have. Of Meridian's **480** requirements, **374: 77.92 %** were marked "must have" on first pass. A structured re-test, in which each originator was asked to **name the consequence of releasing without it**, left **148: 30.83 %** standing as genuine must-haves. The discretionary pool that prioritisation can actually act on therefore grows from **106** requirements (**22.08 %**) to **332 (69.17 %)**, a **3.13-fold** increase in the scope available to trade. That number is the whole argument for the re-test: at 77.92 %

must-have the classification carries almost no information and prioritisation is theatre, because four requirements in five are outside its reach.

**Two further failure modes worth naming.** *Prioritising by cost rather than by value per unit of cost*, which systematically favours cheap low-value work and is how a release ships twelve small conveniences and no capability. And *prioritising without the dependency map*, which produces sequences that cannot be built: the subject of the caution in 5.3.3 and of Domain 6's logic networks.

### 5.3.3 Capacity-constrained selection

**The constraint decides the method.** Domain 2, KA 2.2.3 established the principle at portfolio level and it holds unchanged here: **greedy ranking by value per unit of the scarce resource is a heuristic, not an optimum, and lumpy candidates require enumerating the feasible sets.** What follows is that result worked at requirement level, where the effect is larger because requirement bundles are lumpier than projects and the capacity window is shorter.

#### WORKED EXAMPLE

#### Worked example 5.3.3 – where the ranking loses to the enumeration.

1. **Setup.** Meridian's Release 1 has **20 development-weeks** of capacity before the first clinic wave must begin, a hard constraint set by the clinical training calendar, not by money. Five candidate requirement bundles, with their attributed annual benefit and their effort:

| BUNDLE | REQUIREMENT BUNDLE                          | EFFORT<br>(DEV-<br>WEEKS) | ANNUAL<br>BENEFIT<br>(USD) | VALUE<br>PER<br>WEEK |
|--------|---------------------------------------------|---------------------------|----------------------------|----------------------|
| A      | Unified patient record and clinical search  | 13                        | 299,000                    | <b>23,000</b>        |
| B      | e-Prescribing with decision support         | 10                        | 228,000                    | <b>22,800</b>        |
| C      | Appointments, referrals and clinic workflow | 10                        | 225,000                    | <b>22,500</b>        |
| D      | Automated national reporting                | 5                         | 110,000                    | <b>22,000</b>        |
| E      | Analytics dashboards                        | 5                         | 95,000                     | <b>19,000</b>        |

Total effort **43** weeks against 20 available; total attributed benefit **957,000** of the 979,200 full potential. 2. **Formula.** Greedy: rank by value per week, take each bundle that fits the remaining capacity. Enumeration: evaluate every subset whose total effort  $\leq$  capacity and take the highest total value. Then value the difference:  $\text{annual difference} \times \text{AF}(r, n)$ , with  $\text{AF}(0.07, 8) = 5.971299$

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from Domain 2. 3. **Substitution.** Greedy takes **A** (23,000/week, 13 weeks, 7 remaining), skips B and C (10 weeks each, do not fit), takes **D** (5 weeks, 2 remaining), and cannot fit E. Enumeration compares every feasible subset, of which the material ones are  $A+D = 18 \text{ weeks} / 409,000$ ,  $A+E = 18 / 394,000$ ,  $B+D+E = 20 / 433,000$ ,  $C+D+E = 20 / 430,000$  and  $B+C = 20 / 453,000$ . 4. **Result.** Greedy selects **A + D: USD 409,000** a year, using **18 of 20 weeks — 90.0 %** capacity utilisation, with **2 weeks idle and no bundle small enough to use them**. Enumeration selects **B + C: USD 453,000** a year, using **20 of 20 weeks, 100 %** utilisation. The difference is **USD 44,000** a year, **10.76 %** more benefit, worth  $44,000 \times 5.971299 = \text{USD } 262,737$  of present value over the eight-year appraisal period at 7 %. 5. **Interpretation.** The mechanism generalises to every constrained selection a leader will make, so it is worth understanding rather than memorising. Greedy fails **because the highest-ratio bundle is lumpy**: A leads on value per week by only 200 (23,000 against B's 22,800, a **0.88 %** advantage), and taking it strands two weeks that nothing can use. A 0.88 % ratio advantage bought a 10.76 % value loss, and the closer the ratios, the likelier that is. The rule follows directly: **where the constraint binds and the candidate set is small enough to enumerate, enumerate; otherwise rank greedily and then test whether the residual capacity can be filled, because stranded capacity is the signature of the failure**. Three cautions, the first severe. The enumeration assumes the bundles are **independent**; if C's clinic-workflow requirements depend on A's unified record then  $B+C$  is infeasible, so the dependency map (Domain 6, KA 6.1) is a **precondition** for this arithmetic, not a refinement of it, and without it the enumeration yields a confident answer that cannot be built. Second, a 44,000 difference between figures of 409,000 and 453,000 sits well inside the uncertainty of any benefit estimate, so the defensible claim is not "B+C is worth 44,000 more" but "the ranking method systematically strands capacity and the enumeration does not". The structural argument survives uncertainty that the point figure does not. Third, deferring A has a cost the table cannot show: the unified record is the bundle clinicians associate with the programme's purpose, and adoption is behavioural (Domain 2's 70 %). A sponsor may defensibly overrule the arithmetic on adoption grounds, and should then record that they are buying adoption confidence for 44,000 a year, which is a decision, whereas an unexamined greedy ranking is not.

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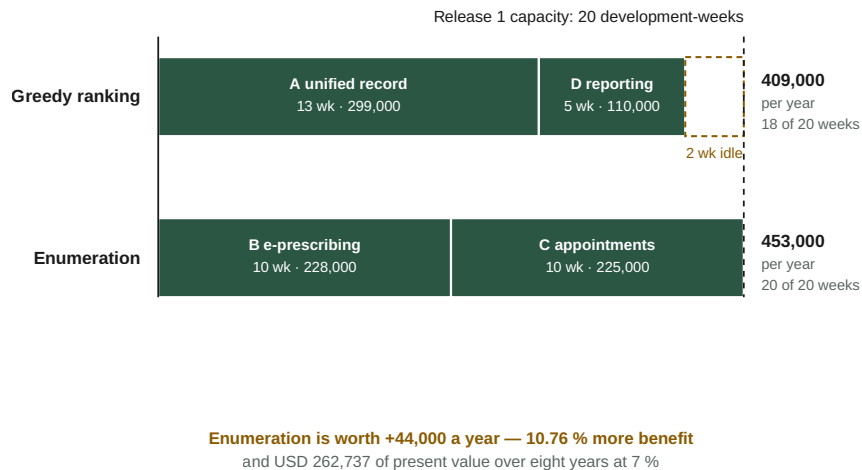


Fig 5.3.1 — Greedy ranking strands capacity; enumeration does not

**What no prioritisation method does.** It does not decide. Ranking and enumeration order candidates against one stated constraint and one stated value measure; they cannot see option value, sequencing dependencies, adoption behaviour, political commitment or the strategic cost of not doing something: the same limits Domain 2, KA 2.2.3 recorded at portfolio level. A release board that treats an enumeration output as its decision has automated its own accountability, which is Domain 1's principle failing quietly.

### AI in this KA

**Where it earns its place.** Enumerating feasible subsets under a stated constraint is exactly the work this KA shows humans getting wrong, and it is deterministic: correct enumeration removes the greedy failure entirely and the answer is verifiable by re-computation. Constructing the dependency graph among bundles from the traceability matrix and design register, and reporting which enumerated sets it makes infeasible. Running the sum test against the business case and reporting the shortfall. Running the must-have re-test as a first pass by flagging every "must have" whose stated consequence of omission is absent or circular: how a 77.92 % figure becomes informative. And sensitivity analysis: how far the benefit estimates must move before the selection changes, which is more decision-relevant than the point answer.

**Where it must not go.** Attributing benefit. A benefit figure is a claim about the organisation's future behaviour, owned by a named benefit owner (Domain 2), and a model producing plausible attributions manufactures the precision the worked example warns is absent. Ruling on must-have status, which is a stakeholder consequence judgement. Selecting release content, which is a governance decision with adoption, political and sequencing dimensions no ranking captures. And no model re-weights the value measure, because whoever controls the weights controls the outcome.

**Verification, concretely.** Both the greedy and the enumerated answer are computed and reported, because the comparison is the finding: a single reported optimum hides whether the constraint bound at all. Every enumerated set is checked against the dependency graph, itself confirmed by the technical owner. The value-per-week arithmetic is reproduced by hand for the leading candidates. Attributed benefits are confirmed with their owners and their ranges stated. And the discounting reuses Domain 2's registered  $AF(r, n)$ , cited rather than re-derived.

## KEY TERMS — KA 5.3

| TERM                                        | MEANING                                                                                                                 |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Requirement bundle</b>                   | The smallest set of requirements that together delivers a measurable benefit, the unit of value attribution.            |
| <b>Sum test</b>                             | Attributed benefit across bundles must not exceed the business case, and any shortfall is explained, not distributed.   |
| <b>Counterfactual test</b>                  | What happens to the benefit if this bundle is not delivered? A zero answer means zero attributed value.                 |
| <b>Must-have inflation</b>                  | Originator-assigned categories collapsing into near-universal "must have", destroying the scheme's information content. |
| <b>Value per unit of constrained effort</b> | Attributed benefit ÷ effort in the scarce resource: the ranking that engages the constraint.                            |
| <b>Greedy ranking</b>                       | Taking candidates in ratio order while they fit; a heuristic that strands capacity when candidates are lumpy.           |
| <b>Enumeration</b>                          | Evaluating every feasible subset under the constraint; the optimum where the candidate set is small enough.             |
| <b>Stranded capacity</b>                    | Unused constrained resource that no remaining candidate is small enough to use, the signature of greedy failure.        |

## SAMPLE MCQs — KA 5.3

**MCQ 5.3-A [5.3.3 · Application]** With 20 development-weeks available and bundles A (13 wk, 299,000), B (10 wk, 228,000), C (10 wk, 225,000), D (5 wk, 110,000) and E (5 wk, 95,000), greedy ranking by value per week selects:

- A. A + D, worth 409,000 a year CORRECT
- B. B + C, worth 453,000 a year
- C. A + B, worth 527,000 a year
- D. B + D + E, worth 433,000 a year

*Rationale:* Greedy takes A first (23,000/week), then D, stranding 2 weeks — 409,000 (5.3.3). B is the enumerated optimum, not the greedy result; C exceeds capacity at 23 weeks; D is a feasible set greedy never reaches.

**MCQ 5.3-B** [5.3.3 · Analysis] In that selection, greedy loses to enumeration because:

- A. value per week is the wrong measure
- B. the highest-ratio bundle is lumpy and strands 2 of 20 weeks that nothing can fill CORRECT
- C. bundle A's benefit is overstated
- D. the capacity constraint is not binding

*Rationale:* A's ratio advantage is under 1 % and its size wastes 10 % of capacity (5.3.3). The measure is not wrong; it is a heuristic, and the constraint is precisely what makes it fail.

**MCQ 5.3-C** [5.3.3 · Evaluation] Before the enumeration above can be relied upon, the indispensable additional input is:

- A. a probability distribution on each benefit estimate
- B. the dependency map among bundles, since an infeasible set must not be selected CORRECT
- C. the sponsor's ranking preference
- D. a re-test of the must-have classification

*Rationale:* If C depends on A, B+C cannot be built and the answer is confidently wrong (5.3.3); dependency feasibility is a precondition, not a refinement. A and D are valuable and not indispensable.

**MCQ 5.3-D** [5.3.2 · Analysis] 374 of 480 requirements are marked "must have". The consequence for prioritisation is that:

- A. the project is unusually well specified
- B. the discretionary pool is 106 requirements — 22.08 % — so prioritisation can act on a fifth of the scope, and the classification carries almost no information CORRECT
- C. the must-haves should be delivered first in ranked order
- D. capacity must be increased

*Rationale:* At 77.92 % must-have the category no longer discriminates; the re-test that asks for the consequence of omission raises the actionable pool to 332, a 3.13-fold increase (5.3.2).

**MCQ 5.3-E** [5.3.1 · Comprehension] Regulatory and safety requirements should be:

- A. scored with the highest weight in the value model
- B. removed from the value ranking and held as constraints on capacity

CORRECT

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C. prioritised last, since they generate no benefit

D. attributed a nominal benefit for completeness

*Rationale:* They are not prioritisable; scoring them produces a statutory obligation ranking eleventh and forces the process to override itself, discrediting the ranking (5.3.1).

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SELF-CHECK — KA 5.3

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1. *Why is value attributed to bundles rather than to individual requirements?* — Because benefit arises from usable capability; attributing at requirement level manufactures precision and invites delivery of half a capability, which returns nothing at full cost.
  2. *What is the signature of a greedy prioritisation failure?* — Stranded capacity: 2 of 20 weeks idle with no candidate small enough to fill them, costing 44,000 a year and 262,737 of present value.
  3. *What must be true before an enumeration is trustworthy?* — The sets enumerated must be feasible against the dependency map; otherwise the optimum cannot be built.
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KNOWLEDGE AREA 5.4

## Scope change, creep and verification/acceptance

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*Topics:* 5.4.1 change versus creep · 5.4.2 acceptance criteria and testability · 5.4.3 verification, validation and acceptance.

### 5.4.1 Change versus creep

**The distinction, stated precisely.** A **change** is a movement of the scope baseline that has been raised, screened, assessed, decided by an authority and recorded — Domain 4's flow, and Domain 4 quantified what its accumulation costs. **Creep** is a movement of the delivered content with **no corresponding baseline movement and no record**. The two are not degrees of the same thing. Change is a control operating; creep is the absence of one, and it is invisible to every instrument built to monitor change, including Domain 4's cumulative test, which reads a change log that by definition does not contain it.

**The three routes creep takes**, all of them familiar and none of them dishonest at the point of entry. *Misclassification as clarification:* the requester says the baseline already covered it, the screening step agrees without testing the claim against the requirement text, and scope grows with a paper trail that records no growth, the mirror error Domain 4, KA 4.4.1 named. *Acceptance-criteria expansion:* the require-



ment text is unchanged and its acceptance criterion grows, which is a scope change wearing a testing costume and is the commonest route where criteria are weak (5.4.2). *Undocumented addition*: someone with access and good intentions adds capability, a clinician asks a developer directly, or a developer improves something while in the code. This last is "gold-plating", and 5.2.3's reverse traceability test is the only cheap way to see it.

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#### WORKED EXAMPLE

##### Worked example 5.4.1 – the movement that left no trace in the change log.

1. **Setup.** Meridian's requirements baseline at design freeze was **480** approved requirements. Through the following 13 months, integrated change control approved **12** requirement additions. At entry to acceptance testing the traceability matrix carries **531** requirements. The average direct build cost of an uncontrolled requirement is **USD 4,200**, and **16** of them consumed critical-path time averaging **0.25 weeks** each. Cost of delay **USD 14,280** per week; baseline **USD 2,400,000**. Domain 4, KA 4.3.3 separately established **USD 291,176** of *authorised* baseline drift over the programme's first year, so the two figures set beside each other are Meridian's cumulative baseline movement to date rather than two readings of one identical window.
2. **Formula. Requirement-count reconciliation:** baseline count + approved additions = expected traced count; actual – expected = unexplained requirements. Creep cost = (unexplained × average direct cost) + (affected count × average weeks × cost of delay). Then combine with the authorised drift and compute what share of total movement the change log captured.
3. **Substitution.**  $480 + 12 = 492$ ;  $531 - 492$ ;  $39 \times 4,200$ ;  $16 \times 0.25 \times 14,280$ ;  $291,176 + 220,920$ ;  $291,176 / 512,096$ .
4. **Result.** **39** requirements are present in the delivered content and absent from every record: about **3 a month**, and **3.25 times** the number that came through change control. They cost **USD 163,800** direct plus **4.0** weeks of critical path at **USD 57,120**, a total of **USD 220,920**: **9.20 %** of the baseline, at **USD 5,664.62** each. Combined with Domain 4's authorised drift, Meridian's total baseline movement is **USD 512,096**, or **21.34 %** of the approved baseline, of which the change log accounts for only **56.86 %**.
5. **Interpretation.** The decisive number is **56.86 %**, and it should change how a leader reads a change report. Meridian had a working change process, a decision log and (after Domain 4) a cumulative test derived from its own change rate, and all of that machinery was monitoring a little over half the movement it existed to control. The other **43 %** was not hidden; it was simply never a change, because nobody raised it. Each crept requirement carries **USD 4,200** of direct cost (**0.175 %** of the baseline, the basis on which a delegation threshold reads a single item), so no threshold in Domain 3's delegation schedule can see one, and no cumulative test can aggregate what its log does not contain. Which gives the instrument,

and it is almost embarrassingly cheap: **count the requirements**. Baseline count plus approved additions must equal the traced count, every reporting period, one addition and one subtraction against two registers that already exist, and the only control in this book that detects creep. Two further readings. Against Domain 2's approved NPV of **+USD 1,332,898**, total scope movement of 512,096 consumed **38.42 %** of the programme's whole justification, leaving **USD 820,802**, still positive, which is exactly why nobody stopped: creep is survivable right up until it is not. And a caution on the arithmetic: 4,200 is an average over requirements of very different size, and the 16-of-39 critical-path attribution comes from the schedule rather than the change log, carrying the schedule's own uncertainty (Domain 6). Report the 220,920 as an estimate with a stated basis and the **39** as a fact; because the count is a fact, and the count is what produces the action.

#### 5.4.2 Acceptance criteria and testability

**Definition.** An acceptance criterion states the **condition** under which the deliverable will be judged to meet the requirement, in terms of an observable and measurable result. It is written when the requirement is written, by the requirement's owner with the tester present, because a criterion written later is written by whoever needs the deliverable accepted.

**The three states a criterion can be in**, and the middle one is the dangerous one:

| STATE             | APPEARANCE                                                                                                      | CONSEQUENCE                                                          |
|-------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Measurable</b> | States a condition, an action and an observable result with a threshold                                         | Acceptance is a test                                                 |
| <b>Restated</b>   | Reads like a criterion, restates the requirement in other words ("the system shall retrieve records correctly") | Acceptance is an argument, and the gap is invisible until it happens |
| <b>Absent</b>     | No criterion at all                                                                                             | Acceptance is an argument, and the gap is visible now                |

#### WORKED EXAMPLE

##### Worked example 5.4.2 — what it is worth to write a testable criterion.

1. **Setup.** Of Meridian's **480** requirements, **382** carry a measurable acceptance criterion, **71** carry a criterion that restates the requirement, and **27** carry none. Writing a measurable criterion, with a reviewer, costs **USD 320** of analyst and tester time. The organisation's history is that a deficient criterion produces a formal acceptance dispute in about **1 case in 4**; a dispute costs about **USD 9,600** in rework, re-test and clinical re-review; and about **1 dispute in 3** delays a clinic go-live by **0.5 weeks** at the cost of delay of **USD 14,280** per week.

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2. **Formula.** Remediation cost = deficient count × cost per criterion. Expected cost per dispute = rework + (share delaying × weeks × cost of delay). Expected dispute cost = deficient count × P(dispute) × cost per dispute. Breakeven probability = remediation cost ÷ (deficient count × cost per dispute).
  3. **Substitution.**  $(71 + 27) \times 320; 9,600 + (1/3 \times 0.5 \times 14,280); 98 \times 0.25 \times 11,980; 31,360 / (98 \times 11,980)$ .
  4. **Result.** 98 requirements are deficient — **20.42 %** of the baseline, comprising **14.79 %** restated and **5.62 %** absent (the two class rates rounding to 20.41 against the combined 20.42 — display rounding, as in 5.2.3, and one more reason the classes and not the total are the report). Remediating all of them costs **USD 31,360**. Expected cost per dispute is **USD 11,980** (9,600 of rework plus **USD 2,380** of expected delay). Expected disputes are **24.5**, at an expected cost of **USD 293,510**. Remediation therefore nets **USD 262,150**, a return of **9.36 times**. And the **breakeven dispute probability is 2.67 %** against an observed **25 %**.
  5. **Interpretation.** The breakeven settles the argument this work always loses. Writing testable acceptance criteria pays for itself if deficient criteria cause disputes more than **2.67 %** of the time; the observed rate is nine times that, and an organisation would have to believe its acceptance process almost frictionless before declining to spend 320 a requirement. Note next which class is worse, because the count says the opposite of the truth. The **27 absent** criteria are a visible gap: they appear on every completeness report and someone raises them. The **71 restated** criteria look complete, pass every completeness check, and are discovered at acceptance (deliverable built, clinic scheduled, supplier invoicing), which is Domain 4's "recorded state differs from actual state" failure moved from configuration to acceptance, and the class WE 5.2.3's nine unverified acceptances came from. **A restated criterion is worse than a missing one, and it is commoner.** Three cautions. The 24.5 is an expectation, not a forecast of 24 or 25 events, and must not be reported as a count of disputes that will happen. The rates and costs are the organisation's own history; a programme without that history should say it is using assumed rates and test the breakeven's sensitivity rather than the point result, and here the breakeven sits so far below the observed rate that the conclusion survives any plausible error in either. And remediation is not free of schedule: 98 criteria at four a day is about five weeks of one analyst's time, which has to be planned rather than absorbed, or it will not happen.
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### 5.4.3 Verification, validation and acceptance

**The distinction that Case study B turns on.** **Verification** asks *did we build what we specified?* **Validation** asks *does what we built produce the outcome we needed?* They are different questions with different evidence and, critically, different failure signatures: a project can pass verification completely and fail validation completely, and it will report itself as a success throughout, because verification is the test it has

instrumented. Validation requires the receiving organisation and real use, which is why it belongs to handover and benefits realisation (Domain 16) as well as here, and why Domain 1 insisted the leader is answerable for the outcome and not only the output.

**Acceptance** is the authorised act of taking a deliverable as meeting its criteria. It requires a named acceptor with authority, evidence against each criterion, and a recorded decision. Two provisions make it real rather than ceremonial: the acceptor is not the producer, and the evidence is retained; because the nine requirements of WE 5.2.3 were "accepted" and the record could not say by whom, on what evidence, or against what criterion.

**"Retained" needs three fields, or it is a wish.** Acceptance evidence is the primary evidence in any later dispute, audit or defect claim, and those arrive years after the acceptance meeting, so the record carries a **record class**, a **named custodian role** and a **retention period with the source of that period named**: set at the longest of the contractual limitation period, any retention requirement the organisation is subject to, the benefits-realisation horizon and the organisation's records policy. Domain 3, KA 3.3.4 sets the custody machinery and Domain 16, KA 16.4.4 works the economics; the applicable retention requirements are jurisdiction- and sector-specific and come from the records and legal functions, not from this book. Custody transfers to a named continuing role at closure, because the acceptor's project will not exist when the evidence is asked for.

**Conditional acceptance, and its price.** Deliverables are routinely accepted with open items, and this is often correct: waiting for perfection forgoes benefit at Meridian's 14,280 a week. But deferring an item moves it up the correction ladder of 5.2.1, and that movement is computable.

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#### WORKED EXAMPLE

##### Worked example 5.4.3 — deferring six severity-one items.

1. **Setup.** At Meridian's Release 1 acceptance there are **41** open items, of which **6** are severity one, a defect against a requirement that materially affects clinical use. Fixing them before acceptance costs **USD 6,400** each on the correction ladder (build stage) and delays go-live by **1.5 weeks**. Deferring them to live service costs **USD 102,400** each. Cost of delay **USD 14,280** per week.
2. **Formula.** Fix-first cost = items × build-stage cost + delay weeks × cost of delay.  
Defer cost = items × live-service cost. Breakeven go-live delay = (defer cost – items × build-stage cost) ÷ cost of delay.
3. **Substitution.**  $6 \times 6,400 + 1.5 \times 14,280; 6 \times 102,400; (614,400 - 38,400) / 14,280.$

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4. **Result.** Fixing first costs **USD 59,820** (38,400 of correction plus **USD 21,420** of delay). Deferring costs **USD 614,400: 10.27 times** as much, a difference of **USD 554,580**. Go-live would have to be delayed by **more than 40.34 weeks** before deferral became the cheaper option.
  5. **Interpretation.** The 40.34-week breakeven makes this decision arithmetic rather than temperament, which is why it is worth computing in the room: the instinct at acceptance is that a 1.5-week delay is unaffordable, and the tolerable delay is about twenty-seven times that. But the result holds **per severity class, not per item**, and generalising it is the error to avoid. The **35** severity-two and severity-three items may be entirely correct to defer, because the live-service figure on the ladder prices correcting a *requirement* defect in a running clinical service (regression, re-release, re-training, clinical re-approval), and a cosmetic or low-frequency item does not incur most of that. A programme that applies the top of the ladder to all 41 items will refuse to go live at all, forgoing 14,280 a week for items worth a fraction of it. Two further points. The go-live delay already prices forgone benefit through the registered cost of delay (Domain 1), so a separate benefit-loss term would double-count. And the arithmetic assumes the six items *can* be fixed in 1.5 weeks; if they cannot, the question is not fix-versus-defer but what Release 1 contains, which belongs to KA 5.3, not to an acceptance meeting under time pressure.
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## AI in this KA

**Where it earns its place.** The requirement-count reconciliation of WE 5.4.1, run every period and producing the **list** of unexplained requirements rather than only the number. This is the domain's highest-value automation: the check is trivially cheap, nobody runs it, and it catches the 43 % of movement no change-log instrument can see. Classifying acceptance criteria as measurable, restated or absent: a linguistic test with a definite answer, and how 71 invisible criteria become visible before acceptance rather than during it. Detecting acceptance-criteria expansion by diffing criterion text against its baselined version, which is the creep route that leaves the requirement text untouched. Assembling the evidence pack per criterion and listing criteria with none. And applying the correction ladder by severity class to an open-item list to produce the fix-versus-defer comparison for decision.

**Where it must not go.** Accepting a deliverable: an authorised act by a named human who is not the producer. Ruling that a request is a clarification rather than a change, the misclassification route one of creep depends on, and a determination with commercial consequences. Deciding which open items may be deferred, which here is a clinical, safety and commercial judgement. And no AI-generated text may serve as acceptance evidence: a model asserting that a criterion is met is not evidence that it is, and treating it as such recreates the defect WE 5.2.3 found nine instances of.

**Verification, concretely.** Every unexplained requirement is investigated by a human before it is reported as creep, because some prove to be recorded changes with a broken reference, itself a finding, of a different kind. Criterion classifications are sampled and confirmed by the tester, particularly the restated class, since that judgement is the one with money attached. The fix-versus-defer arithmetic is reproduced by hand with the severity class stated for each item, because applying the live-service figure to a cosmetic defect is the error a tool makes silently. And every acceptance decision names its human acceptor, with any AI contribution recorded as analysis input.

## KEY TERMS — KA 5.4

| TERM                                    | MEANING                                                                                                                          |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Change</b>                           | An assessed, authorised, recorded movement of the scope baseline (Domain 4, KA 4.4).                                             |
| <b>Scope creep</b>                      | Movement of delivered content with no baseline movement and no record; invisible to every change-log instrument.                 |
| <b>Requirement-count reconciliation</b> | Baseline count + approved additions = traced count; the only cheap control that detects creep.                                   |
| <b>Acceptance-criteria expansion</b>    | Creep entering through a growing criterion while the requirement text is unchanged.                                              |
| <b>Restated criterion</b>               | A criterion that repeats the requirement instead of stating a measurable condition: worse than an absent one, and commoner.      |
| <b>Breakeven dispute probability</b>    | The dispute rate at which writing testable criteria exactly pays for itself; <b>2.67 %</b> at Meridian against an observed 25 %. |
| <b>Verification</b>                     | Did we build what we specified?                                                                                                  |
| <b>Validation</b>                       | Does what we built produce the outcome we needed? A project can pass the first completely and fail the second completely.        |
| <b>Conditional acceptance</b>           | Acceptance with open items; priced by moving each item up the correction ladder <b>by severity class</b> .                       |

## SAMPLE MCQs — KA 5.4

**MCQ 5.4-A** [5.4.1 · Application] A baseline of 480 requirements received 12 approved additions; the traceability matrix at acceptance-test entry carries 531. Uncontrolled requirements average 4,200 direct, and 16 of them consumed 0.25 weeks of critical path each at a delay cost of 14,280 per week. Total creep cost is:

- A. USD 163,800
- B. USD 220,920 CORRECT
- C. USD 214,200

D. USD 57,120

*Rationale:*  $531 - (480 + 12) = 39$  unexplained;  $39 \times 4,200 = 163,800$  plus  $16 \times 0.25 \times 14,280 = 57,120$  gives 220,920 (5.4.1). A omits the schedule impact; C prices all 51 additions including the 12 that were properly approved; D is the schedule impact alone.

**MCQ 5.4-B** [5.4.1 · Evaluation] Domain 4 established USD 291,176 of authorised baseline drift over the programme's first year, giving cumulative movement of USD 512,096 against a USD 2,400,000 baseline. The most important consequence of the creep figure is that:

- A. total movement is 21.34 % of the baseline
- B. the change log captures only 56.86 % of the movement it exists to control, and no cumulative test on that log can see the rest CORRECT
- C. the average crept requirement cost 5,664.62
- D. the delegation threshold should be lowered below 4,200

*Rationale:* Creep leaves no change record, so change-log instruments, including Domain 4's cumulative test, monitor a little over half the movement (5.4.1); the remedy is the count reconciliation. A and C are true and subordinate; D cannot work, since no change is ever raised.

**MCQ 5.4-C** [5.4.2 · Application] Remediating 98 deficient criteria costs 320 each. A deficient criterion causes a dispute 25 % of the time, at an expected cost of 11,980 per dispute. The breakeven dispute probability is:

- A. 25.00 %
- B. 2.67 % CORRECT
- C. 9.36 %
- D. 20.42 %

*Rationale:*  $31,360 / (98 \times 11,980) = 2.67 \%$  (5.4.2). A is the observed rate; C is the return multiple; D is the share of the baseline that is deficient.

**MCQ 5.4-D** [5.4.3 · Evaluation] Six severity-one items cost 6,400 each to fix before acceptance, with a 1.5-week go-live delay at 14,280 per week, or 102,400 each in live service. The result and its limit:

- A. defer; the go-live delay is unaffordable
- B. fix first — it costs 59,820 against 614,400, and the breakeven delay is 40.34 weeks — but the ladder must be applied by severity class, not to all 41 open items CORRECT
- C. fix first, and apply the same reasoning to all 41 open items
- D. the comparison cannot be made without the benefit forgone during the delay

*Rationale:* Fix-first is cheaper by 554,580 and would remain so up to a 40.34-week delay (5.4.3); applying the live-service figure to minor items would prevent go-live altogether. D double-counts: the cost of delay already prices forgone benefit (Domain 1).

**MCQ 5.4-E** [5.4.3 · Comprehension] A programme reports 100 % of requirements verified and its benefits are 34 % of forecast. This is:

- A. a verification failure
- B. a validation failure — the specified thing was built, and it did not produce the outcome CORRECT
- C. a benefits-measurement error
- D. evidence of scope creep

*Rationale:* Verification asks whether the specified thing was built and validation whether it produces the outcome; the two fail independently (5.4.3), which is Case study B.

#### SELF-CHECK — KA 5.4

1. *Why can no cumulative change test detect creep?* — Because creep generates no change record; the test reads a log that by definition does not contain it. Only a count reconciliation catches it.
2. *Which acceptance-criterion class is most dangerous, and why?* — The restated criterion: it passes completeness checks and is discovered at acceptance, with the deliverable built and the supplier invoicing.
3. *What does the 40.34-week breakeven in WE 5.4.3 tell an acceptance meeting?* — That the instinct "we cannot afford 1.5 weeks" is wrong by a factor of about twenty-seven for severity-one items, and that the same arithmetic does not license fixing everything.

## Advanced topics — Domain 5

### 5.A.1 Scope in adaptive and hybrid delivery

The obvious objection to this domain is that adaptive delivery deliberately does not fix scope, so a scope baseline is the wrong instrument. The objection is half right and the conclusion is wrong. What adaptive delivery declines to fix is the **item list**; what it must still fix (more strictly than predictive delivery) is the **capacity, the cadence and the value envelope**. A backlog is not an absent baseline but a baseline on different variables, and Domain 13 works the mechanics.

Three of this domain's controls become *more* important, and they are the three adaptive programmes most often drop as belonging to a predictive method. **Value**



**attribution** (KA 5.3.1), because iteration decisions are prioritisation decisions taken every few weeks and an unattributed backlog is prioritised on advocacy. **Acceptance criteria** (KA 5.4.2), because the definition of done is the only acceptance instrument available and a restated criterion fails inside a sprint exactly as it fails at a formal gate, only faster. And **backward traceability** (KA 5.2.3), because the shorter the path from idea to build, the easier it is for an item nobody approved to be delivered.

What genuinely changes is the creep instrument: a requirement-count reconciliation is meaningless where the count is supposed to move. The equivalent is a **value-envelope reconciliation**, attributed benefit of delivered items against the business case's benefit commitment, period by period. It detects the failure adaptive programmes actually suffer, which is not item growth but a backlog that has drifted away from the benefits it was funded to produce: delivering steadily, validating badly. Hybrid delivery needs both, applied to the parts that are each kind, with a named boundary between them, a hybrid programme running one control regime has silently chosen one method.

### 5.A.2 Requirements across a contractual boundary

Where the specification is written by one party and built by another, requirement quality becomes an allocation of risk, and the allocation is usually made accidentally, by the choice of specification style. An **output specification** states what the supplier must deliver, and the buyer retains the risk that delivering it produces no outcome: verification passes, validation fails, and there is no claim because the supplier built what was specified. An **outcome specification** transfers more of that risk to a supplier who will price it and who cannot control the buyer's own enabling change, so an outcome specification carrying no buyer obligations is either unpriceable or priced with a large margin. Neither is right in general; what is always wrong is not knowing which one has been signed.

Four provisions follow, and they belong in the contract rather than a plan (Domain 10; PFL-AI Domain 11). Requirement defects and their remedy are **allocated explicitly**, who bears the cost when a requirement turns out to be wrong rather than badly built, because on the ladder of 5.2.1 that difference is 6,400 against 102,400 and it will be argued. Acceptance criteria are **agreed and baselined before signature**, since a criterion negotiated after award is negotiated by the party who wants payment. **Traceability is a deliverable**, with a stated format and an audit right, or the buyer cannot run either test in KA 5.2.3 against work it did not do. And the buyer's own obligations (data, access, decisions within stated times, where Domain 3's *E[wait]* becomes a contractual exposure) are stated with the specificity demanded of the supplier, because an unstated buyer obligation is a common root of a claim.

Two boundaries on this section. Whether a given specification style, remedy or acceptance provision has the effect described depends on the contract's governing law and on the form of agreement used, and those differ materially between jurisdic-

tions and between standard forms; nothing here states the law of any jurisdiction and none of it is legal advice. Drafting, risk allocation and any remedy question go to qualified counsel (Domain 10). What is portable is the *method*: decide which specification style you have signed, allocate requirement-defect risk deliberately rather than by default, baseline the acceptance criteria before signature, make traceability a deliverable, and state the buyer's own obligations as precisely as the supplier's.

### 5.A.3 The reviewer's scope eye

Invariants to test on any scope and requirements position, each cheap and each diagnostic.

The scope statement's extents are **counted, not categorised**, and every category noun has a number against it. The exclusion list exists and is **non-empty**: an empty exclusion list means the boundary questions have not been asked, not that there are none. Every assumption has an owner and a resolution date. The **WBS dictionary has content**, not only titles, and carries the acceptance basis. Every requirement carries a **class**, and the non-functional and transition classes are **present at all**; their total absence is the single fastest defect to spot and among the most expensive. Every requirement is **singular and testable**, and the acceptance criteria have been classified as measurable, restated or absent: with the restated count reported, because it is the one nobody has. **Traceability is audited in both directions**, reported **by class and never totalled**, with the orphan list produced; a programme that has never run the reverse test does not know what it is building. Every requirement bundle has an **attributed benefit with a named owner**, the attribution **passes the sum test** against the business case, and regulatory requirements are **held outside the value ranking** as constraints. Where a capacity constraint binds, both the greedy and the enumerated selections are computed, the **dependency map** has been used to exclude infeasible sets, and any **stranded capacity** is reported as a finding. The **requirement-count reconciliation** is run every reporting period (baseline plus approved additions against the traced count), and its unexplained list is investigated, not just counted. Criterion text is **diffed against its baselined version**, because that is where creep hides. And on any conditional acceptance, the correction ladder has been applied **by severity class** with the fix-versus-defer breakeven stated, and the acceptor is not the producer.

## Industry variations — Domain 5

- **Construction and infrastructure.** Scope is employer's requirements plus a design-responsibility allocation, and the highest-value line states **who owns the design risk for each element**: a defect in a specification the contractor was told to design to is a claim, so the ladder of 5.2.1 becomes a commercial argument rather than an internal cost. Physical scope also carries an irreversibility the software cases lack: a wall in the wrong place is not a build-stage defect.

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- **Software and digital.** The backlog carries the scope and the definition of done carries the criteria, so 5.A.1's transfers apply and the top of the correction ladder genuinely compresses under continuous deployment, which is the honest argument for it. What does not compress is the cost of a *requirement* being wrong: a fast cadence ships the wrong thing sooner.
  - **Regulated manufacturing (pharmaceutical, medical devices, aerospace).** Bidirectional traceability from requirement through design and verification to release evidence is a regulatory obligation with an inspection consequence, so the nine unverified acceptances of WE 5.2.3 are a finding against the quality system rather than a project defect. Requirements are frozen at defined points, making late elicitation expensive by design, which raises the KA 5.2.1 return well above Meridian's.
  - **Public-sector programmes.** Requirements originate partly in policy and statute, so a material share of the baseline is non-prioritisable (5.3.1) and the discretionary pool is smaller than the count suggests. Consultation obligations make elicitation a legal process with a published record; the systematic omission is the **external user**, who attends no internal meeting, Case study B.
  - **Defence and complex systems.** Requirements are verified against a formal cross-reference of requirement to verification method and evidence, over lifecycles long enough that the author has left. Traceability is the institutional memory, so a 4 % orphan rate is a configuration problem rather than an untidiness.
  - **Healthcare.** Clinical safety requirements are bound to a safety case and to specific system versions (Domain 4's configuration point), so acceptance-criteria expansion is a safety-case change and the restated-criterion class of 5.4.2 carries clinical rather than merely commercial consequence, which makes Meridian's 71 restated criteria a clinical-governance finding.
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## Case study — Domain 5: the requirement nobody could test (health, Meridian)

**Situation.** Meridian's Release 1 acceptance ran eight weeks against a planned three. The programme reported "acceptance complexity" and requested a schedule change; the supplier reported that all delivered functionality met the specification. Both statements were true, and the assurance review found the mechanism.

**What the review found.** Of the 480 approved requirements, **98**, **20.42 %**, carried acceptance criteria that could not settle a dispute: **71** restated the requirement and **27** had none. Twenty-two of the 98 reached formal dispute during acceptance, against the **24.5** the organisation's own dispute rate predicted, and six of those delayed a clinic go-live. The traceability audit found the corollaries: **34** requirements with no design element, **21** with no test case and **9** recorded as accepted with no

verification evidence, meaning the release recommendation had been assembled partly from a record that stated something untrue. The reverse test, run for the first time, found 17 of the 425 design elements tracing to no approved requirement.

**What the arithmetic had already said, and nobody had computed.** Remediating all 98 criteria would have cost  $98 \times 320 = \text{USD } 31,360$  against an expected dispute cost of **USD 293,510**: a **9.36-times** return on a **breakeven dispute probability of 2.67 %** against an observed 25 %. The 71 restated criteria had passed every completeness check the programme ran, which is why the deficiency reached acceptance intact: the programme measured whether criteria **existed**, never whether they **said anything**.

**How it resolved.** Acceptance was suspended for four weeks ( $4 \times 14,280 = \text{USD } 57,120$ ), while criteria were rewritten for the 41 requirements still in dispute or awaiting test, at  $41 \times 320 = \text{USD } 13,120$ . Six severity-one open items were fixed rather than deferred, at **USD 59,820** against a deferral cost of **USD 614,400** (WE 5.4.3). The nine unverified acceptances were reopened and verified. Of the 17 orphans, 11 were retained as retrospectively approved changes and 6 removed. And the requirement-count reconciliation was run for the first time, exposing the **39** crept requirements and **USD 220,920** of unrecorded movement that WE 5.4.1 quantifies.

**What the domain teaches here.** A criterion that exists and says nothing is worse than one that is missing, because the missing one appears on a report and the empty one does not: the programme's own completeness metric was the instrument that concealed it. Finding out at acceptance cost **57,120** of suspension plus 22 disputes; preventing it would have cost **31,360**, decided a year earlier by an analyst nobody would have escalated to. The second lesson is procedural: three controls (criterion classification, reverse traceability, count reconciliation) had never been run once, and each cost less than a day.

## Case study B — Domain 5: one hundred per cent complete and thirty-four per cent useful (public sector)

**Situation.** A national licensing authority replaced its permit-application platform. At closure the programme reported **612** requirements delivered, **100 %** verified, on budget and two weeks early. Fourteen months later the benefits review found realised annual benefit of **USD 391,000** against a forecast **USD 1,150,000**: **34.0 %**. Nothing in the delivery record was untrue.

**Why.** Requirements had been elicited thoroughly, over four months, from a working group of **9** internal officers. External applicants (about **4,200** a year, the people whose behaviour the entire benefit case depended on) were not consulted: they were not employees and there was no mechanism to convene them. Of the 612 requirements, **47**, **7.68 %**, addressed the applicant's own journey; the rest described the officers' work. The platform therefore automated the authority's processing

beautifully and left applicants submitting on paper, so the digital-channel adoption on which **78 %** of the forecast benefit rested barely arrived. The decomposition is worth stating, because it is why the shortfall was 66 points and not 78: **USD 897,000** of the forecast depended on digital adoption and **USD 253,000** did not, so a realised 391,000 implies that (taking the non-digital benefit as landing in full) only **USD 138,000** of the digital-dependent benefit, **15.4 %** of it, was ever delivered. Verification was complete because the specification was internally coherent; validation had never been attempted, because nobody had defined what a validated outcome looked like from outside the organisation.

**How it resolved.** Re-elicitation with applicant panels and observation of real submission attempts produced **63** further requirements, of which **41** were implemented as changes costing **USD 285,000**. Realised benefit reached **USD 816,500** (**71.0 %** of forecast, an uplift of **USD 425,500** a year, a payback of **0.67 years**) **about 8.0 months**. The authority's standing corrections: every benefit must name the population whose behaviour produces it, and that population must appear in the elicitation record; and closure requires a **validation** statement, not only a verification one.

**What the domain teaches here.** A hundred per cent verification rate against a specification nobody outside the organisation contributed to measures internal consistency, and it is routinely reported as success. The requirement worth 425,500 a year was available for the price of a conversation before the specification was frozen and cost 285,000 to add afterwards, entirely consistent with the ladder of KA 5.2.1. And the warning generalises past this sector: **the stakeholder group that produces the benefit is frequently the one with no seat in the requirements process**, because elicitation follows the organisational chart and benefits do not.

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## Executive perspective — Domain 5

What a programme director cannot delegate in this domain:

- **The boundary, stated as counts.** Every extent in your scope statement carries a number, and the exclusion list is non-empty. Meridian's undrawn boundary was worth **143,000**. And the right answer was to include the eleven sites, at a nine-month payback, which only the written boundary would have surfaced in time (5.1.3).
- **Who was in the room.** Requirement loss is not random: it clusters on operators, supporters, external users and whoever owns the enabling change. A licensing platform verified at **100 %** delivered **34 %** of its benefit because **4,200** applicants a year had no seat (Case study B).
- **The correction ladder, and that you have bought elicitation against it.** One defect moved out of live service saves **102,000** and pays for an **84,000** programme; the delay objection breaks even at **36.89 weeks** (5.2.1).

- **That the reverse traceability test has been run.** It costs one query, it is the only cheap detector of work nobody asked for, and Meridian's first run found **17** orphans worth **108,800** (5.2.3).
- **That capacity is not stranded.** Where a constraint binds, require both the ranked and the enumerated answer. The intuitive method left **2 of 20** weeks idle and lost **44,000** a year, **262,737** of present value (5.3.3).
- **The count reconciliation, monthly.** Baseline requirements plus approved additions against the traced count. Meridian's change log was monitoring **56.86** % of the movement it existed to control; the missing **39** requirements cost **220,920** and no threshold could ever have seen one (5.4.1).
- **A validation statement at closure, not only a verification one.** They fail independently, and only one of them is the question the organisation asked (5.4.3).

## Calculation exercises — Domain 5

**Exercise 5.1** A traceability audit covers 640 approved requirements and a 578-element design register. It finds 51 requirements with no design element, 38 with no test case, and 12 recorded as accepted with no verification evidence; 23 design elements trace to no approved requirement. Compute the forward defect rate overall and by class, and the reverse defect rate. State which class is most serious and why. *Solution.* Forward non-conformances  $51 + 38 + 12 = 101$ , a rate of  $101/640 = 15.78\%$ , comprising  $7.97\%$  with no design,  $5.94\%$  with no test and  $1.88\%$  accepted without evidence. Reverse rate  $23/578 = 3.98\%$ . The 12 are most serious: the acceptance record states something untrue, so every decision taken on it, including release, rested on nothing. *Common error:* combining the forward and reverse rates, or presenting the forward classes as a single  $15.78\%$  figure. The denominators differ (640 requirements against 578 design elements) and the classes fail differently, so neither total is informative.

**Exercise 5.2** An organisation's observed correction ladder is 500 / 2,000 / 8,000 / 32,000 / 128,000 per requirement defect by stage (definition, design, build, test, live). A review costing USD 62,000 is expected to move 11 build-stage, 8 test-stage and 2 live-service defects to the definition stage. Cost of delay is USD 11,500 per week. Compute the saving, the net, the return, and the definition-phase extension at which the investment stops paying. *Solution.* Saving  $11 \times 7,500 + 8 \times 31,500 + 2 \times 127,500 = 82,500 + 252,000 + 255,000 = \text{USD } 589,500$ . Net **USD 527,500**; return **9.51 times**. Breakeven extension  $527,500 / 11,500 = 45.87 \text{ weeks}$ . *Common error:* computing the saving on the gross later-stage costs ( $11 \times 8,000 + 8 \times 32,000 + 2 \times 128,000 = 600,000$ ), which forgets

that each defect still costs 500 to correct at definition and overstates the saving by 10,500.

**Exercise 5.3** A release has 24 development-weeks of capacity. Four candidate requirement bundles: P (16 weeks, USD 496,000 a year), Q (12 weeks, 366,000), R (12 weeks, 360,000), S (6 weeks, 168,000). Compute the greedy selection by value per week and the enumerated optimum, their capacity utilisation, the annual difference, and its present value over 6 years at 8 % ( $AF(0.08, 6) = 4.622880$ ). *Solution.* Ratios: P 31,000/week, Q 30,500, R 30,000, S 28,000. Greedy takes P (16 weeks), cannot fit Q or R, takes S (6 weeks) — **P + S = USD 664,000** using **22 of 24** weeks (**91.7 %**), with 2 weeks stranded. Enumeration finds **Q + R = USD 726,000** at **24 of 24** weeks (**100 %**). Difference **USD 62,000** a year — **9.34 %**: worth  $62,000 \times 4.622880 = \text{USD } 286,619$  of present value. *Common error:* ranking by value per week and stopping. The greedy answer is defensible on the ratio and wrong on the outcome; the diagnostic is the stranded capacity, and the check is one enumeration over four candidates. A second error is enumerating without a dependency map: if R requires P, **Q + R** is not a feasible set.

**Exercise 5.4** At design freeze a project's requirements baseline held 560 requirements. Integrated change control subsequently approved 19 additions. At acceptance-test entry the traceability matrix carries 638. Uncontrolled requirements cost an average of USD 3,800 direct, and 21 of them consumed 0.2 weeks of critical path each. Cost of delay is USD 11,500 per week; the baseline is USD 3,100,000. Compute the unexplained count, the direct and schedule cost, the total, its share of the baseline, and each crept requirement's individual share. *Solution.* Expected traced count  $560 + 19 = 579$ ; actual 638, so **59** requirements are unexplained. Direct  $59 \times 3,800 = \text{USD } 224,200$ ; schedule  $21 \times 0.2 = 4.2$  weeks at 11,500 = **USD 48,300**; total **USD 272,500**: **8.79 %** of the baseline, while each crept requirement is **0.12 %** of it. *Common error:* reconciling cost and never counting requirements. The change log can balance to the penny while 59 requirements enter without ever appearing in it; only the count reconciliation detects them, and no delegation threshold can, because 0.12 % of a baseline is below every threshold anyone sets.

**Exercise 5.5** Of 720 requirements, 126 carry acceptance criteria that restate the requirement and 41 carry none. Writing a measurable criterion costs USD 280. A deficient criterion produces a dispute 20 % of the time; a dispute costs USD 8,400 in rework, and 1 dispute in 4 delays delivery by 0.5 weeks at a cost of delay of USD 11,500 per week. Compute the remediation cost, the expected cost per dispute, the expected dispute cost, the net, and the breakeven dispute probability. *Solution.* Deficient  $126 + 41 = 167$ ; remediation  $167 \times 280 = \text{USD } 46,760$ . Expected delay per dispute  $0.25 \times 0.5 \times 11,500 = \text{USD } 1,437.50$ , so cost per



dispute USD 9,837.50. Expected disputes  $167 \times 0.20 = 33.4$ , expected cost USD 328,572.50. Net USD 281,812.50. Breakeven probability  $46,760 / (167 \times 9,837.50) = 2.85\%$ . *Common error*: two, and both understate the case. Comparing the remediation cost against the rework cost alone omits the delay component (1,437.50 of the 9,837.50). And applying the 20 % dispute rate to all 720 requirements rather than to the 167 deficient ones inflates the expected cost by more than four times: an error that happens to favour the right decision, which is why it survives.

## Practitioner's toolkit — Domain 5

*Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.*

### Toolkit 5.T.1 — Requirements traceability matrix

One row per requirement: reference · statement · **class** (functional / non-functional / constraint / data / transition / regulatory) · source, naming the **stakeholder or obligation**, not the document · benefit served and the **bundle** it belongs to (Toolkit 5.T.2) · design element · test case · deliverable · **acceptance criterion classification** (measurable / restated / absent) · verification evidence reference · acceptance decision, acceptor by name and date · baseline version approved at, and the change reference if added later.

Four reports every period carry the whole value of the matrix: rows missing a design, test or deliverable link, **by class and never totalled**; rows accepted with no verification evidence, the class that invalidates the record; **orphans**, from the reverse query over the design and test registers; and criteria classified restated or absent, with the restated count shown separately because it is the number nobody has. A matrix maintained and never queried is a cost with no control attached.

### Toolkit 5.T.2 — Value attribution and constrained selection sheet

*Attribution side.* One row per **requirement bundle**: reference · requirements included · benefit delivered, with its **named owner** (Domain 2's benefits register) · attributed annual value with its range · effort in the **binding constrained resource**, named explicitly (usually a team, not money) · dependencies on other bundles · **value per unit of constrained effort** · and a flag for regulatory or safety bundles, removed from the ranking and held as constraints. Foot the attribution column and run the **sum test** against the business case, stating any shortfall rather than distributing it.

*Selection side.* State the constraint and its quantity. Compute and record **both** the greedy result and the enumerated optimum over feasible sets, with the dependency map applied; report capacity utilisation for each and flag any **stranded capacity** as



a finding. Where the two differ, state the annual difference and its present value at the appraisal rate. Then record the decision, its maker by name, and (where it departs from the enumeration) what is being bought and at what annual cost, because overruled arithmetic is a decision and an unexamined ranking is not.

### Toolkit 5.T.3 — Scope reconciliation and acceptance-criteria pattern set

*Reconciliation side*, run every reporting period and published on one page: baselined requirement count · plus approved additions (with change references) · minus approved deletions · **equals expected traced count** · against the **actual traced count** · equals **unexplained requirements**, listed individually and investigated, not merely counted. Then: criterion text **diffed against its baselined version**, with every expansion listed; the priced creep total, direct and schedule, at the cost of delay; and the cumulative total set beside Domain 4's authorised drift, so the report states **what share of total scope movement the change log actually captured**.

*Acceptance-criterion patterns*, four forms that make a criterion testable. **Threshold:** given *[state]*, when *[action]*, the result is *[observable]* within *[limit]* (for performance, capacity and time). **Enumeration:** the deliverable handles each of *[n]* listed cases, each individually verifiable (for functional coverage, and it forces the case list into the open). **Negative:** given *[invalid input or failure condition]*, the result is *[defined behaviour]* (the pattern that catches the exception handling nobody specified). **Evidence:** acceptance requires *[named artefact]* reviewed by *[named role]* (for requirements whose satisfaction is a judgement: clinical safety, accessibility, regulatory conformance), which is how a judgement becomes auditable rather than an opinion. Prohibit four constructions outright: "correctly", "appropriately", "as required", and any comparative without a referent.

## Exam preparation — Domain 5

**What is assessed.** The three registers and why scope is the deliverable register; decidable scope; the scope baseline's three artefacts and the primacy of the WBS dictionary; **pricing an undrawn boundary on both cost and forgone benefit**; elicitation technique selection and the stakeholder groups that cause systematic loss; **the correction ladder, the saving computed on stage differences, the return on elicitation and the breakeven delay**; requirement quality properties and the six requirement classes; solution capture and requirement inflation; **traceability in both directions, defect rates by class on their own denominators, and which class invalidates the record**; value attribution to bundles with the sum and counterfactual tests; **must-have inflation quantified; value per unit of constrained effort, greedy against enumeration, stranded capacity and the dependency precondition**; change against creep and the **count reconciliation**; acceptance-criterion states and the **breakeven dispute probability**; verification against validation; and **conditional acceptance priced by severity class**.

**The calculations to be able to do under time pressure.** Per-unit boundary exposure and its payback from forgone benefit. Expected correction cost from a stage profile and a ladder; the saving from moving  $k$  defects between stages, **always on the difference**; the return on an elicitation spend and the breakeven delay. Forward defect rate overall and by class, and the reverse rate on the design-register denominator. Value per unit of constrained effort; the greedy selection; the enumerated optimum over a small candidate set; capacity utilisation; and the annual difference converted to present value with  $AF(r, n)$ . The requirement-count reconciliation, and creep priced as direct plus schedule at the cost of delay, with its share of baseline and of total movement. Remediation cost, expected cost per dispute including its delay component, and the breakeven dispute probability. Fix-versus-defer against the ladder, and the breakeven go-live delay.

**The traps.** Scope written in the activity register, so no boundary can be tested (5.1.1) · reading a boundary statement as a way of saying no when the priced answer is to include (5.1.3) · computing an elicitation saving on gross later-stage costs rather than stage differences (5.2.1, Exercise 5.2) · assuming every late defect was detectable at definition (5.2.1) · totalling traceability classes, or combining forward and reverse rates on different denominators (5.2.3, Exercise 5.1) · running only the forward test and never seeing the orphans (5.2.3) · attributing value to individual requirements rather than bundles, and delivering half a capability (5.3.1) · scoring regulatory requirements in a value ranking (5.3.1) · accepting a 78 % must-have classification as information (5.3.2) · ranking greedily and stopping, ignoring stranded capacity (5.3.3, Exercise 5.3) · enumerating without a dependency map (5.3.3) · monitoring change and never counting requirements, so 43 % of movement stays invisible (5.4.1, Exercise 5.4) · measuring whether acceptance criteria exist rather than whether they say anything (5.4.2) · applying the dispute probability to all requirements rather than the deficient ones (Exercise 5.5) · applying the top of the ladder to every open item and so refusing to go live (5.4.3) · and reporting verification as though it answered the validation question (5.4.3, Case study B).

**How the domain connects.** Domain 1 supplies accountability for outcome rather than output and the cost of delay every calculation is priced at. Domain 2 supplies the benefits map requirements attach to, the ramped adoption profile the attributions inherit, the  $AF(r, n)$  used to discount a prioritisation difference, and the capacity-constrained selection principle KA 5.3.3 works at requirement level. Domain 3 supplies the decidability test the boundary discipline applies to content, and the thresholds that cannot see a 0.175 % crept requirement. Domain 4 supplies the WBS the scope baseline includes, the change flow that distinguishes change from creep, the authorised drift the creep figure is set beside, and the "record states something untrue" failure class recurring here as unverified acceptance and as the restated criterion. Forward: Domain 6 schedules these bundles and supplies the dependency map the enumeration requires; Domain 7 costs them; Domain 8 attaches uncertainty to the attributions; Domain 9 owns verification and acceptance in depth; Domain 10 carries the contractual face of specification risk (5.A.2); Domain 13 works the

adaptive equivalents of 5.A.1; and Domain 16 answers the validation question at hand-over.

## Domain 5 summary

Part One produced a project and assumed the content of its scope was known. It was not, and the expensive failure is not incompleteness but a scope that is complete, agreed, delivered, verified and valueless: the condition Case study B reports as **612 requirements, 100 % verified, 34.0 %** of forecast benefit, because the **4,200** external applicants whose behaviour produced the benefit had no seat in a requirements process built from the organisational chart. Only **47** of the 612 requirements, **7.68 %**: addressed them. The correction cost **USD 285,000** and returned **USD 425,500** a year, a payback of about **8.0 months**, for a conversation that had been available free eighteen months earlier.

Scope is the deliverable register, bounded by outcomes and decomposed into activities, and it is decidable only when its extents are counted. Meridian's boundary ("all clinics in the region" against an approved 40) carried an exposure of **USD 143,000** at **USD 13,000** a site; and the professional result is the inversion, because those 11 sites would return **USD 188,496** a year at realistic adoption, a payback of about **9.1 months**. The boundary statement's value is not that it refuses; it is that it forces the question while the answer is still cheap to act on.

Requirement defects cost four times more at every stage they survive: **400 · 1,600 · 6,400 · 25,600 · 102,400**. On Meridian's detection profile the unimproved correction bill is **USD 1,286,400** (**USD 13,400** a defect), and an **USD 84,000** elicitation programme returns **USD 610,800**, **7.27 times** its cost, computed on stage *differences* rather than gross later-stage costs, which would have overstated it by 10,000. Two sentences carry the whole argument: **one defect moved out of live service saves USD 102,000 and pays for the entire programme**, and the delay objection does not break even until the definition phase is extended by **36.89 weeks**. Traceability then audits the chain in both directions and reports **by class, never as a total: 13.33 %** forward non-conformance across **7.08 %** with no design, **4.38 %** with no test and **1.88 %** accepted with no verification evidence (that last class being the one that makes every decision taken on the record worthless) plus a **4.00 %** reverse rate, **17** orphans worth **USD 108,800**, invisible to the forward test almost everyone runs alone.

Value is attributed to bundles, not requirements, and prioritisation only bites once the categories mean something: **77.92 %** of Meridian's requirements were marked must-have, and asking each originator to name the consequence of omission left **30.83 %**, raising the tradeable pool **3.13-fold**. Against a hard **20 development-week** constraint, greedy ranking by value per week takes the highest-ratio bundle, strands **2 of 20** weeks and returns **USD 409,000** a year; enumeration returns **USD**

**453,000 (10.76 % more, USD 262,737** of present value at 7 % over eight years), and the ratio advantage that caused the loss was **under 1 %**. Stranded capacity is the signature; a dependency map is the precondition; and no ranking is a decision.

Finally, scope moves without anyone approving anything. Meridian's baseline of **480** requirements plus **12** approved additions should have traced **492**; it traced **531**, so **39** requirements (about three a month, **3.25 times** the approved additions) entered with no record at all, costing **USD 163,800** direct and **4.0** weeks of critical path, **USD 220,920** in total, **9.20 %** of the baseline, at **USD 4,200** of direct cost each: **0.175 %** of baseline, below every threshold any delegation schedule will ever set. Set beside Domain 4's **USD 291,176** of authorised drift, total movement is **USD 512,096 (21.34 % of baseline and 38.42 % of the approved NPV)** of which the change log captured **56.86 %**. No cumulative test on a change log can find the rest; **only counting the requirements can**, and it is one addition and one subtraction a month. On the way out, two more numbers worth keeping: writing testable acceptance criteria pays for itself above a **2.67 %** dispute rate against an observed **25 %**, and the **71** criteria that restate their requirement are more dangerous than the **27** that are missing, because completeness reports cannot see them. And fixing six severity-one defects before acceptance costs **USD 59,820** against **USD 614,400** deferred, a breakeven go-live delay of **40.34 weeks**, applied by severity class and not to all 41 open items.

DOMAIN

# 6

## Planning, Scheduling and Delivery Flow

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## Why this domain exists

A project leader's authority rests on one repeated act: making a credible statement about the future, when the work will finish, what could move that date, and what it would cost to move it back. This domain builds the machinery behind that credibility. It starts where planning actually starts, with levels of plan and the logic that binds work together (KA 6.1). The critical path method is then built in full, forward and backward passes and both kinds of float (KA 6.2), before the realities the textbook network ignores — resources, constraints, milestones and rolling-wave elaboration — are added (KA 6.3). The domain finishes with delivery flow across predictive, agile and hybrid worlds: recovery, compression economics, and forecasting under uncertainty (KA 6.4). Cost joins schedule in Domain 7 (earned value); risk quantification deepens in Domain 8. A leader who cannot read a network diagram is hostage to whoever can; this domain removes the hostage-taking.

**Learning objectives.** After this domain a candidate can: choose the right planning level for an audience and a decision; build a logic network with correct dependency types; **model an approval as a dated event against a governance calendar and compute what mis-modelling it costs**; run forward and backward passes and derive total and free float; identify the critical path and near-critical paths, **enumerate every path with its float, choose a near-critical band and price the monitoring decision**; explain why float is not spare time and **why activity floats must never be summed**; plan resources against the schedule, **produce a resource-feasible schedule under a hard cap, distinguish the critical chain from the critical path and price the peak that binds**; apply rolling-wave elaboration honestly and **price the commitment exposure a ranged planning package carries**; build a compression menu with cost slopes, **locate the least-cost duration and state the range of week-values over which it holds**; state the **non-financial ceiling on each compression lever and the second signature a shift, night-working or simultaneous-operations move requires**; price fast-tracking as a probabilistic decision with a **breakeven rework probability**; distinguish predictive, agile and hybrid scheduling, **translate a throughput forecast into a required rate through the backward pass**, and govern each; **size and manage a schedule buffer from aggregated safety, and forecast from buffer consumption**; **convert a repetitive-delivery date into a takt, a crew count and a benefit integral**; and use three-point estimates and scenario analysis to forecast completion with stated uncertainty, with any AI-generated schedule subject to the family's verification rule.

**The master worked project.** One project runs through this domain and returns in Domains 7 and 8: **Project Auriga**, a control-systems upgrade for a regional utility, planned in weeks:

| ID | ACTIVITY                       | DURATION (WK) | PREDECESSORS |
|----|--------------------------------|---------------|--------------|
| A  | Mobilise                       | 2             | —            |
| B  | Detailed design                | 6             | A            |
| C  | Procure control hardware       | 8             | B            |
| D  | Civil and cabling works        | 7             | B            |
| E  | Installation                   | 5             | C, D         |
| F  | Testing and commissioning      | 4             | E            |
| G  | Operator training and handover | 2             | D            |

Every calculation below uses this network. Auriga's **BAC** is **USD 4,000,000** and the value of a week (the client's early-completion bonus, and equally the cost of a week's delay) is **USD 45,000**, the figure Domains 7, 8 and 9 use unchanged. An engineer-week costs **USD 5,225** (Domain 7, KA 7.4.1).

**The second scale.** Scheduling questions change shape between one project and a rolling programme, so this domain also works the family's programme case, **Meridian Care Records**: the clinical-records rollout to **40 clinics**, approved cost **USD 2,400,000**, benefit **USD 979,200** a year at full potential and **USD 685,440** at the realistic 70 % adoption, with a **cost of delay of USD 14,280 per week** derived in Domain 1 and a steering committee whose expected decision latency Domain 3 computes as  $E[\text{wait}] = M/2 + L = 4 \text{ weeks}$  from a four-weekly meeting and a two-week paper deadline. Meridian supplies what Auriga cannot: a governance calendar the network has to obey (KA 6.1.2), a ranged planning package (KA 6.3.3) and forty near-identical units of repetitive work whose schedule is a *rate* rather than a network (6.A.5). A deployment specialist-week costs **USD 4,200** (Domain 1, KA 1.3.3b).

#### KNOWLEDGE AREA 6.1

### Planning levels and logic networks

*Topics: 6.1.1 levels of plan · 6.1.2 dependencies and logic · 6.1.3 network rules and quality.*

#### 6.1.1 Levels of plan

**The principle.** One schedule cannot serve a board, a site supervisor and a lender at once. Mature delivery organisations maintain a **hierarchy of schedules**, each derived from the one below, each honest with the one above:

| LEVEL | NAME                | AUDIENCE & DECISION                                          | TYPICAL GRANULARITY  |
|-------|---------------------|--------------------------------------------------------------|----------------------|
| L1    | Executive summary   | Board, sponsor, go/no-go, milestones                         | 10–30 bars           |
| L2    | Management schedule | Steering, PMO (stage gates, interfaces)                      | 50–300 activities    |
| L3    | Control schedule    | Project team (the CPM network; progress and float live here) | 300–5,000 activities |
| L4    | Execution/lookahead | Site and squads (2–6 week detail)                            | daily/shift detail   |

The **control schedule (L3)** is where this domain's machinery operates: it carries the logic, the floats and the critical path, and it is the version under change control (Domain 4, KA 4.3). An L1 milestone that cannot be traced to L3 logic is decoration, and steering committees should treat it as such (Domain 3, KA 3.2).

Fig 6.1.1 — The schedule hierarchy: one logic, four honest views

### 6.1.2 Dependencies and logic

**Definitions.** Activities link through four dependency types, finish-to-start (**FS**, the default: design finishes, then procurement starts), start-to-start (**SS**: cabling can start once civils have started, often with a lag), finish-to-finish (**FF**: commissioning documentation finishes when commissioning finishes), and the rare start-to-finish (**SF**). A **lag** delays a successor ( $FS+2$  = start two weeks after the predecessor finishes); a **lead** (negative lag) overlaps it. Logic expresses *physics and choice* (what must follow, and what the team has chosen to sequence), and the two should be distinguishable in the schedule's notes, because only chosen logic can be re-chosen when recovery demands it (KA 6.4).

**Common pitfall (the lag that hides work.** A 3-week lag labelled "curing" is physics; a 3-week lag labelled nothing is often an activity somebody didn't model) unresourced, uncosted, invisible to progress measurement. Audit rule: every lag has a stated reason, and long lags (> 10 % of project duration) are activities in disguise.

#### WORKED EXAMPLE

#### Worked example 6.1.2 — re-choosing a dependency.

1. **Setup.** Survey work P (4 weeks) precedes report drafting Q (6 weeks), currently linked  $FS+2$  (two weeks of data cleaning after all surveying ends). The team proposes drafting in parallel as survey sections complete:  $SS+1$ . Compute Q's dates both ways (P starts week 0).
2. **Formula.**  $FS+lag: ES(Q) = EF(P) + lag.$   $SS+lag: ES(Q) = ES(P) + lag.$



3. **Substitution.** FS+2:  $ES(Q) = 4 + 2 = 6$ ,  $EF(Q) = 12$ . SS+1:  $ES(Q) = 0 + 1 = 1$ ,  $EF(Q) = 7$ .
4. **Result.** The chain finishes week **12** under FS+2 and week **7** under SS+1, five weeks earlier from one logic choice.
5. **Interpretation.** Nothing was crashed and nobody works faster: the saving comes from overlapping, and its price is rework risk if early sections change after later surveying. This is why chosen logic must be visibly chosen (not fossilised as physics). It is the cheapest recovery currency the project owns (KA 6.4.2), and spending it is a risk decision, not a scheduling trick.

**The risk is priceable, and the breakeven is reassuring.** Suppose the worst credible consequence is that late survey sections invalidate the early drafting and Q must be rewritten in full (six weeks lost) with probability  $p$ . The expected saving is  $5 - 6p$  weeks, so the overlap stops paying at  $p = 5/6 = \mathbf{83.33\%}$ . At a more plausible  $p = 30\%$  the expected saving is  $5 - 1.8 = \mathbf{3.20\text{ weeks}}$ . Re-choosing a dependency is therefore an unusually favourable trade: the saving is banked immediately and in full, while the loss is contingent, which is exactly the asymmetry KA 6.4.2 exploits again for fast-tracking, and exactly the asymmetry that makes overlap the first move in any recovery.

**Two things break that comfortable arithmetic, and a reviewer should ask about both.** The first is **detectability**. The model above assumes the rework is discovered while the report is still in draft. If the invalidated early sections reach a published report unnoticed, the consequence is not six schedule weeks but an escaped defect in a delivered product, priced by Domain 9's containment economics on a ladder an order of magnitude steeper. Overlap is safe in proportion to the strength of the check that sits between the overlapping activities. The second is that  $p$  is not a constant but a function of *how much* overlap is taken: SS+1 drafts on one week of survey data, SS+3 on three. Where a team can state the relationship it should optimise the overlap; where it cannot, the usual case, it should take the overlap that a downstream check can catch and say so, rather than pretending a single  $p$  describes every possible degree of overlap.

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WORKED EXAMPLE

**Worked example 6.1.2b – the approval that was drawn as a lag.**

1. **Setup.** Meridian's wave-2 clinic fit-out design must be approved by the steering committee before the estate contractor mobilises. The planner has drawn this as FS+3 on the design → mobilise link, labelled "approval". The committee meets every  $M = 4$  weeks (programme weeks 20, 24 and 28), and papers are due  $L = 2$  weeks before each meeting (weeks 18, 22, 26). The design's forecast finish is **week 23**. The path carrying the approval has **2 weeks** of total float. Cost of delay **USD 14,280** per week (Domain 1, KA 1.3.3).

2. **Formula.** A submission finishing at  $t$  reaches the first meeting  $m$  for which  $m - t \geq L$ ; the wait is  $m - t$ . Its range across arrival times is  $L$  (best) to  $M + L$  (worst), and its mean for a decision arriving at a uniformly random point is  $E[\text{wait}] = M/2 + L$  (derived in Domain 3, KA 3.2.3, not re-derived here).
3. **Substitution.** Finish week 23: the week-24 meeting closed its papers at week 22, so the first reachable meeting is **week 28** and the wait is  $28 - 23 = 5$  weeks. Finish week 22 instead: papers for the week-24 meeting are still open, so the wait is  $24 - 22 = 2$  weeks.
4. **Result.** The scheduled wait is **5 weeks**, not the 3 weeks drawn and not the 4-week expectation, so the approval lands in **week 28**. The 2-week understatement **exactly consumes the path's float**:  $TF$  falls from 2 to 0. Pulling the design finish one week earlier, to week 22, cuts the wait to **2 weeks** and the approval to **week 24**, so the successor starts **4 weeks** earlier for **one week** of design work. Note the asymmetry: the *wait* shortens by 3 weeks ( $5 \rightarrow 2$ ) while the *approval date* moves by 4 ( $28 \rightarrow 24$ ), because the design's own week is saved as well. Four weeks are worth  $4 \times 14,280 = \text{USD } 57,120$ , or **USD 45,120** net of a USD 12,000 compression cost.
5. **Interpretation. A mis-modelled lag makes a path critical while every duration is still "as planned".** No activity slipped, no estimate changed, nobody was late; and the path went from two weeks of protection to none, purely because a three-week label stood in for a five-week event. This is the pitfall above made arithmetical: the schedule was not optimistic, it was *silent*, and silence is not conservative. The audit habit follows directly, for every lag, ask what event it represents, then ask what that event's own logic is.

**Once the network exists, the wait is a choice, not a random variable.**

Domain 3's  $E[\text{wait}] = M/2 + L = 4$  weeks is the correct planning figure for a decision arriving at an unknown time, which is what an escalation is. A *planned* approval is not that: its arrival date is a date the planner controls, so the honest model is deterministic against the meeting calendar, and the wait ranges from **2 to 6 weeks** depending on where the finish lands. Substituting the expectation for a planned submission is a category error in both directions: it flatters a submission that will miss its window and penalises one that will make it.

**The highest-leverage week in a governance-bound schedule is the one immediately before a paper deadline.** One week of compression bought four weeks of schedule here: a **4:1** return, and it arises entirely from the discreteness of the calendar. Compare it with Domain 3's own lever, shortening the paper lead time  $L$  by one week saves one week of expected wait, worth **USD 14,280**. The scheduling lever is four times the governance lever *on this submission*, costs nothing to exercise, and needs nobody's permission. It is also the more fragile: it works only if the submission actually lands before the deadline, so the planner who claims it must also protect it, which means treating the paper deadline as a milestone with float, not as an administrative detail.

**The cautions.** The 4:1 leverage is a property of this arrival date, not of the calendar. A design finishing at week 21 gains nothing from compression because it already makes the week-24 meeting, and a leader who reads "compress the design" as a general rule will buy weeks that convert into nothing. The gain is realised only if the approval sits on a binding path; on a path with four weeks of float the whole analysis is worth zero and the compression money should be spent elsewhere. And the wait is the *committee's* latency only, a deferral, a request for more information or a conditional approval restarts the cycle, which is why the register in toolkit 6.T.3 records approval events with their meeting dates rather than with nominal durations.

### 6.1.3 Network rules and quality

A network the passes can trust obeys checkable rules: every activity except the first has a predecessor and except the last a successor (**no dangles**); no loops; durations estimated by the owning discipline, not the scheduler; no date constraints doing logic's job (a "must start on" pin that overrides logic converts the schedule from a model into a poster); and logic density in a healthy range: too few links and float is fiction, too many and nothing can move. These rules are the schedule-quality gate a leader can run without scheduling software: ask for the dangle count, the constraint count and the reasons for both.

**Two of those rules are countable, and counting them is the whole skill.** *Logic density* is  $\text{links} \div \text{activities}$ . Auriga carries 7 dependency links across 7 activities, a density of **1.00**, and the number a reviewer compares it against is structural rather than conventional: binding  $n$  activities into one connected network takes at least  $n - 1$  links, so the floor here is **6** and Auriga sits exactly one link above it. That one surplus link is the second predecessor of E, which is to say **the surplus over  $n - 1$  is the count of genuine convergences in the network**: a directly meaningful figure, and the reason a density materially below 1 on a large network is proof of dangles rather than evidence of elegant simplicity. At the other extreme a density far above 2 usually means the same constraint has been expressed several times, which makes float unresponsive because every activity is held by a redundant link. *Float distribution* is the share of activities at or near zero float. Auriga shows  $5/7 = 71.4\%$  at zero float, which on a real network would be a warning that the logic is over-tight or the constraints are doing the work.

Here the professional caution matters more than the metric: **Auriga is a seven-activity teaching network, and neither number means anything at that size.** Both measures are properties of a population of activities, and both take their meaning from an organisation's own calibrated history: a construction contractor's healthy density is not a software programme's. The transferable habit is therefore the arithmetic and the direction of the inference, not a benchmark: compute density and the zero-float share on your own portfolio of accepted schedules, and treat a new schedule that sits far outside your own distribution as a question to be answered

rather than a defect to be asserted. A reviewer quoting a universal threshold for either is quoting something nobody derived.

### AI in this KA

Scheduling tools now draft networks from scope statements and historical libraries. Useful (the draft logic is often 80 % right), and dangerous in a specific way: generated logic *looks* authoritative while encoding assumptions nobody made. The governed workflow: AI proposes the network; the discipline leads walk every link of their scope ("does B truly need all of A finished?"); the scheduler runs the quality rules above; and the leader signs the logic as a decision record (Domain 3, KA 3.3.4). **AI proposes; the professional verifies, decides and remains accountable.**

#### KEY TERMS — KA 6.1

| TERM                              | MEANING                                                                                                              |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Schedule hierarchy (L1–L4)</b> | Nested plans for different audiences, derived from one logic.                                                        |
| <b>Control schedule</b>           | The L3 CPM network under change control; home of float and progress.                                                 |
| <b>FS / SS / FF / SF</b>          | The four dependency types.                                                                                           |
| <b>Lag / lead</b>                 | Imposed delay / overlap on a dependency.                                                                             |
| <b>Dangle</b>                     | An activity missing a predecessor or successor; a network defect.                                                    |
| <b>Date constraint</b>            | A pinned date overriding logic; each one weakens the model.                                                          |
| <b>Logic density</b>              | Links ÷ activities; the surplus over $n - 1$ counts the network's convergences.                                      |
| <b>Governance calendar</b>        | The meeting and paper-deadline dates an approval event must be scheduled against, rather than approximated by a lag. |

#### SAMPLE MCQs — KA 6.1

**MCQ 6.1-A** [6.1.2 · Application] Cabling may begin one week after civil works begin. The correct dependency is:

- A. FS+1
- B. SS+1 CORRECT
- C. FF+1
- D. FS–1

*Rationale:* The condition binds the two *starts* with a one-week lag; start-to-start plus one. A would wait for all civils to finish; C binds the finishes; D (a lead on FS) overlaps the finish, which is not what was stated.

**MCQ 6.1-B** [6.1.3 · Analysis] A schedule shows 14 "must finish on" constraints, all on milestones reported to the board. The most accurate reading is:

- A. the schedule is well-governed, because board dates are protected
- B. the milestones will be met, because the tool will honour the constraints
- C. the schedule may no longer model reality — pinned dates can mask logic-driven slippage that float analysis would otherwise reveal CORRECT
- D. constraints are neutral scheduling hygiene with no analytical effect

*Rationale:* Pins override logic: the network can be slipping while pinned milestones hold still, hiding negative float until it is unrecoverable. A and B mistake suppression for control; D is false, every pin degrades the model's predictive value.

**MCQ 6.1-E** [6.1.1 · Recall] The schedule level that carries the logic, the floats and the critical path — and sits under formal change control — is:

- A. L1, because the board owns the schedule
- B. L2, because the PMO maintains it
- C. L3, the control schedule CORRECT
- D. L4, because it has the most detail

*Rationale:* The L3 control schedule is the analytical model; L1/L2 summarise it and L4 elaborates near-term execution from it. Detail (D) is not the same as control, L4 churns weekly by design and is never the baselined network.

**MCQ 6.1-C** [6.1.2 · Application] Survey P (4 wk) feeds report Q (6 wk). Under FS+2 the chain completes week 12. Re-linked SS+1, it completes week:

- A. 7 CORRECT
- B. 11
- C. 12
- D. 5

*Rationale:*  $ES(Q) = ES(P) + 1 = 1$ ,  $EF(Q) = 7$ . B subtracts one lag week from 12; C assumes logic changes nothing; D forgets Q's own duration must still run.

**MCQ 6.1-D** [6.1.3 · Recall] A "dangle" in a schedule network is:

- A. an activity with negative float
- B. an activity missing a predecessor or successor link CORRECT
- C. a milestone with zero duration
- D. a lag longer than its predecessor

*Rationale:* Dangles are unbound activity ends, the passes cannot constrain them, so their dates and float are unreliable. A describes constraint-driven lateness; C describes every milestone; D is unusual but legal when justified.

**MCQ 6.1-F** [6.1.2 · Analysis] A design finishing in week 23 needs approval from a committee meeting in weeks 20, 24 and 28 whose papers are due two weeks before each meeting. The planner has drawn the approval as a three-week lag and the path carries two weeks of total float. The correct reading is:

- A. the lag is conservative, since the average wait for this committee is four weeks
- B. the approval will actually take five weeks, the two-week understatement consumes the path's float, and the path is critical with no duration having changed CORRECT
- C. the wait is four weeks, so one week of float remains
- D. the lag is irrelevant because approvals are not project work

*Rationale:* Week 23 misses the week-24 meeting (papers closed at week 22), so the first reachable meeting is week 28 and the wait is 5 weeks against 3 drawn (6.1.2b). A calls a three-week lag conservative against a five-week event; C substitutes Domain 3's expectation for a *planned* submission whose date is known, which is the category error 6.1.2b names; D is the lag-that-hides-work pitfall stated as a principle.

**MCQ 6.1-G** [6.1.3 · Analysis] A 400-activity network reports 310 dependency links. Before looking at anything else, a reviewer can conclude that:

- A. the logic is admirably lean
- B. the network cannot be fully connected, so it contains dangles CORRECT
- C. the network is over-linked and float will be unresponsive
- D. nothing at all — link counts carry no information

*Rationale:* Connecting 400 activities needs at least 399 links, so 310 proves unbound activities exist (6.1.3). A mistakes a structural impossibility for economy; C is the opposite defect, which appears at densities far above 2; D denies a countable check that costs one division.

#### SELF-CHECK — KA 6.1

1. *Why must chosen logic be distinguishable from physical logic?* — Because recovery (KA 6.4) works by re-choosing choices; physics cannot be renegotiated.
2. *What three numbers give a fast schedule-quality read?* — Dangle count, constraint count, unexplained-lag count. Each should be at or near zero, with reasons for the rest.
3. *When is Domain 3's  $E[\text{wait}] = M/2 + L$  the wrong figure for a schedule?* — Whenever the submission date is planned rather than random: the wait is then a deterministic function of the finish date against the meeting calendar, ranging from  $L$  to  $M + L$  (2 to 6 weeks on Meridian).
4. *What does a logic density of exactly 1.00 on a seven-activity network tell you?* — That the network has one link more than the six needed to connect it, so it

contains exactly one convergence. And and that the metric is meaningless at this size.

## KNOWLEDGE AREA 6.2

### The critical path and float

*Topics: 6.2.1 the forward and backward passes · 6.2.2 total and free float · 6.2.3 reading the critical path.*

#### 6.2.1 The forward and backward passes

**Definitions.** For each activity: **ES/EF** — earliest start/finish (forward pass, left to right:  $ES = \text{latest } EF \text{ of predecessors}$ ;  $EF = ES + \text{duration}$ ); **LS/LF** — latest start/finish that do not delay the project (backward pass, right to left:  $LF = \text{earliest } LS \text{ of successors}$ ;  $LS = LF - \text{duration}$ ). The project duration is the largest **EF** in the network.

**Worked example 6.2.1 — Auriga, both passes.** The passes are inherently tabular; the labelled table replaces the five-step skeleton, and the interpretation follows (pattern rule).

| ACT | DUR | ES | EF | LS | LF | TF | FF |
|-----|-----|----|----|----|----|----|----|
| A   | 2   | 0  | 2  | 0  | 2  | 0  | 0  |
| B   | 6   | 2  | 8  | 2  | 8  | 0  | 0  |
| C   | 8   | 8  | 16 | 8  | 16 | 0  | 0  |
| D   | 7   | 8  | 15 | 9  | 16 | 1  | 0  |
| E   | 5   | 16 | 21 | 16 | 21 | 0  | 0  |
| F   | 4   | 21 | 25 | 21 | 25 | 0  | 0  |
| G   | 2   | 15 | 17 | 23 | 25 | 8  | 8  |

*Interpretation.* Project duration **25 weeks**; the **critical path is A–B–C–E–F** ( $2+6+8+5+4 = 25$ ), every activity on it with zero float. D can slip one week without moving the end date; G can slip eight. The passes turn a drawing into an instrument: every schedule statement a leader makes in this domain ("we finish in week 25", "D matters more than G", "procurement drives the job") is read directly off this table.

Three checks confirm the table before anything is read off it, and they are the same three a reviewer runs on any pass, human or machine. **The critical path's durations sum exactly to the project duration:  $2 + 6 + 8 + 5 + 4 = 25$ . Every**

**critical activity shows**  $ES = LS$  and  $EF = LF$ : A, B, C, E and F all do; D and G do not, and their gaps are precisely their floats. **Every activity's TF equals both**  $LS - ES$  and  $LF - EF$ : for G,  $23 - 15 = 8$  and  $25 - 17 = 8$ . A pass failing any of the three is arithmetically broken, and the failure is visible in seconds without re-running anything.

What the table also shows, and what the eye misses, is that **the second path is one week from binding**. A-B-D-E-F runs 24 weeks (4.00 % below the project duration), so the network's whole protection against a slip in civil works is a single week. That is the observation KA 6.2.3 turns into a monitoring decision, and it is why float is the number to report rather than the critical path's membership list.

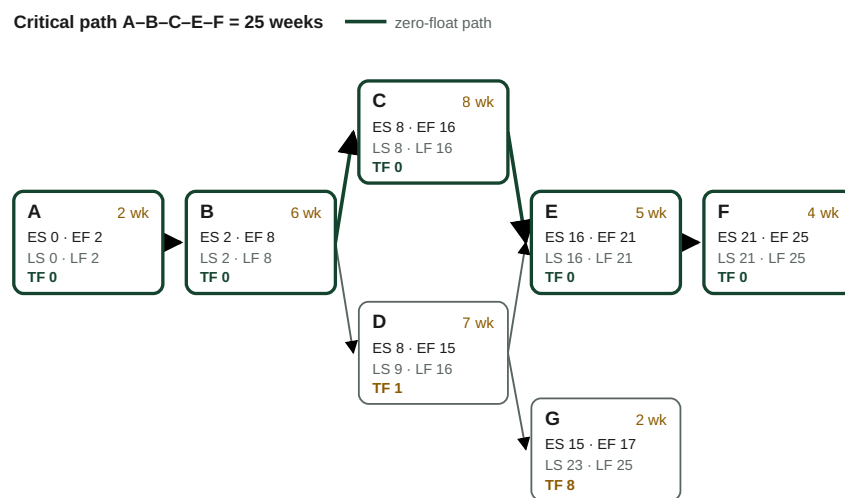


Fig 6.2.1 — The Auriga network with both passes

## 6.2.2 Total and free float

### Definitions.

$TF = LS - ES$  — how far an activity can slip without delaying the project  
 $FF = \min(ES \text{ of successors}) - EF$  — how far it can slip without delaying ANY successor

$FF \leq TF$  always. The Auriga table carries the domain's subtlest lesson: **D has TF = 1 but FF = 0**, D can slip a week without moving the end date, but the moment it slips at all, G's earliest start moves. Total float is a project-level commodity; free float is a courtesy to your successors. A leader who grants a subcontractor "the float" without saying which kind has just given away someone else's schedule.

**Float is not spare time.** Float belongs to the *path*, not the activity: consume D's week and every activity sharing that path inherits zero. Contract regimes differ on who owns float (Domain 10, KA 10.3, treats float ownership clauses); the delivery



answer is simpler: float is a risk buffer under the leader's governance, spent deliberately or not at all.

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#### WORKED EXAMPLE

##### Worked example 6.2.2 — why activity floats must never be added.

1. **Setup.** A float report for Auriga lists two activities with float: D ( $TF = 1$ ) and G ( $TF = 8$ ). A subcontract manager reads the report as "the network carries nine weeks of slack" and proposes to spend it. How much cumulative slip can the network actually absorb on that chain without moving week 25, and what does the difference cost?
2. **Formula.** Float is a property of a **path**, not an activity. For a chain of activities whose only route to the end is one path of length  $L_p$ , the absorbable slip is the *path* float  $PD - L_p$ , shared among every activity on it. Where an activity also lies on a tighter path, its own slip is additionally capped by that path's float.
3. **Substitution.** Path A–B–D–G runs  $2 + 6 + 7 + 2 = 17$  weeks against  $PD = 25$ , so its path float is **8**. Let D slip  $d$  and G slip  $g$ : the path finishes at  $17 + d + g$ , so  $d + g \leq 8$ . D also sits on A–B–D–E–F (24 weeks), which caps  $d \leq 1$ .
4. **Result.** The maximum cumulative slip the chain can absorb is **8 weeks** (for example  $d = 1$  and  $g = 7$ , or  $d = 0$  and  $g = 8$ ) against the **9** weeks the report's arithmetic implies. The report overstates the network's absorption capacity by **one week**, which at Auriga's USD 45,000 a week is **USD 45,000** of false comfort.
5. **Interpretation.** **The overstatement factor is the number of activities on the path.** For  $n$  activities lying on a single non-critical path with path float  $f$ , every one of them reports  $TF = f$ , so the float register sums to  $n \cdot f$  while the path can absorb only  $f$ . A ten-activity chain with five weeks of path float shows **50 weeks** of "float" and holds **five**. Summing an activity float column is therefore not an approximation but a category error, and it is a mistake that scales with the size of the schedule, which is why it is commonest on exactly the large networks where it does most damage.

The practical consequence of float being shared is that two parties can each spend "their" float in good faith and jointly overrun, which is why float is a *governed* commodity (the float policy of the Executive perspective) rather than an entitlement recorded against each activity.

**The reporting rule that follows is specific.** A float report is per **path**, not per activity: list each path with its length, its float and the activities on it, and state the float *remaining* after what has already been spent. Auriga's report has three lines, not seven. A report with one line per activity cannot be read correctly even by someone who knows all this, because the shared structure has been discarded before the reader sees it.

**The caution against over-applying it.** Free float *is* additive in one narrow sense (an activity's **FF** is a genuinely local quantity, spendable without consult-

ing anyone downstream), and this is exactly why **FF** is the right float to grant to a subcontractor and **TF** the wrong one. But **FF** is usually zero (D's is), so the honest answer to "how much float may we have?" is most often "none that is yours alone; here is what the path holds and who else is drawing on it". A counsel pointer belongs here, because float ownership is enforceability-sensitive: whether float is the contractor's, the employer's or the project's, and what a grant of it concedes about later delay claims, is a matter of the contract and the jurisdiction, varies between regimes and delay-analysis protocols, and should be taken from qualified legal advice (Domain 10, KA 10.3) rather than from a scheduling convention. The arithmetic above is contract-neutral; the answer given to the counterparty is not.

**Common pitfall: managing only the critical path.** D sits one week from critical. A team watching only A-B-C-E-F will discover D's importance the week it becomes too late to matter. Near-critical analysis (all paths within, say, 10 % of project duration) is standing practice; KA 6.4's scenario work builds on it, and KA 6.2.3 prices it.

### 6.2.3 Reading the critical path

The critical path is the *longest* path and therefore the *shortest possible duration*: both statements are the same fact, and fluency means holding both. Three leader-grade readings of the Auriga table: **(1)** procurement (C) is the single most schedule-driving activity — an expediting conversation is worth more than any amount of site pressure; **(2)** the C/D pair converge on E, so E's start is hostage to the later of two chains — convergence points are where slippage compounds and where progress review should focus (Domain 8, KA 8.1); **(3)** G's eight weeks of float make it the natural donor of resources in any recovery (KA 6.4).

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#### WORKED EXAMPLE

**Worked example 6.2.3 — choosing the near-critical band, and paying for it.**

- 1. Setup.** Auriga's team must decide what to monitor weekly. Enumerate every path, choose a near-critical band, and price the decision. Extending the weekly review to a second path costs about **0.5 engineer-weeks** per cycle for the **12** remaining cycles at **USD 5,225** an engineer-week (Domain 7, KA 7.4.1). A week of lateness costs **USD 45,000**.
- 2. Formula.** Path float = PD – path length. A path is near-critical if its float ≤ the chosen band, conventionally a percentage of PD. Monitoring pays if **cost ≤ weeks of lateness avoided × cost of delay**.
- 3. Substitution.** Paths: A-B-C-E-F = 2+6+8+5+4 = 25; A-B-D-E-F = 2+6+7+5+4 = 24; A-B-D-G = 2+6+7+2 = 17. Band at 10 %: 0.10 × 25 = 2.5 weeks. Monitoring cost 0.5 × 12 × 5,225.

#### 4. Result.

| PATH      | LENGTH<br>(WK) | FLOAT<br>(WK) | FLOAT AS % OF<br>PD | INSIDE A 10 %<br>BAND? |
|-----------|----------------|---------------|---------------------|------------------------|
| A-B-C-E-F | 25             | 0             | 0.00 %              | yes (critical)         |
| A-B-D-E-F | 24             | 1             | 4.00 %              | <b>yes</b>             |
| A-B-D-G   | 17             | 8             | 32.00 %             | no                     |

Two of three paths fall inside the band, and because they share A, B, E and F they cover **6 of 7** activities, **85.71 %** of the network. Monitoring cost  $0.5 \times 12 \times 5,225 = \text{USD } 31,350$ , recovered by avoiding  $31,350 / 45,000 = 0.6967$  weeks of lateness. 5. **Interpretation. The band captures almost everything, and that is the point.** On a well-formed network the near-critical set is not a short annexe to the critical path but most of the schedule, here 85.71 % of activities. "Manage the critical path" is therefore not a management system; it is a *reporting* convention that survives because it fits on a slide. What the leader actually needs is the float column sorted ascending, with a line drawn where the band falls.

**\*\*The band must be wider than the largest float you are unwilling to lose.\*\*** A-B-D-E-F's float is 1 week, which is **\*\*4.00 %\*\*** of 25, so any band of 4 % or more captures it and any band below 4 % does not. A team that had adopted a 2 % band (0.5 weeks) would monitor the critical path alone and would be blind to the path that is one week from binding. The percentage is not a convention to be inherited; it is a decision made by asking *\*which paths would I want to hear about?\** and then setting the band to include them. Quoting a band without saying what it captures is the same defect as quoting a confidence level without a date (Domain 8, KA 8.A.2).

**\*\*The monitoring economics are not close.\*\*** Watching a second path to handover costs USD 31,350 and pays for itself if it prevents **\*\*0.70 weeks\*\*** of lateness once. If D slips two weeks undetected, E cannot start until week 17 and F finishes week 26: a full week late, **\*\*USD 45,000\*\***, or **\*\*1.44 times\*\*** the entire cost of monitoring. Near-critical analysis is one of the few schedule controls whose case is so lopsided that it should not need making; it is omitted for want of the arithmetic rather than for want of the money.

**\*\*The cautions.\*\*** Path enumeration is trivial on seven activities and combinatorially hopeless on five thousand, where the practical instrument is the *\*float histogram\** (count activities by float band and read the shape) rather than a path list. Path float is computed against the *\*current\** network, so it changes with every crash and every slip (KA 6.4.2's path migration); a band chosen once and never revisited will be monitoring last quarter's near-critical set. And float measured against logic is not float against **\*\*capacity\*\***: KA 6.3.1c shows an activity with eight weeks of logical float extending the project once a resource cap binds, so a watch-list drawn only from the float column can still miss what will actually make the project late.

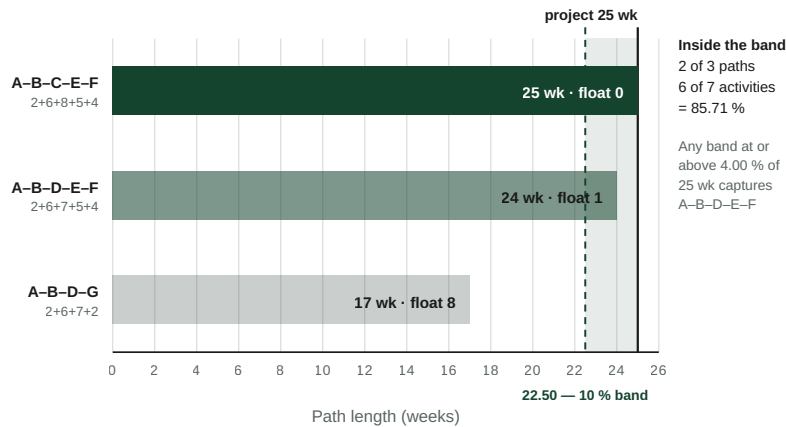


Fig 6.2.2 — Auriga's three paths, their floats, and the near-critical band

### AI in this KA

CPM arithmetic is deterministic, ideal for machine computation and instant human verification. The two checks that catch nearly every pass error: the critical path's durations must **sum exactly** to the project duration, and every critical activity must show  $TF = 0$  with  $ES = LS$ . An AI-produced schedule that fails either is broken; one that passes both may still have wrong *logic*, which is why KA 6.1's link-walking discipline precedes trust in any pass, human or machine.

#### KEY TERMS — KA 6.2

| TERM                           | MEANING                                                                                       |
|--------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Forward / backward pass</b> | Left-to-right ES/EF computation; right-to-left LS/LF computation.                             |
| ES EF LS LF                    | Earliest/latest start and finish per activity.                                                |
| <b>Total float TF</b>          | $LS - ES$ ; slip available without delaying the project.                                      |
| <b>Free float FF</b>           | Slip available without delaying any successor; $FF \leq TF$ .                                 |
| <b>Critical path</b>           | The zero-float longest path; the project's shortest possible duration.                        |
| <b>Convergence point</b>       | A node where paths merge; slippage compounds there.                                           |
| <b>Path float</b>              | $PD$ – path length; the slip a whole chain can absorb, shared by every activity on it.        |
| <b>Near-critical band</b>      | The float threshold below which a path joins the monitored set; a decision, not a convention. |

## SAMPLE MCQs — KA 6.2

**MCQ 6.2-A** [6.2.1 · Application] In the Auriga network, activity D (duration 7, predecessor B finishing week 8, successors E and G) has  $LS = 9$  and  $ES = 8$ . Its total float is:

- A. 0
- B. 1 CORRECT
- C. 8
- D. 7

*Rationale:*  $TF = LS - ES = 9 - 8 = 1$ . A confuses D with the critical activities; C is G's float; D is the activity's duration, not its float.

**MCQ 6.2-B** [6.2.2 · Analysis] An activity shows  $TF = 1$  and  $FF = 0$ . Delaying it by one week will:

- A. delay the project by one week
- B. delay nothing, because total float absorbs the slip
- C. delay at least one successor's earliest start while leaving the project end date unchanged CORRECT
- D. be impossible — free float can never be below total float

*Rationale:* Positive  $TF$  protects the end date (not A); zero  $FF$  means some successor moves immediately (not B). D inverts the invariant,  $FF \leq TF$  always.

**MCQ 6.2-C** [6.2.3 · Application] Auriga's procurement activity C is crashed from 8 weeks to

1. The new project duration is:
2. A. 23 weeks
3. B. 24 weeks CORRECT
4. C. 25 weeks
5. D. 21 weeks

*Rationale:* After one week of crashing, path A–B–D–E–F ( $2+6+7+5+4$ ) becomes co-critical at 24; the second crashed week buys nothing, duration stays 24. A assumes both weeks convert to project weeks; C forgets the crash entirely; D subtracts from the wrong baseline. (The full economics: KA 6.4.)

**MCQ 6.2-F** [6.2.1 · Application] Auriga's procurement C slips from 8 weeks to 9 (all else unchanged). The new project duration is:

- A. 25 weeks
- B. 26 weeks CORRECT
- C. 27 weeks
- D. 24 weeks

*Rationale:* C was critical with zero float, so its extra week passes straight through: E runs 17–22, F finishes week 26. A assumes float absorbed a critical activity's slip; C adds the week twice; D subtracts it.

**MCQ 6.2-D** [6.2.1 · Application] Auriga's training activity G runs ES 15–EF 17 with LF = 25. Its total float is:

- A. 8 CORRECT
- B. 10
- C. 2
- D. 0

*Rationale:*  $TF = LF - EF = 25 - 17 = 8$  (equivalently  $LS - ES = 23 - 15$ ). B ignores G's duration ( $25 - 15$ ); C is G's duration; D confuses G with the critical path.

**MCQ 6.2-E** [6.2.3 · Analysis] A programme's binding path shows  $TF = -3$  after a constraint-honest pass. The professional reading is:

- A. the software has malfunctioned — float cannot be negative
- B. the plan, as constrained, is already three weeks late; the number sizes the recovery problem and must be reported now CORRECT
- C. delete the constraints so the float returns to zero
- D. the project will finish three weeks early

*Rationale:* Negative float is the gap between what logic needs and what constraints allow, information, not error (A) and not good news (D). C is the pin-and-squeeze suppression this domain's Case B dissects; it hides the problem until it is unrecoverable.

**MCQ 6.2-G** [6.2.2 · Analysis] A float report for Auriga lists D at  $TF = 1$  and G at  $TF = 8$ . The cumulative slip the D–G chain can absorb without moving week 25 is:

- A. 9 weeks — the sum of the two floats
- B. 8 weeks, because both activities draw on the same path float of 8 CORRECT
- C. 1 week, the smaller of the two floats
- D. 7 weeks, since D must retain a week

*Rationale:* Path A–B–D–G runs 17 weeks against a 25-week project, so  $d + g \leq 8$  (6.2.2). A sums a shared quantity, the error that scales with schedule size, a ten-activity chain with five weeks of path float reports fifty. C takes the minimum instead of the shared total; D invents a reservation rule that no pass produces.

**MCQ 6.2-H** [6.2.3 · Evaluation] A team adopts a near-critical band of 2 % of project duration on Auriga. The consequence is:

- A. none — 2 % is a standard threshold

- B. path A–B–D–E–F, whose float is 4.00 % of the 25-week duration, falls outside the band and goes unmonitored CORRECT
- C. all three paths are monitored, since 2 % is stricter
- D. the band is invalid because bands must be at least 10 %

*Rationale:* The band is 0.5 weeks and A–B–D–E–F's float is 1 week, so the second path (one week from binding) is excluded (6.2.3). A appeals to a convention nobody derived; C inverts what "stricter" does to a threshold; D invents a rule, when the correct test is whether the band captures the paths the leader wants to hear about.

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#### SELF-CHECK — KA 6.2

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1. *State both definitions of the critical path and why they are equivalent.* — Longest path through the network; shortest possible project duration: the longest chain of unavoidable work is precisely what bounds how soon the whole can finish.
  2. *Why is  $FF \leq TF$  always?* — Delaying any successor is one of the ways of delaying the project; the constraint set defining  $FF$  is stricter.
  3. *Auriga slips: D takes 10 weeks, not 7. New duration?* — 27 weeks: D finishes week 18, E waits for it ( $18 > 16$ ), E–F add 9  $\rightarrow$  the critical path re-routes through D (KA 6.4 works the recovery).
  4. *Why can a float column never be summed?* — Because every activity on a path reports that path's float:  $n$  activities on a path with float  $f$  report  $n \cdot f$  and the path holds  $f$ . Auriga's report implies 9 weeks and the chain absorbs 8.
  5. *Three checks that confirm a pass in seconds?* — Critical durations sum to the project duration; critical activities show  $ES = LS$  and  $EF = LF$ ; every  $TF$  equals both  $LS - ES$  and  $LF - EF$ .
- 

#### KNOWLEDGE AREA 6.3

### Resources, constraints, milestones and rolling wave

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*Topics:* 6.3.1 resource planning and levelling · 6.3.2 constraints and milestones · 6.3.3 rolling-wave planning.

#### 6.3.1 Resource planning and levelling

**The principle.** The passes of KA 6.2 assume infinite resources; reality staffs the network. Loading resources onto the schedule produces a **histogram**, demand per period, and the histogram almost always spikes. Two responses: **resource smoothing** uses float to flatten demand *without* moving the end date (slip D and G inside their float to pull crews off the peak); **resource levelling** accepts a later end date if the resource cap is hard. The order of operations is a leadership decision in disguise:



smoothing spends float (the project's risk buffer) to save money, and someone accountable should price that trade explicitly, not discover it in a progress meeting.

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#### WORKED EXAMPLE

##### Worked example 6.3.1 — smoothing Auriga's field crews.

1. **Setup.** Weeks 9–15: civil works D needs 3 crews; early installation staging inside C needs 2; the site cap is 4 crews. Overlap weeks 9–15 demand 5, one over cap.
  2. **Formula.** Smoothing test: can a demanding activity slip within **TF** (and, if successors matter, **FF**) to clear the peak?
  3. **Substitution.** D has **TF** = 1, **FF** = 0: slipping D one week clears one peak week but consumes D's entire float and immediately moves G (**FF** = 0). The staging work inside C has discretionary timing across weeks 9–16.
  4. **Result.** Re-sequence the staging within C to weeks 14–16 where D's demand tails off: peak demand falls to 4, the cap holds, the end date holds, and **no float is spent**.
  5. **Interpretation.** The cheapest capacity is re-sequencing; the second cheapest is float; the expensive ones are overtime, second shifts and extension. A leader asks for the histogram *with the options priced*, not just the spike.
- 

#### WORKED EXAMPLE

##### Worked example 6.3.1b — when the cap is hard: levelling, priced.

1. **Setup.** Now suppose the site cap is **3 crews** (a permit condition, not a preference). D needs all 3 for weeks 9–15; the 4-week staging package (2 crews) must finish before E can start (week 16). Delay costs USD 45,000 per week (the client's bonus forgone); a second-shift waiver for the staging crew costs USD 20,000 per week. What does the leader do?
2. **Formula.** Compare total cost of each feasible plan: pure levelling (extend) vs paid capacity (second shift) vs any float-funded re-sequence.
3. **Substitution.** Levelling: staging cannot start until D releases crews (week 16); it runs weeks 16–19, E starts week 20, project 25 → 29, cost  $4 \times 45,000 = 180,000$ . Second shift: staging runs weeks 12–15 off-shift, E starts on time — cost  $4 \times 20,000 = 80,000$ . No float-funded option exists: D's **TF** = 1 cannot host a 4-week move.
4. **Result.** **Buy the second shift: USD 80,000** against USD 180,000 of extension, and the end date holds.
5. **Interpretation.** A hard cap converts a scheduling problem into a procurement problem. The histogram's job is to surface that conversion early enough for the cheap option to exist: second shifts, pre-assembly, off-site staging all have lead

times of their own. Levelling that silently extends the project is a decision someone should have been asked to make.

**The breakeven is wide, and saying so is what makes the recommendation robust.** The shift wins for any premium below **USD 45,000** a week (the cost of delay itself), so at USD 20,000 the shift is bought at **44.4 %** of the price at which the leader would become indifferent, and the quoted premium could rise **2.25 times** before extension became the better plan. That is the sentence for the steering paper, because it answers the only question worth asking about a quoted price: how wrong can it be before the decision changes? The corollary is equally useful in the other direction, where the cost of delay is small, levelling into an extension is the *correct* answer and paying a premium to hold a date is the error.

**The comparison is only valid because both plans deliver the same scope.** Levelling to week 29 and second-shifting to week 25 produce the same works; if the second shift also carried a quality consequence (night-shift error rates are not day-shift error rates) the comparison would need Domain 9's containment economics on the other side of the ledger, and USD 80,000 would not be the whole price. A reviewer's question here is always *what else changes when the shift pattern changes?*

**Two cautions on the arithmetic.** The USD 45,000 assumes the extension actually costs four weeks of delay cost, which holds only because E is on the critical path with zero float: the same four-week levelling on a path with four weeks of float would cost nothing at all, and the histogram cannot tell you which case you are in. And the permit condition that fixes the cap at three crews is a genuine external constraint, not a preference: the option of simply exceeding it does not exist, and a plan that quietly assumes a waiver will be granted is a plan with an unpriced risk in it (Domain 8's register, not the schedule's float).

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#### WORKED EXAMPLE

##### Worked example 6.3.1c — the resource-feasible schedule, and why the critical path is not the critical chain.

1. **Setup.** Both examples above levelled one activity against a cap. This one levels the **whole network** against a hard cap, which is the problem a planner actually faces. Auriga's specialist **engineering pool** (a different resource from the field crews above, and levelled separately, which is itself part of the difficulty) is capped at **6 engineers**. Each activity's engineer demand while it runs:

| ACTIVITY      | A | B | C | D | E | F | G |
|---------------|---|---|---|---|---|---|---|
| Duration (wk) | 2 | 6 | 8 | 7 | 5 | 4 | 2 |
| Engineers     | 2 | 4 | 1 | 3 | 5 | 6 | 2 |

| ACTIVITY      | A | B | C | D | E | F | G |
|---------------|---|---|---|---|---|---|---|
| TF (KA 6.2.1) | 0 | 0 | 0 | 1 | 0 | 0 | 8 |

Activities may not be split. What is the shortest resource-feasible duration, what does the standard priority rule give, and what should the leader buy? 2. **Formula.** Load the early-start schedule and read the histogram: demand in week  $t = \sum \text{engineers of activities running in } t$ . Where demand exceeds the cap, defer activities by priority (conventionally **minimum total float first**), and re-check. Total demand =  $\sum (\text{duration} \times \text{engineers})$ ; average utilisation =  $\text{total demand} \div (\text{cap} \times \text{duration})$ . 3. **Substitution.** Total demand  $2 \times 2 + 6 \times 4 + 8 \times 1 + 7 \times 3 + 5 \times 5 + 4 \times 6 + 2 \times 2 = 4 + 24 + 8 + 21 + 25 + 24 + 4 = 110$  engineer-weeks against  $6 \times 25 = 150$  of capacity. Early-start profile: weeks 0–2 A alone (2); 2–8 B (4); 8–15 C+D ( $1+3 = 4$ ); 15–16 C+G ( $1+2 = 3$ ); **16–17 E+G** ( $5+2 = 7$ ); 17–21 E (5); 21–25 F (6). 4. **Result.** Average utilisation  $110/150 = 73.33\%$ , and the cap is still breached, by **one engineer, in the single week 16–17**, where E (5) overlaps G (2). Three plans follow.

| PLAN                            | WHAT IT DOES                                                                  | DURATION     | COST CONSEQUENCE VS THE 25-WEEK PLAN           |
|---------------------------------|-------------------------------------------------------------------------------|--------------|------------------------------------------------|
| Minimum-float priority rule     | G (TF 8) yields to E (TF 0); G then finds no 2-engineer window until F ends   | <b>27 wk</b> | 2 weeks of delay = <b>USD 90,000</b>           |
| Best resource-feasible schedule | G runs weeks 15–17; <b>E is deferred one week</b> to 17–22; F 22–26           | <b>26 wk</b> | 1 week of delay = <b>USD 45,000</b>            |
| Buy the peak out                | Hire <b>one</b> engineer for the single week 16–17; G 15–17, E 16–21, F 21–25 | <b>25 wk</b> | one engineer-week purchased = <b>USD 5,225</b> |

2. **Interpretation. Float against logic is not float against capacity.** G carries **eight weeks** of total float and is nevertheless the activity that makes the project late: under the cap it has nowhere to go, because every week between its earliest start and the project end is already spending five or six of the six available engineers. The set of activities that determines a resource-constrained duration is the **critical chain** (the binding sequence once both logic *and* capacity are honoured), and it need not be the critical path. Any statement of the form "G doesn't matter, it has eight weeks of float" is a statement about the logic network alone, and the logic network is not the project.

**The standard priority rule loses money.** Protecting the zero-float activity is the instinctive and usually correct move, and here it produces **27 weeks** where deliberately slipping E by one week produces **26**: a difference of **USD 45,000** left on the table by following the rule. The reason is structural: deferring G instead

pushes it past a **nine-week** block (E's five weeks then F's four) in which no capacity is ever free, so the rule's refusal to defer the critical activity by one week costs the project two. Whether protecting the critical path is right depends on what lies behind it, which no priority rule inspects. Resource-constrained scheduling is a combinatorial optimisation, not a rule application; priority rules are **heuristics**, they are known to be beatable, and a planner who presents a levelled schedule should say which rule produced it and whether anything better was sought. This is precisely where an optimiser earns its place (see *AI in this KA*), and it is the honest reason to use one: not speed, but the USD 45,000.

**Averages cannot answer capacity questions.** The pool is **73.33 % utilised on average** across a schedule it cannot execute, and in the feasible 26-week plan it is only **70.51 %** utilised. Any resourcing conversation conducted in averages ("we have plenty of engineering capacity, we're only three-quarters loaded") is a category error, because feasibility is a property of the **peak** and the peak is a property of the logic. The histogram is not a presentational nicety; it is the only instrument in the domain that answers the question the average appears to answer.

**The whole overrun traces to one engineer-week, and that ratio is the lesson.** The breach is one engineer for one week. Buying it costs **USD 5,225** and saves **USD 45,000** (a return of **8.61 times**), because a single week of excess demand at the wrong point in the network converts into a full project week. The general habit: price the *excess area* of the histogram in resource-weeks, compare it with the cost of delay, and expect the ratio to be startling. The breakeven price for that engineer-week is **USD 45,000**, or **8.61 times** the internal rate, which is why hiring in at a premium is so often right and so rarely proposed.

**And optimise before you buy, because a bad baseline inflates what capacity looks worth.** Measured against the priority rule's 27 weeks, that one engineer-week appears to save **two** weeks and to be worth up to **USD 90,000: 17.22 times** the internal rate. Measured against the correct 26-week baseline it saves one and is worth up to **USD 45,000**. A leader negotiating a hire-in rate against the un-optimised schedule would rationally pay **twice** the defensible price, which is a general property of buying capacity against a heuristic baseline rather than a feature of this network.

**Three cautions.** The no-splitting rule is doing real work here: allowing G to run as two separate single weeks changes the answer, and whether splitting is permissible is a physical and contractual question, not a software setting. The example levels one resource; real projects level several simultaneously, which is why the problem is hard rather than tedious, and why a plan feasible on engineers can be infeasible on cranes. And the 6-engineer cap must be *real*: a cap that is actually a budget line is negotiable, and levelling against a negotiable cap manufactures a delay that a conversation would have removed.

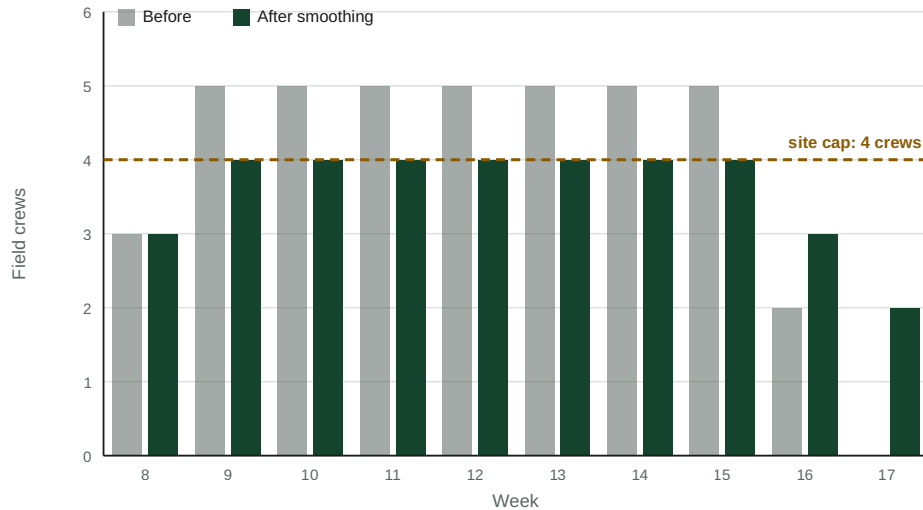


Fig 6.3.1 — Auriga field-crew histogram, before and after smoothing

### 6.3.2 Constraints and milestones

**Constraints** are externally imposed dates, an outage window, a regulatory deadline, a seasonal limit. Modelled honestly they are logic (an outage window is a predecessor with a date); modelled lazily they are pins that falsify float (KA 6.1.3). **Milestones** are zero-duration events that carry meaning: contractual obligations (Domain 10), gate decisions (Domain 3), integration points with other projects (Domain 15). A milestone's date is an *output* of the network, never an input. The moment a milestone is typed rather than computed, the schedule has stopped forecasting and started decorating.

**Milestone hygiene:** each has an owner, an unambiguous done-definition (Domain 5's acceptance criteria), traceable L3 logic, and (for contractual milestones) a stated float position reported honestly to the counterparty (Domain 11's reporting ethics).

### 6.3.3 Rolling-wave planning

**The principle.** Detail decays with distance: estimating month 18's tasks at daily granularity today manufactures precision, not knowledge. **Rolling wave** plans near-term work at execution detail (L4) and far work at planning packages (L3 summary), elaborating each wave as it approaches under change control.

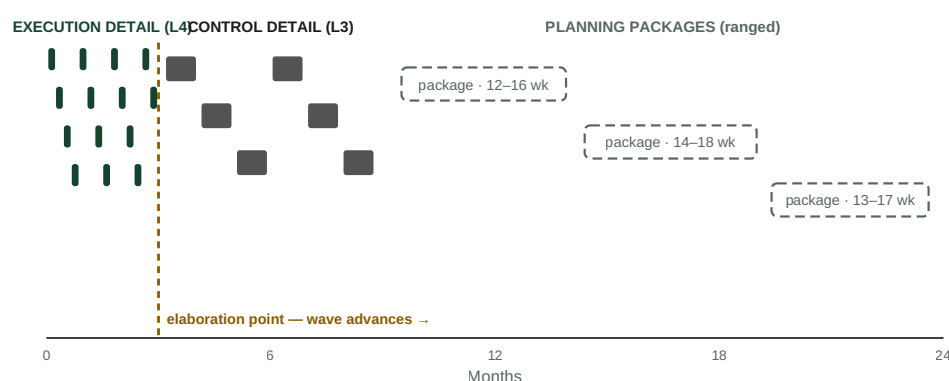


Fig 6.3.2 — The rolling-wave horizon

## WORKED EXAMPLE

**Worked example 6.3.3 — what a ranged planning package may be committed to.**

1. **Setup.** Meridian's wave-3 clinic tranche is a far-wave planning package with a ranged duration of **14–18 weeks**, starting at programme **week 40**. The sponsor asks for a go-live date for the tranche. Elaborating the package (a workshop with the estate, clinical and training leads, **3 planner-weeks** at **USD 4,200**) would narrow the range to **15.5–16.5 weeks**. Cost of delay **USD 14,280** per week. What date may be committed, and is the elaboration worth buying first?
2. **Formula.** Milestone range = start + duration range. Commitment exposure = range width × cost of delay. Elaboration is worth buying if the exposure it removes exceeds its cost: noting that elaboration changes the *precision* of the promise, not the duration of the work.
3. **Substitution.** Milestone range  $40 + 14 = \text{week } 54$  to  $40 + 18 = \text{week } 58$ ; mid-point week 56. Exposure before =  $4 \times 14,280$ ; after =  $1 \times 14,280$ . Elaboration cost =  $3 \times 4,200$ .
4. **Result.** The tranche completes somewhere in **weeks 54–58**, mid-point **56**. Commitment exposure is **USD 57,120** before elaboration and **USD 14,280** after, a reduction of **USD 42,840** for **USD 12,600** of planning effort, a ratio of **3.40**. Committing to the mid-point and missing by two weeks costs **USD 28,560**.
5. **Interpretation.** **The mid-point is a coin toss, not a commitment.** Week 56 is the centre of a range, so a commitment to it is a commitment with roughly even odds. And the *appearance* of a firm date is what makes it dangerous, because nobody downstream will treat "week 56" as a 50 % statement. Exactly three honest options exist: commit to the upper bound (week 58, and hold the two weeks internally as the buffer they are), commit to the mid-point with the range stated beside it, or decline to commit until the package is elaborated. Which of the three is chosen is a governance decision (Domain 3), and Domain 8's methods set the confidence at which a buffered date should sit. What is not

available is the fourth option everyone reaches for: quoting the mid-point as though it were the upper bound.

**Elaboration is a purchasable reduction in commitment error, and it can be priced.** USD 12,600 buys a **3.40-fold** return in exposure terms, and the arithmetic is available before the workshop is convened, which is what turns "we should elaborate wave 3" from an aspiration into a funded activity with an owner and a date. The general rule: elaborate when the exposure removed exceeds the elaboration cost, and elaborate *before* committing rather than after; the reverse order is how a programme acquires a public date it then has to defend.

**Ranges must not be rolled up by adding mid-points.** Three consecutive tranches each ranged 14–18 weeks do not make a 48-week programme with a 12-week range; the aggregate spread is narrower than the sum of the spreads where the tranches are independent, and wider where they share crews, estate or approvals. Domain 8, KA 8.2.4 supplies the aggregation, and this is the single most common quantitative error in rolling-wave reporting.

**The warning.** Elaboration does not shorten the work, and a leader who reads the USD 42,840 as a saving has misread it: nothing was saved, a promise merely became more accurate. Nor does narrowing the range make the mid-point right, the elaborated range 15.5–16.5 could sit anywhere inside the original 14–18, so the elaborated *date* may be later than the date the sponsor was informally given. That is the moment the honesty of a rolling-wave regime is actually tested, and it is why the elaboration event needs a re-baselining rule agreed in advance (Domain 4, KA 4.3) rather than negotiated in the meeting where the number appears.

## AI in this KA

Resource optimisation is a genuine AI strength: levelling across thousands of activities and dozens of calendars is combinatorial work machines do better than planners. Governance holds the same shape: the optimiser *proposes* a levelled plan; the planner verifies the constraint set was real (crew interchangeability, shift rules, site logistics); the leader owns the trade the optimiser cannot see; whether spending float, money or scope is the right currency this month. An optimiser given a wrong constraint produces a beautifully efficient fiction.

**Why the case is unusually strong here, and what still has to be checked.** KA 6.3.1c is the argument in miniature: the priority rule every planner is taught produced 27 weeks where 26 was available, and USD 45,000 turned on a search nobody had time to run by hand. Resource-constrained scheduling is genuinely hard (the search space grows combinatorially with activities, resources and calendars), so this is one of the few places in delivery where a machine does not merely accelerate a human process but reaches answers humans systematically do not. The verification duties are correspondingly specific. **Check feasibility, not optimality:** re-derive

the resource histogram from the proposed schedule and confirm no period exceeds any cap, which is one addition per period and catches the commonest failure: an optimiser that has quietly relaxed a constraint it could not satisfy. **Check that the objective was the right one:** minimising duration, minimising peak demand and minimising cost give three different schedules, and a tool asked for the wrong one returns an excellent answer to a question nobody had. **Check what was allowed to move:** splitting, overtime, calendar exceptions and crew substitution are assumptions with contractual and physical consequences, and an optimiser silently permitted to split an activity has produced a plan the site cannot run. And **compare against a stated baseline** (the priority-rule schedule), so that the improvement is a number rather than a claim.

## KEY TERMS — KA 6.3

| TERM                              | MEANING                                                                                                         |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Resource histogram</b>         | Demand per period from the loaded schedule.                                                                     |
| <b>Smoothing / levelling</b>      | Flattening demand within float / accepting a later date under a hard cap.                                       |
| <b>Constraint</b>                 | An externally imposed date; modelled as logic, not as a pin.                                                    |
| <b>Milestone</b>                  | Zero-duration event with an owner and a done-definition; its date is computed.                                  |
| <b>Rolling wave</b>               | Near-term detail, far-term packages, elaborated under change control.                                           |
| <b>Planning package</b>           | A far-wave summary activity with ranged duration awaiting elaboration.                                          |
| <b>Resource-feasible schedule</b> | A schedule in which no period's demand exceeds any resource cap.                                                |
| <b>Critical chain</b>             | The binding sequence once logic <i>and</i> capacity are honoured; need not be the critical path.                |
| <b>Priority rule</b>              | The heuristic (commonly minimum total float first) used to break resource conflicts; beatable, and known to be. |
| <b>Commitment exposure</b>        | Range width × cost of delay, what a date quoted from a ranged package risks.                                    |

## SAMPLE MCQs — KA 6.3

**MCQ 6.3-A [6.3.1 · Application]** A planner clears a resource peak by delaying an activity with  $TF = 1$ ,  $FF = 0$  by one week. The immediate consequences are:

- A. no schedule effect of any kind
- B. project delay of one week



- C. the end date holds, the activity's path float is exhausted, and at least one successor's earliest start moves CORRECT
- D. the resource peak worsens

*Rationale:* One week inside TF protects the end date (not B), but zero FF moves a successor and the path's buffer is now spent (not A). The smoothing was legal but not free.

**MCQ 6.3-B** [6.3.3 · Analysis] A 30-month programme shows daily-level tasks throughout, including month 30. The strongest inference is:

- A. the planning team is unusually diligent
- B. far-horizon detail is manufactured precision that will churn on every update, obscuring real variance CORRECT
- C. the schedule will need no re-baselining
- D. rolling-wave planning has been correctly applied

*Rationale:* Detail beyond the knowable horizon creates update churn and false confidence, the opposite of diligence in effect (A) and the opposite of rolling wave (D); constant churn makes re-baselining more likely, not less (C).

**MCQ 6.3-E** [6.3.1 · Application] D needs 3 crews in weeks 9–15 and concurrent staging needs 2; the site cap is 4. The excess demand the histogram must clear is:

- A. 1 crew for 7 weeks CORRECT
- B. 5 crews for 7 weeks
- C. 2 crews for 7 weeks
- D. nothing —  $3 + 2 = 5$  is within a 4-crew cap across two activities

*Rationale:* Peak demand 5 against cap 4 leaves one excess crew-week in each of the seven overlap weeks. The precise quantity smoothing must relocate. B is total demand, not excess; C is the staging demand itself; D misreads the cap as per-activity when it is per-site.

**MCQ 6.3-C** [6.3.1 · Application] A hard 3-crew cap forces either a 4-week project extension (delay cost USD 45,000/week) or a second-shift waiver at USD 20,000/week for the same 4 weeks (end date held). The value-maximising choice and its saving are:

- A. extend: saves USD 100,000
- B. second shift: saves USD 100,000 CORRECT
- C. second shift: saves USD 25,000
- D. they cost the same

*Rationale:* Extension costs  $4 \times 45,000 = 180,000$ ; the shift premium  $4 \times 20,000 = 80,000$  — choosing the shift saves USD 100,000 and keeps the date. A

picks the dearer plan; C compares one week only; D ignores the 65,000-per-week difference... which compounds four times.

**MCQ 6.3-D** [6.3.3 · Recall] A legitimate planning package in a rolling-wave schedule must carry:

- A. daily-level task detail for its whole span
- B. a ranged duration consistent with its estimate class and a dated elaboration event under change control CORRECT
- C. zero float, to keep pressure on the team
- D. a pinned finish date agreed with the sponsor

*Rationale:* Far-wave honesty is ranged duration plus a governed elaboration point. A is the manufactured precision rolling wave exists to avoid; C fabricates criticality; D is the typed milestone anti-pattern (Case study B).

**MCQ 6.3-F** [6.3.1 · Analysis] Auriga's engineering pool is capped at 6. Activity G needs 2 engineers for 2 weeks and carries 8 weeks of total float, but every week from its earliest start to the project end already commits 5 or 6 engineers. The correct conclusion is:

- A. G is irrelevant to the end date, because it has eight weeks of float
- B. G determines the resource-constrained duration despite its logical float, because float against logic is not float against capacity CORRECT
- C. the cap must be wrong, since average utilisation is only 73.33 %
- D. G should be deleted from the network and tracked separately

*Rationale:* Under the cap G has nowhere to go, so it joins the critical chain even though its total float is 8 (6.3.1c). A applies a logic-network statement to a capacity problem; C reasons from an average when feasibility is a property of the peak; D removes the activity that is about to make the project late from the model that would show it.

**MCQ 6.3-G** [6.3.1 · Evaluation] A levelled Auriga schedule produced by the minimum-total-float priority rule runs 27 weeks; a search finds a feasible 26-week schedule that deliberately defers the critical activity E by one week. The professional reading is:

- A. the 26-week schedule is invalid, because critical activities must never be deferred
- B. the priority rule is a heuristic and here costs USD 45,000; a levelled schedule should state the rule used and whether better was sought CORRECT
- C. both schedules are equally good, since both respect the cap
- D. the rule is wrong and should never be used

*Rationale:* Deferring E by one week releases G two weeks earlier, so the "protect the critical path" instinct loses a week worth 45,000 (6.3.1c). A elevates a heuristic to a constraint; C ignores a one-week difference in duration; D over-corrects; the rule is a sound default whose output needs a stated baseline, not abolition.

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#### SELF-CHECK — KA 6.3

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1. *Why is smoothing "spending the risk buffer"?* — It consumes float, and float is the project's absorption capacity for the unknown (KA 6.2.2).
  2. *What makes a milestone honest?* — Computed date, named owner, unambiguous done-definition, traceable logic.
  3. *What must a planning package carry to be legitimate?* — A ranged duration consistent with its estimate class and a dated elaboration event under change control.
  4. *Why can average utilisation never establish that a resource plan is feasible?* — Because feasibility is a property of the peak: Auriga breaches a 6-engineer cap while only 73.33 % loaded on average.
  5. *What is the difference between the critical path and the critical chain?* — The critical path binds on logic alone; the critical chain binds once capacity is honoured too, and on Auriga it includes G, an activity with eight weeks of logical float.
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#### KNOWLEDGE AREA 6.4

## Delivery flow: predictive, agile and hybrid; recovery and forecasting

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*Topics:* 6.4.1 flow across delivery modes · 6.4.2 schedule compression and recovery · 6.4.3 forecasting and scenario analysis.

### 6.4.1 Flow across delivery modes

**Predictive** scheduling (this domain so far) plans the whole network and measures variance against baseline. **Agile** delivery (Domain 13 in full) fixes cadence and capacity and lets scope flow: the schedule questions become *throughput* (items per iteration), *cycle time* and *forecast by velocity*. **Hybrid** programmes, the commonest reality, run engineered streams predictively and product streams by cadence, joined at **integration milestones** that appear in the CPM network as fixed-capacity deliveries. The leader's job is translation: a velocity forecast ("the reporting module completes in 4–6 sprints") enters the network as a ranged duration on the integration activity, and the network returns what the programme needs from the agile stream: the latest acceptable delivery, read from **LS**. Neither mode is senior; the

network prices *time*, the cadence protects *focus*, and governance (Domain 3, KA 3.1.3) holds both to evidence.

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WORKED EXAMPLE

**Worked example 6.4.1 — the network's answer to an agile stream is a rate, not a date.**

1. **Setup.** Auriga's operator-interface configuration is delivered by a cadence team working from a **72-item** backlog. Over its last eight iterations the team's throughput has run between **3.6** and **4.5** items a week, averaging **4.0**. The package must be complete before testing and commissioning **F** begins, and F's backward pass gives  $LS(F) = 21$  (KA 6.2.1). The stream cannot start until the design freeze at **week 5**. Domain 13 supplies the throughput forecasting method; this example does the translation.
2. **Formula.** Forecast duration = backlog  $\div$  throughput, computed across the observed throughput range to give a ranged duration. Available window = LS of the successor – stream start. **Required throughput = backlog  $\div$  available window**. Deliverable scope at a given throughput = throughput  $\times$  window.
3. **Substitution.** Durations:  $72/4.5 = 16.0$ ,  $72/4.0 = 18.0$ ,  $72/3.6 = 20.0$  weeks. Window =  $21 - 5 = 16$  weeks. Required throughput =  $72/16$ . At the mean rate the window delivers  $4.0 \times 16 = 64$  items.
4. **Result.** The team's forecast duration is **16–20 weeks**, mean **18**. The window is **16 weeks**, so the required throughput is **4.5 items a week**: the team's **best observed iteration, sustained without variation for sixteen weeks**. At its mean rate the stream delivers **64** of 72 items, leaving **8 items (11.11 % of the package) undelivered**, and finishing the full package two weeks late costs  $2 \times 45,000 = \text{USD } 90,000$ . Starting at week 3 instead widens the window to 18 weeks and drops the required rate to **4.0** (the mean, and therefore roughly even odds); starting at week 1 widens it to 20 and drops the requirement to **3.6**: the worst rate the team has recorded.
5. **Interpretation.** **The backward pass does not ask the team when it will finish; it tells the team what rate it must hold.** "Deliver by week 21" is unactionable inside a cadence system and invites the team to agree to something it cannot forecast. "Sustain 4.5 items a week from week 5" is a statement the team can test against its own record, and in this case reject, which is the whole value of putting it that way. The conversion runs in both directions: the team's ranged forecast enters the network as a ranged duration on the integration activity, and the network returns a required rate read off LS. A hybrid programme that cannot perform this conversion has two plans and no plan.

**A requirement equal to the best-ever observation is a plan with no margin, and saying so is the professional act.** 4.5 is not impossible (it happened), but a rate achieved once is not a rate sustained for eight iterations, and the honest statement to the steering committee is that the current plan

requires the team's historical maximum with zero allowance for variation. Domain 13, KA 13.2.4 sets out how to express such a forecast as a range with a stated meaning rather than as a single number; what belongs here is that the *network* is what turned an ambitious plan into a checkable claim.

**Of the three levers in the arithmetic (the start date, the rate and the scope) only the rate belongs to the team.** Moving the start and deferring items are the leader's to pull, which is why "the team must go faster" is at once the weakest available response and the most frequently chosen. It is also the one lever Domain 13 shows cannot be pulled by starting more work in parallel.

**The cautions.** Backlog items are being treated as interchangeable units of work, which is acceptable for forecasting a large package and wrong for a small one: 72 items averages out, 7 does not.  $LS(F) = 21$  is read off the *current* network, so a crash or a slip anywhere on the critical path moves the required rate without the team touching anything, and the required rate must therefore be re-derived every cycle rather than fixed at planning. And the 16-week window assumes the integration itself is instantaneous: where the package needs a hardening or acceptance period before F can use it, that period is an activity in the network with its own duration, not a rounding allowance.

#### 6.4.2 Schedule compression and recovery

**The two levers.** **Crashing** buys duration with money (more resources, overtime, expediting); **fast-tracking** buys it with risk (overlapping activities that logic preferred in sequence). Both obey the same law: work the critical path, and re-run the passes after every move, because **the critical path migrates**.

##### WORKED EXAMPLE

#### Worked example 6.4.2 — the crash that stopped paying.

1. **Setup.** Auriga's client offers a bonus of USD 45,000 ( $\approx$  SAR 168,750) per week saved. Expediting C costs USD 30,000 per week, up to two weeks. Should the leader buy one week, two, or none?
2. **Formula.** Crash only while (weeks actually saved  $\times$  value per week) > crash cost: checked against the *re-run passes*, not the original network.
3. **Substitution.** Crash C by 1 (8 $\rightarrow$ 7): duration 25 $\rightarrow$ 24; net gain 45,000 – 30,000 = +15,000. Crash C by 2 (8 $\rightarrow$ 6): path B–D–E–F (2+6+7+5+4) is now co-critical at 24 — duration stays 24; second week's net 0 – 30,000 = –30,000.
4. **Result. Crash one week only:** +USD 15,000. The second week is pure loss.
5. **Interpretation.** Compression is subject to sharply diminishing returns because every crash promotes the next-longest path. The disciplined sequence: identify the binding path  $\rightarrow$  price the cheapest week on it  $\rightarrow$  re-run the passes  $\rightarrow$

repeat until the next week costs more than it is worth. Leaders who mandate "crash procurement by two weeks" from the original network pay for weeks that no longer exist.

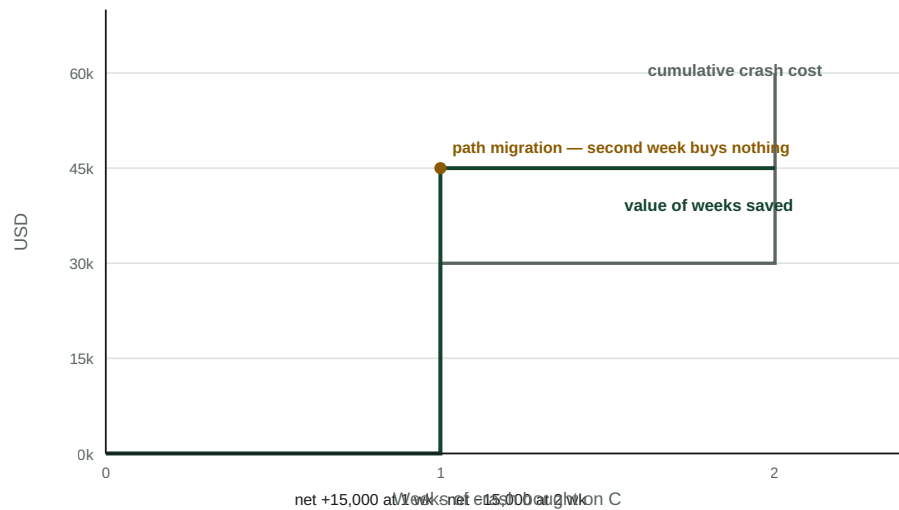


Fig 6.4.1 — The economics of crashing Auriga's procurement

**What bounds the menu, before any of it is priced.** A cost slope says what a week *costs*. It does not say whether the week is **available**, and the menu below would be professionally misleading if read as though it did. Every compression lever in it (overtime, a second shift, night working, crew addition, overlapping two disciplines on the same plant) carries a **non-financial ceiling** that sits above its cost:

- **Working time, rest and fatigue.** Limits on hours, rest periods and consecutive shifts, and the duty to manage fatigue as a hazard rather than as a productivity variable.
- **The safety case and the permit regime.** What the plant, the site or the system is authorised to have done to it, and under what conditions: permits to work, isolations, competency requirements and the conditions attached to any approval the asset holds.
- **Agreement terms.** Collective and individual employment agreements, and sub-contract terms, which may set shift patterns, notice, rates and maximum hours independently of anything the schedule wants.

These ceilings are **jurisdiction-, sector- and site-specific**, they differ substantially, and they are taken from the safety function, from human resources and from qualified counsel, never from a cost slope, and never from this book, which states no legal position and characterises no arrangement as permitted or otherwise. The professional consequence is a hard one and it is the point of the paragraph: **a lever whose ceiling binds is not available at any price**, and it is struck from the menu

before the optimisation is run rather than priced into it. A compression plan that contains a lever nobody asked the safety function about is an unpriced one, not an aggressive plan.

**Which makes compression a two-signature decision.** Where a compression move changes a shift pattern, adds night working, or overlaps disciplines on the same plant, the decision requires the **named safety approver** in addition to the authority spending the money. The two are different people and the second cannot speak for the first. The same holds for a change that touches a condition of any approval the asset or the system holds: the approval's owner decides whether the move is available, and that decision sits outside the project's authority. The recovery plan (Toolkit 6.T.2) records **both** authorities, and a recovery plan naming only the funder has recorded half the decision. Worked example 6.4.2c's overlap of installation and commissioning is exactly this case: it is a simultaneous-operations question before it is an arithmetic one, and in many sectors it requires a separate review and approval before the overlap can be bought at all.

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#### WORKED EXAMPLE

#### Worked example 6.4.2b – the whole compression menu, and the least-cost duration.

1. **Setup.** The example above priced **one** lever. A leader planning a compression programme prices the **menu**. Auriga's available buys, each quoted by the owning discipline as a cost per week with a technical limit:

| ACTIVITY                       | ON PATHS               | COST SLOPE (USD/<br>WK) | MAX<br>WEEKS |
|--------------------------------|------------------------|-------------------------|--------------|
| C Procure hardware             | A–B–C–E–F              | 30,000                  | 2            |
| B Detailed design              | all three              | 40,000                  | 2            |
| D Civil and cabling            | A–B–D–E–F, A–B–<br>D–G | 35,000                  | 1            |
| E Installation                 | both long paths        | 55,000                  | 1            |
| F Testing and<br>commissioning | both long paths        | 65,000                  | 1            |

A week is worth **USD 45,000**. What is the least-cost project duration? 2. **Formula.** For each target duration, find the **cheapest set of crashes that shortens every path** to that duration, not the cheapest activity, which is the standard error. Then net saving = weeks saved × 45,000 – cumulative crash cost, and the optimum is the duration maximising it. Equivalently: buy the next week while its **marginal** cost is below 45,000, and stop at the first week where it is not. 3. **Substitution.** 25 → 24 needs only A–B–C–E–F shortened: crash C × 1, **30,000**. 24 → 23 needs *both* long paths shortened (they are co-critical at 24), so

the buy must be on a shared activity: B at **40,000** beats  $C \times 1 + D \times 1$  at  $30,000 + 35,000 = 65,000$ . 23  $\rightarrow$  22: B's second week, **40,000**. 22  $\rightarrow$  21: B is exhausted, so E at **55,000** (again beating  $C + D$  at 65,000). 21  $\rightarrow$  20: F at **65,000**, tying with  $C + D$ . 20  $\rightarrow$  19:  $C \times 1 + D \times 1$ , **65,000**, exhausting the menu. 4. **Result.**

| DURATION<br>(WK) | CHEAPEST<br>PLAN                 | MARGINAL<br>COST | CUMULATIVE<br>CRASH COST | WEEKS<br>SAVED $\times$<br>45,000 | NET<br>SAVING  |
|------------------|----------------------------------|------------------|--------------------------|-----------------------------------|----------------|
| 25               | —                                | —                | 0                        | 0                                 | 0              |
| 24               | $C \times 1$                     | 30,000           | 30,000                   | 45,000                            | +15,000        |
| 23               | $+ B \times 1$                   | 40,000           | 70,000                   | 90,000                            | +20,000        |
| <b>22</b>        | <b><math>+ B \times 2</math></b> | <b>40,000</b>    | <b>110,000</b>           | <b>135,000</b>                    | <b>+25,000</b> |
| 21               | $+ E \times 1$                   | 55,000           | 165,000                  | 180,000                           | +15,000        |
| 20               | $+ F \times 1$                   | 65,000           | 230,000                  | 225,000                           | (5,000)        |
| 19               | $+ C \times 2, D$<br>$\times 1$  | 65,000           | 295,000                  | 270,000                           | (25,000)       |

The least-cost duration is **22 weeks**, three weeks compressed, for **USD 110,000** of crash spend and a net **USD 25,000** better than the 25-week plan. The shortest *technically* feasible duration is **19 weeks**, which destroys **USD 25,000**. 5. **Interpretation. The optimum is interior, and it is set by the marginal week.** The cost slopes rise: 30,000, 40,000, 40,000, 55,000, 65,000, 65,000; because each crash promotes the next-longest path and forces the buy onto a shared, dearer activity. Compression is therefore an optimisation with an interior solution, and the objective is not "as fast as possible" but "as fast as pays". The same shape appears in Domain 9's cost-of-quality minimum for the same underlying reason: the last increments of any good are the expensive ones.

**Averages mislead here and marginal figures do not.** At 22 weeks the *average* crash cost is  $110,000/3 = \text{USD } 36,666.67$  a week, comfortably below the 45,000 a week is worth; and a manager reasoning from that average will buy the fourth week too, at a marginal 55,000, and lose **USD 10,000**. The rule is always marginal, never average, and the two diverge precisely because the slope rises.

**The range of optimality is what makes the plan defensible.** Since only the marginal cost matters, the optimal duration as a function of the value of a week  $v$  is a step function:

| VALUE OF A WEEK $v$ | LEAST-COST DURATION         |
|---------------------|-----------------------------|
| below 30,000        | 25 weeks (compress nothing) |
| 30,000–40,000       | 24 weeks                    |



| VALUE OF A WEEK $v$  | LEAST-COST DURATION |
|----------------------|---------------------|
| <b>40,000–55,000</b> | <b>22 weeks</b>     |
| 55,000–65,000        | 21 weeks            |
| above 65,000         | 19 weeks            |

Auriga's 45,000 sits **12.50 %** above the lower bound of its band and **18.18 %** below the upper, so the three-week plan survives a substantial error in the cost of delay, which is the sentence to put in the steering paper. Note also what the table says about *other* projects: the identical network with a week worth 25,000 should be compressed not at all, and with a week worth 80,000 should be compressed to its technical limit. **Compression policy is not a property of a schedule; it is a property of a schedule and a cost of delay together.** Where the project also carries a time-related indirect cost (site establishment, supervision, financing) that cost is *added* to the value of a week: an extra USD 18,000 a week takes  $v$  to 63,000 and moves the optimum from 22 weeks to **21**.

**The optimum is a property of the option set, not of the network, and option sets decay.** KA 6.4.2 concluded "crash one week only" and was right, because only C's expediting was on offer. With the full menu the answer is three weeks. The same effect runs the other way as delivery proceeds: 6.A.4 recovers two weeks late in the project, when B is complete and only C and E remain buyable, for **USD 85,000**; where the same two weeks bought from this menu at planning time cost **USD 70,000**. The **USD 15,000** difference is the price of the options that expired, and it is the quantitative case for pricing the compression menu at baseline rather than in the crisis. A compression menu is a perishable asset and should be reviewed on the same cycle as the risk register.

**The cautions.** Cost slopes are treated as linear within each activity's limit, which is a simplification: the second crashed week of an activity is often dearer than the first, and where a discipline can say so the table should carry two slopes rather than one. The limits are technical claims and belong to the disciplines that made them, a "maximum two weeks" that is really a preference will hide a cheap week. Every crash is a *risk* decision as well as a cost one: more people on the same work raises coordination load (Domain 12) and error rates (Domain 9), neither of which appears in a cost slope. And the arithmetic assumes the crashes are independent; two crashes competing for the same scarce specialist are not, which is a resource-feasibility question (KA 6.3.1c) to be re-run after the plan is chosen, not before.

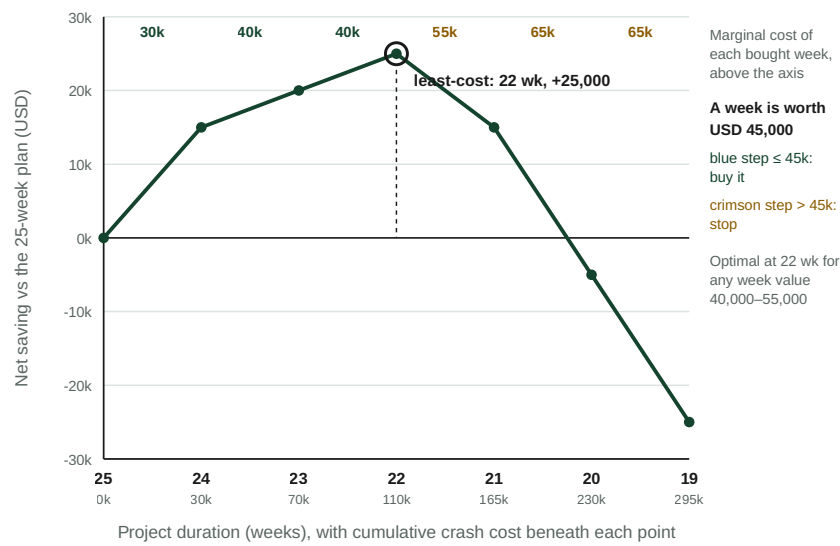


Fig 6.4.2 — The compression menu and the least-cost duration

#### WORKED EXAMPLE

##### Worked example 6.4.2c — fast-tracking priced as the probabilistic decision it is.

- Setup.** The alternative to buying weeks with money is buying them with overlap. Auriga can begin testing and commissioning F when installation E is 60 % complete (commissioning the completed sections while the last are installed) an overlap of **2 weeks** (40 % of E's five), taking the project from 25 weeks to **23**. The risk is specific: if the last-installed sections change the commissioning basis, the affected sections must be re-commissioned, costing **USD 140,000** and **1.5 weeks**. The commissioning lead assesses the probability at **25 %**. A week is worth **USD 45,000**. The crash alternative from the menu above buys the same two weeks for a certain **USD 70,000**.
- Formula.** Value each outcome against the do-nothing baseline: good outcome = weeks saved × cost of delay; bad outcome = weeks saved after rework × cost of delay – rework cost. Then  $EV = \text{good} - p \times (\text{good} - \text{bad})$ , and the break-even probability is  $p^* = \text{good} \div (\text{good} - \text{bad})$ . Against a certain alternative worth K, the indifference probability is  $p^{**} = (\text{good} - K) \div (\text{good} - \text{bad})$ .
- Substitution.** Good:  $2 \times 45,000 = 90,000$ . Bad: duration  $23 + 1.5 = 24.5$  weeks, so  $0.5 \times 45,000 - 140,000 = -117,500$ . Swing  $90,000 - (-117,500) = 207,500$ .  $p^* = 90,000/207,500$ ;  $p^{**} = (90,000 - 20,000)/207,500$  where the crash's net is  $90,000 - 70,000 = 20,000$ . EV at  $p = 0.25$ :  $90,000 - 0.25 \times 207,500$ .

4. **Result.** The fast-track is worth **+USD 90,000** if the overlap holds and **-USD 117,500** if it does not, a swing of **USD 207,500**. It beats doing nothing while  $p < 43.37\%$  and beats crashing while  $p < 33.73\%$ . At the assessed 25 % its expected value is **+USD 38,125**, which is **USD 18,125** better than the crash plan's certain **+USD 20,000**.
5. **Interpretation. Expected value ranks the two options; variance is why a leader may still choose the other one.** The fast-track wins on expectation by 18,125 and carries a **USD 137,500** worse outcome in the quarter of futures where the overlap fails: the gap between a certain +20,000 and a possible -117,500. Nothing in the arithmetic settles which to take: that is a risk-appetite decision belonging to whoever carries the consequence, and Domain 8, KA 8.4.1 sets out when an expected-value comparison is the wrong test altogether. What the arithmetic does settle is that the decision must be *made* rather than drifted into, and by someone with the authority to accept the downside.

**Two thresholds, not one, and their ratio is the useful part.**  $p^* = 43.37\%$  answers "is overlapping better than accepting the date?";  $p^{**} = 33.73\%$  answers "is it better than paying cash?". Their ratio is  $70,000/90,000 = 0.7778$ . The crash option removes 22.22 % of the probability headroom the fast-track had against doing nothing. Whenever a certain alternative exists, the relevant threshold is the lower one, and a leader who computes only  $p^*$  will overlap in cases where writing a cheque was better.

**Both remedies can be mixed, and the mixture is usually right.** One week of overlap plus one crashed week costs 30,000 with a smaller exposure than two weeks of overlap; the reason to prefer a hedge is that  $p$  is a **function of the overlap taken**, rising as the overlap deepens, and almost no team can state that function. Where  $p(w)$  is unknown, the usual case, the defensible practice is to take the smallest overlap that achieves the needed weeks, buy the remainder, and state both thresholds for the overlap actually taken.

**The cautions.** The outcome is modelled as binary when reality is a distribution of partial rework, so the two thresholds are guides rather than boundaries; the honest reading of  $p^* = 43.37\%$  is "this has room to be wrong", not "43.37 % is the answer". The 25 % is an expert judgement, and it should be recorded in Domain 8's register with its owner and its basis rather than embedded in a schedule. Overlap also consumes coordination and supervision that the cost slope of a crash does not (two disciplines working the same plant at once), so the fast-track is cheaper on paper than in the field. And note that even the bad outcome finishes at **24.5 weeks**, still inside the original date: fast-tracking that fails is usually not a schedule disaster but a cost one, which is exactly why the decision belongs to whoever owns the cost.

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**Recovery.** When the network slips (self-check 6.2.3: D at 10 weeks  $\rightarrow$  27), recovery options rank by cost-of-time: re-sequencing and logic re-choice (cheapest, KA 6.1's

chosen logic); float harvesting from non-critical paths (G's 8 weeks fund nothing on the new critical path, but its resources might); crashing the *new* critical path (now through D); fast-tracking E against D's tail with an explicit rework risk (Domain 8's risk register prices it); and scope or acceptance re-negotiation (Domain 5) as the honest last resort. A **recovery plan** states the target date, the moves, their costs and risks, each move's **non-financial ceiling**, and the decision authorities: the one spending the money and, where a move changes shift patterns, adds night working, overlaps disciplines on the same plant or touches a condition of an approval, the **named safety or approval-holding authority** whose agreement the money cannot substitute for. The template is toolkit 6.T.2.

### 6.4.3 Forecasting and scenario analysis

**Three-point estimates.** Uncertain durations are ranges, not points. The PERT expected duration and spread for an activity estimated optimistic  $o$ , most-likely  $m$ , pessimistic  $p$ :

$$t_e = (o + 4m + p) / 6 \qquad \sigma = (p - o) / 6$$

#### WORKED EXAMPLE

##### Worked example 6.4.3 – Auriga's installation risk.

1. **Setup.** Installation E is estimated  $o = 4$ ,  $m = 5$ ,  $p = 12$  weeks (the long tail: a legacy-system integration problem discovered late).
2. **Formula.**  $t_e = (o + 4m + p)/6$ ;  $\sigma = (p - o)/6$ .
3. **Substitution.**  $t_e = (4 + 20 + 12)/6 = 36/6$ ;  $\sigma = (12 - 4)/6 = 8/6$ .
4. **Result.**  $t_e = 6.0$  weeks (not 5);  $\sigma = 1.33$  weeks.
5. **Interpretation.** The deterministic network used 5 weeks; the risk-weighted expectation is 6. The skewed tail alone adds a week to the honest forecast, pushing expected completion toward 26 weeks before any risk event "happens". Deterministic dates are the *mode*, not the *mean*; boards deserve the mean and the spread. Full quantitative schedule risk analysis (correlating activities, simulating paths, criticality indices) is Domain 8 (KA 8.2), built on exactly this input.

**The bias has a closed form, and it is worth memorising.** Subtracting  $m$  from  $t_e$  gives

$$t_e - m = (o - 2m + p) / 6$$

– the deterministic bias is one sixth of the estimate's **skew**,  $\frac{(p - m) - (m - o)}{6}$ . On E:  $\frac{(4 - 10 + 12)}{6} = 1.0000$  week. On the o = 6, m = 8, p = 16 estimate of Exercise 6.4:  $\frac{(6 - 16 + 16)}{6} = 1.0000$  week again, despite quite different numbers. Three consequences. A **symmetric** estimate has zero bias, so the deterministic plan is unbiased and PERT adds nothing, which is why applying it indiscriminately wastes effort. The bias depends only on how **lop-sided** the range is, not on how **wide** it is: a wide symmetric estimate (5, 9, 13) has a large  $\sigma$  of 1.3333 and no bias at all, so  $\sigma$  and bias answer different questions and neither substitutes for the other. And the bias is **positive** whenever the tail is on the pessimistic side, which is nearly always in delivery, because durations are bounded below by physics and unbounded above by circumstance. Right-skew is the normal condition, so the deterministic plan is normally optimistic: not because planners are optimists but because the mode of a right-skewed distribution is below its mean.

**What this does and does not license.** It does not license adding a week to every activity: doing that on all seven of Auriga's activities would inflate the plan by far more than the network's expected completion moves, because only the activities on the binding path contribute, and because path-level aggregation is not the sum of activity-level expectations. That aggregation (variances adding, merge bias at convergence points, criticality indices) is Domain 8, KA 8.A.2, worked on this same network, and it finds that E's start is only **13.16 %** likely in week 16. Nor does it license treating  $\sigma = (p - o)/6$  as a derived quantity: it is a convention that assumes the range spans roughly six standard deviations, it understates spread on strongly skewed estimates, and probabilities computed from it should be read to the nearest percentage point at best.

**The reviewer's question is where the three points came from.**  $\sigma$ ,  $m$  and  $p$  elicited from the same person in the same breath are usually one number with decorations: the range is anchored on the most likely value and adjusted by a habitual percentage, which produces symmetric estimates and therefore zero bias by construction. The tail that matters (the legacy-integration

problem in E's  $\text{'p'} = 12$ ) comes from asking a different question ("what would have to go wrong, and how long would *that* take?"), which is a risk-identification act (Domain 8, KA 8.1) rather than an estimating one. An estimate whose  $\text{'p'}$  cannot be traced to a named scenario is not a three-point estimate.

**Scenario analysis.** Between the single date and the full simulation sits the leader's workhorse: three coherent scenarios (base / threat / opportunity), each a *complete re-run of the passes* under stated assumptions, each with owners for the assumptions that differ. Scenarios are cheap, auditable, and, because each is a real network, they expose path migration that a percentage-confidence number hides.

**The earned-schedule bridge.** Once cost joins schedule (Domain 7), progress data yields a time-based forecast: **earned schedule ES** asks *when the value now earned was planned to have been earned*, and  $SPI(t) = ES / AT$  (actual time) measures schedule efficiency in time units. If Auriga's week-22 status shows work that the baseline expected by week 20,  $SPI(t) = 20/22 = 0.91$ . The programme is running at 91 % of planned tempo, a signal the currency-based SPI famously loses late in a project. On Auriga's own week-13 position the value earned was planned to have been earned by week 12, so  $ES = 12.0000$  and  $SPI(t) = 12/13 = 0.9231$ : identical to the currency-based SPI of  $1,920,000/2,080,000$ , because the planned-value curve is locally linear there. Domain 7 (KA 7.3 and 7.A.1) builds the full machinery, including the point at which the two measures separate; it is flagged here because schedule forecasting belongs to whoever owns the network, not only to the cost engineer.

AI in this KA

Machine forecasting earns trust in exactly one way: calibration. A model that ingests progress data and produces completion forecasts must show its record (forecast vs actual across past periods), before its numbers enter a board pack (Domain 14, KA 14.3's verification workflow). The leader's questions are the auditor's: what data trained it, what does it do when the data thins, who re-ran its arithmetic, and what would make it wrong? Fluent confidence without a calibration record is the machine version of the site manager who is "sure it'll be fine".

KEY TERMS — KA 6.4

| TERM                     | MEANING                                                              |
|--------------------------|----------------------------------------------------------------------|
| Crashing / fast-tracking | Buying duration with money / with overlap risk.                      |
| Path migration           | The critical path moving as durations change; why passes are re-run. |

| TERM                                | MEANING                                                                                                                                                                                                                                            |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Recovery plan</b>                | Target date, ranked moves, costs, risks, non-financial ceilings, and <b>both</b> decision authorities: the one spending the money and the safety or approval-holding authority where one is engaged.                                               |
| <b>Non-financial ceiling</b>        | The working-time, rest, fatigue, safety-case, permit or agreement limit bounding a compression lever. Supplied by the safety function, HR or counsel, jurisdiction- and sector-specific, and where it binds the lever is unavailable at any price. |
| <b>Simultaneous operations</b>      | Two disciplines working the same plant at once; a safety decision requiring its own review and approval before it is an arithmetic one.                                                                                                            |
| <b>Three-point / PERT estimate</b>  | $t_e = (o+4m+p)/6$ , $\sigma = (p-o)/6$ .                                                                                                                                                                                                          |
| <b>Scenario analysis</b>            | Complete network re-runs under coherent stated assumptions.                                                                                                                                                                                        |
| <b>Integration milestone</b>        | The CPM node where an agile stream's delivery enters the network.                                                                                                                                                                                  |
| <b>Cost slope</b>                   | An activity's crash cost per week, with a technical limit on weeks available.                                                                                                                                                                      |
| <b>Least-cost duration</b>          | The duration maximising net saving; found where the marginal week's cost first exceeds the value of a week.                                                                                                                                        |
| <b>Required throughput</b>          | Backlog ÷ (successor's <i>LS</i> – stream start): what the network asks a cadence team to sustain.                                                                                                                                                 |
| <b>Breakeven rework probability</b> | $p^* = \text{good outcome} \div (\text{good} - \text{bad})$ ; the probability at which an overlap stops paying.                                                                                                                                    |
| <b>Deterministic bias</b>           | $t_e - m = (o - 2m + p)/6$ , one sixth of the estimate's skew.                                                                                                                                                                                     |

## SAMPLE MCQS — KA 6.4

**MCQ 6.4-A** [6.4.2 · Application] Saving a week is worth USD 45,000; crashing the critical activity costs USD 30,000 per week for up to two weeks; after one week of crashing, a parallel path becomes co-critical. The value-maximising decision is:

- A. crash two weeks (net +USD 30,000)
- B. crash one week (net +USD 15,000) CORRECT
- C. crash nothing (avoid all cost)
- D. crash two weeks and fast-track the parallel path at no cost

*Rationale:* Week one converts to a project week: +15,000. Week two is absorbed by the co-critical path: –30,000. A prices weeks that don't materialise; C leaves +15,000 unclaimed; D invents a free fast-track, overlap always carries rework risk.

**MCQ 6.4-B** [6.4.3 · Application] An activity is estimated  $o = 4$ ,  $m = 5$ ,  $p = 12$  weeks. Its PERT expected duration is:

- A. 5.0 weeks
- B. 7.0 weeks
- C. 6.0 weeks CORRECT
- D. 6.5 weeks

*Rationale:*  $(4 + 4 \times 5 + 12)/6 = 36/6 = 6.0$ . A is the most-likely value (the mode); B is the unweighted mean of the three points; D miscounts the weighting.

**MCQ 6.4-C** [6.4.1 · Analysis] A hybrid programme needs a software module from an agile team for an integration milestone. The schedule-sound way to join the two worlds is:

- A. impose the CPM date on the team as a sprint deadline
- B. enter the team's velocity-based forecast as a ranged duration and read the latest acceptable delivery from the backward pass CORRECT
- C. exclude the module from the network since agile work cannot be scheduled
- D. convert the agile team to predictive planning for the integration period

*Rationale:* B translates in both directions, evidence-based forecast in, latest-start requirement out. A dictates without evidence; C leaves the network blind at a convergence point; D destroys the team's delivery system to decorate the network.

**MCQ 6.4-F** [6.4.3 · Application] An activity is estimated optimistic 3, most-likely 4, pessimistic 8 weeks. Its PERT expected duration and standard deviation are:

- A.  $t_e = 4.5$ ,  $\sigma = 0.83$  CORRECT
- B.  $t_e = 4.0$ ,  $\sigma = 0.83$
- C.  $t_e = 5.0$ ,  $\sigma = 1.67$
- D.  $t_e = 4.5$ ,  $\sigma = 5.0$

*Rationale:*  $t_e = (3 + 16 + 8)/6 = 4.5$ ;  $\sigma = (8 - 3)/6 = 0.83$ . B reports the mode as the mean; C is the unweighted three-point average with a doubled spread; D confuses  $\sigma$  with the pessimistic-minus-optimistic range.

**MCQ 6.4-D** [6.4.3 · Application] At week 22, a programme has earned the value its baseline planned to earn by week 20. Its time-based schedule performance index  $SPI(t)$  is:

- A. 1.10
- B. 0.91 CORRECT
- C. 0.80
- D. 20.0



*Rationale:*  $SPI(t) = ES/AT = 20/22 = 0.91$ , the programme delivers at 91 % of planned tempo. A inverts the ratio; C subtracts the two weeks from the wrong base; D reports earned schedule itself, not the index.

**MCQ 6.4-E** [6.4.3 · Analysis] Compared with quoting "78 % confidence of the date", giving the board three fully re-run schedule scenarios is stronger because:

- A. three numbers always beat one
- B. each scenario is an auditable network with named assumptions, showing how the date moves and where the path migrates CORRECT
- C. percentages are always statistically invalid
- D. scenarios eliminate the need for risk analysis

*Rationale:* The scenario's power is auditability and mechanism, assumptions with owners, visible path migration. A is numerology; C overclaims (a calibrated percentage is legitimate, Domain 8); D reverses the relationship: scenarios are inputs to risk analysis, not substitutes.

**MCQ 6.4-G** [6.4.2 · Evaluation] A compression menu offers successive weeks at marginal costs of 30,000, 40,000, 40,000, 55,000, 65,000 and 65,000. A week is worth 45,000. The least-cost plan buys:

- A. one week, because only the first step is clearly cheap
- B. three weeks, stopping at the first marginal cost above 45,000 CORRECT
- C. six weeks, because the average cost of 49,167 is close to 45,000
- D. four weeks, because the average cost of the first four (41,250) is below 45,000

*Rationale:* Buy while the marginal step is at or below 45,000, 30,000, 40,000, 40,000; and stop at 55,000, giving 22 weeks and a net +25,000 (6.4.2b). C and D both reason from averages, the standard error: the fourth week costs 55,000 whatever the average of the first four is, and buying it loses 10,000. A stops at the single-lever answer of 6.4.2, which was correct only for a menu of one.

**MCQ 6.4-H** [6.4.2 · Application] An overlap saves 2 weeks worth 45,000 each; if it fails it costs 140,000 and gives back 1.5 weeks, and a certain crash plan would buy the same 2 weeks for 70,000. The probability of failure above which the overlap is worse than crashing is:

- A. 43.37 %
- B. 33.73 % CORRECT
- C. 50.00 %
- D. 25.00 %

*Rationale:* The swing is  $90,000 - (-117,500) = 207,500$  and the crash's net is 20,000, so  $p^{**} = 70,000/207,500 = 33.73\%$  (6.4.2c). A is the threshold against *doing nothing*, which is the wrong comparator once a certain alternative exists; C

treats the choice as a symmetric coin toss: the answer if the two outcomes were equal in size, which they are not; D is the assessed probability, the input rather than the threshold.

**MCQ 6.4-I** [6.4.1 · Application] A cadence team must clear a 72-item package between week 5 and its successor's LS of week 21. Its observed throughput has ranged 3.6–4.5 items a week, averaging 4.0. The network's requirement is:

- A. delivery by week 21, which the team should commit to
- B. 4.5 items a week — the team's best observed rate, sustained for the full 16 weeks CORRECT
- C. 4.0 items a week, since the mean is the fair planning figure
- D. 3.6 items a week, since planning should use the worst observation

*Rationale:*  $72 \div (21 - 5) = 4.5$  items a week (6.4.1). C and D quote the team's own forecast rates rather than deriving the requirement from the window, the mean fits an 18-week window and the worst rate a 20-week one, neither of which exists here. A restates the date instead of translating it, which is what leaves the requirement untested.

#### SELF-CHECK — KA 6.4

1. *Why does compression show diminishing returns?* — Each crash promotes the next-longest path; saved weeks stop converting to project weeks (path migration).
2. *What makes a scenario more useful to a board than a confidence percentage?* — It is a complete, auditable network with named assumptions. It shows *how* the date moves, not just that it might.
3. *When may a machine forecast enter a board pack?* — With a calibration record, verified arithmetic, stated data lineage and a named human owner.
4. *Why is the average crash cost the wrong figure?* — Because the decision is marginal: Auriga's average at 22 weeks is 36,666.67 against a week worth 45,000, yet the next week costs 55,000 and buying it loses 10,000.
5. *What does the network give a cadence team, and in what units?* — A required throughput,  $\text{backlog} \div (\text{successor's LS} - \text{start})$ , which on Auriga's interface package is 4.5 items a week, the team's best observed rate.
6. *When does a three-point estimate add nothing?* — When it is symmetric: the bias is  $(o - 2m + p)/6$ , which is zero, so  $t_e = m$  and only the spread is new information.
7. *What does a cost slope not tell you about a compression lever?* — Whether the week is available. Working-time, rest and fatigue limits, safety-case and permit conditions and agreement terms each set a ceiling above the price; where a ceiling binds, the lever is struck from the menu rather than priced into it, and the ceilings come from the safety function, HR and counsel (6.4.2).

- 
8. *Who signs a compression that changes a shift pattern or overlaps disciplines on live plant?* – The authority spending the money **and** the named safety or approval-holding authority. The recovery plan records both; money cannot substitute for the second signature.
- 

## Advanced topics – Domain 6

### 6.A.1 The drum: constraint-aware flow

Where one resource is the system's constraint (a single commissioning team, one test rig), throughput scheduling subordinates everything to that drum: buffer it, never starve it, and let non-constraint efficiency go. This reframes utilisation worship, a 100 %-busy non-constraint builds inventory, not progress. In schedule terms: protect the constraint's feed path with deliberate float placement rather than spreading buffer evenly.

### 6.A.2 Schedule baselines and the update cycle

The control schedule lives on a cycle: status (actuals and remaining durations in), re-run passes, variance vs baseline, forecast, and (rarely, formally) re-baseline under Domain 4's change control. Two integrity rules carried from the family's controls tradition: never edit history (actualised dates are records), and never re-baseline to hide variance, a re-baseline is a governance decision with an audit trail, not a cosmetic one.

### 6.A.3 PDM edge cases: float under SS and FF links

Precedence-diagram (PDM) links change how float behaves, and two edge cases bite reviewers. **(1) The SS-linked short successor.** P (10 weeks) **SS+2** Q (3 weeks): Q can run weeks 2–5, finishing seven weeks before its predecessor, legal, and correct if the dependency truly binds only the starts; but if Q *also* needs P's output to finish, the network is missing an FF link, and Q's float is fiction. Paired **SS+FF** links express "starts together, finishes together" honestly. **(2) Duration-driven float.** Under an FF link the successor's float depends on its own duration: lengthen Q and its **LS** moves *earlier*, planners who "add duration for safety" on FF-linked activities silently destroy their float. The audit habit: on every SS/FF link, ask which end of each activity the logic genuinely binds, and whether the unbound end needs its own link. A network is only as honest as its least-considered link type.

### 6.A.4 Negative float and the honest schedule

A constraint-bound network can compute **TF** < 0: the plan, as constrained, is late already. Negative float is information, the size of the recovery problem, and suppressing it (deleting constraints, shortening durations by fiat) is the scheduling

equivalent of cooking the books. The professional response mirrors KA 6.4.2: quantify it, trace the binding path, price the recovery options, and escalate the decision to whoever owns the trade (Domain 3's escalation design).

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#### WORKED EXAMPLE

##### Worked example 6.A.4 – buying back two weeks of negative float.

1. **Setup.** A newly imposed outage constraint requires Auriga to finish by **week 23**; logic says 25, so  $TF = -2$  on A–B–C–E–F. Available buys: crash C at USD 30,000/week (max 2); crash E at USD 55,000/week (max 1). Find the cheapest feasible recovery to week 23.
2. **Formula.** Recover week by week on the *currently binding* path(s), re-running the passes after each buy (path migration, 6.4.2).
3. **Substitution.** Week one: crash C by 1  $\rightarrow$  duration 24, but B–D–E–F is now co-critical at 24. Week two must shorten *both* paths: the only shared activity available is E  $\rightarrow$  crash E by 1 (USD 55,000)  $\rightarrow$  duration 23. (C's second week alone would leave B–D–E–F at 24, spent money, no schedule.)
4. **Result.** Feasible recovery: **crash C  $\times$  1 + crash E  $\times$  1 = USD 85,000**, finishing week 23.
5. **Interpretation.** Negative float is bought back on the binding path *as it migrates*: the second-cheapest week on the original critical path (C's second week, USD 30,000) is worthless, while the dearest activity (E, on both paths) is the only week-two purchase that works. This is why recovery plans list options *in re-run order*, not in unit-cost order (toolkit 6.T.2).

Note also what the restricted option set costs. Only C and E are buyable here because design B is complete and commissioning F cannot be accelerated at this stage; the same two weeks bought from the full planning-time menu of 6.4.2b (where B was still open at a 40,000 slope) cost **USD 70,000** against this **USD 85,000**. The **USD 15,000** gap is the price of options that expired, and it is the argument for pricing the compression menu at baseline and reviewing it on the risk register's cycle rather than assembling it in the crisis.

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#### 6.A.5 Repetitive delivery: takt, crews and the benefit integral

A rollout is not a network. Meridian installs the same clinic forty times, and a forty-times-repeated activity-on-node diagram teaches nothing: the schedule's content is a **rate**. **Takt** is the interval between successive completions (window  $\div$  units), and it is the quantity that converts a programme date into a crew count, because the rate a crew can sustain is a productivity figure the delivery team already owns. The arithmetic is three divisions, and the reason to do it is that the *value* of a rate behaves quite differently from the value of a date.

## WORKED EXAMPLE

**Worked example 6.A.5 – Meridian's rollout rate, and what accelerating it is worth.**

1. **Setup.** Meridian's **40** clinics are installed by a crew of **6** deployment specialists, each completing **0.1** clinics a week (Domain 1, KA 1.3.3b), so **0.6** a week in total. The estate and training window requires all 40 live within **50 weeks** of rollout start. Specialists cost **USD 4,200** a week; per-clinic work content is **10** specialist-weeks. Adding specialists incurs Domain 1's ramp: **4 weeks** at **25 %** productivity, each newcomer absorbing **50 %** of an existing specialist's time. Benefit accrues at **6 hours** × **USD 85** = **USD 510** a week per adopting clinic and adoption runs at **70 %**, so **USD 357** a week per installed clinic. What crew does the window need, and does it pay?
2. **Formula.** Required takt = window ÷ units; required rate = units ÷ window; crew = required rate ÷ per-head rate. With reinforcement: ramp rate = base – newcomers × supervision × per-head + newcomers × ramp productivity × per-head; post-ramp rate = base + newcomers × per-head; duration = ramp + (units – ramp × ramp rate) ÷ post-ramp rate. For a linear rollout over a fixed horizon, the **unit-weeks** gained by finishing  $\Delta T$  sooner =  $(\text{units} \div 2) \times \Delta T$ .
3. **Substitution.** Required takt  $50/40 = 1.25$  weeks; required rate  $40/50 = 0.8$  a week; crew  $0.8/0.1 = 8$ . With 2 newcomers: ramp rate  $0.6 - 2 \times 0.5 \times 0.1 + 2 \times 0.25 \times 0.1 = 0.55$ ; post-ramp  $0.6 + 0.2 = 0.8$ ; duration  $4 + (40 - 4 \times 0.55)/0.8$ . Baseline  $40/0.6$ . Clinic-weeks gained  $20 \times \Delta T$ .
4. **Result.** The 50-week window needs a takt of **1.25 weeks** and therefore **8** specialists. With the ramp, the eight-person crew finishes in **51.2500 weeks**, an actual takt of **1.2813 weeks**, against **66.6667 weeks** for the crew of six, a saving of **15.4167 weeks**. Effort is nearly invariant:  $6 \times 66.6667 = 400.0$  specialist-weeks against  $8 \times 51.25 = 410.0$ , so the reinforcement costs **10** extra specialist-weeks, or **USD 42,000**. The acceleration yields  $20 \times 15.4167 = 308.3333$  clinic-weeks at **USD 357 (USD 110,075)** for a net **+USD 68,075**.
5. **Interpretation.** **Effort is invariant to the rate; only the benefit integral moves.** Forty clinics at 10 specialist-weeks each is **400 specialist-weeks** whether delivered by six people over 67 weeks or eight over 50. That is the structural reason repetitive-work acceleration is so often nearly free and so rarely attempted: the labour bill is set by the *work content*, not by the *rate*, and the only incremental cost is the transient of changing the rate, here 10 specialist-weeks of ramp. On a network, buying weeks costs a cost slope every time; on a rollout it costs a one-off. The two are different economies and must not be priced with the same instinct.

**Accelerating a linear rollout is worth exactly half the fully-adopted cost of delay per week saved.** The clinic-weeks gained are  $(N/2) \times \Delta T$ , so the value per week saved is  $(40/2) \times 357 = \text{USD } 7,140$ : precisely half of Meridian's **USD 14,280**, and the identity holds for any linear ramp because the

average number of live units during the ramp is half the final number. This matters because the temptation is to value rollout acceleration at the steady-state cost of delay, which **doubles** it. The reconciling check is worth noting:  $40 \times 357 = \text{USD } 14,280$  exactly, Domain 1's figure, because 70 % of 40 clinics is the 28 adopting clinics it was derived from.

**Where benefit only switches on at completion the full figure applies, and here the decision is robust to which regime holds.** If the shared records system delivers nothing until coverage is complete, the 15.4167 weeks are worth  $15.4167 \times 14,280 = \text{USD } 220,150$ , exactly twice the ramp-integral figure, and the net becomes **+USD 178,150**. Both readings say the same thing: buy the two specialists. That is the professionally important observation, a factor-of-two ambiguity in the valuation basis need not be resolved when the decision is unchanged either way, and the leader who notices this stops arguing about the benefits model and gets on with the rollout. Where the decision is sensitive to the factor of two, it is a benefits-realisation question (Domain 16) and must be settled by whoever owns the benefit, not by the planner.

**The threshold is a volume, and it explains why Domain 1 reached the opposite verdict.** Domain 1, KA 1.3.3b added three specialists to the *last eight* clinics and destroyed **USD 23,333.33**. The reason is now visible: the extra specialist-weeks a reinforcement consumes are **constant** (10 for two newcomers, 15 for three, independent of how much work remains), while the weeks saved grow linearly with the volume remaining,  $0.416667N - 1.25$  for two newcomers. On the full cost-of-delay basis two newcomers break even at  $N = 10.0588$  clinics and on the ramp-integral basis at  $N = 17.1176$ . Domain 1's eight clinics sit below both; forty sits far above. **Reinforcement pays above a computable remaining volume, so the same arithmetic gives opposite answers at the two ends of one rollout**, which is why the decision is re-taken as the programme runs down rather than settled once.

**And the rate is capped by the drum, not by the crew.** Suppose one data-migration specialist can handle only **0.7** clinics a week. The achievable rate is then  $\min(0.8, 0.7) = 0.7$ , the rollout takes  $40/0.7 = 57.1429$  weeks, and the eighth specialist buys nothing: the **5.8929 weeks** the drum costs, worth  $20 \times 5.8929 \times 357 = \text{USD } 42,075$ , can be recovered only by relieving the drum (6.A.1). This is the most expensive error available in repetitive scheduling (sizing the crew from the takt while a subordinate step sets the actual rate), and the only defence is to compute the sustainable rate of **every** step, not of the one whose people are most numerous.

**Two further cautions.** The 0.1-clinics-per-specialist-week rate is locally calibrated and clinics are not identical, so a takt computed from an average unit will be missed by the difficult units; the honest plan carries a unit-difficulty distribution and a recovery rule rather than a single takt. And a takt is a commitment to a *cadence*, which requires estate readiness, trained staff and

approvals to arrive at the same cadence, which is where rollouts actually fail, the crew rarely being the binding constraint by the third month.

### 6.A.6 Buffer management: aggregating safety and reading its consumption

Every duration in the Auriga network contains protection. An eight-week procurement estimate is not anyone's honest expectation of eight weeks; it is an expectation with a margin folded in, sized so that the estimator is comfortable committing to it. That is rational behaviour with an expensive property: **safety held inside each activity cannot be shared, while safety held at the end of a chain can be**. A week of protection inside C is available only to C, and if C finishes early the protection is usually consumed anyway, since there is no reward for finishing early when the successor is not ready. Aggregating the protection into an explicit **buffer** converts private, unusable margin into a shared, managed and visible reserve. Buffer sizing and buffer-consumption control belong to the **critical-chain** tradition in schedule management, whose vocabulary KA 6.3.1c already uses; the arithmetic below is worked from first principles on Auriga, and the conventions are named as conventions rather than as results.

#### WORKED EXAMPLE

#### Worked example 6.A.6 — sizing and then managing Auriga's project buffer.

1. **Setup.** Auriga's disciplines are asked for two figures per critical-path activity: the duration in the network, and the duration they would give with an even chance of meeting it.

| ACTIVITY                    | NETWORK<br>DURATION | 50 %<br>DURATION | EMBEDDED<br>SAFETY |
|-----------------------------|---------------------|------------------|--------------------|
| A Mobilise                  | 2                   | 1.5              | 0.5                |
| B Detailed design           | 6                   | 5                | 1.0                |
| C Procure control hardware  | 8                   | 6.5              | 1.5                |
| E Installation              | 5                   | 4                | 1.0                |
| F Testing and commissioning | 4                   | 3                | 1.0                |
| <b>Chain</b>                | <b>25</b>           | <b>20</b>        | <b>5.0</b>         |

Size a project buffer, then manage against it. A week is worth **USD 45,000**. 2.

**Formula.** Chain =  $\sum$  50 % durations. Two sizing conventions: **half the aggregated safety**,  $\sum s_i / 2$ , or **root-sum-square**,  $\sqrt{(\sum s_i^2)}$ , the same



aggregation Domain 8 uses when variances add (KA 8.A.2). Committed duration = chain + buffer. In control: consumption ratio = (buffer consumed ÷ buffer) ÷ (chain complete ÷ chain); projected final consumption = buffer × consumption ratio. 3. **Substitution.** Chain  $1.5 + 5 + 6.5 + 4 + 3 = 20$ . Safety  $0.5 + 1 + 1.5 + 1 + 1 = 5.0$ . Half-safety buffer  $5.0/2$ . Root-sum-square  $\sqrt{(0.25 + 1 + 2.25 + 1 + 1)} = \sqrt{5.5}$ . At a status date with 45 % of the chain complete and 30 % of the buffer consumed: ratio  $0.30/0.45$ . 4. **Result.** The half-safety method commits to  $20 + 2.5 = 22.5000$  weeks, 2.5 weeks inside the 25-week network and worth USD 112,500. The root-sum-square buffer is 2.3452 weeks, a commitment of 22.3452 weeks, 2.6548 weeks inside the network and worth USD 119,465.65. In control, the consumption ratio is 0.6667: projected final buffer use  $2.5 \times 0.6667 = 1.6667$  weeks, a forecast chain completion at 21.6667 weeks with 0.8333 weeks of buffer unused, USD 37,500 still in hand, and 3.3333 weeks inside the original network date. 5. **Interpretation.** Aggregation is where the weeks come from, and it is arithmetic rather than pressure. Nothing was crashed, no estimate was challenged, nobody was told to work faster: the plan is 2.5 weeks shorter because five weeks of private protection became 2.5 weeks of shared protection, and shared protection covers more contingencies per week held. The two conventions differ by only 0.1548 weeks here, and the root-sum-square form is the more defensible because it is the variance-addition result Domain 8 derives, which means it holds under independence, so where the activities share a cause (one subcontractor, one site condition, one approval body) the correct buffer is larger and approaches the simple sum of 5.0. A root-sum-square buffer across correlated activities is the schedule version of the aggregation error Domain 8, KA 8.2.4 prices for contingency.

**Buffer consumption forecasts earlier than variance against baseline.** A ratio below 1 means protection is being consumed more slowly than the chain is being completed; above 1, faster. Auriga's 0.6667 projects a finish with a third of the buffer unspent: information available at 45 % completion, from two numbers, without re-running a pass. It is the schedule counterpart of Domain 7's EAC family: an index observed to date, projected over what remains. The control regime should state thresholds in advance (below 0.5, no action; 0.5 to 1.0, prepare a recovery plan; above 1.0, execute it), set and published before the first status date rather than derived from the first uncomfortable reading.

**The linear projection is only as good as the resemblance between what is done and what remains.** A chain whose risky activities sit at the end shows a comfortable ratio for months and consumes its whole buffer in the final quarter. That is the standard misuse, and it is detectable, because the same three-point estimates that sized the buffer say which activities carry the spread: where the remaining work is riskier than the completed work, weight the projection by remaining variance rather than remaining duration, and say that you have.



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**The method fails completely if the 50 % durations are not 50 % durations.**

Halving network durations by fiat produces the same table with none of the meaning: the buffer then protects estimates nobody believes, it is consumed in the first third of the chain, and the team correctly concludes that the plan was a compression exercise wearing a technique's clothing. Eliciting an even-chance duration requires that missing it half the time carries no penalty, which is a leadership commitment (Domain 12) rather than a scheduling one. Where that commitment cannot honestly be made, the aggregation should not be attempted and the schedule should run with visible float and a float policy instead.

**And the buffer is a governed reserve, not a hiding place.** It appears in the schedule as an explicit activity with an owner, its consumption is reported every cycle in the same pack as float, and it is never silently topped up; that is 6.A.2's re-baselining integrity rule applied to a buffer. A buffer nobody reports is indistinguishable from padding and will be removed by the first cost review that finds it.

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## Industry variations — Domain 6

The passes are universal; what counts as a schedule, and what discipline it needs, is sectoral:

- **Construction and EPC.** Deep L3/L4 hierarchies, thousand-activity networks, contractual float-ownership clauses (Domain 10) and delay-analysis protocols make float a legal quantity: the honesty settings of the Executive perspective are contract compliance here, not just culture.
- **Technology and product.** Hybrid is the default (KA 6.4.1): cadence-based streams joined to a thin predictive network at integration milestones; the commonest defect is a network so sparse that convergence points are invisible until they slip.
- **Shutdowns and turnarounds.** The network runs in *hours*; near-critical analysis widens to every path within minutes of critical, and the drum logic of 6.A.1 governs a single shared crane or permit desk. Compression economics (6.4.2) turn over in shifts, not weeks.
- **Pharmaceutical and regulated development.** Milestones are regulatory events with submission-window physics: genuine date constraints (6.3.2), modelled as logic, with the pin-and-squeeze temptation at its most dangerous because windows are unforgiving.
- **Rollouts and repetitive delivery:** retail estates, network upgrades, clinic and branch programmes. The schedule is a **rate**, not a network: takt, crew count and the sustainable rate of every step, with the drum found before the crew is sized, and acceleration priced as a benefit integral rather than a cost slope (6.A.5). The

binding constraint is almost never the installation crew; it is the upstream supply of site readiness, trained staff and approvals arriving at the same cadence.

- **Public programmes.** Announced dates precede networks (Case study B); the leader's protection (milestones only after the backward pass) is hardest exactly where it matters most, and negative-float honesty (6.A.4) is the difference between an early re-plan and a public failure.

## Case study — Domain 6: recovering Auriga (utilities / technology)

**Situation.** End of week 13. Civil and cabling works D, planned at 7 weeks from week 8, hits contaminated-ground remediation; the discipline lead's honest re-estimate takes D to 10 weeks (finish week 18). The client's outage window for final commissioning opens at week 25 and closing the window costs USD 45,000 per week of delay. The team re-runs the passes.

**Analysis.** With  $D = 10$ : E starts week 18 (D now later than C's week 16), F finishes week 27, **two weeks into the penalty window**, and the critical path has migrated to **A–B–D–E–F**. Procurement C, critical all project, now carries  $TF = 2$ . G still holds float.

**The recovery decision.** Options priced by the team:

| OPTION                                     | WEEKS<br>RECOVERED | COST                            | NET VS DOING<br>NOTHING           |
|--------------------------------------------|--------------------|---------------------------------|-----------------------------------|
| (1) Crash the <i>old</i> critical path (C) | 0                  | 30,000/wk                       | worthless: C is no longer binding |
| (2) Second civil crew on the remediation   | 1                  | 35,000                          | +10,000                           |
| (3) Fast-track E against D's tail          | 1                  | 12,000 expected (20 % × 60,000) | +33,000                           |
| (4) Do nothing                             | 0                  | 90,000 penalty exposure         | baseline                          |
| <b>(2) + (3) taken together</b>            | <b>2</b>           | <b>47,000 expected</b>          | <b>+43,000</b>                    |

The leader takes (2) + (3):  $2 \times 45,000 = 90,000$  of penalty avoided for **USD 47,000** of expected cost, a net **USD 43,000**, and week 25 with the window intact.

**What the arithmetic settles, and what it does not.** Three figures decide the case and are worth extracting, because each is a general instrument rather than a feature of this story.

The **ranking is by net, not by cost**. Option (3) is the cheapest and option (2) the dearer, yet both are bought because each recovers a week worth 45,000 and both

clear that bar; a leader who stops at the cheapest option buys one week and pays a 45,000 penalty to save 35,000. The stopping rule is 6.4.2b's: buy while the marginal week costs less than a week is worth.

The **fast-track's breakeven probability** is  $45,000/60,000 = 75.00\%$ : the rework would have to be three times likelier than the assessed 20 % before the overlap stopped paying. That is the sentence that makes the recommendation defensible to an assurance reviewer, and it is a stronger statement than the expected cost, because the assessed probability is the softest number in the analysis. Note also what the full treatment of 6.4.2c would add: the assessed 20 % gives an expected cost of 12,000, but the *downside* is a certain 60,000 against a certain 45,000 penalty avoided, so the bad outcome leaves the project 15,000 worse off than doing nothing: a small enough exposure that the expected-value test is the right test here, which is not always so.

The **new float table is the deliverable, not the new date**. With D at 10 weeks, procurement C (critical for the whole project until this week) carries **TF = 2**, and the critical path is A–B–D–E–F. Every subsequent expediting conversation, every progress question and every recovery option must be aimed at the new binding path; the float table is what redirects the team, and issuing it is a separate act from announcing the recovered date. Teams that announce the date and not the table spend the following month managing the path that used to matter.

**What the domain teaches here.** Re-run the passes before spending a cent (the instinctive "crash procurement" would have bought nothing); price recovery in expected cost including risk, and state the breakeven for the probability you are least sure of; harvest overlap where the physics genuinely allows it; and report the new float positions to the client honestly: the schedule survived because the remediation estimate was honest a full five weeks before the window (Domain 11's reporting culture, applied to time).

## Case study B – Domain 6: the milestone that was typed, not computed (public programme)

**Situation.** A government services programme publishes a go-live milestone eighteen months out (programme **week 78**), set in a ministerial announcement before the L3 schedule existed. When the network is finally built, the test environment cannot exist before **week 69** and user-acceptance testing needs **12 weeks**, so honest logic finishes at week 81: the backward pass returns **TF = -3** from day one. To show the pinned date, the plan carries a **9-week** UAT window, **75.00 %** of the tested requirement, and calls it a plan.

**What happened, in numbers.** For six monthly reporting cycles (**24 weeks**) the programme's L1 view showed the milestone green. The date was pinned, so each upstream slip was absorbed by silently shortening the testing window: by cycle seven the environment date had moved to **week 73** and the window to **5 weeks**, or

**41.67 %** of the 12 weeks the test plan required. Two consequences were accumulating while the reporting said nothing.

**The recovery problem grew by a factor of 2.33.** At cycle one the gap between honest logic and the pinned date was 3 weeks; at cycle seven it was  $73 + 12 - 78 = 7$  **weeks**. Negative float does not sit still while it is being concealed. The upstream slips that the squeeze absorbed were real, so the pin converted a three-week problem into a seven-week one and spent 24 weeks of available runway doing it.

**And the concealment moved the risk from schedule to product.** If acceptance-test detection is taken as roughly proportional to the share of the test plan actually executed (a stated simplification, and exactly the kind of asserted detection rate Domain 9, KA 9.2.2 warns must be measured rather than assumed) then a regime designed to detect 75 % of defects reaching it detects  $0.75 \times 5/12 = 31.25$  % when only five twelfths of the plan is run. The escape fraction rises from 0.25 to 0.6875: **2.75 times** as many defects reach live service. The squeeze did not remove the programme's problem; it converted a visible schedule variance into an invisible quality one, priced not by the schedule but by Domain 9's external-failure economics and paid by users.

In cycle seven an assurance review (Domain 3, KA 3.3) ran the passes without the pin, surfaced the negative float, and forced the choice the pin had deferred: descope the first release (Domain 5), or move the date. The minister moved the date. What that cost politically is not quantified here (it cannot honestly be), but the two figures above can be, and they are the ones that would have made the case at cycle one.

**What the domain teaches here.** A typed milestone is a promise without a plan. Negative float is not embarrassment to be formatted away; it is the earliest, cheapest warning the programme will ever get (6.A.4), and its value decays because the problem it names grows. The tell was available every cycle and needed no assurance review to find: **a test window shorter than the test plan is a negative-float report written in a different notation**, and any reviewer who compares the two numbers finds it in one line. And the leader's protection is procedural, not heroic: milestones enter public commitments only *after* the backward pass says they exist.

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## Executive perspective — Domain 6

What a project leader cannot delegate in this domain:

- **The logic sign-off.** Discipline leads own their links; the leader owns the assembled claim that this network is how the project actually works.
- **Float policy.** Who may spend float, in what increments, reported how — decided before the first progress update, not during the first crisis.

- **The honesty settings.** No pins doing logic's work; negative float reported the cycle it appears; re-baselining only through governance. Schedules fail morally before they fail mathematically.
- **The feasibility question.** A schedule is approved as a logic network and executed with people. The leader asks one question before signing: *has this been levelled, against which caps, and what was the peak?* Auriga's plan was infeasible on engineering capacity while only 73.33 % loaded on average (6.3.1c), and no amount of logical float protected it. A baseline signed without a resource histogram is a baseline signed without knowing whether it can be run.
- **The compression chequebook.** Crash and fast-track decisions are investments under uncertainty: the leader signs the expected-value arithmetic (6.4.2) and the risk acceptance (Domain 8), never a bare instruction to "go faster". The chequebook has a price list: the compression menu with its cost slopes, priced at baseline while every option is still open, because options expire and the same two weeks cost USD 70,000 at planning and USD 85,000 in the crisis (6.A.4).
- **The translation duty.** Boards hear dates; teams live in passes, floats and scenarios. The leader is the honest converter between the two languages, both directions.

## Calculation exercises — Domain 6

**Exercise 6.1** Using the Auriga network, compute **ES/EF** for every activity and the project duration, showing the convergence at E. *Solution.* A 0–2 · B 2–8 · C 8–16 · D 8–15 · E  $\max(16,15)=16$ –21 · F 21–25 · G 15–17. Duration **25 weeks**; E's start is set by C (16), not D (15). The convergence rule takes the later predecessor. Common error: taking the earlier predecessor at a merge (giving E 15–20 and a false 24-week project).

**Exercise 6.2** Complete the backward pass and state **TF** and **FF** for D and G. *Solution.* From **LF** = 25: F 21/25 · E 16/21 · C 8/16 · G 23/25 · D **LF** =  $\min(E\text{ LS }16, G\text{ LS }23) = 16$ , **LS** = 9. **D: TF** = 1, **FF** =  $\min(16,15) - 15 = 0$ . **G: TF** = 8, **FF** =  $25 - 17 = 8$ . Common error: computing D's **FF** against only E (giving 1) — free float tests *every* successor.

**Exercise 6.3** The client offers USD 45,000 per week saved; crashing C costs USD 30,000/week (max 2 weeks); crashing E costs USD 55,000/week (max 1 week). Find the value-maximising plan. *Solution.* Crash C by 1: 25→24, +15,000. Second week of C: co-critical B–D–E–F holds 24 – –30,000, reject. Crash E by 1 (E is on *both* paths): 24→23,  $45,000 - 55,000 = -10,000$ , reject. **Plan: crash C by one week; net +USD 15,000.** Common error: crashing E first because it is "on more paths", path count doesn't change its negative unit economics. Note the option set: with only C and E on offer the answer is one week, and Exercise 6.8 shows

the same network reaching **22 weeks** once design B and commissioning F are also buyable; the optimum is a property of the menu, not of the network.

**Exercise 6.4** Activity durations:  $o = 6$ ,  $m = 8$ ,  $p = 16$  weeks. Compute  $t_e$  and  $\sigma$ , and state the deterministic bias. *Solution.*  $t_e = (6 + 32 + 16)/6 = \mathbf{9.0 \text{ weeks}}$ ;  $\sigma = (16 - 6)/6 = \mathbf{1.67 \text{ weeks}}$ . The deterministic plan at  $m = 8$  understates the expectation by a full week: right-skewed tails always pull the mean above the mode.

**Exercise 6.5** A programme's network shows  $TF = -2$  on the binding path. List, in cost order, the recovery families and the governance step each requires. *Solution.* (1) Logic re-choice/re-sequencing — planner's proposal, leader's sign-off; (2) float and resource harvesting from non-binding paths — float-policy authority; (3) crashing the binding path — expected-value case, budget authority; (4) fast-tracking — risk acceptance recorded in the register; (5) scope/acceptance renegotiation: sponsor and change control. Reporting the  $-2$  honestly precedes all five (6.A.4).

**Exercise 6.6** Enumerate Auriga's paths with their floats. Then state the cumulative slip the D–G chain can absorb without moving week 25, and the error a float report makes by summing D's and G's total floats. *Solution.* A–B–C–E–F **25**, float **0**; A–B–D–E–F **24**, float **1**; A–B–D–G **17**, float **8**. On the D–G chain  $d + g \leq 8$  with  $d \leq 1$  from the tighter path, so the absorbable cumulative slip is **8 weeks**, not the **9** implied by  $1 + 8$ , an overstatement of one week, or **USD 45,000** of false comfort at Auriga's cost of delay. Common error: adding an activity float column. The general form is that  $n$  activities on a path with float  $f$  report  $n \cdot f$  while the path holds  $f$ , so the overstatement factor is the number of activities on the path.

**Exercise 6.7** Auriga's engineering pool is capped at **6**. Engineer demands: A 2, B 4, C 1, D 3, E 5, F 6, G 2. Compute total demand, average utilisation against the 25-week plan, and find the week in which the early-start schedule breaches the cap. Then state the cheapest fix and its return. *Solution.* Total demand  $4 + 24 + 8 + 21 + 25 + 24 + 4 = \mathbf{110}$  engineer-weeks against  $6 \times 25 = \mathbf{150}$  of capacity, **73.33 %** average utilisation. The early-start profile is 2, 4, 4 (C+D), 3 (C+G), then **7 in week 16–17** where E (5) overlaps G (2), a breach of **one engineer for one week**. Hiring one engineer for that week costs **USD 5,225** and preserves the 25-week duration worth **USD 45,000**: a return of **8.61 times**. Common error: concluding from the 73.33 % utilisation that capacity is adequate; feasibility is a property of the peak, not the average.

**Exercise 6.8** From the compression menu — C 30,000/wk (max 2), B 40,000/wk (max 2), D 35,000/wk (max 1), E 55,000/wk (max 1), F 65,000/wk (max 1) — find

the least-cost duration when a week is worth 45,000, and state the range of week-values over which that answer holds. *Solution.* Marginal costs of successive weeks: **30,000** ( $C \times 1$ ), **40,000** ( $B \times 1$ , the first buy that must shorten *both* long paths), **40,000** ( $B \times 2$ ), **55,000** (E), **65,000** (F), **65,000** ( $C \times 2 + D$ ). Buy while the step is at or below 45,000: three weeks, cumulative **USD 110,000**, **22 weeks**, net **+USD 25,000**. The answer holds for any week-value between **40,000 and 55,000**; below 40,000 the optimum is 24 weeks and above 55,000 it is 21. Common error: comparing the *average* crash cost at 22 weeks (**USD 36,666.67**) with 45,000 and buying a fourth week that costs 55,000, a **USD 10,000** loss.

**Exercise 6.9** An overlap saves 2 weeks at 45,000 each; if the overlap fails it costs 140,000 and returns 1.5 weeks of the saving. A certain crash plan buys the same 2 weeks for 70,000. Compute both breakeven probabilities and the expected value at an assessed 25 %. *Solution.* Good outcome **+90,000**; bad outcome  $0.5 \times 45,000 - 140,000 = -117,500$ ; swing **207,500**. Against doing nothing,  $p^* = 90,000 / 207,500 = 43.37\%$ . Against the crash plan's certain net of  $90,000 - 70,000 = 20,000$ ,  $p^{**} = 70,000 / 207,500 = 33.73\%$ . At  $p = 0.25$  the expected value is  $90,000 - 0.25 \times 207,500 = +38,125$ , beating the crash plan by **18,125** while carrying a **137,500** worse downside. Common error: computing only  $p^*$  and overlapping in cases where a certain alternative was better, once a paid alternative exists,  $p^{**}$  is the relevant threshold.

**Exercise 6.10** A critical chain's 50 % durations are 1.5, 5, 6.5, 4 and 3 weeks against network durations of 2, 6, 8, 5 and 4. Size the buffer both conventional ways, then forecast completion from a status showing 45 % of the chain complete and 30 % of the buffer consumed. *Solution.* Chain **20.0** weeks; embedded safety 0.5, 1, 1.5, 1, 1 = **5.0**. Half-safety buffer **2.5** → commitment **22.5000** weeks; root-sum-square  $\sqrt{5.5} = 2.3452 \rightarrow 22.3452$  weeks, **2.6548** weeks inside the 25-week network, worth **USD 119,465.65**. Consumption ratio  $0.30 / 0.45 = 0.6667$ ; projected buffer use  $2.5 \times 0.6667 = 1.6667$  weeks; forecast chain completion **21.6667** weeks with **0.8333** weeks of buffer unused. Common error: applying root-sum-square to correlated activities, which understates the buffer, under shared causes the correct figure approaches the simple sum of 5.0 (Domain 8, KA 8.2.4).

**Exercise 6.11** Meridian must install 40 clinics in 50 weeks. A specialist completes 0.1 clinics a week. Compute the required takt, the crew, and the value of the acceleration from the existing crew of six on both valuation bases. *Solution.* Takt  $50 / 40 = 1.25$  weeks; required rate **0.8** clinics a week; crew  $0.8 / 0.1 = 8$ . With Domain 1's ramp (4 weeks, 25 % productivity, 50 % supervision) two newcomers give a ramp rate of **0.55** and a post-ramp rate of **0.8**, so duration is  $4 + (40 - 2.2) / 0.8 = 51.2500$  weeks against  $40 / 0.6 = 66.6667$ : **15.4167** weeks saved for **10** extra specialist-weeks, **USD 42,000**. On the ramp-integral basis the gain is



$20 \times 15.4167 \times 357 = \text{USD } 110,075$  (net +68,075); on the switch-on-at-completion basis  $15.4167 \times 14,280 = \text{USD } 220,150$  (net +178,150). Common error: valuing rollout acceleration at the fully-adopted cost of delay when benefit accrues per clinic. That doubles it, because the value per week of a linear ramp is  $(N/2) \times$  per-unit benefit, exactly half the steady-state figure.

## Practitioner's toolkit – Domain 6

*Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.*

### Toolkit 6.T.1 – Schedule quality gate (run before trusting any network)

- ☐ Zero dangles; zero loops; every lag has a stated reason.
- ☐ Date constraints listed, each justified as genuine external physics, none doing logic's job.
- ☐ Durations owned by the executing discipline, estimate class stated (far waves: ranged).
- ☐ Passes re-run after every change; critical-path durations sum exactly to project duration.
- ☐ Near-critical paths listed with their floats, and the band stated with what it captures.
- ☐ Float report is **per path**, not per activity; no summed float column anywhere.
- ☐ Float report distinguishes **TF** and **FF**; float policy names who may spend it.
- ☐ Milestones computed, owned, done-defined; no typed dates.
- ☐ Every approval modelled against the governance calendar, not approximated by a lag.
- ☐ Resource histogram attached, caps named as real or negotiable, peak stated with the excess area in resource-weeks; levelling rule and any better schedule sought both recorded.
- ☐ Where a buffer is used: 50 % durations genuinely elicited, sizing convention stated, correlation considered, buffer visible with an owner and reported with float.
- ☐ AI-drafted logic, levelling or forecasts marked, verified against a stated baseline, and owned by a named human.

### Toolkit 6.T.4 – Compression menu (priced at baseline, reviewed on the risk cycle)

Per candidate activity: current duration · which paths it lies on · cost slope per week · technical maximum weeks · owning discipline and the basis of the quote · risk consequence of taking it (coordination, quality, rework) · **non-financial ceiling**: the working-time, rest, fatigue, safety-case, permit or agreement limit that bounds this lever, with the function that supplied it · **the approval it needs** beyond the money,



naming the safety or approval-holding authority where one is engaged · expiry: the date after which the option no longer exists. A lever whose ceiling binds is struck from the menu, not discounted in it (6.4.2). Then, on one line: the value of a week (cost of delay plus any time-related indirect cost), the resulting least-cost duration, and the range of week-values over which that duration stays optimal. Re-priced whenever the network changes, because the cheapest week is a property of the current binding paths.

### Toolkit 6.T.2 — Recovery plan (one page)

Slip statement (activity, cause, size, date detected) · re-run pass results (new duration, new critical path, float table) · options priced in expected cost (re-sequence / harvest / crash / fast-track / renegotiate), each with risk and owner, **and each with its non-financial ceiling and the approval it needs** (6.4.2) · selected plan and **both** its decision authorities: the authority spending the money and, where a shift pattern, night working, an overlap of disciplines or a condition of an approval is engaged, the **named safety or approval-holding authority** · revised commitments and the stakeholder notice list · review date.

### Toolkit 6.T.3 — Milestone register

Per milestone: computed date and current TF · owner · done-definition (acceptance reference) · contractual status and counterparty · L3 logic trace · reporting treatment (internal / contractual / public) · change history.

## Exam preparation — Domain 6

**The calculation traps.** Taking the earlier predecessor at a convergence (Exercise 6.1) · computing FF against one successor instead of all (Exercise 6.2) · quoting TF after it has been spent · **summing an activity float column, which overstates absorption by the number of activities on the path** (6.2.2) · crashing from the original network instead of re-running passes (path migration, 6.4.2) · **comparing the average crash cost with the value of a week instead of the marginal one** (6.4.2b) · **inferring resource feasibility from average utilisation when it is set by the peak** (6.3.1c) · **valuing a rollout acceleration at the fully-adopted cost of delay when benefit accrues per unit, which doubles it** (6.A.5) · **rolling up ranged planning packages by adding mid-points** (6.3.3) · **substituting  $E[\text{wait}] = M/2 + L$  for a planned approval whose submission date is known** (6.1.2b) · treating the PERT mode as the mean, and forgetting that a symmetric estimate has no bias at all (6.4.3) · reading a pinned milestone as a forecast · confusing project-level float with activity-level slack in reports.

### Reflection questions.

1. Your programme's board pack shows every milestone green. What three questions establish whether that is information or formatting? (*Computed or typed? Passes re-run this cycle? Where is the float table?*)
2. A subcontractor asks for "the float" on their package. What must you settle before answering? (*Which float (TF or FF; the float policy; contractual float ownership) Domain 10; and whether anyone else is already drawing on the same path float, 6.2.2.)*)
3. When did you last see money spent on compression that the passes would have shown was worthless, and which governance step was missing? (*6.4.2; toolkit 6.T.2.)*)
4. Has your last approved baseline been levelled against a real resource cap, and do you know its peak? (*6.3.1c; toolkit 6.T.1. If the answer is an average utilisation figure, it is not an answer.*)
5. Which of your approvals is currently modelled as a lag rather than as a date against a meeting calendar, and how much float is that costing you? (*6.1.2b.*)

### Domain 6 summary

This domain turned "when will it finish?" from an opinion into an instrument. Planning levels give each audience a true view of one logic; dependency discipline makes the network a model rather than a drawing, and an approval modelled against its governance calendar rather than as a three-week label was worth four weeks of Meridian's schedule from one week of design work. The forward and backward passes yield the dates, the two floats and the critical path: with the master project showing why free float can vanish while total float survives, why convergence points compound slippage, why an activity float column can never be summed (nine weeks reported, eight available), and why a near-critical band of 10 % captures 85.71 % of the network rather than a footnote to it. Resources, constraints, milestones and rolling waves connect the network to the real world's crews, windows and horizons; and connect it hard: a plan only 73.33 % loaded on average was infeasible on its peak, an activity with eight weeks of logical float set the resource-constrained duration, and one engineer-week of excess demand was worth USD 45,000. Delivery flow closes the loop: a compression menu with rising cost slopes whose least-cost duration is 22 weeks and holds for any week worth 40,000–55,000 (each lever carrying a non-financial ceiling above its price, and a shift-pattern, night-working or simultaneous-operations move carrying a second signature the money cannot buy) fast-tracking priced with two breakeven probabilities rather than a hope, recovery ranked from re-choice to renegotiation, buffers sized by aggregating safety and managed by their consumption ratio, rollouts scheduled as a takt and valued as an integral, and agile streams joined to the network through a required rate read off the backward pass. The leadership spine throughout: schedules fail morally before they fail mathematically, computed dates, visible float, reported negative float, resource

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feasibility tested before signature, and machine output verified against a stated baseline by named humans are what make a programme's word worth something.

DOMAIN

# 7

## Cost, Resources and Commercial Awareness

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## Why this domain exists

A leader who can defend a date but not a number is half-equipped. Domain 6 built the schedule; this domain builds the money that pays for it and the commercial arrangements that bind other people to deliver it. It starts with estimating and budgeting: how a credible number is constructed and what its accuracy claim actually means (KA 7.1). From there it establishes the cost baseline and the forecasting that keeps it honest (KA 7.2), and builds **earned value** in full, because no other technique answers cost and schedule performance in one integrated language (KA 7.3). The close is the commercial awareness a leader cannot delegate: resource economics, procurement strategy, contract models and cash flow (KA 7.4). Risk quantification deepens in Domain 8; contracts and supply networks get their own treatment in Domain 10. What belongs here is the leader's own numeracy: enough to know when a forecast is arithmetic and when it is hope.

**Learning objectives.** After this domain a candidate can: select an estimating method appropriate to the available definition and state its accuracy class; **read an accuracy class as a distribution and say what percentile the approved number sits at**; build a three-point cost estimate and distinguish a tail that is a spread from a tail that is a scenario; assemble a cost baseline separating contingency from management reserve, **and trend contingency draw against progress to a required draw efficiency**; measure **AC** so that it covers the same work as **EV**, **and compute what omitting accruals does to every index and forecast built on it**; distinguish commitments from actuals and **compute the uncommitted exposure of a forecast**; compute and interpret the full earned-value set (**CV**, **SV**, **CPI**, **SPI**); **show how much of a reported index is an artefact of the earning rule chosen, including the level-of-effort distortion identity**; forecast with each **EAC** method and choose the one matching the variance's cause; **state where each forecast's funding runs out if efficiency persists**; compute **VAC** and **TCPI**, **derive the identity family linking TCPI to every EAC, and convert a recovery-to-budget promise into the descope or the efficiency gain it actually requires**; read resource economics through blended rates, **burdened cost per productive hour and the priced cost of a crew-week**; distinguish the main contract models by who carries cost risk and **compute the hours breakeven between time-and-materials and a fixed price**; compute a cost-incentive fee and the point of total assumption, **and show how the share ratio and the ceiling move it**; explain why cash flow and profit differ on a project **and compute the working capital a project absorbs**; **price the governance latency on a reserve release**; **verify an automated accrual, ledger or commitment feed against the ledger, the following period's reversal and a hand-reproduced coverage figure, and name the four determinations no tool may make**; and govern AI-produced cost forecasts under the family verification rule.

**The master worked project.** Project Auriga continues from Domain 6 (the 25-week control-systems upgrade for a regional utility) now with money attached. Its approved cost baseline is **BAC = USD 4,000,000**, assembled from **USD 3,640,000** of control-account budgets plus **USD 360,000** of contingency, with **USD 240,000** of management reserve outside it for a total funding requirement of **USD 4,240,000** (KA 7.1.3). The delivering organisation's fixed price to the utility is **USD 4,400,000**, so the bid margin is **USD 400,000**, 9.09 % of price (KA 7.4.4). At the **data date, end of week 13**, the baseline says **PV = USD 2,080,000**; measurement gives **EV = USD 1,920,000** and **AC = USD 2,120,000**. Every calculation in KA 7.2–7.3 uses those four numbers, and Auriga's engineering blended rate of **USD 130.625 per hour** (an engineer-week of **USD 5,225**, the figure Domain 9 costs rework with) carries the resource arithmetic of KA 7.4. Domain 6's **cost of delay of USD 45,000 per week** is the price of time throughout. **Meridian Care Records** (the 40-clinic programme of Domains 1, 2, 15 and 16, approved cost **USD 2,400,000**, cost of delay **USD 14,280 per week** at its planned 70 % adoption) returns in Case study C, because reserve architecture and the latency of the authority that releases it are programme-scale cost questions a single project cannot show.

**Reference points.** The body most often named in connection with the cost-engineering material of KA 7.1 is **AACE International**, referenced for the existence and purpose of the cost-estimate classification progression in its Total Cost Management framework (7.1.1). It is a professional association's framework (voluntary, describing practice, neither legislation nor a certifiable requirement), and it obliges nobody of itself unless an organisation or a contract adopts it. It is named here and not reproduced: no class table, range or definition from it appears in this volume, the treatment in 7.1.1 is written in this book's own words, and a reader who wants the framework should obtain the current version from its publisher. Naming it implies no endorsement in either direction.

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## KNOWLEDGE AREA 7.1

# Estimating and budgeting

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*Topics: 7.1.1 estimating methods and accuracy classes · 7.1.2 three-point estimating · 7.1.3 from estimate to budget.*

## 7.1.1 Estimating methods and accuracy classes

**The principle.** An estimate's accuracy is governed by how well the work is *defined*, not by how much effort went into the arithmetic. The three standard methods trade definition against speed:

| METHOD                      | HOW IT WORKS                                           | NEEDS                               | TYPICAL USE                   |
|-----------------------------|--------------------------------------------------------|-------------------------------------|-------------------------------|
| <b>Analogous (top-down)</b> | Scale a comparable past project                        | A true analogue and a scaling basis | Concept, screening (Domain 2) |
| <b>Parametric</b>           | Cost = rate × parameter (per metre, per point, per kW) | A calibrated rate from real history | Early design, repetitive work |
| <b>Bottom-up</b>            | Estimate each work package, then sum                   | A WBS at package level (Domain 4)   | Baseline setting, control     |

**Accuracy classes.** Mature cost practice publishes an estimate's *class* alongside its number: a range that narrows as definition matures (the AACE Total Cost Management framework's class-5-to-class-1 progression is the reference treatment, described here in this book's own words). The professional discipline is simple and widely broken: **an estimate is never a single number without a range and a class.** "USD 4 million" is not an estimate; "USD 4 million, -15 %/+30 %, at a class consistent with 30 % design completion" is.

That second formulation contains far more than a caveat, and almost nobody reads what it says.

#### WORKED EXAMPLE

##### Worked example 7.1.1 – what Auriga's accuracy class actually claims.

1. **Setup.** Auriga's sanction estimate is **USD 4,000,000** with a declared class range of **-15 %/+30 %** at 30 % design completion. The programme board approves that number as the baseline. Read the range as a **triangular distribution** with the point estimate as its most likely value (the least assumption-heavy reading of a three-point statement), and ask three questions: how wide is the claim, where does the approved number sit inside it, and what would funding at a stated confidence have cost?
2. **Formula.** Bounds  $lo = P(1 - d^-)$ ,  $hi = P(1 + d^+)$ . For a triangular distribution on (a, m, b): mean =  $(a + m + b)/3$ ; the cumulative probability at the mode is  $(m - a)/(b - a)$ ; median =  $b - \sqrt{((b - a)(b - m)/2)}$ ; the p-quantile above the mode is  $b - \sqrt{((1 - p)(b - a)(b - m))}$ ; standard deviation  $\sigma = \sqrt{((a^2 + m^2 + b^2 - ab - am - bm)/18)}$ . For comparison, the PERT weighting of the same three points is  $(a + 4m + b)/6$ .
3. **Substitution.**  $lo = 4,000,000 \times 0.85$ ;  $hi = 4,000,000 \times 1.30$ ; mean =  $(3,400,000 + 4,000,000 + 5,200,000)/3$ ;  $F(\text{mode}) = 600,000/1,800,000$ ;  $P80 = 5,200,000 - \sqrt{(0.20 \times 1,800,000 \times 1,200,000)}$ ; median =  $5,200,000 - \sqrt{(1,800,000 \times 1,200,000/2)}$ .

4. **Result.** Range USD 3,400,000 to USD 5,200,000: a band of USD 1,800,000, or 45.0 % of the point estimate.  $\sigma$  = USD 374,166. The mean is USD 4,200,000; the median USD 4,160,770; the P80 USD 4,542,733. The approved USD 4,000,000 sits at the 33.33rd percentile of the estimate's own declared range. The PERT weighting of the same three points gives USD 4,100,000.
5. **Interpretation.** The arithmetic turns a routine caveat into four statements a board can be held to, and the first one is uncomfortable.

**Auriga was funded at a number it had a two-in-three chance of exceeding.** Not because anybody was careless, but because an asymmetric range with the point estimate at its mode puts only  $(m - a) / (b - a)$  of the distribution at or below that point: here exactly one third. The general result is worth carrying: **whenever the upside tail is longer than the downside tail, the point estimate is below the mean and below the median**, so approving the point estimate is approving a sub-50 % confidence position. It is a legitimate thing to do; it is not a legitimate thing to do unknowingly, and the sentence that makes it knowing is "this baseline is a P33".

**The mean of the declared range is USD 4,200,000, which is exactly the week-13 EAC that Auriga's own data produces under method (a) (KA 7.3.3).** That coincidence is worth pausing on, because it is the shape of most cost surprises: the eventual outturn was inside the estimate's stated accuracy from the first day. What went wrong was not the estimate but the **funding decision taken against it**: approving the mode and reporting the variance against the mode. A reviewer's question follows directly: *is the current forecast outside the original class range, or merely above the point?* Outside the range is an estimating failure; above the point but inside the range is a funding-level decision coming home.

**Confidence has a price and it is computable.** Funding at the P80 rather than the point requires USD 542,733 more; funding at the mean requires USD 200,000 more. Auriga's baseline in fact carries USD 360,000 of contingency inside the 4,000,000 (KA 7.1.3), which is a different structure again, because that contingency was sized bottom-up against identified risks rather than read off a distribution, and the two figures answer different questions. Where both are available they should be reconciled explicitly: a risk-based contingency far below the distribution-based number means either the register is incomplete (Domain 8, KA 8.2.4 aggregates it properly) or the range was drawn too wide.

**Two cautions, both of which a reviewer should raise.** The distribution shape is an assumption, not a fact: triangular gives a mean 200,000 above the point, the PERT weighting 100,000, so the *direction* is robust and the *magnitude* is not, and no more than "the mean is 100,000 to 200,000 above the point, 2.5 % to 5.0 % of the estimate" should be claimed. And a class range is a statement about **definition maturity**, not a probability distribution anybody measured; treating it as one is a modelling convenience that must be labelled. What it is *not* is dec-



oration. The honest use is the one above: convert the range into the percentile the approved number occupies, and say so out loud.

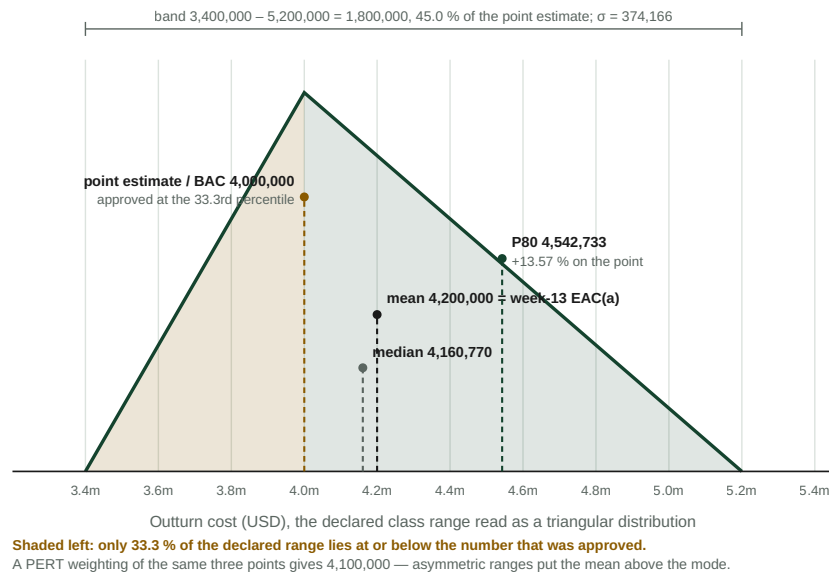


Fig 7.1.1 — What an accuracy class actually says

**Common pitfall: precision mistaken for accuracy.** A bottom-up estimate summing 400 packages to USD 4,183,662 looks authoritative and is no more accurate than its worst assumption. Leaders should be more suspicious of an over-precise number than a rounded one.

### 7.1.2 Three-point estimating

Cost, like duration (Domain 6, KA 6.4.3), is a distribution. The same PERT weighting applies:

$$C_e = (o + 4m + p) / 6 \quad \sigma = (p - o) / 6$$

#### WORKED EXAMPLE

##### Worked example 7.1.2 — Auriga's control-hardware package.

1. **Setup.** Procurement of control hardware is estimated **optimistic USD 680,000**, **most-likely USD 750,000**, **pessimistic USD 1,000,000** (the tail: a single-source controller with volatile lead-time pricing).
2. **Formula.**  $C_e = (o + 4m + p) / 6$ ;  $\sigma = (p - o) / 6$ .
3. **Substitution.**  $C_e = (680,000 + 3,000,000 + 1,000,000) / 6 = 4,680,000 / 6$ ;  $\sigma = 320,000 / 6$ .
4. **Result.**  $C_e = \text{USD } 780,000$ ;  $\sigma \approx \text{USD } 53,333$ .

5. **Interpretation.** The most-likely figure is 750,000, but the *expected* cost is 780,000. The right-skewed tail adds USD 30,000 before anything goes wrong. Budgeting at the mode systematically under-funds a portfolio of such packages; the difference between mode and mean, summed across a project, is a large part of what contingency exists to cover (7.1.3, and Domain 8's quantification).

**Budget the mean, hold the tail centrally.** Auriga's control account CA-20 is budgeted at **USD 780,000**: the  $C_e$ , not the mode (7.1.3's table). Budgeting the mode would have loaded **USD 30,000** of arithmetically certain under-funding into the baseline of this one package, and a project of forty such packages inherits forty such gaps that no manager can be held accountable for. The professionally right structure is the one used here: **each control account budgeted at its expected value, the dispersion held once, centrally, as contingency.**

**The mode-to-mean gap is 0.5625  $\sigma$ , and that ratio is the tell.** It is 4.00 % of the mode here: small enough to be dismissed in a meeting and large enough to matter forty times over.

**The pessimistic value is 4.125  $\sigma$  above the mean, and that is a warning, not a spread.** The optimistic value sits 1.875  $\sigma$  below it. A tail four standard deviations out is not the upper end of a continuum of outcomes; it is a **discrete event** (the single-source controller not being available at the quoted price), which has its own probability and its own response. The professional treatment is to carry it in the risk register with that probability and cost (Domain 8, KA 8.2.2), *not* to smear it into a  $\sigma$  that then implies a symmetric spread nobody believes. Three-point estimating is the wrong instrument for a bimodal outcome, and asymmetry of this severity is the diagnostic.

**Sizing from  $\sigma$  needs its approximation stated.** Treating  $C_e$  and  $\sigma$  as a normal approximation, a P80 for this package is  $780,000 + 0.8416 \times 53,333 = \text{USD } 824,885$ , so **USD 44,885** of contingency above the mean, or USD 74,885 above the mode. That is a defensible working figure and it is an approximation twice over. The PERT  $\sigma$  is itself a rule of thumb, and a distribution with a 4  $\sigma$  tail is not normal. Quote it as an order of magnitude, aggregate properly at project level (Domain 8, KA 8.2.4, where correlation between packages does the real work), and never present a package-level P80 as though it were measured.

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### 7.1.3 From estimate to budget

**The structure.** An estimate becomes a controllable budget through a deliberate hierarchy:

|                                |                       |
|--------------------------------|-----------------------|
| work-package estimates         |                       |
| → control-account budgets      | (WBS × organisation – |
| the management-control points) |                       |
| → + contingency reserve        | → the COST BASELINE   |
| (BAC): the PM's authority      |                       |
| → + management reserve         | → the total project   |
| funding requirement            |                       |

Two rules carry the discipline. **Contingency is inside the baseline**, sized against identified risks (Domain 8) and spent by the project manager under a stated protocol. **Management reserve is outside the baseline**, held for unknown-unknowns, and released only by the sponsor or change authority (Domain 3's decision rights) – releasing it *changes* the baseline through change control (Domain 4, KA 4.4). Blurring the two is how projects appear to be "on budget" while consuming their own funding runway.

**Time-phasing.** The baseline is spread across the schedule to produce the **cumulative cost curve** (the S-curve), and that curve *is* Planned Value (KA 7.3). A budget with no phasing cannot be measured against; the schedule (Domain 6) is therefore a precondition of cost control, not a parallel activity.

#### WORKED EXAMPLE

##### Worked example 7.1.3 – assembling Auriga's baseline, and reading the contingency trend.

1. **Setup.** Auriga's six control accounts, each budgeted at its expected value (7.1.2's discipline), with contingency sized bottom-up against the identified risks of Domain 8's register and management reserve set by the sponsor:

| CONTROL ACCOUNT                          | BUDGET (USD)     |
|------------------------------------------|------------------|
| CA-10 Project management and engineering | 620,000          |
| CA-20 Control hardware procurement       | 780,000          |
| CA-30 Civil and enabling works           | 540,000          |
| CA-40 Installation                       | 1,020,000        |
| CA-50 Testing and commissioning          | 480,000          |
| CA-60 Training and handover              | 200,000          |
| <b>Control-account total</b>             | <b>3,640,000</b> |

Contingency is **USD 360,000**; management reserve is **USD 240,000**. At the week-13 data date the project is **48.00 %** complete by **EV/BAC** (KA 7.3.2) and has drawn **USD 190,000** of contingency: USD 150,000 for the ground-remediation risk and USD 40,000 for control-hardware price escalation. 2. **Formula.** **BAC** =

control accounts + contingency. Total funding requirement = BAC + management reserve. Contingency draw index = (draw ÷ contingency) ÷ percent complete. Draw per point of progress = draw ÷ percent complete. Projected total draw = that rate × 100. Required draw efficiency for the remainder = remaining contingency ÷ (demonstrated rate × remaining points of progress). 3. **Substitution.** BAC = 3,640,000 + 360,000; funding = 4,000,000 + 240,000. Index = (190,000/360,000)/0.48 = 0.5278/0.48. Rate = 190,000/48.00. Required efficiency = 170,000/(3,958.33 × 52.00). 4. **Result.** BAC = USD 4,000,000; total funding requirement USD 4,240,000. Contingency is 9.89 % of the control-account total and 9.00 % of BAC; management reserve is 6.00 % of BAC. Contingency is 52.78 % drawn at 48.00 % complete, a draw index of 1.0995. At the demonstrated rate of USD 3,958.33 per point of progress the full job needs USD 395,833, a shortfall of USD 35,833 (9.95 % of the contingency, and 14.93 % of the management reserve). The remaining USD 170,000 must cover 52.00 points, i.e. USD 3,269.23 per point: a **required draw efficiency of 0.8259**, meaning the remaining work must consume contingency 17.41 % more slowly than the first half did. 5. **Interpretation.** The third of the readings below is a reviewer's invariant most cost reports would fail.

**Contingency has a TCPI of its own, and it should be reported.** The structure is identical to KA 7.3.4's: a demonstrated rate, a remaining allowance, and the efficiency the remainder must achieve. Reporting "contingency 52.8 % drawn" invites a shrug; reporting "the remaining work must draw contingency 17.41 % more slowly than the work to date, and there is no stated reason why it would" invites a decision. The threshold to watch is the **draw index**: above 1.00 the reserve is depleting faster than the work is being done, and the projected shortfall (here USD 35,833) is the number that eventually becomes a management-reserve request. It is visible now, at 48 % complete, which is the whole point of trending it.

**The two reserves are not interchangeable and the arithmetic shows why.** The 35,833 projected shortfall is 14.93 % of the management reserve, so the reserve can absorb it; but absorbing it is a **sponsor** act that changes the baseline through change control (Domain 4, KA 4.4), not a project-level top-up. A project that quietly funds a contingency overrun from management reserve has converted an early warning into an invisible one, and MCQ 7.1-B's erosion pattern is what that looks like three periods later.

**The draw must reconcile to the cost variance, and here it does not.** CV at week 13 is (USD 200,000) while contingency drawn is USD 190,000: a gap of USD 10,000. That gap is not a rounding artefact; it means either a risk was funded without a draw being recorded, or USD 10,000 of overrun has no identified cause. Both are findings. The invariant is worth carrying into every review: **identified-risk draws plus unattributed variance must equal the cost variance**, and the unattributed component is the one to interrogate, because a

variance with no cause cannot be forecast (KA 7.2.2) and therefore cannot be recovered.

The caution: none of this arithmetic is valid if contingency was sized as a percentage rather than against a register. A flat "10 % contingency" has no risks behind it, so a draw against it cannot be reconciled to anything, and the trend above degenerates into an observation about spending speed. Domain 8, KA 8.2.4 is where the number comes from; this KA is where it is governed.

## AI in this KA

Estimating is where AI assistance is most seductive and most in need of provenance. A model can produce a plausible parametric rate instantly, and it cannot tell you whether that rate came from comparable work, a different market, or nowhere at all. The governed workflow: **AI proposes** a method, a rate and a range; the estimator supplies the *calibration evidence* (which projects, which years, which escalation basis, the source discipline of the shared registry); the range and class are stated; and a named human owns the number. **AI proposes; the professional verifies, decides and remains accountable.** An estimate whose basis cannot be produced on request is not an estimate.

Two placements are worth being exact about, because 7.1.1 and 7.1.3 create work that machines do well and work they must not touch. Converting a declared class range into percentiles, means and a funding-confidence comparison (and repeating it across every package and every candidate distribution shape) is mechanical, exactly checkable and almost never done by hand; that is a good use. **Choosing the shape is not**, because the shape encodes a belief about how the tail behaves, and 7.1.2's four-sigma pessimistic value shows what happens when the wrong shape is applied to a discrete event: a scenario is quietly turned into a spread. Likewise, computing the contingency draw index and the required draw efficiency each period is routine; deciding whether the draw pattern means the register was incomplete or the work is simply front-loaded on its risks is a judgement about the project, and belongs to whoever will have to ask for the management reserve.

## KEY TERMS — KA 7.1

| TERM                                      | MEANING                                                                 |
|-------------------------------------------|-------------------------------------------------------------------------|
| <b>Analogous / parametric / bottom-up</b> | Scale an analogue · rate × parameter · sum the packages.                |
| <b>Accuracy class</b>                     | The definition-linked range accompanying an estimate; never optional.   |
| <b>Control account</b>                    | The WBS × organisation point where scope, budget and actuals integrate. |

| TERM                             | MEANING                                                                                                                                                     |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Contingency reserve</b>       | Inside the baseline; funds identified risks; PM-controlled.                                                                                                 |
| <b>Management reserve</b>        | Outside the baseline; unknown-unknowns; sponsor-controlled.                                                                                                 |
| <b>BAC</b>                       | Budget at Completion: the time-phased cost baseline's total.                                                                                                |
| <b>Total funding requirement</b> | <b>BAC</b> + management reserve: the cash the sponsor must have available; Auriga's is USD 4,240,000.                                                       |
| <b>Approval percentile</b>       | The position of the approved number inside its own declared range; Auriga's baseline is a P33.                                                              |
| <b>Contingency draw index</b>    | $(\text{Contingency drawn} \div \text{contingency}) \div \text{percent complete}$ ; above 1.00 the reserve is depleting faster than the work is being done. |
| <b>Required draw efficiency</b>  | $\text{Remaining contingency} \div (\text{demonstrated draw rate} \times \text{remaining progress}) - \text{the TCPI of the reserve.}$                      |

## SAMPLE MCQs — KA 7.1

**MCQ 7.1-A** [7.1.2 · Application] A package is estimated  $o = 680,000$ ,  $m = 750,000$ ,  $p = 1,000,000$ . Its PERT expected cost is:

- A. USD 750,000
- B. USD 780,000 CORRECT
- C. USD 810,000
- D. USD 840,000

*Rationale:*  $(680,000 + 4 \times 750,000 + 1,000,000)/6 = 780,000$ . A is the mode; C is the unweighted three-point mean; D over-weights the pessimistic value.

**MCQ 7.1-B** [7.1.3 · Analysis] A project reports "on budget" while having consumed 60 % of its management reserve at 40 % complete. The correct reading is:

- A. genuinely on budget — management reserve exists to be spent
- B. the baseline is intact but the project's total funding is eroding faster than progress; the trend belongs in the next report to the sponsor CORRECT
- C. a baseline breach requiring immediate re-baselining
- D. an accounting error, since management reserve is inside the baseline

*Rationale:* Management reserve sits *outside* the baseline (so D is wrong and A is technically true but misleading), and its release is a sponsor-level signal, not a project-level convenience. It is not yet a breach (C). It is the early warning that precedes one.

**MCQ 7.1-C** [7.1.1 · Recall] Which statement about a bottom-up estimate summing to USD 4,183,662 is soundest?

- A. its precision indicates high accuracy

- B. its accuracy is bounded by its assumptions and definition maturity, and it must still carry a range and class CORRECT
- C. rounding it would reduce its accuracy
- D. bottom-up estimates do not need accuracy classes

*Rationale:* Precision (digits) and accuracy (closeness to outturn) are independent; definition maturity governs the latter. Rounding changes no information (C), and every estimate carries a class (D).

**MCQ 7.1-D** [7.1.1 · Analysis] A point estimate of USD 4,000,000 carries a declared class range of -15 %/+30 %. Read as a triangular distribution, its expected cost is:

- A. USD 4,000,000
- B. USD 4,100,000
- C. USD 4,200,000 CORRECT
- D. USD 4,300,000

*Rationale:*  $(3,400,000 + 4,000,000 + 5,200,000)/3 = 4,200,000$  (7.1.1). A treats the point estimate as the mean, which only holds for a symmetric range; B applies the PERT weighting, which weights the mode four times and is a different shape from the one asked for; D averages the two range endpoints and ignores the mode entirely.

**MCQ 7.1-E** [7.1.3 · Application] Control accounts total 3,640,000; contingency is 360,000; management reserve is 240,000. BAC and the total funding requirement are:

- A. 3,640,000 and 4,000,000
- B. 4,000,000 and 4,240,000 CORRECT
- C. 4,240,000 and 4,240,000
- D. 4,000,000 and 4,000,000

*Rationale:* Contingency is inside the baseline and management reserve outside it (7.1.3). A excludes contingency from BAC, which would leave the project's own risk funding outside its authority; C puts management reserve inside the baseline, the error MCQ 7.1-B's misreading rests on; D omits the reserve from the funding requirement, so the sponsor holds no cash for it.

**MCQ 7.1-F** [7.1.3 · Analysis] Contingency of 360,000 is 190,000 drawn at 48.00 % complete. The draw efficiency the remaining work must achieve is:

- A. 0.4722
- B. 0.8259 CORRECT
- C. 0.9095
- D. 1.0995

*Rationale:*  $170,000 / (3,958.33 \times 52.00) = 0.8259$ , the remainder must draw 17.41 % more slowly than demonstrated (7.1.3). A is the remaining balance as a share of the total, which is a level not an index; C inverts the two shares; D is the draw index itself: the rate at which contingency is being consumed relative to progress, not the efficiency required to recover.

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SELF-CHECK — KA 7.1

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1. *Where do contingency and management reserve sit, and who spends each?* — Contingency inside the baseline, PM-controlled; management reserve outside it, sponsor-controlled.
  2. *Why is the time-phased baseline a precondition for cost control?* — Because the phased cumulative curve is Planned Value; without it there is nothing to measure performance against.
  3. *What two things must accompany every estimate?* — A range and an accuracy class tied to definition maturity.
  4. *What percentile of its own declared range was Auriga's baseline approved at, and why?* — The 33.33rd: an asymmetric -15 %/+30 % range puts only  $(m - a) / (b - a)$  of the distribution at or below the mode, so the approved number had a two-in-three chance of being exceeded.
  5. *What does a contingency draw index above 1.00 mean, and what number does it eventually produce?* — The reserve is depleting faster than the work is being done; on Auriga it projects a USD 35,833 shortfall, which becomes a management-reserve request unless the draw rate changes.
  6. *What must the contingency draw reconcile to?* — The cost variance. Auriga's USD 190,000 of draws against a CV of (USD 200,000) leaves USD 10,000 of variance with no identified cause, which is a finding because an unexplained variance cannot be forecast.
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KNOWLEDGE AREA 7.2

## The cost baseline, actuals and forecasting

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*Topics:* 7.2.1 measuring actual cost · 7.2.2 the forecasting question · 7.2.3 baseline integrity.



### 7.2.1 Measuring actual cost

**The principle.** AC must cover the same work as EV, in the same period, or every index built from them is fiction. Three mechanics decide whether it does:

- **Accruals.** Work received but not yet invoiced belongs in this period's AC. Omit accruals and cost performance looks excellent until the invoices land: the commonest cause of a "sudden" overrun that was months old.
- **Commitment vs actual.** A purchase order is a *commitment*, not a cost. Both matter: AC drives performance, commitments drive the funding forecast. Confusing them double-counts or hides money.
- **Open-commitment hygiene.** Stale purchase orders left open inflate the forecast; cleansing them is a standing month-end task.

Each is usually stated and left; the size of what they do to a report is not intuitive.

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#### WORKED EXAMPLE

##### Worked example 7.2.1 – the accrual that made Auriga look perfect.

1. **Setup.** At week 13 Auriga has genuinely incurred USD 2,120,000 against EV of USD 1,920,000. The ledger, however, shows only USD 1,920,000 of invoiced cost: the installation subcontractor's work for weeks 11–13 has been received and accepted but not yet invoiced, and USD 200,000 of it has not been accrued. Weeks 14–17 then run at a true EV of USD 480,000 for a true cost of USD 520,000, and the accrual backlog is invoiced in that period. Compare what the report says with what is true, in both periods and cumulatively.
2. **Formula.**  $CPI = EV/AC$ ;  $CV = EV - AC$ ;  $EAC = BAC/CPI$ ;  $TCPI = (BAC - EV)/(BAC - AC)$ . Period index = period EV ÷ period AC. Cumulative index = cumulative EV ÷ cumulative AC.
3. **Substitution.** Reported  $CPI = 1,920,000/1,920,000$ ; true  $CPI = 1,920,000/2,120,000$ . Reported  $EAC(b) = 4,000,000/1.0000$ ; true =  $4,000,000/0.905660$ . Period reported =  $480,000/(520,000 + 200,000)$ ; period true =  $480,000/520,000$ . Cumulative after week 17 =  $2,400,000/2,640,000$ .
4. **Result.**

| AT WEEK 13  | REPORTED  | TRUE      |
|-------------|-----------|-----------|
| AC          | 1,920,000 | 2,120,000 |
| CV          | 0         | (200,000) |
| CPI         | 1.0000    | 0.9057    |
| EAC(b)      | 4,000,000 | 4,416,667 |
| TCPI to BAC | 1.0000    | 1.1064    |

The single omission understates the forecast by **USD 416,667: 10.42 % of BAC**. In the catch-up period the reported **AC** is  $520,000 + 200,000 = \text{USD } 720,000$ , so the reported period **CPI** is **0.6667** against a true period **CPI** of **0.9231**: an error of **0.2564**, and a reported figure implying the period's work cost **50.0 %** more than budget. Cumulatively after week 17 both versions converge on **EV 2,400,000** against **AC 2,640,000**, **CPI 0.9091**. 5. **Interpretation.** The invariant is the useful part: **an accrual omission is a timing error, so it cancels cumulatively and misstates every period in between.**

**The direction of the distortion reverses, and the second half is the one that gets investigated.** Weeks 1–13 report perfection; week 17 reports a collapse. The true story is that performance *improved* (0.9057 to 0.9231) in exactly the period the report shows falling apart. Anyone acting on the period reading will investigate the wrong thing, and the operational explanations offered in that meeting will be inventions, because there is nothing operational to explain.

**The plausibility test is the detection mechanism.** A single-period **CPI** of 0.6667 implies costs 50 % above budget for four weeks. Ask what physically happened to cause that; if the answer is thin, the cause is measurement, not production, and the ledger is where to look. This is MCQ 7.2-A's step change with the arithmetic attached: **a step in CPI with no operational counterpart is a measurement event.**

**The forecast error is the real damage, not the index.** Reported **TCPI** to **BAC** was 1.0000 ("no recovery required"), while the true figure was 1.1064. A board told the project needs par performance to finish at budget, when it in fact needs an 11 % efficiency gain, has been given the opposite of a warning. And the reported **EAC(b)** of exactly **BAC** is the tell: **when CPI reads exactly 1.0000, check the accruals before celebrating**, because a real project rarely lands on the integer.

**The cure is a control, not diligence.** **AC** must never be below invoices plus receipted-but-uninvoiced value, the reconciliation must be a signed month-end step, and the accrual basis must be stated in the report. The caution runs the other way too: an *over-accrual* flatters the following period exactly as symmetrically, so a project whose **CPI** improves in every period after a bad month should be asked whether the bad month was over-accrued.

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#### WORKED EXAMPLE

### Worked example 7.2.1b – commitments, coverage and the exposure nobody reports.

1. **Setup.** Auriga at week 13: **AC USD 2,120,000**; open purchase orders and subcontract commitments for work not yet received total **USD 1,180,000**. Of those open commitments, **USD 120,000** relates to scope already delivered and invoiced: stale lines nobody has closed. The forecast in use is method (b), **EAC USD 4,416,667** (KA 7.3.3). Twelve weeks remain.
2. **Formula.** Uncommitted balance of the baseline = **BAC – AC – open commitments**. Commitment coverage = **(AC + open commitments) ÷ EAC**. Uncommitted balance of the *forecast* = **EAC – AC – open commitments**. Escalation exposure = uncommitted forecast balance × price movement.
3. **Substitution.** Baseline uncommitted =  $4,000,000 - 2,120,000 - 1,180,000$ . Coverage =  $3,300,000 / 4,416,667$ ; after cleansing =  $3,180,000 / 4,416,667$ . Forecast uncommitted =  $4,416,667 - 3,300,000$ .
4. **Result.** The baseline leaves **USD 700,000** uncommitted; the forecast leaves **USD 1,116,667**. Commitment coverage is **74.72 %** of the method-(b) forecast (78.57 % of the method-(a) forecast). Cleansing the USD 120,000 of stale lines moves coverage to exactly **72.00 %** and raises the baseline's uncommitted balance to **USD 820,000**; while changing **AC**, **CPI**, **CV** and every **EAC** by nothing at all. The USD 1,116,667 still to be committed over twelve weeks is **USD 93,056 per week** of new commitment, and a 5 % adverse price movement on it is **USD 55,833**: **27.92 %** of the entire cost variance the week-13 review is about to discuss.
5. **Interpretation.** The identity is the thing to remember, and it is exact: **USD 1,116,667 – USD 700,000 = USD 416,667 = |VAC(b)|**. The gap between what the baseline leaves to be bought and what the forecast says will actually have to be bought is the forecast variance at completion, seen from the funding side rather than the performance side. Two people arguing about whether the overrun is real are arguing about the same number twice, and the funding view is usually the one a sponsor accepts, because it is a statement about money not yet spent rather than a criticism of money already spent.

**Commitment coverage answers a question CPI cannot: how much of the forecast is still exposed to prices?** At 74.72 % coverage, a quarter of the outturn is unfixed. That is the number to report beside **CPI** when a market is moving, and its sensitivity (5 % on the remainder being more than a quarter of the variance to date) is what makes escalation an *active* commercial matter rather than an estimating footnote. Long-lead items are where coverage is bought (Domain 10, KA 10.1.3 prices the lead time itself).

**Cleansing commitments makes the picture look worse and be more true, which is why it does not happen.** The stale USD 120,000 was silently claiming that scope was already secured. Removing it moves coverage down by 2.72 points and leaves an extra USD 120,000 visibly still to buy. Nothing in any performance index rewards the exercise; the whole return is a funding forecast

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that is not lying. That is precisely why it belongs in a standing month-end checklist with a named owner (Toolkit 7.T.1) rather than in anyone's judgement.

The caution: commitment coverage is not a measure of performance and must never be read as one. A project can be 100 % committed and badly over budget, indeed committing early at bad prices is one way to get there. Coverage bounds *uncertainty* in the forecast, not its level.

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### 7.2.2 The forecasting question

A forecast answers one question (*what will this cost when it is done?*), and the honest answer depends entirely on **why** the current variance exists. That judgment is the leader's; the arithmetic is KA 7.3.3's EAC family. The rule to internalise now: **a forecast is a statement about the remaining work, not an extrapolation ritual**. A variance caused by a closed, non-recurring event says nothing about what is left; a variance caused by a systemic productivity shortfall says everything.

### 7.2.3 Baseline integrity

The baseline's authority rests on its stability. Four standing rules: budgets of completed or open work packages are never retrospectively adjusted (except to correct an authorised error); re-baselining happens only through change control, with an audit trail (Domain 4, KA 4.3); transfers between control accounts are logged, not silent; and the baseline never absorbs a variance by moving budget to where the money went. A baseline edited to match reality has stopped measuring anything: the cost analogue of Domain 6's pinned milestone.

## AI in this KA

**Where it earns its place.** This knowledge area is where the most valuable automation in the whole cost discipline sits, because everything here is reconciliation and nobody has time to do it by hand. **Reconciling ledger actuals to the cost report** line by line, and producing the *list* of differences rather than only the total, since the total is what a project already knows and the list is what it can act on. **Identifying periods with no accrual posted against evidence that work was received** (goods-receipt notes, timesheets, site records, progress certificates), which is exactly the omission Worked example 7.2.1 costs at USD 416,667 of forecast error, and which is invisible in any index until the invoice lands. **Sweeping open purchase orders** to compute commitment coverage and to flag lines with no receipt movement for a stated period, which is the stale-commitment hygiene of 7.2.1b that never happens because nothing in a performance index rewards it. And **flagging control-account transfers not carried by a change reference**, which is 7.2.3's baseline integrity rule expressed as a query rather than as an aspiration.

**Where it must not go.** Four determinations, and each of them moves money or moves the measurement basis. **Determining whether work has been received:** a

measurement judgement belonging to a named human with the standing to make it, and the input on which every accrual rests. **Setting or adjusting an accrual:** an accrual is an assertion about value received, and a model that estimates one from a run rate has invented the very number the control exists to test. **Authorising a control-account transfer**, which is a change decision under Domain 4, KA 4.3. And **re-baselining** in any form, since a baseline edited to match reality has stopped measuring anything. A tool may propose all four and may compute what each would do; the authorisation is a person's, recorded.

**Verification, concretely.** Three checks, each cheap and each reproducible by hand. **Recompute the AC-plus-accrual total against the ledger** for the period and confirm it reconciles to the signed month-end position: the control 7.2.1 names, run as arithmetic rather than as an assurance. **Confirm the reversal in the following period:** an accrual is a timing entry, so it must reverse, and an accrual that does not reverse is either a real cost that has been double-counted or an adjustment wearing an accrual's clothing. That test is what makes 7.2.1's invariant (an accrual omission cancels cumulatively and misstates every period in between) checkable rather than merely stated. And **reproduce commitment coverage by hand for one period**, including the stale-line cleanse, because a coverage figure that nobody has ever recomputed is the number most likely to be quietly wrong in the direction that flatters. Where any of these feeds a report a board reads, the verification card of Domain 1, KA 1.4.3 applies: state who verified it, how, and what the check would have caught.

#### KEY TERMS — KA 7.2

| TERM                         | MEANING                                                                                                                                                                                |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AC                           | Actual Cost of the work performed, including period accruals.                                                                                                                          |
| Accrual                      | Cost of work received but not yet invoiced, recognised in the period.                                                                                                                  |
| Commitment                   | A contractual obligation (e.g. a PO); a funding fact, not yet a cost.                                                                                                                  |
| Commitment coverage          | $(AC + \text{open commitments}) \div EAC$ : the share of the forecast whose price is already fixed; 74.72 % on Auriga.                                                                 |
| Uncommitted forecast balance | $EAC - AC - \text{open commitments}$ ; its excess over the baseline's uncommitted balance equals  VAC .                                                                                |
| Re-baselining                | Domain 4's instrument (KA 4.3.3) seen from earned value: it resets the PV curve every index is measured against, which is exactly why it cannot be used to retire an adverse variance. |

## SAMPLE MCQs — KA 7.2

**MCQ 7.2-A** [7.2.1 · Analysis] A project's **CPI** has read 1.02 for four months; then one month it drops to 0.91 with no change in productivity. The likeliest explanation is:

- A. genuine sudden inefficiency
- B. accruals were not being recognised, so earlier **AC** understated cost and this month absorbed the catch-up CORRECT
- C. the baseline was too generous
- D. **EV** was over-claimed this month

*Rationale:* A step change in **CPI** without an operational change points at measurement, and missing accruals are the classic cause, earlier periods flattered, one period punished. D would raise, not lower, the earlier readings' credibility, and C would show as a stable, not stepped, pattern.

**MCQ 7.2-B** [7.2.3 · Recall] Which action preserves baseline integrity?

- A. moving budget from an underspent control account to cover an overspend, without record
- B. re-baselining through change control with an audit trail when scope genuinely changes CORRECT
- C. adjusting a completed package's budget to match its actual cost
- D. reducing remaining budgets so the total still equals **BAC**

*Rationale:* Only governed change preserves the measurement. A is a silent transfer, C is retrospective adjustment, D is the classic "make the numbers add up" manoeuvre, each destroys comparability.

**MCQ 7.2-C** [7.2.1 · Application] **EV** is 1,920,000; invoiced cost is 1,920,000; a further 200,000 of work has been received and accepted but neither invoiced nor accrued. Reported and true **CPI** are:

- A. 1.00 and 0.91 CORRECT
- B. 0.91 and 1.00
- C. 1.00 and 1.00
- D. 0.91 and 0.91

*Rationale:* The report divides **EV** by the understated **AC** of 1,920,000 (1.00); the truth divides by 2,120,000 (0.9057, shown as 0.91) — 7.2.1. B reverses the two; C treats the receipted work as a commitment rather than a cost, which is the substantive error the accrual rule exists to prevent; D assumes the accrual was recognised, in which case there would be nothing to detect.

**MCQ 7.2-D** [7.2.1 · Analysis] In the month a 200,000 accrual backlog is finally invoiced, the reported period **CPI** is 0.67 while true period performance was 0.92. The soundest reading is:

- A. productivity collapsed by a third in that period
- B. the period absorbed a timing correction: the cumulative index is now right and the period reading is not CORRECT
- C. **EV** was over-claimed in the period
- D. the baseline for that period was too generous

*Rationale:* Accrual omissions cancel cumulatively and misstate the periods in between (7.2.1), so the cumulative **CPI** of 0.9091 is now correct while the period figure is an artefact. A takes the artefact at face value and would launch an investigation into production that has nothing to find; C would depress the *earlier* periods too, not this one alone; D would show as a stable bias rather than a single-period step.

**MCQ 7.2-E** [7.2.1 · Application] **AC** is 2,120,000, open commitments are 1,180,000 and the **EAC** in use is 4,416,667. The balance of the forecast still to be committed is:

- A. USD 700,000
- B. USD 1,116,667 CORRECT
- C. USD 1,316,667
- D. USD 3,236,667

*Rationale:*  $4,416,667 - 2,120,000 - 1,180,000 = 1,116,667$  (7.2.1b). A measures against **BAC** rather than the forecast, and the 416,667 difference between A and B is exactly  $|\text{VAC}(b)|$ ; C uses **EV** in place of **AC**, double-counting the cost variance; D omits **AC** altogether.

#### SELF-CHECK — KA 7.2

1. *Why must **AC** include accruals?* — So it covers the same work as **EV** in the same period; otherwise every index is misstated.
2. *Is a purchase order a cost?* — No: a commitment. It informs the funding forecast, not performance.
3. *What single question does a forecast answer?* — What the remaining work will cost, given why the variance exists.
4. *State the accrual invariant and what it implies for period reporting.* — An accrual omission is a timing error: it cancels cumulatively and misstates every period in between, so a period index with no operational counterpart is a measurement event, not a performance event.

5. *What does commitment coverage tell you that CPI cannot?* — How much of the forecast is still exposed to price movement: at Auriga's 74.72 %, a 5 % movement on the uncommitted remainder is USD 55,833, or 27.92 % of the cost variance to date.
6. *Why does cleansing stale commitments never improve a report?* — Because it changes no performance index and makes the funding forecast look worse while making it true, which is why it needs to be a checklist step with an owner rather than a matter of judgement.

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### KNOWLEDGE AREA 7.3

## Earned value: measurement, variances and forecasting

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*Topics:* 7.3.1 the three measures · 7.3.2 variances and indices · 7.3.3 the EAC family · 7.3.4 VAC and TCPI.

### 7.3.1 The three measures

**Definitions.** All three are expressed in the same budget currency so they are directly comparable:

- **Planned Value (PV)**, the budgeted cost of the work *scheduled* by the data date: the time-phased baseline of KA 7.1.3.
- **Earned Value (EV)**, the budgeted cost of the work *actually performed*: physical progress **valued at the budget rate**, never at what it cost.
- **Actual Cost (AC)**: the cost actually incurred for that work, accruals included (KA 7.2.1).

**The single most important conceptual point:** EV is measured *at budget*. That is what lets EV be compared with PV (both at budget → schedule progress) and with AC (budget vs actual for the same work → cost efficiency). Confusing "value earned" with "cost incurred" collapses the method.

**Earning rules.** How EV is claimed decides how much it can be gamed: **0/100** (nothing until complete, objective, good for short packages); **50/50**; **percent complete** (needs an objective basis); **units completed** (best where output is countable); **weighted milestones**; and **level of effort**, where EV is set equal to PV by the calendar, so it can *never* show a schedule variance. Level of effort dilutes whatever it is mixed into, formal practice segregates it and caps its share of a control account.

How much difference the choice makes is the question that decides whether earning rules are a technicality or a governance matter.



## WORKED EXAMPLE

**Worked example 7.3.1 – one physical state, three earned values.**

- Setup.** A **USD 400,000** group of ten equal work packages inside Auriga's installation control account CA-40. At the data date: **4** packages complete, **3** in progress at an objectively measured **30 %** physical completion, **3** not started. The baseline expected **6** complete by now, so **PV = USD 240,000**. Actual cost for the group is **USD 210,000**. Compute **EV**, **SPI** and **CPI** under **0/100**, **50/50** and objective percent-complete earning.
- Formula.** **0/100:** **EV** = complete packages × package budget. **50/50:** complete × budget + in-progress × budget/2. Percent complete: complete × budget + Σ(in-progress physical % × budget). Then **SPI** = **EV/PV**, **CPI** = **EV/AC**.
- Substitution.** **0/100** = **4 × 40,000**. **50/50** = **160,000 + 3 × 20,000**. Percent complete = **160,000 + 3 × 12,000**.
- Result.**

| EARNING RULE                       | EV      | SPI           | CPI           | READS AS                           |
|------------------------------------|---------|---------------|---------------|------------------------------------|
| 0/100                              | 160,000 | <b>0.6667</b> | <b>0.7619</b> | badly late, badly over             |
| 50/50                              | 220,000 | <b>0.9167</b> | <b>1.0476</b> | slightly late, <b>under budget</b> |
| Percent complete (objective, 30 %) | 196,000 | <b>0.8167</b> | <b>0.9333</b> | late and over                      |

The spread in **EV** is **USD 60,000 (15.00 %** of the group's budget), and the **SPI** spread is **0.25**, on a physical state that did not change between rows. 5. **Interpretation.** The decisive observation is in the third column of the middle row. **The earning rule alone flips the account from over-running (CPI 0.9333) to under-running (CPI 1.0476).** Nothing was misreported, nothing was manipulated, and the two readings support opposite management decisions. That is why the earning rule is a control, not a preference.

**The direction of each rule's bias is predictable, so a reviewer can compute it.** **50/50** overstates by **USD 24,000 (12.24 %** of the objective figure), because the in-progress work is only 30 % done; **0/100** understates by **USD 36,000**, or 18.37 %. The breakeven is exact and worth carrying: **50/50 is unbiased precisely when in-progress work averages 50 % physical completion**, and at genuine 50 % it and objective percent-complete agree at **USD 220,000**; while **0/100** still reads 160,000, because it credits nothing until a package closes. So **50/50** flatters a control account whose in-progress work is young, which is the normal condition of an account that has just started, and exactly when a project most wants reassurance.

**Which rule to choose follows from package duration, not from taste.**

0/100 is objective and cheap and its understatement is harmless where packages are short relative to the reporting period: a package that starts and finishes inside a month never sits half-earned at a data date. For long packages 0/100 produces a sawtooth that hides real progress, and objective percent complete earns its administrative cost. 50/50 is a convenience that should be reserved for packages short enough that the error cannot matter, and its use on long packages is the single commonest source of an over-stated EV.

**The rule must be fixed before the period it measures.** A rule chosen after the data date is not a measurement convention but a selection among answers, and the 15 % spread above is the size of the discretion involved. This is the cost analogue of Domain 6's pinned milestone: the defect is not arithmetic but the loss of a fixed reference. State the rule per package in the plan, disclose any change, and, the reviewer's test, recompute one account under a different rule to see how much of the reported position is the convention rather than the work.

**The level-of-effort distortion, computed.** Because level of effort sets  $EV \equiv PV$ , a control account that mixes it with discrete work reports a blend, and the blend has a closed form. With  $s$  the level-of-effort share of the account's budget and  $SPI_d$  the discrete work's own schedule performance index:

$$\begin{aligned} \text{reported SPI} &= s \times 1 + (1 - s) \times SPI_d = 1 - (1 - s)(1 - SPI_d) \\ \text{distortion} &= \text{reported SPI} - SPI_d = s \times (1 - SPI_d) \end{aligned}$$

An account 70 % level of effort whose discrete work is running at  $SPI_d$  0.60 reports 0.88: a distortion of 0.28, which is to say the report conceals nearly all of the problem. The same account under a 20 % level-of-effort cap would report 0.68, a distortion of 0.08. And the share required to make a discrete SPI of 0.60 report as 0.95 or better is  $s \geq 0.875$ , so **87.5 % level of effort makes a two-fifths schedule shortfall disappear into a figure a steering committee would wave through.** The distortion is *linear in the level-of-effort share*, which is precisely why the counter-measure is a cap expressed as a share of budget, and why the share must be disclosed per control account rather than for the project as a whole.

### 7.3.2 Variances and indices

$$\begin{aligned} CV &= EV - AC & SV &= EV - PV \\ CPI &= EV / AC & SPI &= EV / PV \end{aligned}$$

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#### WORKED EXAMPLE

**Worked example 7.3.2 – Auriga at week 13.**

1. **Setup.**  $BAC = 4,000,000$ ; at the data date  $PV = 2,080,000$ ,  $EV = 1,920,000$ ,  $AC = 2,120,000$ .
2. **Formula.** As above; variances in currency, indices as ratios (two decimals).
3. **Substitution.**  $CV = 1,920,000 - 2,120,000$ ;  $SV = 1,920,000 - 2,080,000$ ;  $CPI = 1,920,000/2,120,000$ ;  $SPI = 1,920,000/2,080,000$ .
4. **Result.**  $CV = (\text{USD } 200,000)$  ·  $SV = (\text{USD } 160,000)$  ·  $CPI = 0.91$  ·  $SPI = 0.92$ . Auriga is **48.0 % complete** ( $EV/BAC$ ) having spent **53.0 %** of budget ( $AC/BAC$ ).
5. **Interpretation.** Behind and over: for every dollar spent the project is producing 91 cents of budgeted work, and it has delivered 92 % of what it planned to by now. The percent-complete versus percent-spent pair (48 vs 53) is the same story in the form a board absorbs fastest. Note this is the *cost* view of the very slippage Domain 6's case study recovered on the schedule side. The two domains are describing one project.

**The percent-complete/percent-spent gap is the cost variance, exactly.**  $\%spent - \%complete = (AC - EV)/BAC$ , so the 5.00-point gap between 53.00 % and 48.00 % is the USD 200,000, and the identity holds always. That makes the two percentages a legitimate board summary rather than a simplification: nothing is lost in the translation, and a report where the gap and the *CV* disagree has an arithmetic error in it.

**Express the overrun as a rate, because a rate can be projected and a total cannot.** The work performed so far cost **10.42 %** more than it was budgeted to:  $1/CPI - 1 = 1/0.905660 - 1$ . That is the same 10.42 % that appears again as the ratio between removed scope and saved cash in KA 7.3.4, and it is the number that scales: USD 200,000 is a fact about the past, 10.42 % is a claim about the production process.

**Express the schedule variance in time, because currency conceals the size of it.** Auriga's *SV* of (USD 160,000) is exactly one week of planned value at this point on the curve, where the baseline is rising at USD 160,000 per week. So *SV* (160,000) means **one week behind**, which is both more useful and more challengeable than a currency figure, and it is the bridge to earned schedule in 7.A.1. The general caution: *SV* in currency is not comparable between projects or even between phases of one project, because it is scaled by the local slope of the *PV* curve.

**Both indices are cumulative, so they lag, and neither says whether things are improving.** A cumulative *CPI* of 0.91 at 48 % complete cannot distinguish a project that has stabilised from one still deteriorating; only the period index can, and 7.2.1 shows how easily the period index is corrupted. The reviewer's pairing is therefore cumulative index for level and period index for direction, with the accrual reconciliation done before either is believed. And the whole set rests on the earning rules of 7.3.1: on the arithmetic there, a 15 % *EV* swing was available

from the convention alone, which is larger than the variance being discussed here.

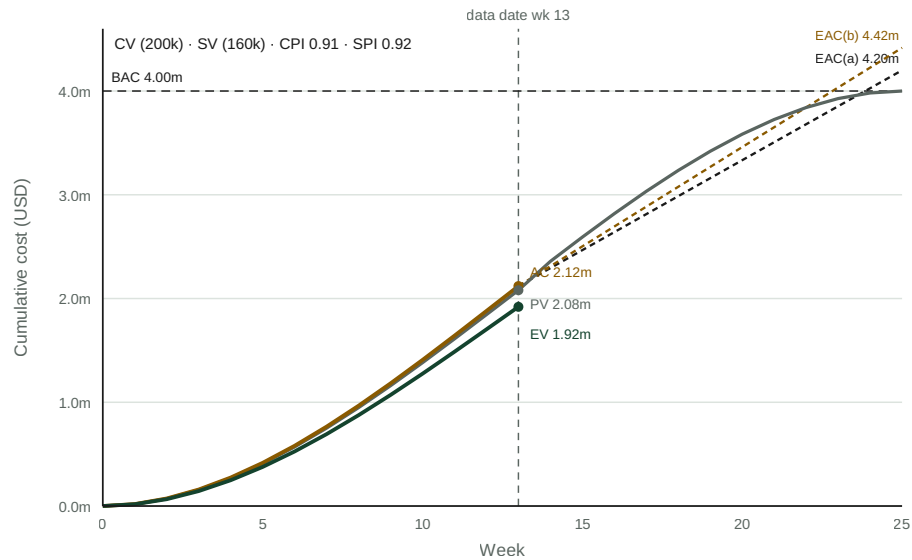


Fig 7.3.1 — Auriga's earned-value S-curves at week 13

### 7.3.3 The EAC family — forecasting

|                                          |                                        |
|------------------------------------------|----------------------------------------|
| ETC = the remaining work's forecast cost | EAC = AC + ETC                         |
| (a) EAC = AC + (BAC - EV)                | remaining work at the BUDGETED rate    |
| (b) EAC = BAC / CPI                      | current cost efficiency PERSISTS       |
| (c) EAC = AC + (BAC - EV) / (CPI × SPI)  | cost AND schedule pressure compound    |
| (d) EAC = AC + bottom-up ETC             | re-estimate the remainder from scratch |

#### WORKED EXAMPLE

#### Worked example 7.3.3 — the same project, four futures.

- Setup.** Auriga's week-13 figures, forecast to completion by each method.
- Formula.** (a)–(c) above;  $CPI = 0.905660$ ,  $SPI = 0.923077$  at full precision.
- Substitution.** (a)  $2,120,000 + (4,000,000 - 1,920,000)$ ; (b)  $4,000,000 / 0.905660$ ; (c)  $2,120,000 + 2,080,000 / (0.905660 \times 0.923077)$ .
- Result.** (a) USD 4,200,000 · (b) USD 4,416,667 · (c) USD 4,608,056.

5. **Interpretation.** The same four measured numbers support forecasts spanning **USD 408,056**: over 10 % of **BAC**. The spread is not uncertainty in the arithmetic; it is the *assumption* about the remaining work, made explicit. Choosing among them is the leader's judgment, and it must be stated: if Auriga's overrun was the contaminated-ground remediation of Domain 6 (a discrete, now-closed event) method (a) is right and the others over-forecast. If it reflects a systemic productivity shortfall that will persist, (b). If schedule pressure is now driving overtime and expediting, (c). A forecast presented without its assumption named is a number, not a forecast.

**Scale the spread against the remaining work, not against BAC.** USD 408,056 is 10.20 % of **BAC** and **19.62** % of the USD 2,080,000 still to be earned. The second figure is the honest one, because the disagreement is entirely about the remainder, the methods agree exactly about the past. A disagreement of a fifth about the work still to do is a large disagreement, and describing it as "about ten per cent" understates it by half.

**Every method is an implied efficiency for the remaining work, and stating it that way ends most arguments.** Method (a) assumes the remainder runs at **1.0000**: budgeted rate, i.e. a 10.42 % improvement on demonstrated performance. Method (b) assumes **0.9057**, today's **CPI** exactly. Method (c) assumes **0.8360**, the product **CPI × SPI**. So the choice is not between three formulae but between three claims about productivity, and each is falsifiable. The gap between (b) and (c) — **USD 191,389** — is the price the forecast puts on schedule pressure alone; asking "who is claiming that recovering the date will cost a further **6.97 points** of efficiency on top of the **9.43** already lost — **16.40** % below the budgeted rate in total, not on top of it — and through what mechanism?" is a better question than comparing formula names.

**Method (c) is a different hypothesis, not more conservative.** **CPI × SPI** compounds two indices that are not independent and were never designed to be multiplied, and it is defensible only where a specific, describable mechanism links lateness to cost — overtime, expediting premiums, out-of-sequence work, extended site establishment. Where that mechanism exists, (c) is right and (b) under-forecasts. Where it does not, (c) is pessimism with an equation on it, and it does the same damage as optimism: it destroys the credibility of the next forecast.

**Method (d) is the only one whose basis can be checked, and the only one that can be lower than (a).** A bottom-up re-estimate of the remainder can legitimately land below USD 4,200,000 (because the remaining scope may be less exposed than the scope that overran), and no index-extrapolation method can ever produce that. Its cost is real (it takes a team a week or more, and it consumes the same engineers who are delivering) and it is the only method that survives cross-examination at a funding gate. The professional practice is therefore: report the index methods weekly as a *trend*, and commission (d) whenever the decision at stake is money.

## WORKED EXAMPLE

**Worked example 7.3.3b – where each forecast's funding actually runs out.**

1. **Setup.** The board must authorise funding, and it will authorise the **EAC** it is given. Assume the demonstrated **CPI** of **0.905660** persists. For each forecast, how much budgeted work does the money it authorises actually buy, and at what point does the project stop?
2. **Formula.** Funded **ETC** = **EAC** – **AC**. Budgeted work that **ETC** buys at efficiency **e** = **ETC** × **e**. **EV** at exhaustion = **EV** + that amount; unfunded budgeted work = **BAC** – **EV** at exhaustion. Weeks short = unfunded work ÷ the baseline's average planned rate over the remaining weeks.
3. **Substitution.** Method (a): **ETC** =  $4,200,000 - 2,120,000 = 2,080,000$ ; buys  $2,080,000 \times 0.905660$ . Remaining planned rate =  $2,080,000/12$  over weeks 14–25.
4. **Result.**

| FORECAST         | FUNDED<br>ETC | BUDGETED WORK<br>IT BUYS AT CPI<br>0.9057 |           | EV<br>REACHED | POSITION                                           |
|------------------|---------------|-------------------------------------------|-----------|---------------|----------------------------------------------------|
|                  |               |                                           |           |               |                                                    |
| (a)<br>4,200,000 | 2,080,000     | 1,883,774                                 | 3,803,774 |               | <b>95.09 % complete<br/>(money gone)</b>           |
| (b)<br>4,416,667 | 2,296,667     | 2,080,000                                 | 4,000,000 |               | <b>exactly complete</b>                            |
| (c)<br>4,608,056 | 2,488,056     | 2,253,333                                 | 4,173,333 |               | over-funded by<br>173,333 of budget-<br>equivalent |

Method (a) leaves **USD 196,226** of budgeted work unfunded: **4.91 %** of **BAC**, and at the baseline's average remaining rate of USD 173,333 per week, **1.13 weeks** of work.

5. **Interpretation.** The result is a sentence a sponsor cannot misunderstand: **on its own demonstrated performance, the method-(a) forecast funds Auriga to about one and a bit weeks short of the finish line.** That is a far more useful statement than "the forecast may be USD 216,667 light", because it names when the problem arrives and what it will feel like, a second funding request at 95 % complete, which is the worst moment in a project's life to ask for money, since every alternative to paying has already expired. Method (b)'s exactness is not a coincidence but the definition of the method: **BAC/CPI** is precisely the funding that completes the scope at today's efficiency, which is the same identity KA 7.3.4 finds from the **TCPI** side.

**\*\*The asymmetry of forecast error is the governance point.\*\***  
 Over-forecasting costs the organisation the opportunity value of committed but unspent funds, real, and recoverable at the next portfolio review (Domain 15). Under-forecasting costs a mid-project funding crisis at the moment of maximum sunk cost and minimum optionality, which is where escalation of commitment (Domain 2, KA 2.4.2) does its worst work. **\*\*The two errors are not symmetric, so a forecast should not be chosen as though they were\*\***, and a leader who genuinely cannot distinguish between causes should say so and fund nearer (b) than (a) while commissioning method (d).

**\*\*Method (c)'s "over-funding" is only over-funding if efficiency does not deteriorate.\*\*** Its USD 2,488,056 buys USD 2,253,333 of budgeted work at `CPI` alone (an apparent surplus of USD 173,333), but at the compounded rate of 0.8360 it completes the scope exactly, to the dollar. That is the test of whether (c) is right: it is the correct number if and only if the compounded efficiency materialises. Presenting (c) without that conditional is how a project acquires a reputation for hoarding contingency, and presenting (a) without the conditional above is how it acquires a reputation for surprises.

#### 7.3.4 VAC and TCPI

|                                        |                     |
|----------------------------------------|---------------------|
| VAC = BAC – EAC                        | the forecast over-  |
| run/underrun                           |                     |
| TCPI = (BAC – EV) / (BAC – AC)         | efficiency the      |
| REMAINING work must achieve to hit BAC |                     |
| TCPI = (BAC – EV) / (EAC – AC)         | ...to hit a revised |
| EAC                                    |                     |

#### WORKED EXAMPLE

##### Worked example 7.3.4 – what recovery would actually require.

1. **Setup.** Auriga's week-13 figures; test recovery to BAC, and to the method-(b) forecast.
2. **Formula.** VAC = BAC – EAC; TCPI as above.
3. **Substitution.** VAC(a) = 4,000,000 – 4,200,000; VAC(b) = 4,000,000 – 4,416,667. TCPI(BAC) = 2,080,000 / (4,000,000 – 2,120,000) = 2,080,000/1,880,000. TCPI(EAC b) = 2,080,000 / (4,416,667 – 2,120,000).

4. **Result.**  $VAC(a) = (\text{USD } 200,000) \cdot VAC(b) = (\text{USD } 416,667)$ .  $TCPI$  to  $BAC = 1.11$ ;  $TCPI$  to the (b) forecast = **0.91**.

5. **Interpretation.** To finish at budget, the remaining 52 % of work must run at **1.11**: an 11 % efficiency *gain* by a team currently achieving 0.91. That is a 22-point swing, and the honest question in the room is what specifically would deliver it. Meanwhile  $TCPI$  to the (b) forecast is 0.91, exactly today's  $CPI$ , which is the arithmetic identity worth internalising: **forecasting with  $BAC/CPI$  is precisely the assumption that nothing changes**.  $TCPI$  is the reality check on every recovery promise: a required index far above demonstrated performance is a plan that needs a mechanism, not encouragement.

**State the swing as a ratio, not a difference.** "1.11 against 0.91" understates the demand, because the two are a ratio:  $1.106383 / 0.905660 = 1.2216$ , so the remaining work must run **22.16** % more efficiently than everything done so far. A 22 % productivity improvement, mid-project, on the same scope with the same team and the same suppliers, is not a management intention; it is an engineering change, a scope change or a commercial change, and the recovery plan should name which.

**The identity generalises, and once it is seen the  $EAC$  family documents itself.**  $TCPI$  computed to any  $EAC$  equals the efficiency that  $EAC$  assumes for the remaining work:

| TARGET                | $TCPI = (BAC - EV) \div (TARGET - AC)$ | EQUALS                                     |
|-----------------------|----------------------------------------|--------------------------------------------|
| $BAC$ 4,000,000       | $2,080,000 / 1,880,000 = 1.1064$       | the recovery demand                        |
| $EAC(a)$<br>4,200,000 | $2,080,000 / 2,080,000 = 1.0000$       | budgeted rate, method (a)'s assumption     |
| $EAC(b)$<br>4,416,667 | $2,080,000 / 2,296,667 = 0.9057$       | $CPI$ , method (b)'s assumption            |
| $EAC(c)$<br>4,608,056 | $2,080,000 / 2,488,056 = 0.8360$       | $CPI \times SPI$ , method (c)'s assumption |

**Choosing an  $EAC$  is choosing a  $TCPI$** , so the two should always be reported as a pair; and any  $EAC$  whose  $TCPI$  differs from the efficiency its author claims to be assuming contains an arithmetic error. That single check catches more defective cost reports than any other in this domain, and it takes one division.

The caution about  $TCPI$  runs the other way as well. A  **$TCPI$  below** demonstrated  $CPI$  is not good news. It means the target has already been relaxed to something easier than current performance, which is what re-baselining to the forecast does, and it is how a project comes to report favourable indices against a target nobody would have approved at sanction (7.2.3).



**Worked example 7.3.4b – what TCPI 1.11 costs in scope, not in exhortation.**

1. **Setup.** The board will not authorise more than the original **USD 4,000,000** of cost. Auriga's team is achieving **CPI 0.905660** and can offer no mechanism to improve it. The only remaining lever is scope: how much budgeted work must come out for the recovery to be arithmetic rather than aspiration?
2. **Formula.** With **D** the budgeted value of scope removed and the cost target **T**:  

$$TCPI = (BAC - D - EV) / (T - AC).$$
 Setting  $TCPI = CPI$  and solving,  $D = (BAC - EV) - CPI \times (T - AC).$
3. **Substitution.**  $D = 2,080,000 - 0.905660 \times (4,000,000 - 2,120,000)$   
 $= 2,080,000 - 1,702,641.$
4. **Result.**  $D = \text{USD } 377,358$  (exactly  $20,000,000/53$ ), which is **9.43 %** of **BAC** and **18.14 %** of the work remaining. For comparison, completing the full scope at the demonstrated efficiency needs **USD 416,667** of additional funding (that is  $EAC(b) - BAC$ , i.e.  $|VAC(b)|$ ).
5. **Interpretation.** The identity between those two figures is the one to carry out of this domain:  $416,667/377,358 = 1.104167 = 1/CPI$ . **At a CPI of 0.9057, one dollar of budgeted scope removed saves USD 1.10 of forecast cash;** because the scope you do not do is scope you do not do inefficiently. That converts a vague recovery conversation into a single exchange rate the sponsor can act on, and it explains why descoping is a more powerful lever than its size suggests: the leverage is exactly  $1/CPI$ , and it grows as performance worsens.

**The number reframes the meeting.** "We need a 22 % efficiency improvement" invites optimism. "Recovery to the approved 4,000,000 requires USD 377,358 of budgeted scope to come out, and here are the three candidate packages" invites a decision, with a named owner, at the authority that owns scope. **TCPI** is what makes the second sentence available.

**Three cautions, and the first is the one that gets skipped.** Removing 9.43 % of budgeted scope does not remove 9.43 % of the benefit, and it may remove far more: on a control-systems upgrade the cheap-to-build packages are frequently the ones the operating case depends on. Every descope must be priced against benefit (Domain 2, KA 2.3) and tested against acceptance criteria (Domain 5, KA 5.4.2) before its cost saving is booked. Second, a "descope" that is really a **deferral** does not save the money at all. It moves it to a later period and usually a higher price, which makes it a portfolio decision (Domain 15) rather than a project one, and the honest report says so. Third, the arithmetic assumes the removed scope carries its budgeted cost at the same demonstrated efficiency; removing scope from a package that was running *well* saves less than the formula says, and the sequence for choosing candidates therefore starts from the accounts with the worst performance, not the ones easiest to argue for.

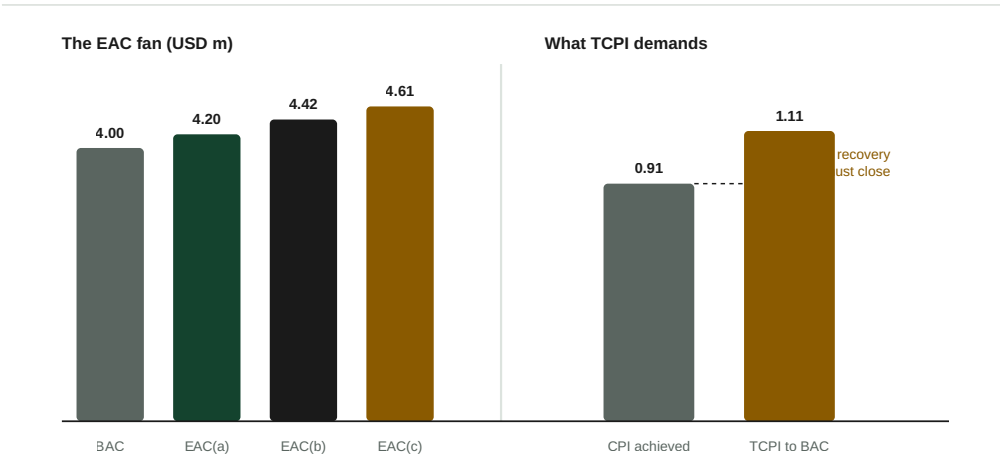


Fig 7.3.2 — The EAC fan and what TCPI demands

AI in this KA

Earned value is arithmetic on four numbers, so machine computation is safe and instant verification is cheap, which makes unverified AI output indefensible here rather than merely risky. The deterministic checks:  $EV/BAC$  must equal the claimed percent complete;  $CV = EV - AC$  and  $CPI = EV/AC$  recomputed independently;  $TCPI$  to  $BAC/CPI$  must equal  $CPI$  (7.3.4's identity);  $TCPI$  to each of the other  $EACs$  must equal that method's own implied efficiency (1.0000,  $CPI$ ,  $CPI \times SPI$ ), which is a four-division audit of the whole forecast set; the percent-spent-minus-percent-complete gap must reproduce  $CV/BAC$ ; and every  $EAC$  must state its assumption. A model asked to "check the earned-value numbers" will confirm arithmetic and miss the two things that actually go wrong: an earning rule applied inconsistently between periods, and a level-of-effort share that makes the reported  $SPI$  meaningless (7.3.1). Both are questions about convention rather than computation, so both stay with the professional. Where AI forecasts from trend data, the calibration rule of Domain 6 (KA 6.4) applies unchanged: show the record of past forecasts against outturn before the number enters a board pack, and name the human who owns it.

KEY TERMS — KA 7.3

| TERM     | MEANING                                                                   |
|----------|---------------------------------------------------------------------------|
| PV EV AC | Budgeted cost of work scheduled · performed · the cost actually incurred. |
| CV SV    | $EV - AC$ · $EV - PV$ , in currency.                                      |
| CPI SPI  | $EV/AC$ · $EV/PV$ , as ratios.                                            |
| EAC ETC  | Estimate at / to complete; $EAC = AC + ETC$ .                             |
| VAC      | $BAC - EAC$ , forecast variance at completion.                            |
| TCPI     | Efficiency the remaining work must achieve to hit a stated target.        |

| TERM                              | MEANING                                                                                                                                                                           |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Level of effort</b>            | Earning by calendar ( $EV \equiv PV$ ); can never show schedule variance.                                                                                                         |
| <b>Earning rule</b>               | The convention by which physical progress becomes <i>EV</i> ; fixed before the period it measures. On Auriga's ten-package group the choice moved <i>EV</i> by 15.00 % of budget. |
| <b>Level-of-effort distortion</b> | $s \times (1 - SPI_d)$ : the amount a level-of-effort share <i>s</i> adds to a reported <i>SPI</i> ; 0.28 at a 70 % share with discrete <i>SPI</i> 0.60.                          |
| <b>Implied ETC efficiency</b>     | The efficiency an <i>EAC</i> assumes for the remaining work; equal to the <i>TCPI</i> computed to that <i>EAC</i> .                                                               |
| <b>Descope exchange rate</b>      | $1/CPI$ , the forecast cash saved per unit of budgeted scope removed; USD 1.10 per dollar on Auriga.                                                                              |

## SAMPLE MCQs — KA 7.3

**MCQ 7.3-A** [7.3.2 · Application] BAC 4,000,000; PV 2,080,000; EV 1,920,000; AC 2,120,000. CPI and SPI are:

- A. 0.91 and 0.92 CORRECT
- B. 1.10 and 1.08
- C. 0.92 and 0.91
- D. 0.91 and 1.02

*Rationale:*  $CPI = 1,920,000 / 2,120,000 = 0.91$ ;  $SPI = 1,920,000 / 2,080,000 = 0.92$ . B inverts both ratios; C swaps them (dividing *EV* by the wrong denominator); D miscalculates *SPI* against *BAC*-derived progress.

**MCQ 7.3-B** [7.3.3 · Analysis] The overrun was caused by a one-off ground-remediation event, now closed, and the remaining work is expected to run to budget. The appropriate *EAC* is:

- A.  $AC + (BAC - EV) = 4,200,000$  CORRECT
- B.  $BAC/CPI = 4,416,667$
- C.  $AC + (BAC - EV) / (CPI \times SPI) = 4,608,056$
- D.  $BAC = 4,000,000$

*Rationale:* A discrete, closed cause makes the variance **atypical**, so remaining work is forecast at the budgeted rate. B assumes the inefficiency persists and C that it compounds with schedule pressure, both contradict the stated cause. D ignores money already spent above budget.

**MCQ 7.3-C** [7.3.4 · Application] With BAC 4,000,000, EV 1,920,000 and AC 2,120,000, the *TCPI* required to complete at BAC is:

- A. 0.91

B. 1.00

C. 1.11 CORRECT

D. 1.08

*Rationale:*  $(4,000,000 - 1,920,000)/(4,000,000 - 2,120,000) = 2,080,000/1,880,000 = 1.11$ . A is the demonstrated CPI; B assumes recovery needs only par performance; D uses PV in place of AC in the denominator:  $2,080,000/1,920,000 = 1.08$ ; and so makes recovery look nearly free (Exercise 7.3).

**MCQ 7.3-D** [7.3.1 · Analysis] A control account is 70 % level-of-effort by budget. Its reported SPI of 1.00 most likely means:

A. the account is exactly on schedule

B. little about schedule: level of effort earns by the calendar, so  $EV \equiv PV$  for most of the account regardless of progress CORRECT

C. the discrete work is ahead, offsetting a delay

D. the earning rules were misapplied

*Rationale:* LOE sets EV equal to PV by construction, so a heavily-LOE account reads 1.00 whatever happens, which is why practice segregates and caps it. C invents an offset the data cannot show; the rules may have been applied entirely correctly (D), and that is the problem.

**MCQ 7.3-E** [7.3.1 · Application] Ten equal work packages of 40,000: four complete, three at an objectively measured 30 % physical completion, three not started. EV under 50/50 and under objective percent-complete earning is:

A. 160,000 and 196,000

B. 220,000 and 196,000 CORRECT

C. 220,000 and 220,000

D. 196,000 and 220,000

*Rationale:* 50/50 credits half of each in-progress package ( $160,000 + 60,000$ ); percent complete credits the measured 30 % ( $160,000 + 36,000$ ) — 7.3.1. A gives the 0/100 figure for the first rule; C assumes the in-progress work is genuinely 50 % done, which is the only condition under which 50/50 is unbiased; D transposes the two.

**MCQ 7.3-F** [7.3.1 · Analysis] A control account is 70 % level of effort by budget and its discrete work is running at SPI 0.60. The account's reported SPI is:

A. 0.60

B. 0.88 CORRECT

C. 1.00

D. 0.42

*Rationale:*  $1 - (1 - 0.70)(1 - 0.60) = 0.88$ , a distortion of  $0.70 \times 0.40 = 0.28$  (7.3.1). A ignores that the level-of-effort portion reads 1.00 by construction; C assumes it swamps the account entirely; D multiplies the share by the discrete index instead of blending them.

**MCQ 7.3-G** [7.3.4 · Analysis] For which target is TCPI exactly 1.00 on Auriga's week-13 figures?

- A.  $BAC = 4,000,000$
- B.  $AC + (BAC - EV) = 4,200,000$  CORRECT
- C.  $BAC/CPI = 4,416,667$
- D.  $AC + (BAC - EV)/(CPI \times SPI) = 4,608,056$

*Rationale:* TCPI to any EAC equals the efficiency that EAC assumes, and method (a) assumes the budgeted rate (7.3.4). A gives 1.1064, the recovery demand; C gives CPI 0.9057; D gives  $CPI \times SPI$  0.8360: each the assumption of its own method.

**MCQ 7.3-H** [7.3.3 · Evaluation] The board authorises the method-(a) forecast of 4,200,000 and the demonstrated CPI of 0.9057 persists. The funding is exhausted at:

- A. 100.00 % complete
- B. 95.09 % complete CORRECT
- C. 90.57 % complete
- D. 52.00 % complete

*Rationale:* The funded ETC of 2,080,000 buys  $2,080,000 \times 0.905660 = 1,883,774$  of budgeted work, taking EV to 3,803,774 (7.3.3b). A is the position only if the remainder runs at the budgeted rate, which is the assumption being tested; C applies CPI to the whole of BAC rather than to the funded remainder; D is the share of work remaining, not a completion point.

#### SELF-CHECK — KA 7.3

1. *Why is EV measured at budget, not at cost?* — So it is comparable with PV (schedule) and with AC (efficiency); at cost it would collapse into AC.
2. *What does  $TCPI = 1.11$  against  $CPI = 0.91$  tell a leader?* — Recovery to budget requires a 22-point efficiency swing; the plan needs a named mechanism, not optimism.
3. *State the identity linking TCPI and BAC/CPI.* — TCPI to an EAC of BAC/CPI equals the current CPI: that forecast is the assumption that nothing changes.
4. *Generalise that identity.* — TCPI to any EAC equals the efficiency that EAC assumes for the remaining work: 1.0000 for method (a), CPI for (b),  $CPI \times SPI$  for (c). Choosing an EAC is choosing a TCPI, so report them as a pair.

5. *How much scope must come out for Auriga to finish at 4,000,000 without improving?* — USD 377,358 of budgeted work, 9.43 % of BAC; and because a dollar of budget removed saves  $1/CPI = \text{USD } 1.10$  of cash, that is equivalent to the USD 416,667 of extra funding method (b) requires.
6. *When is 50/50 earning unbiased, and when does it flatter?* — Unbiased when in-progress work averages exactly 50 % physical completion; it flatters whenever that average is lower, which is the normal state of a recently started account.

## KNOWLEDGE AREA 7.4

## Resource economics, procurement strategy and cash

Topics: 7.4.1 resource economics and blended rates · 7.4.2 contract models and cost risk · 7.4.3 incentive fees and the point of total assumption · 7.4.4 cash flow versus profit.

## 7.4.1 Resource economics and blended rates

Most project cost is people. A leader who reasons in headcount rather than **cost per unit of capability** will mis-price every option in Domain 6's compression decisions.

## WORKED EXAMPLE

## Worked example 7.4.1 — Auriga's engineering blended rate.

1. **Setup.** The engineering pool is 40 technicians at USD 95/hour, 25 engineers at USD 140/hour, 15 senior specialists at USD 210/hour.
2. **Formula.** Blended rate =  $\Sigma(\text{count} \times \text{rate}) / \Sigma(\text{count})$ .
3. **Substitution.**  $(40 \times 95 + 25 \times 140 + 15 \times 210) / 80 = (3,800 + 3,500 + 3,150) / 80 = 10,450 / 80$ .
4. **Result.** **USD 130.63 per hour** (130.625 exactly;  $\approx$  SAR 490 indicatively).
5. **Interpretation.** The blended rate makes options commensurable: a week of schedule compression staffed from this pool costs a computable amount, and *shifting the mix* changes the price as much as changing the headcount, adding five specialists moves the blend more than adding five technicians. This is the number that turns Domain 6's "crash it" instruction into an expected-value decision.

**The mix moves the price more than the size does, and the arithmetic ranks the moves.** Re-grading five technicians into specialists (same 80 people) takes the blend to **USD 137.8125, +5.50 %**. Adding five specialists to make 85 gives USD 135.2941, **+3.57 %**. Adding five technicians gives USD 128.5294, **-1.60 %**. So the ordering is re-grading, then growing at the top, then growing at the bottom, and a leader who negotiates headcount while the mix moves underneath

them has not negotiated the cost at all. The practical control is to fix the *mix* in the plan, not just the number, and to report it.

**The engineer-week that follows is the unit the rest of the book uses.**  
 $130.625 \times 40 = \text{USD } 5,225$  per engineer-week ( $\approx$  SAR 19,594 indicatively), which is the figure Domain 9, KA 9.2.3 uses to cost rework capacity and Domain 8 uses in response pricing. Registering the rate once and reusing it is what makes those numbers comparable across domains.

**The blend is only valid for work drawn in the pool's proportions, and the error when it is not is large.** A task performed entirely by senior specialists costs USD 210 per hour: **60.77 %** above the blend, an under-pricing of USD 79.375 an hour if the blend is used. That is the commonest misuse of a blended rate: applying it to a small, senior-heavy activity such as commissioning support or an interface investigation, where it can understate cost by more than half again. The discipline is to blend within a *grade-homogeneous* scope and to price senior-only work at senior rates.

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#### WORKED EXAMPLE

##### Worked example 7.4.1b – what an hour of capacity actually costs.

1. **Setup.** The USD 95 per hour in the estimate above is a charge-out rate. Auriga's finance team builds the underlying cost of a technician hour: base pay **USD 55.00** per paid hour; statutory and employer on-costs **32 %**; **15 %** of paid hours are non-productive (leave, training, standby, weather, travel); recoverable overhead — site establishment, supervision, tools, information systems, **18 %**. What does one *productive* hour cost, and what does the estimate's rate actually recover?
2. **Formula.** Paid-hour cost =  $\text{base} \times (1 + \text{on-costs})$ . Cost per productive hour =  $\text{paid-hour cost} \div (1 - \text{non-productive share}) \times (1 + \text{overhead})$ . The paid-to-productive multiplier is  $1/(1 - n)$ .
3. **Substitution.**  $55.00 \times 1.32 = 72.60$ ;  $72.60/0.85 = 85.4118$ ;  $85.4118 \times 1.18$ .
4. **Result.** Paid-hour cost **USD 72.60**; per productive hour before overhead **USD 85.4118**; **full cost per productive hour USD 100.7859**. The paid-to-productive multiplier is **1.17647**. Against the USD 95.00 charge-out rate the estimate **under-recovers by USD 5.7859 an hour, 5.74 % of cost**.
5. **Interpretation.** The first duty is to know which rate a number is. **A rate quoted per paid hour understates the cost of a productive hour by  $1/(1 - n)$  (17.65 % here), and the two are confused constantly**, because both are called "the labour rate". An estimate built on paid-hour rates against a schedule built on productive hours is under-funded by that multiplier before any risk materialises, and the discrepancy is invisible because both documents look right.

**Utilisation is the number with the most leverage on the sheet, and nobody owns it.** Cost per productive hour scales as  $1/\text{utilisation}$ , so losing five points from 85 % to 80 % raises it to **USD 107.0850: +6.25 %**, exactly the ratio  $0.85/0.80$ . On Auriga, where labour is 62 % of BAC (**USD 2,480,000**) that is **USD 155,000**, which is **77.50 %** of the entire USD 200,000 cost variance the week-13 board meeting exists to discuss. **Five points of utilisation is three-quarters of the variance everyone is arguing about**, and it is lost in ways nobody records: waiting for access, waiting for a permit, waiting for a decision (Domain 3's governance latency, priced in Case study C), re-mobilising after a stand-down.

**The 5.74 % under-recovery is a finding, not an error.** It may be deliberate (a rate agreed commercially below full cost to win work, with the shortfall funded from elsewhere), or it may be an unnoticed drift as on-costs rose. Which of those it is matters, and the only way to know is to maintain the build-up and re-derive it annually. A charge-out rate carried unchanged for three years is almost certainly recovering less than it did.

**A jurisdiction caution that must not be skipped.** The composition of on-costs (statutory contributions, leave entitlements, end-of-service or severance provisions, insurance), and their accounting and tax treatment differ substantially between jurisdictions, and so does what may be recovered as overhead in a reimbursable contract. The 32 % and 18 % here are illustrative figures for a fictitious project. Build the rate with your own payroll and finance functions, and where a contract permits cost recovery, confirm the allowable basis with qualified advisers before relying on it.

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#### WORKED EXAMPLE

##### Worked example 7.4.1c — the price of a crew-week, four ways.

1. **Setup.** Domain 6, KA 6.3.1 establishes the ladder of capacity options (re-sequence, spend float, overtime, second shift, extend), and ranks it by cost without pricing it. Price it. An Auriga field crew is **4 technicians × 40 hours** at the USD 95 charge-out rate. The options for adding one crew-week of capacity: re-sequence within float; overtime at a **1.5×** premium; an agency crew at **USD 128** per hour achieving **85 %** of in-house productivity; or accept the delay, on a zero-float activity, at Domain 6's cost of delay of **USD 45,000 per week**.
2. **Formula.** Straight-time crew-week =  $4 \times 40 \times \text{rate}$ . Overtime crew-week =  $\text{straight} \times \text{premium}$ . Agency cost per *productive* crew-week =  $4 \times 40 \times \text{agency rate} \div \text{relative productivity}$ . Delay cost per week is given. Breakeven relative productivity between agency and overtime:  $\text{agency nominal} \div \text{overtime cost}$ .



3. **Substitution.**  $\text{Straight} = 4 \times 40 \times 95 = 15,200$ .  $\text{Overtime} = 15,200 \times 1.5$ .  $\text{Agency} = 20,480/0.85$ .  $\text{Breakeven} = 20,480/22,800$ , equivalently  $128/142.50$ .

4. **Result.**

| OPTION                            | COST PER CREW-WEEK OF<br>CAPACITY | × STRAIGHT<br>TIME |
|-----------------------------------|-----------------------------------|--------------------|
| Re-sequence within float          | 0 cash (float consumed)           | —                  |
| Straight-time crew (if available) | 15,200                            | 1.0000             |
| Overtime at 1.5×                  | 22,800                            | 1.5000             |
| Agency crew, 85 % productivity    | 24,094                            | 1.5851             |
| Accept the delay (zero float)     | 45,000                            | 2.9605             |

The agency crew beats overtime only if its relative productivity is at least **89.82 %**; symmetrically, overtime beats the agency crew only while sustained-overtime productivity holds above **94.63 %**. The dearest capacity option is **53.54 %** of the delay cost, and USD 45,000 buys **2.96** straight-time crew-weeks, **1.97** overtime crew-weeks or **1.87** agency crew-weeks. 5. **Interpretation.** The ladder Domain 6 asserts is confirmed and (more usefully) the gaps between its rungs are now numbers.

**Any of these beats the delay, so the decision is which to buy, not whether.** On a zero-float activity every purchased option costs between **33.78 %** and **53.54 %** of doing nothing. That is the general shape once a cost of delay has been computed, and it is why computing one changes behaviour: without USD 45,000 on the table, "we cannot afford overtime" sounds prudent.

**The two breakevens are the real content, because they are the arguments that actually happen.** Nobody disputes that an agency crew costs more per hour; the dispute is about how productive it will be, and 89.82 % is the number that dispute is about. Likewise the case against sustained overtime is not that it costs 1.5× but that its productivity decays: **the moment sustained overtime productivity falls below 94.63 %, the agency crew is the cheaper purchase**, at 90 % productivity overtime effectively costs USD 25,333 per crew-week and has lost. Both breakevens are checkable against the project's own timesheet and output records, which converts a preference into a measurement.

**Capacity only buys time where the activity is resource-limited.** Adding a crew to an activity constrained by logic, curing time, a possession window or a single test facility buys nothing at all (Domain 6, KA 6.2 and 6.3.2), and the four prices above are then all wasted. Establish that the constraint is capacity before pricing capacity.

**What the table excludes should be stated when it is used.** Onboarding, induction and supervision load for new crews; the lead time each option needs before it can start (Domain 6's point, and often decisive); the quality consequence of fatigue or unfamiliarity, which Domain 9 prices as escaped defects at USD 12,000 each; and the float that the free option consumes, which is the project's risk buffer and has a value Domain 8 can quantify. **"Free" is the option whose price is hardest to see**, and a leader who spends float without saying so has bought capacity on credit.

#### 7.4.2 Contract models and cost risk

Every contract model is an answer to one question: **who carries cost risk?**

| MODEL                                 | BUYER PAYS                                              | COST RISK SITS WITH                          | SUITS                                                      |
|---------------------------------------|---------------------------------------------------------|----------------------------------------------|------------------------------------------------------------|
| <b>Firm fixed price (FFP)</b>         | An agreed price, whatever it costs                      | <b>Seller</b>                                | Well-defined scope; priced-in risk premium                 |
| <b>Fixed price incentive (FPIF)</b>   | Target cost/fee with a share ratio, capped by a ceiling | <b>Shared, then seller above the ceiling</b> | Definable scope with real uncertainty                      |
| <b>Cost plus fixed fee (CPFF)</b>     | Allowable cost + a fixed fee                            | <b>Buyer</b>                                 | Development, unclear scope                                 |
| <b>Cost plus incentive fee (CPIF)</b> | Allowable cost + a fee varying with performance         | <b>Shared</b>                                | Scope where effort is uncertain but effort quality matters |
| <b>Time and materials (T&amp;M)</b>   | Rates × quantities                                      | <b>Buyer</b>                                 | Staff augmentation, short or open-ended work               |

The leader's discipline: **risk transferred is risk priced**. An FFP contract for ill-defined scope does not remove the risk. It converts it into a premium plus a claims exposure when the scope moves (Domain 10, KA 10.4). Conversely a cost-plus contract on well-defined work pays for uncertainty that no longer exists. Domain 10, KA 10.3.2 works the full outturn arithmetic: what each model pays the two parties at the same actual cost, and why an expected-value comparison of fixed price against cost-plus buys variance rather than money. What belongs *here* is the calculation a delivery leader does before that conversation starts: the point at which the simplest two options cross.

#### WORKED EXAMPLE

**Worked example 7.4.2 – fixed price or time and materials: where they cross.**

1. **Setup.** A defined configuration-and-testing package on Auriga. A supplier quotes a firm fixed price of **USD 480,000**. The alternative is time and materials from the same engineering pool at the blended **USD 130.625** per hour (KA 7.4.1). The team's own three-point estimate of the hours is **optimistic 2,900, most likely 3,200, pessimistic 4,100**.
2. **Formula.** Breakeven hours =  $\text{fixed price} \div \text{hourly rate}$ . T&M cost =  $\text{hours} \times \text{rate}$ . Premium =  $\text{fixed price} - \text{T\&M cost at the estimate}$ . PERT expected hours =  $(o + 4m + p)/6$ ; triangular mean =  $(o + m + p)/3$ . For a triangular distribution, the probability of exceeding a value  $x$  above the mode is  $(b - x)^2 / ((b - a)(b - m))$ .
3. **Substitution.** Breakeven =  $480,000/130.625$ . T&M at the mode =  $3,200 \times 130.625$ . PERT hours =  $(2,900 + 12,800 + 4,100)/6$ . Exceedance =  $(4,100 - 3,674.6411)^2 / (1,200 \times 900)$ .
4. **Result.** Breakeven **3,674.64 hours**. At the most likely 3,200 hours, T&M costs **USD 418,000**, so the fixed price carries a premium of **USD 62,000: 14.83 %** over the estimate (12.92 % of the fixed price itself). At the pessimistic 4,100 hours T&M costs **USD 535,563**, USD 55,563 worse than the fixed price; at the optimistic 2,900 it costs **USD 378,813**, USD 101,188 better. PERT expected hours are **3,300** (T&M **USD 431,063**, a premium of **11.35 %**; the triangular mean is **3,400 hours**) T&M **USD 444,125**, a premium of **8.08 %**. The probability that the hours exceed the breakeven is **16.75 %**.
5. **Interpretation.** The identity is exact and worth keeping: **the fixed-price premium expressed as a percentage of the estimate is the hours overrun at which the two options break even**. 14.83 % premium, 14.83 % overrun tolerance. So the question at the table is never "is USD 62,000 a lot?" but "how confident are we that this package will not run more than 14.83 % over its estimated hours?": a question the estimating team can actually answer, and one whose answer here is *not very*, since a 14.83 % overrun on 3,200 hours is well inside the declared range.

**State the premium against expectation, not against the mode.** Against expected hours the premium is **8.08 % to 11.35 %** depending on the distribution shape, not 14.83 %. Quoting the larger figure makes the fixed price look worse than it is, and quoting no figure at all is how fixed prices get accepted or rejected on temperament. What the buyer receives for that USD 35,875–48,938 is the removal of a 16.75 % chance of paying more than 480,000 and of a tail that reaches USD 535,563 — the variance-for-money trade Domain 10, KA 10.3.2 quantifies properly.

**The distributions are not the same distribution, and this is the error the arithmetic hides.** Under time and materials nobody is paid to be efficient: the supplier's revenue rises with the hours, so the hours estimate that was drawn for an in-house or fixed-price case does not transfer. The correct comparison prices

T&M against a *shifted* distribution, and since the size of the shift is unknowable in advance, the practical countermeasures are structural, an hours cap or not-to-exceed envelope, a fixed rate schedule, an agreed productivity or output measure, and approval gates at stated cumulative hours. **A T&M arrangement without a cap is an open account, not a contract model.**

**And the fixed price only covers the scope specified.** The 14.83 % tolerance is worthless if the scope is 30 % defined, because the variations will arrive priced without competition (MCQ 7.4-C, and Domain 10, KA 10.4.1 on entitlement). The order of operations is therefore fixed: define the scope, then choose the model, then compute the breakeven, never the reverse.

### 7.4.3 Incentive fees and the point of total assumption

**The mechanics.** Under an incentive contract, buyer and seller agree a **target cost**, a **target fee** and a **share ratio** (buyer/seller) for over- and under-runs, with a **ceiling price** capping the buyer's exposure.

#### WORKED EXAMPLE

#### Worked example 7.4.3 – the incentive that stops incentivising.

1. **Setup.** Auriga's installation subcontract: target cost **USD 2,000,000**, target fee **USD 150,000**, share ratio **70/30** (buyer 70 %, seller 30 %), ceiling price **USD 2,450,000**. The seller finishes at an actual cost of **USD 2,300,000**.
2. **Formula.** Fee = target fee – (overrun × seller share). Buyer pays = actual cost + fee (subject to the ceiling). Point of total assumption **PTA** = **target cost + (ceiling – target price)/buyer share**, where target price = target cost + target fee.
3. **Substitution.** Overrun  $2,300,000 - 2,000,000 = 300,000$ ; fee  $150,000 - 300,000 \times 0.30 = 150,000 - 90,000$ . Buyer pays  $2,300,000 + 60,000$ . Target price  $2,150,000$ ; **PTA** =  $2,000,000 + (2,450,000 - 2,150,000)/0.70$ .
4. **Result.** Fee **USD 60,000** (down from 150,000); buyer pays **USD 2,360,000**. **PTA** = **USD 2,428,571.43**; and at that cost the buyer pays exactly the USD 2,450,000 ceiling.
5. **Interpretation.** The share ratio does its job up to the **PTA**: both parties lose money on overrun, so both want efficiency. **Above the PTA the seller bears 100 %** of further cost, which is the moment the incentive inverts: a seller heading past it has no financial reason to spend more on your project and every reason to argue that the extra cost is *your* scope change. Knowing where the **PTA** sits is therefore a **delivery** insight, not an accounting one: it predicts when a commercial relationship will turn adversarial (Domain 10, KA 10.4; Domain 11's negotiation).

**Verify the PTA by walking it, because the check is one line and it catches sign errors.** At a cost of 2,428,571.43 the overrun is 428,571.43, so the fee is  $150,000 - 0.30 \times 428,571.43 = \text{USD } 21,428.57$ , and the buyer pays  $2,428,571.43 + 21,428.57 = \text{USD } 2,450,000$ : the ceiling, exactly. Any PTA that fails to reproduce the ceiling this way has been computed with the wrong share.

**The fee is not yet exhausted at the PTA, and that boundary has a rule.** The seller's fee reaches zero only at  $2,000,000 + 150,000/0.30 = \text{USD } 2,500,000$ , above the PTA. So on this structure the *ceiling* binds first, which is what makes the PTA the meaningful inflection. The general condition is worth carrying: **the ceiling binds before the fee is exhausted only while the ceiling is below target price + target fee × buyer share ÷ seller share**, here  $2,150,000 + 150,000 \times (0.70/0.30) = \text{USD } 2,500,000$ . At a ceiling of exactly 2,500,000 the PTA and the zero-fee cost coincide at 2,500,000; above it the PTA is a nominal figure, because the seller has already lost its whole fee before the buyer's cap engages. A leader shown a PTA should therefore also be shown the zero-fee cost, and told which comes first.

**The share ratio moves the PTA, and it moves it the way nobody expects.**

| BUYER / SELLER<br>SHARE | PTA              | FEE AT THE<br>PTA | COMES BEFORE THE ZERO-FEE<br>COST? |
|-------------------------|------------------|-------------------|------------------------------------|
| 50 / 50                 | 2,600,000        | (150,000)         | No: zero-fee cost is 2,300,000     |
| 60 / 40                 | 2,500,000        | (50,000)          | No: zero-fee cost is 2,375,000     |
| <b>66.67 / 33.33</b>    | <b>2,450,000</b> | <b>0</b>          | Exactly coincides, at the ceiling  |
| 70 / 30 (Auriga)        | <b>2,428,571</b> | 21,429            | Yes: zero-fee cost is 2,500,000    |
| 80 / 20                 | 2,375,000        | 75,000            | Yes                                |
| 90 / 10                 | 2,333,333        | 116,667           | Yes                                |

**The more of the overrun the buyer absorbs, the *earlier* the inflection arrives:** moving from 70/30 to a superficially more generous 90/10 brings the PTA forward by **USD 95,238**, because the buyer's fixed headroom to the ceiling is consumed faster. A buyer who concedes share in negotiation to appear reasonable has bought an earlier adversarial turn, and should know it. Note too the boundary in the third row: on this structure a buyer share below **66.67 %** means the ceiling never binds before the fee is gone.

**Sensitivity to the ceiling is 1/buyer share, so the ceiling is the stronger lever.** Each dollar added to the ceiling moves the PTA by **1.4286** dollars at a 70 % buyer share. And the figure to monitor monthly is the *headroom*: with the seller's cost forecast at 2,300,000, the distance to the PTA is **USD 128,571**, **5.59 %** of that

forecast. **A single 6 % cost movement puts this relationship past its inflection**, which is why toolkit 7.T.3 asks for the cost trend against the **PTA** at every commercial checkpoint rather than the **PTA** alone. Domain 10, KA 10.3.2 reads the same figure as a statement about the seller's marginal dollar.

**A counsel pointer, because this is an enforceability-sensitive area.** Ceilings, share ratios, fee adjustment mechanics and their interaction with variation, notice and claims provisions vary by contract form and by jurisdiction, and so does the treatment of a seller trading past a ceiling. The arithmetic here is contract-neutral; the drafting is not. Take qualified legal advice on the instrument before relying on any of it commercially (Domain 10, KA 10.4).

#### 7.4.4 Cash flow versus profit

A project can be profitable and still fail for lack of cash: the same truth PFL-AI establishes for financings (its Domain 1). For a delivery leader the mechanism is payment terms: cost is incurred as work happens; cash arrives when invoices are approved and paid. On Auriga, with **AC** = USD 2,120,000 at week 13 and 60-day terms, roughly **USD 742,000** of incurred cost (about 35 %) is still unpaid: an exposure carried by whoever is funding the work. The leader's obligations are practical: know the terms in both directions (client and subcontractors), front-load nothing you cannot fund, and never let a retention or milestone-payment structure be agreed without someone computing its cash profile. Where the project sits inside a financed asset, this is precisely the CFADS conversation PFL-AI Domain 10 has with lenders.

That last obligation is the one routinely left undone, so here is the computation.

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#### WORKED EXAMPLE

##### Worked example 7.4.4 — Auriga's cash, its profit, and the gap between them.

1. **Setup.** Auriga is delivered under a fixed price to the utility of **USD 4,400,000** against a **BAC** of 4,000,000, a bid margin of USD 400,000. Work is valued monthly (weeks 4, 8, 12 ...) at the contract rate, **5 % retention** is withheld from each payment, and the client's terms are **60 days** from invoice. Half the retention is released at practical completion and half after the defects period. At week 13: **EV 1,920,000**, of which **1,760,000** had been earned by the week-12 valuation, **880,000** by the week-8 valuation and **320,000** by the week-4 valuation; **AC 2,120,000**, of which **USD 742,000** sits in supplier invoices within terms and accruals not yet paid. By week 13 only the week-4 invoice has been paid. What is the position?

2. **Formula.** Revenue recognised =  $(\text{price}/\text{BAC}) \times \text{EV}$ . Margin to date = revenue – AC. Cash paid out = AC – payables. Cash received = net-of-retention value of invoices whose terms have expired. Funding absorbed = cash paid out – cash received. Days of spend = funding absorbed  $\div$  (AC  $\div$  elapsed days). Client exposure = receivables + retention held + unbilled work in progress.
3. **Substitution.** Price factor =  $4,400,000/4,000,000 = 1.10$ . Revenue =  $1.10 \times 1,920,000$ . Certified to week 12 =  $1.10 \times 1,760,000 = 1,936,000$ , retention = 96,800. Week-4 payment =  $1.10 \times 320,000 \times 0.95$ . Cash out =  $2,120,000 - 742,000$ . Days of spend =  $1,043,600 \div (2,120,000/91)$ .

#### 4. Result.

| POSITION AT WEEK 13                                       | USD              |
|-----------------------------------------------------------|------------------|
| Cost incurred (AC)                                        | 2,120,000        |
| less payables (35.00 % of AC)                             | (742,000)        |
| <b>Cash paid out</b>                                      | <b>1,378,000</b> |
| less cash received (the week-4 invoice, net of retention) | (334,400)        |
| <b>Funding absorbed</b>                                   | <b>1,043,600</b> |
| Revenue recognised at $1.10 \times \text{EV}$             | 2,112,000        |
| <b>Margin to date</b>                                     | <b>(8,000)</b>   |
| Receivables outstanding                                   | 1,504,800        |
| Retention held                                            | 96,800           |
| Unbilled work in progress (week 13, at 1.10)              | 176,000          |
| <b>Total client exposure</b>                              | <b>1,777,600</b> |

The funding absorbed is **26.09 %** of BAC and **44.80 days** of spend at the current rate of USD 23,297 a day. 5. **Interpretation.** Set the two headline numbers side by side. **The cost report's headline is a variance of USD 200,000. The bank account's headline is USD 1,043,600: 5.22 times** larger, payable now, and absent from every document in the week-13 pack.

**The working-capital identity ties the two statements together and should be used as a check.** Funding absorbed = client exposure – payables + loss to date:  $1,777,600 - 742,000 + 8,000 = 1,043,600$ , exactly. So cash is not a separate subject from performance; it is performance plus the timing of two sets of payment terms. A cash forecast that cannot be reconciled to the cost report through that identity has an error in one of them.

**Express the requirement in days of spend, because that is the number that transfers.** USD 1,043,600 means nothing across a portfolio; **44.80 days of**

**spend** is immediately comparable between projects of different sizes and immediately actionable by a treasurer. A project running at 45 days of spend has a month and a half of its own cost outstanding at all times, and every week of client payment delay adds directly to it.

**Payment terms are the largest single lever, and they are agreed by people who never see this table.** Moving the client's terms from 60 days to 30 brings the week-8 invoice inside the data date (a further **USD 585,200** received), and cuts the funding absorbed to **USD 458,400**, a **56.08 %** reduction. Nothing about the work changes. That is the whole of the case for treating payment terms as a delivery parameter rather than a procurement formality, and for computing the profile *before* signature: after signature the same improvement has to be bought.

**Retention is margin that is not cash, and the proportion is startling.** Retention at completion will be  $5\% \times 4,400,000 = \text{USD } 220,000$  (**55.00 %** of the entire bid margin) with **USD 110,000 (27.50 % of margin)** held beyond practical completion into the defects period. A project can therefore be complete, profitable and still waiting on more than a quarter of its profit, which is why release criteria and their evidence belong in the closeout plan (Domain 16) rather than being discovered afterwards.

**The forecast method decides whether this contract makes money at all.** At the fixed price of 4,400,000: method (a)'s **EAC** of 4,200,000 leaves a margin of **USD 200,000 (+4.55 % of price)**; method (b)'s 4,416,667 leaves (**USD 16,667**) (**-0.38 %**); method (c)'s 4,608,056 leaves (**USD 208,056**) (**-4.73 %**). So KA 7.3.3's choice of assumption is not a reporting preference. It is the difference between a profitable contract and a loss-making one, and preserving the bid margin of 400,000 requires an **EAC** of exactly **BAC**, which is **TCPI 1.11** (KA 7.3.4). **The TCPI conversation and the margin conversation are the same conversation**, and a commercial manager and a project controller who do not know that will present two irreconcilable papers to the same board.

**An accounting caution.** How revenue is recognised on a contract, and how retentions, accruals and unbilled work in progress are presented, are matters for the applicable financial reporting framework and the entity's own accounting policy, and they vary by jurisdiction. The  $1.10 \times \text{EV}$  basis used above is a transparent illustrative convention for a fictitious project, not a statement of how any entity must report. Agree the basis with the finance function before any figure of this kind leaves the project.



**Auriga at week 13: the cost report says 200,000; the bank account says 1,043,600**

Revenue recognised 2,112,000 against cost 2,120,000 — a margin of (8,000).

Client exposure 1,777,600 = receivables 1,504,800 + retention 96,800 + unbilled 176,000.

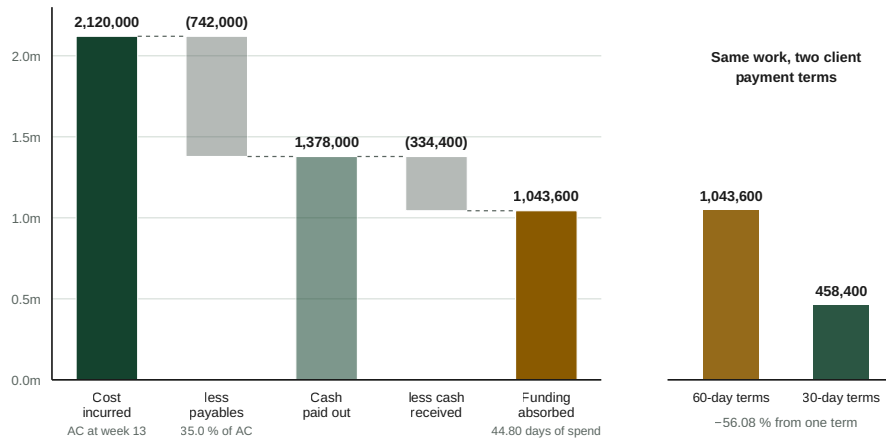


Fig 7.4.1 — Auriga at week 13: the cost report says 200,000; the bank account says 1,043,600

**AI in this KA**

Contract analytics (extracting terms, comparing models, flagging inconsistent clauses) is real and useful AI assistance, and it is also decision support rather than legal or commercial advice. Two boundaries: an AI-produced reading of a clause is verified against the clause itself before anyone relies on it (the document-against-summary check), and commercial and legal positions go to qualified counsel (Domain 10). Fee arithmetic and PTA computations are deterministic: they get the same golden-answer treatment as the earned-value set.

Three further placements are worth being specific about, because the temptation in this KA is to delegate the wrong half. Assembling a **rate build-up** from payroll and overhead data across many grades and cost centres, the mechanical part of 7.4.1b, is a strong application, and the resulting cost per productive hour is exactly checkable. Sweeping a portfolio of contracts for payment terms, retention percentages and release criteria and producing the aggregate cash profile of 7.4.4 is work most organisations do not do at all because it is tedious across dozens of documents; a model does it quickly and every extracted term is verifiable against its clause. What must **not** be delegated is the **utilisation** assumption and the **relative productivity** figure of 7.4.1c: both are judgements about a specific workforce in a specific setting, both drive their answers almost entirely, and a model asked for either will supply a confident, well-formatted number with no provenance: the failure mode Domain 9, KA 9.1 describes for detection rates. Where the evidence does not exist, the correct output is the breakeven — 89.82 %, 94.63 %: plus the statement that nobody has measured which side of it the project sits on.

## KEY TERMS — KA 7.4

| TERM                                      | MEANING                                                                                                                                                                                 |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Blended rate</b>                       | Weighted average cost per hour of a mixed resource pool.                                                                                                                                |
| <b>FFP / FPIF / CPFF / CPIF / T&amp;M</b> | Contract models ordered by who carries cost risk.                                                                                                                                       |
| <b>Share ratio</b>                        | The agreed buyer/seller split of over- and under-run.                                                                                                                                   |
| <b>Ceiling price</b>                      | The cap on the buyer's total exposure under an incentive contract.                                                                                                                      |
| <b>Point of total assumption (PTA)</b>    | The cost above which the seller bears 100 % of further overrun.                                                                                                                         |
| <b>Zero-fee cost</b>                      | $\text{target cost} + \text{target fee} \div \text{seller share}$ , where the incentive fee is exhausted; on Auriga USD 2,500,000, above the PTA, which is why the ceiling binds first. |
| <b>PTA headroom</b>                       | The distance from the seller's current cost forecast to the PTA; 5.59 % on Auriga, and the figure to trend monthly.                                                                     |
| <b>Cost per productive hour</b>           | $\text{Burdened pay} \div (1 - \text{non-productive share}) \times (1 + \text{overhead})$ ; USD 100.79 against a USD 95.00 charge-out rate.                                             |
| <b>Paid-to-productive multiplier</b>      | $1 / (1 - n)$ , 1.17647 at a 15 % non-productive share; the commonest silent under-funding in a labour estimate.                                                                        |
| <b>Cost per crew-week of capacity</b>     | The priced rungs of Domain 6's capacity ladder, from 15,200 at straight time to 45,000 for accepting the delay (7.4.1c).                                                                |
| <b>T&amp;M/fixed-price breakeven</b>      | $\text{fixed price} \div \text{hourly rate}$ ; the premium as a percentage of the estimate equals the hours overrun at which the two options cross.                                     |
| <b>Retention</b>                          | A withheld percentage of payment, released on completion criteria; Auriga's 5 % is 55.00 % of the bid margin at completion.                                                             |
| <b>Funding absorbed</b>                   | Cash paid out – cash received; equals client exposure – payables + loss to date. Best stated in days of spend (44.80 on Auriga).                                                        |

## SAMPLE MCQs — KA 7.4

**MCQ 7.4-A** [7.4.3 · Application] Target cost 2,000,000; target fee 150,000; share 70/30; actual cost 2,300,000. The seller's fee is:

- A. USD 150,000
- B. USD 60,000 CORRECT
- C. USD 90,000
- D. USD 45,000

*Rationale:* The seller absorbs 30 % of the 300,000 overrun:  $150,000 - 90,000 = 60,000$ . A ignores the incentive; C states the fee reduction rather than the fee; D applies the buyer's share to the fee.

**MCQ 7.4-B** [7.4.3 · Analysis] Target cost 2,000,000, target fee 150,000, ceiling 2,450,000, buyer share 70 %. The PTA is 2,428,571, and its delivery significance is that above it:

- A. the contract becomes void
- B. the buyer absorbs all further cost
- C. the seller bears 100 % of further cost, so the incentive inverts and cost growth becomes a scope-change argument CORRECT
- D. the fee becomes negative but risk-sharing continues unchanged

*Rationale:* Beyond the PTA the ceiling binds the buyer, so every further dollar is the seller's, which predictably redirects the seller's effort from efficiency to entitlement. B reverses the exposure; A is fiction; D misses that sharing has *stopped*.

**MCQ 7.4-C** [7.4.2 · Analysis] A leader lets an FFP contract for scope that is only 30 % defined. The most likely outcome is:

- A. cost risk is genuinely eliminated
- B. a priced-in risk premium plus a claims-and-variations exposure as the scope is defined CORRECT
- C. the seller absorbs all scope growth at no cost to the buyer
- D. the contract converts automatically to cost-plus

*Rationale:* Fixed price transfers risk *at a price* and only for the scope actually specified; undefined scope returns as variations. A and C mistake the contractual form for the underlying uncertainty; D invents a mechanism.

**MCQ 7.4-D** [7.4.1 · Application] A pool of 40 at USD 95/h, 25 at USD 140/h and 15 at USD 210/h has a blended rate of:

- A. USD 148.33
- B. USD 130.63 CORRECT
- C. USD 112.31
- D. USD 140.00

*Rationale:*  $10,450/80 = 130.63$ . A averages the three rates unweighted (445/3); C blends only the two cheapest grades, dropping the specialists (7,300/65); D takes the middle rate as representative.

**MCQ 7.4-E** [7.4.1 · Application] Base pay USD 55.00 per paid hour; on-costs 32 %; 15 % of paid hours are non-productive; recoverable overhead 18 %. The full cost of one productive hour is:

- A. USD 85.41
- B. USD 100.79 CORRECT
- C. USD 98.52

D. USD 72.60

*Rationale:*  $55.00 \times 1.32 \div 0.85 \times 1.18 = 100.7859$  (7.4.1b). A stops before overhead; C adds 15 % rather than dividing by 0.85, understating by the difference between an uplift and a divisor; D is the paid-hour cost, which is what most rate cards actually contain.

**MCQ 7.4-F** [7.4.1 · Evaluation] An agency crew charges USD 128 per hour against an in-house USD 95 with overtime at 1.5×. The agency crew is the cheaper purchase per productive crew-week only if its relative productivity is at least:

- A. 74.22 %
- B. 89.82 % CORRECT
- C. 100.00 %
- D. 66.67 %

*Rationale:*  $128/142.50 = 0.8982$ , the agency rate against the overtime rate (7.4.1c). A compares the agency rate with straight time and so ignores the overtime premium being avoided; C assumes any productivity shortfall disqualifies the option, which the arithmetic refutes; D inverts the 1.5 premium.

**MCQ 7.4-G** [7.4.2 · Application] A firm fixed price of USD 480,000 is offered against time and materials at USD 130.625 per hour, with the package estimated at 3,200 hours. The breakeven hours and the premium over the estimate are:

- A. 3,674.64 hours and 14.83 % CORRECT
- B. 3,674.64 hours and 12.92 %
- C. 3,428.57 hours and 14.83 %
- D. 3,200 hours and nil

*Rationale:*  $480,000/130.625 = 3,674.64$ ; the premium is  $62,000/418,000 = 14.83$  %, which equals the hours overrun at breakeven (7.4.2). B expresses the premium against the fixed price rather than the estimate, understating the tolerance the buyer is purchasing; C uses the mid-grade rate of 140 instead of the blend; D assumes the estimate is the crossing point.

**MCQ 7.4-H** [7.4.4 · Application] The contract price is USD 4,400,000 against a BAC of 4,000,000. At week 13 EV is 1,920,000 and AC is 2,120,000. The margin recognised to date is:

- A. USD 400,000
- B. (USD 8,000) CORRECT
- C. USD 200,000
- D. (USD 200,000)

*Rationale:* Revenue at  $1.10 \times 1,920,000 = 2,112,000$  against cost of 2,120,000 (7.4.4). A is the bid margin, which assumes nothing has gone wrong; C and D read

the cost variance as the margin, which omits the 10 % mark-up earned on the work performed and so mis-states the position in both directions.

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SELF-CHECK — KA 7.4

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1. *State the one question every contract model answers.* — Who carries cost risk.
  2. *Why is the PTA a delivery concern, not just a commercial one?* — Above it the seller bears all further cost, so behaviour shifts from efficiency to entitlement.
  3. *How can a profitable project run out of cash?* — Cost is incurred as work happens; cash arrives on payment terms. The gap must be funded.
  4. *Which way does the PTA move when the buyer takes a larger share of the overrun, and why?* — Earlier: fixed headroom to the ceiling is consumed faster, so 70/30 to 90/10 brings it forward by USD 95,238. Below a 66.67 % buyer share this structure's ceiling never binds before the fee is exhausted.
  5. *What is the difference between a paid-hour rate and a productive-hour rate?* — The factor  $1/(1 - n)$  (17.65 % at Auriga's 15 % non-productive share), and confusing them under-funds a labour estimate before any risk materialises.
  6. *State Auriga's capacity ladder in money.* — Per crew-week: re-sequencing free but paid in float, straight time 15,200, overtime 22,800, agency at 85 % productivity 24,094, accepting the delay 45,000, so on a zero-float activity every purchased option beats doing nothing.
  7. *What does the funding absorbed at week 13 equal, and how is it best expressed?* — USD 1,043,600 (client exposure 1,777,600 less payables 742,000 plus the 8,000 loss) best stated as **44.80 days of spend**, and halved to 458,400 by 30-day rather than 60-day client terms.
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## Advanced topics — Domain 7

### 7.A.1 Earned schedule — closing SPI's late-project blind spot

SPI converges on 1.00 as a project finishes, whatever its lateness, because EV and PV both approach BAC. **Earned schedule** restates progress in time: ES is the date at which the value now earned was *planned* to have been earned, and  $SPI(t) = ES / AT$ . This is the bridge flagged in Domain 6 (KA 6.4.3), now with its cost-side companions. And the size of the blind spot it closes is worth computing rather than asserting.

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WORKED EXAMPLE

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**Worked example 7.A.1 – the same project, read twice, at week 13 and at week 27.**

- Setup.** Auriga's baseline PV curve reaches **1,920,000 at week 12** and **2,080,000 at week 13**; a local slope of USD 160,000 a week. At week 13 the project has earned exactly **1,920,000**. Now project forward: suppose the work is not complete at week 25 and at **week 27 EV** stands at **3,920,000**, against a baseline whose PV was **3,880,000 at week 24** and reaches **BAC 4,000,000 at week 25** (after which PV stays at BAC, having nothing left to plan). Planned duration PD = **25 weeks**. Read the schedule position both ways at both dates.
- Formula.** ES = the baseline time at which the EV now earned was planned to be earned, interpolated between the bracketing periods:  $ES = t + (EV - PV_t) / (PV_{t+1} - PV_t)$ .  $SPI(t) = ES/AT$ , with AT the actual time elapsed. Time forecast  $IEAC(t) = PD/SPI(t)$ . For comparison  $SPI = EV/PV$ .
- Substitution.** Week 13:  $ES = 12$  exactly, so  $SPI(t) = 12/13$ ;  $SPI = 1,920,000/2,080,000$ . Week 27:  $ES = 24 + (3,920,000 - 3,880,000) / (4,000,000 - 3,880,000) = 24 + 1/3$ ;  $SPI(t) = 24.3333/27$ ;  $SPI = 3,920,000/4,000,000$ ;  $IEAC(t) = 25/0.901235$ .

**4. Result.**


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|         | SPI           | ES      | SPI(t)        | TIME FORECAST               |
|---------|---------------|---------|---------------|-----------------------------|
| Week 13 | <b>0.9231</b> | 12.0000 | <b>0.9231</b> | 27.0833 weeks (2.0833 late) |
| Week 27 | <b>0.9800</b> | 24.3333 | <b>0.9012</b> | 27.7397 weeks (2.7397 late) |

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At week 13 the two indices agree exactly. At week 27 they differ by **0.0788**: SPI reports the project **2 %** behind while it is in fact **2.74 weeks, 10.96 %**, late. 5. **Interpretation.** The week-13 agreement is not a coincidence and it is the more instructive half. **SPI and SPI(t) coincide wherever the PV curve is locally linear**, because  $SV/slope$  is then exactly the number of weeks of lateness: Auriga's SV of (160,000) is one week of planned value at 160,000 a week, so ES is 12 and  $SPI(t)$  is 12/13, the same 0.9231 that  $EV/PV$  gives. That is why SPI is perfectly serviceable mid-project on a smooth S-curve, and it is worth saying so rather than presenting earned schedule as a correction to a broken measure.

**\*\*The divergence appears where the curve flattens, which is exactly where the decisions get expensive.\*\*** Past the planned finish,  $\text{'PV'}$  is pinned at  $\text{'BAC'}$  and has nothing left to compare against, so  $\text{'SPI'}$  is arithmetically compelled towards 1.00 and reports 0.98 for a project nearly three weeks late.  $\text{'SPI(t)'}$  keeps working because its denominator is *time*, which does not run out. The rule for practice: **\*\*quote  $\text{'SPI'}$  while the  $\text{'PV'}$  curve is still rising steeply, and  $\text{'SPI(t)'}$  from the point at which it flattens\*\***, and never quote  $\text{'SPI'}$  in the last 15 % of a project without the time-based figure beside it (7.A.2's limitation list; the exam-preparation trap is the same one).

**\*\*The time forecast is what makes the index worth computing, because it converts into money at a known rate.\*\***  $\text{'PD/SPI(t)'}$  gives 27.08 weeks at the data date (2.0833 weeks late), which at Domain 6's cost of delay of USD 45,000 per week is **\*\*USD 93,750\*\*** of delay cost, a figure directly comparable with the compression options Domain 6 prices and with the recovery spend the week-13 case study argues about. Note the consistency check: at week 27 the remaining baseline work is  $25 - 24.3333 = 0.6667$  weeks, which at the demonstrated time-efficiency of 0.9012 takes 0.7397 weeks, completion at 27.7397, exactly the  $\text{'IEAC(t)'}$ . Any earned-schedule presentation should reproduce that agreement.

**\*\*Two cautions, both about the baseline rather than the technique.\*\***  $\text{'ES'}$  is read *off the  $\text{'PV'}$  curve*, so it inherits every defect of that curve, a curve phased by straight-line spread rather than by the schedule gives a confidently wrong  $\text{'ES'}$ , and the interpolation above assumes the curve is meaningful between period points. And re-baselining resets  $\text{'ES'}$  exactly as it resets  $\text{'SPI'}$  (7.2.3): a project that re-phases its  $\text{'PV'}$  curve to match its actual progress will report  $\text{'SPI(t)'}$  of 1.00 while remaining as late as it was the day before, which is why the integrity rules of KA 7.2.3 are a precondition for every index in this domain and not only for the cost ones.

### 7.A.2 EVM's limitations, stated plainly

Earned value measures conformance to *plan*, not value to the *customer*: a project can score 1.00 on both indices while building the wrong thing (Domain 5's acceptance

criteria are the defence). It says nothing about quality (Domain 9). It is only as honest as its earning rules and its accruals. And it is blind to risk that has not yet materialised: Domain 8's contingency analysis, not **CPI**, tells you whether what remains is adequately funded. EVM is a measurement system, not a management philosophy; leaders who treat the indices as targets get optimised indices.

### 7.A.3 The reviewer's cost eye

Invariants a reviewer runs before trusting a cost report: **EV/BAC** equals the claimed percent complete;  $CV = EV - AC$  and  $SV = EV - PV$  reconcile to the stated indices;  $EAC = AC + a$  stated **ETC**, with the assumption named; **TCPI** to **BAC/CPI** equals **CPI**; sum of control-account budgets + contingency equals **BAC**; management reserve sits outside that total; no completed package's budget has moved since last period; accruals are present in the current period; and level-of-effort share is disclosed per control account. Any violation is a defect somewhere, find it before the board builds on it.

Seven more, each of which this domain has now derived, and each a single line of arithmetic:

- **The percent-spent minus percent-complete gap must equal  $CV/BAC$ .**  
Auriga:  $53.00\% - 48.00\% = 5.00\%$ , and  $200,000/4,000,000 = 5.00\%$  (7.3.2).
- **TCPI to every reported **EAC** must equal that method's own implied efficiency:** 1.0000 for method (a), **CPI** for (b),  $CPI \times SPI$  for (c). A mismatch means the forecast and its stated assumption are not the same document (7.3.4).
- **Contingency drawn plus unattributed variance must equal  $CV$ .** Auriga's USD 190,000 of draws against a **CV** of 200,000 leaves USD 10,000 with no named cause (7.1.3).
- **AC must not be below invoices plus receipted-but-uninvoiced value**, and a cumulative **CPI** of exactly 1.0000 is a prompt to check that before it is a cause for satisfaction (7.2.1).
- **Commitment coverage must be stated wherever prices are moving**, and the excess of the forecast's uncommitted balance over the baseline's must equal  $|VAC|$  (7.2.1b).
- **The earning rule per package must be the same rule as last period**, and one account should be recomputed under an alternative rule to size the discretion involved: 15 % of budget on Auriga's ten-package group (7.3.1).
- **SPI must not be quoted alone in the last 15 % of a project**, where it is compelled towards 1.00; the time-based figure belongs beside it (7.A.1).

And two outside the earned-value set that decide whether the report describes a viable project at all: the **funding absorbed**, stated in days of spend (44.80 on Auriga), and the **PTA headroom** on every incentive package (5.59 %). Neither appears in a conventional cost report, and both change decisions.



## Industry variations — Domain 7

Each variation below changes a *specific* number in this domain, not merely its emphasis.

- **Construction and EPC.** Full EVM against a resource-loaded baseline, with EV claimed from measured-work valuations, so the earning rule is effectively units-completed and the earning-rule discretion of 7.3.1 is largely designed out, at the cost of a monthly measurement effort. Retention is the binding cash term: on Auriga's structure a 5 % retention is **USD 220,000**, or **55.00 %** of the bid margin, and half of it is held past completion. Variation and claims machinery under the standard international forms makes the PTA conversation routine rather than exceptional, so the headroom figure of 7.4.3 belongs on the monthly commercial report.
- **Government and defence programmes.** Formal earned-value management-system compliance with surveillance, control-account discipline and (the operative difference) a **hard cap on the level-of-effort share**, because 7.3.1's distortion identity  $s \times (1 - SPI\_d)$  is exactly what a surveillance regime exists to bound: a 20 % cap holds the distortion to 0.08 where a 70 % share allows 0.28. Forecasting is auditable rather than discretionary, which in practice means method (d) is commissioned at defined points rather than when someone asks for it.
- **Technology and product delivery.** Cost is overwhelmingly people, so the numbers that move the outturn are the blended rate, the mix and **utilisation**: 7.4.1b's result (five points of utilisation being 6.25 % of the labour bill, USD 155,000 on Auriga's labour content) is the whole game, and materials barely register. Where cadence replaces a fixed baseline, throughput and cost-per-increment stand in for CPI (Domain 13), and hybrid programmes report both; the commercial default is time and materials, which makes 7.4.2's cap-and-rate-schedule discipline the difference between a contract and an open account.
- **Energy and resources.** Long-lead equipment and commodity exposure move the centre of gravity to **commitment coverage** and escalation: 7.2.1b's arithmetic (a 5 % movement on the uncommitted remainder being 27.92 % of the cost variance to date) is why the coverage figure is reported monthly here and rarely elsewhere, and why early commitment is bought deliberately at a price (Domain 10, KA 10.1.3; PFL-AI Domain 3 for escalation and currency machinery). Contingency is sized quantitatively as a matter of course, so the draw-index trend of 7.1.3 has a register behind it.
- **Public services and transformation.** Benefits, not cost variance, are the accountability currency (Domain 16), and the cost-side consequence is the one Case study C computes: contingency gets sized against **delivery** risk while the exposure sits on the **benefit** side, where Meridian's unreserved annual gap of USD 293,760 was **2.4480 times** its entire contingency. The leader's task is keeping cost reporting honest while the value case is what gets debated, and

extending reserve thinking to the risks that actually decide whether the money was worth spending.

### Case study — Domain 7: the forecast the board actually needed (utilities)

**Situation.** Auriga's week-13 review. The cost report shows **CPI** 0.91, **SPI** 0.92, **CV** (200,000), **SV** (160,000). The programme director's paper proposes reporting **EAC USD 4,200,000**, method (a), on the grounds that the overrun was the contaminated-ground remediation, a discrete event now closed and remediated (Domain 6's case study).

**The challenge.** The assurance reviewer asks three questions. *Is the cause genuinely closed?* Yes for the remediation, but the recovery bought a second civil crew and a fast-track of installation, both of which continue for six more weeks, and neither is in method (a)'s "remaining work at budgeted rate" assumption. *What does recovery to BAC require?* **TCPI** 1.11 against a demonstrated 0.91. *What is the mechanism?* There isn't one: the recovery spend increases cost to protect the date.

**The outcome.** The board is given a **range with named assumptions**: 4,200,000 if the closed event is the whole story; 4,416,667 if current efficiency persists; and a bottom-up (method d) re-estimate of the remaining 52 %, commissioned that week, as the number they will actually manage to. Contingency draw is authorised against the identified risk; management reserve is not touched. The minute records the **TCPI** gap and the explicit finding that **no recovery-to-budget plan exists**, so nobody later claims one was promised.

**What the domain teaches here.** A single-number forecast hides the only thing that matters: the assumption. **TCPI** converts optimism into an arithmetic claim someone has to defend, and the honest report is the one that survives the question "what would have to be true?"

### Case study B — Domain 7: past the point of total assumption (technology)

**Situation.** A systems-integration subcontract ran on an incentive structure: target cost USD 2,000,000, target fee 150,000, 70/30 share, ceiling 2,450,000, so **PTA** USD 2,428,571. By month eight the supplier's internal cost was tracking to 2,600,000, well past the **PTA**.

**What happened.** The supplier's behaviour changed abruptly and, in hindsight, rationally: fresh change requests on work previously treated as in-scope, slower responses on defect fixes that earned nothing, and a claim that the buyer's late environment provision had caused the growth. The buyer's team read it as bad faith. It was arithmetic: past the **PTA** every additional engineer-hour came out of the sup-

plier's own margin, and the only route back to profit was re-characterising cost as buyer-caused scope.

**The outcome.** The parties re-set commercially, a re-baselined target cost recognising the genuine environment delay (documented, and partly the buyer's), a revised ceiling, and a time-and-materials envelope for the disputed remainder. Delivery recovered within two months of the reset. The retrospective's finding: the buyer had never computed the **PTA**, so a foreseeable inflection was experienced as a betrayal.

**What the domain teaches here.** Commercial structures create behaviour. Computing the **PTA** at signature (and watching the supplier's cost trend against it) turns an adversarial surprise into a managed conversation held early, while both parties still have options (Domain 10's supplier governance; Domain 11's negotiation).

### Case study C — Domain 7: the reserve that could not be reached in time (public health)

**Situation.** Meridian Care Records (the 40-clinic shared clinical-records programme of Domains 1, 2, 15 and 16) is at month 14. Its approved cost of **USD 2,400,000** is structured as **USD 2,160,000** of control-account budgets plus **USD 120,000** of contingency inside a baseline of **USD 2,280,000**, with **USD 120,000** of management reserve outside it. Contingency is **5.56 %** of the control-account total; management reserve is **5.26 %** of the baseline. Installation is running well (this is the programme that will deliver 40 of 40 clinics), and **USD 74,000** of contingency has been drawn against installation risks, leaving **USD 46,000**.

Adoption is not running well. Sixteen clinics are using the system where the case assumed twenty-eight (Domain 1, KA 1.3.2). The clinical lead brings a costed **adoption-support package** (floor-walking, workflow coaching and local super-users in the twelve clinics not yet using the system) at **USD 168,000**, expected to move adoption from 40 % to the planned 70 %.

**The arithmetic the programme had not done.** Twelve clinics at Domain 1's benefit rate of **USD 510 a week** each are worth **USD 6,120 a week**, or **USD 293,760** a year over a 48-week operating year. So the package pays for itself in **27.4510 weeks** of operation. That is not the problem.

The problem is funding it. Remaining contingency is **USD 46,000**, so **USD 122,000** must come from elsewhere; and **USD 122,000** exceeds the *entire* **USD 120,000** management reserve by **USD 2,000**. This is therefore not a reserve draw at all but a request for additional funding, decided by the programme board through change control (Domain 4, KA 4.4). Meridian's board sits on a four-week cycle with a two-week paper deadline, so Domain 3's governance latency applies unchanged:  $E[\text{wait}] = M/2 + L = 4/2 + 2 = 4 \text{ weeks}$  (Domain 3, KA 3.2.3).

**Which cost of delay applies, and the error that is easy to make here.** Meridian's registered cost of delay is **USD 14,280 a week**, and using it would price the wait at  $4 \times 14,280 = \text{USD } 57,120$ . That would be wrong by a factor of **2.3333**. The registered figure is the benefit of the *whole* 28-clinic adoption the case assumed; this decision unlocks only the **12** additional clinics, worth **USD 6,120 a week**. **The cost of delay applicable to a decision is the benefit that decision unlocks, not the programme's headline figure**, and applying a portfolio-level rate to a package-level choice inflates every business case built on it. So the four-week wait costs  $4 \times 6,120 = \text{USD } 24,480$ .

**The outcome.** The board approved the package at its next meeting, and the four-week wait cost **14.57 %** of the package's own price, a whole month of benefit from twelve clinics, bought by nobody and noticed by no one. The retrospective made three findings, each a number rather than a sentiment.

*The reserve was pointed entirely at the half of the programme that was succeeding.* All USD 120,000 of contingency was sized against installation risks. The benefit-side exposure (the gap between 40 % and 70 % adoption) is worth **USD 293,760** a year, **2.4480 times** the whole contingency, and carried no reserve at all. Contingency sized only against delivery risk on a benefit-driven programme funds the risks that will not decide the outcome.

*Decision-rights design is a cost parameter.* Had the sponsor pre-authorized a delegated benefit-protection band of USD 150,000 to the programme director, decided in the director's own weekly cycle with no paper lead time ( $E[\text{wait}] = 0.5$  weeks) the latency cost would have been **USD 3,060** instead of USD 24,480, a saving of **USD 21,420** on this single decision. The delegation is free to create and costs nothing when unused; one decision pays for it many times over. Note the sizing constraint: the decision here is the **USD 122,000** of additional funding, so any band below that figure would have accelerated nothing, a delegation set below the size of the decisions it exists to accelerate is decorative, which is Domain 3, KA 3.2.3's threshold-design point measured in money. The USD 150,000 band above clears it with room for the next one.

*The package was late because it was unbudgeted, not because it was unaffordable.* At a 27.4510-week payback it would have been approved at sanction had anyone computed it. What the funding structure lacked was not money but a *line*: enabling change was outside the programme's cost baseline entirely, which is Domain 2, KA 2.3's enabling-change point seen from the cost side.

**What the domain teaches here.** Reserve architecture is not bookkeeping. Which risks a reserve is sized against decides what the programme can respond to; where the release authority sits decides how fast; and both are priced by the same cost of delay that prices everything else. **A reserve that cannot be reached inside the window in which the decision matters is a provision, not a reserve.** The instruments are Domain 3's delegation thresholds and Domain 8's aggregation: this

domain's contribution is insisting that the latency be costed and that contingency be sized against the risks that decide the benefit, not only those that decide the delivery.

## Executive perspective — Domain 7

What a project leader cannot delegate in this domain:

- **The forecast's assumption.** Analysts compute **EAC**; the leader owns *which* method and why, in one sentence a board can challenge. Reporting a number without its assumption is the domain's cardinal sin.
- **The TCPI reality test.** Before endorsing any recovery-to-budget plan: what index does the remaining work require, what is being demonstrated, and what specific mechanism closes the gap?
- **Reserve discipline.** Contingency and management reserve kept distinct, spent under stated authority, with consumption trended against progress: the erosion in MCQ 7.1-B is a leadership signal, not a bookkeeping detail.
- **Measurement integrity.** Earning rules fixed in advance, accruals present, level-of-effort share disclosed, no retrospective budget edits. Cost systems fail morally before they fail arithmetically: the same sentence as Domain 6, and the same reason.
- **The commercial shape.** Which model, why, and where the **PTA** sits. A leader who cannot say who carries cost risk on their largest package is not yet in control of it.
- **The confidence the baseline was approved at.** Not the number, the percentile. Auriga's 4,000,000 was a P33 of its own declared range, and a leader who does not know that figure for their own baseline is managing to a target whose difficulty nobody has stated.
- **The cash position, in days of spend.** USD 1,043,600, 44.80 days, against a cost variance of 200,000. The variance is the subject of the meeting and the cash is the thing that can stop the project, and it is the leader who has to insist both appear in the same pack. Payment terms and retention structures are delivery parameters, priced before signature, not procurement detail.
- **Whether the reserve can be reached in time.** Case study C's finding generalises: a reserve behind a four-week decision cycle cost USD 24,480 to open on a decision worth USD 6,120 a week, and no amount of the reserve being adequate compensates for its being slow. Delegated bands for benefit-protecting spend are free to create, must be set above the size of the decisions they exist to accelerate, and are a leadership decision rather than an administrative one: as is insisting that a cost of delay be the benefit *this* decision unlocks and not the programme's headline rate.

## Calculation exercises — Domain 7

**Exercise 7.1** BAC 4,000,000; PV 2,080,000; EV 1,920,000; AC 2,120,000. Compute CV, SV, CPI, SPI, percent complete and percent spent. *Solution.* CV (200,000); SV (160,000); CPI 0.91; SPI 0.92; complete  $1,920,000/4,000,000 = 48.0\%$ ; spent  $2,120,000/4,000,000 = 53.0\%$ . Common error: computing percent complete from AC/BAC; that is percent *spent*, and reporting it as progress overstates the project.

**Exercise 7.2** Same data. Compute EAC by methods (a), (b) and (c), and the corresponding VAC. *Solution.* (a)  $2,120,000 + 2,080,000 = 4,200,000$ , VAC (200,000). (b)  $4,000,000/0.905660 = 4,416,667$ , VAC (416,667). (c)  $2,120,000 + 2,080,000/(0.905660 \times 0.923077) = 4,608,056$ , VAC (608,056). Common error: rounding CPI to 0.91 before dividing —  $4,000,000/0.91 = 4,395,604$ , USD 21,062 adrift. Indices are display; arithmetic is full precision.

**Exercise 7.3** Same data. Compute TCPI to BAC and to the method-(b) EAC, and interpret. *Solution.* To BAC:  $2,080,000/1,880,000 = 1.11$ . To (b):  $2,080,000/(4,416,667 - 2,120,000) = 2,080,000/2,296,667 = 0.91$ . The second equals the current CPI, the identity of 7.3.4: BAC/CPI forecasts *are* "nothing changes". Common error: using PV rather than AC in the denominator (giving 1.08) and concluding recovery is nearly free.

**Exercise 7.4** Incentive subcontract: target cost 2,400,000; target fee 180,000; share 80/20; ceiling 2,900,000. The seller finishes at 2,650,000. Find the fee, what the buyer pays, and the PTA. *Solution.* Overrun 250,000; fee  $180,000 - 250,000 \times 0.20 = 130,000$ ; buyer pays  $2,650,000 + 130,000 = 2,780,000$  (below the 2,900,000 ceiling). Target price 2,580,000; PTA  $= 2,400,000 + (2,900,000 - 2,580,000)/0.80 = 2,400,000 + 400,000 = 2,800,000$ . Common error: applying the buyer's 80 % share to the fee reduction.

**Exercise 7.5** A pool of 30 at USD 105/h, 20 at USD 150/h and 10 at USD 220/h. Compute the blended rate, then the cost of adding four weeks of a 5-person crew at 40 h/week. *Solution.*  $(30 \times 105 + 20 \times 150 + 10 \times 220)/60 = (3,150 + 3,000 + 2,200)/60 = 8,350/60 = \text{USD } 139.17/\text{h}$ . Crew cost  $5 \times 40 \times 4 \times 139.17 = \text{USD } 111,333$ . Common error: using the blended rate for a crew drawn entirely from one grade. The blend applies to a representative mix, not to any arbitrary subset.

**Exercise 7.6** A point estimate of USD 4,000,000 carries a declared class range of -15 %/+30 %. Treating it as a triangular distribution, find the bounds, the mean,

the median, the P80, and the percentile at which the point estimate sits. Then state the PERT-weighted mean of the same three points. *Solution.* Bounds **3,400,000** and **5,200,000**: a band of 1,800,000, **45.0** % of the point. Mean  $(3,400,000 + 4,000,000 + 5,200,000)/3 = \mathbf{4,200,000}$ . Median  $5,200,000 - \sqrt{(1,800,000 \times 1,200,000/2)} = \mathbf{4,160,770}$ . P80  $5,200,000 - \sqrt{(0.20 \times 1,800,000 \times 1,200,000)} = \mathbf{4,542,733}$ . Percentile at the point =  $(4,000,000 - 3,400,000)/1,800,000 = \mathbf{33.33\%}$ . PERT  $(3,400,000 + 16,000,000 + 5,200,000)/6 = \mathbf{4,100,000}$ . Common error: reporting the point estimate as the expected cost, with an asymmetric range the mean always exceeds the mode, so the approved figure had a **66.67** % chance of being exceeded.

**Exercise 7.7** Control accounts total 3,640,000; contingency 360,000; management reserve 240,000. At 48.00 % complete, 190,000 of contingency has been drawn while CV is (200,000). Compute BAC, the total funding requirement, the draw index, the projected total draw and the draw efficiency the remainder must achieve, then state the reconciliation finding. *Solution.* BAC **4,000,000**; funding requirement **4,240,000**. Draw index  $(190,000/360,000)/0.48 = \mathbf{1.0995}$ . Rate  $190,000/48.00 = \mathbf{3,958.33}$  per point, so projected total draw **395,833** and a shortfall of **35,833** (14.93 % of the management reserve). Required efficiency  $170,000/(3,958.33 \times 52.00) = \mathbf{0.8259}$ . The remainder must draw **17.41** % more slowly. Finding: draws of 190,000 against a CV of 200,000 leave **10,000** of variance with no identified cause. Common error: reading the remaining balance  $(170,000/360,000 = 0.4722)$  as the required efficiency; that is a level, not an index, and it says nothing about the rate.

**Exercise 7.8** EV 1,920,000; invoiced cost 1,920,000; a further 200,000 received and accepted but neither invoiced nor accrued. The next period earns 480,000 for a true cost of 520,000 and the backlog is invoiced in it. Compute reported and true CPI and EAC(b) at the data date, the reported and true period CPI, and the cumulative CPI afterwards. *Solution.* Reported CPI **1.0000**, EAC(b) **4,000,000**; true CPI **0.9057**, EAC(b) **4,416,667**: a forecast understatement of **416,667**, 10.42 % of BAC. Period reported  $480,000/720,000 = \mathbf{0.6667}$  against a true **0.9231**, an error of **0.2564**. Cumulative afterwards  $2,400,000/2,640,000 = \mathbf{0.9091}$  either way. Common error: treating the 0.6667 as a production collapse. It implies costs 50.0 % above budget for the period, and the true movement was an *improvement* from 0.9057 to 0.9231.

**Exercise 7.9** Ten equal packages of 40,000 in one account: four complete, three at 30 % measured physical completion, three not started. PV 240,000; AC 210,000. Compute EV, SPI and CPI under 0/100, 50/50 and objective percent-complete earning. Then compute the reported SPI of a separate account that is 70 % level of effort whose discrete work runs at SPI 0.60. *Solution.* 0/100 EV **160,000** (SPI



0.6667, CPI 0.7619); 50/50 **220,000** (0.9167, **1.0476**); percent complete **196,000** (0.8167, 0.9333), an EV spread of **60,000**, 15.00 % of budget, on one physical state. The level-of-effort account reports  $1 - (1 - 0.70)(1 - 0.60) = \mathbf{0.88}$ , a distortion of **0.28**. Common error: reading the 50/50 CPI of 1.0476 as good news, the rule, not the work, produced it, and the same account read objectively is over-running at 0.9333.

**Exercise 7.10** Auriga's week-13 data. Compute TCPI to BAC and to each of the three EACs, and state what each equals. *Solution.* To BAC  $2,080,000/1,880,000 = \mathbf{1.1064}$  (the recovery demand). To EAC(a)  $2,080,000/2,080,000 = \mathbf{1.0000}$  = the budgeted rate. To EAC(b)  $2,080,000/2,296,667 = \mathbf{0.9057} = \text{CPI}$ . To EAC(c)  $2,080,000/2,488,056 = \mathbf{0.8360} = \text{CPI} \times \text{SPI}$ . **TCPI to any EAC equals the efficiency that EAC assumes**, so choosing a forecast is choosing a TCPI. Common error: believing TCPI measures something independent of the forecast. It does not, and a TCPI that disagrees with the author's stated assumption is an arithmetic error in the report.

**Exercise 7.11** The board will fund no more than the approved 4,000,000 and no efficiency improvement is available. How much budgeted scope must be removed, and what is the exchange rate between scope removed and cash saved? *Solution.*  $D = (\text{BAC} - \text{EV}) - \text{CPI} \times (\text{BAC} - \text{AC}) = 2,080,000 - 0.905660 \times 1,880,000 = \mathbf{377,358}$  (exactly 20,000,000/53): **9.43 % of BAC** and **18.14 %** of the remaining work. Completing the full scope instead needs **416,667** of extra funding, and  $416,667/377,358 = \mathbf{1.104167} = \mathbf{1/\text{CPI}}$ : **one dollar of budgeted scope removed saves USD 1.10 of forecast cash**. Common error: booking the saving without pricing the benefit lost, 9.43 % of budget is not 9.43 % of benefit, and on an integrated system it is frequently much more.

**Exercise 7.12** A crew is 4 technicians  $\times$  40 h at USD 95/h. Price one crew-week of extra capacity by overtime at 1.5 $\times$ , by an agency crew at USD 128/h achieving 85 % relative productivity, and by accepting a week's delay at USD 45,000. Then find the two productivity breakevens. *Solution.* Straight time **15,200**; overtime **22,800** (1.50 $\times$ ); agency **24,094** per productive crew-week (1.5851 $\times$ ); delay **45,000** (2.9605 $\times$ ). The agency crew beats overtime at a relative productivity of  $128/142.50 = \mathbf{89.82 \%}$  or better; overtime beats the agency crew only while sustained-overtime productivity holds above  $22,800/24,094 = \mathbf{94.63 \%}$ . Common error: comparing the agency's USD 128 with the in-house USD 95 and rejecting it; the relevant comparison is with the USD 142.50 overtime rate it displaces, and the 24,094 is still 53.54 % of the cost of doing nothing.

**Exercise 7.13** A firm fixed price of USD 480,000 against time and materials at USD 130.625/h, with hours estimated optimistic 2,900 / most likely 3,200 / pessimistic 4,100. Find the breakeven hours, the premium over the most-likely



estimate, the T&M cost at each of the three points, and the probability (triangular) that the hours exceed the breakeven. *Solution.* Breakeven  $480,000/130.625 = 3,674.64$  h. T&M at 3,200 h **418,000**, so a premium of **62,000 = 14.83 %**: identical to the hours overrun at breakeven. At 2,900 h **378,813**; at 4,100 h **535,563**, which is 55,563 worse than the fixed price. Exceedance  $(4,100 - 3,674.6411)^2 / (1,200 \times 900) = 16.75 \%$ . Common error: expressing the premium against the fixed price (12.92 %) rather than against the estimate. That understates the overrun tolerance the buyer is actually purchasing.

**Exercise 7.14** Contract price 4,400,000 against BAC 4,000,000. At week 13 EV 1,920,000, AC 2,120,000, payables 742,000, cash received 334,400. Compute the margin to date, the funding absorbed, the funding in days of spend, and the margin at completion under each of the three EACs. *Solution.* Revenue  $1.10 \times 1,920,000 = 2,112,000$ , so margin to date **(8,000)**. Cash out  $2,120,000 - 742,000 = 1,378,000$ ; funding absorbed  $1,378,000 - 334,400 = 1,043,600 = 26.09 \%$  of BAC. Daily spend  $2,120,000/91 = 23,297$ , so **44.80 days of spend**. Margin at completion: method (a) **+200,000** (+4.55 % of price); (b) **(16,667)** (-0.38 %); (c) **(208,056)** (-4.73 %). Common error: assuming the bid margin of 400,000 survives. It requires an EAC of exactly BAC, which is TCPI 1.11, so the margin conversation and the recovery conversation are one conversation.

**Exercise 7.15** Target cost 2,000,000; target fee 150,000; ceiling 2,450,000. Compute the PTA at buyer shares of 50 %, 70 % and 90 %, the fee at the PTA in each case, and the zero-fee cost. State which of the PTA and the zero-fee cost binds first at 70/30. *Solution.* PTA =  $2,000,000 + 300,000/s$ : at 50 % **2,600,000**, at 70 % **2,428,571**, at 90 % **2,333,333**. Fee at the PTA =  $150,000 - (PTA - 2,000,000)(1 - s)$ : **(150,000)**, **21,429** and **116,667**. Zero-fee cost =  $2,000,000 + 150,000/(1 - s)$ : 2,300,000, **2,500,000**, 3,500,000. At 70/30 the PTA (2,428,571) arrives **before** the fee is exhausted (2,500,000), so the ceiling binds first and the PTA is the real inflection; at 50/50 it does not, and the PTA there is nominal. Common error: assuming a more generous buyer share pushes the PTA further away, it brings it **closer**, by 95,238 between 70/30 and 90/10, because fixed headroom to the ceiling is consumed faster.

**Exercise 7.16** Auriga's baseline reaches PV 1,920,000 at week 12 and 2,080,000 at week 13, and EV at week 13 is 1,920,000. Compute ES, SPI(t) and the time forecast. Then, at week 27 with EV 3,920,000 against baseline PV of 3,880,000 at week 24 and BAC at week 25, compute SPI, ES, SPI(t) and the time forecast, and explain the divergence. *Solution.* Week 13: ES **12.0000**, SPI(t)  $12/13 = 0.9231$  — equal to SPI, because the PV curve is locally linear and the SV of (160,000) is exactly one week of planned value. Forecast  $25/0.923077 = 27.0833$  weeks, 2.0833 late, which at USD 45,000 per week is **93,750** of delay cost. Week 27: ES =  $24 + 40,000/120,000 = 24.3333$ , SPI(t) **0.9012**, forecast  $25/0.901235 =$

**27.7397 weeks** (2.7397 weeks or **10.96 %** late), while **SPI** reads **0.98**. Common error: quoting **SPI** late in a project: past the planned finish **PV** is pinned at **BAC**, so **SPI** is arithmetically driven to 1.00 and reported 2 % lateness for an 11 % overrun.

## Practitioner's toolkit — Domain 7

*Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.*

### Toolkit 7.T.1 — Cost-report integrity checklist (run before publishing the month)

- ☐ Earning rules fixed in advance per package and unchanged this period; any change disclosed.
- ☐ **EV** claims sampled against physical evidence, not accepted on assertion.
- ☐ Accruals recognised for work received but not invoiced; open commitments cleansed.
- ☐ **PV** read from the current controlled baseline, phased over the approved schedule.
- ☐ No retro-fitted budgets; transfers between control accounts logged.
- ☐ **EV/BAC** reconciles to the reported percent complete; indices recompute from the four numbers.
- ☐ Every **EAC** states its method **and its assumption**; **TCPI** to **BAC** reported beside it.
- ☐ Contingency and management reserve shown separately, with consumption vs progress.
- ☐ Level-of-effort share disclosed per control account.
- ☐ AI-produced forecasts marked, calibration record attached, human owner named.

### Toolkit 7.T.2 — Estimate basis sheet (one per estimate)

Method (analogous/parametric/bottom-up) · definition maturity and accuracy class · range (–x %/+y %) · rate sources with dates and escalation basis · quantities and their source · exclusions stated explicitly · risks feeding contingency (link to the register) · reviewer and date. An estimate whose basis sheet cannot be produced is not releasable.

### Toolkit 7.T.3 — Commercial one-pager (per major package)

Model (FFP/FPIF/CPFF/CPIF/T&M) and why · who carries cost risk · target cost, fee, share ratio, ceiling · **PTA, the zero-fee cost, which binds first, and the current cost trend against the PTA expressed as headroom in per cent** · for T&M, the hours cap, the rate schedule and the breakeven against any fixed-price alternative · payment terms both directions and the cash profile · retention and release criteria ·

variation and claims route · escalation contacts. Reviewed at every commercial checkpoint, not filed at signature.

### Toolkit 7.T.4 — Cash and commitment page (one per period, beside the cost report)

A single page, six lines, none of which appears in a conventional cost report and each of which has changed a decision on the projects in this domain:

- **Cash paid out** =  $AC - \text{payables}$ , with the payables figure and its share of  $AC$ .
- **Cash received**, and the date of the oldest unpaid certified invoice.
- **Funding absorbed** = the difference: reported both in currency and in **days of spend**, so it is comparable across the portfolio.
- **Client exposure** =  $\text{receivables} + \text{retention held} + \text{unbilled work in progress}$ , with the reconciliation check  $\text{funding absorbed} = \text{client exposure} - \text{payables} + \text{loss to date}$ .
- **Retention**: held now, due at completion, held beyond completion; and each as a percentage of the contract margin.
- **Commitment coverage** =  $(AC + \text{open commitments}) \div EAC$ , with the uncommitted forecast balance, the weekly rate at which it must now be committed, and the exposure to a stated price movement. Stale open commitments cleansed this period are stated explicitly, since cleansing them worsens the page and improves its truth.

### Exam preparation — Domain 7

**The calculation traps.** Percent spent reported as percent complete (Exercise 7.1) · rounding  $CPI$  before dividing (Exercise 7.2) ·  $PV$  instead of  $AC$  in the  $TCPI$  denominator (Exercise 7.3) · applying the wrong party's share to an incentive fee (Exercise 7.4) ·  $EV$  valued at actual cost rather than budget · reading a level-of-effort-heavy  $SPI$  as schedule performance (MCQ 7.3-D) · quoting  $SPI$  late in a project where it must converge on 1.00 (7.A.1) · confusing commitments with actuals · treating contingency and management reserve as one pot · reporting a point estimate as though it were the expected cost of an asymmetric range (Exercise 7.6) · reading a remaining reserve *balance* as the reserve's required *efficiency* (Exercise 7.7) · taking a single-period index at face value in the month an accrual backlog clears (Exercise 7.8) · using 50/50 earning on long packages and reading the resulting favourable  $CPI$  as performance (Exercise 7.9) · assuming  $TCPI$  says something independent of the  $EAC$  it is computed to (Exercise 7.10) · booking a descope saving without pricing the benefit removed (Exercise 7.11) · comparing an agency hourly rate with straight time rather than with the overtime rate it displaces (Exercise 7.12) · expressing a fixed-price premium against the price rather than the estimate (Exercise 7.13) · assuming a bid margin survives a forecast overrun (Exercise 7.14) · assuming a more generous

share ratio pushes the **PTA** further away (Exercise 7.15) · adding a non-productive time allowance as an uplift rather than dividing by utilisation (MCQ 7.4-E).

**The identities worth memorising.**  $\% \text{spent} - \% \text{complete} = \text{CV}/\text{BAC} \cdot 1/\text{CPI} - 1$  is the rate at which the work performed exceeded its budget · **TCPI** to any **EAC** equals that **EAC**'s implied efficiency ( $1.0000 / \text{CPI} / \text{CPI} \times \text{SPI}$ ) · a dollar of budgeted scope removed saves  $1/\text{CPI}$  dollars of forecast cash · reported  $\text{SPI} = 1 - (1 - s)(1 - \text{SPI}_d)$ , so level-of-effort distortion is  $s \times (1 - \text{SPI}_d)$  · paid-hour to productive-hour cost is  $1/(1 - n)$ , and cost per productive hour scales as  $1/\text{utilisation}$  · a fixed-price premium in per cent equals the hours overrun at which T&M breaks even · funding absorbed = client exposure – payables + loss to date ·  $\text{E}[\text{wait}] = M/2 + L$  prices the latency of the authority that releases a reserve.

### Reflection questions.

1. Your cost report shows one **EAC**. What must accompany it before you will sign it? (*The method and the assumption about remaining work; TCPI to BAC beside it.*)
2. On your largest package: who carries cost risk, where is the **PTA**, and how close is the supplier's cost trend to it? (7.4.2–7.4.3; toolkit 7.T.3.)
3. Which invariant in 7.A.3 would have caught the last cost surprise you experienced, and why wasn't it running?
4. What percentile of its own declared range was your current baseline approved at, and who in the governance chain knows that number? (7.1.1.)
5. How many days of spend is your project absorbing, and what would 30-day rather than 60-day client terms be worth? (7.4.4; toolkit 7.T.4.)
6. If your board refused any further funding tomorrow, how much budgeted scope would have to come out, and which packages, priced against benefit? (7.3.4b.)
7. Where does the authority to release your largest reserve sit, what is  $\text{E}[\text{wait}]$  for it, and what does that wait cost per week? (*Case study C.*)

## Domain 7 summary

Cost leadership begins with an honest number, and honesty here is arithmetic rather than character: a method matched to definition maturity, a range and class always stated; and *read*, since Auriga's approved USD 4,000,000 sat at the 33.33rd percentile of its own declared range, whose mean of 4,200,000 is the very forecast week 13 produced. Three-point thinking where tails are real, with a  $4.125 \sigma$  tail recognised as a scenario for Domain 8 rather than a spread; a budget built through control accounts into a time-phased baseline with contingency inside it (USD 360,000) and management reserve outside (USD 240,000) for a funding requirement of USD 4,240,000; and the contingency trend read as an index of its own, where a draw of 52.78 % at 48.00 % complete demands that the remainder run 17.41 % more frugally than the first half.

Measurement then has to be earned (*AC* with accruals, *EV* at budget under earning rules fixed in advance), and both halves of that sentence carry a number. One unrecognised USD 200,000 accrual makes a project reporting *CPI* 0.91 report 1.00 and understates its forecast by USD 416,667, then punishes the innocent period in which the backlog clears. One choice of earning rule moves *EV* by 15.00 % of a control account's budget and flips it from over-running to under-running. Everything downstream is arithmetic on numbers that fragile: *CV*, *SV*, *CPI*, *SPI* at Auriga's week 13 (0.91 and 0.92; 48 % complete against 53 % spent, whose 5-point gap is the USD 200,000); the *EAC* family spanning USD 408,056 (a fifth of the remaining work), because each encodes a different implied efficiency, and funding the cheapest of them buys the project only to 95.09 % complete; and *VAC* and *TCPI* turning recovery talk into a defensible claim, since *TCPI* to any *EAC* is exactly the efficiency that forecast assumes, and recovery to budget here means either 22.16 % more productivity or USD 377,358 of scope out, at an exchange rate of  $1/CPI$ , USD 1.10 of cash saved per dollar of budget removed.

The commercial half is the same discipline pointed outward. Blended rates make options commensurable and the mix moves the price more than the headcount does; behind the rate sits a cost per *productive* hour of USD 100.79 where five points of utilisation are worth USD 155,000: three-quarters of the variance being debated; Domain 6's capacity ladder prices out at 15,200 straight, 22,800 overtime, 24,094 agency and 45,000 to do nothing, with the arguments living at breakevens of 89.82 % and 94.63 %. Contract models allocate cost risk at a price whose crossing point is computable (a 14.83 % fixed-price premium is exactly a 14.83 % tolerance on hours), and the point of total assumption predicts when a supplier's incentives invert, moving *closer* as the buyer concedes share and becoming nominal once the ceiling passes the zero-fee cost. Beneath all of it runs cash: a project whose reported problem is USD 200,000 is absorbing USD 1,043,600, 44.80 days of spend, half of which one payment term would return, holding retention worth 55.00 % of its margin, and whose profitability is decided by which *EAC* is believed. At programme scale Meridian adds the last lesson: a reserve sized entirely against delivery risk while the exposure sat on the benefit side, behind an authority whose four-week wait cost USD 24,480 — 14.57 % of the decision it was blocking, and a reminder that the cost of delay applicable to a decision is the benefit that decision unlocks, not the programme's headline rate.

Throughout, the leadership rule holds: the number is only as good as the assumption named beside it, and machine-produced forecasts reach a board only with a calibration record and a human owner. Domain 8 quantifies the risk this domain reserves for; Domain 9 costs the rework this domain's engineer-week prices; Domain 10 takes the commercial relationships into their own depth.

DOMAIN

# 8

## Risk, Uncertainty and Resilience

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## Why this domain exists

Domains 6 and 7 built plans and budgets; both left an obligation outstanding. Domain 6 forecast with ranges but did not say where the ranges come from. Domain 7 put a contingency reserve inside the baseline and said it is "sized against identified risks" without showing how. This domain closes both gaps and adds the part neither covers: what a leader does when the unexpected happens anyway. It builds risk identification honestly, including opportunities (KA 8.1); works the analysis from qualitative screening through EMV, decision trees and aggregated contingency (KA 8.2); sizes and governs reserves (KA 8.3); and turns to resilience, crisis leadership, cognitive bias and AI-enabled risk sensing (KA 8.4). The discipline throughout is that **risk management is a decision-support activity, not a documentation activity**: a register that changes no decision has cost money and bought nothing.

Beneath that lies the domain's central quantitative claim, and it is the one most often missed: **aggregate risk is not the sum of its parts, and the term that decides the difference is variance**. Every number a leader is actually asked for (a contingency, a confidence level, a buffer, a date the board can commit to) is a statement about the *spread* of an aggregate distribution, not about its average. Expected monetary value, which is where most training stops, fixes only the average; it is silent on the shape, and the shape is what breaks projects. Three consequences run through the rest of this domain. Two registers with identical total EMV can need reserves that differ by a third, because one concentrates its exposure in a few large, unlikely events and the other spreads it. Two responses of identical cost that remove identical EMV can differ by a third in the reserve they release, because one attacks a probability and the other an impact. And a reserve labelled P80 can be a P73 reserve the moment a shared driver is admitted: the label failing before the money does. A leader who can compute those three things is doing risk management; a leader who can only multiply probability by impact is doing arithmetic.

**Learning objectives.** After this domain a candidate can: distinguish risk from uncertainty and issue; write a risk statement that supports a decision; identify opportunities as rigorously as threats; **estimate how complete a register is from the overlap between two independent identification methods, and use the answer to size resilience rather than to reassure**; run qualitative screening, **compute the span of expected values a single matrix cell can conceal and derive the band ratio a defensible scale needs**; compute EMV for individual risks and interpret the total, including the sensitivity of its ranking to the probabilities assumed; build and solve a decision tree, including the value of information; **value imperfect information, show that the value depends on how the sample is designed as much as on what it costs, and compute the prior probability below which buying it destroys value**; aggregate risks to a mean and a confidence level rather than summing point estimates, **and check the normal approximation against the exact enumerated distribution**; **quantify what one shared**



driver does to an aggregate, and state the confidence an independence-based reserve actually buys; rank responses by the confidence-level reduction they buy rather than by EMV reduction, and explain from the algebra why an impact lever removes between 1.5 and 3 times the variance of a probability lever at equal EMV cost; convert a risk-appetite statement into an operable tolerance, a confidence level and an escalation threshold; size a contingency reserve defensibly, distinguish it from management reserve, and test a part-consumed reserve against the register still open; select and cost responses across the response families with their secondary risks priced, and show where omitting the secondary risks selects the wrong response; explain resilience as distinct from prediction and price a resilience measure on both its expected value and its tail; recognise the biases that corrupt estimates and reviews, and reconcile a bottom-up register against a reference class of comparable out-turns; lead in a crisis: specifying who declares it, what authorities the declaration changes and who stands it down, building the notification map before it is needed, keeping a contemporaneous decision log from the declaration, and pricing the drift between an early notification and a late one; mark the register rows where a duty is engaged and take them out of the expected-value comparison before it is run; compute merge bias at a convergence point, correct it for shared predecessors, and state the date range over which it stops mattering; and govern AI-produced risk analysis, including deriving the base rate at which a sensitive monitor beats a specific one.

**The master worked project (and its programme.** Project Auriga continues from Domains 6 and 7) the 25-week, BAC USD 4,000,000 control-systems upgrade. Its risk register, used throughout KA 8.2–8.3, carries the ground-conditions risk that actually materialised in Domain 6's case study, so the reader sees the same event before and after the fact. Its five-risk register has a total EMV of USD 278,000, an independence-based P80 of USD 490,624, and, once one shared subcontractor is admitted, a  $\sigma$  of USD 340,339 that turns that same reserve into a 73.4 % reserve.

Single-project arithmetic is not, however, where most contingency is lost, so this domain also works **Meridian Care Records** at programme scale, the fictional public-health programme of Domains 1, 2, 5, 15 and 16: 40 clinics, approved cost USD 2,400,000, a cost of delay of USD 14,280 per week, and a board risk appetite that this domain converts into a number. Its five quantified risks total USD 119,000 of EMV against a 5 % tolerance of USD 120,000, which is the whole problem in one comparison, and the reason KA 8.3 exists.

**Reference points.** The international document most often named in connection with this subject is ISO 31000, which addresses risk management principles and guidance on managing risk. It is voluntary guidance describing practice (not legislation, and not a certifiable requirement), and it obliges nobody of itself unless an organisation, a contract or a regulator adopts it. It is named here and not reproduced: no clause, table, matrix or process diagram from it appears in this volume, every method below is derived rather than cited, and a reader who wants



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## KNOWLEDGE AREA 8.1

# Threats, opportunities and identification

*Topics: 8.1.1 risk, uncertainty and issues · 8.1.2 writing a usable risk statement · 8.1.3 identification methods and their blind spots.*

## 8.1.1 Risk, uncertainty and issues

**Definitions.** A **risk** is an uncertain event that, if it occurs, affects an objective — threats reduce achievement, **opportunities** improve it. **Uncertainty** in the broader sense covers what cannot be enumerated with probabilities at all: the conditions where scenario thinking and resilience substitute for calculation. An **issue** is a risk that has occurred; it is managed, not analysed. Keeping these apart matters practically: registers clogged with issues stop being forward-looking, and treating deep uncertainty as if it were quantifiable produces false precision (KA 8.4.1).

**Risk appetite and thresholds** convert judgment into something operable: how much exposure the organisation accepts, expressed as thresholds that trigger escalation or response, connecting this domain to Domain 3's decision rights. Appetite that is never expressed numerically cannot be applied consistently by different people, which is the usual reason two projects in one portfolio treat the same exposure differently. The conversion is arithmetic and it has a result worth anticipating: a well-formed appetite statement **determines the confidence level at which contingency is held**, so P80 is not a convention to inherit but a consequence to derive. Worked example 8.3.2 does that conversion on Meridian, because the reserve is where the number lands.

## 8.1.2 Writing a usable risk statement

Most registers fail at the sentence level. A usable statement has three parts (**cause** → **event** → **consequence**), and names the affected objective:

*Because the controller is single-sourced with a volatile lead time (**cause**), it may be that delivery slips beyond the installation window (**event**), resulting in a delay to commissioning and additional preservation cost (**consequence**).*

Compare "supplier risk", which supports no decision: it identifies no cause to attack, no event to monitor and no consequence to size. The three-part form is not a formatting preference, each part maps to a different response type (attack the cause,

monitor the event, mitigate the consequence), which is why KA 8.3's response selection depends on it.

### 8.1.3 Identification methods and their blind spots

Methods: structured workshops, checklists and prompt lists from prior projects, assumption analysis (every assumption is a risk in disguise — Domain 2's assumption register), interviews for what people will not say in a group, and lessons from comparable work (Domain 9's lessons-learned). Each has a blind spot, and knowing them is the professional part:

- **Workshops** surface what the group already shares and suppress what a junior member suspects, so run them with an explicit invitation to dissent and a route for private input.
- **Checklists** find known risks and, by their comfort, discourage looking for novel ones.
- **Assumption analysis** is the highest-yield method and the least used, because it requires admitting how much of the plan is assumed.
- **Historical data** encodes the past's risks, not this project's novelties.

**Opportunities are systematically under-identified**, most registers are 90 % threats, because the process is framed defensively and because nobody is rewarded for one. A leader who wants them must ask for them separately and size them the same way (Auriga's R5 below is one).

**How complete is the register?** Every method above leaves something out, and the resilience argument of KA 8.4.1 rests on the claim that the register is *always* incomplete. That claim can be measured rather than asserted, and the measurement costs nothing beyond running two identification methods separately and comparing their lists. If method 1 finds  $n_1$  risks, method 2 finds  $n_2$ , and  $m$  risks appear on both, then (treating the two methods as independent samples from the same underlying population of risks) the population is estimated by the standard capture–recapture estimator:

$$\begin{aligned} \hat{N} &= (n_1 \times n_2) / m && \text{(Lincoln-Petersen)} \\ \hat{N}_C &= ((n_1 + 1)(n_2 + 1)) / (m + 1) - 1 && \text{(Chapman, less biased at small } m) \end{aligned}$$

The logic is the same as estimating the size of a fish population by tagging and re-sampling: the smaller the overlap between two independent samples, the larger the population they were drawn from. Applied to a risk register it converts an uncomfortable feeling into a number a leader can act on.

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#### WORKED EXAMPLE

**Worked example 8.1.3 – how much of Meridian's register is missing?**

1. **Setup.** Meridian's identification ran two methods separately and deliberately. A structured workshop with clinical, technical and operational representatives produced **34** distinct risks. Independently, and without seeing the workshop output, an assumption-analysis review of the programme plan (against Domain 2's assumption register) produced **22**. Comparing the lists, **14** risks appear on both. Each register entry is written in the three-part form of 8.1.2, so "the same risk" means the same cause-and-event pair, not similar wording. The assumption review cost **USD 12,000** of reviewer time.
2. **Formula.**  $\hat{N} = n_1 n_2 / m$ ; distinct identified =  $n_1 + n_2 - m$ ; estimated unidentified =  $\hat{N} - (n_1 + n_2 - m)$ ; coverage = distinct  $\div \hat{N}$ . Cross-check with Chapman's  $\hat{N}_C = ((n_1 + 1)(n_2 + 1)) / (m + 1) - 1$ .
3. **Substitution.**  $\hat{N} = (34 \times 22) / 14 = 748 / 14$ ; distinct =  $34 + 22 - 14$ ;  $\hat{N}_C = (35 \times 23) / 15 - 1 = 805 / 15 - 1$ .
4. **Result.** Estimated population **53.4286** risks (Chapman: **52.6667**). Distinct identified **42**. Estimated unidentified **11.4286** risks, **21.39 %** of the estimated population, so coverage is **78.61 %** (Chapman: 10.6667 missing, **79.75 %** coverage). The workshop *on its own* covered **63.64 %** of the estimated population. The assumption review added **8** risks the workshop had missed, at **USD 1,500** each.
5. **Interpretation.** Four things follow, and the fourth is the one that changes a decision.

**The workshop is not "the risk identification".** It found under two-thirds of the estimated population, and it is routinely the only method run. The eight risks the assumption review added are, by construction, exactly the ones the workshop's shared frame of reference could not see: the failure mode named at the top of this topic, now with a size attached. At USD 1,500 per new risk the review pays for itself if the average newly found risk carries more than USD 1,500 of avoidable exposure, which on a USD 2,400,000 programme is close to a certainty and is why the second method is not optional.

**The bias runs one way, so the estimate is a floor.** Capture–recapture assumes the two methods are independent and that every risk is equally detectable. Neither holds exactly. Both methods were run by people inside the same programme, so they share some blind spots; shared blind spots raise  $m$ , and a higher  $m$  lowers  $\hat{N}$ . Unequal detectability (obvious risks found by both methods, subtle ones by neither) pushes the same way. So **11.43 is a lower bound on what is missing**, and a leader should present it as "at least eleven risks we have not written down", never as a precise count. The Chapman cross-check moving the estimate by less than one risk is evidence that the arithmetic is stable, not that the assumptions are true.

**The estimate does not license adding EMV for imaginary risks.** The unidentified 11.43 have no cause, no event, no consequence and no owner; they cannot be quantified, responded to or escalated. What they justify is **management**

**reserve** (8.3.2) and **resilience** (8.4.1) — money and capacity held against exposure that has no line item. That is precisely the distinction between the two reserve types, now with a quantitative basis rather than a shrug, and it is why an organisation that folds management reserve into contingency has no way to fund what this arithmetic finds.

**And the number is a decision rule, not a report line.** Meridian's programme office set a coverage threshold: below 85 %, commission a third independent method before the tranche gate. At 78.61 % the threshold fired, and a supplier-side review was run. The professional caution is the mirror image: **coverage above the threshold is not evidence that identification is finished**. It is evidence that two methods agree, which on a novel programme is as consistent with a shared blind spot as with completeness. Deep uncertainty (8.1.1) is invisible to this arithmetic by definition, because it cannot be enumerated at all.

## AI in this KA

Generating candidate risk lists is a real AI strength: broad, fast, and free of the group's shared blind spot, genuinely useful against the workshop failure mode above. It is equally a machine for **plausible generic risk**, producing items that fit any project and support no decision. The governed workflow: use AI to *widen* identification (especially by asking it to challenge the plan's assumptions), then require every surviving item to pass the three-part statement test of 8.1.2 with a named owner. **AI proposes; the professional verifies, decides and remains accountable**, and here the verification is specifically that the risk is *this* project's, not a template's.

## KEY TERMS — KA 8.1

| TERM                               | MEANING                                                                                                                                                    |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Risk / opportunity</b>          | Uncertain event affecting an objective, adversely / favourably.                                                                                            |
| <b>Uncertainty</b>                 | Conditions that cannot be meaningfully enumerated with probabilities.                                                                                      |
| <b>Issue</b>                       | A risk that has occurred; managed, not analysed.                                                                                                           |
| <b>Risk appetite / threshold</b>   | Accepted exposure, expressed operably enough to apply consistently.                                                                                        |
| <b>Cause → event → consequence</b> | The three-part risk statement; each part maps to a response type.                                                                                          |
| <b>Capture–recapture estimate</b>  | Register completeness inferred from the overlap of two independent identification methods: $\hat{N} = n_1 n_2 / m$ .                                       |
| <b>Coverage</b>                    | Distinct risks identified as a share of the estimated population; an optimistic figure, since shared blind spots deflate the population and so inflate it. |

## SAMPLE MCQS — KA 8.1

**MCQ 8.1-A** [8.1.2 · Analysis] Which register entry best supports a decision?

- A. "Supplier risk — high"
- B. "Because the controller is single-sourced with volatile lead times, delivery may slip beyond the installation window, delaying commissioning and adding preservation cost" CORRECT
- C. "Delay to commissioning — probability 35 %"
- D. "Supplier may cause problems — owner: procurement"

*Rationale:* Only B names a cause to attack, an event to monitor and a consequence to size (8.1.2). C states a consequence with a probability but no cause, so no response can be designed; A and D support nothing at all.

**MCQ 8.1-B** [8.1.1 · Recall] The ground contamination has been discovered and remediation is under way. In the register this is:

- A. a risk with probability 1.0
- B. an issue — it has occurred, and is managed rather than analysed CORRECT
- C. an opportunity, since remediation was funded
- D. removed entirely, with no further record

*Rationale:* Occurred risks become issues (8.1.1). A is the common fudge that clogs registers; D loses the audit trail and the lesson; C is nonsense.

**MCQ 8.1-C** [8.1.3 · Analysis] A register of 60 items contains 3 opportunities. The soundest inference is:

- A. this project genuinely has few opportunities
- B. the identification process is framed defensively; opportunities must be asked for separately and sized the same way CORRECT
- C. opportunities do not belong in a risk register
- D. the register is too long and should be cut

*Rationale:* A 95 % threat ratio reflects process framing rather than reality (8.1.3). C contradicts the definition of risk; D addresses a different problem.

**MCQ 8.1-D** [8.1.3 · Application] A workshop finds 34 risks and an independent assumption review finds 22, with 14 common to both. The estimated risk population and the estimated number still unidentified are:

- A. 56 and 14
- B. 53.43 and 11.43 CORRECT
- C. 42 and nil — the two methods between them found everything
- D. 78.61 and 36.61

*Rationale:*  $\hat{N} = (34 \times 22)/14 = 53.4286$ , and 42 distinct entries were identified, leaving **11.4286** (8.1.3). A adds the two lists without removing the 14 duplicates and then treats the overlap as the missing count; C assumes two methods exhaust the population, which is the assumption the estimator exists to test; D takes the coverage percentage (78.61) as a population count and subtracts the 42 from it.

**MCQ 8.1-E** [8.1.3 · Analysis] Two identification methods overlap heavily, giving a high coverage estimate. The soundest professional reading is:

- A. identification is complete and can be closed
- B. a large overlap deflates the population estimate, so high coverage is as consistent with a shared blind spot as with completeness CORRECT
- C. one of the two methods was run incorrectly
- D. the estimate is invalid whenever the overlap exceeds half of either list

*Rationale:*  $\hat{N} = n_1 n_2 / m$  falls as  $m$  rises, so methods sharing a frame of reference produce reassuring coverage; the estimate is a floor on what is missing (8.1.3). A is the error the arithmetic is meant to prevent; C infers a process fault from a statistic that has other explanations; D invents a threshold the estimator does not have.

#### SELF-CHECK — KA 8.1

1. *State the three parts of a usable risk statement.* — Cause, event, consequence, against a named objective.
2. *Why is assumption analysis the highest-yield identification method?* — Every assumption is a risk in disguise, and plans rest on more of them than teams admit.
3. *What is the register consequence of treating an occurred risk as a 100 % risk?* — It clogs the forward-looking register; occurred risks are issues.
4. *What does a capture–recapture coverage estimate justify, and what does it not?* — It justifies management reserve and resilience sized against unidentified exposure; it does not justify adding EMV for risks that have no cause, event, consequence or owner.
5. *Which way does the bias run in a register-completeness estimate?* — Shared blind spots raise the overlap and lower the population estimate, so the count of missing risks is a floor.

## KNOWLEDGE AREA 8.2

## Analysis: from screening to quantification

*Topics: 8.2.1 qualitative screening and its limits · 8.2.2 expected monetary value · 8.2.3 decision trees and the value of information · 8.2.4 aggregating to a contingency.*

### 8.2.1 Qualitative screening and its limits

Probability-and-impact scoring on a matrix is a **screening** tool: cheap, fast, and adequate for ranking a long list into an attention order. Its limits are structural and must be stated wherever it is used. Ordinal scales cannot be arithmetic, a "4" is not twice a "2", so scores must not be multiplied and summed as if they were money. Band boundaries create false distinctions (a risk at the top of "medium" outranks one at the bottom of "high" in reality but not on the matrix). And the matrix is blind to **correlation**: five risks driven by one supplier are one risk wearing five costumes.

The professional practice: screen qualitatively, then **quantify what matters**; the items that could breach a threshold, plus anything a decision now depends on.

How coarse a screen is it? That is answerable, and the answer is the argument for the quantification threshold rather than a matter of taste. A matrix cell is a rectangle in probability-by-impact space, so the range of expected values it contains follows directly from the widths of its two bands.

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#### WORKED EXAMPLE

#### Worked example 8.2.1 — what one matrix cell conceals.

1. **Setup.** A delivery organisation screens on a  $3 \times 3$  matrix. Probability bands: **Low** 0.01–0.10, **Medium** 0.10–0.35, **High** 0.35–0.70. Cost-impact bands: **Low** USD 10,000–60,000, **Medium** USD 60,000–250,000, **High** USD 250,000–800,000. Two risks are both scored **Medium probability** / **Medium impact** and therefore receive the same attention, the same review cadence and the same response budget. How different can they actually be?
2. **Formula.** Within one cell, EMV ranges from  $p_{\min} \times I_{\min}$  to  $p_{\max} \times I_{\max}$ , so the span factor is  $(p_{\max}/p_{\min}) \times (I_{\max}/I_{\min})$ , the product of the two band ratios. Compare that against the step between adjacent cells, taken at band midpoints.
3. **Substitution.** Cell floor  $0.10 \times 60,000$ ; cell ceiling  $0.35 \times 250,000$ . Span factor  $(0.35/0.10) \times (250,000/60,000) = 3.5 \times 4.1667$ . Midpoint of Medium/Medium  $0.225 \times 155,000$ ; of High/High  $0.525 \times 525,000$ .

4. **Result.** The Medium/Medium cell contains expected values from **USD 6,000** to **USD 87,500**: a span factor of **14.5833**, exactly  $3.5 \times 4.1667$ . The step from the Medium/Medium midpoint (**USD 34,875**) to the High/High midpoint (**USD 275,625**) is a factor of **7.9032**. So the range *inside* one cell is **1.8452 times** the whole step from that cell to the one a full band higher on **both** axes.
5. **Interpretation.** The identity is the useful part, and it converts a familiar complaint into a design specification.

**Within-cell span = probability band ratio × impact band ratio.** Nothing about the shading, the wording of the bands or the number of colours changes it; only the ratios do. A matrix whose bands span factors of 3.5 and 4.17 cannot resolve two risks that differ by a factor of fourteen, which means **its internal resolution is coarser than its own band structure implies**. Two risks in the same cell here can differ by more than the distance the matrix draws between "medium" and "high". That is why ordinal scores must not be multiplied and summed (the point above), and it is also why "we manage all amber items the same way" is a funding error rather than a simplification: at the extremes of this cell, a response budget set by cell allocates the same money to an exposure of 6,000 and one of 87,500.

**The design rule falls straight out.** To hold the within-cell span below a factor of 4, each axis needs a band ratio below 2. Covering probabilities from 0.01 to 0.70 (a range of **70**) in bands of ratio 2 requires 7 bands ( $\log_2 70 = 6.1293$ ); covering impacts from 10,000 to 800,000, a range of **80**, requires 7 as well ( $\log_2 80 = 6.3219$ ). A  $7 \times 7$  matrix on ratio-constant bands is a defensible screen; a  $3 \times 3$  matrix on the same total range is not, and no amount of careful colouring repairs it. Note what this also says about **linear** bands, which are commoner and worse: equal-width bands have enormous ratios at the bottom of the scale (a band from 0.01 to 0.10 is a ratio of 10) and negligible ones at the top, so a linear matrix is least reliable exactly where low-probability high-impact risks live: the class 8.3.1 says must be handled structurally.

**What screening is still for.** The conclusion is not that matrices are useless; it is that they are a triage instrument whose output is an *attention order*, and that they must be paired with a stated quantification threshold. Meridian's threshold was set at the Medium/Medium floor: any risk whose assessed impact reaches USD 60,000 is quantified individually, **plus every identified opportunity whatever its size**, because the under-identification of 8.1.3 means an impact threshold applied symmetrically would remove the few opportunities a register contains. That took its 42-item register down to the five entries 8.2.2b aggregates, four threats above the floor and one opportunity of USD 40,000 below it. The reviewer's test is simply whether that threshold is written down: a screen with no threshold is a screen that has replaced quantification instead of ordering it. One caution. Recalibrating the bands to reduce span makes the matrix harder to complete quickly, which is the property that made it useful. So so the trade-off should be made deliberately at the organisational level, not per



project. (Nothing above touches the correlation blindness named at the top of this topic; band width cannot repair it.)

### 8.2.2 Expected monetary value

EMV = probability × impact (per risk; threats positive, opportunities negative cost)

#### WORKED EXAMPLE

#### Worked example 8.2.2 — Auriga's risk register, quantified.

1. **Setup.** Four threats and one opportunity, each with an assessed probability and cost impact (BAC USD 4,000,000).
2. **Formula.**  $EMV = p \times \text{impact}$ ; total exposure =  $\Sigma EMV$ , opportunities carried negative.
3. **Substitution and result.**

| ID                             | Risk                                         | P    | IMPACT<br>(USD) | EMV<br>(USD)    |
|--------------------------------|----------------------------------------------|------|-----------------|-----------------|
| R1                             | Controller lead-time slip                    | 0.35 | 240,000         | 84,000          |
| R2                             | Ground conditions worse than surveyed        | 0.50 | 180,000         | 90,000          |
| R3                             | Integration rework                           | 0.25 | 320,000         | 80,000          |
| R4                             | Permit delay                                 | 0.15 | 400,000         | 60,000          |
| R5                             | Early-delivery rebate ( <i>opportunity</i> ) | 0.30 | (120,000)       | <b>(36,000)</b> |
| <b>Total expected exposure</b> |                                              |      |                 | <b>278,000</b>  |

4. **Interpretation.** Total EMV is USD 278,000: 6.95 % of BAC, and USD 314,000 if the opportunity is ignored, an overstatement of 12.95 %, which is why opportunities must be in the same arithmetic rather than mentioned in prose. Five things the table teaches, in the order a reviewer should test them.

EMV is an average of outcomes that will not happen. No single risk will cost 84,000, R1 costs 240,000 or nothing. So EMV is the right basis for *funding a portfolio* of risks and the wrong basis for deciding whether one specific risk is survivable. The strongest form of the point is arithmetic: the probability that all four threats occur together, producing the 1,140,000 worst case, is  $0.35 \times 0.50 \times 0.25 \times 0.15 = 0.65625$  %, while the probability that the register's net cost is **zero or better** (no threat occurs, or only the opportunity does) is **20.72** % (enumerated in 8.2.4). One future in five costs nothing; the register's central estimate describes neither of those futures.

**Ranking by EMV reorders the register.** R<sub>4</sub> has the largest impact (400,000) but the smallest threat EMV (60,000), while R<sub>3</sub>'s lower impact carries more expected cost. A leader managing by impact alone would attend to them in the wrong order, though see 8.3.1, because impact still governs *survivability*.

**That ranking is fragile, and the fragility is computable.** R<sub>4</sub>'s EMV overtakes R<sub>1</sub>'s as soon as its probability reaches  $84,000/400,000 = 0.21$ , and tops the whole register at  $90,000/400,000 = 0.225$ . So a six-point revision in one probability (well inside the precision of any real assessment) reverses the top of the order. Likewise a plausible  $\pm 0.05$  error on R<sub>2</sub>'s probability moves its EMV by  $\pm$ USD 9,000 and the total by  $\pm 3.24\%$ . The professional consequence is a presentation rule: quote total exposure to the nearest thousand at most, and where a response decision turns on a rank, state the probability at which the rank flips rather than defending the rank itself.

**The total is a mean, and means are additive whatever else is true.**  $\Sigma$ EMV is the mean of the aggregate distribution under *any* dependence structure between the risks: correlation moves the spread and not the average (8.2.4b proves it on this register, and the AI note at the end of this KA turns it into a hand-check).

**And the sign convention has to be enforced, not assumed.** R<sub>5</sub>'s EMV is carried as (36,000), a reduction in expected cost, because it is an opportunity against the same objective (cost) as the four threats. Mixing objectives inside one total is the commoner and more damaging version of the same error, and 8.2.2b takes it up.

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#### WORKED EXAMPLE

**Worked example 8.2.2b — Meridian's programme register, and the objective a total belongs to.**

1. **Setup.** Meridian's 42-item register (8.1.3) has five entries above the quantification threshold of 8.2.1. All five are assessed against a **single objective (cost against the approved USD 2,400,000)**, and the programme separately holds benefit risks, schedule risks and reputational risks in their own aggregations.

| ID | RISK (CAUSE → EVENT →<br>CONSEQUENCE, ABBREVIATED)                             | P    | IMPACT<br>(USD) | EMV<br>(USD) |
|----|--------------------------------------------------------------------------------|------|-----------------|--------------|
| M1 | Legacy extract quality → second data run<br>needed at clinics                  | 0.40 | 90,000          | 36,000       |
| M2 | Clinician availability → extended parallel<br>running at the 8 largest clinics | 0.35 | 120,000         | 42,000       |
| M3 | National records-format revision →<br>interface rebuild                        | 0.20 | 150,000         | 30,000       |

| ID                                  | RISK (CAUSE → EVENT →<br>CONSEQUENCE, ABBREVIATED)                           | P    | IMPACT<br>(USD) | EMV<br>(USD)    |
|-------------------------------------|------------------------------------------------------------------------------|------|-----------------|-----------------|
| M4                                  | Trainer attrition → contracted backfill for the training programme           | 0.30 | 70,000          | 21,000          |
| M5                                  | Shared-licence rebate on completing the rollout early ( <i>opportunity</i> ) | 0.25 | (40,000)        | <b>(10,000)</b> |
| <b>Total expected cost exposure</b> |                                                                              |      |                 | <b>119,000</b>  |

2. **Formula.**  $EMV = p \times impact$ , summed within one objective. Express the total against the objective's tolerance, not against the whole approved cost, because the tolerance is what governance actually controls (8.3.2).
3. **Substitution.**  $0.40 \times 90,000, 0.35 \times 120,000, 0.20 \times 150,000, 0.30 \times 70,000, 0.25 \times (40,000)$ . Tolerance =  $5\% \times 2,400,000$ .
4. **Result.** Total expected cost exposure **USD 119,000**, which is **4.96 %** of the approved USD 2,400,000; and **99.17 %** of the board's 5 % tolerance of **USD 120,000**.
5. **Interpretation.** The comparison in step 4 is the whole finding, and it is invisible if the total is expressed as a percentage of approved cost.

**The mean alone consumes 99.17 % of the tolerance.** Because a mean is roughly the 50th percentile of an aggregate, that means Meridian has approximately a **coin-flip** chance of breaching its stated tolerance before any confidence level is discussed. No contingency figure can repair this, because contingency covers the risks; the tolerance is the space the risks have to fit inside. Only three things can: reduce the exposure by response (KA 8.3), widen the tolerance through governance, or reduce scope. Presenting a P80 contingency without saying this would be technically correct and professionally negligent, and 8.3.2 completes the arithmetic.

**Why one objective per total, always.** M2's consequence is cost, paying for parallel running. Had it been written as "adoption below plan", its consequence would be *benefit*, measured against Meridian's USD 685,440 a year (Domain 1, KA 1.3.2) and against a different tolerance owned by a different body. Summing a cost impact and a benefit impact produces a number that is compared with nothing, cannot be funded from any reserve, and is the commonest reason a programme register's "total exposure" is quietly ignored by the board that receives it. The register may hold every objective; a **total** may hold only one. Where a single event hits two objectives (a delay that costs money *and* defers benefit at USD 14,280 a week) it appears in both aggregations with the relevant consequence, and the double appearance is disclosed rather than netted.

**Scale changes which comparison matters, not the arithmetic.** Auriga's 278,000 against a 4,000,000 **BAC** reads as comfortable at 6.95 %; Meridian's 119,000 against a 2,400,000 approved cost reads as comfortable at 4.96 % and is

not, because the governing constraint is a 120,000 tolerance rather than the budget. The reviewer's question is therefore never "what percentage of the budget is the exposure?" but "**what is the exposure as a share of the room we have?**", and if nobody can state the room, that is the finding.

### 8.2.3 Decision trees and the value of information

A decision tree makes a choice under uncertainty explicit: decision nodes (what we control), chance nodes (what we do not), and outcomes valued and rolled back to the present decision.

#### WORKED EXAMPLE

#### Worked example 8.2.3 — should Auriga survey before excavating?

1. **Setup.** Excavation may hit contamination: probability **0.40**, cost if hit **USD 300,000**. Option A: proceed directly. Option B: commission a **USD 25,000** ground survey first, which reliably reveals the condition; if contamination is present, a planned mitigation costs **USD 90,000** instead of the reactive 300,000.
2. **Formula.** Roll back each branch:  $EV(\text{cost}) = \sum \text{probability} \times \text{outcome cost}$ . Value of information =  $EV(\text{cost without}) - EV(\text{cost with})$ .
3. **Substitution.** A:  $0.40 \times 300,000$ . B:  $25,000 + 0.40 \times 90,000 = 25,000 + 36,000$ .
4. **Result.** A **USD 120,000** expected cost; B **USD 61,000**. **Value of the survey = USD 59,000**, against its USD 25,000 price.
5. **Interpretation.** The survey is worth buying, and the arithmetic shows *why*: it converts a reactive 300,000 into a planned 90,000 in the 40 % of futures where the problem exists. This is the general shape of **buying information**, worth it when it changes what you would do, worth nothing when it does not, however reassuring it feels. **Information is valuable only in proportion to the better action it enables.** (Domain 6's case study is the counterfactual: Auriga did not survey, met the condition reactively, and paid the recovery this tree prices.)

Three breakevens make the conclusion defensible rather than merely correct, and each is one division.

**How much could the survey have cost?** Branch B beats branch A while the survey price stays below  $120,000 - 0.40 \times 90,000 = \text{USD } 84,000$ : **3.36 times** the actual price. That is the sentence for a procurement conversation: a survey quoted at three times the budget figure would still have been the right purchase, so the decision was never close and the cost-reduction pass that cut it (see the case study) was not making a marginal call.

**At what prior does information stop paying?** The survey pays while  $25,000 < p \times (300,000 - 90,000)$ , i.e.  $p > 25,000/210,000 = 11.90\%$ . Below a roughly one-in-eight belief in contamination, the survey destroys value.

This is the number that governs whether the same tree is worth building on the *next* site, and it explains why an apparently identical decision can go the other way on ground everyone considers clean.

**And what would make the informed action worthless?** If mitigation once forewarned still cost  $M$ , branch B is  $25,000 + 0.40M$ , which reaches branch A's 120,000 at  $M = \text{USD } 237,500$ . So the entire value of knowing rests on a planned response costing less than 237,500; at the 250,000 the text above uses as an illustration, B's EV is  $25,000 + 100,000 = 125,000$  and the survey destroys 5,000 of value. **Information's value is bounded by the response it enables**, which is why "we should get more data" is not a proposal until the responder and the response are named.

One further property is worth naming because 8.2.3b relaxes it. This survey is assumed to **reveal the condition reliably**, a perfect signal, so its gross value equals the value of perfect information,  $120,000 - 36,000 = \text{USD } 84,000$ , and it captures 100 % of what could possibly be known. Real signals are imperfect, and their value falls in a way that depends less on their price than on how they are designed.

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#### WORKED EXAMPLE

**Worked example 8.2.3b — Meridian's pilot, and the value the sample design decides.**

1. **Setup.** Meridian's migration approach may be unfit for clinics running the **legacy patient index**; **12 of the 40** clinics run it, and which twelve is known from the asset data. Whether the approach fails on that configuration is *not* known: the programme assesses the probability at **0.30**. Discovered only at full rollout, the failure costs **USD 480,000** of rework plus **6 weeks** of programme delay at the cost of delay of **USD 14,280 per week**. Discovered in a pilot, the approach is redesigned for **USD 130,000** plus **2 weeks** of delay. A two-clinic pilot costs **USD 45,000**. Three options: no pilot; a pilot at two clinics chosen at random from the 40; a pilot at two clinics chosen so that at least one runs the legacy index.
2. **Formula.** Late total = rework + 6 × cost of delay. Early total = redesign + 2 × cost of delay. For a random pair, the flaw is detected only if at least one pilot clinic runs the legacy index:  $P(\text{detect}) = 1 - C(28, 2)/C(40, 2)$ .  $EV(\text{cost}) = \text{pilot price} + p \times [\text{detect} \times \text{early} + (1 - \text{detect}) \times \text{late}]$ . Breakeven prior: pilot price ÷ (late – expected cost given the pilot).
3. **Substitution.** Late  $480,000 + 6 \times 14,280$ ; early  $130,000 + 2 \times 14,280$ . Random-pair detection  $1 - (28 \times 27)/(40 \times 39) = 1 - 756/1,560$ . Designed pair detection = 1.
4. **Result.** Late total **USD 565,680**; early total **USD 158,560**.

| OPTION                          | DETECTION<br>GIVEN A FLAW | EXPECTED COST (USD)                                        | VALUE VS NO<br>PILOT |
|---------------------------------|---------------------------|------------------------------------------------------------|----------------------|
| No pilot                        | —                         | $0.30 \times 565,680 =$<br><b>169,704.00</b>               | —                    |
| Random two<br>clinics           | <b>51.5385 %</b>          | $45,000 + 0.30 \times$<br>$355,856.62 =$ <b>151,756.98</b> | <b>17,947.02</b>     |
| <b>Designed two<br/>clinics</b> | <b>100 %</b>              | $45,000 + 0.30 \times$<br>$158,560 =$ <b>92,568.00</b>     | <b>77,136.00</b>     |

The **design choice alone is worth USD 59,188.98** and costs nothing: both pilots run two clinics at the same USD 45,000. Perfect free information would cost  $0.30 \times 130,000 =$  **USD 39,000**, so the value of perfect information is **USD 130,704** and the designed pilot captures **59.02 %** of it; the random pilot captures **13.73 %**. 5. **Interpretation.** The finding is not that pilots are valuable (everyone believes that), but that **the value of an imperfect signal is set by its design, and a badly designed pilot is roughly a quarter as valuable as a well designed one at identical cost.**

**Where the loss comes from.** A random pair of clinics misses every legacy-index site nearly half the time (**48.4615 %**), and in those futures the programme pays the full 565,680 *plus* the 45,000 it spent finding nothing. A pilot chosen to include the configuration under suspicion cannot miss. Nothing about the money changed; only the question the pilot was capable of answering. This is the same lesson Domain 9 (KA 9.3.2) states for acceptance sampling (a convenience sample of the accessible items establishes nothing), arriving here through the value of information rather than through a confidence bound, and it is worth recognising as one idea in two instruments.

**The breakevens diverge sharply, which is the decision rule.** The designed pilot pays while the prior exceeds  $45,000 / (565,680 - 158,560) =$  **11.0533 %**; the random pilot needs  $45,000 / 209,823.38 =$  **21.4466 %**: a prior **1.9403 times** higher. So there is a whole band of beliefs, from 11.05 % to 21.45 %, in which **the right decision is to pilot and the wrong decision is to pilot badly**: a programme that runs the random pilot in that band destroys value while appearing prudent. The professional habit that follows is to state, of any proposed investigation, *which uncertainty it can resolve and with what probability*, not merely what it costs.

**Two cautions and one governance point.** The arithmetic treats the flaw as a single binary condition; if the approach could fail *partially*, or fail on a configuration nobody has thought of, the pilot's real detection probability is lower than 1 even when designed, and the honest figure is a range. The 2-week and 6-week delays are themselves estimates, and because the cost of delay is a rate, an error in the *weeks* propagates linearly into the answer: one week either way on the late case moves the no-pilot option by  $0.30 \times 14,280 =$  USD 4,284. And delib-

erately selecting the pilot sites has a stakeholder consequence that no expected value captures. The clinics chosen are those most likely to have a bad experience, which is a Domain 11 conversation that has to be held before the arithmetic is acted on, not after.

Survey worth buying: it converts a reactive 300,000 into a planned 90,000 in the 40 % of futures where the problem exists

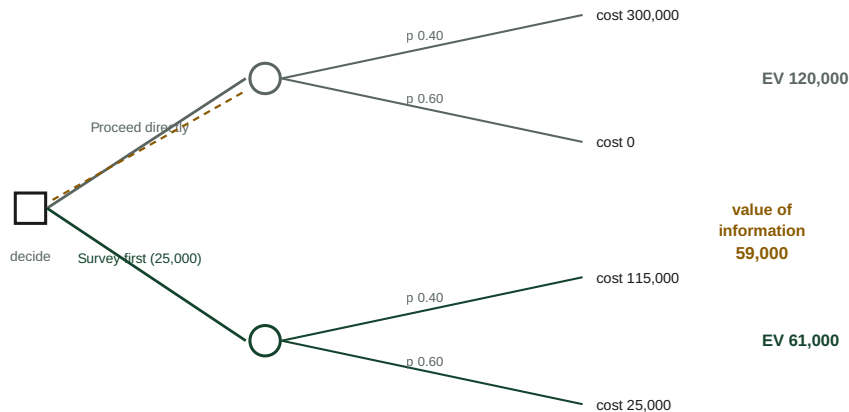


Fig 8.2.1 — The survey decision tree

#### 8.2.4 Aggregating to a contingency

**The error to avoid.** Summing worst cases produces a number nobody will fund ( $240,000 + 180,000 + 320,000 + 400,000 = 1,140,000$ , or 28.5 % of BAC); summing EMVs alone (278,000) funds the *average*, which by construction is exceeded about half the time. Neither is a contingency. What a leader needs is a **confidence level**: an amount that covers the aggregate outcome with stated probability.

##### WORKED EXAMPLE

#### Worked example 8.2.4 — Auriga's contingency at P80.

1. **Setup.** The five register risks of 8.2.2, treated as independent (an assumption examined below).
2. **Formula.** For independent risks:  $\text{mean} = \sum p \times \text{impact}$ ;  $\text{variance} = \sum p(1-p) \times \text{impact}^2$ ;  $\text{standard deviation} = \sqrt{\text{variance}}$ . A P80 amount  $\approx \text{mean} + 0.8416 \times \sigma$  (the 80th percentile of a normal approximation).
3. **Substitution.** Mean **278,000** (from 8.2.2). Variance  $0.35 \times 0.65 \times 240,000^2 + 0.50 \times 0.50 \times 180,000^2 + 0.25 \times 0.75 \times 320,000^2 + 0.15 \times 0.85 \times 400,000^2 + 0.30 \times 0.70 \times 120,000^2 = 63,828,000,000$ ;  $\sigma = 252,642$ . P80 =  $278,000 + 0.8416 \times 252,642$ .
4. **Result.**  $\sigma$  USD 252,642; P80  $\approx$  USD 490,624. Against the worst-case sum of 1,140,000 and the EMV sum of 278,000; and against the "10 % of BAC" rule of thumb, which would give 400,000 with no reasoning attached.

5. **Interpretation.** The P80 figure is defensible in a way the other three are not: it states what confidence it buys. Five things belong beside it, and the first two are checks a reviewer can run in five minutes.

**What confidence does the rule of thumb actually buy?** The "10 % of BAC" convention gives 400,000, which on this distribution sits at  $z = (400,000 - 278,000) / 252,642 = 0.4829$ , i.e. the **68.54th** percentile. So the rule of thumb is not "conservative" or "prudent"; it is a **P68.5** reserve, and it happens to be one on this register only by coincidence: on a register with the same mean and half the  $\sigma$  it would be a **P83** reserve. The convention's real defect is not that it is wrong but that it **conceals which question it is answering**, and the two-line calculation above converts it into a number that can be argued with.

**Check the approximation against the exact distribution, because with five risks you can.** Five independent binary risks have  $2^5 = 32$  outcomes; enumerating them gives the aggregate exactly with no normal assumption at all. Doing so shows that **USD 490,624 in fact covers 78.60 %** of futures, not 80.00 %. The approximation overstates the confidence by **1.40 percentage points**, which is small enough to use and large enough to disclose. The exact distribution also shows why a percentile is a slippery object on a lumpy aggregate: the outcomes cluster at 32 discrete levels, and the smallest level whose cumulative probability reaches 0.80 is **USD 500,000**, at which point the cumulative probability has already jumped to **83.43 %**. There is no outcome at all whose probability of non-exceedance is exactly 80 %. The professional habit: quote the confidence level as an approximation with its method named, and never defend the fourth digit of a percentile on a register of five risks. (The rule of thumb fares similarly: 400,000 covers **68.58 %** exactly against the approximation's 68.54 %.)

**The price of confidence is computable, and it is the conversation governance should be having.** The same register gives P50 = the mean = **USD 278,000** (6.95 % of BAC), P80 = **USD 490,624** (12.27 %), and P90 =  $278,000 + 1.2816 \times 252,642 = \text{USD } 601,786$  (15.04 %). Moving from P80 to P90 therefore costs **USD 111,163**, or about **USD 11,116 per percentage point** of confidence. That is the number a sponsor needs in order to choose, and it is why "the confidence level is a policy choice" is a statement about a priced trade rather than a disclaimer.

**Independence is an assumption, usually optimistic;** if R1 and R3 share a supplier they are correlated, the variance is larger, and the true P80 is higher; correlation is what turns a bad month into a crisis. Worked example 8.2.4b prices it on this register rather than leaving it as a warning.

**And the normal approximation is a convenience** for a handful of Bernoulli risks; proper practice runs a Monte Carlo simulation over the register (and over the schedule, extending Domain 6's three-point durations) to produce a distribution rather than a formula. Note what the 32-outcome enumeration establishes about that: the approximation's error here is a point and a half, so simulation on a five-risk register buys presentation rather than accuracy. Its value appears



when the register is large, when impacts are themselves distributions rather than points, or when correlations must be modelled; and the enumeration above is the check that a simulation is doing what its owner thinks, because both must return the same mean of 278,000.

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#### WORKED EXAMPLE

#### Worked example 8.2.4b – what one shared subcontractor does to Auriga's reserve.

1. **Setup.** Auriga's register is re-examined and one fact emerges from the procurement schedule: **R1** (controller lead time), **R3** (integration rework) and **R4** (permit delay, whose application package the same firm prepares) all depend on the same specialist subcontractor. The risk team assesses a pairwise correlation of  $\rho = 0.5$  among the three: a judgment, and one that must be recorded as such. R2 and R5 remain independent of everything.
2. **Formula.** For correlated risks,  $\text{Var}(\Sigma X) = \Sigma \text{Var}(X_i) + 2 \Sigma_{i < j} \rho_{ij} \sigma_i \sigma_j$ . The mean is unchanged:  $E(\Sigma X) = \Sigma E(X_i)$  whatever the dependence. Per-risk  $\sigma_i = \sqrt{(p_i(1-p_i))} \times \text{impact}_i$ .
3. **Substitution.** Per-risk  $\sigma$ : R1 **114,472.70**, R2 **90,000.00**, R3 **138,564.06**, R4 **142,828.57**, R5 **54,990.91**. Correlation term  $2 \times 0.5 \times (\sigma_1 \sigma_3 + \sigma_1 \sigma_4 + \sigma_3 \sigma_4) = 15,861,803,176.18 + 16,349,972,477.04 + 19,790,907,002.96$ .
4. **Result.** Added variance **52,002,682,656**, taking total variance from 63,828,000,000 to **115,830,682,656** and  $\sigma$  from 252,642 to **USD 340,339.07**: a factor of **1.3471**. The correlated P80 is  $278,000 + 0.8416 \times 340,339.07 = \text{USD } 564,429.36$ , **USD 73,805.82** (or **15.04 %**) above the independence figure, and **14.11 %** of BAC against 12.27 %. And the reserve already approved at 490,624 now covers  $z = 0.6247$  of the correlated distribution, a **73.39 %** reserve.
5. **Interpretation.** Three results, and the third is the one to take to a board.

**The mean did not move.** Total EMV is still exactly 278,000, because correlation redistributes probability without changing an average. This is why a register can pass every EMV check, reconcile perfectly against a simulation's mean, and still be under-reserved: **the error lives entirely in the second moment, where no EMV review looks.** It is also the reason  $\Sigma \text{EMV}$  remains a valid hand-check on a simulation even when correlation is modelled.

**The correlation term is large because it scales with  $\sigma$  products, not with EMVs.** Three risks (60 % of the register by count) added 81.5 % as much variance again as all five risks contributed independently. The general form of that, and why concentration hurts faster than counting suggests, is 8.A.1's. A programme with twenty risks on one scarce resource is not twenty risks.

**And the label fails before the money does.** Nobody wrote down a wrong reserve; they wrote down a reserve that was P80 under an assumption, and the assumption was never tested. The board approved "an 80 % confidence contingency" and holds a 73.4 % one, a 6.6-point shortfall in confidence that appears nowhere in the financial statements and can only be found by asking which risks share a driver. Restoring 80 % costs the **USD 73,806** computed above; that is the priced version of 8.A.1's warning and the exact question the Executive perspective says a leader must not delegate.

One caution on the arithmetic itself.  $\rho = 0.5$  is a judgment, and the answer is sensitive to it: the correlation term scales linearly in  $\rho$ , so halving the assumed correlation to 0.25 halves the added variance and the honest presentation is a range with the assumption named, not a single figure. Where the shared driver is identifiable at all, 8.A.1 argues for a different model altogether, and Case study B computes what that model does to the answer.

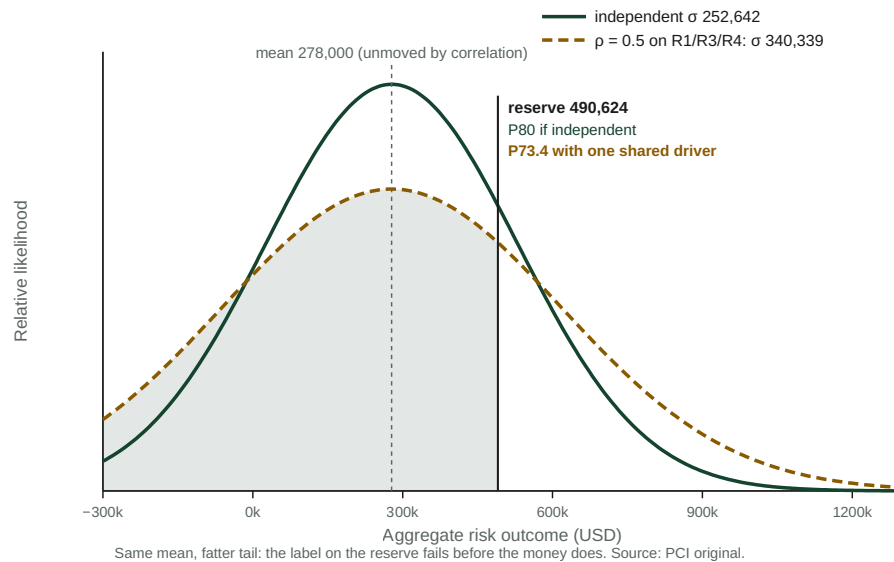


Fig 8.2.2 — Independence is worth 6

#### WORKED EXAMPLE

##### Worked example 8.2.4c — the response that buys variance, on Meridian's register.

- Setup.** Meridian's five quantified risks (8.2.2b) give a mean of **USD 119,000**, a variance of **10,149,000,000** and  $\sigma$  of **USD 100,742.25**, so the P90 the board's appetite implies (8.3.2) is  $119,000 + 1.2816 \times 100,742.25 = \text{USD } 248,111.26$ . Two responses are proposed, each costing **USD 25,000**. **Response A** halves the probability of M2 from 0.35 to 0.175 by seconding clinical backfill so that parallel running is less likely to extend. **Response B** cuts the impact of M3 from USD 150,000 to USD 60,000 by building the interface behind an abstrac-

tion layer, so a records-format revision is a configuration change rather than a rebuild; the probability stays 0.20 because the national timetable is outside the programme's control.

2. **Formula.**  $EMV \text{ contribution} = p \times I$ ;  $\text{variance contribution} = p(1-p)I^2$ ; recompute mean, variance,  $\sigma$  and P90 after each response and compare the **P90 reduction per USD spent**.
3. **Substitution.** A:  $EMV \ 42,000 \rightarrow 0.175 \times 120,000 = 21,000$ ; variance  $3,276,000,000 \rightarrow 0.175 \times 0.825 \times 120,000^2 = 2,079,000,000$ . B:  $EMV \ 30,000 \rightarrow 0.20 \times 60,000 = 12,000$ ; variance  $3,600,000,000 \rightarrow 0.20 \times 0.80 \times 60,000^2 = 576,000,000$ .

#### 4. Result.

|                                | EMV<br>REMOVED | VARIANCE<br>REMOVED  | NEW<br>MEAN | NEW $\sigma$ | NEW P90           | P90<br>REDUCTION |
|--------------------------------|----------------|----------------------|-------------|--------------|-------------------|------------------|
| A halve<br>M2's<br>probability | <b>21,000</b>  | 1,197,000,000        | 98,000      | 94,615.01    | <b>219,258.60</b> | <b>28,852.67</b> |
| B cut M3's<br>impact           | 18,000         | <b>3,024,000,000</b> | 101,000     | 84,409.72    | <b>209,179.49</b> | <b>38,931.77</b> |

A removes **USD 3,000 more EMV**; B removes **USD 10,079.11 more of the P90 requirement**: a reduction **34.93 %** larger, for the same USD 25,000. Per dollar spent, A returns **1.1541** of P90 reduction and B returns **1.5573**. 5. **Interpretation.** The two rankings disagree, and which one is right depends on what the organisation is short of.

**If the binding constraint is the reserve, buy variance; if it is the expected cost, buy the mean.** Meridian's constraint is the tolerance (8.2.2b), so B is the correct purchase even though it looks worse on the metric most registers publish. A programme whose problem is a thin margin rather than a thin contingency would choose A. The professional error is not choosing wrongly but **choosing without stating which constraint is binding**: at which point the **EMV** column decides by default, and the default is wrong whenever a confidence level is the thing under pressure.

**The algebra says why, and it generalises.** **EMV** is linear in both  $p$  and  $I$ , so halving either halves the **EMV** identically. Variance goes as  $p(1-p)I^2$ , which is **quadratic in impact and non-monotonic in probability**. Halving an impact therefore removes exactly **75 %** of that risk's variance contribution, always. Halving a probability leaves  $(1 - p/2)/(2(1 - p))$  of it (0.5278 at  $p = 0.10$ , 0.6346 at  $p = 0.35$ , 0.75 at  $p = 0.50$ ), so it removes between **25 %** and **50 %**. Dividing gives the invariant: **for any risk with  $p \leq 0.5$ , halving the impact removes between 1.5 and 3 times as much variance as halving the probability, at identical EMV cost** (1.5882 at  $p = 0.10$ , 1.7143 at 0.20, 2.0526 at

0.35, exactly 3 at 0.50). Impact reduction is the reserve lever; probability reduction is the margin lever. This is also the arithmetic behind the intuition that "cap the downside" beats "make it less likely", and it is the reason **modularity and staged commitment (8.4.1) release contingency**, since both are impact levers.

**Two cautions.** Impact reduction is frequently harder to engineer than probability reduction and sometimes impossible: an abstraction layer exists to be built, but no design change makes a regulator less likely to publish. The invariant tells a leader where to *look* first, not what is available. And these responses were compared at equal cost precisely so that the levers could be isolated; in practice each option's own cost, secondary risks and schedule effect enter as well, which is the fuller comparison worked example 8.3.1 runs.

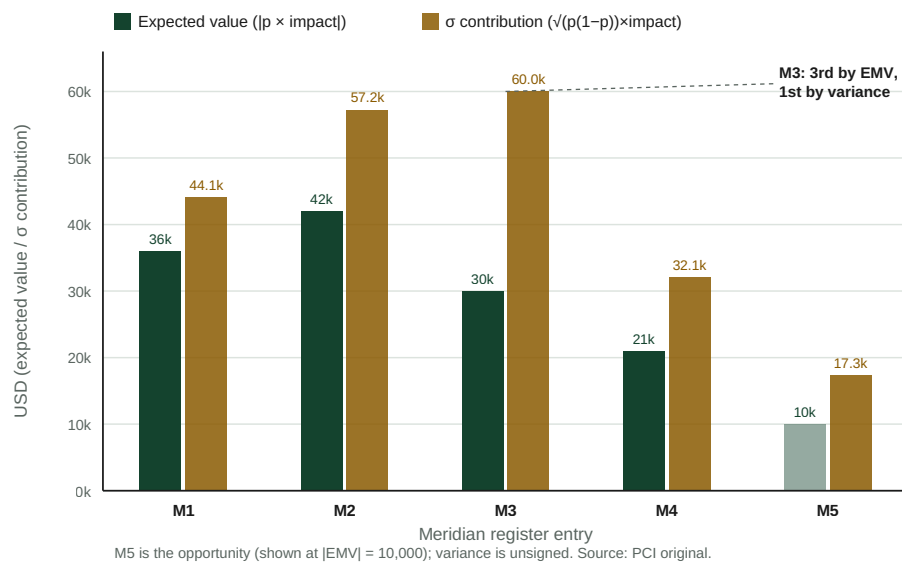


Fig 8.2.3 — Expected value and variance rank the same register differently

### AI in this KA

Simulation, correlation analysis and scenario generation are legitimate machine work, and the outputs are unusually seductive because a distribution *looks* like evidence. Three checks before any simulated number reaches a board. **Interrogate the inputs:** a simulation is an amplifier of its assumed distributions and correlations, and those are judgments, garbage in, precisely-shaped garbage out. **Recompute a landmark by hand:** the mean should match  $\Sigma EMV$ ; if it does not, the model is not doing what you think. **Ask what would change the decision:** a P8o quoted to the dollar with no sensitivity is decoration. The register's inputs remain human judgments with named owners, whatever produced the output.

## KEY TERMS — KA 8.2

| TERM                         | MEANING                                                                                                              |
|------------------------------|----------------------------------------------------------------------------------------------------------------------|
| EMV                          | probability × impact; the average of outcomes that will not individually occur.                                      |
| Decision tree                | Explicit rollback of decisions and chance events to a present choice.                                                |
| Value of information         | The reduction in expected cost that knowing enables; zero if it changes no action.                                   |
| P50 / P80                    | Confidence levels, amounts covering the aggregate outcome with stated probability.                                   |
| Correlation                  | Shared drivers that raise aggregate variance; independence is usually optimistic.                                    |
| Monte Carlo simulation       | Repeated sampling to produce an outcome distribution rather than a point.                                            |
| Within-cell span             | The EMV range a single matrix cell contains: the product of its two band ratios.                                     |
| Quantification threshold     | The impact above which a screened risk is quantified individually; a screen without one has replaced quantification. |
| Variance contribution        | $p(1-p)I^2$ for one risk; quadratic in impact, so it ranks a register differently from EMV.                          |
| Value of perfect information | The most any signal could be worth: expected cost without information less expected cost knowing the truth free.     |
| Price of confidence          | The cost per percentage point of moving a reserve between confidence levels.                                         |

## SAMPLE MCQs — KA 8.2

**MCQ 8.2-A** [8.2.2 · Application] A risk has probability 0.15 and impact USD 400,000. Its EMV is:

- A. USD 400,000
- B. USD 60,000 CORRECT
- C. USD 340,000
- D. USD 26,667

*Rationale:*  $0.15 \times 400,000 = 60,000$ . A is the impact; C is impact less EMV; D divides instead of multiplying.

**MCQ 8.2-B** [8.2.2 · Analysis] R4 has the register's largest impact (400,000) but its smallest threat EMV (60,000). The correct managerial reading is:

- A. R4 should be ignored, having the lowest EMV

- B. **EMV** sets the funding priority, while impact still governs whether the event is survivable — both readings are needed CORRECT
- C. impact alone should drive priority
- D. the assessment must be wrong, since impact and **EMV** disagree

*Rationale:* **EMV** is the right basis for funding a portfolio of risks; a single large impact may still be existential regardless of probability (8.2.2, 8.3.1). A and C each discard half the information; D misunderstands that the two measure different things.

**MCQ 8.2-C [8.2.3 · Application]** Proceeding directly has an expected cost of  $0.40 \times 300,000$ . A USD 25,000 survey reduces the mitigated cost to USD 90,000. The value of the information is:

- A. USD 25,000
- B. USD 59,000 CORRECT
- C. USD 84,000
- D. nil — the survey costs more than it saves

*Rationale:*  $120,000 - (25,000 + 36,000) = 59,000$ . A is the survey's price, not its value; C omits the survey cost from branch B; D reverses the conclusion.

**MCQ 8.2-D [8.2.4 · Analysis]** A register's **EMV** sum is 278,000 and its worst-case sum is 1,140,000. Setting contingency at 278,000 means:

- A. an appropriately funded reserve
- B. funding the average outcome, which by construction is exceeded roughly half the time CORRECT
- C. a conservative reserve, since not all risks will occur
- D. the same as a P80 reserve

*Rationale:* The mean is the ~50th percentile of the aggregate (8.2.4), so it is exceeded about half the time, the reason a confidence level is chosen explicitly. C mistakes the mean for conservatism; D confuses two different statistics (490,624 here).

**MCQ 8.2-E [8.2.1 · Analysis]** Why must ordinal probability-impact scores not be multiplied and summed as money?

- A. the matrix is only for threats
- B. ordinal bands are ranks, not quantities — a "4" is not twice a "2", so the arithmetic is meaningless CORRECT
- C. multiplication requires more than five bands
- D. scores may be summed provided they are weighted

*Rationale:* Ordinal scales support ordering, not arithmetic (8.2.1). Weighting (D) does not repair a scale that never carried magnitude.

**MCQ 8.2-F** [8.2.1 · Application] A matrix's medium probability band runs 0.10–0.35 and its medium impact band USD 60,000–250,000. The span of expected values inside that single cell is a factor of:

- A. 3.5
- B. 14.58 CORRECT
- C. 4.17
- D. 2.5

*Rationale:* Within-cell span is the product of the two band ratios,  $3.5 \times 4.1667 = 14.5833$  (8.2.1). A gives the probability ratio alone and C the impact ratio alone; each answers half the question; D subtracts one from the probability band ratio ( $3.5 - 1$ ), the slip that reads a ratio as an increment.

**MCQ 8.2-G** [8.2.4 · Analysis] A register's contingency was set at a P80 of USD 490,624 on an independence assumption. Three of its risks are then found to share a subcontractor, raising  $\sigma$  from 252,642 to 340,339. The reserve is now:

- A. still a P80 reserve, because the mean has not changed
- B. a 73.4 % reserve — correlation widens the distribution without moving the mean, so the same amount covers fewer futures CORRECT
- C. inadequate by USD 73,806 of expected cost
- D. unaffected, since correlation is a modelling choice rather than a fact

*Rationale:*  $z = (490,624 - 278,000) / 340,339 = 0.6247$ , i.e. **73.4 %** (8.2.4b). A confuses the mean's invariance with the percentile's; C misreads the USD 73,806. It is the extra reserve needed to restore 80 % confidence, not an increase in expected cost, which is unchanged at 278,000; D treats a shared supplier as an opinion.

**MCQ 8.2-H** [8.2.3 · Analysis] Twelve of 40 clinics run the configuration under suspicion. A two-clinic pilot chosen at random detects the problem with probability 0.5154; a two-clinic pilot chosen to include a suspect configuration detects it with certainty. Both cost USD 45,000. The right conclusion is:

- A. the pilots are equivalent, since the cost and sample size are identical
- B. the design choice is worth USD 59,189 at no additional cost, because the value of information depends on what the sample is capable of detecting CORRECT
- C. the random pilot is preferable because it is unbiased
- D. neither pilot is worthwhile, since 45,000 exceeds the redesign saving

*Rationale:* Expected costs are 151,756.98 random against 92,568.00 designed (8.2.3b). A prices inputs rather than the information bought; C applies a sampling virtue that is irrelevant when a specific condition is being tested and Domain 9 (KA 9.3.2) warns about in its own instrument; D compares the pilot price with the wrong quantity.

The comparison is against the 565,680 late case, and the designed pilot pays above an 11.05 % prior.

**MCQ 8.2-I** [8.2.4 · Application] Two responses cost the same and each removes the same proportion of a risk. Response A halves the probability; Response B halves the impact. Compared on the reserve they release:

- A. they are equivalent, since both halve the EMV
- B. B releases more, because variance goes as impact squared — halving an impact removes 75 % of that risk's variance while halving a probability removes between 25 % and 50 % CORRECT
- C. A releases more, because probability drives whether the event happens at all
- D. neither releases reserve, since contingency is set from EMV

*Rationale:* EMV is linear in both, so A's premise is right and its conclusion wrong; variance is  $p(1-p)I^2$  (8.2.4c). C reverses the result; D confuses a mean-based reserve with a confidence-based one, which is the error MCQ 8.2-D addresses.

#### SELF-CHECK — KA 8.2

1. *Why is EMV wrong for judging whether one risk is survivable?* — It averages outcomes that will not occur; the actual event is impact-or-nothing.
2. *When is information worth nothing however reassuring?* — When it would not change the action taken.
3. *Which assumption in the P80 calculation is usually optimistic, and why does it matter?* — Independence; correlated risks raise variance and the true confidence amount.
4. *What determines the span of expected values inside one matrix cell?* — The product of its probability and impact band ratios; nothing else.
5. *What does correlation change and what does it leave alone?* — It widens the distribution and raises every percentile; the mean is unchanged, which is why EMV reviews cannot detect it.
6. *Which lever should a leader short of contingency pull, and why?* — Impact, because variance is quadratic in impact: at equal EMV reduction it removes 1.5 to 3 times as much variance.
7. *Why express total exposure against a tolerance rather than against the budget?* — The tolerance is the room governance controls; Meridian's 119,000 is 4.96 % of approved cost and 99.17 % of the tolerance.



## Responses, reserves and governance

*Topics: 8.3.1 selecting and costing responses · 8.3.2 reserves and their authority · 8.3.3 monitoring and the register that earns its keep.*

### 8.3.1 Selecting and costing responses

The response families — **avoid** (remove the cause or the exposure), **reduce** (lower probability or impact), **transfer** (insurance, contract terms — a *price*, not a disappearance; Domain 7, KA 7.4.2), **accept** (with or without a fallback), and for opportunities **exploit**, **enhance**, **share**, **ignore**. Two rules make selection professional rather than reflexive.

**A response is an investment.** It costs money and time and reduces EMV; if the reduction is smaller than the cost, the response destroys value and acceptance is the correct answer. Auriga's fast-track decision from Domain 6 is exactly this arithmetic: one week of client bonus (USD 45,000) against an accepted rework risk of  $0.20 \times 60,000 = 12,000$ , a net **+USD 33,000**, which is why the recovery plan took it.

**Impact governs survivability even at low probability.** A 3 % chance of an event the project cannot survive is not managed by its EMV; it is avoided, transferred, or the project's viability is reconsidered. EMV funds portfolios; existential risks are handled by structure, and confusing the two is how organisations optimise their way into single points of failure.

**Where a duty is engaged, the menu is shorter.** Domain 1, KA 1.4.3 states the boundary and it governs here unchanged: **where a duty arising under law or regulation, an operating licence or a safety obligation is engaged, the duty is not transferable and acceptance is not an available response.** Insurance and contract terms move *money*; they do not move the duty or the accountability, and they move the money only so far as the wording and the applicable law allow, which is why the recovery percentage in the transfer arithmetic below comes from qualified counsel rather than from the register. This is not a refinement of the expected-cost rule; it removes rows from it.

The practical consequence for the comparison that follows is procedural, and it is the part most often skipped. **Rows whose realisation would engage such a duty are marked and taken out of the EMV comparison before it is run.** What remains for those rows is a narrower question (*how* the duty is met, at what cost, on what evidence), and it is answered by choosing among compliant options, not by weighing compliance against its expected cost. A register that prices "accept" against a duty-engaged row has not made an aggressive commercial judgement; it has answered a question that was never open, and the arithmetic beside it lends the answer a false respectability. The register therefore carries a **duty-engaged flag** as a mandatory field, mirroring the non-prioritisable regulatory class of Domain 5 and Domain 1's

mandatory – not an expected-value decision marking on the verification card. Whether any duty is in fact engaged, by what, and on whom, is a question for qualified counsel and the relevant regulator in the relevant jurisdiction; nothing here states the position anywhere, and the flag records that the question was asked and by whom it was answered.

**Secondary risks.** Every response creates its own: fast-tracking creates rework risk, a transfer creates counterparty risk, mitigation creates delivery risk on the mitigation itself. A register whose responses have no secondary entries has not been thought through. That is usually asserted as a completeness rule; it is in fact a selection rule, because the secondary risks can reverse which response is cheapest.

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#### WORKED EXAMPLE

##### Worked example 8.3.1 – four responses to Auriga's R1, priced against each other.

1. **Setup.** R1 (controller lead-time slip) carries  $p$  0.35 and an impact of **USD 240,000**, so its **EMV** is **USD 84,000** and its impact is **6.00 %** of **BAC** (survivable, so the **EMV** logic below is legitimate; see the caution in step 5). Four options are costed:
2. **Accept**, with the slip absorbed by contingency.
3. **Reduce**: dual-qualify a second controller supplier for **USD 22,000**, cutting  $p$  to 0.15. Secondary risk: the alternate's first article fails inspection,  $p$  0.10, impact USD 40,000.
4. **Transfer**: a delivery-guarantee term in the supply contract priced at **USD 18,000** that recovers **60 %** of the impact. Secondary risk: the supplier is a single-site specialist whose ability to pay is impaired by the very disruption that triggers the claim, assessed at **0.25** conditional on the event.
5. **Avoid**: redesign to a commodity controller for **USD 95,000**, removing R1 entirely. Secondary risk: the redesign introduces integration rework,  $p$  0.20, impact USD 120,000.
6. **Formula.** Total expected cost of an option = response cost + residual **EMV** +  $\Sigma$  secondary **EMV**. Choose the minimum. For the transfer, residual **EMV** =  $p \times \text{impact} \times (1 - \text{recovery})$  and the counterparty secondary =  $p \times P(\text{cannot pay}) \times \text{recovered amount}$ .
7. **Substitution.** Reduce  $22,000 + 0.15 \times 240,000 + 0.10 \times 40,000$ . Transfer  $18,000 + 0.35 \times 240,000 \times 0.40 + 0.35 \times 0.25 \times (0.60 \times 240,000)$ . Avoid  $95,000 + 0 + 0.20 \times 120,000$ .
8. **Result.**

| OPTION        | RESPONSE<br>COST | RESIDUAL<br>EMV | SECONDARY<br>EMV | TOTAL          | TOTAL<br>IGNORING<br>SECONDARIES |
|---------------|------------------|-----------------|------------------|----------------|----------------------------------|
| Accept        | 0                | 84,000          | 0                | <b>84,000</b>  | 84,000                           |
| <b>Reduce</b> | 22,000           | 36,000          | 4,000            | <b>62,000</b>  | 58,000                           |
| Transfer      | 18,000           | 33,600          | 12,600           | <b>64,200</b>  | 51,600                           |
| Avoid         | 95,000           | 0               | 24,000           | <b>119,000</b> | 95,000                           |

With secondary risks priced, **reduce wins at USD 62,000**. Ignoring them, transfer appears best at 51,600 (better than reduce by 6,400), when in truth it is worse by 2,200: a **swing of USD 8,600** in the comparison, and a change of decision. 5. **Interpretation.** Four conclusions, and the first is the reason this example exists.

**Omitting secondary risks does not make a response look slightly better; it changes which response is chosen.** The transfer's secondary is large precisely because it is *correlated with the event it covers*: the supplier's insolvency and the delivery failure have the same cause. That is the general case rather than an exotic one, and it is why "transfer is a price, not a disappearance" (Domain 7, KA 7.4.2; Domain 10, KA 10.3) needs a number beside it. The breakeven is checkable: the transfer ties the reduce option when the counterparty's probability of being unable to pay reaches  $(62,000 - 18,000 - 33,600) / (0.35 \times 144,000) = 20.63\%$ . Below that, transfer; above it, dual-qualify. A leader who cannot form a view on that one number has no basis for choosing between the two, and asking for it is a better use of a review than re-reading the register. One pointer, because this is an enforceability-sensitive area: whether a guarantee or indemnity is enforceable, what it actually reaches, and where the claim ranks if the counterparty fails are legal questions that turn on the wording and on the jurisdiction. Take the 60 % recovery this table multiplies from qualified counsel (Domain 10, KA 10.3), not from the register.

**Avoidance is not the safe answer.** It is the most expensive option here by a wide margin (**USD 35,000 worse than doing nothing**), and it does not even eliminate exposure, because the redesign carries 24,000 of secondary EMV of its own. Avoidance moved the risk from procurement to engineering and paid 95,000 for the move. The instinct that the strongest-sounding response family is the most responsible one is the commonest failure in response selection, and the arithmetic is the only reliable corrective.

**Acceptance is a priced position, not a default.** Reduce creates **USD 22,000** of value against accept (a **26.19 %** improvement on the 84,000), so acceptance is wrong here. But it becomes right the moment the reduce option's cost rises above 44,000 ( $84,000 - 36,000 - 4,000$ ), which a single change in the alternate supplier's qualification scope could do. Acceptance and response are

the two ends of one calculation, and the register should record the cost at which the answer flips, not just the answer.

**Three boundaries on all of it.** First, this comparison is on **expected cost only**. Worked example 8.2.4c's result applies here too: the reduce option is a probability lever and the transfer is effectively an impact lever, so if the binding constraint were the confidence level rather than the expected cost, the ranking would need recomputing on variance: dual-qualifying leaves  $0.15 \times 0.85 \times 240,000^2 = 7,344,000,000$  of variance where the transfer leaves  $0.35 \times 0.65 \times 96,000^2 = 2,096,640,000$ . Second, and overriding: **if R1's impact were existential rather than 6 % of BAC, the avoid option's 119,000 would be cheap** and the whole table would be the wrong instrument. That is a limit on this rule rather than a refinement of it. Third, and equally overriding: the table is the right instrument for R1 **because R1 is a commercial lead-time exposure with no duty attached to it**. Had the same slip engaged a safety obligation or a condition of the utility's operating licence, the accept row would not appear on the table at all and the transfer row would be priced as money only, with the duty and the accountability remaining exactly where they were. Marking the duty-engaged rows and lifting them out is therefore the *first* step in building this comparison, not a caveat on its result.

### 8.3.2 Reserves and their authority

Domain 7 (KA 7.1.3) established the structure; this domain supplies the sizing and the governance:

| RESERVE                   | COVERS                            | SIZED BY                                   | SPENT BY                                       |
|---------------------------|-----------------------------------|--------------------------------------------|------------------------------------------------|
| <b>Contingency</b>        | Identified risks in the register  | Aggregation to a stated confidence (8.2.4) | Project manager, under a published protocol    |
| <b>Management reserve</b> | Unknown-unknowns and scope change | Judgment and organisational policy         | Sponsor / change authority, via change control |

Three governance rules. **The draw protocol is published in advance** (which risk, what evidence, what approval), because a reserve released ad hoc is indistinguishable from an overrun. The protocol also names who may not approve: a draw approved by the person whose overspend it covers is not a controlled reserve whatever the paperwork says, so the approver's interest in the draw is declared before the approval and an interested approver stands aside, with the record naming who approved instead (Domain 1, KA 1.2.2a). This is the commonest live conflict in cost control, because the person with the best information about the draw is usually the person it relieves. **Consumption is trended against risk retirement**, not against time: burning 60 % of contingency while 20 % of the register has been retired is the

signal MCQ 7.1-B describes. And **contingency released by retired risks is returned, not reallocated** to convenient overspends; otherwise the reserve silently becomes a slush fund and the next real risk is unfunded.

Two questions remain, and both are arithmetic rather than procedural: *at what confidence should the reserve be held*, and *is the reserve still adequate now that some of it has been spent*. The first is answered from the appetite, not from convention.

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#### WORKED EXAMPLE

#### Worked example 8.3.2 – deriving Meridian's confidence level from its appetite.

1. **Setup.** Meridian's board has expressed its appetite in one sentence: it will accept **no more than a 10 % chance** that the programme's outturn cost exceeds the approved **USD 2,400,000** by more than **5 %**. The quantified register (8.2.2b) has a mean of **USD 119,000**,  $\sigma$  of **USD 100,742.25**. No confidence level has yet been chosen for contingency.
2. **Formula.**  $\text{Tolerance} = 5\% \times \text{approved cost}$ . "No more than a 10 % chance of exceeding" is the definition of a **P90** requirement, so the appetite requires  $\text{mean} + 1.2816 \sigma \leq \text{tolerance}$ . Rearranged, the maximum  $\sigma$  the appetite permits at a given mean is  $\sigma_{\text{max}} = (\text{tolerance} - \text{mean})/1.2816$ .
3. **Substitution.**  $\text{Tolerance } 0.05 \times 2,400,000$ . Required **P90 119,000 + 1.2816 × 100,742.25**. Permitted  $\sigma$   $(120,000 - 119,000)/1.2816$ .
4. **Result.** Tolerance **USD 120,000**. The register's P90 is **USD 248,111.26: 2.0676 times** the tolerance, and **10.34 %** of approved cost against an appetite of 5 %. The  $\sigma$  the appetite permits at the current mean is **USD 780.27**, against an actual  $\sigma$  of 100,742.25: the appetite is not merely breached but **unreachable by any contingency decision**.
5. **Interpretation.** Three findings, each of which is a different conversation.

**The confidence level was never a choice.** The board said "10 % chance of exceeding", which is P90; nobody needed to prefer P80 or P90, and a programme that holds contingency at P80 while its board has stated a 10 % tolerance is inconsistent with its own governance. This is the general result: **a well-formed appetite statement determines the confidence level**. The corollary is more useful in practice, where a project has inherited "P80 by convention", the honest reconstruction is to ask what appetite statement P80 implies (a 20 % acceptance of exceedance) and put *that* sentence in front of the sponsor. Most sponsors have never seen it written down, and some do not agree with it.

**The appetite constrains  $\sigma$ , not just the mean, and that is where it becomes unreachable.** With the mean at 119,000 the tolerance leaves USD 1,000 of room, so the permitted  $\sigma$  is 780, a register with essentially no uncertainty at all. No reserve fixes this, because contingency is money held *inside* the tolerance; the tolerance is the room. The three real options are the

ones 8.2.2b named, and the arithmetic now sizes them: reduce the exposure (responses, per 8.2.4c and 8.3.1), widen the tolerance (a governance decision the board must take knowingly), or reduce scope. The professional obligation is to present the choice, not to select the confidence level that makes the paper pass.

**And the gap is the number that makes the case.** Restoring the appetite requires taking the P90 from 248,111 to 120,000 (a reduction of **USD 128,111**), which on the evidence of 8.2.4c is bought mainly by impact levers. Response B there removed USD 38,932 of P90 for USD 25,000, so the order of magnitude is three to four such responses, and that is a fundable programme of work rather than an argument. Two cautions. The 1.2816 factor is the normal approximation again, and on a five-risk register it carries the sort of error 8.2.4 measured, so the target should be expressed as a range. And an appetite expressed against *cost* says nothing about schedule, benefit or reputation; each objective needs its own statement, and a board that has expressed only one has expressed less than it thinks (Domain 15, KA 15.4).

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#### WORKED EXAMPLE

##### Worked example 8.3.2b — is Auriga's reserve still adequate at week 13?

- Setup.** Auriga's contingency was set at the P80 of **USD 490,624** (8.2.4). At the week-13 review — *PV* 2,080,000, *EV* 1,920,000, *AC* 2,120,000, *CPI* 0.91, *SPI* 0.92 (Domain 7, KA 7.3) — three things have happened. **R2** (ground conditions) has **occurred** and is now an issue. **USD 232,000** of contingency has been drawn under the published protocol: **USD 195,000** against R2, whose outturn exceeded its 180,000 assessment, and **USD 37,000** between two register entries that sat below the quantification threshold of 8.2.1 and occurred. **R4** (permit delay) has been **retired**: the consenting window closed without incident. **R1**, **R3** and **R5** remain open at unchanged assessments. Is the remaining reserve still a P80 reserve?
- Formula.** Requirement = P80 of the **open** register (mean + 0.8416  $\sigma$ , recomputed from the open entries only). Remaining reserve = original reserve – drawn. Adequacy ratio = remaining  $\div$  requirement. Implied confidence =  $\Phi((\text{remaining} - \text{open mean}) / \text{open } \sigma)$ . Release from a retired risk = original P80 – P80 recomputed without it.
- Substitution.** Open register: R1  $0.35 \times 240,000$ , R3  $0.25 \times 320,000$ , R5  $0.30 \times (120,000)$ ; variance  $13,104,000,000 + 19,200,000,000 + 3,024,000,000$ . Remaining  $490,624 - 232,000$ .
- Result.** Open mean **USD 128,000**, open variance **35,328,000,000**, open  $\sigma$  **USD 187,957.44**, so the P80 **requirement is USD 286,184.98**. Remaining reserve **USD 258,623.54**: **47.29 %** of the reserve has been drawn. Adequacy ratio **0.9037**, a **shortfall of USD 27,561.44**, and the remaining reserve in fact covers  $z = 0.6950$  of the open distribution: a **75.65 %** reserve, not an 80 % one.

Separately, retiring R<sub>4</sub> releases more than its EMV. The register without R<sub>4</sub> has mean 218,000,  $\sigma$  208,393.86 and P80 USD 393,384.27, so R<sub>4</sub>'s retirement releases  $490,624 - 393,384 = \text{USD } 97,239.27$  against an EMV of only 60,000, a ratio of 1.6207. 5. **Interpretation.** Two results, and together they replace the usual reserve test with a better one.

**The naive test passes and the real test fails.** On the conventional comparison, 47.29 % of the reserve is drawn while R<sub>4</sub>'s retirement alone accounts for 21.58 % of the original EMV and R<sub>2</sub>'s occurrence for a further 32.37 %: drawdown and register progress look broadly in step, and a status report saying so would not be challenged. Recomputing the requirement from the open register shows the reserve is 90.37 % of what its own policy requires and is buying 75.65 % confidence rather than 80 %. The discipline that follows: **a part-consumed reserve is tested against the register still open, at the stated confidence, not against elapsed time or against the original register.** The number to report each period is the adequacy ratio, and the trigger (a ratio below 1.00) is a request to top up from management reserve or to reduce exposure, taken deliberately while USD 27,561 is a small decision.

**Retiring a low-probability, high-impact risk releases far more than its EMV.** R<sub>4</sub> carried 32.0 % of the original variance on 21.6 % of the EMV, because variance is quadratic in impact (8.2.4c). A leader who returns only the retired risk's EMV of 60,000 leaves USD 37,239 sitting in the reserve doing nothing — money that the "return, do not reallocate" rule above was never meant to trap. The corollary is uncomfortable and worth stating: **an unretired risk whose window has quietly passed is expensive**, because it holds variance-weighted reserve against an exposure that no longer exists. Register hygiene is a funding activity, which is not how it is usually presented.

**Two cautions.** Recomputing the requirement each period tempts a project into re-deriving its contingency downwards whenever the arithmetic is convenient; the protocol should fix the *confidence level* and the *method* in advance, so that only the register changes. And the whole calculation still assumes independence among R<sub>1</sub>, R<sub>3</sub> and R<sub>5</sub>: on 8.2.4b's finding, the true requirement is higher and the adequacy ratio worse, which is the first question a reviewer should ask of this table rather than the last.

### 8.3.3 Monitoring and the register that earns its keep

A live register has movement: probabilities revised as evidence arrives, risks retired when their window closes, new entries as the project learns, owners who report rather than merely appear. The tests of whether it is working are behavioural, not documentary: **has a decision changed because of it this month?, are the top items the ones the team actually worries about?, and did anything that hurt us appear in it beforehand?** A register that answers no to all three is theatre, and the honest response is to fix the process rather than reformat the document. **Early-**



**warning indicators** are the register's operational edge: leading measures tied to specific risks (supplier confirmations slipping, defect discovery rate rising, permit queue lengthening) that fire before the event, which is what makes a response affordable (Domain 6's lead-time point).

### AI in this KA

**Where it earns its place.** The recomputation work, which is mechanical, tedious and therefore skipped. **Recomputing the reserve requirement each period from the register still open** (mean, variance,  $\sigma$  and the P-level at the stated confidence), which is Worked example 8.3.2b's whole finding and the reason the naive drawdown test passes while the real test fails. **Recomputing the adequacy ratio and the implied confidence** the remaining reserve actually buys, and reporting the trend rather than the point. **Assembling response-option comparisons** at the scale of a full register: response cost, residual EMV, secondary EMV and total expected cost per option, with the breakeven at which the ranking flips, which is the number 8.3.1 says the register should carry and almost never does. **Listing register hygiene defects** (risks whose window has passed but which remain open, responses with no secondary entry, entries whose early-warning indicator has never moved) each reported with its location so a human can confirm it. And **sweeping draws against the published protocol** for the two conditions that make a reserve a reserve: that the approver was entitled to approve, and that the interest declaration is present with a nil return where there was none.

**Where it must not go. No model output may authorise a reserve draw.** A draw is an attributable decision under the published protocol of 8.3.2, taken by a named person who is entitled to take it and who is not the person the draw relieves; a tool that computes the draw, drafts the justification and populates the approval has produced everything except the only part that matters. Nor may a model **choose the confidence level** (that is derived from a board's appetite statement and is a governance act), **set or revise a probability or an impact** without the named owner's assessment, or **retire a risk**, which is an evidenced judgement that a window has closed. And no model may decide **whether a duty is engaged** on a register row: that determination comes from qualified counsel and the relevant regulator, and a tool's confident answer to it is the most dangerous single output this domain could accept, because it looks like analysis and removes a row from a comparison it should never have entered.

**Verification, concretely.** Reproduce the reserve requirement by hand for one period against the open register, because the failure mode is a stale register rather than bad arithmetic. The recomputation is only as honest as the open/retired flags feeding it. Confirm that every retired risk carries retirement *evidence* and a date, since retirement releases variance-weighted reserve well above the risk's EMV and is therefore worth getting wrong in someone's favour. Check every recomputed  $\sigma$  against the independence assumption stated on the contingency derivation sheet (Toolkit 8.T.2), because a tool will assume independence silently and 8.2.4b shows



what that costs. And test each proposed response comparison for its secondary entries before reading its ranking: a comparison with no secondaries is not a cheaper analysis, it is a different and wrong one, by USD 8,600 in the example above.

## KEY TERMS — KA 8.3

| TERM                                      | MEANING                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Avoid / reduce / transfer / accept</b> | The threat-response families; transfer is a price, not a disappearance.                                                                                                                                                                                                                                                                                                                      |
| <b>Secondary risk</b>                     | Risk created by a response; a register without them is incomplete.                                                                                                                                                                                                                                                                                                                           |
| <b>Draw protocol</b>                      | The published rules for releasing contingency.                                                                                                                                                                                                                                                                                                                                               |
| <b>Risk retirement</b>                    | Closing a risk whose window has passed; frees contingency for return, often far more than the risk's EMV.                                                                                                                                                                                                                                                                                    |
| <b>Early-warning indicator</b>            | A leading measure tied to a risk, firing before the event.                                                                                                                                                                                                                                                                                                                                   |
| <b>Tolerance</b>                          | The amount by which an objective may be exceeded before appetite is breached; the room inside which contingency sits.                                                                                                                                                                                                                                                                        |
| <b>Adequacy ratio</b>                     | Remaining reserve ÷ the confidence-level requirement of the register still open; below 1.00 is a governance trigger.                                                                                                                                                                                                                                                                         |
| <b>Counterparty risk (priced)</b>         | The secondary EMV of a transfer: $p \times P(\text{cannot pay}) \times \text{amount recovered}$ , usually correlated with the event transferred.                                                                                                                                                                                                                                             |
| <b>Duty-engaged flag</b>                  | A mandatory register field marking a row whose realisation would engage a duty arising under law or regulation, an operating licence or a safety obligation. Such rows leave the EMV comparison before it is run: the duty is not transferable and acceptance is not available. Whether a duty is engaged is a question for counsel and the regulator, and the flag records who answered it. |

## SAMPLE MCQS — KA 8.3

**MCQ 8.3-A** [8.3.1 · Application] A mitigation costs USD 50,000 and reduces a risk's EMV from 84,000 to 20,000. The decision and its basis are:

- A. reject — 50,000 is a large outlay
- B. accept — the EMV reduction of 64,000 exceeds the 50,000 cost CORRECT
- C. accept — any reduction in EMV justifies a response
- D. indifferent — cost and benefit are equal

*Rationale:* Responses are investments:  $84,000 - 20,000 = 64,000$  of reduction for 50,000 of cost is value-creating. A prices the outlay without the benefit; C would justify unlimited spend; D miscalculates.

**MCQ 8.3-B** [8.3.1 · Analysis] A risk has probability 0.03 and an impact the project could not survive. The correct treatment is:

- A. accept it — the EMV is small
- B. treat it structurally: avoid, transfer, or reconsider viability, because EMV funds portfolios while survivability is governed by impact CORRECT
- C. fund its EMV in contingency and monitor
- D. exclude it from the register as improbable

*Rationale:* Existential exposure is not an averaging problem (8.3.1). A and C both apply portfolio logic to a single point of failure; D removes the entry that most needs governance attention.

**MCQ 8.3-C** [8.3.2 · Analysis] Contingency freed by a retired risk is used to cover an unrelated overspend. This is:

- A. efficient reserve management
- B. a governance failure: the reserve silently becomes a slush fund and the next genuine risk is unfunded CORRECT
- C. acceptable if the total baseline is unchanged
- D. required, since contingency is inside the baseline

*Rationale:* Contingency is tied to identified risks; reallocating it to overspends destroys the link between reserve and exposure (8.3.2). C is the reasoning that makes the failure invisible.

**MCQ 8.3-D** [8.3.1 · Analysis] On one risk, transferring appears cheaper than reducing — USD 51,600 against 58,000 — until secondary risks are priced, after which reducing is cheaper at 62,000 against 64,200. The correct reading is:

- A. the second calculation is double-counting, since the transfer already covers the risk
- B. secondary risks are a selection rule, not only a completeness rule: here the transfer's counterparty exposure is correlated with the very event transferred, and pricing it changes the decision CORRECT
- C. the difference is immaterial at USD 2,200 and either option may be taken
- D. reducing is always preferable to transferring

*Rationale:* The transfer's secondary EMV of 12,600 arises because the supplier's ability to pay is impaired by the disruption that triggers the claim (8.3.1). A misses that the counterparty may not perform; C ignores that the 2,200 is a *reversal* of a 6,400 apparent advantage and that the breakeven default probability, 20.63 %, is the thing to form a view on; D generalises one case into a rule.

**MCQ 8.3-E** [8.3.2 · Application] A risk with  $p = 0.15$  and impact USD 400,000 is retired when its window closes. Its EMV was USD 60,000, and recomputing the register's P80 without it falls by USD 97,239. The amount to return to the sponsor is:

- A. USD 60,000 — the risk's EMV
- B. USD 97,239, because the reserve was sized on a confidence level and the retired risk carried 31.96 % of the variance on 21.58 % of the EMV CORRECT
- C. nil, since contingency stays inside the baseline
- D. USD 37,239 — the difference between the two figures

*Rationale:* A confidence-based reserve releases what recomputing it releases; variance is quadratic in impact, so a low-probability high-impact risk carries disproportionate reserve (8.3.2b). A returns the mean-based amount and leaves USD 37,239 idle; D returns only that idle remainder; C confuses where contingency sits with whether it may be released.

**MCQ 8.3-F** [8.3.2 · Analysis] A board states it will accept no more than a 10 % chance of exceeding approved cost by more than 5 %. The confidence level for contingency is therefore:

- A. P80, the industry convention
- B. P90, because "no more than a 10 % chance of exceeding" is the definition of the 90th percentile, and the tolerance is 5 % of approved cost CORRECT
- C. P95, to be prudent
- D. undetermined — appetite statements are qualitative

*Rationale:* The appetite determines the confidence level (8.3.2); the leader's job is to derive it, not to select it. A inherits a convention that implies a 20 % acceptance of exceedance the board did not state; C substitutes caution for the board's decision; D is the position the arithmetic refutes.

#### SELF-CHECK — KA 8.3

1. *When is acceptance the professionally correct response?* — When the response's cost exceeds the EMV reduction it buys, and the risk is survivable.
2. *What does transfer actually achieve?* — It prices the risk to a counterparty and creates counterparty risk; it does not remove the exposure.
3. *What should contingency consumption be trended against?* — Risk retirement, not elapsed time.
4. *State the adequacy test for a part-consumed reserve.* — Remaining reserve divided by the stated confidence level's requirement recomputed on the register still open; below 1.00 is a trigger.
5. *Where does an appetite statement's confidence level come from?* — The statement itself: "no more than a 10 % chance of exceeding" is P90.

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6. *Why can avoidance be the most expensive response?* — It is paid in full whether or not the event would have occurred, and it creates its own secondary risks: auriga's avoid option costs USD 35,000 more than accepting.
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## KNOWLEDGE AREA 8.4

## Resilience, bias and crisis leadership

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*Topics: 8.4.1 resilience versus prediction · 8.4.2 cognitive bias · 8.4.3 crisis leadership · 8.4.4 AI-enabled risk sensing.*

### 8.4.1 Resilience versus prediction

Quantification (KA 8.2) handles risks you can name. **Resilience** is the capability to absorb what you cannot: it assumes the register is incomplete, because it always is. Its levers are structural rather than analytical — **buffers** (float placed deliberately, contingency, capacity headroom), **optionality** (staged commitments, alternative suppliers qualified in advance, reversible decisions — Domain 1, KA 1.3.3), **modularity** (a failure contained rather than propagated across interfaces), **redundancy** where failure is intolerable, and **fast detection** (early-warning indicators, short feedback loops), because response cost rises with delay.

The trade-off is real and should be stated rather than finessed: resilience costs efficiency. A project optimised to the last dollar and week has no capacity to absorb surprise, and one buffered everywhere is uncompetitive. The leadership judgment is *where* to be resilient (at the points where failure propagates or cannot be tolerated) not whether.

That judgment is usually left qualitative, and it need not be. A resilience measure has a price, a probability of being needed and a recovery time it shortens, which is enough to price it twice: once on expected value and once on the tail. The two answers frequently disagree, and knowing what to do when they do is the substance of this topic.

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#### WORKED EXAMPLE

#### Worked example 8.4.1 — Meridian's standby trainer retainer, on the average and on the tail.

1. **Setup.** Meridian's rollout depends on a single training provider. If that provider's capacity fails (attrition, a competing contract) clinics cannot go live, and the programme's cost of delay is **USD 14,280 per week**. The probability of a capacity failure over the rollout is assessed at **0.25**. Recovering unaided means re-tendering: **7 weeks**. A **USD 18,000** annual retainer with a second, pre-qualified provider reduces recovery to **2 weeks**. The programme separately notes the tail:

unaided, the 95th-percentile recovery is **12 weeks**, which would miss the winter demand window and defer the remaining **40 %** of the benefit by a full **13-week** quarter; with the retainer the 95th-percentile recovery is **3 weeks** and the window holds.

2. **Formula.** Expected delay cost =  $P(\text{failure}) \times \text{recovery weeks} \times \text{cost of delay}$ ; add the retainer where taken. Breakeven probability =  $\text{retainer} \div (\text{weeks saved} \times \text{cost of delay})$ . Tail cost =  $95\text{th-percentile weeks} \times \text{cost of delay}$ , plus any window-miss deferral.
3. **Substitution.** Without:  $0.25 \times 7 \times 14,280$ . With:  $18,000 + 0.25 \times 2 \times 14,280$ . Breakeven  $18,000 / (5 \times 14,280)$ . Tail deferral  $0.40 \times 14,280 \times 13$ .
4. **Result.** Expected cost **without** the retainer **USD 24,990.00**; **with** it **USD 25,140.00**. The retainer is a **net loss of USD 150.00**. It breaks even at a failure probability of  $18,000 / 71,400 = 25.2101 \%$ , against the assessed 25.00 %: it misses by **0.2101 of a percentage point**.

On the tail the picture inverts. Without the retainer the 95th-percentile outcome costs  $12 \times 14,280 = \text{USD } 171,360$  of delay **plus USD 74,256** of deferred benefit — **USD 245,616**. With it,  $18,000 + 3 \times 14,280 = \text{USD } 60,840$ . The retainer removes **USD 184,776** of tail exposure and reduces it by a factor of **4.0371**. 5. **Interpretation.** The two calculations disagree, and the professional content of this example is what a leader does about that.

**When the breakeven sits inside the estimating error, expected value has stopped being the deciding test.** The breakeven probability is 25.2101 % and the assessment is 25 %. Nobody can estimate a provider-capacity failure to a fifth of a percentage point; the difference is noise. The correct reading is not "reject, it loses USD 150" (that is false precision dressed as rigour), but **"on expected value this decision does not matter, so decide it on the tail and on survivability"**. That is exactly the boundary 8.3.1 draws between portfolio logic and structural logic, arriving here from the other direction: the arithmetic hands the decision back to judgment, and names why it is doing so.

**On the tail it is not close.** The retainer costs USD 18,000 and removes USD 184,776 of 95th-percentile exposure, a ratio of **10.27 to 1**. And the tail is not a smooth extension of the average, because a 12-week recovery crosses a **threshold**: it misses the winter window, at which point a further 74,256 of deferred benefit appears that no expected-value calculation containing only a weekly rate would ever show. **Resilience is bought where the consequence function has a step in it**, which is a sharper statement of "where failure cannot be tolerated" and a testable one: find the thresholds, then ask what recovery time clears them.

**The efficiency cost is real and should be named, not hidden in the tail argument.** USD 18,000 buys nothing at all in the 75 % of futures where the provider performs; that is what "resilience costs efficiency" means in cash, and it

is the reason the lever has to be aimed rather than applied everywhere. The discriminator is the step: dependencies whose failure crosses a threshold get structural treatment, dependencies whose failure is priced linearly are accepted and funded through contingency. Two cautions. The 95th-percentile recovery times are themselves estimates with wider errors than the mean, so the tail argument should be presented as an order of magnitude (10 to 1) rather than as USD 184,776 to the dollar. And a retainer is itself a contract with a counterparty, so it carries the secondary risk 8.3.1 priced: a second provider that cannot mobilise in two weeks when called has sold an option that does not exercise, which is a procurement-terms question (Domain 10, KA 10.3) and not a risk-register one.

#### 8.4.2 Cognitive bias

Estimates and reviews are produced by people, and the predictable distortions are worth naming because each has a countermeasure:

| BIAS                                        | EFFECT                                          | COUNTERMEASURE                                                                |
|---------------------------------------------|-------------------------------------------------|-------------------------------------------------------------------------------|
| <b>Optimism bias / planning fallacy</b>     | Systematic underestimation of duration and cost | Reference-class forecasting against comparable completed projects             |
| <b>Anchoring</b>                            | First number dominates subsequent estimates     | Independent estimates before any figure is shared                             |
| <b>Availability</b>                         | Recent or vivid events overweighted             | Structured checklists and historical base rates                               |
| <b>Groupthink / social pressure</b>         | Dissent suppressed in workshops                 | Pre-mortem, private input channels, explicit dissent invitation               |
| <b>Sunk-cost / escalation of commitment</b> | Continuing because of what is spent             | Decision framed on remaining cost and benefit only (Domain 2's kill criteria) |
| <b>Confirmation bias</b>                    | Evidence sought that supports the plan          | Assign someone to argue the opposite; test the disconfirming case             |

The **pre-mortem** deserves particular mention as the highest-yield, cheapest technique in this domain: before committing, ask the team to imagine the project has failed and to write down why. It licenses the dissent a status meeting suppresses, and it reliably surfaces risks the workshop of 8.1.3 missed.

**Reference-class forecasting** is the countermeasure with arithmetic behind it, and the arithmetic does something the table above cannot: it prices the risks nobody identified. The method is to stop asking "what will this project cost?" and instead ask "what did comparable completed projects cost against what they were approved for?", then apply that distribution of *outturn ratios* to the current estimate. The word

doing the work is **completed**: the reference class contains the overruns, the surprises and the unidentified risks of every project in it, which is precisely the material a bottom-up register cannot contain.

---

#### WORKED EXAMPLE

#### Worked example 8.4.2 – Meridian against its reference class.

1. **Setup.** Meridian's approved cost is **USD 2,400,000**, built bottom-up, with a quantified risk register whose P80 contingency requirement is **USD 203,784.67** (8.2.2b, at the 0.8416 factor). The sponsoring authority holds records of **12** comparable completed clinical-system rollouts. Each is expressed as an **outturn ratio** (final cost ÷ approved cost at the equivalent gate), and sorted: 0.95, 1.02, 1.05, 1.08, 1.12, 1.15, 1.18, 1.22, 1.28, 1.35, 1.48, 1.72.
2. **Formula.** Mean ratio =  $\Sigma \text{ratios} \div n$ . Median = average of the two central values at even  $n$ . The 80th percentile is read from the order statistics at position  $(n + 1) \times 0.80$ , interpolating between neighbours. Reference-class forecast = approved cost  $\times$  ratio; uplift = forecast – approved cost.
3. **Substitution.**  $\Sigma = 14.60$ , so mean =  $14.60/12$ . Median =  $(1.15 + 1.18)/2$ . Order position  $13 \times 0.80 = 10.40$ , so P80 =  $1.35 + 0.40 \times (1.48 - 1.35)$ .
4. **Result.**

| STATISTIC           | RATIO    | FORECAST<br>(USD) | UPLIFT ON<br>2,400,000 | UPLIFT<br>%    |
|---------------------|----------|-------------------|------------------------|----------------|
| Best case in class  | 0.95     | 2,280,000         | (120,000)              | (5.00 %)       |
| Median              | 1.1650   | 2,796,000         | 396,000                | 16.50 %        |
| Mean                | 1.216667 | <b>2,920,000</b>  | 520,000                | 21.67 %        |
| P80                 | 1.4020   | <b>3,364,800</b>  | <b>964,800</b>         | <b>40.20 %</b> |
| Worst case in class | 1.72     | 4,128,000         | 1,728,000              | 72.00 %        |

**11 of the 12** projects (**91.67 %**) exceeded their approved cost. The reference-class P80 uplift of **USD 964,800** is **4.7344 times** the register-based P80 of 203,785, a gap of **USD 761,015.33**. 5. **Interpretation.** The gap is the finding, and what to do with it is the professional skill.

**Do not add the two numbers.** They are two estimates of the same quantity by different methods, not two separate exposures, and summing them double-counts every risk the register identified. The reference class already contains ground-condition surprises, interface rebuilds and trainer attrition, because the twelve projects in it experienced them.

---

**Use the reference class as a challenge on the total, and treat the gap as an estimate of what the register cannot see.** Two causes account for it, and only one is measurable from the data here. The register-completeness arithmetic of 8.1.3 estimated **11.4286** unidentified risks; if each resembled the average quantified entry at  $119,000/5 = \text{USD } 23,800$  of **EMV**, they would add **USD 272,000.00: 35.74 %** of the gap. The residual **USD 489,015.33** cannot be decomposed from the evidence available and must be presented as such: part of it is **optimism in the impacts already assessed** (8.4.2's subject), and part is **scope growth** rather than risk at all, because projects in a reference class typically had less well-defined scope at the equivalent gate and grew into their outturns (Domain 5). Presenting the residual as a single cause would be exactly the false precision this topic warns against.

**What the numbers change.** Not the approved cost, which is a governance artefact, but three things: the **management reserve**, which now has a defensible basis (8.1.3's unidentified risks and the residual gap, held outside the baseline under sponsor control); the **funding requirement** presented to the authority, which should carry the reference-class range rather than a single figure; and the **tolerance conversation** of 8.3.2, since a 5 % tolerance against a class whose median outturn is 16.5 % is a statement about this programme being materially better managed than its peers: a claim that may be true and must then be evidenced, not assumed.

**Three cautions on the method itself.** The class must be genuinely comparable: twelve rollouts of clinical systems, not twelve public-sector programmes of any kind, and the leader should be able to say what makes them comparable and what makes each different. Twelve is a small class, so the P80 read from order statistics is coarse: the interpolation between the 10th and 11th values moves the answer by USD 312,000 across that single interval, which is the honest measure of its precision. And the class's own approved costs were set under whatever appraisal regime applied at the time; a class drawn from an era of weaker scope definition overstates the uplift a well-defined project needs, which is a reason to prefer recent comparators and to say so.

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### 8.4.3 Crisis leadership

When a serious risk materialises, the leader's job changes shape. The sequence that works: **stabilise** (stop the harm; secure safety, then the situation), **establish facts** (separate what is known from assumed — crises run on rumour), **decide with a clock** (a good decision now beats an optimal one later, and say explicitly which decisions are reversible), **communicate early and honestly** (bad news does not improve with age; Domain 11's stakeholder work is now executing under compressed time), **protect the team** (crises consume people, and exhausted teams make the second mistake), and **capture the lesson while it is vivid** (Domain 9's lessons-learned). Domain 6's recovery machinery is the schedule expression of the



same posture: re-run the passes, price the options, escalate the trade with named authority.

That is posture. Posture is necessary and it is not an instrument, and the three parts below are what turn it into one. Each is written in advance, because none of them can be designed at the moment it is needed.

**Declaration and authority, who may say the word, and what the word changes.**

A crisis that is never declared is managed by whoever happens to be in the room, at ordinary latency, under ordinary authority, which is the failure 3.A.1 describes. So the design states four things. **Who may declare**, by named role, with at least one alternate reachable outside working hours; the threshold should be low enough that declaring is not itself a career act, because a declaration threshold set by embarrassment is set too high. **What the declaration changes**, stated as authorities rather than as urgency: a named **stop-work authority** and the scope of what it may stop; a stated **fund** the crisis lead may commit against, with the reserve it draws on named and the ordinary draw protocol of 8.3.2 expressly modified rather than ignored: an emergency that suspends the protocol silently has removed the control at the moment it matters; and a **standing meeting cadence** with a fixed information pack, which is what stops the day dissolving into briefings. **Who stands it down**, which is normally the declaring role or its superior and never the crisis lead alone, with a stated hand-back of the temporary authorities. And **what does not change**: the accountability for the decisions taken, the duty rule of 8.3.1, and the obligation to record. This is deliberately modelled on the emergency change route of Domain 4, KA 4.4.1. The two instruments should be consistent, because they will be used together and a project cannot operate two different definitions of "urgent".

**Notification: the map that has to exist before the day it is needed.**

A serious materialised risk frequently brings obligations that run *outside* the organisation and on a clock the project does not control: safety and environmental authorities, data-protection authorities, market or listing disclosure, insurer notification conditions, and client or counterparty notification provisions in the contract. The professional obligation is to establish, in advance, **which of these could apply to this project's sector, territory and contract, who determines that they apply, to whom the notification goes, within what period, and who is entitled to make it**: recorded as a one-page **notification map** held with the risk register and reviewed whenever the register is.

**The standing caution, and it is the point of the paragraph.** Obligations of this kind arise under law, regulation, licence conditions or contract rather than at the project's discretion; but whether any of them reaches this organisation, this incident, this territory and this timetable is a question for the relevant authority and for qualified counsel. **Nothing here states the position in any jurisdiction, states what any regime requires, or characterises any act or omission as compliant**

**or otherwise.** The professional obligation this domain does impose is narrower and unconditional: build the map before the incident, name the person who takes the question to counsel, take it promptly, and record that you did. A leader who is discovering on the day *whether* an obligation exists has already lost the part of the response that could not be recovered.

Two further provisions belong on the map. The **insurer** entry is the one most often missing, and notification conditions in a policy are frequently tighter than anything else on the page. And the map records **who may speak**; because an unco-ordinated notification made by the wrong person, in the wrong terms, to the right recipient is a problem the organisation then owns twice.

#### **Record, from the moment of declaration, not from the moment of recovery.**

The declaration starts a **contemporaneous decision log**: reference, **time** rather than date, decision-maker by name and role, the decision in words permitting one reading, the information relied on with its version, and what was known to be unknown at the time. That is Domain 3, KA 3.3.4's record with a finer clock, and the reason for the finer clock is that a crisis produces several decisions an hour and the sequence is exactly what a later reader will want to reconstruct.

Alongside it, an instruction that is easy to state and hard to follow: **preserve rather than tidy**. From the declaration, the working papers, logs, messages, readings, drafts and superseded versions are retained as they are, not cleaned up, not consolidated, not selectively kept. The instinct under pressure is to produce a neat account and discard the mess it came from; the mess is the evidence that the account was reasonable on what was known at the time, and it is unrecoverable once overwritten. Custody, retention and access follow Domain 3, KA 3.3.4's provisions, and where records name identified individuals the data-protection considerations of Domain 11, KA 11.1.2 and Domain 16, KA 16.4.4 apply from the first entry.

The three parts interlock with the machinery the volume has already built: the escalation route of Domain 3, KA 3.3.3 is what carries the declaration upward at a stated latency, and the emergency change route of Domain 4, KA 4.4.1 is what carries the resulting scope and baseline moves: a crisis that bypasses both has not been fast, it has been undocumented.

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#### WORKED EXAMPLE

##### **Worked example 8.4.3 — the price of a notification that drifts.**

1. **Setup.** A record-integrity incident on Meridian is confirmed at the start of week 0, and the programme's notification map (above) identifies an external recipient with a **72-hour** clock. Notifying inside the window costs an assessment and notification pack (specialist time, counsel and the evidence bundle) of **USD 26,000**, and a **2-week** hold on clinic onboarding while the affected records are re-verified. If instead the decision drifts and the notification is made at **week 6**, the same 26,000 pack and the same 2-week hold are incurred anyway, and two

further costs appear: the affected records stay in use for six more weeks, so records created in that period must also be re-verified, assessed at **USD 9,500 a week**; and the 2-week hold, and with it the resumption of onboarding, moves six weeks later at Meridian's cost of delay of **USD 14,280 a week**. What did the drift cost?

2. **Formula.** Cost of notifying early = pack + hold weeks × cost of delay. Cost of notifying late = that, + drift weeks × (propagation rate + cost of delay). Avoidable cost = drift weeks × (propagation rate + cost of delay). Weeks of drift that cost as much as the whole notification pack = pack ÷ (propagation rate + cost of delay).
3. **Substitution.** Early 26,000 + 2 × 14,280. Late 26,000 + 2 × 14,280 + 6 × 9,500 + 6 × 14,280. Avoidable 6 × (9,500 + 14,280). Pack-equivalent drift 26,000 ÷ 23,780.
4. **Result.** Notifying inside the window costs **USD 54,560**. Notifying at week 6 costs **USD 197,240**. The six weeks of drift therefore cost **USD 142,680: 2.6151 times** the entire cost of notifying on time. The avoidable cost accrues at **USD 23,780 a week**, so **1.0934 weeks** of drift costs as much as the whole notification pack.
5. **Interpretation.** Four readings, and the last is a prohibition rather than a lesson.

**The run rate is the argument, not the total.** USD 142,680 is a number that arrives after the fact and persuades nobody in advance. **USD 23,780 a week** is a number a leader can put in front of a sponsor on day one, and it has the same shape as Worked example 1.2.2's escalation timing: the cost of an early notification is fixed and small, the cost of a late one grows linearly with the delay, so the benefit-to-cost ratio of acting early is bounded only by how long the drift would otherwise have run. That is why the instrument is a map and a named owner rather than an exhortation to move quickly.

**The pack pays for itself in about a week.** At 1.0934 weeks, the whole USD 26,000 of assessment, counsel and evidence bundling is recovered by avoiding one week of drift, which disposes of the objection that a full assessment is disproportionate to an incident of uncertain size. Standing an assessment team up immediately is close to free in expectation and expensive only if the incident turns out to be nothing, which is the one outcome nobody regrets paying for.

**The propagation term is what distinguishes this from an ordinary delay.** An ordinary schedule slip costs the cost of delay. This costs the cost of delay **plus** a rate of contamination: every week the affected records remain in use enlarges the population that has to be re-verified. Look for that second term whenever a materialised risk involves data, materials, medication, a shared platform or anything else that is *consumed downstream while the decision is pending*, because it is the term that makes early notification arithmetically obvious and it is the term nobody puts in the paper.

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**The prohibition, and it overrides everything above.** *This arithmetic prices drift; it never decides whether a notification is due.* Whether an obligation applies, to whom, and on what clock is determined by the relevant authority and by qualified counsel, and it is not an expected-value question: the rule of 8.3.1 applies here at its sharpest. A leader who uses figures of this kind to argue that a notification can wait because the weekly rate looks modest has inverted the example: the numbers exist to fund the fast response, never to price the option of not making one. Two cautions on the figures themselves. The 9,500 a week is an assessed propagation rate and should be recorded in the register with its owner and basis, like any other assessment. And the two components are not independent (a longer drift makes both the re-verification population and the onboarding restart worse), so the linear model here is a floor, not a forecast.

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#### 8.4.4 AI-enabled risk sensing

The genuine capability is **detection at scale**: anomaly detection across cost, schedule, quality and supplier data; pattern recognition against historical projects; and sentiment or volume signals in issue logs and correspondence that precede formal escalation. Used well, this shortens the detection half of 8.4.1's fast-detection lever, which is where its value actually lies.

The honest boundaries: models detect **patterns like the past**, so novel risk is invisible to them; they produce **false positives** that erode attention if unfiltered; and a **calibration record is mandatory** before their alerts influence decisions (Domain 6, KA 6.4; Domain 14). The failure mode to watch for is not a wrong alert but **displaced judgment**: a team that stops thinking about risk because a dashboard is watching. The register's owners remain human, the analysis stays challengeable, and the principle holds: AI proposes; the professional verifies, decides and remains accountable.

"Tune the thresholds" is the standard remedy for false positives, and it is incomplete advice because it does not say **which way**. The direction is decided by the base rate and by the ratio of what a caught problem saves to what an alert costs to investigate: both of which a programme can measure, and neither of which is a matter of opinion.

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#### WORKED EXAMPLE

##### Worked example 8.4.4 — which way to tune Meridian's risk monitor.

1. **Setup.** Meridian runs an anomaly monitor over the weekly rollout data from its **40** clinics. In a given clinic-week a genuine emerging problem is present with probability **0.01**: the measured base rate over the first two quarters, i.e. **0.4** genuine problems a week across the estate. Caught early, a problem is remediated for USD 4,000; caught late it costs USD 26,000, so catching one **saves USD 22,000**.

Investigating an alert costs **USD 900**: half a day of a delivery manager. The vendor offers two calibrations: **A, sensitive** (sensitivity 0.85, specificity 0.90) and **B, specific** (sensitivity 0.60, specificity 0.98).

2. **Formula.** With  $N$  clinic-weeks, base rate  $b$ , sensitivity  $s$  and specificity  $k$ : true positives =  $Nbs$ ; false positives =  $N(1-b)(1-k)$ ; alerts = their sum; precision =  $TP \div \text{alerts}$ . Net value =  $TP \times \text{saving} - \text{alerts} \times \text{investigation cost}$ . An alert is worth investigating while  $\text{precision} \times \text{saving} \geq \text{cost}$ , so the breakeven precision is  $\text{cost} \div \text{saving}$ . Setting the two configurations' net value equal gives the crossover base rate.
3. **Substitution.** A:  $TP = 40 \times 0.01 \times 0.85$ ;  $FP = 40 \times 0.99 \times 0.10$ . B:  $TP = 40 \times 0.01 \times 0.60$ ;  $FP = 40 \times 0.99 \times 0.02$ . Breakeven precision  $900/22,000$ . Net value as a function of  $b$ :  $A = 748,000b - (30b + 4) \times 900 = 721,000b - 3,600$ ;  $B = 528,000b - (23.2b + 0.8) \times 900 = 507,120b - 720$ .

#### 4. Result.

|                                             | TP/<br>WEEK | FP/<br>WEEK | ALERTS | PRECISION      | MISSED/<br>WEEK | NET<br>VALUE/<br>WEEK   |
|---------------------------------------------|-------------|-------------|--------|----------------|-----------------|-------------------------|
| <b>A</b><br>sensitive,<br>at $b = 1.0$<br>% | 0.3400      | 3.9600      | 4.3000 | <b>7.91 %</b>  | 0.0600          | <b>USD<br/>3,610.00</b> |
| <b>B</b> specific,<br>at $b = 1.0$<br>%     | 0.2400      | 0.7920      | 1.0320 | <b>23.26 %</b> | 0.1600          | <b>USD<br/>4,351.20</b> |
| A<br>sensitive,<br>at $b = 7.5$<br>%        | 2.5500      | 3.7000      | 6.2500 | 40.80 %        | 0.4500          | USD<br>50,475.00        |
| B specific,<br>at $b = 7.5$<br>%            | 1.8000      | 0.7400      | 2.5400 | 70.87 %        | 1.2000          | USD<br>37,314.00        |

The breakeven precision is  $900/22,000 = 4.0909\%$ . The crossover base rate is  $(3,600 - 720)/(721,000 - 507,120) = 1.3465\%$ : below it the **specific** configuration wins, above it the **sensitive** one. Configuration A stops paying at all below a base rate of **0.4993 %**; B keeps paying down to **0.1420 %**. Over Meridian's 26-week rollout at its measured 1.0 % base rate, B returns **USD 113,131.20** against A's **USD 93,860.00** (a difference of **USD 19,271.20**) at the cost of **2.60** more missed problems across the rollout (4.16 against 1.56). 5. **Interpretation.** Three results, and they are not the ones the received wisdom predicts.

**A monitor whose alerts are wrong 92.09 % of the time still pays.** At configuration A's 7.91 % precision the monitor returns USD 3,610 a week, because precision only has to clear  $\text{cost} \div \text{saving} = 4.0909 \%$ , and 7.91 % is **1.9328 times** that threshold. So "we get too many false alarms" is not, by itself, an economic argument for anything. It is an **attention** argument, and attention is the resource the calibration should be protecting; the arithmetic makes clear that the choice between A and B is about how a scarce delivery manager's time is spent, not about whether the monitor is worth running.

**The direction of tuning depends on the base rate, and it flips.** At Meridian's 1.0 % base rate the specific configuration is better by USD 741.20 a week; at a 7.5 % base rate the sensitive one is better by USD 13,161 a week. The crossover, **1.3465 %**, is uncomfortably close to Meridian's measured 1.0 %, which means the recommendation is **fragile** and must be presented as such: a modest deterioration in the estate would reverse it. Two governing quantities, one ratio, one crossover; that is the whole of it, and it is why calibration is a measurement duty rather than a preference. The corollary for practice is that the **base rate must be measured before the threshold is set**, and re-measured when the delivery environment changes, which is the calibration record the paragraph above requires, now with a stated purpose.

**And the expected-value answer can be overruled, for a nameable reason.** B wins at Meridian's base rate by missing more: **4.16** genuine problems across the rollout against A's 1.56. If a missed problem is a USD 22,000 remediation, that trade is already priced and B is right. If any single missed problem could be a clinical-safety event or an irreversible data loss, then 8.3.1's rule binds (**existential exposure is not managed by its expected value**), and the sensitive configuration is correct regardless of the USD 19,271 it forgoes. The professional obligation is to state which regime applies before quoting the arithmetic, and the failure to do so is how an optimisation becomes an incident. Two cautions on the model. Sensitivity and specificity are themselves estimates from a calibration sample and drift as the estate changes (Domain 14, 14.A.2 on revalidation). And the arithmetic assumes alerts are independent draws; a monitor that fires repeatedly on one underlying condition inflates the alert count without adding information, which is the correlation problem of 8.A.1 appearing in the detection layer.

KEY TERMS — KA 8.4

| TERM       | MEANING                                                                                               |
|------------|-------------------------------------------------------------------------------------------------------|
| Resilience | Capability to absorb unidentified risk; buffers, optionality, modularity, redundancy, fast detection. |

| TERM                               | MEANING                                                                                                                                                                                                                                                                                            |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reference-class forecasting</b> | Estimating from comparable completed projects to counter optimism bias.                                                                                                                                                                                                                            |
| <b>Pre-mortem</b>                  | Imagining failure before committing, to license dissent and surface missed risk.                                                                                                                                                                                                                   |
| <b>Escalation of commitment</b>    | Continuing because of sunk cost; countered by forward-looking framing.                                                                                                                                                                                                                             |
| <b>Displaced judgment</b>          | Ceasing to think because a tool is watching.                                                                                                                                                                                                                                                       |
| <b>Outturn ratio</b>               | Final cost ÷ approved cost for a comparable completed project; the unit of a reference class.                                                                                                                                                                                                      |
| <b>Consequence threshold</b>       | A point where the consequence function steps rather than scales; where resilience is bought.                                                                                                                                                                                                       |
| <b>Base rate</b>                   | The prevalence of the condition a monitor looks for; with the saving-to-investigation ratio it decides how a monitor should be tuned.                                                                                                                                                              |
| <b>Precision</b>                   | True alerts ÷ all alerts; worth investigating while precision exceeds investigation cost ÷ saving.                                                                                                                                                                                                 |
| <b>Crisis declaration</b>          | The named role's act that changes stated authorities (a stop-work authority, a stated fund against a named reserve, a standing cadence), and which someone other than the crisis lead stands down.                                                                                                 |
| <b>Notification map</b>            | The one-page record, written before any incident, of which external notifications could fall due, who determines that they do, to whom, within what period and who may make them. Whether any obligation applies is for the relevant authority and qualified counsel, never for the project alone. |
| <b>Preserve rather than tidy</b>   | From declaration, working papers, logs, messages and superseded drafts are retained as they are; the mess is the evidence that the account was reasonable on what was known.                                                                                                                       |
| <b>Propagation rate</b>            | The weekly cost of a materialised risk continuing to contaminate downstream work while a decision is pending; the term that makes early notification arithmetically obvious and is usually omitted.                                                                                                |

## SAMPLE MCQS — KA 8.4

**MCQ 8.4-A** [8.4.2 · Application] A programme's estimates have run 25 % low across four comparable past projects. The most effective countermeasure is:

- A. instruct estimators to be more careful
- B. reference-class forecasting — estimate from the distribution of comparable completed projects rather than from the plan CORRECT
- C. add a 25 % contingency and continue the same process
- D. escalate to the sponsor for a larger budget

*Rationale:* Optimism bias is systematic, so the fix is methodological (8.4.2). A relies on exhortation against a structural effect; C treats the symptom while preserving the cause; D funds it without correcting it.

**MCQ 8.4-B** [8.4.1 · Analysis] A project has been optimised to eliminate all float and buffers. Its risk position is:

- A. improved — waste has been removed
- B. degraded — it now has no capacity to absorb the risks not in the register, and resilience is what covers those CORRECT
- C. unchanged, provided contingency is funded
- D. improved, since efficiency reduces exposure time

*Rationale:* Resilience assumes register incompleteness (8.4.1); removing all absorption capacity maximises fragility. C confuses money with time and capacity, funded contingency cannot buy back a schedule with nowhere to move.

**MCQ 8.4-C** [8.4.4 · Analysis] An AI monitor has raised 40 alerts this month, 3 of which mattered. The correct response is:

- A. disable the monitor
- B. tune thresholds and require a calibration record, because unfiltered false positives erode the attention the tool exists to direct CORRECT
- C. investigate all 40 equally
- D. accept the ratio as inherent to anomaly detection

*Rationale:* False positives consume the scarce resource (attention) that detection is meant to focus (8.4.4). A discards real capability, C guarantees the erosion, D abandons the calibration duty.

**MCQ 8.4-D** [8.4.3 · Recall] In the first hour of a crisis, the leader's priority order is:

- A. communicate, then investigate, then stabilise
- B. stabilise and secure safety, establish facts, then decide with a clock and communicate early CORRECT
- C. establish blame, then stabilise
- D. wait for complete information before acting

*Rationale:* Stabilisation and factual grounding precede decisions, which are taken against a clock (8.4.3). D is the failure the sequence exists to prevent; C is never a first-hour activity.



**MCQ 8.4-E** [8.4.1 · Analysis] A resilience measure costs USD 18,000, is needed with probability 0.25 and saves five weeks of a USD 14,280-per-week delay. Its breakeven probability is 25.2101 %. The correct professional conclusion is:

- A. reject it — the expected loss is USD 150
- B. the breakeven lies inside the estimating error of the probability, so expected value cannot decide it; decide on the tail and on survivability CORRECT
- C. accept it — a resilience measure is always worth buying
- D. re-estimate the probability to three decimal places and re-run the comparison

*Rationale:* No one estimates a capacity failure to a fifth of a percentage point, and the tail comparison is decisive at roughly 10 to 1 (8.4.1). A treats a noise-level difference as a result; C abandons the arithmetic entirely; D is false precision. The input cannot support the digits.

**MCQ 8.4-F** [8.4.4 · Application] A monitor's alerts are correct 7.91 % of the time. Investigating one costs USD 900 and catching a genuine problem saves USD 22,000. Investigating every alert is:

- A. wasteful — over 92 % of investigations find nothing
- B. worthwhile: the breakeven precision is  $900/22,000 = 4.09\%$ , so 7.91 % returns 1.93 times the investigation cost CORRECT
- C. worthwhile only if precision exceeds 50 %
- D. impossible to assess without the monitor's sensitivity

*Rationale:* An alert is worth investigating while precision exceeds cost ÷ saving (8.4.4). A counts the failures rather than valuing the successes, the arithmetic's whole point; C invents a threshold; D confuses what is needed to compare two calibrations with what is needed to decide whether to investigate a given alert.

**MCQ 8.4-G** [8.4.2 · Analysis] A bottom-up register gives a P80 contingency of USD 203,785 while a reference class of twelve comparable completed projects gives a P80 uplift of USD 964,800. The professional treatment is:

- A. add them, giving USD 1,168,585 of required funding
- B. treat them as two estimates of one quantity: use the reference class to challenge the total, attribute part of the gap to unidentified risks and the rest to optimism and scope growth, and hold the difference as management reserve CORRECT
- C. discard the reference class, since the register is specific to this project
- D. discard the register, since the reference class is evidence and the register is judgment

*Rationale:* The reference class already contains the register's risks, so adding double-counts (8.4.2). C loses the only measurement of what the register cannot see; D

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discards the only basis for response and ownership: the reference class names no risk, no owner and no action.

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SELF-CHECK — KA 8.4

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1. *Why is resilience not a substitute for quantification, and vice versa?* — Quantification handles named risks; resilience absorbs the unnamed. Each fails at the other's job.
  2. *What is a pre-mortem for?* — To license dissent before commitment and surface risks a workshop suppresses.
  3. *What is AI risk sensing's structural blind spot?* — Novel risk: it detects patterns like the past.
  4. *Where is resilience bought, expressed as a test?* — Where the consequence function has a step in it: find the thresholds, then ask what recovery time clears them.
  5. *Why must a bottom-up contingency and a reference-class uplift never be added?* — They estimate the same quantity; the reference class already contains the register's risks.
  6. *What two quantities decide whether a monitor should be tuned sensitive or specific?* — The base rate and the ratio of the saving from a caught problem to the cost of investigating an alert.
  7. *What does a crisis declaration actually change?* — Stated authorities, not urgency: a named stop-work authority, a stated fund against a named reserve with the draw protocol expressly modified rather than suspended, and a standing cadence. The declaring role or its superior stands it down; the crisis lead does not (8.4.3).
  8. *What is a notification map, and when is it written?* — A one-page record of which external notifications could fall due, who determines that they do, to whom, within what period and who may make them, written before any incident and held with the risk register. Whether any obligation applies is a question for the relevant authority and qualified counsel.
  9. *What is the run rate in Worked example 8.4.3, and why is it the number to lead with?* — **USD 23,780 a week**, the propagation rate plus the cost of delay. The pack and the hold are incurred either way and cancel from the comparison, so drift is the whole decision, and the pack is repaid by **1.0934 weeks** of avoided drift. The arithmetic prices drift; it never decides whether a notification is due.
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## Advanced topics — Domain 8

### 8.A.1 Correlation and why aggregate risk is worse than it looks

Independence made 8.2.4's arithmetic tractable and understated the answer; worked example 8.2.4b prices the understatement on Auriga's own register,  $\sigma$  rising by a factor of 1.3471 and a P80 reserve degrading to 73.4 % on one shared subcontractor. This topic takes the organisational form of the same error, which is larger and less visible.

The mechanism is worth stating once in its general form. When risks share a driver (one supplier, one technology, one regulator, one labour market) they move together, and the correlation term  $2 \sum_{i < j} \rho_{ij} \sigma_i \sigma_j$  grows with the *number of pairs*,  $k(k-1)/2$ , while the independent term grows only with  $k$ . **Concentration therefore hurts quadratically**: doubling the number of risks on one driver roughly quadruples the variance it contributes. That is the arithmetic behind the observation that a bad month becomes a crisis, and it is why counting entries is not a measure of exposure.

Three practical implications follow. Identify **common drivers** explicitly and give each a column in the register, so that grouping by driver is a sort rather than a project. Model correlation in simulation rather than assuming it away, and disclose the coefficients as the judgments they are. And where a driver is identifiable, prefer to **model the driver itself as a single risk with a structural response** (8.3.1) over modelling its symptoms as correlated line items: a correlation coefficient is a poor description of a subcontractor that either performs or does not, and it understates the extreme tail where all the symptoms occur together.

The organisational version of this error is believing a portfolio of projects is diversified when every one depends on the same scarce engineering resource, and it is the subject of Case study B, where the arithmetic is done.

### 8.A.2 Schedule risk analysis and the merge-bias trap

Applying this domain's methods to Domain 6's network produces **quantitative schedule risk analysis**: three-point durations (KA 6.4.3) simulated across the logic to yield a completion distribution and, valuably, a **criticality index** (how often each activity lands on the critical path across simulations), which is more decision-useful than a single deterministic critical path. The trap it exposes is **merge bias**: where paths converge (Domain 6's node E), the merged event waits for the *latest* predecessor, so the probability of being on time is the product of the paths' probabilities, not the best of them. Two paths each 80 % likely to meet a date give the merge point 64 %. Deterministic CPM cannot see this, which is why convergence points are systematically optimistic and why Domain 6 flagged them for attention.

That much is standard. What is usually left out is that the simple product is itself wrong when the paths share predecessors (as converging paths almost always do),

and that merge bias **fades as the date becomes less aggressive**, so quoting it without a date is meaningless. Both are worth working on the real network.

#### WORKED EXAMPLE

##### Worked example 8.A.2 — merge bias at Auriga's node E, three ways.

- Setup.** Domain 6's network reaches installation E through two paths: A–B–C (mobilise, design, procure hardware) and A–B–D (mobilise, design, civil and cabling). Deterministic durations put A–B–C at 16 weeks and A–B–D at 15, so  $ES(E) = 16$  with one week of float on D. The estimating team supplies three-point durations:

| ACTIVITY                   | O   | M | P   | $T_e = (O+4M+P) / 6$ | $\Sigma = (P-O) / 6$ |
|----------------------------|-----|---|-----|----------------------|----------------------|
| A Mobilise                 | 1.5 | 2 | 3.5 | 2.166667             | 0.333333             |
| B Detailed design          | 5   | 6 | 10  | 6.500000             | 0.833333             |
| C Procure control hardware | 6   | 8 | 14  | 8.666667             | 1.333333             |
| D Civil and cabling works  | 5   | 7 | 12  | 7.500000             | 1.166667             |

What is the probability that E can actually start in week 16? 2. **Formula.** Path  $t_e = \Sigma \text{ activity } t_e$ ; path  $\sigma = \sqrt{(\Sigma \sigma_i^2)}$  (variances add). Probability of meeting a date T on one path =  $\Phi((T - t_e)/\sigma)$ . For a merge of **independent** paths, multiply. For paths sharing predecessors, decompose:  $start_E = AB + \max(C, D)$ , and condition on the shared part,  $P = \int f_{AB}(t) \cdot \Phi((T-t-t_eC)/\sigma C) \cdot \Phi((T-t-t_eD)/\sigma D) dt$ . 3. **Substitution.** A–B–C:  $t_e = 2.166667 + 6.5 + 8.666667$ , variance  $0.111111 + 0.694444 + 1.777778$ . A–B–D:  $t_e = 2.166667 + 6.5 + 7.5$ , variance  $0.111111 + 0.694444 + 1.361111$ . Shared A+B:  $t_e = 8.666667$ ,  $\sigma = \sqrt{0.805556}$ . 4. **Result.** Path A–B–C  $t_e$  **17.3333** weeks,  $\sigma$  **1.6073**; path A–B–D  $t_e$  **16.1667**,  $\sigma$  **1.4720**; shared A+B  $\sigma$  **0.8975**.

| READING OF THE SAME NETWORK                    | P(E CAN START IN WEEK 16)         |
|------------------------------------------------|-----------------------------------|
| Deterministic CPM                              | 100 % (a date, not a probability) |
| Critical path alone                            | <b>20.34 %</b>                    |
| Independent-path product (textbook merge bias) | <b>9.25 %</b>                     |
| <b>Correct, sharing A and B</b>                | <b>13.16 %</b>                    |

And the effect decays with the date:

| TARGET FOR ES (E)                        | DOMINANT PATH<br>ALONE | CORRECT<br>MERGE | REDUCTION FROM<br>THE MERGE |
|------------------------------------------|------------------------|------------------|-----------------------------|
| Week 16                                  | 20.34 %                | 13.16 %          | <b>35.31 %</b>              |
| Week 17                                  | 41.79 %                | 34.33 %          | 17.85 %                     |
| Week 18                                  | 66.08 %                | 61.53 %          | 6.90 %                      |
| Week 18.686 (the<br>dominant path's P80) | 80.00 %                | <b>77.61 %</b>   | 2.99 %                      |
| Week 20                                  | 95.15 %                | 94.80 %          | <b>0.37 %</b>               |

The date at which E's start is 80 % likely is **week 18.8093** (a buffer of **2.8093 weeks** over the deterministic 16) against **week 18.6860** if only the dominant path is considered. So ignoring the merge understates the buffer by **0.1233 weeks**, which at Auriga's cost of delay of USD 45,000 per week is **≈ USD 5,550**: quoted to the nearest fifty deliberately, because it is the difference between two numerically integrated dates built on PERT's  $\sigma = (p - o)/6$  convention, and a figure stated to the cent would claim a precision the method has not got. 5. **Interpretation.** Four conclusions, and the last two are the ones that stop merge bias being misapplied.

**The deterministic date is not a forecast.** Week 16 has a **13.16 %** chance of being met. Nothing is wrong with the network; the date is the *mode* of a distribution and is being read as a commitment (Domain 6, KA 6.4.3 makes the same point on a single activity). The instrument that fixes it is a buffer chosen at a stated confidence, exactly as contingency is (8.2.4): the schedule version of the same discipline, and the reason a project with a P80 cost reserve and a Po schedule date is internally inconsistent.

**The textbook product overstates the bias.** Multiplying 20.34 % by 45.49 % gives 9.25 %, but the two paths share A and B: a good mobilisation and design help *both* paths simultaneously, so the failures are correlated and the true figure is **13.16 %**, 42.2 % higher than the naive product. The ordering is an invariant worth remembering: **naive product ≤ correct merge ≤ dominant path alone**. A leader given a merge-bias number should ask which of the three it is; a schedule simulation that honours the logic computes the middle one automatically, while a spreadsheet multiplying path probabilities computes the lower bound and calls it the answer.

**Merge bias is a property of a date, not of a network.** At week 16 the merge removes 35.31 % of the probability; at week 20 it removes 0.37 %. The reason is structural: as the date becomes generous, the dominant path's probability approaches 1 and the secondary path's approaches 1 faster, so the product approaches the dominant path's own figure. **Convergence points are dangerous at aggressive dates and almost irrelevant at conservative ones**, which is why the buffer understatement above is only 0.12 weeks even though the

week-16 distortion is severe. Quoting "merge bias adds X weeks" without the confidence level attached is therefore not a result.

**And the practical consequence is where to look, not how much to add.** The decision-useful output of this analysis is not the 2.81-week buffer but the **criticality index**: D reaches E only 1.17 weeks behind C on expectation with a  $\sigma$  of 1.47, so D lands on the critical path in a substantial minority of futures despite showing one week of float deterministically. That is what makes it worth monitoring, and it is the same finding Domain 6 reached by re-running its passes after a crash, here obtained without crashing anything. Two cautions. PERT's  $\sigma = (p - o)/6$  is a convention rather than a derivation and understates spread on strongly skewed estimates, so the probabilities above should be read to the nearest point. And the arithmetic treats C and D as independent given A and B, which fails if the same site conditions or the same subcontractor drive both: in which case 8.2.4b's correlation treatment applies to the schedule exactly as it did to the cost aggregation.

### 8.A.3 The reviewer's risk eye

Invariants worth testing in an hour: every entry has cause, event and consequence and a named owner; no occurred risks parked as high-probability entries; opportunities present in credible proportion;  $\Sigma EMV$  reconciles to the register; contingency stated at a named confidence level with its independence assumption disclosed; contingency consumption trended against risk retirement, not time; responses costed against  $EMV$  reduction, with secondary risks recorded; existential risks handled structurally rather than by  $EMV$ ; early-warning indicators defined for the top items; and at least one decision this month traceable to the register. Failure of the last one matters most: it means the process is documentation, not decision support.

Seven further tests, each of which this domain has now made arithmetic, and each of which takes one division:

- **Does every total belong to one objective?** A "total exposure" mixing cost and benefit impacts is compared with nothing and can be funded from nothing (8.2.2b).
- **Is total exposure stated against the tolerance, not the budget?** Meridian's 4.96 % of approved cost is 99.17 % of the room (8.2.2b).
- **Does the confidence level follow from an appetite statement, or from habit?** If the latter, write down the appetite that the inherited level implies and put it to the sponsor (8.3.2).
- **Is the reserve adequate against the register still open?** Remaining  $\div$  recomputed requirement; a ratio below 1.00 is a trigger, and Auriga's was 0.9037 while the naive drawdown test passed (8.3.2b).

- **Have retired risks released their variance share, not just their EMV?** R4's retirement released USD 97,239 against an EMV of 60,000 (8.3.2b).
- **Is there a quantification threshold, written down?** A screen with no threshold has replaced quantification rather than ordering it, and one matrix cell can conceal a 14.6-fold span of expected values (8.2.1).
- **Has register completeness been estimated rather than assumed?** Two independent identification methods and their overlap give a floor on what is missing, and that floor, not a percentage convention, is the basis for management reserve (8.1.3).

## Industry variations — Domain 8

- **Construction and infrastructure.** Ground conditions, weather and permits dominate, and they are **shared drivers** rather than independent risks: one wet season moves every external activity together, so the correlation term of 8.A.1 is the dominant part of the variance and an independence-based reserve is furthest from the truth here. Quantitative schedule risk analysis is contractually expected on major programmes, which makes 8.A.2's distinction (naive product against correct merge) a commercial point and not an academic one. Risk allocation is largely executed through contract terms (Domain 10), so the counterparty arithmetic of 8.3.1 is the profession's central skill in this sector: a transfer to a thinly capitalised subcontractor is priced at its breakeven default probability or it is not priced at all.
- **Energy and resources.** Commodity price and currency exposure sit alongside delivery risk and are managed with financial instruments as well as project responses (PFL-AI Domains 3, 11). Two consequences for this domain's methods: price risk is **continuous** rather than binary, so the Bernoulli variance of 8.2.4 is the wrong model and simulation over a price distribution is the right one; and because hedges are impact levers rather than probability levers, they release reserve at the favourable end of 8.2.4c's 1.5-to-3 range, which is a large part of why they are bought.
- **Technology and digital.** Requirement volatility and integration risk dominate; resilience is bought through modularity, staged release and reversibility rather than buffers, so 8.4.1's levers apply in a different mix. The reason is 8.2.4c's identity: modularity and reversibility are **impact** levers, and impact levers are what release contingency. Reversibility also makes information cheap to buy, so the value-of-information arithmetic of 8.2.3 favours acting and observing over surveying and deciding: the opposite conclusion to the utilities case study, from the same tree with a different reactive cost.
- **Pharmaceutical and regulated development.** Low-probability/high-consequence events and regulatory outcomes make structural treatment (8.3.1) the norm; risk files are auditable artefacts, not internal management tools. The

characteristic quantitative failure is presenting an **EMV**-optimal response for an exposure that a regulator treats as unacceptable at any probability: the same error 8.4.4 names in monitor tuning, where a missed detection cannot be traded against investigation cost. The correct output is the optimum among admissible options, which is a smaller set than the arithmetic alone would suggest.

- **Public programmes.** Optimism bias is explicitly recognised in appraisal guidance, and reference-class adjustment may be mandated; where such guidance applies, the arithmetic of 8.4.2 is a submission requirement rather than an option, and the professional skill is the reconciliation, never adding a mandated uplift to a bottom-up contingency. Political and reputational risk carries weight that no **EMV** captures well, and it typically has the **step consequence function** of 8.4.1: a threshold crossed rather than a cost incurred, which is why public programmes buy resilience that looks uneconomic on expected value.
- **Financial services and operational change.** Regulatory deadlines make the consequence function almost purely a step, so 8.4.1's threshold test governs and buffer economics dominate expected-value economics. Detection is heavily instrumented, which puts 8.4.4's crossover arithmetic at the centre, and it lands the other way. Base rates for genuine control failures are very low, which favours the specific configuration, but the cost of an escaped failure is very high, which favours the sensitive one, and the second effect wins because the crossover base rate falls roughly in proportion to the saving-to-investigation ratio. On Meridian's two calibrations, raising that ratio tenfold (an escaped problem costing USD 220,000 rather than 22,000) moves the crossover from **1.3465 %** to **0.1313 %**, so the sensitive configuration becomes correct at base rates ten times lower. Same two quantities, opposite answer, and it is the ratio rather than the rarity that decides.

## Case study — Domain 8: the survey Auriga did not commission (utilities)

**Situation.** At planning, Auriga's team considered a USD 25,000 ground survey ahead of the civil works. The register carried R2 (ground conditions worse than surveyed, p 0.50, impact USD 180,000), and the survey was cut in a cost-reduction pass as "nice to have". The decision was not analysed; it was traded against a round-number savings target.

**What happened.** At week 13 the contamination was found. The reactive response cost more than the planned one would have, a second civil crew at USD 35,000, a fast-track carrying  $0.20 \times 60,000 = 12,000$  of expected rework, and the exposure to a USD 45,000-per-week client penalty window that made the whole recovery urgent (Domain 6's case study; Domain 7's forecast consequence, **EAC** USD 4.2m against **BAC** 4.0m).



**The analysis, done afterwards.** On the decision-tree arithmetic of 8.2.3 (0.40 probability at the time of the decision, USD 300,000 reactive versus USD 90,000 planned) the survey was worth **USD 59,000** against its USD 25,000 price. Cutting it saved 25,000 and cost the project substantially more, and the register had contained the information needed to see that. Two further figures from the same tree show how far from marginal the call was: the survey would still have paid at any price up to **USD 84,000 (3.36 times** what it was quoted at), and it would only have stopped paying had the team's belief in contamination fallen below **11.90 %**, against the 0.40 the register itself carried. The cost-reduction pass therefore did not make a close judgment badly; it made a decision whose arithmetic was not close in either direction, without doing the arithmetic.

**What the domain teaches here.** Two things, both uncomfortable. **A cost-reduction pass that does not consult the register is a risk decision taken blind:** savings targets and risk exposure must be traded explicitly. And **the register was not the failure; its irrelevance to the decision was** (8.3.3's test: did a decision change because of it?). The corrective adopted afterwards was procedural: any proposed saving above a threshold must state which register entries it affects and by how much.

### Case study B – Domain 8: the diversified portfolio that wasn't (technology programme)

**Situation.** A transformation programme's board reviewed 22 project-level risks spread across six workstreams and concluded exposure was well diversified. The register held **14** risks each assessed at  $p$  0.25 with an impact of **USD 60,000**, and **8** each at  $p$  0.20 with an impact of **USD 80,000**: a total **EMV** of **USD 338,000** in which no single item exceeded **4.73 %** of the aggregate. Contingency was set at **USD 380,000**: the mean plus a **12.43 %** margin, assuming independence. On that assumption the reserve covers **62.41 %** of futures; the independence-based P80 would have been **USD 449,784**, so even on its own terms the reserve was thin, and nobody had computed that either.

**What happened.** Fourteen of the 22 risks depended on the same six-person integration team. When two members left within a month, **11 of the 14** materialised or worsened *together*, along with 2 of the 8 independent risks: an outcome of  $11 \times 60,000 + 2 \times 80,000 = \text{USD } 820,000$ , or **USD 440,000** beyond the reserve. The programme's recovery required a re-baseline and a management-reserve release.

**The arithmetic, done afterwards.** The concentration was priced three ways, and the ladder is the lesson.

| MODEL OF THE<br>14 SHARED<br>RISKS | $\sigma$ OF THE<br>AGGREGATE | P80            | CONFIDENCE THE<br>380,000 RESERVE<br>BUYS | WHERE THE<br>820,000<br>OUTCOME SITS  |
|------------------------------------|------------------------------|----------------|-------------------------------------------|---------------------------------------|
| Independent (as<br>assumed)        | 132,823                      | 449,784        | 62.41 %                                   | 99.99th<br>percentile (1 in<br>7,026) |
| Pairwise $\rho = 0.6$              | 302,245                      | 592,369        | 55.53 %                                   | 94.46th<br>percentile (1 in<br>18)    |
| One driver, all 14<br>together     | <b>374,823</b>               | <b>653,451</b> | <b>54.46 %</b>                            | 90.08th<br>percentile (1 in<br>10)    |

The mean is **USD 338,000** in all three columns: correlation moved nothing about the average and everything about the answer (8.2.4b). The correlation term at  $\rho = 0.6$  is  $2 \times 0.6 \times 91 \text{ pairs} \times 675,000,000 = \text{USD } 73,710,000,000$  of variance (4.18 times the entire independent variance of the 22-risk register), because 14 risks generate 91 pairs and the correlation term grows with pairs while the independent term grows with entries (8.A.1).

**What the board had actually been told.** That the outcome was a one-in-seven-thousand event. It was a **one-in-ten** event, and the difference between those two statements is a single untested assumption. Note also that the ladder's third rung is the *lowest* confidence figure: modelling the integration team as one risk with an impact of  $14 \times 60,000 = \text{USD } 840,000$  produces the widest distribution of the three, because a perfectly shared driver is the limiting case of correlation. That is the honest model of a six-person team, and it makes the exposure legible in a way fourteen line items never did.

**What was done differently.** The register was restructured by **common driver** rather than by workstream, immediately revealing the concentration; the integration capability was treated as a single structural risk with a structural response (cross-training, a retained partner, staged scope) rather than as fourteen separate **EMV** line items; and contingency was re-derived by simulation with correlation modelled explicitly. The structural response is also the cheaper one on 8.2.4c's logic: cross-training and staged scope are **impact** levers on the driver, and impact levers release reserve at up to three times the rate of probability levers.

**What the domain teaches here.** Counting entries measures paperwork; counting **drivers** measures exposure. A register with many risks and few drivers is concentrated, and independence is the assumption most likely to be both convenient and wrong. The one-sentence test the Executive perspective demands (*which of these risks share a driver?*) would have moved this programme's stated confidence by eight percentage points and its P80 by **USD 203,667** before a single event occurred.

## Executive perspective — Domain 8

What a project leader cannot delegate in this domain:

- **The confidence level, and where it comes from.** Whether contingency is set at P50 or P80 is a risk-appetite decision belonging to governance, and the leader ensures it is chosen, stated and understood, not inherited from a spreadsheet default. The stronger version of the duty is to *derive* it: an appetite statement of the form "no more than a 10 % chance of exceeding by more than 5 %" fixes the level at P90 and the tolerance at a number, and a leader who has never seen that sentence written down should write it and take it to the sponsor (8.3.2).
- **The independence question.** Asking, of any aggregate number, which risks share a driver. It is one sentence and it is where Case study B's programme lost control, worth 7.95 points of stated confidence and USD 203,667 of P80 before anything happened.
- **Which constraint the responses are buying against.** Expected cost and confidence level are different objectives, and the same USD 25,000 buys different amounts of each: on Meridian, an impact lever returned 34.93 % more P90 reduction than a probability lever that removed more EMV (8.2.4c). The leader states which constraint binds before the register's EMV column decides by default.
- **The room, not the budget.** Insisting that total exposure be presented against the objective's tolerance rather than as a percentage of cost: the comparison that showed Meridian's mean consuming 99.17 % of its 5 % tolerance, and that no percentage-of-budget statement would ever have surfaced.
- **The structural exposures.** Existential risks handled by structure, never by EMV, and named personally by the leader, because low probability makes them easy to leave in the register.
- **The register's relevance.** Insisting on the 8.3.3 test: which decision changed because of it this month? A register that changes nothing gets fixed or stopped.
- **The bias countermeasures.** Running the pre-mortem, commissioning independent estimates, framing continuation decisions on remaining cost and benefit; because the leader is the only person positioned to make dissent safe.
- **The AI boundary.** Detection is delegable; judgment is not, and a team that has stopped thinking because a dashboard is watching is the outcome to prevent.

## Calculation exercises — Domain 8

**Exercise 8.1** Risks: A p 0.40 impact 150,000; B p 0.20 impact 500,000; C p 0.60 impact 80,000; D (opportunity) p 0.25 impact (200,000). Compute each EMV and the total exposure. *Solution.* A **60,000**; B **100,000**; C **48,000**; D **(50,000)**. Total USD

**158,000.** Common error: omitting the opportunity, giving 208,000 and overstating required contingency by 31.6 %.

**Exercise 8.2** Using Exercise 8.1's four risks and assuming independence, compute  $\sigma$  and a P80 contingency (mean + 0.8416  $\sigma$ ). *Solution.* Variances:

$0.40 \times 0.60 \times 150,000^2 = 5.40e9$ ;  $0.20 \times 0.80 \times 500,000^2 = 40.0e9$ ;  
 $0.60 \times 0.40 \times 80,000^2 = 1.536e9$ ;  $0.25 \times 0.75 \times 200,000^2 = 7.50e9$ . Total  
**54.436e9**;  $\sigma = \text{USD } 233,315$ ; P80 =  $158,000 + 0.8416 \times 233,315 = \text{USD } 354,358$ . Common error: taking  $\sqrt{(\Sigma\sigma)}$  rather than  $\sqrt{(\Sigma\text{variance})}$ .

**Exercise 8.3** A risk (p 0.30, impact 400,000) can be mitigated for USD 70,000, reducing probability to 0.10. Is the mitigation worthwhile? *Solution.* EMV before  $0.30 \times 400,000 = 120,000$ ; after  $0.10 \times 400,000 = 40,000$ ; reduction **80,000** against a cost of 70,000  $\rightarrow$  **net +USD 10,000, accept** (marginally; and the leader should note that a 400,000 impact may warrant response on survivability grounds regardless). Common error: comparing the mitigation cost against the *impact* rather than the EMV reduction.

**Exercise 8.4** Two independent paths each have a 0.80 probability of meeting a milestone date and converge on it. What is the probability the milestone is met, and what is the effect called? *Solution.*  $0.80 \times 0.80 = 0.64$ . **Merge bias.** The merged event waits for the later path, so convergence points are systematically optimistic in deterministic CPM (8.A.2). Common error: answering 0.80, the best of the two. *Second common error, and the subtler one:* multiplying when the two paths **share predecessors**, which correlates them and makes 0.64 a lower bound rather than the answer, on Auriga's node E the naive product gives 9.25 % where the correct figure is 13.16 %.

**Exercise 8.5** Three risks: X p 0.30 impact 200,000; Y p 0.40 impact 150,000; Z p 0.20 impact 250,000. X and Z depend on the same supplier, with an assessed correlation of  $\rho = 0.6$ ; Y is independent of both. Compute the P80 contingency assuming independence, the P80 with the correlation modelled, and the confidence the independence-based figure actually buys. *Solution.* EMVs **60,000, 60,000, 50,000**; mean **USD 170,000**. Variances  $0.30 \times 0.70 \times 200,000^2 = 8.40e9$ ;  $0.40 \times 0.60 \times 150,000^2 = 5.40e9$ ;  $0.20 \times 0.80 \times 250,000^2 = 1.00e10$ ; total **23.80e9**,  $\sigma$  **USD 154,272.49**, so P80 =  $170,000 + 0.8416 \times 154,272.49 = \text{USD } 299,835.72$ . Correlation term  $2 \times 0.6 \times \sigma_x \sigma_z = 1.2 \times 91,651.51 \times 100,000 = 10,998,181,667.89$ , giving total variance **34,798,181,667.89**,  $\sigma$  **USD 186,542.71** and a corrected P80 of **USD 326,994.34**: **9.06 %** higher. The independence figure of 299,836 sits at  $z = (299,836 - 170,000) / 186,542.71 = 0.6960$  of the correlated distribution, i.e. it is a **75.68 %** reserve, not an 80 % one. *Common error:* adding the two  $\sigma$  values (154,272 + a correlation adjustment) instead of adding variances, or (more damaging),

reporting the reserve as P80 after discovering the shared supplier, which leaves the label attached to the wrong number.

**Exercise 8.6** A risk carries  $p$  0.20 and an impact of USD 300,000. Two responses cost the same: one halves the probability, the other halves the impact. Compare their effect on EMV and on variance, and state the general rule. *Solution.* Base EMV **60,000**; base variance  $0.20 \times 0.80 \times 300,000^2 = 14.40e9$ . Halving the probability: EMV **30,000**, variance  $0.10 \times 0.90 \times 300,000^2 = 8.10e9$ , so **6.30e9** removed. Halving the impact: EMV **30,000**, variance  $0.20 \times 0.80 \times 150,000^2 = 3.60e9$ , so **10.80e9** removed, a factor of **1.714286** more, at identical EMV reduction. The rule: EMV is linear in both  $p$  and  $I$ , so either halving removes the same expected cost; variance is  $p(1-p)I^2$ , so halving the impact always removes 75 % of a risk's variance while halving the probability removes between 25 % and 50 %, giving a ratio between **1.5** and **3** for any  $p \leq 0.5$  (8.2.4c). *Common error:* concluding the two responses are equivalent because their EMV reduction is identical, true, and irrelevant whenever the binding constraint is a confidence level rather than an expected cost.

**Exercise 8.7** A risk has  $p$  0.25 and an impact of USD 400,000. Three responses are offered. **Reduce:** USD 28,000, cutting  $p$  to 0.10, with a secondary risk of  $p$  0.15 and impact 40,000. **Transfer:** USD 26,000 premium recovering 70 % of the impact, with the counterparty unable to pay with probability 0.30 given the event. **Avoid:** USD 150,000, removing the risk, with a secondary risk of  $p$  0.20 and impact 90,000. Rank the four options including acceptance, first ignoring the secondary risks and then including them, and compute the counterparty default probability at which the transfer ties the best alternative. *Solution.* Accept **USD 100,000**. Ignoring secondaries: transfer  $26,000 + 0.25 \times 400,000 \times 0.30 = 56,000$ ; reduce  $28,000 + 0.10 \times 400,000 = 68,000$ ; avoid **150,000** – transfer appears best by 12,000. Including secondaries: reduce  $68,000 + 0.15 \times 40,000 = 74,000$ ; transfer  $56,000 + 0.25 \times 0.30 \times 280,000 = 77,000$ ; avoid  $150,000 + 0.20 \times 90,000 = 168,000$ , so **reduce wins**, and the ranking reversed. Reduce creates **USD 26,000** of value against acceptance, a **26.00 %** improvement. The transfer ties reduce at  $(74,000 - 26,000 - 30,000) / (0.25 \times 280,000) = 25.71$  % of counterparty default. *Common error:* omitting the secondary risks, which here selects the wrong response; and treating the transfer's counterparty exposure as independent of the event, when the disruption that triggers the claim is usually what impairs the counterparty.

**Exercise 8.8** A risk workshop identifies 28 risks; an independent checklist-and-interview review identifies 19; 11 appear on both lists. Estimate the risk population, the number still unidentified and the coverage achieved, cross-check with the Chapman estimator, and state what the answer may and may not be used for. *Solution.*  $\hat{N} = (28 \times 19) / 11 = 532 / 11 = 48.3636$ . Distinct identified

$= 28 + 19 - 11 = 36$ , so **12.3636** risks are estimated unidentified, **25.56 %** of the population, and coverage is **74.44 %**. Chapman:  $(29 \times 20)/12 - 1 = 47.3333$ , giving **11.3333** missing, the two estimators differ by **1.0303**, which is the estimated missing count divided by  $m + 1$  ( $12.3636/12$ ), the identity Toolkit 8.T.4 uses as its divergence test. The workshop alone covered **57.89 %**. The estimate justifies **management reserve and resilience** sized against unidentified exposure and a decision on whether to run a third identification method; it does **not** justify adding **EMV**, because the missing risks have no cause, event, consequence or owner (8.1.3). *Common error*: reading a high overlap as evidence of completeness, a large  $m$  deflates  $\hat{N}$ , so shared blind spots produce reassuring coverage; and adding the estimated missing count to contingency, which funds risks that cannot be responded to or retired.

**Exercise 8.9** A monitor watches 60 sites weekly. A genuine emerging problem is present in a site-week with probability 0.03. Catching one early saves USD 15,000; investigating an alert costs USD 750. Configuration A has sensitivity 0.90 and specificity 0.88; configuration B has sensitivity 0.65 and specificity 0.97. Compute each configuration's weekly net value and A's precision, the breakeven precision for investigating an alert, and the base rate at which the two configurations are equally good. *Solution.* At  $b = 0.03$ , A: true positives  $60 \times 0.03 \times 0.90 = 1.62$ , false positives  $60 \times 0.97 \times 0.12 = 6.984$ , alerts **8.604**, precision **18.83 %**; net  $1.62 \times 15,000 - 8.604 \times 750 = \text{USD } 17,847.00$ . B: true positives **1.17**, false positives  $60 \times 0.97 \times 0.03 = 1.746$ , alerts **2.916**; net  $1.17 \times 15,000 - 2.916 \times 750 = \text{USD } 15,363.00$ , so **A wins by USD 2,484**. Breakeven precision  $750/15,000 = 5.00 \%$ , which A's 18.83 % clears by a factor of 3.77. As functions of the base rate,  $\text{Net}_A = 774,900b - 5,400$  and  $\text{Net}_B = 557,100b - 1,350$ ; equating gives  $b^* = 4,050/217,800 = 1.8595 \%$ . Below that the specific configuration wins; above it the sensitive one. A stops paying below **0.6969 %**, B below **0.2423 %**. *Common error*: choosing the configuration with the higher precision, B's precision is better at every base rate and its net value is worse above 1.86 %; and applying the crossover from one programme to another, since it moves with both the base rate and the saving-to-investigation ratio.

## Practitioner's toolkit — Domain 8

*Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.*

### Toolkit 8.T.1 — Risk register columns that earn their keep

ID · **cause** → **event** → **consequence** statement · affected objective · **common driver** (8.A.1) · **duty-engaged flag**, whether realisation would engage a duty arising under law or regulation, an operating licence or a safety obligation, with who determined that and when (8.3.1); a flagged row leaves the **EMV** comparison before it is run, because the duty is not transferable and acceptance is not available ·

probability · impact (cost and time) · **EMV** · qualitative band (screening only) · response family and description · **response cost and expected EMV reduction** · **secondary risks created** · owner (a person) · early-warning indicator · **notification map reference**, where realisation could bring an external notification (8.4.3) · review date · status (open/retired/**issue**) · retirement evidence. Omit any column and the register loses a specific capability; the two most commonly missing (common driver and response cost) are the two that make it decision-useful, and the duty-engaged flag is the one whose absence lets an arithmetic answer stand where no question was open.

### Toolkit 8.T.2 — Contingency derivation sheet (one per baseline)

Register version and date · **objective the total belongs to**, with the other objectives' aggregations cross-referenced ·  $\Sigma EMV$  · **the objective's tolerance, and  $\Sigma EMV$  as a percentage of it** · variance and  $\sigma$  by risk, ranked by **variance contribution** as well as by **EMV** · **independence assumption stated and challenged** (which risks share drivers, with the assumed  $\rho$  and the P80 both ways) · method (formula approximation or simulation, with correlation treatment) · **the appetite statement the confidence level is derived from**, and who owns it · resulting contingency · **the price of confidence** (cost per percentage point between the adjacent levels) · management reserve separately, with its basis including the **register-completeness estimate** of 8.1.3 · draw protocol reference · and, from the second period onwards, the **adequacy ratio**: remaining reserve ÷ the requirement recomputed on the register still open. A contingency figure without this sheet is a number without a defence; a contingency figure without the last line is a number that was defensible once.

### Toolkit 8.T.3 — Pre-mortem session guide

Before commitment, one hour: (1) state the plan and the commitment about to be made; (2) each participant writes privately for ten minutes, "it is twelve months on and this has failed; why?"; (3) round-robin with no challenge permitted during collection; (4) cluster into causes and identify **common drivers**; (5) convert each into a three-part risk statement with an owner; (6) identify the three cheapest early-warning indicators. Rule: the most senior person speaks last, always.

### Toolkit 8.T.4 — Register-completeness check (one per gate)

A twenty-minute calculation that turns "the register is probably incomplete" into a funded position. (1) Run two identification methods **separately**, by different people, with no sight of each other's output: a workshop and an assumption-analysis review are the standard pair (8.1.3). (2) Reconcile the two lists on **cause-and-event identity**, not wording; record  $n_1$ ,  $n_2$  and the overlap  $m$ . (3) Compute  $\hat{N} = n_1 n_2 / m$  and the Chapman cross-check  $((n_1 + 1)(n_2 + 1)) / (m + 1) - 1$ . The two differ by exactly the estimated missing count divided by  $m + 1$ , so the divergence needs no threshold of its own: read it as a fraction of the missing count, and where it approaches that



count  $m$  is too small to support the estimate and a third method is the answer rather than a third decimal place. (4) Record distinct identified, estimated unidentified and coverage. (5) State the coverage threshold that triggers a further method, and whether it fired. (6) Carry the estimated unidentified count into the management-reserve basis on Toolkit 8.T.2, **never** into contingency. Standing caution to print on the sheet: the missing count is a floor, not an estimate (8.1.3).

### Toolkit 8.T.5 — Response option comparison (one per quantified risk)

One row per option (accept, avoid, reduce, transfer, and for opportunities exploit, enhance, share, ignore) with five columns that must all be filled before a choice is recorded: **response cost** · **residual EMV** ( $p$  and impact after the response, both restated) · **secondary risks created**, each with its own  $p$  and impact and its own owner · **total expected cost** · and **variance after**,  $p(1-p)I^2$  on the residual, because a response is also a reserve decision (8.2.4c). Two lines beneath the table complete it: the **cost at which the ranking flips** to the next-best option, and, for any transfer, the **counterparty default probability at which the transfer ceases to be preferred**, noting whether that default is correlated with the event transferred. A comparison without the secondary-risk column selects the wrong option often enough to be a known defect (8.3.1); one without the flip point cannot be reviewed, because a reviewer cannot tell whether the choice was close.

## Exam preparation — Domain 8

**The traps.** Multiplying and summing ordinal scores as if they were money (8.2.1) · omitting opportunities from total exposure (Exercise 8.1) · setting contingency at  $\Sigma EMV$  and calling it funded (MCQ 8.2-D) ·  $\sqrt{\Sigma\sigma}$  instead of  $\sqrt{\Sigma\text{variance}}$  (Exercise 8.2) · comparing response cost against impact rather than **EMV** reduction (Exercise 8.3) · managing an existential low-probability risk by its **EMV** (8.3.1) · assuming independence without testing common drivers (8.A.1) · answering the best path probability at a merge point (Exercise 8.4) · leaving occurred risks in the register as  $p = 1.0$  · treating a transfer as removing exposure.

**The traps this domain's later topics add.** Adding a correlation adjustment to  $\sigma$  rather than to the variance, and (worse), leaving the "P80" label on a reserve after a shared driver has been found (Exercise 8.5, MCQ 8.2-G) · treating two responses as equivalent because their **EMV** reduction is equal, when one is an impact lever and the other a probability lever (Exercise 8.6) · omitting secondary risks from a response comparison, which reverses the ranking whenever a transfer's counterparty default is correlated with the event (Exercise 8.7, MCQ 8.3-D) · returning only a retired risk's **EMV** when the reserve was sized on a confidence level (MCQ 8.3-E) · testing a part-consumed reserve against elapsed time or the original register instead of the register still open (8.3.2b) · choosing a confidence level by convention when the appetite statement determines it (MCQ 8.3-F) · summing a bottom-up contingency and a reference-class uplift (MCQ 8.4-G) · reading a large overlap between two identifica-



tion methods as evidence of completeness (MCQ 8.1-E) · comparing a monitor's precision against a target instead of against  $\text{cost} \div \text{saving}$ , or choosing the higher-precision calibration without computing net value (MCQ 8.4-F, Exercise 8.9) · multiplying path probabilities at a merge point when the paths share predecessors, and quoting a merge-bias figure without the date it applies to (8.A.2) · treating a breakeven that falls inside the estimating error as a result (MCQ 8.4-E) · and expressing total exposure as a percentage of budget when the governing constraint is a tolerance (8.2.2b).

**The calculations to be able to do under time pressure.**  $EMV$  and  $\Sigma EMV$  with opportunities signed correctly · variance  $p(1-p)I^2$ ,  $\sigma = \sqrt{(\Sigma \text{variance})}$ , and a percentile as  $\text{mean} + z\sigma$  · the correlation term  $2\Sigma p_i\sigma_i\sigma_j$ , and inverting a reserve into the confidence it buys · decision-tree rollback and the value of information, with its breakeven price and breakeven prior ·  $p^*$  for a response: the cost at which acceptance becomes correct · the adequacy ratio · an appetite statement converted into a tolerance and a confidence level ·  $\hat{N} = n_1 n_2 / m$  · PERT  $t_e$  and  $\sigma$ , path  $\sigma$ , and a merge probability both ways · and a monitor's precision against  $\text{cost} \div \text{saving}$ .

### Reflection questions.

1. Take your project's contingency figure: what confidence level does it represent, who chose it, and what independence assumption sits underneath?
2. Group your register by common driver rather than by workstream. How concentrated is it actually?
3. Which decision changed last month because of your register, and if none, is the process decision support or documentation?
4. Write down the appetite statement your confidence level implies ("we accept an X % chance of exceeding by more than Y"), and consider whether your sponsor would sign it.
5. Take your last response decision. What were its secondary risks, and would pricing them have changed the choice?
6. Rank your register by variance contribution rather than by  $EMV$ . Which entry moves furthest, and is it being managed as though it matters that much?

### Domain 8 summary

Risk management earns its place only when it changes decisions. That starts at the sentence level: cause  $\rightarrow$  event  $\rightarrow$  consequence, against a named objective, with opportunities identified as deliberately as threats because defensive framing otherwise buries them. And and it can be tested for completeness rather than assumed complete, two independent methods and their overlap putting a floor under what is missing (Meridian's **11.43** unidentified risks, **78.61** % coverage) and giving management reserve a basis at last. Analysis then progresses from qualitative screening, useful for ordering, invalid as arithmetic, and coarser than it looks, since

one matrix cell conceals an **EMV** span equal to the product of its band ratios (**14.5833** here, **1.8452 times** the step between cells two bands apart), to quantification: **EMV** for each risk (Auriga's register totalling **USD 278,000**, 6.95 % of **BAC**, and USD 314,000 if the opportunity is ignored, an overstatement of 12.95 %), with the total always belonging to **one objective** and always stated against that objective's **tolerance** rather than its budget, which is how Meridian's comfortable-looking 4.96 % of approved cost turns out to consume **99.17 %** of the room the board allowed.

Decision trees price choices and the **value of information** (the survey worth **USD 59,000** against its 25,000 price, still worth buying at any price up to **USD 84,000**, and worthless if it changed no action), and, once the signal is imperfect, the value depends on the **design** of the enquiry as much as its price: Meridian's designed two-clinic pilot was worth **USD 77,136** against the random pilot's **USD 17,947** at identical cost, and needed only an **11.05 %** prior to pay against the random pilot's 21.45 %. Aggregation then converts a register into a **stated confidence**: auriga's P80 of **USD 490,624** from a mean of 278,000 and  $\sigma$  of 252,642, defensible precisely because it says what confidence it buys, unlike a worst-case sum, an **EMV** sum or a 10 % rule of thumb that turns out on enumeration to be a **P68.5** reserve. Two refinements matter more than the base calculation. **Correlation moves the spread and not the mean**: one shared subcontractor takes Auriga's  $\sigma$  to **USD 340,339** and makes that same 490,624 a **73.4 %** reserve, the label failing before the money does. And because variance goes as  $p(1-p)I^2$ , **impact levers release between 1.5 and 3 times the reserve of probability levers at equal EMV cost**, which is why Meridian's second response bought **34.93 %** more P90 reduction for the same USD 25,000 while removing less **EMV**.

Responses are investments judged against the reduction they buy, priced across the families with their **secondary risks** included, because omitting them is not an incompleteness but a selection error: on Auriga's R1 the transfer looks best at 51,600 and is worst-but-one at 64,200 once a counterparty default correlated with the event is priced, and the reduce option wins at 62,000. Existential exposures are handled structurally rather than by averaging. Reserves are sized by aggregation at a confidence level **derived from the appetite statement** rather than inherited ("no more than a 10 % chance of exceeding by more than 5 %" is P90, and on Meridian it is unreachable by any contingency decision) separated by authority, and tested each period against the register still open, which showed Auriga's part-consumed reserve at an adequacy ratio of **0.9037** and a real confidence of **75.65 %** while the conventional drawdown test passed. Retirement releases variance, not **EMV**: R4's closure freed **USD 97,239** against an **EMV** of 60,000.

Beyond what can be named sits resilience, buffers, optionality, modularity, redundancy and fast detection, bought at an efficiency cost that should be chosen rather than stumbled into, and bought specifically where the consequence function has a **step** in it: Meridian's trainer retainer loses USD 150 on expected value, misses its breakeven by two-tenths of a percentage point, and removes **USD 184,776** of tail exposure, which is what decides it. Alongside sit the bias countermeasures (pre-

mortem, independent estimates) and **reference-class forecasting**, whose uplift is never added to a bottom-up contingency but used to challenge it: meridian's class of twelve giving a P80 uplift of **USD 964,800** against a register-based **USD 203,785**, a gap of which about a third is attributable to the unidentified risks the completeness estimate found and the rest to optimism and scope growth that must be named as undecomposable. AI-enabled detection is governed by the same arithmetic: an alert is worth investigating while precision exceeds  $\text{cost} \div \text{saving}$  (**4.09 %** on Meridian, cleared nearly twice over by a monitor wrong 92 % of the time), and the direction of tuning is decided by a crossover base rate (**1.3465 %**), not by preference. Applied to the schedule, the same tools give merge bias its honest form: Auriga's node E has a **13.16 %** chance of its deterministic date, not the 20.34 % of its critical path nor the 9.25 % of a naive product, and the effect fades from a 35.31 % distortion at week 16 to 0.37 % by week 20, so a merge-bias figure without a date is not a result. Crisis leadership closes the domain, and it is an instrument rather than a posture: stabilise, establish facts, decide against a clock and communicate early, under a **declaration** by a named role that changes stated authorities and that someone else stands down, a **notification map** written before any incident because an obligation cannot be discovered on the day, and a contemporaneous decision log kept from the declaration under an instruction to preserve rather than tidy. The drift between an early notification and a late one is priceable in the project's own terms: on Meridian, an avoidable **USD 23,780 a week**, repaying the whole USD 26,000 notification pack in **1.0934 weeks** — arithmetic that funds a fast response and never decides whether a notification is due, which is a question for the relevant authority and for qualified counsel. The same boundary governs response selection: where a duty arising under law or regulation, an operating licence or a safety obligation is engaged, the duty is not transferable and acceptance is not an available response, so the row is flagged and lifted out of the **EMV** comparison before it is run. Independence is the assumption most likely to be convenient and wrong: count drivers, not entries. Domain 9 turns to quality and the assurance that catches what risk analysis missed.

DOMAIN

# 9

## Quality, Assurance and Continuous Improvement

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## Why this domain exists

Domain 5 defined what the project must produce and how acceptance would be tested. Domain 6 built the schedule, Domain 7 the money, Domain 8 the risk. Every one of them assumed something this domain has to supply: **that the work will be right, and that somebody has priced being wrong.**

Quality is the discipline most often reduced to a virtue. Organisations declare a commitment to "the highest quality", write a plan nobody reads, and discover in commissioning that quality is a budget line, a capacity constraint and a schedule driver: all three set months earlier by people who did not know they were setting them. The recurring failure is not indifference. It is the belief that quality is free at the top and infinitely expensive to give up, so the only two available positions are "we do not compromise on quality" and an unadmitted compromise. Neither is a management position, because neither can be costed.

The domain's central claim is that **quality is an optimisation with an interior solution, and the optimum is not zero defects.** Prevention, appraisal, internal failure and external failure move against each other; their sum has a minimum; that minimum sits at a positive escape rate; and where it sits is fixed by one ratio: what an escaped defect costs against what finding it costs. An organisation that cannot compute that ratio is not choosing its quality regime but inheriting one. KA 9.1 computes it. KA 9.2 shows where the money actually goes (into layers of containment whose economics differ by an order of magnitude), and why rework is a *capacity* problem before it is a cost problem, with a schedule consequence no cost-of-quality model contains. KA 9.3 turns to the decisions: accepting work, disposing of nonconformance, and the arithmetic of sampling, where the central and uncomfortable result is that **a passing sample routinely fails to exclude the defect rate at which its own plan is the wrong plan.** KA 9.4 closes on improvement and on the two data problems that now dominate delivery quality: the quality of the data a project runs on, and the quality of the output an AI system hands a professional who must sign for it.

**Learning objectives.** After this domain a candidate can: distinguish quality from grade and both from fitness for purpose, and write a quality objective that is measurable rather than aspirational; decompose the cost of quality into prevention, appraisal, internal failure and external failure and explain why internal failure cost *rises* before it falls; **compute total cost of quality across candidate regimes, locate the interior minimum and state the range of external-failure unit cost over which it holds;** explain why zero defects is not the objective and what would make it one; design a containment chain and compute its escape fraction as the product of layer escape rates; **compute the expected containment cost per introduced defect and use it as the breakeven price of prevention;** show arithmetically why equal money on prevention beats equal money on appraisal, and name the

condition under which it does not; **convert a rework share of capacity into a duration multiplier and a delay cost and explain why the relationship is hyperbolic**; run a formal acceptance decision and select a nonconformance disposition with its authority; **make the cumulative-effect test operable by fixing its threshold source, derivation, period, relatedness test, owner and consequence on breach**, and recognise where a disposition additionally engages an approval body outside the project; assign acceptance and disposition records a class, a custodian and a retention period with the source of that period named; compute what a clean acceptance sample does and does not establish, derive the breakeven defective fraction from the ratio of verification cost to failure cost, and test an acceptance plan for self-consistency; conduct root-cause analysis to the level at which the cause can be removed and price removal against expected recurrence; **compute rolled throughput yield as the product of stage yields, target improvement at the binding step and derive the uniform step yield an end-to-end target requires**; assess data quality across dimensions and compute composite fitness; and specify a verification regime for AI-produced work whose sample size is derived from the consequence of an escaped error rather than from convenience.

**The master worked project.** Project Auriga continues from Domains 6, 7 and 8: the 25-week control-systems upgrade for a regional utility, **BAC USD 4,000,000**, at week 13 showing **CPI** 0.91 and **SPI** 0.92 (Domain 7, KA 7.3), with a **cost of delay of USD 45,000 per week** and a critical path through installation and testing and commissioning at zero float (Domain 6). This domain attaches quality economics to it: a cost of quality of **USD 364,000** at the chosen regime (9.1 % of **BAC**), a containment chain that lets **8 of 80** introduced control-logic defects escape at **USD 12,000** each, a commissioning phase whose 30 % rework share turns a 4-week activity into **4.7619 weeks**, and an acceptance sample of 20 instrument loops that cannot exclude a defect rate of **13.91 %**. Auriga's engineer-week costs **USD 5,225** throughout, from Domain 7's blended engineering rate of USD 130.625 per hour over a 40-hour week.

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## KNOWLEDGE AREA 9.1

### Quality planning

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*Topics: 9.1.1 quality, grade and fitness for purpose · 9.1.2 the cost of quality and its interior minimum · 9.1.3 quality objectives, metrics and tolerances.*

#### 9.1.1 Quality, grade and fitness for purpose

**Definition.** **Quality** is the degree to which delivered work conforms to its stated requirements. **Grade** is the level of capability, feature richness or material specification those requirements ask for. The two are independent, and confusing

them is the commonest conceptual error in delivery: a low-grade product built exactly to a low-grade specification is high quality, and a high-grade product riddled with nonconformances is not. A leader may legitimately reduce grade; that is a scope decision under Domain 5, taken with the sponsor and recorded. A leader may never legitimately reduce quality, because quality is conformance to what was agreed, and reducing it is delivering something other than the thing promised while saying nothing.

**Fitness for purpose** is the third and hardest test: whether the conforming, correctly graded output serves the use it was built for. Domain 5's Case study B (a public-sector system one hundred per cent complete against specification and thirty-four per cent useful) is the canonical failure, and the reason quality management cannot stop at conformance. Conformance is verifiable against a document; fitness for purpose only against a user, an operating context and a benefit measure (Domain 2). A regime that tests only the first will pass work that fails the second with complete documentary integrity.

The reference points are worth naming precisely and describing in the book's own words. The **ISO 9000 family** supplies the vocabulary and the management-system requirements for quality (ISO 9001 being the certifiable requirements standard); it evidences that an organisation has defined and follows processes, not that any deliverable is correct. **Statistical process control** and the **Six Sigma** family of improvement methods supply measurement machinery for repetitive processes; projects borrow their arithmetic (yields, sampling, capability) with care, because a project is not a repeating process and its samples are small. Neither tradition answers the question this domain treats as primary: *how much conformance is it worth buying?*

**Three distinctions to hold.** *Quality assurance* is process-directed and preventive. *Quality control* is product-directed and detective. *Quality improvement* is directed at the causes of the answers the first two give. They have different owners, cadences and costs, and an organisation funding only the second will pay for the same defect indefinitely.

### 9.1.2 The cost of quality and its interior minimum

**Definition.** The **cost of quality (CoQ)** is the total cost incurred because the work might not conform, plus the total cost incurred because it did not. It has four categories, and every serious quality decision is a trade among them:

| CATEGORY          | WHAT IT BUYS             | AURIGA EXAMPLES                                                                                      | BEHAVIOUR AS QUALITY RISES |
|-------------------|--------------------------|------------------------------------------------------------------------------------------------------|----------------------------|
| <b>Prevention</b> | Fewer defects introduced | Design standards, supplier qualification, training, model-based design review, configuration control | Rises                      |

| CATEGORY                | WHAT IT BUYS                      | AURIGA EXAMPLES                                                               | BEHAVIOUR AS QUALITY RISES |
|-------------------------|-----------------------------------|-------------------------------------------------------------------------------|----------------------------|
| <b>Appraisal</b>        | Defects found before the customer | Peer review, factory acceptance test, site acceptance test, inspection, audit | Rises                      |
| <b>Internal failure</b> | Nothing: it is loss               | Rework, retest, scrap, document re-issue, disruption inside the project       | Rises, then falls          |
| <b>External failure</b> | Nothing (loss plus reputation)    | Field defects, warranty attendance, outage credits, client disruption, claims | Falls                      |

The first two are the **cost of conformance**, the last two the **cost of nonconformance**. Two features of that table are routinely got wrong. Internal failure cost is **not monotone**: an organisation that improves testing before it improves engineering finds *more* defects internally and its internal failure cost rises, a movement almost always misread as deterioration, and the commonest reason quality programmes are abandoned in their first period. And external failure is the only category whose unit cost the project does not control, because it is set by the operating context: the same defect costs a trifle in a marketing website and a great deal in a live utility control system.

#### WORKED EXAMPLE

##### Worked example 9.1.2 – Auriga's cost of quality, and where its minimum actually is.

1. **Setup.** Auriga's control-logic and configuration scope can be delivered under five candidate regimes. For each, the estimating team has stated prevention spend, appraisal spend, the defects the regime would allow to be **introduced**, the fraction it would **detect** before handover, and the observed average internal correction cost per defect found (which rises as detection reaches deeper, later layers). An escaped defect in live operation costs Auriga's client **USD 12,000** on average — diagnosis, a site attendance inside an outage window, retest and the contractual service credit. **BAC** is **USD 4,000,000**.

| REGIME             | PREVENTION | APPRAISAL | INTRODUCED | DETECTED | INTERNAL UNIT COST |
|--------------------|------------|-----------|------------|----------|--------------------|
| R0 test at the end | 12,000     | 48,000    | 160        | 75.00 %  | 1,000              |
| R1 basic           | 40,000     | 60,000    | 120        | 85.00 %  | 1,300              |
| R2 planned         | 96,000     | 64,000    | 80         | 90.00 %  | 1,500              |



| REGIME         | PREVENTION | APPRAISAL | INTRODUCED | DETECTED | INTERNAL<br>UNIT<br>COST |
|----------------|------------|-----------|------------|----------|--------------------------|
| R3<br>enhanced | 170,000    | 80,000    | 64         | 93.75 %  | 1,650                    |
| R4<br>maximum  | 285,000    | 100,000   | 50         | 96.00 %  | 1,750                    |

2. **Formula.** Per regime: internal failures = introduced  $\times$  detected; escapes = introduced – internal failures; internal failure cost = internal failures  $\times$  internal unit cost; external failure cost = escapes  $\times$  external unit cost; **CoQ** = prevention + appraisal + internal failure + external failure. Choose the regime minimising **CoQ**.

3. **Substitution.** R2: internal failures  $80 \times 0.90 = 72$ ; escapes 8; internal failure cost  $72 \times 1,500 = 108,000$ ; external failure cost  $8 \times 12,000 = 96,000$ ; **CoQ** =  $96,000 + 64,000 + 108,000 + 96,000$ .

4. **Result.**

| REGIME    | ESCAPES  | PREVENTION | APPRAISAL | INTERNAL<br>FAILURE | EXTERNAL<br>FAILURE | TOTAL<br>CoQ   | %<br>OF<br>BAC |
|-----------|----------|------------|-----------|---------------------|---------------------|----------------|----------------|
| R0        | 40       | 12,000     | 48,000    | 120,000             | 480,000             | <b>660,000</b> | 16.5 %         |
| R1        | 18       | 40,000     | 60,000    | 132,600             | 216,000             | <b>448,600</b> | 11.2 %         |
| <b>R2</b> | <b>8</b> | 96,000     | 64,000    | 108,000             | 96,000              | <b>364,000</b> | <b>9.1 %</b>   |
| R3        | 4        | 170,000    | 80,000    | 99,000              | 48,000              | <b>397,000</b> | 9.9 %          |
| R4        | 2        | 285,000    | 100,000   | 84,000              | 24,000              | <b>493,000</b> | 12.3 %         |

The minimum is **USD 364,000** at regime **R2**, with **8 escaped defects**, not at the lowest attainable defect count. 5. **Interpretation.** Five things follow, and a leader who can state all five commands the quality budget rather than spectating.

**The optimum is interior, and not zero.** R4 costs **USD 129,000** more than R2 to remove six further escapes: **USD 21,500 per escape avoided** against damage of USD 12,000, or **1.79 times** the harm it prevents. The last defects are the expensive ones, so an organisation treating zero defects as the objective is committing to buy protection at roughly twice its value. The honest objective is

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not "no defects" but **"no defect whose prevention costs less than its consequence"**: a computable instruction, and the one this table executes.

**The marginal test is the decision, not the total.**  $R_1 \rightarrow R_2$  costs **USD 60,000** more in conformance and returns **USD 24,600** of internal and **USD 120,000** of external failure, a net **USD 84,600** gain.  $R_2 \rightarrow R_3$  costs **USD 90,000** more and returns **USD 9,000** and **USD 48,000**: a net **USD 33,000** loss. The rule: spend the next increment while its conformance cost is below the failure cost it removes, and stop at the first increment where it is not.

**The range of optimality is what makes the recommendation defensible.**  $R_2$  stays optimal for any external-failure unit cost between **USD 3,540** (below which  $R_1$  wins) and **USD 20,250** (above which  $R_3$  wins), a **5.72-fold** band. Auriga's assumed **USD 12,000** sits **3.39 times** above the lower bound and at **59.3 %** of the upper. That is the sentence for a steering committee, because it answers the only question worth asking about an estimated input: how wrong can it be before the recommendation changes? In a safety-related or heavily regulated context the escaped-defect cost is a multiple of **USD 12,000**: above **USD 20,250** the optimum moves to  $R_3$  and above **USD 60,000** to  $R_4$ . Same arithmetic, different context, different regime, which is the quantitative statement of why quality regimes are not transferable between sectors.

**Internal failure cost rises before it falls, and this must be said in advance.**  $R_0 \rightarrow R_1$  raises it from 120,000 to **132,600** while total **CoQ** falls by **USD 211,400**: better detection finds defects that were escaping, and the cost moves from the external column to the internal one at a fifth of the unit price. A programme reported on internal failure cost alone will look like a failure in its first period, which is where such programmes are cancelled.

**The cautions.** The introduced-defect counts and detection rates are **estimates from comparable work** and must be labelled as such; the conclusion is robust to the external unit cost across a wide band but not to a detection rate that has been asserted rather than measured. The model prices only what is in it: it excludes the **schedule** consequence of rework (KA 9.2.3 computes that separately and it is material), the tail risk of a single catastrophic escape rather than an average one, and any licence or reputational consequence not expressible per defect. And the four categories must be **defined once and counted consistently**, because the easiest way to make a quality programme look successful is to reclassify internal failure as appraisal.

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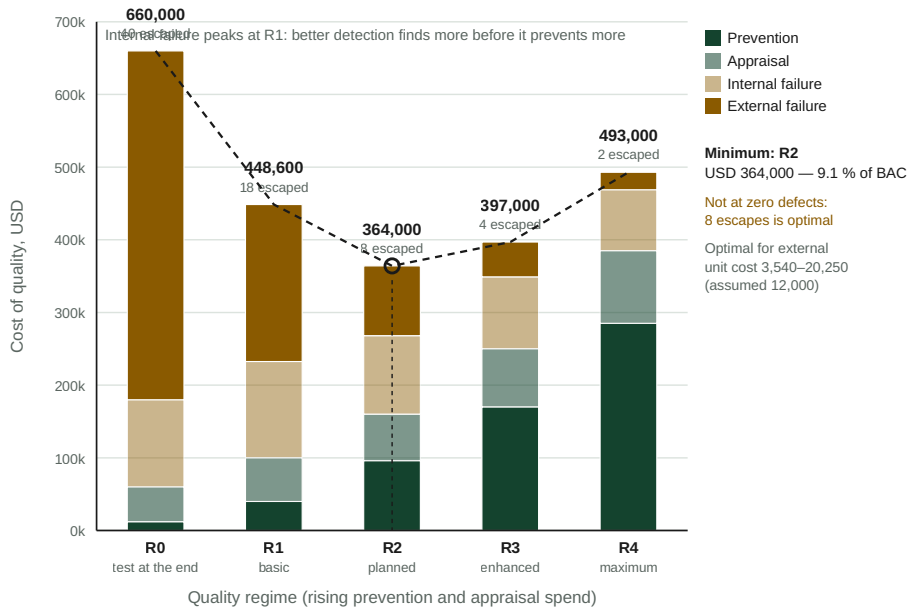


Fig 9.1.1 — Auriga's cost of quality and its interior minimum

### 9.1.3 Quality objectives, metrics and tolerances

**The plan's job.** A quality management plan is the record of five decisions, each with an owner and a number. *What conformance means* for each deliverable class: the acceptance criteria of Domain 5, KA 5.4.2, referenced not restated. *What tolerance applies*, because a specification without a tolerance is untestable and will be argued about at acceptance. *Which containment layers exist*, with their coverage and expected detection rates (KA 9.2). *Who accepts, and on what authority* (KA 9.3). And *what it costs*: the four-category budget of 9.1.2, reconcilable to Domain 7's cost baseline rather than a parallel document.

**Metrics that survive contact with delivery.** A quality metric earns its place if a named person changes a decision when it moves. Four do, on Auriga: **escape rate** (defects reaching the next stage or the customer — the primary outcome measure, and what 9.1.2 optimises); **first-time-right rate** per step and rolled end to end (KA 9.4.2 — the primary process measure); **rework share of capacity** (KA 9.2.3: the primary *schedule* measure, and the one most often absent); and **cost of quality as a percentage of BAC** split four ways, of which Auriga's 9.1 % is the figure the sponsor should know.

**Tolerance and the two errors.** Every tolerance trades rejecting conforming work against accepting nonconforming work. It is therefore the same class of decision as Domain 3's delegation threshold (a certain, recurring, invisible cost against an uncertain, occasional, highly visible one), and organisations reliably optimise against the visible one. The countermeasure is the same: compute both, state the breakeven, and let the accountable person choose knowing the price.

## AI in this KA

**Where it earns its place.** Assembling a four-category cost-of-quality picture from accounting data never coded for it (mapping rework time codes, inspection orders, warranty claims and service credits into the four categories) is a classification task over messy records and the single largest practical obstacle to doing cost-of-quality work at all. Testing acceptance criteria and tolerances for measurability and flagging those that cannot be tested. Sweeping a specification set for missing tolerances. Running the 9.1.2 model across dozens of regime and unit-cost combinations to produce the range-of-optimality band rather than a point estimate.

**Where it must not go.** Setting the external-failure unit cost, which is a judgement about consequence in an operating context and, where safety or licence is involved, not an expected-value judgement at all. Choosing the regime, a risk-appetite decision belonging to the sponsor and, on a safety-related system, to the accountable engineer. And supplying detection rates or introduced-defect counts from plausibility: a model asked for these returns confident, well-formatted numbers with no provenance which then drive a real budget. Where the history does not exist, the correct output is that it does not exist, plus the range over which the conclusion holds.

**Verification, concretely.** Reproduce the whole CoQ table by hand or in a spreadsheet whose formulae are visible (four multiplications and a sum per regime), and state every input's source beside it. Publish the marginal step analysis and the range of optimality, not the point minimum, because those tell a board whether the recommendation is robust or fragile. And where an AI system has classified historical costs into the four categories, sample-check the classification on a stated sample size: category misclassification makes the model wrong in a direction nobody notices.

### KEY TERMS — KA 9.1

| TERM                          | MEANING                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------|
| <b>Quality</b>                | The degree to which delivered work conforms to its stated requirements.                    |
| <b>Grade</b>                  | The level of capability or specification the requirements ask for; independent of quality. |
| <b>Fitness for purpose</b>    | Whether conforming, correctly graded output serves the use it was built for.               |
| <b>Cost of quality (CoQ)</b>  | Prevention + appraisal + internal failure + external failure.                              |
| <b>Cost of conformance</b>    | Prevention + appraisal: money spent so defects do not occur or do not escape.              |
| <b>Cost of nonconformance</b> | Internal + external failure: money lost because they did.                                  |

| TERM                              | MEANING                                                                                           |
|-----------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Escape</b>                     | A defect reaching the next stage, or the customer, undetected.                                    |
| <b>External-failure unit cost</b> | The average cost of one escaped defect in the operating context; set by context, not the project. |
| <b>Range of optimality</b>        | The band of an uncertain input over which the recommended option stays recommended.               |
| <b>Tolerance</b>                  | The range within which output is conforming; a specification without one is untestable.           |

SAMPLE MCQS — KA 9.1

**MCQ 9.1-A** [9.1.2 · Application] A regime allows 80 defects to be introduced and detects 90 % before handover; internal correction averages 1,500 and an escaped defect costs 12,000. Prevention is 96,000 and appraisal 64,000. Total cost of quality is:

- A. USD 268,000
- B. USD 364,000 CORRECT
- C. USD 204,000
- D. USD 460,000

*Rationale:*  $96,000 + 64,000 + (72 \times 1,500) + (8 \times 12,000) = 364,000$  (9.1.2). A omits external failure; C counts nonconformance only and drops conformance; D applies the 12,000 external unit cost to all 80 introduced defects rather than to the 8 that escape.

**MCQ 9.1-B** [9.1.2 · Evaluation] Moving to the next-stricter regime costs 90,000 more in conformance and removes 9,000 of internal failure and 4 escaped defects. At what external-failure unit cost does the stricter regime become worth buying?

- A. USD 12,000
- B. USD 22,500
- C. USD 20,250 CORRECT
- D. USD 81,000

*Rationale:* The step pays when  $90,000 \leq 9,000 + 4u$ , so  $u \geq 81,000/4 = 20,250$  (9.1.2). B divides the conformance cost by escapes avoided and forgets the internal-failure saving; D is the numerator, not the unit cost; A is the assumed unit cost, which is the thing being tested.

**MCQ 9.1-C** [9.1.2 · Analysis] An organisation strengthens testing and its internal failure cost rises in the first period while total cost of quality falls. The correct reading is that:

- A. the testing programme is failing and should be reversed

- B. defects have moved from the external column to the internal one at a much lower unit cost CORRECT
- C. internal and external failure have been misclassified
- D. prevention spending was set too low

*Rationale:* Internal failure cost is not monotone in quality, better detection finds defects that were previously escaping (9.1.2). Reading the rise as deterioration is the standard reason such programmes are cancelled in their first period.

**MCQ 9.1-D** [9.1.1 · Comprehension] A supplier delivers a product built exactly to a deliberately basic specification. It is best described as:

- A. low grade and low quality
- B. low grade and high quality CORRECT
- C. high grade and low quality
- D. not assessable until the client uses it

*Rationale:* Grade is the specified level; quality is conformance to it (9.1.1). D confuses quality with fitness for purpose, which is a third and separate test.

**MCQ 9.1-E** [9.1.2 · Evaluation] Removing the last 6 escaped defects costs 129,000 more in total cost of quality, and each escape would have cost 12,000. The strongest professional statement is that:

- A. the spend is justified because defects should be eliminated
- B. the spend buys protection at 21,500 per defect, 1.79 times the harm it prevents, so it is not justified on these figures CORRECT
- C. the spend is justified because 129,000 is small against a 4,000,000 budget
- D. the comparison cannot be made without a Monte Carlo simulation

*Rationale:*  $129,000/6 = 21,500$  against a 12,000 consequence, a ratio of 1.79 (9.1.2). A is the zero-defects fallacy; C is affordability, not value; D over-reaches. The expected-value comparison is valid on stated averages, though a safety case would also require the tail to be examined.

#### SELF-CHECK — KA 9.1

1. *Why is the cost-of-quality minimum not at zero defects?* — Because the marginal conformance cost of removing the last defects exceeds their consequence; on Auriga the last six escapes cost 21,500 each to remove against 12,000 of harm.
2. *What single input determines where the optimum sits, and who sets it?* — The external-failure unit cost, set by the operating context and not by the project; Auriga's optimum holds for 3,540–20,250 and moves at 20,250 and again at 60,000.

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3. *Which cost category is non-monotone, and why does it matter?* — Internal failure: it rises as detection improves before falling as introduction falls, and misreading that rise is why quality programmes get cancelled in their first period.
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## KNOWLEDGE AREA 9.2

## Assurance and control

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*Topics: 9.2.1 assurance and control as different instruments · 9.2.2 containment layers and the economics of detection · 9.2.3 rework, capacity and the schedule consequence.*

### 9.2.1 Assurance and control as different instruments

*Quality assurance* acts on the **process** and is preventive: it asks whether the way the work is being done is capable of producing conforming output, and its evidence is process evidence, competence records, controlled documents, a managed configuration, a followed method. *Quality control* acts on the **product** and is detective: it asks whether this output conforms, against its criteria. Both are the first line of the assurance model built in Domain 3, KA 3.3.2; the second and third lines form opinions about whether the first line works, and Domain 3's failure modes (duplication, gap, capture, and treating a favourable opinion as a transfer of accountability), apply here unchanged and are not restated.

The distinction is operationally load-bearing because control cannot make output conform; it can only report that it does not. Every unit of control spend therefore buys **information**, and information is worth what the decision it changes is worth. An inspection whose result cannot change anything (carried out after the only feasible correction window has closed, or reported to someone with no authority to act) is pure cost, and the commonest wasted quality spend in delivery is not too little inspection but inspection positioned too late. Domain 3, KA 3.3.1 computes the same result for a governance gate and establishes that gate value is destroyed by elapsed time and by weak detection; the containment arithmetic below is its product-level counterpart, and the two should be read together.

### 9.2.2 Containment layers and the economics of detection

**Definition.** A **containment layer** is a review, test or inspection with a stated coverage, a stated **detection rate** (the share of the defects reaching it that it finds) and a stated **unit correction cost**. A delivery process is a chain of such layers, and the property that matters is the **escape fraction**: the product of the layers' individual escape rates.

$$\text{escape fraction} = \prod (1 - d_i) \quad \text{over layers } i, \text{ with } d_i \text{ the detection rate of layer } i$$

$$\text{escapes} = \text{defects introduced} \times \text{escape fraction}$$

The product form is the lesson. Layers are multiplicative, not additive, so three unimpressive layers can outperform one strong one; and, symmetrically, removing one layer does not degrade the result by its own detection rate but multiplies the escape fraction by  $1/(1 - d_i)$ , which for a 50 % layer is a doubling.

**The correction-cost ladder.** Correction cost rises with the stage at which a defect is found, because later stages have built more on top of it and involve more parties. Auriga's observed ladder for a control-logic or configuration defect:

| LAYER | WHAT IT IS              | DETECTION RATE $d_i$ | CORRECTION COST PER DEFECT | RATIO TO THE FIRST LAYER |
|-------|-------------------------|----------------------|----------------------------|--------------------------|
| L1    | Design peer review      | 0.50                 | USD 800                    | ×1.000                   |
| L2    | Factory acceptance test | 0.60                 | USD 2,000                  | ×2.500                   |
| L3    | Site acceptance test    | 0.50                 | USD 3,500                  | ×4.375                   |
| —     | Live operation (escape) | —                    | USD 12,000                 | ×15.000                  |

Domain 5, KA 5.2.1 establishes the same shape for *requirement* defects on a much steeper ladder (a factor of four per stage, from USD 400 at definition to USD 102,400 in live service), because a requirement defect propagates through design, build and test before it is visible. The two ladders are not interchangeable and must not be averaged: a single "cost per defect" figure that mixes them will mis-price every containment decision made from it.

#### WORKED EXAMPLE

##### Worked example 9.2.2 — Auriga's containment chain, and the price of one avoidable defect.

- Setup.** Under regime R2 (9.1.2), **80** control-logic defects are introduced. They pass through the three layers above, with detection rates 0.50, 0.60 and 0.50 and correction costs of 800, 2,000 and 3,500. An escape costs USD 12,000.
- Formula.** Escape fraction =  $\prod (1 - d_i)$ . Defects found at layer  $i$  = defects reaching it  $\times d_i$ . Internal failure cost =  $\sum (\text{found at } i \times \text{unit cost at } i)$ . Expected containment cost per **introduced** defect = (internal + external failure cost)  $\div$  defects introduced.



3. **Substitution.**  $(1 - 0.50)(1 - 0.60)(1 - 0.50) = 0.50 \times 0.40 \times 0.50$ . Flow: L1 finds  $80 \times 0.50 = 40$ , leaving 40; L2 finds  $40 \times 0.60 = 24$ , leaving 16; L3 finds  $16 \times 0.50 = 8$ , leaving 8. Internal cost  $40 \times 800 + 24 \times 2,000 + 8 \times 3,500$ .
4. **Result.** Escape fraction **0.100** (detection **90.0 %**); escapes **8**. Internal failure cost  $32,000 + 48,000 + 28,000 = \text{USD } 108,000$ ; an average of **USD 1,500** per defect found internally, reconciling to 9.1.2. External failure cost  $8 \times 12,000 = \text{USD } 96,000$ . Total nonconformance **USD 204,000**, which over 80 introduced defects is an **expected containment cost of USD 2,550 per introduced defect**.
5. **Interpretation.** That last figure is the most useful number in the domain and is almost never computed. **USD 2,550 is what one avoidable Auriga defect costs, whoever eventually finds it:** the weighted average of being caught cheaply at peer review, expensively at site acceptance and catastrophically in service. It converts every prevention proposal into a one-line test: *does this remove a defect for less than 2,550?* It also prices the layers. Removing L3 raises the escape fraction to  $0.50 \times 0.40 = 0.200$ , escapes double from 8 to 16, adding  $8 \times 12,000 = 96,000$  of external failure while saving only  $8 \times 3,500 = 28,000$  of L3 correction, so **L3 must cost more than USD 68,000 to run** before its removal is arguable. That is a checkable claim rather than an opinion about whether site testing is worth it. Two cautions. The model assumes layers are **independent detectors**, and they are not: two layers staffed by the same team reading the same document with the same blind spot miss the same defects, so the real escape fraction is worse than the product. The countermeasure is to make consecutive layers *methodologically different* (a review, then an executed test, then a test in the target environment) rather than more of the same. And detection rates are estimates; where no history exists, state a range and compute the decision across it.

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WORKED EXAMPLE

**Worked example 9.2.2b — prevention or appraisal, on the same USD 40,000.**

1. **Setup.** Auriga has USD 40,000 of uncommitted quality budget at the end of design and two proposals. **(a) Appraisal:** extend site acceptance test coverage, raising L3's detection rate from 0.50 to 0.75. **(b) Prevention:** a design-standards and supplier-qualification package cutting defects introduced from 80 to 60, detection rates unchanged. The baseline is the R2 chain: nonconformance USD 204,000.
2. **Formula.** Re-run the chain under each proposal; compare (added spend + internal + external failure) against the baseline 204,000.

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3. **Substitution.** (a) 80 introduced; L1 finds 40, L2 finds 24, L3 finds  $16 \times 0.75 = 12$ , escapes 4; internal  $32,000 + 48,000 + 12 \times 3,500 = 122,000$ ; external  $4 \times 12,000 = 48,000$ . (b) 60 introduced; L1 finds 30, L2 finds 18, L3 finds 6, escapes 6; internal  $24,000 + 36,000 + 21,000 = 81,000$ ; external  $6 \times 12,000 = 72,000$ .
4. **Result.** (a)  $40,000 + 122,000 + 48,000 = \text{USD } 210,000 - \text{USD } 6,000$  worse than doing nothing. (b)  $40,000 + 81,000 + 72,000 = \text{USD } 193,000 - \text{USD } 11,000$  better than doing nothing, and **USD 17,000 better than (a)** on identical money.
5. **Interpretation.** Prevention wins **even though it removes fewer escapes**: (a) halves escapes from 8 to 4, (b) only cuts them from 8 to 6. The reason is the containment cost per introduced defect. Option (b) removes 20 defects before any layer has to pay to find them, worth  $20 \times 2,550 = \text{USD } 51,000$  against USD 40,000 of spend, a net **USD 11,000**, exactly the result above and the identity worth remembering. Option (a) removes no defect; it **relocates** detection to a later and dearer layer, so its saving on external failure is partly consumed by internal failure rising from 108,000 to 122,000. Both rules are computable: **prevention pays whenever it removes a defect for less than the expected containment cost per introduced defect** (USD 2,550 here, so a 40,000 programme must avoid at least  $40,000/2,550 = 15.69$ , i.e. **16 defects**), and **appraisal pays only when the layer it strengthens is cheaper than the layer or escape it displaces work from**. Raising detection at the *earliest* layer would have been a different and better proposition.

Two observations belong in the record. Option (b) takes total CoQ to  $160,000 + 40,000 + 81,000 + 72,000 = \text{USD } 353,000$ , below the USD 364,000 that 9.1.2 called optimal. That is not a contradiction: 9.1.2 optimised along a ladder of **pre-costed regimes**, and prevention more efficient than the ladder assumed shifts the whole curve down, so a cost-of-quality optimum is conditional on the prevention technology available and should be recomputed when that changes. And option (a) reaches 4 escapes for a total CoQ of **USD 370,000** where regime R3 reaches 4 escapes at **USD 397,000**: two routes to the same escape count, USD 27,000 apart, because R3 got there partly by introducing fewer defects. **The escape count is not a sufficient description of a quality regime.**

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### 9.2.3 Rework, capacity and the schedule consequence

**The principle no cost-of-quality model contains.** Rework is not only money; it is **capacity**. Correcting a defect consumes the same people, equipment and test facility that would otherwise produce new output, so a rework share of capacity is a direct reduction in throughput; and where the activity is on the critical path, the reduction converts straight into elapsed time. The relationship is not linear:

effective capacity = nominal capacity  $\times$  (1 - r)      r =  
 rework share of capacity  
 duration = first-pass work content  $\div$  effective  
 capacity  
 duration multiplier = 1 / (1 - r)

$1/(1 - r)$  is hyperbolic. The tenth percentage point of rework costs far less than the fortieth, which is why a team absorbs a modest rate almost invisibly and then loses control of a phase over what looks like a small further deterioration.

#### WORKED EXAMPLE

#### Worked example 9.2.3 – Auriga's commissioning, and what 30 % rework costs.

1. **Setup.** Auriga's activity **F, testing and commissioning**, is planned at **4 weeks** with **6 engineers**, a nominal capacity of **24 engineer-weeks**. First-pass work content is **20 engineer-weeks**, so the plan carries a rework allowance of **4 engineer-weeks**, or **16.67 %** of capacity. F is on the critical path with zero float (Domain 6, KA 6.2.1); Auriga's cost of delay is **USD 45,000 per week**; an engineer-week costs **USD 5,225** (Domain 7, KA 7.4.1). At the two-week point, rework is running at **30 %** of capacity.
2. **Formula.** duration = content  $\div$  (crew  $\times$  (1 - r)); overrun = duration - planned duration; delay cost = overrun  $\times$  cost of delay; rework consumed = r  $\times$  crew  $\times$  duration; labour above allowance = (rework consumed - allowance)  $\times$  engineer-week cost.
3. **Substitution.**  $20 \div (6 \times 0.70) = 20 \div 4.2$ . Rework consumed  $0.30 \times 6 \times 4.7619$ .
4. **Result.**

| REWORK<br>SHARE R       | MULTIPLIER 1/<br>(1-R) | DURATION<br>(WEEKS) | OVERRUN<br>(WEEKS) | DELAY<br>COST<br>(USD) |
|-------------------------|------------------------|---------------------|--------------------|------------------------|
| 16.67 % (as<br>planned) | 1.2000                 | 4.0000              | 0.0000             | 0                      |
| 25 %                    | 1.3333                 | 4.4444              | 0.4444             | 20,000                 |
| <b>30 % (actual)</b>    | <b>1.4286</b>          | <b>4.7619</b>       | <b>0.7619</b>      | <b>34,286</b>          |
| 40 %                    | 1.6667                 | 5.5556              | 1.5556             | 70,000                 |
| 50 %                    | 2.0000                 | 6.6667              | 2.6667             | 120,000                |

At 30 %: duration **4.7619 weeks**, overrun **0.7619 weeks**, delay cost **USD 34,286**. Rework consumes **8.5714 engineer-weeks** against a 4-engineer-week allowance – **4.5714 engineer-weeks** of unbudgeted labour at 5,225, or **USD**

**23,886. Total consequence: USD 58,171. 5. Interpretation. The cost is mostly schedule, and the schedule cost is invisible in the quality report.** Of the 58,171, 59 % is delay and 41 % labour; a quality function reporting rework hours reports the smaller half. On a critical-path activity the rework share of capacity is a schedule metric that happens to be collected by the quality system, and it belongs in the same report as float.

**The convexity is the management lesson.** Moving from 30 % to 40 % costs a further **USD 35,714** of delay and 40 % to 50 % costs **USD 50,000**, where 10 % to 20 % would have cost only **USD 20,833**. Every additional point is dearer than the last, so intervene early (while the numbers still look tolerable), and trigger on the *rate*, not the accumulated slip.

**The allowance is a decision and should be visible as one.** A 16.67 % allowance is legitimate and honest, and it is also a forecast that can be wrong; a plan with **no** allowance is not a plan without rework but a plan whose first defect creates a slip. State the allowance, measure against it weekly, and treat the variance as a leading indicator. The rework share reveals itself two weeks before the milestone does.

**The easy-to-miss point:** the natural response, reduce test coverage to recover the schedule, is a decision to raise the escape fraction, and 9.2.2 prices it. Cutting L3's detection from 0.50 to 0.25 raises escapes from 8 to 12, adding  $4 \times 12,000 = \text{USD } 48,000$  of external failure against  $4 \times 3,500 = \text{USD } 14,000$  of internal correction saved. Any schedule recovery must beat that net USD 34,000, computed, before it is taken.

## AI in this KA

**Where it earns its place.** Detecting defects at scale: static analysis of control logic and code, cross-checking a configuration set against a design baseline, comparing an as-installed instrument schedule against the design register, sweeping test evidence for missing or contradictory results. These have checkable right answers, and a model doing them is functioning as a **containment layer**, and should be characterised as one, with a measured detection rate and a unit cost, so it can be placed in the chain of 9.2.2 rather than trusted qualitatively. Also: classifying a rework log by cause and computing the rework share of capacity per team and per week, which most organisations cannot report because the time data is not coded for it.

**Where it must not go.** It must not be the **only** layer for anything consequential, because 9.2.2's independence caution applies with unusual force: a model's misses are systematic, so two checks by the same model are one check, and a model that generated the artefact is not an independent reviewer of it. It must not issue a test result, sign an inspection record or grant a release. Those are attributable acts under Domain 3, KA 3.A.2. And its detection rate must not be *asserted*: a claim that an automated check finds "most" defects is not a number, and a chain computed from

asserted rates gives a false escape fraction that is then used to justify removing a human layer.

**Verification, concretely.** Measure the model's detection rate against a known, held-back defect set before relying on it, and re-measure when the model, the prompt or the artefact type changes: a detection rate is a property of a *configuration*, not of a tool. Record its false-positive rate too, because a layer raising many false findings consumes the engineering attention the next layer needs. Keep a human layer **methodologically different** from the automated one. And reproduce the arithmetic by hand: the escape fraction is a product of three numbers and the duration multiplier is one division.

KEY TERMS — KA 9.2

| TERM                                                   | MEANING                                                                                           |
|--------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Quality assurance</b>                               | Process-directed, preventive activity: is the way we work capable of producing conforming output? |
| <b>Quality control</b>                                 | Product-directed, detective activity: does this output conform?                                   |
| <b>Containment layer</b>                               | A review, test or inspection with a stated coverage, detection rate and unit correction cost.     |
| <b>Detection rate (<math>d_i</math>)</b>               | The share of defects reaching a layer that the layer finds.                                       |
| <b>Escape fraction</b>                                 | $\prod (1 - d_i)$ across the chain, the share of introduced defects reaching the customer.        |
| <b>Correction-cost ladder</b>                          | The rising cost of correcting a defect by the layer at which it is found.                         |
| <b>Expected containment cost per introduced defect</b> | (Internal + external failure) ÷ defects introduced; the breakeven price of prevention.            |
| <b>Rework share of capacity (<math>r</math>)</b>       | The fraction of a team's available effort consumed correcting output.                             |
| <b>Duration multiplier</b>                             | $1 / (1 - r)$ , the hyperbolic effect of rework on elapsed time.                                  |
| <b>Rework allowance</b>                                | The capacity a plan reserves for rework; a forecast, and a decision.                              |

SAMPLE MCQs — KA 9.2

**MCQ 9.2-A** [9.2.2 · Application] Three containment layers have detection rates 0.50, 0.60 and 0.50. Of 80 introduced defects, how many escape?

- A. 8 CORRECT
- B. 3

C. 12

D. 27

*Rationale:* Escape fraction  $0.50 \times 0.40 \times 0.50 = 0.10$ , so  $80 \times 0.10 = 8$  (9.2.2). B adds the detection rates to 1.60 and treats the chain as detecting everything with a remainder; C uses two layers only; D applies the *average* detection rate (0.5333) three times as if each layer saw all 80.

**MCQ 9.2-B** [9.2.2 · Analysis] On a chain where 80 defects are introduced, internal failure costs 108,000 and external failure 96,000. The expected containment cost per introduced defect is:

A. USD 1,500

B. USD 2,550 CORRECT

C. USD 12,000

D. USD 2,000

*Rationale:*  $(108,000 + 96,000)/80 = 2,550$  (9.2.2). A is the average cost per defect found *internally*, omitting escapes; C is the escape unit cost; D divides internal failure alone by 80 and rounds: all three understate the breakeven price of prevention.

**MCQ 9.2-C** [9.2.2 · Evaluation] USD 40,000 will either raise the last layer's detection rate from 0.50 to 0.75, or cut defects introduced from 80 to 60. On the figures of 9.2.2, the better choice and its reason are:

A. raising detection, because it halves escapes from 8 to 4

B. cutting introduction, because it removes 20 defects at an expected containment cost of 2,550 each — USD 51,000 of value for USD 40,000 CORRECT

C. either, since both cost the same

D. raising detection, because internal correction is cheaper than external failure

*Rationale:* Prevention returns 51,000 against 40,000 (a net 11,000), and beats the appraisal option by 17,000 (9.2.2b). A counts escapes rather than cost and misses that raising late-layer detection moves work to a dearer layer, pushing internal failure from 108,000 to 122,000; D states a true premise that does not reach the conclusion.

**MCQ 9.2-D** [9.2.3 · Application] A 4-week activity with 6 engineers has 20 engineer-weeks of first-pass content. Rework runs at 30 % of capacity. The activity will take:

A. 4.00 weeks

B. 4.76 weeks CORRECT

C. 5.20 weeks

D. 4.29 weeks

*Rationale:*  $20 \div (6 \times 0.70) = 4.7619$  weeks (9.2.3). C adds 30 % to the 4-week plan, the linear error; D mishandles the allowance by scaling the 3.33 weeks of pure content; A ignores the overrun.

**MCQ 9.2-E** [9.2.3 · Analysis] Why does the marginal cost of rework rise as the rework share rises?

- A. because rework is charged at a premium rate
- B. because the duration multiplier  $1/(1 - r)$  is convex, so equal increments of  $r$  add increasing amounts of elapsed time CORRECT
- C. because defects found later cost more to fix
- D. because float is consumed first and then lost

*Rationale:* The convexity is the mechanism: on Auriga's figures 10 %→20 % adds 0.4630 weeks while 40 %→50 % adds 1.1111 (9.2.3). C is the correction-cost ladder, a different effect (9.2.2); A and D are not general.

**MCQ 9.2-F** [9.2.2 · Analysis] An automated check and a manual review are placed as consecutive containment layers, both driven from the same design document by the same team. The escape fraction computed as the product of their detection rates will be:

- A. correct
- B. too optimistic, because the layers are not independent detectors CORRECT
- C. too pessimistic, because two layers always find more than one
- D. correct only if their detection rates are equal

*Rationale:* The product form assumes independence; layers sharing a source, a method or a blind spot miss the same defects, so the true escape fraction is worse than the product (9.2.2). The countermeasure is to make consecutive layers methodologically different.

SELF-CHECK — KA 9.2

1. *What does removing a 50 % containment layer do to the escape fraction?* — Doubles it: the layer's escape rate of 0.50 leaves the product, multiplying escapes by  $1/(1 - 0.50)$ .
2. *State the breakeven test for a prevention proposal.* — It pays if it removes a defect for less than the expected containment cost per introduced defect: USD 2,550 on Auriga, so a USD 40,000 programme must avoid 16 defects.
3. *Why is a rework share of capacity a schedule metric?* — Because effective capacity is  $\text{nominal} \times (1 - r)$  and duration is content  $\div$  effective capacity, so on a zero-float activity every point of rework is elapsed time: 30 % turned Auriga's 4-week commissioning into 4.7619 weeks and USD 34,286 of delay.

## KNOWLEDGE AREA 9.3

## Acceptance, nonconformance and root-cause analysis

*Topics: 9.3.1 the acceptance decision and nonconformance disposition · 9.3.2 acceptance sampling and what a passing sample establishes · 9.3.3 root-cause analysis and recurrence.*

### 9.3.1 The acceptance decision and nonconformance disposition

**Definition. Acceptance** is a formal decision by a named authority that a deliverable meets its acceptance criteria and may pass to the next stage, to the client, or into service. Domain 5, KA 5.4.3 established the criteria and the verification/validation distinction; what belongs here is the *decision*, its authority, and what happens when the answer is no.

Acceptance goes wrong in four recognisable ways, each with a countermeasure that costs nothing. **Acceptance by silence.** Nobody rejects, so the work is deemed accepted, usually by the expiry of a contractual review period; the countermeasure is a positive acceptance record with a named signatory, since the absence of a rejection is not a decision. **Acceptance by exhaustion.** The deliverable is accepted at the fourth submission because the reviewers have no more time, not because it changed; the countermeasure is to record the open nonconformances at acceptance, converting exhaustion into a visible conditional pass. **Acceptance without authority.** The signatory is not entitled to bind the receiving organisation, which is Domain 3's single-A defect at deliverable level. And **acceptance against criteria written after the evidence**, where the criteria were adjusted to what the deliverable does, so the test is guaranteed to pass and evidences nothing.

**Nonconformance disposition.** When output does not conform, exactly five dispositions exist, each with a different authority and a different downstream obligation:

| DISPOSITION                   | WHAT IT MEANS                                     | AUTHORITY REQUIRED                                                                       | DOWNSTREAM OBLIGATION                    |
|-------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------|
| <b>Rework</b>                 | Bring it into full conformance                    | Delivery management                                                                      | Retest to the same criteria              |
| <b>Repair</b>                 | Make it fit for use without full conformance      | Design authority                                                                         | The as-built record must show the repair |
| <b>Use as is (concession)</b> | Accept the nonconformance as it stands            | The party bearing the consequence (client or design authority, never the producer alone) | Register it; assess cumulative effect    |
| <b>Regrade</b>                | Use it for a lesser purpose for which it conforms | Design authority plus the owner of that purpose                                          | Traceability so it cannot migrate back   |



| DISPOSITION           | WHAT IT MEANS | AUTHORITY REQUIRED  | DOWNSTREAM OBLIGATION                               |
|-----------------------|---------------|---------------------|-----------------------------------------------------|
| <b>Reject / scrap</b> | Do not use it | Delivery management | Root-cause analysis if value or recurrence warrants |

Two disciplines make this table work rather than decorate a procedure. **The producer never grants its own concession:** a concession transfers a consequence, so its authority sits with whoever will carry it, and a process in which the delivery team accepts its own nonconformances has no acceptance function at all. And **concessions must be counted, not merely recorded:** individually minor concessions accumulate into a delivered product materially different from the specified one, which is Domain 3's Case study B at deliverable level. The cumulative-effect test (related concessions aggregating above a threshold within a period require the authority appropriate to the aggregate) is the countermeasure, and it is the instrument Domain 4, KA 4.4 applies to change.

**The authority column stops at the design authority, and sometimes the decision does not.** In sectors operating under an approval regime (where a certifying body, a notified assessor or a regulator has a standing in the item's fitness for use, as in aerospace, rail signalling, medical devices, pressure systems and nuclear) a repair, a concession or a regrade may engage that body as well, potentially **before the item is used**. The professional position is therefore that such a disposition is **treated as unavailable until the external position has been established**, and that establishing it sits *outside* the project's authority: no internal signature substitutes for it, and it is not something a commercial pressure can be allowed to shorten. Whether any external notification or acceptance is engaged, by whom, for what and in what form differs by sector and by jurisdiction, and is established with the organisation's regulatory or technical-compliance function and with the body itself. **Nothing here states the position anywhere, and nothing here characterises any disposition as acceptable, or unacceptable, to any body.** The obligation this domain does impose is that the disposition record names, for each disposition class, **whether an external acceptance is engaged and who established that:** recorded before the disposition is granted, not discovered at inspection.

**Making the cumulative-effect test operable.** As stated, the test has three undefined parameters and no owner, which is why it is widely written into procedures and nowhere enforced. Five decisions make it real, and all five belong in the quality management plan of 9.1.3 rather than in the judgement of whoever is granting the concession:

- **Who sets the threshold, and when.** The **sponsor or design authority sets it at baseline:** the party that carries the consequence of an altered product, never the delivery team that would trip it. Setting it late, or setting it by whoever is asked first, produces a threshold shaped by the concessions already granted.

- **Where the threshold comes from.** It is derived, not chosen round: the same window logic Domain 3, KA 3.3.4c derives for a delegation's cumulative rule applies unchanged. The threshold sits comfortably **above** the base-rate aggregate of ordinary concession traffic in that class per period, and comfortably **below** the aggregate at which the delivered product would differ materially from the specified one. A threshold below the base rate re-centralises every disposition and will be abandoned within two months; a threshold above the material-difference point is decorative.
- **The aggregation period.** Stated explicitly (a quarter, a release, or the deliverable's whole production run), and chosen so that a slow accumulation cannot escape by straddling two periods. Where a deliverable is produced once, the period is the deliverable.
- **The test for relatedness.** By **what the concessions touch** (the same deliverable, the same interface, the same assured control, the same requirement) never by who requested them or which budget carried them, because the exposure the rule exists to catch is a coherent change to one thing arriving in instalments. Widening the class is the natural error and it multiplies false trips.
- **The named owner of the running total.** One person maintains it (normally the quality function), and reports it monthly against the threshold whether or not anything moved, because a total reported only when it is interesting cannot be distinguished from a total nobody kept.

**And what a breach triggers, stated in advance.** On breach, further concessions **in that class** are suspended pending a decision by the authority appropriate to the aggregate, taken through Domain 3's escalation machinery at its stated latency rather than by informal referral. The decision before that authority is not "may this one be granted?" but the harder and correct question: *given what has already been conceded, is the delivered product still the specified one, and if not, what is now being accepted?* Suspension is the part that gives the rule teeth, and it is also the part most often omitted: a cumulative test with no consequence on breach is a counter, and a counter is not a control.

**The record, and how long it lives.** Acceptance records, concession authorisations, disposition decisions and the verification evidence behind them are the primary evidence in any later dispute, audit, defect claim or inspection; and all of those arrive years after the signature, when the people are gone and the memory is not admissible. So each is assigned a **record class**, with a **named custodian role**, a **retention period** and a stated **disposal authority**, on the custody machinery of Domain 3, KA 3.3.4. The period is set by the applicable regime and by contract rather than by the project's convenience, it differs by jurisdiction and by sector, and it is taken from the organisation's records, regulatory and legal functions — **nothing in this domain states a legal minimum or maximum, and nothing here should be relied on as stating one.** Domain 16, KA 16.4.4 works the economics and reaches a conclusion worth carrying here: retaining contract and technical evidence

pays as insurance at a very low probability of ever needing it, so the reflex to purge for storage cost is almost always wrong, and retention is **a schedule by class, never a single rule**. Custody transfers to a named continuing role at closure, because the quality function that signed will not exist in the organisation that is asked.

### 9.3.2 Acceptance sampling and what a passing sample establishes

Verifying every item is often uneconomic, so acceptance is taken on a sample. The arithmetic is elementary, almost never done, and produces two results practitioners find genuinely surprising.

For a population with true defective fraction  $p$ , the probability that a sample of  $n$  contains **no** defective item is  $(1 - p)^n$ . Turning that round gives the **confidence bound**: if a sample of  $n$  is clean, the largest defective fraction consistent with that observation at confidence  $1 - \alpha$  is

$$p_{\text{upper}} = 1 - \alpha^{(1/n)} \quad (\alpha = 0.05 \text{ for a 95 \% upper bound})$$

And the economics of how much to verify reduce to a single ratio. With  $c$  the cost of verifying one item and  $u$  the cost of one escaped defective item, full verification of  $N$  items costs  $cN$  while verifying  $n$  and accepting the rest costs  $cn + u \cdot p \cdot (N - n)$ . Setting them equal, the  $(N - n)$  terms cancel and

$$p^* = c / u$$

**Full verification pays exactly when the defective fraction exceeds the ratio of verification cost to failure cost, and that breakeven is independent of the sample size.** It is the number to put on the table before anyone argues about sample sizes.

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#### WORKED EXAMPLE

**Worked example 9.3.2 – Auriga's instrument loops: what a clean sample of 20 proves.**

1. **Setup.** Auriga's installation includes **120** instrument loops. Verifying one costs **USD 1,400** (a two-engineer point-to-point check and signal injection). A defective loop reaching service costs **USD 12,000**. The subcontractor proposes verifying **20** loops and accepting the batch if none is defective. The sample is drawn and is clean. What has been established?
2. **Formula.**  $P(\text{clean sample}) = (1 - p)^n$ ; 95 % bound  $p_{\text{upper}} = 1 - 0.05^{(1/n)}$ ; breakeven  $p^* = c/u$ ; minimum self-consistent sample  $n^* = \ln(0.05)/\ln(1 - p^*)$ .
3. **Substitution.**  $(1 - 0.05)^{20}$ ;  $1 - 0.05^{(1/20)}$ ;  $1,400/12,000$ ;  $\ln(0.05)/\ln(1 - 0.116667)$ .

- 
4. **Result.** A population 5 % defective passes this plan **35.85 %** of the time (at 2 % defective, **66.76 %**; at 10 %, **12.16 %**). The 95 % upper bound after a clean sample of 20 is **13.91 %**: on 120 loops, up to **16.69**, i.e. as many as 16 defective loops. The breakeven defective fraction is  $1,400/12,000 = 11.67 \%$ . The smallest sample whose 95 % bound falls below that breakeven is **25** (the bound is 11.29 % at  $n = 25$  and 11.73 % at  $n = 24$ ), costing **USD 7,000** more than the proposed 20.
5. **Interpretation.** The decisive observation compares two of those numbers. **The plan's own confidence bound (13.91 %) is higher than the defective fraction at which its strategy is wrong (11.67 %).** A clean sample of 20 cannot exclude the very defect rate at which the correct decision would have been to verify all 120 loops. The plan is **not self-consistent**, and no amount of it passing changes that. Five more loops (USD 7,000 on a USD 4,000,000 project) repairs it, and that is the cheapest quality decision anywhere in this domain.

What the sample *does* establish is worth stating precisely. It establishes that the population is probably not grossly defective: rates above about 14 % are unlikely given a clean 20. It does **not** establish conformance, and it does not establish any tighter rate, bounding  $p$  at 5 % with 95 % confidence needs **59** items ( $\ln 0.05 / \ln 0.95 = 58.404$ , and the familiar rule-of-three approximation  $3/p$  gives 60 as a mental check).

The economics cut against instinct. At a true 5 % defective rate the sample-of-20 plan is the **cheapest** of the three options:  $1,400 \times 20 + 12,000 \times 0.05 \times 100 = \text{USD } 88,000$ , against **USD 119,200** for a 59-loop plan and **USD 168,000** for verifying all 120. Sampling is not laziness and full inspection is not automatically responsible, below  $p^* = 11.67 \%$  it destroys value. But the plan must be able to *see* whether  $p$  is below the breakeven, and the 20-loop plan cannot. Hence the discipline: **choose  $n$  from the confidence bound the economics require, not from what feels proportionate.** Three cautions. The  $(1 - p)^n$  form treats draws as independent, a good approximation when  $n$  is small relative to  $N$  and conservative here. Defects are often **clustered** (one mis-trained crew, one bad batch), so the sample must be random, and a convenience sample of the accessible loops establishes nothing. And zero-defect acceptance interacts badly with measurement error, so the disposition rules of 9.3.1 must cover a contested sample result before the sampling starts.

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### 9.3.3 Root-cause analysis and recurrence

**The purpose, stated to exclude what usually happens.** Root-cause analysis exists to reach the level at which a cause can be **removed**, so the defect class does not recur. It is not an explanation, not an apology and not a search for a person. The standard techniques (iterative "why" questioning, cause-and-effect decomposition, fault-tree reasoning backwards from an undesired event, and change analysis comparing a working case with a failing one) share one failure mode: **stopping at the level at which somebody can be blamed**, which is always shallower than the

level at which something can be fixed. "The engineer loaded the wrong parameter set" describes the event. "There was no controlled baseline against which a parameter set could be checked, and no process step that would have detected the mismatch" is a cause, because it names something a manager can change.

**The three tests of a completed analysis.** A root cause has been reached when removing it would have prevented this occurrence; removing it prevents the whole **class**, not just the instance; and the removal is within somebody's authority to enact. If the third test fails, the analysis has produced a finding for a higher level, and Domain 3's escalation machinery is how it travels.

**Recurrence is the metric.** The **recurrence rate** (repeat nonconformances as a share of all) measures whether the programme works; a stable or rising rate means causes are being described rather than removed. **Cause concentration** (the share attributable to the largest single cause) tells you where to spend, and is almost always higher than intuition suggests.

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WORKED EXAMPLE

**Worked example 9.3.3 — Auriga's dominant cause, priced.**

1. **Setup.** Auriga's 8 escaped control-logic defects (9.2.2) trace to three root causes: 5 to an uncontrolled supplier firmware configuration baseline (two field devices shipped with a superseded parameter set and nothing in the process compared them to a controlled reference), 2 to an ambiguous interlock specification, 1 to a one-off wiring error. Removing the dominant cause (imposing configuration control on the supplier's firmware, re-baselining, and adding a baseline comparison to the factory acceptance test) costs **USD 18,000**. Over the remaining installation programme and the 12-month warranty period the same cause is expected to produce 5 further occurrences at **USD 12,000** each.
2. **Formula.** Cause concentration = defects from the largest cause ÷ total. Net value of removal = expected recurrences × unit consequence – removal cost. Breakeven recurrences = removal cost ÷ unit consequence.
3. **Substitution.**  $5/8$ ;  $5 \times 12,000 - 18,000$ ;  $18,000/12,000$ .
4. **Result.** Cause concentration **62.5 %**. Expected avoided cost **USD 60,000** against a removal cost of **USD 18,000**: a net **USD 42,000** and a payback ratio of **3.33 times**. Removal breaks even at **1.5**, so **2** recurrences.
5. **Interpretation.** The payback ratio is not the interesting number; the **break-even of 1.5 recurrences** is, because it changes what the analysis must prove. Nobody needs to defend a forecast of five recurrences; the investment is justified if the cause recurs **twice**, which for a cause that has already recurred five times is an observation rather than a forecast. That is the general shape of a root-cause business case, and it is why such cases are easier to make than they are made: arguing about the expected recurrence count is arguing about the wrong number.

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Three professional points. **Cause concentration is the budget allocator:** with 62.5 % on one cause, a programme treating all three equally spends most of its effort on 37.5 % of the problem, and this concentration is the normal finding once causes are classified consistently rather than described individually. **The one-off must be tested, not assumed:** a single occurrence is what the fifth firmware defect also looked like at the time, so the honest test is whether a mechanism exists that would catch a recurrence. And **the removal must be verified as effective, on a date:** an action closed in a corrective-action log is not a cause removed, and the evidence is the absence of recurrence over a stated observation window, which is why the recurrence rate is reported and not merely the closure rate.

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### AI in this KA

**Where it earns its place.** Clustering a nonconformance log by cause: the task producing the concentration figure above, that humans do badly because recurrences are weeks apart and worded differently every time, and that has a checkable answer. Extracting a structured acceptance and concession register from correspondence and minutes, flagging deliverables accepted without a named signatory or with open nonconformances unrecorded. Computing sampling plans and confidence bounds across candidate  $n$  values, including the self-consistency test. Drafting the first pass of a root-cause narrative from evidence, for a human to interrogate and own.

**Where it must not go.** It must not accept a deliverable, grant a concession or close a corrective action: all three are attributable decisions with a consequence-bearing owner (9.3.1). It must not *conclude* a root-cause analysis: a plausible causal narrative is exactly what a language model produces most readily and what is least verifiable, and a fluent wrong cause is worse than no analysis because it terminates the enquiry. And it must not estimate the recurrence count, the defective fraction or the escaped-defect cost: the three inputs on which the calculation turns and for which a model has no evidence.

**Verification, concretely.** For a clustered log, re-read a stated sample of records and confirm the cause assignment, because the concentration figure that will drive the budget is only as good as the classification. For any proposed cause, apply the three tests of 9.3.3 explicitly and in writing, and reject a cause failing the second. Reproduce every sampling number by hand. The bound is one power, the breakeven one division. And put the self-consistency comparison in the acceptance paper, since it is the line that tells a reviewer whether the plan can support the decision it is being used to make.

KEY TERMS — KA 9.3

| TERM                                                   | MEANING                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Acceptance</b>                                      | A formal decision by a named authority that a deliverable meets its criteria and may pass on.                                                                                                                                                                                                                                                                   |
| <b>Acceptance by silence</b>                           | Deemed acceptance through the expiry of a review period; not a decision.                                                                                                                                                                                                                                                                                        |
| <b>Nonconformance</b>                                  | Output that does not meet its specified requirement.                                                                                                                                                                                                                                                                                                            |
| <b>Concession (use as is)</b>                          | Acceptance of a nonconformance as it stands, authorised by the party bearing the consequence.                                                                                                                                                                                                                                                                   |
| <b>Regrade</b>                                         | Use of nonconforming output for a lesser purpose for which it conforms, with traceability.                                                                                                                                                                                                                                                                      |
| <b>Cumulative-effect test</b>                          | The rule that related concessions aggregating above a threshold require the authority appropriate to the aggregate. Operable only once five things are fixed in the quality plan: who sets the threshold (sponsor or design authority, at baseline), how it is derived, the aggregation period, the relatedness test, and the named owner of the running total. |
| <b>Relatedness class (concessions)</b>                 | The set over which the cumulative total is summed, defined by what the concessions touch, never by who requested them; widening it multiplies false trips.                                                                                                                                                                                                      |
| <b>Suspension on breach</b>                            | The consequence that gives the cumulative test teeth: further concessions in that class stop pending a decision by the authority appropriate to the aggregate, taken through the escalation machinery.                                                                                                                                                          |
| <b>External acceptance</b>                             | Where an approval regime applies, a body outside the project may need to be notified of, or to accept, a repair, concession or regrade before use. It sits outside the project's authority and differs by sector and jurisdiction; the record names whether it is engaged and who established that.                                                             |
| <b>Record class, custodian, retention period</b>       | The three fields that make "the evidence is retained" real: what class of record this is, the named role accountable for it existing, and how long it is held with the source of that period named. Periods come from the applicable regime and the contract, not from this book.                                                                               |
| <b>Acceptance sampling</b>                             | Accepting or rejecting a population on the evidence of a sample.                                                                                                                                                                                                                                                                                                |
| <b>Confidence bound (<math>p_{upper}</math>)</b>       | $1 - \alpha^{(1/n)}$ , the largest defective fraction consistent with a clean sample of $n$ .                                                                                                                                                                                                                                                                   |
| <b>Breakeven defective fraction (<math>p^*</math>)</b> | $c/u$ : verification cost ÷ escaped-defect cost; above it, full verification pays. Independent of $n$ .                                                                                                                                                                                                                                                         |
| <b>Self-consistent sampling plan</b>                   | One whose confidence bound lies below its own breakeven defective fraction.                                                                                                                                                                                                                                                                                     |
| <b>Cause concentration</b>                             | The share of nonconformances attributable to the single largest root cause.                                                                                                                                                                                                                                                                                     |

| TERM                   | MEANING                                                                                 |
|------------------------|-----------------------------------------------------------------------------------------|
| <b>Recurrence rate</b> | Repeat nonconformances as a share of all; the test of whether causes are being removed. |

## SAMPLE MCQs — KA 9.3

**MCQ 9.3-A** [9.3.2 · Application] A sample of 20 items is drawn and none is defective. At 95 % confidence, the largest defective fraction consistent with that result is closest to:

- A. 0 %
- B. 5 %
- C. 13.9 % CORRECT
- D. 2.5 %

*Rationale:*  $1 - 0.05^{(1/20)} = 0.1391$  (9.3.2). A treats a clean sample as proof of conformance; B is the fraction a 59-item sample would bound; D halves the 5 % significance level.

**MCQ 9.3-B** [9.3.2 · Analysis] Verifying one item costs 1,400; an escaped defective item costs 12,000. Above what defective fraction does verifying the whole population beat sampling?

- A. 8.57 %
- B. 11.67 % CORRECT
- C. it depends on the sample size
- D. 1.4 %

*Rationale:*  $p^* = c/u = 1,400/12,000 = 11.67 \%$ , and the  $(N - n)$  terms cancel so the breakeven is independent of  $n$  (9.3.2). A inverts the ratio; C is the intuition the algebra refutes.

**MCQ 9.3-C** [9.3.2 · Evaluation] A 20-item zero-defect acceptance plan has a 95 % bound of 13.91 % and a breakeven defective fraction of 11.67 %. The most serious criticism of the plan is that:

- A. 20 items is too few to be statistically valid
- B. its confidence bound exceeds its own breakeven, so a clean sample cannot exclude the defect rate at which full verification would have been correct CORRECT
- C. it should use a one-defect acceptance number instead of zero
- D. sampling is inappropriate for safety-related work



*Rationale:* Self-consistency is the precise defect, repaired by raising  $n$  to 25 (9.3.2). A is an unquantified assertion; C would weaken the plan further; D is a different argument these figures do not establish.

**MCQ 9.3-D** [9.3.1 · Comprehension] A delivery team decides to accept its own nonconforming output as fit for use without correction. The defect in that process is that:

- A. rework is always preferable to concession
- B. the authority for a concession belongs to the party that will bear the consequence, not the producer CORRECT
- C. concessions are never permissible
- D. the disposition should have been regrade

*Rationale:* A concession transfers a consequence, so its authority sits with whoever carries it (9.3.1). A and C overstate, concession is a legitimate disposition; D presumes a lesser purpose exists.

**MCQ 9.3-E** [9.3.3 · Application] Removing a root cause costs 18,000; each recurrence costs 12,000. The strongest way to present the case is that removal:

- A. has a payback ratio of 3.33 times on five expected recurrences
- B. breaks even at 1.5 recurrences, so needs the cause to recur only twice CORRECT
- C. saves 60,000
- D. reduces the cause concentration from 62.5 %

*Rationale:* The breakeven recurrence count is the robust argument because it does not depend on forecasting the recurrence rate (9.3.3). A and C are true but rest on the five-recurrence estimate; D describes a measure, not a value.

**MCQ 9.3-F** [9.3.3 · Analysis] Which statement identifies a root cause rather than an event description?

- A. the engineer loaded the wrong parameter set
- B. the device was delivered with a superseded configuration
- C. there was no controlled baseline against which a parameter set could be checked, and no process step that would have detected the mismatch CORRECT
- D. the site acceptance test did not detect the fault

*Rationale:* Only C names something within a manager's authority to change whose removal prevents the whole class (9.3.3). A stops where a person can be blamed; B and D describe the occurrence and a layer's miss.

## SELF-CHECK — KA 9.3

1. *What does a clean sample of 20 establish?* — That the defective fraction is probably below 13.91 % at 95 % confidence; neither conformance nor any tighter rate, and bounding  $p$  at 5 % would need 59 items.
2. *State the breakeven for full verification and what it depends on.* —  $p^* = c/u$ , 11.67 % on Auriga, and it is independent of the sample size.
3. *Who may authorise a concession, and what must accompany it?* — The party bearing the consequence, never the producer alone; plus a register entry and a cumulative-effect test, since individually minor concessions aggregate into a different product.
4. *What five things must be fixed before a cumulative-effect test is a control rather than a sentence?* — Who sets the threshold and when (sponsor or design authority, at baseline); how it is derived (above the base-rate aggregate of the class, below the material-difference point); the aggregation period; the relatedness test, defined by what the concessions touch; and the named owner of the running total. Plus the consequence on breach: suspension of further concessions in that class pending the aggregate authority's decision (9.3.1).
5. *When does a disposition need something the project cannot give it?* — Where an approval regime applies, a repair, concession or regrade may need notification to or acceptance by a certifying body or regulator before use. That sits outside the project's authority, differs by sector and jurisdiction, and is established with the regulatory function and the body itself.
6. *What three fields turn "the evidence is retained" into a retention obligation?* — Record class, named custodian role, and retention period with the source of that period named: the source being the applicable regime and the contract, not the project's convenience. Nothing in this domain states a legal minimum (Domain 3, KA 3.3.4; Domain 16, KA 16.4.4).

## KNOWLEDGE AREA 9.4

## Lessons learned, continuous improvement, data quality and AI-output quality

*Topics: 9.4.1 lessons that change behaviour · 9.4.2 first-time-right and the compounding of stage yields · 9.4.3 data quality as a delivery constraint · 9.4.4 AI-output quality and the verification that earns its cost.*

### 9.4.1 Lessons that change behaviour

Lessons-learned processes are near-universal and largely ineffective, for structural rather than cultural reasons. A lesson is captured at the end of a project, by the

people leaving it, into a repository consulted by nobody, as a narrative rather than an instruction. Nothing in that chain contains a mechanism by which a future decision changes.

**The four properties of a lesson that works**, each a test the entry passes or fails. It is **specific to a decision**: it names the decision class it should alter, not a theme. It has an **owner in the standing process**, so the change lands in a template, checklist, gate criterion, estimating parameter or contract clause rather than in a document. It carries a **number** where one exists — Auriga's expected containment cost per introduced defect (USD 2,550), its rework allowance (16.67 %), the self-consistent sample size for a zero-defect plan — because a parameter propagates and a paragraph does not. And it is **captured when it is learned**, not at closeout, since late capture loses both fidelity and the chance to apply it on the same project.

**The improvement loop, and its only honest measure.** Continuous improvement is the cycle of measuring a process, changing it and re-measuring: the plan-do-check-act discipline from quality management, applied to delivery processes. Its integrity rests entirely on the last step, which is the one usually skipped: an improvement implemented but never re-measured is a change, not an improvement, and organisations accumulate large numbers of them. The measure that keeps the loop honest is the **recurrence rate** of 9.3.3, because it is the only figure that cannot be satisfied by activity.

#### 9.4.2 First-time-right and the compounding of stage yields

**Definition.** The **first-time-right yield** of a step is the share of items passing it without rework. The **rolled throughput yield (RTY)** of a chain of sequential steps is the product:

$$RTY = \prod y_i \quad y_i = \text{first-time-right yield of step } i$$

Sequential steps compound multiplicatively, which is why chains of individually respectable steps produce disreputable end-to-end results, and why step-level metrics can all look healthy while delivery does not.

#### WORKED EXAMPLE

**Worked example 9.4.2 — Auriga's handover chain: six good steps, one coin flip.**

1. **Setup.** Auriga's 60 handover packages (one per plant area) pass through six sequential steps with observed first-time-right yields: design package issue **0.95**, supplier document review **0.92**, factory acceptance **0.90**, site installation **0.94**, site acceptance test **0.88**, client handover documentation **0.85**. A first-pass failure at any step triggers a rework loop costing on average **USD 2,600**.

- 
2. **Formula.**  $RTY = \prod y_i$ . First-pass failures at step  $i$  = items reaching it first time  $\times (1 - y_i)$ ; total rework loops =  $\Sigma$  those failures. Uniform yield required for a target  $RTY$  over  $k$  steps =  $RTY^{1/k}$ .
  3. **Substitution.**  $0.95 \times 0.92 \times 0.90 \times 0.94 \times 0.88 \times 0.85$ . Flow from 60: failures 3.0000, 4.5600, 5.2440, 2.8318, 5.3237, 5.8561.
  4. **Result.**  $RTY = 55.3074\%$ . The arithmetic mean of the six step yields is **90.6667%**: a figure describing nothing anyone experiences. Of 60 packages, **33.18** pass the whole chain first time and **26.82** require at least one rework loop, costing **USD 69,720**.
  5. **Interpretation.** The gap between 90.67 % and 55.31 % is the point, and it is the arithmetic behind a complaint heard in every delivery organisation: *every function reports good numbers and the handover is still chaos*. Both figures are correct; only one is relevant, because a package must survive all six steps and the customer experiences the product, not the mean.

**Where to improve is determined by the product form, not by judgement.**

Raising the weakest step (handover documentation,  $0.85 \rightarrow 0.95$ ) multiplies  $RTY$  by  $0.95/0.85$  to **61.8142%**, a gain of **6.507 percentage points**, cutting rework loops from 26.82 to 22.91 and saving **USD 10,151** on this chain alone. Raising the strongest step,  $0.95 \rightarrow 0.98$ , gives **57.0540%**: a gain of **1.747 points**, or **3.73 times less**, for effort that is usually harder because the step is already good. The binding step is identifiable from the yields with no judgement at all.

**The target arithmetic should be said out loud to sponsors.** An end-to-end first-time-right target of **80%** over six steps requires **96.35%** at every single step ( $0.80^{1/6}$ ); a 90 % target requires **98.26%**. That is why "we are about 90 % right" organisations fail at handover, and why a target set without this calculation is an instruction nobody can execute.

Three cautions. The failure counts are **first-pass failures**, counted on the flow reaching each step first time; a package failing twice contributes two loops in reality, and the model states its convention rather than hiding it.  $RTY$  assumes steps are **sequential and each must be passed**: parallel or optional steps need a different structure, and a chain drawn wrongly gives a confidently wrong number. And a step yield is meaningful only where the step has a **defined pass criterion**: yields measured against an undefined standard record the reviewer's mood, so Domain 5, KA 5.4.2's testability requirement is the prerequisite for this whole measure.

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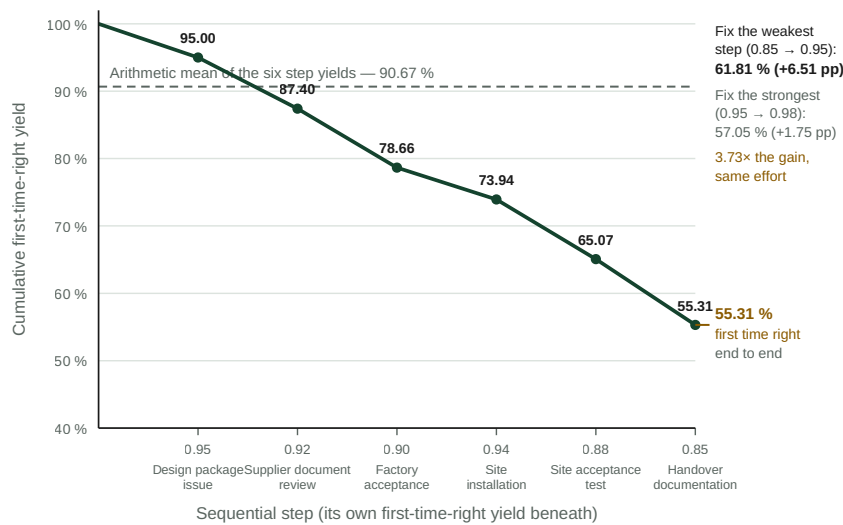


Fig 9.4.1 — Rolled throughput yield: six good steps, one coin flip

### 9.4.3 Data quality as a delivery constraint

A project's decisions are made on its data (the asset register, the requirements baseline, the cost ledger, the risk register, progress measurement), and data with defects produces decisions with defects, at a scale and speed no deliverable inspection catches. Data quality is also what determines whether the analytics, dashboards and AI systems of Domain 14 are usable at all, and the standard sequencing error in digital delivery is to buy the analytics before fixing the data.

**The dimensions**, each measurable as a pass rate over records: **completeness** (required fields populated), **accuracy** (values correspond to reality), **validity** (values conform to defined format, type and range), **consistency** (the same fact agrees across records and systems), **timeliness** (current enough for the decision), **uniqueness** (no unintended duplicates). International reference points may be cited by name (the ISO 8000 series on data quality and the ISO/IEC 25012 data quality model), and both are frameworks for defining and assessing dimensions rather than sources of a target number; the target is a project decision.

#### WORKED EXAMPLE

#### Worked example 9.4.3 — Auriga's asset register, and why 96 % is 78 %.

- Setup.** Auriga must hand over an asset register of **3,200** equipment records to the utility's maintenance system. Sampling gives per-dimension conformance of completeness **0.960**, accuracy **0.930**, validity **0.980**, consistency **0.910**, timeliness **0.990**, uniqueness **0.995**. A record is fit for use only if it satisfies **all six**.
- Formula.** Composite fitness =  $\prod$  dimension conformance rates (treating dimensions as independent). Conforming records = composite × record count.

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3. **Substitution.**  $0.960 \times 0.930 \times 0.980 \times 0.910 \times 0.990 \times 0.995$ .
  4. **Result.** Composite fitness **78.43** % against an arithmetic mean of the six dimensions of **96.08** %. Of 3,200 records, approximately **2,510** are fit for use and **690** are not.
  5. **Interpretation.** This is 9.4.2's compounding applied to data, producing the same surprise for the same reason: a dashboard reporting six dimensions all above 91 % is reporting a register in which nearly a quarter of records fail somewhere. The remediation logic is also the same, attack the weakest dimension. Raising **consistency** from 0.910 to 0.980 lifts the composite to **84.46** %, a gain of **6.03 points** and roughly **193** more usable records, for work that is usually reconciliation rather than re-survey. Three professional points. Independence across dimensions is an **assumption and usually optimistic**: records that are incomplete are often also inconsistent, so for correlated defects the true composite tends to be better than the naive product, and the figure should be sample-verified end to end rather than only dimension by dimension. The composite must be computed against **fitness for a named decision**: a register good enough for statutory reporting may be inadequate for condition-based maintenance, and a single "data quality percentage" divorced from a use is a number without a meaning. And data quality is an **acceptance criterion**, not a wish: the 690 non-conforming records are a nonconformance under 9.3.1, with a disposition, an owner and a date, or they will be discovered by the receiving organisation after Domain 16's transition, when the project team has dispersed.
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#### 9.4.4 AI-output quality and the verification that earns its cost

When an AI system produces work a professional must sign for (drafted commissioning procedures, generated test cases, extracted requirements, a summarised evidence pack) the professional's obligation is unchanged: they are accountable for the output (Domain 1; *AI proposes; the professional verifies, decides and remains accountable*). The practical question is therefore not whether to verify but **how much**, and that is a sampling question with an answer, which is where 9.3.2's arithmetic becomes the most useful tool in this domain.

Two properties of AI output make the answer differ from a supplier's. Errors are **systematic rather than random**: a model that misunderstands an instruction misunderstands it consistently, so defects cluster by type and stratified sampling across item types is essential. And output is **fluent regardless of correctness**, which removes the informal signal reviewers rely on with human work: an unsure human writes hesitantly, and a wrong model does not.

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#### WORKED EXAMPLE

### Worked example 9.4.4 – verifying 240 AI-drafted commissioning steps.

1. **Setup.** Auriga's commissioning team uses an AI assistant to draft **240** commissioning steps from the design documents. A reviewer checks a random **20** and finds no error. Reviewing one step costs **USD 40** of engineer time; an erroneous step reaching execution costs **USD 1,800** (an aborted test, a re-scheduled outage slot, a re-issued procedure). What verification is defensible?
2. **Formula.** As 9.3.2: 95 % bound  $1 - 0.05^{(1/n)}$ ; breakeven error fraction  $p^* = c/u$ ; sample to bound the error rate at a target  $p$ :  $n = \ln(0.05)/\ln(1 - p)$ .
3. **Substitution.**  $1 - 0.05^{(1/20)}$ ;  $40/1,800$ ;  $\ln(0.05)/\ln(0.98)$ .
4. **Result.** The clean sample of 20 bounds the error rate at **13.91 %**: up to **33.39** of the 240 steps, i.e. as many as 33 erroneous steps. The breakeven error fraction is  $40/1,800 = 2.22$  %. Bounding the error rate at **2 %** would require **149** items: **62.1 %** of the population, at a review cost of **USD 5,960** against **USD 9,600** to review all 240. If the error rate really were at the bound, the 220 unreviewed steps would carry an expected consequence of  $0.1391 \times 220 \times 1,800 = \text{USD } 55,087$ .
5. **Interpretation.** The conclusion is uncomfortable and it is arithmetic, not conservatism. **Because the breakeven error fraction (2.22 %) is far below what a small sample can bound (13.91 %), and because the sample needed to get below the breakeven is 62 % of the population, the defensible regime here is full review:** USD 9,600 against a potential exposure of USD 55,087 on the unreviewed remainder. And that is precisely where the claimed productivity gain goes: **the saving from generating a deliverable is real only to the extent that the verification it requires is cheaper than producing it correctly in the first place.** A leader who accepts a time saving on generation without pricing the verification has not made a saving; they have moved a cost somewhere nobody is measuring it, and accepted an accountability whose evidence base is a sample of 20.

Two refinements make this practical rather than merely discouraging. The ratio  $c/u$  is a **design variable**: reducing  $u$  (by placing an independent, cheap containment layer *after* the AI-drafted step and before execution, as 9.2.2 would) raises the breakeven error fraction and can make sampling defensible where full review was not. The economics reward **building a containment chain around AI output**, not reviewing harder. And the argument is **per item class, not per tool**: a low-consequence output may be sampled lightly and legitimately, so a verification policy applying one sample size to everything is wrong in both directions at once. The governance instrument is a **verification standard proportional to consequence** (Domain 3, KA 3.A.2's requirement with the arithmetic attached), and Domain 14 develops the wider frame, including the standards that may be cited by name (ISO/IEC 42001 for AI management systems, ISO/IEC 23894 for AI risk management).

## AI in this KA

**Where it earns its place.** Reading a large lessons repository and clustering entries into recurring themes with counts, which is what makes a repository usable. Computing rolled throughput yields and their sensitivity to each step, identifying the binding step and the uniform yield a target implies. Profiling a data set across the six dimensions of 9.4.3 (a strong application, because dimension profiling is mechanical, high-volume and exactly checkable), and proposing the remediation rules that would fix the largest number of records. Generating candidate improvement actions from a nonconformance pattern, for human triage.

**Where it must not go.** It must not decide which lessons enter the standing process; that is a change to how the organisation works and needs an owner who can be held to it. It must not be the sole judge of its own output quality: a model's self-assessed confidence is not a detection rate, and a second pass by the same model is not an independent layer (9.2.2). It must not remediate data silently: an inferred value that fills a completeness gap improves the metric and can degrade the register, the worst combination available. And it must not set the verification sample size for its own output, because that is the decision whose independence the whole argument of 9.4.4 rests on.

**Verification, concretely.** Characterise every AI containment layer with a **measured** detection rate against a held-back defect set, and re-measure on any change of model, prompt or artefact type. Compute the verification sample from the consequence ratio  $c/u$  and record that computation in the quality plan, so the sample size has a stated basis rather than being a habit. Where data has been machine-remediated, hold an untouched control sample and compare, since a plausible wrong fill is detectable only against ground truth. And reproduce the yields, composites and bounds by hand: every calculation in this KA is a product, a power or a division.

### KEY TERMS — KA 9.4

| TERM                                             | MEANING                                                                               |
|--------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>First-time-right yield (<math>y_1</math>)</b> | The share of items passing a step without rework.                                     |
| <b>Rolled throughput yield (RTY)</b>             | $\prod y_1$ , the share of items passing every sequential step first time.            |
| <b>Binding step</b>                              | The step whose yield improvement raises RTY most; identifiable from the yields alone. |
| <b>Uniform yield requirement</b>                 | $RTY^{1/k}$ : the per-step yield an end-to-end target implies over $k$ steps.         |
| <b>Data quality dimensions</b>                   | Completeness, accuracy, validity, consistency, timeliness, uniqueness.                |



| TERM                                                     | MEANING                                                                                                                    |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Composite fitness</b>                                 | The product of dimension conformance rates; the share of records fit for use.                                              |
| <b>Recurrence rate</b>                                   | Repeat nonconformances as a share of all; the honest measure of an improvement loop.                                       |
| <b>Verification standard proportional to consequence</b> | A rule setting review depth from the cost of an escaped error, not from convenience.                                       |
| <b>Systematic error</b>                                  | An error repeating consistently across similar items, the characteristic shape of AI error, requiring stratified sampling. |

SAMPLE MCQS — KA 9.4

**MCQ 9.4-A** [9.4.2 · Application] Six sequential steps have first-time-right yields 0.95, 0.92, 0.90, 0.94, 0.88 and 0.85. The rolled throughput yield is closest to:

- A. 90.7 %
- B. 55.3 % CORRECT
- C. 44.7 %
- D. 85.0 %

*Rationale:* RTY is the product, 0.55307 (9.4.2). A is the arithmetic mean of the six yields, which describes nothing anyone experiences; C is the complement of the product; D is the worst single step.

**MCQ 9.4-B** [9.4.2 · Evaluation] On that chain, improvement effort is best directed at the step with yield 0.85 rather than the one with 0.95 because:

- A. the 0.85 step is later in the chain
- B. raising 0.85 to 0.95 multiplies RTY by 0.95/0.85 and gains 6.51 points, against 1.75 points for raising 0.95 to 0.98 CORRECT
- C. late steps are always cheaper to improve
- D. the 0.95 step is already compliant

*Rationale:* The multiplicative form makes the gain proportional to the ratio of new to old yield, so the weakest step gives the largest lift, here 3.73 times as much (9.4.2). A confuses position with leverage; C is unsupported.

**MCQ 9.4-C** [9.4.2 · Analysis] An end-to-end first-time-right target of 80 % across six sequential steps requires a per-step yield of about:

- A. 80.0 %
- B. 96.4 % CORRECT
- C. 93.3 %
- D. 88.9 %

*Rationale:*  $0.80^{(1/6)} = 0.9635$  (9.4.2). A applies the end-to-end figure per step; C divides 80 % by 6 and adds it back; D treats the losses as additive.

**MCQ 9.4-D** [9.4.3 · Application] Six data quality dimensions score 0.960, 0.930, 0.980, 0.910, 0.990 and 0.995, and a record is fit for use only if it satisfies all six. Composite fitness is closest to:

- A. 96.1 %
- B. 78.4 % CORRECT
- C. 91.0 %
- D. 21.6 %

*Rationale:* The product is 0.7843 (9.4.3). A is the arithmetic mean; C is the weakest dimension; D is the complement of the product.

**MCQ 9.4-E** [9.4.4 · Evaluation] Reviewing one AI-drafted item costs 40; an escaped erroneous item costs 1,800; a clean sample of 20 has been taken from 240 items. The defensible conclusion is:

- A. the sample is clean, so the output may be accepted
- B. the breakeven error fraction is 2.22 % while a clean sample of 20 bounds the rate only at 13.91 %, and reaching the breakeven needs 149 items — so full review at 9,600 is indicated CORRECT
- C. the sample should be increased to 59 items, bounding the rate at 5 %
- D. the output should not be used at all

*Rationale:* The bound must lie below the breakeven for the plan to support its own decision (9.4.4, applying 9.3.2). A accepts on a sample that cannot exclude 33 erroneous items; C bounds at 5 %, still more than double the breakeven; D is not supported, the arithmetic supports full review, not abandonment.

**MCQ 9.4-F** [9.4.4 · Analysis] Which measure most raises the breakeven error fraction and so makes sampling AI output defensible?

- A. increasing the sample size
- B. adding an independent, cheap containment layer after the AI step and before execution, which lowers the cost of an escaped error CORRECT
- C. asking the model to check its own output
- D. improving the prompt

*Rationale:* The breakeven is  $c/u$ , so lowering  $u$  raises it (9.4.4 with 9.2.2). A changes the bound, not the breakeven; C is not an independent layer; D may lower the error rate but does not change the economics of verification.

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SELF-CHECK — KA 9.4

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1. *Why does a chain of six steps averaging 90.67 % deliver 55.31 %?* — Because sequential yields compound multiplicatively; the mean describes no item's experience, and every item must pass all six.
  2. *What per-step yield does an 80 % end-to-end target require over six steps?* — 96.35 % ( $0.80^{(1/6)}$ ), which is why targets set without this calculation cannot be executed.
  3. *How is the verification sample for an AI-produced deliverable set?* — From the consequence ratio: the sample must bound the error rate below  $p^* = c/u$ ; on Auriga's commissioning steps that is 2.22 %, needing 149 of 240 items, so full review is the defensible regime.
- 

## Advanced topics — Domain 9

### 9.A.1 Quality across a contractual boundary

Where a supplier produces the work, every mechanism in this domain must survive a commercial boundary, and each acquires a failure mode a leader should assume present until disproved.

**Containment layers become contractual artefacts.** A layer exists only if the contract obliges the supplier to run it, entitles the client to witness or audit it, and defines what its evidence must show. Absent those, the client's chain is one layer shorter than the plan says and its escape fraction is worse by a factor of  $1/(1 - d_i)$  for the layer that turns out not to be real. **Detection rates become commercial information:** a supplier has no incentive to disclose a weak internal detection rate, so the client's chain arithmetic runs on the supplier's optimism unless the client's own layer is independently staffed. **Nonconformance dispositions become negotiations:** a concession the client's design authority should grant is routinely proposed by the supplier alongside a schedule benefit, and 9.3.1's rule about who bears the consequence has to hold against commercial pressure. And **cost of quality relocates without falling:** a fixed-price contract moves prevention and appraisal into the supplier's cost base, where the client cannot see them and can only infer them from escapes, and the supplier's optimum, computed on *their* external-failure unit cost (a warranty obligation) rather than the client's (an operating consequence), is systematically lower. That divergence is the most important quality fact about fixed-price delivery, and the countermeasures are contractual: specified containment layers with witness rights, defect-liability terms that move  $u$  closer to the client's true figure, and acceptance criteria and sampling plans agreed before award rather than argued at delivery. Domain 10 handles the contracting machinery; the quality point is that a contract not specifying the containment chain has not specified the quality.

### 9.A.2 Governing quality in adaptive and hybrid delivery

Iterative delivery changes where quality is decided, not whether it is. **The definition of done replaces the inspection:** quality is built into the step rather than inspected at the end, so the definition of done is the containment layer and must be stated with the specificity of an acceptance criterion and measured with the discipline of a detection rate, or it is a slogan. **Technical debt is deferred internal failure, and it compounds:** work knowingly left non-conforming raises the introduced-defect count of every subsequent increment, so the honest treatment is a quantified backlog item that 9.2.2's arithmetic can see, not a cultural complaint. **Rework capacity is the flow constraint:** KA 9.2.3's multiplier  $1/(1 - r)$  is exactly how velocity degrades (a 30 % rework share is a 1.4286-times duration multiplier on everything, presenting as a team that has become slower rather than as a quality problem), and the measure that surfaces it is the rework share of capacity per iteration, reported alongside throughput.

The hybrid case, which is most real programmes, adds one requirement: parts of the work under different methods have **different containment chains and different escape fractions**, and the escape fraction of a deliverable is the product across the chain it actually traverses, not an average of the two regimes. A component built iteratively with a strong definition of done and then installed under a sequential commissioning regime inherits both chains; a leader reasoning about the average will under-estimate escapes on the path that matters.

### 9.A.3 The reviewer's quality eye

Invariants to test on any quality regime, each cheap and each diagnostic.

The **four cost-of-quality categories are defined and counted consistently**, with internal failure not quietly reclassified as appraisal. There is a **computed total cost of quality** with a stated percentage of budget and a **marginal analysis**, not merely a chosen regime. The **external-failure unit cost is stated with its source**, and the recommendation carries a **range of optimality** rather than a point. The **escape fraction is computed as a product** of layer detection rates, and consecutive layers are **methodologically different** rather than repetitions. Every layer has a **measured** detection rate, a coverage statement and a unit correction cost, including automated and AI layers, which are characterised, not trusted. There is an **expected containment cost per introduced defect**, and every prevention proposal is tested against it. The plan states a **rework allowance** as a percentage of capacity, actual rework share is measured against it weekly, and the **schedule** consequence is priced at the cost of delay on any zero-float activity. Every acceptance is a **positive record with a named signatory** and a list of open nonconformances. Every concession is authorised by the **party bearing the consequence** and subject to a **cumulative-effect test**. Every sampling plan is **self-consistent**: its confidence bound below its own breakeven  $c/u$ . Root causes pass the **three tests**, and the **recurrence rate** is reported rather than the closure rate. **RTY** is computed as a product, improvement is

directed at the **binding step**, and any end-to-end target carries the **uniform per-step yield** it implies. Data quality is expressed as a **composite against a named use**, not an average of dimensions. And AI-produced work carries a **verification sample derived from  $c/u$** , recorded in the quality plan. The test that catches the commonest new defect in modern delivery, which is an accountability accepted on the evidence of a sample of twenty.

## Industry variations – Domain 9

- **Utilities, energy and process industries.** External failure is dominated by outage windows and availability obligations, so  $u$  is high,  $p^* = c/u$  is low, and full verification is far more often the economic answer than elsewhere, which is why Auriga's regime sits at the strict end. Commissioning is the concentrated rework risk and is on the critical path almost by construction.
- **Regulated life sciences and medical devices.** Quality is a *licence* condition, not an optimisation: certain containment layers and record requirements are mandatory whatever the arithmetic says, and the cost-of-quality model runs *inside* that constraint. The professional error is presenting an economic optimum that breaches a regulatory requirement; the correct output is the optimum among compliant options.
- **Software and digital services.** Correction is cheap and deployment reversible, so  $u$  is low and rapid detection with fast rollback frequently beats heavy prevention. The characteristic failure is applying that logic to the irreversible parts (data migrations, external interfaces, published records), where a rollback does not undo the consequence.
- **Construction and infrastructure.** Physical rework is expensive and often infeasible, so prevention and early appraisal dominate and the correction-cost ladder is steeper than Auriga's. Concession management is central because a physical nonconformance frequently cannot be reworked at any acceptable cost, which puts real pressure on 9.3.1's rule about who may grant one.
- **Aerospace, rail and safety-critical transport.** Escapes carry consequences not expressible as an average cost, so 9.1.2's expected-value logic is bounded by a safety case and traceability obligations; sampling plans are typically fixed by an approval regime rather than derived, and the leader's task is to know the arithmetic well enough to see when a mandated plan is *weaker* than the economics would require.
- **Public services and administrative programmes.** Fitness for purpose dominates conformance, because the deliverable is a service experienced by citizens who did not write the specification; and data quality (9.4.3) is frequently the binding constraint on the whole benefit, since the records were assembled for a different purpose over decades.

## Case study — Domain 9: the fortnight in commissioning (utilities, Auriga)

**Situation.** Auriga entered testing and commissioning (activity F, 4 weeks, 6 engineers, on the critical path with zero float), after the week-13 review that had already shown **CPI** 0.91 and **SPI** 0.92 (Domain 7). Two weeks into F, the commissioning manager reported that the team was "working through a lot of retests" and asked for a decision by the end of the week. The programme had no rework metric, so nobody could say how much.

**What the arithmetic showed.** The project leader had the time coding re-cut and computed four numbers. Rework was consuming **30 %** of commissioning capacity against a plan that assumed **16.67 %**: 4 engineer-weeks of allowance in a 24-engineer-week phase. At 30 %,  $20 \div (6 \times 0.70)$  gives **4.7619 weeks**, an overrun of **0.7619 weeks** worth **USD 34,286** at Auriga's cost of delay of 45,000 per week, plus **4.5714 engineer-weeks** of unbudgeted labour at USD 5,225 — **USD 23,886**. Total consequence: **USD 58,171**, of which 59 % was schedule and therefore invisible in every report the programme produced. The trend mattered more than the level: at 40 % the phase would run **5.5556 weeks** and the delay cost **USD 70,000**, a further **USD 35,714** for ten points, against **USD 20,833** for the ten points between 10 % and 20 %.

**The proposal, and why it was rejected.** The commissioning manager recommended halving site acceptance test coverage to recover the schedule. Priced against 9.2.2's chain, dropping L3's detection rate from 0.50 to 0.25 raises escapes from **8** to **12**, adding  $4 \times 12,000 = \text{USD } 48,000$  of external failure against  $4 \times 3,500 = \text{USD } 14,000$  of internal correction saved, a net loss of **USD 34,000** before any schedule benefit. The recovery on offer could not close that gap, and the leader was able to say so in one sentence with a number attached, which is the difference between a governance conversation and an argument about commitment to quality.

**What was done instead.** Two decisions. **Buy capacity.** With 11.6 engineer-weeks of first-pass content remaining, adding two engineers on an off-shift basis takes the remaining duration from 2.7619 to **2.0714 weeks**, a total phase duration of **4.0714 weeks** and an overrun of only **0.0714 weeks**, or **USD 3,214** of delay instead of USD 34,286, a saving of **USD 31,071** for 4.1429 engineer-weeks of extra capacity costing USD 21,646 plus a 25 % off-shift premium of USD 5,412, or **USD 27,058** in total. Net **USD 4,013**: thin, worth doing, and worth recognising as thin. **Remove the cause.** Root-cause analysis showed **5 of 8** escapes, a cause concentration of **62.5 %**, traced to an uncontrolled supplier firmware configuration baseline. Removal cost **USD 18,000** against **5** expected recurrences at 12,000 — **USD 60,000** avoided, a net **USD 42,000**, breaking even at **1.5** recurrences.

**What the domain teaches here.** The most valuable calculation was the counterfactual. Had Auriga taken option (b) of 9.2.2b (USD 40,000 of prevention at the end of

design, cutting introduced defects from 80 to 60) site acceptance would have found 6 defects rather than 8, removing **1.5 engineer-weeks** of rework, shortening F to **4.5119 weeks** and saving **USD 11,250** of delay *on top of* the USD 11,000 of cost-of-quality saving the model had already shown. The prevention option was worth **USD 22,250**, not USD 11,000, and the missing half was schedule. **The cost-of-quality model systematically understates prevention, because it contains no clock.** The practical consequence is that any prevention proposal touching a critical-path activity should be priced twice, once in the CoQ model and once at the cost of delay. Note also which decisions were still available at week 15: buying capacity at a net USD 4,013, and removing a cause for the *next* occurrences. The decision worth USD 22,250 had expired at the end of design, and nobody had been asked to make it.

## Case study B – Domain 9: the clean sample (rail signalling, transport)

**Situation.** A metropolitan signalling renewal programme accepted a supplier's batch of **480** axle-counter installations on the evidence of a **30-item** zero-defect sample. Verifying one installation cost **USD 900**: a possession slot, two technicians and a test set. A defective installation reaching service cost the programme an estimated **USD 21,000**: an emergency possession, retest, re-commissioning and the operator's performance regime. The sample was drawn at random, was clean, and the batch was accepted. Over the following eleven months, **21** of the 450 unsampled installations were found defective in service (an observed defective fraction of **4.667 %**) at a realised cost of **USD 441,000**.

**What the arithmetic showed afterwards.** The plan had been unsound before it was run, and the test that would have shown it takes two lines. The breakeven defective fraction was  $p^* = 900/21,000 = 4.286 \%$ . The 95 % upper confidence bound after a clean sample of 30 is **9.50 %**: more than twice the breakeven. The plan could not exclude a defect rate at which verifying all 480 installations would have been correct; its bound exceeded its own breakeven by **5.22 percentage points**. The self-consistent sample size, the smallest  $n$  whose 95 % bound falls below 4.286 %, is **69** (the bound is 4.249 % at 69 and 4.310 % at 68), costing  $69 \times 900 = \text{USD } 62,100$  against the **USD 27,000** actually spent: an extra **USD 35,100**. At the defect rate that in fact obtained, a 30-item plan passes the batch **23.84 %** of the time and a 69-item plan **3.70 %**, so the extra 39 items would have converted a one-in-four chance of missing the problem into a one-in-twenty-seven chance: an expected saving of  $(0.2384 - 0.0370) \times 441,000 = \text{USD } 88,838$  for USD 35,100 of additional testing, a net **USD 53,738**.

**How it resolved.** The realised escape cost of USD 441,000 slightly exceeded the **USD 432,000** that verifying all 480 installations would have cost, a coincidence worth noticing, because it means even the most conservative option would have been only marginally justified after the event and was genuinely hard to justify before it. That is why the correction was not "inspect everything". Three changes went into the



framework contract for the following batches. Acceptance plans acquired a **self-consistency test**: the plan's 95 % bound must lie below the breakeven  $c/u$ , with both figures stated in the acceptance paper. The escaped-defect unit cost was **defined contractually**, including the operator's performance regime, so  $p^*$  could be computed rather than argued. And the supplier's own containment layers were **specified with witness rights**, on the reasoning of 9.A.1. The client's escape fraction had been computed on the assumption that the supplier ran an internal check the contract had never required.

**What the domain teaches here.** A passing sample is not evidence of conformance; it is a bound, and the bound has to be compared with something. The something is  $p^* = c/u$ , and the comparison is the whole of acceptance-sampling judgement: **a plan whose confidence bound exceeds its own breakeven defective fraction cannot support the decision it is being used to make, however cleanly it passes.** The remedy was cheap, available before any test was run, and needed no statistical sophistication beyond one power and one division.

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## Executive perspective — Domain 9

What a programme director cannot delegate in this domain:

- **The external-failure unit cost, and therefore the quality regime.**  $u$  sets where the cost-of-quality minimum sits and what  $p^* = c/u$  is; it is a judgement about consequence in an operating context, and every quality decision below it inherits it (9.1.2, 9.3.2).
- **The decision that zero defects is not the objective, and the version that is.** "No defect whose prevention costs less than its consequence" is computable; a commitment to zero defects buys the last escapes at, on Auriga's figures, 1.79 times their harm (9.1.2).
- **The rework share of capacity as a schedule metric.** Require it weekly against a stated allowance on every critical-path activity, priced at the cost of delay. Its absence is why commissioning overruns arrive as surprises (9.2.3).
- **Who may grant a concession, and whether concessions are counted.** Never the producer alone, never without a cumulative-effect test; this is Domain 3's Case study B at deliverable level and it is how a delivered product ends up materially different from the specified one (9.3.1).
- **The self-consistency of every acceptance plan you rely on.** Two numbers (the 95 % bound and  $c/u$ ), and the plan either supports its decision or does not. Ask for both in the acceptance paper (9.3.2, Case study B).
- **The verification standard for AI-produced work, derived rather than habitual.** The sample must bound the error rate below  $c/u$ ; where it cannot, the productivity gain was never real and the accountability has been accepted on the evidence of a sample of twenty (9.4.4).



## Calculation exercises – Domain 9

**Exercise 9.1** A programme with a budget of USD 6,000,000 can adopt one of four quality regimes. An escaped defect costs **USD 9,000**.

| REGIME | PREVENTION | APPRAISAL | INTRODUCED | DETECTED | INTERNAL<br>UNIT COST |
|--------|------------|-----------|------------|----------|-----------------------|
| Q1     | 15,000     | 45,000    | 150        | 80 %     | 1,000                 |
| Q2     | 55,000     | 65,000    | 110        | 90 %     | 1,200                 |
| Q3     | 120,000    | 85,000    | 80         | 95 %     | 1,400                 |
| Q4     | 220,000    | 110,000   | 50         | 96 %     | 1,500                 |

Compute the total cost of quality for each, identify the optimum, and state the external-failure unit cost at which the optimum would move one regime stricter. *Solution.* Escapes 30, 11, 4, 2. Internal failure  $120 \times 1,000 = 120,000$ ;  $99 \times 1,200 = 118,800$ ;  $76 \times 1,400 = 106,400$ ;  $48 \times 1,500 = 72,000$ . External failure 270,000; 99,000; 36,000; 18,000. Totals **USD 450,000** (7.50 % of budget), **USD 337,800** (5.63 %), **USD 347,400** (5.79 %), **USD 420,000** (7.00 %). The optimum is **Q2**. The  $Q2 \rightarrow Q3$  step costs  $205,000 - 120,000 = 85,000$  more in conformance and returns 12,400 of internal failure and 7 escapes, so it pays once  $u \geq (85,000 - 12,400)/7 = \text{USD } 10,371.43$ . The assumed 9,000 sits only **13.22 %** below that flip point, so the recommendation is **fragile** and the paper must say so. (For completeness,  $Q1 \rightarrow Q2$  pays above **USD 3,094.74** and  $Q3 \rightarrow Q4$  above **USD 45,300**.) *Common error:* choosing the regime with the fewest escapes, or reporting the minimum without the range of optimality, which converts a fragile recommendation into an apparently settled one.

**Exercise 9.2** A process introduces **150** defects. Three containment layers have detection rates **0.40**, **0.50** and **0.60**, with correction costs of **USD 700**, **USD 2,200** and **USD 5,000**. An escaped defect costs **USD 16,000**. Compute the escape fraction, the escapes, the internal and external failure costs, and the expected containment cost per introduced defect. Then evaluate a prevention programme costing **USD 95,000** that would avoid **30** defects. *Solution.* Escape fraction  $0.60 \times 0.50 \times 0.40 = 0.120$  (detection 88.0 %). Flow: L1 finds  $150 \times 0.40 = 60$  (90 remain), L2 finds  $90 \times 0.50 = 45$  (45 remain), L3 finds  $45 \times 0.60 = 27$ , escapes **18**. Internal failure  $60 \times 700 + 45 \times 2,200 + 27 \times 5,000 = 42,000 + 99,000 + 135,000 = \text{USD } 276,000$ . External failure  $18 \times 16,000 = \text{USD } 288,000$ . Total nonconformance **USD 564,000**, so the expected containment cost per introduced defect is  $564,000/150 = \text{USD } 3,760$ . The prevention programme returns  $30 \times 3,760 = \text{USD } 112,800$  for USD 95,000, a net **USD**

**17,800**; it breaks even at  $95,000/3,760 = 25.27$ , i.e. **26 defects avoided**. *Common error*: computing the escape fraction as  $1 -$  the sum of the detection rates, or applying each detection rate to the original 150 rather than to the defects reaching that layer.

**Exercise 9.3** A test phase is planned at **6 weeks** with **8 engineers** (48 engineer-weeks of capacity) against **40** engineer-weeks of first-pass work content. Cost of delay is **USD 30,000** per week and an engineer-week costs **USD 5,000**. Compute the planned rework allowance as a share of capacity; then the duration, overrun, delay cost and unbudgeted labour if rework runs at **35 %**; and the rework share at which the phase would take twice its planned duration. *Solution*. Allowance  $48 - 40 = 8$  engineer-weeks = **16.67 %** of capacity (equivalently  $1 - 40/48$ ). At  $r = 0.35$ : duration  $40 \div (8 \times 0.65) = 7.6923$  weeks, overrun **1.6923 weeks**, delay cost **USD 50,769**. Rework consumed  $0.35 \times 8 \times 7.6923 = 21.5385$  engineer-weeks, **13.5385** above allowance, costing **USD 67,692**. Total consequence **USD 118,462**. For a 12-week duration,  $40 \div (8(1 - r)) = 12$  gives  $1 - r = 0.41667$ , so  $r = 58.33 \%$ . *Common error*: adding 35 % to the 6-week plan (giving 8.1 weeks) instead of dividing by  $(1 - r)$ , and forgetting that the plan already contained a 16.67 % allowance, so the *variance* is 18.33 points, not 35.

**Exercise 9.4** A population of **300** welds is to be accepted. Verifying one weld costs **USD 260**; a defective weld reaching service costs **USD 6,500**. The proposed plan verifies **30** welds and accepts on zero defects. Compute (a) the probability the plan passes a population that is 4 % defective, (b) the 95 % upper confidence bound after a clean sample of 30, (c) the breakeven defective fraction, and (d) the smallest self-consistent sample size and its cost. *Solution*. (a)  $0.96^{30} = 29.39 \%$ . (b)  $1 - 0.05^{(1/30)} = 9.50 \%$  — on 300 welds, up to **28.51** defective. (c)  $p^* = c/u = 260/6,500 = 4.00 \%$ . (d)  $n = \ln(0.05)/\ln(0.96) = 73.385$ , so **74** (the bound is 3.967 % at 74 and 4.021 % at 73), costing  $74 \times 260 = \text{USD } 19,240$  against **USD 7,800** for the 30-weld plan and **USD 78,000** for verifying all 300 — the self-consistent plan costs only **24.7 %** of full verification. Note that the proposed plan's bound (9.50 %) is more than twice its breakeven (4.00 %): it cannot support the decision it is being used to make. *Common error*: treating a clean sample as evidence of conformance, or comparing the bound with the *target* defect rate instead of with  $p^* = c/u$ . The target is an aspiration, the breakeven is the economics.

**Exercise 9.5** A document chain has five sequential steps with first-time-right yields **0.98, 0.94, 0.91, 0.96** and **0.89**. **200** items enter and each first-pass failure costs **USD 1,900**. Compute the rolled throughput yield, the number of first-pass failures, the gain from raising the weakest step to 0.96 against raising the strongest to 0.99, and the uniform per-step yield a 90 % end-to-end target would require. *Solution*.  $RTY = 0.98 \times 0.94 \times 0.91 \times 0.96 \times 0.89 = 71.6237 \%$

(the arithmetic mean of the five yields is 93.60 %, which describes nothing). First-pass failures, computed on the flow reaching each step:  $4.0000 + 11.7600 + 16.5816 + 6.7063 + 17.7047 = 56.7527$ , costing **USD 107,830; 143.2473** items pass clean. Raising the weakest step  $0.89 \rightarrow 0.96$  multiplies RTY by  $0.96/0.89$  to **77.2570 %**, a gain of **5.633 points**, cutting failures to **45.4860** and saving **USD 21,407**. Raising the strongest  $0.98 \rightarrow 0.99$  gives **72.3545 %**, a gain of **0.731 points: 7.71 times less**. A 90 % end-to-end target requires  $0.90^{(1/5)} = 97.91$  % at every step. *Common error*: averaging the yields rather than multiplying them, and (more damaging in practice), directing improvement at the step that is easiest to change rather than at the binding one, which here costs a factor of 7.71 in return.

## Practitioner's toolkit – Domain 9

*Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.*

### Toolkit 9.T.1 – Cost-of-quality and containment sheet

One page per deliverable class, completed at planning and revisited at each gate. **Top block, the four categories**: prevention, appraisal, internal failure and external failure, each with a budget, an actual, and a definition of what is counted in it, the definitions matter more than the numbers, because inconsistent classification is the commonest way the model is made to lie. **Middle block, the containment chain**: one row per layer with its coverage, its *measured* detection rate and its source, its unit correction cost, and a note of whether the next layer is methodologically different; a computed escape fraction  $\Pi(1 - d_i)$ ; escapes; and the **expected containment cost per introduced defect**, which is what every prevention proposal is then tested against. **Bottom block, the decision**: total cost of quality with its percentage of budget, the marginal step analysis for one regime stricter and one looser, and the **range of optimality** on the external-failure unit cost. A sheet whose bottom block is blank has recorded a quality regime rather than chosen one.

### Toolkit 9.T.2 – Acceptance and disposition record with the self-consistency test

Per deliverable: reference and version · acceptance criteria referenced to the requirements baseline · the verification evidence · **the sampling plan, if any, with n, the acceptance number, the 95 % bound  $1 - 0.05^{(1/n)}$ , the verification unit cost c, the escaped-defect unit cost u, the breakeven  $p^* = c/u$ , and a pass/fail on the test that the bound lies below the breakeven** · the acceptance decision with a **named signatory and role** · the open nonconformances at acceptance, each with its disposition (rework / repair / concession / regrade / reject), the authority that granted it, the party bearing the consequence and **whether an external acceptance is engaged**, naming the body and who established that (9.3.1)

· and the running **cumulative concession total** against its threshold, with the threshold's **source** (sponsor or design authority, set at baseline), its **relatedness class**, its **aggregation period** and the **named owner of the running total** printed on the same sheet, so the test can be audited rather than merely quoted. Three record fields close it: **record class** · **custodian role** · **retention period with the source of the period named**, that source being the applicable regime and the contract rather than the project: the periods themselves come from the organisation's records, regulatory and legal functions, and nothing in this template states a legal minimum (Domain 3, KA 3.3.4; Domain 16, KA 16.4.4). Monthly integrity counts, each a number: deliverables accepted without a named signatory; concessions granted by the producer; sampling plans failing the self-consistency test; cumulative concession value against threshold, **with the classes currently suspended on breach**; and records with no custodian or no retention period. Those counts will find a quality function's defects before an auditor does.

### Toolkit 9.T.3 — Improvement register: rework, yield and cause

Three linked sections, all numeric, reported at the same cadence as cost and schedule. **Rework and capacity**: per activity, the planned rework allowance as a share of capacity, the measured share this period, the implied duration multiplier  $1/(1 - r)$ , and (for any activity with zero or minimal float) the overrun priced at the cost of delay. **Yield**: per process chain, each step's first-time-right yield, the rolled throughput yield as their product, the identified binding step, and the uniform per-step yield the current end-to-end target implies. **Cause**: per nonconformance, the root cause tested against the three tests (prevents this occurrence, prevents the class, within someone's authority to remove), the removal cost, the breakeven recurrence count  $\text{removal cost} \div \text{unit consequence}$ , the owner in the **standing process** the change lands in (template, checklist, gate criterion, estimating parameter, contract clause) and the date effective; plus the two portfolio measures, **cause concentration** and **recurrence rate**, which together answer the only question that matters about an improvement programme: are causes being removed, or described?

## Exam preparation — Domain 9

**What is assessed.** Quality, grade and fitness for purpose as three distinct tests; the four cost-of-quality categories and their opposing behaviour, including the non-monotonicity of internal failure; computing total cost of quality across regimes, locating the interior minimum and stating the range of optimality; why zero defects is not the objective; assurance versus control; containment layers, the multiplicative escape fraction and the independence caveat; the correction-cost ladder and its distinction from Domain 5's requirement ladder; the expected containment cost per introduced defect as the breakeven price of prevention; prevention against appraisal at equal spend; rework as a share of capacity, the duration multiplier and its

convexity; acceptance decisions, their four failure modes and the five nonconformance dispositions with their authorities; acceptance sampling, the confidence bound, the breakeven defective fraction  $c/u$  and the self-consistency test; root-cause analysis, the three tests, cause concentration and breakeven recurrence; lessons that change behaviour; rolled throughput yield, the binding step and the uniform yield requirement; data quality dimensions and composite fitness; and the derivation of a verification regime for AI-produced work.

**The calculations to be able to do under time pressure.** Total cost of quality for a regime and the marginal step test between two regimes, including the breakeven external-failure unit cost. Escape fraction as  $\prod(1 - d_i)$  and the defect flow layer by layer. Expected containment cost per introduced defect, and the breakeven number of defects a prevention spend must avoid. Duration from  $\text{content} \div (\text{crew} \times (1 - r))$ , the overrun, and its cost at a cost of delay.  $P(\text{clean sample}) = (1 - p)^n$ ; the 95 % bound  $1 - 0.05^{(1/n)}$ ; the breakeven  $c/u$ ; and the smallest  $n$  satisfying  $1 - 0.05^{(1/n)} \leq c/u$ . Cause concentration and breakeven recurrence count. RTY as a product, the gain from a step improvement as a ratio of yields, and the uniform yield  $\text{RTY}^{(1/k)}$ . Composite data fitness as a product of dimension rates.

**The traps.** Treating zero defects as the objective and ignoring that the last escapes cost more than they save (9.1.2, MCQ 9.1-E) · reporting the cost-of-quality minimum without the range of optimality, which turns a fragile recommendation into a settled one (Exercise 9.1) · reading a rise in internal failure cost as deterioration when quality is improving (9.1.2, MCQ 9.1-C) · adding detection rates instead of multiplying escape rates (Exercise 9.2, MCQ 9.2-A) · applying each layer's detection rate to the original defect count rather than to the defects reaching it (Exercise 9.2) · treating two layers with a shared method or source as independent detectors (MCQ 9.2-F) · adding a rework percentage to a duration instead of dividing by  $(1 - r)$  (Exercise 9.3, MCQ 9.2-D) · forgetting that a plan already contains a rework allowance, so the variance is the difference and not the whole share (Exercise 9.3) · comparing escape counts rather than costs when choosing between prevention and appraisal (9.2.2b, MCQ 9.2-C) · treating a clean sample as evidence of conformance (9.3.2, MCQ 9.3-A) · believing the breakeven defective fraction depends on the sample size (MCQ 9.3-B) · comparing the confidence bound with a target rate instead of with  $c/u$  (Exercise 9.4) · letting the producer grant its own concession, or recording concessions without a cumulative test (9.3.1, MCQ 9.3-D) · stopping root-cause analysis where a person can be blamed (9.3.3, MCQ 9.3-F) · averaging stage yields instead of multiplying them (9.4.2, MCQ 9.4-A) · improving the strongest step rather than the binding one (Exercise 9.5, MCQ 9.4-B) · setting an end-to-end yield target without computing the per-step yield it requires (MCQ 9.4-C) · reporting a data quality average rather than a composite against a named use (9.4.3, MCQ 9.4-D) · and accepting AI-produced work on a sample whose confidence bound exceeds  $c/u$  (9.4.4, MCQ 9.4-E).

**How the domain connects.** Domain 3 supplies the assurance-line model and the gate economics whose product-level counterpart is the containment chain, and the

escalation machinery that carries a root cause exceeding the project's authority. Domain 4's configuration management is the control whose absence produced this domain's dominant cause, and its cumulative test is the instrument reused for concessions. Domain 5 supplies acceptance criteria, testability and the requirement-defect ladder this domain deliberately does not duplicate. Domain 6 supplies the float that determines whether a rework overrun is a cost or a delay, and Domain 7 the blended rate and cost of delay that price both. Domain 8's risk register is where an escape probability belongs once quantified. Domain 10 carries quality across the contract boundary, where the supplier's optimum and the client's diverge. Domain 14 develops the AI governance frame that 9.4.4's sampling arithmetic serves. And Domain 16 receives the consequences: a handover chain's rolled yield and an asset register's composite fitness are readiness measures before they are quality measures.

## Domain 9 summary

Quality is conformance to what was agreed; grade is what was agreed; fitness for purpose is whether the conforming, correctly graded thing works for its use. All three are testable, and only the first is testable against a document.

The domain's contribution is to make quality an **optimisation with an interior solution**. Auriga's cost of quality across five candidate regimes runs **660,000 · 448,600 · 364,000 · 397,000 · 493,000**, and the minimum (USD 364,000, 9.1 % of BAC) sits at a regime that lets **8 of 80** introduced defects escape, not at the strictest one. Removing the last six escapes would cost **USD 129,000**, or **USD 21,500 each**, against a consequence of USD 12,000, **1.79 times** the harm prevented. The honest objective is therefore not zero defects but *no defect whose prevention costs less than its consequence*, and the regime holds for any escaped-defect cost between **USD 3,540** and **USD 20,250**, which is what makes it defensible. Internal failure cost rises before it falls (132,600 at R1 against 120,000 at Ro) because better detection finds what was escaping: the movement that gets quality programmes cancelled in their first period.

Where the money goes is a **chain of containment layers** whose escape fraction is the *product* of their escape rates: Auriga's  $0.50 \times 0.40 \times 0.50 = 0.100$ , giving 8 escapes, **USD 108,000** of internal failure at an average of 1,500 and **USD 96,000** of external failure — **USD 2,550 of expected containment cost per introduced defect**, which is the breakeven price of prevention and the most useful single number in the domain. It settles the prevention-or-appraisal question arithmetically: on the same USD 40,000, cutting introduced defects from 80 to 60 is worth **USD 11,000** while raising the last layer's detection from 0.50 to 0.75 is worth **minus USD 6,000**: prevention wins by USD 17,000 despite removing fewer escapes, because appraisal relocates detection to a dearer layer while prevention removes the cost entirely. And rework is capacity before it is cost: a 30 % rework share against a 16.67 % allowance turned Auriga's 4-week commissioning into **4.7619 weeks**, costing **USD 34,286** of delay and **USD 23,886** of unbudgeted labour, with the multiplier  $1/$



$(1 - r)$  convex enough that the next ten points cost **USD 35,714** and the ten after that USD 50,000.

Acceptance is a decision with a named signatory, and the five dispositions carry different authorities: a concession belongs to whoever bears the consequence, never to the producer, and must be counted cumulatively, under a test whose threshold, derivation, period, relatedness class, owner and consequence on breach are all fixed at baseline rather than asserted in a procedure. Two things sit outside the table: where an approval regime applies, a repair, concession or regrade may additionally need a body outside the project to be told or to accept it before use; and the records this KA creates (acceptances, concessions, dispositions and their evidence) are the primary evidence of any later dispute or inspection, so each carries a class, a custodian and a retention period whose source is named, the periods themselves coming from the applicable regime, the contract and the organisation's legal and records functions rather than from this book. Sampling yields the domain's sharpest result: a clean sample of 20 instrument loops bounds the defective fraction only at **13.91 %**, while the breakeven at which full verification pays is  $p^* = c/u = 1,400/12,000 = 11.67 \%$ , independent of the sample size, so **the plan cannot exclude the defect rate at which its own strategy is wrong**, and five more loops (USD 7,000) repairs it. Case study B is the same defect realised: a 30-item plan with a 9.50 % bound against a 4.286 % breakeven passed a batch that was 4.667 % defective and cost **USD 441,000**, when **USD 35,100** of additional testing carried an expected saving of **USD 88,838**. Root causes are reached only when removal prevents the class and lies within somebody's authority; Auriga's dominant cause carried **62.5 %** of escapes and its removal broke even at **1.5 recurrences**.

Improvement compounds, and so does its absence. Six handover steps yielding 0.95, 0.92, 0.90, 0.94, 0.88 and 0.85 average **90.67 %** and deliver **55.31 %** (26.82 of 60 packages needing rework at **USD 69,720**), and the binding step, not the weakest-looking process, is where effort belongs: fixing the 0.85 step gains **6.51 points** against **1.75** for fixing the 0.95 step, **3.73 times** the return. An 80 % end-to-end target requires **96.35 %** at every one of six steps. The same arithmetic governs data: six dimensions averaging **96.08 %** leave a composite fitness of **78.43 %**, about **690** of 3,200 asset records unfit for use. And it governs AI output, where the breakeven error fraction of **2.22 %** sits far below the **13.91 %** a clean sample of 20 can bound and the sample needed to reach it is **149 of 240 items**, so full review at **USD 9,600** is the defensible regime, and the productivity gain claimed for generation was never real until the verification was priced.

The through-line: **quality is a computable trade, its optimum is not zero, and every claim of conformance is a bound that must be compared with the ratio of what checking costs to what being wrong costs**. Compute that ratio, and quality stops being a virtue and becomes a decision.

DOMAIN

# 10

## Procurement, Contracts and Supply Networks

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## Why this domain exists

Domains 5 to 9 assumed that the organisation would have the capability to do the work. On most real projects it does not, and buys a substantial part of it. Auriga's own baseline says so: of its USD 4,000,000, the installation package alone is a **USD 2,000,000** subcontract, and the controllers, the civils and the commissioning support are all bought. More than half the project is delivered by people the project leader cannot instruct, on terms someone else negotiated, against a specification written before the design was finished.

That is the gap this domain closes, and the reason it sits last in Part Two rather than first. Every technique in the preceding five domains (the requirement, the critical path, the earned-value forecast, the risk response) becomes something the leader can only *ask for* once the boundary of the contract is crossed. The instrument that converts an intention into an obligation is the contract, and the instrument that decides who is obliged is the procurement process. Both are usually treated as functions belonging to somebody else, and both are among the least reversible decisions a project makes. A poorly chosen supplier can be managed; a poorly chosen contract structure cannot, because it is the thing you would have to use to manage it.

The domain's central claim is that **procurement decisions are delivery decisions with a computable price, and almost all of them are decided on the wrong number**. Make-or-buy is decided on unit price when the answer lies in total cost of ownership including transition and exit. Tenders are decided on a weighting that nobody has tested, when the weighting itself picks the winner. Contract types are chosen by habit, when their incentive arithmetic determines what the supplier will do at month eight. Sourcing is called resilient because there are two suppliers, when the exposure sits in a sub-tier neither of them will mention. In each case the right number exists, is small enough to compute in a meeting, and changes the decision.

**Learning objectives.** After this domain a candidate can: compute a procurement chain's total lead time including governance latency and treat it as a schedule predecessor; build a make-or-buy comparison on total cost of ownership including transition-in, operating and exit costs, compute the breakeven volume, and show how pricing the capability stand-up delay moves it; select a route to market and state what each route buys; construct a price/quality evaluation model, compute the weighting at which the winner changes, and state the professional obligation to publish the model before bids are opened; test a bid for abnormal pricing against a named benchmark; compute buyer outturn and supplier margin under fixed-price, cost-plus and target-cost structures at the same actual cost, state the marginal-dollar allocation each implies, and say where the **PTA** binds; demonstrate that an incentive structure only creates value if it changes behaviour, and compute the behaviour change the supplier needs to accept it; size a performance regime so that its cap

deters rather than prices non-performance; assemble a claim's heads of cost and quantify what a notice provision puts at risk; price a dispute against its irrecoverable cost; assess an ethical-sourcing programme honestly, including where the expected-value case fails and obligation governs instead; compare single, dual-split and dual-qualified sourcing on expected cost, compute the breakeven disruption probability and the premium resilience is worth, and show why a shared sub-tier destroys the calculation; and govern AI-assisted contract and supplier analysis without letting it become commercial or legal advice.

**The master project.** Project Auriga continues. This domain buys its **84 substation controllers**, lets its **USD 2,000,000 installation subcontract** on the target-cost structure Domain 7 priced (target fee 150,000, share 70/30, ceiling 2,450,000, *PTA* **USD 2,428,571.43**), decides whether to build or buy its remote-terminal-unit configuration capability, evaluates three installation bids whose ranking depends entirely on a weighting, prices a variation claim at the blended engineering rate of **USD 130.625 per hour** established in KA 7.4.1, and tests three sourcing structures against the controller lead-time risk that Domain 8 carries as R1 (p 0.35, impact USD 240,000). Every figure in this domain is either computed here or cited from the domain that derived it.

## KNOWLEDGE AREA 10.1

# Make-or-buy and the procurement lifecycle

*Topics: 10.1.1 what procurement is for · 10.1.2 make-or-buy on total cost of ownership · 10.1.3 the procurement lead time as a delivery constraint.*

## 10.1.1 What procurement is for

**Definition.** Project procurement is the process by which a project **converts a need it cannot meet internally into an enforceable obligation held by another organisation**, and then manages that obligation to delivery and closure. The definition is deliberately narrow at one end and wide at the other: it excludes nothing about performance management, and it insists on *enforceability*, because a supplier relationship that rests only on goodwill has no mechanism for the week goodwill runs out.

**The lifecycle, and what each stage decides.** Six stages, and the value of naming them is that each one closes options the next cannot reopen.

1. **Define the need:** the specification, and the choice between specifying an *output* (what it must achieve) and an *input* (how it must be built). This choice, not the contract type, determines who is answerable for fitness for purpose, and it is made before anyone in procurement is involved.

2. **Decide make-or-buy:** treated in 10.1.2, and the only stage at which the answer "we should not buy this at all" is still available.
3. **Choose the route to market and the contract structure:** 10.2.1 and 10.3.1.
4. **Tender, evaluate and award:** KA 10.2. The stage with the most process and the least remaining freedom, which is why it absorbs attention disproportionate to its leverage.
5. **Administer the contract:** KA 10.3. Where nearly all of the value is won or lost, and where staffing is nearly always thinnest.
6. **Close out:** final account, retention release, warranties, knowledge and data transfer, and the lessons that inform the next specification. Routinely under-resourced because the project is over, which is precisely why final accounts settle badly.

**The three questions a leader must answer before procurement starts**, none of which a procurement function can answer for them. *What must this supplier be accountable for, in words that will still mean the same thing in a dispute? What will I do if they fail, and is that response available under the structure I am about to sign? What information am I entitled to, on what cadence, and is it enough to run Domain 7's earned-value system across the boundary?* A project that reaches award without those three answers has bought a supplier rather than a capability.

**The boundary problem.** Domain 3 established that governance must span the contract boundary and that decision rights are most often left undefined exactly there (KA 3.1.2). The specific asymmetry to watch is **threshold mismatch**: the client's change-approval threshold and the supplier's are set independently, by different organisations, and are almost never equal. Where the supplier may commit to a change at 25,000 and the client's project leader may approve one only at 10,000, every change between those figures arrives already committed on one side and unapproved on the other: a design failure at the moment the contract was signed, not a governance failure at the moment it appears.

### 10.1.2 Make-or-buy on total cost of ownership

**The principle.** A make-or-buy decision compares **total cost of ownership over the whole period the capability is needed, including the cost of getting into the arrangement and the cost of getting out of it**. Almost every make-or-buy decision that is later regretted was made on unit price, and unit price is the one component that systematically favours the option with the higher fixed cost, because it hides that cost.

$$\begin{aligned} \text{TCO(option)} &= \text{transition-in} + (\text{unit cost} \times \text{volume}) + \text{transition-out} \\ \text{Breakeven volume } Q^* &= (F_{\text{make}} - F_{\text{buy}}) / (v_{\text{buy}} - v_{\text{make}}) \end{aligned}$$

where  $F$  is the option's total fixed cost (transition-in plus exit) and  $v$  is its cost per unit of output.  $Q^*$  exists only where the option with the higher fixed cost also has the lower unit cost, which is the normal case, and the reason the decision is a genuine choice rather than an arithmetic one.

**The four costs the unit price hides.** *Transition-in*: recruitment, training, tooling, test environments, licences, and the management attention consumed while none of it works yet. *Exit*; and this is the one most often scored at zero: redeployment or severance, decommissioning, data extraction in a usable form, knowledge transfer, re-qualification of whoever takes over, and the cost of running both arrangements in parallel during handover. *Stand-up elapsed time*, priced at the project's cost of delay wherever the capability is on the critical path. And *residual management cost*, which does not vanish when work is outsourced. It changes shape, from supervising people to administering a contract, and a buy case that assumes it falls to zero is understating by the cost of the contract manager it will need.

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#### WORKED EXAMPLE

##### Worked example 10.1.2 — Auriga's remote-terminal-unit configuration: make or buy?

- Setup.** Auriga must configure and commission remote terminal units. Building the capability in-house costs **USD 420,000** to stand up (two recruits, training, a test rig, licences), **USD 3,600** per unit configured, and **USD 60,000** to wind down at the end (rig decommissioning, redeployment). Buying it from a specialist costs **USD 95,000** to transition in (specification, tender, mobilisation, interface definition), **USD 5,400** per unit, and **USD 145,000** to transition out (data extraction into a supportable format, knowledge transfer, re-qualification of the utility's own team at handover). Standing the in-house capability up takes **14 weeks** against the supplier's **5 weeks**, and the capability is on the critical path. Auriga requires **84 units**.
- Formula.**  $TCO = F + vQ$ , with  $F$  = transition-in + exit; breakeven  $Q^* = (F_{\text{make}} - F_{\text{buy}}) / (v_{\text{buy}} - v_{\text{make}})$ ; and, where the stand-up is on the critical path, add (weeks of difference  $\times$  cost of delay) to the make case.
- Substitution.**  $F_{\text{make}} = 420,000 + 60,000 = 480,000$ ;  $F_{\text{buy}} = 95,000 + 145,000 = 240,000$ .  $Q^* = (480,000 - 240,000) / (5,400 - 3,600) = 240,000 / 1,800$ . At  $Q = 84$ : make  $480,000 + 3,600 \times 84$ ; buy  $240,000 + 5,400 \times 84$ . Delay term  $(14 - 5) \times 45,000 = 9 \times 45,000$ .
- Result.** Breakeven volume  $Q^* = 133.33$  units. At the required 84 units, make costs **USD 782,400** and buy costs **USD 693,600**: **buy is USD 88,800 cheaper**, 11.35 % of the make case. With the 9-week stand-up delay priced at **USD 405,000**, the make case rises to **USD 1,187,400** and the breakeven volume moves to **358.33 units**.

5. **Interpretation.** On unit price alone, making looks decisively cheaper: 3,600 against 5,400 is a **33.33 %** saving per unit, and that is the number a make-or-buy paper usually leads with. It is the wrong test, and the arithmetic shows exactly why: the unit-price advantage of 1,800 has to recover a fixed-cost disadvantage of 240,000, which needs 133.33 units, and Auriga has 84. The decision is therefore **a bet on volume, and the honest question is not "is making cheaper?" but "do we believe the volume will exceed 134 units?"** That reframing is the deliverable. It also yields the sensitivity that matters: a second phase of 60 further units takes the total to 144 and *reverses* the answer (make USD 998,400 against buy USD 1,017,600, a **USD 19,200** advantage to making), so a decision to buy at 84 units should be written so that it can be revisited if the second phase is authorised, with the exit cost of 145,000 understood as the price of that reversibility rather than as an overhead. Two further cautions belong in the paper, and both are about what the number cannot carry. First, **the schedule term dominates everything else here**: pricing the 9-week difference at Auriga's cost of delay adds 405,000 and moves the breakeven from 133.33 to 358.33 units, a factor of **2.69**, so on a project with a critical-path constraint the make case is not marginal, it is unavailable, and no volume Auriga will ever see makes it competitive. Second, the calculation is silent on **capability strategy**. A utility that intends to own its control systems for twenty years may rationally build a capability that loses money on this project, and that is a legitimate decision, but it must be taken as a deliberate investment with the 88,800 (or 493,800 with delay) named as its price, not smuggled in behind a unit-price comparison. The failure mode is not choosing to make; it is choosing to make while believing it is cheaper.

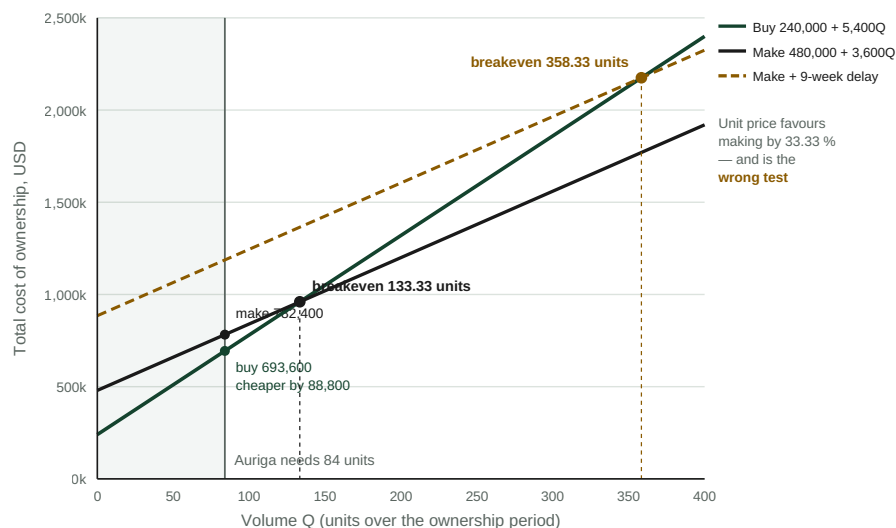


Fig 10.1.1 — Make-or-buy on total cost of ownership: the breakeven the unit price hides

### 10.1.3 The procurement lead time as a delivery constraint

**The principle.** A procurement process has a duration, that duration is a predecessor in the schedule, and it is systematically absent from plans because it is nobody's activity. Domain 6's critical path is computed over the work; the procurement chain that must complete before the work can start is computed, if at all, by a different function on a different document.

**The chain, and its two kinds of time.** Every procurement contains *process* time (specification, tender, evaluation, approval, execution) and *physical* time (manufacture, transport, acceptance). Practitioners assume the physical time dominates. It usually does not.

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#### WORKED EXAMPLE

#### Worked example 10.1.3 — the controller procurement chain against a 25-week project.

1. **Setup.** Auriga's 84 substation controllers are a long-lead item. The chain: write and internally review the specification **3 weeks**; obtain authority to go to market (a decision for a body meeting every 4 weeks with a 1-week paper deadline; tender period **4 weeks**; evaluation and clarification **3 weeks**; award approval) the same body again; contract execution and mobilisation **2 weeks**; manufacture **11 weeks**; delivery, site acceptance and functional test **2 weeks**.
2. **Formula.** Total lead time =  $\Sigma$  process legs +  $\Sigma$  governance latency +  $\Sigma$  physical legs, with governance latency  $E[\text{wait}] = M/2 + L$  per approval (Domain 3, KA 3.2.3).
3. **Substitution.** Each approval:  $E[\text{wait}] = 4/2 + 1 = 3.0$  weeks, twice. Total  $3 + 3 + 4 + 3 + 3 + 2 + 11 + 2$ .
4. **Result.** **31 weeks** of total procurement lead time, against a **25-week** project, an excess of **6 weeks**. Governance latency accounts for **6 weeks**, **19.35 %** of the chain, worth **USD 270,000** at Auriga's cost of delay. Process time totals **18 weeks** and physical time **13 weeks**, so **58.06 %** of the chain is administration.
5. **Interpretation.** The headline is uncomfortable and completely ordinary: **the procurement chain for a critical component is longer than the project that needs it**. Three consequences follow, and a leader who has done this arithmetic behaves differently at three specific moments. First, long-lead items must be committed **before the baseline is approved**, which means committing on an incomplete specification and accepting the variation exposure that follows, an explicit, priced trade, not an oversight, and the origin of Domain 8's R1 (controller lead-time slip, p 0.35, impact 240,000). Second, the largest single compressible block is not the manufacture but the **process**: 18 of 31 weeks, of which 6 are pure governance latency. Domain 3's result applies unchanged and is worth more here than anywhere, cutting the paper deadline from 1 week to nothing saves a full week per approval, so two approvals give back **2 weeks and**

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**USD 90,000** for the price of an administrative decision, while collapsing the two approvals into one delegated award authority gives back **3 weeks and USD 135,000**. Third, and least popular: the honest schedule shows procurement on the critical path from day one. A plan that starts at "contract awarded" is not optimistic, it is incomplete by 18 weeks, and the difference will be discovered at the worst possible moment; when the site team is standing ready and the controllers are in manufacture. The professional caution: the 11-week manufacture figure is a *supplier's* number, and it is the one figure in the chain that a supplier has an incentive to state optimistically at bid stage and revise after award. It should be tested against the supplier's actual recent performance, not against its quotation.

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### AI in this KA

**Where it earns its place.** Assembling total-cost-of-ownership comparisons from disparate sources (payroll, licence schedules, historical supplier invoices, exit provisions buried in incumbent contracts) is a data-collation task with a clear right answer, and the exit column is precisely the one humans leave blank because the evidence is scattered. Extracting stated lead times from a set of bids and comparing them with the same suppliers' delivery history. Modelling the breakeven volume across ranges of unit cost, transition cost and delay, which is deterministic and cheap to sweep. Reading an incumbent contract for its termination, transition-out, data-return and intellectual-property clauses and listing the exit obligations, which is where the exit cost actually comes from.

**Where it must not go.** It must not supply the *estimates* it is asked to compare. A model asked "what does it cost to stand up an RTU configuration team?" will produce a confident figure with no provenance, and that figure will then anchor a real decision: the same defect Domain 3 identified for critical-path shares. It must not decide capability strategy, which is a decision about what the organisation intends to be able to do, and belongs to an accountable executive. And its reading of an exit clause is a summary, not the clause: the obligation is what the contract says.

**Verification, concretely.** Every input to a make-or-buy paper carries a named source and a date. The breakeven volume is recomputed by hand (it is one subtraction and one division), and stated in the paper alongside the point estimate, because the breakeven is what tells the decision-maker whether the recommendation is robust. Where an AI tool has listed exit obligations, each one is confirmed against the clause before it enters the cost.

## KEY TERMS — KA 10.1

| TERM                                  | MEANING                                                                                                                                             |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Procurement</b>                    | Converting a need the project cannot meet internally into an enforceable obligation held by another organisation, and managing it to closure.       |
| <b>Total cost of ownership (TCO)</b>  | Transition-in + operating cost over the period of need + transition-out; the only valid basis for make-or-buy.                                      |
| <b>Transition-in / transition-out</b> | The cost of entering and of leaving an arrangement; the exit column is the one most often scored at zero.                                           |
| <b>Breakeven volume (Q*)</b>          | The volume at which two options cost the same: $(F_{\text{make}} - F_{\text{buy}}) / (v_{\text{buy}} - v_{\text{make}})$ .                          |
| <b>Output vs input specification</b>  | Specifying what must be achieved against how it must be built; determines who is answerable for fitness for purpose.                                |
| <b>Threshold mismatch</b>             | Client and supplier change-approval thresholds set independently, so changes between them arrive committed on one side and unapproved on the other. |
| <b>Long-lead item</b>                 | A component whose procurement chain exceeds the time available after baseline approval, and must therefore be committed earlier.                    |
| <b>Procurement lead time</b>          | Process legs + governance latency + physical legs; a schedule predecessor, not an administrative overhead.                                          |

## SAMPLE MCQS — KA 10.1

**MCQ 10.1-A** [10.1.2 · Application] In-house provision costs 420,000 to stand up, 3,600 per unit and 60,000 to exit; buying costs 95,000 to transition in, 5,400 per unit and 145,000 to exit. The breakeven volume is:

- A. 66.67 units
- B. 133.33 units CORRECT
- C. 180.56 units
- D. 227.78 units

*Rationale:*  $(480,000 - 240,000) / (5,400 - 3,600) = 240,000 / 1,800 = 133.33$  (10.1.2). A divides the fixed-cost difference by the *make* unit cost instead of the unit-cost difference ( $240,000 / 3,600 = 66.67$ ). C omits both exit costs ( $325,000 / 1,800 = 180.56$ ). D assigns each option's exit cost to the other option ( $410,000 / 1,800 = 227.78$ ).

**MCQ 10.1-B** [10.1.2 · Analysis] At 84 units the make option costs 782,400 and the buy option 693,600, yet the make option's unit cost is 33.33 % lower. The correct reading is that:

- A. the unit costs must have been miscalculated



- B. the fixed-cost difference of 240,000 has not been recovered at this volume, so the decision is a bet on volume exceeding 133.33 units CORRECT
- C. unit cost is the more reliable comparison because it excludes one-off items
- D. the two options are equivalent because the difference is under 15 %

*Rationale:* The unit-price advantage must recover a fixed-cost disadvantage, and 84 units does not (10.1.2). C inverts the principle, excluding the one-off items is the error. D substitutes a tolerance for an answer.

**MCQ 10.1-C** [10.1.2 · Evaluation] Standing the in-house capability up takes 9 weeks longer than mobilising the supplier, and the capability is on the critical path at a cost of delay of 45,000 per week. The effect on the breakeven volume is to move it from 133.33 units to:

- A. 133.33 units — elapsed time does not affect a cost comparison
- B. 225.00 units
- C. 358.33 units CORRECT
- D. 491.67 units

*Rationale:*  $(480,000 + 405,000 - 240,000)/1,800 = 645,000/1,800 = 358.33$  (10.1.2). A omits the delay entirely. B prices the delay alone and forgets the fixed-cost difference ( $405,000/1,800 = 225.00$ ). D omits the buy option's fixed cost from the numerator ( $885,000/1,800 = 491.67$ ).

**MCQ 10.1-D** [10.1.3 · Application] A procurement chain has 12 weeks of process legs, 13 weeks of manufacture and delivery, and two approvals by a body meeting every 4 weeks with a 1-week paper deadline. Total lead time is:

- A. 25 weeks
- B. 29 weeks
- C. 31 weeks CORRECT
- D. 33 weeks

*Rationale:* Each approval adds  $E[\text{wait}] = 4/2 + 1 = 3.0$  weeks, so  $12 + 13 + 6 = 31$  (10.1.3, Domain 3 KA 3.2.3). A omits governance latency altogether; B counts only half of each interval and omits the paper deadlines; D adds the whole meeting interval twice.

**MCQ 10.1-E** [10.1.1 · Comprehension] Which stage of the procurement life-cycle is the last at which the option "we should not buy this at all" remains available?

- A. define the need
- B. decide make-or-buy CORRECT
- C. choose the route to market
- D. tender and evaluate

*Rationale:* After the make-or-buy decision, subsequent stages choose *how* to buy, not *whether* (10.1.1). The stage with the most process (tender and evaluate) has the least remaining freedom.

#### SELF-CHECK — KA 10.1

1. *Why does unit price systematically favour the wrong option in make-or-buy?* — It excludes fixed transition-in and exit costs, which are exactly the costs the low-unit-cost option carries.
2. *What single figure converts a make-or-buy recommendation into a testable proposition?* — The breakeven volume: the decision becomes "do we believe volume will exceed  $Q^*$ ?"
3. *Which part of a procurement chain is usually most compressible, and why is that surprising?* — The process and governance legs, 18 of Auriga's 31 weeks, not the manufacture, which is where attention goes.

#### KNOWLEDGE AREA 10.2

## Tendering and evaluation

*Topics:* 10.2.1 routes to market · 10.2.2 the evaluation model and the weighting that chooses the winner · 10.2.3 probity, and fixing the model before the bids are opened.

### 10.2.1 Routes to market

**The choice.** A route to market is a decision about **how much competition to buy and what to pay for it**, and competition is not free: it costs elapsed time, bid cost on both sides, and the goodwill of suppliers who lose repeatedly. The main routes, with what each actually purchases:

| ROUTE                                        | WHAT IT BUYS                                                | WHAT IT COSTS                                                                                     | SUITS                                                |
|----------------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <b>Open competition</b>                      | The widest price discovery and the strongest probity record | The longest chain (10.1.3), high evaluation effort, high aggregate bid cost                       | Novel requirements, unknown market, regulated buyers |
| <b>Selective (pre-qualified) competition</b> | Price discovery among capable bidders only                  | Pre-qualification effort; the risk of excluding a capable newcomer                                | Most substantial project packages                    |
| <b>Framework call-off</b>                    | Speed: the tender is already done                           | The framework's rates, which may have drifted from the market; and the aggregation illusion below | Repeat purchases of definable scope                  |

| ROUTE                            | WHAT IT BUYS                                                            | WHAT IT COSTS                                                           | SUITS                                                    |
|----------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|----------------------------------------------------------|
| <b>Single-source negotiation</b> | Speed and continuity of a known capability                              | The absence of price discovery; a documented justification is mandatory | Genuine sole source, proprietary interfaces, emergency   |
| <b>Two-stage</b>                 | Early supplier involvement on buildability while price is settled later | Weak price tension at stage two, when the supplier is embedded          | Complex or undefined scope needing supplier design input |

**Three cautions worth stating.** *Competition is not price tension.* Three bidders who all believe one of them is favoured produce three prices and no competition; price tension requires that each bidder believes it can win, which is a function of how the model is published (10.2.3), not of the number of bidders. *The aggregation illusion in frameworks:* a call-off placed without any comparison inside the framework buys the framework's rate, not the market's, so framework rates need periodic benchmarking against a live test: a framework that has never been benchmarked is a single-source arrangement with paperwork. *Two-stage arrangements* fail predictably at stage two, when the supplier is the only party who understands the design it helped produce and the buyer's walk-away option has quietly disappeared; the countermeasure is to fix the stage-two commercial terms (fee percentage, overhead recovery, basis of preliminaries) at stage *one*, while that option still exists.

### 10.2.2 The evaluation model, and the weighting that chooses the winner

**The principle.** An evaluation model converts a set of bids into a ranking. It is a piece of arithmetic, it is designed by people, and **the design determines the outcome at least as much as the bids do**. A leader who does not understand that is not evaluating bids; they are ratifying whatever the model was set up to produce.

$$S_i(w) = w \times P_i + (1 - w) \times Q_i$$

where  $S_i$  is bidder  $i$ 's weighted score,  $P_i$  its normalised price score,  $Q_i$  its quality score, and  $w$  the weight given to price. Because each  $S_i(w)$  is linear in  $w$ , two bidders' scores cross at exactly one weighting:

$$w^* = (Q_b - Q_a) / [(P_a - Q_a) - (P_b - Q_b)]$$

#### WORKED EXAMPLE

**Worked example 10.2.2 — three bids, three winners.**

1. **Setup.** Auriga's installation package attracts three compliant bids. Alpha USD 2,000,000, quality **62**/100; Beta USD 2,200,000, quality **78**; Gamma USD 2,480,000, quality **92**. Price is normalised on the ratio convention: the lowest price scores 100, others score  $\text{lowest} \div \text{own} \times 100$ . The evaluation panel has not yet fixed the weighting.
2. **Formula.**  $S_i(w) = w P_i + (1 - w) Q_i$ ; crossover  $w^* = (Q_b - Q_a) / [(P_a - Q_a) - (P_b - Q_b)]$ .
3. **Substitution.** Price scores: Alpha  $2,000,000 / 2,000,000 \times 100 = 100.000000$ ; Beta  $2,000,000 / 2,200,000 \times 100 = 90.909091$ ; Gamma  $2,000,000 / 2,480,000 \times 100 = 80.645161$ . At  $w = 0.70$ : Alpha  $0.70 \times 100 + 0.30 \times 62$ ; Beta  $0.70 \times 90.909091 + 0.30 \times 78$ ; Gamma  $0.70 \times 80.645161 + 0.30 \times 92$ . Then at  $w = 0.40$ , and the crossovers.
4. **Result.**

| BIDDER | PRICE<br>(USD) | QUALITY | PRICE<br>SCORE | S AT<br>70/30 | S AT<br>60/40 | S AT<br>40/60 |
|--------|----------------|---------|----------------|---------------|---------------|---------------|
| Alpha  | 2,000,000      | 62      | 100.00         | <b>88.60</b>  | 84.80         | 77.20         |
| Beta   | 2,200,000      | 78      | 90.91          | 87.04         | <b>85.75</b>  | 83.16         |
| Gamma  | 2,480,000      | 92      | 80.65          | 84.05         | 85.19         | <b>87.46</b>  |

**Three different winners from one bid set.** The crossovers are Beta/Gamma at a price weight of **57.70 %** and Alpha/Beta at **63.77 %**: Gamma wins below 57.70 %, Beta in the band between, and Alpha above 63.77 %.

5. **Interpretation.** The first and most important reading is procedural, not commercial: **with the bids in hand, a panel could choose a weighting to produce any of the three outcomes it liked.** That is why the weighting must be published before bids are opened (10.2.3), and it is not a formality. It is the only thing standing between an evaluation and a rationalisation. The second reading is about the model's fragility. Beta wins in a band **6.07 percentage points** wide, which is narrower than the uncertainty in the quality scores themselves; a panel whose consensus moved Beta's quality score by two points would relocate the boundary. A model whose output is knife-edge is telling the buyer something useful: that the three bids are, on this model, close to indistinguishable, and the decision should turn on something the model does not capture rather than on the third decimal place. The third reading is the commercial one, and it is the test that should discipline any quality weighting. Gamma's premium over Alpha is **USD 480,000: 24.00 %**, or USD 16,000 per quality point. What does that buy? Suppose the panel's own risk assessment maps the quality scores to the probability of the integration rework that Domain 8 carries as R3 (impact USD 320,000): Alpha 0.30, Beta 0.18, Gamma 0.10, giving expected rework of **96,000, 57,600 and 32,000**. Risk-

adjusted cost is then Alpha **2,096,000**, Beta **2,257,600**, Gamma **2,512,000**; and Alpha is still cheapest by a wide margin, because Gamma's 480,000 premium buys only  $96,000 - 32,000 = 64,000$  of avoided expected cost. The premium is **7.5 times** the risk it removes. Add a generous allowance for schedule as well (say Gamma's quality also avoids three weeks of integration delay at 45,000 a week, worth 135,000), and the total avoided is 199,000, still **281,000** short. Expressed as a breakeven: Gamma's premium is worth paying only if its quality avoids more than 480,000 of expected cost, which at Auriga's cost of delay is **10.67 weeks** of delay. That is the professional obligation in a sentence: **a quality weighting is a statement about how much money quality is worth, and if the number it implies cannot be defended against the risks it is supposed to remove, the weighting is a preference, not a judgement.** Two cautions. Quality may buy things that are genuinely outside this arithmetic (safety performance, regulatory standing, workforce capability retention, the ability to be relied on in a crisis), and where it does, those must be **named and, so far as possible, priced**, not asserted as a reason the arithmetic does not apply. And the risk mapping above is the panel's own assessment; it must be recorded as an assumption with its basis, because the whole conclusion is proportional to it.

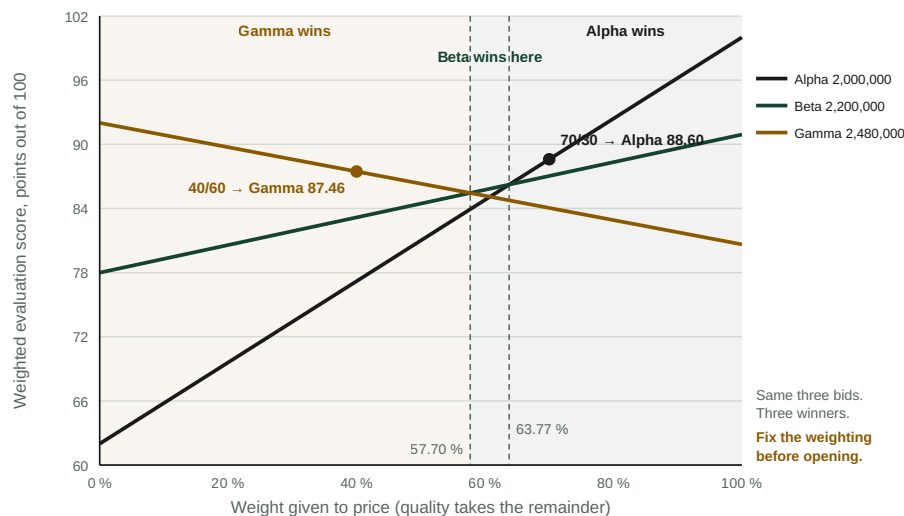


Fig 10.2.1 — The weighting chooses the winner

**Normalisation is part of the model, and it moves the answer too.** The ratio convention above is one of several. A linear convention,  $100 \times (1 - (\text{bid} - \text{lowest}) / \text{lowest})$ , gives Alpha 100, Beta **90.00** and Gamma **76.00**, because it penalises price differences more steeply. Under it the winner still changes with the weighting, but the boundaries move: Beta/Gamma crosses at **50.00 %** and Alpha/Beta at **61.54 %**. The Beta/Gamma boundary has shifted by **7.70 percentage points** purely because of a choice about arithmetic that most invitation documents do not

state. The obligation is therefore wider than publishing the weights: **publish the normalisation formula, the scoring scale, the moderation process and the treatment of non-compliance**, because each of them is a lever on the outcome.

**Testing a bid for abnormal pricing.** A price far below the others may be an efficiency the buyer should welcome, or a bid that cannot be delivered and will return as claims (Domain 7, KA 7.4.2). The test is quantitative and must specify its benchmark, because the benchmark changes the answer. Against the *bid spread*, the mean of the three bids is **USD 2,226,666.67** and Alpha is **10.18 %** below it, enough to trigger a clarification duty under most published thresholds. Against the *buyer's own estimate*, Alpha at 2,000,000 sits exactly on Auriga's target cost, and is not low at all. Both readings are true, and the professional response is neither to disqualify nor to ignore: it is to seek a structured explanation of the price build-up, satisfy the panel that the scope and the risk are actually covered, and record the answer; because if the bid is genuinely under-priced, the buyer will meet the difference again as variations, and the record is what makes that argument winnable.

### 10.2.3 Probity, and fixing the model before the bids are opened

**The obligation.** The evaluation model (weights, normalisation, scoring scale, sub-criteria and their weights, the treatment of non-compliant bids) must be **fixed, documented and disclosed to bidders before bids are opened**, and not changed afterwards. This is a professional obligation independent of whatever a particular jurisdiction's procurement law requires, and 10.2.2 is the reason: the model selects the winner, so a model chosen after the prices are known is not an evaluation method but a means of choosing a supplier while appearing not to.

**The mechanics that make it real**, in the order they occur. Publish the model in the invitation, with sub-criteria weights, not merely the top-level split. Receive bids to a single controlled point with a recorded receipt time, and hold them unopened. **Separate the price and quality submissions** and score quality *before* any price is seen. This is the single most effective anchoring control available, and its absence is the commonest defect in otherwise diligent processes. Have each panel member score independently and record those scores before moderation, so that the moderation is a documented convergence rather than a consensus with no history. Record the reason for every score, especially the low ones, since that is what a debrief and any challenge will test. Manage conflicts of interest by declaration and exclusion, before the panel is constituted. And where a change to the model becomes genuinely unavoidable, a criterion turns out to be unassessable, **re-run the evaluation under the original model as well**, disclose both results, and have the decision taken by an authority senior to the panel, because the only defence against a post-hoc change is transparency about what it altered.

**Debriefing, and why it is a commercial instrument.** A specific, evidence-based debrief to unsuccessful bidders costs a few hours and buys three things: better bids next time, a market that continues to bid, and a documented rationale that has

already survived contact with the party most motivated to attack it. A vague debrief buys a challenge.

### AI in this KA

**Where it earns its place.** Compliance checking a bid against a long requirements schedule and listing omissions and qualifications: high-volume, low-judgement, and exactly where human panels tire and miss things. Extracting a comparable price build-up from bids submitted in different formats, so that like is compared with like. Sweeping the evaluation model across weightings and normalisation conventions to reveal how sensitive the outcome is: the arithmetic of 10.2.2 run over hundreds of combinations, which is deterministic, verifiable and genuinely informative *if it is done before bids are opened, as model design*. Drafting the clarification questions that a price build-up implies.

**Where it must not go.** It must not score quality. A quality score is an expert judgement for which a named human must be answerable, and an AI-generated score cannot be defended in a debrief or a challenge because no one can explain it in the terms the criterion was written in. It must not be used to sweep weightings **after** bids are opened. The same computation that is good model design before opening is the mechanism of a rigged evaluation after it, and the difference is entirely one of timing. And it must not decide non-compliance, which is a contractual judgement with legal consequences.

**Verification, concretely.** Any AI-produced compliance list is confirmed against the bid text before it affects a score, with the confirmation recorded against the bidder. Any AI-assembled price comparison is reconciled to the bidders' own totals to the cent. The evaluation arithmetic itself (weighted scores and crossovers) is reproduced by hand for the top two bidders, because it is four multiplications and the whole award rests on it. And the audit trail must show that any weighting sensitivity analysis was performed and locked **before** the receipt deadline, with its date.

#### KEY TERMS — KA 10.2

| TERM                        | MEANING                                                                                                                                |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Route to market</b>      | The competitive structure chosen for a procurement; each route buys a different amount of price discovery at a different cost in time. |
| <b>Aggregation illusion</b> | Treating a framework call-off as competitively priced because the framework was competed; the rate may have drifted from the market.   |
| <b>Evaluation model</b>     | Weights, normalisation, scoring scale and moderation rules; a piece of arithmetic that materially selects the winner.                  |
| <b>Price normalisation</b>  | The formula converting prices into scores; the ratio and linear conventions give different scores and different crossovers.            |

| TERM                                          | MEANING                                                                                                                         |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Crossover weighting (<math>w^*</math>)</b> | The price weight at which two bidders' weighted scores are equal.                                                               |
| <b>Quality premium test</b>                   | Comparing a higher-quality bid's price premium with the expected cost it demonstrably avoids.                                   |
| <b>Abnormally low tender</b>                  | A bid materially below its benchmark; the benchmark must be stated, since the bid spread and the buyer's estimate can disagree. |
| <b>Two-envelope evaluation</b>                | Scoring quality before price is revealed, to remove price anchoring from quality judgement.                                     |

## SAMPLE MCQs — KA 10.2

**MCQ 10.2-A** [10.2.2 · Application] Bids: Alpha 2,000,000 quality 62; Beta 2,200,000 quality 78; Gamma 2,480,000 quality 92. Price is scored  $\text{lowest} \div \text{own} \times 100$ . At a 70/30 price/quality weighting the winner and score are:

- A. Gamma, 84.05
- B. Beta, 87.04
- C. Alpha, 88.60 CORRECT
- D. Alpha, 81.00

*Rationale:*  $0.70 \times 100 + 0.30 \times 62 = 88.60$ , against Beta 87.04 and Gamma 84.05 (10.2.2). A and B name the other bidders' correct scores but not the winner at this weighting; D is Alpha's score at a 50/50 weighting.

**MCQ 10.2-B** [10.2.2 · Analysis] For the same three bids, the price weight at which Beta overtakes Gamma is closest to:

- A. 50.00 %
- B. 57.70 % CORRECT
- C. 60.78 %
- D. 63.77 %

*Rationale:*  $w^* = (92 - 78) / [(90.909091 - 78) - (80.645161 - 92)] = 14 / 24.263930 = 57.70\%$  (10.2.2). A is the Beta/Gamma crossover under the *linear* normalisation convention, not the ratio convention. C is the Alpha/Gamma crossover, which lies inside Beta's winning band and is therefore not a boundary at all. D is the Alpha/Beta crossover.

**MCQ 10.2-C** [10.2.2 · Evaluation] Gamma's price premium over Alpha is 480,000. On the panel's own risk mapping, Gamma's higher quality reduces expected integration rework from 96,000 to 32,000. The strongest statement a leader can make about the premium is that it is:

- A. justified, because Gamma scores 30 quality points higher



- B. 7.5 times the expected cost it avoids, so it must be justified by something the risk assessment does not capture — named and priced CORRECT
- C. unjustifiable in all circumstances
- D. justified, because 480,000 is only 24 % of Alpha's price

*Rationale:*  $480,000 / (96,000 - 32,000) = 7.5$  (10.2.2). A and D restate inputs as conclusions. C overreaches: quality may buy safety, regulatory standing or capability the mapping omits, but those must be named, not assumed.

**MCQ 10.2-D** [10.2.3 · Analysis] A panel opens the priced envelopes, then confirms the price/quality weighting at 40/60. The defect is that:

- A. 40/60 is too low a price weighting for a construction package
- B. the weighting should have been an odd split to avoid ties
- C. with the bids known, the weighting can be selected to produce any of three winners, so a model fixed after opening is not an evaluation method

CORRECT

- D. the panel should have used the linear normalisation convention

*Rationale:* The bid set in 10.2.2 has three different winners across the weighting range, so post-hoc weighting selects the supplier (10.2.3). A and D are matters of judgement, not defects; B is invented.

**MCQ 10.2-E** [10.2.2 · Analysis] Alpha's 2,000,000 bid is 10.18 % below the mean of the three bids but exactly equal to the buyer's own target cost. The correct professional response is to:

- A. disqualify Alpha as abnormally low
- B. ignore the test, since the bid matches the buyer's estimate
- C. seek a structured explanation of the price build-up, satisfy the panel that scope and risk are covered, and record the answer CORRECT

- D. average the two benchmarks and apply the threshold to the result

*Rationale:* The two benchmarks disagree, which is information rather than a tie to be broken; the response is clarification and a record, because an under-priced bid returns as variations (10.2.2, Domain 7 KA 7.4.2). D invents a procedure.

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SELF-CHECK — KA 10.2

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1. *Why must the evaluation weighting be published before bids are opened?* — Because the weighting selects the winner: one bid set can yield three different winners across the weighting range.
2. *What does a very narrow winning band tell the buyer?* — That the bids are close to indistinguishable on this model, so the decision should turn on something the model does not capture.

- 
3. *Name the control that removes price anchoring from quality scoring.* — Two-envelope evaluation: score quality before any price is revealed.
- 

#### KNOWLEDGE AREA 10.3

## Contract strategy and supplier governance

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*Topics: 10.3.1 contract strategy as risk allocation · 10.3.2 incentive arithmetic: buyer and supplier outturn · 10.3.3 supplier governance and performance regimes.*

### 10.3.1 Contract strategy as risk allocation

**The starting point, cited not re-derived.** Domain 7 KA 7.4.2 sets out the contract models (firm fixed price, fixed-price incentive, cost plus fixed fee, cost plus incentive fee, time and materials) ordered by who carries cost risk, together with the discipline that **risk transferred is risk priced**. That taxonomy is assumed here. What this KA adds is the arithmetic of what each model does to the two parties' money at the same actual cost, and what that predicts about behaviour.

**The four questions a contract strategy must answer.** *Who carries which risk*, and the test of a sound allocation is whether the party carrying a risk can actually **influence** it; a risk allocated to a party with no control over it is not managed, it is priced, and priced badly. *What the payment mechanism rewards*: the marginal-dollar question of 10.3.2. *What happens when things go wrong*: the change, notice, suspension, step-in and termination machinery of KA 10.4, which must exist before it is needed. *What the buyer is entitled to know*: access, records, audit and progress data granular enough to run Domain 7's earned-value system across the boundary, because a supplier reporting percentage complete on its own definition supplies comfort, not control.

**Reference points, characterised strictly.** A candidate should be able to name where the practice of contract form lives, and the naming has to be exact, because this is an area where a familiar acronym is routinely mistaken for an authority.

Several families of **standard-form contract** are published and widely used: **NEC**, **FIDIC** and **JCT** in the English-language tradition, with established equivalents in civil-law systems. Each is a set of **privately published model forms**. None of them is legislation, a regulation or a standard in the sense KA 9.1.1 uses the word; none has any standing of its own. A standard form matters to a project because **the parties chose it and wrote it into their agreement**, and what it then means between them is a question of that contract and of the applicable law, which is a question for qualified counsel and not one this book answers. The professional value of knowing the families is that each embeds a *default risk allocation and a default administration rhythm*, so choosing one is choosing a starting position on the four

questions above; the professional error is to treat "we are using a standard form" as though it settled an allocation that the parties have in fact amended. None of the publishers has reviewed or endorsed this volume, no form is reproduced or paraphrased here, and no clause is cited anywhere in it.

On the collaboration side, **ISO 44001** is the document usually named as addressing the management of collaborative business relationships. It is voluntary guidance (it obliges nobody of itself unless an organisation or a contract adopts it) it is named here and not reproduced, and the alliance and collaborative-contracting material in this KA is written in this book's own words. A reader who wants it should obtain the current edition from its publisher.

**Packaging, and the interface cost of splitting work.** How work is divided between contracts is a contract-strategy decision with a computable cost, and Domain 4's result applies directly:  $n$  packages let by a buyer who integrates them itself create up to  $n(n-1)/2$  pairwise interfaces, all of which the buyer owns by default (Domain 4, KA 4.2.3). Five packages create ten interfaces; a single main-contract-plus-subcontracts structure creates five to one integrator. Splitting work buys price tension in each package and sells the integration risk to yourself, and the trade should be stated in those terms: **the price saving from splitting must exceed the cost of the interfaces it creates**, and on a technically integrated scope it frequently does not.

### 10.3.2 Incentive arithmetic: buyer and supplier outturn

**The principle.** A contract's payment mechanism determines, for every additional dollar of cost, who pays it. That allocation is what the supplier's management responds to at month eight, long after the strategy paper has been filed.

**The marginal-dollar allocation identity.** For any payment mechanism, at any cost level,

$$\begin{array}{l} \text{buyer's share of the next dollar of cost} + \text{supplier's share} \\ = 1 \end{array}$$

The identity is trivial and its consequences are not: **a contract cannot make a dollar of cost disappear; it can only decide who holds it.** Every claim that a structure "removes cost risk" is therefore a claim about allocation, and the allocation can be read off the mechanism:

| MECHANISM                                      | BUYER'S SHARE<br>OF THE<br>MARGINAL<br>DOLLAR | SUPPLIER'S<br>SHARE | WHAT THE SUPPLIER IS<br>REWARDED FOR                                                           |
|------------------------------------------------|-----------------------------------------------|---------------------|------------------------------------------------------------------------------------------------|
| Firm fixed price                               | 0.00                                          | 1.00                | Efficiency, and, unless specification and inspection are tight, for reducing quality and scope |
| Cost plus fixed fee                            | 1.00                                          | 0.00                | Nothing; cost recovery is certain and effort is not rewarded                                   |
| Target cost, 70/30 share, <b>below</b> the PTA | 0.70                                          | 0.30                | Efficiency, at 30 cents in the dollar: both parties lose on overrun                            |
| Target cost, <b>above</b> the PTA              | 0.00                                          | 1.00                | Entitlement, not efficiency: the marginal incentive is a fixed price the supplier never priced |

The last row is Domain 7's point of total assumption read as a behavioural statement rather than a commercial one, and Auriga's installation subcontract puts it at **USD 2,428,571.43** (Domain 7, KA 7.4.3 — target cost 2,000,000, target fee 150,000, share 70/30, ceiling 2,450,000). Nothing about that figure is re-derived here; what matters here is that it is the cost level at which the supplier's marginal share jumps from 0.30 to 1.00, and therefore the level at which a cooperative relationship becomes an adversarial one for reasons that have nothing to do with the people involved.

#### WORKED EXAMPLE

##### Worked example 10.3.2 — the same scope, three mechanisms, one cost distribution.

- Setup.** Auriga's installation scope. The supplier's own assessment of its outturn cost, disclosed in negotiation: **1,850,000** with probability **0.20**, **2,000,000** with **0.40**, **2,300,000** with **0.30**, **2,600,000** with **0.10**. The supplier requires an expected margin of **USD 150,000**. Three candidate mechanisms: a firm fixed price; cost plus a fixed fee of 150,000; and the target-cost structure of Domain 7 (target cost 2,000,000, target fee 150,000, share 70/30, ceiling 2,450,000, PTA 2,428,571.43).
- Formula.**  $E[\text{cost}] = \sum p \times \text{cost}$ . Fixed price (risk-neutral) =  $E[\text{cost}] + \text{required margin}$ . Cost-plus buyer outturn =  $\text{cost} + \text{fee}$ . Target-cost fee =  $\text{target fee} - \text{overrun} \times \text{supplier share}$  (negative overrun increases the fee), buyer outturn =  $\text{cost} + \text{fee}$ , capped at the ceiling. Supplier margin = buyer outturn – cost.

3. **Substitution.**  $E[\text{cost}] = 0.20 \times 1,850,000 + 0.40 \times 2,000,000 + 0.30 \times 2,300,000 + 0.10 \times 2,600,000 = 370,000 + 800,000 + 690,000 + 260,000$ . Fixed price =  $2,120,000 + 150,000$ . Target-cost fee at 1,850,000:  $150,000 + 150,000 \times 0.30 = 195,000$ ; at 2,300,000:  $150,000 - 300,000 \times 0.30 = 60,000$ ; at 2,600,000 the uncapped outturn 2,570,000 exceeds the ceiling, so the buyer pays 2,450,000 and the supplier absorbs the rest.
4. **Result.**  $E[\text{cost}] = \text{USD } 2,120,000$ ; risk-neutral firm fixed price **USD 2,270,000**.

| ACTUAL<br>COST  | FIRM FIXED PRICE:            | COST PLUS FIXED<br>FEE: BUYER / MARGIN | TARGET COST:<br>BUYER / MARGIN |
|-----------------|------------------------------|----------------------------------------|--------------------------------|
|                 | BUYER / SUPPLIER<br>MARGIN   |                                        |                                |
| 1,850,000       | 2,270,000 / <b>420,000</b>   | 2,000,000 / 150,000                    | 2,045,000 / 195,000            |
| 2,000,000       | 2,270,000 / 270,000          | 2,150,000 / 150,000                    | 2,150,000 / 150,000            |
| 2,300,000       | 2,270,000 / (30,000)         | 2,450,000 / 150,000                    | 2,360,000 / 60,000             |
| 2,600,000       | 2,270,000 / <b>(330,000)</b> | 2,750,000 / 150,000                    | 2,450,000 /<br>(150,000)       |
| <b>Expected</b> | <b>2,270,000 / 150,000</b>   | <b>2,270,000 / 150,000</b>             | <b>2,222,000 /<br/>102,000</b> |

The two expected buyer outturns for fixed price and cost-plus are **identical at USD 2,270,000**. The target-cost structure gives the buyer **USD 2,222,000** and the supplier **USD 102,000**: the buyer's **48,000** saving is exactly the supplier's **48,000** shortfall. 5. **Interpretation.** Three results, each of which changes how a leader argues about contract type.

**First: at risk-neutral pricing, transferring risk is free in expectation and buys only variance.** The fixed price and the cost-plus arrangement cost the buyer the same 2,270,000 on average, because the fixed-price premium of  $2,120,000 - 2,000,000 = 120,000$  over the efficient cost is the expected cost of the risk. What the buyer gains is certainty: the standard deviation of its outturn falls from **USD 230,434** under cost-plus to **zero** under fixed price. That is the honest case for fixed price (not that it is cheaper, but that it converts a distribution into a number), and it is worth paying a real premium for where the buyer's own funding or covenants cannot absorb the variance. A risk-averse supplier will charge one: a 25 % loading on the 120,000 risk element takes the price to **USD 2,300,000**, and that 30,000 is the visible, arguable price of certainty. The corollary is the part buyers forget: the transfer is only as good as the supplier's balance sheet. At the 2,600,000 outcome the fixed-price supplier loses **330,000**, and a supplier that cannot absorb that does not deliver a fixed price: it delivers an insolvency and a re-procurement, at which point the risk returns to the buyer with the delay added (KA 10.4.4).

**Second: an incentive structure does not create value, it reallocates it; unless behaviour changes.** As specified, the target-cost arrangement is simply 48,000 transferred from supplier to buyer, and a supplier that can do this arithmetic will respond by raising the target cost, raising the target fee, or declining. The reason is structural and worth stating precisely: **the target fee is the fee at the target, not the expected fee.** Any real probability of overrun makes the expected fee less than the target fee, so a supplier facing the distribution above expects 102,000 against a requirement of 150,000. Negotiations that stall over "the fee" are usually stalling over this, unnamed.

**Third: the behaviour change required can be computed, and it is the number the negotiation is actually about.** Suppose the incentive causes the supplier to shift probability towards the underrun outcome, the remaining mass staying in its original 3:1 ratio between the 2,300,000 and 2,600,000 branches. The supplier's expected margin is then  $187,500q + 64,500$  where  $q$  is the probability of the 1,850,000 outcome, so it reaches 150,000 at  $q = 0.456$ : the underrun probability must rise from 0.20 to 0.456, a factor of **2.28**. At that point expected cost falls from 2,120,000 to **USD 1,985,600**, so the incentive has created **USD 134,400** of genuine value; the buyer's expected outturn is **USD 2,135,600**, which is 134,400 below the fixed price; and the supplier is exactly indifferent. In other words, **at the point where the supplier will just accept the structure, the buyer captures the entire value created.** That is the real subject of the negotiation: not the fee in the abstract, but how the 134,400 is split, and a buyer who understands this can concede part of it knowing precisely what it is buying, a supplier with an actual reason to be efficient. The professional cautions are three. The distribution is the *supplier's* assessment and it has every incentive to shade it pessimistically, so it must be tested against comparable outturns rather than accepted. The behaviour shift is a hypothesis, not a fact; the structure should therefore carry measurable efficiency commitments rather than resting on the incentive alone. And none of this arithmetic survives contact with a badly defined scope: on undefined scope every mechanism converges on cost-plus with a dispute attached, whatever the contract says (Domain 7, KA 7.4.2).

### 10.3.3 Supplier governance and performance regimes

**The principle.** A contract is a document until someone administers it. Supplier governance is the machinery that turns obligations into observed performance, and its design errors are the same three every time: it measures what is easy rather than what matters, it meets on a cadence slower than the decisions it must take, and its sanctions are too small to change anyone's behaviour.

**Governing across the boundary.** Domain 3's structures apply, with three additions specific to a contract. Decision rights must be **mapped on both sides**, with the threshold mismatch of 10.1.1 resolved explicitly. The governance cadence must be set against the decision rate, not the reporting rhythm: a supplier review every eight

weeks with a two-week paper deadline imposes  $E[\text{wait}] = 8/2 + 2 = 6$  **weeks** on every decision it owns (Domain 3, KA 3.2.3), which is longer than most delivery problems will politely wait, so an out-of-cycle route with a named decision-maker on each side is not a refinement but a requirement. And the buyer must specify the **data** it receives (cost and progress at a granularity that lets its own earned-value system run across the boundary), because a governance body reading the supplier's own summary is governing the summary.

**Performance regimes: what a cap actually does.** Service credits, liquidated sums for late delivery and KPI-linked fee adjustments all work by making failure expensive. Whether they work at all depends on a comparison almost nobody makes: **the cap against the supplier's cost of compliance.**

Auriga's five-year support contract runs at **USD 2,200,000** a year with service credits capped at 5 %: **USD 110,000**. Meeting the availability target requires the supplier to hold a standby crew costing **USD 180,000** a year. The supplier's arithmetic is immediate: fail, pay the cap, and save **USD 70,000**. The regime has not deterred non-performance; it has **priced** it, and at a discount. Two different caps follow from two different purposes, and confusing them is the design error. For the credits to *deter*, the cap must exceed the cost of compliance, at least  $180,000/2,200,000 = 8.18$  % of contract value. For the credits to *compensate*, they must cover the buyer's own loss from unavailability, assessed at **USD 320,000** a year, which needs  $320,000/2,200,000 = 14.55$  %. At the 5 % cap actually agreed, the credits recover **34.38** % of the buyer's loss and deter nothing. The professional discipline is to state which purpose the regime serves, compute the cap that purpose requires, and (where the required cap is unobtainable in negotiation), record that the residual exposure is the buyer's, so that it appears in the risk register at its real size rather than being assumed away because "there are service credits".

**Relationship management, and why it is not softness.** Suppliers allocate their best people by judgement, not by contract, and they allocate them away from clients who are slow to pay, slow to decide, late with access and inclined to treat every issue as a breach. The behaviours that keep a supplier's A-team on a project are concrete and cheap: decide inside the contractual period, pay to terms, give access and information when promised, escalate issues before they become entitlements, and keep the commercial conversation separate from the delivery one so that neither poisons the other. **A buyer who is difficult to work with pays for it in the next tender and in every discretionary decision the supplier takes in between:** a real cost that appears in no report.

## AI in this KA

**Where it earns its place.** Contract analytics at scale: extracting obligations, notice periods, liability caps, change mechanisms and termination rights from a portfolio of agreements into a structured obligations register, and flagging inconsistencies between the main agreement and its schedules. Modelling the payment mechanism

across cost outcomes (the whole of 10.3.2, swept over share ratios, ceilings and distributions), which is deterministic and tedious. Monitoring supplier performance data for trend breaks. Comparing a supplier's reported progress with its own historical reporting patterns to flag optimistic drift.

**Where it must not go.** It must not give legal or commercial advice, and it must not draft a clause that is signed without qualified review; the boundary Domain 7 states holds here in full. It must not set a liability cap, a share ratio or a target cost. Those are risk-appetite decisions belonging to accountable people. It must not generate the supplier's cost distribution: a model asked for probabilities will supply them, and they will look exactly like data. And an AI reading of a clause is a summary; the obligation is the clause.

**Verification, concretely.** Every extracted obligation is cited to a clause number and spot-checked at a stated sample rate before the register is relied on. Fee, share and PTA arithmetic is reproduced by hand for at least three cost points, including one above the PTA, because that is where the mechanism changes shape. Any distribution used in an incentive model carries a named source, and the paper states the result at the pessimistic and optimistic ends, not only at the mean.

#### KEY TERMS — KA 10.3

| TERM                                       | MEANING                                                                                                                                                                  |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Marginal-dollar allocation identity</b> | For any payment mechanism, buyer share + supplier share of the next dollar of cost = 1; a contract allocates cost, it does not remove it.                                |
| <b>Point of total assumption (PTA)</b>     | The cost above which the supplier bears 100 % of further overrun (Domain 7, KA 7.4.3); read here as the point at which behaviour turns from efficiency to entitlement.   |
| <b>Risk-neutral fixed price</b>            | Expected cost plus the required margin; at this price, transferring risk costs the buyer nothing in expectation and buys variance reduction only.                        |
| <b>Risk loading</b>                        | The premium a risk-averse supplier adds above the risk-neutral price; the arguable, visible price of certainty.                                                          |
| <b>Target fee vs expected fee</b>          | The fee at the target cost, against the probability-weighted fee actually expected; any real overrun risk makes the second lower than the first.                         |
| <b>Influence test</b>                      | A risk should be allocated to the party that can influence it; a risk allocated elsewhere is priced, not managed.                                                        |
| <b>Deterrence cap vs compensation cap</b>  | The service-credit cap needed to exceed the supplier's cost of compliance, against the cap needed to cover the buyer's loss; they differ.                                |
| <b>Out-of-cycle route</b>                  | Domain 3's mechanism (KA 3.3.3) applied across a contract boundary: named authorities on <b>both</b> sides, since a route that exists for only one party is not a route. |



## SAMPLE MCQS — KA 10.3

**MCQ 10.3-A** [10.3.2 · Application] A supplier's cost outcomes are 1,850,000 (0.20), 2,000,000 (0.40), 2,300,000 (0.30) and 2,600,000 (0.10), and it requires a 150,000 expected margin. The risk-neutral firm fixed price is:

- A. USD 2,150,000
- B. USD 2,270,000
- C. USD 2,120,000
- D. USD 2,450,000

CORRECT

*Rationale:*  $E[\text{cost}] = 2,120,000$ , so the price is  $2,120,000 + 150,000 = 2,270,000$  (10.3.2). A adds the margin to the *target* cost rather than the expected cost, the commonest error, and it understates by 120,000. C is the expected cost with no margin. D is the ceiling of the target-cost alternative.

**MCQ 10.3-B** [10.3.2 · Analysis] Under a target-cost contract with a 70/30 share, above the point of total assumption the buyer's and supplier's shares of the next dollar of cost are:

- A. 0.70 and 0.30, unchanged
- B. 0.00 and 1.00
- C. 1.00 and 0.00
- D. 0.50 and 0.50

CORRECT

*Rationale:* Above the PTA the ceiling binds the buyer, so the supplier carries every further dollar (Domain 7 KA 7.4.3; 10.3.2). A misses that sharing has stopped; C reverses the exposure; D invents a split.

**MCQ 10.3-C** [10.3.2 · Evaluation] The expected buyer outturn is 2,270,000 under both firm fixed price and cost plus fixed fee, while the outturn standard deviation is 230,434 under cost-plus and zero under fixed price. The correct conclusion is that at risk-neutral pricing, fixed price:

- A. is cheaper in expectation and therefore always preferable
- B. costs the same in expectation and buys variance reduction, whose value depends on the buyer's ability to absorb variance — and on the supplier's solvency
- C. is more expensive in expectation by the risk premium
- D. eliminates the cost risk from the project

CORRECT

*Rationale:* The premium equals the expected cost of the risk, so nothing is gained in expectation except certainty (10.3.2). D contradicts the marginal-dollar identity. The risk is allocated, not removed, and returns if the supplier fails.

**MCQ 10.3-D** [10.3.2 · Analysis] As specified, the target-cost structure gives the buyer an expected outturn 48,000 below the fixed price and the supplier an expected margin 48,000 below its requirement. The most useful inference is that:

- A. the target-cost structure is superior for the buyer and should be used
- B. an incentive structure reallocates value unless it changes behaviour; the supplier will raise the target cost or fee, or decline CORRECT
- C. the supplier has mis-stated its cost distribution
- D. the share ratio should be 50/50

*Rationale:* The buyer's gain is exactly the supplier's loss, so no value is created (10.3.2). A treats a transfer as an improvement and will not survive negotiation; C is possible but not inferable; D is an unmotivated fix.

**MCQ 10.3-E** [10.3.3 · Evaluation] Service credits are capped at 5 % of a 2,200,000 contract. Compliance costs the supplier 180,000 a year and non-performance costs the buyer 320,000 a year. The regime:

- A. deters non-performance, because 110,000 is a material sum
- B. prices non-performance at a 70,000 discount to compliance, and recovers only 34.38 % of the buyer's loss CORRECT
- C. is adequate because service credits are a secondary remedy
- D. should be capped at 5 % of the buyer's loss instead

*Rationale:*  $0.05 \times 2,200,000 = 110,000 < 180,000$ , so failing is the supplier's cheaper course; and  $110,000/320,000 = 34.38\%$  of the buyer's loss (10.3.3). Deterrence needs 8.18 %, compensation 14.55 %.

#### SELF-CHECK — KA 10.3

1. *State the marginal-dollar allocation identity and its consequence.* — Buyer share + supplier share of the next dollar of cost equals 1; a contract allocates cost, it cannot remove it.
2. *Why does a supplier expect less than the target fee?* — Because the target fee is the fee at the target cost; any real probability of overrun reduces the probability-weighted fee below it.
3. *What two different caps can a service-credit regime need, and why?* — A deterrence cap exceeding the supplier's cost of compliance (8.18 % on Auriga's support contract) and a compensation cap covering the buyer's loss (14.55 %); they serve different purposes and rarely coincide.

## KNOWLEDGE AREA 10.4

## Claims, change and disputes; ethical sourcing; supply resilience

*Topics: 10.4.1 claims awareness and the anatomy of an entitlement · 10.4.2 dispute avoidance and the escalation ladder · 10.4.3 ethical and sustainable sourcing · 10.4.4 supply-chain resilience: single, dual-split and dual-qualified sourcing.*

### 10.4.1 Claims awareness and the anatomy of an entitlement

**The professional position.** A project leader is not a lawyer and should not act as one. What a leader must be able to do is recognise a claim situation as it forms, preserve the record that will decide it, and quantify the exposure well enough to escalate it accurately. **Claims are decided on contemporaneous records, and the records are made (or not made) long before anyone uses the word claim.**

**The four elements of a claim,** all of which must be present. **Entitlement,** an identified contractual basis: an instruction, a variation, a relevant event, a breach. Without it the merits do not matter. **Causation.** A demonstrated link from that basis to the effect claimed, which is where most claims are actually won or lost, because concurrent causes and the claimant's own delays are pleaded against it. **Quantum.** The money, built up by head of cost and evidenced. **Notice.** Given in the form and within the period the contract requires, which in many contracts is a precondition rather than a courtesy.

**The heads of cost, and what each requires.** *Direct cost:* additional labour, plant and materials, evidenced by records at the time. *Prolongation:* time-related site and management costs for the period of extended duration; requires a demonstrated critical-path effect (Domain 6), not merely a late finish. *Disruption,* lost productivity on work that was not itself changed, the hardest head to prove and usually established by a **measured mile:** comparing productivity in an unimpacted period with the impacted one. *Finance cost* on the funded difference, where the contract or the governing law allows it. *Overhead and profit,* at the contractual percentage. Acceleration and mitigation costs, where instructed or reasonably incurred.

## WORKED EXAMPLE

**Worked example 10.4.1 — a variation claim, and what a notice provision puts at risk.**

1. **Setup.** Auriga's installation supplier is instructed to relocate two remote-terminal-unit cabinets. It claims: **240 hours** of additional labour at the blended engineering rate of **USD 130.625** per hour (Domain 7, KA 7.4.1); **USD 18,400** of materials; **1.5 weeks** of prolongation at **USD 14,000** per week of site establishment; and disruption on **1,204** planned hours of unaffected work whose productivity fell to a factor of **0.86** of the measured-mile baseline. Overhead and

profit is **12 %** by the contract. The instruction issued on day 1; the supplier gave notice on **day 41**, against a contractual requirement of **28 days**, and under this contract the prolongation and disruption heads depend on valid notice.

2. **Formula.** Claim = (direct + prolongation + disruption) × (1 + OH&P), with disruption = (planned hours ÷ productivity factor – planned hours) × rate. Notice-bar exposure = (heads dependent on notice) × (1 + OH&P).
3. **Substitution.** Labour 240 × 130.625; disruption hours 1,204/0.86 = 1,400, so extra 1,400 – 1,204 = 196 hours at 130.625; prolongation 1.5 × 14,000. Subtotal then × 1.12.
4. **Result.** Labour **USD 31,350.00**; materials 18,400; prolongation **USD 21,000**; disruption **USD 25,602.50**. Subtotal **USD 96,352.50**; overhead and profit **USD 11,562.30**; **total claim USD 107,914.80** (≈ SAR 404,681 indicatively). The heads exposed to the notice bar total **USD 52,194.80**, **48.37 %** of the claim, leaving **USD 55,720.00** unaffected.
5. **Interpretation.** Nearly half of a properly constructed claim is at risk from a diary entry nobody made, and that is the single most useful thing a delivery leader can know about claims: **notice provisions destroy more value than negotiation does, and they do it silently**. The lesson runs in both directions, which is what makes it a professional rather than a tactical point. As a *seller*, the discipline is a notice register: every instruction, every relevant event, logged on the day with its notice deadline computed, because the 28 days runs from the event and not from the moment the cost becomes obvious. As a *buyer*, resisting a genuine entitlement purely on a time bar buys a number and sells a relationship: the supplier's response is predictable and expensive, its A-team moves, its pricing of every subsequent variation stiffens, and its own notices thereafter arrive on day one for everything, generating administrative load out of all proportion to the 52,194.80. Three further cautions. **Whether a notice provision of this kind operates as a condition precedent varies by jurisdiction and by the law of the contract**; in some legal systems such bars are enforced strictly, in others they may be read down or fall foul of statutory controls, and nothing here is legal advice or a statement of any particular jurisdiction's law. The measured-mile figure is only as good as the comparison period, an unpacted baseline drawn from a different crew, season or work type is not a measured mile, and this is the head most often successfully challenged. And the productivity arithmetic must be done the right way round: dividing by 0.86 (giving 196 extra hours) is correct, whereas multiplying by 0.14 gives 168.56 hours and understates the claim by 27.44 hours, or **USD 3,584.35** before overhead and profit.

### 10.4.2 Dispute avoidance and the escalation ladder

**The principle.** Disputes are expensive in a way that is systematically mis-estimated, because the irrecoverable cost of resolving one is compared with the amount in issue only after the decision to fight has already been taken.

**The ladder, and what each rung costs.** Direct negotiation between the people accountable for delivery; then structured negotiation between executives outside the project; then a neutral evaluation or mediation; then a determinative but interim process: adjudication, expert determination or a dispute board, depending on the contract and the jurisdiction; then arbitration or litigation. Cost and elapsed time rise by roughly an order of magnitude across the ladder, and **the relationship's usable life ends somewhere around the fourth rung**, which matters when the supplier still has work to do.

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#### WORKED EXAMPLE

#### Worked example 10.4.2 — the arithmetic of settling.

1. **Setup.** A **USD 400,000** claim is in dispute at week 30 of Auriga. Settlement now at **55 %** (USD 220,000) is achievable, with **USD 15,000** of internal and advisory cost and **3 weeks** to conclude. The alternative is arbitration: the buyer's advisers assess the outcomes as the claim largely upheld (400,000) with probability **0.55**, partly upheld (180,000) with **0.30**, and dismissed (nil) with **0.15**; the buyer's own **irrecoverable** costs are **USD 340,000** whatever the result, and the process takes **78 weeks**.
2. **Formula.** Expected award =  $\sum p \times \text{award}$ . Expected total = expected award + irrecoverable cost. Compare with the negotiated total.
3. **Substitution.**  $0.55 \times 400,000 + 0.30 \times 180,000 + 0.15 \times 0 = 220,000 + 54,000 + 0$ . Expected total =  $274,000 + 340,000$ . Negotiated total =  $220,000 + 15,000$ .
4. **Result.** Expected award **USD 274,000**: **68.50 %** of the claim, above the 55 % on offer. Expected total cost of arbitrating **USD 614,000** against **USD 235,000** for settling: a saving of **USD 379,000**, or **61.73 %**, and 75 weeks of elapsed time.
5. **Interpretation.** The decisive figure is not the 379,000 but the structural one behind it: **the irrecoverable cost of the formal route, USD 340,000, exceeds the entire negotiated settlement of 235,000**. On these numbers the buyer cannot come out ahead by arbitrating *even if it wins outright*, because a total victory saves the 235,000 it would have paid and spends 340,000 to do it, a net loss of 105,000 in the best case. That is the arithmetic that should be on the table before anyone instructs anybody, and it is very often not, because the comparison usually made is between the settlement and the *expected award* (274,000 against 220,000), which favours fighting and omits the cost of doing so. Three cautions keep this honest. Costs recovery, interest and the availability of interim determinative processes **vary substantially by jurisdiction and by**

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**forum**, and the 340,000 irrecoverable assumption must be replaced with advice specific to the contract's governing law. This is illustrative arithmetic, not a rule. Some disputes should be fought despite the arithmetic: where a point of principle will otherwise be conceded across a portfolio of contracts, where a regulator or auditor requires the matter tested, or where settling would create a precedent worth more than 379,000 to the other side. And the 78 weeks carries its own cost beyond money: the management attention consumed, and the supplier's behaviour on the remaining work while the matter is live.

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**The avoidance machinery, which is cheaper than any rung of the ladder.** A single agreed set of records, maintained jointly, so that the facts are not themselves in dispute. Early-warning obligations running both ways with short periods, so that problems arrive as information rather than as entitlements. A change mechanism fast enough to be used: Domain 4's integrated change control across the boundary, with the threshold mismatch of 10.1.1 resolved. A named commercial lead on each side with authority to settle at a stated value, and an out-of-cycle route above it (Domain 3, KA 3.3.3). And a scheduled joint review of open commercial items, so that they are closed at fifty rather than argued at five hundred.

#### 10.4.3 Ethical and sustainable sourcing

**The obligation, stated carefully.** Buyers increasingly carry duties in respect of conditions in their supply chains: labour standards and forced labour, health and safety, environmental impact, anti-bribery and sanctions compliance, and the accuracy of what they publish about all of it. Several jurisdictions have enacted human-rights, modern-slavery or supply-chain due-diligence legislation, and their scope, thresholds and reporting requirements **differ materially**; the applicable regime depends on where the buyer is established, where it sells, and the contract's governing law. Nothing in this section states the requirements of any particular jurisdiction, and none of it is legal advice. What is portable is the *method*: risk-tier the supply base, apply proportionate diligence, contract for the standard and for the right to verify it, and act on what verification finds.

**Reference points, and the distinction that must not be blurred.** The *method* named above is not this book's invention, and a candidate should be able to say where it comes from. Two international instruments are the usual reference points: the **UN Guiding Principles on Business and Human Rights** and the **OECD Guidelines for Multinational Enterprises**, together with the OECD's associated due-diligence guidance. Both are **non-binding international instruments**: they are addressed to states and, through them, to enterprises as expectations of conduct, and they impose no obligation of themselves on any project company. What they supply is the shape of the method: a risk-tiered rather than uniform diligence, diligence proportionate to severity and likelihood of harm rather than to contract value, a **remediation-first** response, and attention to the people affected rather than

only to the buyer's exposure. And that shape is why the paragraph above is written as it is.

**They are a different kind of thing from the national due-diligence legislation this section has already mentioned, and conflating the two is the error to avoid.** Legislation applies on its own terms within its own jurisdiction; a non-binding instrument does not, however widely it is cited or however often a contract refers to it. Which regimes reach a given buyer, on what thresholds, with what reporting, remains a question for qualified counsel: nothing here states the position anywhere, and adopting the method described below is a professional and commercial choice rather than a compliance conclusion.

On the procurement side, **ISO 20400** is the document usually named as addressing sustainable procurement. It is **voluntary guidance and not a certifiable requirements standard**. There is no conformity to be claimed against it in the way there is against a requirements standard, and an organisation stating that it "follows" it has described an intention rather than evidenced an outcome. All three documents are named here and not reproduced; none of their publishers has reviewed or endorsed this volume; and a reader who wants any of them should obtain the current version from its publisher.

**What "contracting for it" means concretely.** A stated standard the supplier must meet, not an aspiration. A right of audit (including of sub-tiers) with a duty to disclose them, which is also the control that makes 10.4.4's resilience analysis possible. A duty to notify adverse findings. A remediation-first response, because terminating a supplier where a labour abuse is found removes the buyer's leverage and frequently harms the workers concerned; termination belongs at the end of an escalation, not the start. And flow-down obligations, since the risk concentrates where the buyer's visibility ends.

**The arithmetic, and its honest limits.** Consider a proportionate programme across Auriga's supply base: **34** first-tier suppliers desktop-screened at **USD 1,200** each, and the **6** assessed as highest risk audited on site at **USD 14,000** each, a programme cost of **USD 124,800**. Against it: the assessed probability of a material breach in the supply chain over a three-year window is **0.08**, with a consequence (remediation, re-procurement, contractual and reputational cost) assessed at **USD 1,900,000**, giving an expected exposure of **USD 152,000** (EMV, Domain 8 KA 8.2.2). If the programme is **70 %** effective at detecting and preventing, it avoids  $0.70 \times 152,000 = \text{USD } 106,400$ , which is **USD 18,400 less** than it costs.

That result is uncomfortable and it is stated deliberately, because pretending otherwise is worse. Three things follow. First, the breakevens are useful and should be in the paper: the programme pays at a detection effectiveness above **82.11 %**, or at a breach probability above **9.38 %**, both entirely plausible in higher-risk categories and geographies, which is precisely why the programme should be **risk-tiered rather than uniform**, concentrating spend where the probability is highest instead of screening everyone equally. Second, **EMV** is a poor instrument here for the reason



Domain 8 gives: it is an average of outcomes that will not individually occur, and the consequence distribution for a serious supply-chain finding is heavily skewed, the 1,900,000 is a mean whose upper tail includes losing a licence, a listing or a market. A decision taken on the mean of a fat-tailed distribution is a decision taken on the wrong statistic. Third, and decisively: **where legal duties apply, this is not an expected-value decision at all.** Compliance is not optional because the arithmetic is marginal, and where the obligation is one of values rather than law, the organisation should say so plainly rather than construct a business case it does not believe. The arithmetic's proper job is to allocate the programme's effort, not to decide whether to have one.

#### 10.4.4 Supply resilience: single, dual-split and dual-qualified sourcing

**The principle.** Resilience is bought, and the thing being bought is the ability to keep going when a supplier cannot deliver. The mistake made almost universally is to treat "two suppliers" as the product being purchased, when what actually reduces exposure is **a qualified, exercisable alternative**; and those are not the same purchase.

$$\text{Expected cost of disruption} = \sum P(\text{state}) \times \text{consequence}(\text{state})$$

$$\text{Total expected cost} = \text{certain cost} + \text{expected cost of disruption}$$

$$\text{Breakeven disruption probability } p^* = \frac{\text{extra certain cost}}{\text{reduction in consequence}}$$

#### WORKED EXAMPLE

##### Worked example 10.4.4 — Auriga's 84 controllers: three sourcing structures.

- Setup.** Domain 8 carries controller lead-time slip as R1. Three structures are on the table. **Option 1, single source:** supplier A at **USD 9,600** per unit, one supplier qualified at **USD 40,000**. **Option 2, dual source with a 60/40 volume split:** A at 9,600 and B at **USD 10,600**, both qualified (**USD 80,000**), plus **USD 20,000** to hold two build configurations to a common specification. **Option 3, single award with a qualified alternate:** full volume to A at 9,600, both suppliers qualified (80,000), and **USD 25,000** a year to keep the alternate exercisable, specification maintenance and an annual sample build. Each supplier has an independent **0.18** probability of a disruption in the delivery window. Consequences: a single supplier disrupted with no qualified alternate requires a 14-week re-source ( $14 \times 45,000 = 630,000$  of delay plus 120,000 of re-qualification and rework plus a 150,000 contractual step) **USD 900,000**; under the dual split, one supplier disrupted means a 3-week surge ramp on the other,  $3 \times 45,000 = 135,000$  plus a 60,000 surge premium (**USD 195,000**; with a qualified alternate, activation takes 5 weeks)  $5 \times 45,000 = 225,000$  plus 70,000 of switching and re-qualification — **USD 295,000**.



2. **Formula.** Certain cost = units  $\times$  unit price + qualification + holding cost. Expected disruption =  $\sum P(\text{state}) \times \text{consequence}$ , with  $P(\text{exactly one of two}) = 2p(1 - p)$  and  $P(\text{both}) = p^2$  under independence. Breakeven  $p^* = \text{extra certain cost} \div \text{consequence reduction}$ .

3. **Substitution.** Option 1:  $84 \times 9,600 + 40,000$ , disruption  $0.18 \times 900,000$ . Option 2: blended unit  $0.60 \times 9,600 + 0.40 \times 10,600 = 10,000$ , so  $84 \times 10,000 + 80,000 + 20,000$ ; disruption  $2 \times 0.18 \times 0.82 \times 195,000 + 0.18^2 \times 900,000$ . Option 3:  $84 \times 9,600 + 80,000 + 25,000$ , disruption  $0.18 \times 295,000$ .

#### 4. Result.

|                                     | CERTAIN COST<br>(USD) | EXPECTED<br>DISRUPTION (USD) | TOTAL EXPECTED<br>(USD) |
|-------------------------------------|-----------------------|------------------------------|-------------------------|
| Option 1 (single source)            | 846,400               | 162,000                      | <b>1,008,400</b>        |
| Option 2 (dual source, 60/40 split) | 940,000               | 86,724                       | <b>1,026,724</b>        |
| Option 3 (qualified alternate)      | 911,400               | 53,100                       | <b>964,500</b>          |

Option 3 is best: **USD 43,900** better than single sourcing and **USD 62,224** better than the dual split. Its breakeven disruption probability against Option 1 is  $65,000 / (900,000 - 295,000) = 10.74\%$ ; the dual split's, from  $510,000p^2 - 510,000p + 93,600 = 0$ , is **24.22 %**. The maximum premium worth paying for Option 3's resilience (the whole reduction in expected disruption cost) is  $162,000 - 53,100 = \text{USD } 108,900$ , or **12.87 %** of the single-source cost. 5. **Interpretation.** The result that carries the teaching is the ranking: **the resilience is in the qualified alternate, not in the split volume**. Splitting the order does two things the buyer rarely accounts for. It pays a price premium (the blended unit cost rises from 9,600 to 10,000, worth 33,600 on this volume), and, under independence, it **doubles the exposure surface**: the probability that *something* goes wrong rises from 0.18 to  $1 - 0.82^2 = 0.3276$ , and although each individual event is now far cheaper, the arithmetic has to earn back the premium before it helps. Dual qualification, by contrast, retains the full volume with the single supplier (and its discount) and buys only the option, at 65,000 of extra certain spend against a 108,900 reduction in expected exposure. That leads to the general design rule: **qualify two, award one, and keep the second exercisable**, with the emphasis on *exercisable*, because a second supplier that has not built the part for three years is a name on a list, not an alternative, and the 25,000 a year is what distinguishes them. The breakevens are the negotiating numbers: Option 3 pays whenever the disruption probability exceeds **10.74 %**, while the dual split needs more than **24.22 %**, so a colleague arguing for

split sourcing on resilience grounds is implicitly asserting a disruption probability more than twice as high as the one that justifies the cheaper structure, and should be asked for it.

**\*\*The caution that overturns everything above: correlation.\*\*** All three calculations assume the two suppliers fail independently. Suppose instead that both draw a critical module from a single sub-tier source, so that the joint probability of disruption is **\*\*0.12\*\*** rather than 0.0324 (each supplier's marginal probability still 0.18, leaving  $0.18 - 0.12 = 0.06$  of idiosyncratic risk each). Then  $P(\text{exactly one}) = 2 \times 0.06 = 0.12$ , and the expected disruption costs become  $0.12 \times 195,000 + 0.12 \times 900,000 = \text{USD } 131,400$  for Option 2, total **\*\*1,071,400\*\***, and  $0.06 \times 295,000 + 0.12 \times 900,000 = \text{USD } 125,700$  for Option 3, total **\*\*1,037,100\*\***. Both are now **\*\*worse than single sourcing's 1,008,400\*\***: the buyer has paid for resilience and bought exposure. The rule to take away is blunt (**\*\*two suppliers with one sub-tier is one supplier with two invoices\*\***), and the correct response is not to choose between the options but to attack the sub-tier: require disclosure of the critical sub-tier sources as a contractual obligation (10.4.3), qualify a second module design, or hold buffer stock sized to the re-source interval. This is Domain 8's correlation result (KA 8.A.1) in a commercial setting, and it is the reason a supply network must be mapped rather than assumed.

### AI in this KA

**Where it earns its place.** Building the notice register of 10.4.1 (extracting every instruction, relevant event and correspondence item from a project's document set, computing each notice deadline and flagging what is approaching or missed) is high-volume, deterministic work with an unambiguous right answer, and it is exactly the work that goes undone. Assembling a chronology from thousands of documents for a delay analysis, for expert verification. Screening a supply base against sanctions, adverse-media and insolvency signals, and flagging changes. Reading a set of supplier disclosures to build a sub-tier map and identify shared dependencies: the concentration that 10.4.4 shows destroys a resilience case. Modelling sourcing options and their breakevens, as computed above.

**Where it must not go.** It must not assess entitlement, causation or the strength of a claim; those are legal judgements on which qualified advice is required and on which a confident, unattributable opinion is actively dangerous. It must not generate a delay analysis presented as evidence: a chronology is an input to an expert's opinion,

not the opinion. It must not decide a supplier's ethical standing from inference: a screening or adverse-media hit is a prompt to investigate, and treating it as a finding is both unfair and, in some jurisdictions, restricted, take advice before any such inference is acted on. And it must not supply the probabilities in 10.4.4; those come from delivery history and market intelligence, and a model asked for them will produce numbers indistinguishable from data.

**Verification, concretely.** Every notice deadline the tool computes is checked against the clause and the event date before anyone relies on it, and the notice register is reconciled to the correspondence log at a stated sample rate. Every chronology entry cited in a claim is verified against the source document. Every screening hit is human-reviewed before any action, with the review recorded. And the sourcing arithmetic is reproduced by hand at two probabilities, the point estimate and the breakeven, because the breakeven is the number that decides whether the recommendation survives an argument.

#### KEY TERMS — KA 10.4

| TERM                              | MEANING                                                                                                                                       |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Entitlement</b>                | The contractual basis for a claim; without it, merits are irrelevant.                                                                         |
| <b>Causation</b>                  | The demonstrated link from the contractual basis to the effect claimed; where most claims are decided.                                        |
| <b>Quantum</b>                    | The claim's money, built up by head of cost and evidenced contemporaneously.                                                                  |
| <b>Notice provision</b>           | The required form and period for notifying a claim; in some contracts and jurisdictions a condition precedent.                                |
| <b>Prolongation</b>               | Time-related costs of extended duration; requires a demonstrated critical-path effect.                                                        |
| <b>Disruption / measured mile</b> | Lost productivity on unchanged work, established by comparing an unimpacted period's productivity with the impacted period's.                 |
| <b>Notice-bar exposure</b>        | The value of the heads of claim that depend on valid notice, grossed up for overhead and profit.                                              |
| <b>Irrecoverable cost</b>         | The share of a formal dispute's cost that cannot be recovered whatever the outcome; the figure that usually decides whether fighting can pay. |
| <b>Risk-tiered diligence</b>      | Concentrating supply-chain assurance effort where breach probability is highest, rather than screening uniformly.                             |
| <b>Qualified alternate</b>        | A second supplier approved and kept exercisable, receiving no volume; the structure that actually buys resilience.                            |
| <b>Sub-tier concentration</b>     | A shared dependency below the first tier that correlates suppliers' failures and destroys a dual-sourcing case.                               |

| TERM                               | MEANING                                                                                                                                |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Expected cost of disruption</b> | $\Sigma P(\text{state}) \times \text{consequence}(\text{state})$ ; compared against the certain cost of the structure that reduces it. |

## SAMPLE MCQS — KA 10.4

**MCQ 10.4-A** [10.4.1 · Application] Disruption is claimed on 1,204 planned hours whose productivity fell to 0.86 of the measured-mile baseline, at USD 130.625 per hour. The disruption cost is:

- A. USD 22,018.15
- B. USD 25,602.50 CORRECT
- C. USD 29,770.35
- D. USD 157,272.50

*Rationale:* Hours required =  $1,204 / 0.86 = 1,400$ , so extra hours = 196 and cost =  $196 \times 130.625 = 25,602.50$  (10.4.1). A multiplies the planned hours by  $(1 - 0.86)$  instead of dividing by 0.86: 168.56 hours rather than 196, understating by USD 3,584.35. C applies the correct division to 1,400 hours instead of 1,204 (227.91 extra hours). D prices all 1,204 hours rather than the extra ones.

**MCQ 10.4-B** [10.4.1 · Analysis] A USD 107,914.80 claim comprises labour and materials of 49,750 and prolongation and disruption of 46,602.50, each grossed up by 12 % overhead and profit. Notice was given on day 41 against a 28-day requirement, and the prolongation and disruption heads depend on notice. The amount at risk is:

- A. USD 46,602.50
- B. USD 52,194.80 CORRECT
- C. USD 55,720.00
- D. USD 107,914.80

*Rationale:*  $46,602.50 \times 1.12 = 52,194.80$ , 48.37 % of the claim (10.4.1). A omits the overhead and profit attaching to those heads; C is the surviving amount; D assumes the whole claim falls, which the direct heads do not.

**MCQ 10.4-C** [10.4.2 · Evaluation] A 400,000 claim can be settled for 220,000 plus 15,000 of costs. Arbitration has an expected award of 274,000 and irrecoverable costs of 340,000. The strongest argument for settling is that:

- A. the expected award of 274,000 exceeds the 220,000 settlement
- B. arbitration takes 78 weeks
- C. the irrecoverable cost of 340,000 exceeds the entire negotiated settlement of 235,000, so arbitrating cannot pay even on a total win CORRECT
- D. settlements are always cheaper than awards

*Rationale:* A actually argues *against* settling and is the comparison usually made; it omits the cost of obtaining the award. B is real but secondary. C is decisive: a total victory saves 235,000 and spends 340,000 (10.4.2). D is an unsupported generalisation.

**MCQ 10.4-D** [10.4.4 · Application] Single sourcing costs 846,400 with a 0.18 probability of a 900,000 disruption. A qualified alternate adds 65,000 of certain cost and cuts the consequence to 295,000. The breakeven disruption probability is:

- A. 7.22 %
- B. 10.74 % CORRECT
- C. 18.00 %
- D. 22.03 %

*Rationale:*  $p^* = 65,000 / (900,000 - 295,000) = 65,000 / 605,000 = 10.74\%$  (10.4.4). A divides by the 900,000 consequence rather than the reduction; C restates the assumed probability; D divides by 295,000.

**MCQ 10.4-E** [10.4.4 · Analysis] Two suppliers each have a 0.18 disruption probability, but both draw a critical module from one sub-tier source, so the joint probability is 0.12 rather than 0.0324. The effect on a dual-split sourcing case is that expected disruption cost:

- A. falls, because two suppliers are still better than one
- B. is unchanged, since the marginal probabilities are unchanged
- C. rises from 86,724 to 131,400, making the dual split worse than single sourcing CORRECT
- D. rises, but the dual split remains the best option

*Rationale:*  $P(\text{exactly one}) = 2(0.18 - 0.12) = 0.12$ , so expected cost is  $0.12 \times 195,000 + 0.12 \times 900,000 = 131,400$ , taking the total to 1,071,400 against single sourcing's 1,008,400 (10.4.4). B mistakes marginals for the joint distribution, which is the error the whole example exists to expose.

**MCQ 10.4-F** [10.4.3 · Evaluation] A 124,800 supply-chain diligence programme avoids an expected 106,400 of exposure. The correct professional conclusion is that:

- A. the programme should be cancelled, since it fails an expected-value test
- B. the programme should be risk-tiered to raise effectiveness where probability is highest, the breakevens (82.11 % effectiveness, 9.38 % probability) stated, and any legal duty met regardless of the arithmetic CORRECT
- C. the consequence figure should be raised until the business case works
- D. expected monetary value is the correct basis for this decision

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*Rationale:* The arithmetic allocates effort; it does not decide whether a legal or values obligation applies, and the consequence distribution is fat-tailed so its mean is the wrong statistic (10.4.3, Domain 8 KA 8.2.2). C is the manipulation the honest presentation of breakevens is designed to prevent.

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SELF-CHECK — KA 10.4

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1. *Name the four elements of a claim and the one on which most claims are decided.* — Entitlement, causation, quantum and notice; causation decides most of them.
  2. *Which figure usually decides whether a formal dispute can pay, and why is it missed?* — The irrecoverable cost, because the comparison usually made is between the settlement and the expected award, which omits the cost of obtaining it.
  3. *Why is dual-splitting an order a weaker resilience purchase than dual qualification?* — It pays a price premium and, under independence, doubles the exposure surface, while the option value comes from having an exercisable alternate, which dual qualification buys without either cost.
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## Advanced topics — Domain 10

### 10.A.1 Supply-network mapping and the visibility that stops at tier one

Almost every organisation's supplier data stops at the parties it pays. Almost every serious supply disruption originates below that line. The consequence is a systematic error in the direction of optimism: exposure is assessed on the first tier, where it is diversified, and materialises in the lower tiers, where it is concentrated.

Mapping is therefore a deliberate exercise with a defined scope, and the scope should be set by criticality rather than by spend: the cheapest component on a critical path can stop a project that a major package cannot. For each critical item, the questions are: which sub-tier sources does each of my suppliers depend on; do any of my suppliers share one; what is the single-source geography, qualification or intellectual-property constraint that makes substitution slow; and what is the **resource interval**: the elapsed time from a failure to an alternative in production, which is the figure that converts a supply risk into a schedule risk at the project's cost of delay. The contractual enabler is a disclosure obligation with an audit right (10.4.3), because a supplier will not volunteer a dependency that is also a commercial vulnerability.

Two structural insights are worth carrying. Domain 4's interface arithmetic applies to supply networks:  $n$  suppliers who must interoperate create up to  $n(n-1)/2$  pairwise relationships, and a buyer who splits packages to gain price tension acquires all of them (Domain 4, KA 4.2.3). And Domain 8's correlation result governs the

aggregate: a register with many supplier risks and few underlying drivers is concentrated, however it is presented (Domain 8, KA 8.A.1). Restructuring a supplier risk register **by shared driver** rather than by supplier is a half-day exercise that regularly reveals a single point of failure behind a dozen entries, and, as 10.4.4 computes, a single shared sub-tier is enough to make a paid-for dual-sourcing arrangement worse than the single source it replaced.

### 10.A.2 Contracting for AI-delivered and outcome-based services

Where a supplier's deliverable is produced with, or consists of, an AI-enabled service, several contractual assumptions that normally go unexamined stop holding, and the gaps must be closed deliberately rather than left to a standard schedule.

**The deliverable can change without a variation.** A model updated by the supplier can alter output quality, latency and behaviour while the contract's description of the service remains satisfied. Insist on a defined baseline of behaviour with measurable acceptance criteria, notice of material model or version changes, and a right to re-test against the baseline after one: otherwise the buyer has bought a moving target and has no mechanism to notice it moving.

**Data rights need four separate answers, not one.** Who owns the input data; what the supplier may do with it, specifically including whether it may train models serving other customers; what happens to derived data, embeddings and model improvements at termination; and which sub-processors and jurisdictions are involved. "Confidentiality" answers none of these.

**Performance obligations must be honest about probabilistic output.** A service whose output is correct with some probability cannot be warranted as correct, and a supplier accepting such a warranty has either mispriced it or will resist it in practice. The workable structure specifies accuracy against a defined test set, a human-verification obligation on the buyer's side for consequential uses, and an allocation of responsibility for errors that reflects who was required to verify: the contractual expression of the family principle that **AI proposes; the professional verifies, decides and remains accountable**. If the contract does not say who the professional is, no one is.

**Outcome-based structures inherit the same problem commercially.** Paying for an outcome shifts the burden onto a definition that must now bear weight: what is measured, who measures it, what happens when the measure is disputed, and what proportion of the outcome the supplier actually controls. Where a supplier controls only part of it, an outcome-based payment is a lottery with a margin attached, and it will be priced as one.

### 10.A.3 The reviewer's procurement eye

Invariants to test on any procurement or contract arrangement, each cheap and each diagnostic.

Every make-or-buy decision states a **total cost of ownership including a non-zero exit cost** and a **breakeven volume**, and where the capability is on the critical path, the stand-up delay is priced at the cost of delay. The procurement chain for every long-lead item has a **computed total lead time** including governance latency, and is a predecessor in the schedule rather than an assumption behind it. The evaluation model (weights, sub-weights, normalisation formula, scoring scale, moderation and non-compliance treatment) was **fixed and disclosed before the receipt deadline**, with a dated record, and quality was scored before price was seen. Any quality premium has been tested against the expected cost it avoids, and the ratio is stated. Every bid materially below its benchmark has a recorded explanation, with the benchmark named. Every payment mechanism has its **marginal-dollar allocation** written down, and every target-cost contract has its **PTA** computed and known to the delivery team before mobilisation, not after. Every performance regime states whether its cap is set to deter or to compensate, and shows the arithmetic for whichever it claims. Every contract's notice periods are in a live **notice register** with computed deadlines. Every critical item has a mapped sub-tier and a stated **resource interval**. And the resilience claim for every dual-sourced item is tested for **correlation**, because a resilience arrangement that has not been tested for a shared sub-tier is an assertion, and 10.4.4 shows what the assertion costs when it is wrong.

## Industry variations — Domain 10

- **Public sector and regulated buyers.** Procurement route, timescales, publication, evaluation disclosure, standstill and challenge rights may be prescribed by law, and those rules (which differ by jurisdiction) override commercial preference. The practical effects are that the chain of 10.1.3 is longer and largely incompressible, that the model-fixing discipline of 10.2.3 is a legal requirement rather than good practice, and that the leader's remaining levers are the paper deadline, the delegation of award authority, and early market engagement before the formal process begins.
- **Construction and infrastructure.** Standard-form contracts dominate, with well-developed variation, extension-of-time, notice and dispute machinery, and interim determinative processes available in some jurisdictions. Notice discipline (10.4.1) carries more value here than anywhere else, retention and payment cycles drive cash (Domain 7, KA 7.4.4), and multi-party governance follows the contract structure rather than the organisation chart (Domain 3, KA 3.1.2).
- **Technology and software services.** Scope is defined late by nature, so fixed price on undefined scope is the characteristic error and the AI-service gaps of 10.A.2 are the characteristic gap. Exit cost is systematically understated because data extraction, re-implementation and licence discontinuity are discovered only at termination, which is exactly why the make-or-buy exit column matters most in this sector.



- **Energy, utilities and process industries.** Long-lead, engineered-to-order equipment with limited qualified sources; the re-source interval of 10.A.1 is measured in quarters, so 10.4.4's qualified alternate and buffer strategies dominate, and the procurement chain frequently starts before the design is complete. Auriga is this shape.
- **Pharmaceutical and life sciences.** Supplier qualification is a regulated activity: changing a supplier or a manufacturing site may require re-validation and regulatory notification, which makes the alternate expensive to keep exercisable and the switching consequence far larger than a commercial calculation suggests. Dual qualification must be planned at development, not at supply.
- **Consumer manufacturing and retail.** High volumes make the unit-price term dominant, so the make-or-buy breakeven of 10.1.2 is often comfortably passed, but the ethical-sourcing exposure of 10.4.3 is at its most acute, sub-tiers are numerous and opaque, and reputational consequence distributions are the most heavily skewed of any sector.

### Case study – Domain 10: the weighting confirmed after the envelopes (utilities, Auriga)

**Situation.** Auriga's installation package attracted three compliant bids: Alpha USD 2,000,000, Beta 2,200,000, Gamma 2,480,000. The invitation stated a price/quality weighting of **70/30**. The panel, under time pressure and conscious that the previous installation contractor had performed poorly, opened the priced envelopes alongside the technical submissions, scored quality at 62, 78 and 92, and then, recording it as a "confirmation", applied a **40/60** weighting on the grounds that quality mattered more on this scope than the invitation had implied. Gamma won with **87.46** against Alpha's **77.20**, and the award recommendation went to the utility's procurement board.

**What happened.** Alpha requested a debrief, was told that quality had been weighted at 60 %, and challenged the award. The utility's internal audit function reconstructed the evaluation and reached three findings the panel had not anticipated. Under the **published** 70/30 model the same scores gave Alpha **88.60**, Beta 87.04 and Gamma 84.05, **Alpha won**. The bid set had **three** possible winners across the weighting range, with boundaries at **57.70 %** and **63.77 %** price weight, so the choice of weighting had selected the supplier. And the panel's own risk mapping (integration rework probabilities of 0.30, 0.18 and 0.10 against a 320,000 impact) made Gamma's **USD 480,000** premium worth only  $96,000 - 32,000 = \text{USD } 64,000$  of avoided expected cost, a ratio of **7.5 to 1**, so the quality case the re-weighting was meant to express did not survive its own numbers.

**How it resolved.** The award was withdrawn and re-made under the published model. Alpha was appointed at 2,000,000 with two additions the audit finding made

possible: a supplier-development condition funding 90,000 of integration-team secondment against the identified rework risk, and a target-cost structure with the PTA computed and shared with the delivery team before mobilisation (Domain 7, KA 7.4.3). The process cost **11 weeks** (3 weeks of probity review, 3 weeks of re-evaluation, 3 weeks of re-approval latency at  $E[\text{wait}] = 4/2 + 1$ , and 2 weeks to execute) worth **USD 495,000** at Auriga's cost of delay, plus **USD 62,000** of external review: **USD 557,000** in total.

**What the domain teaches here.** The arithmetic is brutal and simple: **the probity failure cost USD 557,000, more than the entire USD 480,000 price gap the re-weighting was arguing about, and 1.16 times it.** The panel was not corrupt and was not even wrong about quality mattering; it was wrong about *when* that judgement may be made. Had the same conviction been expressed in the invitation as a 40/60 weighting, published before bids were received, the award to Gamma would have been unimpeachable; and the bidders would have priced and pitched differently, which is the point of publishing it. Two further lessons. Scoring quality **after** seeing price is not a procedural nicety: the panel's quality scores were never tested for the anchoring the sequence introduced, and could not be defended in the debrief. And the 7.5-to-1 ratio is the discipline a quality weighting needs before it is set, not after it is challenged: a panel that had computed it during model design would have chosen 70/30 with conviction rather than 40/60 with hindsight.

## Case study B — Domain 10: two suppliers, one foundry (rail rolling stock)

**Situation.** A rolling-stock refurbishment programme needed **1,200** traction converters. Wary of single-source exposure after an earlier disruption, the sourcing team dual-sourced on a **55/45** split: supplier V at **USD 14,800** per unit and supplier W at **USD 15,600**, a blended **USD 15,160** and a contract value of **USD 18,192,000**. Awarding the full volume to V would have secured a **3 %** volume discount (**USD 14,356** a unit, **USD 17,227,200**), so the programme paid **USD 964,800** for resilience. The board minute recorded the arrangement as "dual-sourced, exposure mitigated". Each supplier was assessed at a **0.15** probability of disruption in the delivery window, treated as independent, giving an assumed expected disruption cost of  $0.2550 \times 294,000 + 0.0225 \times 1,297,000 = \text{USD } 104,152.50$ : with a one-supplier disruption costing 2 weeks of fleet unavailability at USD 128,000 a week plus a 38,000 surge premium (**294,000**), and both suppliers disrupted costing 9 weeks plus 145,000 of re-engineering (**1,297,000**).

**What happened.** Both suppliers slipped in the same quarter. Their converters used the same power module, and that module had a single qualified foundry, which reallocated capacity. Neither supplier had disclosed the dependency and neither had been asked. Reconstructed with the true joint probability of **0.11**, the arrangement's expected disruption cost was  $0.12 \times 294,000 + 0.11 \times 1,297,000 = \text{USD}$

**166,190**, not 104,152.50 (the independence assumption had understated exposure by **USD 62,037.50**, or **59.56 %**), and the realised outcome was the 9-week case.

**How it resolved.** A sub-tier mapping exercise to tier three found the single foundry behind both suppliers and two further shared dependencies. The programme then did the arithmetic it had never done: qualifying a **second power-module design** would cost **USD 380,000**, against the **USD 964,800** already spent on a volume split that had bought nothing, a ratio of **2.54 to 1**. The arrangement was restructured to a single award to V at the discounted 14,356, plus the 380,000 second module design, plus a contractual sub-tier disclosure and audit obligation. With the module risk reduced (joint probability 0.02, leaving V's idiosyncratic 0.04, so 0.06 in total) and the recovery shortened to 4 weeks plus 90,000, expected disruption fell to **USD 36,120** and the total position to **USD 17,643,320** against the dual-split arrangement's **USD 18,358,190**: better by **USD 714,870**, with the actual exposure reduced rather than relabelled.

**What the domain teaches here.** **Two suppliers with one sub-tier is one supplier with two invoices**, and the money spent on the split is not resilience, it is a price premium with a resilience narrative attached. Three transferable points. The independence assumption is the one most likely to be both convenient and wrong, and it understated this programme's exposure by nearly 60 %: Domain 8's correlation lesson (KA 8.A.1) arriving through a purchase order. Resilience spending must be directed at the **actual single point of failure**, which requires knowing where it is: 380,000 spent on the module would have delivered what 964,800 spent on the split did not. And a sub-tier disclosure obligation is not administrative box-ticking; it is the contractual instrument that makes the analysis possible at all, and it costs nothing to include at tender and cannot be obtained afterwards.

## Executive perspective — Domain 10

What a programme director cannot delegate in this domain:

- **The make-or-buy basis, and its exit column.** Insist on total cost of ownership with a non-zero exit cost and a stated breakeven volume, and require the stand-up delay to be priced at the cost of delay wherever the capability is on the critical path. A unit-price comparison is not a make-or-buy paper (10.1.2).
- **The evaluation model, fixed and dated before the receipt deadline.** Weights, sub-weights, normalisation and the sequence in which quality and price are seen. One bid set can have three winners; whoever chooses the model after opening has chosen the supplier, and Case study A shows the cost of learning that afterwards (10.2.2, 10.2.3).
- **The marginal-dollar allocation of every material contract, and where its PTA sits.** You should be able to say, without notes, who pays the next dollar of

cost under each of your major contracts, and at what cost level that changes (10.3.2; Domain 7, KA 7.4.3).

- **Whether your performance regimes deter or merely price non-performance.** Compare every cap with the supplier's cost of compliance and with your own loss. A 5 % cap against a 180,000 compliance cost is a discount voucher for failure (10.3.3).
- **The notice register.** Your suppliers keep one, or their advisers do. Nearly half of a properly constructed claim can turn on a diary entry, in both directions (10.4.1).
- **The correlation test on every resilience claim.** Ask, of every dual-sourced critical item, which sub-tier sources the two suppliers share; and treat "we don't know" as the answer that it is. Two suppliers with one sub-tier is one supplier (10.4.4, Case study B).

## Calculation exercises — Domain 10

**Exercise 10.1** In-house provision of a support capability costs 260,000 to stand up, 1,450 per unit and 40,000 to exit. Outsourcing costs 55,000 to transition in, 1,900 per unit and 85,000 to transition out. The requirement is 520 units. Compute the breakeven volume and the better option; then recompute with a 7-week capability stand-up delay on the critical path at a cost of delay of 18,000 per week.

*Solution.*  $F_{\text{make}} = 300,000$ ,  $F_{\text{buy}} = 140,000$ , so  $Q^* = 160,000 / (1,900 - 1,450) = 160,000 / 450 = 355.56$  units. At 520 units: make  $300,000 + 754,000 = 1,054,000$ ; buy  $140,000 + 988,000 = 1,128,000$  — **make is 74,000 cheaper**, because 520 exceeds the breakeven. Adding the delay:  $7 \times 18,000 = 126,000$ , so make becomes **1,180,000** and **buy is now 52,000 cheaper**; the breakeven moves to  $(300,000 + 126,000 - 140,000) / 450 = 635.56$  units. Common error: omitting the exit costs from both sides, which gives a breakeven of  $205,000 / 450 = 455.56$  units. It happens to reach the right answer at 520 units, for the wrong reason, and would reach the wrong one at any volume between 356 and 456.

**Exercise 10.2** Two bids: P at 1,600,000 with quality 70, and Q at 1,840,000 with quality 88. Price is scored  $\text{lowest} + \text{own} \times 100$ . Determine the winner at a 70/30 and a 50/50 price/quality weighting, compute the crossover weighting, and state the implied price per quality point. *Solution.* Price scores: P **100.00**, Q  $1,600,000 / 1,840,000 \times 100 = 86.96$ . At 70/30: P  $70 + 21 = 91.00$ , Q  $60.87 + 26.4 = 87.27$ , **P wins**. At 50/50: P  $50 + 35 = 85.00$ , Q  $43.48 + 44 = 87.48$ , **Q wins**. Crossover  $w^* = (88 - 70) / [(100 - 70) - (86.956522 - 88)] = 18 / 31.043478 = 57.98\%$ . Q's premium is 240,000: **15.00 %**: for 18 quality points, **13,333.33 per point**, which must be tested against the expected

cost those points avoid. Common error: scoring price as a share of the *highest* bid or on a linear scale without saying so; the normalisation convention is part of the model and moves the crossover, as 10.2.2 shows by 7.70 percentage points on a three-bid set.

**Exercise 10.3** A target-cost subcontract has a target cost of 1,500,000, a target fee of 120,000, a 60/40 share (buyer 60 %) and a ceiling of 1,800,000. The actual cost is 1,700,000. Compute the fee, buyer outturn and supplier margin, the *PTA*, and compare with a firm fixed price of 1,650,000 and with cost plus a fixed fee of 120,000. *Solution.* Overrun  $1,700,000 - 1,500,000 = 200,000$ ; fee  $120,000 - 200,000 \times 0.40 = 40,000$ ; buyer pays **1,740,000** (below the ceiling); supplier margin **40,000**. Target price 1,620,000, so  $PTA = 1,500,000 + (1,800,000 - 1,620,000)/0.60 = 1,500,000 + 300,000 = 1,800,000$ : here the *PTA* coincides with the ceiling, and at a cost of 1,800,000 the fee is exactly nil. Firm fixed price: buyer pays **1,650,000**, supplier margin (**50,000**). Cost plus fixed fee: buyer pays **1,820,000**, margin **120,000**. The buyer's outturn spans **170,000** across the three mechanisms on the *same* actual cost, and the marginal-dollar allocation is 0/1.00 under fixed price, 1.00/0 under cost-plus, and 0.60/0.40 under the target cost below the *PTA*. Common error: computing the fee as *target fee - overrun × buyer share* ( $120,000 - 120,000 = 0$ ), which reverses the share ratio; the supplier bears its own share of the overrun, not the buyer's.

**Exercise 10.4** A single-source arrangement has a certain cost of 1,240,000 and a 0.22 probability of a disruption costing 640,000. Qualifying an alternate adds 58,000 of certain cost and cuts the consequence to 210,000. Compute both total expected costs, the breakeven disruption probability and the maximum premium the resilience is worth. Then recompute assuming a shared sub-tier makes the alternate unavailable in 0.14 of the 0.22. *Solution.* Single:  $1,240,000 + 0.22 \times 640,000 = 1,380,800$ . Alternate:  $1,298,000 + 0.22 \times 210,000 = 1,344,200$  — better by **36,600**. Breakeven  $p^* = 58,000 / (640,000 - 210,000) = 58,000 / 430,000 = 13.49\%$ . Maximum premium worth paying =  $140,800 - 46,200 = 94,600$ . With the shared sub-tier: expected disruption  $(0.22 - 0.14) \times 210,000 + 0.14 \times 640,000 = 16,800 + 89,600 = 106,400$ , total **1,404,400** — **worse than single sourcing**, and the correct response is to attack the sub-tier rather than to choose between the two structures. Common error: computing the breakeven by dividing the extra certain cost by the full consequence ( $58,000/640,000 = 9.06\%$ ) rather than by the *reduction* in consequence, which overstates the case for the alternate.

**Exercise 10.5** A variation claim comprises 320 hours of labour at 118.50 per hour, 12,600 of materials, 2 weeks of prolongation at 16,500 per week, and disruption on 900 planned hours whose productivity fell to 0.90 of the measured-mile baseline. Overhead and profit is 10 %. Notice was late, and the prolongation and disruption

heads depend on valid notice. Compute the total claim, the amount at risk and the percentage lost. *Solution.* Labour  $320 \times 118.50 = 37,920$ ; materials **12,600**; prolongation  $2 \times 16,500 = 33,000$ ; disruption — hours required  $900/0.90 = 1,000$ , extra **100** hours, so  $100 \times 118.50 = 11,850$ . Subtotal **95,370**; with 10 % overhead and profit, **total claim 104,907.00**. At risk:  $(33,000 + 11,850) \times 1.10 = 49,335.00$ , which is **47.03 %** of the claim; surviving  $(37,920 + 12,600) \times 1.10 = 55,572.00$ . Common error: computing disruption as  $900 \times 0.10 \times 118.50 = 10,665$ , which multiplies by the productivity shortfall instead of dividing by the productivity factor and understates the head by 1,185 before overhead and profit.

## Practitioner's toolkit — Domain 10

*Adoption-ready artefacts; adapt the headings to your organisation, then keep them stable.*

### Toolkit 10.T.1 — Make-or-buy total-cost-of-ownership sheet

One page per decision, completed before any market engagement. Two columns, make and buy, and these rows: transition-in cost, itemised (recruitment, training, tooling, environments, licences, specification and tender effort, mobilisation); unit cost with the volume basis stated; residual management cost, **never zero on either side; exit cost**, itemised (redeployment or severance, decommissioning, data extraction in a supportable format, knowledge transfer, re-qualification, parallel running); stand-up elapsed time in weeks and whether the capability is on the critical path; and the delay term, priced at the project's cost of delay. Below the columns, three computed figures: the **breakeven volume  $Q^*$** , the outturn at the required volume, and the outturn at the plausible upper volume. The sheet's purpose is to make the decision a proposition about volume rather than a preference about sourcing; a paper that cannot fill in the exit row has not done the analysis, and a paper with a zero in it is asserting something the incumbent's contract will disprove.

### Toolkit 10.T.2 — Evaluation model lock sheet

One page, completed, dated, signed by the accountable authority and **issued with the invitation**. Contents: the criteria and their weights; the sub-criteria and their weights; the **price normalisation formula written out in full**; the quality scoring scale with what each score means in words; the sequence of evaluation, stating explicitly that quality is scored before price is disclosed; the moderation process and how individual scores are recorded before it; the treatment of non-compliant, qualified and abnormally low bids, with the benchmark and threshold for the last of these named; the panel members and their conflict declarations; and (the row that turns the sheet into a control) the **pre-award sensitivity analysis**: the winner at

three weightings and the crossover weightings between the leading bidders, computed before the receipt deadline and dated. Any change after issue is recorded on the sheet with its author, its reason and the result under **both** the original and the amended model.

### Toolkit 10.T.3 — Supplier resilience and sub-tier register

One row per critical item, not per supplier, and criticality set by schedule effect rather than by spend. Columns: item; why it is critical (which path, which milestone); first-tier supplier(s) and volume allocation; **disclosed sub-tier sources for the critical component**, to the depth the disclosure obligation reaches; **shared dependencies flagged across rows**. This is the column the register exists for; qualification status of any alternate (qualified / qualified and exercised in the last 12 months / named only); **re-source interval** in weeks; consequence of disruption, priced at the cost of delay plus switching cost; assessed disruption probability with its source; expected cost of disruption; the annual cost of keeping the alternate exercisable; and the **breakeven disruption probability** at which that cost is justified. Reviewed quarterly, with two standing questions: has any shared dependency appeared since the last review, and has any alternate lapsed from *exercisable* to *named only*?

## Exam preparation — Domain 10

**What is assessed.** The procurement lifecycle and what each stage forecloses; make-or-buy on total cost of ownership and its breakeven volume; the procurement chain as a schedule constraint including governance latency; routes to market; the evaluation model, price normalisation, the crossover weighting and the obligation to fix the model before opening; abnormally low tender testing against a named benchmark; contract strategy as risk allocation and the influence test; buyer and supplier outturn under fixed-price, cost-plus and target-cost mechanisms; the marginal-dollar allocation identity and the *PTA* as a behavioural boundary; why an incentive reallocates value unless behaviour changes; performance-regime caps for deterrence against compensation; the four elements of a claim, the heads of cost and notice-bar exposure; the dispute ladder and irrecoverable cost; risk-tiered ethical sourcing and the honest limits of its arithmetic; and single, dual-split and dual-qualified sourcing with correlation.

**The calculations to be able to do under time pressure.**  $TCO = F + vQ$  and  $Q^* = (F_{\text{make}} - F_{\text{buy}}) / (v_{\text{buy}} - v_{\text{make}})$ , with and without a delay term. A procurement chain's total lead time including  $E[\text{wait}] = M/2 + L$  per approval. Price normalisation on both conventions,  $S_i(w) = wP_i + (1 - w)Q_i$ , and the crossover  $w^*$ . Target-cost fee, buyer outturn and supplier margin at a stated actual cost, plus the *PTA*. Expected cost, risk-neutral fixed price and expected supplier margin from a cost distribution. A service-credit cap against a compliance cost and



against a buyer loss. Claim heads including measured-mile disruption, overhead and profit, and notice-bar exposure. Expected cost of disruption under independence and under a stated joint probability, with the breakeven probability  $p^*$  and the maximum resilience premium.

**The traps.** Deciding make-or-buy on unit price, which systematically favours the high-fixed-cost option (10.1.2) · scoring exit cost at zero (Toolkit 10.T.1) · dividing the fixed-cost difference by a unit cost rather than by the unit-cost *difference* (MCQ 10.1-A) · omitting governance latency from a procurement chain (Exercise implied by 10.1.3; Domain 3 KA 3.2.3) · treating the evaluation weighting as a detail rather than as the thing that selects the winner (10.2.2, Case study A) · failing to state the normalisation convention, which moves a crossover by up to 7.70 percentage points (10.2.2) · testing an abnormally low bid against an unnamed benchmark (10.2.2) · adding the required margin to the *target* cost instead of the *expected* cost when pricing a fixed-price bid (MCQ 10.3-A) · computing an incentive fee with the buyer's share instead of the supplier's (Exercise 10.3) · assuming sharing continues above the *PTA* (MCQ 10.3-B; Domain 7 KA 7.4.3) · reading a buyer's expected saving under an incentive as value created when it is value transferred (10.3.2) · comparing a service-credit cap with nothing at all (10.3.3) · computing measured-mile disruption by multiplying by the productivity shortfall instead of dividing by the productivity factor (Exercise 10.5) · comparing a settlement with the expected award and omitting irrecoverable cost (10.4.2) · dividing an extra certain cost by the full consequence rather than by the consequence *reduction* when computing a breakeven probability (Exercise 10.4) · and assuming supplier independence, which understated Case study B's exposure by 59.56 %.

**How the domain connects.** Domain 3 supplies the governance latency that makes the procurement chain 31 weeks long and the escalation design that a dispute ladder rests on. Domain 4 supplies the interface arithmetic that prices a packaging decision and the change control that must run across the contract boundary. Domain 5's specification determines whether any contract mechanism can work, on undefined scope they all converge on cost-plus with a dispute attached. Domain 6 supplies the critical path that makes a stand-up delay or a re-source interval expensive, and the cost of delay of USD 45,000 a week at which everything here is priced. Domain 7 supplies the contract-model taxonomy, the blended rate, the cash-flow consequence of payment terms and the *PTA* — all cited here, none re-derived. Domain 8 supplies *EMV*, the correlation result that Case study B demonstrates commercially, and the R1 and R3 risks these sourcing and evaluation decisions act on. Domain 11 handles the negotiation behaviour that determines where inside a bargaining range these arithmetic results actually land, and Domain 13 the contracting problem for adaptive delivery, where scope is deliberately not fixed. PFL-AI Domain 11 treats the risk-allocation face of the same problem from the lender's side.



## Domain 10 summary

Procurement decisions are delivery decisions, and almost all of them are taken on the wrong number.

**Make-or-buy** is a proposition about volume, not about unit price. Auriga's remote-terminal-unit capability looks 33.33 % cheaper to build per unit and is USD 88,800 more expensive to own at the 84 units required, because the unit-price advantage of 1,800 has to recover a fixed-cost disadvantage of 240,000, a breakeven of **133.33 units**. Price the 9-week capability stand-up at the project's cost of delay and the breakeven moves to **358.33 units**, which is not a marginal case but an unavailable one. The exit column is where make-or-buy papers fail, and it is never zero. The procurement chain that delivers the controllers runs to **31 weeks against a 25-week project**, of which 18 weeks are process and 6 are pure governance latency worth **USD 270,000**, which is why long-lead items are committed before the baseline and why the compressible time is administrative, not industrial.

**Tender evaluation** is arithmetic that selects a supplier. One bid set (Alpha 2,000,000/62, Beta 2,200,000/78, Gamma 2,480,000/92) produces **three different winners**: Gamma below a 57.70 % price weight, Beta between 57.70 % and 63.77 %, Alpha above. Changing only the normalisation convention moves the Beta/Gamma boundary by **7.70 percentage points**. That is the whole case for fixing and publishing the model, in full, before bids are opened, and for scoring quality before price is seen; Case study A paid **USD 557,000** to learn it, against the **USD 480,000** price gap the re-weighting was arguing about. The discipline that should precede any quality weighting is the premium test: Gamma's 480,000 premium bought **USD 64,000** of avoided expected rework, a ratio of **7.5 to 1**.

**Contract strategy** is the allocation of the marginal dollar, and the allocation always sums to one: a contract cannot make cost disappear. On the same cost distribution (expected cost USD 2,120,000), a risk-neutral firm fixed price of **USD 2,270,000** and cost plus a 150,000 fee cost the buyer *the same in expectation*; what fixed price buys is the collapse of a **USD 230,434** standard deviation to zero, and it is only as good as the supplier's balance sheet, at the 2,600,000 outcome that supplier loses 330,000. The target-cost structure as specified moves **USD 48,000** from supplier to buyer and creates nothing: the supplier's expected margin is 102,000 against a 150,000 requirement, because the target fee is the fee at the target and not the expected fee. Value appears only when behaviour changes; the underrun probability must rise from 0.20 to **0.456** for the supplier to accept the structure, at which point expected cost falls by **USD 134,400**, all of it accruing to the buyer, which is what the negotiation is really about. Above Auriga's **PTA of USD 2,428,571.43** (Domain 7, KA 7.4.3) the supplier's marginal share jumps from 0.30 to 1.00 and cooperation turns to entitlement for structural reasons. And a performance regime capped at 5 % of a 2,200,000 contract against a 180,000 compliance cost prices failure at a **USD 70,000 discount**: deterrence needs **8.18 %**, compensation **14.55 %**, and confusing the two is the design error.

**Claims, disputes and resilience** are where records earn their keep. A properly built USD 107,914.80 variation claim had **48.37 %** of its value (USD 52,194.80) exposed to a notice provision, which is why a notice register is worth more than any negotiating skill, in both directions. In a 400,000 dispute settling at 235,000 against an expected arbitration cost of 614,000, the decisive figure is that the **USD 340,000 of irrecoverable cost exceeds the entire settlement**, so fighting cannot pay even on a total win. An ethical-sourcing programme costing 124,800 against 106,400 of avoided expected exposure fails an expected-value test at 70 % effectiveness; and the honest response is to risk-tier it, state the breakevens (**82.11 %** effectiveness, **9.38 %** probability), recognise the mean of a fat-tailed consequence as the wrong statistic, and meet any legal duty regardless. Resilience, finally, is bought in the wrong form almost universally: for Auriga's 84 controllers, single sourcing costs **1,008,400** in expectation, a 60/40 dual split **1,026,724**, and a **qualified alternate 964,500**, so qualify two, award one, keep the second exercisable, at a breakeven disruption probability of **10.74 %** against the split's **24.22 %**. Then test the assumption that matters: introduce a shared sub-tier at a joint probability of 0.12 and both resilience options become **worse than single sourcing**, which Case study B lived, at a cost of 964,800 spent on a split that bought nothing while 380,000 spent on the actual single point of failure would have.

The through-line: **every procurement decision has a number that decides it, the number is usually not the one on the front of the paper, and the breakeven is more useful than the point estimate**. Compute the breakeven volume, the crossover weighting, the marginal dollar, the deterrence cap and the breakeven disruption probability, and a procurement conversation stops being about preference and becomes about evidence.

PART THREE

# Leading people and organisations

Domains 11–13 — stakeholders and influence, leadership and teams, and adaptive delivery: the work that is done through other people.

DOMAIN

# 11

## Stakeholders, Communication and Influence

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## Why this domain exists

Part Two built the delivery machinery: scope, schedule, cost, risk, quality, supply. Every one of those disciplines assumed that the people whose agreement the project needs would give it, roughly when asked, on roughly the terms proposed. That assumption is the largest unpriced item in most delivery plans, and this domain prices it.

Two facts sit behind the domain's central claim. The first is Domain 1's arithmetic, which nobody disputes and few programmes act on: Meridian releases **USD 24,480 a year** for each clinic that actually uses the system, so **moving adoption creates more value than moving the installation date**; and adoption is a stakeholder outcome, not a delivery one. The second is that stakeholder work is nevertheless the first thing cut, because it has no unit of account. A schedule has weeks, a budget has dollars, and engagement has "relationships", which are not measurable and therefore not defensible when capacity is short. The domain's central claim is the correction: **engagement is a resource-allocation problem with a computable return, and communication is an architecture with a computable load**. Both can be designed, priced and reviewed like any other part of the delivery system, and where they genuinely cannot be (legitimacy, trust, the moral weight of a community's objection) the professional obligation is to say so plainly rather than to manufacture a metric that survives challenge better than the judgement it replaced.

The domain proceeds from the map to the mechanism. KA 11.1 treats the stakeholder set as a *system* rather than a list, assesses it on influence, interest, attitude and **consent risk**, and allocates a finite engagement capacity against those four things. KA 11.2 designs communication as an architecture: it applies Domain 4's interface result to people, computes the load a mesh imposes, and then computes something almost no reporting regime measures, **how old the number is when an executive decides on it**. KA 11.3 makes negotiation preparation quantitative through reservation values, the priced alternative and the zone of possible agreement, then shows that most of the value in a multi-issue negotiation is created by trading rather than claimed by conceding. KA 11.4 handles the stakeholders who cannot be managed (the public, communities, and audiences across languages and decision conventions), and the newest failure mode in the domain: communication that a machine wrote and nobody checked.

The through-line: **the stakeholder objection you have not paid for in engagement, you will pay for in delay, at the cost of delay, with interest**.

**Learning objectives.** After this domain a candidate can: distinguish a stakeholder system from a list and name the parties a register habitually omits; assess stakeholders on influence, interest, attitude and consent risk, and state what each dimension predicts that the others do not; **treat the register as what it is, a record about identified people, with an owner, an access list, attitude recorded as an**

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observed act with its source, deletion of departed role holders at refresh and a deletion point at closure; price a late objection at the cost of delay and compute the earlier engagement spend that would have avoided it, with the breakeven probability that justifies it; allocate a finite engagement capacity as a three-term design (value-led core, consent-risk floors, salience-proportional residual), and demonstrate what a salience-only allocation forgoes; apply Domain 4's interface arithmetic to communication and state the maximum parties a mesh and a routed design each support within a stated capacity; **compute the age of the information an executive decides on from the reporting period, consolidation time and paper lead time, and price the reduction a shorter cut buys**; write an honest status and an escalation-grade message; **build a reservation value from a fully priced alternative, bound the zone of possible agreement, and compute what preparation is worth in transferred value**; compute the joint gain a multi-issue trade produces against a split-the-difference compromise; diagnose conflict by source rather than by personality; price a consultation against the consent risk it reduces while stating what the arithmetic cannot settle; communicate across decision conventions without assuming one is universal; and govern AI-assisted communication so that no external statement is issued unverified and no synthetic voice speaks for the programme.

**The master programme.** Meridian Care Records continues from Domains 1–4: the clinical-records rollout to **40 clinics**, approved cost **USD 2,400,000**, full-potential benefit **USD 979,200** a year, realistic benefit **USD 685,440** a year at **70 %** adoption, and a **cost of delay of USD 14,280 per week** (Domain 1, KA 1.3.2–1.3.3). Three inherited results do the work in this domain. Each adopting clinic is worth **USD 24,480** a year, so adoption is the largest lever available to the leader. Meridian's steering committee imposes a governance latency of  $E[\text{wait}] = M/2 + L = 4/2 + 2 = 4.0 \text{ weeks}$  (Domain 3, KA 3.2.3), which is what makes a late objection expensive. And one interface costs **USD 18,000** to specify, build, test and document (Domain 4, WE 4.2.3), which is what makes a late objection expensive *twice*. This domain adds the stakeholder register (**62** identified parties, of which **14** require an individually managed relationship), and spends the programme's **480 hours** of annual engagement capacity three different ways to show what the choice is worth.

**Reference points.** Stakeholder engagement has no single international standard of its own; the nearest reference point is **ISO 21502**, which addresses guidance on project management and treats stakeholder engagement as one of its practices: voluntary guidance describing practice, not legislation and not a certifiable requirement, obliging nobody of itself unless an organisation, a contract or a regulator adopts it. It is named here and not reproduced, and everything below is derived rather than cited. Where this domain touches records about identified individuals (11.1.2), the governing considerations are data-protection obligations that vary by territory and belong to whoever holds data-protection accountability in the organisation and to qualified counsel, not to any standard, and not to this book.

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## Stakeholder systems and engagement strategies

*Topics: 11.1.1 the stakeholder system · 11.1.2 assessment (influence, interest, attitude and consent risk · 11.1.3 engagement strategy under a capacity constraint.)*

### 11.1.1 The stakeholder system

**Definition.** A **stakeholder** is any party whose decisions, consent, resources, behaviour or objection can materially affect the project's outputs, outcomes or legitimacy, or who is materially affected by them. A **stakeholder system** is that set of parties together with the relationships *between* them, who influences whom, who speaks for whom, and which pairs will align or divide under pressure.

The distinction is not pedantry. A list supports a communication schedule; only a system supports prediction, because the events that damage projects are relational, a supplier and a regulator reaching a shared view of a specification the client has not seen, a user group and a professional body forming a coalition, a funder deferring to an operational director whose interests diverge from its own. A leader working from a list is repeatedly surprised by conclusions formed in conversations they were not part of, and each surprise arrives late.

**What registers habitually omit**, consistently enough to be a checklist. *Parties with no organisational representation* (future users, patients, tenants, passengers, the public), whose interests are real, whose consent may have to be obtained under obligations that vary by jurisdiction, and who have no one on the distribution list. *Internal parties whose work the project changes without commissioning them*: finance, procurement, legal, information security, service desks, training functions. *Parties visible only at handover*, where Domain 16's readiness failures originate. *The supplier's own subcontractors*, whose interests are not the supplier's (Domain 10). *Predecessors and successors*: the sponsor of the previous attempt, the owner of the system being replaced, the programme inheriting the dependency. And *the ones who left*: registers record roles, but relationships attach to people, a point 11.A.2 makes arithmetic.

**The four questions that make a register useful.** Not "who are the stakeholders?" but: what does this party **decide, consent to, resource, or do** that the project needs? What does the project change **for them**, and is that a gain or a loss in their own terms? What is their **relationship to the other parties** — who do they follow, and who follows them? And what is the **cost if their agreement arrives late**? The last question converts a register from an administrative artefact into a planning input, because it produces a number, and 11.1.2 computes it.

Note the vocabulary. Stakeholders are not managed; work is managed, and stakeholders are engaged, consulted, negotiated with and occasionally opposed. "Stakeholder management" quietly licenses the posture (that the party's view is an obstacle to be handled) that guarantees the late objection this domain prices. An objection is

*information*, sometimes about the stakeholder and sometimes about the project, and the leader's first task is to establish which.

#### 11.1.2 Assessment — influence, interest, attitude and consent risk

**The four dimensions, and what each predicts.** Most assessment practice uses two; four are needed, because two of the failure modes are invisible in the first pair.

- **Influence:** the party's capacity to affect the project's decisions, resources or permissions. Predicts *how much damage a disagreement does*.
- **Interest:** how much the party cares, measured by attention rather than by stake. Predicts *how quickly they will notice*, and it is emphatically not a measure of how much is at stake for them: a party can have an enormous stake and no attention, and that combination is the most dangerous in the register.
- **Attitude:** supportive, neutral, sceptical or opposed, assessed on evidence rather than on courtesy. Predicts *what an engagement contact will achieve*. A supportive party needs information; a sceptical one needs argument; an opposed one needs negotiation or containment, and sending each of them the same monthly bulletin serves none of them.
- **Consent risk**, whether the party holds a veto, a statutory approval, a licence, a signature, a clinical or safety authority, or a practical ability to withhold cooperation. Predicts *the consequence of being wrong about the other three*, and it is a property of the party's position rather than of its disposition. A helpful regulator still holds the veto.

**What the register is, as a record, and it is not a spreadsheet.** Before any of the four dimensions is scored, a professional obligation attaches to the artefact itself. A register that names **individuals**, scores their influence, records an assessment of their **attitude**, and maps who they follow and who follows them is a file of information about identified living people, held by the organisation, containing recorded opinions about them. That is personal data, and the domain's own AI treatment already states the governing principle for the machine-generated case, *an assessment of a colleague's motives generated by a tool and stored in a project record is an employment and dignity problem regardless of its accuracy* (KA 11.3). The principle does not soften because a human typed it. Five provisions follow, and none of them is expensive:

- **A stated owner and an access list.** One named owner for the register, and a list of who may read it. It is not a general-circulation document, it does not belong in a status pack, and it is not attached to an email to a distribution group. Where a party's entry names their own staff, assume the entry may be seen by them and write it accordingly.
- **Attitude recorded as an observed position with its source, never as a characterisation of the person.** "Objected to the interface scope in the 14 March minute" is a fact with a reference, and it is more useful to a successor



than any adjective. "Obstructive", "empire-building" or "will never agree" are conclusions about a person that the register is not entitled to draw, that cannot be evidenced, and that survive in a file long after the situation that produced them. The discipline is the same one Domain 12 applies to a performance record: **record the observation, not the verdict.**

- **Deletion of departed role holders' assessments at the refresh.** When a role holder changes, the previous holder's attitude and influence assessment is **deleted**, not archived "for context". The register exists to manage a live relationship; a personal assessment of somebody who has left the role has no purpose and every risk. 11.A.2's refresh is the natural point, and its cadence is therefore a data provision as well as an accuracy one.
- **A deletion point at project closure.** The register carries a stated end: what is deleted, what is retained and for how long, and who decides. Domain 3, KA 3.3.4 supplies the custody machinery and Domain 16, KA 16.4.4 the economics, which point in opposite directions for these two record classes, since contract and technical evidence is cheap insurance while personal data is the reverse trade.
- **The obligations themselves come from elsewhere.** Data-protection obligations (including any entitlement of the individual to see what is held about them, which changes how an attitude entry should be written before it changes anything else), apply on their own terms in their own territories, differ substantially, and are settled with whoever holds data-protection accountability in the organisation and with qualified counsel. **Nothing here states the position in any jurisdiction**; this is a description of a professional discipline, illustrative of what care looks like, and the register itself remains a non-optional instrument. Domain 14, KA 14.4.4 treats the wider privacy estate.

The high-influence, low-interest, high-consent-risk quadrant is where projects are damaged. Such a party is quiet for months, is therefore assessed as low priority by any allocation rule that reads interest, and then reads the specification properly at the point of approval. What happens next has a price.

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#### WORKED EXAMPLE

##### Worked example 11.1.2 – the objection that arrived in week 34.

1. **Setup.** Meridian's national reporting body is assessed influence **5**, interest **2**, attitude *neutral*, consent risk **high** (it approves the statutory data extract). It received the monthly bulletin and nothing else. In **week 34** of the rollout it objects that the extract's record-level structure does not meet its submission requirement. Resolving the objection requires a specification change approved by the steering committee, Domain 3's E[wait] of **4.0 weeks**, followed by **5 weeks** of rework and re-verification, all on the critical path. The direct work is re-specification and re-verification of the reporting-gateway interface and the two interfaces that depend on it, at Domain 4's unit cost of **USD 18,000** each, plus **USD 42,000** of application rework. Cost of delay **USD 14,280 per week**. A

technical pre-consultation at design stage would have taken **24 hours** of a data specialist's time and **6 hours** of the project leader's, at the blended internal rate of **USD 110 per hour**.

2. **Formula.** Assessed total = (governance latency + rework weeks) × cost of delay + interface re-verification + application rework. Avoidance cost = hours × rate. Breakeven probability = avoidance cost ÷ assessed total.
3. **Substitution.** Delay (4.0 + 5) × 14,280. Direct 3 × 18,000 + 42,000. Avoidance (24 + 6) × 110. Breakeven 3,300 ÷ 224,520.
4. **Result.** Delay **9.0 weeks**, costing **USD 128,520**; direct cost **USD 96,000**; **assessed total USD 224,520** (≈ SAR 841,950 indicatively). Avoidance cost **USD 3,300**. The consequence is **68.0 times** the avoidance cost, and the pre-consultation pays whenever the probability of the objection exceeds **1.47 %**.
5. **Interpretation.** The 68-fold ratio is arresting and it is not the useful number; **1.47 %** is. A leader arguing for engagement with a ratio is making a rhetorical case, and a sceptical sponsor will answer that the objection might not have happened. A leader arguing with the breakeven probability has made that answer impossible: the sponsor must now assert that the chance of a statutory approver objecting to a specification it has never been shown is *below one and a half per cent*, which nobody will say out loud. The asymmetry generalises far beyond this case, because engagement costs are hours and consequences are weeks of critical path. So so the breakeven probability for engaging a high-consent-risk party is almost always in the low single figures. Three cautions belong with it. The arithmetic prices the *avoidable* objection; a pre-consultation that discovers a genuine incompatibility does not eliminate the cost, it moves it earlier, where it is cheaper (Domain 5's design-maturity argument, and Case study B's 7.45-fold escalation). The 5 weeks of rework is an estimate and the 4.0 weeks of governance latency is not; it is a computed property of Meridian's committee design, which means **a governance design and an engagement design are the same conversation**: at Domain 3's redesigned latency of 3.0 weeks the same objection costs 210,240 rather than 224,520. And the pre-consultation must be a *technical* engagement with the people who will apply the requirement, not a courtesy briefing of their director, because the director did not know either.

### 11.1.3 Engagement strategy under a capacity constraint

**The constraint, stated honestly.** Engagement capacity is finite and small. Meridian's is **40 hours a month** (the project leader's own engagement time plus two part-time engagement leads), giving **480 hours** over the twelve months of rollout. Every engagement strategy is an allocation of those 480 hours whether or not anyone writes it down, and the strategies that are not written down allocate by whoever asks most persistently, a default that correlates with neither value nor risk.

**The conventional allocation and its defect.** Standard practice scores each party on influence and interest, multiplies to a **salience** score, and allocates attention in proportion. It is a reasonable first cut with one structural defect: it allocates by *power and attention*, and neither is the same as *value at stake* or *consequence of refusal*. So it over-serves the loud and the powerful and under-serves two groups at once: the users who determine whether any benefit is realised, and the quiet approver of

**The three-term design.** The correction is to allocate in three terms, in this order:

1. A **value-led core**: the effort that demonstrably moves the largest benefit lever, sized from the benefit arithmetic rather than from the score.
2. **Consent-risk floors**: a minimum allocation to each party that can stop the work, set by what the relationship actually requires, not by its interest score.
3. A **salience-proportional residual**: what is left, distributed by influence  $\times$  interest.

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#### WORKED EXAMPLE

##### Worked example 11.1.3 — allocating Meridian's 480 engagement hours.

1. **Setup.** Seven groups, scored influence  $\times$  interest: regional health authority  $5 \times 3 = 15$ ; clinic directors  $4 \times 5 = 20$ ; clinicians  $2 \times 5 = 10$ ; practice administrators  $2 \times 4 = 8$ ; national reporting body  $5 \times 2 = 10$ ; patient representatives  $3 \times 3 = 9$ ; records-system supplier  $2 \times 4 = 8$ . Total salience **80**. Pilot evidence supports a structured adoption intervention, a clinical-workflow session plus two follow-ups, costing **8 hours per clinic** and raising expected adoption by **12.5 percentage points**, from 70 % to 82.5 %. Each adopting clinic is worth **USD 24,480** a year (Domain 1). Consent-risk floors are set at **60 hours** for the regional health authority (which holds the funding), **40** for the national reporting body and **30** for the patient representative group.
2. **Formula.** Salience allocation = capacity  $\times$  salience  $\div \Sigma$  salience. Three-term allocation = value core (8 h  $\times$  clinics) +  $\Sigma$  floors + residual apportioned by salience. Benefit gain = uplift in adopting clinics  $\times$  annual value per clinic. Value per engagement hour = gain  $\div$  core hours.
3. **Substitution.** Salience:  $480 \div 80 = 6$  hours per point. Core  $8 \times 40$ . Residual  $480 - 320 = 160$ , apportioned across salience  $8 + 8 = 16$ . Gain  $40 \times 0.125 \times 24,480$ .

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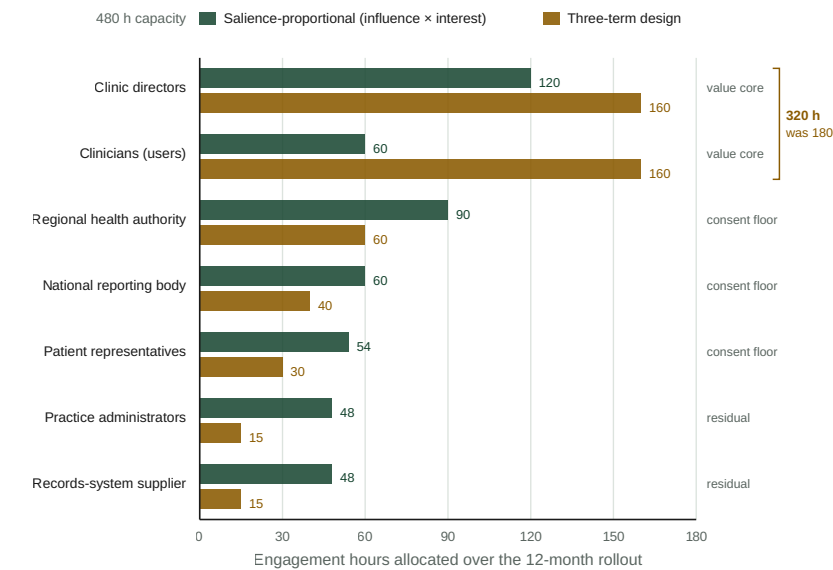
| GROUP                     | SALIENCE | SALIENCE-<br>PROPORTIONAL | THREE-TERM<br>DESIGN | BASIS         |
|---------------------------|----------|---------------------------|----------------------|---------------|
| Regional health authority | 15       | 90 h                      | <b>60 h</b>          | consent floor |

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| GROUP                   | SALIENCE  | SALIENCE-<br>PROPORTIONAL | THREE-TERM<br>DESIGN | BASIS         |
|-------------------------|-----------|---------------------------|----------------------|---------------|
| Clinic directors        | 20        | 120 h                     | <b>160 h</b>         | value core    |
| Clinicians (users)      | 10        | 60 h                      | <b>160 h</b>         | value core    |
| Practice administrators | 8         | 48 h                      | <b>15 h</b>          | residual      |
| National reporting body | 10        | 60 h                      | <b>40 h</b>          | consent floor |
| Patient representatives | 9         | 54 h                      | <b>30 h</b>          | consent floor |
| Records-system supplier | 8         | 48 h                      | <b>15 h</b>          | residual      |
| <b>Total</b>            | <b>80</b> | <b>480 h</b>              | <b>480 h</b>         |               |

1. **Result.** The value core is **320 hours**; floors take **130**; the residual is **30**. The core delivers **5** additional adopting clinics and **USD 122,400** a year of recurring benefit, at **USD 382.50 of benefit per engagement hour** in year one alone. The salience allocation gives the two adoption-determining groups only **180 hours**, enough for **22** of the 40 clinics (an uplift of **6.875 percentage points**, **2.75** clinics and **USD 67,320** a year), so it forgoes **USD 55,080 a year, permanently**. The core's opportunity cost is  $320 \times 110 = \text{USD } 35,200$  of existing capacity, against which the breakeven uplift is  $35,200 \div 24,480 = 1.4379$  clinics, or **3.59 percentage points** of adoption.
2. **Interpretation.** The decisive figure is the **3.59-point breakeven**, because it converts an argument about the value of engagement into a question a clinical lead can answer: does a workflow session plus two follow-ups, run in every clinic, move programme adoption by more than three and a half points? Almost certainly; and if it does not, the intervention should be redesigned rather than defended. Note also what the reallocation costs: nothing. The 480 hours are already paid for, so the 55,080 forgone is not an underspend but a **misallocation of capacity that was going to be spent anyway**, which is why this is the cheapest large improvement in the domain. Two second-order results matter more than they look. First, raising adoption to 82.5 % raises the **cost of delay** itself, because the benefit rate rises with adopting clinics: 33 clinics at 510 a week is **USD 16,830 per week**, up **17.86 %**. So engagement makes every schedule lever more valuable — Domain 1's 8-week compression for 60,000, worth a net **+USD 54,240** at 70 % adoption, becomes worth  $8 \times 16,830 - 60,000 = \text{+USD } 74,640$  at 82.5 %. **Sequence engagement before acceleration** is therefore an arithmetic ordering, not a preference. Second, the floors cannot be justified by this arithmetic and must not be: their return is the avoided 224,520 of 11.1.2, which is a *risk* return with a breakeven probability, not a benefit return

with a rate. A design that computed only the value core would fund adoption brilliantly and be stopped in week 34.



Saliency alone funds the 8-hour adoption intervention at 22 of 40 clinics — 6.875 pp, USD 67,320 a year.

The three-term design funds all 40 — 12.5 pp, USD 122,400 a year. Forgone by saliency: USD 55,080 a year.

Fig 11.1.1 — Two allocations of the same 480 hours

**From allocation to plan.** An allocation is not yet a strategy. Each group needs a stated **objective** as a dated future state ("the reporting body has confirmed the extract specification in writing by week 12", not "maintain good relations"), a **named owner** at a level the party will engage with, a **method** matched to attitude, a **cadence**, and a **test of whether it worked**. Objectives written as states rather than activities are what make an engagement plan reviewable: "held four meetings" cannot fail, and therefore cannot inform.

## AI in this KA

**Where it earns its place.** Building a first-pass register from documents the project already has (contract schedules, distribution lists, minutes, planning submissions, service-desk records), and listing parties named in them that the register omits: a set-difference task over unstructured text, the omission checklist of 11.1.1 mechanised, and one that routinely finds internal functions nobody thought to list. Classifying a large volume of consultation responses into themes with counts, so that a leader reads a structured summary of two thousand comments rather than a sample of twenty. Drafting a saliency and consent-risk assessment for human challenge, useful precisely because a draft is easier to argue with than a blank page. Modelling allocations across capacity and floor assumptions, which is arithmetic.

**Where it must not go.** Assigning **attitude** or **influence**. A model reads text; influence is political and attitude is often deliberately unstated, so a fluent assessment of either will be confidently wrong in the direction the source documents are polite.

Inferring a party's negotiating position from its published statements as though that settled it. And any "stakeholder sentiment score" entering a report as a governance input, a number with no measurement behind it displaces the judgement it was meant to support, permanently, because numbers survive committee meetings better than qualifications do.

**Verification, concretely.** Every register entry carries a named human owner who has confirmed the party's role, authority and consent risk against a source (the contract, the statutory instrument, the delegation schedule) not against the model's summary of it. Classified consultation responses are re-read on a stated sample (10 % is a defensible floor) with the disagreement rate recorded, and the classification is rejected above a pre-set tolerance. The allocation arithmetic is reproduced by hand: it is a division and two multiplications.

## KEY TERMS — KA 11.1

| TERM                              | MEANING                                                                                                                                                                         |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Stakeholder</b>                | A party whose decisions, consent, resources, behaviour or objection can materially affect the project's outputs, outcomes or legitimacy, or who is materially affected by them. |
| <b>Stakeholder system</b>         | The stakeholder set together with the relationships between its members, who influences whom, and which pairs align under pressure.                                             |
| <b>Influence</b>                  | Capacity to affect the project's decisions, resources or permissions; predicts the damage a disagreement does.                                                                  |
| <b>Interest</b>                   | Attention paid, not stake held; predicts how quickly a party notices.                                                                                                           |
| <b>Attitude</b>                   | Supportive, neutral, sceptical or opposed, assessed on evidence; predicts what an engagement contact will achieve.                                                              |
| <b>Consent risk</b>               | Whether the party holds a veto, approval, licence, signature or practical ability to withhold cooperation; a property of position, not disposition.                             |
| <b>Salience</b>                   | Influence × interest, a useful first cut and an inadequate allocation rule.                                                                                                     |
| <b>Value-led core</b>             | Engagement effort sized from the benefit arithmetic rather than from the salience score.                                                                                        |
| <b>Consent-risk floor</b>         | A minimum engagement allocation to a party that can stop the work, justified by breakeven probability rather than by benefit rate.                                              |
| <b>Engagement objective</b>       | A stated future <i>state</i> of a relationship with a date and a test, never an activity count.                                                                                 |
| <b>Basis of an attitude entry</b> | The observed act and its source, recorded instead of a characterisation of the person; the form an assessment takes in a record that names identified individuals.              |
| <b>Register access list</b>       | The stated set of people who may read the stakeholder register; it is not a general-circulation document.                                                                       |

| TERM                  | MEANING                                                                                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Deletion point</b> | The stated end of the register: what is deleted at closure, what is retained, for how long and on whose authority, with departed role holders' assessments deleted at each refresh. |

## SAMPLE MCQS — KA 11.1

**MCQ 11.1-A** [11.1.1 · Comprehension] The practical difference between a stakeholder list and a stakeholder system is that only the system:

- A. records contact details and communication preferences
- B. captures the relationships between parties, and so supports prediction of positions formed in conversations the project is not part of CORRECT
- C. is approved by the sponsor
- D. includes external as well as internal parties

*Rationale:* The system's added content is the relationships between members (11.1.1). A is administrative; C is governance; D is a completeness property a list can also have.

**MCQ 11.1-B** [11.1.2 · Application] An objection causes 4.0 weeks of governance latency and 5 weeks of rework on the critical path at a cost of delay of 14,280 per week, plus three interface re-verifications at 18,000 each and 42,000 of application rework. The assessed total is:

- A. USD 96,000
- B. USD 128,520
- C. USD 167,400
- D. USD 224,520 CORRECT

*Rationale:*  $9.0 \times 14,280 + 54,000 + 42,000 = 224,520$  (11.1.2). A is the direct cost alone; B is the delay alone; C omits the 4.0 weeks of governance latency and counts only the rework weeks.

**MCQ 11.1-C** [11.1.2 · Evaluation] A 30-hour pre-consultation costing 3,300 would have addressed the objection above. The most persuasive argument to a sceptical sponsor is that:

- A. the consequence is 68.0 times the avoidance cost
- B. the pre-consultation pays whenever the probability of the objection exceeds 1.47 % CORRECT
- C. engaging regulators early is good practice
- D. 3,300 is immaterial against a 2,400,000 budget

*Rationale:* The breakeven probability forces the sceptic to assert something indefensible, whereas a ratio invites the answer "it might not have happened" (11.1.2). D is true and irrelevant, since immateriality is not a reason to spend.

**MCQ 11.1-D** [11.1.3 · Analysis] Allocating engagement capacity in proportion to influence × interest systematically under-serves:

- A. the most powerful stakeholders
- B. the parties with the highest interest
- C. the user group that determines benefit realisation and the quiet party holding a veto CORRECT
- D. the project's own delivery team

*Rationale:* Salience measures power and attention, not value at stake or consequence of refusal (11.1.3), which is why the design adds a value-led core and consent-risk floors.

**MCQ 11.1-E** [11.1.3 · Application] A 320-hour adoption core raises adoption by 12.5 percentage points across 40 clinics, each adopting clinic being worth 24,480 a year. The benefit per engagement hour in year one is:

- A. USD 76.50
- B. USD 382.50 CORRECT
- C. USD 3,060.00
- D. USD 2,142.00

*Rationale:*  $40 \times 0.125 \times 24,480 = 122,400$ ;  $122,400 \div 320 = 382.50$  (11.1.3). A divides one clinic's annual value by the 320 hours ( $24,480 \div 320$ ); C divides the gain by the 40 clinics rather than by hours; D divides the whole 685,440 annual benefit by 320 hours instead of the uplift.

#### SELF-CHECK — KA 11.1

1. *Name the assessment quadrant that damages projects, and why an allocation rule misses it.* — High influence, low interest, high consent risk: any rule that reads interest scores it low, and it then reads the specification at the point of approval.
2. *State the three terms of an engagement allocation, in order.* — Value-led core sized from the benefit arithmetic; consent-risk floors for parties that can stop the work; salience-proportional residual.
3. *Why does raising adoption make schedule compression more valuable?* — The cost of delay is the benefit rate, and the benefit rate rises with adopting clinics: 14,280 a week at 70 % adoption, 16,830 at 82.5 %.
4. *What kind of record is a stakeholder register, and what follows?* — Information about identified living people, containing recorded opinions about them. It needs a named owner and an access list rather than open circulation; attitude recorded



as an observed act with its source rather than as a characterisation; departed role holders' assessments deleted at refresh; and a stated deletion point at closure. The obligations themselves come from whoever holds data-protection accountability and from qualified counsel, not from this book (11.1.2).

## KNOWLEDGE AREA 11.2

# Executive communication and reporting

*Topics: 11.2.1 communication as a designed architecture · 11.2.2 the executive report and the age of its numbers · 11.2.3 honest status and the escalation-grade message.*

## 11.2.1 Communication as a designed architecture

**The claim.** A communication plan is usually a table of meetings and distribution lists. It should be an **architecture**, because communication has the same combinatorial property as integration, and for the same reason: what costs money is the **channels between them**, not the parties.

Domain 4, KA 4.2.3 established the arithmetic and this domain does not re-derive it. For  $n$  parties each communicating directly with every other, the number of channels is  $n(n - 1)/2$ ; routing every party through a single integrating point gives  $n$ . Applied to people it explains something practitioners experience and rarely diagnose: a programme that has added four stakeholder groups has not added four relationships but a great many, and the engagement plan written for the original count is still in use.

## WORKED EXAMPLE

### Worked example 11.2.1 — the communication load Meridian was carrying.

- Setup.** Meridian's register holds **62** parties, of which **14** require an individually managed relationship. In the design as found, each of those 14 dealt directly with each of the others and with the programme, and maintaining one such channel (preparation, contact, follow-up, recording) costs **1.5 hours per month**. The alternative routes every party through the programme's engagement function as the integrating point, at a hub overhead of **12 hours a month** (two stakeholder forums, all-in). Engagement capacity is **40 hours a month** (11.1.3).
- Formula.** Mesh channels  $n(n - 1)/2$ ; routed channels  $n$  (Domain 4, KA 4.2.3). Load = channels  $\times$  hours per channel (+ hub overhead). Sustainable party count = the largest  $n$  whose load fits the capacity.
- Substitution.** Mesh  $14 \times 13/2 = 91$ , load  $91 \times 1.5$ . Routed 14, load  $14 \times 1.5 + 12$ . Sustainable mesh count: largest  $n$  with  $n(n - 1)/2 \times 1.5 \leq 40$ .

- 
4. **Result.** 91 channels demanding 136.5 hours a month, 341.25 % of capacity, against 14 channels demanding 33.0 hours, or 82.5 %. An unrouted design sustains at most 7 parties (21 channels, 31.5 hours); the eighth takes it to 42.0 hours and over capacity. A routed design sustains 18 (39.0 hours). Routing saves 103.5 hours a month, 1,242 hours a year, worth USD 136,620 at the blended rate.
5. **Interpretation.** The number to carry away is seven. An unrouted design supports seven actively managed parties within Meridian's capacity and Meridian has fourteen, so the design was never feasible, and infeasible communication designs do not fail by anyone reporting a shortfall. They fail silently: contacts are skipped, the skipped ones are the low-interest parties because they do not chase, and 11.1.2's objection is the result. The overload is therefore not a workload complaint but the **mechanism** by which the late objection is generated, which is the professional insight this arithmetic buys. Three cautions. Routing is not free of distortion: a hub is a single point through which every message is paraphrased, and the parties who must not be paraphrased (the sponsor, the statutory approver, the clinical authority) need bilateral channels regardless of load. Meridian's honest design is therefore **hybrid**: a deliberate three-party mesh among those three plus all 14 routed to the hub, giving  $3 + 14 = 17$  channels, 37.5 hours a month, 93.75 % of capacity. That is tight and it is the truth; a plan showing 82.5 % would understate the design actually chosen. Next, the 1.5-hour unit is an average over relationships of very different weight, and a leader quoting one figure for a regulator and a practice administrator should say so. And the hub overhead is real work that must be resourced as work: an engagement function that is nobody's named job is a mesh with a diagram on top of it.
- 

### 11.2.2 The executive report and the age of its numbers

**What an executive report is for.** Not to describe the project: to enable a decision the reader has authority to take, or to confirm that none is required. Everything serving neither purpose consumes the scarcest resource in the governance system, the attention of people who can decide (Domain 3, Fault 3). The structural consequence is a fixed order, the reverse of how most reports are written: **the decision or confirmation first**, then the two or three facts that drive it, then the forecast with its assumptions, then exceptions, then the rest by reference. A report that reaches its decision on page four was written for the author's comfort.

**The property nobody measures.** A report's usefulness depends on more than its content: it depends on **how old the facts in it are when the decision is taken**. Three quantities determine that, and all three are known at design time. Let

|                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="border-left: 1px solid black; padding-left: 10px;"> <p>P = the reporting period (e.g. 4 weeks for a monthly pack)</p> <p>C = consolidation time – data cut-off to pack issue</p> <p>L = the paper lead time – how far before the meeting submissions close (Domain 3)</p> </div> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Then, measuring back from the decision meeting:

$$\begin{aligned}\text{Age of the newest fact} &= C + L \\ \text{Mean age of the facts} &= C + L + P/2 \\ \text{Maximum blindness} &= P + C + L\end{aligned}$$

The third is the important one: **maximum blindness** is how long a problem arising just after a cut-off remains invisible to the decision body. It is the number that should be quoted when anyone proposes a longer reporting cycle, and it is almost never computed.

#### WORKED EXAMPLE

#### Worked example 11.2.2 — how old is the number Meridian's committee decides on?

1. **Setup.** Meridian reports monthly:  $P = 4$  **weeks**, data cut-off at period end, consolidation and internal review  $C = 1$  **week** from cut-off to pack issue, and the steering committee's paper lead time  $L = 2$  **weeks** (Domain 3, KA 3.2.3). The proposed redesign keeps the monthly committee but cuts data **weekly** into a standing dashboard ( $P = 1$ ,  $C = 0.5$ ,  $L = 1$ ), so the pack presents the latest weekly cut rather than a monthly consolidation. Cost of delay **USD 14,280 per week**.
2. **Formula.** Newest =  $C + L$ ; mean =  $C + L + P/2$ ; maximum blindness =  $P + C + L$ . Saving priced at the cost of delay.
3. **Substitution.** As found:  $1 + 2$ ;  $1 + 2 + 4/2$ ;  $4 + 1 + 2$ . Redesigned:  $0.5 + 1$ ;  $0.5 + 1 + 0.5$ ;  $1 + 0.5 + 1$ .
4. **Result.**

|                                     | NEWEST FACT | MEAN FACT AGE     | MAXIMUM BLINDNESS |
|-------------------------------------|-------------|-------------------|-------------------|
| Monthly pack ( $P\ 4, C\ 1, L\ 2$ ) | 3.0 weeks   | <b>5.0 weeks</b>  | <b>7.0 weeks</b>  |
| Weekly cut ( $P\ 1, C\ 0.5, L\ 1$ ) | 1.5 weeks   | <b>2.0 weeks</b>  | <b>2.5 weeks</b>  |
| Weeks saved                         | 1.5         | 3.0               | 4.5               |
| Priced at 14,280/week               | USD 21,420  | <b>USD 42,840</b> | <b>USD 64,260</b> |

1. **Interpretation.** Meridian's committee has been deciding on a picture whose average fact is **five weeks old**, and a problem arising in the first week of a period can run **seven weeks** before the body that could stop it sees it: at 14,280 a week, **USD 99,960** consumed before the decision system is aware of it. That reframes the usual argument: the objection to monthly reporting is not that it is "not current enough", which is taste, but that its **maximum blindness is 7.0 weeks and can be 2.5**, which is arithmetic. Note the ordering of the levers, mirroring Domain 3's result about paper deadlines. Shortening the reporting

*period* is the strongest lever on maximum blindness and costs almost nothing once the data layer exists; consolidation time is next and is usually pure process; the paper lead time is Domain 3's lever, shared with the governance design. The professional caution is that these savings are **conditional and not additive across the year**: they accrue only where a faster picture would have produced a faster decision *and* the delay was on the critical path. The honest claim is therefore a per-occurrence one, 42,840 on the mean and 64,260 in the worst case, with the number of qualifying occurrences stated as an assumption. Presenting it as an annual saving is the error that gets a good proposal rejected by a finance director who can see the double count.

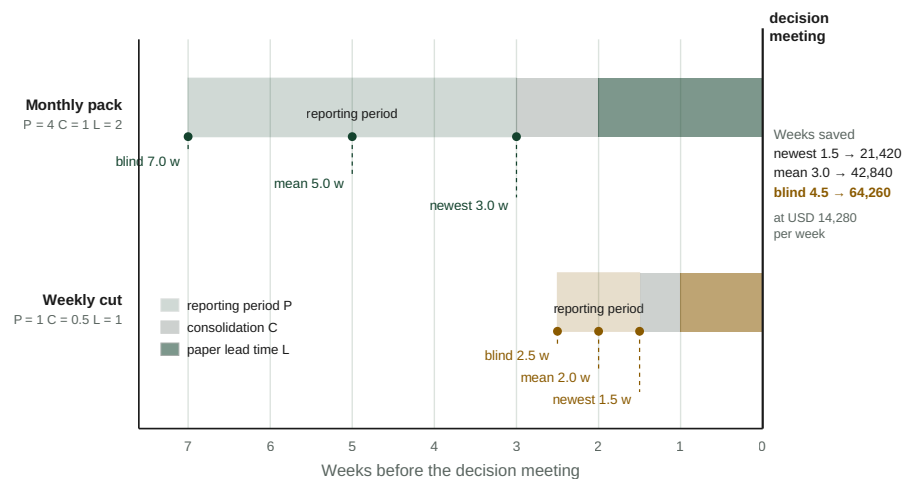


Fig 11.2.1 — The age of the number an executive decides on

### 11.2.3 Honest status and the escalation-grade message

**The status problem.** Traffic-light status is universal and structurally corrupt, for a reason worth stating precisely: the light is set by the person whose performance it evaluates, against undefined thresholds, for an audience that reacts to colour before content. The predictable result is **amber compression**: nearly everything is amber, red is reserved for the unrecoverable, and the status carries almost no information at the point it is most needed. The fix is not more honesty but **pre-agreed objective thresholds** set at baseline and applied mechanically: "red where forecast completion exceeds baseline by more than four weeks or forecast cost exceeds budget by more than 5 %" removes the author's discretion and with it the incentive that corrupted the signal, and it makes the status arguable on the facts, which is what a governance body needs.

**What a red status must carry**, and this is the difference between reporting a problem and escalating one. The **exception**, quantified against its threshold. The **cause**, distinguished from the symptom. The **impact** if nothing changes, priced. The **options**, each with cost, consequence and risk. The **recommendation**, made by

name. The **decision required**, from whom, **by when**, with the consequence of missing that date. And the **date the leader knew**, which is the field that makes Domain 3's escalation lead time measurable.

**Bad news.** Three rules earn their place. *Escalate the forecast, not the event*: a leader who waits for a variance to occur has converted a decision into a report. *Never let a governance body learn something material from a third party first*: once it happens, the leader's reporting is discounted for the remainder of the project. And *separate the news from the plea*, because a message mixing an exception with a request for sympathy invites a response to the second. Domain 12 handles the personal dimension; the reporting obligation here is narrow and absolute. The material fact reaches the accountable decision-maker, in writing, on the date it is known.

**The tailoring rule.** Different audiences need different *depth*, never different *facts*. Tailoring selects what a reader needs to decide and expresses it in their vocabulary; it does not mean the board's version and the delivery team's version disagree. Where two versions of one status exist, the project has created a discovery problem for itself, and Domain 3's decision record is where the discrepancy will be found.

## AI in this KA

**Where it earns its place.** Assembling a report to a fixed skeleton from the data layer (figures, variances, trends, exception lists), which removes the copy-and-paste error class that infects manually built packs. **Consistency checking**, the highest-value use in this KA and the least used: reading a drafted report against its own numbers and flagging where the narrative and the arithmetic disagree, where a status contradicts its stated threshold, or where a figure differs from the previous pack without explanation. Rewriting a technical explanation for a non-specialist reader, for human approval. Producing audience-specific depth variants from one factual base: the tailoring rule mechanised.

**Where it must not go.** Setting the status, which is an accountable judgement against a threshold. Writing the recommendation. And. The failure that is easy to walk into and hard to see afterwards, **smoothing**. A model asked to make a report "clearer", "more balanced" or "less alarming" will reliably produce a calmer document, because calm prose is what fluent business writing looks like in the material it learned from, and the qualifier carrying the warning is the first thing to go. Any instruction to soften a report is an instruction to alter its meaning, and must be treated as an editorial decision taken by a person, on the record.

**Verification, concretely.** The status and the recommendation are entered by the accountable human and excluded from any automated drafting step. Every figure traces to a query against the data layer, available on request rather than reconstructed after a challenge. A named person signs the pack as issued, and that signature means they have read it, which implies a report short enough to read. Every consistency flag is resolved and the resolution recorded, because an unresolved flag

in an issued report is documented knowledge of an inconsistency, which is worse than no checker at all.

## KEY TERMS — KA 11.2

| TERM                              | MEANING                                                                                                                                                                                                                                                                                                      |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Communication architecture</b> | The designed set of channels between parties (mesh, routed or hybrid) with its load computed against capacity.                                                                                                                                                                                               |
| <b>Channel load</b>               | Channels × hours per channel per period; compared with engagement capacity to give the sustainable party count.                                                                                                                                                                                              |
| <b>Integrating point (hub)</b>    | The function through which routed communication passes; carries its own overhead and its own distortion risk.                                                                                                                                                                                                |
| <b>Reporting period (P)</b>       | The interval of activity a report covers.                                                                                                                                                                                                                                                                    |
| <b>Consolidation time (C)</b>     | Elapsed time from data cut-off to pack issue.                                                                                                                                                                                                                                                                |
| <b>Maximum blindness</b>          | $P + C + L$ , how long a problem arising just after a cut-off stays invisible to the decision body.                                                                                                                                                                                                          |
| <b>Amber compression</b>          | The collapse of traffic-light status into a single uninformative colour, caused by author discretion over undefined thresholds.                                                                                                                                                                              |
| <b>Escalation-grade message</b>   | Exception, cause, impact priced, options, recommendation by name, decision required by whom and by when, and the date the leader knew.                                                                                                                                                                       |
| <b>Tailoring (of a message)</b>   | Varying depth and vocabulary by audience while holding the facts identical. <b>Context flag:</b> distinct from <i>tailoring of method</i> (Domain 4, KA 4.1.3), which is a recorded decision to adapt process; the shared word carries two different concepts and neither reading substitutes for the other. |

## SAMPLE MCQs — KA 11.2

**MCQ 11.2-A** [11.2.1 · Application] Fourteen parties each communicate directly with every other. The number of channels is:

- A. 14
- B. 91 CORRECT
- C. 182
- D. 196

*Rationale:*  $n(n - 1)/2 = 14 \times 13/2 = 91$  (11.2.1, citing Domain 4 KA 4.2.3). A is the routed count; C counts each pair twice; D is  $n^2$ .

**MCQ 11.2-B** [11.2.1 · Analysis] At 1.5 hours per channel per month and 40 hours of engagement capacity, the largest number of parties an unrouted design sustains is:

- A. 7 CORRECT
- B. 8
- C. 14
- D. 26

*Rationale:* Seven parties give 21 channels and 31.5 hours; eight give 28 channels and 42.0 hours, which exceeds capacity (11.2.1). C is the actual party count, which is why the design was infeasible; D divides capacity by the unit cost and ignores the combinatorial term.

**MCQ 11.2-C** [11.2.2 · Application] A monthly report has a 4-week period, 1 week of consolidation and a 2-week paper lead time. The mean age of the facts at the decision meeting is:

- A. 2.0 weeks
- B. 3.0 weeks
- C. 5.0 weeks CORRECT
- D. 7.0 weeks

*Rationale:*  $C + L + P/2 = 1 + 2 + 2 = 5.0$  weeks (11.2.2). B is the newest fact only; D is maximum blindness; A counts only the period's half-life.

**MCQ 11.2-D** [11.2.2 · Evaluation] The strongest argument for cutting data weekly rather than monthly is that it:

- A. makes the pack look more current to the committee
- B. reduces maximum blindness from 7.0 to 2.5 weeks — 4.5 weeks, or USD 64,260 at the cost of delay CORRECT
- C. saves consolidation effort
- D. reduces the paper lead time

*Rationale:* Maximum blindness is the decision-relevant quantity and the redesign is arguable arithmetically (11.2.2). A is presentational; C is false, since more frequent cuts add consolidation events; D is a separate governance lever (Domain 3).

**MCQ 11.2-E** [11.2.3 · Analysis] The root cause of amber compression is that:

- A. project leaders are optimistic by disposition
- B. the status is set by the person it evaluates, against undefined thresholds CORRECT
- C. boards react badly to red status
- D. reporting cycles are too long

*Rationale:* The corruption is structural (discretion plus undefined thresholds), so the remedy is pre-agreed objective thresholds applied mechanically (11.2.3). A and C are contributing conditions, not the cause.

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SELF-CHECK — KA 11.2

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1. *Why is an overloaded communication architecture a risk mechanism rather than a workload complaint?* — Skipped contacts fall on the parties who do not chase, which are the low-interest, high-consent-risk parties whose late objection 11.1.2 prices.
  2. *State the three components of report age and which one to quote against a longer cycle.* —  $C + L$  (newest),  $C + L + P/2$  (mean),  $P + C + L$  (maximum blindness); quote maximum blindness.
  3. *What must a red status carry beyond the exception?* — Cause, priced impact, options, a named recommendation, the decision required from whom by when, and the date the leader knew.
- 

KNOWLEDGE AREA 11.3

## Negotiation and conflict

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*Topics:* 11.3.1 preparation (reservation value, the priced alternative and the zone of possible agreement · 11.3.2 creating value before claiming it · 11.3.3 conflict) sources, modes and the resolution route.

### 11.3.1 Preparation — reservation value, the priced alternative and the zone of possible agreement

**The three quantities.** Negotiation in a delivery context is not mainly a matter of technique. It is almost entirely a matter of three numbers, two of which are knowable before the meeting.

- Your **best alternative**: what happens if no agreement is reached, **fully priced**, including delay at the cost of delay, internal effort, and the risk the alternative carries, valued as an expected monetary value (EMV, Domain 8).
- Your **reservation value**: the worst terms you would rationally accept, derived from the priced alternative. For a buyer it is a maximum; for a seller a minimum. A reservation value that is not derived from a priced alternative is a preference, and preferences move under pressure.
- The **zone of possible agreement (ZOPA)**: the interval between the two parties' reservation values. If it is empty there is no deal at any level of skill, and the



professional act is to stop and change the alternatives rather than to keep meeting.

The single most consequential preparation failure is pricing the alternative on its **invoice** rather than its **consequences**. A competitive re-tender does not cost the difference in licence fees; it costs that plus the weeks it takes at the cost of delay, plus internal effort, plus risk. Omit those and the reservation value is set too low. And a buyer whose reservation value is too low walks away from deals it should accept, which is the opposite of the error everyone worries about.

---

#### WORKED EXAMPLE

##### Worked example 11.3.1 – the Meridian support-and-enhancement extension.

1. **Setup.** Meridian must extend the records-system supplier's support and enhancement contract across the rollout year. **Meridian's alternative** is to re-tender and transition to another supplier: licence and transition **USD 640,000**, **6 weeks** of transition delay at the cost of delay of **14,280** per week, **USD 74,320** of internal transition effort, and transition risk assessed at an **EMV** of **USD 45,000**. **The supplier's position**, as estimated by Meridian from the tender history and published rate cards: cost to serve **USD 520,000**, and a best alternative use of the same team worth **USD 95,000** of contribution. The supplier opens at **900,000**. Assume, as a stated benchmark, that a settlement splits the zone equally.
2. **Formula.**  $RV(\text{buyer})$  = fully priced alternative.  $RV(\text{seller})$  = cost to serve + forgone contribution. ZOPA width =  $RV(\text{buyer}) - RV(\text{seller})$ . Midpoint settlement = mean of the two reservation values. Surplus = own reservation value less price (buyer) or price less own reservation value (seller). Share of ZOPA = surplus ÷ width.
3. **Substitution.**  $RV(\text{buyer}) = 640,000 + 6 \times 14,280 + 74,320 + 45,000$ .  $RV(\text{seller}) = 520,000 + 95,000$ . Width  $845,000 - 615,000$ . Midpoint  $(845,000 + 615,000)/2$ .
4. **Result.** The alternative's delay component is **USD 85,680**, giving  $RV(\text{buyer}) = \text{USD } 845,000$ .  $RV(\text{seller}) = \text{USD } 615,000$ . The ZOPA runs from **615,000** to **845,000**, a width of **USD 230,000**, and the supplier's 900,000 opening is **outside it**. There is no deal at that price. The midpoint settlement is **USD 730,000**, giving each side **USD 115,000** of surplus. Had Meridian not priced its alternative and instead anchored on the supplier's ask, settling at **780,000**, its surplus would be **USD 65,000** (28.3 % of the zone against the supplier's 71.7 %), and **USD 50,000** of value would have transferred purely through the absence of preparation.
5. **Interpretation.** Fifty thousand dollars is the price of an unpriced alternative, available to whichever party has done the arithmetic, roughly fifteen times the 3,300 of 11.1.2, which is this domain's general pattern: preparation is always

cheap relative to consequence. Now the subtler result. Suppose Meridian spends **USD 25,000** genuinely pre-qualifying a second supplier, cutting the transition delay from 6 weeks to **2** (saving 57,120) and the transition risk **EMV** from 45,000 to **12,000** (saving 33,000). Its alternative now costs **USD 754,880**, so its reservation value **falls** to that figure, the ZOPA narrows to **USD 139,880**, and the mid-point settlement falls to **USD 684,940**: **USD 45,060** less cash, a net gain of **USD 20,060** after the spend, with **USD 33,000** less risk carried; the breakeven pre-qualification spend is therefore **USD 45,060**. And here is the trap: Meridian's *surplus* has fallen from 115,000 to **USD 69,940**, so a negotiator measured on surplus captured will report the improved position as a worse outcome. Surplus is measured against your own reservation value, which the investment deliberately moved. **The measures that govern are cash paid and risk carried, not surplus captured**: worth insisting on, because surplus-based incentives reward negotiators for keeping their alternatives bad. Three cautions. The counterparty's reservation value is an *estimate* from observable evidence and must be presented as a range with the assumptions that move it; a single confident figure invites a concession plan built on a guess. The equal-split benchmark is a benchmark, not a prediction: real splits turn on patience, alternatives, internal deadlines and who is measured on what. And a ZOPA computed on price alone is the smaller half of the problem, which is 11.3.2's subject.

### 11.3.2 Creating value before claiming it

**The distinction.** Claiming value moves a fixed quantity between the parties. Creating value increases the quantity available, and it is possible whenever the parties value the issues **differently**, which, across any real contract, they almost always do. A negotiation conducted on price alone forgoes all of it, and a negotiation conducted by conceding equally on every issue forgoes most of it.

#### WORKED EXAMPLE

#### Worked example 11.3.2 — the trade that created 82,000.

1. **Setup.** Beyond price, three issues are open in the Meridian extension. **A**: priority-incident response time cut from 4 hours to 2, worth **USD 84,000** to Meridian, costing the supplier **USD 30,000**. **B**: two named engineers dedicated on site through the rollout — worth **USD 36,000** to Meridian, costing the supplier **USD 60,000**. **C**: extending the licence to 8 additional clinics in year 2 — worth **USD 40,000** to Meridian, costing the supplier **USD 12,000**. Compare conceding half of each issue with trading whole issues.
2. **Formula.** Joint gain on a set of issues =  $\Sigma(\text{value to buyer}) - \Sigma(\text{cost to seller})$ . Compromise: half of every issue. Trade: take the issues with positive joint value, drop the rest. Share the joint gain through price.

- 
3. **Substitution.** Compromise  $0.5 \times (84,000 + 36,000 + 40,000) - 0.5 \times (30,000 + 60,000 + 12,000)$ . Trade  $(84,000 + 40,000) - (30,000 + 12,000)$ .
  4. **Result.** The compromise gives Meridian **USD 80,000** of value at a cost to the supplier of **USD 51,000**: a joint gain of **USD 29,000**. Trading A and C in full and dropping B gives Meridian **USD 124,000** at a cost of **USD 42,000**: a joint gain of **USD 82,000**, which is **2.83 times** the compromise and **USD 53,000** better in absolute terms. Issue B has a joint value of  $36,000 - 60,000 = \text{USD } 24,000$ : it destroys value, so any agreement containing it is jointly worse than the same agreement without it. Sharing the 82,000 equally means compensating the supplier its 42,000 of cost plus **USD 41,000**, so the price rises **USD 83,000** (from the 684,940 of 11.3.1 to **USD 767,940**), and each side is **USD 41,000** better off than under the compromise-free baseline.
  5. **Interpretation.** The compromise looks fair, feels reasonable and destroys **53,000** of joint value, which is the most useful thing here: **"splitting the difference" is not a neutral procedure, it is an expensive one.** Two further results are worth carrying. Issue B is the one a buyer instinctively pushes hardest, because on-site presence feels like commitment, and it is the one issue that should be traded away; the arithmetic says so before any relationship is spent on it. A leader who cannot compute joint value will spend negotiating capital on B and concede on C, which is worth 3.33 times its cost to the other side. And the price rise of 83,000 is what makes the trade acceptable to the supplier: a buyer capturing the whole 82,000 by refusing to move on price has converted a value-creating trade into a claiming exercise and will be met accordingly at renewal (Domain 10 makes the same point about gainshare). The cautions are real. These valuations are the buyer's own estimates of both sides' numbers, and the trade only survives if both parties disclose *relative priorities*, which is not disclosing reservation values, and the distinction is the whole of negotiation ethics in practice: revealing that response time matters more to you than on-site presence is cooperative, revealing the maximum you would pay is simply losing. Where a relationship has no history of reciprocity, an early unilateral disclosure of priorities will be used to claim rather than to create, which is why value creation is easier in a second contract than in a first.
- 

### 11.3.3 Conflict — sources, modes and the resolution route

**Diagnose by source, not by personality.** Conflict on projects is routinely attributed to individuals and is usually produced by structure. Five sources, each with a different remedy, and misdiagnosis wastes the intervention:

- **Structural:** two people have been given incompatible objectives, or a matrix has no precedence rule (Domain 3, KA 3.1.2). No amount of facilitation resolves this; only a decision about authority does.

- **Resource:** two workstreams need the same scarce person. Resolved by priority at the level that owns both, and prevented by Domain 6's resource planning.
- **Information:** the parties hold different facts, or the same facts at different ages (11.2.2). The cheapest conflict to resolve if diagnosed early and the most damaging if left, because it teaches each party that the other is unreliable.
- **Interest:** the parties want genuinely different outcomes. This is a negotiation (11.3.1–11.3.2), not a misunderstanding, and treating it as one is condescending as well as ineffective.
- **Interpersonal:** genuine relational breakdown. Real, less common than attributed, and Domain 12's territory.

**Choosing a response.** The five familiar modes (competing, accommodating, avoiding, compromising, collaborating) are situational rather than ranked, and the skill is selection. Collaboration is the highest-value mode and the most expensive in elapsed time, so it fits a large stake and a continuing relationship and not a decision needed on Friday. Competing is right where safety, legality or a non-negotiable requirement is at stake, and a leader who cannot compete on those will be rolled on them. Accommodating is right where the issue matters little to you and much to the other party. It is how reciprocity is built and the mode most under-used by anxious leaders. Avoiding is right only where events will genuinely resolve the issue, which is rarer than it is invoked. Compromising is the fast default whose cost 11.3.2 has just computed: acceptable for small single-issue matters, expensive everywhere else.

**The resolution route, and its cost.** Escalation is not a mode; it is what happens when the chosen mode has failed, and it has a price. Domain 3, KA 3.3.3 computed escalation latency and Domain 10, KA 10.4 the dispute ladder for contractual conflict; neither is re-derived here. The point specific to this KA is the **threshold**: an unresolved conflict holding a critical-path decision consumes the cost of delay every week, so the comparison is not "facilitation versus toughing it out" but the intervention's cost against the delay it stops, divided by the probability it works. Exercise 11.5 computes one such case, and the shape of the answer is that breakeven success probabilities for cheap facilitation sit in the low single figures, so the intervention is almost always worth attempting and almost never attempted early.

Two obligations bind a leader who is party to the conflict. Separate the *position* from the *person* in the written record as well as in the room, because the record is what the organisation reads later. And hand the resolution to someone who is not a party: a leader chairing a conflict they are in has converted a resolvable dispute into a legitimacy problem, and legitimacy is not recovered by being right.

## AI in this KA

**Where it earns its place.** Preparation, which is where negotiations are decided. Building the reservation-value model of 11.3.1 from its components and testing which assumptions move it: a sensitivity task with a clear right answer. Enumerating

trade combinations: with three issues there are few and with twelve there are thousands, and finding the sets with positive joint value is genuine combinatorial work humans do badly. Rehearsing counter-arguments, so that a leader has met the strongest version of the other side's case before the room. Summarising a long contract and correspondence history into a chronology with the source document cited for each entry.

**Where it must not go.** Estimating the counterparty's reservation value from plausibility: asked for that number a model will supply a confident one with no provenance, it will enter the concession plan as a fact, and real money will move against it. Conducting the negotiation, or drafting messages sent without a human deciding every commitment they contain: a sentence that concedes an issue concedes it. Generating a "fair" outcome, which is a judgement about legitimacy, not a computation. And any use in a live conflict between named individuals: an assessment of a colleague's motives generated by a tool and stored in a project record is an employment and dignity problem regardless of its accuracy.

**Verification, concretely.** Every input to a reservation value carries a source and a date, and the alternative's delay component is priced at the project's own cost of delay rather than a rate someone remembered. The counterparty's estimated reservation value is recorded as a range with the evidence for each bound, and the concession plan is tested at both. Nothing model-derived is stated to the counterparty as fact. And the ZOPA arithmetic is reproduced by hand before the meeting: two additions and a subtraction, and the only number in the room that must be exactly right.

#### KEY TERMS — KA 11.3

| TERM                                     | MEANING                                                                                                                                     |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Best alternative</b>                  | What happens absent agreement, priced fully: invoice cost plus delay at the cost of delay plus internal effort plus risk as an <i>EMV</i> . |
| <b>Reservation value</b>                 | The worst terms a party would rationally accept, derived from its priced alternative; a maximum for a buyer, a minimum for a seller.        |
| <b>Zone of possible agreement (ZOPA)</b> | The interval between the two reservation values; empty means no deal at any level of skill.                                                 |
| <b>Surplus</b>                           | Own reservation value less the price agreed (buyer), or price less own reservation value (seller).                                          |
| <b>Joint gain</b>                        | $\Sigma(\text{value to one party}) - \Sigma(\text{cost to the other})$ over the issues traded; negative for issues that destroy value.      |
| <b>Logrolling (trading whole issues)</b> | Conceding entirely on issues valued asymmetrically, rather than partially on all of them.                                                   |
| <b>Conflict source</b>                   | Structural, resource, information, interest or interpersonal, each with a different remedy.                                                 |

| TERM                    | MEANING                                                                                |
|-------------------------|----------------------------------------------------------------------------------------|
| <b>Resolution route</b> | The chosen mode, its cost, and the escalation path with its latency if the mode fails. |

## SAMPLE MCQs — KA 11.3

**MCQ 11.3-A** [11.3.1 · Application] A supplier's cost to serve a contract is 520,000 and the best alternative use of the same team would earn 95,000 of contribution. Its reservation value is:

- A. USD 425,000
- B. USD 520,000
- C. USD 615,000 CORRECT
- D. USD 95,000

*Rationale:*  $520,000 + 95,000 = 615,000$  (11.3.1). B omits the forgone contribution, which is the commonest error and would put the ZOPA floor 95,000 too low; A subtracts it.

**MCQ 11.3-B** [11.3.1 · Analysis] With a ZOPA of 615,000–845,000, a buyer that fails to price its alternative and settles at 780,000 rather than the 730,000 midpoint has:

- A. captured 71.7 % of the zone
- B. transferred USD 50,000 of value to the seller through the absence of preparation CORRECT
- C. made an error of USD 165,000
- D. exceeded its reservation value

*Rationale:*  $780,000 - 730,000 = 50,000$ ; the buyer's share falls to 28.3 % (11.3.1). A is the seller's share; C is the seller's surplus; D is false, since 780,000 is inside the zone.

**MCQ 11.3-C** [11.3.1 · Evaluation] Pre-qualifying a second supplier for 25,000 cuts the buyer's reservation value from 845,000 to 754,880 and the midpoint settlement from 730,000 to 684,940. The correct reading is that the buyer is:

- A. worse off, because its surplus falls from 115,000 to 69,940
- B. better off by USD 20,060 in cash after the spend, and carrying USD 33,000 less risk CORRECT
- C. unaffected, since both reservation values moved
- D. better off by USD 45,060, ignoring the pre-qualification cost

*Rationale:* Surplus is measured against a reservation value the investment deliberately moved; cash paid and risk carried govern (11.3.1). A is the trap; D omits the 25,000.

**MCQ 11.3-D** [11.3.2 · Application] Three issues are worth 84,000, 36,000 and 40,000 to the buyer and cost the seller 30,000, 60,000 and 12,000. Conceding half of each yields a joint gain of 29,000. Trading the two value-creating issues in full yields:

- A. USD 58,000
- B. USD 82,000 CORRECT
- C. USD 124,000
- D. USD 160,000

*Rationale:*  $(84,000 + 40,000) - (30,000 + 12,000) = 82,000$  (11.3.2). A trades all three issues, including the one whose joint value is -24,000, giving  $160,000 - 102,000 = 58,000$ ; C is the buyer's value on the traded pair with no deduction of the seller's cost; D is the buyer's value across all three issues.

**MCQ 11.3-E** [11.3.3 · Analysis] Two workstream leads are in open conflict; each has been given an objective the other's success would prevent. Facilitation has failed twice. The diagnosis is:

- A. interpersonal conflict requiring coaching
- B. structural conflict requiring a decision about objectives and authority CORRECT
- C. information conflict requiring a single authoritative source
- D. interest conflict requiring negotiation

*Rationale:* Incompatible assigned objectives are structural, and no facilitation resolves a structure (11.3.3), the repeated failure of facilitation is itself the diagnostic.

#### SELF-CHECK — KA 11.3

1. *What is the commonest error in setting a reservation value, and which way does it bite?* — Pricing the alternative on its invoice rather than its consequences, which sets the reservation value too low, so a buyer walks away from deals it should accept.
2. *Why is splitting the difference expensive?* — It concedes partially on every issue, including those with negative joint value; trading whole asymmetrically valued issues produced 82,000 against the compromise's 29,000.
3. *What may be disclosed in a value-creating negotiation, and what may not?* — Relative priorities may; reservation values may not.

## Public and community stakeholders, cross-cultural communication and AI-generated communication risk

*Topics: 11.4.1 public and community stakeholders and consent · 11.4.2 cross-cultural and multilingual communication · 11.4.3 misinformation and AI-generated communication risk.*

### 11.4.1 Public and community stakeholders and consent

**What is different about them.** Four properties change the method. They are **unbounded**: there is no membership list, so representativeness is a judgement rather than a fact. They are **heterogeneous**, so a single "community view" is usually an artefact of who attended. Their engagement is often **statutory**, with prescribed forms, periods and grounds of objection that vary entirely by jurisdiction; and nothing here should be read as describing the requirements of any particular one. And their influence is **discontinuous**: low for long periods, then decisive through a planning objection, a legal challenge, a political intervention or simple non-cooperation.

**Consultation, and the distinction that decides its value.** Consultation before a decision is engagement: it can change the design, so participants have a reason to invest in it. Consultation after a decision is notification with a comment box, and it reliably produces objections rather than absorbing them, because obstruction is the only remaining way to influence the outcome. Case study B computes what that distinction was worth on one project, a factor of **3.90** on total cost, and Domain 5's design-maturity argument explains the mechanism: accommodating a concern costs what it costs to change the design at the maturity you have reached.

**Pricing consent risk.** Consent risk can be priced with Domain 8's **EMV**, provided the probabilities are presented as judgements and the breakeven is stated.

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#### WORKED EXAMPLE

#### Worked example 11.4.1 — is Meridian's public consultation worth holding?

1. **Setup.** Meridian's records system creates a regional data-sharing capability with a legitimate public interest in it. Without early public consultation, the programme assesses the probability of a formal privacy complaint that suspends the data-sharing element at **0.35**; with a structured consultation costing **USD 48,000** it assesses **0.10**. A complaint would cost **12 weeks** of delay at the cost of delay of **14,280** per week plus **USD 60,000** of legal and response cost.
2. **Formula.** **EMV** = probability × consequence (Domain 8). Compare **EMV** without consultation against consultation cost + **EMV** with it. Breakeven consultation cost = the difference in the two **EMV** terms. Required probability reduction = consultation cost ÷ consequence.



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3. **Substitution.** Consequence  $12 \times 14,280 + 60,000$ . Without  $0.35 \times 231,360$ . With  $48,000 + 0.10 \times 231,360$ . Breakeven  $80,976 - 23,136$ . Required reduction  $48,000 \div 231,360$ .
  4. **Result.** The consequence is **USD 231,360** (delay **171,360** plus response **60,000**). **EMV** without consultation **USD 80,976**; with consultation **USD 71,136**. The consultation is worth **USD 9,840**, it remains worth holding while it costs less than **USD 57,840**, and it must reduce the probability of complaint by at least **20.75 percentage points** (from 0.35 to **0.1425**) to break even.
  5. **Interpretation.** The point estimate of 9,840 is the weakest number here and should never be the one presented: it is the difference of two products of judged probabilities, so it is fragile in both directions and a reviewer can argue it away in a sentence. The **20.75-point required reduction** is the number to present, because it asks a question consultation practitioners can answer from experience (does a structured, pre-decision consultation cut the likelihood of a formal complaint by more than a fifth?), and because it exposes how much of the case rests on judgement rather than concealing it. The breakeven cost of **57,840** answers a different and often more useful question: how much *more* consultation could be justified on risk grounds alone. Two limits must then be stated plainly, and stating them is part of the professional standard rather than a caveat on it. First, **the probabilities are not measurements**. They are structured judgements, to be recorded with their basis and their author; a programme reporting 0.35 without saying who judged it and on what has manufactured a number. Second, and more important: **consultation is not only a risk control**. A community's entitlement to be heard about a change to its own services does not depend on whether hearing it reduces an expected cost, and a programme that would abandon consultation because the arithmetic came out at -9,840 has misunderstood what the arithmetic is for. The **EMV** supports the case for consultation where it is resisted; it is not a licence to withdraw it where it is not. Legitimacy is not quantifiable, and this book will not pretend otherwise.
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#### 11.4.2 Cross-cultural and multilingual communication

**What actually varies.** Language is the visible difference and rarely the expensive one. Four others cost more, and each is *observable* rather than stereotypical. The discipline is to observe how a specific organisation behaves rather than to predict it from nationality, which is both unreliable and offensive.

- **Directness.** Whether disagreement is stated or signalled. Where it is signalled, a project that hears only stated objections has a silent risk register, and the counter-move is to seek disagreement in private and in writing rather than to ask for it in a meeting.
- **Decision convention.** Whether a meeting decides or ratifies a decision reached beforehand, and whether authority is individual or collective. A leader who

brings a decision to a meeting in a consensus-forming organisation gets agreement in the room, no decision, and the discovery weeks later.

- **Attitude to hierarchy.** Whether bad news may travel directly to the accountable person or must pass through the chain, which determines the *actual* escalation latency, potentially several times the designed one of Domain 3, KA 3.3.3.
- **Time and commitment.** Whether a date is a commitment, an intention or an aspiration; and whether a written record is the agreement or a note about it.

**Working method.** Establish the convention explicitly and early ("a decision is recorded in the log and takes effect then; agreement in a meeting is not yet a decision"), substituting an agreed project convention for a clash of unexamined ones. Provide material in the language of the people who must act on it, not merely of those who sign; for Meridian that means clinical-workflow material in the language clinicians work in. Use **back-translation** for anything carrying a commitment, instruction or safety implication. Confirm understanding by asking the recipient to state the action, not to confirm receipt. And allow in the schedule for the elapsed time consensus formation actually takes: a plan assuming decisions are made in meetings, in an organisation where they are not, carries a systematic and invisible underestimate in every approval duration.

#### 11.4.3 Misinformation and AI-generated communication risk

**The exposure.** A programme of any public significance operates in an information environment where claims about it circulate faster than it can respond: some mistaken, some deliberate, and increasingly some machine-generated. Three risks matter to a delivery leader, and none is a communications-department problem.

**Misinformation about the project.** Incorrect claims about cost, purpose, data handling or safety gather support because they arrive first and are more interesting than the truth. Corrections propagate less well than the original claim, a real and well-observed asymmetry, and one this book will not attach a multiplier to, because no honest general figure exists. The countermeasures are structural rather than rhetorical: publish the authoritative facts *before* the questions arrive, so a correction is a reference rather than a rebuttal; maintain a single named source of truth that journalists and community groups can check; respond to the substance and never to the motivation; and monitor deliberately, because the first a programme usually hears of a claim is when a governance member forwards it.

**Impersonation and synthetic content.** A programme's name, branding, spokespeople and letterhead can be reproduced convincingly at negligible cost, including synthetic voice and video of real individuals. Two controls carry most of the weight: a published, stable statement of the channels through which the programme communicates officially, so a recipient can verify; and an internal rule that no commitment is ever created by an inbound communication alone, whatever it appears to be, a fraud control as much as a communications one.

**AI-drafted communication that nobody verified.** The risk most within the leader's control and most often mishandled, because the failure does not look like a failure. A model drafting a stakeholder bulletin produces fluent, plausible text containing dates, commitments and attributions it inferred rather than knew. A bulletin stating that a clinic will go live in March, when no such commitment exists, has created an expectation, possibly a representation, and certainly a dispute. And and it will be quoted back to the programme by the party it misled. **That is a governance defect, not a copy-editing one:** it is an unauthorised commitment, and it warrants the seriousness Domain 3 attaches to a decision taken without authority.

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#### WORKED EXAMPLE

##### Worked example 11.4.3 – the verification step, priced.

1. **Setup.** Meridian issues **96** external communications a year: 48 weekly bulletins, 24 clinic-specific notices, 12 authority updates and 12 public updates. AI-assisted drafting saves **1.2 hours** each. A mandatory verification step (every factual claim, date and commitment checked against a source by a named person) adds **0.5 hours** each. Blended rate **USD 110** per hour. Without verification, the programme assesses that **4 %** of items would carry a material misstatement, each costing **USD 6,000** in correction, clarification and relationship repair.
2. **Formula.** Net hours = items × (drafting saving – verification time). Verification cost = verification hours × rate. Unverified exposure = items × error rate × cost per error. Breakeven error rate = verification cost ÷ (items × cost per error).
3. **Substitution.** Net  $96 \times (1.2 - 0.5)$ . Verification  $96 \times 0.5 \times 110$ . Exposure  $96 \times 0.04 \times 6,000$ .
4. **Result.** Drafting saves **115.2 hours**; verification consumes **48.0**; the net saving is **67.2 hours**, worth **USD 7,392**. Verification costs **USD 5,280** and stands against an unverified exposure of **USD 23,040** across an expected **3.84** defective items, a return of **4.36 times**. The verification step pays at any error rate above **0.92 %**.
5. **Interpretation.** The result contradicts the argument usually made in both directions. AI-assisted drafting **does** save time (67.2 hours a year net, **1.4 hours a week** across the 48-week operating year, returned to the work 11.1.3 showed is worth 382.50 an hour), and the verification step **is not** what destroys the saving: it consumes **41.7 %** of it and returns 4.36 times its cost. The decisive figure is the **0.92 % breakeven error rate**, because nobody will claim that unverified machine-drafted communications about a health programme are accurate more than 99.08 % of the time at the level of individual dates and commitments, so the verification step is not a judgement call. Three qualifications. The 4 % error rate and the 6,000 per-error cost are assumptions and should be replaced with measured values as soon as the programme has any; the verification log produces the error rate as a by-product, which is the cheapest measurement available and almost never kept. The exposure prices correction and relationship

repair only: a misstatement about patient data would not be a 6,000 event, so the highest-consequence categories warrant a heavier check than 0.5 hours. And the saving is only real if drafting replaces work rather than generating *more* communication: 96 items a year is a designed number, and a programme issuing 150 because drafting became cheap has not saved 67 hours, it has spent 40 more and added 54 items of exposure.

AI in this KA

**Where it earns its place.** Drafting in multiple languages with **back-translation** as a check: a genuine capability improvement for programmes that previously could not afford multilingual material at all. Monitoring public and social channels at volume for claims about the programme, so the first a leader hears of a misstatement is not a governance member's forwarded message. Readability and plain-language testing, which is objective and tedious. Producing accessible formats (summaries, alternative text, simplified versions) from one approved factual base. And drafting the first version of the factual-claims register that 11.4.3's verification step needs.

**Where it must not go.** External publication without human verification, for the reasons priced above. Any **synthetic voice or likeness** of a real person, however convenient and however disclosed: a programme cannot credibly campaign against impersonation while practising it. Automated replies to community objections, which read exactly as they are and convert a substantive objection into a legitimacy grievance. Personalised persuasive targeting of identified individuals, a manipulation risk no delivery benefit justifies. And translation of safety-critical or legally operative text without qualified human review, where a plausible error is worse than an obvious one.

**Verification, concretely.** A **two-person rule** on every external communication: a drafter and a named verifier, recorded, the verifier accountable for every date, figure and commitment in the text. A **factual-claims register** listing each claim and its source document, retained with the issued item so that a challenge is answered from the record rather than from memory. Back-translation on anything carrying a commitment or an instruction. A published statement of where the programme uses AI in its communications, because asking a community to trust you while concealing how you speak is the wrong economy. And the verification log's error rate reported monthly, since it is the only measurement that says whether the controls work.

KEY TERMS — KA 11.4

| TERM                  | MEANING                                                                                                                                                |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Community stakeholder | An unbounded, heterogeneous group with a legitimate interest in a project's effects, often with statutory engagement rights that vary by jurisdiction. |

| TERM                             | MEANING                                                                                                                            |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pre-decision consultation</b> | Consultation capable of changing the design, engagement; as distinct from post-decision notification.                              |
| <b>Consent risk pricing</b>      | EMV of a consent failure, presented with its breakeven cost and required probability reduction, never as a point estimate alone.   |
| <b>Decision convention</b>       | Whether a meeting decides or ratifies, and whether authority is individual or collective.                                          |
| <b>Back-translation</b>          | Independent re-translation to the source language as a check on anything carrying a commitment, instruction or safety implication. |
| <b>Correction asymmetry</b>      | The observed tendency of corrections to propagate less well than the original claim; real, and not reduced here to a multiplier.   |
| <b>Synthetic content risk</b>    | Convincing reproduction of a programme's identity, spokespeople, voice or documents at negligible cost.                            |
| <b>Factual-claims register</b>   | Each claim in an external communication with its source document, retained with the issued item.                                   |
| <b>Two-person rule</b>           | A named drafter and a named verifier for every external communication.                                                             |

## SAMPLE MCQS — KA 11.4

**MCQ 11.4-A** [11.4.1 · Application] A consent failure would cost 231,360. Its probability is 0.35 without consultation and 0.10 with a consultation costing 48,000. The consultation is worth:

- A. USD 57,840
- B. USD 9,840 CORRECT
- C. USD 80,976
- D. USD 23,136

*Rationale:*  $0.35 \times 231,360 = 80,976$  against  $48,000 + 0.10 \times 231,360 = 71,136$ ; the difference is 9,840 (11.4.1). A is the breakeven consultation cost; C and D are the two EMV terms.

**MCQ 11.4-B** [11.4.1 · Evaluation] For the same case, the figure a leader should present to a sceptical board is:

- A. the expected saving of 9,840
- B. the required probability reduction of 20.75 percentage points CORRECT
- C. the consequence of 231,360
- D. the consultation cost of 48,000

*Rationale:* The point estimate is the difference of two judged probabilities and is fragile; the required reduction states exactly what must be believed and can be answered from experience (11.4.1).

**MCQ 11.4-C** [11.4.2 · Analysis] A programme translates all its material accurately but assumes that agreement reached in a meeting is a decision. In a consensus-forming counterpart organisation the predictable consequence is:

- A. the translation will be misunderstood
- B. agreement in the room, no decision, and the difference discovered weeks later
- CORRECT**
- C. escalation will be faster than designed
- D. the counterpart will refuse to meet

*Rationale:* Decision convention, not language, is the expensive variable (11.4.2); the plan's approval durations are systematically understated as a result.

**MCQ 11.4-D** [11.4.3 · Application] Ninety-six external communications a year, verification at 0.5 hours each and USD 110 an hour, against an assessed 4 % error rate at USD 6,000 per error. The return on the verification step is:

- A. 2.18 times
- B. 4.36 times **CORRECT**
- C. 0.23 times
- D. 3.12 times

*Rationale:* Exposure  $96 \times 0.04 \times 6,000 = 23,040$  against verification  $96 \times 0.5 \times 110 = 5,280$ , a ratio of 4.36 (11.4.3). A uses 1.0 hour per item rather than 0.5 ( $23,040 \div 10,560$ ); C inverts the ratio ( $5,280 \div 23,040$ ); D uses the 7,392 net hourly saving as if it were the verification cost.

**MCQ 11.4-E** [11.4.3 · Analysis] An AI-drafted bulletin states a go-live month to which the programme has not committed. This is best treated as:

- A. a copy-editing error, corrected in the next issue
- B. a governance defect — an unauthorised commitment made outside the decision system **CORRECT**
- C. a supplier performance issue
- D. an acceptable risk given the drafting time saved

*Rationale:* The defect is the creation of an expectation and possibly a representation without authority, which is Domain 3's category of a decision taken without authority (11.4.3).

#### SELF-CHECK — KA 11.4

1. *What distinguishes consultation from notification, and why does it change the cost?* — Consultation can change the design, so accommodating a concern costs what a change costs at that design maturity; after the decision, the only remaining influence is obstruction.

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2. *Name the cross-cultural variable that most distorts a schedule, and how.* — Decision convention: where meetings ratify rather than decide, every approval duration in the plan is understated.
  3. *Why is an unverified AI-drafted commitment a governance matter?* — It is an unauthorised commitment, created outside the decision system, and it will be quoted back to the programme by the party it misled.
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## Advanced topics — Domain 11

### 11.A.1 Influence without authority, and the arithmetic of a blocking minority

Project leaders routinely need agreement from people they cannot instruct, which is why influence is a professional competence rather than a personality trait. Four sources require no authority: **expertise** (being the person who knows, demonstrated rather than asserted); **information** (holding the integrated picture nobody else holds — a by-product of the leader's position, squandered by hoarding it); **reciprocity** (having accommodated others on issues that mattered to them and not to you — 11.3.3's under-used mode, and the reason accommodating is an investment rather than a weakness); and **coalition** (the support of parties the target respects). All four are built before they are needed, which is the practical difficulty: influence is a stock accumulated in quiet periods and spent in loud ones, and a leader who begins building it when the difficult decision arrives has begun too late.

Coalition work has an arithmetic worth doing, because voting rules make the target countable. In a nine-member body requiring a two-thirds majority, **6 of 9** votes carry a motion, so the smallest blocking coalition is  $9 - 6 + 1 = 4$ . Those two numbers answer different questions and are habitually confused: a leader seeking approval must secure **six** supporters (not five, and not "a majority"), while a leader assessing risk must ask whether any **four** parties share an interest in refusal. Domain 3, KA 3.1.2 showed that unanimity is indistinguishable from an inability to decide; this is the same result at finer resolution, and the consequence is that an influence plan for a supermajority body targets a *named list of six*, with the four most likely refusers engaged first, rather than a general campaign of goodwill.

### 11.A.2 The stakeholder register as a decaying asset

A register is treated as a document and behaves like a perishable one, because the entries are roles and the relationships are with people. Meridian's register holds **62** entries; over twelve months **19** changed role holder, an annual turnover of **30.6** %. The consequence is computable from the refresh cadence: with an annual refresh, entries are on average half a year old, so about **15.3** % of the register is wrong at any moment; with a quarterly refresh, about **3.8** %. Fifteen per cent of 62 means roughly nine relationships the programme believes it has and does not, and the nine are not



distributed evenly, because turnover is highest in exactly the junior and operational roles that carry the day-to-day cooperation.

Four practices follow. Refresh on a **cadence tied to observed turnover** rather than annually by habit; the arithmetic above is the justification and it costs an hour a quarter. Record, for each key relationship, **who holds it on the programme side**, because the programme's own turnover does the same damage in the other direction and a departing engagement lead takes the relationship with them unless it is deliberately transferred. Treat a change of role holder as a **re-engagement trigger**, not an administrative update: a new incumbent has agreed nothing, has not seen the specification, and is the single most common source of the 11.1.2 objection arriving in week 34.

And (the practice that turns a refresh into a discipline rather than an update) **delete the departing holder's assessment rather than carrying it forward**. The nineteen changed role holders leave behind nineteen influence scores, attitude judgements and relationship maps about individuals who no longer hold the role, and none of those has any further engagement purpose (11.1.2). Carrying them forward is not diligence; it accumulates recorded opinions about identified people, and it does it silently, at 30.6 % a year. The successor's entry starts from the position they have actually taken, which is also the only assessment that would have been any use. Deletion at refresh, plus the register's stated deletion point at closure, are what stop a live management instrument becoming a long-lived file of personal judgements, and what applies to any of it under data-protection obligations is for whoever holds that accountability in the organisation and for qualified counsel, not for this book.

### 11.A.3 The reviewer's stakeholder eye

Invariants to test on any engagement and communication design, each cheap and each diagnostic.

Every register entry names a **decision, consent, resource or action** the project needs from that party, and the **cost if it arrives late**. Every party is scored on **four** dimensions, not two, with **consent risk** scored from position rather than disposition. The engagement allocation is **explicit and sums to a stated capacity**, with a value-led core sized from the benefit arithmetic, consent-risk floors and a residual (not a single proportional rule), and every floor is justified by a stated **breakeven probability**. The communication architecture's **channel load is computed** against capacity, and the design names which relationships are deliberately bilateral and why. Report **P, C and L are stated and maximum blindness computed**, and any proposal to lengthen the cycle quotes the new figure. Status thresholds are **objective and set at baseline**, and the report records **the date the leader knew**. Every negotiation of material value has a **written reservation value derived from a priced alternative** (including delay at the cost of delay and risk as an **EMV**) plus an estimated counterparty range with its evidence, and multi-issue negotiations carry an



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**issue-by-issue value and cost table** so that value-destroying issues are identified before capital is spent on them. Every external communication has a **named verifier** and a factual-claims register. The register's **refresh cadence is derived from observed turnover**. And the test that subsumes several others: for each of the last three stakeholder surprises, ask **which dimension of the assessment would have predicted it**, a surprise no dimension would have predicted indicates a missing dimension, and one all four would have predicted indicates an allocation that was never made.

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## Industry variations — Domain 11

- **Public sector and government.** Consultation is often statutory in form, period and grounds, so the engagement plan is partly a compliance instrument and the discretion sits in *how early* rather than *whether*; political stakeholders change on an electoral cycle unrelated to the project, making 11.A.2's re-engagement trigger a scheduled event rather than a contingency.
  - **Healthcare.** Clinical authority is not delegable to a project body (Domain 3), so clinical stakeholders hold consent risk regardless of their interest score, which is why Meridian's design floors them; and the benefit is almost entirely adoption-determined, so 11.1.3's value core dominates.
  - **Construction and infrastructure.** Community and landowner objection rights convert directly into programme delay, so consent risk is the dominant dimension; the interface-to-people mapping of 11.2.1 is at its most literal, with parties numerous, physically co-located and contractually separated.
  - **Energy and resources.** Community consent is frequently the critical path rather than a risk to it (Case study B), long asset lives mean the engagement outlives the project team, and consultation held after a layout decision is the standard and expensive error.
  - **Technology and product organisations.** Users are numerous, unrepresented and reachable directly, which makes 11.1.3's value core computable from telemetry rather than judgement; the characteristic failure is over-weighting the loud internal stakeholder whose interest score is high and whose consent risk is nil.
  - **Financial services and regulated industries.** The regulator is the archetypal high-influence, low-interest, high-consent-risk party of 11.1.2, and engagement is bounded by rules about what may be discussed and when, so the floor is spent on formal, documented pre-submission engagement rather than relationship building, and the record of that engagement is itself a regulatory artefact.
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## Case study — Domain 11: the stakeholder who was only informed (health, Meridian)

**Situation.** At week 36 of Meridian's rollout, 31 of 40 clinics were installed and the programme was reported amber. Adoption among installed clinics was running at 62 %, below the 70 % the business case assumed. The national reporting body had objected in week 34 to the structure of the statutory data extract, and the objection was being handled as a technical issue by the integration team. The steering committee had been told none of this in those terms: the week-32 pack described adoption as "building" and the extract as "under discussion with the regulator".

**What the arithmetic showed.** The project leader computed four numbers before the next governance review. The **objection**: 4.0 weeks of governance latency plus 5 weeks of rework, three interface re-verifications at 18,000 and 42,000 of application rework — **USD 224,520**, against a technical pre-consultation at design stage costing **USD 3,300**, a breakeven probability of **1.47 %**. The **communication load**: 14 actively managed parties, unrouted, 91 channels at 1.5 hours a month (**136.5 hours** against a capacity of 40, or **341.25 %**), which explained without anyone needing to be blamed why the reporting body had received a bulletin and nothing else. The **report age**: **P 4, C 1, L 2**, a mean fact age of **5.0 weeks** and a maximum blindness of **7.0 weeks**, so the extract issue could have run seven weeks before the committee saw it, and had. And the **allocation**: the plan's 480 hours, distributed in proportion to influence × interest, gave the two adoption-determining groups **180 hours** — enough to run the 8-hour adoption intervention in 22 of 40 clinics, **USD 67,320** a year of the available **USD 122,400**, forgoing **USD 55,080** a year permanently.

**How it resolved.** Four changes, none requiring new money. The communication architecture became **hybrid**: all 14 parties routed through a named engagement function, with a deliberate three-party bilateral mesh for the sponsor, the clinical authority and the reporting body — 17 channels, **37.5 hours a month, 93.75 %** of capacity, stated as tight in the plan rather than presented as comfortable. Data cut **weekly** into a standing dashboard with the monthly pack drawing the latest cut, taking maximum blindness from **7.0 to 2.5 weeks** and mean fact age from 5.0 to 2.0: worth **USD 42,840** on the mean and **USD 64,260** in the worst case per qualifying occurrence. The 480 hours were **reallocated** on the three-term design: a 320-hour adoption core, 130 hours of consent-risk floors including 40 for the reporting body, and a 30-hour residual. And status thresholds were fixed at baseline, so "amber" acquired a definition and the week-32 pack's language became impossible to repeat.

**The outcome, and the part that mattered.** The extract objection took 9 weeks to clear as computed; nothing recovered that. Adoption among installed clinics reached 79 % by month 12: short of the 82.5 % the intervention was designed for, and the shortfall traced to four clinics where the workflow sessions were run by the supplier rather than by clinical peers, a finding the programme would not have had without an allocation explicit enough to audit. The durable change was elsewhere. "We need more stakeholder engagement" had sat in the risk register for two quarters and

produced nothing. "Our communication design requires 341 % of our engagement capacity, our committee decides on five-week-old facts, and our allocation forgoes 55,080 a year" was approved in one meeting.

**What the domain teaches here.** Engagement failures present as behaviour and are usually capacity and allocation failures with an arithmetic signature. The overloaded architecture is not a workload complaint; it is the *mechanism* that produced the week-34 objection, because the contacts an overloaded design drops are precisely the low-interest, high-consent-risk ones. And note which correction was largest for the least money: the 480 hours were already being spent.

## Case study B — Domain 11: the consultation that was held twice (energy)

**Situation.** A regional utility developed a 90 MW onshore wind project with a new grid connection across farmland. Its financial model priced the cost of delay at **USD 62,000 per week** in forgone contracted revenue net of variable cost. The development team ran a full public consultation costing **USD 180,000**; after the turbine layout and connection corridor had been fixed, on the reasonable view that consulting on an incomplete design would confuse people.

**What happened.** The consultation produced 340 responses, the substantive ones concentrating on two turbine positions and one section of the corridor. Because the layout was fixed, the only available responses were to refuse the objection or to change the design at detailed-design maturity. Two objections proceeded formally. The developer ran a **second consultation** at **USD 240,000**, conceded a layout change costing **USD 410,000** at detailed design, and lost **22 weeks** of planning determination — **USD 1,364,000** at the project's cost of delay. Total: **USD 2,194,000**.

**How it resolved, and what the alternative would have cost.** The post-project review modelled the counterfactual with the same objections arising. Consultation at **options** stage, on three candidate layouts, would have cost **USD 260,000**: 80,000 more, because two rounds and three options are more work than one round and one answer. The same two turbine positions and the same corridor section would have been changed while the layout was still a set of options, at **USD 55,000** rather than 410,000, a **7.45-fold** escalation avoided, and Domain 5's design-maturity argument in money. Determination would have run 4 weeks over plan rather than 22: **USD 248,000**. Total: **USD 563,000**. The early programme costs **USD 80,000** more in consultation and **USD 1,711,000** less in everything else, a net saving of **USD 1,631,000**, a total cost ratio of **3.90**, and **USD 21.39** avoided downstream for each additional dollar of early consultation.

**What the domain teaches here.** Consultation before a decision is engagement; after a decision it is notification, and it reliably produces objections because obstruction is the only influence left. The mechanism is not goodwill but **design maturity**: the identical concern cost 55,000 to accommodate at options stage and

410,000 at detailed design, and the 22 weeks of delay followed from having nothing to offer. Note what the developer's reasoning got right (consulting on an incomplete design *is* harder), and where it went wrong: it treated that difficulty as a reason to defer, when the arithmetic says the difficulty is the price of the option. And note the number without which none of this is arguable: without 62,000 a week, the 22 weeks are an inconvenience and the whole case dissolves into opinion.

## Executive perspective — Domain 11

What a programme director cannot delegate in this domain:

- **The engagement allocation, in hours, summing to your stated capacity.** Not a communication plan, an allocation, with a value-led core sized from the benefit arithmetic and a breakeven stated for every consent-risk floor. Meridian's misallocation forgave 55,080 a year of capacity it was spending anyway (11.1.3).
- **The channel load of your own communication design.** Compute it. An architecture demanding 341 % of capacity is not a workload problem, it is the mechanism that generates your next late objection, and it fails silently on the parties who do not chase (11.2.1).
- **Maximum blindness.**  $P + C + L$ . Know the number, quote it whenever anyone proposes a longer cycle, and be able to say how old the average fact in your last pack was when the committee decided (11.2.2).
- **The status definition, set at baseline, and the date you knew.** Discretion over an undefined threshold is what corrupted the signal; a definition and a knew-date restore it, and the knew-date is what makes your escalation lead time measurable (11.2.3, Domain 3).
- **Your reservation value on every material negotiation, in writing, before the meeting.** Derived from a fully priced alternative including delay at your cost of delay and risk as an EMV. An unpriced alternative cost Meridian 50,000 on one contract; and never let anyone measure your negotiators on surplus captured, which rewards them for keeping the alternative bad (11.3.1).
- **That no external communication leaves the programme unverified.** A named verifier and a factual-claims register on every item. The verification step returns 4.36 times its cost and pays at any error rate above 0.92 %, and an AI-drafted commitment nobody checked is an unauthorised commitment, not a typographical one (11.4.3).

## Calculation exercises — Domain 11

**Exercise 11.1** A programme actively manages 11 parties. Each maintained channel costs 2.0 hours a month; engagement capacity is 45 hours a month; a routed design carries a hub overhead of 10 hours a month. Compute the mesh and routed loads and their utilisation, the largest number of parties each design sustains, and the annual hour saving from routing. *Solution.* Mesh channels  $11 \times 10/2 = 55$ , load  $55 \times 2 = 110$  hours a month, or  $110/45 = 244.4\%$  of capacity. Routed  $11 \times 2$

•  $10 = 32$  hours,  $71.11\%$ . Largest sustainable unrouted count: 7 parties give 21 channels and 42 hours (within 45); 8 give 28 channels and 56 hours (over). So 7. Routed:  $2n + 10 \leq 45$  gives  $n \leq 17.5$ , so 17 (44 hours). Annual saving  $(110 - 32) \times 12 = 936$  hours. Common error: computing  $n^2(121)$  or  $n(n-1)$  (110 channels rather than 55). The second is doubly seductive here because it coincides numerically with the correct hour figure.

**Exercise 11.2** A programme has 600 engagement hours over 15 months and five stakeholder groups scored influence  $\times$  interest: A  $5 \times 4$ , B  $3 \times 5$ , C  $4 \times 2$ , D  $2 \times 4$ , E  $3 \times 3$ . Compute the salience-proportional allocation. Then rebuild it as a three-term design in which C (the statutory approver) takes a consent floor of 120 hours and D (the adopting user group across 50 sites) takes a value-led core of 260 hours, with the residual apportioned by salience among A, B and E. Each adopting site is worth USD 19,200 a year and the core is expected to raise adoption by 10 percentage points. Compute the annual gain, the benefit per engagement hour and the core's hours per site. *Solution.* Salience: A 20, B 15, C 8, D 8, E 9, total 60;  $600/60 = 10$  hours per point  $\rightarrow$  A 200, B 150, C 80, D 80, E 90. Three-term: C 120, D 260, residual  $600 - 120 - 260 = 220$  across salience  $20 + 15 + 9 = 44$  at 5 hours a point  $\rightarrow$  A 100, B 75, E 45; total 600. Gain  $50 \times 0.10 \times 19,200 = \text{USD } 96,000$  a year; per engagement hour  $96,000/260 = \text{USD } 369.23$ ; core intensity  $260/50 = 5.20$  hours per site. Common error: allocating by salience alone, which gives the statutory approver 80 hours against a required floor of 120 and the adopting group 80 against a required 260, underserving simultaneously the party that can stop the work and the party that determines whether any benefit exists.

**Exercise 11.3** A monthly report has a 4-week period, 1.5 weeks of consolidation and a 1-week paper lead time; cost of delay is 9,500 per week. Compute the newest fact age, mean fact age and maximum blindness, then the same three under a fortnightly cycle with 0.5 weeks of consolidation and a 0.5-week paper lead time, and price each saving. *Solution.* As found: newest  $C + L = 1.5 + 1 = 2.5$  weeks; mean  $C + L + P/2 = 2.5 + 2 = 4.5$  weeks; maximum blindness  $P + C + L = 6.5$  weeks. Redesigned: newest  $0.5 + 0.5 = 1.0$ ; mean 1.0

+ 1.0 = 2.0; blindness 2 + 0.5 + 0.5 = 3.0. Savings 1.5, 2.5 and 3.5 weeks, priced at USD 14,250, USD 23,750 and USD 33,250. Common error: reporting the newest fact age as the age of the picture, which understates the mean by  $P/2$ , 2.0 weeks here, or 19,000 at this cost of delay.

**Exercise 11.4** A buyer's fully priced alternative to renewing a contract costs USD 1,200,000. The incumbent's cost to serve is USD 820,000 and its best alternative use of the same team would earn USD 130,000 of contribution. Compute both reservation values, the ZOPA and its width, the midpoint settlement and each side's surplus there. Then compute the buyer's surplus and share of the zone if it anchors on the incumbent's opening ask and settles at USD 1,090,000, and the value transferred relative to the midpoint. *Solution.*  $RV(\text{seller}) = 820,000 + 130,000 = \text{USD } 950,000$ ;  $RV(\text{buyer}) = \text{USD } 1,200,000$ . ZOPA 950,000 – 1,200,000, width USD 250,000. Midpoint USD 1,075,000, surplus USD 125,000 each. At 1,090,000 the buyer's surplus is  $1,200,000 - 1,090,000 = \text{USD } 110,000$ , 44.0 % of the zone against the seller's 56.0 %, and USD 15,000 has transferred relative to the midpoint. Common error: treating the seller's cost to serve as its reservation value, which puts the ZOPA floor at 820,000 (130,000 too low), and leads a buyer to pursue prices the seller will never accept, or to conclude wrongly that a workable deal is unavailable.

**Exercise 11.5** An unresolved priority conflict between two workstreams has held a critical-path decision for 3 weeks at a cost of delay of 14,280 a week, and escalation preparation has consumed 4 people × 5 hours at the blended rate of 110. A facilitated resolution session would take 6 hours of a neutral facilitator at 150 an hour plus 14 hours of participant time at 110. Compute the cost already incurred, the cost of the facilitated route, the ratio between them, and the minimum probability of success that would have justified the facilitation. *Solution.* Incurred: delay  $3 \times 14,280 = \text{USD } 42,840$  plus preparation  $20 \times 110 = \text{USD } 2,200$ , total USD 45,040. Facilitation:  $6 \times 150 + 14 \times 110 = 900 + 1,540 = \text{USD } 2,440$ . Ratio 18.46. Breakeven success probability  $2,440/45,040 = 5.42$  %. Common error: comparing the facilitator's fee alone (900) with the delay and concluding the intervention is trivially cheap, participant time of 1,540 is the larger part of its cost and is the real reason facilitation is resisted; conversely, omitting the 2,200 already spent on escalation preparation understates what the conflict has cost by 5 % of the total.

## Practitioner's toolkit — Domain 11

*Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.*

### Toolkit 11.T.1 — Stakeholder register and engagement allocation sheet

One row per party: reference · party and named individual · **what the project needs from them** (decide / consent / resource / do) · **cost if their agreement arrives late**, priced at the cost of delay · influence (1–5) · interest (1–5) · attitude, recorded as the **basis of the entry: the observed act and its source** ("objected to the interface scope in the 14 March minute"), never as a characterisation of the person · **consent risk** (none / withholds cooperation / holds approval or veto, with the instrument) · relationships (who they follow, who follows them) · engagement objective as a **dated future state** · named owner on the programme side · method · cadence · **allocated hours** · test of whether it worked · date last verified.

**The record block, above the rows, because this sheet holds information about identified people** (11.1.2): named register owner · **access list**, who may read it, this not being a general-circulation document · the refresh cadence at which **departed role holders' assessments are deleted** rather than archived (11.A.2) · and the register's **deletion point**, stating what is deleted at closure, what is retained, for how long and on whose authority (Domain 3, KA 3.3.4; Domain 14, KA 14.4.4; Domain 16, KA 16.4.4). Data-protection obligations, including any entitlement of an individual to see what is held about them, are settled with whoever holds that accountability in the organisation and with qualified counsel before the register is circulated; nothing in this template states the position in any jurisdiction. Two monthly counts belong with the others below: attitude entries carrying no observed act and source, and entries for role holders who have left.

Beneath the rows, the allocation summary: stated capacity in hours; the **value-led core** with the benefit arithmetic that sizes it; each **consent-risk floor** with its breakeven probability; the **residual** and its apportionment; and a total that must equal capacity. Report monthly: parties with no named programme owner, floors underspent, objectives past their date, and entries unverified within the refresh cadence derived from observed turnover (11.A.2). An allocation that does not sum to a stated capacity is a wish list.

### Toolkit 11.T.2 — Communication architecture and report-age sheet

*Part one, the architecture.* Actively managed parties  $n$  · channels under a mesh  $n(n - 1)/2$  and under routing  $n$  · hours per channel per period, with the basis · hub overhead · computed load and utilisation against stated capacity · the **sustainable party count** under each design · and an explicit list of **deliberately bilateral relationships** with the reason each cannot be intermediated. *Part two, report age.* For each recurring report:  $P \cdot C \cdot L$  · computed newest fact age, mean fact age and **maximum blindness** · the cost of delay · and the priced saving from each candidate reduction. The purpose is that both quantities are **visible at design time**; a communication plan that cannot fill this in is a distribution list.



### Toolkit 11.T.3 — Negotiation preparation sheet

*Our position.* The alternative, itemised: invoice or transition cost · elapsed weeks × cost of delay · internal effort · risk as an **EMV**, with its basis · **total = our reservation value**. What would improve the alternative, at what cost, and the **breakeven spend**. The shift it produces in the reservation value. *Their position, estimated.* Cost to serve · forgone contribution · **estimated reservation value as a range**, with the evidence for each bound. *The zone.* ZOPA low and high, width, and the settlement at the equal-split benchmark. *The issues.* One row per non-price issue: value to us · cost to them (estimated) · **joint value** · trade or drop, with every negative-joint-value issue flagged as not to be pursued. *Conduct.* Concession sequence and what each concession buys · what we will disclose (relative priorities) and what we will not (reservation value) · who speaks · what requires escalation before agreement. Reviewed against outcome afterwards, because the estimate of the counterparty's reservation value is the input that most needs calibrating and is almost never checked against what happened.

### Exam preparation — Domain 11

**What is assessed.** The stakeholder system and the parties registers omit; four-dimensional assessment and what each dimension predicts; **pricing a late objection and its breakeven probability; the three-term engagement allocation under a capacity constraint** and what a salience-only rule forgoes; communication as an architecture, with **channel load and sustainable party count; report age: newest, mean and maximum blindness**; honest status, objective thresholds and the escalation-grade message; **reservation values from priced alternatives, the ZOPA, surplus and share; joint gain from trading whole issues against splitting the difference**; conflict diagnosis by source; consultation before versus after a decision, and **consent risk priced as an EMV with its required probability reduction**; cross-cultural decision conventions; and the verification of AI-drafted communication.

**The calculations to be able to do under time pressure.** Assessed cost of an objection from governance latency plus rework at the cost of delay, plus interface and rework components, and its breakeven probability. A salience-proportional allocation, then a three-term allocation to the same capacity; benefit per engagement hour; the breakeven adoption uplift. Mesh and routed channel counts and loads, and the sustainable party count under a stated capacity.  $C + L$ ,  $C + L + P/2$  and  $P + C + L$ , each saving priced. Reservation values from a fully priced alternative (including delay and **EMV**), ZOPA width, midpoint, surplus, share of zone and value transferred against a benchmark. Joint gain for a compromise and for a trade, and the price rise that shares it. **EMV** with and without a mitigation, the breakeven mitigation cost and the required probability reduction. Verification cost against exposure, and the breakeven error rate.



**The traps.** Computing  $n^2$  or  $n(n - 1)$  for channels (Exercise 11.1) · quoting the newest fact age as the age of the picture, understating the mean by  $P/2$  (Exercise 11.3) · omitting governance latency when pricing an objection (MCQ 11.1-B) · allocating capacity by influence  $\times$  interest alone, which under-serves the benefit-determining group and the quiet approver simultaneously (11.1.3, Exercise 11.2) · treating a seller's cost to serve as its reservation value and putting the ZOPA floor too low (Exercise 11.4) · pricing an alternative on its invoice rather than its consequences, which sets a reservation value too low and forfeits deals that should be accepted (11.3.1) · reading a fall in surplus as a worse outcome when the reservation value was deliberately moved (MCQ 11.3-C) · splitting the difference across issues including those with negative joint value (11.3.2) · presenting an **EMV** point estimate rather than the required probability reduction (11.4.1) · annualising a report-age saving that accrues only per qualifying occurrence (11.2.2) · and treating an unverified AI-drafted commitment as a copy-editing error (11.4.3).

**How the domain connects.** Domain 1 supplies the cost of delay every calculation here is priced at, and the adoption arithmetic that makes engagement the largest available lever. Domain 3 supplies the governance latency  $E[\text{wait}]$  that makes a late objection expensive and the paper lead time  $L$  that enters report age. Domain 4 supplies the interface arithmetic  $n(n - 1)/2$  applied here to people, and the 18,000 interface unit cost. Domain 5 supplies the design-maturity result behind Case study B. Domain 6 consumes engagement durations as schedule input, since a consensus-forming approval is a longer predecessor than the plan assumes (11.4.2). Domain 8 supplies **EMV**, used for the priced alternative and for consent risk. Domain 10 supplies the procurement and dispute machinery this domain's negotiation behaviour operates inside, and defers negotiation itself to here. Domain 12 takes the interpersonal and team dimension of conflict and influence; Domain 14 the systematic treatment of AI governance whose communication face appears in 11.4.3; Domain 15 the same arithmetic at portfolio scale; and Domain 16 the adoption measurement this domain's value core exists to move.

## Domain 11 summary

Stakeholder work has no natural unit of account, which is why it is cut first and why it is the largest unpriced item in most delivery plans. This domain gives it units. Meridian's national reporting body (influence 5, interest 2, consent risk high, and sent a bulletin) objected in week 34 at an assessed cost of **USD 224,520**: 9.0 weeks of critical path (4.0 of them Domain 3's governance latency) at 14,280 a week, plus three interface re-verifications and application rework. The pre-consultation that would have surfaced it cost **USD 3,300**, so engagement paid at any probability of objection above **1.47 %**. And and that breakeven, not the 68-fold ratio, is what wins the argument, because a sponsor can dismiss a ratio and cannot assert one and a half per cent. The register that carries all this is not a spreadsheet: it names individuals and

records opinions about them, so it has an owner and an access list, its attitude entries are observed acts with their sources rather than characterisations of people, departed role holders' assessments are deleted at each refresh rather than archived (at **30.6 %** annual turnover that is nineteen deletions a year), and it carries a stated deletion point at closure. What any of that attracts under data-protection obligations is for whoever holds that accountability in the organisation and for qualified counsel, never for this book.

Engagement is an allocation of a stated capacity. Meridian's 480 hours, distributed in proportion to influence × interest, gave the two adoption-determining groups 180 hours and forgave **USD 55,080 a year, permanently**, of capacity that was being spent anyway; the three-term design (a **320-hour** value-led core, **130 hours** of consent-risk floors, a **30-hour** residual) delivers **5** more adopting clinics and **USD 122,400** a year at **USD 382.50 per engagement hour**, on a breakeven uplift of **3.59 percentage points**. It also raises the cost of delay to **USD 16,830** a week, which makes every schedule lever more valuable and settles the sequencing question: engagement before acceleration.

Communication is an architecture with a computable load. Fourteen actively managed parties admit **91** channels unrouted (**136.5 hours a month** against a capacity of 40, **341.25 %**), while routing gives 14 channels and 33.0 hours; an unrouted design sustains 7 parties and a routed one **18**, and Meridian's honest hybrid runs 17 channels at **37.5 hours, 93.75 %** of capacity. Reporting has an age: **C + L** for the newest fact, **C + L + P/2** for the mean, **P + C + L** for **maximum blindness** — 3.0, 5.0 and **7.0 weeks** as found, 1.5, 2.0 and **2.5** after weekly cuts, saving up to **USD 64,260** per qualifying occurrence. Status is corrupted by discretion over undefined thresholds and repaired by thresholds set at baseline plus a recorded date the leader knew.

Negotiation is preparation. A reservation value derived from a fully priced alternative (640,000 plus 6 weeks at 14,280 plus 74,320 of effort plus a 45,000 **EMV**), put Meridian's at **USD 845,000** against the supplier's **USD 615,000**, a **USD 230,000** zone whose midpoint is **730,000**; failing to price the alternative and settling at 780,000 would have transferred **USD 50,000** for nothing. Improving the alternative for 25,000 cut the settlement to **USD 684,940** and the risk carried by 33,000 while *reducing* surplus to 69,940; the trap that reveals which measures govern. And most of the value was never in the price: trading two asymmetrically valued issues produced a joint gain of **USD 82,000** against the **USD 29,000** of a split-the-difference compromise, with issue B destroying **24,000** and therefore never worth pursuing.

Public stakeholders cannot be managed, only engaged, and consultation before a decision is engagement while consultation after it is notification: Case study B's project paid **USD 2,194,000** for the second where **USD 563,000** would have bought the first, a ratio of **3.90**, because the identical concern cost 55,000 at options stage and **410,000** at detailed design. Consent risk prices as an **EMV** (80,976 against 71,136, worth 9,840, breaking even at a **20.75-point** reduction in probability) with the plain

statement that legitimacy is not quantifiable and consultation is not only a risk control. And the newest failure mode is the cheapest to close: verifying AI-drafted external communication costs **USD 5,280** against an exposure of **USD 23,040**, returns **4.36 times**, pays at any error rate above **0.92 %**, and still leaves **67.2 hours** of net saving; while an unverified machine-written commitment is not a typographical error but a decision taken without authority.

The through-line: **the objection you have not paid for in engagement, you will pay for in delay, at the cost of delay**; and every quantity in that sentence is computable at design time, from numbers a programme already has.

DOMAIN

# 12

## Leadership, Teams and Organisational Behaviour

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## Why this domain exists

Domains 1 to 11 built a delivery system: an accountable leader, an honest business case, decision rights with a computable latency, an integrated architecture, a scope, a schedule, a cost model, a risk register, a quality regime, a supply chain and a stakeholder strategy. Every one is executed by people who can leave, disengage, misunderstand each other, fear speaking up, or run out of the one resource nobody baselines, attention. **The system is only as good as the human capacity available to run it**, and that capacity is not a constant the leader inherits but a variable the leader designs.

This is the most abused domain in the discipline, and the abuse takes a specific form: leadership content asserting effects it cannot demonstrate. Confident claims about what motivates people, what a personality instrument predicts, what a culture programme returns circulate widely and are mostly unfalsifiable as stated. So this domain carries two obligations it does not carry elsewhere. The first is **honesty about the evidential status of every claim**: some of what follows is arithmetic, which is checkable; some is widely used vocabulary, useful without being predictive; some is professional judgement, labelled as such. The second is to find the parts of the domain that genuinely *are* quantitative, because they exist and are neglected: team size, turnover, span of control, delegation and time-zone overlap all have arithmetic, and each turns a conversation about commitment into a conversation about design, which is the only kind that gets a decision.

Hence the central claim: **the human capacity of a delivery organisation is a designed quantity with a computable cost, and most of what is described as a leadership or culture problem is a structural problem that arithmetic can locate**. A leader who can compute the coordination overhead of a team, the coaching capacity of a span, the cost of a departure and the latency of a distributed handoff can change those things; a leader who cannot will be asked to exhort people instead, and exhortation moves none of them. The domain proceeds outward from the individual: KA 12.1 handles leadership itself and the leader's attention; KA 12.2 the group, its size arithmetic, its safety and its turnover; KA 12.3 what a leader does with individuals; KA 12.4 distance, culture, ethics and leading people whose work is increasingly mediated by AI.

**Learning objectives.** After this domain a candidate can: distinguish leadership from management by what each supplies; state what each leadership theory family is useful for and what it does not establish, and select behaviour from situation rather than preference; apply emotional intelligence as four practised operations rather than a trait claim; **audit a leader's week and compute the discretionary attention actually available**; explain the motivation frameworks with their evidential status stated, and design conditions rather than incentives alone; **compute communication paths from  $n(n - 1)/2$ , convert them into coordination**

overhead as a share of team capacity, identify the size at which a further member adds no net capacity, and price the capacity a structured design releases; define psychological safety and specify observable signals without overclaiming its effects; compute the full cost of one departure as recruitment plus ramp-up plus lost productivity, and state which components are real costs and which are transfers; compute coaching time per report against span of control and derive the maximum sustainable span from a stated sufficiency convention; make delegation an economic decision with a breakeven, citing Domain 3's threshold arithmetic rather than repeating it; structure and time a difficult conversation and price its deferral; compute working-hour overlap from a common time reference, price the round-trip latency of a zero-overlap pair, and design a shared and rotated overlap window; lead multicultural teams without substituting stereotype for enquiry; hold the ethical obligations that cannot be delegated and say what to do when the decision goes the other way: the dissent record, the signature declined in writing, and the escalation route beyond the accountable person where harm is in question; specify the custodian, access, copy, retention and disposal of a conversation record without stating any jurisdiction's requirements; and govern AI's use in people decisions, where the constraints are tighter than anywhere else in the book.

**The master threads.** Meridian Care Records supplies the programme case: the rollout to 40 clinics, benefits USD 685,440 a year at 70 % adoption, cost of delay USD 14,280 per week (Domain 1), and a steering committee whose latency  $E[\text{wait}] = 4/2 + 2$  is 4.0 weeks (Domain 3, KA 3.2.3). This domain follows that team as it grows from 12 people to 40 and asks what the growth cost. Project Auriga supplies the project case: the 25-week control-systems upgrade, BAC USD 4,000,000, cost of delay USD 45,000 per week, a blended engineering rate of USD 130.625 per hour — an engineer-week of USD 5,225 (Domain 7, KA 7.4.1) — and the six-engineer commissioning team of Domain 9. This domain audits that leader's week, prices a resignation from that team, and prices the decision to delegate.

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## KNOWLEDGE AREA 12.1

# Leadership theories in practice and emotional intelligence

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*Topics: 12.1.1 what leadership supplies that management does not · 12.1.2 the theory families and what each is honestly good for · 12.1.3 emotional intelligence as an operational discipline · 12.1.4 the leader's attention as the binding constraint.*

### 12.1.1 What leadership supplies that management does not

**Definition.** In a delivery context, **management** is the allocation and control of resources against a plan; **leadership** is the production of *willing, aligned and informed effort* under conditions the plan did not anticipate. These are different outputs, not a hierarchy of seniority, and a project needs both from the same person, often in the same hour.

The test is what fails when each is absent. Without management, work is enthusiastic and uncoordinated. Without leadership, work is coordinated and inert: people do what the plan says after the plan has stopped being true, escalate nothing, and wait. The second failure is far harder to detect, because **a project failing for want of leadership produces perfectly compliant status information until the moment it produces a disaster.**

Leadership supplies four things a plan cannot. **A reason the work matters that survives difficulty:** not a slogan but a specific account connected to Domain 1's benefits chain (KA 1.3.2); Meridian's is *twenty-eight clinics of clinicians spending six hours a week on patients instead of paperwork*, a fact with a number attached. **Interpretation under ambiguity:** when plan and situation diverge, someone must say what the team should do before governance can rule, and that act is done in public and is the main way a team learns what the leader values. **The conditions in which people tell the truth,** on which every control in this book depends, and which a team defeats once it learns that bad news is punished (KA 12.2.3). And **named, bounded discretion:** Domain 3's bounded envelope (KA 3.1.3) at individual scale, priced in KA 12.3.

Leadership does not substitute for authority (a leader without decision rights is eventually defeated by the delegation schedule), for capacity (no inspiration converts 40 hours into 60 sustainably), or for competence in the work: a leader who cannot follow the technical argument cannot tell a confident answer from a correct one, which is exactly the judgement Domain 9's assurance depends on.

### 12.1.2 The theory families and what each is honestly good for

Leadership theory is a century of overlapping vocabularies, and the professional skill is knowing what each family is *for* rather than which is true. **No claim is made here about effect sizes, and none should be made from this material.**

**Trait approaches** ask what leaders are like. Their contribution is vocabulary for individual differences; their limitation is that describing a difference is not explaining an outcome. The practical caution: personality instruments are widely used for team composition and selection, and their predictive claims are much weaker than their marketing, use them, if at all, as a **conversation aid inside a team**, never as a selection filter or a performance judgement. **Behavioural approaches** ask what leaders do, usually along attention to the task against attention to the relationship; this is useful because behaviour is observable, coachable and changeable, and it

underpins the instruction most delivery leaders need. Your team reads what you *spend time on*.

**Situational and contingency approaches** vary direction against support according to the competence and confidence of the person on the task. As predictive theory the evidence is mixed; as a **discipline for choosing behaviour deliberately** it is the most useful family here, because it forces the question a leader otherwise skips: *what does this person, on this task, this week, need from me?* A competent engineer on familiar work needs a boundary and to be left alone; the same engineer on unfamiliar work needs direction they will not ask for. Getting that backwards is the commonest leadership error in technical delivery, and it is a failure to diagnose, not a character flaw.

**Transformational and transactional approaches** distinguish leadership that changes what people want from leadership that trades reward for performance; both are needed, and transformational language is easily counterfeited: a team can tell within a fortnight whether the vision matches the decisions. **Servant and steward framings** name obligations that otherwise go unstated: removing impediments, absorbing external pressure rather than transmitting it, defending the team's ability to work honestly; their failure mode is being read as licence to avoid difficult decisions. **Distributed leadership** observes that leadership acts are performed by many people, not only the title-holder. The family that matters most on a programme like Meridian, and one that connects to Domain 3: distributed leadership without distributed decision rights is a slogan, and distributed rights without the information to use them are abdication (KA 3.2.3).

These are lenses, not a menu. A defensible leader takes a few operational habits from the list: diagnose before behaving; watch what you spend time on because your team does; state boundaries explicitly; make the reason for the work truthful and specific; and treat obligations to the team as obligations rather than preferences.

### 12.1.3 Emotional intelligence as an operational discipline

**Definition.** Emotional intelligence, as used here, is the practised ability to notice emotional information, one's own and others', and act on it deliberately rather than reactively. It is treated as **four operations that can be practised**, not a trait or a score. The popular literature makes strong claims about its predictive power; those claims are contested, and nothing here depends on them, because each operation is defensible as professional craft.

**Self-awareness** is knowing which situations reliably degrade your judgement. The delivery-specific ones recur: being late, being contradicted in front of your sponsor, and receiving bad news you suspect was withheld. A leader who knows the third makes them punitive can suppress the first reaction for an hour, and that hour is often the difference between a team that escalates next time and one that does not. **Self-regulation** is having a rule for the predictable bad moments, decided in advance; the most valuable in delivery is *first reaction to bad news is questions only*,



*no judgements*, because it directly protects the information flow every control consumes. **Social awareness** is noticing who has stopped contributing, whose agreement was compliance, and which silence is consent rather than resignation, then **testing the reading rather than trusting it**, since confident misreading of others is common and is made worse by believing oneself emotionally intelligent. **Relationship management** is acting on the reading: the conversation held early (KA 12.3.3), the disagreement surfaced rather than absorbed, the repair after a bad exchange.

The concept is misused three ways: as a **label** applied to people, which converts a practicable skill into a fixed attribute; as a **substitute for candour**, where avoidance is called sensitivity; and as a **selection instrument** with claimed predictive validity: the same category error as the trait instruments above.

#### 12.1.4 The leader's attention as the binding constraint

Every behaviour above consumes the same resource, and it is the only resource on a project never baselined, levelled or reported. A schedule assuming a leader will coach six people, hold two difficult conversations, prepare a steering paper, manage a supplier and absorb a client escalation in one week has made a resource assumption as material as any in Domain 6, and, unlike Domain 6's, nobody has checked whether it is feasible. The check takes twenty minutes.

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#### WORKED EXAMPLE

##### Worked example 12.1.4 – auditing the Auriga project leader's week.

1. **Setup.** Auriga's project leader works to a **45.0-hour** planning week (the project's stated assumption; it is not a claim about what is healthy or contractual, and where norms differ the number and every result below change). Four weeks of diary and interruption logging give the recurring commitments: governance and reporting **6.0 h**; client and regulator liaison **5.5 h**; supplier and commercial **4.0 h**; technical review and approvals **7.5 h**; team ceremonies **5.0 h**; unplanned interruption and escalation **9.0 h** (observed, not planned). The leader intends to reserve **1.5 h** for personal planning.
2. **Formula.** Discretionary attention = working week –  $\Sigma$  recurring commitments. People-work capacity = discretionary attention – personal reserve. Shares are taken against the working week.
3. **Substitution.**  $6.0 + 5.5 + 4.0 + 7.5 + 5.0 + 9.0$ ; then  $45.0 - 37.0$ ; then  $8.0 - 1.5$ .
4. **Result.** Recurring commitments **37.0 hours**, **82.2 %** of the week. Discretionary attention **8.0 hours**, **17.8 %**. Capacity for people work (coaching, difficult conversations, one-to-ones, recruitment, team design) **6.5 hours**, or **14.4 %** of the week.

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5. **Interpretation.** Six and a half hours is the entire leadership budget of this project, and every commitment in KA 12.3 is paid from it. Three consequences follow, and the third changes behaviour. **The largest single line is not a decision anyone made:** 9.0 hours of interruption is **20.0 %** of the week, more than governance and technical review combined, and much of it is the absence of Domain 3's bounded envelope (KA 3.1.3), so decisions arrive that need not have. Recovering a third of it returns 3.0 hours and lifts people-work capacity from 6.5 to 9.5, a **46.2 %** increase in the leadership budget from a change that removes nothing. **The 7.5-hour technical-review line is a trade-off, not waste:** it is what lets the leader tell a confident answer from a correct one, and cutting it to buy coaching time trades one competence for another. And **6.5 hours cannot be spread thinly across an arbitrary number of people:** the arithmetic KA 12.3.2 completes, and the reason span of control is a leadership question rather than an organisational-chart question. The professional caution: this audit is only worth doing on **observed** data. A leader who fills it in from intention will produce 15 discretionary hours and will not know why the coaching never happens.

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### AI in this KA

**Where it earns its place.** Classifying a quarter of a leader's meetings into the commitment categories above and computing the shares: high-volume, low-judgment work nobody does by hand. Rehearsing a difficult conversation by generating the likely responses and the leader's weakest answers, since the failure mode of these conversations is being unprepared for the second sentence. Drafting from the project record the factual chronology a performance conversation must rest on. Summarising team-survey free text into themes with the source comments retained, so the leader reads the actual words.

**Where it must not go.** Three prohibitions, firmer here than anywhere else in this book because the subject is people. **No inference about an individual's emotional state, engagement, personality or likelihood of leaving** from behavioural telemetry (keystrokes, message sentiment, camera analysis, attendance: such inference is unreliable, restricted or prohibited in many jurisdictions) a question for qualified advice before any such tool is even considered, and corrosive of the trust every control in this domain depends on, and a leader who deploys it loses the truthful upward information of 12.1.1 permanently and will not be told why. **No AI-generated assessment of a person** entering a performance, promotion or termination decision. And **no AI-authored message purporting to be the leader's own voice** in a sensitive conversation: the counterfeit is detectable, and the damage when detected exceeds any time saved.

**Verification, concretely.** Any AI classification of the leader's diary is checked against the raw entries for one week in four, with categories defined by the leader, not the model. Any factual chronology is verified line by line against the source

record, because a single wrong date destroys the conversation. Where AI is used for rehearsal, no personal data about the other party is entered. The rehearsal works on the situation, not the person.

## KEY TERMS — KA 12.1

| TERM                                 | MEANING                                                                                                                           |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>Leadership (delivery context)</b> | The production of willing, aligned and informed effort under conditions the plan did not anticipate.                              |
| <b>Management (delivery context)</b> | The allocation and control of resources against a plan.                                                                           |
| <b>Situational diagnosis</b>         | Deciding what direction and support this person needs on this task before choosing a behaviour.                                   |
| <b>Emotional intelligence</b>        | Four practised operations (self-awareness, self-regulation, social awareness, relationship management) skills, not a trait score. |
| <b>Discretionary attention</b>       | The leader's working week less recurring commitments; the resource all leadership acts are paid from.                             |
| <b>People-work capacity</b>          | Discretionary attention less the personal planning reserve: 6.5 h/week on Auriga.                                                 |
| <b>Interruption load</b>             | Observed unplanned escalation and interruption time; the largest and most designable line in the attention audit.                 |

## SAMPLE MCQS — KA 12.1

**MCQ 12.1-A** [12.1.4 · Application] A leader's 45.0-hour planning week contains 37.0 hours of recurring commitments. The share of the week available as discretionary attention, before any personal planning reserve, is:

- A. 14.4 %
- B. 17.8 % CORRECT
- C. 20.0 %
- D. 82.2 %

*Rationale:*  $(45.0 - 37.0) / 45.0 = 17.8\%$  (12.1.4). A is the people-work share after the 1.5-hour reserve, the answer to a different question; C is the interruption load's share of the week; D is the committed share.

**MCQ 12.1-B** [12.1.1 · Analysis] A project reports faithfully on plan for four months, then discloses a six-month overrun in one meeting. The failure this most strongly suggests is:

- A. inadequate management, because variance control was absent

- B. inadequate leadership, because the conditions for truthful upward information were absent CORRECT
- C. inadequate governance, because the committee met too rarely
- D. inadequate scope definition

*Rationale:* Compliant reporting is evidence that management existed; what was missing was the condition under which people report what the reports do not ask for (12.1.1, and Domain 3's escalation-lead-time indicator at 3.3.3).

**MCQ 12.1-C** [12.1.2 · Evaluation] A delivery organisation proposes using a personality instrument to select project leaders. The soundest professional position is that the instrument:

- A. should be used, because it is widely adopted
- B. may be useful as a conversation aid inside a team, but its predictive claims are too weak to support a selection decision CORRECT
- C. should be used only for senior appointments
- D. is invalid and has no legitimate use

*Rationale:* Describing an individual difference is not explaining an outcome, so the legitimate use is internal dialogue, not selection (12.1.2). D overstates in the other direction, which is also a failure of judgement.

**MCQ 12.1-D** [12.1.2 · Application] A competent engineer is put on genuinely unfamiliar work and does not ask for help. The situational diagnosis indicates that the leader should:

- A. leave them alone, since they are competent
- B. increase direction on this task while keeping the relationship supportive CORRECT
- C. reduce their scope permanently
- D. treat the silence as confidence

*Rationale:* The diagnosis is per person **per task** (12.1.2); competence on familiar work does not transfer, and the absence of a request for help is not evidence of not needing it.

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SELF-CHECK — KA 12.1

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1. *What does leadership supply that a plan cannot?* — A truthful reason the work matters, interpretation under ambiguity, the conditions for truthful upward information, and named bounded discretion.
2. *Why is the Auriga leader's interruption load the most important line in the attention audit?* — It is the largest single line at 9.0 hours (20.0 % of the week), nobody decided it, and much of it is the absence of a bounded envelope, so it is the most designable.

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3. *State the honest limit of trait and personality approaches.* — Describing an individual difference is not explaining an outcome; their predictive claims are too weak for selection or performance decisions.
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## KNOWLEDGE AREA 12.2

## Motivation, team formation and psychological safety

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*Topics: 12.2.1 motivation without the folklore · 12.2.2 team formation and the arithmetic of team size · 12.2.3 psychological safety as a design variable · 12.2.4 retention and the cost of turnover.*

### 12.2.1 Motivation without the folklore

**Definition.** Motivation, for a delivery leader's purposes, is the direction, intensity and persistence of effort a person applies to work. It is not a quantity a leader installs; it is a response to conditions, some of which the leader controls. The frameworks in general use are worth knowing with their evidential status attached, because they are quoted constantly and interrogated rarely.

**Need hierarchies** (the family associated with Maslow) propose that needs are ordered and that higher needs become salient once lower ones are met. The ordering has not held up as a predictive model; the framework is best used as **vocabulary** for noticing that a person whose employment feels insecure will not be reached by an appeal to professional growth. **Two-factor accounts** (associated with Herzberg) separate factors whose absence causes dissatisfaction (pay, conditions, supervision, security) from those whose presence produces satisfaction — achievement, recognition, responsibility, growth. The empirical separation is contested; the **operational insight is sound**: fixing a dissatisfier removes a complaint, it does not produce engagement. A team with a broken build pipeline and no career conversations will not be engaged by fixing the pipeline; it will stop complaining about the pipeline. This is the most frequently misapplied idea in the domain, because organisations reliably spend on dissatisfiers and count the spending as motivation work.

**Self-determination framings** make intrinsically motivated effort depend on autonomy, competence and relatedness. This family has the strongest current standing and is the most actionable in project work, because all three levers are the leader's: autonomy is Domain 3's bounded envelope at individual scale, competence is whether people get work they can succeed at with the support to do so, relatedness is whether they belong to a team rather than a resource pool. **Expectancy and goal accounts** make effort follow from believing that effort leads to performance, that performance leads to an outcome, and that the outcome is worth having; and hold that specific, difficult, accepted goals draw more effort than vague ones. This family explains most demotivation in troubled projects without reference to

character: a team on a plan it privately believes impossible has a broken **effort** → **performance** link, and no incentive attached to the outcome can repair it. That is why an honest re-baseline is often the most motivating act available to a leader of a late project, and why "stretch" targets nobody believes demotivate.

For practice (professional judgement, not measured effect), design the **conditions** and be sceptical of designing the **incentives**: a credible plan, work people can succeed at, an explicit boundary of their own judgement, visible connection between their work and the outcome, specific and accepted goals, and recognition specific enough to be believed. Financial incentives have a real but narrow role and two known hazards: they attach effort to the measure rather than the objective, which is Domain 9's measurement problem in another form, and they are hard to withdraw without a cost exceeding their original benefit.

### 12.2.2 Team formation and the arithmetic of team size

**Formation as a described sequence.** The forming–storming–norming–performing vocabulary describes how a new team settles: politeness and dependence on the leader; conflict as real differences about approach and status surface; the settling of working norms; productive work. Two qualifications. It is a **description, not a mechanism**, and teams do not traverse it once, a change of membership, sponsor or circumstance returns a team to conflict. And the leader's task differs by stage: in forming supply structure and purpose; in storming make the conflict about the work rather than the people and *do not suppress it*, because a team that never storms has usually concluded that disagreement is unsafe; in norming make the emerging norms explicit so they can be chosen; in performing protect the team from disruption, which mainly means from your own organisation.

**The part that is arithmetic.** Domain 4 established that components grow linearly while interfaces grow combinatorially, and registered the count (KA 4.2.3):

Communication paths =  $n(n - 1)/2$  — every person to every other

That is the same formula Domain 4 used to count 66 technical interfaces across 12 components; it counts 66 communication paths across a 12-person team, and the two are one problem seen from two sides. What turns the count into a management number is a **cost per path**. Let

$c$  = hours of total team time consumed per active pairwise relationship per week  
 $h$  = productive hours per person per week  
 $n$  = team size

Then coordination overhead is  $c \cdot n(n - 1)/2$  hours a week, and as a share of gross capacity  $h \cdot n$ :

$$\text{Coordination share} = c(n - 1) / (2h)$$

With  $c = 0.5$  hours and  $h = 40$  hours the share collapses to the identity  $(n - 1)/160$ : linear in team size and independent of everything else. **The parameter  $c$  is a locally calibrated planning figure, not a constant**: 0.5 hours of total team time per relationship per week is roughly 15 minutes each from two people, plausible for colleagues who genuinely need to stay aligned, and the figure this book uses illustratively. Calibrate it from your own data if you have any; the *shape* of the result, not the parameter, is the lesson.

#### WORKED EXAMPLE

##### Worked example 12.2.2 — what Meridian's team growth cost, and what structure recovers.

1. **Setup.** Meridian's programme team grew from **12** people to **40** as the rollout scaled from pilot to national, with no change in how it worked: one team in which anyone might coordinate with anyone. Take  $c = 0.5$  hours per pairwise relationship per week and  $h = 40$  productive hours. The alternative is **five pods of eight**, each owning a cluster of clinics end to end with one representative per pod in a weekly forum: the human form of Domain 4's layered architecture (KA 4.2.3). Compare all three states.
2. **Formula.** Paths  $n(n - 1)/2$ ; overhead  $c \times \text{paths}$ ; share  $= c(n - 1)/(2h)$ . For pods, paths  $= (n/s) \times s(s - 1)/2 + (n/s)(n/s - 1)/2$  with pod size  $s$ , and share  $= c \times \text{paths} / (h \cdot n)$ .
3. **Substitution.** At 12:  $12 \times 11/2 = 66$ ; overhead  $0.5 \times 66 = 33.0$  h; share  $33.0/(40 \times 12)$ . At 40:  $40 \times 39/2 = 780$ ; overhead  $390.0$  h; share  $390.0/1,600$ . Pods:  $5 \times (8 \times 7/2) + (5 \times 4/2) = 140 + 10 = 150$ ; overhead  $75.0$  h; share  $75.0/1,600$ .
4. **Result.** At 12 people: **66** paths, **33.0** hours a week, **6.875 %** of capacity, **0.825** of a full-time person. At 40: **780** paths, **390.0** hours, **24.375 %**, **9.75** full-time people. In five pods of eight: **150** paths, **75.0** hours, **4.6875 %**, **1.875** full-time people. The pod design releases **630** paths, **315.0** hours a week, or **7.875 full-time-equivalent people**, cutting coordination by **19.6875 percentage points** of capacity.
5. **Interpretation.** The programme tripled its headcount and, fully connected, spent nearly a quarter of the result on talking about the work. **It was invisible in every report**: no line in Meridian's cost model says "coordination", the hours were charged to the work packages people were coordinating about, and the effect presented as unexplained productivity decline, exactly how it presents in practice. **The leverage is structural, not personal**: the pod design reduces nobody's workload and adds nobody; it removes 630 relationships that were not needed by giving each pod an end-to-end scope and one named path outward, and 7.875 FTE is more capacity than most improvement programmes deliver in a

year. **Structure barely degrades with scale while a mesh degrades quadratically:** the pod share moves only from 4.375 % at 8 people to 5.156 % at 88, while the mesh share passes **50.0 %** at **81**. That is also the exact size at which net capacity  $hn - c \cdot n(n - 1)/2$  stops rising — **1,620 hours (40.5 FTE)** at both 80 and 81 people — so **the 81st member of a fully connected team adds nothing**, and members beyond it subtract. That is the arithmetic behind the familiar observation that adding people to a late project can make it later, with the qualification that this is a *model of a fully connected team*: the honest reading is not "teams cannot exceed 80" but "**a team not given structure has a hard capacity ceiling, and it is lower than anyone expects**". The sensitivity decides how seriously to take any of it. At  $c = 0.25$  the 40-person mesh costs **12.1875 %** and the 50 % point moves to 161 people; at  $c = 0.8$ , **39.0 %** and 51; at  $c = 1.0$ , **48.75 %** and 41. So the *conclusion* (structure your team) is robust across the plausible range while the *specific ceiling* is not, and a leader who quotes 81 as a fact has misused the model. Quote your own team's share at your own calibrated  $c$ , as an estimate.

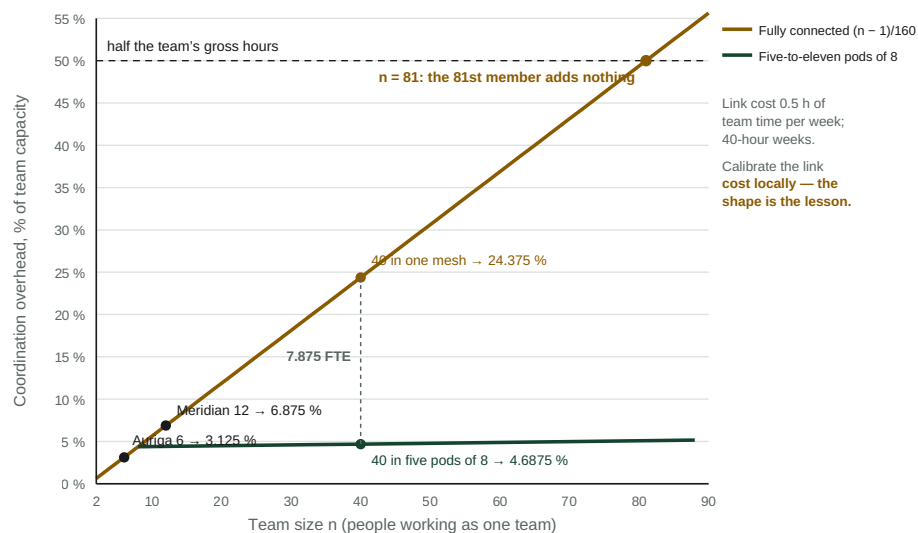


Fig 12.2.1 — The coordination ceiling, and what structure buys

**At Auriga's scale.** The six-engineer commissioning team of Domain 9 has **15** paths, **7.5** hours a week of coordination, **3.125 %** of capacity — **USD 979.69** a week at USD 130.625 an hour, or **USD 24,492.19** across the 25-week project. That is small, and it should be: small teams are cheap to coordinate, which is the actual reason to prefer them. The instructive number is marginal. A seventh member brings 40 hours and 6 new relationships costing 3.0 hours, so the net addition is **37.0 hours**: plainly worth having; the same calculation at the 81st member returns zero. **The marginal capacity of the  $n$ th member is  $h - c(n - 1)$** , and a leader who knows that will never again argue about team size without a number.



### 12.2.3 Psychological safety as a design variable

**Definition. Psychological safety** is a shared belief that the team is safe for interpersonal risk-taking. That a person can ask a question, admit an error, disagree with a senior colleague or report bad news without penalty to their standing. The term comes from organisational-behaviour research on team learning; this book uses it in the registered sense ([registries/TERMINOLOGY.md](#)).

**What it is not.** Not comfort, not niceness, not the absence of disagreement: a team with high safety and low standards is pleasant and produces nothing, and that conflation is the main reason the concept is dismissed by experienced delivery people. The useful frame pairs safety with **accountability for standards**: safety governs whether people can speak, standards govern what they are held to. Low safety with high standards produces anxiety and concealment; high safety with low standards produces comfortable underperformance; the combination the concept exists to name is high safety with high standards.

**On evidential status.** This book does **not** claim a measured effect size for psychological safety on delivery outcomes, and a professional should be sceptical of anyone quoting one from a project context. What can be asserted is narrower and sufficient: **every control system in this book consumes voluntary upward information**, and a team that has learned that raising problems is penalised will not supply it. Domain 3's escalation-lead-time indicator, Domain 8's risk identification, Domain 9's nonconformance reporting and Domain 4's interface issues all fail silently in the same way. That is an argument from the architecture of the delivery system, not from a research finding, and it is strong enough.

**Observable signals, because belief is not measurable and behaviour is.** Presence: junior people ask questions in front of senior people; errors are reported by the person who made them, before they are found; bad news arrives with time to act on it; disagreement is expressed in the meeting rather than after it; someone says "I don't know". Absence: pre-meetings where the real conversation happens; problems that arrive already solved and slightly too late; unanimous agreement followed by inaction; and a rising share of issues discovered by assurance rather than reported by the team, a *countable* signal available from data Domain 9 already produces.

**What a leader does**, in order of effect and all of it cheap. **Respond well to the first piece of bad news.** The climate is set by a handful of visible early responses, which is 12.1.3's self-regulation rule doing real work. **Report your own errors** with the specificity you would ask of others. **Ask a question you do not know the answer to, in public.** **Separate the issue from the person** when escalating, which is Domain 3's escalation-culture provision written into governance. And **name the standard alongside the safety**, so nobody mistakes candour for tolerance of poor work.

#### 12.2.4 Retention and the cost of turnover

Turnover is how every failure in this knowledge area becomes a cost, and it is almost never costed. Organisations know the recruitment fee, treat the vacancy as a saving, and never compute the rest. The build-up below is deliberately conservative, and its most important feature is what it **excludes**.

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##### WORKED EXAMPLE

#### Worked example 12.2.4 – the full cost of one resignation from Auriga's team of six.

1. **Setup.** One of the six engineers on Auriga's commissioning team resigns, giving notice in week 14 of the 25-week project. The blended engineering rate is **USD 130.625** an hour, so an engineer-week is **USD 5,225** (Domain 7, KA 7.4.1). Recruitment: an agency fee of **USD 24,000** and **20 hours** of leader and team time in shortlisting and interviewing. The post is vacant **6 weeks**; the work is covered by overtime at a **50 % premium** on the affected hours. The replacement's effectiveness ramps linearly from **30 %** in week 1 to **100 %** in week 8, with **4 hours a week** of colleague mentoring across those 8 weeks. The leaver's last two weeks run at **60 %** effectiveness, and handover consumes **12 hours** across the team. Every one of those is a stated planning assumption for this project, not a measured constant; the ramp shape in particular is a professional judgement and the result is sensitive to it.
2. **Formula.** Total = recruitment + ramp-up + lost productivity, where recruitment = fee + (selection hours × rate); ramp-up = (weeks –  $\Sigma$  effectiveness) × engineer-week + (mentoring hours × rate); lost productivity = leaver disengagement + handover + the **incremental** cost of covering the vacancy.
3. **Substitution.** Recruitment  $24,000 + 20 \times 130.625$ . Ramp effectiveness  $0.30 + 0.40 + \dots + 1.00 = 5.20$ , so lost effective weeks  $8 - 5.20 = 2.80$ ; ramp-up  $2.80 \times 5,225 + 32 \times 130.625$ . Lost productivity:  $0.8 \times 5,225$ ,  $12 \times 130.625$ , and  $0.50 \times 6 \times 5,225$ .
4. **Result.** Recruitment **USD 26,612.50**; ramp-up **USD 18,810.00** (14,630 of lost effectiveness plus 4,180 of mentoring); lost productivity **USD 21,422.50** (4,180 + 1,567.50 + 15,675). **Total USD 66,845.00: 12.79 engineer-weeks, 26.65 %** of the post's annual charge-out value of 250,800 over 48 weeks, and **1.671 %** of Auriga's **BAC** of 4,000,000. Two departures in a year cost **USD 133,690**.
5. **Interpretation.** Start with the exclusion, because it makes the number defensible. **The vacancy does not cost the salary; the salary is saved.** What it costs is the work not done and whatever is paid to get it done anyway, here a 50 % overtime premium of 15,675 on six engineer-weeks, not the 31,350 those weeks are worth. Counting the saved pay as a loss inflates the figure to 82,520, is the commonest error in turnover costing, and destroys the credibility of the calculation with any finance function that reads it. **The overtime decision is then not close.** Uncovered, six engineer-weeks missing from a six-person crew

is exactly **1.0 week** of elapsed delay on a zero-float activity, costing **USD 45,000** at Auriga's cost of delay, **2.87 times** the premium. The breakeven premium is  $45,000 / (6 \times 5,225) = 143.54\%$ : the project could pay two-and-a-half times basic rate and still beat the delay, worth knowing before the negotiation. **The number also sets the retention budget a leader may argue for, and only that.** At 66,845 per departure, an intervention costing 20,000 a year breaks even by preventing **0.2992** of a departure, roughly one every three and a third years: a low bar. But **the arithmetic tells you what a prevented departure is worth; it does not tell you that any intervention prevents one.** No claim is made here that a named practice reduces turnover by a stated amount, and a leader presenting such a claim without their own data is doing the thing this domain exists to stop. The defensible proposal is "this is the value at risk, here is what we propose to try, here is how we will know", Domain 2's discipline applied to a people decision. **Finally, note where the cost sits:** recruitment is only **39.8 %** of the total and **60.2 %** is ramp-up and lost productivity, the components nobody records; the **28.1 %** that is ramp-up is incurred *after* the vacancy is filled and the problem considered closed, which is why a team that has replaced three people stays slower than its headcount suggests for a quarter after the last one started.

## AI in this KA

**Where it earns its place.** Computing the coordination arithmetic across candidate structures (pod sizes and boundary options, each deterministic and checkable), and finding the design that minimises required paths given a stated dependency set, which is a genuine combinatorial problem. Extracting the actual communication graph from collaboration metadata **in aggregate** to test whether the assumed structure matches reality. Analysing free-text survey responses into themes with the source comments retained. Building the turnover model and its sensitivities.

**Where it must not go.** Individual-level inference from communication data. The aggregate question (*are our pods real, or is everyone still talking to everyone?*) is legitimate; the same data at individual level becomes surveillance, is unreliable as a measure of anything worth knowing, is restricted in many jurisdictions, and converts truthful upward information into managed information permanently. There is no version of *inferring who is disengaged* that this book endorses, and none of predicting which individuals will resign and acting on it: the prediction is weak, the action is visible, and the effect on those not selected is corrosive.

**Verification, concretely.** Reproduce  $n(n - 1)/2$  and the coordination share by hand, two operations each. State **c** with its basis and present the result across at least three values, because the conclusion is robust and the ceiling is not. For any aggregate communication analysis, confirm with the teams that the graph matches how they actually work: a low observed link count can mean a well-structured team or a team that has stopped talking, and only the people in it can tell you which.

## KEY TERMS — KA 12.2

| TERM                                                    | MEANING                                                                                                             |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Communication path</b>                               | One pairwise relationship in a team; counted by $n(n - 1)/2$ (Domain 4, KA 4.2.3).                                  |
| <b>Link cost <math>c</math></b>                         | Hours of total team time consumed per active pairwise relationship per week; a locally calibrated planning figure.  |
| <b>Coordination share</b>                               | $c(n - 1)/(2h)$ , coordination overhead as a fraction of team capacity; $(n - 1)/160$ at $c = 0.5$ , $h = 40$ .     |
| <b>Net team capacity</b>                                | $hn - c \cdot n(n - 1)/2$ ; maximised at 80–81 people under the illustrative parameters.                            |
| <b>Marginal capacity of the <math>n</math>th member</b> | $h - c(n - 1)$ ; zero at $n = 81$ under the same parameters.                                                        |
| <b>Pod (structured team)</b>                            | A sub-team with end-to-end ownership and one named path outward: the human form of Domain 4's layered architecture. |
| <b>Psychological safety</b>                             | A shared belief that the team is safe for interpersonal risk-taking, candour without penalty.                       |
| <b>Safety–standards pairing</b>                         | Safety governs whether people can speak; standards govern what they are held to. Both are required.                 |
| <b>Turnover cost</b>                                    | Recruitment + ramp-up + lost productivity, excluding the saved pay of a vacant post.                                |
| <b>Ramp-up loss</b>                                     | Weeks of a replacement's tenure minus the sum of their weekly effectiveness, valued at a person-week.               |

## SAMPLE MCQS — KA 12.2

**MCQ 12.2-A** [12.2.2 · Application] A 12-person team works as a single group. Each pairwise relationship consumes 0.5 hours of total team time per week; people are productive 40 hours a week. Coordination overhead as a share of team capacity is:

- A. 3.4375 %
- B. 6.875 % CORRECT
- C. 13.75 %
- D. 82.5 %

*Rationale:* Paths  $12 \times 11/2 = 66$ ; overhead  $0.5 \times 66 = 33.0$  h; capacity  $40 \times 12 = 480$  h;  $33.0/480 = 6.875$  % (12.2.2). A counts only one side of each relationship, halving the cost; C omits the division by two in  $n(n - 1)/2$ , counting each pair twice; D divides the 33 coordination hours by a single 40-hour week, attributing all coordination to the leader.

**MCQ 12.2-B** [12.2.2 · Analysis] Under the same parameters, the team size at which one further member adds no net capacity at all is:

A. 41

B. 81 CORRECT

C. 160

D. 161

*Rationale:* Marginal capacity is  $h - c(n - 1) = 40 - 0.5(n - 1)$ , zero at  $n = 81$  (12.2.2). A is the answer if the link cost is doubled to 1.0 hours; C mistakes  $n - 1 = 160$  for  $n$ ; D is the size at which coordination consumes the team's *entire* gross capacity, a different and later point.

**MCQ 12.2-C** [12.2.4 · Evaluation] A departure costs 26,612.50 in recruitment, 18,810.00 in ramp-up, and — in lost productivity — 4,180 of leaver disengagement, 1,567.50 of handover, and a 50 % overtime premium of 15,675 to cover a six-week vacancy worth 31,350 in engineer-weeks. The defensible total is:

A. USD 26,612.50

B. USD 45,422.50

C. USD 66,845.00 CORRECT

D. USD 82,520.00

*Rationale:*  $26,612.50 + 18,810.00 + 21,422.50 = 66,845.00$  (12.2.4). A counts recruitment only; B omits lost productivity; D substitutes the vacant post's 31,350 of pay for the 15,675 premium actually incurred: the saved-salary double count.

**MCQ 12.2-D** [12.2.2 · Evaluation] Restructuring a 40-person team from one group into five pods of eight, each with a single representative in a coordination forum, changes coordination from 780 paths to

1. At 0.5 hours per path per week, the capacity released is:

2. A. 630

3. B. 7.875 FTE CORRECT

4. C. 9.75 FTE

5. D. 19.6875 FTE

*Rationale:*  $0.5 \times (780 - 150) = 315.0$  hours a week  $\div 40 = 7.875$  FTE (12.2.2). A is the reduction in paths; C is the mesh design's total coordination cost, not the saving; D is the saving in percentage points of capacity, mislabelled as FTE.

## SELF-CHECK — KA 12.2

1. *State the coordination-share identity and its one calibrated parameter.* —  $\text{Share} = c(n - 1) / (2h)$ , which is  $(n - 1) / 160$  at  $c = 0.5$  hours per path per week and  $h = 40$  hours;  $c$  is calibrated locally and the ceiling result is sensitive to it.
2. *Why does a vacant post not cost its salary?* — The salary is saved; the cost is the work not done and whatever is paid to get it done anyway: on Auriga a 50 % overtime premium of 15,675 rather than the 31,350 the vacant weeks are worth.
3. *What claim does this book make about psychological safety, and what claim does it refuse?* — It claims that every control in the book consumes voluntary upward information which low safety suppresses; it refuses to quote a measured effect size on delivery outcomes.

## KNOWLEDGE AREA 12.3

## Delegation, coaching and difficult conversations

*Topics: 12.3.1 delegation as an economic decision · 12.3.2 span of control and coaching capacity · 12.3.3 the difficult conversation · 12.3.4 performance management honestly.*

### 12.3.1 Delegation as an economic decision

**Definition.** Delegation is the transfer of authority to decide and act on a defined class of work, together with the information and criteria needed to do so, with the leader retaining accountability. The last two clauses separate it from its counterfeits: transferring the *work* without the authority is allocation, and transferring the authority without the information is abdication (Domain 3, KA 3.2.3). Delegation is habitually discussed as a development activity and decided as a matter of trust. It is also an economic decision with a computable breakeven, and the arithmetic reliably points the other way from instinct.

## WORKED EXAMPLE

#### Worked example 12.3.1 — should Auriga's leader delegate the weekly interface coordination?

1. **Setup.** Weekly coordination of Auriga's technical interfaces (Domain 4's interface register) consumes **3.5 hours** of the project leader's week and will do so for the remaining **12 weeks**. The leader's management charge rate is **USD 185 per hour**; the senior engineer who would take the task is charged at **USD 140** (Domain 7's engineer grade). Delegated, the engineer needs **5.0 hours** a week for the first 4 weeks and **3.5 hours** thereafter, and the leader retains **1.0 hour** a week of oversight for 4 weeks and **0.5 hours** thereafter. The leader's people-work capacity is **6.5 hours** a week (KA 12.1.4).

- 
2. **Formula.** Cost of each option =  $\Sigma(\text{hours} \times \text{rate})$ . Released leader hours = status-quo leader hours – retained oversight. Breakeven value of a released leader hour = incremental direct cost ÷ released hours.
  3. **Substitution.** Status quo  $3.5 \times 12 = 42.0$  leader hours  $\times 185$ . Delegated: engineer  $5.0 \times 4 + 3.5 \times 8 = 48.0$  h  $\times 140$ ; oversight  $1.0 \times 4 + 0.5 \times 8 = 8.0$  h  $\times 185$ .
  4. **Result.** Status quo **USD 7,770.00**. Delegated **USD 8,200.00** ( $6,720 + 1,480$ ) – delegation is **USD 430.00** more expensive in direct labour. It releases **34.0** leader hours. The breakeven value of a released leader hour is  $430.00/34.0 = \text{USD } 12.65$ , or **6.836 %** of the leader's own charge rate.
  5. **Interpretation.** The decision is not close, and the arithmetic shows why the instinctive framing gets it wrong. Priced as a labour comparison, delegation looks like a 430-dollar mistake; priced correctly it is the purchase of 34 leader hours for 430 dollars, and it pays if a leader hour is worth more than **USD 12.65**: under 7 % of what the organisation charges for it. **The denominator makes it urgent:** the 3.5 hours the task consumes is **53.85 %** of the leader's entire discretionary people-work capacity of 6.5 hours, and the 34 released hours are **5.23 weeks** of that capacity. One recurring coordination task was consuming more than half of the only resource from which coaching, difficult conversations and team design are paid. **The risk side.** The objection is not cost but error: if the delegate mishandles one interface, Domain 4 prices re-verification at **USD 6,000** (KA 4.4.2). Value the 34 released hours at the leader's own rate ( $34 \times 185 = \text{USD } 6,290$ ), and net benefit is **USD 5,860**, so the delegation fails only if the probability of a 6,000-dollar incident exceeds  $5,860/6,000 = 97.67 \%$ . That is structurally Domain 3's threshold result (KA 3.2.3) and for the same reason, worth naming rather than re-deriving: **the cost of not delegating is certain, recurring and invisible, while the cost of a delegated error is uncertain, occasional and highly visible**, and organisations optimise against the visible cost. Two cautions. The 6,290 valuation holds **only if the released hours are actually spent on something worth 185 dollars an hour**: delegation that produces a busier leader rather than a coaching leader creates none of this value, which is why the released time should be reallocated at the moment of delegation, in writing. And the delegate must receive the information and criteria, not just the task: without them the 97.67 % breakeven is not the relevant number, because the error probability is no longer small.
- 

### 12.3.2 Span of control and coaching capacity

Span of control is usually argued as a structural convention, six to eight, or whatever the organisation last used. The better framing is capacity: **a leader can supervise many people and can coach few, and the binding constraint is the second**. Let  $R$  be people-work capacity in hours a week,  $n$  the direct reports,  $b$  the team size at which  $R$  was measured, and  $e$  the additional fixed load each report above  $b$  adds to



the leader's week: more one-to-ones, more interfaces held, more escalations absorbed. Then

$$\text{Coaching time per report} = (R - e(n - b)) / n$$

The second term is what most span discussions omit, and omitting it is not conservative; it overstates the sustainable span.

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#### WORKED EXAMPLE

#### Worked example 12.3.2 – how many people can Auriga's leader actually lead?

1. **Setup.** From KA 12.1.4,  $R = 6.5$  hours a week, measured at  $b = 6$  direct reports. Each report above six adds  $e = 0.5$  hours of fixed load, a planning judgement calibrated from the observed interruption log, not a constant. The project's stated sufficiency convention is that meaningful weekly coaching of a delivery professional requires at least **45 minutes** of undivided attention; this too is a deliberately adopted convention, not a measured threshold, and an organisation with different practices should set its own and re-run the arithmetic.
2. **Formula.** Per-report time =  $(R - e(n - b))/n$ . Maximum sustainable span: solve  $(R - e(n - b))/n \geq T$ , giving  $n \leq (R + eb)/(T + e)$ .
3. **Substitution.**  $(6.5 - 0.5(n - 6))/n = (9.5 - 0.5n)/n$ . Threshold  $n \leq (6.5 + 0.5 \times 6)/(0.75 + 0.5) = 9.5/1.25$ .
4. **Result.** Coaching time per report: **65.00** minutes at 6 reports, **51.43** at 7, **41.25** at 8, **33.33** at 9, **27.00** at 10, **21.82** at 11 and **17.50** at 12. The threshold is  $n \leq 7.6$ , so the maximum sustainable span is **7 direct reports**. Coaching capacity reaches exactly **zero** at **19** reports.
5. **Interpretation.** Three results, in ascending order of usefulness. **The obvious one:** doubling the span roughly halves each person's time, and at 12 reports each receives 17.5 minutes a week, which is a status check with a friendly tone, not coaching. **The non-obvious one:** because the leader's own fixed load grows with the team, the decline is faster than  $1/n$ , and ignoring that changes the answer. On the naive curve  $6.5/n$  the test is met up to  $n \leq 8.667$ , a maximum span of **8** (48.75 minutes at 8, 43.33 at 9), so **omitting the leader's growing fixed load overstates the sustainable span by one report**, exactly the margin by which most spans are set too wide. **The decisive one:** the sustainable span is a function of  $R$ , and  $R$  is designable. Recovering the 3.0 hours of interruption identified in KA 12.1.4 raises  $R$  to 9.5 and moves the threshold to  $n \leq 12.5/1.25 = 10$  reports. **The way to widen a span is to raise the leader's people-work capacity, not to divide a fixed budget more thinly.** And an organisation that loads its leaders with governance and interruption has made a span decision without knowing it. Two cautions: this bounds *coaching*, not supervision, a leader can hold accountability for 30 people through team



leads, which is what KA 12.2.2's pod design does, and the coaching arithmetic then applies to the leads; and a wide span is not always wrong, since experienced people on familiar work need less coaching. What is always wrong is a wide span *assumed to be coachable*, because that is a commitment the arithmetic says cannot be met.

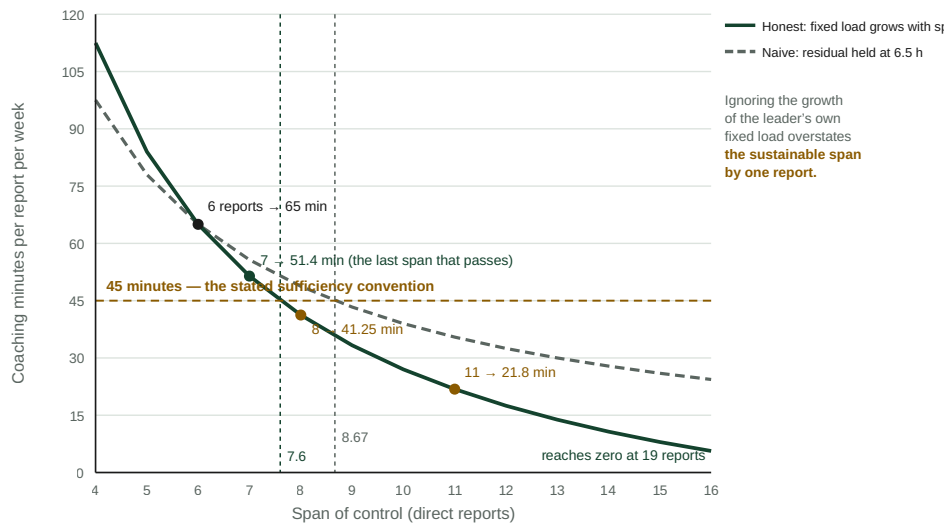


Fig 12.3.1 — Coaching minutes per report against span of control

### 12.3.3 The difficult conversation

**Definition.** A difficult conversation here is one in which the leader must tell someone something about their work, conduct or prospects that they will not want to hear, in a way that leaves the working relationship and the person's dignity intact. It is the highest value-per-hour act available to a delivery leader and the one most reliably postponed, and both facts have the same cause: the cost of holding it is immediate, personal and certain, while the cost of not holding it is deferred, diffuse and borne mostly by other people. Correcting that requires either changing the asymmetry — which is what pricing the deferral does (Case study B computes it) — or removing the structural excuse, which is what the span arithmetic does, since a leader with 21.8 minutes a report per week has a genuine reason the conversation never happens.

**The structure that works**, in five parts; the failure mode is losing control of the second minute. **The factual account, assembled beforehand.** Specific, dated, verifiable observations, not characterisations: "three of the last five interface specifications were submitted after the agreed date, on the 4th, the 18th and the 2nd" is a fact, while "you have a problem with deadlines" is a judgement dressed as one, and the conversation will be spent litigating it. **The consequence, in the project's terms.** What it costs the work and the other people, using the numbers of Domain 4

and Domain 9 rather than the leader's displeasure. **The enquiry, genuinely open**, because a material proportion of under-performance has caused the leader does not know about (an unstated dependency, an absent skill nobody provided, a health or caring situation, a conflicting instruction), and a conversation with no room for this is a verdict, resented in proportion to how right it was. **The specific expectation and support**: what will be different, by when, measurably, and what the leader will supply. **The record and the follow-up date**, since Domain 3's decision-record discipline (KA 3.3.4) applies unchanged, an unrecorded conversation was, for every practical purpose afterwards, a chat.

**The timing rule**. Hold it at the **first clear instance**, not the third. The first is a short, low-stakes, factual conversation about one event; the third is about a pattern, which is about the person, which is the one that fails. That is a claim about what the available evidence lets you discuss, not about psychology. **What not to do**, each common and each with a consequence: **the sandwich** teaches that compliments precede bad news and devalues both; **feedback by proxy** tells the person the leader would not say it to them; **the deferred aggregate** (a year of observations saved for an annual review) denied them any chance to correct, so any consequence attached is unfair on its face; and **the written-only conversation** removes the enquiry entirely.

### 12.3.4 Performance management honestly

Formal performance-management systems vary enormously across organisations and jurisdictions, and their legal requirements are jurisdiction-specific: what constitutes fair process, adequate notice, reasonable adjustment or lawful termination differs by country and by employment relationship, and **nothing in this domain should be read as advice on any jurisdiction's employment law**. Where a performance issue may lead to a formal consequence, involve the organisation's human-resources and legal functions early, follow the applicable process precisely, and document contemporaneously.

Four things generalise, and they are the leader's obligations rather than the system's. **Contemporaneous, specific records**: the difference between a defensible process and an indefensible one is almost always whether the record was made at the time (Domain 3, KA 3.3.4). **The distinction between capability, conduct and circumstance**: a person who cannot do the work needs training or a different role, one who will not needs a consequence, one temporarily unable needs support; and treating any as another is both unjust and ineffective. **A stated standard the person knew about before they were measured against it**, the same fairness principle Domain 9 applies to acceptance criteria. And **proportionality**: the first response to an isolated lapse by a good performer is a conversation, not a process.

On rating and ranking systems this book takes no position, and notes two consequences. Ratings that determine reward draw effort toward the measured behaviours: Domain 9's measurement problem applied to people, and why a scheme

ignoring collaboration produces less of it. And a distribution imposed across small teams produces outcomes the team can see are arbitrary, a psychological-safety cost paid for an administrative convenience. Two habits belong to the leader's own performance: ask for feedback in a form that makes it easy to give (a specific question about a specific recent decision), and **act visibly on one piece of it**, because the second time you ask depends on what happened the first time.

### AI in this KA

**Where it earns its place.** Assembling from the project record the factual chronology a difficult conversation requires (dates, submissions, agreed commitments) the part leaders skimp and the part that decides whether the conversation holds. Rehearsal: generating likely responses and testing the leader's answers, on the situation rather than the person. Computing the delegation and span arithmetic across options. Checking written objectives for specificity, measurability and a date. Drafting the reallocation plan for the hours a delegation releases, so released time is committed rather than absorbed.

**Where it must not go.** No AI-generated assessment of a person's performance, potential or conduct in any record, and no AI recommendation on a promotion, reward or termination decision. The prohibition is not squeamishness: these decisions carry consequences that differ by jurisdiction, they require the accountable human's judgement on evidence largely not in the system, and an organisation that lets a model's output stand as the basis of one has created accountability without a holder, Domain 1's defect at the highest stakes. Nor should the words of a difficult conversation be model-authored: the person will be able to tell, and what they learn is that the leader would not compose it themselves.

**Verification, concretely.** Every fact in an AI-assembled chronology is checked against the source record line by line, because one wrong date hands the conversation to the other party. The delegation and span arithmetic is reproduced by hand. Where AI has helped draft objectives, the leader confirms each is something *they* would hold themselves to. And no personal data about a named individual is entered into any tool outside the organisation's stated and lawful boundary for it (Domain 3, KA 3.A.2).

### KEY TERMS — KA 12.3

| TERM              | MEANING                                                                                                          |
|-------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Delegation</b> | Transfer of authority to decide and act, with the information and criteria to do so; accountability is retained. |
| <b>Abdication</b> | Transferred authority without the information or criteria to exercise it.                                        |

| TERM                                       | MEANING                                                                                                                      |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Released leader hours</b>               | Status-quo leader hours less retained oversight; the actual product a delegation buys.                                       |
| <b>Breakeven value of a released hour</b>  | Incremental direct cost ÷ released hours, USD 12.65 on Auriga, 6.836 % of the leader's rate.                                 |
| <b>Coaching time per report</b>            | $(R - e(n - b))/n$ ; 65.00 minutes at 6 reports on Auriga, 17.50 at 12.                                                      |
| <b>Sufficiency convention</b><br>T         | The minimum weekly coaching time an organisation states as meaningful; 45 minutes on Auriga, a convention not a measurement. |
| <b>Maximum sustainable span</b>            | $n \leq (R + eb)/(T + e)$ ; 7.6 on Auriga, so 7 reports.                                                                     |
| <b>First-instance rule</b>                 | Hold the conversation at the first clear instance, when it is about an event rather than a person.                           |
| <b>Capability / conduct / circumstance</b> | The three causes of under-performance, requiring training, consequence and support respectively.                             |

## SAMPLE MCQs — KA 12.3

**MCQ 12.3-A** [12.3.2 · Application] A leader has 6.5 hours a week of people-work capacity at six direct reports, and each additional report adds 0.5 hours to the leader's fixed load. Against a 45-minute weekly coaching convention, the maximum sustainable span is:

- A. 6 reports
- B. 7 reports CORRECT
- C. 8 reports
- D. 19 reports

*Rationale:*  $n \leq (6.5 + 0.5 \times 6)/(0.75 + 0.5) = 9.5/1.25 = 7.6$ , so 7 (12.3.2), giving 51.43 minutes each; 8 gives 41.25 and fails. C is the answer from the naive curve  $6.5/n$ , which ignores the growth of the leader's fixed load and overstates the span by one; D is the span at which coaching capacity reaches zero.

**MCQ 12.3-B** [12.3.1 · Analysis] Delegating a task costs USD 430 more in direct labour over 12 weeks and releases 34.0 hours of the leader's time. The breakeven value of a released leader hour is:

- A. USD 10.24
- B. USD 12.65 CORRECT
- C. USD 185.00
- D. USD 430.00

*Rationale:*  $430.00/34.0 = 12.65$  (12.3.1). A divides by the leader's original 42.0 hours rather than the 34.0 released; C is the leader's charge rate, which is the value the released hour *may* have, not the breakeven; D reads the incremental cost as an hourly figure.

**MCQ 12.3-C** [12.3.1 · Evaluation] The strongest argument for delegating in MCQ 12.3-B, given that a mishandled interface would cost USD 6,000 to re-verify and the released hours are worth USD 6,290 at the leader's rate, is that:

- A. the senior engineer is capable
- B. the leader is too busy
- C. the delegation fails only if the probability of a 6,000-dollar incident exceeds 97.67 % CORRECT
- D. delegation develops people

*Rationale:* The decisive argument is the breakeven error probability,  $(6,290 - 430)/6,000 = 97.67\%$  (12.3.1). A and D are assertions, however true; B is a capacity claim, real but not the comparison.

**MCQ 12.3-D** [12.3.3 · Analysis] A leader raises a recurring lateness issue for the first time at its third occurrence. The principal cost of the delay is that:

- A. the person will be more defensive
- B. the conversation is now about a pattern, and therefore about the person, rather than about a single event CORRECT
- C. the leader has lost credibility with the team
- D. the record is now incomplete

*Rationale:* The first-instance rule is about what the available evidence permits you to discuss: one event supports a short factual conversation, a pattern forces a conversation about the person, which is the one that fails (12.3.3). A is a likely symptom of B rather than the cause.

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SELF-CHECK — KA 12.3

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1. *What does a delegation actually buy, and what is the breakeven?* — Released leader hours; on Auriga 34.0 hours for USD 430, breaking even at USD 12.65 per released hour, 6.836 % of the leader's charge rate.
  2. *Why does omitting the leader's growing fixed load matter in span arithmetic?* — It overstates the sustainable span by one report, 8 instead of 7 on Auriga's figures, and one report is the usual margin of error.
  3. *Why hold a difficult conversation at the first instance?* — Because one event supports a factual conversation about the event; a pattern forces a conversation about the person.
-

## Remote and multicultural teams, ethical leadership

*Topics: 12.4.1 distributed teams and the arithmetic of overlap · 12.4.2 culture without stereotype · 12.4.3 ethical leadership and the leader's own conduct · 12.4.4 leading people alongside AI.*

### 12.4.1 Distributed teams and the arithmetic of overlap

**The definition that matters.** A distributed team is not one that is "remote"; it is one whose members' working hours do not fully coincide. Distance in space is mostly a solved problem. Distance in *time* is not, and it governs how long a decision takes, because a question asked outside the other party's working window waits. Of the two failure modes only one is discussed: the discussed one is communication quality, the undiscussed one is arithmetic. A pair of teams with **zero** overlapping working hours cannot complete an exchange in less than a working day, so a clarification requiring three exchanges takes three working days whatever anyone's intentions are. This is the same latency logic as Domain 3's `E[wait]` (a periodic opportunity to be heard, imposing a wait), and it should be computed at design time for the same reason.

#### WORKED EXAMPLE

**Worked example 12.4.1 — the cost of Meridian's time-zone design, and the price of fixing it.**

1. **Setup.** Meridian's programme runs across three groups, each working **09:00–17:00 local**, eight productive hours: the **core programme team** at **UTC+3**, the **development supplier** at **UTC+5:30**, and the **clinical-informatics advisers** at **UTC–5**. The first programme year logged **26** clarifications between the core team and the advisers that sat on the critical path of a clinic go-live and required **three** exchanges each. Meridian's cost of delay is **USD 14,280 per week**, or **USD 2,856** per working day over a five-day week.
2. **Formula.** Convert every window to a common reference (UTC), then  $\text{overlap} = \max(0, \min(\text{end}) - \max(\text{start}))$ . For a zero-overlap pair, round-trip latency is one working day per exchange.  $\text{Cost} = \text{exchanges} \times \text{days} \times \text{cost of delay per day}$ .
3. **Substitution.** Core  $09:00 - 3 = 06:00\text{--}14:00$  UTC; supplier  $09:00 - 5.5 = 03:30\text{--}11:30$  UTC; advisers  $09:00 + 5 = 14:00\text{--}22:00$  UTC. Core  $\cap$  supplier  $\min(14:00, 11:30) - \max(06:00, 03:30)$ . Core  $\cap$  advisers  $\min(14:00, 22:00) - \max(06:00, 14:00)$ .

- 
4. **Result.** Core and supplier overlap **5.5 hours** a day. Core and advisers overlap **0.0 hours**; supplier and advisers **0.0 hours**; all three together **0.0 hours**. Each core–adviser clarification therefore costs **3 working days (USD 8,568)**, and the 26 of them cost **USD 222,768** in the first year.
5. **Interpretation.** Two hundred and twenty-two thousand dollars was spent on a fact about the Earth's rotation that nobody had written down. **The fix and its exact cost are the valuable part.** Shifting the core team's day **two hours later** (11:00–19:00 local, 08:00–16:00 UTC) and the advisers' **one hour earlier** (08:00–16:00 local, 13:00–21:00 UTC) creates a **3.0-hour** daily overlap, inside which a three-exchange clarification completes in **one** working day. The 26 then cost **USD 74,256**, a saving of **USD 148,512** a year, or **USD 5,712** each. **But the redesign is not free, and the arithmetic shows where the cost lands.** Moving the core team two hours later reduces its overlap with the *supplier* from 5.5 hours to **3.5**: a consequence nobody would have noticed until it appeared as supplier friction, and the reason a distributed design must be computed as a whole rather than fixed one pair at a time. Here 3.5 hours still supports one substantial daily synchronisation, so the trade is worth making; the point is that it must be **a decision, priced, rather than a side effect. And there is a human cost that must be allocated explicitly, which is where this topic meets 12.4.3.** The 3.0 hours of overlap are bought with 3.0 hours a day of unsocial working. Split as designed (two hours on the core team, one on the advisers) the burden is **10 hours a week** for the core team and **5** for the advisers. Pushed entirely onto whichever group has least power, that group carries **15 hours a week** indefinitely, which is the default outcome when a supplier or an offshore team is the junior party. **Rotating the shift monthly equalises it at 7.5 hours a week each**, and the choice between those three allocations is an ethical decision made by a leader, not a scheduling detail. A leader who takes the 148,512 and does not decide who pays the 15 hours has not finished the calculation.
- 

**What else follows for practice.** Design the team so that **the dependencies with the most exchanges sit inside a single time zone**, the same principle as Domain 4's interface minimisation and KA 12.2.2's pods: put the boundary where the traffic is thin. Convert exchanges into **asynchronous artefacts** wherever the exchange is informational rather than deliberative, since a complete written specification with the anticipated questions already answered removes exchanges rather than accelerating them, and removing one is worth a full working day at zero overlap. Make the **handover artefact** formal where work passes between zones: what was done, what is in flight, what is blocked, what the next party must decide. And publish the overlap map, because the commonest cause of a three-day answer is that the asker did not know the other party had already gone home.

### 12.4.2 Culture without stereotype

Cross-cultural difference in delivery teams is real, consequential and routinely handled badly in both directions: ignored, or reduced to a table of national generalisations applied to individuals. This book's position, stated as a position rather than a finding: **national-culture dimension frameworks are useful for anticipating the *existence* of differences and useless for predicting the behaviour of the person in front of you.** They earn their place as a checklist of what may vary; they fail when a leader uses a country average to predict an individual, because within-group variation in every dimension exceeds the between-group differences described.

Four differences matter most in project work. **How disagreement is expressed:** in some working cultures a direct "no, that will not work" is expected; in others the same message is carried by a qualified yes, a silence or a change of subject, and the leader who hears only the words hears agreement. The countermeasure is not a national rule but a **changed question:** replace "does everyone agree?" (which almost always returns agreement) with "what is the strongest argument against this?", asked of a named person, plus written confirmation so a misread silence surfaces within a day. **How hierarchy is treated;** whether a junior engineer will contradict a senior one varies enormously and interacts directly with psychological safety, so a leader who concludes there is no safety problem because nobody objected may be observing a hierarchy norm; the countermeasure is structural, asking for input in a form that does not require public contradiction and never punishing the first person who does. **How commitment is talked about,** whether "we will try to have it by Thursday" is a commitment or an aspiration is not universal, and many cross-boundary schedule failures are this ambiguity rather than any failure of intent; the countermeasure is Domain 4's discipline, commitments written, dated and confirmed by the committing party in their own words. **How much is left implicit,** some working cultures carry context implicitly and find exhaustive specification faintly insulting, others treat anything unwritten as unagreed; make the convention explicit for the project, and on a distributed programme make it written-and-explicit, because that is the one that survives being read at 03:00 by someone who cannot ask a follow-up until tomorrow.

Where the working language is a second language for part of the team: slow the pace and reduce idiom rather than raising the volume; put the substance in writing, so reading (easier than listening) carries the load; and never mistake fluency for competence in either direction, since the most competent engineer on a team is frequently not the most fluent speaker in the room. **Inclusion is a delivery variable as well as an ethical one:** a team member who cannot contribute (for reasons of language, time zone, hierarchy norm, disability, or simply never being asked) is capacity the project is paying for and not receiving. That is the honest business case, and it does not replace the ethical case, which stands on its own.



### 12.4.3 Ethical leadership and the leader's own conduct

**Definition.** Ethical leadership in delivery is the consistent subordination of the leader's own short-term interest (reputation, comfort, bonus, the appearance of control) to the obligations owed to the sponsor, the team, the users and the public. Domain 1 established those obligations (KA 1.2.2); this topic is about the leader's conduct when honouring them is costly.

**Reporting what is true when it is unwelcome.** The pressure is to report amber, absorb the variance personally and hope; the response is Domain 1's standard of care and Domain 3's record: report the forecast you believe, with its basis, in writing, and keep the paper, because a leader who reports optimism they do not hold has transferred a risk to people who do not know they are carrying it. **Being asked to commit to a date you do not believe in** is the most common ethical situation in delivery and is rarely recognised as one; the response is neither refusal nor silent acceptance but stating the date you can commit to, the conditions under which the requested date becomes possible, and the risk of the requested date, in writing, then letting the accountable person decide with the information (Domain 3, KA 3.2.1). **Protecting the team from unreasonable demand:** sustained overtime is a loan against future capacity at a poor rate, and KA 12.2.4's 66,845 per departure is the argument. **Safety and harm;** where the work can hurt someone, the obligation to stop is absolute and personal and is not subject to the delegation schedule. **Fairness in the things people watch:** who gets the visible work, who is in the room, who is credited, who is believed when two accounts conflict — the decisions from which a team infers what the leader actually values, made quickly and observed closely. And **the small compromises** (supplier hospitality, a friend's firm on a shortlist, a favour that would help the schedule) for which the reliable test is whether you would be comfortable if the whole decision, including your reasoning, were in the decision record with your name on it.

**When the decision goes the other way.** Every path above ends in the same place: *state it in writing, then let the accountable person decide with the information.* That is right, and it is incomplete, because it says nothing about the case the leader actually fears: the accountable person, having been told in writing, directs the misstatement anyway, or proceeds despite a stated safety objection. A professional obligation asserted without a mechanism for that case is an exhortation. The mechanism has four parts.

**Record a dissent.** Not a complaint and not a resignation letter, a dated record of five things: the fact as you stated it, the date and forum in which you stated it, to whom, the decision that was taken, and **the risk you believe remains.** It is filed in the decision record under Domain 3, KA 3.3.4, which is the register that answers *was this reasonable on what was known at the time?*, and **a copy goes to a party other than the decision-maker** (the assurance function, the programme office, the sponsor's own line, whichever the governance provides) for the plain reason that a record held only by the person it concerns is a record that can cease to exist. The

tone is factual and the content is specific. A dissent that characterises the decision-maker rather than the decision is weaker, not stronger, and it converts a professional record into a personal dispute.

**Decline the signature you do not believe.** The leader does not sign an acceptance, a readiness certificate, a completion statement, a forecast or a final account they do not believe to be true; and says **in writing why**, naming what would have to change for the signature to be available. This is the point at which the obligation stops being about reporting and starts being about attestation: a signature is a representation to everyone downstream who relies on it, and Domain 14, KA 14.4.3 makes the same point about being asked to verify an output without the time or information to do it. Refusing a signature is a professional act with consequences, and the mechanism is designed so that the refusal is documented, reasoned and reversible rather than dramatic.

**Escalate beyond the accountable person where the matter is safety or harm.** For an ordinary commercial disagreement, the accountable person's decision stands and the dissent is the record. For a risk of harm to people, **it does not**: the obligation to escalate beyond the accountable person is absolute and personal, it is not subject to the delegation schedule, and the route is stated in advance so that nobody has to invent one under pressure: the next governance tier (Domain 3, KA 3.3.3, at its stated latency), then independent assurance or internal audit, then the board or its audit or risk committee. Each step is taken in writing, each is recorded, and the leader keeps their own copy of what was sent and when.

**The neutral closing note, and it is expressly not legal advice.** Arrangements protecting a person who raises a concern in good faith (often described as protected-disclosure or whistleblowing regimes) are understood to exist in many jurisdictions. Their scope, the conditions attached to them, who is covered, which routes qualify and what protection follows differ materially between them, and **nothing here states the position in any jurisdiction, states that any such arrangement applies to any person or organisation, or characterises any response to a disclosure as lawful or otherwise.** Two practical consequences follow, and they are the whole of what this book will say. The organisation's own policy and its legal function are engaged **before** any step outside the organisation, because the order in which steps are taken can matter and is not something to work out alone. And the professional obligation that does not depend on any of it is narrower and unconditional: **raise the concern internally, promptly, in writing, keep your own copy, and take advice early.** Where a leader believes they are being placed in an untenable position, that is a matter for qualified advice in their own jurisdiction, and this book is not a substitute for it.

Two closing observations. Ethical failure in delivery is rarely a single dramatic choice; it is an accumulation of small deferrals (an unreported slippage, an unraised concern, a number rounded favourably) each defensible alone and collectively a misrepresentation, and the countermeasure is the cumulative test Domain 3 applies

to changes below a threshold. And the leader's conduct is the only reliable instrument of culture available: values statements are read, decisions are studied, and a leader who wants candour and punishes the first candid person has settled the matter whatever the poster says.

#### 12.4.4 Leading people alongside AI

Four leadership problems are immediate. **Skill formation:** where a tool performs the tasks on which junior practitioners used to learn judgement, the pipeline of competence is affected; the honest position is that the effect is plausible and not yet measurable, so act conservatively: deliberately reserve some work for human development even where a tool would be faster, and make the *review* of AI output a taught skill rather than an assumed one, since reviewing a plausible wrong answer is harder than producing a rough right one. **Honest attribution:** a team member presenting model output as their own analysis has misled colleagues about the basis of a recommendation, which matters because the verification obligation attaches to the basis, state the norm before it is needed: **say what was AI-generated, say what you verified and how.** **Fear, and the conversation nobody is having:** where people believe a tool threatens their role they will not say so, and they will adopt it visibly while routing around it privately, which is the worst outcome because the organisation gets neither the productivity nor the truth; the response is a direct, early, honest conversation about what will and will not change, in which "I don't know yet" is acceptable and false reassurance is not. And **accountability, unchanged: AI proposes; the professional verifies, decides and remains accountable.** Applied here, a leader may not attribute a decision about a person to a system, may not accept a model's assessment of a person as evidence, and may not delegate to a tool a conversation that is theirs to hold.

#### AI in this KA

**Where it earns its place.** Time-zone and overlap computation across a large distributed programme (every pair, every proposed shift pattern, every consequence) the deterministic arithmetic of 12.4.1 at a scale nobody would attempt by hand, and where the second-order effect on the core-supplier overlap is easily missed. Producing the rotation roster that equalises unsocial hours, with the fairness constraint stated as an input. Translation and plain-language support for second-language colleagues, with the caution that a translated commitment must be confirmed by the committing party in their own words. Summarising a long asynchronous thread into decisions taken and questions still open.

**Where it must not go.** No cultural inference about individuals, a model asked "how should I communicate with a colleague from country X" returns a confident national generalisation, precisely the error 12.4.2 identifies, and receiving it from a system makes it feel like data. No monitoring of remote workers' activity as a proxy for productivity: the measure is poor, the jurisdictional position varies, and the effect on trust is severe and permanent. And no automated allocation of unsocial hours

without a human deciding the fairness rule, because the fairness rule is the whole ethical content of the decision.

**Verification, concretely.** Recompute the overlaps that matter by hand, in UTC, and check each proposed shift against **every** pair rather than the one being fixed. Confirm any AI-summarised decision list against the thread with the people who took the decisions. And where a rotation roster is generated, verify that the unsocial-hours totals per group are equal to the stated tolerance, a one-line check, and the only part of the output that carries the ethics.

#### KEY TERMS — KA 12.4

| TERM                           | MEANING                                                                                                                                                                                                   |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Distributed team</b>        | A team whose members' working hours do not fully coincide; time distance, not spatial distance, is the governing variable.                                                                                |
| <b>Working-hour overlap</b>    | $\max(0, \min(\text{end}) - \max(\text{start}))$ with all windows converted to a common reference.                                                                                                        |
| <b>Zero-overlap pair</b>       | Two groups with no common working hours; each exchange costs at least one working day.                                                                                                                    |
| <b>Overlap window</b>          | A deliberately created shared period, bought with unsocial hours that must be explicitly allocated.                                                                                                       |
| <b>Unsocial-hours burden</b>   | The hours per group per week spent outside normal local working time; equalised by rotation.                                                                                                              |
| <b>Asynchronous artefact</b>   | A written product complete enough to remove an exchange rather than accelerate it.                                                                                                                        |
| <b>Handover artefact</b>       | What was done, what is in flight, what is blocked, what the next party must decide.                                                                                                                       |
| <b>Within-group variation</b>  | The reason national-culture averages cannot predict an individual's behaviour.                                                                                                                            |
| <b>Honest attribution (AI)</b> | Stating what was AI-generated and what was verified, and how.                                                                                                                                             |
| <b>Dissent record</b>          | The dated record of the fact stated, when and to whom, the decision taken, and the risk the leader believes remains — filed in the decision record, with a copy to a party other than the decision-maker. |
| <b>Declining the signature</b> | The refusal to sign an acceptance, readiness certificate, forecast or account the leader does not believe, stated in writing with what would have to change.                                              |
| <b>Route of last resort</b>    | For a risk of harm to people: next governance tier, then independent assurance, then the board or its audit or risk committee — stated in advance, each step in writing.                                  |

| TERM                                 | MEANING                                                                                                                                                                                                                                                                    |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Employment-records provisions</b> | Custodian, access list, copy to the individual, retention period from the organisation's policy, and disposal — settled with human resources, legal and the data-protection accountability holder, whose requirements differ by territory and are not stated in this book. |

## SAMPLE MCQS — KA 12.4

**MCQ 12.4-A** [12.4.1 · Application] Team A at UTC+3 and team B at UTC-5 both work 09:00–17:00 local. Their daily overlap is:

- A. 0.0 hours CORRECT
- B. 2.0 hours
- C. 3.0 hours
- D. 8.0 hours

*Rationale:* In UTC, A works 06:00–14:00 and B works 14:00–22:00, so  $\min(14:00, 22:00) - \max(06:00, 14:00) = 0$  (12.4.1). B results from taking the offset as 6 hours rather than 8; C is the overlap after the shift redesign; D compares local clock times without converting to a common reference. The error the topic exists to prevent.

**MCQ 12.4-B** [12.4.1 · Analysis] Twenty-six critical-path clarifications a year each need three exchanges. At zero overlap each exchange costs one working day; the cost of delay is USD 2,856 per working day. Creating a three-hour overlap lets each clarification complete in one day. The annual saving is:

- A. USD 5,712
- B. USD 74,256
- C. USD 148,512 CORRECT
- D. USD 222,768

*Rationale:*  $26 \times 3 \times 2,856 = 222,768$  before and  $26 \times 1 \times 2,856 = 74,256$  after, so the saving is **148,512** (12.4.1). A is the saving per clarification; B is the post-fix cost; D is the pre-fix cost.

**MCQ 12.4-C** [12.4.1 · Evaluation] Shifting the core team's day two hours later to create the overlap above reduces its overlap with a supplier from 5.5 hours to 3.5. The correct professional response is to:

- A. abandon the redesign, since it damages an existing relationship
- B. proceed, having computed the effect on every pair and judged 3.5 hours sufficient for one daily synchronisation CORRECT
- C. proceed without recomputing, since the adviser problem is larger
- D. ask the supplier to shift instead, since they are the supplier

*Rationale:* A distributed design must be computed as a whole, not fixed one pair at a time, and the trade must be a priced decision rather than a side effect (12.4.1). D allocates the burden by power rather than by reason, the failure 12.4.1 and 12.4.3 both identify.

**MCQ 12.4-D** [12.4.2 · Analysis] A leader asks a distributed team "does everyone agree?" and receives agreement, then discovers a fortnight later that two members had serious objections. The best countermeasure is to:

- A. learn the national communication norms of the members concerned
- B. ask a named person for the strongest argument against the proposal, and confirm decisions in writing CORRECT
- C. require objections to be raised in the meeting
- D. reduce the team's size

*Rationale:* Changing the question and confirming in writing works regardless of which norm is operating, whereas a national generalisation cannot predict an individual (12.4.2). C restates the requirement that has just failed.

#### SELF-CHECK — KA 12.4

1. *How is working-hour overlap computed, and what is the trap?* — Convert every window to a common reference, then  $\max(0, \min(\text{end}) - \max(\text{start}))$ ; the trap is comparing local clock times, which invents overlap that does not exist.
2. *What does a three-hour overlap window actually cost, and who decides?* — Three hours a day of unsocial working; the allocation between groups (10/5, 15/0 or 7.5/7.5 rotated) is an ethical decision the leader must make explicitly.
3. *What are national-culture frameworks good for, and not good for?* — Anticipating that differences exist; not predicting an individual's behaviour, because within-group variation exceeds the between-group differences described.
4. *The accountable person has been told in writing and directs the misstatement anyway. What does the leader do?* — Record a dissent (fact stated, date, to whom, decision taken, risk remaining) in the decision record with a copy to a party other than the decision-maker; decline to sign what they do not believe, in writing, saying what would have to change; and where the matter is a risk of harm to people, escalate beyond the accountable person by the stated route: next governance tier, independent assurance, then the board or its audit committee (12.4.3).
5. *What does this book say about protected-disclosure regimes?* — That such arrangements are understood to exist in many jurisdictions, that their scope, conditions and protections differ materially, and that nothing here states the position anywhere. The organisation's policy and its legal function are engaged before any external step; the unconditional obligation is to raise the concern internally, promptly, in writing, keep a copy, and take advice early.

## Advanced topics — Domain 12

### 12.A.1 Leading a team through loss of confidence

Domain 3 designed recovery *governance*: raised authority, short cadence, narrowed membership, a single information source, a log kept in the room (KA 3.A.1). That structure does nothing about the condition that determines whether a recovery works: a team that has stopped believing the work can succeed.

The behavioural signature is specific and often mistaken for its opposite. Effort does not fall. It frequently rises and becomes defensive. People work longer on their own scope and stop attending to the whole. Estimates lengthen and stop being negotiable, because the estimate has become a shield rather than a forecast. Escalations become formal and documentary, a rational response where blame is expected, and Domain 3's escalation-lead-time indicator will be shortening at the same time. And the best people leave first: they have the most options and the least need to tolerate it. At KA 12.2.4's USD 66,845 per departure, a recovery that loses three people has spent **USD 200,535** before any recovery action, and has lost the three people whose judgement it most needed.

What works, as professional judgement rather than measured effect. **Re-establish a credible plan before asking for effort**, because the broken link is effort → performance and no exhortation repairs it: an honest re-baseline is a leadership act, not an admission. **Shorten the horizon** to something the team can succeed at within a fortnight, and let them succeed visibly. **Reduce the count of things in flight** rather than adding people, since KA 12.2.2's arithmetic says the second worsens coordination exactly when coordination is failing. **Be present and specific about what is not known**, since ambiguity filled by the team is filled pessimistically. **Protect the escalation route**, because a recovery in which bad news stops arriving cannot be steered. And **decide team changes early and transparently**, since a recovery in which people suspect unannounced changes spends the whole period on that question. Two cautions: raised authority must remain **bounded and time-limited** or the directive style it licenses becomes permanent, which is how an organisation loses the people it kept through the crisis; and replacing the project leader mid-recovery imposes a ramp-up cost on **every** member of the team simultaneously, not on one post, KA 12.2.4's ramp-up component multiplied by the team.

### 12.A.2 Team design as an architecture decision

The organisational and technical architectures are not independent. There is a long-standing observation in systems engineering that a delivered system tends to resemble the communication structure of the organisation that built it. This book treats that as an **observation worth designing against**, not a law, and notes that Domain 4's arithmetic gives it teeth: if the team structure and the interface structure



disagree, one of them will be paid for. The design rule follows from KA 12.2.2 and KA 4.2.3 together: **put the team boundary where the interface traffic is thin**. A pod owning a cluster of clinics end to end has few external interfaces and needs few external relationships; a pod defined by technical speciality (all the testers, all the integrators) has an interface with every other pod on every deliverable, and will generate exactly the coordination load the pod structure was meant to remove.

**The cost of double membership** is where this fails in practice. Meridian's five pods of eight carry **150** paths, **4.6875** % of capacity, **1.875 FTE**. Put **ten** of the forty people in a second pod as well, and each acquires the seven relationships of the additional pod: **70** extra paths, **220** in total, **110.0** hours a week, **6.875** % of capacity, **2.75 FTE**. The extra **35.0** hours a week is **0.875 FTE** (nearly one full-time person consumed by a staffing convenience), and the resulting share is exactly that of a **12-person fully connected team**, which is a useful sentence in a resourcing discussion: *a quarter of our people double-hatted means our forty-person programme coordinates like a twelve-person one, and pays for the privilege twice*.

Three further consequences. **Long-lived teams are cheaper than re-formed ones**, because the relationships in  $n(n-1)/2$  have a formation cost as well as a running cost, and a team reassembled each phase pays it repeatedly while passing through the conflict stage again. **Ownership must include the consequence**: a pod that delivers a clinic but does not own its defects will optimise for delivery, which is Domain 9's problem arriving through an organisational door. And **the representative in the coordination forum is a role, not a courtesy**; it carries both pods' context and is the single path the layered design depends on, so it needs allocated time and a named deputy, or the pods quietly revert to a mesh and the 7.875 FTE returns as unexplained slowness.

### 12.A.3 The reviewer's leadership eye

Invariants to test on any delivery organisation's people design; each cheap, each countable, each diagnostic.

The project leader's week has been **audited from observed data**, not intention, and people-work capacity is a stated number. Every team has a **computed coordination share** at a stated link cost, presented across at least three values of  $c$ . No team above roughly a dozen people works as a **single fully connected group** without a stated reason. Team boundaries sit where the **interface traffic is thin**, and the double-membership count is known and priced. Every leader's span is **within the sustainable span** implied by their own people-work capacity and the organisation's stated coaching convention, and the convention is written down. Coaching time per report is computed with the leader's **growing fixed load** included, not from a constant residual. Every recurring task consuming more than a stated share of a leader's discretionary capacity has a **delegation decision on record**, with the released hours **explicitly reallocated**. Turnover cost is computed on the three-component build-up with the **saved pay of a vacant post excluded**, and the resulting value at



risk is what any retention proposal is measured against, with no claimed effect size attached to the proposal. Psychological safety is tracked through **countable signals** (issues found by assurance versus reported by the team, escalation lead time) not a survey score alone. Every distributed pair has a **computed overlap in a common time reference**, every proposed shift is checked against **every** pair, and the unsocial-hours burden is **allocated explicitly and rotated**. Difficult conversations are held at the **first instance** and recorded contemporaneously. And no AI system is inferring anything about a **named individual's** state, engagement, personality or likelihood of leaving: the single test in this domain that admits no proportionality argument.

## Industry variations — Domain 12

- **Construction and infrastructure.** The leader's span includes people they do not employ, so coaching capacity applies mainly to a supervisory tier and the coordination arithmetic must count paths across contractual boundaries. Span is frequently fixed externally by safety supervision ratios (the one context where the number is not the leader's to design), and safety leadership is a personal, non-delegable obligation that overrides schedule pressure absolutely.
- **Healthcare.** Clinical professional autonomy means a project leader has authority over the project and none over clinical judgement: Domain 3's point about Meridian's clinical sign-off (KA 3.1.2) reappears as a leadership constraint, influence, not instruction. Multi-professional teams with parallel hierarchies put KA 12.4.2's hierarchy-norm problem inside one building.
- **Technology and product organisations.** Long-lived small teams with end-to-end ownership make 12.A.2's design rule native rather than a reform. Retention economics dominate because ramp-up is long and largely tacit: KA 12.2.4's eight-week curve is optimistic in a complex codebase, and the ramp-up component should be expected to exceed recruitment by a wide margin.
- **Public sector.** Staff rotation is often a feature of the career system rather than a failure, so the problem is **knowledge continuity** rather than retention: handover artefacts, written decision records and deliberate overlap between outgoing and incoming post-holders carry more value than any retention effort. Pay levers are usually unavailable, making KA 12.2.1's condition design the only approach on offer.
- **Energy and utilities.** Rotational and remote-site working (Auriga's commissioning crew is a mild version) makes shift handover a formal artefact with safety consequences and KA 12.4.1's unsocial-hours fairness question continuous rather than occasional. Turnover of specialist commissioning staff mid-campaign lands on zero-float activities, which is why KA 12.2.4's delay comparison matters more here than its labour figure.

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- **Financial and professional services.** Utilisation targets compete directly with KA 12.1.4's people-work capacity, since coaching time is by construction non-chargeable: the span arithmetic collides with the business model and the collision must be made explicit rather than resolved by pretending. Matrix double-membership is endemic, which makes 12.A.2's 0.875 FTE the most useful number in this domain for these organisations.
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## Case study — Domain 12: the programme that tripled (health, Meridian)

**Situation.** Meridian Care Records moved from a 12-clinic pilot to the 40-clinic national rollout, and the programme team grew from **12** people to **40** over two quarters. No structural change accompanied the growth: the team continued as a single group with a weekly all-hands, a shared backlog and an expectation that anyone would coordinate with anyone. Nine months in, the rollout was running **six weeks** behind, individual productivity was reported as falling, and the quarterly report attributed the slippage to "clinic readiness and team ramp-up". Five people had left during the year. The steering committee, whose latency is 4.0 weeks (Domain 3, KA 3.2.3), was asked to approve eight additional staff.

**What the arithmetic showed.** The programme director computed three numbers before the paper went in. **Coordination.** At a link cost of 0.5 hours of team time per relationship per week on 40-hour weeks, the 12-person team had carried  $12 \times 11/2 = 66$  paths, **33.0** hours a week, **6.875 %** of capacity, **0.825** of a person. The 40-person team carried **780** paths, **390.0** hours, **24.375 %**: **9.75** full-time people, none visible in any cost line, all charged to the work packages people were coordinating about. **Structure.** Five pods of eight, each owning a cluster of eight clinics end to end with one representative in a weekly forum, would carry  $5 \times 28 + 10 = 150$  paths, **75.0** hours, **4.6875 %**: **1.875** people. The restructure would release **7.875 FTE**, more capacity than the eight staff being requested, at no additional cost. **Turnover.** On KA 12.2.4's three-component build-up with the same parameter set (a 6-week vacancy covered at a 50 % overtime premium, 2.80 person-weeks of ramp-up loss, 32 hours of mentoring, 0.8 person-weeks of leaver disengagement, 12 hours of handover), but Meridian's own figures, a USD 15,000 agency fee and a person-week of **USD 3,840** at USD 96 an hour, each departure had cost **USD 46,488**: recruitment 16,920, ramp-up 13,824, lost productivity 15,744. The five departures had therefore cost **USD 232,440**, of which **USD 147,840 — 63.6 %** — was ramp-up and lost productivity, recorded nowhere, and **USD 69,120** of it incurred after the last vacancy was filled and the recruitment problem considered closed.

**What changed, and how it resolved.** The recruitment request was withdrawn and replaced by a restructure into five pods, with the representative role given time, a named deputy and a place in its own pod's workload. The all-hands moved from weekly to fortnightly, the intervening slot given to pods, a small change that

removed the main mechanism by which everyone stayed connected to everyone, which was precisely the point. And the five exit conversations were re-read against KA 12.2.1: three cited the same broken **effort** → **performance** link, a rollout plan nobody in the team believed, which the programme had been treating as a communication problem. The programme then delivered the remaining rollout **six weeks earlier** than the re-forecast made at restructure, worth  $6 \times 14,280 = \text{USD } 85,680$  at Meridian's cost of delay. Read that alongside the theoretical figure: 7.875 FTE at a person-week of 3,840 is **USD 30,240** a week, or **USD 1,451,520** over a 48-week year, **16.94 times** what the programme actually banked. The gap is not an arithmetic error but the difference between **capacity released** and **value realised**: released capacity becomes value only where it is redeployed onto work that matters and where the rest of the system (clinic availability, training slots, the 4.0-week governance latency) does not immediately become the binding constraint instead. It did.

**What the domain teaches here. The cost of an unstructured team is real, large and invisible**, and it presents as declining individual productivity, the one explanation that guarantees the wrong response. **The instinctive remedy makes it worse**: eight more people in a fully connected 40-person team would have added  $8 \times 40 = 320$  hours of capacity and, from  $h - c(n - 1)$  summed across the additions,  $0.5 \times (40 + 41 + \dots + 47) = 174.0$  hours of new coordination; and the request was made in the same paper that reported falling productivity. And **a structural saving must be traced to a realised outcome or it is not a saving**: the honest claim was 85,680, not 1,451,520.

### Case study B — Domain 12: the conversation that cost sixty-two times its price (rail systems design, transport)

**Situation.** A design consultancy was producing the systems-design package for a rail re-signalling scheme. The design manager had 11 direct reports. One of them, a lead engineer, had been producing 9 drawings a week against a plan of 12, with 18 % of drawings requiring rework against a 6 % allowance. The manager was aware from week 2. The conversation happened in week 16.

**What the arithmetic showed.** Computed afterwards in a lessons review, which is when this arithmetic is usually done and far too late to be useful. Over the 14 weeks between the first clear instance and the conversation: an output shortfall of  $3 \times 14 = 42$  drawings, each taking 4.5 hours at a design rate of **USD 115** an hour,  $42 \times 4.5 \times 115 = \text{USD } 21,735$  of work displaced onto other people. Excess rework on the 126 drawings actually delivered:  $(0.18 - 0.06) \times 126 = 15.12$  drawings reworked at 2.0 hours each,  $15.12 \times 2 \times 115 = \text{USD } 3,477.60$ . And the package sat on the critical path of the client's design programme, so the 42-drawing shortfall converted at the planned rate of 12 a week into **3.5 weeks** of delay, exposed at **USD 28,000** a week under the consultancy's delay provisions — **USD 98,000**. Total consequence **USD 123,212.60**. The conversation itself, held in week 2, would have cost 3

**hours** of the manager's time plus **1.5 hours a week for 6 weeks** of supported improvement: 12 hours at **USD 165** an hour, or **USD 1,980**. The ratio is **62.23 to 1**.

**Why it did not happen, which is the instructive part.** The manager was not avoidant by disposition; the structure had made the conversation unavailable. At 11 direct reports, with a people-work capacity of 6.5 hours a week measured at six reports and 0.5 hours of additional fixed load per report above that, coaching capacity was  $6.5 - 0.5 \times 5 = 4.0$  hours a week across 11 people, **21.82 minutes** each (KA 12.3.2). Twenty-two minutes a week is enough to ask how someone is and not enough to hold a documented, evidenced, five-part conversation about their performance. The manager had also never been given a sufficiency convention, a factual chronology, or any figure for what deferral cost.

**How it resolved.** Two structural changes and one conversation. The eleven reports were split into two teams of five under newly created team-lead roles, taking the design manager's span to **6 (65.00 minutes a report a week)**, and giving the two leads spans of five. The consultancy adopted a written **45-minute** weekly coaching convention and required every span above the resulting sustainable maximum to be approved with a stated reason, which is the span arithmetic turned into a governance rule. And the conversation was held in week 16 on KA 12.3.3's five-part structure with a dated factual account. Its outcome was neither a dismissal nor a recovery to plan: the engineer's difficulty was substantially capability (a modelling technique the package needed and nobody had provided), and they moved to a role where their strengths applied, with the technique taught to two others. That is a common and under-described ending, and it was available in week 2 at a fraction of the price.

**What the domain teaches here.** First, the honest caveat, because a 62-to-1 ratio is the kind of number that gets misused: it attributes the **entire** consequence to the absent conversation, which overstates it. Halve the attribution and the deferral still cost **USD 61,606.30** against a conversation costing 1,980, a ratio of **31.11 to 1**, and the conclusion is unchanged, which is what makes it worth quoting *with* the caveat. Second, **the asymmetry that causes deferral is neutralised by pricing it**: the manager was trading an hour of certain personal discomfort against a diffuse cost borne by other people, and only a number on the other side changes that trade. Third, **this was a design failure before it was a leadership failure**: an organisation that sets spans without the coaching arithmetic has decided that difficult conversations will be late. It has simply not written the decision down.

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## Executive perspective — Domain 12

What a programme director cannot delegate in this domain:

- **Your own attention budget, audited from observed data.** Recurring commitments, discretionary attention, people-work capacity and the interrup-

tion line — 82.2 %, 17.8 %, 6.5 hours and 9.0 hours on Auriga. A leadership commitment made without this number is a resource assumption nobody has checked (12.1.4).

- **The coordination cost of every team you own, at a stated link cost and as a range.** 24.375 % for 40 people in one group against 4.6875 % in five pods, a difference of 7.875 FTE. Adding people to an unstructured team is the one intervention the arithmetic says will not work (12.2.2, and D4 KA 4.2.3).
- **The span of every leader reporting to you, tested against a written coaching convention.** Not the chart's convention, the arithmetic: 7 reports at Auriga's capacity, not 8, and the way to widen it is to raise the leader's capacity, not divide a fixed budget more thinly (12.3.2).
- **The value at risk in turnover, computed honestly.** USD 66,845 per departure on Auriga's team of six and USD 46,488 on Meridian's, with the vacant post's saved pay excluded. Approve retention spending against that number, and refuse any proposal claiming an effect size it cannot evidence (12.2.4).
- **The allocation of the unsocial hours a distributed design requires.** Whether that is 10/5, 15/0 or 7.5/7.5 rotated is a decision you make, and if you do not make it, relative power will (12.4.1, 12.4.3).
- **The prohibition on AI inference about individuals.** No model output about a named person's state, engagement, personality or likely departure enters a decision, a record or a conversation: the one line in the domain with no proportionality argument, because the trust it protects cannot be re-baselined once spent.

## Calculation exercises — Domain 12

**Exercise 12.1** A 25-person team works as a single group. Each pairwise relationship consumes 0.5 hours of total team time per week; people are productive 40 hours a week. Compute the paths, the coordination hours and the share of capacity; then the same for five pods of five with one representative each, and the capacity released; then re-run both shares at a link cost of 0.8 hours. *Solution.*

Mesh: paths  $25 \times 24/2 = 300$ ; overhead  $0.5 \times 300 = 150.0$  hours; capacity  $40 \times 25 = 1,000$  hours; share **15.0 %** (equal to  $(25 - 1)/160$ ). Pods:  $5 \times (5 \times 4/2) + (5 \times 4/2) = 50 + 10 = 60$  paths; overhead **30.0** hours; share **3.0 %**. Released **120.0** hours a week, **3.0 FTE**, a reduction of **12.0** percentage points. At  $c = 0.8$ : mesh **24.0 %**, pods **4.8 %**; the conclusion is unchanged across the range, which is the point of running it. *Common error:* using  $n^2$  (625) instead of  $n(n - 1)/2$  (300), or omitting the division by two, both of which double the answer; and quoting a single link cost as though it were measured.

**Exercise 12.2** A leader has 7.0 hours a week of people-work capacity measured at 5 direct reports, and each report above five adds 0.4 hours to their fixed load. The stated sufficiency convention is 40 minutes of weekly coaching per report. Compute coaching minutes per report at 5, 8 and 9 reports, the maximum sustainable span, and the span at which coaching capacity reaches zero. *Solution.* Per report =  $(7.0 - 0.4(n - 5))/n = (9.0 - 0.4n)/n$ . At 5:  $7.0/5 = 1.4$  h = **84.00** minutes. At 8:  $5.8/8 = 0.725$  h = **43.50** minutes. At 9:  $5.4/9 = 0.6$  h = **36.00** minutes. Threshold  $n \leq (7.0 + 0.4 \times 5)/(0.6667 + 0.4) = 9.0/1.0667 = \mathbf{8.4375}$ , so the maximum sustainable span is **8**. Capacity reaches zero where  $9.0 - 0.4n = 0$ , at  $n = \mathbf{22.5}$ , so from 23 reports the leader's fixed load exceeds their entire people-work capacity. *Common error:* dividing the whole 7.0-hour budget by the report count and ignoring the growth of fixed load, which gives  $7.0/9 = \mathbf{46.67}$  minutes at nine reports and wrongly passes the 40-minute test.

**Exercise 12.3** A five-person team loses one member. The person-week is 40 hours at USD 92 an hour. Recruitment: a USD 12,500 fee and 16 hours of selection time. The post is vacant 4 weeks, covered by overtime at a 40 % premium. The replacement ramps linearly from 40 % to 100 % effectiveness over 6 weeks with 3 hours a week of mentoring; the leaver's last 3 weeks run at 70 %; handover consumes 10 hours. Compute the total, its equivalent in person-weeks, and its share of the post's annual charge-out value over 46 weeks. Then compare the overtime decision with accepting the delay, given a cost of delay of USD 22,000 a week and a five-person zero-float activity. *Solution.* Person-week  $40 \times 92 = \mathbf{USD\ 3,680}$ . Recruitment  $12,500 + 16 \times 92 = \mathbf{USD\ 13,972}$ . Ramp effectiveness  $0.40 + 0.52 + 0.64 + 0.76 + 0.88 + 1.00 = \mathbf{4.20}$ , so lost  $6 - 4.20 = \mathbf{1.80}$  person-weeks; ramp-up  $1.80 \times 3,680 + 18 \times 92 = 6,624 + 1,656 = \mathbf{USD\ 8,280}$ . Lost productivity: disengagement  $0.9 \times 3,680 = 3,312$ , handover  $10 \times 92 = 920$ , overtime premium  $0.40 \times 4 \times 3,680 = 5,888 - \mathbf{USD\ 10,120}$ . **Total USD 32,372**, or **8.80** person-weeks, or **19.12 %** of the post's annual charge-out value of  $46 \times 3,680 = 169,280$ . Uncovered, 4 person-weeks missing from a five-person crew is  $4/5 = \mathbf{0.8}$  weeks of delay costing **USD 17,600**: against a premium of 5,888, so covering is right by a wide margin, and the breakeven premium is  $17,600/(4 \times 3,680) = \mathbf{119.57\ %}$ . *Common error:* counting the vacant post's four person-weeks of pay (14,720) as a cost. The pay is saved; the cost is the work not done and the 5,888 premium paid to get it done anyway. Including it inflates the total to 41,204.

**Exercise 12.4** Three groups each work 09:00–17:00 local: P at UTC+2, Q at UTC+8, R at UTC–6. Compute all three pairwise overlaps. For the P–R pair, an average clarification needs 4 exchanges and 18 such clarifications a year sit on the critical path; the cost of delay is USD 22,000 a week over a five-day week. Compute the annual cost, then the cost and saving if P shifts 1.5 hours later and R



1.5 hours earlier. State the effect on the other pairs and the unsocial-hours burden. *Solution.* In UTC: P 07:00–15:00, Q 01:00–09:00, R 15:00–23:00. Overlaps: P–Q 2.0 hours; P–R 0.0; Q–R 0.0. Cost of delay per working day  $22,000/5 = \text{USD } 4,400$ . As-is each P–R clarification costs  $4 \times 4,400 = \text{USD } 17,600$ , and 18 of them **USD 316,800**. Shifted: P 08:30–16:30 UTC, R 13:30–21:30 UTC, overlap 3.0 hours, so a clarification completes in one day at **USD 4,400**; 18 cost **USD 79,200**, a saving of **USD 237,600**. The consequence elsewhere: P–Q falls from 2.0 hours to 0.5. The shift very nearly destroys the only other overlap in the programme, which must be decided rather than discovered. Unsocial burden: 1.5 hours a day each, 7.5 hours a week per group, already equal without rotation. *Common error:* computing overlap from local clock times (both groups "work nine to five", so they must overlap), which invents eight hours of overlap that does not exist; and fixing the target pair without recomputing every other pair.

**Exercise 12.5** A recurring task consumes 4.0 hours of a leader's week for the remaining 15 weeks. Delegated, the delegate needs 6.0 hours a week for 5 weeks and 4.0 thereafter, with leader oversight of 1.5 hours a week for 5 weeks and 0.5 thereafter. The leader is charged at USD 175 an hour, the delegate at USD 120. Compute both options, the released leader hours, the breakeven value of a released leader hour, and (valuing released hours at the leader's rate) the breakeven probability of a single USD 9,000 error incident. *Solution.* Status quo  $4.0 \times 15 = 60.0$  leader hours  $\times 175 = \text{USD } 10,500.00$ . Delegated: delegate  $6.0 \times 5 + 4.0 \times 10 = 70.0$  h  $\times 120 = 8,400.00$ ; oversight  $1.5 \times 5 + 0.5 \times 10 = 12.5$  h  $\times 175 = 2,187.50$ ; total **USD 10,587.50**. Incremental cost **USD 87.50**; released hours **47.5**. Breakeven value of a released leader hour  $87.50/47.5 = \text{USD } 1.8421$ , which is **1.053 %** of the leader's charge rate. Valuing the released hours at 175 gives  $47.5 \times 175 = \text{USD } 8,312.50$ , a net **USD 8,225.00**, so the delegation fails only if the probability of a 9,000 incident exceeds  $8,225.00/9,000 = \textbf{91.39 \%}$ . *Common error:* comparing only the two labour totals and concluding that delegation costs 87.50 more and is therefore worse, which ignores the 47.5 released hours that are the entire product being bought. The paired error is claiming the 8,312.50 without reallocating the released hours to anything, in which case none of the value exists.

## Practitioner's toolkit – Domain 12

*Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.*

### Toolkit 12.T.1 – Leader's attention and span sheet

One page per leader, completed from **four weeks of observed data** and reviewed quarterly. Top block: the working-week assumption, then each recurring commitment with its hours (governance and reporting, external stakeholders, supplier and

commercial, technical review, team ceremonies, observed interruption and escalation) the total, the discretionary residual, the personal reserve, and the resulting **people-work capacity**  $R$ . Second block: direct reports  $n$ ; the stated **sufficiency convention**  $T$ ; the per-report fixed load  $e$ ; computed coaching time per report  $(R - e(n - b))/n$ ; and the maximum sustainable span  $(R + eb)/(T + e)$ . Third block: the three largest recurring tasks consuming discretionary capacity, each with a delegation decision (delegate, retain with a reason, or eliminate), and, where delegated, the **released hours and their committed reallocation**. A span approved without this sheet is a commitment the arithmetic may not permit.

### Toolkit 12.T.2 — Team structure and coordination sheet

One page per team or programme, completed at formation and whenever headcount changes by more than a stated threshold. Columns per candidate design: team size  $n$ ; sub-team structure; required paths (computed, and separately the **needed** paths from the dependency map. The honest count is of required relationships, not the formula's maximum, exactly as Domain 4's interface register is built from need); link cost  $c$  with its basis; coordination hours; share of capacity; FTE consumed; and the same three columns at  $c \times 0.5$  and  $c \times 1.6$  as a stated sensitivity. Below: the double-membership count with its extra paths and FTE; each sub-team's external interface count against the dependency map (the test of whether the boundary is in the right place); and the named representative and deputy for each sub-team, with time allocated. A headcount increase proposed without this sheet should be refused.

### Toolkit 12.T.3 — Difficult-conversation preparation and record

Two sides of one page, used for every performance, conduct or expectation conversation. **Before:** the factual account — dated, specific, verifiable observations only, each traceable to a source in the project record; the consequence in the project's terms with a figure where one exists; the three questions you will ask and genuinely want answered; the specific expectation, its measure and its date; the support you will provide; and the anticipated responses with your answers, rehearsed. **After, within 24 hours:** what was said, the other party's account, what was agreed, the review date, and their confirmation that the record is accurate. **Standing checks,** reported monthly: the interval between the first clear instance and the conversation, per case. The single metric that tells an organisation whether its leaders can afford to have these conversations; conversations held without a contemporaneous record; and expectations past their review date. Where a formal consequence is possible, the applicable process and the human-resources and legal functions are engaged before the conversation, and the jurisdiction's requirements govern.

**The record as a record: the block this template is incomplete without.** What the two sides above produce is a dated file of observations about a named individual's work, together with that individual's own account. It is not the leader's



private aide-mémoire, and treating it as one is the commonest failure in the artefact. Six provisions, and they are cheap:

- **A named custodian.** One role accountable for the file existing and being retrievable, normally the leader while the matter is live, and the human-resources function once it is not.
- **A limited access list.** The leader, the leader's manager, and human resources. Not the team, not the successor by default, not a shared drive, and not an attachment on a thread.
- **A copy to the individual.** The template already asks for *their confirmation that the record is accurate*; that confirmation is meaningless unless they hold a copy of what they are confirming. Give it at the same time, and record that you did.
- **A retention period taken from the organisation's employment-records policy,** not set by the project and not left open-ended, with **disposal at the end of it** by a stated authority. A note about a conversation held four years ago, retained because nobody decided otherwise, serves nobody and is the version most likely to surface at the wrong moment.
- **A stated distinction between the leader's working note and the file.** The preparation side of this sheet is a working note; the "after" side, once shared and confirmed, is a document that may be produced in a formal process later. Where the matter may feed such a process, its retention, access and custody belong to the **human-resources and legal functions**, not to the leader, and they should be told the file exists.
- **Nothing here states the position in any jurisdiction.** What may be recorded about an individual, for how long, who may see it, and what entitlement the individual has to see or correct it differ substantially between territories and employment relationships, and are settled with the human-resources and legal functions and with whoever holds data-protection accountability in the organisation (Domain 11, KA 11.1.2 makes the same point about the stakeholder register). Keeping a contemporaneous record remains a non-optional professional obligation (12.3.4); how it is held is not the project's decision alone.

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## Exam preparation – Domain 12

**What is assessed.** Leadership against management by what each supplies; the theory families with their honest uses and limits; emotional intelligence as four practised operations; **the attention audit and people-work capacity**; the motivation frameworks with their evidential status, and condition design over incentive design; team formation stages and the leader's task at each; **the arithmetic of team size: paths, coordination share, net capacity, marginal capacity and structured designs**; psychological safety, the safety-standards pairing and its countable signals; **turnover cost on the three-component build-up with the**

**saved-pay exclusion; delegation economics and its breakevens; span of control and coaching capacity;** the structure and timing of a difficult conversation; performance management's generalisable obligations and jurisdictional limits; **working-hour overlap, round-trip latency and the fairness of an overlap window;** culture without stereotype; the ethical situations of delivery leadership; and the prohibitions on AI inference about individuals.

**The calculations to be able to do under time pressure.**  $n(n-1)/2$  paths and the coordination share  $c(n-1)/(2h)$ ; net capacity  $hn - c \cdot n(n-1)/2$  and marginal capacity  $h - c(n-1)$  with the size at which it reaches zero; the path count, share and released FTE for a pod design; the attention audit's totals and shares; coaching time per report  $(R - e(n-b))/n$  and the maximum sustainable span  $(R + eb)/(T + e)$ ; the three-component turnover cost, in person-weeks and as a share of a post's annual value, plus the breakeven overtime premium against a delay; delegation cost comparison, released hours, the breakeven value of a released hour and the breakeven error probability; and working-hour overlaps in a common time reference with the latency, annual cost, saving and unsocial-hours allocation that follow.

**The traps.** Using  $n^2$  or omitting the division by two in the path count, both of which double coordination (Exercise 12.1) · counting only one side of a communication link, which halves it (MCQ 12.2-A) · attributing all coordination hours to the leader (MCQ 12.2-A) · confusing the size at which marginal capacity reaches zero (81) with the size at which coordination consumes all capacity (161) (MCQ 12.2-B) · quoting a single link cost as measured rather than presenting a range (12.2.2) · counting a vacant post's saved pay as a turnover cost (MCQ 12.2-C, Exercise 12.3) · reporting a turnover total without the ramp-up and lost-productivity components, which are 60.2 % of it (12.2.4) · dividing a leader's whole discretionary budget by the report count and ignoring the growth of fixed load, which overstates the sustainable span by one (Exercise 12.2, MCQ 12.3-A) · confusing the zero-coaching span with the sustainable span (MCQ 12.3-A) · comparing only the labour totals in a delegation decision and missing the released hours (Exercise 12.5, MCQ 12.3-B) · claiming the value of released hours without reallocating them (12.3.1) · computing time-zone overlap from local clock times without converting to a common reference (MCQ 12.4-A, Exercise 12.4) · fixing one distributed pair without recomputing the others (MCQ 12.4-C, Exercise 12.4) · quoting an effect size for a leadership or culture intervention that the evidence does not support (12.2.1, 12.2.3, 12.2.4) · and using a national-culture average to predict an individual (MCQ 12.4-D).

**How the domain connects.** Domain 1 supplies the accountability principle, the benefits chain that makes the work's purpose truthful, and Meridian's cost of delay at which every latency here is priced. Domain 3 supplies the bounded envelope this domain's interruption arithmetic depends on, the  $E[\text{wait}]$  latency logic KA 12.4.1 reuses, the threshold economics KA 12.3.1 parallels rather than repeats, the decision-record discipline behind the conversation record, and the recovery structure 12.A.1 completes. Domain 4 supplies  $n(n-1)/2$  and the layered-versus-mesh choice this

domain applies to people, plus the interface re-verification cost that prices the delegation risk. Domain 6's resource levelling is the schedule form of the same capacity problem. Domain 7 supplies the blended rate and engineer-week every figure here is denominated in. Domain 9 supplies the countable psychological-safety signal (issues found versus reported), and the Auriga commissioning team this domain prices a resignation from. Domain 11 supplies the stakeholder and communication skills this domain assumes when the audience is outside the team. Domain 13's adaptive delivery depends on the safety and bounded discretion designed here. Domain 14 carries the systematic responsible-AI treatment whose people-specific prohibitions this domain states. And Domain 15 applies KA 12.2.2 and 12.A.2 at enterprise scale, where the coordination arithmetic is the dominant cost.

## Domain 12 summary

Leadership in delivery supplies what a plan cannot: a truthful reason the work matters, interpretation under ambiguity, the conditions in which people tell the truth, and named bounded discretion. The theory families are lenses with honest uses and real limits, trait and personality approaches describe differences without predicting outcomes; situational diagnosis is the most useful discipline among them; emotional intelligence is four practised operations, not a score. **This domain refuses to quote effect sizes it cannot evidence, and finds instead the parts of the subject that are genuinely computable.**

The leader's attention is the binding constraint and is never baselined. Auriga's leader carries **37.0 hours** of recurring commitments in a 45.0-hour week: **82.2 %**: leaving **8.0 hours** of discretionary attention and **6.5 hours** of people-work capacity, **14.4 %** of the week, from which every coaching conversation, delegation and team-design decision is paid. The largest single line, **9.0 hours** of interruption, is **20.0 %** of the week and nobody decided it.

Team size is arithmetic. Communication paths grow as  $n(n - 1)/2$  (Domain 4, KA 4.2.3), and at a link cost of 0.5 hours of team time per week on 40-hour weeks the coordination share is the identity  $(n - 1)/160$ : **3.125 %** for Auriga's six, **6.875 %** for Meridian's original twelve, and **24.375 % (9.75 full-time people)** for the forty it became. Five pods of eight carry **150** paths instead of **780**, **4.6875 %** instead of 24.375 %, releasing **7.875 FTE** at no cost. Net capacity  $hn - c \cdot n(n - 1)/2$  is **1,620 hours** at both 80 and 81 people, so the 81st member of a fully connected team adds exactly nothing, a model, not a law, whose conclusion (structure your teams) is robust across every plausible link cost while its specific ceiling is not. Meridian banked **USD 85,680** of realised acceleration against **USD 1,451,520** of theoretical released capacity, a ratio of **16.94**: the honest distance between capacity released and value realised.

Turnover is arithmetic. One resignation from Auriga's team of six costs **USD 66,845** (recruitment 26,612.50, ramp-up 18,810.00, lost productivity 21,422.50) **12.79 engineer-weeks** and **26.65 %** of the post's annual charge-out value, of which recruitment is only **39.8 %** and the **28.1 %** that is ramp-up falls after the vacancy is filled and the problem thought closed. The vacant post's pay is **not** a cost: it is saved, and counting it inflates the figure to 82,520. The overtime cover that avoided a **1.0-week** delay worth **USD 45,000** cost **USD 15,675**, and would have been worth paying up to a **143.54 %** premium. That number sets the retention budget a leader may argue for; it does not establish that any intervention earns it.

Span of control is arithmetic. Coaching time per report is  $(R - e(n - b))/n$  (**65.00** minutes at six reports, **41.25** at eight, **21.82** at eleven), and against a stated 45-minute convention the maximum sustainable span is 7, not the **8** the naive  $6.5/n$  curve gives, because the leader's own fixed load grows with the team; the way to widen a span is to raise R, and recovering 3.0 hours of interruption moves it from 7 to **10**. Delegation is arithmetic too: handing over Auriga's interface coordination costs **USD 430** more in labour and releases **34.0** leader hours, **5.23 weeks** of discretionary capacity, so it pays if a leader hour is worth more than **USD 12.65** (**6.836 %** of its own charge rate), and fails only if a 6,000-dollar error is more than **97.67 %** likely. Structurally Domain 3's threshold result, for the same reason.

Distance is arithmetic. Meridian's core team and clinical advisers, both working nine to five, overlap for **0.0 hours**, so each three-exchange clarification takes three working days and the 26 on the critical path cost **USD 222,768** a year; a **3.0-hour** window created by a two-hour and a one-hour shift saves **USD 148,512**, costs 2.0 hours of the supplier's overlap, and is bought with 3.0 hours a day of unsocial working whose allocation (10/5, 15/0, or 7.5/7.5 rotated) is an ethical decision a leader must make explicitly. Culture varies in ways worth anticipating and not in ways that predict an individual, and the ethical obligations (reporting what is true, refusing a date you do not believe, protecting the team, stopping for safety, being fair in the decisions people watch) are the ones from which a team infers what a leader actually values. And where the accountable person, told in writing, decides the other way, the obligation has a mechanism rather than a sentiment: a **dissent** recorded in the decision log with a copy held outside the decision-maker's reach, a **signature declined** in writing with what would have to change, and (where the risk is harm to people) an **escalation route** stated in advance that runs past the accountable person to the next tier, to independent assurance, and to the board or its audit committee. Arrangements protecting a person who raises a concern are understood to exist in many jurisdictions and differ materially; nothing here states the position in any of them, and the organisation's policy and legal function are engaged before any external step.

The through-line: **the human capacity of a delivery organisation is a designed quantity with a computable cost**. Case study B is the domain in one sentence, a conversation worth USD 1,980 deferred fourteen weeks at a cost of USD 123,212.60, by a manager whose eleven reports left them 21.82 minutes each per week. Compute

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it, and what looked like a culture problem becomes a design decision: the only kind of problem an organisation reliably fixes.

DOMAIN

# 13

## Agile, Adaptive and Hybrid Delivery

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## Why this domain exists

Part Three has been about people: who is engaged (Domain 11), who leads and how a team is built (Domain 12). Both assumed a **way of working** (a cadence, a unit of delivery, a rhythm at which decisions arrive), and neither specified it. Part Two specified one: a network, a baseline, a variance. That specification is right for a great deal of project work and wrong for a great deal more, and the gap between the two is where this domain lives.

The gap is usually discussed as a preference, which is why the discussion goes nowhere. Practitioners declare for a method, defend it, and attribute failures to the other camp's contamination. The professional position is narrower and duller: **adaptive delivery is a set of mechanisms with measurable properties, and so is predictive delivery, and a leader's job is to know which properties each supplies and what each costs.** That is the domain's central claim. It has a consequence a candidate should expect to be tested on: almost every complaint made about agile delivery in large organisations ("the team keeps changing the date", "we can't get a commitment", "governance is constantly bypassed", "the contract doesn't fit") is not a cultural complaint at all. Each is an arithmetic statement about throughput, work in progress, latency or commercial exposure, and each has a number attached that nobody has computed.

This domain computes them. KA 13.1 strips product ownership back to the decision right it is and prices the sequencing decision that prioritisation *is*: worth **USD 155,040** on Meridian's release, and routinely taken by whoever argues best. KA 13.2 builds the flow arithmetic and applies it to the commonest management instruction in delivery, *start more work*, which on Meridian buys nothing and costs **USD 180,379**; it then replaces the forecast date with a forecast range. KA 13.3 quantifies the mismatch Domain 3 named, a monthly committee governing a fortnightly cadence parks **20.0 %** of a team's work in progress and makes the affected items take **2.667 times** as long as the rest. KA 13.4 compares a capacity-based and a scope-based commercial model on the only test that discriminates between them, what a change costs the buyer, and closes on metrics that survive incentives and the anti-patterns that do not.

The through-line: **adaptive delivery is not a licence to stop measuring; it is a different set of things to measure, and every one of them is a number.**

**Learning objectives.** After this domain a candidate can: state the conditions under which adaptive delivery outperforms predictive delivery and the conditions under which it does not, in terms of feedback cost and requirement volatility rather than preference; specify product ownership as a decision right with a named holder, and apply the decidability test of Domain 3 to it; **state the two bounds on the ordering right (the value envelope, and the obligation items the right does not reach), and exclude those items from the priced ranking**; rank a backlog by

**delay-cost density** and compute the delay cost of a chosen sequence, its saving against the intuitive ordering, and the breakeven estimate error that would change the ranking; state and apply **Little's Law** in both directions: deriving cycle time from work in progress and throughput, and deriving the throughput implied by a cycle-time change at fixed work in progress; compute what raising work in progress does to throughput, cycle time, flow efficiency and the delivery date, and price it; forecast from throughput history **as a range with a stated meaning**, adjusted for discovery arrivals, and test a committed date against the best rate the team has ever sustained; compute governance latency against iteration length using Domain 3's  $E[\text{wait}] = M/2 + L$ , express the mismatch as blocked work in progress and as a cycle-time decomposition, and price the remedy; design hybrid delivery with a named boundary, two honest control regimes and a translation rule for reporting; compare capacity-based and scope-based commercial models on buyer exposure under a change, including the breakeven number of changes; select flow metrics that resist gaming and explain why velocity is a planning input and never a performance measure; diagnose the standard anti-patterns with their arithmetic signature; and govern AI-assisted backlog work and forecasting without letting a model's output become a commitment.

**The master programme.** Meridian Care Records continues from Domains 1–4 and 11–12: the clinical-records rollout to **40 clinics**, approved cost **USD 2,400,000**, benefit **USD 685,440** a year at the realistic 70 % adoption (against **USD 979,200** at full potential), and the **cost of delay of USD 14,280 per week** derived in Domain 1 from 28 adopting clinics at 6 hours a week and USD 85 an hour. Meridian's shape is exactly the shape this domain is about: an **iteratively developed records application**, a **sequentially rolled-out** clinic programme with immovable estate and training dependencies, and a regulatory approval that is a genuine gate (Domain 3, KA 3.1.3). The application build carries a **240-item** release backlog delivered by a team whose measured throughput is **6 items a week**, governed by a steering committee whose expected decision latency Domain 3 computed at **4.0 weeks** from  $M = 4$ ,  $L = 2$ . Those five numbers (240, 6, 4.0, 14,280 and the 2-week iteration) generate almost every result in this domain. Project Auriga, the 25-week control-systems upgrade of Domains 6–8, reappears in the exercises where a physical commissioning queue makes the same flow arithmetic behave differently.

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#### KNOWLEDGE AREA 13.1

### Agile principles and product ownership

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*Topics: 13.1.1 what adaptive delivery actually is · 13.1.2 product ownership as a decision right · 13.1.3 prioritisation as a priced sequencing decision · 13.1.4 the increment and the definition of done.*



### 13.1.1 What adaptive delivery actually is

**Definition.** Adaptive delivery is a delivery mode in which **capacity and cadence are fixed and scope is variable**, direction is corrected from working output at short intervals, and commitment to detail is deliberately deferred until the cost of deciding is lowest. Predictive delivery fixes scope and varies capacity and time to achieve it. Neither is a philosophy; each is a choice about **which variable absorbs uncertainty**.

That framing does the work that value statements cannot. The historical literature of agile methods — the collaborative statement of values published by a group of software practitioners in 2001, and the named frameworks that followed it (Scrum, Kanban, DSDM, and the various scaling frameworks) — is worth reading and is not reproduced here; what a leader needs is the decision rule underneath it. And the decision rule is economic. Adaptive delivery outperforms predictive delivery when three conditions hold together, and a leader should be able to test each:

1. **Requirements are genuinely volatile or genuinely unknown.** Not "poorly documented": unknowable in advance, because they depend on what users do with something that does not exist yet. Where requirements are stable and knowable, deferring commitment buys nothing and costs coordination.
2. **Feedback is cheap and fast.** An increment can be put in front of someone who can judge it, within the iteration, at a cost small relative to the work. Where feedback requires a commissioned plant, a regulatory submission or a construction season, the iteration cannot close its loop and the method loses its engine (13.A.1).
3. **The cost of a wrong decision falls with delay, or the cost of reversal is low.** Deferring commitment is only valuable if information arrives; if the decision is irreversible and the information will not improve, deferring it is procrastination with a vocabulary.

**What adaptive delivery does not do.** It does not remove the baseline; it moves it onto capacity, cadence and the value envelope (Domain 5, KA 5.A.1). It does not remove governance; it changes the governance question from variance to continuation (Domain 3, KA 3.1.3). It does not remove the date; it changes what an honest date looks like, from a point to a range with a stated basis (13.2.4). And it does not make a project cheaper by itself: the case for it rests on **not building the wrong thing**, which is a value argument, so a programme that adopts adaptive delivery while retaining a fixed scope has kept the cost and discarded the benefit.

**The honest sentence.** Every method is a rule about which of scope, time and cost floats, and the useful professional habit is to say out loud which is floating on *this* work: "capacity and date fixed, scope floats" (adaptive); "scope and date fixed, cost floats" (a crash decision, Domain 6 KA 6.4.2); "scope and cost fixed, date floats" (most infrastructure). An organisation that cannot say the sentence has not chosen a method; it has chosen to be surprised.

### 13.1.2 Product ownership as a decision right

**Definition.** The product owner is the single individual accountable for the **order** of the backlog (for deciding what is built next, and therefore what is not built next) within a value envelope set by the sponsor. It is a decision right, not a facilitation role, and Domain 3's decidability test (KA 3.1.1) applies to it unchanged: exactly one accountable holder, with sufficient authority, or the decision is not made but taken by whoever is present.

The role fails in four recognisable ways, and each has a test a leader can run in an afternoon.

**The proxy owner** must consult a committee before ordering the backlog, so the effective authority and the latency are the committee's, the whole subject of KA 13.3.2. *Test:* count the ordering decisions made alone in the last quarter, and the elapsed time on the rest.

**The unfunded owner** holds authority over order but not over the value envelope, so they can sequence but not decline, and backlogs grow monotonically because refusal has no owner. *Test:* count items removed in the last quarter. A backlog from which nothing is ever removed is a wish list with a burndown chart.

**The absent owner** is available for ceremonies and unavailable between them, so the team decides by default and the ordering drifts to whatever is technically convenient. *Test:* the team's median wait for a product-owner decision, a number worth putting on the wall.

**The specifier owner** writes solutions rather than ordering outcomes, becoming a bottleneck on design and destroying the team's ability to propose. *Test:* the share of items stating a required outcome and an acceptance criterion rather than an implementation.

**What the role must be able to do.** Four capabilities, and the absence of any one is a design defect rather than a personal failing: state the value of an item in the terms of the benefits map (Domain 2), so that ordering is arguable on evidence; decline an item and record the refusal; accept or reject an increment against pre-agreed criteria (Domain 5, KA 5.4.2); and escalate a decision that exceeds their envelope, within a stated latency (Domain 3, KA 3.3.3). Meridian's product owner holds an envelope of **USD 25,000** per change (the threshold Domain 3's KA 3.2.3 arithmetic recommended), and the 15 % of items that exceed it are what KA 13.3.2 prices.

**What the ordering right does not reach.** The value envelope is one boundary on the role and it is not the only one, and the second is the one that causes damage when it is left unstated. **The product owner orders discretionary value.** Items that carry a safety, regulatory, statutory, contractual or security-remediation obligation are **constraints on the order, not candidates within it**. Three con-

sequences follow, and they are stated here rather than a thousand lines later because this is where the decision right is defined:

- **They are scheduled to their required date by the authority accountable for the obligation** (the clinical-safety authority, the information-governance authority, the security owner, the contract manager) not by the product owner, and not by the team's estimate of when it is convenient.
- **They are excluded from the delay-cost-density ranking of 13.1.3.** A priced ranking answers the question *what is most valuable to build next?*, and that question is not open for an obligation. Ranking such an item alongside discretionary features does not produce a rigorous answer; it produces a computed number that will sometimes place a clinical-safety mitigation below a reporting enhancement, which is not a trade-off but a category error. Where the obligation itself admits options (*how* it is met, at what cost, in what sequence among other obligations) the arithmetic returns, applied to the options rather than to whether.
- **A product owner who deprioritises one has exceeded the right rather than exercised it.** This is the same distinction Domain 3 draws between a decision class and its authority, and Domain 8, KA 8.3.1 draws between a priced response and a duty: authority over the order of value is not authority over what must be done. The countermeasure is a **classification on the backlog item itself**, applied when the item is created and owned by the accountable authority (obligation or discretionary), because a distinction that has to be remembered at ranking time will not be.

Whether an obligation of this kind is in fact engaged, by what, and on whom, is established with the accountable authority for that class and, where it is a question of law or regulation, with qualified counsel: not by the product owner and not by this book, which states no position anywhere. The healthcare Industry variation below shows the governance half of the same rule in operation: clinical sign-off sits with clinical authority and those classes cannot be resolved by a written-resolution route.

### 13.1.3 Prioritisation as a priced sequencing decision

**The definition and the point.** Prioritisation is not ranking by importance; it is choosing a **sequence**, and a sequence has a computable cost because value forgone accrues per unit of time. The instrument is **delay-cost density**: the cost of delay of an item divided by the effort required to deliver it.

$$\text{delay-cost density} = \text{cost of delay per week} \div \text{effort in team-weeks}$$

Sequencing in decreasing order of that ratio minimises the total cost of delay across a set of items delivered one at a time by a single team. This is not a framework opinion but a classical result in single-machine scheduling theory (the weighted-

shortest-processing-time ordering, which minimises total weighted completion time), and ratio rules of this shape appear under various names in the scaling frameworks. What matters for a leader is that the rule is *derivable*, so its assumptions are visible: one delivery resource, items independent, each item releasable on completion, and effort and cost of delay both estimated. Every one of those assumptions is a place the rule can fail, and 13.1.3 closes by pricing the most important failure.

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#### WORKED EXAMPLE

#### Worked example 13.1.3 — sequencing Meridian's four release epics.

1. **Setup.** Meridian's 240-item release comprises four epics. Domain 2's benefits map apportions all six hours per clinic per week of clinical time saved across the four capabilities, so the four epics between them carry the whole of the programme's **USD 14,280 per week** cost of delay (Domain 1). One team delivers them sequentially; total effort is **34 team-weeks**.

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| EPIC                    | COST OF<br>DELAY<br>(USD/<br>WEEK) | SHARE OF<br>PROGRAMME | EFFORT<br>(TEAM-<br>WEEKS) | DENSITY (USD/<br>WEEK PER<br>TEAM-WEEK) |
|-------------------------|------------------------------------|-----------------------|----------------------------|-----------------------------------------|
| E2 clinical notes       | 6,120                              | 42.9 %                | 14                         | 437.14                                  |
| E1 appointment booking  | 4,080                              | 28.6 %                | 6                          | 680.00                                  |
| E3 regulatory reporting | 2,720                              | 19.0 %                | 3                          | 906.67                                  |
| E4 patient portal       | 1,360                              | 9.5 %                 | 11                         | 123.64                                  |

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2. **Formula.** Density = cost of delay ÷ effort. Total delay cost of a sequence =  $\Sigma$  (cost of delay of each epic × the week in which that epic completes).
3. **Substitution.** Density order is E3, E1, E2, E4, completing in weeks 3, 9, 23 and 34:  $2,720 \times 3 + 4,080 \times 9 + 6,120 \times 23 + 1,360 \times 34$ . The intuitive order, largest cost of delay first, is E2, E1, E3, E4, completing in weeks 14, 20, 23 and 34:  $6,120 \times 14 + 4,080 \times 20 + 2,720 \times 23 + 1,360 \times 34$ .
4. **Result.** Density order **USD 231,880**. Cost-of-delay-first order **USD 276,080**. The density order saves **USD 44,200**. The worst available order (E4, E2, E1, E3: lowest density first) costs **USD 386,920**, so the spread between the best and worst sequences of the *same scope, same effort and same team* is **USD 155,040**.

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5. **Interpretation.** The first thing to take from this is that sequencing is a **USD 155,040 decision**, and in most organisations it is taken in a room by whoever advocates most effectively, with no number in front of anyone. The second is that the intuitive rule (do the biggest thing first) is wrong by **USD 44,200**, and wrong for a reason worth internalising: E2 carries the largest benefit but also fourteen weeks of effort, during which the three smaller epics each accrue their own cost of delay. Doing the cheap, high-value work first is arithmetic, not opportunism. Third, the ranking is **robust to moderate estimation error**, which is what makes it usable with imperfect inputs: E2 would have to carry a cost of delay above  $680 \times 14 = \text{USD } 9,520$  per week (a 55.6 % increase on its estimate), before it belonged at the front, so a leader can defend the order without defending the third decimal place. Fourth, and most important, the entire **USD 155,040** is created by **releasability**. If nothing can go live until all four epics are complete, every epic completes at week 34 and the total delay cost is  $14,280 \times 34 = \text{USD } 485,520$  regardless of order, so incremental release is worth  $485,520 - 231,880 = \text{USD } 253,640$  on this release, which is a larger number than the sequencing decision it enables and is usually treated as a technical detail. Two cautions. The rule assumes independence, and where E1 must precede E3 technically, the dependency binds and the sequence is constrained: price the constraint rather than ignoring the rule. And a cost of delay estimated from advocacy rather than from the benefits map will produce a confidently wrong order; the density is only as good as its numerator, which is why Domain 2's value attribution is a prerequisite and not a nicety.

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#### 13.1.4 The increment and the definition of done

**Definition.** An increment is a slice of the product that is **complete against a pre-agreed standard**, the definition of done, and therefore capable of being judged, released or rejected on its own. The definition of done is that standard, written once, applied unchanged to every item.

The increment is the unit that makes adaptive governance possible, because it is the only thing a governance body can inspect that is not a claim. A percentage complete is a claim; a working, tested, documented slice of service is evidence. This is why Domain 3's KA 3.1.3 puts the increment review at the centre of iterative governance and why the definition of done is a **governance artefact** rather than a team convention: it defines what the word "done" means in every report the programme issues, and a definition that is quietly relaxed under schedule pressure corrupts every subsequent number.

**What belongs in it.** The standard should be short, binary and testable: the acceptance criteria met (Domain 5, KA 5.4.2); the tests written and passing; the security and privacy checks completed, which for Meridian's clinical records is a regulatory obligation and not a preference; the documentation and operational handover material produced (Domain 16); no known defects above an agreed severity; and the

item demonstrable in an environment representative of production. Each is either true or it is not.

**The two failure modes.** The **relaxed definition** (items counted as done with tests or documentation deferred) creates a debt that Domain 9's KA 9.4 quantifies as a rework share of capacity, degrading throughput at the multiplier  $1/(1 - r)$  established there. The **inflated definition** (a standard so demanding that no item ever meets it) produces a team with a full board and no throughput, which the flow measures of KA 13.2 detect immediately as a cycle time rising with no change in work in progress.

### AI in this KA

**Where it earns its place.** Splitting a large item into candidate thin slices that each satisfy the definition of done, for the team to accept, amend or reject, a generative task with cheap verification, because a bad slice is obvious to the people who would build it. Drafting acceptance criteria from an outcome statement, then being checked against Domain 5's criteria tests. Computing delay-cost density and the total delay cost of alternative sequences across a large backlog, with the sensitivity analysis of 13.1.3 swept automatically: deterministic arithmetic over many combinations, which is exactly what a machine is for and no human has time to do. Detecting backlog items that duplicate one another or that carry no stated value, which is a text-comparison task humans perform badly at scale.

**Where it must not go.** It must not set the order. Ordering a backlog allocates the organisation's capacity between competing claims on it, and that is a decision right held by a named accountable person under 13.1.2; a model that produces an order has produced a recommendation, and a product owner who adopts it unexamined has vacated the role rather than automated it. It must not estimate a cost of delay from plausibility: asked for one, a model will produce a confident number with no provenance, and because the density ranking is proportional to that numerator the fabrication propagates straight into a sequencing decision worth six figures. And it must not write the definition of done, which encodes the organisation's quality and regulatory obligations.

**Verification, concretely.** Every AI-produced sequence is re-derived by hand for the top five items, five divisions, and the cost-of-delay inputs are traced to a benefits-map line with an owner, not to a model output. Every AI-drafted acceptance criterion is tested against a real example and a real counter-example before it is accepted. Where a model has split an item, the team confirms that each slice independently satisfies the definition of done, because a slice that cannot be released alone has re-created the big-bang release the arithmetic above shows to be worth **USD 253,640** to avoid.

## KEY TERMS — KA 13.1

| TERM                                      | MEANING                                                                                                                                                                                                                                                                                         |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Adaptive delivery</b>                  | A delivery mode fixing capacity and cadence and varying scope, correcting direction from working output at short intervals.                                                                                                                                                                     |
| <b>Hybrid delivery</b>                    | A deliberate combination of predictive and adaptive methods under one governance frame, with a named boundary between them.                                                                                                                                                                     |
| <b>Product owner</b>                      | The single individual accountable for the order of the backlog within a value envelope set by the sponsor.                                                                                                                                                                                      |
| <b>Value envelope</b>                     | The bounded authority within which a product owner may commit capacity without escalation.                                                                                                                                                                                                      |
| <b>Obligation item</b>                    | A backlog item carrying a safety, regulatory, statutory, contractual or security-remediation obligation. A constraint on the order, not a candidate within it: scheduled to its required date by the authority accountable for the obligation and excluded from the delay-cost-density ranking. |
| <b>Ordering right, and its two bounds</b> | The product owner orders <i>discretionary</i> value, within a value envelope above which decisions escalate and outside the obligation items the ordering right does not reach. Deprioritising an obligation item exceeds the right rather than exercising it.                                  |
| <b>Delay-cost density</b>                 | Cost of delay per week ÷ effort in team-weeks; sequencing in decreasing order minimises total delay cost.                                                                                                                                                                                       |
| <b>Increment</b>                          | A slice of product complete against the definition of done and therefore judgeable, releasable or rejectable alone.                                                                                                                                                                             |
| <b>Definition of done</b>                 | The pre-agreed, binary, testable standard that gives the word "done" one meaning across every report.                                                                                                                                                                                           |
| <b>Releasability</b>                      | The property that an increment can go live alone; it is what converts sequencing into value.                                                                                                                                                                                                    |

## SAMPLE MCQS — KA 13.1

**MCQ 13.1-A** [13.1.3 · Application] Four epics carry costs of delay of 6,120, 4,080, 2,720 and 1,360 per week and efforts of 14, 6, 3 and 11 team-weeks respectively. The sequence that minimises total delay cost is:

- A. 6,120 first, then 4,080, 2,720, 1,360 — largest cost of delay first
- B. 2,720 first, then 4,080, 6,120, 1,360 — decreasing cost of delay per team-week
- CORRECT
- C. 2,720 first, then 4,080, 1,360, 6,120 — shortest effort first
- D. 1,360 first, then 6,120, 4,080, 2,720 — lowest density first

*Rationale:* Densities are 906.67, 680.00, 437.14 and 123.64, so the order is 2,720 → 4,080 → 6,120 → 1,360, costing **231,880** (13.1.3). A is the intuitive largest-first error, costing **276,080**: 44,200 worse. C ranks on effort alone, which promotes the low-

value 1,360 epic ahead of the 6,120 one and costs **280,160**. D reverses the density rule and is the worst available order at **386,920**.

**MCQ 13.1-B** [13.1.3 · Evaluation] For the same four epics, if no epic can be released until all four are complete at week 34, the total delay cost is:

- A. USD 231,880, unchanged — sequencing still helps
- B. USD 485,520, and sequencing is worth nothing CORRECT
- C. USD 155,040
- D. USD 253,640

*Rationale:* With a single release every epic completes at week 34, so the cost is  $14,280 \times 34 = 485,520$  whatever the order, the whole value of sequencing comes from releasability (13.1.3). C is the best-to-worst sequencing spread; D is the value of releasability itself,  $485,520 - 231,880$ .

**MCQ 13.1-C** [13.1.2 · Analysis] A named product owner must obtain a steering committee's agreement before changing the backlog order. The most accurate description is that:

- A. the arrangement strengthens governance by adding scrutiny
- B. the effective decision right, and therefore the latency, belongs to the committee CORRECT
- C. the product owner remains accountable and the arrangement is sound
- D. the team should escalate less often

*Rationale:* Authority is where the binding agreement sits, so the product owner is a proxy and the committee's latency applies to every ordering decision (13.1.2, and Domain 3 KA 3.1.1's decidability test). A confuses scrutiny with authority; C mistakes the title for the right.

**MCQ 13.1-D** [13.1.1 · Analysis] A programme adopts adaptive delivery but keeps its scope fixed by contract. The predictable consequence is that:

- A. delivery becomes faster because iterations are short
- B. it retains the coordination cost of the method and forgoes its benefit, which comes from not building the wrong thing CORRECT
- C. quality improves because increments are tested
- D. the cost of delay falls

*Rationale:* The economic case for adaptive delivery rests on varying scope in response to feedback; fixing scope removes the mechanism and leaves the overhead (13.1.1).



**MCQ 13.1-E** [13.1.4 · Comprehension] A team, under schedule pressure, begins counting items as done with tests deferred. The measure that detects this soonest is:

- A. velocity, which will fall
- B. the count of items completed per iteration, which will rise
- C. the rework share of capacity in later iterations, which degrades throughput at  $1/(1 - r)$  CORRECT
- D. stakeholder satisfaction

*Rationale:* A relaxed definition of done initially *raises* reported completion and later shows as rework consuming capacity, at Domain 9 KA 9.2.3's multiplier (13.1.4). A and B move the wrong way; D is a lagging and non-specific signal.

SELF-CHECK — KA 13.1

1. *State the three conditions under which adaptive delivery outperforms predictive delivery.* — Requirements genuinely volatile or unknowable; feedback cheap and fast enough to close inside the iteration; and deferral genuinely valuable because information arrives or reversal is cheap.
2. *Why does ranking by cost of delay alone give the wrong sequence?* — Because a large, slow item blocks several smaller ones that are each accruing their own cost of delay; the correct ranking is cost of delay per unit of effort.
3. *What single property creates the value of sequencing, and what happens without it?* — Releasability. Without it every item completes at the end and the total delay cost is the same for every order: on Meridian, **USD 485,520** instead of **USD 231,880**.
4. *Name the two bounds on a product owner's ordering right.* — The **value envelope** set by the sponsor, above which the decision escalates; and the **obligation items** (safety, regulatory, statutory, contractual, security-remediation), which are constraints on the order rather than candidates within it, scheduled by the authority accountable for the obligation and excluded from the delay-cost-density ranking. Deprioritising one exceeds the right rather than exercising it (13.1.2).

KNOWLEDGE AREA 13.2

## Backlogs, iteration planning, flow and Kanban

*Topics:* 13.2.1 the backlog as an instrument · 13.2.2 iteration planning and the commitment fallacy · 13.2.3 flow: throughput, work in progress and cycle time · 13.2.4 forecasting from throughput history.

### 13.2.1 The backlog as an instrument

**Definition.** A backlog is an **ordered** list of items, each **attributed** to a benefit, each **sized**, and each carrying an acceptance criterion. Remove any one of those four properties and it stops being an instrument: an unordered backlog is a wish list, an unattributed one is prioritised on advocacy (Domain 5, KA 5.3.1), an unsized one cannot be forecast, and one without criteria cannot be accepted.

Three disciplines keep it usable, and each is cheap.

**Refinement is a capacity cost and should be budgeted as one.** Items near the front must be understood well enough to start; items far back must not be, because elaborating them wastes effort on work that may never be built and creates a sunk-cost pull towards building it, Domain 6's rolling-wave principle, applied to a backlog: elaborate to an explicit horizon.

**The value-envelope reconciliation is the creep control.** A requirement count means nothing where the count is supposed to move, so the adaptive equivalent of Domain 5's creep test is the reconciliation Domain 5 KA 5.A.1 specifies, attributed benefit of delivered items against the business case's commitment, period by period. It detects the failure adaptive programmes actually suffer: delivering steadily against a backlog that has drifted from the benefits it was funded to produce. Case study B is that failure, costed.

**Discovery arrivals must be counted.** Items found necessary during build are the method working, not scope creep. The defect is failing to count them, because an uncounted arrival rate makes every forecast optimistic by exactly its magnitude. Meridian's measured rate is **1.5 items per week**, and KA 13.2.4 shows what ignoring it does to a date.

### 13.2.2 Iteration planning and the commitment fallacy

Iteration planning answers one question (what will the team take on in the next iteration), and it is routinely corrupted by being asked to answer a different one: what will the team promise. The difference matters because the two have different arithmetic.

**Capacity is an observation, not an aspiration.** Meridian's team completes **6 items a week**, so a 2-week iteration has a capacity of **12 items**. That is a measurement, and it is the only defensible input to a plan.

**The commitment fallacy.** Committing to the mean is committing to fail about half the time. Meridian's last twelve weeks, paired into six iterations, produced **12, 10, 13, 13, 13 and 11** items. The mean is exactly **12.00**, and **2 of the 6 iterations: 33.3 %**: delivered fewer than 12. A team held to the mean therefore misses its commitment in a third of iterations while performing exactly to its own average, and the organisational response is invariably to question the team rather than the arithmetic. Committing instead to the **lowest observed total, 10 items**, would have been met

in every observed iteration, at the cost of forecasting less throughput than the team actually has. The professional resolution is to stop conflating the two numbers: **plan with the mean and commit with the low end**, and report both. A commitment is a statement about reliability; a forecast is a statement about expectation; and an organisation that has one number is using it for both purposes and being misled at least once.

**What the iteration boundary is actually for.** Three things, none of which is a status report: a **re-ordering opportunity**, the point at which new information legitimately changes the sequence. The mechanism that makes KA 13.4.1's commercial comparison come out as it does; an **acceptance point**, where the increment meets the definition of done or does not; and a **continuation decision** in Domain 3 KA 3.1.3's sense: is the next iteration's cost worth its expected value? A programme whose iteration boundaries produce reports rather than decisions has bought the cadence and left the value on the table.

### 13.2.3 Flow: throughput, work in progress and cycle time

This is the quantitative core of the domain, and it rests on one relationship.

**Definitions.** **Throughput** ( $T$ ) is completed items per unit of time. **Work in progress** ( $W$ ) is the number of items started and not finished. **Cycle time** ( $C$ ) is the elapsed time from an item starting to its finishing. For a system at a steady state, **Little's Law** relates the three:

$$\begin{array}{l} W = T \times C \quad \text{equivalently} \quad C = W / T \\ \text{and} \quad T = W / C \end{array}$$

Little's Law is a theorem, not a heuristic. It holds for any queueing system in steady state, independent of the arrival distribution, the service distribution or the queue discipline. Two consequences follow that make it the most useful single relationship in delivery flow. First, **cycle time is a consequence of a management decision, not an attribute of the work**: at a given throughput, the elapsed time an item takes is determined by how much work is in progress alongside it. Second, because the relationship can be read in either direction, a reduction in cycle time at unchanged work in progress is **arithmetically identical** to an increase in throughput, which is the identity KA 13.3.2 uses to price a governance change.

**Flow management, and where Kanban fits.** A flow-managed system (the approach the Kanban method names) does not commit to an iteration's contents at all. Work is made visible, work in progress is explicitly limited at each stage, new work is **pulled** when capacity frees rather than pushed on a cadence, and the system is managed by the three measures above rather than by iteration commitments. It is therefore the natural regime wherever arrivals are genuinely unpredictable (support, operations, regulatory response), and it can be combined with an iteration cadence, in which case the cadence serves review and release while the work-in-progress limit

governs flow. The arithmetic below applies to both: Little's Law does not care whether the work was committed or pulled.

**Meridian's team at its working limit.** With  $W = 18$  items and  $T = 6$  items a week,  $C = 18 / 6 = 3.00$  weeks. If a typical item requires about **0.9 weeks** of active work, the **flow efficiency** (active time as a share of elapsed time) is  $0.9 / 3.00 = 30.0\%$ , meaning seven-tenths of an item's life is spent waiting. That is not unusual and it is the number most worth knowing, because it says where the improvement is: not in working faster, but in waiting less.

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#### WORKED EXAMPLE

##### Worked example 13.2.3 – the instruction to start more work.

1. **Setup.** Three clinics are waiting on the reporting module. Meridian's steering committee instructs the team to start it in parallel with existing work, raising work in progress from **18** to **30** items. Nothing else changes: same nine people, same cadence, same definition of done. The team's product owner objects and cannot price the objection. Assume the stated switching model, calibrated from the team's own records and to be recalibrated in any other setting: at or below the capacity point  $W^* = 18$  the team is the constraint, so throughput is  $6W/18$  and cycle time sits on its floor; above it, **each additional concurrent item consumes 2 % of the team's productive capacity** in switching, handover and re-familiarisation, so  $T(W) = 6 \times (1 - 0.02 \times (W - 18))$ . Meridian's cost of delay is **USD 14,280 per week** (Domain 1) and the remaining release backlog is **240 items**.
2. **Formula.**  $T(W)$  as stated;  $C = W / T$  (Little's Law); time to clear the backlog = backlog ÷ throughput; cost = additional weeks × cost of delay.
3. **Substitution.**  $T(30) = 6 \times (1 - 0.02 \times 12) = 6 \times 0.76$ .  $C(30) = 30 / 4.56$ .  $240 / 4.56$  against  $240 / 6.00$ .
4. **Result.** Throughput falls from **6.00** to **4.56** items a week (**-24.0 %**). Cycle time rises from **3.00** to **6.58 weeks** (**+119.3 %**, a multiple of **2.193**). Flow efficiency falls from 30.0 % to **13.7 %**. The 240-item backlog goes from **40.00 weeks** to **52.63 weeks**, and the extra **12.63 weeks** cost **USD 180,379**.
5. **Interpretation.** The headline is the one every delivery leader should be able to say in a sentence: **raising work in progress by 66.7 % bought 0 % more throughput and more than doubled the time any individual item takes**. Nothing arrived sooner. The reporting module the committee wanted accelerated arrives *later* than it would have if the team had finished its current work first, because it is now competing with that work for the same capacity, and so is everything else. The arithmetic also explains why the instruction is so persistently attractive: starting work is **visible and immediate**, while the cost lands weeks later, distributed across every item, and is attributed to the team's pace rather than to the decision. Three professional cautions, each of which a reviewer should press on. The 2 % switching coefficient is a **stated assumption**

**calibrated locally**, not a law; the *shape* of the result (throughput flat then falling, cycle time rising more than proportionally) is robust to the coefficient, but the 180,379 is not, and a paper quoting it must quote the assumption beside it. The model is linear and therefore predicts zero throughput at  $W = 68$ , which is obviously false and is a useful reminder that the relationship is only calibrated near the observed range: **calibrate, do not extrapolate**. And Little's Law itself assumes a steady state, so it describes a team over several iterations and not a single week; a leader who applies it to one week's numbers will find it noisy and conclude wrongly that it does not work. The practical control that follows is a **work-in-progress limit** (an explicit number, visible, changed only deliberately), and the case for it is not tidiness. It is USD 180,379.

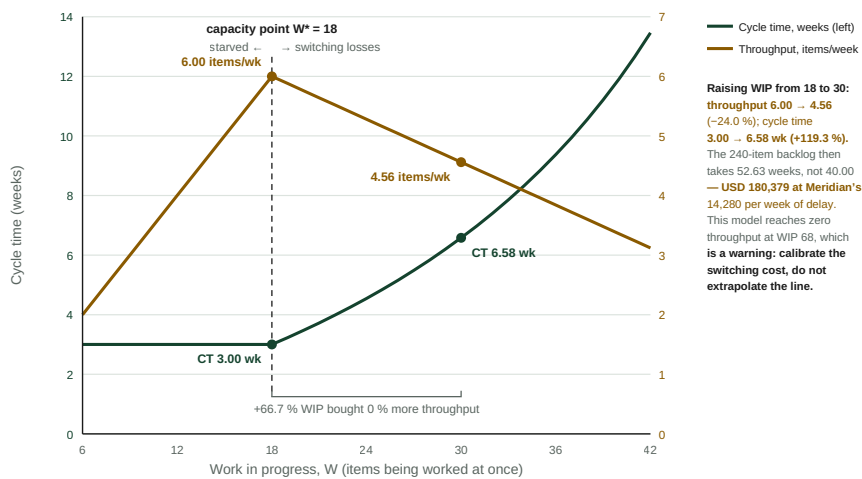


Fig 13.2.1 — Little's Law and the cost of starting more work

### 13.2.4 Forecasting from throughput history

**The principle.** A forecast produced from throughput history is a **range**, and the range means something precise that must be stated: it is the span of completion times that the team's **own recent behaviour** would have produced, on the assumptions that the future resembles the sampled past, that item sizes are drawn from the same distribution, that the team composition is unchanged and that the arrival of new work continues at the observed rate. It is **not** a confidence interval, it carries no probability, and presenting it as though it did is a misrepresentation that will be discovered.

**Why a range and not a date.** A single date drawn from an average is wrong roughly half the time by construction (13.2.2), and its wrongness is not symmetric in its consequences: the organisation plans irreversible things (training, clinic closures, communications) against it. A range with a stated basis lets those things be planned

against the slow end and celebrated at the fast end, which is the only sequence that does not require an apology.

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#### WORKED EXAMPLE

##### Worked example 13.2.4 – Meridian's release date, honestly.

1. **Setup.** Twelve weeks of completions: 5, 7, 4, 6, 8, 5, 7, 6, 5, 8, 6, 5 – total 72, mean 6.00 items a week. Remaining release backlog 240 items. Measured discovery arrivals 1.5 items a week. The programme plan carries a committed date 34 weeks away. Cost of delay USD 14,280 per week.
2. **Formula.** Smooth the history into **four-week rolling totals** to remove single-week noise, and take the observed minimum and maximum of those as the sustained-rate bounds. Net drain = gross throughput – arrival rate. Forecast = remaining backlog ÷ net drain. Required rate = remaining backlog ÷ target weeks, plus the arrival rate.
3. **Substitution.** Rolling four-week totals: 22, 25, 23, 26, 26, 23, 26, 25, 24, i.e. sustained rates of 5.50 to 6.50 items a week with a median of 6.25. Net drains  $6.50 - 1.5 = 5.00$  and  $5.50 - 1.5 = 4.00$ . Forecasts  $240 / 5.00$  and  $240 / 4.00$ . Naive figure  $240 / 6.00$ . Required rate  $240 / 34 + 1.5$ .
4. **Result.** The naive forecast (mean rate, arrivals ignored) is 40.00 weeks. At the mean rate net of arrivals it is 53.33 weeks, so ignoring discovery understates the date by 13.33 weeks and USD 190,400. The honest range is 48.00 to 60.00 weeks. The plan's 34 weeks requires a net drain of 7.06 items a week and therefore a gross throughput of 8.56 items a week: 31.7 % above the highest four-week rate the team has ever sustained and 42.6 % above its twelve-week mean. Against the fast end of the range the plan is 14.00 weeks and USD 199,920 optimistic; against the slow end, 26.00 weeks and USD 371,280.
5. **Interpretation.** The decisive sentence is not "the plan is optimistic" but "**the plan requires a throughput the team has never achieved, by 31.7 %, in its best four weeks on record**"; and that sentence changes a conversation, because it is not an opinion and it does not accuse anyone. Note the structure of the error, which is general: two independent optimisms compound. Using the mean rather than a range hides a 12-week spread; ignoring arrivals adds 13.33 weeks and USD 190,400 on its own, and it is the more damaging of the two because it is invisible; the arrival rate appears in no burndown chart, and a team whose gross throughput is perfectly stable will still miss a date built on it. Three cautions belong in any paper carrying this range. The range is **empirical, not probabilistic**: it says what the last twelve weeks would have produced, and if the remaining items are systematically larger than the delivered ones the range is optimistic (KA 13.3.4 quantifies exactly that). Four-week smoothing is a **stated choice**; a two-week window widens the range and a six-week window narrows it, so the window must be declared and held constant rather than selected after seeing the answer. And a range is only honest if the organisation is allowed to

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receive one: where the culture demands a date, the range will be reduced to its midpoint by someone, and the professional protection is to publish the required rate (8.56 items a week) alongside it, because that number cannot be averaged away.

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**What to do with the answer.** The 34-week date is not recoverable by exhortation, and the arithmetic says so precisely: even the governance remedy of KA 13.3.2, which lifts gross throughput to **7.06** items a week, leaves **43.17 weeks** once arrivals are subtracted. Meeting 34 weeks would require arrivals of zero, which for a clinical records build is not a credible commitment. A **40-week** forecast, by contrast, is achievable and can be stated as a conditional undertaking: it requires the governance remedy *and* discovery arrivals held at or below **1.06 items a week**, a **29.4 %** reduction on the observed rate, with a named owner for each condition. That is what a defensible re-plan looks like, not a new date, but a date with its two conditions attached and both of them measurable weekly.

### AI in this KA

**Where it earns its place.** Computing flow measures from a work-item history and flagging the anomalies a human eye misses: items with cycle times several times the median, items that have been started and abandoned, and the aged work in progress that no report shows. Sweeping the throughput history across window lengths to show how the range depends on the smoothing choice, which is the honest way to present 13.2.4's stated choice. Running the arrival-rate arithmetic across scenarios. Detecting a relaxed definition of done from the pattern of items reopened after acceptance, a correlation search over a large log, which is genuinely beyond manual capacity.

**Where it must not go.** It must not produce the forecast that goes to the board without the range's meaning attached, and this is the specific and serious hazard in this KA: a model asked "when will this finish?" will answer with a date, in confident prose, and that date will be quoted. The failure mode is not inaccuracy (the arithmetic may be perfect) it is the **silent loss of the basis**, so a range with four stated assumptions becomes a number with none. It must not be given a target date and asked to justify it, which is the same fabrication risk in a more flattering costume. And it must not size items, because sizing is the team's judgement about work it will do and a model's size estimate carries the authority of a measurement with none of the substance.

**Verification, concretely.** Every forecast that leaves the team carries its window length, its arrival rate, its assumption list and the required rate implied by any date it is compared against; and the arithmetic behind all four is reproducible on one page. Little's Law results are checked in both directions ( $C = W/T$  and  $T = W/C$ ) as a consistency test, since the two must agree. And any model-supplied flow anomaly is

confirmed against the underlying items before it is raised, because a false positive here accuses a person.

## KEY TERMS — KA 13.2

| TERM                          | MEANING                                                                                          |
|-------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Throughput (T)</b>         | Completed items per unit of time, a measurement, never an aspiration.                            |
| <b>Work in progress (W)</b>   | Items started and not finished; a managed quantity with an explicit limit.                       |
| <b>Cycle time (C)</b>         | Elapsed time from an item starting to finishing; a consequence of W at a given T.                |
| <b>Little's Law</b>           | $W = T \times C$ for a system in steady state, independent of arrival and service distributions. |
| <b>Flow efficiency</b>        | Active work time as a share of cycle time; typically low, and where the improvement lies.        |
| <b>Work-in-progress limit</b> | An explicit cap on W, changed only deliberately; the control the flow arithmetic justifies.      |
| <b>Discovery arrivals</b>     | New items found necessary during build; subtracted from throughput to give the net drain.        |
| <b>Net drain</b>              | Gross throughput minus arrival rate, the rate at which a backlog actually shrinks.               |
| <b>Range forecast</b>         | Backlog ÷ the observed high and low sustained net drains, with its assumptions stated.           |
| <b>Required rate</b>          | Backlog ÷ target weeks, plus arrivals, the throughput a committed date presupposes.              |

## SAMPLE MCQs — KA 13.2

**MCQ 13.2-A** [13.2.3 · Application] A team completes 6 items a week with 18 items in progress. Its average cycle time is:

- A. 0.33 weeks
- B. 3.00 weeks CORRECT
- C. 6.00 weeks
- D. 108 weeks

*Rationale:*  $C = W/T = 18/6 = 3.00$  weeks (13.2.3). A inverts the ratio; C reads the throughput as a duration; D multiplies instead of dividing.



**MCQ 13.2-B** [13.2.3 · Analysis] The same team is instructed to raise work in progress from 18 to 30 items. Each concurrent item above 18 costs 2 % of capacity. Throughput and cycle time become:

- A. 6.00 items a week and 5.00 weeks
- B. 4.56 items a week and 6.58 weeks CORRECT
- C. 4.56 items a week and 3.00 weeks
- D. 7.20 items a week and 4.17 weeks

*Rationale:*  $T = 6 \times (1 - 0.24) = 4.56$  and  $C = 30/4.56 = 6.58$  weeks (13.2.3). A is the common error of holding throughput constant, giving  $30/6 = 5.00$ ; C forgets that cycle time depends on both; D assumes work in progress raises throughput.

**MCQ 13.2-C** [13.2.4 · Application] A 240-item backlog, sustained rates of 5.5 to 6.5 items a week, and measured discovery arrivals of 1.5 items a week. The honest forecast range is:

- A. 36.9 to 43.6 weeks
- B. 48.0 to 60.0 weeks CORRECT
- C. 40.0 weeks
- D. 24.0 to 30.0 iterations, reported as weeks

*Rationale:* Net drains are  $6.5 - 1.5 = 5.0$  and  $5.5 - 1.5 = 4.0$ , giving  $240/5.0 = 48.0$  and  $240/4.0 = 60.0$  (13.2.4). A ignores arrivals, the single most common forecasting error here, and it understates the date by 11 weeks. C is the naive mean-rate point estimate. D is the correct range expressed in two-week iterations and mislabelled as weeks, which halves it.

**MCQ 13.2-D** [13.2.4 · Evaluation] For the same team, a plan commits to 34 weeks. The most useful single statement to put in front of the sponsor is that the plan requires:

- A. more effort from the team
- B. a gross throughput of 8.56 items a week, 31.7 % above the team's best sustained four-week rate CORRECT
- C. a throughput of 7.06 items a week
- D. the backlog to be reduced

*Rationale:*  $240/34 = 7.06$  net, plus 1.5 arrivals gives **8.56** gross, which is 31.7 % above the 6.5 best observed rate (13.2.4). C omits arrivals and so understates the requirement; A and D are responses, not the diagnosis.

**MCQ 13.2-E** [13.2.2 · Analysis] Six iterations delivered 12, 10, 13, 13, 13 and 11 items, a mean of exactly 12. A team committed to 12 items an iteration will:

- A. meet the commitment, since 12 is the mean

B. miss it in 33.3 % of iterations while performing exactly to its own average

**CORRECT**

C. miss it in 50 % of iterations

D. exceed it in every iteration

*Rationale:* Two of the six totals are below 12, so the commitment fails a third of the time at unchanged performance (13.2.2). C assumes a symmetric distribution the data does not have; the resolution is to plan with the mean and commit with the low end of 10.

**MCQ 13.2-F [13.2.3 · Comprehension]** A team's cycle time is 3.00 weeks and a typical item takes 0.9 weeks of active work. Flow efficiency is:

A. 30.0 %

**CORRECT**

B. 70.0 %

C. 3.33 %

D. 333 %

*Rationale:*  $0.9/3.00 = 30.0 \%$  (13.2.3). B is the waiting share; C and D invert the ratio. The low figure is normal and locates the improvement in waiting, not in working faster.

#### SELF-CHECK — KA 13.2

1. *State Little's Law in all three forms and say what each is used for.* —  $W = T \times C$ ,  $C = W/T$ ,  $T = W/C$ : the second derives the elapsed time a management decision about work in progress has caused; the third converts a cycle-time improvement into a throughput improvement at unchanged work in progress, which is how KA 13.3.2 prices governance.
2. *Why is ignoring discovery arrivals worse than using a mean rather than a range?* — Because it is invisible: a team with perfectly stable gross throughput still misses the date, and on Meridian the error is **13.33 weeks** and **USD 190,400** on its own.
3. *What must accompany a range forecast for it to be honest?* — The smoothing window, the arrival rate, the four assumptions (sampled past, item-size distribution, team composition, arrival continuity) and the statement that it is empirical and carries no probability.

## Scaling considerations and hybrid governance

*Topics: 13.3.1 what scaling actually costs · 13.3.2 governance latency against iteration length · 13.3.3 designing hybrid delivery honestly · 13.3.4 reporting flow to a predictive governance body.*

### 13.3.1 What scaling actually costs

**The principle.** Adaptive delivery scales badly by default and well by design, and the difference is entirely a matter of how many things must agree. Two costs rise as team count rises, and both have already been computed in this book, so they are cited rather than re-derived.

**Coordination overhead within and between teams** rises with the square of the number of people working as one group: Domain 12's KA 12.2.2 computes the fully connected case as  $(n - 1)/160$  of team capacity on its stated link-cost assumption, giving **6.875 %** at Meridian's original twelve people and **24.375 %** if forty were run as one team, against **4.6875 %** for the same forty in five structured groups of eight, a release of **7.875 full-time equivalents**. **Interface count between delivery streams** behaves the same way: Domain 4's KA 4.2.3 gives  $n(n-1)/2$  possible pairwise interfaces against  $n$  to an integration layer, so five streams carry **10** potential interfaces against **5**, and nine carry **36** against **9**.

**What this implies for scaling design** is short, because the arithmetic is decisive. Prefer **fewer, larger increments of independence** to more teams: a team that can release without agreement costs  $n$ , one that must agree with every other costs  $n(n-1)/2$ . Where teams must integrate, buy the **integration layer** (an architecture, an interface contract, a shared platform) rather than paying the mesh, and price it against the interface count it removes (Domain 4's method). And accept the honest consequence: **scaling frameworks are largely mechanisms for reducing the number of things that must agree**, and any scaling design should be assessed on exactly that question. A framework adopted without reducing agreement requirements has added ceremony to a mesh.

**The dependency that matters most is on a shared scarce resource.** Domain 6's KA 6.A.1 drum applies unchanged: where several streams need the same constrained capability (a security review, one test environment, one clinical safety officer) that capability sets programme throughput, and every stream's local flow measures look healthy while the programme's do not. The diagnostic is to compute cycle time for items **requiring** the shared resource separately, which is the same decomposition KA 13.3.2 performs for governance decisions.

### 13.3.2 Governance latency against iteration length

Domain 3 named this mismatch and left it as a design defect: a monthly steering committee governing two-week sprints is always behind, so the team either waits or proceeds and seeks retrospective approval (Domain 3, KA 3.1.3). This topic quantifies it, and the quantification is what converts the complaint into a proposal.

**The two numbers, and their ratio.** Domain 3's KA 3.2.3 gives the expected wait for a committee decision as  $E[\text{wait}] = M/2 + L$ : half the meeting interval plus the whole paper lead time. Meridian's steering committee has  $M = 4$  and  $L = 2$ , so  $E[\text{wait}] = 4.0$  weeks. The iteration is **2 weeks**. The ratio is therefore **2.00**: every escalated decision costs **two entire iterations of waiting**, which is the precise sense in which the governance body cannot keep up with the work it governs. Note that this is not a criticism of the committee's diligence: the number is a property of two administrative parameters and would be the same if every member were faultless.

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#### WORKED EXAMPLE

#### Worked example 13.3.2 – what governance latency does to Meridian's flow.

- Setup.** Meridian's team runs at  $T = 6$  items a week with  $W = 18$  and therefore  $C = 3.00$  weeks (13.2.3). Reviewed history shows **15 %** of items require a decision above the product owner's **USD 25,000** envelope and therefore go to the steering committee at  $E[\text{wait}] = 4.0$  weeks. What does that cost in flow terms, and what would a one-week out-of-cycle written-resolution route (Domain 3, KA 3.3.3) be worth?
- Formula.** Blocked arrival rate = share  $\times$  throughput. **Blocked work in progress = blocked arrival rate  $\times E[\text{wait}]$ :** little's Law applied to the blocked sub-population. Decompose the average cycle time:  $C = (1 - s) \times C_u + s \times (C_u + E[\text{wait}])$ , so  $C_u = C - s \times E[\text{wait}]$ . Then recompute  $C$  at the shorter wait and convert to throughput by  $T = W / C$ .
- Substitution.** Blocked arrivals  $0.15 \times 6 = 0.90$  items a week; blocked work in progress  $0.90 \times 4.0$ . Unblocked cycle time  $3.00 - 0.15 \times 4.0$ . New average  $0.85 \times 2.40 + 0.15 \times 3.40$ . New throughput  $18 / 2.55$ .
- Result.** **1.80** of every 12-item iteration requires a committee decision. **3.60 items** of work in progress are permanently parked waiting for one: **20.0 %** of the team's entire work in progress. Unblocked items take **2.40 weeks**; blocked items take **6.40 weeks**, a multiple of **2.667**. With a one-week out-of-cycle route, parked work in progress falls to **0.90 items (5.0 %)**, average cycle time falls from 3.00 to **2.550 weeks (-15.0 %)**, and at unchanged work in progress throughput rises from 6.00 to **7.0588 items a week: +17.6 %**.
- Interpretation.** Three results are worth stating separately because they persuade different audiences. To a delivery manager, **one item in five that the team is holding is not being worked on at all**: it is waiting for a room, and

no report shows it, because a blocked item and a work-in-progress item look identical on a board. To a product owner, **the items that need a decision take 2.667 times as long as the items that do not**, which is why the roadmap's high-value items (the ones large enough to exceed the envelope) are systematically the late ones; the governance design is quietly selecting against value. And to a sponsor, the remedy is a **17.6 % throughput increase for the cost of an administrative procedure**, achieved without hiring anyone, working faster, or removing any scrutiny that anyone can name. The decision is still made by the same authority, merely in writing and within five working days. Note the identity that makes the last claim exact rather than rhetorical: at fixed work in progress, a cycle-time reduction *is* a throughput increase, because Little's Law can be read in either direction (13.2.3), and  $18 / 2.55 = 7.0588$  is the same operation as  $240 / 34$ . Two cautions. The 15 % share must come from a counted history, not an impression; the whole result scales linearly with it, so a paper quoting 17.6 % must show how the share was counted. And the remedy is only available where the decision *class* can legitimately be decided out of cycle: a clinical safety approval or a regulatory submission cannot be written-resolved, and Domain 3's KA 3.3.1 rule stands, keep the gates that carry real optionality, remove the ones that re-approve a decision already taken. The general form of the lesson: **governance latency is a flow parameter, and it belongs in the flow arithmetic rather than in the culture chapter.**

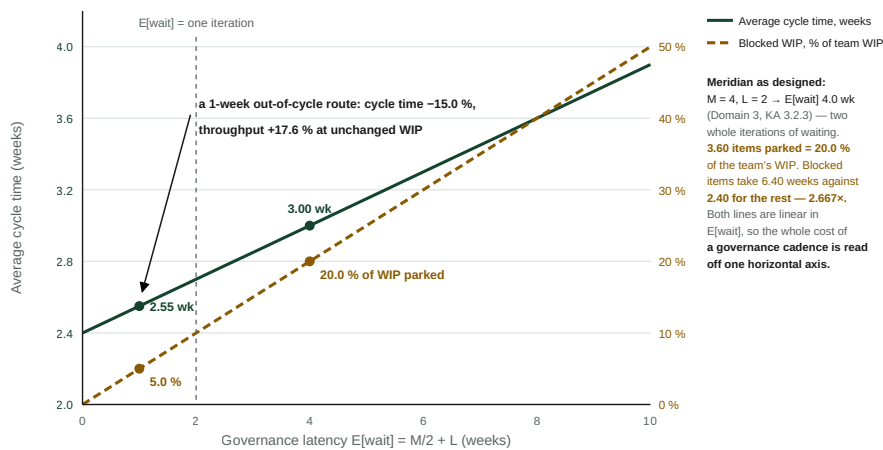


Fig 13.3.1 — Governance latency as a flow parameter

### 13.3.3 Designing hybrid delivery honestly

**Definition.** Hybrid delivery is a deliberate combination of predictive and adaptive methods under one governance frame, **with a named boundary**. The words that matter are "deliberate" and "named": most programmes are hybrid by circumstance and manage the whole of it with one control regime, which means they have silently chosen a method for work it does not suit.

**The three design decisions.** First, **where the boundary runs.** It runs where the three conditions of 13.1.1 change, where feedback stops being cheap and fast. Meridian's boundary is unusually clear: the records application is adaptive (feedback within an iteration from clinicians using a working increment), the clinic rollout is predictive (estate works, training cohorts, immovable dependencies, feedback measured in months), and the regulatory approval is a genuine gate. Second, **what crosses the boundary and in what form.** Domain 6's KA 6.4.1 gives the translation rule and it is not repeated here: a throughput-based forecast enters the network as a **ranged duration on an integration activity**, and the network returns the **latest acceptable delivery** read from the backward pass. Third, **which control regime governs which side**, stated explicitly, so that neither side is measured by the other's instruments: the adaptive side on flow and value delivered, the predictive side on variance against baseline, and the boundary itself on the integration milestone's date range.

**The four hybrid failure modes**, each with its test. **One regime for both.** A single percentage-complete report covering an adaptive stream, forcing the team to invent a denominator (13.3.4); *test*: ask what it is derived from. **A boundary nobody owns**, an integration milestone with no accountable holder on either side, so it slips without a decision, which is Domain 4's interface failure in a new costume; *test*: name the owner. **Cadence imposed across the boundary.** The predictive side's monthly reporting cycle becomes the adaptive side's decision cadence, re-creating 13.3.2's latency; *test*: compare `E[wait]` for each side against its own cycle. **Hybrid as a euphemism.** A predictive programme that holds stand-up meetings, carrying the cost of both regimes and the benefit of neither; *test*: ask which variable floats on the adaptive side.

### 13.3.4 Reporting flow to a predictive governance body

A governance body accustomed to variance against baseline will ask an adaptive stream for a percentage complete, and the team will supply one by dividing items delivered by items known. That number is almost always wrong in the same direction, and the error is computable.

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#### WORKED EXAMPLE

#### Worked example 13.3.4 — the count view and the size view.

1. **Setup.** 96 of Meridian's 240 release items are complete, delivered over 16 weeks. On re-sizing, the delivered items average 1.2 size units and the remaining 144 average 2.0: the later items being the integration-heavy and regulatory ones deliberately deferred.
2. **Formula.** Percentage complete by count = items done ÷ total items. By size = size delivered ÷ total size. Remaining duration on the count view = remaining items ÷ item throughput; on the size view = remaining size ÷ delivered size rate.

- 
3. **Substitution.**  $96/240$  against  $(96 \times 1.2) / (96 \times 1.2 + 144 \times 2.0)$ . Delivered size rate  $115.2 / 16$ . Remaining duration  $144 / 6.00$  against  $288.0 / 7.20$ .
  4. **Result.** By count the release is **40.0 %** complete; by size, **28.6 %**: an overstatement of **11.4 percentage points**. The count view says **24.0 weeks** remain; the size view says **40.0 weeks**, an understatement of **16.0 weeks** and **USD 228,480**.
  5. **Interpretation.** The count view is not a lie and the team is not manipulating anything; it is simply the wrong denominator, and it fails in a **predictable direction** because teams rationally sequence the tractable work first, which is precisely what 13.1.3's density rule tells them to do. So the two disciplines of this domain interact: **optimal sequencing guarantees that an item count will overstate progress**. A leader who understands that will report count and size together, or report size alone, and will never let an item count travel to a board unlabelled. The defensive habit that follows is to report **three numbers and no percentage**: items and size delivered against items and size remaining; the net drain and the range it implies (13.2.4); and the conditions under which the range holds. Two cautions. Re-sizing remaining work is legitimate and must be **dated and recorded**, because a re-size that quietly moves a denominator is indistinguishable from a re-baseline without governance (Domain 6, KA 6.A.2). And size units are a team-relative measure with no meaning across teams, so they must never be aggregated across streams into a programme total, which is the single most common misuse of adaptive metrics at scale, and the reason KA 13.4.2 insists on throughput and cycle time as the portfolio-level measures.
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## AI in this KA

**Where it earns its place.** Sweeping governance-design options (cadences, paper lead times, delegation envelopes), and reporting the resulting  $E[\text{wait}]$ , blocked work in progress and throughput for each, which is deterministic arithmetic over a large combination space. Classifying a decision history to establish the 15 % share honestly from records rather than impression. Detecting cross-stream dependencies from work-item text and commit metadata, producing a candidate interface list for human confirmation against Domain 4's interface register. Reconciling a count view against a size view automatically each period, so that 13.3.4's gap is a standing number rather than a discovery.

**Where it must not go.** It must not decide which decisions may be taken out of cycle: that is a risk-appetite and, for clinical or regulatory classes, a legal judgement, and it belongs to the accountable authority. It must not aggregate size units across teams, however plausible the resulting programme total looks. The operation is meaningless and a model will perform it without complaint. And it must not be the source of the hybrid boundary: where feedback is cheap and fast is a matter of fact about the work, established by people who know the work.

**Verification, concretely.**  $E[\text{wait}]$  is two operations and is reproduced by hand for every design that reaches a paper. The blocked share is traced to counted decisions with dates in and out. Any cross-stream dependency a model proposes is confirmed with both stream leads before it enters the interface register, because a false dependency creates real coordination cost, at Domain 12's rates.

## KEY TERMS — KA 13.3

| TERM                            | MEANING                                                                                                                                  |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Blocked work in progress</b> | Blocked arrival rate $\times E[\text{wait}]$ : the items a team holds but cannot work on, invisible on a board.                          |
| <b>Cycle-time decomposition</b> | Splitting average cycle time into its blocked and unblocked populations to locate the delay.                                             |
| <b>Out-of-cycle route</b>       | Domain 3's mechanism (KA 3.3.3) applied where governance latency exceeds the iteration length, the specific mismatch KA 13.3 quantifies. |
| <b>Integration layer</b>        | An architecture or interface contract that replaces $n(n-1)/2$ pairwise agreements with $n$ .                                            |
| <b>Hybrid boundary</b>          | The named line where feedback stops being cheap and fast, and the control regime changes.                                                |
| <b>Integration milestone</b>    | The boundary event where an adaptive stream's ranged forecast enters a predictive network.                                               |
| <b>Count view / size view</b>   | Progress by item count against progress by size delivered; the first overstates when sequencing is optimal.                              |

## SAMPLE MCQs — KA 13.3

**MCQ 13.3-A** [13.3.2 · Application] A team runs at 6 items a week with 18 in progress; 15 % of items need a committee decision with  $E[\text{wait}] = 4.0$  weeks. The work in progress parked waiting for a decision is:

- A. 1.80 items
- B. 3.60 items, 20.0 % of the team's work in progress CORRECT
- C. 0.90 items
- D. 2.70 items

*Rationale:* Blocked arrivals are  $0.15 \times 6 = 0.90$  a week, and by Little's Law the blocked population is  $0.90 \times 4.0 = 3.60$  items (13.3.2). A is the blocked items per iteration ( $0.15 \times 12$ ), not the parked population; C is the weekly arrival rate; D applies the 15 % share to the work in progress of 18 instead of to the arrival rate, which omits the wait entirely.



**MCQ 13.3-B** [13.3.2 · Analysis] For the same team, average cycle time is 3.00 weeks. The cycle times of the unblocked and blocked populations are:

- A. 3.00 and 3.00 weeks
- B. 2.40 and 6.40 weeks
- C. 3.00 and 7.00 weeks
- D. 2.55 and 6.55 weeks

CORRECT

*Rationale:*  $C_u = 3.00 - 0.15 \times 4.0 = 2.40$ , and blocked items add the whole 4.0-week wait, giving 6.40, a multiple of 2.667 (13.3.2). C adds the wait to the average rather than to the unblocked figure, double-counting the 0.60 weeks the blocked items already contribute; D uses the post-remedy average.

**MCQ 13.3-C** [13.3.2 · Evaluation] Replacing the 4.0-week committee wait with a 1.0-week written-resolution route changes cycle time to 2.550 weeks. At unchanged work in progress of 18, the throughput effect is:

- A. none — throughput depends on capacity, not on cycle time
- B. an increase from 6.00 to 7.06 items a week, +17.6 %
- C. an increase of 15.0 %, matching the cycle-time reduction
- D. an increase from 6.00 to 7.20 items a week

CORRECT

*Rationale:*  $T = W/C = 18/2.55 = 7.0588$ , an increase of 17.6 % (13.3.2). A forgets that Little's Law reads both ways; C confuses the proportional fall in C with the proportional rise in T, which are not equal because the relationship is reciprocal ( $1/0.85 = 1.176$ , not 1.15); D rounds the cycle time to 2.50 before dividing, and rounding before the division moves the answer by 0.14 items a week.

**MCQ 13.3-D** [13.3.4 · Analysis] 96 of 240 items are done, averaging 1.2 size units, and the 144 remaining average 2.0. Reporting 40.0 % complete:

- A. is correct, since  $96/240 = 40\%$
- B. overstates progress by 11.4 percentage points, because optimal sequencing delivers the tractable items first
- C. understates progress
- D. is correct provided the team's velocity is stable

CORRECT

*Rationale:* By size the release is  $115.2/403.2 = 28.6\%$  complete (13.3.4). The direction of the error is systematic, not accidental: the density rule of 13.1.3 tells teams to do the cheap high-value work first, which guarantees that a count overstates.

**MCQ 13.3-E** [13.3.1 · Application] Nine delivery streams must integrate. The reduction in potential pairwise interfaces from adopting an integration layer is:

- A. from 36 to 9
- B. from 81 to 9

CORRECT

C. from 45 to 9

D. from 36 to 18

*Rationale:*  $n(n-1)/2 = 36$  against  $n = 9$  (Domain 4, KA 4.2.3, cited at 13.3.1). B squares  $n$ ; C uses  $n(n+1)/2$ .

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SELF-CHECK — KA 13.3

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1. *How is blocked work in progress computed, and why does no board show it?* — Blocked arrival rate  $\times E[\text{wait}]$ . A board shows an item as in progress whether it is being worked on or waiting for a committee, so 20.0 % of Meridian's work in progress is invisible.
  2. *Why does an item count systematically overstate progress on an adaptive stream?* — Because optimal sequencing delivers the cheap, high-value items first, leaving the larger ones, on Meridian, 40.0 % by count against 28.6 % by size.
  3. *What is the test for whether a programme is genuinely hybrid rather than nominally so?* — Name the boundary, name the person accountable for the integration milestone, and state which variable floats on the adaptive side. If nothing floats, it is not adaptive.
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KNOWLEDGE AREA 13.4

## Contracting for adaptive delivery, metrics and anti-patterns

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*Topics:* 13.4.1 capacity-based against scope-based commercial models · 13.4.2 what the buyer must control instead of price · 13.4.3 metrics that survive contact with incentives · 13.4.4 the anti-patterns and their arithmetic signatures.

### 13.4.1 Capacity-based against scope-based commercial models

**The two models.** A **scope-based** contract fixes a price for a defined body of work; the supplier bears the risk that the work takes more effort than expected. A **capacity-based** contract buys a funded, stable delivery capability for a period (a team, at a rate, for an agreed cadence), and the buyer bears the risk of how much gets done. Domain 10's KA 10.3.1 establishes the general frame: a contract is a risk allocation, and the only useful question about a mechanism is what it rewards at the margin. This topic applies that frame to the one test that actually discriminates between the two models in adaptive delivery, because on expected price at award they can be made identical: **what does a change cost the buyer?**

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WORKED EXAMPLE

**Worked example 13.4.1 — Meridian's build, priced two ways, then changed.**

1. **Setup.** The 240-item release, two commercial offers designed to be equivalent at plan. *Scope-based:* **USD 1,200,000** firm for the 240 items — **USD 5,000** an item. *Capacity-based:* **USD 60,000** per 2-week iteration for the supplier's squad; the plan of 20 iterations (40 weeks) totals **USD 1,200,000**, which at the planned 12 items an iteration is also **USD 5,000** an item. At week 14 a regulatory reporting requirement emerges. The buyer wants **30 items added** and 30 lower-value items **dropped**, a net-zero change in item count. The scope contract's change schedule prices additions at **USD 7,500** an item (a 50 % premium on the bid rate, because change work is not competed) and credits omissions at **USD 3,000** an item (60 % of the bid rate, the supplier retaining margin and recovered overhead). A scope change must go to the change board at Domain 3's  $E[wait]$  = **4.0 weeks**; a capacity re-prioritisation happens at the next iteration boundary, an average wait of **1.0 week**. Cost of delay **USD 14,280** per week.
2. **Formula.** Scope exposure = (items added × change rate) – (items dropped × omission credit) +  $E[wait]$  × cost of delay. Capacity exposure = boundary wait × cost of delay. Then the buyer's quantity exposure under the capacity model = (worst-case iterations × iteration price) – the fixed price. Breakeven change count = quantity exposure ÷ the per-change difference.
3. **Substitution.** Additions  $30 \times 7,500 = 225,000$ ; credit  $30 \times 3,000 = 90,000$ ; latency  $4.0 \times 14,280 = 57,120$ . Capacity  $1.0 \times 14,280$ . Duration range from 13.2.4's honest forecast, 48 to 60 weeks, i.e. **24 to 30 iterations**:  $24 \times 60,000$  to  $30 \times 60,000$ .
4. **Result.** The net-zero change moves the scope-based price by **USD 135,000** and costs a further **USD 57,120** in decision latency: total buyer exposure **USD 192,120** for a change that adds no items at all. Under the capacity model the same change costs **USD 14,280** (the one week to the next boundary) a difference of **USD 177,840** per change. Against that, the capacity model exposes the buyer to duration: **USD 1,440,000 to USD 1,800,000** across the honest range, so worst-case quantity exposure above the fixed price is **USD 600,000**. The break-even is  $600,000 / 177,840 = 3.37$ , so from the **fourth** material re-prioritisation the capacity model is cheaper even at the worst end of the observed throughput range: 4 changes cost the scope buyer  $1,200,000 + 4 \times 192,120 = \text{USD } 1,968,480$  against the capacity model's worst case of **USD 1,800,000**. Measured against the capacity model's *expected* cost (53.33 weeks, 26.67 iterations, **USD 1,600,000**) the breakeven falls to **2.25** changes.
5. **Interpretation.** The result to carry away is the shape, not the numbers: **a scope-based contract converts every change into a priced commercial event with a governance latency attached, and a capacity-based contract converts every change into a free re-ordering at the next boundary; while transferring the quantity risk to the buyer.** Neither is better; they price different uncertainties, and the choice follows from which uncertainty the work actually has. The decisive diagnostic is therefore a *count*: how many mater-

ial re-prioritisations does this work expect? Below about three, the fixed price is the cheaper instrument; above it, the capacity model is, and the gap widens by **USD 177,840** a change. Four cautions, and they matter more than the arithmetic. First, the **asymmetry in the change schedule** is where scope-based exposure really lives: additions at 7,500 against credits at 3,000 means a net-zero change moves the price by 135,000, and a buyer who negotiates the addition rate without negotiating the omission credit has secured half a protection. Second, the **fixed price presupposes specifiability**. Where the 240 items genuinely cannot be specified at bid (which is the premise of adaptive delivery) a competent supplier prices the ambiguity into the bid, so the premium is paid whether or not a change occurs; the observed 1,200,000 is then not a bargain but a contingency the buyer cannot see or reclaim. Third, the capacity model's **600,000 of quantity exposure is real** and it is not managed by the contract; it is managed by the continuation decision of 13.4.2, and a buyer who adopts the model without that control has bought an open-ended commitment, which is Case study B. Fourth, none of this is a legal analysis: the enforceability of change schedules, omission credits and termination-for-convenience rights varies by jurisdiction and by contract form, and the arithmetic above informs a negotiation rather than settling one.

| TEST                                | SCOPE-BASED (FIRM PRICE)                            | CAPACITY-BASED (FUNDED TEAM)                            |
|-------------------------------------|-----------------------------------------------------|---------------------------------------------------------|
| Expected price at award             | USD 1,200,000                                       | USD 1,200,000 at plan                                   |
| Buyer exposure to a net-zero change | <b>USD 192,120</b> (135,000 price + 57,120 latency) | <b>USD 14,280</b> (one week to the boundary)            |
| Buyer exposure to throughput risk   | none (supplier bears it)                            | <b>up to USD 600,000</b> across the 48–60 week range    |
| What the buyer must control         | change control and the change schedule's asymmetry  | the continuation decision and the value envelope        |
| Breaks even against the other at    | fewer than ~3 material changes                      | 4 or more material changes (2.25 against expected cost) |

### 13.4.2 What the buyer must control instead of price

Under a capacity model the price is no longer the control, and a buyer who does not replace it has replaced a bounded commitment with an unbounded one. Three controls do the work, and all three already exist in this book.

**The continuation decision at a stated cadence.** Domain 3's KA 3.1.3 posture (is the next increment's cost worth its expected value) applied at a real interval with a real authority to stop. The mechanism is a **bounded funding envelope**: authority to spend a stated number of iterations, after which continuation requires an explicit

decision. Meridian's would be four iterations (**USD 240,000** at 60,000 an iteration), which is small enough to be a genuine decision and large enough not to be a ceremony.

**The value-envelope reconciliation.** Domain 5's KA 5.A.1 test, run every funding cycle: attributed benefit of delivered items against the business case's commitment. This is the control whose absence Case study B costs at **USD 3,360,000**, and its price there is **USD 36,000**.

**The exit provision that is actually usable.** A termination or step-down right is only a control if exercising it leaves the buyer with something: the code, the data, the environments, the documentation and a supplier obligation to support transition, all of which belong in the definition of done (13.1.4) rather than in a schedule nobody reads. An exit right without a transferable asset is a theoretical right, and Domain 10's KA 10.3.3 supplier-governance treatment applies.

### 13.4.3 Metrics that survive contact with incentives

**The principle.** Goodhart's law (that a measure adopted as a target tends to cease being a good measure) is not a reason to avoid measurement; it is a reason to choose measures whose gaming is either impossible or visible. Four flow measures satisfy that test and one popular measure does not.

**The four that hold. Throughput:** completed items per period, where "completed" means the definition of done. **Cycle time,** ideally as a distribution rather than a mean, because the tail is where the failures are and the mean conceals it. **Work in progress,** including its aged tail, which is where blocked items hide (13.3.2). And **flow efficiency,** which is the only one of the four that points directly at the improvement. All four are counts of real events with timestamps, and all four are cross-team comparable, which is what makes them the right measures at portfolio level.

**The one that does not: velocity as a performance measure.** Velocity is throughput weighted by team-relative size units, and its weights are set by the team being measured. The arithmetic of gaming it is trivial: Meridian's team delivers 12 items an iteration at an average size of 1.2, so **14.4 size units**; inflate the sizing by 25 % and the same 12 items become **18.0 size units**, a reported 25 % improvement with no change whatever in delivery. The inflation is not usually dishonest (it is the predictable response to being measured on a number one controls), and it is undetectable from the metric itself. Velocity is therefore a legitimate and useful **planning input inside a team** and an illegitimate performance measure across teams, and the two uses must be separated explicitly rather than by convention. Size units must never be aggregated into a programme total (13.3.4).

**What the governance body should actually receive,** and it fits on one page: throughput and its trend; the cycle-time distribution with its tail; work in progress against its limit, and the blocked share; the net drain and the range forecast with its

conditions (13.2.4); and the value-envelope reconciliation. Five numbers, none of them a percentage complete, all of them auditable.

#### 13.4.4 The anti-patterns and their arithmetic signatures

Each of the recurring failures of adaptive delivery has a signature in the measures above, which is what makes them diagnosable rather than debatable.

**Adaptive delivery with fixed scope.** The overhead of the method without its benefit (13.1.1). *Signature:* a backlog from which nothing is ever removed; zero refusals by the product owner.

**Starting more to go faster.** *Signature:* work in progress rising, throughput flat or falling, cycle time rising more than proportionally — worth **USD 180,379** on Meridian (13.2.3). This is the most expensive and most common instruction in delivery.

**The proxy product owner.** *Signature:* a median product-owner decision wait of the same order as the committee's  $E[\text{wait}]$ , and blocked work in progress well above the 5.0 % a delegated envelope produces (13.1.2, 13.3.2).

**Governing a fortnightly cadence monthly.** *Signature:* **20.0 %** of work in progress parked; blocked items at **2.667×** the cycle time of the rest (13.3.2).

**Reporting an item count as progress.** *Signature:* a widening gap between the count view and the size view, **11.4 percentage points** and **16.0 weeks** on Meridian (13.3.4).

**Velocity as a target.** *Signature:* size units per iteration rising while throughput in items is flat (13.4.2).

**The relaxed definition of done.** *Signature:* completion rising, then reopened items rising, then throughput falling as rework consumes capacity at Domain 9's  $1/(1 - r)$  (13.1.4).

**Hybrid as a euphemism.** *Signature:* no named boundary, no floating variable, and an integration milestone with no accountable owner (13.3.3).

**A capacity contract without a continuation decision.** *Signature:* healthy velocity reports and no value-envelope reconciliation, Case study B, at **169.2 %** of planned cost for **41.0 %** of committed benefit.

#### AI in this KA

**Where it earns its place.** Modelling the commercial comparison of 13.4.1 across change counts, change rates, omission credits and throughput ranges, which is exactly the deterministic sweep a machine should perform and a human should not. Producing the cycle-time distribution and its tail from a work-item log, including the aged work in progress that no standard report surfaces. Reconciling the count and size views each period automatically. Screening a supplier's iteration reports for the

signatures above (sizing inflation against flat item throughput, for example), and raising a candidate finding for a human to confirm.

**Where it must not go.** It must not select the commercial model, set a rate, or draft a change schedule: those are risk-appetite and legal judgements belonging to the accountable buyer, and Domain 10's KA 10.3 position stands. It must not evaluate individual or team performance from flow data. The measures of 13.4.2 are properties of a **system**, and applying them to people converts every one of them into a target and destroys it. The gaming arithmetic above is the demonstration, and a model that ranks teams on velocity has industrialised the error. And it must not be the source of the anti-pattern finding itself: a signature is evidence for a conversation, not a verdict.

**Verification, concretely.** Every figure in a commercial comparison is reproduced by hand at one change count, and the change schedule's rates are read from the contract rather than from a summary. Every flow anomaly is traced to the underlying items before it is reported. And any model-produced comparison states its assumed throughput range and its source, because the whole of 13.4.1's conclusion turns on the 48-to-60-week range and a fabricated range produces a confident and wrong contracting recommendation.

#### KEY TERMS — KA 13.4

| TERM                            | MEANING                                                                                                                          |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>Scope-based contract</b>     | A firm price for a defined body of work; the supplier bears effort risk, the buyer pays for change.                              |
| <b>Capacity-based contract</b>  | A funded team at a rate for a cadence; the buyer bears quantity risk, change is free at the boundary.                            |
| <b>Omission credit</b>          | The rate at which removed scope is credited back, typically below the bid rate, and the asymmetry that prices a net-zero change. |
| <b>Bounded funding envelope</b> | Authority to spend a stated number of iterations before continuation requires an explicit decision.                              |
| <b>Continuation decision</b>    | The periodic judgement that the next increment's cost is worth its expected value; the capacity model's brake.                   |
| <b>Velocity</b>                 | Throughput weighted by team-relative size units, a planning input inside a team, never a performance measure.                    |
| <b>Flow metric set</b>          | Throughput, cycle-time distribution, work in progress with its blocked and aged shares, flow efficiency.                         |

## SAMPLE MCQs — KA 13.4

**MCQ 13.4-A** [13.4.1 · Application] A firm-price contract of 1,200,000 for 240 items prices additions at 7,500 and credits omissions at 3,000. The buyer adds 30 items and drops 30. The price movement is:

- A. zero — the change is net-zero in item count
- B. an increase of USD 135,000 CORRECT
- C. an increase of USD 225,000
- D. a decrease of USD 90,000

*Rationale:*  $30 \times 7,500 - 30 \times 3,000 = 135,000$  (13.4.1). A is the error the item count invites; C counts only the additions; D only the credit. The asymmetry between the two rates is where the exposure lives.

**MCQ 13.4-B** [13.4.1 · Evaluation] The same change costs 57,120 in change-board latency, against 14,280 under a capacity model; the capacity model's worst-case quantity exposure is 600,000. The number of material changes at which the capacity model becomes cheaper is:

- A. 2
- B. 3
- C. 4 CORRECT
- D. it never becomes cheaper

*Rationale:* The per-change difference is  $192,120 - 14,280 = 177,840$ , and  $600,000/177,840 = 3.37$ , so the fourth change tips it:  $1,200,000 + 4 \times 192,120 = 1,968,480$  against 1,800,000 (13.4.1). A and B are below the breakeven; against the *expected* capacity cost of 1,600,000 the breakeven is 2.25, which is a different and clearly-labelled comparison.

**MCQ 13.4-C** [13.4.2 · Analysis] A buyer adopts a capacity-based contract and keeps its existing change-control process as the principal control. The defect is that:

- A. change control is unnecessary in adaptive delivery
- B. price is no longer the binding constraint, so the control must be the continuation decision and the value-envelope reconciliation CORRECT
- C. the iteration price should be renegotiated
- D. the supplier will slow down

*Rationale:* A capacity contract has no per-change price to control, so retaining change control leaves the quantity exposure unmanaged (13.4.2). A overstates; the change *record* still matters (Domain 3, KA 3.3.4); the missing controls are the brake and the value test.



**MCQ 13.4-D** [13.4.2 · Application] A team delivers 12 items an iteration at an average size of 1.2 units. Sizing is inflated by 25 %. The reported velocity and the actual throughput become:

- A. 18.0 units and 15 items
- B. 18.0 units and 12 items — a 25 % reported improvement with no change in delivery CORRECT
- C. 14.4 units and 12 items
- D. 14.4 units and 15 items

*Rationale:*  $12 \times 1.5 = 18.0$  units against an unchanged 12 items (13.4.2). This is why velocity is a planning input inside a team and not a performance measure: the team being measured sets the weights, and the inflation is invisible in the metric itself.

**MCQ 13.4-E** [13.4.4 · Analysis] A supplier's reports show size units per iteration rising steadily while items completed per iteration are flat. The most likely diagnosis is:

- A. the team is improving
- B. sizing inflation under a velocity target CORRECT
- C. the definition of done has been relaxed
- D. work in progress has been raised

*Rationale:* Rising units with flat item throughput is the specific signature of sizing inflation (13.4.2, 13.4.4). A relaxed definition of done would show completion rising then reopened items rising; raised work in progress would show cycle time rising.

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SELF-CHECK — KA 13.4

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1. *What single question discriminates between a capacity-based and a scope-based model?* — How many material re-prioritisations the work expects. Below about three the fixed price is cheaper; above it the capacity model is, by **USD 177,840** a change.
  2. *What replaces price as the buyer's control under a capacity contract?* — The continuation decision at a bounded funding envelope, the value-envelope reconciliation, and an exit right that leaves a transferable asset.
  3. *Why must size units never be aggregated across teams?* — They are team-relative weights set by the team being measured, so the sum has no meaning; throughput and cycle time are the portfolio-level measures.
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## Advanced topics — Domain 13

### 13.A.1 Adaptive delivery where the increment cannot be released

The honest limit of this domain's methods is reached when condition two of 13.1.1 fails: feedback is neither cheap nor fast, because the increment cannot be put in front of anyone. A partially commissioned substation, a half-migrated clinical dataset, a drug trial at interim, an aircraft modification awaiting airworthiness sign-off: in each case the loop the iteration exists to close cannot close inside it, and a team running two-week iterations against such work is performing the ceremony of adaptive delivery without its engine.

What survives, and it is a great deal, is the flow arithmetic. Little's Law does not require releasability; it requires a steady state. Project Auriga's commissioning queue is the illustration worked in Exercise 13.1: test packs flow through a single commissioning team, work in progress can be limited, cycle time can be measured, and raising work in progress damages throughput exactly as it does on a software team: auriga's numbers give **4.94 weeks** of delay worth **USD 222,353**. So the practitioner's rule is to separate the two families of technique rather than adopting or rejecting them together: **flow control travels everywhere; iterative release travels only where feedback closes**.

What must be replaced is the feedback mechanism: a **representative environment** (a simulator, a pilot clinic, a digital twin — Domain 14, KA 14.2), a **partial-scope release to a limited population**, or an explicit acceptance that the work is predictive and should be controlled as such. The failure to avoid is the third option adopted implicitly: keeping the cadence and the vocabulary while quietly abandoning the feedback, which is 13.1.1's warning in its most expensive form.

### 13.A.2 AI-assisted adaptive delivery, and the forecast that becomes a commitment

Where AI is used inside the delivery loop (generating candidate backlog items, drafting acceptance criteria, producing code, forecasting throughput) three obligations arise that existing adaptive practice does not express, and they must be added deliberately (the same conclusion Domain 3's KA 3.A.2 reaches for governance).

**The definition of done must extend to AI-produced work.** An AI-generated item is not done because it compiles; it is done when it meets the same binary standard as anything else, including review by a person who can be answerable for it. The specific hazard is **volume**: a model can produce candidate items and code faster than a team can review them, which raises work in progress without raising capacity, and 13.2.3 has already computed what that does. AI-assisted generation without a matched increase in review capacity is a work-in-progress decision in disguise.

**Throughput history must be segmented when the tooling changes.** A team adopting a substantial new tool has changed its own service-time distribution, so the twelve-week history of 13.2.4 is no longer a sample of one system. State the change date, hold both histories, and widen the range until enough post-change observations exist, rather than claiming the improvement in advance, which is the commonest error and the one that produces the next missed date.

**The forecast must not become the commitment.** This is the most serious hazard in the domain and it is sociological rather than technical. A machine-produced forecast arrives with more apparent authority than a human one: it is precise, it is fast, and it does not hedge unless asked to. A range with four stated assumptions becomes a date; the date is quoted upward; and by the time it is challenged it has been repeated in three papers. The countermeasure is procedural and cheap: **no forecast leaves the team without its required rate attached** (13.2.4's 8.56 items a week), because a required rate cannot be rounded into a promise, it either exceeds the team's demonstrated capability or it does not, and the question is answerable by anyone.

### 13.A.3 The reviewer's flow eye

Invariants to test on any adaptive or hybrid delivery design, each cheap and each diagnostic.

Every backlog is **ordered, attributed, sized and criterion-bearing**, and items have been *removed* from it within the last quarter. Exactly **one** named product owner holds the ordering right, with a stated value envelope, and the median wait for their decisions is measured. Prioritisation is on **delay-cost density**, not on cost of delay or effort alone, and the cost-of-delay numerator traces to a benefits-map line with an owner. The **definition of done is written, binary and unchanged** under schedule pressure, and reopened-item counts are reported. A **work-in-progress limit exists as a number**, is visible, and every change to it is a recorded decision. **Throughput, cycle-time distribution, work in progress with its blocked and aged shares, and flow efficiency** are all measured; velocity, if used at all, is confined to planning inside one team and never aggregated. Every date is a **range with its window, its arrival rate and its assumptions stated**, and every committed date is accompanied by its **required rate**. The **net drain** is computed: arrivals are counted, not assumed away.  $E[\text{wait}]$  is computed for **each** decision body against **its own** cycle, the blocked share of work in progress is reported, and out-of-cycle routes exist for every class that may legitimately use one. Hybrid designs name the **boundary**, name the **integration milestone's owner**, and state which variable floats on each side. Progress reaches governance as **count and size**, never as an unlabelled percentage. And the commercial model is matched to the expected **change count**, with the capacity model's continuation decision and value-envelope reconciliation actually in place; because a capacity contract without a brake is the most expensive single arrangement in this domain.

## Industry variations — Domain 13

- **Healthcare.** Clinical safety and information-governance approvals are decision classes that cannot be written-resolved, so 13.3.2's remedy applies to the commercial and functional classes only; Meridian's design must delegate those it may and place clinical sign-off with clinical authority (Domain 3's healthcare variation). The compensating move is a standing clinical reviewer inside the cadence rather than a committee outside it.
  - **Financial services and other regulated releases.** Release is gated by change-management and audit requirements, so the increment can be *complete* far more often than it can be *released*; the honest design separates a fortnightly definition of done from a monthly or quarterly release train, and the flow measures must then track both cycle time to done and cycle time to live. The gap between them is the regulatory cost, and it should be a stated number rather than a grievance.
  - **Construction and infrastructure.** Physical works are predictive and the design and digital workstreams around them are frequently adaptive, so these programmes are hybrid by nature and fail at 13.3.3's boundary rather than inside either regime. Flow control nonetheless travels: snagging, commissioning and inspection queues are pure Little's Law problems, and work-in-progress limits on them are among the cheapest interventions available (13.A.1).
  - **Public sector.** Statutory approvals and annual appropriation cycles set `E[wait]` and the funding cadence from outside, so the leader's available levers are the paper lead time and the delegated envelope, exactly as Domain 3's public-sector variation concludes. The characteristic risk is the capacity contract without a continuation decision, because appropriation encourages spend-to-budget, which is Case study B's mechanism.
  - **Energy and utilities.** Control-systems and operational-technology work of Auriga's kind cannot release increments into a live network at will, so iterations produce tested, staged increments that are cut over in windows; the flow measure that matters is the cycle time from "ready to cut over" to "cut over", which is usually dominated by outage-window availability rather than by the team.
  - **Software, digital and product organisations.** The natural home of the methods and the natural home of their anti-patterns: the characteristic failures are velocity as a performance measure across teams (13.4.2), scaling by adding teams rather than reducing the number of things that must agree (13.3.1), and light governance on genuinely irreversible architectural and data-model choices, which do not present as gates (Domain 3's technology variation).
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## Case study — Domain 13: the parallel start (health, Meridian)

**Situation.** At week 26 of Meridian's application build, three clinics were waiting on the regulatory reporting module and the programme was under visible pressure. The steering committee, meeting on its four-week cycle, instructed the team to start the reporting module in parallel with existing work, taking work in progress from **18** to **30** items. The team's product owner objected. Asked why, they said the team would slow down, which was heard as a preference. The instruction stood.

**What the arithmetic showed.** The delivery manager computed four numbers before the following governance review, all from records the programme already held. **The parallel start.** On the team's own switching coefficient of 2 % of capacity per extra concurrent item, throughput fell from **6.00** to **4.56** items a week and cycle time rose from **3.00** to **6.58 weeks**; the 240-item backlog moved from **40.00** to **52.63 weeks**, a cost of USD **180,379** at the programme's 14,280 a week; and the reporting module the committee wanted accelerated arrived later than it would have if the team had finished its current work first. **Governance latency.** With  $E[\text{wait}] = 4/2 + 2 = 4.0$  weeks (Domain 3, KA 3.2.3) and 15 % of items exceeding the product owner's envelope, **3.60 items** (20.0 % of the team's work in progress) were parked waiting for the committee at any moment, and those items took **6.40 weeks** against **2.40** for the rest. **The forecast.** The plan's 34-week date required a gross throughput of **8.56** items a week, **31.7 %** above the best four-week rate the team had ever sustained. **The combined position.** Left as instructed (work in progress at 30, arrivals at 1.5 a week, governance at four weeks) the net drain was **3.06** items a week and the release would take **78.43 weeks**. On the corrected design it would take **40.00**. The difference was **38.43 weeks** and USD **548,800**.

**How it resolved.** Three changes, none of which reduced any scrutiny anyone could name. The work in progress limit was set at **18** and made visible, with any change to it recorded as a decision, which also settled the reporting module, now sequenced rather than parallelised. A **written-resolution route with a five-working-day turnaround** was adopted for commercial and functional decision classes, excluding clinical safety and information governance, taking blocked work in progress from 3.60 to **0.90 items** and average cycle time from 3.00 to **2.55 weeks**: a **17.6 %** throughput increase at unchanged staffing. And the 34-week date was replaced with a **40-week conditional forecast**: conditional on the governance route and on discovery arrivals being held at or below **1.06 items a week**, a **29.4 %** reduction, with a named owner for each condition and both reported weekly.

**What the domain teaches here.** The product owner's objection was correct and unpriced, and an unpriced objection loses to an urgent instruction every time. The same sentence in two forms: "starting the reporting module in parallel will slow us down" is a preference, while "it costs USD 180,379, delivers the reporting module later than sequencing it would, and takes the release from 40 to 52.63 weeks" is a proposal; and the second was accepted at the first meeting at which it was made. Note also which of the four numbers moved the room: not the 180,379, which was

contested, but the observation that **one item in five the team was holding was not being worked on at all**. Flow arithmetic persuades because it describes something the audience can verify by walking to the board.

## Case study B — Domain 13: the capacity contract that lost its brake (public sector)

**Situation.** A public-service digital programme contracted two supplier squads on a capacity basis at **USD 60,000** per squad per two-week iteration (**USD 120,000** an iteration) with a planned twelve months of **26 iterations**, or **USD 3,120,000**. The commercial rationale was sound and matched 13.4.1's analysis: the requirement was genuinely unspecifiable at award and material re-prioritisation was expected. The governance body received an iteration report every cycle showing velocity, items completed and a burn-up chart. Velocity was stable and slightly improving throughout.

**What had happened.** The contract had been let for capacity and controlled as though it had been let for scope. There was no bounded funding envelope, so continuation was never a decision: each iteration followed the last because nothing stopped it. There was no value-envelope reconciliation, so nobody compared what had been delivered against what the business case had promised. And the reports the body received measured **activity that was genuinely healthy**: the squads were delivering, the definition of done was holding, and no anti-pattern of the delivery kind was present. By **iteration 44** the programme had spent **USD 5,280,000 (169.2 % of the planned cost, an overrun of USD 2,160,000)**, when an assurance review finally ran Domain 5's KA 5.A.1 reconciliation. Against a business-case benefit commitment of **USD 1,850,000** a year, the attributed benefit of everything delivered was **USD 758,500** a year: **41.0 %**. The programme had bought **169.2 %** of its cost for **41.0 %** of its benefit, and had done so while every delivery metric it monitored was green.

**How it resolved.** The reconciliation was made a standing control on a four-iteration cycle, alongside a **bounded funding envelope of four iterations (USD 480,000)**, requiring an explicit continuation decision by a named authority at each boundary (13.4.2). The divergence between delivered value and committed benefit had first become computable at around iteration 12; had the two controls been in place from then, the natural stop point was **iteration 16** at **USD 1,920,000**, so the controls were worth **USD 3,360,000**. Their cost was three days of analyst time per cycle (about **USD 4,500**, or **USD 36,000** across the eight cycles) a protection ratio of **93.3 times**. The programme was re-scoped to the two capabilities carrying most of the attributed benefit and closed nine iterations later.

**What the domain teaches here.** A capacity-based contract is the right instrument for unspecifiable work and it has no brake of its own: the price does not stop rising when the value stops arriving, because price is a function of time and not of scope.

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Whatever else changes, **a capacity model must be paired with a continuation decision and a value test, or it is an open-ended commitment with a delivery report attached.** Note the specific trap, which is the reason this failure is so hard to see from inside: every metric the governance body received was accurate, favourable and about the wrong thing. Velocity, items completed and a burn-up chart measure whether a team is delivering; none of them can measure whether what is being delivered is worth buying, and no amount of improvement in the first answers the second. That is the whole case for the fifth number on 13.4.2's one-page report.

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## Executive perspective — Domain 13

What a programme director cannot delegate in this domain:

- **The work-in-progress limit, and the authority to instruct past it.** "Start it in parallel" is the most expensive sentence available to an executive in adaptive delivery: **USD 180,379** on Meridian, for nothing delivered sooner. If you may override the limit, you own the arithmetic (13.2.3).
  - **The product owner's authority, in writing.** One named holder, a stated value envelope, and a measured decision wait. A product owner who must consult a committee is a proxy, and the committee's latency is now your delivery cycle time (13.1.2, 13.3.2).
  - **Your own governance cadence, computed against the work's cadence.**  $E[\text{wait}] = M/2 + L$  against the iteration length. Four weeks against a fortnight parks **20.0 %** of a team's work in progress and makes the high-value items the late ones (13.3.2).
  - **Whether a date is a forecast or a commitment, and which one you are quoting.** Ask for the range, the arrival rate and the **required rate**. A committed date requiring 31.7 % more throughput than the team's best ever four weeks is not ambitious; it is unevidenced (13.2.4).
  - **The commercial model against the expected change count.** Below about three material changes take the fixed price; above it take capacity, and if you take capacity, install the continuation decision and the value-envelope reconciliation before the first iteration, not after the forty-fourth (13.4.1, 13.4.2, Case study B).
  - **The refusal to measure people with flow data.** Throughput, cycle time and work in progress are properties of a system you designed. Used as individual or team targets they become useless within a quarter, and velocity fastest of all, a 25 % sizing inflation reports a 25 % improvement and delivers nothing (13.4.2).
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## Calculation exercises — Domain 13

**Exercise 13.1** Project Auriga's commissioning team completes 5 test packs a week with 20 in progress. The site manager adds 10 packs, taking work in progress to 30; each pack in progress above 20 costs 1.5 % of the team's capacity. The remaining queue is 140 packs and Auriga's schedule value is USD 45,000 a week (Domain 6, KA 6.4.2). Compute cycle time before and after, and the cost of the decision. *Solution.* Before:  $C = 20/5 = 4.00$  weeks. After:  $T = 5 \times (1 - 0.015 \times 10) = 5 \times 0.85 = 4.25$  packs a week, so  $C = 30/4.25 = 7.06$  weeks. Queue:  $140/5 = 28.00$  weeks against  $140/4.25 = 32.94$  weeks, a delay of 4.94 weeks costing USD 222,353. Note that the physical setting changes nothing about the arithmetic: releasability is what does not travel to Auriga, not Little's Law. Common error: holding throughput constant and reporting cycle time as  $30/5 = 6.00$  weeks with no effect on the queue, which treats work in progress as free and understates both the cycle time and the whole of the cost.

**Exercise 13.2** A team's last ten weeks of completions are 7, 5, 8, 6, 9, 6, 7, 5, 8, 9. The remaining backlog is 180 items, discovery arrivals run at 1.25 items a week, and the cost of delay is USD 9,500 a week. Using four-week rolling totals, compute the honest range, the mean-rate forecast, and the error made by ignoring arrivals. *Solution.* Total 70, mean 7.00 items a week. Four-week rolling totals 26, 28, 29, 28, 27, 26, 29: sustained rates 6.50 to 7.25. Net drains  $7.25 - 1.25 = 6.00$  and  $6.50 - 1.25 = 5.25$ , giving 30.00 to 34.29 weeks. At the mean rate net of arrivals,  $180/5.75 = 31.30$  weeks. Ignoring arrivals gives  $180/7.00 = 25.71$  weeks, understating by 5.59 weeks and USD 53,106. Common error: forecasting from the single best week ( $180/9 = 20.00$  weeks), which uses a rate the team has sustained for one week in ten and ignores arrivals as well, compounding two optimisms into a figure 11 weeks below the fast end of the honest range.

**Exercise 13.3** A board meets every 6 weeks with a 2-week paper lead time and governs a team running 3-week iterations at 10 items a week with 40 items in progress; 12 % of items require a board decision. Compute  $E[\text{wait}]$ , the blocked work in progress, the cycle-time decomposition, and the effect of a one-week written-resolution route. *Solution.*  $E[\text{wait}] = 6/2 + 2 = 5.0$  weeks: 1.67 iterations.  $C = 40/10 = 4.00$  weeks. Blocked arrivals  $0.12 \times 10 = 1.20$  a week, so blocked work in progress  $1.20 \times 5.0 = 6.00$  items, 15.0 % of the total. Unblocked  $4.00 - 0.12 \times 5.0 = 3.40$  weeks; blocked 8.40 weeks, a multiple of 2.471. With a one-week route,  $C = 0.88 \times 3.40 + 0.12 \times 4.40 = 3.52$  weeks, a 12.0 % reduction, and at fixed work in progress  $T = 40/3.52 = 11.36$  items a week, up 13.6 %. Common error: computing blocked work in progress from the iteration length rather than from  $E[\text{wait}]$  ( $1.20 \times 3 = 3.60$  items, 9.0 %), which understates the parked population by 40.0 % and hides most of the problem.



**Exercise 13.4** Three features carry costs of delay of **3,600**, **5,400** and **1,800** a week and efforts of **4**, **9** and **6** team-weeks. Compute the delay cost of the density order, of the cost-of-delay-first order and of the shortest-first order, and the estimate error that would change the ranking. *Solution.* Densities **900.00**, **600.00**, **300.00**, so the density order is F1, F2, F3, completing in weeks 4, 13 and 19:  $3,600 \times 4 + 5,400 \times 13 + 1,800 \times 19 = \text{USD } 118,800$ . Cost-of-delay-first (F2, F1, F3), completing in weeks 9, 13, 19: **USD 129,600** — **USD 10,800** worse. Shortest-first (F1, F3, F2), completing in weeks 4, 10, 19: **USD 135,000** — **USD 16,200** worse. The worst order (F3, F2, F1) costs **USD 160,200**, a spread of **USD 41,400**. F2 would need a cost of delay above  $900 \times 9 = \text{USD } 8,100$  a week (**50.0** % above its estimate) to belong first, so the ranking is robust to moderate error. Common error: ranking on effort alone, which looks like a flow heuristic ("smallest first") and is **USD 16,200** worse here than ranking on the ratio; the ratio is the rule and either component alone is a special case of it.

**Exercise 13.5** A firm price of **USD 900,000** covers **180** items; the alternative is capacity at **USD 45,000** per two-week iteration over a planned 20 iterations. The change schedule prices additions at **USD 7,500** and credits omissions at **USD 3,000**; the change board's  $E[\text{wait}]$  is **4.0** weeks against a **1.0**-week iteration boundary, and the cost of delay is **USD 9,500** a week. A net-zero change swaps **24** items. The honest duration range is **22 to 27 iterations**. Compute both buyer exposures and the breakeven change count. *Solution.* Both models price at **USD 900,000** ( $900,000/180 = 5,000$  an item;  $45,000 \times 20 = 900,000$ ). Scope-based change:  $24 \times 7,500 = 180,000$  less  $24 \times 3,000 = 72,000$ , a net **USD 108,000**, plus latency  $4.0 \times 9,500 = \text{USD } 38,000$  — exposure **USD 146,000**. Capacity-based:  $1.0 \times 9,500 = \text{USD } 9,500$ . Difference **USD 136,500** a change. Capacity cost range  $22 \times 45,000 = \text{USD } 990,000$  to  $27 \times 45,000 = \text{USD } 1,215,000$ , so worst-case quantity exposure is **USD 315,000** and the breakeven is  $315,000/136,500 = 2.31$ : from the **third** change the capacity model is cheaper even at the worst end ( $900,000 + 3 \times 146,000 = \text{USD } 1,338,000$  against 1,215,000). Common error: comparing the change order's **USD 108,000** against zero and concluding the capacity model is free. It is not free: it transfers up to **USD 315,000** of quantity risk to the buyer, and that risk is controlled by the continuation decision, not by the contract.

## Practitioner's toolkit — Domain 13

*Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.*

### Toolkit 13.T.1 — Flow sheet (one page, one team, weekly)

Rows, all counts or timestamps and none of them a judgement: **throughput** this week and its four-week rolling total; **work in progress** against its stated limit, split into *being worked*, *blocked* (with what each is waiting for) and *aged beyond twice the median cycle time*; **cycle time** as a distribution (median, 85th percentile and longest) never as a mean alone; **flow efficiency** from a sampled touch-time estimate; **discovery arrivals** this week and their four-week rate; and the derived **net drain**. Below the rows, three computed lines: the **range forecast** from the highest and lowest observed four-week net drains, with the window length stated; the **required rate** implied by any date the organisation is currently quoting; and  $E[\text{wait}]$  for each decision body the team depends on, with the blocked share it produces. The sheet's purpose is that every one of these is a number the team can verify by walking to the board, and a delivery design that cannot fill it in is not being measured.

### Toolkit 13.T.2 — Backlog ordering sheet with delay-cost density

Columns: item or epic · outcome statement · **benefits-map line and owner** (the numerator's provenance — without it the density is advocacy with arithmetic on top) · cost of delay per week · effort in team-weeks · **density** · dependency constraints · **releasable alone? (yes/no)** · acceptance criterion · **classification — obligation or discretionary**, applied when the item is created and owned by the accountable authority for that class, since an **obligation item** is scheduled to its required date by that authority and is excluded from the density ranking altogether (13.1.2) · product-owner decision (order / defer / decline) with date, available only on discretionary items. Two standing checks: the **breakeven test** for the top three items. The cost-of-delay increase that would reorder them, which tells you whether the order is robust or fragile; and the **releasability count**, because items that cannot be released alone forfeit the sequencing value the sheet exists to capture (on Meridian, **USD 253,640**). One rule of use: the "decline" cell must be used. A sheet with no declined items records a backlog that only grows, which is 13.4.4's first anti-pattern.

### Toolkit 13.T.3 — Adaptive commercial and continuation control sheet

Part A, **the model choice**: expected count of material re-prioritisations · addition rate and omission credit under a scope model, with the net movement on a representative net-zero change ·  $E[\text{wait}]$  for the change board · the resulting per-change buyer exposure for each model · the honest duration range and the quantity exposure it implies under capacity · the **breakeven change count** · the recommendation and the one assumption it is most sensitive to. Part B, **the brake** (mandatory whenever the model is capacity): the **bounded funding envelope** in iterations and currency · the named authority for each continuation decision and the date of the next one · the **value-envelope reconciliation**: attributed benefit of delivered items against the business-case commitment, with its cycle and its owner · and the exit provision with the list of assets that must transfer for it to be exercisable. Part B is the artefact whose absence cost Case study B **USD 3,360,000** at a control cost of

**USD 36,000**; it should be completed before the first iteration is funded, and reviewed at every envelope boundary.

## Exam preparation — Domain 13

**What is assessed.** The conditions under which adaptive delivery outperforms predictive delivery, and which variable floats in each mode; product ownership as a decision right with its four failure modes; **delay-cost density and the cost of a sequence**, including the role of releasability; the backlog's four properties and the value-envelope reconciliation; capacity as an observation and the commitment fallacy; **Little's Law in all three forms**, work-in-progress effects on throughput, cycle time and flow efficiency, and the price of raising work in progress; **range forecasting from throughput history, net of discovery arrivals, and the required rate implied by a committed date**; the coordination and interface arithmetic of scaling (cited from Domains 12 and 4); **governance latency as a flow parameter: blocked work in progress, the cycle-time decomposition, and the throughput identity at fixed work in progress**; hybrid design with a named boundary and its four failure modes; the count view against the size view; **capacity-based against scope-based commercial models on buyer exposure under a change, and the breakeven change count**; what replaces price as the buyer's control; flow metrics that resist gaming and why velocity is not one; and the anti-patterns with their arithmetic signatures.

**The calculations to be able to do under time pressure.**  $C = W/T$ ,  $T = W/C$  and  $W = T \times C$ . Throughput under a stated switching coefficient, and the resulting cycle time, flow efficiency and backlog duration, priced at a cost of delay. Delay-cost density, the total delay cost of a sequence ( $\Sigma$  cost of delay  $\times$  completion week), the saving against an alternative order, and the breakeven cost-of-delay change that reorders it. Four-week rolling throughput, the net drain, a range forecast, and the required gross rate implied by a target date.  $E[\text{wait}] = M/2 + L$ , blocked arrival rate, blocked work in progress, the cycle-time decomposition into blocked and unblocked populations, and the throughput uplift from a shorter wait at fixed work in progress. Percentage complete by count against by size, and the remaining duration each implies. Buyer exposure per change under both commercial models, the quantity exposure of the capacity model, and the breakeven change count.

**The traps.** Holding throughput constant when work in progress rises, so cycle time is computed as  $W/T_0$  and the backlog looks unaffected (Exercise 13.1) · forecasting from a mean rather than a range, and from gross throughput rather than the net drain. The second is worth **13.33 weeks** and **USD 190,400** on Meridian on its own (13.2.4) · forecasting from the single best week (Exercise 13.2) · computing blocked work in progress from the iteration length instead of  $E[\text{wait}]$ , understating it by 40 % (Exercise 13.3) · adding  $E[\text{wait}]$  to the *average* cycle time rather than to the *unblocked* population, double-counting the blocked contribution (MCQ 13.3-B) ·

assuming a proportional cycle-time fall gives an equal proportional throughput rise, when the relationship is reciprocal (MCQ 13.3-C) · ranking a backlog on cost of delay alone or on effort alone rather than on the ratio (Exercise 13.4) · forgetting that sequencing value requires releasability, so a big-bang release costs **USD 485,520** whatever the order (MCQ 13.1-B) · reading a net-zero change as a zero-cost change, when the asymmetry between addition rate and omission credit moves the price by **USD 135,000** (MCQ 13.4-A) · treating a capacity contract as free of change cost while leaving its quantity exposure unmanaged (Exercise 13.5, Case study B) · reporting an item count as progress when optimal sequencing guarantees it overstates (13.3.4) · and aggregating team-relative size units into a programme total (13.4.2).

**How the domain connects.** Domain 1 supplies the cost of delay every result here is priced at and the accountability principle that keeps a forecast owned by a person; Domain 2 the benefits map that makes delay-cost density defensible and the kill criteria the continuation decision re-tests; Domain 3  $E[\text{wait}] = M/2 + L$ , the bounded envelope and the out-of-cycle route, which this domain quantifies rather than re-derives; Domain 4 the interface arithmetic and the change-control machinery the scope-based model runs on; Domain 5 the acceptance criteria, value attribution and value-envelope reconciliation; Domain 6 the translation rule that lets a throughput forecast enter a CPM network, and the drum that explains why a shared scarce resource sets programme throughput; Domain 9 the rework multiplier that prices a relaxed definition of done; Domain 10 the risk-allocation frame; and Domain 12 the coordination arithmetic that makes scaling expensive. Forward: Domain 14 takes the flow measures into dashboards and the AI-use register, Domain 15 aggregates them at portfolio level (throughput and cycle time, never size units), and Domain 16 receives the definition of done as the handover standard.

## Domain 13 summary

Adaptive delivery fixes capacity and cadence and lets scope float; predictive delivery does the reverse; hybrid delivery does both, on named sides of a named boundary. None of that is a preference, and the whole of this domain's contribution is to show that the arguments practitioners have about it are arithmetic arguments in disguise.

Prioritisation is a sequencing decision with a price. Ranked by **delay-cost density**, cost of delay per team-week, Meridian's four release epics cost **USD 231,880** in delay; ranked by cost of delay alone, **USD 276,080**; in the worst available order, **USD 386,920**. The sequencing decision is therefore worth **USD 155,040**, and the ranking is robust to a **55.6 %** error in the largest epic's estimate. All of that value depends on **releasability**: a single release at week 34 costs **USD 485,520** whatever the order, so incremental release is worth **USD 253,640** on this release alone. And the ranking has a boundary as well as a formula: the product owner orders *discretionary* value, so items carrying a safety, regulatory, statutory, contractual or security-remediation

obligation are constraints on the order rather than candidates within it: scheduled to their required date by the authority accountable for the obligation, excluded from the density ranking, and classified as such when the item is created. A product owner who deprioritises one has exceeded the ordering right rather than exercised it.

Flow rests on one theorem. **Little's Law** ( $W = T \times C$ ) says that cycle time is a consequence of a management decision about work in progress, and that at fixed work in progress a cycle-time reduction is a throughput increase. Meridian's team at  $W = 18$  and  $T = 6$  has a cycle time of **3.00 weeks** and a flow efficiency of **30.0 %**. Instructed to raise work in progress to 30, throughput falls to **4.56 (-24.0 %)**, cycle time rises to **6.58 weeks (+119.3 %)**, flow efficiency falls to **13.7 %**, and the 240-item backlog moves from **40.00** to **52.63 weeks** at a cost of **USD 180,379**, a **66.7 %** increase in work in progress that bought **0 %** more throughput and delivered nothing sooner.

Forecasts are ranges with stated meanings. Meridian's twelve-week history of **72** completions smooths into four-week sustained rates of **5.50 to 6.50** items a week; net of **1.5** items a week of discovery arrivals the honest range is **48.00 to 60.00 weeks**. Ignoring arrivals alone understates the date by **13.33 weeks** and **USD 190,400**. The plan's committed 34 weeks requires a gross throughput of **8.56** items a week (**31.7 %** above the best four weeks the team has ever had), which is why the defensible answer was not a new date but a **40-week** forecast with two named, measured conditions.

Governance latency is a flow parameter. Domain 3's  $E[\text{wait}] = M/2 + L$  gives Meridian's committee **4.0 weeks**, which is **two entire iterations**; with **15 %** of items exceeding the product owner's envelope, **3.60 items, 20.0 % of the team's work in progress, are parked at any moment**, and blocked items take **6.40 weeks** against **2.40** for the rest, a multiple of **2.667**. A one-week written-resolution route takes average cycle time to **2.55 weeks** and, at unchanged work in progress, throughput to **7.06** items a week: a **17.6 %** increase for the price of an administrative procedure. Reported to a predictive body, the same release is **40.0 %** complete by item count and **28.6 %** by size, an overstatement of **11.4 percentage points** and **16.0 weeks**, and the direction is systematic, because optimal sequencing delivers the tractable items first.

Commercially, the two models price different uncertainties. On Meridian's build both cost **USD 1,200,000** at plan; a net-zero change of 30 items exposes the scope-based buyer to **USD 192,120** (a **USD 135,000** price movement from the asymmetry between a 7,500 addition rate and a 3,000 omission credit, plus **USD 57,120** of decision latency) against **USD 14,280** under capacity; but capacity carries up to **USD 600,000** of quantity exposure across the honest duration range. The breakeven is **four** material changes against the worst case and **2.25** against the expected cost. And a capacity model has no brake of its own: Case study B's programme reached **169.2 %** of planned cost for **41.0 %** of committed benefit while every delivery metric

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it monitored was green, and the two controls that would have stopped it at **USD 1,920,000** cost **USD 36,000** against **USD 3,360,000** protected, a ratio of **93.3 times**.

The through-line: **adaptive delivery does not remove the numbers, it changes which numbers**. Throughput, work in progress, cycle time, net drain, blocked share, required rate, delay-cost density, buyer exposure per change: every complaint about agile delivery in a large organisation is one of these eight quantities in ordinary language, and a leader who can compute them stops having the argument and starts making the proposal.

PART FOUR

# Enterprise delivery and the digital future

Domains 14–16 — digital delivery and responsible AI, programmes and portfolios, and transition and benefits realisation: delivery at enterprise scale.

DOMAIN

# 14

## Digital Delivery, Data and Responsible AI

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## Why this domain exists

Thirteen domains have now used AI in the way a professional actually uses it: as an assistant to a specific task, bounded by a specific prohibition. Domain 3 refused it a governance decision; Domain 4 refused it a baseline; Domain 5 refused it the authorship of a requirement; Domain 9 refused it a test result and an inspection signature; Domain 10 refused it a quality score and a liability cap; Domains 11 and 12 refused it inference about people. Each refusal was correct and each was local. What none of them could supply, standing alone, is the thing an organisation actually needs: **one system in which those refusals are consistent, resourced, auditable and priced.**

That is what is missing in practice, and its absence has a characteristic shape. Organisations adopt digital tools and AI assistance tool by tool, each justified on a productivity claim, each verified by whoever happens to be nervous, none priced. The result is an estate in which the verification effort is simultaneously excessive on low-consequence outputs and absent on high-consequence ones; in which a data defect rate is reported as a single percentage that no decision can use; in which an automation is celebrated for the labour it removed and never charged for the errors it lets through; and in which the most dangerous artefact in the whole system (a fluent, plausible, wrong number) passes every review that was designed to catch an obviously wrong one.

Hence the domain's central claim: **digital delivery, data quality and AI assistance are all economic quantities with computable breakevens, and the professional obligation is to compute them rather than to adopt or resist on principle.** A leader who can state the consequence at which verification becomes worth its cost, the volume at which an automation starts paying, the exposure that each class of data defect carries and the ratio by which a plausible error out-damages an obvious one has a governable digital estate. A leader who cannot will oscillate between two equally indefensible postures: blanket prohibition, which forfeits real value, and blanket adoption, which transfers an unmeasured cost to whoever inherits the output.

This is therefore the most rigorous domain in the volume rather than the softest, which is the opposite of how responsible-AI material is usually written. KA 14.1 builds the substrate: the digital environment, the governance of data, the common data environment, and the defect rate by data class that determines whether anything built on top of it is usable. KA 14.2 covers what organisations build on that substrate: dashboards, analytics, digital twins and automation, each tested against the decision it claims to serve and the volume at which it pays. KA 14.3 is the domain's core: where AI earns its place across the lifecycle, how a prompt becomes professional practice, and the derivation of the verification standard proportional to consequence, followed by the arithmetic of why the plausible wrong number is the

one to fear. KA 14.4 closes with the four obligations that cannot be computed away: explainability that supports a decision, differential error, human accountability, and the security and privacy of a delivery estate that now holds everything.

**Learning objectives.** After this domain a candidate can: describe a digital delivery environment as a set of systems with a countable interface problem and apply Domain 4's interface arithmetic to it; specify data governance as ownership, definition, lineage and access rather than as a policy document; state what a common data environment must guarantee and test whether a claimed one does; **compute a defect rate by data class, convert it into consequence-weighted exposure, rank remediation by exposure per record, and demonstrate arithmetically that a uniform data-quality target both reduces defect counts and increases expected cost**; test a dashboard against the decision it claims to inform and **compute the expected age of a fact at the moment of decision using the registered  $M/2 + L$  identity, priced at the cost of delay**; place an analytics initiative on the descriptive-to-prescriptive ladder and state what each rung requires and returns; define what makes a digital twin a twin and **compute the fidelity at which it breaks even**; **compute an automation breakeven volume, adjust it for the differential error-escape rate, and identify the cases in which no volume justifies the automation**; state where AI earns its place at each stage of the lifecycle and cite the domain that governs each; write and govern a prompt as a professional artefact with grounding, provenance and a recorded version; **derive a tiered verification standard from the marginal cost of a review tier, its marginal detection rate and the consequence of an escaped error, and price the tiered standard against uniform alternatives**; **compute why a plausible wrong number costs an order of magnitude more than an obviously wrong one, and price the reperformance step that closes the gap**; specify explainability as a decision requirement rather than a model property; **compute differential error rates across groups, price the remediation, and state where expected-value reasoning stops applying**; maintain an AI use register that satisfies Domain 1's accountability test; **state the failure route for an output found materially wrong after reliance: stop and suspend, enumerate the decisions that cited it, notify the same day, reopen above a pre-set threshold with the original authority, record whether anyone outside must be told and on whose determination, and re-measure before resumption**; and **compute the economics of security controls, distinguishing probability reduction from impact reduction and demonstrating why control benefits are sub-additive**.

**The master threads.** Meridian Care Records supplies the programme case: the clinical-records rollout to **40 clinics**, approved cost **USD 2,400,000**, full-potential benefit **USD 979,200** a year, realistic benefit **USD 685,440** a year at 70 % adoption, **cost of delay USD 14,280 per week** (Domain 1) — **USD 2,040 per day** — and a steering committee whose latency  $E[\text{wait}] = 4/2 + 2$  is **4.0 weeks** (Domain 3, KA 3.2.3). Internal effort is valued at the blended programme rate of **USD 110 per hour** established in Domain 11. This domain audits Meridian's common data

environment across **14,100** records in six classes, derives its verification standard across **941** AI-assisted outputs a month, and prices its security controls. Project Auriga supplies the project case: the 25-week control-systems upgrade, **BAC USD 4,000,000**, cost of delay **USD 45,000 per week**, a blended engineering rate of **USD 130.625 per hour** (Domain 7, KA 7.4.1), and at week 13 **PV 2,080,000 / EV 1,920,000 / AC 2,120,000**, giving **CPI 0.91** and **SPI 0.92**. This domain prices Auriga's data-check automation, its control system's digital twin, and the forecast error that its week-13 numbers make possible.

#### KNOWLEDGE AREA 14.1

## Digital project environments, data governance and the common data environment

*Topics: 14.1.1 the digital delivery environment · 14.1.2 data governance · 14.1.3 the common data environment · 14.1.4 defect rate by data class and the exposure it carries.*

### 14.1.1 The digital delivery environment

**Definition.** A project's **digital delivery environment** is the set of systems that hold its authoritative information and mediate its work: the schedule tool, the cost ledger, the requirements and change registers, the document and model repository, the test and defect systems, the procurement and contract records, the risk register, and the reporting layer assembled over them.

The environment is not a technology question, and treating it as one is the first error. It is an **information architecture** question with a shape Domain 4 already priced. Each system holds facts that other systems need, so each pair of systems is a candidate interface, and the count of possible pairwise interfaces over **n** systems is the registered  $n(n - 1) / 2$  (Domain 4, KA 4.2.3). Meridian's environment has eleven systems of record; the mesh count is  $11 \times 10 / 2 = 55$  possible interfaces against **11** connections to a single integration layer, and the practical consequence is the one Domain 4 states: the mesh is cheaper for the first two or three systems and catastrophically more expensive by the eleventh, because each interface is a thing to build, test, version, secure and re-verify after every change on either side.

Three properties determine whether the environment helps or hinders, and all three are testable before anything is procured. **Authority.** For each material fact, exactly one system is the source of record, and the others hold copies marked as copies. An environment in which two systems each believe they own the baseline dates has not integrated anything; it has arranged for a disagreement. **Lineage.** For any number on a report, the path from source record to displayed value is reconstructable. **Latency:** the age of the information at the point of use, which 14.2.1 computes.

**What the leader owns here.** Not the tool selection, which is usually an enterprise decision, but four things that are never in a procurement document: which system is authoritative for which fact; who may change each class of record and under what authority (which is Domain 3's delegation schedule expressed in system permissions); what the environment must produce for the governance body and by when; and what happens to the information at closeout, which Domain 16, KA 16.4 governs and which is decided far too late in almost every programme.

### 14.1.2 Data governance

**Definition.** **Data governance** is the allocation of ownership, definition, quality accountability and access rights over an organisation's data: the same four questions Domain 3 asked about decisions, asked about facts.

It is routinely written as a policy and therefore routinely ineffective. The workable form is a **data class schedule** with one row per class of data the project relies on and five columns that are each individually testable:

| COLUMN                      | THE TEST IT MUST PASS                                                                                            |
|-----------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Owner</b>                | A named person, not a function, accountable for the class being fit for its stated use.                          |
| <b>Definition</b>           | One written definition per field, with units and permitted values, agreed across every system that holds it.     |
| <b>Source of record</b>     | The one system whose value prevails; every other holding is a copy.                                              |
| <b>Quality standard</b>     | The conformance rate required, stated <b>per class and per use</b> , with the consequence that sets it (14.1.4). |
| <b>Access and retention</b> | Who may read, who may change, how long it is kept, and on what basis it is destroyed.                            |

The **definition** column is the one that repays attention out of all proportion to its cost. A field whose definition differs between two systems produces a reconciliation problem that presents as a data quality problem and is actually a governance problem: nothing is wrong with either value. PFL-AI's treatment of **CFADS** makes the same point in a financial register, a defined term whose striking point changes every ratio built on it. In delivery the classic instances are "forecast completion date" (contractual? current best estimate? with or without float consumption?), "committed cost" (purchase order raised? goods received? invoiced?), and "percentage complete" (Domain 7's **PoC** conventions). Each has more than one defensible meaning and exactly one permissible meaning per organisation, and an organisation that has not chosen will reconcile the difference for the life of the programme.

International reference points may be named accurately: the ISO 8000 series on data quality and ISO/IEC 25012's data quality model give vocabulary and assessment structure for dimensions and characteristics; ISO 19650's information-management

series is the widely used frame in construction and infrastructure; and ISO/IEC 38507 addresses the governance implications for an organisation of using AI. None of them supplies a target number. The target is a project decision, and 14.1.4 derives it.

### 14.1.3 The common data environment

**Definition.** A **common data environment (CDE)** is a single agreed information store, with defined states and transitions, through which project information is shared — such that every party draws on the same version of the same fact, and the state of each item (work in progress, shared, published, archived) is explicit.

The concept comes from information management in construction and infrastructure, where ISO 19650 frames it, and it generalises cleanly to any multi-party delivery. Its value is not storage; it is the elimination of a specific failure that costs real money and is otherwise almost undetectable: **two parties working correctly from different versions**. Domain 3's decision record made the same move for decisions with its versioned information reference; the CDE makes it for the information itself.

A claimed CDE is worth testing against five guarantees, and most fail at least one. **Single instance.** There is one copy, referenced, not distributed as attachments. **Explicit state:** an item's status is a property of the item, so nobody has to ask whether a drawing is issued. **Controlled transition.** Moving an item between states is an authorised act with a record, which is what makes the environment auditable. **Access by role.** Read and change rights follow the data class schedule of 14.1.2, not convenience. **Retained history.** Superseded versions remain retrievable, because the question asked afterwards is always what was current on a given date.

Two cautions belong here because they are the ones that bite. A CDE is where a **project's** shared information lives, and it should not become a repository for information that has a stricter home: Meridian deliberately holds no patient clinical data in its delivery CDE, keeping the delivery estate and the clinical estate separate, which is both a privacy decision (14.4.4) and a simplification of the security problem. And a CDE **enforces nothing about quality**: it enforces consistency of version. A single authoritative copy of a defective record is still a defective record, propagated more efficiently, which is the subject of the rest of this KA.

### 14.1.4 Defect rate by data class and the exposure it carries

Domain 9, KA 9.4.3 established that a record satisfying six quality dimensions independently is less conforming than any dimension suggests, because the dimensions compound: auriga's asset register scored **96.08 %** on the arithmetic mean of its dimensions and **78.43 %** on the product, so roughly **690** of **3,200** records were unfit for use. That result is about one class of record. The question this KA answers is the next one, and it is the question a leader actually faces: **across many classes of data, where should remediation effort go, and what quality target should each class carry?**

The wrong answer, and the one almost universally adopted, is a **single organisational target** ("98 % data quality") applied uniformly. It is wrong for a reason that is arithmetic rather than philosophical, and the reason is worth stating precisely before the example demonstrates it: a uniform rate target treats a defect as a defect, while the organisation's actual loss is  $\text{defects} \times \text{consequence}$ , and consequence varies across classes by two or three orders of magnitude.

**The two quantities.** For each data class  $i$  holding  $n_i$  records with observed defect rate  $d_i$  and a consequence per defect  $u_i$  (the internally assessed cost of one defect reaching use):

$$\begin{aligned} \text{class defects} &= n_i \cdot d_i \\ \text{class exposure} &= n_i \cdot d_i \cdot u_i \\ \text{total exposure} &= \sum n_i \cdot d_i \cdot u_i \\ \text{exposure per record} &= d_i \cdot u_i && \leftarrow \text{the quantity that} \\ &\text{ranks remediation} \end{aligned}$$

$u_i$  is an internally assessed remediation-and-consequence cost, calibrated locally and stated as an assumption; it is not a market price and it is not comparable between organisations.

#### WORKED EXAMPLE

##### Worked example 14.1.4 — Meridian's common data environment, by class.

- Setup.** Meridian's delivery CDE holds **14,100** records in six classes. Sampling gives the defect rate for each, and the programme has assessed the cost of one defect of each class reaching use:

| DATA CLASS                              | RECORDS<br>$n_i$ | DEFECT RATE<br>$d_i$ | COST PER<br>DEFECT $u_i$ |
|-----------------------------------------|------------------|----------------------|--------------------------|
| Clinician accounts and role assignments | 4,800            | 1.5 %                | USD 250                  |
| Device and network asset records        | 1,600            | 4.0 %                | USD 180                  |
| Interface field-mapping records         | 900              | 6.0 %                | USD 1,200                |
| Training and competency records         | 3,600            | 2.5 %                | USD 90                   |
| Clinic readiness checklist records      | 1,200            | 3.0 %                | USD 400                  |
| Migration reconciliation records        | 2,000            | 0.8 %                | USD 3,500                |

- Formula.** Class defects  $n_i d_i$ ; class exposure  $n_i d_i u_i$ ; total exposure  $\sum n_i d_i u_i$ ; weighted mean defect rate  $\sum n_i d_i \div \sum n_i$ ; exposure per record  $d_i u_i$ . For the uniform-target comparison, exposure at a common rate  $d$  is  $d \cdot \sum n_i u_i$ .

3. **Substitution.** Defects 72, 64, 54, 90, 36, 16. Exposures 18,000 · 11,520 · 64,800 · 8,100 · 14,400 · 56,000. Uniform 2 % exposure  $0.02 \times 10,372,000$ .
4. **Result.** 332 defective records, a weighted mean defect rate of **2.3546 %**, carrying a total exposure of **USD 172,820** (indicatively SAR 648,075). Exposure per record, which is the ranking that matters: interface mappings **USD 72.00**, migration reconciliation **USD 28.00**, readiness checklists **USD 12.00**, device records **USD 7.20**, accounts **USD 3.75**, training records **USD 2.25**. A uniform 2 % target across all six classes produces **282 defects (15.06 % fewer)**, and an exposure of **USD 207,440**, which is **USD 34,620 or 20.03 % worse**.
5. **Interpretation.** The last line of the result is the whole KA, and it should be read twice: the uniform target **reduces the defect count by 15 % and increases the expected cost by 20 %**. It does so by mechanism, not by accident. It loosens the two classes whose defects are expensive (migration reconciliation from 0.8 % to 2 %, mappings implicitly permitted at 2 %), and tightens the classes whose defects are cheap, spending effort on training records at USD 2.25 of exposure per record while permitting reconciliation defects at USD 3,500 each. Any target expressed as a rate, with no consequence attached, has this property.

**The remediation ranking inverts the rate ranking, and that is the practical payoff.** Ranked by defect rate the priority order is mappings (6.0 %), devices (4.0 %), readiness (3.0 %), training (2.5 %), accounts (1.5 %), reconciliation (0.8 %), which puts the most expensive class last. Ranked by exposure per record it is mappings, reconciliation, readiness, devices, accounts, training. Test the two candidates that the two rankings disagree about most. Re-verifying all **900** interface mappings at **USD 22** per record costs **USD 19,800** and cuts the rate from 6.0 % to 1.0 %, removing 45 defects worth **USD 54,000**: a net **USD 34,200**, and it would remain worthwhile up to a remediation cost of **USD 60.00** per record. Re-reconciling the **2,000** migration records at **USD 12** each costs **USD 24,000** and cuts 0.8 % to 0.2 %, removing 12 defects worth **USD 42,000**: a net **USD 18,000**, breakeven at **USD 21.00** per record, from the class with the *lowest* defect rate in the estate. Now the class the rate ranking would have reached fourth: re-checking all **3,600** training records at **USD 8** each costs **USD 28,800** to remove 72 defects worth **USD 6,480**, a net **loss of USD 22,320**, and it only becomes worthwhile at a remediation cost below **USD 1.80** per record. Three interventions, two of them strongly positive and one strongly negative, and nothing in the defect rates alone distinguishes them.

Four professional cautions, each of which changes the answer if ignored. The consequence figures **u<sub>i</sub>** are **assessed, not observed**, and the whole ranking is proportional to them, so they belong in the data class schedule with a named owner and a basis, and a class whose **u<sub>i</sub>** nobody will own is a class whose quality target is arbitrary. This model treats a defect as **one discrete event with one cost**, which understates classes where a single defect propagates: a wrong interface mapping corrupts every record that passes through it, so its true **u<sub>i</sub>**



should be assessed per *mapping*, not per record that suffers, and Meridian's USD 1,200 is exactly such a propagated figure. Where a consequence is not cost-compensable (patient safety, a licence condition, a statutory notification) **expected value is the wrong test entirely**, the class carries an absolute standard, and the arithmetic here is used only to size the effort of meeting it, never to trade it away (the position Domain 9, KA 9.1 takes on safety-related requirements). And a defect rate is only meaningful against a **defined pass criterion** for the class, which is Domain 5, KA 5.4.2's testability requirement arriving as a prerequisite: rates measured against an undefined standard record the sampler's judgement.

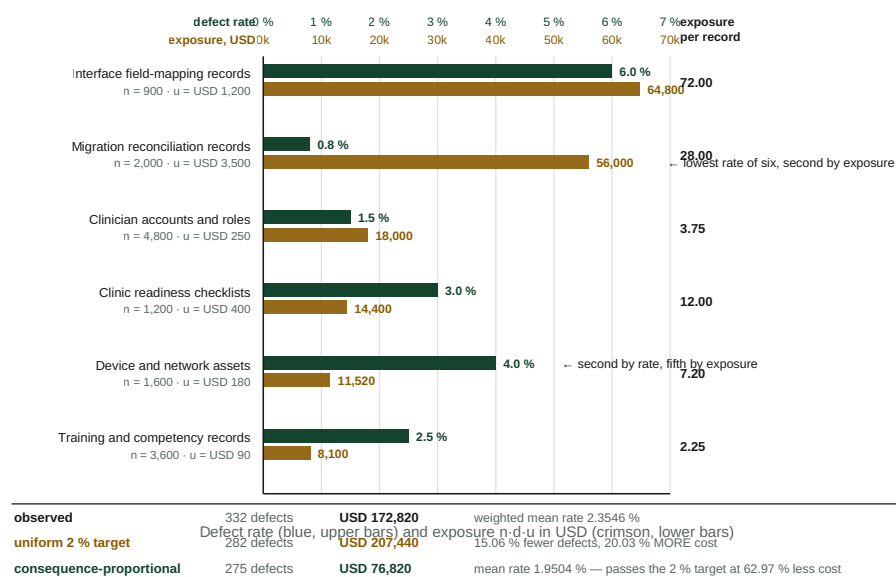


Fig 14.1.1 — Defect rate against exposure: why a uniform data-quality target costs more

## AI in this KA

**Where it earns its place.** Profiling a data class against its defined pass criteria at full population scale, which is mechanical, high-volume and exactly checkable — Domain 9, KA 9.4's strongest AI application, and the input this KA's arithmetic consumes. Reconciling field definitions across systems and producing the list of fields whose definitions differ, which is a document-comparison task humans do badly because it is tedious and enormous. Proposing the remediation rule that would fix the largest number of records in a class, for human ruling. Detecting duplicate and near-duplicate records, where fuzzy matching genuinely outperforms exact rules. Assembling the lineage path for a reported number across systems, so that a challenged figure can be traced in minutes rather than days.

**Where it must not go.** It must not set a class's quality target or its consequence figure `u.i`. Those are risk-appetite judgements owned by the data owner and the



sponsor, and a model asked for them returns confident numbers with no provenance that then drive real remediation spending, the defect Domain 3, KA 3.2.3 and Domain 10, KA 10.1 both identify. It must not **remediate silently**: an inferred value that fills a completeness gap improves the metric and can degrade the register, which is the worst available combination, and Domain 9, KA 9.4's prohibition applies unchanged here. It must not be the source of record for anything: a generated value is a copy with no provenance. And it must not decide which system is authoritative for a fact, which is a governance allocation.

**Verification, concretely.** Any machine profiling result is confirmed on a human-checked sample with the sample size and its basis stated (Domain 9, KA 9.3.2). Where records have been machine-remediated, an untouched control sample is held and compared, because a plausible wrong fill is detectable only against ground truth: the general principle 14.3.4 prices. Definition discrepancies are ruled on by the named field owner, not accepted as found. And the exposure arithmetic is reproduced by hand: every figure in this KA is a multiplication of three numbers.

#### KEY TERMS — KA 14.1

| TERM                                             | MEANING                                                                                                                 |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Digital delivery environment</b>              | The set of systems holding a project's authoritative information and mediating its work.                                |
| <b>Source of record</b>                          | The one system whose value for a fact prevails; all other holdings are copies.                                          |
| <b>Lineage</b>                                   | The reconstructable path from source record to a displayed value.                                                       |
| <b>Data class schedule</b>                       | One row per data class: owner, definition, source of record, quality standard, access and retention.                    |
| <b>Common data environment (CDE)</b>             | A single agreed information store with explicit item states and authorised, recorded transitions.                       |
| <b>Defect rate by class (<math>d_i</math>)</b>   | The share of records in a class failing that class's defined pass criteria.                                             |
| <b>Consequence per defect (<math>u_i</math>)</b> | The internally assessed cost of one defect of a class reaching use; assessed, owned and stated.                         |
| <b>Consequence-weighted exposure</b>             | $\sum n_i d_i u_i$ : the expected cost of a data estate's defects.                                                      |
| <b>Exposure per record</b>                       | $d_i u_i$ , the quantity that ranks remediation effort across classes.                                                  |
| <b>Uniform-target defect</b>                     | Setting one rate target across classes of unequal consequence, which can cut defect counts while raising expected cost. |

## SAMPLE MCQs — KA 14.1

**MCQ 14.1-A** [14.1.4 · Application] A data class holds 900 records with a 6.0 % defect rate and an assessed consequence of USD 1,200 per defect. Its consequence-weighted exposure is:

- A. USD 54.00
- B. USD 72.00
- C. USD 64,800 CORRECT
- D. USD 1,080,000

*Rationale:*  $900 \times 0.06 \times 1,200 = 64,800$  (14.1.4). A is the defect count; B is the exposure *per record* ( $d_i u_i$ ), which is the remediation ranking key, not the class exposure; D omits the defect rate and prices every record as defective.

**MCQ 14.1-B** [14.1.4 · Evaluation] Six data classes with a weighted mean defect rate of 2.3546 % carry a total exposure of USD 172,820. Imposing a uniform 2 % target on all six would produce 282 defects and an exposure of USD 207,440. The correct conclusion is that the uniform target:

- A. improves quality and reduces cost, since 282 is fewer than 332
- B. reduces the defect count by 15.06 % and raises expected cost by 20.03 %, because it loosens the low-rate, high-consequence classes and tightens the cheap ones CORRECT
- C. is acceptable because it is simpler to administer
- D. is equivalent to the observed position, since the mean rate is close to 2 %

*Rationale:* Cost is  $\text{defects} \times \text{consequence}$ , and a rate target is blind to consequence (14.1.4). A counts defects and ignores their price; D confuses a mean rate with an expected cost.

**MCQ 14.1-C** [14.1.4 · Analysis] Across six classes, remediation effort should be ranked by:

- A. defect rate  $d_i$
- B. record count  $n_i$
- C. exposure per record  $d_i u_i$  CORRECT
- D. total class exposure  $n_i d_i u_i$

*Rationale:* Remediation cost scales with records touched, so the return per record touched is  $d_i u_i$  (14.1.4). D ranks by size of prize without regard to the effort of claiming it, and would direct effort to a large cheap class ahead of a small expensive one; A puts the lowest-rate, highest-consequence class last.

**MCQ 14.1-D** [14.1.2 · Comprehension] Two systems report different "committed cost" for the same purchase order, and both values are correct within their own system. The defect is:

- A. a data accuracy failure
- B. a definition failure — the field has more than one meaning across systems
- C. an interface failure
- D. a completeness failure

CORRECT

*Rationale:* Nothing is wrong with either value; the field has no single agreed definition (14.1.2), which is why the definition column of the data class schedule repays attention out of proportion to its cost. It will present as a reconciliation problem indefinitely until a meaning is chosen.

**MCQ 14.1-E** [14.1.3 · Analysis] A programme stores every document in one repository with full version history, but item status is recorded in a separate spreadsheet and changing status requires no authorisation. Against the five CDE guarantees this arrangement fails on:

- A. single instance and retained history
- B. explicit state and controlled transition
- C. access by role only
- D. nothing — it is a valid CDE

CORRECT

*Rationale:* Status is not a property of the item and transitions are unauthorised and unrecorded (14.1.3). Single instance and retained history are satisfied; the two failures are precisely the ones that make an environment unauditably.

#### SELF-CHECK — KA 14.1

1. *Why can a uniform data-quality target reduce defect counts and increase expected cost?* — Because loss is  $\text{defects} \times \text{consequence}$  and a rate target is blind to consequence: it loosens low-rate, high-consequence classes and tightens cheap ones. Meridian's case: 15.06 % fewer defects, 20.03 % more cost.
2. *What ranks remediation across classes, and why is it not the defect rate?* — Exposure per record, **diui**, because remediation cost scales with records touched. Meridian's lowest-rate class (0.8 %) is its second-highest by exposure per record (USD 28.00).
3. *What does a common data environment guarantee, and what does it not?* — Consistency of version through explicit states and authorised, recorded transitions; it guarantees nothing about quality, so a defective record is simply propagated more efficiently.

## Dashboards, analytics, digital twins and automation

Topics: 14.2.1 dashboards and the age of a fact · 14.2.2 the analytics ladder · 14.2.3 digital twins and the fidelity that pays · 14.2.4 automation economics.

### 14.2.1 Dashboards and the age of a fact

**Definition.** A **dashboard** is a presentation of measures selected to support a defined set of decisions at a defined cadence. Every word in that constrains it, and the standard failure is to build a presentation of the measures that are *available* to an audience with no decision to make.

Domain 11, KA 11.2 governs what an executive report must contain and prohibits the smoothing of its warnings. This topic addresses the property Domain 11 did not price and that determines whether a dashboard can inform anything at all: **the age of the information at the moment of decision.**

The arithmetic is already in the registry. Domain 3, KA 3.2.3 established that a decision arising at a uniformly random point in a cycle of length  $M$ , with a paper lead time  $L$ , waits  $E[\text{wait}] = M/2$

- $L$ . A *fact* arising at a uniformly random point in a reporting cycle of period  $R$ , where production takes a lag  $G$  from data cut-off to publication, is by the identical argument expected to be  $R/2 + G$  old when it is published. The derivation is Domain 3's and is not repeated; only the interpretation changes. The quantity is now the staleness of information rather than the wait for a decision, and its two levers behave in the same asymmetric way.

#### WORKED EXAMPLE

#### Worked example 14.2.1 — how old is the fact on Meridian's report?

1. **Setup.** Meridian's rollout status is reported monthly: data cut-off at month end, then extraction, validation, review and commentary before publication, a production lag of **9 days**. The programme is considering weekly reporting with a compressed production lag of **4 days**. Cost of delay **USD 14,280 per week**, i.e. **USD 2,040 per day**.
2. **Formula.** Expected age of a fact at publication =  $R/2 + G$ , the registered  $E[\text{wait}]$  identity (Domain 3, KA 3.2.3) applied to information. Cost of acting that late on a detectable slip = age  $\times$  cost of delay per day.
3. **Substitution.** Monthly  $30/2 + 9$ . Weekly  $7/2 + 4$ . Costs  $24 \times 2,040$  and  $7.5 \times 2,040$ .

- 
4. **Result.** On the monthly cycle a fact is on average **24 days** old when the report is published, **3.4286 weeks**, worth **USD 48,960** at the programme's cost of delay. On the weekly cycle with a 4-day lag it is **7.5 days** old, worth **USD 15,300**. The change is worth **USD 33,660** per detectable slip.
5. **Interpretation.** Three things follow, and the third is the one that changes behaviour. First, the *monthly* report that everybody treats as current describes a world nearly three and a half weeks gone, which is why a slip is so often reported as new when the people doing the work have known about it for a month; the report is not concealing it, the cycle is. Second, the lever asymmetry carries over exactly from Domain 3: a one-day cut in the **production lag G** saves a full day of staleness, while a one-day cut in the **reporting period R** saves only half a day. Compressing the production process is twice as effective per day as reporting more often, and it is the cheaper of the two to change: the same result Domain 3 obtained for paper lead times, and for the same reason. Third, and decisively: **a dashboard whose information age exceeds the interval within which the decision must change cannot inform that decision, at any level of presentation quality.** Meridian's clinic readiness interventions must be triggered at least two weeks before a scheduled go-live to be effective; a 24-day-old fact arrives after the window has closed, so the monthly report is, for that decision, decoration. The professional response is not a better chart. It is either a faster cycle for that specific measure, or an exception trigger that fires on the source system rather than waiting for the report, and the honest test to apply to any dashboard is to name the decision, name the window, compute  $R/2 + G$ , and compare.

Two cautions.  $R/2 + G$  assumes facts arise uniformly through the cycle, which is a reasonable default and wrong where events cluster at period boundaries (invoicing, sprint ends, gate submissions), for those, model the actual arrival pattern rather than the average. And a shorter cycle imposes real production cost: at the blended rate of USD 110 per hour, a report taking 14 person-hours to produce costs **USD 1,540** each time, so moving from monthly to weekly adds **3.2857** productions a month ( $30/7 - 1$ ) and **USD 5,060** of monthly effort. That is worth paying against USD 33,660 of avoided lateness on a single slip, and it must still be paid rather than assumed away.

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#### 14.2.2 The analytics ladder

Analytics initiatives are usually described by their technology and should be described by the question they answer, because the four questions differ in what they require and what they are worth. **Descriptive:** what happened? Requires only clean data and correct aggregation, and is where the great majority of realised value in delivery analytics sits, unglamorously. **Diagnostic:** why did it happen? Requires the ability to decompose a measure into contributions, which in turn requires the data class schedule of 14.1.2 to be coherent. **Predictive.** What is likely to happen?

Requires history that is genuinely comparable to the future being predicted, which is a strong assumption on projects and a fatal one on novel work. **Prescriptive:** what should be done? Requires a model of the levers *and* of their costs, and it is where the accountability question becomes acute, because a recommendation presented in a system's voice is very hard for a governance body to interrogate (Domain 3, KA 3.A. 2).

**The test each rung must pass, and the ordering rule.** For each rung: *is there a named decision that would change if this output changed?* An analytics product with no such decision is a cost. And the ordering rule is unglamorous and reliable: **the rungs must be climbed in order, because each depends on the one below being right.** A predictive model built on a register with a 6 % mapping defect rate predicts the defects. The standard sequencing error in digital delivery, stated plainly, is to buy the analytics before fixing the data, which is why 14.1 precedes 14.2 in this domain and should precede it in a programme's plan.

### 14.2.3 Digital twins and the fidelity that pays

**Definition.** A **digital twin** is a model of a physical or process asset that is kept synchronised with the real asset through a data connection, such that questions can be asked of the model instead of the asset. The synchronisation is what distinguishes a twin from a design model or a simulation; a model that is not updated from the asset is a drawing, whatever it is called, and much of what is marketed as a twin is a drawing.

Twins earn their place in delivery for three uses: **rehearsal** (testing a sequence off-line before committing an irreversible operation), **handover** (a validated as-built information set the receiving organisation can operate from, which is Domain 16, KA 16.1's readiness problem), and **training** (operators competent before go-live rather than during it). Their value in all three depends on one property, and it is the property least often measured: **fidelity**, the share of questions on which the twin's answer matches the asset's behaviour within a stated tolerance.

**Auriga's control-system twin, priced.** The twin costs **USD 180,000** to build and **USD 3,500** a month to keep synchronised over the **9** months to handover — **USD 211,500** all in. Its two benefit streams, both assessed locally: avoiding **3** of the 8 live outage windows required for sequence testing at **USD 62,000** each, worth **USD 186,000**; and operator training plus the as-built information handover, assessed at **USD 120,000**. Gross benefit at perfect fidelity is therefore **USD 306,000**, and the **breakeven fidelity** is  $211,500 / 306,000 = 69.12$  %. Measured against the commissioned plant on **40** test cases, the twin agreed within tolerance on **34**: a measured fidelity of **85.00** %, giving an expected net value of  $0.85 \times 306,000 - 211,500 = \text{USD } 48,600$ .

That is a positive answer with a thin margin, and the thinness is the lesson. Measured fidelity sits only **15.88 percentage points** above breakeven, and it was measured on 40 cases. If the next 40 cases produce 10 failures, pooled fidelity is

$64/80 = 80.00\%$  and expected net value falls to **USD 33,300**; a pooled fidelity of 69.12 % takes it to zero. Two structural cautions must accompany any such calculation. Scaling benefit linearly with fidelity assumes the twin's failures are **distributed across decisions in proportion to their value**, and they are not: a twin typically fails on the novel, coupled, poorly instrumented cases, which are exactly the cases worth rehearsing, so linear scaling is optimistic and the failures should be examined individually rather than counted. And a twin's fidelity **decays**: the asset is modified, the model is not, and the synchronisation that defines a twin is the first budget line cut after handover, which is why 14.A.2's revalidation cadence applies to twins as much as to models.

#### 14.2.4 Automation economics

**Definition. Automation** here means replacing a repeated human task with a machine process: rule-based scripting, robotic process automation, or an AI-assisted step. The decision is an economic one with a well-known form and a routinely omitted term.

The familiar form is a breakeven volume. With a fixed build-and-maintain cost  $F$  over the horizon considered, a manual unit cost  $m$  and an automated unit cost  $a$ :

$$n^* = F / (m - a)$$

The omitted term is quality. An automated step and a manual step do not have the same **error escape rate**, and errors that escape cost money. Writing  $e_m$  and  $e_a$  for the share of units on which each route lets a material error through, and  $u$  for the cost of one escaped error, the honest unit costs are  $m + e_m u$  and  $a + e_a u$ , and the **quality-adjusted breakeven volume** is:

$$n^* = F / [ (m + e_m u) - (a + e_a u) ]$$

Three consequences follow immediately and are worth stating before the example. If automation **raises** the escape rate, the breakeven volume rises and may exceed any volume the organisation will ever process. If automation **lowers** it (which deterministic rule-checking often does, since machines do not tire) the breakeven volume falls below the naive figure, and the naive calculation *understates* the case. And if  $(a + e_a u) > (m + e_m u)$ , **no volume justifies the automation**: the breakeven volume does not exist, and a business case expressed in visible unit costs will nonetheless show a healthy one, which is the arithmetic behind Case study B.

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#### WORKED EXAMPLE

##### Worked example 14.2.4 — automating Auriga's handover data checks.

1. **Setup.** Auriga's **60** handover packages each require **12** data-consistency checks against the asset register, **720** checks on this project. A check takes an engineer **24 minutes** at the blended engineering rate of **USD 130.625** per hour. An auto-

mated checker costs **USD 46,000** to build and **USD 1,400** a month to maintain, and processes a check for **USD 4.25** of exception-handling time. Consider a **24-month** horizon. Measured on a held-back set, the manual route lets a material error through on **1.5 %** of checks and the automated route on **2.5 %**; an escaped error costs **USD 1,800** (a re-issued package, a repeated site visit, a corrected handover record). The utility runs **6** comparable projects a year.

2. **Formula.** Naive  $n^* = F/(m - a)$ ; quality-adjusted  $n^* = F/[(m + e_m u) - (a + e_a u)]$ , with  $F$  = build + maintenance over the horizon.
3. **Substitution.**  $m = 130.625 \times 24/60 = 52.25$ ;  $a = 4.25$ .  $F = 46,000 + 1,400 \times 24 = 79,600$ . Naive  $79,600/48.00$ . Adjusted  $m + 0.015 \times 1,800 = 79.25$ ;  $a + 0.025 \times 1,800 = 49.25$ ;  $79,600/30.00$ .
4. **Result.** Build cost alone breaks even at **958.33** → **959** checks. Including 24 months of maintenance the naive breakeven is **1,658.33** → **1,659** checks. The **quality-adjusted** breakeven is **2,653.33** → **2,654** checks, exactly **1.60 times** the naive figure, since the unit saving falls from USD 48.00 to USD 30.00. Auriga alone generates **720** checks: below every one of those thresholds. Across the portfolio the volume is **4,320** a year, **8,640** over 24 months, giving a quality-adjusted net of **USD 179,600** against a naive claim of **USD 335,120**.
5. **Interpretation.** The decision reverses depending on the boundary drawn, and both answers are correct within their boundary, which is why the boundary is the first thing to state. **For Auriga alone the automation is indefensible:** 720 checks against a build-only breakeven of 959, so the project would spend USD 46,000 to save USD 34,560 of visible effort even before quality is considered. **For the portfolio it is strongly positive**, and the case must therefore be made, funded and owned at portfolio level (Domain 15), with the tool treated as an asset with a maintainer rather than as a project deliverable. A project-funded automation with no portfolio owner is the standard way an organisation acquires an unmaintained tool that quietly stops being verified.

**The quality term is the professional content of this example.** A 1.0-percentage-point increase in the escape rate (from 1.5 % to 2.5 %, a difference no business case would notice) raises the breakeven volume by 60 % and removes **USD 155,520** from the 24-month net. Two design responses follow. If the automated route can be made *better* than the human one rather than worse, the arithmetic transforms: at an automated escape rate of **0.5 %** the adjusted unit cost is **USD 13.25**, the unit saving is USD 66.00, and the breakeven falls to **1,206.06** → **1,207** checks, below the naive figure. Machines are genuinely better than tired humans at exhaustive rule-checking, so this is often achievable and almost never measured. Alternatively, place a cheap containment layer after the automated step so that  $u$  itself falls: Domain 9, KA 9.4.4's result that the economics reward **building a containment chain around machine output** rather than reviewing harder. Three cautions: the escape rates must be **measured against a held-back set**, never asserted, and re-measured on any change of rule set, model or artefact type (14.A.2); maintenance is a real recurring cost



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that is systematically omitted, and omitting it here would have made the breakeven 959 instead of 1,659; and the automated route's exception-handling cost *a* must include the human time actually spent on exceptions, which in practice is where an automation's promised saving most often disappears.

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## AI in this KA

**Where it earns its place.** Generating the exception logic for a dashboard (the rules that decide what is worth surfacing) for human ruling, which is a genuine strength because the rules are testable against history. Producing the first draft of an aggregation or transformation, with the lineage stated, for review. Detecting anomalies in high-volume operational data where the anomaly class is defined and the model's output is a *prompt to investigate* rather than a finding. Reconciling a twin's predictions against measured behaviour across a large test set and clustering the disagreements by type, which is how a fidelity measure becomes diagnostic rather than a percentage. Modelling breakeven volumes across dozens of automation candidates, which is deterministic and verifiable.

**Where it must not go.** It must not select the measures on a governance dashboard: that is a statement about what the organisation will manage, and it belongs to the accountable body (Domain 11, KA 11.2). It must not supply the escape rates, defect rates, consequence figures or fidelity tolerances that these calculations consume: asked for any of them a model will return a confident, well-formatted number with no provenance, and it will then anchor a real capital decision. It must not write the commentary that accompanies a status measure, for the smoothing reason Domain 11 established: a model asked to make a report clearer reliably produces a calmer one, and the qualifier carrying the warning is the first casualty. And it must not be the **only** step between a generated output and an irreversible operation, which is what a twin-based rehearsal exists to prevent.

**Verification, concretely.** Every automation carries a **measured** escape rate against a held-back set, re-measured on change, and a named owner for the measurement. Every dashboard measure carries its  $R/2 + G$  information age on the specification, so the age is a published property rather than a discovery. A twin carries its fidelity, the test-case count it was measured on, the date of measurement and the tolerance used. And the breakeven arithmetic in any automation business case is reproduced by hand (it is one division) with the maintenance term and the quality term both visible, since a case presenting neither has not been reviewed.

## KEY TERMS — KA 14.2

| TERM                                     | MEANING                                                                                                                                                                         |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Information age</b>                   | $R/2 + G$ , the expected staleness of a fact at publication, for reporting period $R$ and production lag $G$ ; the registered $E[\text{wait}]$ identity applied to information. |
| <b>Production lag (G)</b>                | Elapsed time from data cut-off to publication; the more powerful of the two staleness levers, day for day.                                                                      |
| <b>Analytics ladder</b>                  | Descriptive $\rightarrow$ diagnostic $\rightarrow$ predictive $\rightarrow$ prescriptive; each rung depends on the one below being right.                                       |
| <b>Digital twin</b>                      | A model kept synchronised with a real asset through a data connection; without synchronisation it is a drawing.                                                                 |
| <b>Twin fidelity</b>                     | The share of test questions on which the twin's answer matches the asset within a stated tolerance.                                                                             |
| <b>Breakeven fidelity</b>                | Twin cost $\div$ gross benefit at perfect fidelity: the fidelity below which the twin destroys value.                                                                           |
| <b>Error escape rate (e)</b>             | The share of units on which a route lets a material error through; differs between manual and automated routes and must be measured.                                            |
| <b>Quality-adjusted breakeven volume</b> | $F / [(m + e_m u) - (a + e_a u)]$ : the volume at which an automation pays once escaped errors are priced.                                                                      |

## SAMPLE MCQs — KA 14.2

**MCQ 14.2-A** [14.2.1 · Application] A programme reports monthly with a 9-day production lag from data cut-off to publication. The expected age of a fact at publication is:

- A. 9 days
- B. 15 days
- C. 24 days CORRECT
- D. 39 days

*Rationale:*  $R/2 + G = 30/2 + 9 = 24$  days (14.2.1, applying Domain 3's  $E[\text{wait}]$  identity). A counts only the lag; B only half the period; D adds the whole period to the lag.

**MCQ 14.2-B** [14.2.1 · Analysis] For that report, which change reduces information age more, and by how much: cutting the production lag by 3 days, or cutting the reporting period by 3 days?

- A. the reporting period, by 3.0 days
- B. the production lag, by 3.0 days — twice the 1.5-day saving from the period

CORRECT

C. both equally, by 3.0 days

D. both equally, by 1.5 days

*Rationale:* Age is  $R/2 + G$ , so a cut of  $x$  in  $G$  saves  $x$  while a cut of  $x$  in  $R$  saves  $x/2$ , the same asymmetry Domain 3, KA 3.2.3 found for paper lead times, and the lag is usually the cheaper lever to move.

**MCQ 14.2-C** [14.2.4 · Application] An automation costs USD 79,600 over the horizon. Manual unit cost is USD 52.25 and automated USD 4.25; escape rates are 1.5 % manual and 2.5 % automated with an escaped error costing USD 1,800. The quality-adjusted breakeven volume is closest to:

A. 959 units

B. 1,659 units

C. 2,654 units CORRECT

D. 1,524 units

*Rationale:* Adjusted unit costs are  $52.25 + 27.00 = 79.25$  and  $4.25 + 45.00 = 49.25$ , so  $79,600/30.00 = 2,653.33 \rightarrow 2,654$  (14.2.4). A uses the build cost alone with unadjusted units; B is the naive figure including maintenance; D divides the fixed cost by the manual unit cost instead of by the saving.

**MCQ 14.2-D** [14.2.4 · Evaluation] An automation's quality-adjusted automated unit cost exceeds its quality-adjusted manual unit cost. The correct conclusion is that:

A. the breakeven volume is very large but attainable at portfolio scale

B. no volume justifies the automation; the breakeven volume does not exist

CORRECT

C. the automation is justified if the visible unit saving is positive

D. the escape rates should be excluded as they are estimates

*Rationale:* With a negative unit saving every additional unit adds loss, so there is no crossover (14.2.4) — the position Case study B describes. C is exactly the error that produces it; D discards the term that decides the answer.

**MCQ 14.2-E** [14.2.3 · Application] A digital twin costs USD 211,500 and would deliver USD 306,000 of benefit at perfect fidelity. Its breakeven fidelity is:

A. 58.82 %

B. 69.12 % CORRECT

C. 85.00 %

D. 113.71 %

*Rationale:*  $211,500/306,000 = 69.12\%$  (14.2.3). A divides the build cost alone by the benefit, omitting nine months of synchronisation; C is the measured fidelity, not the breakeven; D divides by one benefit stream only.

## SELF-CHECK — KA 14.2

1. *How old is a fact on a monthly report with a 9-day production lag, and what is it worth at Meridian's cost of delay?* — 24 days ( $30/2 + 9$ ), worth USD 48,960 at USD 2,040 a day; a weekly cycle with a 4-day lag gives 7.5 days and USD 15,300.
2. *What distinguishes a digital twin from a design model, and what determines its value?* — Synchronisation with the real asset; and fidelity, measured against the asset on a stated number of test cases with a stated tolerance, against a computed breakeven: 69.12 % for Auriga's twin.
3. *Which term is systematically omitted from automation business cases, and what does it do?* — The differential error escape rate. One percentage point of extra escape raised Auriga's breakeven volume by 60 %, and a negative quality-adjusted saving removes the breakeven altogether.

## KNOWLEDGE AREA 14.3

## AI use across the lifecycle, prompting and verification

*Topics: 14.3.1 where AI earns its place across the lifecycle · 14.3.2 prompting, grounding and provenance · 14.3.3 verification-effort economics and the standard it derives · 14.3.4 the plausible wrong number.*

### 14.3.1 Where AI earns its place across the lifecycle

The preceding thirteen domains have each stated, in their own subject matter, what AI may and may not do. Restating those positions here would be a defect. What this topic contributes instead is the **pattern** they collectively describe, because once the pattern is visible it can be applied to a use case nobody has yet written a rule for, which is the situation a leader is actually in.

Across every domain in this volume, the permitted uses fall into five classes and the prohibited uses into four. **The five permitted classes:** *extraction* (pulling structure out of unstructured material — a decision register from minutes, Domain 3, KA 3.3; requirements candidates from interview notes, Domain 5, KA 5.1); *comparison at scale* (reading many documents against each other for inconsistency — decision rights across a contract and a terms of reference, Domain 3, KA 3.1; field definitions across systems, 14.1.2); *enumeration* (generating candidates a human will rule on — risks, test cases, failure modes, improvement actions, Domains 8 and 9); *deterministic modelling* (running arithmetic a human specified across many combinations — latency and threshold models in Domain 3, breakeven volumes in 14.2.4); and *drafting for review* (producing a first version of a document whose author remains the named human, Domains 4, 5, 10, 11). Each is characterised by a verifiable output and a human ruling.

**The four prohibited classes**, and this is the more useful list because it generalises: *attributable acts*, anything that confers authority or discharges an obligation: approving a change, accepting a deliverable, signing a test record, granting a concession, authoring a decision record (Domains 3, 4, 5, 9). *Judgements of consequence and appetite*: thresholds, liability caps, target costs, quality scores, risk appetites, kill criteria; these are risk-appetite decisions belonging to accountable people (Domains 2, 3, 9, 10). *Unprovenanced parameters*: supplying the probability, detection rate, critical-path share, escape rate, reservation value or benefit attribution that a calculation then consumes; the model will supply a confident number and it will enter a real decision as though it were data (Domains 3, 5, 8, 10, 11 and every worked example in this domain). *Inference about people*: emotional state, engagement, personality, likely departure, cultural disposition, performance or potential, whether from telemetry, text or images; Domain 12 states the prohibition in its strongest form in this volume, and this domain does not soften it.

The single test that generates all four prohibitions, and the one to apply to a novel case: **could a named human be held to answer for this output as their own judgement, on evidence they can produce?** If yes, AI assistance is a drafting aid and the human remains the author. If no (because the output is the exercise of an authority, or because it rests on a parameter nobody can source, or because the subject is a person's interior state) the use is not permitted at any level of verification, because no amount of checking creates an accountable author.

**Across the lifecycle**, the same pattern maps to stages. In *definition and selection*, extraction and enumeration are strong and benefit attribution is prohibited (Domain 2; Domain 5, KA 5.3). In *planning*, deterministic modelling and scenario enumeration are strong while estimates and durations are not (Domains 6 and 7). In *execution*, extraction, anomaly detection and drafting are strong while approvals, acceptances and classifications are prohibited (Domains 4, 5, 9, 10). In *closeout and benefits*, clustering and summarisation are strong while the benefit judgement and the archive decision remain human (Domain 16). The prohibitions do not weaken as a project matures; the volume of permitted use grows, which is why the verification standard of 14.3.3 has to be a standing instrument rather than a mobilisation exercise.

### 14.3.2 Prompting, grounding and provenance

A prompt is not a conversational convenience; where its output informs a professional judgement, it is a **method**, and it acquires the obligations of a method: it is written down, versioned, reviewed and reproducible. That single reframing removes most of the practical problems organisations report with AI assistance, and it is resisted mainly because prompts feel like typing.

**The four properties of a professional prompt.** *Grounding*: the material the output must be derived from is supplied and identified, and the instruction states that nothing outside it may be used. An ungrounded prompt asks for recall, and recall is where fabrication lives. *Constraint*: the output's form is specified (fields, units,

allowed values, "state 'not stated' where the source is silent"), because an unconstrained output cannot be checked mechanically. *Refusal permission*: the instruction explicitly permits, and expects, an answer of "the source does not say", which is the single most valuable line in a professional prompt because its absence guarantees that gaps are filled with plausible content. *Provenance*: each material assertion in the output cites the supplied source location it came from, which is what converts verification from re-reading everything to checking citations.

**What must be recorded.** For any AI-assisted output that will be relied on: the prompt version, the model and version, the grounding material and its version, the date, the named human who reviewed it, and the verification tier applied (14.3.3). This is not bureaucracy; it is the minimum that lets the question asked afterwards be answered. Domain 3, KA 3.3.4's finding applies unchanged: the retrospective question is never "was the output right?" but "was the decision reasonable on what was known at the time?", and only a versioned record answers it.

**Two failure modes specific to prompting**, both common and both cheap to prevent. **Instruction drift**: a prompt refined over many sessions in a chat window is not the prompt anyone thinks it is, and its output is not reproducible; the fix is that the operative prompt lives in a file with a version, not in a history. **Context contamination**: material from an unrelated matter remains in a session and influences an output, which is both a correctness problem and, where the material is confidential or personal, a disclosure problem: the boundary Domain 3, KA 3.A.2 requires to be stated before use rather than after an incident.

### 14.3.3 Verification-effort economics and the standard it derives

Domain 3, KA 3.A.2 required a **verification standard proportional to consequence** and asserted it without deriving it. Domain 9, KA 9.4.4 supplied one axis of the derivation: given a population of machine-produced items, how large a sample must be checked, from the breakeven error fraction  $p^* = c/u$  and the bound a clean sample supports. This topic supplies the other axis, and it is the one an organisation must answer first: **for a given class of output, how deep should the check be?**

**The structure.** Verification is available at increasing depths. Each depth  $k$  has a cost per item  $v_k$  and a **detection rate**  $q_k$ : the share of material errors present that it finds. For an output class whose items contain a material error with probability  $p$ , and where an escaped error costs  $u$ :

$$\text{expected cost at tier } k = v_k + p (1 - q_k) u$$

Moving from tier  $k - 1$  to tier  $k$  is worth doing when the extra cost is less than the extra expected loss avoided,  $\Delta v_k < p \cdot \Delta q_k \cdot u$ , which rearranges into the quantity a standard is written from, **the consequence at which the next tier begins to pay**:

$$u^*_k = \Delta v_k / (p \cdot \Delta q_k)$$

Three properties of that expression should be read before any numbers. The thresholds depend on  $\Delta v$  and  $\Delta q$ , the **increments**, not on the totals, so the natural but wrong comparison of a tier's whole cost against the whole consequence over-verifies, because every tier looks cheap against a large consequence. The thresholds are **inversely proportional to p**, so a class's required depth falls as the producing process improves, which is where the productivity gain of AI assistance actually lives. And  $p$  and  $q$  are **measured quantities**: a standard built on asserted ones is a standard built on nothing, which is why 14.A.2 makes their measurement a standing obligation.

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#### WORKED EXAMPLE

#### Worked example 14.3.3 – deriving Meridian's verification standard, and pricing it.

1. **Setup.** Meridian's AI-assisted outputs are reviewed at one of five depths, costed at the blended programme rate of **USD 110 per hour**, with detection rates measured against a held-back set of seeded errors: tier 0 accept unchecked ( $v = 0$ ,  $q = 0$ ); tier 1 author scan, 6 minutes (**USD 11.00**,  $q = 0.35$ ); tier 2 independent review, 24 minutes (**USD 44.00**,  $q = 0.70$ ); tier 3 independent review with source trace, 66 minutes (**USD 121.00**,  $q = 0.90$ ); tier 4 two independent reviewers with reperformance, 150 minutes (**USD 275.00**,  $q = 0.97$ ). The measured material-error rate of the current model-and-prompt configuration is  $p = 0.12$ . Monthly output volumes and assessed consequences per escaped error: internal meeting summaries **620** at **USD 150**; clinic readiness assessments **180** at **USD 480**; configuration field-mapping proposals **95** at **USD 1,200**; migration reconciliation scripts **40** at **USD 3,500**; baseline change impact assessments **6** at **USD 28,560** (two weeks of programme delay at the cost of delay of USD 14,280 per week).
2. **Formula.**  $u_k^* = \Delta v_k / (p \cdot \Delta q_k)$  for each tier step; then, for any assignment of tiers to classes, total cost =  $\sum n_i v_k(i) + \sum n_i p (1 - q_k(i)) u_i$ .
3. **Substitution.**  $11/(0.12 \times 0.35)$ ;  $33/(0.12 \times 0.35)$ ;  $77/(0.12 \times 0.20)$ ;  $154/(0.12 \times 0.07)$ .
4. **Result (the standard.)**

---

| STEP                           | $\Delta v$ | $\Delta q$ | CONSEQUENCE AT WHICH IT PAYS, $u^*$ | APPLIES FROM                                          |
|--------------------------------|------------|------------|-------------------------------------|-------------------------------------------------------|
| 0 → 1 author scan              | USD 11.00  | 0.35       | <b>USD 261.90</b>                   | any output whose escaped error costs more than 261.90 |
| 1 → 2 independent review       | USD 33.00  | 0.35       | <b>USD 785.71</b>                   | " more than 785.71                                    |
| 2 → 3 review with source trace | USD 77.00  | 0.20       | <b>USD 3,208.33</b>                 | " more than 3,208.33                                  |

| STEP                                           | $\Delta v$ | $\Delta Q$ | CONSEQUENCE AT WHICH IT PAYS, $U^*$ | APPLIES FROM          |
|------------------------------------------------|------------|------------|-------------------------------------|-----------------------|
| 3 $\rightarrow$ 4 two reviewers, reperformance | USD 154.00 | 0.07       | <b>USD 18,333.33</b>                | " more than 18,333.33 |

Applying it: summaries (150)  $\rightarrow$  **tier 0**; readiness assessments (480)  $\rightarrow$  **tier 1**; mapping proposals (1,200)  $\rightarrow$  **tier 2**; reconciliation scripts (3,500)  $\rightarrow$  **tier 3**; change impact assessments (28,560)  $\rightarrow$  **tier 4**. Monthly verification cost **USD 12,650**; expected escaped loss **USD 24,300.10**; **total USD 36,950.10**.

The three alternatives, on the same 941 outputs. **No verification**: USD 0 + **USD 72,571.20** = **USD 72,571.20**. **Uniform tier 2 on everything** — the intuitively "consistent" policy: USD 41,404 + USD 21,771.36 = **USD 63,175.36**. **Uniform tier 4 on everything** — the intuitively "safe" policy: USD 258,775 + USD 2,177.14 = **USD 260,952.14**. The tiered standard is the cheapest of the four, beating uniform tier 2 by **USD 26,225.26 (41.51 %)** and uniform tier 4 by a factor of **7.06**. **5. Interpretation**. Start with the comparison that changes minds. Uniform tier 2 spends **3.27 times** as much on verification as the tiered standard (USD 41,404 against USD 12,650), and buys a reduction in escaped loss of **USD 2,528.74**, or **10.41 %**. That is the arithmetic signature of an undifferentiated review policy: most of the money goes to items whose errors are cheap, because that is where most of the items are. The tiered standard is not a relaxation of uniform tier 2; it is a **reallocation**. It removes review from 620 low-consequence items and adds review depth to 46 high-consequence ones, and it is better on both cost and loss because the consequences differ by a factor of **190.4** (28,560 against 150) while the review costs differ by a factor of **25.0** (275.00 against 11.00).

**The uncomfortable feature, stated plainly, because a governance body will find it.** Under the tiered standard, **45.93 %** of all expected escaped loss (USD 11,160 of USD 24,300.10) sits in the one class deliberately left unverified. That looks like negligence and it is not: putting the 620 summaries on tier 1 costs **USD 6,820** and saves **USD 3,906** of escaped loss, a **net loss of USD 2,914**, which is exactly what a consequence of USD 150 sitting below the USD 261.90 threshold means. The correct response to a large aggregate of cheap errors is not to check them; it is to **reduce p or u**: a better prompt, a template, a structural constraint, a cheap containment layer downstream (Domain 9, KA 9.4.4). Checking is the most expensive of the available remedies and the only one organisations reliably reach for.

**The sensitivity that matters most is p, and it points somewhere unexpected.** If the configuration improves so that the measured error rate falls from 0.12 to **0.04**, every threshold triples (to **785.71 · 2,357.14 · 9,625.00 · 55,000.00**), and the assignment shifts down: summaries and readiness assessments to tier 0, mappings to tier 1, scripts to tier 2, impact assessments to tier 3. Total cost falls



from USD 36,950.10 to **USD 16,036.44**, a **56.60 %** reduction, of which the **verification bill** falls by **72.09 %** (USD 12,650 → USD 3,531). This is the honest form of the AI productivity claim: **the return on a better model is mostly a reduction in checking, not a reduction in drafting time**, and it is realisable only by an organisation that measures  $p$ , because an unmeasured error rate leaves the verification standard frozen at whatever it was set to on the day someone was nervous.

Four cautions. Every threshold scales with  $1/p$ , so  $p$  must be **measured per output class and per configuration**, and a single organisational  $p$  is as wrong as a single organisational quality target (14.1.4). Detection rates  $q$  must be measured against **seeded or held-back errors**, and Domain 9, KA 9.2.2's independence rule binds: two passes by the same model are one pass, and a model reviewing its own output is not a tier. Where the consequence is not cost-compensable (a safety-related output, a regulatory submission, a clinical decision support artefact) the class carries an **absolute** standard and this arithmetic sizes the effort rather than choosing the depth. And the standard must be **published and owned**, because its value lies in being applied consistently by people who are not doing the calculation each time; an unpublished standard collapses into individual nervousness, which is the condition it exists to replace.

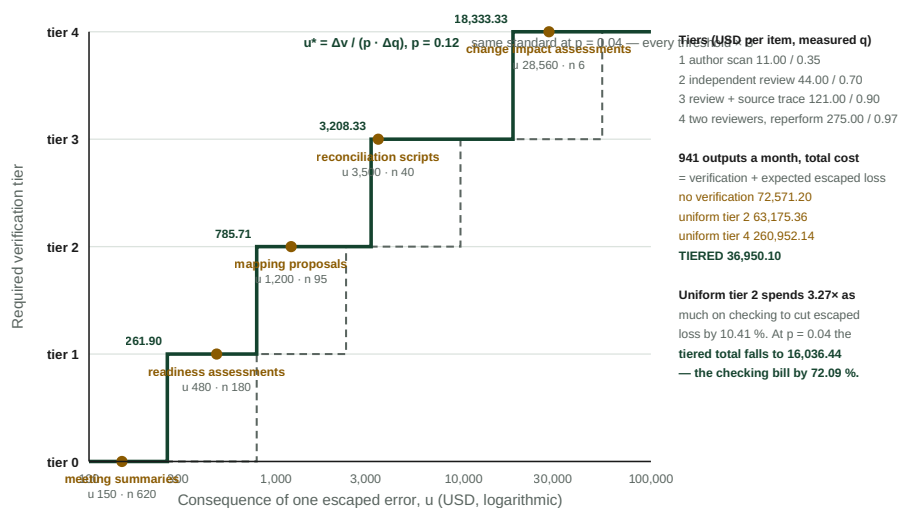


Fig 14.3.1 — The verification standard proportional to consequence

#### 14.3.4 The plausible wrong number

Everything in 14.3.3 assumed a detection rate  $q$ . This topic is about what determines it, because the answer is counter-intuitive and it changes how review effort should be aimed:  $q$  **depends far more on the plausibility of the error than on its size**.

An obviously wrong number (a misplaced decimal, a units confusion, a total that fails to add) is caught by almost anyone who looks, because human review of quantitative material is largely a plausibility check. A plausible wrong number is caught by almost nobody, because there is nothing to notice. And machine-produced output is characteristically plausible: fluent regardless of correctness, internally consistent, correctly formatted, and derived by a process that produces well-shaped answers rather than hesitant ones. Domain 9, KA 9.4.4 made the point qualitatively: an unsure human writes hesitantly and a wrong model does not. Here it is priced.

---

WORKED EXAMPLE

**Worked example 14.3.4 – two wrong forecasts on Auriga, one of them dangerous.**

1. **Setup.** At week 13 Auriga reports PV 2,080,000, EV 1,920,000, AC 2,120,000 against a BAC of 4,000,000, giving CPI 0.91 and SPI 0.92 (Domain 7). The CPI-based forecast is  $EAC = BAC / CPI = \text{USD } 4,416,666.67$ , a VAC of (USD 416,666.67). Governance rule: a forecast overrun above USD 350,000 obliges the sponsor to request supplementary funding. An AI-assisted forecasting step can fail in two ways. *Obvious*: a decimal slip reporting EAC USD 43,200,000, whose consequence if it escaped is assessed at USD 900,000 of misdirected commitment. *Plausible*: a mis-weighted index producing EAC USD 4,320,000, an overrun of USD 320,000, below the threshold, so no funding request is made and the request is instead made 6 weeks later at a cost of delay of USD 45,000 per week, i.e. USD 270,000. Measured on a held-back set, each failure type occurs on 10 % of forecast cycles; review detects the obvious error with probability 0.98 and the plausible one with probability 0.25.
2. **Formula.** Expected escaped cost =  $p (1 - q) u$  per failure type. The asymmetry ratio is  $[(1 - q_p) u_p] \div [(1 - q_o) u_o]$ .
3. **Substitution.** Obvious  $0.10 \times 0.02 \times 900,000$ . Plausible  $0.10 \times 0.75 \times 270,000$ .
4. **Result.** The obvious error carries an expected escaped cost of USD 1,800. The plausible error carries USD 20,250: 11.25 times as much, from a consequence 3.33 times smaller. At equal consequence the ratio would be the ratio of escape probabilities,  $0.75 / 0.02 = 37.50$ .
5. **Interpretation.** The generalisable statement is worth memorising in this form: **the expected damage of an error is its consequence multiplied by its probability of not being caught, and plausibility drives the second term across two orders of magnitude while magnitude drives only the first.** A 10× decimal slip is 3.33 times more consequential and 37.5 times more likely to be caught, so it is 11.25 times less dangerous. Everything a reviewer's instinct rewards (noticing the number that looks wrong) is aimed at the cheap failure mode.

Three practical consequences follow, and they are what this topic is for. **Review must be aimed at plausibility, not at magnitude.** A reasonableness check is a tier-1 control against the obvious class and close to worthless against the plausible one; the control that works is **reperformance**: recomputing the number from its inputs by an independent route. On Auriga, a reperformance step that raises *q\_p* from **0.25** to **0.85** takes about **40 minutes** at the blended engineering rate, **USD 87.08** per forecast cycle, and cuts the expected escaped cost from USD 20,250 to **USD 4,050**: a saving of **USD 16,200**, a return of **186 times** its cost. Its breakeven is a detection improvement of **0.3225 percentage points**, so on this class of output reperformance pays if it works at all. **Presentation determines reperformability.** A forecast published as a single number cannot be reperformed; the same forecast published with *EV*, *AC*, *BAC* and the stated method can be reperformed in two minutes, so *how a number is presented is a control*, and Domain 11, KA 11.2's reporting standard should be read as carrying this requirement. **And the small, plausible error near a threshold is the most dangerous object in a reporting pack**, because its consequence is not proportional to its size: Auriga's plausible error is USD 96,666.67 of forecast (**2.42 %** of *BAC*), and it costs USD 270,000 because it lands on the wrong side of a governance trigger. Any measure with a threshold attached should be reperformed within a band around that threshold regardless of the tier its consequence would otherwise attract.

A related trap worth flagging, since it manufactures plausible errors from nothing: **rounding before dividing.** Computing *EAC* from the *reported CPI* of 0.91 rather than the underlying 0.905660 gives **USD 4,395,604.40** instead of USD 4,416,666.67, understating the overrun by **USD 21,062.27**, **5.05 %** of the variance, from an operation that looks like tidiness. Full precision internally, rounding only at display, is not pedantry; it is the difference between a reperformable number and a plausible one.

## AI in this KA

**Where it earns its place.** Building the tier table itself across many output classes, which is one division repeated and exactly checkable. Maintaining the measured error-rate and detection-rate tables from a seeded-error programme, and flagging classes whose measurements have gone stale (14.A.2). Producing a reperformance of a computed figure by an **independent route** (recomputing *EAC* from *EV*, *AC* and *BAC* where the original came from a different path), which is a legitimate and valuable second pair of eyes precisely because the arithmetic is deterministic. Checking that every material assertion in an AI-assisted document carries a provenance citation to the supplied grounding material, and listing those that do not, which is the mechanical part of verification and the part humans skip. Auditing prompt and model versions against the record.

**Where it must not go.** It must not set the verification tier for its own output: the decision whose independence the entire argument of 14.3.3 rests on, and Domain 9, KA 9.4's prohibition verbatim. It must not supply  $p$ ,  $q$  or  $u$ ; those are measurements and assessments with owners, and a model asked for them returns numbers that look exactly like data. It must not be the reviewer of record at any tier: a tier is a human act with a name attached, and a model-performed check is a *tool used within* a tier, whose contribution is included in the measured  $q$  of that tier and never credited separately. And a model's own expression of confidence must not be read as a detection rate or an error rate. It is neither, and treating it as either silently replaces a measurement with a sentiment.

**Verification, concretely.** Error rates and detection rates come from a **seeded-error programme** with a stated method, sample size and date, re-run on any change of model, prompt, grounding material or output class, and owned by a named person. Every tier-3 and tier-4 review records the reviewer, the date, the prompt and model versions, and what was re-performed. Reperformance is by an **independent route**, not a re-reading. And the tier table is recomputed by hand whenever  $p$  changes materially, since a standard nobody can reproduce will not survive its first challenge.

## KEY TERMS — KA 14.3

| TERM                                        | MEANING                                                                                                                             |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Grounding</b>                            | Supplying and identifying the material an output must be derived from, with nothing outside it permitted.                           |
| <b>Refusal permission</b>                   | An explicit instruction that "the source does not say" is an acceptable and expected answer.                                        |
| <b>Provenance citation</b>                  | Each material assertion in an output referenced to the supplied source location it came from.                                       |
| <b>Detection rate (<math>q</math>)</b>      | The share of material errors present that a given review depth finds; a measured quantity, never asserted.                          |
| <b>Material error rate (<math>p</math>)</b> | The share of items of a class containing a material error before verification; measured per class and per configuration.            |
| <b>Verification tier</b>                    | A defined review depth with a cost per item $v$ and a measured detection rate $q$ , performed by a named human.                     |
| <b>Verification tier threshold</b>          | $u^*_k = \Delta v_k / (p \cdot \Delta q_k)$ : the escaped-error consequence at which the next tier begins to pay.                   |
| <b>Reperformance</b>                        | Recomputing a result from its inputs by an independent route; the only effective control against plausible error.                   |
| <b>Escape-cost asymmetry</b>                | $(1 - q_p)u_p \div (1 - q_o)u_o$ , why a plausible wrong number out-damages an obviously wrong one.                                 |
| <b>Attributable act</b>                     | An output that confers authority or discharges an obligation, and therefore cannot be produced by a tool at any verification depth. |

## SAMPLE MCQS — KA 14.3

**MCQ 14.3-A** [14.3.3 · Application] Tier 2 costs USD 44.00 per item with a detection rate of 0.70; tier 3 costs USD 121.00 with 0.90. The measured material-error rate is 0.12. The consequence at which tier 3 begins to pay is:

- A. USD 712.96
- B. USD 1,120.37
- C. USD 3,208.33 CORRECT
- D. USD 5,041.67

*Rationale:*  $u^* = \Delta v / (p \cdot \Delta q) = 77 / (0.12 \times 0.20) = 3,208.33$  (14.3.3). A divides the increment by the *total* detection rate; B divides the total cost by the total detection rate; D divides the total cost by the increment. Only the increment-over-increment form answers "is the next tier worth it".

**MCQ 14.3-B** [14.3.3 · Evaluation] A tiered standard costs USD 12,650 in verification with USD 24,300 of expected escaped loss. Uniform tier 2 on the same outputs costs USD 41,404 with USD 21,771 of escaped loss. The best characterisation is that uniform tier 2:

- A. is safer, since escaped loss is lower
- B. spends 3.27 times as much on verification to reduce escaped loss by 10.41 %, and is USD 26,225 worse in total CORRECT
- C. is equivalent, since both are defensible policies
- D. is cheaper, because one policy is simpler to administer

*Rationale:* Compare totals: 36,950 against 63,175 (14.3.3). A looks at one term of two; the extra review lands mostly on items whose errors are cheap, which is the signature of an undifferentiated policy.

**MCQ 14.3-C** [14.3.3 · Analysis] The measured material-error rate for a class falls from 0.12 to 0.04. Every verification threshold:

- A. falls by a factor of three
- B. rises by a factor of three CORRECT
- C. is unchanged, since the tiers' costs and detection rates are unchanged
- D. rises by a factor of nine

*Rationale:*  $u^* = \Delta v / (p \cdot \Delta q)$  is inversely proportional to  $p$  (14.3.3), so thresholds triple and more classes fall into lighter tiers, which is where the productivity gain of a better configuration is actually realised. C confuses the tier definitions with the threshold.

**MCQ 14.3-D** [14.3.4 · Analysis] An obvious error occurs on 10 % of cycles, costs USD 900,000 if it escapes and is detected with probability 0.98. A plausible error

occurs equally often, costs USD 270,000 and is detected with probability 0.25. Which is more dangerous, and by how much?

- A. the obvious error, by 3.33 times, because its consequence is larger
- B. the plausible error, by 11.25 times CORRECT
- C. the plausible error, by 37.50 times
- D. they are equally dangerous once probability is taken into account

*Rationale:* Expected escaped costs are  $0.10 \times 0.02 \times 900,000 = 1,800$  and  $0.10 \times 0.75 \times 270,000 = 20,250$  (14.3.4). C is the escape-probability ratio alone, which would be the answer only at equal consequence; A ranks by consequence and ignores detectability.

**MCQ 14.3-E** [14.3.4 · Application] The control that most raises the detection rate for plausible numerical error in an AI-assisted forecast is:

- A. a reasonableness check against expectation
- B. reperformance from the inputs by an independent route CORRECT
- C. asking the model to check its own output
- D. increasing the number of reviewers reading the same document

*Rationale:* A plausible error offers nothing to notice, so a plausibility check is near-worthless against it (14.3.4). C is not an independent layer (Domain 9, KA 9.2.2); D multiplies the same ineffective check.

#### SELF-CHECK — KA 14.3

1. *State the verification tier threshold and what it is proportional to.* —  $u_k^* = \Delta v_k / (p \cdot \Delta q_k)$ : the increment of cost over the increment of detection and the error rate. It is inversely proportional to  $p$ , so a better configuration raises every threshold.
2. *Why is a plausible wrong number more dangerous than an obviously wrong one?* — Expected damage is consequence  $\times$  probability of not being caught, and plausibility moves the second term by two orders of magnitude. On Auriga the plausible error costs 11.25 times the obvious one despite a consequence 3.33 times smaller.
3. *What single test generates every prohibition on AI use in this volume?* — Whether a named human could be held to answer for the output as their own judgement on evidence they can produce. If not (an attributable act, an appetite judgement, an unprovenanced parameter, or inference about a person) no depth of verification makes it permissible.

## KNOWLEDGE AREA 14.4

## Explainability, bias, human accountability, cybersecurity and privacy

*Topics: 14.4.1 explainability as a decision requirement · 14.4.2 bias and differential error · 14.4.3 human accountability and the AI use register · 14.4.4 cybersecurity and privacy in a digital delivery estate.*

### 14.4.1 Explainability as a decision requirement

**Definition. Explainability**, for the purposes of delivery decisions, is the property that a named human can state *why* an output says what it says, in terms the affected party can engage with, sufficiently to defend the decision and to permit it to be contested.

The definition deliberately locates explainability in the **decision**, not in the model. Technical interpretability (which features drove an output, how sensitive it is to inputs) is useful and is not the requirement. The requirement is set by what the decision must survive: an appeal, a debrief, an audit, a regulator's question, a court. Domain 10, KA 10.2 already applies the strict form of this to tender evaluation, where an AI-generated quality score is prohibited precisely because nobody can explain it in the terms the criterion was written in. That is the general rule: **the explanation must be in the vocabulary of the obligation, not of the model.**

**The three levels a leader should distinguish**, because they have different costs and different triggers. *No explanation required.* The output informs an internal, reversible, low-consequence choice; most permitted AI use sits here, and demanding explainability of all of it is how organisations make responsible AI unaffordable. *Explanation required.* A person affected by the decision, or a body reviewing it, is entitled to know its basis: a supplier not shortlisted, a change not approved, a clinic prioritised for support ahead of another. Here the requirement is satisfied by the *human's* reasoning, evidenced, with the AI output as one input, which is the Domain 1 accountability position expressed as a documentation duty. *Contestability required.* The affected party must be able to challenge the decision and have the challenge examined by a human with authority to change it. This is the level at which model-derived scoring becomes untenable, because a challenge that cannot be answered in the criterion's own terms cannot be examined.

Explainability obligations in law and regulation are **jurisdiction-specific and moving**, and nothing in this book should be read as stating them. Regimes may be named accurately as reference points: the European Union's AI Act as a risk-tiered regulatory approach, data-protection regimes such as the General Data Protection Regulation as sources of rights concerning automated decisions, ISO/IEC 42001 as a management-system standard for AI, ISO/IEC 23894 as guidance on AI risk management, and the NIST AI Risk Management Framework as a voluntary framework:



each applying on its own terms, in its own territory, to its own defined scope. The professional discipline is to establish the applicable obligations for the specific decision in the specific place with qualified advice, and to design to the **strictest** applicable level where a programme spans jurisdictions, because the alternative is designing twice.

#### 14.4.2 Bias and differential error

**Definition.** In this context **bias** means a **systematic difference in error rates between identifiable groups**, such that the burden of the model's mistakes falls unevenly. It is a measurable property of a deployed system, and it is measurable only if someone measures error rates *by group* rather than in aggregate.

That last clause is the whole practical difficulty. An aggregate accuracy figure cannot reveal differential error, and it is the figure that is always reported.

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#### WORKED EXAMPLE

##### Worked example 14.4.2 — Meridian's readiness prioritisation, by clinic type.

1. **Setup.** Meridian scores its **40** clinics for risk of failing go-live readiness and sends a support team to those flagged. Of **24 urban** clinics, **9** turned out to be genuinely at risk and the model flagged **8** of them. Of **16 rural** clinics, **11** were at risk and the model flagged **6**. Planned support costs **USD 1,900** per clinic; a clinic that is at risk and *not* flagged needs emergency remediation at **USD 6,400**.
2. **Formula.** Recall by group = flagged at-risk ÷ actual at-risk. Excess cost per missed clinic = emergency – planned. Group burden = missed × excess, and per clinic of the group.
3. **Substitution.** Urban 8/9; rural 6/11; overall 14/20. Excess 6,400 – 1,900 = 4,500. Missed: urban 1, rural 5.
4. **Result.** Overall recall **70.00 %**. Urban recall **88.89 %**; rural recall **54.55 %**: a gap of **34.34 percentage points**. Total excess cost **USD 27,000**, of which rural clinics bear **83.33 %** while being **40.00 %** of the estate. Per clinic of each group the excess is **USD 187.50** urban and **USD 1,406.25** rural, a ratio of **7.50**.
5. **Interpretation.** The aggregate figure, 70 % recall, is the only number that would normally have been reported, and it describes neither group. It is the arithmetic equivalent of Domain 9, KA 9.4.2's mean-of-yields problem: a summary statistic that no member of the population experiences. The measurement that matters is the **disaggregated** one, and the only way to obtain it is to define the groups in advance and measure by group as a standing practice, because the comparison cannot be reconstructed from an aggregate afterwards.

**Correcting it is also, here, economically obvious**, which is worth showing because the debate is usually conducted as though fairness and cost were opposed. Lowering the flagging threshold for rural clinics raises rural recall to **10 of 11 (90.91 %)** at the cost of two additional false positives: two clinics



receiving planned support they did not need. Four fewer missed clinics avoids  $4 \times 4,500 = \text{USD } 18,000$  of emergency remediation for **USD 3,800** of extra planned support: a net **USD 14,200** and a return of **4.74 times** the cost. The differential was not a price the programme was paying for efficiency; it was simply a defect nobody had measured.

Four cautions, and the first is the most important in the book. **These units are clinics.** Where the units are *people*, differential error rates raise legal and ethical questions that are not expected-value questions at all, the applicable obligations differ by jurisdiction and by the characteristic concerned, and Domain 12's prohibition on individual inference applies in full: nothing in this arithmetic licenses building such a model about people, and this volume does not endorse it. Second, the **base rates differ** between the groups here (37.50 % of urban clinics were at risk against 68.75 % of rural ones), and where base rates differ it is mathematically impossible to equalise every error measure at once: equalising recall and equalising the share of flags that prove correct are different targets that cannot both be met, so the choice of which measure to equalise is a **value judgement that must be made explicitly by an accountable person** and recorded, not settled by whichever metric the tool happens to display. Third, the groups must be **defined in advance**: choosing the grouping after seeing the results permits any conclusion, and the honest practice is to name the groupings that matter (geography, size, deprivation, language, service type), before deployment. And fourth, **the cause is usually the data, not the model**: rural clinics were under-represented in the historical records the model learned from, which is a data-coverage defect (14.1.4) presenting as a model defect, and re-tuning a threshold treats the symptom while leaving the coverage gap to reappear in the next model built on the same register.

#### 14.4.3 Human accountability and the AI use register

Domain 1 established the principle and Domain 3, KA 3.A.2 required four governance instruments. This topic makes the first of them operational, because the register is what turns a principle into something an auditor can test.

**The AI use register**, one row per use, with the columns that make it testable rather than descriptive:

| COLUMN                      | WHY IT IS THERE                                                                |
|-----------------------------|--------------------------------------------------------------------------------|
| <b>Use and output class</b> | What the tool produces, specifically enough to attach a consequence to.        |
| <b>Decision informed</b>    | The named decision the output feeds. A use that informs no decision is a cost. |
| <b>Accountable person</b>   | The named human answerable for the output, Domain 1's test, applied to tools.  |

| COLUMN                                                                  | WHY IT IS THERE                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Consequence per escaped error <math>u</math></b>                     | Assessed, owned; the input that sets the verification tier.                                                                                                                                                                                        |
| <b>Measured <math>p</math> and <math>q</math>, with date and method</b> | The measurements that make the tier defensible, and that go stale.                                                                                                                                                                                 |
| <b>Verification tier applied</b>                                        | From 14.3.3's standard, with the reviewer recorded per item at tiers 3–4.                                                                                                                                                                          |
| <b>Model, prompt and grounding versions</b>                             | Reproducibility, and the retrospective question of 14.3.2.                                                                                                                                                                                         |
| <b>Data and confidentiality boundary</b>                                | What may be supplied to the tool, stated before use, not after an incident.                                                                                                                                                                        |
| <b>Prohibited adjacent uses</b>                                         | The uses that must not creep in: the scope creep of AI adoption, and it is real.                                                                                                                                                                   |
| <b>Failure route, who is notified and within what period</b>            | Written before any failure: who is told on the day an output is found to be materially wrong, in what form, and who authorises resumption after re-measurement. A register entry without one has an accountable person and nothing for them to do. |
| <b>Decisions to reopen on failure</b>                                   | The materiality threshold, set in advance, above which decisions that cited this output are reopened by the authority that took them: with any decision not reopened recorded as considered and confirmed, and why.                                |

**Three accountability failures the register exists to prevent.** The **unattributed output**: a document, register or analysis in circulation with no author who can answer for it, which is Domain 1's accountability-without-a-holder defect and which Domain 3 prohibits in anything relied upon. **Scope creep of a permitted use**: a tool approved to extract requirements candidates that begins drafting requirements, or one approved to summarise minutes that begins drafting the decision record, a change of class, not of degree, and the register's last column is the control. And **accountability laundering**, the most corrosive: presenting a model's output as a neutral input in order to avoid owning the judgement it embodies. Its diagnostic is simple and worth asking in a governance meeting: *if this output were wrong, whose judgement was wrong?* If no one can answer, the output is not governed, and Domain 3's duty on the receiving body applies — ask what the input was, who verified it, and what would change the conclusion.

**Two obligations that sit above the register.** Where a person is materially affected by a decision an AI output informed, the affected person's route of challenge must reach a human with authority to change it (14.4.1). And the **workforce obligation**: staff must be told what tools are in use on their work, must not be evaluated by inference from telemetry (Domain 12), and must be able to decline to sign for an output they have not been given the time or the information to verify: the last being the point at which a verification standard either is real or is a document, because a

standard that cannot be met in the time allowed simply relocates accountability onto whoever signs.

**When a relied-upon output turns out to be wrong.** Everything above is preventive, and a control set with no failure path is governance asserted rather than operationalised. The register names an accountable person for every output; this is what that accountability *obliges them to do* when the output fails, when an output relied on in a recorded decision is later found to be materially wrong, or when the differential error of 14.4.2 is detected in a deployed use. It is a sequence, and the order matters:

1. **Stop the use and suspend the register entry.** Before anything is investigated. A use whose error rate is now known to be wrong has no defensible verification tier, because the tier was derived from *p* and *q* that have just been falsified. Suspension is the default and continuing is the exception, taken by the accountable person in writing with a stated reason.
2. **Identify every recorded decision that cited the output.** This is what the register's **decision informed** column is for, and it is the column that repays the whole instrument on the day it is needed. If the question "which decisions rest on this?" cannot be answered from the register in an afternoon, the register is descriptive rather than testable.
3. **Notify in writing, on the day it is known.** The accountable person and the decision-maker are told the same day, in the escalation-grade form of Domain 11, KA 11.2.3 — what is known, what is not yet known, what has already been stopped, what decision is now requested and by when. Not a meeting invitation; not a verbal heads-up followed by a note next week.
4. **Reopen the decisions above a stated materiality threshold, with the same authority that took them.** The threshold is set in advance, in the register, because a threshold chosen after the failure will be chosen to suit it. Reopening is by the original authority (a decision taken by a steering body is reopened by that body, not corrected by the project), and a decision that is not reopened is recorded as *considered and confirmed*, with the reason, so that the absence of action is itself attributable.
5. **Record whether anyone outside must be told, and on whose determination.** An affected person, a counterparty, an insurer or a supervisory body may need to be informed. Whether any such obligation arises, to whom, in what form and on what clock is established with the relevant authority and with qualified counsel: **nothing here states the position in any jurisdiction, and nothing here characterises any act or omission as compliant or otherwise.** What this domain does require is that the question is *asked*, by a named person, promptly, and that the answer and its source are recorded. Domain 8, KA 8.4.3's notification map is the instrument, and an AI-use failure is exactly the class of event it exists for.

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6. **Re-measure before the use resumes, under a named authority.** *p* and *q* are re-measured by the seeded-error method of 14.A.2, using the same method as before so the numbers are comparable, and the verification tier is recomputed from the measurements rather than restored to what it was. A named authority (not the user of the tool) authorises resumption, and the register records the suspension, the re-measurement and the resumption as three dated events.

Two observations that generalise. **The failure route is written before the failure**, in the register, because every step above is one nobody will design well at the moment they need it. And a failure discovered by the organisation itself, handled this way, is a functioning control system; the same failure discovered by someone outside it, with no route, is an accountability defect, which is the same distinction Domain 9 draws between a defect caught by containment and a defect that escaped.

#### 14.4.4 Cybersecurity and privacy in a digital delivery estate

A programme's digital estate now holds its commercial position, its supplier terms, its vulnerabilities, its people's details and, increasingly, whatever has been supplied to third-party tools. Domain 8 treats risk generally and Domain 10 supplier risk; the specific obligations here are the ones a delivery leader owns rather than delegates to a security function.

**Four that are non-delegable.** *Data minimisation by design*: the estate holds what the project needs, not what happened to be available; Meridian's decision to keep patient clinical data out of the delivery CDE (14.1.3) is the model, and it reduces the security problem rather than defending it. *Access on the data class schedule*: 14.1.2's access column implemented as permissions, reviewed on movers and leavers, which is where real breaches begin. *A stated boundary for third-party tools*: what may be supplied to which tool, decided before use; the commonest breach in AI adoption is not an attack but a disclosure by a well-meaning person into a service whose terms nobody read. *Continuity of the environment itself*: the delivery estate is now a single point of failure for the delivery, and its loss is a delivery risk with a cost of delay attached, not merely an IT incident.

Reference points may be named: ISO/IEC 27001 for information security management, ISO/IEC 27701 for privacy information management, IEC 62443 for industrial automation and control system security (directly relevant to Auriga's operational technology), and data-protection regimes that apply on their own terms in their own territories. None of them decides the spend. The arithmetic below does part of it, and its limits matter as much as its results.

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#### WORKED EXAMPLE

**Worked example 14.4.4 — sizing two controls on Meridian's CDE.**

1. **Setup.** Meridian assesses the annual probability of a credential-compromise incident affecting its CDE at **0.08**, with an assessed impact of **USD 480,000** (incident response, remediation, delivery disruption, notification effort). Two controls are proposed. **A — identity hardening** (single sign-on, multi-factor authentication, privileged access review): **USD 26,000** a year, assessed to reduce the probability to **0.02**. **B — segmentation and field-level minimisation:** **USD 14,000** a year, assessed to reduce the *impact* to **USD 260,000** with the probability unchanged.
2. **Formula.** Expected annual loss **EAL** = **P** × **impact**. Total position = control cost + residual **EAL**. For two controls together, the reductions apply to their respective terms and therefore **multiply** rather than add.
3. **Substitution.** None  $0.08 \times 480,000$ . A  $26,000 + 0.02 \times 480,000$ . B  $14,000 + 0.08 \times 260,000$ . Both  $40,000 + 0.02 \times 260,000$ .
4. **Result.**

| OPTION                             | CONTROL<br>COST | RESIDUAL<br>EAL | TOTAL         | LOSS<br>AVOIDED |
|------------------------------------|-----------------|-----------------|---------------|-----------------|
| Do nothing                         | 0               | 38,400          | <b>38,400</b> | —               |
| A identity hardening               | 26,000          | 9,600           | <b>35,600</b> | 28,800          |
| B segmentation and<br>minimisation | 14,000          | 20,800          | <b>34,800</b> | 17,600          |
| A and B together                   | 40,000          | 5,200           | <b>45,200</b> | 33,200          |

**B alone is the lowest total position**, at USD 34,800. Buying both is **worse than buying either**, and worse than doing nothing. 5. **Interpretation.** Three results, each of which is routinely got wrong. First, **control benefits are sub-additive**. Summing the two avoided losses gives USD 46,400; the true combined avoided loss is **USD 33,200**, because the second control acts on a residual the first has already reduced. A business case that adds them overstates the combined benefit by USD 13,200 and, here, converts a value-destroying package into an apparently attractive one. Second, **the impact-reducing control is better value than the probability-reducing one** at these parameters (USD 1.26 of avoided loss per dollar spent against USD 1.11), and this is the general tendency worth carrying: probability reduction is bought against an uncertain frequency, while impact reduction (segmentation, minimisation, tested recovery) works on every incident that does occur, including the ones the probability estimate did not anticipate. A programme that buys only probability reduction has bought the more speculative half.

Third, and this is the professional caution rather than the result: **the answer is fragile in exactly the place the arithmetic is weakest**. Setting the two totals

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equal gives a crossover at an incident probability of **8.57 %**: above it, identity hardening wins; below it, segmentation wins. The assessed probability is 8.00 %, which sits **0.57 percentage points** below a crossover that no organisation can estimate to that precision. The honest conclusion is therefore *not* "buy B"; it is that **the two controls are economically indistinguishable at any probability anyone can defend**, and the choice should be made on grounds the arithmetic cannot see: implementation risk, effect on the ability to detect an incident at all, obligations under applicable regimes, and the fact that minimisation reduces the *scope* of a future incident rather than its price. Doing nothing, by contrast, is beaten by segmentation above an incident probability of **6.36 %** and by identity hardening above **7.22 %**, so at the assessed 8.00 % both dominate it and inaction is not a serious option, which is the robust part of this analysis, and the part worth taking to the board. And the frame itself has a boundary: expected annual loss is a legitimate test where consequences are cost-compensable, and it is the **wrong test** where the consequence is a penalty imposed by an authority, a licence condition, a duty of confidence or harm to a person. There the control is a requirement whose cost is sized, not an investment whose return is computed, and presenting such a control as failing a cost test is a category error a leader should be able to name in the room.

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### AI in this KA

**Where it earns its place.** Measuring error rates by group across a large deployment, once the groups are defined by a human, and flagging where a differential exceeds a stated tolerance — mechanical, high-volume, exactly checkable, and the measurement nobody has time to do. Classifying data holdings against the data class schedule to find minimisation opportunities and misplaced records. Detecting anomalous access patterns as a **prompt to investigate**, never as a finding about a person. Drafting the AI use register's first pass from a tool inventory, for human completion of the accountability and consequence columns. Testing a set of decisions for whether the recorded reasoning would satisfy 14.4.1's explanation level.

**Where it must not go.** It must not generate the explanation for a decision after the fact: a model asked why a decision was made will produce a fluent, plausible rationalisation, and a retrospective rationalisation presented as the decision's basis is worse than no explanation because it is evidence of a reasoning process that did not occur. It must not choose which fairness measure to equalise, which is a value judgement with an accountable owner. It must not make an access, security or privacy decision, and it must not be given the data whose exposure is the risk being managed. The boundary must be stated before use. It must not perform inference about individuals in any of the forms Domain 12 prohibits, and monitoring staff activity as a productivity proxy remains prohibited here as there. And it must not be the route through which an affected person's challenge is answered: contestability requires a human with authority to change the decision.

**Verification, concretely.** Group definitions and the fairness measure chosen are recorded with the accountable person's name and date, before deployment. Differential error measurements state the group sizes, because a 16-clinic group supports a much weaker inference than a 24-clinic one and the gap must be read with that in mind. Explanation levels are assigned per decision class and tested by asking a reviewer to reconstruct the reasoning from the record alone. Security and privacy assessments are performed by qualified people against the applicable regime, with the arithmetic here used to size effort rather than to authorise an exception. And the register is audited on a stated cadence against the actual tool estate, since the failure it exists to catch (a permitted use quietly becoming a prohibited one) is invisible from inside the team using it.

## KEY TERMS — KA 14.4

| TERM                                   | MEANING                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Explainability (decision sense)</b> | The property that a named human can state why an output says what it says, in the vocabulary of the obligation.                                                                                                                                                                                                                                                                |
| <b>Contestability</b>                  | The affected party's ability to challenge a decision and have it examined by a human with authority to change it.                                                                                                                                                                                                                                                              |
| <b>Differential error</b>              | A systematic difference in error rates between identifiable groups; measurable only by disaggregated measurement.                                                                                                                                                                                                                                                              |
| <b>Recall (by group)</b>               | Flagged at-risk cases ÷ actual at-risk cases within a group.                                                                                                                                                                                                                                                                                                                   |
| <b>Base-rate difference</b>            | Unequal prevalence between groups, which makes simultaneous equalisation of all error measures impossible.                                                                                                                                                                                                                                                                     |
| <b>AI use register</b>                 | The row-per-use record of use, decision informed, accountable person, consequence, measurements, tier, versions, boundary, failure route and decisions to reopen on failure.                                                                                                                                                                                                   |
| <b>Failure route</b>                   | The pre-written sequence for an output found materially wrong after reliance: stop and suspend, enumerate the decisions that cited it, notify in writing that day, reopen above the stated materiality threshold with the original authority, record whether anyone outside must be told and on whose determination, and re-measure under a named authority before resumption. |
| <b>Accountability laundering</b>       | Presenting a model's output as a neutral input to avoid owning the judgement it embodies.                                                                                                                                                                                                                                                                                      |
| <b>Expected annual loss (EAL)</b>      | Incident probability × assessed impact; the residual after a control is the basis of comparison.                                                                                                                                                                                                                                                                               |
| <b>Sub-additivity of controls</b>      | Combined avoided loss is less than the sum of individual avoided losses, because reductions multiply.                                                                                                                                                                                                                                                                          |
| <b>Data minimisation</b>               | Holding only what the project needs, which reduces the security and privacy problem rather than defending it.                                                                                                                                                                                                                                                                  |

## SAMPLE MCQs — KA 14.4

**MCQ 14.4-A** [14.4.2 · Application] Of 24 urban clinics, 9 were at risk and 8 were flagged; of 16 rural clinics, 11 were at risk and 6 were flagged. Rural recall is:

- A. 37.50 %
- B. 54.55 % CORRECT
- C. 68.75 %
- D. 70.00 %

*Rationale:*  $6/11 = 54.55\%$  (14.4.2). A divides flagged by all rural clinics; C is the rural base rate  $11/16$ ; D is the aggregate recall  $14/20$ , which describes neither group.

**MCQ 14.4-B** [14.4.2 · Evaluation] Lowering the rural flagging threshold would raise rural recall from 6 of 11 to 10 of 11, adding two false positives at USD 1,900 each and avoiding four emergency remediations whose excess cost is USD 4,500 each. The correct conclusion is:

- A. it should not be done, because false positives rise
- B. it is worth USD 14,200 net, a return of 4.74 times the extra cost CORRECT
- C. it is worth USD 18,000, the emergency cost avoided
- D. it cannot be evaluated economically

*Rationale:*  $4 \times 4,500 - 2 \times 1,900 = 14,200$ , and  $18,000/3,800 = 4.74$  (14.4.2). C omits the extra support cost; the differential here was a defect, not a price paid for efficiency.

**MCQ 14.4-C** [14.4.4 · Analysis] Control A cuts incident probability from 0.08 to 0.02 at USD 26,000 a year; control B cuts impact from USD 480,000 to USD 260,000 at USD 14,000 a year. Which option gives the lowest total position?

- A. both controls, total USD 45,200
- B. control A alone, total USD 35,600
- C. control B alone, total USD 34,800 CORRECT
- D. neither, total USD 38,400

*Rationale:* Totals are control cost plus residual EAL (14.4.4). Both controls cost more than they avoid together because their reductions multiply, the sub-additivity result, so A is the trap the example exists to expose.

**MCQ 14.4-D** [14.4.4 · Evaluation] For those two controls, the individual avoided losses are USD 28,800 and USD 17,600. The combined avoided loss is:

- A. USD 46,400
- B. USD 33,200 CORRECT
- C. USD 28,800



D. USD 38,400

*Rationale:* Combined residual EAL is  $0.02 \times 260,000 = 5,200$ , so avoided is  $38,400 - 5,200 = 33,200$  (14.4.4). A adds the two individual figures and overstates by USD 13,200, which is the error that converts a value-destroying package into an attractive one.

**MCQ 14.4-E** [14.4.1 · Analysis] A supplier not shortlisted asks why. The applicable explainability requirement is best met by:

- A. disclosing the model's feature importances
- B. the evaluator's recorded reasoning against the published criteria, with any AI output identified as one input CORRECT
- C. a statement that the process was automated and consistent
- D. re-running the model with the supplier present

*Rationale:* The explanation must be in the vocabulary of the obligation (the published criteria), and the accountable human's reasoning is what is owed (14.4.1, with Domain 10, KA 10.2). A explains a model, not a decision; C is the answer that converts a substantive objection into a legitimacy grievance.

#### SELF-CHECK — KA 14.4

1. *Why can an aggregate accuracy figure not establish that a model is unbiased?* — Because differential error is invisible in aggregate: Meridian's 70.00 % recall concealed 88.89 % urban and 54.55 % rural, a 34.34-point gap, with rural clinics bearing 83.33 % of the excess cost while being 40.00 % of the estate.
2. *Why is buying two security controls sometimes worse than buying one?* — Because their reductions multiply rather than add: Meridian's combined avoided loss is USD 33,200 against a naive sum of USD 46,400, and the two together total USD 45,200 against USD 34,800 for the cheaper control alone.
3. *What does the AI use register have to contain to satisfy Domain 1's accountability test?* — A named accountable human per use, the decision informed, the assessed consequence, the measured error and detection rates with their dates, the verification tier applied, the model, prompt and grounding versions, the data boundary, the adjacent uses that are prohibited, and the failure route with the decisions to reopen if the output turns out to be wrong.
4. *An output relied on in a recorded decision is found to be materially wrong. State the sequence.* — Stop the use and suspend the register entry; enumerate every recorded decision that cited the output from the register's *decision informed* column; notify the accountable person and the decision-maker in writing the same day, in escalation-grade form; reopen decisions above the pre-set materiality threshold with the authority that took them, recording those not reopened as considered and confirmed; record whether an affected person, counterparty, insurer or supervisory body must be told and on whose determin-

ation, a question for the relevant authority and for qualified counsel, never settled inside the project; and re-measure  $p$  and  $q$  by the same method under a named authority before the use resumes (14.4.3).

## Advanced topics — Domain 14

### 14.A.1 The consolidated position across Domains 1 to 13

The volume's per-domain AI treatments are not thirteen policies; they are one policy applied thirteen times, and consolidating them makes both the consistency and the two genuine tensions visible. The table below is the map, by KA, and it is the artefact to put in front of a governance body that asks "what is our position on AI?"

| DOMAIN                   | PERMITTED CLASS, AND WHERE                                                     | PROHIBITED, AND WHY                                                                                           |
|--------------------------|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 1 The profession         | — (establishes the principle)                                                  | Any delegation of the obligation to answer (KA 1.2, 1.4)                                                      |
| 2 Strategy and selection | Enumeration of options; scoring arithmetic                                     | Benefit attribution; kill criteria (KA 2.3, 2.4)                                                              |
| 3 Governance             | Comparison of authority documents; latency modelling (KA 3.1, 3.2)             | Decisions; thresholds; authorship of the decision record (KA 3.1, 3.3)                                        |
| 4 Integration            | Interface enumeration; register reconciliation (KA 4.2, 4.3)                   | Tailoring approval; baseline entry; schedule impact estimates (KA 4.1, 4.4)                                   |
| 5 Scope and requirements | Extraction of requirement candidates; traceability gap detection (KA 5.1, 5.3) | Authoring requirements; boundary decisions; acceptance (KA 5.2, 5.4)                                          |
| 6 Planning               | Scenario enumeration; deterministic network arithmetic                         | Durations and logic without a planner's ruling                                                                |
| 7 Cost and commercial    | Blended-rate and forecast arithmetic                                           | Estimates; contract terms                                                                                     |
| 8 Risk                   | Risk candidate enumeration; sensitivity modelling                              | Probabilities and impacts from plausibility (KA 8.2)                                                          |
| 9 Quality                | Data profiling; test-case drafting; yield arithmetic (KA 9.4)                  | Test results; inspection records; concessions; root-cause conclusions; its own sample size (KA 9.2, 9.3, 9.4) |
| 10 Procurement           | Bid comparison mechanics; clause comparison (KA 10.1, 10.2)                    | Quality scores; liability caps; entitlement opinions; post-opening reweighting (KA 10.2, 10.3, 10.4)          |

| DOMAIN                  | PERMITTED CLASS, AND WHERE                                     | PROHIBITED, AND WHY                                                                               |
|-------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 11 Stakeholders         | Register structuring; document comparison (KA 11.1)            | Attitude and influence inference; smoothing a report; synthetic voice or likeness (KA 11.2, 11.4) |
| 12 Leadership and teams | Aggregate structural analysis only (KA 12.2)                   | Any individual inference, assessment or people decision (KA 12.1–12.4)                            |
| 13 Adaptive delivery    | Backlog and flow analytics; metric computation (KA 13.2, 13.4) | Prioritisation as a decision; velocity as a performance judgement (KA 13.4)                       |

Two tensions are real and should be named rather than smoothed. The first is between **the value of enumeration and the risk of anchoring**: a model's list of candidate risks, requirements or options genuinely improves coverage, and it also anchors the human list that follows, so the professional practice is to have the human enumerate first and the model second, which costs nothing and is almost never done in that order. The second is between **verification cost and the productivity claim**: 14.3.3's arithmetic shows the gain is real but lands mostly as reduced checking on low-consequence classes, so an organisation whose AI-assisted work is concentrated in high-consequence classes should expect a modest gain and should say so, rather than promising the gain observed in low-consequence work.

#### 14.A.2 Drift, revalidation and the lifecycle of a model in a delivery organisation

Every quantity in this domain is measured, and measurements go stale. Three things drift, and each has a different trigger for re-measurement. **The configuration** drifts when the model, its version, the prompt, or the grounding material changes, an event, not a period, and any of those changes invalidates  $p$  and  $q$  for every class using it. **The population** drifts when the work changes: the same prompt applied to a new clinic type, a new contract form or a new plant area is a new output class, and its error profile is unknown until measured. **The environment** drifts when the downstream containment changes: remove a review step somewhere else and  $u$  rises, which moves every threshold in 14.3.3 without anything about the model having changed.

A workable revalidation régime is therefore **event-triggered with a periodic floor**: re-measure on any configuration change, on any new output class, and on any change to downstream containment; and in the absence of events, re-measure each class on a stated cadence with the cadence set by consequence: the tier-4 classes most often, the tier-0 classes least. The seeded-error method should be specified once and reused, because the comparability of measurements over time is what makes drift detectable at all; a re-measurement by a different method produces a different number and no information.

Three lifecycle obligations close the loop. **Retirement**: a use that is no longer needed is removed from the register and from the estate, because an unmaintained tool whose measurements have expired is worse than no tool. It produces output nobody is verifying to a standard nobody has revisited. **Retention**: the model version, prompt, grounding material and verification record for anything relied on must survive the project, which is Domain 16, KA 16.4's responsible archive and model/data retention obligation, and which must be designed at mobilisation because it cannot be reconstructed at closeout.

And **failure**, which is the one the triggers above imply and do not state. A revalidation trigger that fires because the measurements have gone stale is hygiene; a trigger that fires because an output already relied on has been found materially wrong, or because differential error has been detected in a deployed use, is an **incident with an accountable route**, and it runs through 14.4.3's sequence rather than through a re-measurement diary: stop the use and suspend the entry, enumerate the decisions that cited the output, notify the accountable person and the decision-maker in writing that day, reopen above the stated materiality threshold with the authority that took the decision, record whether anyone outside must be told and on whose determination, and re-measure under a named authority before resumption. The distinction is worth naming because the two arrive looking identical in a monitoring report, a number has moved, and only one of them has a clock on it.

#### 14.A.3 The reviewer's digital and AI eye

Invariants to test on any digital or AI-assisted delivery arrangement. Each is cheap, and each is diagnostic.

Every material fact has **exactly one source of record**, and every other holding is marked as a copy. Every data class has a **named owner, a written definition, and a quality standard expressed with its consequence**: never a single organisational rate target across classes of unequal consequence (14.1.4). Remediation is ranked by **exposure per record**  $d_{i u_i}$ , not by defect rate. Every dashboard measure carries its **information age**  $R/2 + G$  on its specification, and the age is compared against the window of the decision it serves (14.2.1). Every claimed digital twin has a **measured fidelity**, a test-case count, a date and a tolerance, and a computed breakeven (14.2.3). Every automation business case shows **maintenance** and a **measured differential escape rate**, and its breakeven volume is compared against the volume at the *stated organisational boundary* (14.2.4). Every AI use appears in the **register** with a named accountable human, an assessed consequence and a verification tier (14.4.3).  $p$  and  $q$  are **measured, dated and owned**, with a re-measurement trigger (14.A.2). The verification standard is **published**, and the thresholds are reproducible by hand as  $\Delta v / (p \cdot \Delta q)$  (14.3.3). Outputs whose consequences sit near a **governance threshold** are reperformed regardless of tier (14.3.4). Error rates for anything that allocates support, priority or resource are **measured by group**, with the groups defined in advance and the fairness measure chosen by a named person (14.4.2). Security control cases **do not add** avoided losses (14.4.4). **Every**

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**register entry names a failure route, and every AI-informed decision can be found from the output that informed it** (14.4.3): the second half being the test of the first, since a failure route that cannot enumerate the decisions to reopen is a notification list rather than a control. And the question that subsumes several of the others: for any AI-informed output in circulation, *if this were wrong, whose judgement was wrong?*, if nobody can answer, the output is not governed.

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## Industry variations – Domain 14

- **Public sector and government.** Data residency, procurement rules for algorithmic systems and transparency expectations are set externally and vary sharply by jurisdiction; the leader's design freedom is mostly in *where* the human decision sits, not in whether it does. Publication duties can make the explanation level (14.4.1) higher than the internal consequence would justify, and the verification standard must be set to the disclosure obligation rather than to the arithmetic.
- **Regulated life sciences and medical devices.** Computerised systems supporting regulated activity carry validation expectations, and a change to a model or a prompt is a change to a validated system, so 14.A.2's event-triggered revalidation is not a good practice but a condition of use. The practical consequence is that configuration stability has a value here that it does not have elsewhere, and frequent model updates are a cost rather than a benefit.
- **Construction and infrastructure.** The common data environment is the mature instance of 14.1.3, framed by ISO 19650's information-management series and usually contractual; information obligations sit in the contract, so a data class schedule inconsistent with the contract loses to the contract (the general rule Domain 3 states for governance). Digital twins are furthest advanced here and fidelity decay after handover is the characteristic failure.
- **Energy, utilities and process industries.** Operational technology converges with information technology, and the security frame is different: IEC 62443's industrial control-system security applies alongside enterprise standards, availability and safety usually outrank confidentiality, and a control that reduces impact may be unavailable because the plant cannot be segmented as freely as an office network. Auriga's twin and its handover data are the domain's own example.
- **Financial services.** Model risk management is an established discipline with model inventories, independent validation and periodic review: much of 14.4.3's register already exists under another name, and the leader's task is to bring delivery-side AI use into it rather than to invent a parallel regime. Explainability obligations attached to decisions about customers are stricter than anything internal delivery requires.

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- **Healthcare.** Clinical safety obligations run in parallel with project governance and are not delegable to a delivery body (the position Domain 3 takes for Meridian's clinical sign-off). Minimisation is the dominant design move (Meridian's exclusion of clinical data from the delivery CDE), and any output touching clinical decision support crosses from delivery governance into clinical governance, with a different accountable authority and a different standard.
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### Case study – Domain 14: the quality target that cost 34,620 (health, Meridian)

**Situation.** Eighteen weeks into the clinic rollout, Meridian's assurance function reported that the programme's data quality was "below standard at 97.6 % conformance against a 98 % target" and recommended a remediation programme across all six data classes of the delivery CDE. The programme board was minded to approve it. The estate held **14,100** records: **332** were defective, a weighted mean defect rate of **2.3546 %**.

**What the arithmetic showed.** The project leader computed the consequence-weighted exposure by class before the next board. Total exposure **USD 172,820**, and it was distributed nothing like the defect rates: interface field-mapping records, **900** records at a 6.0 % defect rate, carried **USD 64,800**; migration reconciliation records, at the *lowest* defect rate in the estate (**0.8 %**), carried **USD 56,000**. Together, two classes holding **20.57 %** of the records (2,900 of 14,100) carried **69.90 %** of the exposure. Meanwhile training and competency records, **3,600** of them at 2.5 %, carried **USD 8,100**, an exposure of **USD 2.25** per record.

Then the leader priced the recommendation itself. Bringing every class to the **2 %** target would produce **282** defects (**15.06 % fewer** than the observed 332) at an exposure of **USD 207,440**, which is **USD 34,620** or **20.03 % worse** than the position it was correcting. The target achieved its own metric by loosening the two classes whose defects were expensive and tightening the four whose defects were cheap.

**What changed.** The board approved a **consequence-proportional schedule** instead. Interface mappings were re-verified at **USD 22** per record (**USD 19,800**), cutting the rate to 1.0 % and removing **USD 54,000** of exposure. Migration reconciliation records were re-reconciled at **USD 12** each (**USD 24,000**), cutting the rate to 0.2 % and removing **USD 42,000**. The other four classes were left where they were, and the proposed re-check of all 3,600 training records was declined on the arithmetic: it would have cost **USD 28,800** to remove **USD 6,480** of exposure, a net loss of **USD 22,320**, and would only have paid at a remediation cost below **USD 1.80** per record.

**The outcome, and the part that mattered.** Total remediation spend **USD 43,800** against **USD 96,000** of exposure removed (a net **USD 52,200**), leaving the estate at **275** defects and **USD 76,820** of exposure. The number the assurance function found

most uncomfortable is the one worth dwelling on: the new weighted mean defect rate was **1.9504 %**, which *passes* the original 2 % target, the schedule reached a better rate and a **62.97 %** lower expected cost than the uniform target would have produced, for a fraction of the effort, by ignoring the target as a design instruction and using it only as an outcome check.

**What the domain teaches here.** A quality target expressed as a rate, with no consequence attached, is not a weak standard. It is a standard that can be met while making the organisation worse off, and Meridian's recommendation would have spent money to increase expected cost by USD 34,620. Compute exposure by class, rank by exposure per record, and be willing to leave a high-rate, low-consequence class alone. The uncomfortable corollary, which a leader must be prepared to defend, is that the resulting schedule deliberately tolerates the *most* defects where they are cheapest. And that defending it requires the arithmetic, because without it the position is indistinguishable from complacency.

### Case study B – Domain 14: the automation that saved 79,200 a month and lost 86,688 (financial services)

**Situation.** A payments-modernisation programme automated the triage of reconciliation exceptions. The manual process took an analyst **5 minutes** per exception at a blended rate of **USD 112.80** an hour (**USD 9.40** each) across **9,600** exceptions a month. The automation cost **USD 128,000** to build and **USD 2,600** a month to maintain, and handled an exception for **USD 1.15** of residual human exception-handling time. On the visible unit costs the case was overwhelming: a saving of **USD 8.25** per exception, **USD 79,200** a month, a naive breakeven of **21,188** units reached in **2.21 months** against fixed costs of **USD 174,800** over an 18-month horizon. It was approved in one meeting and reported as a success for two quarters.

**What had happened.** A quarterly control review measured the two routes' **error escape rates** for the first time: against a held-back sample of exceptions with known correct dispositions. The manual route mis-triaged **0.8 %**; the automated route mis-triaged **3.5 %**. A mis-triaged exception cost an assessed **USD 640** in customer remediation and rework. Quality-adjusted, the manual unit cost was  $9.40 + 0.008 \times 640 = \text{USD } 14.52$  and the automated unit cost was  $1.15 + 0.035 \times 640 = \text{USD } 23.55$ . The automation was **USD 9.03 per exception more expensive than the process it replaced** (**USD 86,688** a month), so no volume justified it and none ever would: the quality-adjusted breakeven volume did not exist. Against a claimed saving of USD 79,200 a month, the true position was a loss of USD 86,688 a month, a swing of **USD 165,888** a month that had gone unmeasured for eighteen months and cost **USD 1,560,384** in escaped errors on top of the **USD 174,800** of build and maintenance.

**How it resolved.** The programme did not abandon the automation, which would have been the intuitive response and the wrong one. The escape analysis showed the



automated errors were **systematic and concentrated**: they clustered on the **12 %** of exceptions above a value threshold, where the exception patterns were most varied, the characteristic shape of machine error that Domain 9, KA 9.4.4 describes. A rule-based containment check was placed on that 12 % only, at **USD 2.10** per checked item, i.e. **USD 0.25** per exception averaged across the population, and it cut the automated escape rate to **0.9 %**. The automated unit cost became  $1.15 + 0.252 + 0.009 \times 640 = \text{USD } 7.16$ , against the manual **USD 14.52**: a genuine saving of **USD 7.36** per exception, **USD 70,636.80** a month, with a quality-adjusted breakeven of **23,756** units reached in **2.47 months** and an 18-month net of **USD 1,096,662.40**.

**What the domain teaches here.** Three things, in order of how often they are got wrong. An automation business case computed on **visible unit costs alone** is not a conservative estimate of a good decision; it is a different calculation that can have the opposite sign, and the missing term, the differential escape rate, is measurable in a day against a held-back sample and is almost never measured before approval. Where the quality-adjusted unit cost of the automated route exceeds the manual one, **no volume rescues it**, and scale makes it worse rather than better, which is the opposite of the intuition scale usually supports. And the remedy for machine error is rarely to review the output harder. It is to **place a cheap, targeted containment layer where the errors actually concentrate**, which here converted a value-destroying automation into a strongly positive one for USD 0.25 per exception, because machine errors are systematic and therefore locatable in a way human errors are not.

## Executive perspective — Domain 14

What a programme director cannot delegate in this domain:

- **The consequence figures.** Every calculation here (remediation ranking, verification tiers, automation breakevens, control sizing) is proportional to an assessed cost per escaped error. Own the assessments and their owners, or accept that the whole standard is arbitrary (14.1.4, 14.3.3).
- **The verification standard, published and owned.** A tier table derived from measured *p* and *q*, applied consistently by people who are not recomputing it. Unpublished, it collapses into individual nervousness, which is expensive in both directions at once (14.3.3).
- **That *p* and *q* are measured, dated and re-triggered.** A seeded-error programme with a named owner. An unmeasured error rate freezes the standard and forfeits the entire productivity gain, which arrives as reduced checking rather than faster drafting (14.3.3, 14.A.2).
- **The AI use register, and the two questions it answers.** For every AI-informed output in circulation: *if this were wrong, whose judgement was wrong?*; and *what happens on the day it turns out to be?* No answer to the first means no governance; no answer to the second means the accountable person has a title



and no obligation. Every entry carries a failure route and the decisions to reopen (14.4.3).

- **Disaggregated measurement wherever an output allocates support, priority or resource.** Groups defined in advance, error rates measured by group, the fairness measure chosen explicitly and recorded. Aggregate accuracy conceals exactly the failure that will be raised publicly (14.4.2).
- **The refusal to accept a productivity claim without its verification cost.** A saving on generation is a saving only if verification is cheaper than producing the output correctly first time; and an automation's business case that shows neither maintenance nor a measured escape rate has not been reviewed (14.2.4, Case study B).

## Calculation exercises — Domain 14

**Exercise 14.1** A common data environment holds four classes: **A** 2,500 records at a 5.0 % defect rate, consequence USD 260 per defect; **B** 900 records at 1.0 %, USD 4,800; **C** 1,800 records at 3.0 %, USD 150; **D** 600 records at 8.0 %, USD 90. Compute total defects, weighted mean defect rate and total exposure; rank the classes for remediation; and compute the exposure that a uniform 2 % target would produce. *Solution.* Defects  $125 \cdot 9 \cdot 54 \cdot 48 = 236$  of **5,800** records, a weighted mean rate of **4.0690 %**. Exposures  $32,500 \cdot 43,200 \cdot 8,100 \cdot 4,320 = \text{USD } 88,120$ , of which class **B** alone is **49.02 %** despite having the lowest defect rate and the fewest records. Exposure per record, the remediation ranking, is **B USD 48.00 · A USD 13.00 · D USD 7.20 · C USD 4.50**. A uniform 2 % target gives  $0.02 \times 5,800 = 116$  defects, a 50.8 % reduction in count, at an exposure of  $0.02 \times 5,294,000 = \text{USD } 105,880$ ; **USD 17,760** or **20.15 % worse** than the observed position. *Common error:* ranking remediation by defect rate, which puts class D first (the class carrying **USD 4,320** of total exposure, less than a tenth of class B's), and leaves the 1.0 % class, which carries half the estate's exposure, until last.

**Exercise 14.2** An automation costs USD 62,400 to build and USD 900 a month to maintain over a 24-month horizon. The manual unit cost is USD 31.20 and the automated unit cost USD 3.60. Measured escape rates are 1.0 % manual and 4.0 % automated; an escaped error costs USD 720. Volume is 500 units a month. Compute the naive and quality-adjusted breakeven volumes, decide, then re-decide with a containment layer costing USD 1.80 per unit that cuts the automated escape rate to 1.0 %. *Solution.*  $F = 62,400 + 900 \times 24 = \text{USD } 84,000$ . Naive  $84,000 / (31.20 - 3.60) = 84,000 / 27.60 = 3,043.48 \rightarrow 3,044$  units. Quality-adjusted: manual  $31.20 + 7.20 = 38.40$ ; automated  $3.60 + 28.80 = 32.40$ ;  $84,000 / 6.00 = 14,000$  units, **4.60 times** the naive figure.

Volume over the horizon is  $500 \times 24 = 12,000$ , which is below 14,000, so **as designed the automation should be rejected**, despite a naive breakeven it passes four times over. With containment: automated  $3.60 + 1.80 + 7.20 = \text{USD } 12.60$ , unit saving **USD 25.80**, breakeven  $84,000/25.80 = 3,255.81 \rightarrow 3,256$  units, and a 24-month net of  $12,000 \times 25.80 - 84,000 = \text{USD } 225,600$ . *Common error*: computing the breakeven on visible unit costs only. Here it produces a breakeven the volume clears comfortably for a proposal that in fact destroys value, and it is the error Case study B cost a programme USD 1,560,384 to discover.

**Exercise 14.3** Verification tiers for an output class cost USD 14.00 (detection 0.40), USD 42.00 (0.75) and USD 126.00 (0.92) per item, over tier 0 (accept, detection 0). The measured material-error rate is 0.08. Derive the consequence thresholds; assign tiers to outputs whose escaped errors cost USD 900, USD 4,000 and USD 20,000; and state what happens to the thresholds if the error rate doubles to 0.16. *Solution.*  $u^*_k = \Delta v_k / (p \cdot \Delta q_k)$ . Tier 0  $\rightarrow$  1:  $14 / (0.08 \times 0.40) = \text{USD } 437.50$ . Tier 1  $\rightarrow$  2:  $28 / (0.08 \times 0.35) = \text{USD } 1,000.00$ . Tier 2  $\rightarrow$  3:  $84 / (0.08 \times 0.17) = \text{USD } 6,176.47$ . Assignments: USD 900  $\rightarrow$  **tier 1**; USD 4,000  $\rightarrow$  **tier 2**; USD 20,000  $\rightarrow$  **tier 3**. If  $p$  doubles to 0.16 every threshold halves —  $218.75 \cdot 500.00 \cdot 3,088.24$ , so the USD 900 output moves up to tier 2 and the USD 4,000 output to tier 3: a worse configuration requires deeper checking of the same outputs, which is the same result as 14.3.3's improvement case read backwards. *Common error*: comparing a tier's **total** cost against the **total** consequence rather than the increment of cost against the increment of detection. Doing so here would justify tier 3 for the USD 900 output ( $126 < 900$ ), which over-verifies: the step from tier 2 to tier 3 buys 0.17 of detection for USD 84 and is worth only  $0.08 \times 0.17 \times 900 = \text{USD } 12.24$ .

**Exercise 14.4** Two error types occur on 15 % of cycles each. An obvious error costs USD 620,000 if it escapes and is detected with probability 0.99; a plausible error costs USD 180,000 and is detected with probability 0.30. Compute each expected escaped cost and the asymmetry ratio, then evaluate a reperformance step costing USD 145 per output that raises plausible-error detection to 0.88. *Solution.* Obvious  $0.15 \times 0.01 \times 620,000 = \text{USD } 930$ . Plausible  $0.15 \times 0.70 \times 180,000 = \text{USD } 18,900$ : a ratio of **20.32**, which decomposes as an escape-probability ratio of  $0.70/0.01 = 70.00$  divided by a consequence ratio of  $620,000/180,000 = 3.4444$ . Reperformance saves  $0.15 \times (0.88 - 0.30) \times 180,000 = \text{USD } 15,660$  for **USD 145** (a return of **108 times**), and its breakeven detection improvement is  $145 / (0.15 \times 180,000) = \text{0.5370 percentage points}$ . *Common error*: prioritising review by the size of the number. The obvious error is 3.44 times more consequential and 20.32 times less dangerous, so review effort aimed at magnitude is aimed at the cheap failure mode. And the effective control is reperformance, not scrutiny.

**Exercise 14.5** An incident has an assessed annual probability of 0.12 and an impact of USD 750,000. Control X costs USD 34,000 a year and cuts the probability to 0.03; control Y costs USD 21,000 a year and cuts the impact to USD 400,000. Evaluate all four options, compute the combined avoided loss, and find the probability at which X and Y are equally attractive. *Solution.* Baseline EAL =  $0.12 \times 750,000 = \text{USD } 90,000$ . **Nothing** USD 90,000. **X**  $34,000 + 0.03 \times 750,000 = 34,000 + 22,500 = \text{USD } 56,500$ . **Y**  $21,000 + 0.12 \times 400,000 = 21,000 + 48,000 = \text{USD } 69,000$ . **X and Y**  $55,000 + 0.03 \times 400,000 = 55,000 + 12,000 = \text{USD } 67,000$ . X alone is best; buying both is **USD 10,500 worse** than X alone. Combined avoided loss is  $90,000 - 12,000 = \text{USD } 78,000$ , not the naive  $67,500 + 42,000 = \text{USD } 109,500$ : an overstatement of USD 31,500. X and Y are equal at  $(34,000 - 21,000) / (400,000 - 187,500) = 6.1176\%$ , so at the assessed 12 % X is robustly preferred, unlike the marginal case in 14.4.4. *Common error:* adding the two controls' avoided losses. Reductions compound rather than accumulate, and the naive sum here overstates the combined benefit by 40 % and makes a package that is worse than one control alone look like the best option available.

## Practitioner's toolkit – Domain 14

*Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.*

### Toolkit 14.T.1 – AI use register with its verification standard

Two linked tables on two pages, owned by one named person and reviewed at every gate. **The register**, one row per use: use and output class · decision informed · accountable human · assessed consequence per escaped error  $u$  · measured  $p$  and  $q$  with method, sample size and date · verification tier applied · model, prompt and grounding versions · data and confidentiality boundary · prohibited adjacent uses · **failure route, who is notified and within what period** · **decisions to reopen on failure**, with the materiality threshold set in advance · re-measurement trigger and next review date. **The standard**, one row per tier: tier · description · cost per item  $v$  · measured detection rate  $q$  · the derived threshold  $u^* = \Delta v / (p \cdot \Delta q)$  · who may perform it. Three integrity checks run monthly, each a count: uses whose measurements are past their re-measurement trigger, outputs in circulation with no register row, and **register rows with no failure route** (14.4.3). The register's value is that it makes the question *whose judgement was wrong?* answerable in advance rather than after an incident.

### Toolkit 14.T.2 – Data class schedule and quality exposure sheet

One row per data class: class · records  $n_i$  · owner (a named person) · written definition with units and permitted values · source of record · defined pass criteria ·

measured defect rate  $d_i$  with sample size and date · assessed consequence per defect  $u_i$  with its basis · **exposure**  $n_i d_i u_i$  · **exposure per record**  $d_i u_i$  · target rate for *this* class · remediation cost per record and its breakeven. Sort by exposure per record, not by defect rate. A footer carries the three totals a board needs: records, defects, total exposure — plus the exposure a uniform target would produce, which is the number that stops a uniform target being adopted by default. Where a class carries a non-compensable consequence, the target cell records an absolute standard and the arithmetic is used only to size the effort.

### Toolkit 14.T.3 — Quality-adjusted digital business case sheet

One page per automation, analytics product, dashboard or twin, and it refuses to be completed dishonestly. *Scope*: the decision or task, and the **organisational boundary** at which volume is counted (project, portfolio, enterprise). *Costs*: build · maintenance per period × horizon · residual human handling per unit  $a$ . *Manual comparison*: unit time and rate  $m$ . *Quality*: measured escape rate manual  $e_m$  and automated  $e_a$ , with method, sample size and date · assessed cost per escaped error  $u$ . *Results*: naive breakeven  $F/(m - a)$  · **quality-adjusted breakeven**  $F/[(m + e_m u) - (a + e_a u)]$  · volume at the stated boundary · net over the horizon · and, for dashboards, the information age  $R/2 + G$  against the decision window. *Decision*: recommendation, named owner of the tool as an asset, re-measurement trigger, and retirement condition. A sheet returning "no breakeven exists" is a valid and valuable output, and the sheet is designed so that this answer is reached before approval rather than after eighteen months.

## Exam preparation — Domain 14

**What is assessed.** The digital delivery environment and its interface arithmetic; data governance as ownership, definition, source of record, quality standard and access; the common data environment and its five guarantees; **defect rate by class, consequence-weighted exposure, exposure per record and the uniform-target defect**; dashboards tested against a decision, with **information age**  $R/2 + G$ ; the analytics ladder and its ordering rule; digital twins and **breakeven fidelity**; **automation breakeven volume, naive and quality-adjusted**; the five permitted and four prohibited classes of AI use and the single test that generates the prohibitions; prompting as a method with grounding, constraint, refusal permission and provenance; **the derivation of the verification standard**  $u^*_k = \Delta v_k / (p \cdot \Delta q_k)$  and the pricing of tiered against uniform regimes; **the escape-cost asymmetry between plausible and obvious error** and reperformance as its control; explainability as a decision requirement at three levels; **differential error by group** and the base-rate constraint; the AI use register; and **security control economics with sub-additivity**.

**The calculations to be able to do under time pressure.** Class exposure  $n_i d_i u_i$ , total exposure, weighted mean defect rate, exposure per record, and the exposure a uniform rate target would produce. Remediation net value and its breakeven cost per record. Information age  $R/2 + G$  priced at a cost of delay per day, and the two levers' asymmetry. Breakeven fidelity for a twin. Naive and quality-adjusted automation breakeven volumes, and the recognition that no breakeven exists. Verification thresholds  $\Delta v / (p \cdot \Delta q)$ , tier assignment from a consequence, total cost of a regime as verification plus expected escaped loss, and the effect on thresholds of a change in  $p$ . Expected escaped cost  $p(1 - q)u$  and the asymmetry ratio. Recall by group and the economics of correcting a differential.  $EAL$ , residual  $EAL$  after a control, combined  $EAL$  for two controls, and a crossover probability.

**The traps.** Applying a uniform data-quality target across classes of unequal consequence, which can cut defect counts while raising expected cost (Exercise 14.1, Case study A) · ranking remediation by defect rate rather than by exposure per record (14.1.4) · computing an automation breakeven on visible unit costs, omitting maintenance and the differential escape rate (Exercise 14.2, Case study B) · assuming a breakeven volume exists when the quality-adjusted saving is negative (14.2.4) · taking information age as the production lag alone or as half the period alone (MCQ 14.2-A) · comparing a verification tier's total cost against the total consequence instead of increment against increment (Exercise 14.3) · forgetting that verification thresholds scale with  $1/p$ , so a better configuration raises them (MCQ 14.3-C) · prioritising review by the magnitude of a number rather than by its plausibility (Exercise 14.4) · treating a model's self-reported confidence as an error or detection rate (14.3.3) · reading an aggregate accuracy figure as evidence of no differential error (14.4.2) · adding two security controls' avoided losses (Exercise 14.5) · and applying expected-value reasoning to a consequence that is not cost-compensable (14.1.4, 14.3.3, 14.4.4).

**How the domain connects.** Domain 1 supplies the accountability principle every prohibition here rests on, and the cost of delay at which staleness, verification and automation are all priced. Domain 3 supplies the  $E[wait]$  identity this domain applies to information, the delegation logic that becomes system permissions, and the four AI governance instruments this domain makes operational. Domain 4 supplies the interface arithmetic for the systems estate. Domain 5 supplies the testability requirement that data pass criteria depend on. Domain 7 supplies the blended rates and the  $CPI$ -based forecast that 14.3.4 corrupts two ways. Domain 9 supplies the sampling arithmetic, the composite data fitness, the independence rule for layers and the containment-chain result this domain builds on rather than repeats. Domain 10 supplies the strictest explainability case in the volume. Domain 11 supplies the reporting standard the information-age calculation belongs to. Domain 12 supplies the prohibitions on inference about people, which this domain does not soften. Domain 13 supplies the delivery cadence that a dashboard's information age must keep up with. Domain 15 owns the portfolio boundary at which most automation cases actually pay. And Domain 16 owns the retention of

models, prompts, data and verification records after the project that created them has gone.

## Domain 14 summary

Thirteen domains each drew a local line around AI use. This domain makes those lines one system, and its instrument is arithmetic rather than principle; because a principle cannot tell you how deeply to check, and arithmetic can.

The substrate comes first. Meridian's common data environment holds **14,100** records in six classes with **332** defects, a weighted mean rate of **2.3546 %**, carrying **USD 172,820** of consequence-weighted exposure, distributed nothing like the defect rates: the lowest-rate class in the estate (0.8 %) is the second-highest by exposure per record at **USD 28.00**, behind interface mappings at **USD 72.00**. A uniform **2 %** target produces **282** defects, **15.06 % fewer**, at **USD 207,440 (20.03 % worse)**, which is the whole case against rate targets in one comparison. The consequence-proportional schedule that replaced it spent **USD 43,800** to remove **USD 96,000** of exposure and landed at a **1.9504 %** mean rate: it passed the original target while costing **62.97 %** less than the target's own remedy.

What is built on the substrate must be tested against a decision and a volume. A fact on a monthly report with a 9-day production lag is **24 days** old ( $R/2 + G$ , Domain 3's identity applied to information) worth **USD 48,960** at Meridian's cost of delay, against **7.5 days** and **USD 15,300** on a weekly cycle with a 4-day lag; and the production lag is twice the lever the reporting period is, day for day. Auriga's control-system twin costs **USD 211,500** against **USD 306,000** of benefit at perfect fidelity, so it needs **69.12 %** fidelity and has **85.00 %**: a positive **USD 48,600** on a margin of under sixteen points measured on forty cases. Auriga's data-check automation breaks even at **959** checks on build cost, **1,659** with maintenance and **2,654** once a one-point difference in escape rates is priced, **1.60 times** the naive figure, which rejects it for a 720-check project and approves it for an 8,640-check portfolio. Case study B's payments programme shows the other end: an automation reporting **USD 79,200** a month of saving and destroying **USD 86,688** a month, with no breakeven volume at any scale, rescued by a containment layer costing **USD 0.25** per exception.

The core is the verification standard Domain 3 asserted and this domain derives.  $u_k^* = \Delta v_k / (p \cdot \Delta q_k)$  gives Meridian's thresholds as **USD 261.90 · 785.71 · 3,208.33 · 18,333.33**, assigning its five output classes to tiers 0 to 4 at a monthly total of **USD 36,950.10**: verification **USD 12,650** plus expected escaped loss **USD 24,300.10**. Uniform tier 2, the intuitively consistent policy, spends **3.27 times** as much on verification to cut escaped loss by **10.41 %** and costs **USD 63,175.36**; uniform tier 4, the intuitively safe policy, costs **USD 260,952.14**, **7.06 times** the tiered standard; doing nothing costs **USD 72,571.20**. **45.93 %** of the tiered standard's escaped loss sits in the class deliberately left unverified, and checking that class would lose **USD 2,914**: the



correct remedy for a large aggregate of cheap errors is to reduce *p* or *u*, not to check. And when the measured error rate falls from 0.12 to 0.04, thresholds triple and the total falls **56.60 %** while the verification bill falls **72.09 %**: **the return on a better configuration arrives as less checking**, which is the honest form of the productivity claim and is unavailable to anyone who does not measure *p*.

Then the asymmetry that determines where checking should point. On Auriga's week-13 numbers a plausible wrong forecast (**USD 4,320,000** against the correct **USD 4,416,666.67**, landing on the wrong side of a **USD 350,000** funding trigger) carries an expected escaped cost of **USD 20,250**, while a 10× decimal slip with **3.33 times** the consequence carries **USD 1,800**: the plausible error is **11.25 times** more dangerous, and would be **37.50 times** at equal consequence, because expected damage is consequence times the probability of *not* being caught. Reperformance from the inputs raises detection from 0.25 to 0.85 for **USD 87.08** and returns **186 times** its cost, with a breakeven of **0.3225** percentage points of detection. Rounding *CPI* to 0.91 before dividing manufactures a plausible error of **USD 21,062.27** (5.05 % of the variance) from an operation that looks like tidiness.

The obligations that arithmetic cannot discharge close the domain. Explainability is a property of the decision, in the vocabulary of the obligation, at one of three levels. Differential error is invisible in aggregate: Meridian's **70.00 %** recall concealed **88.89 %** urban and **54.55 %** rural, a **34.34**-point gap, with rural clinics carrying **83.33 %** of the excess cost while being **40.00 %** of the estate; and correcting it returned **4.74 times** its cost, so the differential was a defect and not a price. Because the base rates differ (**37.50 %** against **68.75 %**), not every error measure can be equalised at once, and choosing which is a named person's judgement. Security controls are sub-additive: Meridian's two options total **USD 35,600** and **USD 34,800** alone and **USD 45,200** together, so buying both is worse than buying either, the naive sum of avoided losses overstates by **USD 13,200**, and the choice between them flips at an incident probability of **8.57 %** that nobody can estimate to that precision, which is the honest answer, not a preference. And where a consequence is not cost-compensable, every expected-value test in this domain stops applying and the control becomes a requirement to be sized rather than an investment to be justified.

The through-line: **AI proposes; the professional verifies, decides and remains accountable, and this domain is what makes the middle verb affordable**. Verification without arithmetic is either theatre or negligence, and usually both at once in the same organisation: excessive on the outputs that do not matter, absent on the ones that do. Compute the consequence, derive the tier, measure the rates, reperform anything near a threshold, and keep a register that can answer the only question that finally matters (*if this were wrong, whose judgement was wrong?*) together with the question that follows it on the day the answer is needed: *and what do we do now?* Every entry names a failure route: stop and suspend, enumerate the decisions that cited the output, notify the same day, reopen above a pre-set threshold with the authority that decided, record whether anyone outside must be told and on whose determination, and re-measure before resuming.

DOMAIN

# 15

## Programmes, Portfolios and Enterprise Delivery

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## Why this domain exists

Every domain before this one has, quite deliberately, held one thing constant: the boundary of the thing being led. Domain 4 integrated the parts of a project into a whole. Domain 6 scheduled that whole. Domain 8 put ranges round it. Domain 13 chose the delivery method for it. Domain 14 gave it a data and AI estate. In all of them the project had an outside, and the outside was assumed to cooperate.

It does not. The outside is where the other twelve initiatives live, where the same six integration engineers are already committed, where four business cases have each independently promised the same administrative post reduction, and where the decision the project needs will pass through tiers of governance that no one has ever added up. **The central claim of this domain is that a portfolio is not a bigger project, because the arithmetic that governs it is different in kind: at project scale quantities add, and at enterprise scale they multiply, compete and double-count.** A leader who carries project intuitions upward will be confidently wrong in four specific ways, each of which this domain computes. Milestone confidence multiplies down rather than averaging out. Capacity that is sufficient in aggregate is routinely infeasible period by period. Benefits summed across components overstate what the enterprise can bank. And governance latency accumulates tier by tier into a bill that appears in no budget line.

The domain proceeds outward. KA 15.1 builds programme architecture and shows that a programme milestone with several individually likely predecessors is not likely at all (the same multiplicative penalty Domain 8 found at a schedule merge point, now applied to whole components), and identifies **decoupling**, not better estimating, as the lever that pays. KA 15.2 handles portfolio balancing: the benefits register with its double counts eliminated, and allocation under a constraint that binds *per period*, which is where the ranking heuristic Domain 2 warned about fails expensively. KA 15.3 treats enterprise capacity as the real constraint it is, applies Domain 13's flow arithmetic to a portfolio's work in progress, prices protective capacity honestly (it does not always pay), and subjects the enterprise PMO to the same value test it applies to everyone else. KA 15.4 governs transformation: enterprise decision latency summed across tiers using Domain 3's formula, and strategic reporting, where the commonest portfolio report in circulation aggregates performance in a way that reverses its sign.

**Learning objectives.** After this domain a candidate can: distinguish a project, a programme and a portfolio by the decisions each exists to make rather than by size; design a programme architecture in components, tranches and a target operating state, and state what each tranche buys; **compute the probability that a programme milestone is met from the on-time probabilities of its independent predecessors, and show how quickly that collapses as predecessors accumulate**; derive the per-dependency reliability a stated milestone confidence demands,

and use it to demonstrate when better estimating cannot possibly deliver the target; rank the three structural levers (reduce the count, decouple, buffer), and price each; build a portfolio benefits bridge that eliminates shared, same-pool, cascade and already-committed double counts, and show what elimination does to payback; **classify every registered benefit by fund type, name the finance owner who confirms cashability in writing, and disclose the composition of a stream before a cash test is applied to it; allocate a portfolio under a multi-period capacity constraint, demonstrate that aggregate feasibility does not imply period feasibility, and quantify what a ranking heuristic leaves on the table;** apply Domain 13's flow arithmetic to portfolio work in progress and price the benefit deferral that excess concurrency causes; compute the survival probability of a capacity plan and the breakeven failure cost at which protective capacity becomes worth its forgone value; state an enterprise PMO's recurring value test and distinguish it from one-off gains; **sum enterprise decision latency across programme and portfolio tiers, price it at a cost of delay, and rank the available redesigns;** aggregate portfolio performance correctly and detect the unweighted-average trap; and govern AI-assisted portfolio analysis without letting a model select a portfolio.

**The master programme, one tier up.** Meridian Care Records continues from Domains 1 to 4: the clinical-records rollout to **40 clinics**, approved cost **USD 2,400,000**, full-potential benefit **USD 979,200** a year, realistic benefit **USD 685,440** a year at 70 % adoption, and the cost of delay of **USD 14,280 per week** derived in Domain 1. Domain 3 established Meridian's steering-body latency at  $E[\text{wait}] = M/2 + L = 4/2 + 2 = 4.0 \text{ weeks}$ . This domain places Meridian inside the health group's **five-component portfolio** (records, referrals interoperability, clinician scheduling, analytics platform, pharmacy stock automation) with a combined approved cost of **USD 4,880,000**, and inside a **five-tier enterprise decision architecture**. Every figure in the chapter descends from those.

**Reference points.** Two international documents are the usual reference points for the material of this domain: **ISO 21503**, which addresses guidance on programme management, and **ISO 21504**, which addresses guidance on portfolio management — both within the **ISO 21500** family Domain 1 introduced, and both sitting alongside **ISO 21505** on governance, which Domain 3 names. All are voluntary guidance describing practice: none is legislation, none is a certifiable requirement, and none obliges anyone of itself unless an organisation, a contract or a regulator adopts it. They are named here and not reproduced (no clause, table or process model from any of them appears in this volume, and every method below is derived rather than cited), and a reader who wants any of them should obtain the current edition from its publisher. Naming them implies no endorsement in either direction.

## Programme architecture and dependency management

*Topics: 15.1.1 what a programme is, and is not · 15.1.2 architecture, components, tranches and the target state · 15.1.3 dependency arithmetic at programme scale · 15.1.4 decoupling as the primary lever.*

### 15.1.1 What a programme is, and is not

**Definition.** A programme is a temporary organisation created to deliver a **coherent change in outcomes** through a set of components (projects and enabling activities) that cannot deliver the outcome individually. Its unit of decision is the **component set**: what to start, what to stop, what to sequence, and what to fund next.

The definition that matters is comparative, and the comparison is by decision rather than by scale. A **project** decides how to produce a defined output within a mandate. A **programme** decides which components produce the outcome, and repeatedly re-decides it as evidence arrives. A **portfolio** decides which programmes and projects the organisation should be doing at all, given a finite capacity and a strategy, Domain 2's territory, revisited each cycle rather than once.

Three consequences follow, and each is a test a leader can apply on Monday morning.

**A large project is not a programme.** If the components cannot be usefully re-sequenced or stopped independently, the programme layer is adding governance without adding optionality, and the workable structure is one project with a good work breakdown structure (Domain 4, KA 4.2.1). The diagnostic question is direct: *name a component this programme could stop next quarter without abandoning the outcome.* If there is none, dissolve the layer.

**A collection of projects is not a programme.** If the components deliver independent outcomes, the programme layer is a reporting convenience, and the honest structure is a portfolio grouping. The cost of the pretence is real: components acquire a false interdependence, and each waits for programme-level decisions it did not need.

**A programme's benefit is not the sum of its components' benefits.** This is the single most consequential difference and KA 15.2 computes it. Component business cases are written to be approved, and they are written separately, so they claim overlapping benefits as though they were additive. At programme and portfolio level that claim has to be reconciled against reality, and the reconciliation reliably removes a double-digit percentage of the total.

### 15.1.2 Architecture — components, tranches and the target state

**The target operating state.** A programme needs a stated description of the organisation as it will be when the change has landed: the processes, roles, systems, data, locations and measures. Without it the programme has an output list and no test of coherence, and every component optimises locally. The description is a governance artefact, not a design document: its purpose is to make the question "does this component still contribute?" answerable at a gate.

**Components.** Each component carries a named owner, an output, a contribution to the target state, its dependencies in and out, and (this is the part usually missing) a statement of what happens to the outcome if the component is cancelled. A component whose cancellation has no describable consequence is not part of the programme.

**Tranches.** A tranche is a group of components delivered together to reach a **usable intermediate state**: one at which benefit begins and at which the programme could stop with something of value already in operation. Tranche design is therefore the programme's principal option-creating decision, and it is governed by one rule: **a tranche boundary belongs where the organisation can genuinely operate**. A boundary placed for reporting convenience produces the worst of both worlds, a gate with nothing to decide and a partial state nobody can use.

Meridian's tranches illustrate the rule. Tranche 1 delivers the records application and the national patient-index interface to 10 clinics in one region, a genuinely usable state, because those ten clinics can operate on the new record and the remaining thirty continue as they are. Tranche 2 adds the second and third regions and the migration of historical records. Tranche 3 completes the fourth region and retires the legacy system, which is the only irreversible step in the programme and therefore the only place a hard gate is warranted (Domain 3, KA 3.3.1).

**Programme-level dependencies.** Components depend on each other, and they also depend on things outside the programme: an estate refurbishment, a regulatory approval, a supplier's product release, another programme's platform. A programme dependency register records, per dependency, the giver, the receiver, what is needed, the date needed, the date promised, the owner on both sides, and the consequence of a breach. Domain 2's assumption and dependency management (KA 2.3.4) supplies the discipline at project level; the programme addition is that dependencies now cross accountability boundaries, so the register's most important column is *the owner on the giving side*, and a dependency with no owner there is a hope, not a dependency.

### 15.1.3 Dependency arithmetic at programme scale

Here the domain earns its quantitative keep, and the result is one every programme leader should be able to produce on a whiteboard.

**The multiplication rule.** A milestone that requires **all** of **k** independent predecessors to be met is met only if every one of them is met. Its probability is therefore the **product** of theirs, not their average and emphatically not their minimum:

$$P(\text{milestone}) = p_1 \times p_2 \times \dots \times p_k$$

Domain 8 established the schedule form of this at a merge point (8.A.2: two paths each 80 % likely give the merged event 64 %, which is why deterministic critical-path analysis is systematically optimistic at convergence). The programme form is the same theorem applied to whole components rather than to activity paths, and it is more dangerous, because at programme level the predecessors are owned by different people, each of whom reports honestly on their own item and none of whom is looking at the product.

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#### WORKED EXAMPLE

##### Worked example 15.1.3 — Meridian's Region A go-live, and the four-region programme milestone.

1. **Setup.** Meridian's Region A go-live requires six independent predecessors, each assessed by its own owner at the following probability of being met on the committed date: records application release **0.90**; national patient-index interface **0.88**; data migration and reconciliation **0.85**; clinic estate and network works **0.92**; clinician training and competency **0.95**; information-governance approval **0.90**. Every one of them is "on track" by the programme's own reporting convention. The programme milestone committed externally is the **simultaneous** go-live of all four regions, each with its own instance of the same six-dependency structure: 24 dependencies in total. Independence is assumed and is discussed below.
2. **Formula.**  $P(\text{milestone}) = \prod p_i$  over the predecessors. For the four-region milestone,  $P = (\prod p_i)^4$ , the four regions being structurally identical and assumed independent.
3. **Substitution.**  $0.90 \times 0.88 \times 0.85 \times 0.92 \times 0.95 \times 0.90$ ; then that result raised to the fourth power.
4. **Result.** Region A: **0.52953912**, that is **52.95 %**. The four-region programme milestone: **7.86 %**. Expected number of regions on time:  $4 \times 0.52953912 = 2.1182$ , so **1.8818** regions are expected to be late.
5. **Interpretation.** Six components, none worse than 0.85, produce a coin flip; four such regions produce a milestone that is **92.1 % likely to be missed** while every single component is reported as on track. That gap between component reporting and milestone probability is the defining pathology of programme delivery, and it is arithmetic rather than culture. Three consequences a leader should draw. First, **the product is the number to report**: a programme status report that lists twenty-four green components and a committed date is

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internally inconsistent, and the fix is one multiplication. Second, **better estimating cannot rescue the design**. To reach an 80 % confidence on the four-region milestone every one of the 24 dependencies would have to be met with probability  $0.80^{(1/24)} = 99.07\%$ : a standard no estate contractor, training programme or regulator has ever offered, and one the programme has no mechanism to enforce. Even across the six dependencies of a single region, 80 % confidence needs **96.35 %** each. Third, and following directly, **the lever is structural**: change  $k$ , not the  $p_i$ . To see why, note the sensitivity. Improving any single dependency by an absolute 0.03 moves the Region A probability by between **1.67** percentage points (training,  $0.95 \rightarrow 0.98$ ) and **1.87** points (migration,  $0.85 \rightarrow 0.88$ ). The improvement is worth marginally most where the probability is lowest, but the whole range is under two points, and there is no combination of plausible estimating improvements that reaches 80 %. Removing one dependency from the milestone, by contrast, multiplies the probability by  $1/p_i$ , which for the 0.85 migration dependency is a gain of **9.34** points in one decision. **Two professional cautions**. The independence assumption is doing real work here and is usually optimistic: shared causes (one supplier, one scarce team, one regulator, one budget round) correlate the predecessors, and Domain 8's KA 8.A.1 treatment of common drivers explains why a register with many entries and few drivers is concentrated rather than diversified. Correlation makes the *product* less pessimistic than shown for the good outcome and makes clustered failure far more likely, so the honest statement is that **52.95 % is a planning figure derived under an explicit assumption, and the assumption belongs in the report next to the number**. And the  $p_i$  are owner assessments, not measurements; their calibration is a data-quality question (Domain 9, and Domain 14's KA 14.1 on defect rates by class), which is why the register records who assessed each one and on what evidence.

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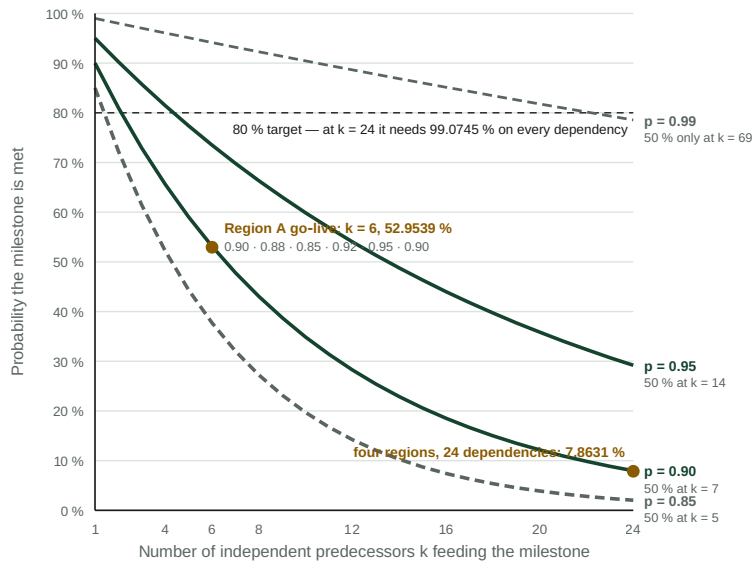


Fig 15.1.1 — Why a programme milestone becomes improbable

#### 15.1.4 Decoupling as the primary lever

If the milestone probability is a product, the only interventions that materially change it are interventions on the *structure* of the product. Three exist, in descending order of value per unit of effort.

**Reduce the count.** Ask of every predecessor whether the milestone genuinely requires it. Many entries are there because someone drew an arrow, not because the outcome depends on the item. Each removal multiplies the probability by  $1/p_i$ , and the removals compound.

**Decouple.** Where a predecessor is genuinely needed but not needed *by that date*, move it off the milestone by designing an operable interim arrangement. Meridian did this twice. Historical **data migration and reconciliation** (0.85) was removed from go-live by operating a documented two-week paper-fallback for historical lookups, at a priced cost of **USD 18,000** in temporary clerical cover, taking Region A from 52.95 % to **62.30 %**. **Information-governance approval** (0.90) was obtained once at programme level for all four regions rather than four times regionally, removing it from each regional milestone and taking Region A to **69.22 %**. The four-region milestone rose from **7.86 %** to **22.96 %**, and the expected number of late regions fell from **1.8818** to **1.2312**.

**Buffer.** Where neither reduction nor decoupling is available, the honest response is to commit to a date that includes a buffer, sized by Domain 8's quantitative methods, and to hold internal dates that are tighter. What is not available is committing to the unbuffered date and calling the components green.

**Why the cost comparison settles the argument.** Programmes under milestone pressure reliably reach for money instead of structure, and the two can be priced



against each other. Meridian's alternative proposal was **USD 240,000** of recovery spend (additional trainers and an accelerated estate package) assessed to raise training from 0.95 to 0.98 and estate works from 0.92 to 0.96. That raises Region A to **57.00 %** and the four-region milestone to **10.56 %**: a gain of **2.69** percentage points on the committed milestone for 240,000, or **USD 89,093** per percentage point. The structural changes gained **15.10** points for 18,000: **USD 1,192** per point, or **1.34 %** of the cost of the money route, a ratio of **74.7 to one**. The general result is worth stating plainly because it is counter-intuitive to sponsors: **on a milestone whose probability is a product, structure is roughly two orders of magnitude cheaper than effort**. Money buys movement in one factor; decoupling removes a factor.

AI in this KA

**Where it earns its place.** Building and reconciling a programme dependency register from component plans, contracts and interface documents, and listing dependencies with no owner on the giving side, no date, or a promised date later than the date needed: a document-comparison task that is tedious, high-volume and has a checkable right answer. Computing the milestone product across hundreds of dependency configurations, and searching for the smallest set of removals that reaches a target confidence, which is a deterministic combinatorial problem and exactly the sort of thing to delegate. Detecting shared causes across component registers by clustering dependency descriptions, which surfaces the correlation that invalidates the independence assumption. Drafting the interim operating arrangement options for a decoupling decision, for human evaluation.

**Where it must not go.** Assessing the **p<sub>1</sub>**. A model asked how likely an estate contractor is to meet a date will produce a confident number with no provenance, and that number will then be multiplied into a board commitment. Probabilities come from owners with evidence, or from calibrated history, and their source is recorded. Nor may a model decide a decoupling: an interim paper-fallback in a clinical setting is a patient-safety decision belonging to clinical authority, and the domain's arithmetic informs it without ever making it.

**Verification, concretely.** Reproduce the product by hand: six multiplications, and a leader quoting 52.95 % who cannot do so should not be quoting it. Check every AI-extracted dependency against its source document on a stated sample, and confirm each flagged defect with a human before reporting it as a finding, per Domain 14's verification tier for outputs of this consequence (KA 14.3.3). State the independence assumption every time the product is printed.

KEY TERMS — KA 15.1

| TERM      | MEANING                                                                                                        |
|-----------|----------------------------------------------------------------------------------------------------------------|
| Programme | A temporary organisation delivering a coherent outcome through components that cannot deliver it individually. |



| TERM                                       | MEANING                                                                                                                             |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Portfolio</b>                           | The set of programmes and projects an organisation chooses to run within a finite capacity and a strategy.                          |
| <b>Target operating state</b>              | The stated description of the organisation once the change has landed; the test of component coherence.                             |
| <b>Component</b>                           | A project or enabling activity within a programme, with a named owner and a stated consequence of cancellation.                     |
| <b>Tranche</b>                             | A group of components delivered together to reach a usable intermediate state at which benefit begins.                              |
| <b>Programme dependency register</b>       | The record of cross-boundary dependencies, whose critical column is the owner on the giving side.                                   |
| <b>Multiplication rule</b>                 | The probability a milestone requiring all of $k$ independent predecessors is met is the product of their probabilities.             |
| <b>Decoupling</b>                          | Removing a predecessor from a milestone by designing an operable interim arrangement, multiplying the probability by $1/p_i$ .      |
| <b>Required per-dependency reliability</b> | The uniform probability each of $k$ dependencies must carry to reach a target milestone confidence: the $k$ -th root of the target. |

## SAMPLE MCQs — KA 15.1

**MCQ 15.1-A** [15.1.3 · Application] A programme milestone requires four independent predecessors, assessed at 0.95, 0.90, 0.90 and 0.85. The probability the milestone is met is closest to:

- A. 0.90
- B. 0.85
- C. 0.65 CORRECT
- D. 0.60

*Rationale:*  $0.95 \times 0.90 \times 0.90 \times 0.85 = 0.654075$ , so 0.65 (15.1.3). A is the arithmetic mean of the four, the commonest error; B is the minimum, the second commonest; D subtracts the summed shortfalls from one ( $1 - 0.40$ ), which treats the shortfalls as additive.

**MCQ 15.1-B** [15.1.3 · Analysis] Meridian's Region A go-live has six independent predecessors giving a milestone probability of 52.95 %. Four structurally identical regions must go live simultaneously. The programme milestone probability is:

- A. 52.95 %, because the regions are identical
- B. 13.24 %, being 52.95 % divided by four
- C. 7.86 % CORRECT
- D. 28.04 %

*Rationale:*  $0.52953912^4 = 7.86\%$  (15.1.3). A ignores that all four must succeed; B divides instead of taking the fourth power; D squares instead of raising to the fourth power ( $0.52953912^2 = 28.04\%$ ), the error of treating four regions as two.

**MCQ 15.1-C** [15.1.3 · Evaluation] A programme needs 80 % confidence in a milestone that depends on 24 independent predecessors. The required probability on each is closest to:

- A. 80.0 %
- B. 96.3 %
- C. 99.1 % CORRECT
- D. 99.9 %

*Rationale:*  $0.80^{(1/24)} = 99.0745\%$  (15.1.3). A applies the target to each component; B is the answer for six dependencies, not 24; D over-corrects. The teaching point is that C is unattainable, which is why the lever is structural.

**MCQ 15.1-D** [15.1.4 · Evaluation] Removing the 0.85 data-migration dependency from Meridian's regional go-live milestone raises the regional probability from 52.95 % to 62.30 %. The mechanism is best described as:

- A. improving the migration team's performance
- B. multiplying the remaining product by  $1/0.85$  CORRECT
- C. adding 0.85 of buffer to the milestone
- D. averaging the remaining five probabilities

*Rationale:* Decoupling divides the product by the removed factor, which is the same as multiplying by its reciprocal,  $0.52953912/0.85 = 0.6229872$  (15.1.4). A describes the money route, which bought 2.69 points on the programme milestone for 240,000; C and D misstate the arithmetic.

**MCQ 15.1-E** [15.1.2 · Analysis] The soundest test that a proposed tranche boundary is real is whether:

- A. it falls at the end of a financial quarter
- B. the organisation can genuinely operate in the intermediate state CORRECT
- C. a governance body is available to meet on that date
- D. it divides the components into groups of similar size

*Rationale:* A tranche exists to reach a usable intermediate state at which benefit begins and the programme could stop with value in operation (15.1.2). A, C and D are reporting conveniences that produce gates with nothing to decide.

SELF-CHECK — KA 15.1

1. *State the multiplication rule and the two wrong answers it displaces.* — The probability a milestone requiring all of  $k$  independent predecessors is met is the

product of their probabilities; the wrong answers are the average and the minimum.

2. *Why can better estimating not deliver an 80 % four-region milestone at Meridian?*  
— It would require 99.07 % on each of 24 dependencies, which no supplier, trainer or regulator offers and the programme cannot enforce; the lever is to reduce or decouple dependencies.
3. *What distinguishes a programme from a large project?* — A programme's components can be usefully re-sequenced or stopped independently; if none can, the programme layer adds governance without optionality.

## KNOWLEDGE AREA 15.2

# Benefits and portfolio balancing

*Topics: 15.2.1 the portfolio benefits register and the four double counts · 15.2.2 balancing under a binding constraint · 15.2.3 the multi-period constraint and the aggregate-capacity illusion · 15.2.4 rebalancing, staging and stopping.*

## 15.2.1 The portfolio benefits register and the four double counts

**Definition.** A portfolio benefits register is the single authoritative statement of what the portfolio will deliver, by benefit rather than by component, with each benefit carrying a measure, a baseline, a receiving-organisation owner, a **fund-type classification** and the components that contribute to it. It is organised by benefit precisely so that two components contributing to one benefit cannot each claim it.

**The classification is mandatory, and it is the field portfolios most often omit.** Every benefit is recorded as exactly one of four types, because a portfolio total that mixes them silently cannot be tested against any investment rule:

| TYPE                     | WHAT IT MEANS                                                                                                                                                                     | WHAT IT CAN BE COMPARED AGAINST                                             |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>Cashable</b>          | A budget line is removed or reduced, and the money is available to spend on something else. The <b>budget line is named</b> in the register.                                      | A cash outlay (payback, cash NPV, funding cases)                            |
| <b>Capacity-released</b> | Time, sessions, space or licences are freed. Nothing leaves anyone's budget; the value is realised only if the freed capacity is redeployed to something the organisation wanted. | Other capacity, or a cash figure only once redeployment is agreed and named |
| <b>Cost-avoidance</b>    | A cost that would otherwise have been incurred is not incurred. Real, and not a saving against today's budget.                                                                    | A future outlay, on a stated counterfactual                                 |

| TYPE                 | WHAT IT MEANS                                                      | WHAT IT CAN BE COMPARED AGAINST           |
|----------------------|--------------------------------------------------------------------|-------------------------------------------|
| <b>Non-financial</b> | Quality, safety, experience, compliance posture, decision quality. | Its own measure, reported in its own unit |

**And the decision right, which is the part that makes the classification more than a label.** The **finance owner of the affected budget** (not the portfolio board, and emphatically not the delivering component) confirms cashability **in writing**, naming the budget line and the period from which it comes off. Until that confirmation exists, **an unconfirmed cashable claim is recorded as capacity-released**. The reason is not bureaucratic: a component cannot remove money from a budget it does not hold, and the commonest way a portfolio ends up with an approval finance cannot honour is that nobody with the authority to remove the budget was ever asked. Domain 2, KA 2.3.2's cash-releasing/non-cash-releasing distinction is the project-level version of the same test; this is it applied where the money is actually held.

Component business cases, written separately to be approved, overstate the portfolio's total for four distinct and separately detectable reasons. Naming them matters because each has a different correction.

**The shared benefit.** Two components claim the same saving because both genuinely contribute to it. Neither case is dishonest; the sum is wrong. Correction: count the benefit once, in the register, with both components recorded as contributors.

**The same-pool over-claim.** Components claim releases from a pool that cannot supply them all: most often headcount, but equally floor space, licences or clinical session time. Correction: cap the aggregate claim at the pool's size and allocate the cap.

**The cascade, or enabler, double count.** A platform or data component claims a share of the benefits that flow through it, which the consuming components have already claimed. Correction: an enabler's own benefit is only what does not pass through another component's claim; the rest is recorded as an enabling contribution with no independent value.

**The already-committed benefit.** A component claims against a baseline that already assumes a saving committed elsewhere, an operational efficiency target, a prior programme's residual, a contractual price reduction. Correction: reset the baseline to the committed position.

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#### WORKED EXAMPLE

**Worked example 15.2.1 – the Meridian portfolio benefits bridge.**

1. **Setup.** Five components, each claiming annual benefit at its own **full potential**: Meridian records **979,200**; referrals interoperability **264,000**; clinician scheduling replacement **318,000**; analytics platform upgrade **180,000**; pharmacy stock automation **240,000**. Review establishes four adjustments. (i) **96,000** of clinician time released from chasing paper referrals is claimed in both records and referrals. (ii) Records claims **5.0** and scheduling **3.4** administrative posts from one pool of **6.0**, valued at **42,000** per post. (iii) **108,000** of the analytics platform's 180,000 is "improved decision-making" realised through records and scheduling, already counted there. (iv) **60,000** of the pharmacy claim rests on a baseline that already assumes savings committed in an operational programme outside the portfolio. Portfolio-wide adoption, per Domain 2's realistic assumption, is **70 %**. Approved portfolio cost is **USD 4,880,000** (records 2,400,000, referrals 640,000, scheduling 890,000, analytics 520,000, pharmacy 430,000). The investment committee applies a **four-year** simple payback rule.
2. **Formula.** Net benefit =  $\Sigma$  claimed at full potential –  $\Sigma$  eliminations, then  $\times$  adoption. Elimination for the same-pool case =  $(\Sigma \text{ claims} - \text{pool cap}) \times \text{unit value}$ . Simple payback = portfolio cost  $\div$  net annual benefit.
3. **Substitution.** Gross  $979,200 + 264,000 + 318,000 + 180,000 + 240,000$ . Eliminations  $96,000 + (8.4 - 6.0) \times 42,000 + 108,000 + 60,000$ . Then  $\times 0.70$ , and  $4,880,000 \div$  the result.
4. **Result.** Gross claimed at full potential **USD 1,981,200**. Eliminations **USD 364,800** – 96,000 shared, **100,800** same-pool (2.4 posts), 108,000 cascade, 60,000 already committed – an **elimination rate of 18.41 %**. Net full potential **USD 1,616,400**; at 70 % adoption, net realistic benefit **USD 1,131,480** a year. The unreconciled figure,  $1,981,200 \times 0.70 = \text{USD } 1,386,840$ , overstates that by **USD 255,360**, which is **22.57 %** of the honest number. Simple payback moves from **3.5188 years** on the unreconciled figure to **4.3129 years** on the honest one, a difference of **0.7941 years**, or **9.53 months**.
5. **Interpretation.** The decision, not the number, is what moved: the unreconciled portfolio passes a four-year payback rule and the reconciled one fails it. That is the whole professional case for doing the bridge before approval rather than discovering it in Domain 16's benefits measurement, and it produces a threshold worth carrying: **the maximum elimination rate this portfolio can absorb and still clear four years is 12.03 %**, so an observed 18.41 % breaks the rule with room to spare, and a portfolio whose bridge has never been done should be assumed to be somewhere in that range. Three further points. First, an identity that makes the bridge auditable: because adoption scales the eliminations too, the **elimination rate is invariant to the adoption assumption** ( $364,800/1,981,200$  and  $255,360/1,386,840$  are both **18.41 %**), so the two adjustments can be argued independently and in either order, and a reviewer can check one without settling the other. Second, eliminating a double count **creates no value**; it prevents a wrong decision. A portfolio report that presents 364,800

of eliminations as a PMO saving is committing the same category error in the opposite direction, and KA 15.3 keeps it out of the PMO's value case for exactly that reason. Third, the same-pool correction is the one most often resisted, because both claimants are right about their own component; the register's answer is that the pool is an enterprise asset and its cap is an enterprise fact, which is why benefits are owned by the receiving organisation and not by the delivering component (Domain 2, KA 2.3.2; Domain 16 measures them).

**Fourth, and it is the disclosure this example would be dishonest without: the payback test above is being applied to a mixed stream, by the committee's own convention.** Simple payback compares a cash outlay against a cash return. The USD 4,880,000 is cash. The benefit stream it is divided into is not homogeneous, and the register's fund-type column is what says so:

6. **Released clinician time** from chasing paper referrals is **capacity-released**. It leaves nobody's budget. It is valuable (it is the largest single element of Meridian's own case), but it repays a cash outlay only to the extent the freed sessions are redeployed to something the organisation would otherwise have bought.
7. The **administrative posts valued at 42,000 each** are **cashable only if the establishment is reduced and the budget line deleted**, which is precisely the confirmation the finance owner of that budget must give in writing. Absent it, they are capacity-released, and the same-pool correction above has already shown that the two components between them claimed more posts than the pool contains.
8. **"Improved decision-making"** is **non-financial**, and it is also the cascade double count eliminated in adjustment (iii), which is not a coincidence. Unclassified benefits are where double counts hide, because a benefit with no fund type has no owner who would notice it being claimed twice.
9. **Pharmacy stock savings** are **cost-avoidance or cashable** depending on whether the stock budget is actually reduced, a question for the budget holder, not for the component.

None of that changes a figure above, and stating it changes what the figures **mean**. The professional obligation is therefore not to abandon the committee's rule but to **disclose the composition of the denominator the rule is being applied to**, so that the committee applies its convention knowingly: *this is a four-year payback test run against a mixed stream, of which the following is confirmed cashable by the named budget holders and the following is not*. A committee told that will sometimes still approve, and it will approve something it understands. A committee not told that has been given a cash-shaped number that finance cannot honour, and the approval it grants is the one Domain 16's benefits review reopens. The register's classification and the finance owner's written confirmation are what make the sentence available; without them the honest payback figure of 4.3129 years is more defensible than the unreconciled 3.5188 and is still a comparison of cash against something that is not all cash.

### 15.2.2 Balancing under a binding constraint

**The principle.** A portfolio decision is an allocation under a constraint, and the first professional act is to **name the constraint honestly**. It is rarely money. It is usually a scarce capability (the integration engineers, the clinical safety reviewers, the one architect who understands the interface layer), and a portfolio balanced against a budget while a capability binds will be approved and will not be delivered.

Domain 2 established the base result at KA 2.2.3, and it is not re-derived here: **ranking candidates by value per unit of the binding constraint is a heuristic, not an optimum**, and with lumpy candidates the feasible sets must be enumerated: the Beta-plus-Gamma outcome that neither the raw value ranking nor the scoring model selected. Domain 15's contribution is the extension that makes this bite at enterprise scale: real constraints bind **per period**, not in aggregate.

**Balance across dimensions, not only value.** Before the arithmetic, a portfolio also has to be balanced against exposures that no single ratio expresses: risk (Domain 8), time to benefit, component novelty, dependence on one supplier or one team, and regulatory exposure. Domain 2 noted that a set of individually sound investments can be collectively unbalanced and that no scoring model detects it. The portfolio-level countermeasure is a small set of stated exposure limits (no more than a given share of the portfolio dependent on one platform, one supplier, or one scarce team) applied as feasibility constraints alongside capacity, so the enumeration returns only sets the organisation can survive.

### 15.2.3 The multi-period constraint and the aggregate-capacity illusion

**The defect.** Portfolio capacity is almost always assessed annually: "we have 24 engineer-quarters next year and the approved set needs 24, so it fits". That statement is not a feasibility test. Work arrives when the components need it, and a set that fits in aggregate can be grossly infeasible in a single period, which is the period in which the portfolio then fails.

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#### WORKED EXAMPLE

**Worked example 15.2.3 — allocating Meridian's portfolio against a quarterly constraint.**

1. **Setup.** One scarce integration team supplies **6 units per quarter** for four quarters: **24 units** in aggregate. Six candidates compete, each with a quarterly demand profile and a net present value computed on PFL-AI Domain 4's methods:

| CANDIDATE                          | Q1 | Q2 | Q3 | Q4 | UNITS | NPV<br>(USD) | NPV<br>PER<br>UNIT |
|------------------------------------|----|----|----|----|-------|--------------|--------------------|
| A records rollout continuation     | 2  | 3  | 3  | 2  | 10    | 1,700,000    | 170,000            |
| B referrals interoperability       | 2  | 2  | 2  | 2  | 8     | 1,360,000    | 170,000            |
| C pharmacy stock automation        | 5  | 1  | 0  | 0  | 6     | 1,140,000    | <b>190,000</b>     |
| D clinician scheduling replacement | 0  | 1  | 2  | 3  | 6     | 1,020,000    | 170,000            |
| E analytics platform upgrade       | 1  | 1  | 1  | 1  | 4     | 640,000      | 160,000            |
| F estate network refresh           | 2  | 0  | 1  | 2  | 5     | 750,000      | 150,000            |

2. **Formula.** Three procedures compared. **Aggregate ranking:** take candidates in descending NPV per unit while total units remain. **Per-unit greedy with a period check:** take in the same order, skipping any candidate that breaches capacity in any quarter. **Enumeration:** evaluate all  $2^6 - 1 = 63$  non-empty subsets, retain those satisfying  $\sum \text{demand} \leq 6$  in **every** quarter, and select the maximum NPV.
3. **Substitution and result.**
4. **Aggregate ranking** takes C (6 units), A (10) and B (8), exactly **24 of 24 units**, NPV USD **4,200,000**, and full utilisation. It is **infeasible**: Q1 demand is  $5 + 2 + 2 = 9$  units against a capacity of 6.
5. **Per-unit greedy** takes C first on its 190,000 ratio, leaving (1, 5, 6, 6); A and B are then both blocked in Q1 and skipped; D and E fit; F is blocked. Result C + D + E, NPV USD **2,800,000**, using **16 of 24** units.
6. **Enumeration** finds **26 feasible sets of 63** and selects **A + B + F**: quarterly demand (6, 5, 6, 6), **23 of 24** units, NPV USD **3,810,000**. The runner-up, A + B + E at **3,700,000** with demand (5, 6, 6, 5), is the set to hold in reserve.
7. **Interpretation.** Two errors, two prices, and they point in opposite directions, which is what makes them hard to see together. The **aggregate illusion** overstates achievable value by  $4,200,000 - 3,810,000 = \text{USD } 390,000$ , or **10.24 %** of the true optimum, and it does so while reporting 100 % utilisation, which is why it survives review: a fully committed annual plan looks like good stewardship. The **greedy heuristic** is feasible but leaves **USD 1,010,000** on the table (**26.51 %** of the optimum, capturing only **73.49 %**), and it fails for a nameable reason: the highest-ratio candidate, C, consumes **five of the six Q1 units** and thereby excludes both of the two largest-value candidates. **A ratio cannot**



see when a demand profile is concentrated in the scarce period, because the ratio has already averaged the period away. The professional conclusion is uncomfortable and correct: the portfolio-selection method most widely used in practice (rank by a value ratio, take until the budget is full) is wrong twice over, and the correction is not sophisticated. Enumerating 63 subsets is a spreadsheet exercise; at 25 candidates it is  $2^{25} = 33,554,432$  subsets, which is a fraction of a second of computation and is the point: **this is not a hard computation, it is an ungoverned one**, and KA 15.3 makes it the enterprise PMO's job. Two cautions. The optimum uses **23 of 24 units — 95.83 % planned utilisation with slack of (0, 1, 0, 0)** — and KA 15.3 shows what that does to the plan's survival probability, so 3,810,000 is the value of a plan, not a forecast. And the NPVs are themselves estimates; a difference of 110,000 between A + B + F and A + B + E is inside anyone's estimating error, so the honest output of an enumeration is the **feasible frontier** (the top few sets and what distinguishes them) rather than a single winner presented as a computation.

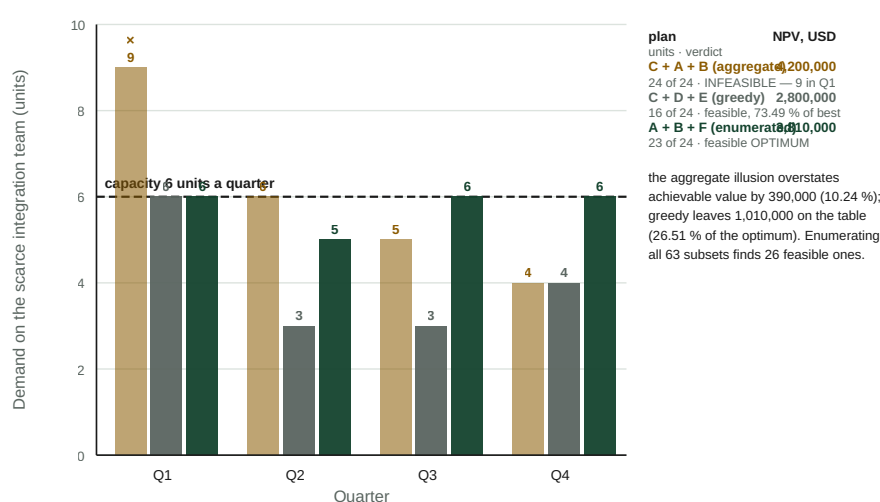


Fig 15.2.1 — Aggregate capacity feasible, period capacity not

### 15.2.4 Rebalancing, staging and stopping

**Rebalancing is the normal state.** A portfolio decided once is a budget, not a portfolio. The enumeration above is a snapshot; capacity, value estimates and strategy all move, so the allocation is re-run on a stated cycle with the components already in flight treated correctly: their remaining demand and remaining value, never their original figures, which is the portfolio form of Domain 2's sunk-cost warning (KA 2.4.2).

**Staging creates option value.** A component admitted as a small first tranche with a decision point attached consumes less of the scarce period and preserves the right

to stop. Meridian's tranche design (15.1.2) is this device applied within a programme; at portfolio level the same move lets a constrained period carry more candidates at lower commitment.

**Stopping is a portfolio capability, not an admission.** The portfolio's hardest arithmetic is that stopping a component releases capacity in the *scarce period*, and that released capacity has a value equal to the best displaced candidate's contribution. Domain 2's kill criteria (KA 2.4.3) supply the test; the portfolio addition is that the case for stopping is strengthened, not weakened, by naming what the released capacity will do next; because a portfolio board that stops something without reallocating has taken a loss without buying anything.

### AI in this KA

**Where it earns its place.** Running the enumeration, including the exposure-limit constraints, and returning the feasible frontier with the binding period identified for each rejected set: a deterministic search that scales far past human patience. Reconciling component business cases against a benefits register and proposing candidate double counts by matching benefit descriptions, measures and receiving owners across cases, which is the highest-yield document task in this domain and one nobody has time to do by hand. Testing whether a same-pool claim exceeds the pool by summing claims against establishment data. Re-running the allocation under alternative capacity, value and adoption assumptions to produce the sensitivity a board should be given.

**Where it must not go.** Selecting the portfolio. The enumeration returns sets; the choice among them involves strategic weight, risk appetite, political feasibility and obligations that no objective function contains, and it belongs to the accountable body. Nor may a model confirm a double count: it proposes candidates, and a human with access to the establishment data and the component owners decides which are real, because a wrongly eliminated benefit removes a component's case as surely as a wrongly counted one inflates it. And AI must not generate the value estimates it then optimises over: a closed loop of plausible numbers is the specific failure Domain 14 calls the plausible wrong number (KA 14.3.4).

**Verification, concretely.** Re-compute the chosen set's per-period demand by hand against capacity, which is a dozen additions and catches every infeasibility. Reconcile the benefits bridge with a signed statement from each receiving-organisation owner that they accept the capped figure. The elimination is only real when the person who must deliver it agrees. Publish the elimination rate and the breakeven elimination rate together, so the board sees whether its decision rule is robust or marginal.

## KEY TERMS — KA 15.2

| TERM                                  | MEANING                                                                                                                                                                                                                                               |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Portfolio benefits register</b>    | The authoritative statement of portfolio benefits, organised by benefit rather than by component, each carrying a measure, a baseline, a receiving-organisation owner, a fund type and its contributing components.                                   |
| <b>Fund type</b>                      | Cashable (a named budget line is removed), capacity-released (time or space freed, no budget moves), cost-avoidance (a future outlay not incurred) or non-financial. Mandatory, because a total that mixes them cannot be tested against a cash rule. |
| <b>Cashability confirmation</b>       | The written confirmation, from the finance owner of the affected budget, that a claimed saving is budget-removable and from which period. Until it exists the claim is carried as capacity-released.                                                  |
| <b>Mixed-stream test</b>              | An investment rule applied to a benefit stream of more than one fund type. Permitted as a committee's own convention, but only where the composition of the denominator is disclosed with the result.                                                 |
| <b>Benefits bridge</b>                | The reconciliation from gross claimed benefit through eliminations and adoption to net realistic benefit.                                                                                                                                             |
| <b>Shared benefit</b>                 | One benefit claimed by two components that both genuinely contribute to it; counted once.                                                                                                                                                             |
| <b>Same-pool over-claim</b>           | Aggregate claims on a resource pool exceeding the pool; capped at the pool and allocated.                                                                                                                                                             |
| <b>Cascade (enabler) double count</b> | An enabling component claiming benefit already claimed by the components it enables.                                                                                                                                                                  |
| <b>Already-committed benefit</b>      | Benefit claimed against a baseline that already assumes a saving committed elsewhere.                                                                                                                                                                 |
| <b>Elimination rate</b>               | $\text{Eliminations} \div \text{gross claimed benefit}$ ; invariant to the adoption assumption.                                                                                                                                                       |
| <b>Breakeven elimination rate</b>     | The elimination rate at which a portfolio just satisfies its investment decision rule.                                                                                                                                                                |
| <b>Binding constraint</b>             | The resource that actually limits the portfolio, usually a scarce capability, rarely money.                                                                                                                                                           |
| <b>Aggregate-capacity illusion</b>    | Judging feasibility on total capacity when the constraint binds period by period.                                                                                                                                                                     |
| <b>Feasible frontier</b>              | The top few feasible sets and what distinguishes them, the honest output of an enumeration.                                                                                                                                                           |

## SAMPLE MCQs — KA 15.2

**MCQ 15.2-A** [15.2.3 · Analysis] A portfolio's scarce team supplies 6 units a quarter for four quarters. Three candidates demanding 24 units in total are selected on the basis that total capacity is 24 units. The most accurate criticism is that:

- A. the portfolio is under-committed
- B. aggregate feasibility does not imply period feasibility; the set demands 9 units in Q1 CORRECT
- C. the team's capacity should have been expressed annually
- D. net present value is the wrong selection measure

*Rationale:* Constraints bind per period; the set is aggregate-feasible and Q1-infeasible by 3 units (15.2.3). A inverts the problem; C names the cause of the error as its cure; D is a different debate.

**MCQ 15.2-B** [15.2.3 · Evaluation] In the Meridian portfolio allocation, ranking by NPV per unit yields 2,800,000 while enumeration yields 3,810,000. The reason the ranking fails is that:

- A. the ratios were computed incorrectly
- B. enumeration uses a different objective function
- C. the highest-ratio candidate consumes five of the six units in the scarce quarter, excluding the two largest-value candidates CORRECT
- D. the candidates have equal ratios, so the ranking is arbitrary

*Rationale:* A ratio averages the period away and cannot see a demand profile concentrated in the binding period (15.2.3). Both methods maximise NPV, so B is wrong; three candidates do share a 170,000 ratio, but that is not what causes the 1,010,000 shortfall.

**MCQ 15.2-C** [15.2.1 · Application] Five components claim 1,981,200 of annual benefit at full potential. Eliminations total 364,800 and portfolio adoption is 70 %. The net realistic annual benefit is:

- A. USD 1,616,400
- B. USD 1,386,840
- C. USD 1,131,480 CORRECT
- D. USD 792,036

*Rationale:*  $(1,981,200 - 364,800) \times 0.70 = 1,131,480$  (15.2.1). A omits the adoption adjustment; B omits the eliminations; D applies adoption twice  $(1,131,480 \times 0.70)$ , the error of taking a component figure that is already adoption-adjusted and adjusting it again.

**MCQ 15.2-D** [15.2.1 · Analysis] Two components claim a combined 8.4 administrative posts from a pool of 6.0, valued at 42,000 each. The elimination is:

- A. USD 352,800
- B. USD 252,000
- C. USD 100,800 CORRECT
- D. USD 42,000

*Rationale:*  $(8.4 - 6.0) \times 42,000 = 100,800$  (15.2.1). A is the gross claim; B is the pool's total value, which is the retained benefit, not the elimination; D is one post.

**MCQ 15.2-E** [15.2.1 · Evaluation] A portfolio's benefits bridge eliminates 18.41 % of gross claimed benefit, and the breakeven elimination rate for its four-year payback rule is 12.03 %. The correct conclusion is that:

- A. the portfolio clears the rule with a 6.4-point margin
- B. the portfolio fails the rule, and would still fail it at any elimination rate above 12.03 % CORRECT
- C. the rule should be relaxed to accommodate the eliminations
- D. the eliminations of 364,800 should be reported as a portfolio saving

*Rationale:* An observed rate above the breakeven rate means the rule is not met (15.2.1). A reads the comparison backwards; C is a decision for the investment committee, not a conclusion from the arithmetic; D commits the error the topic explicitly warns against: eliminating a double count prevents a wrong decision and creates no value.

#### SELF-CHECK — KA 15.2

1. *Name the four portfolio double counts and one correction each.* — Shared benefit (count once, record both contributors); same-pool over-claim (cap at the pool and allocate); cascade or enabler (an enabler's own benefit is only what does not pass through a consumer's claim); already committed (reset the baseline).
2. *Why is the elimination rate invariant to the adoption assumption?* — Adoption scales the eliminations and the gross claim identically, so their ratio is unchanged, which lets the two adjustments be argued and audited independently.
3. *What is the honest output of a portfolio enumeration?* — The feasible frontier: the top few sets, the binding period for each rejected one, and what distinguishes them, not a single winner, since the value estimates cannot support that precision.
4. *Name the four fund types, and who confirms the first of them.* — Cashable, capacity-released, cost-avoidance, non-financial. The **finance owner of the affected budget** confirms cashability in writing, naming the budget line and the period;

not the portfolio board and not the delivering component. An unconfirmed cashable claim is carried as capacity-released (15.2.1).

5. *What must accompany a payback result computed on a mixed benefit stream?* — The composition of the denominator: which elements are confirmed cashable and which are not. A cash test applied to an undisclosed mixture produces a cash-shaped number finance cannot honour, and the approval it supports is the one Domain 16's benefits review reopens.

## KNOWLEDGE AREA 15.3

# Capacity and enterprise PMOs

*Topics: 15.3.1 enterprise capacity as the real constraint · 15.3.2 portfolio work in progress · 15.3.3 protective capacity and its price · 15.3.4 the enterprise PMO and its value test.*

## 15.3.1 Enterprise capacity as the real constraint

**Definition.** Enterprise delivery capacity is the rate at which an organisation can complete change, measured in the unit of its binding capability, at a stated quality. It is a **measured rate**, not a budget and not an aspiration, and the measurement is historical: completions per period over a period long enough to include the organisation's normal disruptions.

Three properties are routinely misunderstood, and each one costs money.

**Capacity is a rate, not a stock.** An organisation with 40 people does not have "40 people of capacity"; it has a completion rate that its structure, dependencies and interruption load produce. Adding people changes the rate by less than proportionally, and by less again where coordination cost rises: Domain 12's coordination arithmetic (KA 12.2.2 and 12.A.2) and Domain 4's interface count  $(n(n-1))/2$  against  $n$  to an integration layer, KA 4.2.3) both bear on this and neither is re-derived here. The practical rule: **treat the completion rate as the primitive and the headcount as one of its inputs.**

**Capacity is capability-specific.** A portfolio does not consume generic effort. It consumes integration engineering, clinical safety review, data migration, change management and testing in component-specific mixes, and the constraint is whichever of those binds first. A capacity model with one number in it will approve a portfolio that is 60 % utilised on average and 190 % utilised on the thing that matters.

**Capacity is consumed by the portfolio's own overheads.** Governance attendance, reporting, onboarding, rework and the interruption load of work already in flight all draw on the same people. Capacity measured from completions already

includes these; capacity estimated from headcount and availability does not, which is why the estimated figure is always higher and always wrong in the same direction.

### 15.3.2 Portfolio work in progress

**The mechanism.** Domain 13 established the flow arithmetic and it is applied, not re-derived, here: **Little's Law**,  $W = T \times C$ , relates work in progress  $W$ , cycle time  $T$  and throughput  $C$  for any system in steady state, independent of arrival and service distributions (KA 13.2.3). Its portfolio reading is uncomfortable and exact: **starting more initiatives does not increase throughput, which capacity sets; it increases cycle time, which is when benefit arrives.**

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#### WORKED EXAMPLE

#### Worked example 15.3.2 – the price of starting everything.

1. **Setup.** The health group's delivery organisation completed **15** portfolio initiatives in the last **three** years, a measured throughput of **5.0 a year**. The portfolio board has approved and started **12**. The portfolio's net realistic benefit run rate, from KA 15.2.1, is **USD 1,131,480** a year across five components: **USD 226,296** an initiative.
2. **Formula.**  $T = W / C$  (Little's Law, KA 13.2.3). Benefit forgone from excess concurrency = annual benefit run rate  $\times$  excess cycle time, where excess cycle time =  $(W_{\text{actual}} - W_{\text{limit}}) / C$ .
3. **Substitution.**  $T = 12/5$  against  $T = 5/5$ ; excess  $(12 - 5)/5$ ; then  $1,131,480 \times 1.4$ .
4. **Result.** Average cycle time **2.40 years** at 12 in flight, against **1.00 year** at 5, an excess of **1.40 years**. The portfolio's entire benefit stream therefore arrives 1.40 years later than it need, worth **USD 1,584,072**: cross-checked per initiative as  $226,296 \times 1.4 \times 5$  = the same figure.
5. **Interpretation.** The number is large, and its character matters more than its size: this is **forgone once and never recovered**. Throughput is unchanged (five initiatives finish per year under either policy), so no annual report will show a loss, no variance will be raised, and the only visible symptom is that everything takes two and a half years. The identity behind it is worth memorising because it prices any concurrency argument in one line: **cost of excess work in progress = annual benefit run rate  $\times$  excess work in progress  $\div$  throughput**. Three professional qualifications. The 1,584,072 is undiscounted and is therefore a **floor**; the rigorous treatment discounts the deferred stream (PFL-AI Domain 3), which reduces the headline and does not change the sign. Little's Law requires a steady state, so a portfolio in the middle of a step change in capacity is outside its scope. The law is a theorem about a system, not a forecast for a transition. And a work-in-progress limit is not a decision to do less; it is a decision to **finish** in a sequence, which requires the portfolio board to choose an order and

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defend it, and that is the reason the limit is resisted:  $w = 12$  is what a board does when it will not choose. The board that says "all twelve are priorities" has, arithmetically, said that none of them is.

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### 15.3.3 Protective capacity and its price

**The problem.** KA 15.2.3's optimum plans **23 of 24 units (95.83 % utilisation) with quarterly slack of (0, 1, 0, 0)**. A plan with no slack in three of four quarters is a plan that survives only if nothing goes wrong in those quarters, and the same multiplicative penalty KA 15.1.3 found for milestones applies here to periods.

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#### WORKED EXAMPLE

#### Worked example 15.3.3 — does protective capacity pay?

- 1. Setup.** Measured over the last twelve quarters, the integration team loses units to unplanned support and absence with the following frequencies: **0 units 0.70, 1 unit 0.22, 2 units 0.06, 3 or more 0.02**. The aggressive plan  $A + B + F$  demands (6, 5, 6, 6) against a capacity of 6, so slack is (0, 1, 0, 0). The alternative is to **reserve one unit a quarter**, plan to 5, and re-run the enumeration at that capacity, which selects  $A + E + F$  with demand (5, 4, 5, 5), slack against the true 6 of (1, 2, 1, 1), and NPV USD 3,090,000. A quarter's capacity breach slips the affected work one quarter: **13 weeks at the programme cost of delay of 14,280, or USD 185,640**.
- 2. Formula.** A plan survives a quarter if the loss is no greater than that quarter's slack; the plan survives the year with probability  $\prod P(\text{loss} \leq \text{slack}_i)$  over the four quarters: the multiplication rule of KA 15.1.3 applied to periods. Expected slippage cost =  $(1 - P(\text{survive})) \times \text{breach cost}$ . Breakeven breach cost =  $\text{NPV forgone} \div \text{the improvement in failure probability}$ .
- 3. Substitution.** Aggressive  $0.70 \times 0.92 \times 0.70 \times 0.70$ ; protective  $0.92 \times 0.98 \times 0.92 \times 0.92$ . Then  $(1 - P) \times 185,640$  for each, and  $(3,810,000 - 3,090,000) \div (0.76311424 - 0.31556)$ .
- 4. Result.** The aggressive plan survives all four quarters with probability **31.56 %**; the protective plan, **76.31 %**. Protective capacity costs **USD 720,000** of NPV, **18.90 %** of the optimum. Expected slippage cost falls from **USD 127,059** to **USD 43,975**, a saving of **USD 83,084**. The breakeven breach cost is **USD 1,608,744**, equivalent to **112.66 weeks** of programme delay, or **8.67 quarters**.
- 5. Interpretation.** On these numbers **protective capacity does not pay**, and a leader should be able to say so, because the received advice (always leave slack) is not a theorem. Spending 720,000 of value to avoid an expected 83,084 of slippage is a poor trade by a factor of nearly nine, and the discipline the arithmetic imposes is to state the condition under which the conclusion reverses: **a breach must cost more than about 1.61 million for the reservation to be worth it**. That condition is not exotic. A breach that lands on a regulatory



submission date, a clinical safety approval, a contractual liquidated-damages milestone or a go-live already announced to patients is easily worth more than eight quarters of ordinary delay, and in that case the answer flips decisively. So the professional output is not "reserve capacity" or "do not"; it is: **price the breach on the specific work in the zero-slack periods, then compare**. Two cautions. The 31.56 % figure is itself the product of four period probabilities, so it inherits KA 15.1.3's independence assumption: correlated losses (one absence wave, one support incident spanning two quarters) make clustered breaches likelier than the product suggests, and where the loss driver is shared the calculation should be run on the pessimistic assumption as well. And a 31.56 % survival probability is not a 68 % chance of disaster: it is a high chance of *some* re-planning, which is normal, and the reason to compute it is to know in advance which quarters will need it, not to be alarmed.

#### 15.3.4 The enterprise PMO and its value test

**Definition.** An enterprise portfolio management office is the function that maintains the portfolio's authoritative data, runs the allocation and prioritisation process, measures capacity and delivery performance, and provides second-line assurance (Domain 3, KA 3.3.2). It has no delivery accountability and no decision rights; it exists to make the accountable bodies' decisions better.

The role has three honest variants and one dishonest one. A **reporting** office collects and publishes; a **standards** office defines and assures method; a **delivery** office also supplies delivery capability. The dishonest variant is the office that reports without improving any decision, which is Domain 3's governance-artefact failure at enterprise scale (KA 3.1.1). The organisation has the portal, the template and the monthly pack, and no allocation decision has ever changed as a result.

**The value test, and why it is usually failed on the second reading.** An enterprise PMO should be subject to the arithmetic it applies to everyone else, and the test that matters distinguishes **recurring** from **one-off** value.

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#### WORKED EXAMPLE

##### Worked example 15.3.4 — does the enterprise PMO pay?

1. **Setup.** The health group's enterprise PMO costs 8 staff at a blended 132,000 plus 124,000 of tooling and other costs. Its claimed year-one contributions: **half** of KA 15.3.2's work-in-progress deferral recovered by imposing a limit; the 517,650 of annual latency saving computed in KA 15.4.1; 186,000 a year of duplicate tooling eliminated across three components; and the 364,800 of benefits eliminations from KA 15.2.1.
2. **Formula.** Cost = staff × rate + other. Separate the claimed value into recurring and one-off, and test the recurring value alone against the annual cost.

- 
3. **Substitution.**  $8 \times 132,000 + 124,000$ . Recurring  $517,650 + 186,000$ . One-off  $1,131,480 \times 0.5$ . The eliminations are excluded, see below.
  4. **Result.** Cost **USD 1,180,000** a year, which is **24.18 %** of the portfolio's 4,880,000 approved cost. Recurring value **USD 703,650**. One-off value **USD 565,740**. Year one totals **USD 1,269,390**: a surplus of **USD 89,390**. Year two, with the one-off gone, is **703,650** against 1,180,000: a **deficit of USD 476,350**.
  5. **Interpretation.** The office pays in year one and fails in year two, and it will be defunded in year three by a finance director who has noticed. That pattern is the single commonest fate of enterprise PMOs and it is a **reporting** failure before it is a value failure: the year-one case was built on a one-off, and one-offs do not repeat. Note also what has been kept out. The **364,800** of benefits eliminations is excluded because eliminating a double count creates no value (KA 15.2.1): it prevented a portfolio from being approved on a false payback, which is decision quality of the highest order and is not a cash benefit, and an office that books it as one has falsified its own case. The constructive output is a target rather than a verdict: the office needs **USD 476,350** of additional recurring value, and KA 15.2.3 identified exactly where it is available — the **1,010,000** allocation gap between the ranking heuristic and the enumerated optimum, which recurs at every allocation cycle. The office therefore breaks even recurringly if it captures **47.16 %** of that gap. That is a testable, monitorable commitment of the right shape: it names the mechanism, it is measurable at the next cycle by comparing the selected set against the enumerated frontier, and it fails visibly. **An enterprise PMO whose value case cannot be written in that form should be resized to the reporting it actually does.**
- 

### AI in this KA

**Where it earns its place.** Measuring capacity honestly from completion history rather than from availability assumptions, including the interruption load that headcount models omit. Maintaining a capability-specific capacity model and flagging the constraint that binds first, per period, across a whole portfolio: high-volume bookkeeping with a right answer. Computing plan survival probabilities across candidate plans and loss distributions, and returning the breakeven breach cost, which is the number a board needs. Detecting duplicate tooling and overlapping capability across component architectures. Producing the enumeration and the feasible frontier each cycle, which is precisely the recurring work the PMO's value case above depends on capturing.

**Where it must not go.** Setting the work-in-progress limit or choosing the sequence, which is a prioritisation decision belonging to the portfolio board, and the decision the board is avoiding when it starts twelve initiatives. Estimating the loss distribution from plausibility rather than from twelve quarters of records. Assessing an enterprise PMO's own value, which is a self-interested judgement no function should automate. And no AI-produced capacity number should reach a board

without its measurement window and its exclusions stated, because a capacity figure without those is not a measurement.

**Verification, concretely.** Recompute throughput from the raw completion list, not from a dashboard, and state the window: five a year from fifteen in three years is one division and it is the foundation of every figure in this KA. Reproduce the survival product by hand for the chosen plan; it is four multiplications. Require the PMO's value case to separate recurring from one-off in the statement itself, and to name the mechanism and the measurement for each recurring item, since that separation is the check that catches the year-two failure a year early.

#### KEY TERMS — KA 15.3

| TERM                                  | MEANING                                                                                                                             |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Enterprise delivery capacity</b>   | The measured rate at which an organisation completes change, in the unit of its binding capability.                                 |
| <b>Capability-specific capacity</b>   | Capacity modelled per scarce capability, since the portfolio consumes them in component-specific mixes.                             |
| <b>Portfolio work in progress</b>     | The number of initiatives in flight; by Little's Law it sets cycle time, not throughput.                                            |
| <b>Excess work in progress cost</b>   | Annual benefit run rate $\times$ excess work in progress $\div$ throughput, forgone once and never recovered.                       |
| <b>Protective capacity</b>            | Capacity deliberately withheld from allocation so that a period's plan can absorb variance.                                         |
| <b>Plan survival probability</b>      | The product over periods of the probability that each period's loss does not exceed its slack.                                      |
| <b>Breakeven breach cost</b>          | Value forgone by reserving capacity $\div$ the improvement in failure probability it buys.                                          |
| <b>Enterprise PMO</b>                 | The function that maintains portfolio data, runs allocation, measures capacity and gives second-line assurance; no decision rights. |
| <b>Recurring versus one-off value</b> | The distinction that decides whether a support function survives its second year.                                                   |

#### SAMPLE MCQS — KA 15.3

**MCQ 15.3-A** [15.3.2 · Application] A delivery organisation completes 5 initiatives a year and has 12 in flight. Average cycle time is:

- A. 0.42 years
- B. 1.00 year
- C. 2.40 years CORRECT
- D. 5.00 years

*Rationale:*  $T = W/C = 12/5 = 2.40$  years (Little's Law, cited from KA 13.2.3). A inverts the ratio; B is the cycle time at a work-in-progress limit of 5; D is the throughput read as a duration.

**MCQ 15.3-B** [15.3.2 · Evaluation] Reducing portfolio work in progress from 12 to 5 at an unchanged throughput of 5 a year, with a net benefit run rate of 1,131,480, is best described as:

- A. increasing throughput by 140 %
- B. bringing the whole benefit stream forward by 1.40 years, worth 1,584,072 once
- C. saving 1,584,072 every year thereafter
- D. reducing the portfolio's benefit, since fewer initiatives are in flight

CORRECT

*Rationale:* Throughput is capacity-set and unchanged; cycle time falls by  $(12-5)/5 = 1.40$  years and the whole stream shifts forward once,  $1,131,480 \times 1.4 = 1,584,072$  (15.3.2). A misreads Little's Law; C converts a one-off shift into an annuity; D confuses starting with finishing.

**MCQ 15.3-C** [15.3.3 · Analysis] A four-quarter plan has slack of 0, 1, 0 and 0 units. Losses of 0, 1 and 2 or more units occur with probabilities 0.70, 0.22 and 0.08. The probability the plan survives all four quarters is closest to:

- A. 70.0 %
- B. 41.5 %
- C. 31.6 %
- D. 24.0 %

CORRECT

*Rationale:*  $0.70 \times 0.92 \times 0.70 \times 0.70 = 31.56 \%$  (15.3.3). A applies the single-quarter figure to the year; B uses 0.92 in two quarters rather than one; D applies 0.70 to all four quarters and ignores the slack quarter's tolerance.

**MCQ 15.3-D** [15.3.3 · Evaluation] Reserving capacity costs 720,000 of NPV and lifts plan survival from 31.56 % to 76.31 %. A quarter's breach costs 185,640. The correct conclusion is:

- A. reserve, because the survival probability more than doubles
- B. do not reserve unless a breach would cost more than about 1.61 million
- C. reserve, because the expected saving of 83,084 is positive
- D. the comparison cannot be made without a discount rate

CORRECT

*Rationale:*  $720,000 / (0.76311424 - 0.31556) = 1,608,744$  is the breach cost at which the trade breaks even (15.3.3). A argues from a ratio without a price; C is true and irrelevant, since 83,084 is far below 720,000; D is a refinement, not an obstacle.

**MCQ 15.3-E** [15.3.4 · Evaluation] An enterprise PMO costing 1,180,000 a year claims 703,650 of recurring value and 565,740 of one-off value. The soundest assessment is that it:

- A. pays, with a year-one surplus of 89,390
- B. pays, because total claimed value exceeds cost
- C. fails from year two by 476,350 and needs a named recurring mechanism

CORRECT

- D. should book the 364,800 of benefits eliminations to close the gap

*Rationale:* One-offs do not repeat, so the recurring test is the one that matters (15.3.4). A and B are true of year one and irrelevant thereafter; D books a prevented wrong decision as a cash benefit, which the domain rejects at KA 15.2.1.

SELF-CHECK — KA 15.3

1. *Why is capacity a rate rather than a stock?* — Completion rate is produced by structure, dependencies and interruption load; headcount is one input to it, and adding people changes the rate by less than proportionally.
2. *State the cost of excess portfolio work in progress in one line.* — Annual benefit run rate  $\times$  excess work in progress  $\div$  throughput; at Meridian,  $1,131,480 \times 7/5 = 1,584,072$ , forgone once.
3. *What makes an enterprise PMO's value case credible?* — Recurring value separated from one-off, each recurring item with a named mechanism and a measurement; for example capturing 47.16 % of the 1,010,000 allocation gap each cycle.

KNOWLEDGE AREA 15.4

## Transformation governance and strategic reporting

*Topics:* 15.4.1 enterprise decision latency · 15.4.2 strategic reporting and the aggregation trap · 15.4.3 the portfolio view and what it must not hide.

### 15.4.1 Enterprise decision latency

**The extension.** Domain 3 established that a decision body's expected latency is  $E[\text{wait}] = M/2 + L$  (half the meeting interval plus the whole paper lead time) that latency sums across escalation tiers, and that a one-week cut in the paper lead time saves a full week while a one-week cut in the meeting interval saves only half of one (KA 3.2.3, KA 3.3.3). None of that is re-derived. The enterprise extension is that a transformation portfolio does not send **one** decision up a path; it sends a **distribution** of decisions up different paths, and the bill is the weighted sum.

## WORKED EXAMPLE

**Worked example 15.4.1 – the annual cost of an enterprise decision architecture.**

1. **Setup.** The health group's five tiers, each with its meeting interval  $M$  and paper lead time  $L$  in weeks: **component board**  $M = 2$ ,  $L = 1$ ; **programme board**  $M = 4$ ,  $L = 2$ ; **portfolio board**  $M = 6$ ,  $L = 2$ ; **investment committee**  $M = 12$ ,  $L = 3$ ; **executive board**  $M = 13$ ,  $L = 4$ . Decisions are routed by value class: class 1 ( $\leq 25,000$ ) **48** a year through 1 tier; class 2 (25,001–150,000) **22** through 2; class 3 (150,001–500,000) **9** through 3; class 4 (500,001–2,000,000) **4** through 4; class 5 ( $> 2,000,000$ , reserved matters) **2** through all 5. Reviewed history, per Domain 3's convention, shows **25 %** of escalated decisions sit on the critical path and convert their wait into delay. Cost of delay **USD 14,280** a week.
2. **Formula.** Tier latency  $M/2 + L$ ; cumulative latency for a  $t$ -tier path is the sum over the first  $t$  tiers (Domain 3, KA 3.3.3). Gross latency-weeks =  $\Sigma$  over classes of count  $\times$  cumulative latency. Delaying weeks = gross  $\times$  critical-path share. Cost = delaying weeks  $\times$  cost of delay.
3. **Substitution.** Tier latencies  $2/2+1 = 2.0$ ,  $4/2+2 = 4.0$ ,  $6/2+2 = 5.0$ ,  $12/2+3 = 9.0$ ,  $13/2+4 = 10.5$ ; cumulative  $2.0$ ,  $6.0$ ,  $11.0$ ,  $20.0$ ,  $30.5$ . Then  $48(2.0) + 22(6.0) + 9(11.0) + 4(20.0) + 2(30.5)$ ,  $\times 0.25$ ,  $\times 14,280$ .
4. **Result.** A decision travelling all five tiers waits **30.5 weeks** and costs **USD 435,540**. Across the portfolio's **85** decisions: **468 gross latency-weeks**, an average of **5.51 weeks** a decision; **117.0** delaying weeks at the 25 % share; a bill of **USD 1,670,760 a year**.
5. **Interpretation.** One million six hundred and seventy thousand dollars a year is the running cost of a committee timetable, and it appears in no budget, is attributed to no decision, and is described in year-end reviews as the portfolio having been slow. Three results follow, ranked by value per unit of organisational pain, and the ranking is the useful part. **First, the paper deadline.** One week off every tier's paper lead time saves one week on every tier traversal, and the portfolio makes **145** traversals a year:  $145 \times 0.25 \times 14,280 = \text{USD } 517,650$ , or **30.98 %** of the entire bill, from an administrative change that costs nothing and removes no scrutiny anyone can name. This is Domain 3's lever result at enterprise scale, and the scale is what makes it a board-level item rather than a housekeeping one. **Second, the out-of-cycle route.** A written-resolution procedure completing classes 4 and 5 in 1.5 weeks saves **USD 471,240**, or **28.21 %**: nearly as much as the first lever, from a mechanism affecting only **6** of the **85** decisions. That is because those 6 decisions (**7.06 %** of the volume) carry **141 of the 468 gross weeks**, or **30.13 %** of the latency. **Seven per cent of the decisions carry thirty per cent of the delay**, which is why "add the executive board to the approval path" is such an expensive sentence and why the out-of-cycle mechanism belongs at the top of the architecture, not at the bottom. **Third,**

**tier removal**, which is the intervention organisations reach for first and which is worth least here: removing the component board from classes 3 to 5 (raised directly, component board informed) saves 30 gross weeks (**USD 107,100**, or **6.41 %**), because the tier removed is the fast one. **The general rule: remove latency where the latency is, which is the slow tier and the paper deadline, not the tier that is easiest to remove.** Two cautions. The 25 % critical-path share is an assumption from history and the whole bill is proportional to it, so it must be stated as an assumption and re-estimated; and latency is not the only property of a governance tier: a reserved matter exists because someone must be answerable for it, and the arithmetic prices the tier without ever deciding whether it is legitimate. What the arithmetic does is force the trade to be explicit, which is a conversation governance reviews almost never have because until the latency is added up there is nothing to weigh the tier against.

#### 15.4.2 Strategic reporting and the aggregation trap

**The principle.** A portfolio report exists to support three decisions and no others: **continue, change or stop** each component; **reallocate** capacity between them; and **escalate** what the portfolio cannot resolve. Every element that serves none of the three is decoration, and Domain 11's executive-communication discipline (KA 11.2) applies unchanged. The portfolio addition is that aggregation itself can invert a conclusion.

**The trap.** Performance indices are **ratios**, and ratios do not average. Averaging component indices without weighting them by the quantity in their denominator gives an answer that is not merely imprecise but frequently of the opposite sign. And this is the most common defect in circulating portfolio reports, because the unweighted average is what a spreadsheet produces by default.

#### WORKED EXAMPLE

##### Worked example 15.4.2 — the portfolio index that reversed its sign.

1. **Setup.** At the end of Q2 the five components report, in USD:

| COMPONENT                  | PV        | EV        | AC        | BAC       | CPI    | SHARE OF<br>PORTFOLIO<br>AC |
|----------------------------|-----------|-----------|-----------|-----------|--------|-----------------------------|
| Meridian records           | 1,300,000 | 1,196,000 | 1,352,000 | 2,400,000 | 0.8846 | 54.67 %                     |
| Referrals interoperability | 380,000   | 361,000   | 372,000   | 640,000   | 0.9704 | 15.04 %                     |
| Clinician scheduling       | 250,000   | 262,500   | 245,000   | 890,000   | 1.0714 | 9.91 %                      |

| COMPONENT           | PV               | EV               | AC               | BAC              | CPI    | SHARE OF        |
|---------------------|------------------|------------------|------------------|------------------|--------|-----------------|
|                     |                  |                  |                  |                  |        | PORTFOLIO<br>AC |
| Analytics platform  | 300,000          | 306,000          | 291,000          | 520,000          | 1.0515 | 11.77 %         |
| Pharmacy automation | 220,000          | 224,400          | 213,000          | 430,000          | 1.0535 | 8.61 %          |
| <b>Portfolio</b>    | <b>2,450,000</b> | <b>2,349,900</b> | <b>2,473,000</b> | <b>4,880,000</b> |        |                 |

2. **Formula.** Portfolio  $CPI = \Sigma EV / \Sigma AC$  and  $SPI = \Sigma EV / \Sigma PV$  (the registered EVM definitions, Domain 7 KA 7.3).  $EAC = BAC / CPI$  by the index method. The defective alternative is the unweighted mean of the component  $CPI$ s.
3. **Substitution.**  $2,349,900/2,473,000$  and  $2,349,900/2,450,000$ ; the mean of 0.8846, 0.9704, 1.0714, 1.0515 and 1.0535; then 4,880,000 divided by each index.
4. **Result.** Portfolio  $CPI$  **0.95** (0.950222) and  $SPI$  **0.96** (0.959143). The unweighted mean of the component indices is **1.01** (1.006308).  $EAC$  at the portfolio  $CPI$  is **USD 5,135,639.81**, giving  $VAC$  of (**USD 255,639.81**); at the unweighted mean it is **USD 4,849,408.40**, giving  $VAC$  of **USD 30,591.60**. The reported variance at completion swings by **USD 286,231** and changes sign. (Cents are carried here only so the swing reconciles exactly; the reporting convention remains whole units.)
5. **Interpretation.** One arithmetic choice moves the portfolio from 30,592 favourable to 255,640 adverse, and the direction of the error is predictable: **the unweighted mean flatters whenever the large components are the troubled ones**, which is the normal case, because scale and difficulty correlate. Here one component holds **54.67 %** of portfolio cost incurred and is the only one below 0.95; four small components each slightly ahead outvote it four to one in an unweighted average and hold 45.33 % of the money. The corrections are two, and both belong in the standard. **Aggregate by summing the numerators and denominators, never by averaging the ratios:** a rule that generalises to every ratio a portfolio reports, including adoption, defect and utilisation rates. And **publish the concentration alongside the aggregate:** the share of portfolio cost sitting in components below the alert threshold, here **54.67 %** below  $CPI$  0.95, which is the figure that tells a board whether one component is carrying the portfolio's problem or the problem is general. Two cautions. A single portfolio  $CPI$  is an aggregate and therefore always masks distribution. It is a headline for the concentration line beneath it, not a substitute for the component table. And  $EAC = BAC/CPI$  assumes performance to date continues, which is one of several defensible forecasts; Domain 7 (KA 7.2) governs the selection, and a portfolio report that presents one  $EAC$  without naming its method has reported an opinion as a measurement.



### 15.4.3 The portfolio view and what it must not hide

**What the one-page view carries.** Per component: the continue/change/stop recommendation with its owner; benefit at stake and benefit realised to date (Domain 16 owns the measurement); performance aggregated correctly with the concentration line; the binding-capability demand in the next two periods; the top dependency with an owner on the giving side; and the decisions required with their dates. At portfolio level: net benefit against the bridge, capacity utilisation by capability and period, work in progress against its limit, and the open decisions with their expected latency.

**Four things it must not hide**, each of which this domain has priced. The **milestone product**, rather than a list of green components against a committed date (15.1.3). The **binding period**, rather than annual capacity (15.2.3). The **elimination rate** on claimed benefits, rather than the gross claim (15.2.1). And the **latency of the open decisions**, rather than a list of items awaiting approval (15.4.1). A portfolio report that omits all four is not a weak report; it is a report that cannot support any of the three decisions it exists for.

**The information-age discipline.** Domain 14 established that a fact's age governs the decisions it can support (KA 14.2.1). At portfolio level the constraint is sharper than at project level, because a portfolio report is assembled from component reports that were themselves assembled earlier: a monthly portfolio pack routinely presents facts six to eight weeks old as current. The remedy is to **date every figure at its own source**, not at the pack, and to refuse to make a reallocation decision on a capacity figure older than the period it constrains.

#### AI in this KA

**Where it earns its place.** Computing the enterprise latency bill across every decision class and candidate architecture, and ranking the redesigns by saving per unit of change, deterministic and tedious in exactly the right proportions. Assembling the portfolio report from component data with each figure carrying its own source date, and flagging figures older than a stated tolerance. Recomputing aggregate indices correctly from the underlying **EV**, **PV** and **AC** sums and detecting where a submitted report has averaged ratios: a mechanical check that catches the KA 15.4.2 defect before a board sees it. Drafting the narrative around a computed exception set, for review.

**Where it must not go.** Writing the continue/change/stop recommendation. That is a judgement with an accountable owner, and a portfolio board receiving a machine-written recommendation is being invited to ratify rather than to decide: Domain 3's decidability failure (KA 3.1.1) reproduced at enterprise scale. Nor may a model set a tolerance, a threshold or a decision class, all of which are risk-appetite decisions. And no AI-assembled portfolio pack goes to a board without the register of AI uses Domain 14 requires (KA 14.4.3), naming for each output what the tool did, whose decision it informs and who verified it.

**Verification, concretely.** Recompute the aggregate indices from the sums by hand (two divisions), and compare against whatever the submitted pack contains; a discrepancy is the averaged-ratio defect and it is present more often than not. Reproduce the latency arithmetic for the two largest decision classes. Confirm every figure's source date against its source system on a stated sample. And require that each recommendation in the pack carries a named human owner, because a recommendation without one is a suggestion nobody has made, not a recommendation.

## KEY TERMS — KA 15.4

| TERM                                    | MEANING                                                                                                                                                                             |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Enterprise decision architecture</b> | The set of governance tiers, the decision classes routed through them, and the resulting latency distribution.                                                                      |
| <b>Gross latency-weeks</b>              | $\Sigma$ over decision classes of count $\times$ cumulative tier latency; the portfolio's total waiting.                                                                            |
| <b>Tier traversal</b>                   | One decision passing one tier; the unit on which a paper-lead-time saving accrues.                                                                                                  |
| <b>Out-of-cycle route</b>               | Domain 3's mechanism (KA 3.3.3) applied at enterprise scale, where it is worth most: the saving is largest at the slowest tier, because $E[\text{wait}] = M/2 + L$ grows with $M$ . |
| <b>Aggregation trap</b>                 | Averaging component ratios instead of summing their numerators and denominators; frequently reverses the sign.                                                                      |
| <b>Concentration line</b>               | The share of portfolio cost sitting in components below the alert threshold; the companion to any aggregate index.                                                                  |
| <b>Portfolio one-page view</b>          | The report supporting exactly three decisions: continue/change/stop, reallocate, escalate.                                                                                          |
| <b>Source-dated figure</b>              | A reported value carrying the date of its own source rather than the date of the pack.                                                                                              |

## SAMPLE MCQs — KA 15.4

**MCQ 15.4-A** [15.4.1 · Application] Five tiers have  $M$  and  $L$  of (2, 1), (4, 2), (6, 2), (12, 3) and (13, 4) weeks. The expected latency of a decision that must pass all five is:

- A. 18.5 weeks
- B. 30.5 weeks CORRECT
- C. 37.0 weeks
- D. 20.0 weeks

*Rationale:*  $(2/2+1) + (4/2+2) + (6/2+2) + (12/2+3) + (13/2+4) = 2.0+4.0+5.0+9.0+10.5 = 30.5$  (15.4.1, formula from KA 3.3.3). A halves the

intervals and omits the paper lead times entirely ( $1.0+2.0+3.0+6.0+6.5$ ); C sums the meeting intervals alone without halving them or adding the lead times ( $2+4+6+12+13$ ); D omits the executive tier.

**MCQ 15.4-B** [15.4.1 · Evaluation] A portfolio's 85 decisions generate 468 gross latency-weeks and 145 tier traversals; 25 % sit on the critical path and delay costs 14,280 a week. Cutting one week from every tier's paper lead time saves:

- A. USD 517,650 CORRECT
- B. USD 1,670,760
- C. USD 303,450
- D. USD 2,070,600

*Rationale:* A one-week cut saves one week per traversal:  $145 \times 0.25 \times 14,280 = 517,650$ , which is 30.98 % of the 1,670,760 total (15.4.1). B is the whole bill; C applies the saving to the 85 decisions rather than the 145 traversals; D omits the critical-path share.

**MCQ 15.4-C** [15.4.1 · Analysis] In that architecture, 6 of the 85 decisions carry 141 of the 468 gross latency-weeks. The strongest implication is that:

- A. those 6 decisions should be delegated downward
- B. an out-of-cycle route at the slow tiers is worth far more per decision affected than removing a fast tier CORRECT
- C. the executive board should meet more often
- D. the 6 decisions are the most important and their latency is justified

*Rationale:* 7.06 % of decisions carry 30.13 % of latency, so a written-resolution route for them saves 471,240 against 107,100 from removing the fast component-board tier (15.4.1). A may breach the reserved-matters rationale; C is the weaker lever, since a week off M saves half a week; D is a legitimate position but not an implication of the arithmetic.

**MCQ 15.4-D** [15.4.2 · Analysis] A portfolio has  $\Sigma EV$  2,349,900 and  $\Sigma AC$  2,473,000. The unweighted mean of its five component CPIs is 1.006. The portfolio CPI is:

- A. 1.01, the mean of the components
- B. 0.95 CORRECT
- C. 0.96
- D. 1.05

*Rationale:*  $CPI = \Sigma EV / \Sigma AC = 0.950222$  (15.4.2). A averages ratios, the defect the topic exists to prevent; C is the portfolio SPI ( $\Sigma EV / \Sigma PV$ ); D inverts the ratio.

**MCQ 15.4-E** [15.4.2 · Evaluation] The unweighted-average CPI flatters this portfolio because:

- A. there are five components and five is too few to average
- B. the component with 54.67 % of cost incurred is the only one below 0.95, and four small ahead-of- budget components outvote it CORRECT
- C. the components have different budgets at completion
- D.  $EAC = BAC/CPI$  is the wrong forecasting method

*Rationale:* An unweighted mean gives each component equal voice regardless of the money behind it, so it flatters whenever the large component is the troubled one (15.4.2). A is not the mechanism; C is true but not the cause, since the weighting that matters is cost incurred; D is a separate question governed by Domain 7.

**MCQ 15.4-F** [15.4.3 · Comprehension] A portfolio report should support exactly three decisions. They are:

- A. plan, monitor and control
- B. continue/change/stop, reallocate capacity, escalate CORRECT
- C. approve, reject and defer
- D. report, review and assure

*Rationale:* Those three are what a portfolio body can actually do; anything serving none of them is decoration (15.4.3).

#### SELF-CHECK — KA 15.4

1. *Which enterprise latency lever pays most, and why?* — One week off every paper lead time: it saves a full week on each of 145 traversals, 517,650 or 30.98 % of the bill, at no cost and with no scrutiny removed.
2. *How is a portfolio performance index aggregated correctly?* — By summing numerators and denominators ( $\Sigma EV / \Sigma AC$ ), never by averaging component ratios, which here reverses the sign of the reported variance at completion.
3. *Name the four things a portfolio view must not hide.* — The milestone product, the binding period, the elimination rate on claimed benefits, and the latency of the open decisions.

## Advanced topics — Domain 15

### 15.A.1 Where the boundary belongs, and what it costs to get wrong

The project/programme/portfolio boundary is treated as a naming question and is in fact a cost question, because each layer added imposes three recurring charges that

can be estimated before the layer exists. It adds **coordination cost**: Domain 4's interface arithmetic and Domain 12's coordination results apply, and neither is re-derived. It adds **latency**, at exactly KA 15.4.1's rates: a tier inserted into a path used by 22 decisions a year at 4.0 weeks of expected wait, with a 25 % critical-path share, costs  $22 \times 4.0 \times 0.25 \times 14,280 = \text{USD } 314,160$  a year before it has improved a single decision. And it adds **reporting load**, which consumes the delivery capacity KA 15.3 has just shown to be the binding constraint.

Against those charges a layer must supply something nameable: optionality (components that can be stopped or re-sequenced), a benefit that no component owns alone, or an authority that no component holds. The test is therefore symmetrical and it can be applied to an existing structure as easily as a proposed one: **name what this layer decides that no other layer can, and compare it with the layer's latency bill**. Layers that fail this test are common, and they are usually created by an organisational event rather than a delivery need (a reorganisation, a merger, an audit finding), which is why nobody remembers what they were for.

The inverse error deserves equal attention. Running a genuine portfolio problem as a programme produces the failure Case study B records: components acquire false interdependence, the allocation is never re-run, and capacity is committed once and never rebalanced. The diagnostic is whether the components' benefits are separable. If they are, it is a portfolio, and it needs the arithmetic of KA 15.2, not the coherence machinery of KA 15.1.

### 15.A.2 Rolling horizons and the option value of deferral

KA 15.2.3's enumeration is a single-shot allocation over four quarters, and real portfolios allocate on a **rolling horizon**: re-optimize each period over the remaining horizon, with committed work treated as fixed and everything else free. Three consequences matter and none of them requires new mathematics.

**Re-optimisation dominates a good one-shot plan**, because information arrives. A plan optimal on January's estimates is not optimal on April's, and the value of re-running the enumeration is the difference, which is precisely the allocation gap KA 15.3.4 asks the enterprise PMO to capture, now recurring quarterly rather than annually.

**Deferral has option value that the enumeration does not price**. A candidate deferred one period is not a candidate lost; it is a candidate whose value estimate will be better next period, and whose demand profile may fit a period that is not yet committed. Staging a candidate into a small first tranche (15.2.4) buys the same option more cheaply. The honest position is that this domain's enumeration maximises value under stated estimates and does **not** value the option; a portfolio board should be told that explicitly, because the un-priced option is the strongest available argument for admitting a candidate at low commitment rather than rejecting it.

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**Commitment discipline is what makes a rolling horizon work.** A portfolio that re-optimises freely each period will churn: components stopped and restarted, teams re-formed, and the coordination cost of Domain 12 paid repeatedly. The countermeasure is a stated commitment horizon (typically one to two periods in which the allocation is fixed absent a material event) beyond which everything is provisional and is described as provisional in every report.

### 15.A.3 The reviewer's portfolio eye

Invariants to test on any programme or portfolio, each cheap and each diagnostic.

Every committed milestone has a stated **product** of its predecessors' probabilities, with the independence assumption disclosed, rather than a list of green components. Every dependency has an owner **on the giving side**, a date needed and a date promised. Every tranche boundary corresponds to a state the organisation can operate in. The benefits register is organised **by benefit**, not by component, and carries a bridge showing the elimination rate and the breakeven elimination rate for the applicable investment rule. Every same-pool claim reconciles to establishment data with the receiving owner's signature. The binding constraint is named as a **capability**, measured as a **rate** from completions, and modelled **per period**; and the selected portfolio's feasibility is demonstrated in the tightest period, not in aggregate. The selected set is compared against an enumerated frontier, and the gap is reported. Work in progress has a stated limit with the excess priced. Every zero-slack period has its breach cost estimated and compared against the value of reserving capacity. Every escalation path has its latency computed as  $\Sigma (M/2 + L)$  and priced, and the paper lead times are the first thing examined. Aggregate ratios are computed from summed numerators and denominators and are published with a concentration line. Every reported figure carries its own source date. Every AI-produced portfolio output appears in the AI use register with its verifier named. And the one test that subsumes several others: **the portfolio's approved benefit total reconciles, line by line, to the receiving organisations' own budgets**; because a benefit no budget holder has accepted is a benefit nobody will deliver.

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## Industry variations — Domain 15

- **Public sector and government.** Annual appropriation cycles impose a hard multi-period constraint that cannot be smoothed by borrowing forward, so KA 15.2.3's period feasibility is binding in law rather than in practice; benefits are frequently non-cash and non-fungible, so the bridge eliminates against policy outcomes rather than budgets, and the same-pool correction is politically the hardest because establishment reductions are announced separately from the programmes that enable them.

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- **Regulated industries (pharmaceutical, nuclear, aviation, financial services).** Some milestone predecessors are external approvals whose probabilities the portfolio cannot influence at any price, which makes decoupling the *only* structural lever available and makes KA 15.1.4's cost comparison decisive; reserved-matters classes are larger, so the latency concentration of KA 15.4.1 is more extreme and the out-of-cycle route correspondingly more valuable.
  - **Construction and infrastructure.** The binding capability is often a physical resource (a tunnelling machine, a crane, a rail possession), whose period constraint is absolute and whose reallocation cost is large, so protective capacity usually *does* pay on KA 15.3.3's test because a breach cascades into a possession window that cannot be re-booked within the year.
  - **Healthcare.** Clinical session time is the binding capability and it is not fungible with money; clinical authority is not delegable to a portfolio body, so Meridian's decoupling decisions run through clinical governance in parallel, and the same-pool over-claim is endemic because clinician time released is the benefit almost every digital component claims.
  - **Energy and utilities.** Regulatory price-control periods create hard multi-period envelopes and a benefits register dominated by avoided costs, which double-count with unusual ease: case study B is this pattern; commissioning crews are the classic capability constraint and their peak-month demand is where portfolios fail.
  - **Technology and product organisations.** Capacity is measured in team-quarters and the portfolio is genuinely re-optimisable, so the rolling-horizon discipline of 15.A.2 is the dominant practice; the characteristic failure is unlimited work in progress, because starting is cheap and visible while finishing is neither, which makes KA 15.3.2's arithmetic the single highest-value conversation available.
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### Case study — Domain 15: the milestone that was never going to happen (health, Meridian)

**Situation.** Meridian's programme board had committed, to its funder and to a patient-facing communications campaign, that all four regions would go live on the same date. Eleven weeks out, the programme's status report showed 24 component dependencies, every one rated green or amber-improving, and a green overall milestone. The programme director asked a question nobody had asked: *what is the probability of the committed date?*

**What the arithmetic showed.** The six dependencies of a single region, at the owners' own assessments (0.90, 0.88, 0.85, 0.92, 0.95 and 0.90) multiply to **52.95 %**. Four structurally identical regions on one date give **7.86 %**: the committed milestone was **92.1 % likely to be missed**, and the expected number of on-time regions was

**2.1182** of four. Nothing in the component reporting was wrong. Every owner had assessed their own item honestly, and no one had multiplied.

**The two proposals, priced against each other.** The delivery directorate proposed **USD 240,000** of recovery spend (additional trainers and an accelerated estate package) assessed to lift training from 0.95 to 0.98 and estate works from 0.92 to 0.96. That takes a region to **57.00 %** and the programme milestone to **10.56 %**: **2.69** percentage points for 240,000, or **USD 89,093 per point**. The programme office proposed two structural changes instead. Historical data migration (0.85) came off the go-live milestone, operated for two weeks on a documented paper fallback for historical lookups at a priced **USD 18,000** of temporary clerical cover, taking a region to **62.30 %**. Information-governance approval (0.90) was obtained **once at programme level** for all four regions rather than four times regionally, taking a region to **69.22 %** and the four-region milestone to **22.96 %**: **15.10** points for 18,000, or **USD 1,192 per point**, **1.34 %** of the cost per point of the money route, a ratio of **74.7 to one**.

**How it resolved.** The board took both structural changes and declined the recovery spend. It then did the thing the arithmetic made unavoidable: at 22.96 % the simultaneous commitment was still not defensible, so the four regional dates became **internal** dates and the external commitment moved to a single programme milestone set after the fourth region with a buffer sized by Domain 8's methods. The paper fallback went through clinical governance before it went into the plan. Expected late regions fell from **1.8818** to **1.2312**, and the programme reported a milestone probability, an independence assumption and a buffer, in place of twenty-four green squares.

**What the domain teaches here.** A programme status report that lists green components against a committed date is internally inconsistent, and the inconsistency is one multiplication away from being visible. Once it is visible, the cheap lever is structural: **on a milestone whose probability is a product, removing a factor is worth roughly seventy times what buying improvement in a factor is worth**. And note what the board could not do; it could not estimate its way to 80 %, because that would have required 99.07 % on each of 24 dependencies, a standard nobody in the supply chain had offered or could.

## Case study B — Domain 15: the portfolio that was fully funded and could not deliver (energy and utilities)

**Situation.** A regional transmission utility, anonymised, approved **11** grid-connection projects for a three-year regulatory period. The scarce capability was high-voltage commissioning crews, of which the utility had **9 crew-weeks a month**: **324** crew-weeks over 36 months. The approved portfolio demanded **311** crew-weeks, **95.99 %** of aggregate capacity, and the investment committee approved it on precisely that basis: the work fitted. Claimed benefits were **USD 24,600,000** a year.



**What had happened.** Two defects, both detectable before approval. **The constraint was never tested by period.** Demand peaked at **17 crew-weeks a month for six consecutive months** in months 14 to 19, **188.89 %** of capacity, against which the annual view was silent. Unmet demand in that window was  $(17 - 9) \times 6 = 48$  crew-weeks, and clearing 48 crew-weeks at the observed throughput of 9 a month takes **5.3333 months** of pure catch-up. **The portfolio was therefore a minimum of five and a third months late in its approved form, before any project encountered a single problem.** And **the benefits were never bridged.** Three projects each claimed the same **3,100,000** of avoided constraint payments and two each claimed the same **1,850,000** of deferred reinforcement: eliminations of **USD 8,050,000** against a claimed 24,600,000, an elimination rate of **32.72 %** and a net portfolio benefit of **USD 16,550,000**.

**How it resolved.** Six projects completed; five slipped past the regulatory window and carried **USD 7,400,000** a year of the net benefit. Benefit forgone over the 5.3333-month structural catch-up was  $7,400,000 \times 5.3333/12 = \text{USD } 3,288,889$ , and the portfolio delivered **USD 9,150,000** of its 16,550,000 on time, **55.29 %**. A re-sequencing exercise then did what the approval process had not: it enumerated feasible sets against the **monthly** constraint and deferred two projects to the following period. Nine projects, carrying **USD 15,000,000**, landed inside the window — **90.63 %**: with the deferred two carrying 1,550,000. **Re-sequencing alone moved USD 5,850,000 of annual benefit from late to on time, and it changed no project's scope, budget or team.**

**What the domain teaches here.** Aggregate capacity feasibility is not feasibility; a portfolio at 95.99 % of annual capacity was at 188.89 % of monthly capacity for half a year, and one afternoon of period-by-period arithmetic before approval would have found it. A benefits register organised by component rather than by benefit will let the same avoided cost be claimed three times, and a 32.72 % elimination rate is not an outlier. And the largest single improvement available to this portfolio was not more crews, more money or better project management: it was **sequence**, which is free.

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## Executive perspective — Domain 15

What a portfolio director cannot delegate in this domain:

- **The product behind every committed milestone.** Insist on  $\Pi p_i$  with its independence assumption stated, not a count of green components. A programme that cannot produce it has not assessed its own commitment (15.1.3).
- **The named binding constraint, measured as a rate and modelled by period.** Not a budget, not an annual total, not a headcount. Feasibility is demonstrated in the tightest period or it is not demonstrated (15.2.3, 15.3.1).

- **The benefits bridge before approval.** The elimination rate, the breakeven elimination rate for your investment rule, and a receiving owner's signature against every capped figure. Meridian's 18.41 % elimination rate broke a four-year payback rule whose breakeven was 12.03 % (15.2.1).
- **The work-in-progress decision, which is a prioritisation decision you are avoiding.** Twelve initiatives in flight at a throughput of five costs 1.40 years of the entire benefit stream: 1,584,072, forgone once. Saying that all twelve are priorities is arithmetically saying none is (15.3.2).
- **Your own decision architecture's annual bill.** 468 gross latency-weeks, 1,670,760 a year, and a free 517,650 available from one week off every paper deadline. Examine the paper deadlines and the slow tiers, not the tier that is easiest to remove (15.4.1).
- **How your portfolio's performance is aggregated.** Sums of numerators and denominators, with a concentration line. An averaged index moved this portfolio from 30,592 favourable to 255,640 adverse, and the flattering direction is the normal one (15.4.2).

## Calculation exercises — Domain 15

**Exercise 15.1** A programme milestone requires five independent predecessors, assessed at 0.96, 0.93, 0.90, 0.88 and 0.80. (a) Compute the milestone probability. (b) Compute it again with the 0.80 dependency decoupled. (c) If instead every dependency were assessed at 0.93, how many could the milestone carry and remain at or above 0.50? (d) What uniform per-dependency probability would deliver 0.85 across five dependencies? *Solution.* (a)  $0.96 \times 0.93 \times 0.90 \times 0.88 \times 0.80 = 56.5678\%$ . (b) Divide by 0.80: **70.7098 %**, a gain of **14.1420** percentage points from one structural change. (c)  $0.93^9 = 52.0411\%$  and  $0.93^{10} = 48.3982\%$ , so at most **nine**. (d)  $0.85^{(1/5)} = 96.8019\%$ . Common error: answering (a) with the minimum, **0.80**, or with the mean of the five, **89.40 %**. Both of which are above the true figure and both of which are what a component-by-component status report implicitly asserts.

**Exercise 15.2** A portfolio's scarce capability supplies **5 units a period** for three periods (15 units in aggregate). Four candidates: P demands (2, 2, 2) for an NPV of 1,020,000; Q demands (2, 2, 1) for 850,000; R demands (4, 1, 0) for 950,000; S demands (1, 1, 2) for 640,000. Compute the selection under (a) ranking by NPV per unit with a period feasibility check, (b) full enumeration, and (c) ranking by NPV per unit against aggregate capacity only. State the cost of each error. *Solution.* Ratios: R **190,000**, P **170,000**, Q **170,000**, S **160,000**. (a) Take R, leaving (1, 4, 5); P and Q are both blocked in period 1 and skipped; S fits. **R + S = USD 1,590,000**,

using 9 of 15 units. (b) Enumerating all  $2^4 - 1 = 15$  subsets, the maximum feasible set is  $P + Q$

•  $S = \text{USD } 2,510,000$  with demand exactly (5, 5, 5). The heuristic leaves  $920,000, 36.65\%$  of the optimum. (c) Aggregate-only ranking takes R, P and S for 15 of 15 units and  $2,610,000$  (apparently the best answer and fully utilised), but period 1 demands  $4 + 2 + 1 = 7$  units against  $5^*$ , so it is infeasible and overstates achievable value by 100,000. Common error: reporting (c) as the answer because it uses 100 % of capacity; full utilisation of an annual total is evidence of nothing.

**Exercise 15.3** Four components claim **1,240,000** of annual benefit at full potential. Review finds: 85,000 of benefit claimed twice; claims of 4.8 posts against a pool of 3.5 valued at 38,000 each; 64,000 of enabler benefit already counted downstream; and 45,000 already committed outside the portfolio. Adoption is 65 %; portfolio cost is 3,150,000; the investment rule is a four-year simple payback. Compute the elimination rate, the net realistic benefit, the payback on both the reconciled and unreconciled figures, and the breakeven elimination rate. *Solution.* Eliminations  $85,000 + (4.8 - 3.5) \times 38,000 + 64,000 + 45,000 = 85,000 + 49,400 + 64,000$

•  $45,000 = \text{USD } 243,400^*$ , an elimination rate of  $19.6290\%^*$ . Net full potential  $996,600^*$ ; at 65 % adoption,  $\text{USD } 647,790^*$ . The unreconciled figure is  $1,240,000 \times 0.65 = 806,000^*$ , overstating by  $158,210^* - 24.42\%^*$  of the honest number. Payback: honest  $4.8627 \text{ years}^*$ ; unreconciled  $3.9082 \text{ years}^*$ , which clears the rule. Breakeven: the rule needs  $3,150,000/4 = 787,500$ , so  $787,500/0.65 = 1,211,538.46$  of net full potential and at most **28,461.54** of eliminations, a breakeven elimination rate of **2.2953 %**. Common error: computing the elimination on the same-pool claim as the pool's value ( $3.5 \times 38,000 = 133,000$ ) rather than as the excess over it; the excess is 1.3 posts.

**Exercise 15.4** Four governance tiers have M and L in weeks of (2, 1), (4, 1), (8, 2) and (12, 4). Decision volumes: **40** through one tier, **18** through two, **7** through three, **3** through four. 30 % of decisions sit on the critical path; cost of delay is **11,500** a week. Compute the gross latency-weeks, the annual delay bill, and the saving from cutting one week from every paper lead time. Then state what proportion of the latency the top class carries. *Solution.* Tier latencies  $2/2+1 = 2.0$ ,  $4/2+1 = 3.0$ ,  $8/2+2 = 6.0$ ,  $12/2+4 = 10.0$ ; cumulative **2.0, 5.0, 11.0, 21.0**. Gross  $40(2.0) + 18(5.0) + 7(11.0) + 3(21.0) = 80 + 90 + 77 + 63 = \mathbf{310 \text{ latency-weeks}}$ . Delaying weeks  $310 \times 0.30 = \mathbf{93.0}$ ; bill  $93 \times 11,500 = \text{USD } \mathbf{1,069,500}$ . Tier traversals  $40 + 36 + 21 + 12 = \mathbf{109}$ , so a one-week cut in every paper lead time saves  $109 \times 0.30 \times 11,500 = \text{USD}$

**376,050: 35.1613 %** of the bill. The top class is **3 of 68** decisions (**4.4118 %**) carrying **63 of 310** weeks (**20.3226 %**). Common error: adding meeting intervals and omitting paper lead times, which gives cumulative 1.0, 3.0, 7.0 and 13.0 and a total of **182** weeks, an understatement of **41.2903 %**.

**Exercise 15.5** A delivery organisation completes **8** initiatives a year at steady state and has **20** in flight. Its portfolio's net benefit run rate is **2,240,000** a year. Compute the average cycle time, the cycle time under a work-in-progress limit of **8**, the benefit consequence of the excess, and the limit that would deliver a 1.5-year cycle time. *Solution.*  $T = W/C = 20/8 = 2.50$  years; at  $W = 8$ ,  $T = 1.00$  year; excess **1.50 years**. The whole benefit stream therefore arrives 1.50 years later than it need, worth  $2,240,000 \times 1.5 = \text{USD } 3,360,000$ , forgone once and never recovered. For  $T = 1.5$ ,  $W = C \times T = 8 \times 1.5 = 12$ . Common error: assuming that cutting work in progress from 20 to 8 cuts throughput in proportion, Little's Law says throughput is set by capacity, not by how much work is started (KA 13.2.3), so the same eight initiatives finish each year under either policy and the only thing that changes is when.

## Practitioner's toolkit – Domain 15

*Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.*

### Toolkit 15.T.1 – Programme dependency register with a computed milestone probability

One row per dependency: reference · milestone it feeds · what is needed · **owner on the giving side** (name, not function) · owner on the receiving side · date needed · date promised · assessed probability of being met on the date needed · **evidence for that assessment** · shared cause or driver · consequence of breach · decoupling option and its priced cost. A footer per milestone computes  $\prod p_i$  across its predecessors, states the independence assumption, lists the shared drivers that violate it, and shows the probability after each candidate decoupling with its cost, so the board sees the product, the assumption and the priced structural options on one page. The register's purpose is to make the difference between component status and milestone probability **impossible to overlook**; a programme whose report shows green components and no product has not filled it in.

### Toolkit 15.T.2 – Multi-period portfolio allocation sheet

A grid: candidates down the side; the binding capability's demand **per period** across the top, plus total units, value, value per unit, and the exposure tags (supplier, platform, team, regulatory) used as feasibility limits. Below it: capacity per period as a measured rate with its measurement window; per-period demand and slack for the

selected set; and the **feasible frontier**, the top three to five feasible sets, their values, and the binding period for each rejected higher-value set. Three mandatory lines: the value of the selected set, the value of the enumerated optimum, and the gap between them; the aggregate-only answer and its infeasible period, recorded so the illusion is visible rather than merely avoided; and the plan's survival probability from the period slacks with the breach cost estimated. Re-run every cycle with committed work fixed and remaining demand and remaining value used for work in flight.

### Toolkit 15.T.3 – Portfolio benefits bridge

A single reconciliation, top to bottom, published with every investment paper: gross claimed benefit at full potential by component, **each line carrying its fund type, cashable (with the budget line named) / capacity-released / cost-avoidance / non-financial**; then a line per elimination, each tagged **shared / same-pool / cascade / already-committed**, with the components affected, the basis, and the receiving owner who has accepted it; then net full-potential benefit and the **elimination rate**; then the adoption adjustment with its source; then net realistic benefit, **subtotaled by fund type**. Beneath it: the investment rule being applied, **the composition of the stream the rule is being applied to**, the result, and the **break-even elimination rate** at which the rule just holds. Four integrity checks, all counts: same-pool claims reconciled against establishment data; benefits accepted in writing by the receiving budget holder; **benefits classified cashable and confirmed in writing by the finance owner of the named budget line**; and **benefits classified cashable but not yet so confirmed**, which are carried as capacity-released until they are (15.2.1). A bridge whose eliminations no receiving owner has signed is a calculation, not a reconciliation; a bridge that applies a cash test to an unclassified stream is a calculation whose denominator nobody has disclosed.

## Exam preparation – Domain 15

**What is assessed.** The project/programme/portfolio distinction by decision rather than scale; programme architecture in components, tranches and a target operating state; the dependency multiplication rule and the structural levers; the four portfolio double counts and the benefits bridge; portfolio balancing under a constraint that binds per period; enterprise capacity as a measured rate; portfolio work in progress and its cost; protective capacity and its breakeven; the enterprise PMO's recurring value test; enterprise decision latency and its redesigns; and correct aggregation of portfolio performance.

**The calculations to be able to do under time pressure.**  $\prod p_i$  for a milestone, and the  $k$ -th root of a target confidence to get the required per-dependency probability. The effect of decoupling, as a multiplication by  $1/p_i$ . Per-period feasibility of a candidate set, and enumeration over a small candidate list. A benefits bridge with all

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four elimination types, the elimination rate, and simple payback before and after.  $T = W/C$  and the excess-work-in-progress cost as benefit run rate  $\times$  excess  $\div$  throughput. Plan survival as a product over periods, and the breakeven breach cost. Gross latency-weeks from decision classes and cumulative tier latency  $\Sigma (M/2 + L)$ , priced at a cost of delay, and the saving from a one-week cut per traversal. Portfolio CPI and SPI from summed EV, AC and PV, and  $EAC = BAC/CPI$ .

**The traps.** Taking a milestone probability as the average or the minimum of its predecessors' rather than their product (Exercise 15.1) · treating aggregate capacity feasibility as feasibility (Exercise 15.2c, Case study B) · ranking by value per unit of constraint and assuming the result is optimal when demand is concentrated in the binding period (15.2.3, Exercise 15.2) · computing a same-pool elimination as the pool's value rather than the excess over it (Exercise 15.3) · omitting the adoption adjustment or applying it twice (MCQ 15.2-C) · booking eliminated double counts as a saving (15.2.1, 15.3.4) · assuming a work-in-progress cut reduces throughput (Exercise 15.5) · converting a one-off benefit shift into an annuity (MCQ 15.3-B) · justifying a support function on one-off value (15.3.4) · adding meeting intervals without paper lead times in a multi-tier path (Exercise 15.4) · applying a per-traversal saving to the decision count instead of the traversal count (MCQ 15.4-B) · averaging component ratios instead of summing numerators and denominators (15.4.2) · reporting a single aggregate index with no concentration line · omitting the independence assumption when a product is printed.

**How the domain connects.** Domain 1 supplies the accountability principle and the cost of delay every figure here is priced at. Domain 2 supplies the business case, the benefits map, the kill criteria and the greedy-versus-enumeration result this domain extends to multiple periods. Domain 3 supplies  $E[\text{wait}] = M/2 + L$ , the paper-deadline lever and the tier-summation this domain scales to a whole decision architecture. Domain 4 supplies the interface arithmetic and the integrated baseline a programme's components must reconcile to. Domain 6 consumes programme dependency dates as schedule inputs. Domain 7 supplies the EVM definitions aggregated in KA 15.4.2. Domain 8 supplies merge bias (the schedule form of this domain's multiplication rule) the common-driver treatment that qualifies independence, and the methods that size a programme buffer. Domain 12 supplies the coordination cost of every layer added. Domain 13 supplies Little's Law, applied here to portfolio work in progress. Domain 14 supplies the AI use register, the verification tier and the information-age discipline. Domain 16 measures whether the benefits this domain reconciled were realised. PFL-AI Domain 4 supplies the appraisal methods behind the candidate values.

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## Domain 15 summary

A portfolio is not a bigger project, because at enterprise scale quantities stop adding. They multiply, they compete for the same period, and they double-count. And each of those has an arithmetic this domain computes.

**They multiply.** A milestone requiring all of  $k$  independent predecessors is met with the **product** of their probabilities. Meridian's Region A go-live, with six predecessors none worse than 0.85, stands at **52.95 %**; four such regions on one date stand at **7.86 %**, a commitment **92.1 %** likely to be missed while all twenty-four components report green. Better estimating cannot fix it: 80 % across 24 dependencies demands **99.07 %** on each. Structure can: decoupling migration and centralising the information-governance approval took a region to **69.22 %** and the programme milestone to **22.96 %** for **USD 18,000**, against **USD 240,000** of recovery spend that bought 2.69 points: **USD 1,192** a point against **USD 89,093**, a ratio of **74.7 to one**.

**They compete for the same period.** Meridian's portfolio has 24 units of scarce integration capacity a year, and the annual view approves a set worth **4,200,000** that demands **9 units in a 6-unit quarter**. Ranking by value per unit is feasible but takes **2,800,000**, because the highest-ratio candidate consumes five of six units in the binding quarter and excludes the two largest; enumerating 63 subsets finds **26 feasible** and an optimum of **3,810,000**. The aggregate illusion overstates by **390,000 (10.24 %)**; the heuristic leaves **1,010,000 (26.51 %)** on the table. That optimum runs at **95.83 %** planned utilisation and survives all four quarters with probability **31.56 %**; reserving a unit a quarter lifts survival to **76.31 %** at a cost of **720,000** of value, which does **not** pay unless a breach costs more than **USD 1,608,744**, a condition a regulatory or go-live date meets easily and an ordinary quarter does not. And starting twelve initiatives against a measured throughput of five gives a **2.40-year** cycle time instead of **1.00**, deferring the portfolio's whole **1,131,480** benefit stream by **1.40 years: USD 1,584,072**, forgone once, invisible in every annual report.

**They double-count.** Five components claiming **1,981,200** at full potential contain **364,800** of shared, same-pool, cascade and already-committed benefit (an **18.41 %** elimination rate, invariant to the adoption assumption), leaving **1,131,480** a year net at 70 % adoption. That moves simple payback from **3.5188** to **4.3129 years** and fails a four-year rule whose breakeven elimination rate was **12.03 %**. Eliminating a double count creates no value; it prevents a wrong decision, which is why it cannot appear in the enterprise PMO's value case: a case that here shows a **89,390** year-one surplus on a one-off and a **476,350** year-two deficit, closable only by capturing **47.16 %** of the recurring allocation gap. And the reconciled figure is still a mixed stream: every registered benefit carries a **fund type** (cashable with its budget line named, capacity-released, cost-avoidance or non-financial) the **finance owner of the affected budget** confirms cashability in writing or the claim is carried as capacity-released, and a cash test applied to the total is disclosed as what it is, a committee convention over a mixture whose composition is printed beside the result.



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**And the decisions themselves have a price.** Five tiers, 85 decisions and **468 gross latency-weeks** cost **USD 1,670,760** a year at Meridian's cost of delay. One week off every paper deadline saves **517,650 (30.98 %)** and costs nothing; a written-resolution route for the **6** decisions that carry **30.13 %** of the latency saves **471,240**; removing the fast component-board tier saves **107,100**. Remove latency where the latency is. Then report it correctly: the same portfolio reads **CPI 0.95** when aggregated as  $\Sigma EV / \Sigma AC$  and **1.01** as an unweighted mean of component indices, swinging variance at completion by **USD 286,231** and reversing its sign, because the component holding **54.67 %** of the cost incurred is the only one in trouble.

The through-line: **project intuitions fail upward in four specific, computable ways (probabilities multiply, constraints bind per period, benefits overlap, and governance tiers accumulate), and a leader who computes all four can defend a portfolio to a board, while one who computes none of them will be asked to commit to a milestone that is 7.86 % likely and to call it green.**



DOMAIN

# 16

## Transition, Closeout and Benefits Realisation

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## Why this domain exists

Fifteen domains have described how to choose work, structure it, plan it, price it, staff it, lead it and control it. Every one of them assumed something that only this domain can supply: **that the project ends, that something else takes it over, and that somebody eventually finds out whether it was worth doing.** Until that happens, a project's entire claim to value rests on an estimate, and the organisation's memory of it rests on whoever is still in post.

Closeout is, on this book's argument, the least respected phase in the delivery lifecycle and the one in which value is most easily lost beyond recovery. The failure modes below are the ones a reviewer meets repeatedly, and they are not primarily about paperwork. Handover happens on a date rather than on a condition, so a receiving organisation that is 90 per cent ready absorbs a system it cannot run. Contracts are left open because closing them requires a conversation nobody wants, and the carrying cost of that avoidance is never added up. Lessons are captured diligently and never retrieved, so the same defect is bought again at full price on the next programme. And benefits (the entire justification for the expenditure) are measured, if at all, against a baseline that was never taken, by an owner who was never appointed, over a period after everyone who could explain the variance has moved on.

That is the domain's central claim: **the closing account is the only honest one, and it is settled after the project ends, not at it.** A leader who cannot state what was realised, against what was promised, decomposed into the reasons for the difference, has not completed the work. They have merely stopped doing it. This domain makes that account computable. KA 16.1 replaces the handover *date* with a handover *condition* and shows arithmetically why partial readiness is not proportional readiness. KA 16.2 takes the receiving organisation's run cost seriously and prices the final account, the retention mechanism and the specific cost of leaving a claim open. KA 16.3 treats knowledge as an asset with a retrieval rate, and shows that the value of a post-project review is destroyed at the retrieval step, not the capture step. KA 16.4 settles Meridian's account, decomposes the variance, and sets the retention and deletion rules for the evidence, data and models the programme leaves behind.

**Learning objectives.** After this domain a candidate can: specify handover as a set of conditions rather than a date, and distinguish what handover transfers from what it cannot; design commissioning and service-acceptance tests that test the service rather than the works; **separate the mandatory preconditions of a transition, which are recorded met or not met and never traded, from the discretionary conditions; compute the probability of a clean go-live as the conjunction of the discretionary conditions, demonstrate why an averaged readiness dashboard systematically overstates it, and rank remediation by proportional**

**rather than absolute shortfall**; price a hold-or-go decision against remediation cost under both independent and correlated failure assumptions; structure hypercare with exit criteria and a reversion plan; transfer a service to a run organisation with its total cost of ownership stated; operate the retention mechanism and strike a final account including variations, claim settlement and defect recovery; **compute the monthly carrying cost of an open claim and the breakeven settlement premium**; run a post-project review whose lessons are retrievable, and compute the breakeven reuse rate that makes it worth holding; build a benefits measurement plan with a named owner, defined measures, a pre-change baseline, a **fund type carried forward from the approval** and a stated duration; **decompose realised benefit against the business case into adoption, benefit-per-unit and timing components, state the interaction term and the convention that assigns it**; test attribution against a comparison cohort established at baseline; set a class-by-class retention and deletion schedule for evidence, personal data and the models used in delivery decisions; and govern AI use in measurement, review and archive without letting a tool author the conclusion it is measuring.

**The master threads, and the figures this domain settles.** Meridian Care Records returns for the last time: the clinical-records rollout to **40 clinics**, an approved cost of **USD 2,400,000**, a business case claiming **USD 979,200** a year against Domain 1's honest **USD 685,440** at 70 per cent adoption, a cost of delay of **USD 14,280** a week, and Domain 2's appraisal over eight years at 7 per cent giving a present value of **USD 3,732,898** on the honest ramped profile and an NPV of **+USD 1,332,898**. This domain measures what actually happened: **27 of 40 clinics** in daily use at **5.4 clinician-hours** released a week, worth **USD 594,864** a year; and decomposes the **USD 90,576** shortfall against the honest case, prices the operating cost the case omitted, and computes the realised NPV of (**USD 304,827**) and the **USD 180,000** of post-project work that turns it positive. Project Auriga returns from Domains 6 to 8 (the 25-week control-systems upgrade for a regional utility, **BAC USD 4,000,000**, **CPI 0.91** and **SPI 0.92** at week 13) to have its commissioning test and final account settled in Case study B. The benefits measure used throughout is the benefits register's benefit measure (**EVA(benefit)** in the shared symbol table, written in words here to avoid the earned-value clash).

**Reference points.** Two areas of this domain have external reference points worth naming, and both are named rather than relied upon. For the archive and retention material of KA 16.4.4, **ISO 15489** is the document usually named as addressing records management: what a record is, and the controls that keep it authentic, reliable and usable over time. For the operational-transition material of KA 16.2, **ISO/IEC 20000** is the document usually named as addressing service management, which is the discipline the receiving organisation is running when it accepts a service into support. Both are voluntary guidance: neither is legislation, neither obliges anyone of itself unless an organisation, a contract or a regulator adopts it, and neither states any retention period applicable to any organisation — retention requirements come from the applicable regime, the contract and the organisation's

records and legal functions, as KA 16.4.4 says throughout. They are named here and not reproduced, and a reader who wants either should obtain the current edition from its publisher. Naming them implies no endorsement in either direction.

#### KNOWLEDGE AREA 16.1

## Handover, commissioning and readiness

*Topics: 16.1.1 what handover transfers · 16.1.2 commissioning and service acceptance · 16.1.3 readiness as a conjunction of conditions · 16.1.4 the go-live decision, hypercare and reversion.*

### 16.1.1 What handover transfers

**Definition.** Handover is the transfer of a deliverable, and of the accountability for operating it, from the delivering organisation to a named receiving organisation, on stated conditions, with the supporting obligations of both parties recorded.

Three things transfer, and they are habitually confused. **Custody** of the asset or service, who holds it, runs it, pays for it. **Accountability for its operation**, who answers when it fails at 02:00. And **the residual obligations of the deliverer**, defects liability, warranty, documentation, training, transition support. Each has a different date, and a handover that names one date for all three has hidden two of them.

Two things do **not** transfer, and pretending otherwise is the commonest and most expensive handover error. **Accountability for the benefit does not transfer at handover, because it was never the project's to begin with:** Domain 2 placed it with a benefits owner in the receiving organisation from the outset, and handover is where that appointment is tested rather than where it is made. And **the deliverer's accountability for what it built does not transfer either:** a defect present at handover is a defect the deliverer owns, and the retention and defects-liability mechanisms of 16.2.2 exist precisely because signature is not absolution.

**Handover on a condition, not a date.** The single structural choice that determines whether a transition works is whether the handover trigger is a calendar date or a satisfied condition set. A date creates an incentive to declare readiness; a condition set creates an incentive to achieve it. The condition set must be agreed early enough that it can still influence the work — which in practice means at the same gate that approves the delivery plan (Domain 3, KA 3.3.1) — and it must be written so that each condition is objectively assessable by someone who is not the person delivering it.

### 16.1.2 Commissioning and service acceptance

**Definition.** Commissioning is the progressive proving of a deliverable in its operating environment, from component test through integrated test to performance demonstration under representative load; service acceptance is the receiving organisation's formal agreement that the service, not merely the works, meets the agreed specification.

The distinction between **works tests** and **service tests** is where most acceptance regimes fail. A works test asks whether the thing was built as specified. A service test asks whether the organisation can run it: whether the runbook is correct, the alerts reach a person, the second-line team can restore from backup within the stated window, and the fallback works when the primary does not. A programme that has passed every works test and no service test has proved that it built the right object and knows nothing about whether it delivered a service. Domain 9's acceptance and nonconformance machinery supplies the discipline for both; the addition here is that the **service test must be witnessed and run by the receiving organisation**, because a service test performed by the project team tests the project team.

**Three commissioning disciplines** carry disproportionate weight. **Test on production configuration**, since a test in a staging environment that differs from production tests the staging environment: the defect most often found on the first live day is a configuration difference nobody listed. **Test the failure paths, not only the success paths**: reversion, degraded mode, manual fallback, and the restoration of service after an outage, because those are the paths that will be used under stress by people who have never rehearsed them. And **record the evidence in a form that survives the team**: witnessed test records, versioned configuration baselines (Domain 4, KA 4.3.2), and the as-built documentation that 16.3.1 shows costs seven times as much to reconstruct later.

### 16.1.3 Readiness as a conjunction of conditions

**Definition.** Readiness is the joint satisfaction of every condition necessary for a clean transition. It has **two blocks**, and only one of them is arithmetic.

**The gate block: mandatory preconditions.** Some conditions of a transition are not assessments of likelihood at all. A **clinical safety case and its hazard log closed by clinical governance**; a **regulatory or licence approval granted** by the body empowered to grant it; a **data-protection or privacy assessment signed** by the accountable authority; **notifications required of the organisation made**; **contractual takeover certificates issued**. Each of these is recorded **met or not met**, with the **approving authority named** and dated. None of them carries a probability, none of them is weighted, and none of them is available for economic trade against the cost of delay. The rule is one line and it has no exceptions clause: **while any item in the gate block is not met, the transition does not proceed, whatever the arithmetic below says**. The gate block is also not the project's to

close. Each item is closed by the authority that owns it, which is the point of naming that authority on the certificate rather than the project role that chased it.

**The discretionary block: conditions that admit degree.** What remains (training coverage, migration reconciliation, device installation, workflow sign-off, champion recruitment, rehearsal of the fallback) is genuinely uncertain at the decision date, genuinely a matter of more or less, and genuinely tradeable against the cost of waiting. Readiness across *these* is a **conjunction**, and its probability is the product of the conditions' probabilities: not their average, not their weighted average, and not the percentage of conditions that are green. Everything that follows in this KA is about the discretionary block; the gate block sits above it and is answered before it is reached.

Keeping the two apart is the professional content of this section, because the failure is not that anyone consciously prices a safety approval. It is that a single readiness template invites every condition to be entered as a  $p_i$ , and a mandatory precondition given a probability at all has been silently converted into a chance that something forbidden is permissible. Two consequences follow. A gate-block item must never appear among the  $p_i$ , because a product that contains it can be raised by improving something else, which is exactly the trade the item exists to forbid. And a gate-block item must never appear in the go/hold economics of 16.1.4, because expressing it as a cost concedes that some cost of delay would be large enough to buy it. Domain 14 states the same boundary for security and privacy controls: expected annual loss is a legitimate test where consequences are cost-compensable and the **wrong test** where the consequence is a penalty imposed by an authority, a licence condition, a duty of confidence or harm to a person (KA 14.4.4); and Domain 9 states it for quality, where the economic optimum is taken among *compliant* options rather than across them. Domain 16 is where the boundary matters most, because this is the decision that is irreversible for the people who did not take it.

The conjunction is then the most consequential piece of arithmetic in the domain, because the artefact that organisations actually use (a readiness dashboard showing a weighted percentage) is not a measurement of readiness at all. The product rule itself is Domain 15's (KA 15.1.3, where a six-predecessor programme milestone with no predecessor assessed worse than 0.85 comes to 52.95 per cent); what this KA adds is the contrast with the averaged dashboard, the remediation ranking that follows from it, and the two correlation bounds the go-live decision must be tested against. With  $k$  conditions each holding with probability  $p_i$ , and treating their failures as independent,

$$P(\text{clean transition}) = \prod p_i \quad \text{for } i = 1 \dots k$$

whereas a dashboard reports  $(\sum p_i)/k$  or some weighted variant of it. Those two numbers diverge quickly and in one direction: the product is always at or below the average, and the gap widens with both the number of conditions and their dispersion. A programme with many conditions, most of them nearly complete, will show

a comfortable average and carry a poor probability of a clean transition, which is the precise mechanism by which "we were 93 per cent ready" and "the go-live failed" are both true statements about the same event.

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#### WORKED EXAMPLE

##### Worked example 16.1.3 — the readiness dashboard that said 90.71 per cent.

1. **Setup.** Meridian's clinic go-live gate carries a gate block and **seven** discretionary conditions. The gate block (the clinical safety case and its hazard log closed by clinical governance, the information-governance assessment signed by the accountable authority, and the clinic's takeover certificate issued) was recorded **met**, authority named and dated, before the readiness assessment was convened; those items carry no probability and are not among the seven, which is why no clinical-safety condition appears in the list below. The transition manager's assessment of the probability that each *discretionary* condition is fully met at the planned go-live date, for a representative clinic, is: clinician training completed to the roster threshold **0.96**; data migration reconciled and signed off **0.90**; interfaces verified in production **0.94**; network and devices installed and load-tested **0.98**; workflow redesign signed off by the clinical lead **0.85**; clinical champion in post and trained **0.80**; fallback and business-continuity procedure rehearsed **0.92**. These are locally calibrated planning figures from the four pilot clinics, not constants. The programme dashboard reports readiness as the equal-weight average.
2. **Formula.** Dashboard reading =  $(\sum p_i)/k$ . Probability of a clean go-live =  $\prod p_i$ .
3. **Substitution.** Average =  $6.35/7$ . Product =  $0.96 \times 0.90 \times 0.94 \times 0.98 \times 0.85 \times 0.80 \times 0.92$ .
4. **Result.** Dashboard reading **90.71** %. Probability of a clean go-live **49.79** %. The gap is **40.92 percentage points**.
5. **Interpretation.** The dashboard is not optimistic; it is measuring a different quantity, and the quantity it measures has no operational meaning. No one is interested in the average completeness of the conditions, only in whether all of them hold. Four consequences follow, and all four are actionable. **First, the running product is the useful display.** Reading the conditions in sequence, the conjunction falls  $96.00 \rightarrow 86.40 \rightarrow 81.22 \rightarrow 79.59 \rightarrow 67.65 \rightarrow 54.12 \rightarrow 49.79$  %: the two conditions that are also the two least likely to be reported as blockers (workflow sign-off and the clinical champion, both owned outside the project, both part of Domain 2's enabling change) together remove **25.47 percentage points**. **Second, remediation must be ranked by proportional shortfall, not absolute gap**, because lifting a condition from  $p$  to  $p'$  multiplies the conjunction by  $p'/p$ . Lifting the champion condition from 0.80 to 0.98 multiplies by 1.2250 and gains **11.20 points**; lifting training from 0.96 to 0.98 multiplies by 1.0208 and gains **1.04**. The weakest link is worth almost eleven times the

strongest, and a remediation plan that works through the conditions in dashboard order will spend its budget on the wrong ones. **Third, certainty cannot be bought.** With all seven conditions at 0.98 the conjunction is only **86.81 %**, so **13.19 %** of go-lives still fail, which is the arithmetic case for the reversion plan of 16.1.4 rather than for more assurance. A **95 %** chance of a clean go-live would require **99.27 %** on every one of the seven, and a programme that claims that has almost certainly not assessed its conditions honestly. **Fourth, and the professional caution: independence is an assumption, and it must be stated rather than assumed.** If the failures share a root cause (an under-resourced change team that starves training, workflow and champion recruitment alike) then the conditions fail together and the conjunction rises towards  $\min(p_i) = 80.00 \%$ . That is the optimistic bound, not the expected case, and 16.1.4 tests the go-live decision against both bounds rather than choosing between them.

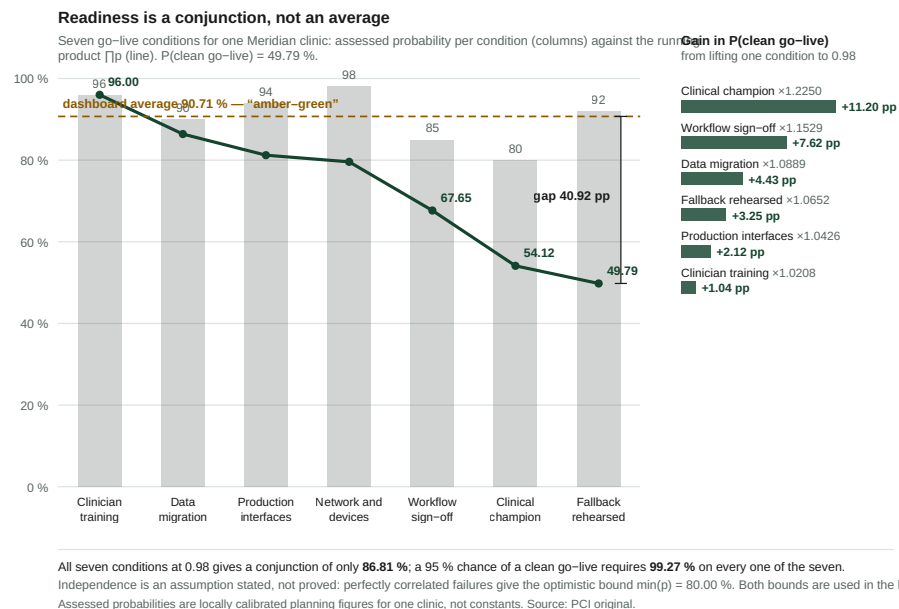


Fig 16.1.1 — Readiness is a conjunction, not an average

#### 16.1.4 The go-live decision, hypercare and reversion

**Definition.** The go-live decision is a gate in Domain 3's sense (a named authority, pre-set criteria, and the power to hold as well as to proceed), whose distinguishing feature is that it is substantially irreversible for the users, even where it is technically reversible for the system.

**The option set exists only above the gate block.** The three options below are available only once **every** mandatory precondition of 16.1.3 is recorded met, with its approving authority named. Where one is not (an unclosed safety case, an ungranted approval, an unsigned privacy assessment, a notification not made, a takeover



certificate not issued) there is no option set and there are no economics: the only available decision is **hold**, and the arithmetic that follows does not apply to it. A paper that prices going live against an open gate-block item has not presented a finely balanced judgement; it has costed a decision that was not the decision-maker's to take, and the correct response to it is to refuse the question rather than to answer it more carefully.

Above that line, the decision has three real options, not two: proceed, hold and remediate, or proceed with a reduced scope of users. Pricing them requires the discretionary conjunction of 16.1.3 and a cost per failed transition, and it is worth doing because the intuitive answer is usually wrong in the expensive direction.

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#### WORKED EXAMPLE

##### Worked example 16.1.4 – hold or go, tested against both correlation bounds.

1. **Setup.** All 40 Meridian clinics face the readiness profile of 16.1.3. A clinic that goes live unready reverts to paper: remediation costs **USD 24,000** in re-training, re-migration and support, and the clinic loses **6 weeks** of benefit at  $6 \times 85 = \text{USD } 510$  a week, so a failed go-live costs **USD 27,060**. Three options: (1) go now; (2) a targeted two-week uplift costing **USD 38,000** that lifts the two weakest conditions (champion and workflow) to 0.98; (3) a blanket three-week uplift costing **USD 96,000** that lifts all conditions to 0.98. Deferred benefit is priced at Domain 1's cost of delay, **USD 14,280** a week. Evaluate each option under both the independent conjunction and the perfectly correlated bound  $\min(p_i)$ .
2. **Formula.** Total cost = (weeks deferred  $\times$  cost of delay) + uplift spend +  $40 \times (1 - P(\text{clean})) \times 27,060$ , with  $P(\text{clean}) = \prod p_i$  in the independent case and  $\min(p_i)$  in the correlated case.
3. **Substitution.** Option 1 independent:  $40 \times (1 - 0.4979) \times 27,060$ . Option 3 independent:  $3 \times 14,280 + 96,000 + 40 \times (1 - 0.98^7) \times 27,060$ . Correlated cases substitute  $\min(p_i) = 0.80, 0.90$  and  $0.98$  respectively.
4. **Result.**

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| OPTION                               | INDEPENDENT ( $\prod p_i$ )               | CORRELATED BOUND ( $\min p_i$ )  |
|--------------------------------------|-------------------------------------------|----------------------------------|
| 1, go now                            | <b>USD 543,445</b> (20.08 failed clinics) | <b>USD 216,480</b> (8.00 failed) |
| 2 (targeted uplift, 2 weeks, 38,000) | <b>USD 387,766</b> (11.87 failed)         | <b>USD 174,800</b> (4.00 failed) |
| 3 (blanket uplift, 3 weeks, 96,000)  | <b>USD 281,581</b> (5.27 failed)          | <b>USD 160,488</b> (0.80 failed) |

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5. **Interpretation.** The blanket uplift is the **dominant** option (best under both bounds), which is the result a leader wants, because the correlation structure is

exactly what nobody knows at the time of the decision. It saves **USD 261,864** against going now if the failures are independent and **USD 55,992** if they share a single root cause, for a total commitment of **USD 138,840** (three weeks of deferred benefit at 42,840 plus the 96,000 spend). Three points generalise beyond Meridian. **The delay cost is small relative to the failure cost, and it is the only one that is visible.** Forty-two thousand dollars of deferred benefit appears in a report as a slipped milestone; half a million dollars of remediation appears as twenty separate operational incidents attributed to change resistance. That asymmetry is a large part of why go-live decisions tend to be taken too early. **The breakeven uplift spend is enormous:** holding remains worthwhile up to **USD 357,864** of uplift cost under the independent case, which is 3.7 times what the blanket remediation actually costs, so the decision is not close, and a leader arguing it should say so rather than presenting it as a fine balance. **And the residual matters more than the improvement.** Even after the blanket uplift, **5.27** clinics are expected to fail, because 0.98 across seven conditions is 86.81 per cent and no amount of readiness work makes a conjunction certain. The professional response is not further assurance but a rehearsed reversion plan and a wave structure that limits the exposure of any single go-live, which is what the remaining cost buys.

**Hypercare** is the bounded period of elevated support immediately after go-live, and it needs three things a project rarely gives it: a **stated duration**, **exit criteria expressed as a measurement rather than a date**, and a **named owner in the receiving organisation from day one** rather than at the end. Meridian's design used an incident rate per clinic per week with a threshold of **0.8** sustained for two consecutive weeks; the observed series over weeks 4 to 7 ran 1.4, 0.9, 0.6, 0.5, so the two-consecutive-week condition was first satisfied at **week 7** and hypercare exited then rather than at the planned week 6, which is the point of a measured exit criterion. Hypercare that ends on a date ends by transferring an unstable service.

**The reversion plan** is the option the go-live arithmetic proves is needed. It must state the decision-maker, the trigger, the maximum time to revert, what happens to data created since go-live, and, the element most often missing, the **latest point at which reversion is still possible**, because after that point the plan is a comfort document rather than a control. A reversion plan that has never been rehearsed is an assumption, and 16.1.2's failure-path testing is where it stops being one.

### AI in this KA

**Where it earns its place.** Assembling the readiness picture, which is a data-collection problem across many sources (training records, migration reconciliation reports, device inventories, test results), and reconciling them into one condition-by-condition status with the source of each assertion named. Computing the conjunction and the proportional-shortfall ranking across dozens of sites, which is

deterministic and tedious. Reading test evidence packs against the agreed acceptance criteria and listing criteria for which no evidence exists: a genuinely valuable gap-finding task, and one humans do badly at scale. Drafting runbooks and reversion procedures from configuration baselines, for verification by the people who will use them.

**Where it must not go.** It must not record a gate-block item as met. A mandatory precondition is closed by the authority that grants it, and an assembled status line evidences that a document exists, never that an approval was given. The distinction that makes the named approving authority, rather than the collated status, the thing the certificate carries. It must not set the readiness probabilities. Those are judgements about specific organisations and specific people, and a model asked for them will supply confident, provenance-free numbers that then drive a real go-live decision, precisely the failure Domain 14 KA 14.3 names. It must not issue a service acceptance, which is an accountable act by the receiving organisation. And it must not author a test result: a generated evidence record cannot evidence that a test was witnessed, and an acceptance regime resting on unwitnessable evidence is worse than none, because it looks complete.

**Verification, concretely.** Every probability carries a named human assessor and a stated basis (pilot data, comparable site, judgement). Every AI-assembled status line carries a link to its source record and is sampled by the transition manager at a stated rate before the gate. The conjunction is reproduced by hand for the go-live paper (it is six multiplications) and the paper states both correlation bounds rather than one, so the decision body can see the robustness rather than a point estimate.

#### KEY TERMS — KA 16.1

| TERM                             | MEANING                                                                                                                                  |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handover</b>                  | Transfer of custody and operational accountability for a deliverable to a named receiving organisation, on stated conditions.            |
| <b>Condition-based handover</b>  | Handover triggered by a satisfied condition set rather than a calendar date.                                                             |
| <b>Commissioning</b>             | Progressive proving of a deliverable in its operating environment, from component test to performance demonstration.                     |
| <b>Works test / service test</b> | Whether the thing was built as specified / whether the organisation can actually run it.                                                 |
| <b>Service acceptance</b>        | The receiving organisation's formal agreement that the service meets specification.                                                      |
| <b>Mandatory precondition</b>    | A go-live condition recorded met or not met, with its approving authority named; it carries no probability and admits no economic trade. |

| TERM                           | MEANING                                                                                                                                            |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Gate block</b>              | The set of mandatory preconditions sitting above the readiness arithmetic; while any item is not met the only available decision is hold.          |
| <b>Discretionary condition</b> | A readiness condition that admits degree, is genuinely uncertain at the decision date, and is therefore assessed as a probability.                 |
| <b>Readiness conjunction</b>   | $P(\text{clean transition}) = \prod p_i$ over the $k$ necessary <b>discretionary</b> conditions; always at or below the averaged dashboard figure. |
| <b>Proportional shortfall</b>  | The ratio $p'/p$ by which lifting a condition multiplies the conjunction: the correct remediation ranking.                                         |
| <b>Hypercare</b>               | A bounded period of elevated support after go-live, with measured exit criteria and a receiving-organisation owner.                                |
| <b>Reversion plan</b>          | The rehearsed route back, with a trigger, a decision-maker, a time limit and a latest-possible-reversion point.                                    |

## SAMPLE MCQs — KA 16.1

**MCQ 16.1-A** [16.1.3 · Application] Seven readiness conditions, whose failures are assessed as independent, hold with probabilities 0.96, 0.90, 0.94, 0.98, 0.85, 0.80 and 0.92. All seven are necessary for a clean go-live. The probability of a clean go-live is closest to:

- A. 90.71 %
- B. 80.00 %
- C. 49.79 % CORRECT
- D. 47.83 %

*Rationale:* The conjunction is the product,  $0.96 \times 0.90 \times 0.94 \times 0.98 \times 0.85 \times 0.80 \times 0.92 = 49.79\%$  (16.1.3). A is the equal-weight average, the dashboard figure, and a different quantity. B is  $\min(p_i)$ , the perfectly correlated bound, which the stem's independence assumption excludes and which is in any case the optimistic case. D rounds the dashboard reading to 0.90 and raises *that* to the seventh power,  $0.90^7 = 47.83\%$ , an answer that both rounds and double-counts the averaging.

**MCQ 16.1-B** [16.1.3 · Analysis] Seven independent readiness conditions, holding at 0.96, 0.90, 0.94, 0.98, 0.85, 0.80 and 0.92, give a conjunction of 49.79 %. The gain in the probability of a clean go-live from lifting one condition is governed by:

- A. the absolute gap to the target, with an equal gain per percentage point of gap closed
- B. the ratio  $p'/p$ , so the same gap closed at a lower  $p$  yields a larger gain CORRECT
- C. the condition's weight on the readiness dashboard

D.  $1/k$ , equally across all seven conditions

*Rationale:* Lifting a condition from  $p$  to  $p'$  multiplies the conjunction by  $p'/p$ , so the gain is  $\Pi p_i \times (p'/p - 1)$  (16.1.3). Closing two points of gap at 0.80 is worth **1.24 pp** while closing the same two points at 0.96 is worth **1.04 pp**, which is exactly what A denies, a common target ranks the conditions in the same order either way, but the magnitudes are set by the ratio. B is therefore the statement that generalises. C substitutes an artefact for the arithmetic. D confuses "a product" with "symmetric in its factors".

**MCQ 16.1-C** [16.1.3 · Evaluation] A programme wants a 95 % probability of a clean go-live across seven independent readiness conditions. The minimum probability required on each condition is closest to:

A. 95.00 %

B. 99.27 % CORRECT

C. 98.00 %

D. 99.90 %

*Rationale:*  $p = 0.95^{(1/7)} = 99.27\%$  (16.1.3). A applies the target to each condition rather than to their conjunction. C gives only  $0.98^7 = 86.81\%$ . D is stricter than required and would be rejected as unachievable, losing the argument the arithmetic wins.

**MCQ 16.1-D** [16.1.4 · Application] Forty sites each face seven independent readiness conditions giving a 49.79 % probability of a clean go-live; a failed go-live costs 27,060. A three-week hold costs 42,840 in deferred benefit plus 96,000 of remediation and lifts all seven conditions to 0.98. The expected saving from holding is closest to:

A. USD 138,840

B. USD 261,864 CORRECT

C. USD 404,605

D. USD 543,445

*Rationale:* Going now costs  $40 \times (1 - 0.4979) \times 27,060 = 543,445$ ; holding costs  $42,840 + 96,000$

$\bullet 40 \times (1 - 0.8681) \times 27,060 = 281,581$ ; the saving is **261,864** (16.1.4). A is the cost of holding, not the saving. C omits the residual remediation after the uplift and so credits the hold with the whole 543,445, giving  $543,445 - 138,840 = 404,605$ . The error that makes readiness work look better than it is. D is the cost of going now with no credit for the hold at all.

**MCQ 16.1-E** [16.1.1 · Comprehension] Which of the following does **not** transfer at handover?

- A. custody of the asset
- B. accountability for operating the service
- C. accountability for realising the benefit CORRECT
- D. responsibility for paying the running costs

*Rationale:* The benefits owner sits in the receiving organisation from the outset (Domain 2, KA 2.3.1), so handover tests that appointment rather than creating it; a benefit "handed over" at closeout has no owner and no baseline (16.1.1).

#### SELF-CHECK — KA 16.1

1. *Why does an averaged readiness dashboard overstate readiness, and in which direction is the error always signed?* — Readiness is a conjunction, so its probability is the product of the conditions' probabilities; a product is always at or below the average, so the dashboard error is always optimistic, and it grows with the number of conditions and their dispersion.
2. *A works test has passed and a service test has not been run. What has been proved?* — That the thing was built as specified, and nothing whatever about whether the organisation can run it.
3. *What must a hypercare period state that a date cannot?* — A measured exit criterion, so that the period ends when the service is stable rather than when the calendar says so.
4. *Which readiness conditions get a probability, and which do not?* — Discretionary conditions, which admit degree, get a p1; mandatory preconditions are recorded met or not met with their approving authority named, and appear neither in the product nor in the go/hold economics (16.1.3).
5. *A safety case is not yet closed and the cost of delay is large. What is the option set?* — Hold. There is no option set below the gate block, and pricing the alternatives would state a figure at which the precondition could be waived (16.1.4).

#### KNOWLEDGE AREA 16.2

### Operational transition and contract closeout

*Topics:* 16.2.1 the run organisation and total cost of ownership · 16.2.2 defects liability, warranty and retention · 16.2.3 the final account · 16.2.4 the cost of leaving a claim open.

### 16.2.1 The run organisation and total cost of ownership

**Definition.** Operational transition is the establishment of a permanent capability to run, support, fund and improve what the project delivered, with the running cost recognised in the receiving organisation's budget before the project closes.

The last clause is where business cases go to die. A project's cost is a capital number, visible, approved and controlled; the service's running cost is an operating number that appears in someone else's budget, in a later year, and is therefore very frequently **not in the business case at all**. Domain 2's whole-life requirement (KA 2.3.3) exists to prevent this, and Meridian's approved case failed it: it carried no operating-cost line whatever. The measured run cost of the records service is **USD 108,000** a year (hosting **42,000**, licences **39,000**, second-line support **27,000**), and over the case's own eight-year appraisal at 7 per cent that is a present value of **USD 644,900**, which is 48.4 per cent of the honest case's entire NPV. A programme that hands over a service without a funded run line has not transferred it; it has abandoned it in a place where someone will find it.

**What the run organisation needs at transition**, stated as a checklist because its items are countable and its gaps are cheap to find: a named service owner and a funded support model with stated hours and response targets; runbooks and as-built documentation at the configuration version actually deployed; a licence and contract register with renewal dates and the notice periods that govern them; an access and privilege model with the project team's own access **removed** on a stated date; a capacity and obsolescence view stating when the platform will next need investment; and the benefits measurement duties of 16.4.1, which are operating duties, not project ones.

### 16.2.2 Defects liability, warranty and retention

**Definition.** Retention is a proportion of the contract sum withheld from payment as security for the contractor's completion and defect-rectification obligations, released in stages against defined events. A **defects-liability period** is the interval after acceptance during which the contractor must rectify defects that appear; a **warranty** is a broader promise about the deliverable's condition or performance, typically longer and narrower in scope.

The mechanism is simple and it is misused in two opposite directions. Held too long or released too loosely, retention becomes either a commercial grievance that poisons the closing relationship or a security that was never there when it was needed. The disciplines are: **release against events, not dates** (takeover, end of defects liability, completion of outstanding documentation); **state what the retention secures**, since a retention held against defects cannot be applied to a performance shortfall unless the contract says so; and **track outstanding items as a priced list from the day of takeover**, because a retention release argued from an unpriced snagging list is argued from nothing. Domain 10's contract mechanisms

supply the drafting; the closeout discipline is that the retention release is a decision with a record (Domain 3, KA 3.3.4), not an accounts-payable event.

The jurisdictional caution is unavoidable and it is real: retention practice, its permissibility, any requirement to hold it in trust, and the limitation periods on latent defects vary by jurisdiction and by contract family. Nothing in this domain states a legal position; it states a management discipline that must be operated within whatever legal frame applies.

### 16.2.3 The final account

**Definition.** The final account is the single agreed statement of everything owed under a contract at its conclusion: the original sum, all approved variations, the settlement of any claims, all deductions and recoveries, and the resulting final payment, signed by both parties as full and final.

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#### WORKED EXAMPLE

#### Worked example 16.2.3 – Meridian's final account and retention release.

1. **Setup.** The systems integrator's contract sum was **USD 1,680,000**. Approved variations total **USD 96,000**. Retention was **5 %** of the contract sum, half released at takeover of the last clinic. A prolongation claim was settled at **USD 74,000** (16.2.4). Three clinics' interface defects were rectified by another supplier at a cost of **USD 18,600**, recoverable from the integrator. Strike the final account, the final retention release and the final payment.
2. **Formula.** Final account = contract sum + variations + claim settlement – recoveries. Final payment = final account – amounts already paid.
3. **Substitution.** Retention  $1,680,000 \times 0.05 = 84,000$ ; certified works  $1,680,000 + 96,000 = 1,776,000$ ; paid on certification  $1,776,000 - 84,000 = 1,692,000$ ; plus the takeover release of  $84,000/2 = 42,000$ , so **1,734,000** paid to date. Final account  $1,680,000 + 96,000 + 74,000 - 18,600$ .
4. **Result.** Final account **USD 1,831,400**: **9.01 %** above the original sum. Final retention release  $42,000 - 18,600 = \text{USD } 23,400$ . Final payment  $1,831,400 - 1,734,000 = \text{USD } 97,400$ , which reconciles as  $42,000 + 74,000 - 18,600 = 97,400$ .
5. **Interpretation.** The reconciliation in the last line is the whole point of the example, and it is the check that a final account is actually closed rather than merely totalled: **the final payment must be explainable as the sum of the specific movements since the last certificate**, and a final payment that cannot be so explained conceals either an unrecorded variation or a double-counted recovery. Three professional observations. The **recovery is only collectable because it was made against the retention that was still held**: had the full 84,000 been released at takeover, the 18,600 would have become a debt to be



pursued from a contractor with no remaining incentive, which is the practical argument for staged release and the reason "release it, they have been good to work with" is a decision and not a courtesy. The **9.01 % growth over the original sum is the number the organisation should carry into its next estimate**, not the change-control record of variations alone: variations were **5.71 %** of the original sum, the claim settlement added **4.40 %** and the defect recovery returned **1.11 %**, netting to 9.01 %, so an organisation that benchmarks only its approved variations will understate contract growth by roughly the net settlement rate every time. And the account must be closed with a **written full and final statement**, because the alternative (a balance agreed in correspondence and never formalised) is the state in which 16.2.4's carrying cost accrues.

#### 16.2.4 The cost of leaving a claim open

**Definition.** The carrying cost of an open claim is the recurring cost, per period, of it remaining unresolved: external advisory fees, internal management attention, and the opportunity cost of the cash locked in retention and provision, none of which appears as a claim cost in any ledger.

Open items are left open because closing them requires accepting a number, and accepting a number feels like a loss while carrying an item feels like prudence. The arithmetic reverses that intuition reliably, and it is worth being able to do in a meeting.

#### WORKED EXAMPLE

**Worked example 16.2.4 – what Meridian's open claim cost per month, and the premium worth paying.**

1. **Setup.** The integrator claimed **USD 148,000** for prolongation. Meridian's quantity assessment put the defensible value at **USD 62,000**, and a provision of the full claimed amount was held. Carrying the claim costs: external commercial advice **USD 3,250** a month; internal management **1.5 days** a month at **USD 700** a day; and the opportunity cost of the **USD 190,000** locked up (42,000 of retention plus the 148,000 provision) at **6 %** a year. Left to formal determination, the claim would close at **month 14**, with outcomes assessed at **USD 40,000** (probability 0.25), **USD 62,000** (0.50) and **USD 110,000** (0.25). The alternative is to settle at **month 2**. What settlement price is worth paying?
2. **Formula.** Carrying cost  $c = \text{advisory} + \text{internal} + \text{locked cash} \times \text{rate}/12$ . Expected cost of determination =  $\text{EMV} + c \times 14$ . Breakeven settlement price at month 2 =  $\text{EMV} + c \times 14 - c \times 2$ . Breakeven premium over the assessed value =  $c \times \Delta\text{months} + (\text{EMV} - \text{assessed})$ .
3. **Substitution.**  $c = 3,250 + 1.5 \times 700 + 190,000 \times 0.06/12 = 3,250 + 1,050 + 950$ .  $\text{EMV} = 0.25 \times 40,000 + 0.50 \times 62,000 + 0.25 \times 110,000$ .

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4. **Result.** Carrying cost **USD 5,250 a month: USD 73,500** over fourteen months, more than the claim's own assessed value. **EMV USD 68,500.** Expected cost of fighting to determination **USD 142,000.** Breakeven settlement price at month 2 **USD 131,500**, a breakeven **premium** over the assessed 62,000 of **USD 69,500**, which the identity  $12 \times 5,250 + (68,500 - 62,000) = 63,000 + 6,500$  reproduces exactly. Meridian settled at **USD 74,000**, a premium of **12,000**, or **17.27 %** of the premium that would still have been worth paying, saving **USD 57,500** against the expected cost of determination.
  5. **Interpretation.** The identity is the transferable result: **the breakeven early-settlement premium is the carrying cost times the months saved, plus the difference between the expected determination and your own assessment.** It says that time, not merit, dominates small and medium claims — twelve months of carrying cost at 5,250 is 63,000, almost exactly the assessed value of the claim itself, so an organisation that fights a 62,000 claim for a year has spent the claim to win the claim. It also identifies which lever to pull first: of the 5,250, **USD 4,300** is advisory and management attention that stops the moment the item closes, and USD 950 is opportunity cost that stops when the cash releases. Four cautions keep this honest. The arithmetic is **not an argument for settling everything**, where a claim raises a point of principle that will recur across a supplier's other contracts, the precedent value can exceed the carrying cost, and that must be quantified rather than asserted. The **internal management figure is real cost and is almost never counted**, which is exactly why the carrying cost is invisible; 1.5 days a month of a senior commercial manager is a modest estimate and doubling it doubles the case for settling. The **assessment must precede the negotiation**, because a settlement premium can only be judged against a defensible assessed value, and an organisation that settles without one is guessing, not paying a premium. And a settlement must be **documented as full and final for defined matters** (drafting that belongs with qualified legal advice, since what a release covers and what it cannot reach is a legal question and not a management one), or the carrying cost simply resumes under a different heading.
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## AI in this KA

**Where it earns its place.** Reconciling a final account against the change register, the payment certificates and the variation approvals, and listing every item that appears in one and not the others: a document-reconciliation task at which it is fast, thorough and genuinely better than a tired human at month-end. Extracting a licence and contract register with renewal and notice dates from a contract set, which is the single most useful artefact the run organisation receives and the one most often missing. Building the carrying-cost model above across a portfolio of open items and ranking them by monthly cost, which turns "we have eleven open

claims" into a prioritised list. Summarising a claim's correspondence history for a settlement paper, with every assertion referenced.

**Where it must not go.** It must not value a claim. A claim valuation is an expert commercial and technical judgement with contractual and evidential foundations, it will be tested by people who are paid to test it, and a model's plausible number carries no provenance and no accountable author. It must not agree a final account or authorise a retention release. Both are accountable decisions with a record. And its summary of a correspondence trail must never be relied on in a dispute without verification against the source documents: the summary is a convenience, the versioned original is the evidence (Domain 3, KA 3.3.4).

**Verification, concretely.** Every reconciliation exception is confirmed against the source certificate before it is reported. The carrying-cost model's components are each traceable to an invoice, a timesheet or a stated cost-of-capital assumption, and the assumption is named in the paper. The valuation on which any settlement rests is signed by the accountable commercial lead, not by the analysis that supported it. And the final account carries a human signature on both sides: the one part of closeout where the signature is the deliverable.

#### KEY TERMS — KA 16.2

| TERM                                 | MEANING                                                                                                                               |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Operational transition</b>        | Establishment of a funded, staffed, permanent capability to run and improve the delivered service.                                    |
| <b>Total cost of ownership</b>       | Whole-life cost of the service, including the operating cost that must appear in the receiving organisation's budget before closeout. |
| <b>Retention</b>                     | A proportion of the contract sum withheld as security, released against defined events.                                               |
| <b>Defects-liability period</b>      | The interval after acceptance in which the contractor must rectify defects that appear.                                               |
| <b>Final account</b>                 | The single agreed statement of the original sum, variations, settlements, deductions and final payment, signed as full and final.     |
| <b>Contract growth</b>               | Final account against original sum, variations <i>and</i> settlements, not variations alone.                                          |
| <b>Carrying cost of an open item</b> | Advisory + internal management + opportunity cost of locked cash, per period; invisible in every ledger.                              |
| <b>Breakeven settlement premium</b>  | $\text{carrying cost} \times \text{months saved} + (\text{EMV of determination} - \text{own assessed value}).$                        |

## SAMPLE MCQs — KA 16.2

**MCQ 16.2-A** [16.2.3 · Application] A contract sum of 1,680,000 carries 96,000 of approved variations, a claim settled at 74,000 and an 18,600 recovery for defects rectified by others. The final account is:

- A. USD 1,776,000
- B. USD 1,831,400 CORRECT
- C. USD 1,850,000
- D. USD 1,757,400

*Rationale:*  $1,680,000 + 96,000 + 74,000 - 18,600 = 1,831,400$  (16.2.3). A is the certified works value, omitting both the settlement and the recovery. C omits the recovery. D omits the settlement.

**MCQ 16.2-B** [16.2.4 · Application] Carrying an open claim costs 5,250 a month. Determination at month 14 has an expected value of 68,500; the alternative is settlement at month 2. The breakeven settlement price is:

- A. USD 68,500
- B. USD 131,500 CORRECT
- C. USD 142,000
- D. USD 73,500

*Rationale:*  $68,500 + 14 \times 5,250 - 2 \times 5,250 = 131,500$  (16.2.4). A ignores carrying cost entirely. C is the expected cost of fighting, without crediting the two months still carried under the settlement option. D is fourteen months of carrying cost alone.

**MCQ 16.2-C** [16.2.4 · Analysis] A claim costing 5,250 a month to carry, with an expected determination of 68,500 at month 14 and settlement available at month 2, is assessed by the client at 62,000 against the contractor's claimed 148,000. The breakeven **premium** over the assessed value is 69,500. The most defensible reading is that:

- A. the claim should be conceded in full at 148,000
- B. up to 69,500 above the assessed value can be paid to close twelve months earlier, because carrying cost times months saved plus the gap between the expected determination and the assessment is  $63,000 + 6,500$  CORRECT
- C. the claim should be fought, since 62,000 is the defensible figure
- D. the premium equals the provision held

*Rationale:* The identity  $c \times \Delta m + (EMV - \text{assessed})$  gives 69,500 (16.2.4). A ignores the assessment altogether. C is the intuition the arithmetic overturns, winning at 62,000 after fourteen months costs 135,500. D confuses a provision with a price.

**MCQ 16.2-D** [16.2.1 · Evaluation] A business case with no operating-cost line hands over a service whose measured run cost is 108,000 a year. Over an eight-year appraisal at 7 % ( $AF = 5.971299$ ), the omission is worth:

- A. USD 864,000
- B. USD 644,900 CORRECT
- C. USD 108,000
- D. USD 540,000

*Rationale:*  $108,000 \times 5.971299 = 644,900$  (16.2.1). A is the undiscounted eight-year total, the commonest error and an overstatement of 34.0 %. C is a single year. D applies a five-year horizon undiscounted.

**MCQ 16.2-E** [16.2.2 · Analysis] A client holds 84,000 of retention, releases half of it at takeover, and later recovers 18,600 for defects that another supplier had to rectify. Releasing the whole retention at takeover rather than in stages would most directly have:

- A. improved the contractor's cash position at no cost to the client
- B. converted an 18,600 recovery into a debt to be pursued from a contractor with no remaining incentive CORRECT
- C. breached the defects-liability period
- D. reduced the final account by 18,600

*Rationale:* Retention is security, and its value is precisely that it is still held when a recovery arises (16.2.2). C confuses a payment mechanism with a liability period; D reverses the direction of the recovery.

#### SELF-CHECK — KA 16.2

1. *Why is the operating cost the business-case line most often missing, and what does that cost Meridian?* — It falls in another budget in a later year, so nobody in the approval chain owns it; at 108,000 a year over eight years at 7 % it is a present value of 644,900, or 48.4 % of the honest case's whole NPV.
2. *What are the three components of the carrying cost of an open claim, and which is largest?* — External advisory, internal management attention and the opportunity cost of locked cash; the first two together are 4,300 of Meridian's 5,250 a month, and both stop the day the item closes.
3. *What check proves a final account is closed rather than merely totalled?* — The final payment must reconcile to the specific movements since the last certificate, here 42,000 of retention plus 74,000 of settlement less 18,600 of recovery equals the 97,400 due.

## Knowledge transfer and post-project review

*Topics: 16.3.1 transferring knowledge to the run organisation · 16.3.2 the post-project review · 16.3.3 the economics of a lesson · 16.3.4 closing the project organisation.*

### 16.3.1 Transferring knowledge to the run organisation

**Definition.** Knowledge transfer is the deliberate conversion of what the project team knows into forms the receiving organisation can use without them: documentation at the deployed configuration version, trained people, and rehearsed procedures.

The failure is rarely a refusal to document; it is that documentation is treated as a deliverable rather than as a **test**. A runbook is adequate when someone who was not on the project can execute it under supervision, and nothing short of that observation proves it. Two measurements make the gap visible cheaply. **Key-person concentration:** of Meridian's 34 operational procedures, 11 were executable by exactly one named individual at handover, a **32.35 %** single-point-of-knowledge rate, and each of those 11 is an availability risk that no risk register had recorded because it was not a project risk. And **documentation currency against the as-built baseline** (Domain 4, KA 4.3.2), since documentation written against the design and never updated against the deployment describes a system that does not exist.

The economics are one-sided and worth stating because they are always resisted at the moment they matter. Three of Meridian's clinics had interface configurations that were changed during commissioning and never recorded. Reconstructing each afterwards (reading the live configuration, re-testing, re-documenting) cost **USD 8,400**, a total of **USD 25,200**, against **USD 1,200** each to document at the time, or **USD 3,600**: a ratio of **7.00**. The multiple is Meridian's own and is not a constant, but its direction follows from a mechanism that does not change with the context, at handover the knowledge is in someone's head and the cost is a note; afterwards it must be re-derived from the artefact by someone who never had it. An organisation that wants the multiple for its own estimating has to measure it, and the measurement is cheap because both costs are already invoiced.

### 16.3.2 The post-project review

**Definition.** A post-project review is a structured examination, after delivery, of what happened and why, conducted to improve the organisation's future decisions rather than to assess its people.

The design faults are predictable and each has a countermeasure. **Held too late**, when the team has dispersed and only the outcome is remembered, not the reasoning; the countermeasure is to hold it within a stated window of handover and to run interim reviews at gates rather than one review at the end. **Held as an attribution**

**exercise**, which guarantees defensive testimony; the countermeasure is an explicit and enforced separation from performance assessment, and the psychological-safety conditions Domain 12 sets out. **Focused on events rather than on decisions**, so it records that the schedule slipped rather than which decision, taken on which information, made the slip likely; the countermeasure is to start from the decision record (Domain 3, KA 3.3.4) and ask of each material decision whether it was reasonable on what was known: a question that produces transferable findings where "what went wrong" produces narrative. And **producing recommendations with no owner**, which is how a review generates activity and no change; every finding needs a named owner, a date and a place in a standing process that will be looked at again.

The most valuable single addition to a review agenda is the **estimate-versus-actual comparison on the organisation's own parameters**: the contract growth of 16.2.3, the adoption ramp of 16.4.2, the readiness probabilities of 16.1.3 against what actually happened. Those numbers are what calibrate the next business case, and they are the only output of a review that improves an estimate rather than a behaviour.

### 16.3.3 The economics of a lesson

**Definition.** A lesson's value is the cost it avoids when it is retrieved and applied, weighted by the probability that it is retrieved and applied at all. Capture without retrieval has a value of zero, and this is not a rhetorical point; it is where almost all of the value is lost.

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#### WORKED EXAMPLE

#### Worked example 16.3.3 — what Meridian's post-project review was worth.

1. **Setup.** The review cost **USD 46,000** in workshop time, analysis and write-up, and produced **34** lessons. From the organisation's own history, a lesson that is retrieved before a comparable decision avoids an average of **USD 26,000** of rework or delay. In the current, unindexed repository, **12 %** of captured lessons are retrieved before a comparable decision. A proposal would spend a further **USD 18,000** indexing the lessons against the specific decision points at which they apply, raising retrieval to an estimated **35 %**.
2. **Formula.** Expected recovery = lessons × retrieval rate × avoided cost per applied lesson. Breakeven retrieval rate = review cost ÷ (lessons × avoided cost).
3. **Substitution.** At 12 %:  $34 \times 0.12 \times 26,000$ . At 35 %:  $34 \times 0.35 \times 26,000$ . Breakeven:  $46,000 / (34 \times 26,000) = 46,000 / 884,000$ .

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4. **Result.** Expected recovery **USD 106,080** at the current retrieval rate and **USD 309,400** with indexing, an increment of **USD 203,320** for **USD 18,000**, a ratio of **11.30**. The review breaks even at a retrieval rate of **5.20 %**; with the index it must reach **7.24 %**. Net value of review plus index at 35 % retrieval: **USD 245,400**.
  5. **Interpretation.** Three results, in ascending order of importance. The review is worth holding, at 2.31 times its cost even with a retrieval rate most organisations would be embarrassed to report. **The breakeven retrieval rate of 5.20 % is the number that should end the recurring argument about whether reviews are worth the time:** they are, comfortably, provided anyone ever reads them, and the honest reading of the widespread belief that they are not is that in many organisations the retrieval rate really is near zero, at which point the belief is correct and the remedy is not to stop reviewing. **And the highest-return intervention available is not better capture but better retrieval:** an 11.30-to-one return on indexing, against which any proposal to improve the workshop is not competitive. This inverts where effort normally goes: organisations invest in the review event, which is visible and social, and not in the retrieval mechanism, which is unglamorous infrastructure. The professional cautions are that the 26,000 avoided cost and both retrieval rates are **organisational parameters that must be measured, not assumed:** the honest way to establish the retrieval rate is to instrument the repository and count, and an organisation that cannot state its retrieval rate is not in a position to argue about the value of its reviews; and that a lesson only avoids cost if it is **specific enough to act on**, since "communicate better" has an avoided cost of zero at any retrieval rate, which is why lesson quality and lesson count are different measurements.
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#### 16.3.4 Closing the project organisation

**Definition.** Closing the project organisation is the deliberate dissolution of the temporary structure: its authority, its accounts, its access rights, its records and its people.

Four items are routinely left undone, and each has a specific consequence. **Authority is not withdrawn**, so a change board with no project continues to approve changes to a live service that now belongs to someone else: the governance defect of Domain 3 KA 3.1.1, arriving after the project has ended. **Cost accounts are left open**, so late charges land in a closed period, the outturn moves after it was reported, and the estimate-versus-actual comparison of 16.3.2 is calibrated on a wrong number. **Access rights persist**, which is a security exposure with a named cause and a trivial fix (Domain 14, KA 14.4). And **people are released without a record of what they did**, which is both an individual injustice and the reason the organisation cannot find, two years later, the person who knows why the interface was configured that way.



The leadership content of this topic is not administrative. A team that has spent two years on a programme experiences its dissolution as an ending, and the manner of it materially affects whether those people will volunteer for the next difficult programme. Recognition that is specific and attributable, honest conversations about what each person did well and what they should develop (Domain 12, KA 12.3), and a visible route to their next role are not courtesies. They are the mechanism by which delivery capability survives the project that consumed it.

### AI in this KA

**Where it earns its place.** Indexing lessons against the decision points at which they apply, which is exactly the intervention 16.3.3 prices at an 11.30-to-one return, and is a classification and retrieval problem of the kind the technology is genuinely good at. Clustering findings across many reviews to expose the recurring cause that no single review can see: the organisational analogue of Domain 3's re-decision count. Drafting as-built documentation from configuration baselines and change records for human verification. Generating candidate review questions from the decision record so the review starts from decisions rather than from recollection. Producing the first draft of a transition-out handbook from existing artefacts.

**Where it must not go.** It must not conduct the review. A post-project review is partly a conversation about judgement under uncertainty, and its value comes from people saying things they would not write down; a transcript-and-summarise exercise gets the record and loses the content. It must not attribute cause, because causal attribution in a delivery failure is contested, consequential for named individuals, and not a text-processing task. And it must not be the author of a lesson that is then relied on: an unattributable lesson cannot be interrogated, and a repository of them decays into plausible advice.

**Verification, concretely.** Every indexed lesson is checked against its source finding on a sampled basis, and the retrieval rate is instrumented and reported rather than estimated. Every AI-drafted document is executed by someone who was not on the project before it is accepted as adequate, the same test as any other runbook (16.3.1). And every finding that names a cause is confirmed by the accountable human reviewer before it leaves the room.

### KEY TERMS — KA 16.3

| TERM                         | MEANING                                                                                                                      |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge transfer</b>    | Conversion of project knowledge into documentation, trained people and rehearsed procedures usable without the project team. |
| <b>Runbook adequacy test</b> | Whether someone who was not on the project can execute it under supervision: the only proof documentation works.             |

| TERM                            | MEANING                                                                                                                              |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>Key-person concentration</b> | The share of procedures executable by exactly one named individual; an availability risk no project register records.                |
| <b>Post-project review</b>      | Structured examination of what happened and why, to improve future decisions, separated from performance assessment.                 |
| <b>Retrieval rate</b>           | The proportion of captured lessons retrieved before a comparable decision; the term in which almost all lesson value is lost.        |
| <b>Breakeven retrieval rate</b> | $\text{review cost} \div (\text{lessons} \times \text{avoided cost per applied lesson})$ : the threshold above which reviewing pays. |
| <b>Transition-out</b>           | Dissolution of the project organisation: authority withdrawn, accounts closed, access revoked, records placed, people released well. |

## SAMPLE MCQs — KA 16.3

**MCQ 16.3-A** [16.3.3 · Application] A review costing 46,000 produces 34 lessons; an applied lesson avoids 26,000 on average. The breakeven retrieval rate is closest to:

- A. 5.20 % CORRECT
- B. 12.00 %
- C. 43.36 %
- D. 52.04 %

*Rationale:*  $46,000 / (34 \times 26,000) = 46,000 / 884,000 = 5.20\%$  (16.3.3). B is the organisation's current retrieval rate, not the breakeven. C is  $46,000 / 106,080$ , the share of the *expected recovery* consumed by the review, which is a cost-recovery ratio and not a rate. D misplaces the decimal, a common slip on a small ratio.

**MCQ 16.3-B** [16.3.3 · Evaluation] A review costing 46,000 produces 34 lessons, an applied lesson avoids 26,000 on average, and 12 % of captured lessons are currently retrieved before a comparable decision. The highest-return improvement is to:

- A. run a longer, better-facilitated review workshop
- B. spend 18,000 indexing the lessons against the decision points where they apply, raising retrieval from 12 % to 35 % CORRECT
- C. capture more lessons per project
- D. shorten the review to reduce its 46,000 cost

*Rationale:* Indexing yields  $34 \times 0.23 \times 26,000 = 203,320$  for 18,000, a ratio of 11.30 (16.3.3). A and C invest in capture, where the value is not being lost. D reduces a cost that is already recovered 2.31 times over.

**MCQ 16.3-C** [16.3.1 · Application] Three undocumented interface configurations cost 8,400 each to reconstruct after handover, against 1,200 each to document at the time. The ratio of avoidable to incurred cost is:

- A. 7.00 CORRECT
- B. 3.00
- C. 21,600 to 1
- D. 2.33

*Rationale:*  $25,200/3,600 = 7.00$ , which is also the per-item ratio  $8,400/1,200$  (16.3.1). B counts the three items rather than the cost ratio. C is the absolute saving,  $25,200 - 3,600 = 21,600$ , mislabelled as a ratio. D divides the *per-item* reconstruction cost by the *total* documentation cost,  $8,400/3,600 = 2.33$ , mixing a unit figure with a total.

**MCQ 16.3-D** [16.3.2 · Analysis] The review practice most likely to produce transferable findings rather than narrative is to:

- A. begin from the schedule variance and work backwards
- B. begin from the decision record and ask of each material decision whether it was reasonable on what was known at the time CORRECT
- C. invite every stakeholder to describe their experience
- D. rank the events of the project by impact

*Rationale:* Decisions are the unit at which an organisation can improve; events are consequences (16.3.2, and Domain 3 KA 3.3.4 on the retrospective question). A and D produce narrative; C produces testimony without a comparison.

#### SELF-CHECK — KA 16.3

1. *What is the only proof that a runbook is adequate?* — That someone who was not on the project can execute it under supervision.
2. *Where is the value of a post-project review actually lost, and what does that imply about where to invest?* — At retrieval, not at capture; so indexing lessons against the decisions they bear on returns far more than improving the workshop, 11.30 to one on Meridian's figures.
3. *Name two consequences of failing to close the project organisation.* — A change board with no project continues to authorise changes to someone else's live service, and open cost accounts let late charges move an already-reported outturn, corrupting the estimating calibration.

## Benefits measurement, responsible archive and model/ data retention

*Topics: 16.4.1 the benefits measurement plan · 16.4.2 the closing account · 16.4.3 attribution and the comparison cohort · 16.4.4 responsible archive and retention · 16.4.5 model and data retention for AI-assisted decisions.*

### 16.4.1 The benefits measurement plan

**Definition.** A benefits measurement plan states, for each benefit in the register, the measure and its exact definition, the baseline value and the date it was taken, the target and its profile over time, the frequency and method of measurement, the named owner accountable for realisation, the **fund type** — cashable (with the budget line named and the finance owner's written confirmation), capacity-released, cost-avoidance or non-financial (Domain 15, KA 15.2.1) — the reporting route, and the date the measurement obligation ends.

**The fund type is not an accounting nicety, and it is the element most often missing.** It decides *what the realised number can be compared against*, and therefore whether the closing account of 16.4.2 is answering the question the approval was granted on. A benefit approved on a cash test and realised as capacity has not failed; it has been measured against the wrong comparator, and the two readings will be argued about for a year. Carrying the classification from the approval into the measurement plan is what stops that, and it also carries the **finance owner's confirmation** with it: where a benefit was approved as cashable, the measurement plan names the budget line that was to come off and the period it was to come off in, so the realisation review can ask the only question that settles it: *did it?* An unconfirmed cashable claim is measured as capacity-released until the confirmation exists, exactly as it is recorded in the portfolio register.

Every element earns its place by a failure it prevents, and three deserve emphasis because they are the ones that make a measured number arguable.

**The measure's exact definition, including its denominator.** Meridian registered "clinics in daily use" as its adoption measure, with "daily use" defined as at least one clinical episode recorded in the system on at least four days of a working week, averaged over a month. At the twelve-month benefits review the figure was **27 of 40 clinics, 67.50 %**. An informal figure of **68 %** had circulated, the number Domain 1's case study records, computed on a *clinician* count rather than a clinic count; the two measures differ because the non-adopting clinics are smaller than average. Neither number is wrong; only one is the registered measure, and the discipline that matters is that the definition was fixed before measurement and the review reports the registered measure with the variant identified. A programme that discovers at

measurement time that its measure has two readings will end up reporting whichever reading is more comfortable.

**The baseline, taken before the change, at the date recorded.** Domain 2 KA 2.3.2 required this and Meridian complied: clinician time on records tasks was sampled in all 40 clinics in the quarter before the first go-live. Without that, every number in 16.4.2 would be a reconstruction, and a reconstruction is an estimate shaped by the result it is used to justify.

**The end date of the measurement obligation.** Benefits measurement that runs indefinitely is not measured at all; the plan should state the period (Meridian's is the eight years of the appraisal, with formal reviews at 12, 24 and 48 months), and what happens at the end, which is normally that the measure either becomes a permanent operational indicator or is retired with a reason.

#### 16.4.2 The closing account

**Definition.** The closing account is the reconciliation of realised value against the value the business case promised, decomposed into the components that explain the difference, and reported to the body that approved the case.

This is the account the whole book has been keeping. Meridian's business case claimed **USD 979,200** a year on an output basis; Domain 1 corrected that to **USD 685,440** at 70 per cent adoption, and Domain 2 appraised the honest ramped profile at a present value of **USD 3,732,898** and an NPV of **+USD 1,332,898**. The measurement is now in.

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#### WORKED EXAMPLE

##### Worked example 16.4.2 — the closing account of Meridian Care Records.

1. **Setup.** At the twelve-month benefits review, with all 40 clinics live: **27 clinics** meet the registered daily-use definition (**67.50 %** adoption), and time sampling in the adopting clinics shows **5.4 clinician-hours** released a week rather than the **6.0** the case assumed. The valuation rate is unchanged at **USD 85** an hour over **48** operating weeks, and the comparison cohort of 16.4.3 shows no material change from other causes. Measured adoption by year, against the case's 40 % / 60 % / 70 % ramp, was **15 %**, **40 %** and **60 %**, reaching the measured steady state of 67.50 % only in year 4: the ramp arrived a full year later than the case assumed. The run cost is **USD 108,000** a year (16.2.1) against no operating-cost line in the case, and the capital outturn was **USD 2,514,000** against **USD 2,400,000** approved.
2. **Formula.** Annual benefit =  $U \times a \times h$ , where  $U = \text{clinics} \times \text{rate} \times \text{weeks} = 40 \times 85 \times 48 = 163,200$  is the benefit per unit of adoption-hours,  $a$  is adoption and  $h$  is hours released per adopting clinic per week. Two-factor variance decomposition: adoption effect =  $U \times h_{\text{plan}} \times (a_{\text{actual}} - a_{\text{plan}})$ ; hours effect =  $U \times a_{\text{actual}} \times (h_{\text{actual}} - h_{\text{plan}})$ .

3. **Substitution.** Case  $163,200 \times 0.70 \times 6.0$ ; measured  $163,200 \times 0.675 \times 5.4$ . Adoption effect  $163,200 \times 6.0 \times (0.675 - 0.700)$ ; hours effect  $163,200 \times 0.675 \times (5.4 - 6.0)$ .

4. **Result.** Measured steady-state benefit **USD 594,864** a year ( $\approx$  SAR 2.23 million indicatively) against the honest case's **USD 685,440**: a shortfall of **USD 90,576**, or **13.21 %**. The realised figure is **86.79 %** of the honest case and exactly **60.75 %** of the flat claim the case actually made. Decomposed, taking adoption first: adoption (**USD 24,480**), hours per adopting clinic (**USD 66,096**), total (**USD 90,576**).

The full present-value bridge, eight years at 7 % ( $AF = 5.971299$ ):

| BRIDGE LINE                                                     | NPV (USD)         |
|-----------------------------------------------------------------|-------------------|
| Flat claim, as approved                                         | <b>+3,447,096</b> |
| Adoption term corrected (Domain 1's term, valued in Domain 2)   | (2,114,198)       |
| <b>Honest ramped case (Domain 2)</b>                            | <b>+1,332,898</b> |
| Level variance (measured hours 5.4 and steady adoption 67.50 %) | (465,015)         |
| Timing variance (the ramp arrived one year late)                | (413,809)         |
| Operating cost omitted from the case (108,000 a year)           | (644,900)         |
| Cost outturn variance (2,514,000 against 2,400,000)             | (114,000)         |
| <b>Realised NPV</b>                                             | <b>(304,827)</b>  |
| Post-project improvement plan                                   | +340,156          |
| <b>Post-plan NPV</b>                                            | <b>+35,329</b>    |

Rows are rounded to the nearest dollar; subtotals are struck at full precision, so a row-by-row addition may differ by USD 1. The two variance lines are counterfactuals and the convention is the same one used on the annual decomposition, first factor at plan, second at actual. The level line values the measured 5.4 hours and 67.50 % steady adoption on the case's own adoption timing (40 % in year 1, 60 % in year 2, steady from year 3), and the timing line is the further loss from the measured ramp of 15 % / 40 % / 60 % reaching steady state only in year 4. Reversing the order would report timing at (479,771) and level at (399,053): the same total, with USD 65,962 of level-and-timing interaction moved between the lines.

1. **Interpretation.** Six readings, and they are the closing lessons of the book. **First, the largest single line is not a delivery failure at all.** The **2,114,198** correction is the adoption term the business case omitted — money that was never available to be realised, and which Domain 1's arithmetic removed before a single clinic was installed. An organisation that measures realisation against the

approved claim will record a **USD 384,336** annual shortfall and conclude the programme failed; measured against the honest case the shortfall is **USD 90,576**, and the difference between those two numbers is a defect in the case, not in the delivery. **Second, the decomposition's order matters and the convention must be stated.** Taking adoption first gives (24,480) adoption and (66,096) hours; taking hours first gives (68,544) hours and (22,032) adoption. Both sum to (90,576); the **USD 2,448** that moves between the lines is the interaction term,  $163,200 \times (-0.025) \times (-0.6)$ , and it exists because both factors moved. The convention used here (first factor at plan, second at actual) assigns it to the second line, and a report that does not say which convention it used has published two of its three numbers ambiguously. **Third, the hours error is nearly three times the adoption error**, which is the opposite of what the programme's own narrative said: everyone discussed adoption, because adoption is visible and Domain 1's case study had made it the headline. The benefit *per adopting unit* was never re-measured until this review, and it was wrong by 10 per cent: a reminder that a benefits case has two quantitative assumptions and organisations habitually monitor one. **Fourth, timing cost almost as much as level:** (413,809) against (465,015), from nothing more than the ramp arriving a year late. A benefit deferred is a benefit partly destroyed, and this is the same arithmetic as the cost of delay from Domain 1 seen from the other end. **Fifth, the realised NPV is negative. And and it is negative because of the line the project never controlled.** Strip the operating cost out and the realised position is **+USD 340,073**; the run cost the case never carried is what turns the account. That is the single most transferable finding in this domain: **the omission of an operating cost line is frequently the whole margin, not a presentational defect.** **Sixth, the account is not closed by the project.** The improvement plan (**USD 180,000** at the end of year 3 to complete the workflow redesign in the thirteen non-adopting clinics and re-run the champion programme, lifting adoption to **80 %** and restoring the full **6.0** hours from year 5) is worth  $188,496 \times 2.584087 = \text{USD } 487,090$  in present value against a present cost of **USD 146,934**: an NPV of **+USD 340,156** and a benefit-cost ratio of **3.32**. It converts the programme's realised NPV from **(304,827)** to **+35,329**. The whole value of a USD 2.4 million programme came to rest on USD 180,000 of post-project adoption work that appeared in no project plan, and the professional caution attached is exact: that plan needs an owner, a budget line and a decision, and at the moment the project closes there is no project to provide any of the three. The breakeven is worth stating for the board: with the plan's USD 180,000 spent, the account needs an effective realisation level of **78.60 %** of full potential in years 5 to 8 to reach zero, against **60.75 %** on the current trajectory and the **80 %** the plan is designed to deliver.

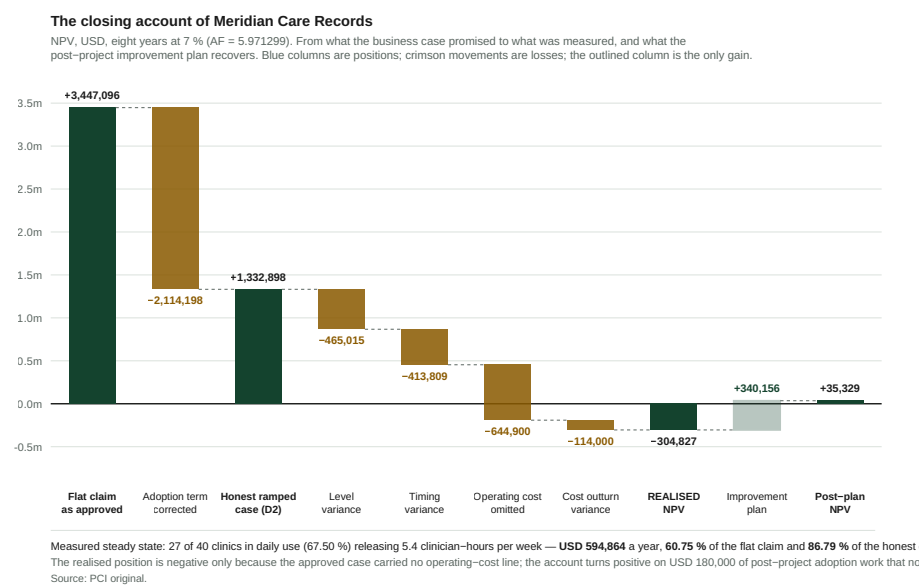


Fig 16.4.1 — The closing account of Meridian Care Records

16.4.3 Attribution and the comparison cohort

**Definition.** Attribution is the determination of how much of an observed change was caused by the project rather than by anything else happening at the same time. A **comparison cohort** is a set of comparable units, measured at baseline and at review but not receiving the change (or receiving it later), against which the observed change is netted.

Attribution is the argument that destroys benefits claims, and it is unwinnable after the fact. Meridian handled it correctly and cheaply: because the rollout ran in waves, the **8 clinics** in the final wave were measured at the same baseline and again at the twelve-month review while still on the old process. Their records-task time changed by **0.0 hours a week**, within the sampling tolerance of **±0.3 hours**, so no attribution adjustment was made and the full 5.4 hours is claimed as attributable.

The value of that cohort is best seen by supposing it absent. A reviewer could then assert that a concurrent national workflow initiative accounted for, say, **0.6** of the 5.4 hours. That assertion is unfalsifiable without a comparison, and it is expensive: at **4.8** attributable hours the steady-state benefit falls to **USD 528,768** a year, the realised present value of benefits falls to **USD 2,536,954**, and the realised NPV falls from **(304,827)** to **(621,946)**. The claim more than doubles the measured loss. A comparison cohort that costs a few days of sampling therefore protects a figure of that order, and the professional rule follows directly: **the attribution question must be settled by design at baseline, not by argument at review**, because at review the only available instruments are assertion and seniority.

Two honest limitations. A wave-based comparison cohort is not a randomised comparison, and the later waves may differ systematically from the earlier ones (Meridian's did, being smaller), so the cohort bounds the attribution question rather



than closing it, and the review should say so. And where no comparison is possible at all, the correct treatment is to report the benefit in its physical unit with the attribution uncertainty stated, not to convert an uncertain quantity into a confident monetary claim (Domain 2, KA 2.3.2 on cash-releasing and non-cash-releasing benefits).

**A correction the closing account forces on the rest of the book.** Every delay calculation in Domains 3, 4 and 6 was priced at Domain 1's cost of delay of **USD 14,280** a week, derived from 28 adopting clinics releasing 6 hours at 85. The realised rate is  $27 \times 5.4 \times 85 = \text{USD } 12,393$  a week: **86.79 %** of it, meaning the figure used throughout was **15.23 %** higher than realisation justifies. The professional obligation is to re-test the conclusions, not to restate them, and the result is reassuring: Domain 3's delegation-threshold analysis saved **USD 171,360** a year at 14,280 and saves **USD 148,716** at 12,393, moving the breakeven value-destruction rate on the delegated band from 84 % to **72.90 %**: still far beyond any plausible delegate. **A conclusion that survives a 15 per cent error in its principal input is a robust conclusion, and one that does not survive should never have been presented as a point estimate.** That test is the most useful thing a closing account gives back to the organisation's future decisions.

#### 16.4.4 Responsible archive and retention

**Definition.** The archive is the retained record of the project: its decisions, contracts, evidence, configurations and data, held for a stated period against stated purposes, with a defined disposal or de-identification action at the end. Retention is a **schedule by class**, never a single rule.

Two forces pull in opposite directions and both are legitimate. Evidence must be kept, because the questions that arrive later (a dispute, an audit, a latent defect, an inquiry) can only be answered from records that exist, at the versions relied on. And personal data must **not** be kept, because holding it creates obligations and exposure that grow with time and volume while its usefulness falls. A single retention rule cannot serve both, which is why the schedule is class-by-class. Meridian's 6.4 TB archive resolves into: contract and commercial evidence **1.9 TB**; the decision and governance record **0.4 TB**; design, configuration and as-built **1.6 TB**; test and commissioning evidence **1.4 TB**; and personal and patient-linked migration logs **1.1 TB**.

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#### WORKED EXAMPLE

##### Worked example 16.4.4 — retention as insurance, and retention as liability.

1. **Setup.** Storing the archive costs **USD 21** per terabyte per month. The programme's assessed probability that a dispute, audit or latent-defect question will require the contractual and technical evidence within seven years is **0.18**, and the assessed cost of being unable to answer it (an adverse determination or an unrecoverable rectification) is **USD 310,000**. Separately, the **1.1 TB** of patient-

linked migration logs can be de-identified for a one-off **USD 34,000**, removing an assessed breach exposure of probability **0.04** and consequence **USD 1,250,000**.

2. **Formula.** Cost of retention = volume × unit rate × months. Value of retention =  $P(\text{need}) \times \text{cost of not having it}$ . Value of de-identification = exposure removed + storage avoided – de-identification cost. Breakeven need probability = retention cost ÷ consequence.
3. **Substitution.**  $6.4 \times 21 = 134.40$  a month,  $\times 12 \times 7.0.18 \times 310,000$ . De-identification:  $0.04 \times 1,250,000 + 1.1 \times 21 \times 12 \times 7 - 34,000$ .
4. **Result.** Seven years of retention costs **USD 11,290**; the expected value of holding the evidence is **USD 55,800**; a ratio of **4.94**, with a breakeven need probability of **3.64 %**. De-identifying the patient-linked logs removes **USD 50,000** of expected exposure and **USD 1,940** of storage for **USD 34,000**, a net gain of **USD 17,940**.
5. **Interpretation.** The two halves of the result point in opposite directions and that is the whole lesson. **Retaining evidence is extraordinarily cheap insurance** (it pays at a need probability above 3.64 %) roughly one chance in twenty-seven (and no serious programme can claim its evidence is less likely than that to be wanted), so the reflex to purge storage for cost reasons is almost always wrong, and the correct argument for deleting is never cost. **Retaining personal data is the reverse trade**: the same volume that costs almost nothing to store carries an exposure that dwarfs its storage cost by a factor of 26 to one, and de-identification pays for itself several times over even before any regulatory dimension is considered. Three cautions. The breakeven is so low that the **decision is really about the consequence estimate**, and a programme that cannot articulate what it would lose by being unable to answer has not thought about the risk. Retention periods for personal and health data are **set by law and by contract in each jurisdiction and by data class**, they differ substantially, and nothing here states a legal minimum or maximum. The schedule must be agreed with whoever holds the data-protection accountability in the organisation, and the figures above are an illustration of the economics, not of the law. And **de-identification must be tested, not asserted**: a log that can be re-identified by joining it to another retained dataset has not been de-identified, and the test belongs in the schedule.

#### 16.4.5 Model and data retention for AI-assisted decisions

**Definition.** Model and data retention is the retained record of what an AI system was asked, what it returned, which model version returned it, who verified the output and what decision relied on it: the minimum that permits an AI-informed decision to be explained after the fact.

Domain 14 established the AI-use register and the verification standard proportional to consequence; this topic is the retention consequence, and it is the one closeout usually discovers too late. Of the **214** AI-assisted outputs that Meridian's programme relied on in recorded decisions, **61** cannot be reproduced because the model identifier and version were not captured at the time: **28.50** % of the AI-informed decision base is therefore unexplainable, and re-deriving those outputs by hand at **USD 1,850** each would cost **USD 112,850**, which nobody will spend. The fix costs nothing and is three fields on a record that was being written anyway: **model identifier, model version, date**, alongside the prompt or input, the output, and the named human verifier.

Retention here has its own class distinctions. Prompts and outputs relied on in a decision belong with the decision record and inherit its retention period (Domain 3, KA 3.3.4). Training or fine-tuning data, where the organisation created any, carries the personal-data considerations of 16.4.4 and usually the strictest treatment in the schedule. Model artefacts themselves may be unretainable (a third-party service's version may simply cease to exist), which is precisely why the *record of what it produced and who verified it* is the durable evidence, and why a retention plan that assumes the model will still be available is not a plan. The closing obligation is a short one, and it is assessable: **for every AI-informed decision in the record, a reader two years later can identify the model, the input, the output, the verifier and the decision-maker**. Anything less means the organisation cannot explain its own decisions, which is the accountability defect Domain 1 named and Domain 14 quantified, arriving at the archive.

### AI in this KA

**Where it earns its place.** Assembling the measurement data: extracting usage counts against the registered definition, reconciling them to the source systems, and flagging units whose classification changed between periods, which is where measured adoption series usually break. Computing the variance decomposition across many benefits and both decomposition orders, and reporting the interaction term explicitly. Classifying an archive by data class against a retention schedule and listing records whose class cannot be determined: the highest-value output, because unclassified records are the ones that get kept by default. Detecting records missing the model-identifier fields of 16.4.5 across a large decision log.

**Where it must not go.** It must not decide what the measured benefit was. The measure's definition, the treatment of edge cases and the attribution judgement are accountable determinations that will be read as the organisation's own statement about its performance. It must not generate the explanation of a variance: a model asked why realisation fell short will produce a fluent, plausible causal account that is indistinguishable in tone from an evidenced one, and a benefits report containing one has substituted narrative for measurement. And it must not authorise a deletion: disposal under a retention schedule is an accountable, usually irreversible act, and it

must carry a human authorisation and a record of what was destroyed and on what authority.

**Verification, concretely.** Every measured figure traces to a source-system query that a second person can re-run, and the query is retained with the figure. The decomposition is reproduced by hand for the report (it is four multiplications) and both orders are computed so the interaction term is visible rather than absorbed. Attribution rests on the comparison cohort's measured data, not on an argument. And every disposal action generates a record identifying the class, the volume, the authority and the date, which is itself retained under the longest applicable period.

#### KEY TERMS — KA 16.4

| TERM                                          | MEANING                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Benefits measurement plan</b>              | Measure definition, baseline and date, target profile, method, frequency, owner, <b>fund type</b> , reporting route and end date, per benefit.                                                                                                                                                |
| <b>Fund type (carried from approval)</b>      | Cashable with the budget line named and the finance owner's written confirmation, capacity-released, cost-avoidance or non-financial (Domain 15, KA 15.2.1). It decides what the realised number may be compared against, and an unconfirmed cashable claim is measured as capacity-released. |
| <b>Registered measure</b>                     | The single definition of a benefit measure fixed before measurement; variants are identified, not substituted.                                                                                                                                                                                |
| <b>Closing account</b>                        | Reconciliation of realised value against the promise, decomposed into its causes and reported to the approving body.                                                                                                                                                                          |
| <b>Benefit per unit of adoption-hours (U)</b> | $\text{clinics} \times \text{valuation rate} \times \text{operating weeks}$ , the constant in $\text{benefit} = U \times a \times h$ .                                                                                                                                                        |
| <b>Level variance</b>                         | The value effect of a wrong steady-state adoption or benefit-per-unit assumption.                                                                                                                                                                                                             |
| <b>Timing variance</b>                        | The value effect of the benefit profile arriving later than the case assumed.                                                                                                                                                                                                                 |
| <b>Interaction term</b>                       | The product of both factor movements; the amount that shifts between decomposition lines when the order changes.                                                                                                                                                                              |
| <b>Comparison cohort</b>                      | Comparable units measured at baseline and review without receiving the change, used to net out other causes.                                                                                                                                                                                  |
| <b>Retention schedule</b>                     | Class-by-class statement of what is kept, for how long, on what authority, and what disposal or de-identification follows.                                                                                                                                                                    |
| <b>Model retention record</b>                 | Model identifier, version and date alongside input, output, verifier and decision; the minimum that makes an AI-informed decision explainable.                                                                                                                                                |

## SAMPLE MCQS — KA 16.4

**MCQ 16.4-A** [16.4.2 · Application] Twenty-seven of 40 clinics are in daily use, each releasing 5.4 hours a week at USD 85 over 48 weeks. The measured annual benefit is:

- A. USD 685,440
- B. USD 594,864 CORRECT
- C. USD 881,280
- D. USD 660,960

*Rationale:*  $27 \times 5.4 \times 85 \times 48 = 594,864$  (16.4.2). A is the honest business-case figure at 70 % adoption and 6.0 hours. C omits the adoption term, counting all 40 clinics at the measured 5.4 hours. D keeps the correct 27 clinics but the case's 6.0 hours, omitting the benefit-per-unit correction.

**MCQ 16.4-B** [16.4.2 · Analysis] With  $U = 163,200$ , planned adoption 0.700 and 6.0 hours, and measured adoption 0.675 and 5.4 hours, the decomposition taking **adoption first** gives:

- A. adoption (24,480); hours (66,096) CORRECT
- B. adoption (22,032); hours (68,544)
- C. adoption (24,480); hours (68,544)
- D. adoption (66,096); hours (24,480)

*Rationale:* Adoption at planned hours is  $163,200 \times 6.0 \times (-0.025) = (24,480)$ ; hours at measured adoption is  $163,200 \times 0.675 \times (-0.6) = (66,096)$ ; both sum to (90,576) (16.4.2). B is the reverse order: correct arithmetic, wrong to the question asked. C mixes the two orders and over-states the total by the 2,448 interaction term. D transposes the labels.

**MCQ 16.4-C** [16.4.2 · Evaluation] Meridian's realised NPV is (304,827). Which single line is most responsible, and what does that imply?

- A. the adoption shortfall; the rollout under-delivered
- B. the operating cost omitted from the case, at (644,900); without it the realised position is +340,073, so the whole margin turned on a line the project never controlled CORRECT
- C. the cost outturn variance of (114,000); the programme overspent
- D. the timing variance; benefits arrived late

*Rationale:* The operating-cost line is the largest realised-period movement and exceeds the whole negative position (16.4.2, 16.2.1). A and D are real but smaller, (465,015) and (413,809). C is the smallest line at 4.75 % of approved cost.

**MCQ 16.4-D** [16.4.3 · Analysis] A reviewer claims a concurrent national initiative accounts for 0.6 of the 5.4 measured hours. The measurement design element that answers the claim is:

- A. a larger time-sampling exercise in the adopting clinics
- B. the comparison cohort of clinics measured at baseline and review without the change CORRECT
- C. a sensitivity analysis on the valuation rate
- D. the benefits owner's professional judgement

*Rationale:* Only a comparison can net out other causes, and it must exist from baseline (16.4.3); without it the assertion is unfalsifiable and, at 4.8 attributable hours, would move the realised NPV from (304,827) to (621,946). A measures the same population more precisely and answers a different question.

**MCQ 16.4-E** [16.4.4 · Application] A 6.4 TB archive costs USD 21 per TB per month. If the probability that the evidence is needed within seven years is 0.18 and the cost of being unable to answer is 310,000, the breakeven need probability is closest to:

- A. 18.00 %
- B. 3.64 % CORRECT
- C. 0.52 %
- D. 36.42 %

*Rationale:*  $6.4 \times 21 \times 12 \times 7 = 11,290$ ;  $11,290/310,000 = 3.64\%$  (16.4.4). A is the assessed probability, not the breakeven. C uses one year of storage. D misplaces the decimal.

**MCQ 16.4-F** [16.4.5 · Comprehension] The minimum retention that makes an AI-informed decision explainable two years later is:

- A. the model artefact itself, retained locally
- B. the prompt and the output
- C. model identifier and version, date, input, output, named verifier and decision-maker CORRECT
- D. a summary of the analysis in the decision record

*Rationale:* A third-party model version may cease to exist, so the durable evidence is the record of what it produced and who verified it (16.4.5). B omits the version and the verifier, which is exactly the omission that made 28.50 % of Meridian's AI-informed decision base unexplainable.

## SELF-CHECK — KA 16.4

1. *Why must the measure's denominator be fixed before measurement?* — Because a measure with two readings will be reported at whichever reading is more comfortable; Meridian's 67.50 % of clinics and the informally circulated 68 % of clinicians are both defensible arithmetic and only one is the registered measure.
2. *Which decomposition line moves when the order is reversed, and by how much?* — The interaction term,  $163,200 \times (-0.025) \times (-0.6) = 2,448$ , which is why the convention must be stated.
3. *State the opposite economics of retaining evidence and retaining personal data.* — Evidence pays as insurance above a 3.64 % need probability, so cost is never the argument for deleting it; personal data carries exposure that dwarfs its storage cost, so de-identification pays for itself, 17,940 net on Meridian's 1.1 TB.
4. *Why does the measurement plan carry the fund type forward from the approval?* — Because the fund type decides what the realised number may be compared against. A benefit approved on a cash test and realised as capacity has not failed; it has been measured against the wrong comparator. Where a benefit was approved as cashable, the plan names the budget line and the period, so the review can ask whether it came off (16.4.1; Domain 15, KA 15.2.1).

## Advanced topics — Domain 16

### 16.A.1 Benefits decay and sustainment

Benefits are not a stock, they are a flow that depends on a behaviour continuing, and behaviours revert. Where the enabling change is not maintained (champions move on, new staff are trained on the old workflow because that is what the induction pack says, a system update degrades a step nobody owns) realisation decays. Meridian's post-review assessment was a **4 % relative decline a year** in the effective realisation level from year 4 in the absence of active sustainment: 0.6075 of full potential falling to 0.5832, 0.5599, 0.5375, 0.5160 and 0.4953 across years 4 to 8. The present value of that decay is **USD 216,824**, against a sustainment programme (refresher training, champion succession, a quarterly workflow review), costing **USD 22,000** a year for those five years, a present value of **USD 73,634**. Net value **USD 143,190**, a ratio of **2.94**, and the breakeven sustainment spend is **USD 64,782** a year, almost three times what the programme proposed.

Two structural points follow. Sustainment is an **operating** activity with a **capital-programme** justification, which is exactly the kind of expenditure that organisations fail to fund because the benefit accrues to a case that is already closed; the closing account of 16.4.2 is the instrument that makes the argument, and it must be handed to the benefits owner with the number attached. And decay is measurable long before it is visible in the money. The leading indicator is the registered adoption

measure, monitored monthly against its own trend rather than against its target, because a measure at 95 per cent of target and falling is a different situation from one at 95 per cent and steady, and a report that shows only the variance against target cannot distinguish them.

### 16.A.2 Closing a project that failed, and closing one that was cancelled

Both are legitimate closures and both are done badly for the same reason: the organisation wants the episode over, and speed is mistaken for decisiveness.

**A cancelled project** (Domain 2 KA 2.4 made the case for stopping) must still be closed properly, and the specific obligations are: secure and value whatever is salvageable, including work in progress that another project can use, licences that can be redeployed and, most often overlooked, **requirements and design work that retains value independently of the delivery**; settle contracts on a termination basis rather than abandoning them, since an unterminated contract accrues the carrying cost of 16.2.4 indefinitely: termination rights, their notice mechanics and the compensation they trigger are contractual and jurisdictional, so take qualified legal advice before serving anything; account for the sunk cost honestly and separately from the decision to stop, because conflating them is how the next cancellation gets delayed; and hold the review, which is the highest-value review the organisation will ever run and the one it is least likely to hold, because the findings are uncomfortable and the participants have dispersed.

**A project that delivered but failed to realise** is the harder case, because there is no moment at which anyone is obliged to say so. Meridian is close to this shape: outputs delivered, adoption short, operating cost unbudgeted, realised NPV negative until a post-project intervention nobody had planned. The closing discipline is to **report the closing account to the body that approved the case**, with the variance decomposed and the improvement plan priced, rather than to close the project and report delivery. That is an uncomfortable paper to write and it is the one that makes the next business case honest, which is the whole return on writing it.

### 16.A.3 The reviewer's closeout eye

Invariants to test on any transition and closeout, each cheap and each diagnostic.

The handover trigger is a **condition set**, not a date, and each condition is assessable by someone who is not delivering it. The condition set is **split into two blocks**: no mandatory precondition (safety case, regulatory or licence approval, privacy assessment, statutory notification, takeover certificate) is represented as a probability, each is recorded met or not met with its approving authority named, and **no gate-block item appears anywhere in the go/hold economics**, since a cost attached to it is a price at which it could be waived. The readiness figure presented to the go-live decision is the **conjunction**  $\prod p_i$  over the *discretionary* conditions, not an average, and the paper shows both the independent and the  $\min(p_i)$  bounds. A **service test** exists and was run by the receiving organisation. Hypercare has a



**measured** exit criterion. A **rehearsed reversion plan** exists with a latest-possible-reversion point. The receiving organisation has a **funded run line** and the whole-life cost appears in the case. Retention releases against **events**, and outstanding items are a priced list from the day of takeover. Every open claim has a **stated monthly carrying cost** and a settlement authority. The final payment **reconciles** to the movements since the last certificate, and the account is signed as full and final. Documentation has passed the **execution test** by someone who was not on the project, and the key-person concentration is counted. The lesson **retrieval rate is instrumented**, not assumed. Every benefit has a **registered measure with a stated denominator**, a **pre-change baseline with a date**, a named owner outside the project, and an end date for the obligation. The closing account is **decomposed** into level, timing, per-unit and cost components with the **interaction term and its convention stated**. A **comparison cohort** was established at baseline. The retention schedule is **class-by-class** with a disposal action and a recorded authority. And for every AI-informed decision, the **model identifier, version, verifier and decision-maker** are all in the record: the test that 28.50 % of Meridian's decision base failed.

## Industry variations – Domain 16

- **Healthcare.** Clinical safety governs the go-live decision and is not delegable to a project authority: a clinical safety case and its hazard log must be closed by clinical governance before transition (the leading item of 16.1.3's gate block, recorded met or not met and never as a probability), and a reversion plan must be viable at any point because the fallback is patient care, not a batch job. Benefits are overwhelmingly capacity rather than cash (Domain 2, KA 2.3.2), which makes the measure definition of 16.4.1 the whole argument.
- **Public sector and government.** Closure reporting is frequently required and published, benefits are audited by an external body against the approved case rather than against the honest one, and the archive's retention period is commonly set by public-records requirements rather than by the economics of 16.4.4. The practical consequence is that the adoption-term correction of 16.4.2 must be made and approved *before* delivery, because after it the published claim is the benchmark.
- **Construction and infrastructure.** Commissioning, takeover, defects liability and the final account are contractually formal, with certificates, dated periods and a defined dispute route; the closeout risk is concentrated in latent-defect exposure that outlives everyone involved, so the as-built documentation and test evidence of 16.1.2 are the retained assets that matter most.
- **Energy and utilities.** Performance demonstration under representative load is the acceptance event, and an availability or efficiency shortfall accepted at commissioning becomes a permanent operating annuity, the arithmetic of Case

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study B. Regulatory witnessing constrains when a retest can occur, so the retest window is a schedule constraint, not a contingency.

- **Technology and product organisations.** There is often no closeout at all, because the product continues: the risk inverts from a botched transition to an **absent** one, with the delivery team becoming a permanent unfunded support function and the run cost of 16.2.1 never surfacing as a decision. The discipline that substitutes for closeout is a funded service-ownership transfer with a stated date and an operating budget.
  - **Financial services and regulated industries.** Model inventory, change evidence and decision auditability drive the archive: the retention schedule of 16.4.4 and the model records of 16.4.5 are supervisory expectations rather than good practice, and an unexplainable AI-informed decision is a finding rather than an inconvenience.
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## Case study — Domain 16: the account nobody asked for (health, Meridian)

**Situation.** Meridian Care Records closed twenty-six months after mobilisation with all 40 clinics live, the systems integrator's final account agreed at **USD 1,831,400**, and a capital outturn of **USD 2,514,000** against **USD 2,400,000** approved, a 4.75 % overspend that the programme board accepted without discussion. The closure report ran to nine pages, recorded delivery of every output, and was noted. No one asked what had been realised, and the governance body that had approved the business case had, by then, been reconstituted twice.

**What the project leader did anyway.** Twelve months after the last go-live, and with no obligation to do so, the outgoing project leader and the clinical directorate's benefits owner produced a closing account against the registered measures. Adoption: **27 of 40** clinics meeting the daily-use definition, **67.50 %**. Time released, by sampling in the adopting clinics: **5.4** hours a week against the case's 6.0. Comparison cohort, the final wave of 8 clinics, measured at the same baseline and still on the old process: **no material change**, so no attribution adjustment. Measured annual benefit **USD 594,864**: **86.79 %** of Domain 1's honest figure and **60.75 %** of the claim the approved case had actually made.

**The decomposition, and what it changed.** The **USD 90,576** shortfall against the honest case split into **(24,480)** of adoption and **(66,096)** of hours per adopting clinic, with the **2,448** interaction term assigned to the second line under the stated convention. That split reversed the programme's own narrative: eighteen months of governance attention had gone to adoption, because adoption was visible and had already been the subject of a public failure, while the benefit *per adopting clinic* (the other half of the case's arithmetic) had never been re-measured and was wrong by 10 per cent. Extending the account to present value produced the harder number: level

variance (465,015), timing variance (413,809) from a ramp that arrived a year late, the omitted operating cost of 108,000 a year worth (644,900), the cost outturn (114,000), and a realised NPV of (USD 304,827) against the +1,332,898 the honest case had promised.

**How it resolved.** The account was sent, unrequested, to the board that had approved the case, with one proposal attached: USD 180,000 to complete the workflow redesign in the thirteen non-adopting clinics and re-run the champion programme, lifting adoption to 80 % and restoring the full 6.0 hours from year 5. Its present value was USD 487,090 against a present cost of USD 146,934: an NPV of +340,156, a benefit-cost ratio of 3.32, and enough to move the programme's realised NPV to +USD 35,329. It was approved in one meeting, funded from operating budget, and given to the benefits owner with a monthly adoption report against the 78.60 % effective level the account identified as the programme's breakeven. Two further changes followed, both about the next programme rather than this one: every business case in the portfolio acquired a mandatory operating-cost line, and the standard benefits register template acquired a **benefit-per-unit** measure alongside the adoption measure.

**What the domain teaches here.** The account nobody asks for is the only one that changes anything. Meridian's largest single bridge line: 2,114,198: was the adoption term the case omitted, money that never existed, and an organisation that had measured realisation against the approved claim would have recorded a 384,336 annual failure and learned nothing about its own case-writing. The second largest was an operating cost the project could not have controlled and should have insisted on. And the entire value of a 2.4 million dollar programme finally rested on 180,000 dollars of adoption work that appeared in no project plan, had no owner at the moment the project closed, and would not have been funded had nobody computed what it was worth.

### Case study B — Domain 16: the 0.8 percentage point that was worth a million (energy, Project Auriga)

**Situation.** Project Auriga, the 25-week control-systems upgrade for a regional utility, reached its witnessed performance demonstration two weeks late. Domain 7's week-13 position had given CPI 0.9057 and an estimate at completion of BAC/CPI = USD 4,416,667; the outturn was USD 4,412,000, 0.11 % below that forecast and 10.30 % above the USD 4,000,000 budget, a useful reminder that a CPI-based forecast made twelve weeks from the end was accurate to within five thousand dollars while the optimistic recovery narrative running alongside it was not.

**What happened at commissioning.** The contract required demonstrated system availability of 98.0 % over the witnessed test period. The system achieved 97.2 %. The contractor offered the contractual reduced-performance deduction of USD 240,000 in lieu of remediation, which was within its liability cap and was recommended for

acceptance by the project team on the grounds that it closed the contract, released the **USD 160,000** of retention held on the 3,200,000 control-systems subcontract, and avoided further delay.

**What the operations director asked for instead.** One number: what does 0.8 of a percentage point of availability cost to operate. The utility's own switching-restriction model priced restricted operation at **USD 1,900** an hour; 0.8 % of 8,760 hours is **70.08** hours a year, or **USD 133,152** annually, and over the asset's 12-year life at 7 % ( $AF = 7.942686$ ) a present value of **USD 1,057,585**. The remediation-and-retest alternative cost **USD 74,000** plus three weeks of deferred cutover at the utility's calibrated **USD 18,000** a week — **USD 128,000** in total. Accepting the deduction was therefore worth  $240,000 - 1,057,585 = \text{(USD 817,585)}$ ; retesting was worth **USD 689,585** more than accepting. The breakeven deduction (the payment at which acceptance would have been the better trade) was **USD 1,185,585**, nearly five times what was offered; equivalently, acceptance would only have been right if the restriction cost were below **USD 661** an hour rather than 1,900.

**How it resolved.** The contractor remediated two controller configurations and one network path and retested, achieving **98.3 %**; the retention was released against the passed test rather than against a deduction; the cutover ran three weeks later than the failed test date. The readiness conjunction for the cutover itself, computed on the five conditions the utility's own gate required (0.97, 0.93, 0.99, 0.88, 0.96), stood at **75.45 %** against an averaged dashboard reading of **94.60 %**; and the remaining exposure was carried by a rehearsed reversion to manual switching rather than by further assurance.

**What the domain teaches here.** A performance shortfall accepted at commissioning is not a one-off concession, it is a permanent annuity of operating cost paid by an organisation that was not in the room. The contractual deduction prices the **contractor's** risk, capped by the contract; the shortfall prices the **operator's** loss, uncapped and lasting as long as the asset; and the two are not comparable quantities, which is why the acceptance decision belongs to the receiving organisation and not to the project that is trying to finish. The transferable discipline is to convert every proposed acceptance concession into an annual operating cost and then into a present value over the asset life before answering, and to state the breakeven (here USD 661 an hour), so that the decision can be argued on its actual sensitivity rather than on the attractiveness of closing the contract.

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## Executive perspective — Domain 16

What a programme director cannot delegate in this domain:

- **The handover condition set, and the arithmetic of readiness.** Insist on the conjunction  $\Pi p_i$  with both correlation bounds, never an averaged dashboard. Meridian's dashboard read 90.71 % where the probability of a clean go-live was

49.79 %, and no other single number in closeout is so routinely misreported (16.1.3).

- **The acceptance of any performance concession.** Convert it to an annual operating cost and a present value over the asset life, and state the breakeven, before agreeing. A deduction prices the supplier's capped risk, not the operator's uncapped loss (Case study B).
- **The funded run line before the project closes.** No transition without a named service owner and an operating budget. Meridian's omitted 108,000 a year was worth 644,900 in present value and turned the whole account negative (16.2.1, 16.4.2).
- **The carrying cost of every open item.** A monthly number per open claim, and a settlement authority. At 5,250 a month, twelve months of avoidance cost Meridian more than the claim's assessed value (16.2.4).
- **The closing account, reported to the body that approved the case.** Decomposed into level, timing, per-unit and cost, with the interaction term and its convention stated, and the improvement plan priced. Nobody will ask for it, and it is the only document that makes the next business case honest (16.4.2, 16.A.2).
- **The retention schedule and the model record.** Class-by-class retention with disposal authority, and the model identifier, version and verifier on every AI-informed decision. Twenty-eight and a half per cent of Meridian's AI-informed decision base is unexplainable for want of three fields (16.4.4, 16.4.5).

## Calculation exercises — Domain 16

**Exercise 16.1** A transition gate carries six conditions, all necessary and assessed as independent, at 0.95, 0.88, 0.92, 0.99, 0.86 and 0.90. Compute the averaged dashboard reading and the probability of a clean transition; identify which single condition, lifted to 0.99, gains most and by how much; and state the per-condition probability needed for a 90 % chance of a clean transition. *Solution.* Average =  $(0.95 + 0.88 + 0.92 + 0.99 + 0.86 + 0.90) / 6 = 5.50 / 6 = 91.67 \%$ . Conjunction =  $0.95 \times 0.88 \times 0.92 \times 0.99 \times 0.86 \times 0.90 = 58.93 \%$ : a gap of **32.73 percentage points**. Lifting the 0.86 condition to 0.99 multiplies the conjunction by  $0.99 / 0.86 = 1.1512$ , giving **67.84 %**, a gain of **8.91 points**: against **2.48** for the 0.95 condition, **7.37** for 0.88, **4.48** for 0.92 and **5.89** for 0.90. For 90 % clean,  $p = 0.90^{(1/6)} = 98.26 \%$  on every condition. *Common error:* treating the gain as proportional to the gap closed (an equal return per percentage point), which understates the value of fixing the weak conditions, since the conjunction is multiplied by  $p'/p$ , so the gain per point closed is proportional to  $1/p$  and a point closed at 0.86 is worth  $0.99 / 0.86 = 1.15$  times a point closed at 0.99. Reading the dashboard average of 91.67 % as a probability loses the ranking entirely.

**Exercise 16.2** A contract sum of USD 2,450,000 carries approved variations of USD 138,000. Retention was 5 % of the contract sum, half released at takeover. A claim settled at USD 55,000 and USD 26,400 of defects were rectified by others and are recoverable. Compute the retention, the amount paid to date, the final account, the final retention release and the final payment. *Solution.* Retention  $2,450,000 \times 0.05 = \text{USD } 122,500$ . Certified works  $2,450,000 + 138,000 = 2,588,000$ ; paid on certification  $2,588,000 - 122,500 = 2,465,500$ ; plus the takeover release of  $122,500/2 = 61,250$ , so **USD 2,526,750** paid to date. Final account  $2,450,000 + 138,000 + 55,000 - 26,400 = \text{USD } 2,616,600$ . Final retention release  $61,250 - 26,400 = \text{USD } 34,850$ . Final payment  $2,616,600 - 2,526,750 = \text{USD } 89,850$ , which reconciles as  $61,250 + 55,000 - 26,400$ . *Common error:* stopping at the certified works value of 2,588,000 and calling it the final account, which omits both the settlement and the recovery and understates contract growth: here by 28,600, or 1.17 % of the original sum.

**Exercise 16.3** An open claim costs USD 2,900 a month in external advice, 1.2 days a month of internal management at USD 750 a day, and the opportunity cost of USD 240,000 of locked cash at 6 % a year. Determination at month 11 has assessed outcomes of 30,000 (probability 0.30), 55,000 (0.45) and 96,000 (0.25); the client's own assessment is 55,000. Settlement is available at month 1. Compute the monthly carrying cost, the expected cost of determination, the breakeven settlement price and the breakeven premium over the client's assessment. *Solution.* Carrying cost  $= 2,900 + 1.2 \times 750 + 240,000 \times 0.06/12 = 2,900 + 900 + 1,200 = \text{USD } 5,000 \text{ a month}$ . EMV  $= 0.30 \times 30,000 + 0.45 \times 55,000 + 0.25 \times 96,000 = 9,000 + 24,750 + 24,000 = \text{USD } 57,750$ . Expected cost of determination  $57,750 + 11 \times 5,000 = \text{USD } 112,750$ . Breakeven settlement price at month 1  $112,750 - 5,000 = \text{USD } 107,750$ . Breakeven premium  $107,750 - 55,000 = \text{USD } 52,750$ , which the identity  $10 \times 5,000 + (57,750 - 55,000)$  reproduces. *Common error:* omitting the internal management time (900 of the 5,000, or 18 %) on the grounds that the staff are paid anyway, which is the reasoning that makes the carrying cost invisible and, at ten months, understates the breakeven premium by 9,000.

**Exercise 16.4** A programme's case assumed 60 sites at 75 % adoption releasing 4.0 hours a week at USD 92 an hour over 46 weeks. Measured: 66 % adoption and 3.6 hours. Compute the planned and measured annual benefits and the total variance, then decompose it both ways and state the interaction term. *Solution.*  $U = 60 \times 92 \times 46 = \text{USD } 253,920$  per unit of adoption-hours. Planned  $253,920 \times 0.75 \times 4.0 = \text{USD } 761,760$ ; measured  $253,920 \times 0.66 \times 3.6 = \text{USD } 603,313.92$ : **79.20 % of plan**, a variance of **(USD 158,446.08)**. Adoption first: adoption  $253,920 \times 4.0 \times (-0.09) = (91,411.20)$ ; hours at measured adoption  $253,920 \times 0.66 \times (-0.4) = (67,034.88)$ . Hours first: hours  $253,920 \times 0.75 \times (-0.4) = (76,176.00)$ ; adoption at measured hours  $253,920 \times 3.6 \times$

$(-0.09) = (82,270.08)$ . Both sum to  $(158,446.08)$ ; the interaction term  $253,920 \times (-0.09) \times (-0.4) = \text{USD } 9,141.12$  is what moves between the lines. *Common error*: computing both variances at plan —  $(91,411.20)$  and  $(76,176.00)$  — which sums to  $(167,587.20)$  and over-states the shortfall by exactly the interaction term, because the same movement has been counted twice.

**Exercise 16.5** A post-project review costs USD 62,000 and produces 41 lessons; an applied lesson avoids USD 19,500 on average; the current retrieval rate is 9 %. Compute the expected recovery and the breakeven retrieval rate, then evaluate spending a further USD 24,000 to raise retrieval to 30 %. *Solution*. Addressable value  $41 \times 19,500 = \text{USD } 799,500$ . At 9 %:  $799,500 \times 0.09 = \text{USD } 71,955$ , the review already returns 1.16 times its cost. Breakeven retrieval  $62,000/799,500 = 7.75\%$ . At 30 %:  $\text{USD } 239,850$ , an increment of  $799,500 \times 0.21 = \text{USD } 167,895$  for 24,000, a ratio of 7.00. Combined breakeven  $86,000/799,500 = 10.76\%$ , and the net value at 30 % retrieval is  $\text{USD } 153,850$ . *Common error*: comparing the review's cost with the *addressable* value of 799,500 rather than with the value weighted by retrieval, which makes every review look overwhelmingly worthwhile and removes the reason to fix retrieval: the one intervention that actually pays here at seven to one.

## Practitioner's toolkit — Domain 16

*Adoption-ready artefacts; adapt headings to your organisation, then keep them stable.*

### Toolkit 16.T.1 — Transition readiness certificate

One page per receiving unit, signed by the receiving organisation, in **two blocks that never share a column**.

**Block A, mandatory preconditions (no probability column)**. One row per item: the precondition, the **external or independent authority** that grants it, the status recorded as **met / not met** only, the date it was granted, and the reference to the granting instrument. The standing rows are the safety case and its hazard log closed by the accountable safety authority, regulatory or licence approval, the data-protection or privacy assessment, required notifications, and contractual takeover certificates, with any further precondition the sector imposes. Beneath Block A, a **signature line for the approving authority itself**, by name and role, distinct from and above the project's and the receiving organisation's signatures: the certificate is not complete on the project's signature alone. A single "not met" in Block A ends the certificate. There is no weighting, no probability and no offsetting entry, and the certificate is issued as a **hold**.

**Block B, discretionary conditions (the arithmetic)**. Rows: each readiness condition, its owner (who must not be the person delivering it), its objective assessment



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method, its assessed probability  $p_i$  with the basis of the assessment named, and its evidence reference. Footer, computed and printed rather than described: the **conjunction**  $\Pi p_i$ , the averaged figure alongside it so the gap is visible, the  $\min(p_i)$  correlated bound, the expected number of failed transitions across the remaining units at the stated cost per failure, and the residual after any proposed uplift. Below that: the hypercare duration with its **measured** exit criterion, the reversion trigger, the reversion decision-maker, the time to revert and the latest point at which reversion remains possible. A certificate that cannot be completed is itself the finding.

### Toolkit 16.T.2 — Closeout and final account pack

Four sections, each a table. **Commercial:** per contract, original sum, approved variations, claims open with their **monthly carrying cost** and settlement authority, claims settled, recoveries, retention held and released with the event that released it, final account, amounts paid, final payment with its reconciliation to the movements since the last certificate, and the date the full and final statement was signed. **Operational:** service owner, funded run line with its annual amount, support model, licence and contract register with renewal and notice dates, access revocation date for the project team, and the obsolescence horizon. **Knowledge:** documentation items with the date each passed the execution test and by whom, key-person concentration count, and the outstanding documentation list with owners and dates. **Organisational:** authority withdrawn on, cost accounts closed on, records placed under the retention schedule, and people released with their contribution recorded.

### Toolkit 16.T.3 — Benefits closing account and retention schedule

Two linked artefacts, owned by the benefits owner and reported to the body that approved the case. The **closing account:** per benefit, registered measure with its exact definition and denominator, baseline value and date, target profile, measured value and date, and the variance decomposed into adoption (or volume), benefit per unit, and timing, with the interaction term shown and the decomposition convention stated in a footnote. Then the present-value bridge from the approved case to the realised position, with an explicit line for any operating cost the case omitted, and the priced improvement plan with its owner and funding source. The **retention schedule:** per data class (volume, retention period, legal or contractual basis, disposal or de-identification action, the authority who may execute it, and the test that de-identification actually worked; plus, for every AI-informed decision, the required fields of 16.4.5) model identifier, version, date, input, output, named verifier, decision-maker, with a monthly count of records missing any of them.

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## Exam preparation – Domain 16

**What is assessed.** What handover transfers and what it cannot; condition-based handover; the distinction between works tests and service tests; **the split between the gate block of mandatory preconditions and the discretionary conditions; readiness as a conjunction and its arithmetic**; proportional-shortfall ranking of remediation; the hold-or-go decision under both correlation assumptions; hypercare exit criteria and the reversion plan; operational transition with a funded run line and whole-life cost; retention, defects liability and warranty; **the final account and its reconciliation; the carrying cost of an open item and the breakeven settlement premium**; knowledge transfer and the runbook execution test; the post-project review and **the economics of retrieval**; the benefits measurement plan and the registered measure; **the closing account and its decomposition**, including the interaction term and the convention that assigns it; attribution and the comparison cohort; class-by-class retention economics; and the model-retention record.

**The calculations to be able to do under time pressure.**  $\Pi p_i$  against  $(\Sigma p_i)/k$ , the gain  $p'/p$  from lifting one condition, and  $p = \text{target}^{(1/k)}$ . Expected failed transitions and the hold-or-go comparison. Final account, retention release and final payment with its reconciliation. Monthly carrying cost, EMV of determination, breakeven settlement price and the identity  $c \times \Delta m + (EMV - \text{assessed})$ . Annual benefit  $U \times a \times h$ , the two-factor decomposition in both orders, and the interaction term. Present value of an operating cost over an appraisal horizon. Breakeven retrieval rate. Retention cost against  $P(\text{need}) \times \text{consequence}$ , and the breakeven need probability.

**The traps.** Entering a mandatory precondition (a safety case, a licence approval, a privacy assessment) as a probability in the readiness product, or pricing it in the hold-or-go comparison, which states a figure at which it could be waived (16.1.3, 16.1.4) · reporting an averaged readiness figure as a probability (16.1.3, Exercise 16.1) · ranking readiness remediation by absolute gap rather than by the ratio  $p'/p$  (Exercise 16.1) · omitting the residual failure rate after an uplift, which makes readiness work look better than it is (MCQ 16.1-D) · treating  $\min(p_i)$  as the expected case rather than the correlated bound (16.1.3) · calling the certified works value the final account (Exercise 16.2) · ignoring internal management time in a carrying cost (Exercise 16.3) · comparing an early-settlement premium with the claim's assessed value rather than with the expected cost of determination (16.2.4) · computing both benefit variances at plan and double-counting the interaction term (Exercise 16.4) · using the undiscounted total for an operating-cost omission (MCQ 16.2-D) · comparing a review's cost with the addressable rather than the retrieval-weighted value of its lessons (Exercise 16.5) · measuring realisation against a claim the case should never have made rather than against the honest figure (16.4.2) · and arguing attribution at review instead of designing a comparison cohort at baseline (16.4.3).

**How the domain connects.** Domain 1 supplies the accountability principle, the outputs-to-benefits chain and the cost of delay that this domain re-tests at the

realised rate. Domain 2 supplies the benefits map, the baseline discipline, the appraisal this account is measured against and the kill criteria that 16.A.2 executes. Domain 3 supplies the gate at which the go-live decision sits and the decision record the post-project review reads. Domain 4 supplies the configuration baseline that as-built documentation must match. Domain 9 supplies the acceptance and nonconformance machinery commissioning uses. Domain 10 supplies the contract mechanisms of retention, defects liability and claim settlement. Domain 12 supplies the conditions under which a review produces candour and a team dissolves well. Domain 14 supplies the AI-use register and verification standard that 16.4.5 retains. Domain 15 supplies the portfolio benefits register that must not double-count what this domain measures. And PFL-AI Domain 4 supplies the appraisal machinery (equivalent annual value and unequal lives) behind the whole-life comparison of 16.2.1 and Case study B.

## Domain 16 summary

The project ends; the account does not. This domain replaces the handover *date* with a handover *condition*, and splits that condition set in two. Mandatory preconditions (the safety case and its hazard log, regulatory and licence approvals, the privacy assessment, required notifications, takeover certificates) are recorded met or not met with their approving authority named, carry no probability and are never traded against the cost of delay; while one is open the only decision is hold. Above that gate block, readiness is a **conjunction**: Meridian's seven discretionary go-live conditions, averaging **90.71 %** on the programme dashboard, gave a probability of a clean go-live of **49.79 %**, a gap of nearly forty-one percentage points between the number that was reported and the number that mattered. The remediation ranking follows the ratio  $p'/p$ , so the clinical champion at 0.80 was worth **11.20 points** against **1.04** for training at 0.96; even all seven conditions at 0.98 leaves **13.19 %** of go-lives failing, which is why the answer to the residual is a rehearsed reversion plan and not more assurance. Holding three weeks and spending **USD 96,000** dominated going now under both the independent conjunction and the perfectly correlated bound, saving **USD 261,864** and **USD 55,992** respectively against a commitment of **USD 138,840**.

Closeout is commercial as well as operational. Meridian's final account came to **USD 1,831,400** (**9.01 %** above the original sum, being 5.71 % of variations plus 4.40 % of claim settlement less 1.11 % of defect recovery), and the final payment of **USD 97,400** reconciled exactly to the movements since the last certificate, which is the check that an account is closed rather than totalled. The open claim cost **USD 5,250 a month** to carry, so fourteen months of it would have cost **USD 73,500**, more than the claim's own assessed value of 62,000; the breakeven early-settlement premium was **USD 69,500**, given by the transferable identity **carrying cost × months saved + (expected determination – own assessment)**, and settling at 74,000 saved **USD 57,500**. Knowledge behaves the same way: three undocumented configurations cost

7.00 times as much to reconstruct as to record, and a post-project review that breaks even at a 5.20 % retrieval rate returns 11.30 to one on the indexing that raises retrieval, which is where the value is lost, and where nobody spends.

The closing account settles what Domains 1 and 2 opened. Measured: 27 of 40 clinics in daily use and 5.4 clinician-hours released, worth USD 594,864 a year — 86.79 % of Domain 1's honest 685,440 and exactly 60.75 % of the 979,200 the approved case claimed. The USD 90,576 shortfall against the honest case decomposes into (24,480) of adoption and (66,096) of benefit per adopting clinic, with USD 2,448 of interaction assigned by a stated convention. And the larger of the two was the assumption nobody was monitoring. In present value the bridge runs +3,447,096 as claimed, (2,114,198) for the adoption term the case omitted, (465,015) of level variance, (413,809) of timing, (644,900) for an operating cost of 108,000 a year that the case never carried, and (114,000) of cost outturn, to a realised NPV of (USD 304,827). Strip out the run cost and the realised position is +340,073: the omission of an operating-cost line was the entire margin. And USD 180,000 of post-project adoption work, worth +USD 340,156 at a benefit-cost ratio of 3.32, moved the account to +USD 35,329: the whole value of a 2.4 million dollar programme, resting on a plan no project document contained. Auriga's closeout makes the same point in a different currency: 0.8 of a percentage point of availability, offered for a 240,000 deduction, was worth USD 1,057,585 over the asset life, and the retest that cost 128,000 was worth 689,585 more than accepting the money.

What survives the project is what was written down. Retaining the evidence pays as insurance above a 3.64 % need probability; retaining personal data is the reverse trade, with de-identification worth USD 17,940 net on 1.1 TB; and 28.50 % of Meridian's AI-informed decision base is permanently unexplainable because three fields (model identifier, version, date) were not recorded on records that were being written anyway. Benefits decay at 4 % a year without sustainment, worth USD 216,824 in present value against a USD 73,634 sustainment programme, so the account has to be handed on rather than closed.

The through-line, and the last of the book: **outputs are delivered, benefits are realised, and the two are separated by an organisation, a period of time and a measurement nobody is obliged to take.** Take it anyway. Decompose it honestly, state the convention, price the recovery, and send it to the people who approved the promise; because the closing account is the only document in the delivery lifecycle that improves the next decision rather than defending the last one.

# Appendix A — Formula and symbol sheet

23 symbols, of which **22 carry a verified worked example** in this volume or its companion. A symbol marked *pending* is registered and used but does not yet have a golden-answer check behind a printed result — treat its appearances with the same care you would any unchecked number. Units are stated because a unit error is the commonest arithmetic failure in practice and the hardest to see in a finished sentence.

| SYMBOL                           | MEANING                                                                       | UNIT     | FIRST HOME                      | VERIFIED |
|----------------------------------|-------------------------------------------------------------------------------|----------|---------------------------------|----------|
| PV (BCWS)                        | Planned Value (EVM context)                                                   | currency | inherited (PCL-AI master table) | yes      |
| EV (BCWP)                        | Earned Value                                                                  | currency | inherited (PCL-AI master table) | yes      |
| AC (ACWP)                        | Actual Cost                                                                   | currency | inherited (PCL-AI master table) | yes      |
| BAC / EAC / ETC / VAC            | Budget / Estimate at Completion, Estimate to Complete, Variance at Completion | currency | inherited (PCL-AI master table) | yes      |
| $CV = EV - AC$<br>$SV = EV - PV$ | Cost / Schedule Variance                                                      | currency | inherited (PCL-AI master table) | yes      |
| $CPI = EV/AC$<br>$SPI = EV/PV$   | Performance indices                                                           | ratio    | inherited (PCL-AI master table) | yes      |
| TCPI                             | To-Complete Performance Index                                                 | ratio    | inherited (PCL-AI master table) | yes      |
| PoC                              | Percentage of completion                                                      | %        | inherited (PCL-AI master table) | yes      |

| SYMBOL                       | MEANING                                                                                                                                                                                                                      | UNIT                | FIRST HOME                               | VERIFIED |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------|----------|
| $r, n$                       | Discount rate per period;<br>number of periods                                                                                                                                                                               | ratio;<br>count     | inherited<br>(PCL-AI<br>master<br>table) | yes      |
| $PV(x)$                      | Present value of amount $x$                                                                                                                                                                                                  | currency            | inherited<br>(PCL-AI<br>master<br>table) | yes      |
| ES                           | Earned Schedule (time-based<br>schedule measure)                                                                                                                                                                             | time                | PML-AI<br>D6                             | yes      |
| $SPI(t)$                     | Schedule Performance Index<br>(time) = $ES/AT$                                                                                                                                                                               | ratio               | PML-AI<br>D6                             | yes      |
| TF, FF                       | Total float, free float                                                                                                                                                                                                      | time                | PML-AI<br>D6                             | yes      |
| EMV                          | Expected monetary value =<br>$\Sigma(\text{probability} \times \text{impact})$                                                                                                                                               | currency            | PML-AI<br>D8                             | yes      |
| $E[\text{wait}]$             | Governance latency = $M/2 + L$<br>(meeting interval $M$ , paper<br>lead time $L$ ) — the expected<br>wait for a committee<br>decision; sums across<br>escalation tiers                                                       | time                | PML-AI<br>D3                             | yes      |
| Cost of delay                | Value forgone per unit of<br>elapsed time; the price at<br>which governance latency,<br>gate duration and escalation<br>paths are all evaluated                                                                              | currency/<br>period | PML-AI<br>D1/D3                          | yes      |
| Gate net value               | $P(\text{defect}) \times \text{build-fix cost}$<br>– [review + elapsed $\times$ cost of<br>delay + $P(\text{defect}) \times$<br>$(P(\text{detect}) \times \text{design-fix} +$<br>$P(\text{miss}) \times \text{build-fix})]$ | currency            | PML-AI<br>D3                             | yes      |
| Committee<br>capacity        | meetings per year $\times$<br>substantive items per<br>meeting; utilisation =<br>demand $\div$ capacity                                                                                                                      | count               | PML-AI<br>D3                             | yes      |
| Mesh / layered<br>interfaces | $n(n-1)/2$ possible pairwise<br>interfaces against $n$ to an<br>integration layer — the<br>architecture choice, priced                                                                                                       | count               | PML-AI<br>D4                             | yes      |
| Hundred-per-<br>cent rule    | $\Sigma \text{children} - \text{parent} = 0$ at<br>every WBS level; a non-zero<br>result is an omission or a<br>duplication                                                                                                  | currency            | PML-AI<br>D4                             | yes      |

| SYMBOL                | MEANING                                                                                                                                                                         | UNIT     | FIRST HOME    | VERIFIED       |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------|----------------|
| Assessed total impact | direct + (schedule weeks × cost of delay) + rework + interface re-verification + regression + documentation (+ risk, + benefit) — the basis a delegation threshold must read on | currency | PML-AI D4     | yes            |
| Baseline drift        | (change count × average direct cost) + (affected count × average weeks × cost of delay); cumulative-test threshold derived from the observed change rate                        | currency | PML-AI D4     | yes            |
| EVA(benefit)          | Benefit measure in benefits register (named in words to avoid EV clash)                                                                                                         | currency | PML-AI D2/D16 | <i>pending</i> |

**Conventions that apply to every formula above.** Financial arithmetic is decimal, never binary floating point. Intermediate values carry full precision; rounding happens only at display, which is why a printed figure may not reproduce if you round as you go. Rates and periods must agree — an annual rate against monthly periods is the error the golden suite catches most often. Datetimes are `YYYY-MM-DD HH:MM:SS` in UTC and are compared as instants, never lexically.

# Appendix B — Notation, terminology and conventions

Terms fixed at programme level, which the chapters use without redefining:

| TERM                                   | AS USED IN BOTH VOLUMES                                                                                                                                                                                                                                    |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PML-AI</b>                          | PCI Project Management Leader – AI: the certification; its BoK is this programme's Book One                                                                                                                                                                |
| <b>PFL-AI</b>                          | PCI AI Project Finance Leader: the certification; its BoK is Book Two                                                                                                                                                                                      |
| <b>Responsible AI principle</b>        | "AI proposes; the professional verifies, decides and remains accountable" — the suite-wide principle; it supersedes the retired formulation "AI proposes, the professional disposes" used in the first PCL-AI printing under its former PCP-AI designation |
| <b>Domain / Knowledge Area / Topic</b> | The three-level content hierarchy <b>D.K.T.</b> , identical to PCL-AI                                                                                                                                                                                      |
| <b>Sponsor</b>                         | The accountable executive owner of the business case (PML-AI); in PFL-AI project-finance contexts, an equity investor promoting the project — the books flag the context at each use                                                                       |
| <b>Special-purpose vehicle (SPV)</b>   | The ring-fenced legal entity created to own, finance and operate a project                                                                                                                                                                                 |
| <b>Bankability</b>                     | The degree to which a project's contracts, risks and cash flows support limited-recourse financing on acceptable terms                                                                                                                                     |
| <b>CFADS</b>                           | Cash flow available for debt service, as defined in the formula registry                                                                                                                                                                                   |
| <b>Cash waterfall</b>                  | The contractually agreed priority order in which project cash is applied each period                                                                                                                                                                       |
| <b>Benefits realisation</b>            | The identification, planning, measurement and sustainment of the outcomes a programme exists to deliver                                                                                                                                                    |
| <b>Decision rights</b>                 | The explicit allocation of which role may take which decision at which threshold                                                                                                                                                                           |
| <b>Psychological safety</b>            | A shared belief that the team is safe for interpersonal risk-taking — candour without penalty                                                                                                                                                              |
| <b>Hybrid delivery</b>                 | A deliberate combination of predictive and adaptive methods under one governance frame                                                                                                                                                                     |
| <b>Model risk</b>                      | The risk of loss from decisions based on flawed, misused or misunderstood models (financial or AI)                                                                                                                                                         |

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## Conventions

**The hierarchy.** Content is addressed as **D.K.T** — Domain, Knowledge Area, Topic. A cross-reference of the form *KA 7.3* or *10.2.2* is precise and is the authority; a page number is not used, because the corpus is regenerated.

**Context flags.** Where one word carries two genuinely different concepts, the later use states the collision explicitly rather than relying on the reader to infer it — *tailoring* of method against *tailoring* of a message, **PV** as present value against **PV** as planned value, **A** as annuity payment against **A** as assets. A flag is a signal that both meanings are live in the corpus, not a warning that one is wrong.

**Citation over restatement.** A result established in one domain is cited from the others rather than re-derived. If two domains appear to derive the same thing, that is a defect and worth reporting.

**Currency and units.** USD throughout, with SAR shown where it aids a Gulf reader at an indicative **USD 1 ≈ SAR 3.75**. That rate is illustrative and is not a forecast or a quotation. Every worked example states its units, its currency, its rounding and its date basis; where one is missing, treat it as a drafting defect.

**Fictitious entities.** Every organisation, project, person and case in this volume is invented. Any resemblance to a real entity is coincidental, and no real project's data is used.

**The responsible-AI principle**, which governs every *AI in this KA* section: *AI proposes; the professional verifies, decides and remains accountable.*



# Appendix C — Verification record

**7,686 golden-answer checks** stand behind this volume. Every number printed as a *result* — in worked examples, in-text calculations, multiple-choice options (all of them, not only the correct one), exercise solutions, case studies and figure specifications — is recomputed independently with decimal arithmetic at 28-digit precision and compared to the printed value. Each domain's module also pins the *invariants* the domain claims: an identity, a breakeven, a bound, an inequality, so that a claim cannot silently stop being true.

| DOMAIN       |                                                | CHECKS       |
|--------------|------------------------------------------------|--------------|
| 1            | The Project Leadership Profession              | 405          |
| 2            | Strategy, Selection and Business Alignment     | 450          |
| 3            | Governance, Organisation and Decision Rights   | 515          |
| 4            | Integration and Delivery Architecture          | 1,947        |
| 5            | Scope, Requirements and Value Definition       | 326          |
| 6            | Planning, Scheduling and Delivery Flow         | 545          |
| 7            | Cost, Resources and Commercial Awareness       | 385          |
| 8            | Risk, Uncertainty and Resilience               | 543          |
| 9            | Quality, Assurance and Continuous Improvement  | 302          |
| 10           | Procurement, Contracts and Supply Networks     | 299          |
| 11           | Stakeholders, Communication and Influence      | 236          |
| 12           | Leadership, Teams and Organisational Behaviour | 263          |
| 13           | Agile, Adaptive and Hybrid Delivery            | 236          |
| 14           | Digital Delivery, Data and Responsible AI      | 295          |
| 15           | Programmes, Portfolios and Enterprise Delivery | 274          |
| 16           | Transition, Closeout and Benefits Realisation  | 341          |
| 17           | Appendix G — integrated capstones              | 324          |
| <b>Total</b> |                                                | <b>7,686</b> |

The Institute's verification suite recomputes this record in full on demand, and can recompute a single domain of it as readily as the whole volume. The suite is fail-closed: a check that cannot be evaluated is recorded as a failure, never passed over as a skip.

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**What this record does and does not establish.** It establishes that the arithmetic is right — that each number is the one its stated method produces from its stated inputs. It does **not** establish that the method chosen was the right method for the situation, that the professional judgements are ones an experienced practitioner would endorse, that the emphasis across topics is well calibrated, or that nothing important is missing. Those are questions for editorial and technical review, and a reader should not read a large number in this appendix as an answer to them.

# Appendix D — Figure index

**51 figures**, all of them drawn as PCI-original artwork generated from source specifications maintained by the Institute. Each is also specified in words in the chapter, stating every plotted value, so that a figure can be read — and redrawn — without the artwork, and so that a reader using assistive technology loses nothing. No figure in either volume reproduces a diagram from another publisher, and every annotated value in a drawn figure is a literal verified by the golden-answer suite.

| FIGURE | SUBJECT                                                                   | ARTWORK |
|--------|---------------------------------------------------------------------------|---------|
| 1.2.1  | The cost of a silent measure                                              | drawn   |
| 1.3.1  | The value chain, and where it leaks                                       | drawn   |
| 1.3.2  | Payback is hyperbolic in adoption, so the first points are worth the most | drawn   |
| 1.3.3  | The reinforcement trough and its crossover                                | drawn   |
| 1.4.1  | Proportionate verification: the breakeven error probability               | drawn   |
| 2.1.1  | The strategy-portfolio alignment index                                    | drawn   |
| 2.2.1  | Two Meridian business cases from identical facts                          | drawn   |
| 2.3.1  | Meridian's benefits map, with the enabling change restored                | drawn   |
| 2.3.2  | Four breakevens from one programme                                        | drawn   |
| 2.4.1  | Why a kill criterion on a weak signal destroys value                      | drawn   |
| 3.1.1  | Why unanimity is a power law                                              | drawn   |
| 3.1.2  | What share of an iterative stream a periodic committee can serve          | drawn   |
| 3.2.1  | Governance latency and its two levers                                     | drawn   |
| 3.3.1  | The price of an escalation path                                           | drawn   |
| 3.3.2  | The conditional pass: how closure rate decides a gate's value             | drawn   |
| 4.2.1  | Interfaces grow combinatorially; components do not                        | drawn   |
| 4.3.1  | Pairwise integrity is the power of item integrity                         | drawn   |
| 4.3.2  | Reconcile gross, never net                                                | drawn   |
| 4.4.1  | What a change actually costs                                              | drawn   |
| 5.2.1  | What a requirement defect costs by the stage it is found                  | drawn   |
| 5.3.1  | Greedy ranking strands capacity; enumeration does not                     | drawn   |
| 6.1.1  | The schedule hierarchy: one logic, four honest views                      | drawn   |

| FIGURE | SUBJECT                                                                          | ARTWORK |
|--------|----------------------------------------------------------------------------------|---------|
| 6.2.1  | The Auriga network with both passes                                              | drawn   |
| 6.2.2  | Auriga's three paths, their floats, and the near-critical band                   | drawn   |
| 6.3.1  | Auriga field-crew histogram, before and after smoothing                          | drawn   |
| 6.3.2  | The rolling-wave horizon                                                         | drawn   |
| 6.4.1  | The economics of crashing Auriga's procurement                                   | drawn   |
| 6.4.2  | The compression menu and the least-cost duration                                 | drawn   |
| 7.1.1  | What an accuracy class actually says                                             | drawn   |
| 7.3.1  | Auriga's earned-value S-curves at week 13                                        | drawn   |
| 7.3.2  | The EAC fan and what TCPI demands                                                | drawn   |
| 7.4.1  | Auriga at week 13: the cost report says 200,000; the bank account says 1,043,600 | drawn   |
| 8.2.1  | The survey decision tree                                                         | drawn   |
| 8.2.2  | Independence is worth 6.6 points of confidence                                   | drawn   |
| 8.2.3  | Expected value and variance rank the same register differently                   | drawn   |
| 9.1.1  | Auriga's cost of quality and its interior minimum                                | drawn   |
| 9.4.1  | Rolled throughput yield: six good steps, one coin flip                           | drawn   |
| 10.1.1 | Make-or-buy on total cost of ownership: the breakeven the unit price hides       | drawn   |
| 10.2.1 | The weighting chooses the winner                                                 | drawn   |
| 11.1.1 | Two allocations of the same 480 hours                                            | drawn   |
| 11.2.1 | The age of the number an executive decides on                                    | drawn   |
| 12.2.1 | The coordination ceiling, and what structure buys                                | drawn   |
| 12.3.1 | Coaching minutes per report against span of control                              | drawn   |
| 13.2.1 | Little's Law and the cost of starting more work                                  | drawn   |
| 13.3.1 | Governance latency as a flow parameter                                           | drawn   |
| 14.1.1 | Defect rate against exposure: why a uniform data-quality target costs more       | drawn   |
| 14.3.1 | The verification standard proportional to consequence                            | drawn   |
| 15.1.1 | Why a programme milestone becomes improbable                                     | drawn   |
| 15.2.1 | Aggregate capacity feasible, period capacity not                                 | drawn   |
| 16.1.1 | Readiness is a conjunction, not an average                                       | drawn   |

| FIGURE | SUBJECT                                      | ARTWORK |
|--------|----------------------------------------------|---------|
| 16.4.1 | The closing account of Meridian Care Records | drawn   |

# Appendix E — Practitioner's toolkit library

**55 adoption-ready artefacts**, indexed here and defined in the Knowledge Areas they belong to. They are indexed rather than reprinted deliberately: a template that exists twice is a template that will disagree with itself, and the version in the chapter is the one the surrounding text explains. Adapt the headings to your organisation, then keep them stable — a checklist whose wording changes every quarter records nothing over time.

| TOOLKIT | ARTEFACT                                                           | WHAT IT IS FOR                                                                                     |
|---------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 1.T.1   | Leader's accountability map (one page)                             | Per outcome: the outcome stated in outcome language (not output language) · the single             |
| 1.T.2   | Benefits-chain test (run on any benefits claim)                    |                                                                                                    |
| 1.T.3   | AI-use and verification record                                     | Per AI-assisted artefact: tool and approved environment · data classification cleared              |
| 1.T.4   | Verification-depth card (one per artefact class, not per artefact) | Six fields, filled once for a class of artefact and reused: the class · what relies on it · the    |
| 2.T.1   | Business-case skeleton (the six questions)                         | do-nothing and one materially cheaper; why each was rejected) · Cost (range, accuracy class,       |
| 2.T.2   | Benefits map and register row                                      | Per benefit: description · the outcome measure that produces it and its owner (outside the         |
| 2.T.3   | Gate decision pack rules                                           | One page of forward-only economics (remaining cost, remaining benefit, forward NPV, and the        |
| 2.T.4   | Assumption register row, and the test-order rule                   | Per assumption: the statement, in one falsifiable sentence · why it is believed, with the evidence |
| 3.T.1   | Governance design sheet                                            | One page, completed before mobilisation and reviewed at every gate. Rows: each decision body, with |
| 3.T.2   | Delegation schedule with a cumulative test                         | Columns: decision class · value threshold · estimated cost to undo and the reversal-cost ratio     |
| 3.T.3   | Decision record entry and its integrity checks                     | Per-decision fields: reference · date · decision-maker (named person and role) · decision (one     |

| TOOLKIT | ARTEFACT                                                          | WHAT IT IS FOR                                                                                        |
|---------|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 3.T.4   | Gate and conditional-pass checklist                               | Two halves, and the second is the one that is usually missing. Before the gate: criteria written and  |
| 4.T.1   | Charter on one page                                               | Fields, in this order, and nothing else: purpose and measurable objectives · named sponsor · named    |
| 4.T.2   | Interface register                                                | One row per required interface (built from need, not from $n(n-1)/2$ ): reference · the two           |
| 4.T.3   | Integrated change request and baseline reconciliation             | (change / clarification / defect, with the ruling recorded) · then the assessment, one line each:     |
| 5.T.1   | Requirements traceability matrix                                  | One row per requirement: reference · statement · class (functional / non-functional / constraint /    |
| 5.T.2   | Value attribution and constrained selection sheet                 | delivered, with its named owner (Domain 2's benefits register) · attributed annual value with its     |
| 5.T.3   | Scope reconciliation and acceptance-criteria pattern set          | count · plus approved additions (with change references) · minus approved deletions · equals expected |
| 6.T.1   | Schedule quality gate (run before trusting any network)           |                                                                                                       |
| 6.T.2   | Recovery plan (one page)                                          | Slip statement (activity, cause, size, date detected) · re-run pass results (new duration, new        |
| 6.T.3   | Milestone register                                                | Per milestone: computed date and current TF · owner · done-definition (acceptance reference)          |
| 6.T.4   | Compression menu (priced at baseline, reviewed on the risk cycle) | Per candidate activity: current duration · which paths it lies on · cost slope per week · technical   |
| 7.T.1   | Cost-report integrity checklist (run before publishing the month) |                                                                                                       |
| 7.T.2   | Estimate basis sheet (one per estimate)                           | Method (analogous/parametric/bottom-up) · definition maturity and accuracy class · range              |
| 7.T.3   | Commercial one-pager (per major package)                          | Model (FFP/FPIF/CPFF/CPIF/T&M) and why · who carries cost risk · target cost, fee, share ratio,       |
| 7.T.4   | Cash and commitment page (one per period, beside the cost report) | A single page, six lines, none of which appears in a conventional cost report and each of which has   |

| TOOLKIT | ARTEFACT                                                         | WHAT IT IS FOR                                                                                          |
|---------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 8.T.1   | Risk register columns that earn their keep                       | ID · cause → event → consequence statement · affected objective · common driver (8.A.1)                 |
| 8.T.2   | Contingency derivation sheet (one per baseline)                  | Register version and date · objective the total belongs to, with the other objectives'                  |
| 8.T.3   | Pre-mortem session guide                                         | Before commitment, one hour: (1) state the plan and the commitment about to be made; (2) each           |
| 8.T.4   | Register-completeness check (one per gate)                       | A twenty-minute calculation that turns "the register is probably incomplete" into a funded position.    |
| 8.T.5   | Response option comparison (one per quantified risk)             | One row per option (accept, avoid, reduce, transfer, and for opportunities exploit, enhance, share,     |
| 9.T.1   | Cost-of-quality and containment sheet                            | One page per deliverable class, completed at planning and revisited at each gate. Top block, the        |
| 9.T.2   | Acceptance and disposition record with the self-consistency test | Per deliverable: reference and version · acceptance criteria referenced to the requirements baseline    |
| 9.T.3   | Improvement register: rework, yield and cause                    | Three linked sections, all numeric, reported at the same cadence as cost and schedule. Rework and       |
| 10.T.1  | Make-or-buy total-cost-of-ownership sheet                        | One page per decision, completed before any market engagement. Two columns, make and buy, and these     |
| 10.T.2  | Evaluation model lock sheet                                      | One page, completed, dated, signed by the accountable authority and issued with the invitation.         |
| 10.T.3  | Supplier resilience and sub-tier register                        | One row per critical item, not per supplier, and criticality set by schedule effect rather than by      |
| 11.T.1  | Stakeholder register and engagement allocation sheet             | One row per party: reference · party and named individual · what the project needs from them            |
| 11.T.2  | Communication architecture and report-age sheet                  | routing n · hours per channel per period, with the basis · hub overhead · computed load and utilisation |
| 11.T.3  | Negotiation preparation sheet                                    | delay · internal effort · risk as an EMV, with its basis · total = our reservation value. What          |
| 12.T.1  | Leader's attention and span sheet                                | One page per leader, completed from four weeks of observed data and reviewed quarterly. Top             |



| TOOLKIT | ARTEFACT                                                            | WHAT IT IS FOR                                                                                         |
|---------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| 12.T.2  | Team structure and coordination sheet                               | One page per team or programme, completed at formation and whenever headcount changes by more than a   |
| 12.T.3  | Difficult-conversation preparation and record                       | Two sides of one page, used for every performance, conduct or expectation conversation. Before:        |
| 13.T.1  | Flow sheet (one page, one team, weekly)                             | Rows, all counts or timestamps and none of them a judgement: throughput this week and its              |
| 13.T.2  | Backlog ordering sheet with delay-cost density                      | Columns: item or epic · outcome statement · benefits-map line and owner (the numerator's               |
| 13.T.3  | Adaptive commercial and continuation control sheet                  | Part A, the model choice: expected count of material re-prioritisations · addition rate and            |
| 14.T.1  | AI use register with its verification standard                      | Two linked tables on two pages, owned by one named person and reviewed at every gate. The              |
| 14.T.2  | Data class schedule and quality exposure sheet                      | One row per data class: class · records $n_i$ · owner (a named person) · written definition with units |
| 14.T.3  | Quality-adjusted digital business case sheet                        | One page per automation, analytics product, dashboard or twin, and it refuses to be completed          |
| 15.T.1  | Programme dependency register with a computed milestone probability | One row per dependency: reference · milestone it feeds · what is needed · owner on the giving          |
| 15.T.2  | Multi-period portfolio allocation sheet                             | A grid: candidates down the side; the binding capability's demand per period across the top,           |
| 15.T.3  | Portfolio benefits bridge                                           | A single reconciliation, top to bottom, published with every investment paper: gross claimed benefit   |
| 16.T.1  | Transition readiness certificate                                    | One page per receiving unit, signed by the receiving organisation, in two blocks that never share a    |
| 16.T.2  | Closeout and final account pack                                     | Four sections, each a table. Commercial: per contract, original sum, approved variations, claims       |
| 16.T.3  | Benefits closing account and retention schedule                     | Two linked artefacts, owned by the benefits owner and reported to the body that approved the case.     |

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**On the one appendix the volume plan lists and this volume does not carry.**

The plan for this family also names a consolidated *self-check answers* appendix. It is not printed, and the reason is worth stating rather than leaving as an apparent omission: each self-check in this volume already carries its own answer beside the question, so a consolidated set would be a verbatim second copy — precisely the duplication this programme's editorial rules forbid. The questions are indexed by their Knowledge Area instead.

The *standards and frameworks referenced* register the plan calls for **is** published, as Appendix F. It states no edition years, deliberately and for a stated reason, and that single outstanding verification is recorded in the Institute's corpus gate report rather than glossed over.

The consolidated question bank the plan calls for is published separately as `QUESTION_BANK.md`, and the global glossary as `GLOSSARY.md`; both are generated from the chapters by the same principle as these appendices.

# Appendix F — Standards and frameworks referenced

This volume references **33 documents, standards or frameworks**. Every one is *named* and *characterised* — never reproduced. No protected text, table, figure, clause, checklist or question from any of them appears in this volume; each is described at the level of its stated scope, in this programme's own words, and the treatment of the underlying subject is this programme's own. Where a chapter needs a number, it derives one rather than citing one.

**Nothing here implies endorsement.** No standards body, institute, professional association or financial institution has reviewed, approved or endorsed this volume, the designation it supports, or this programme. A reference means only that the document exists and is relevant to the topic under discussion.

**Editions are deliberately not stated.** No entry below carries an edition year or revision date. Standards are revised, withdrawn and superseded on their own schedules, and a citation to a superseded edition is worse than no citation, because it reads as authoritative while being wrong. Confirming the current edition of each document against the publisher's catalogue at the date of publication is a review task, and it is recorded as outstanding in `CORPUS_GATE_REPORT.md`. **A reader consulting any document below should obtain the current edition from its publisher.** Standards are copyright works and are not reproduced or redistributed by this programme.

## Project, programme and portfolio management

| REFERENCE | DESCRIBED IN THIS VOLUME AS ADDRESSING                               | DISCUSSED IN |
|-----------|----------------------------------------------------------------------|--------------|
| ISO 21500 | context and concepts for project, programme and portfolio management | D1, D2, D15  |
| ISO 21502 | guidance on project management                                       | D1, D4, D11  |
| ISO 21503 | guidance on programme management                                     | D1, D15      |
| ISO 21504 | guidance on portfolio management                                     | D1, D2, D15  |
| ISO 21505 | guidance on governance of projects, programmes and portfolios        | D1, D3, D15  |
| ISO 10006 | quality management in projects                                       | D1           |

## Risk

| REFERENCE        | DESCRIBED IN THIS VOLUME AS ADDRESSING      | DISCUSSED IN |
|------------------|---------------------------------------------|--------------|
| <b>ISO 31000</b> | risk management — principles and guidelines | D1, D8       |

## Environmental and sustainability quantification

| REFERENCE        | DESCRIBED IN THIS VOLUME AS ADDRESSING                                                                                                 | DISCUSSED IN |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>ISO 14064</b> | the quantification and reporting of greenhouse-gas emissions and removals — voluntary guidance, and not itself a disclosure obligation | D2           |

## Assurance and internal audit

| REFERENCE                             | DESCRIBED IN THIS VOLUME AS ADDRESSING                                                                                                                                                                                                                                                 | DISCUSSED IN |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>Institute of Internal Auditors</b> | an assurance architecture, commonly called the three lines, published by a professional body and revised by it into a model of roles rather than sequential lines of defence — voluntary guidance, neither a standard nor a requirement, and described in this volume in its own words | D3           |

## Quality and data quality

| REFERENCE            | DESCRIBED IN THIS VOLUME AS ADDRESSING                             | DISCUSSED IN |
|----------------------|--------------------------------------------------------------------|--------------|
| <b>ISO 9000</b>      | quality management systems — fundamentals and vocabulary           | D9           |
| <b>ISO 9001</b>      | quality management systems — the certifiable requirements standard | D9           |
| <b>ISO 8000</b>      | data quality                                                       | D9, D14      |
| <b>ISO/IEC 25012</b> | a data-quality model for software and systems                      | D9, D14      |

## Information management in the built environment

| REFERENCE        | DESCRIBED IN THIS VOLUME AS ADDRESSING                                                                                             | DISCUSSED<br>IN |
|------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>ISO 19650</b> | organisation and digitisation of information about buildings and civil engineering works, including building information modelling | D14             |

## Artificial intelligence, information security and privacy

| REFERENCE            | DESCRIBED IN THIS VOLUME AS ADDRESSING                                                                                                                                                                                                                                 | DISCUSSED<br>IN |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>ISO/IEC 42001</b> | artificial-intelligence management systems                                                                                                                                                                                                                             | D1, D9, D14     |
| <b>ISO/IEC 23894</b> | guidance on risk management for artificial intelligence                                                                                                                                                                                                                | D1, D9, D14     |
| <b>ISO/IEC 38507</b> | governance implications of an organisation's use of artificial intelligence                                                                                                                                                                                            | D14             |
| <b>ISO/IEC 27001</b> | information-security management systems                                                                                                                                                                                                                                | D14             |
| <b>ISO/IEC 27701</b> | privacy information management. Its relationship to an ISO/IEC 27001 information-security management system, and whether it can be certified in its own right, differ between editions; establish which edition any claim of conformity refers to before relying on it | D14             |
| <b>NIST</b>          | the United States National Institute of Standards and Technology, referenced for its AI Risk Management Framework — a voluntary, function-based framework, not a standard and not a regulation                                                                         | D14             |

## Project finance, environmental and social frameworks

| REFERENCE   | DESCRIBED IN THIS VOLUME AS ADDRESSING                                                                                                                                                                                                                                                                                                                                                                                          | DISCUSSED<br>IN |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>OECD</b> | the Organisation for Economic Co-operation and Development, referenced for its arrangement on officially supported export credits, for its convention on combating bribery of foreign public officials, and for its Guidelines for Multinational Enterprises and the associated due-diligence guidance — recommendations addressed to adhering governments and, through them, to enterprises, and not obligations of themselves | D10             |

## Standard-form contract families

| REFERENCE        | DESCRIBED IN THIS VOLUME AS ADDRESSING                                                                                                                                                                             | DISCUSSED<br>IN |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>NEC</b>       | a family of privately published standard-form construction and engineering contracts, named for identification only; it has no standing of its own and takes effect only where parties adopt it in their agreement | D10             |
| <b>FIDIC</b>     | a family of privately published standard-form engineering and construction contracts, named for identification only; it has no standing of its own and takes effect only where parties adopt it in their agreement | D10             |
| <b>JCT</b>       | a family of privately published standard-form building contracts, named for identification only; it has no standing of its own and takes effect only where parties adopt it in their agreement                     | D10             |
| <b>ISO 44001</b> | the management of collaborative business relationships — voluntary guidance                                                                                                                                        | D10             |

## Responsible and sustainable procurement

| REFERENCE                                                 | DESCRIBED IN THIS VOLUME AS ADDRESSING                                                                                                       | DISCUSSED<br>IN |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>UN Guiding Principles on Business and Human Rights</b> | business responsibility for human rights — a non-binding international instrument that supplies a method and imposes no obligation of itself | D10             |
| <b>ISO 20400</b>                                          | sustainable procurement — voluntary guidance, and not a certifiable requirements standard                                                    | D10             |

## Records and service management

| REFERENCE            | DESCRIBED IN THIS VOLUME AS ADDRESSING                                                                                                                                                              | DISCUSSED<br>IN |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>ISO 15489</b>     | records management — what a record is, and the controls that keep it authentic, reliable and usable over time; voluntary guidance, and it states no retention period applicable to any organisation | D16             |
| <b>ISO/IEC 20000</b> | service management, the discipline a receiving organisation runs when it accepts a service into support; voluntary guidance                                                                         | D16             |

## Cost engineering

| REFERENCE | DESCRIBED IN THIS VOLUME AS ADDRESSING                                                                                                                                                                                                                                                                      | DISCUSSED IN |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| AACE      | AACE International, referenced for the existence and purpose of the cost-estimate classification progression — in its Total Cost Management framework and in its Recommended Practices on cost-estimate classification; voluntary professional guidance, named for identification only and never reproduced | D7           |

## Industrial control system security

| REFERENCE | DESCRIBED IN THIS VOLUME AS ADDRESSING                 | DISCUSSED IN |
|-----------|--------------------------------------------------------|--------------|
| IEC 62443 | security for industrial automation and control systems | D14          |

## Regulation — named as reference points, not as applicable law

| REFERENCE                          | DESCRIBED IN THIS VOLUME AS ADDRESSING                                                                                                                                     | DISCUSSED IN |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| AI Act                             | the European Union's risk-tiered regulatory approach to artificial intelligence, named as an example of that regulatory shape and binding only within its own jurisdiction | D14          |
| General Data Protection Regulation | the European Union's data-protection regime, named as an example of a rights-based data-protection approach and binding only within its own jurisdiction                   | D14          |

## What this register does not contain

**No bibliography of textbooks or certification-body publications.** This volume is written from first principles and its worked examples are original, so there is no borrowed material to cite. Where a body of practice is discussed — cost-estimate classification, iterative delivery, portfolio governance — the discussion is of the *practice*, described in this programme's own words, and the reader is pointed to the owning body rather than given a paraphrase of its text.

**No legal, tax or accounting authority.** The financial-reporting standards above are named where a chapter explains why a treatment matters, never to state what the treatment is in any jurisdiction. Every such passage carries the volume's standing caution: treatments are jurisdiction-specific, and the applicable position requires written advice for the specific entity and structure. The same applies to every legal question raised in the text.

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## Appendix G — Integrated capstones

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A capstone is not a longer case study. Each domain's cases test whether a candidate can apply one domain's machinery. A capstone tests what the chapters structurally cannot, whether the answers **agree with each other**, and whether the reader can see which of them was actually governing while everyone was watching something else. A programme is not a sequence of sixteen correct disciplines. It is one commitment whose selection, governance, integration, scope, schedule, cost, risk, quality, procurement, stakeholders, leadership, method, data, commitments and closeout all have to be true at the same time, and the instructive failures live in the joints.

**Four capstones, and why the first is different.** Three of the four are new programmes, chosen for different failure shapes: **Calderhall Metro Line 3** (G.2), a metro-rail programme where the binding constraint is physical interface and a political clock; **Sable Plains Energy** (G.3), a grid-scale renewables portfolio where the unit of management is a pipeline rather than a project; and the **National Entitlement Service** (G.4), a national public-service programme where the delivery organisation does not control the operating one. Each is worked from its own arithmetic, and none reuses Meridian's figures, because the point of each is that a different failure shape puts a different quantity in control. Every programme, organisation and person in this appendix is fictitious.

The first capstone is the **Meridian Care Records** programme, the master thread this volume has carried since Domain 1. It is not reworked here: reworking it would be the duplication this programme's editorial rules forbid. What it has never had is an **assembly**. Meridian's numbers are distributed across sixteen chapters, and read in one place they tell a story no chapter tells: a programme that delivered every commitment it made, on time and under budget, and lost more value in its first operating year than it saved in its whole build. Every warning was computable in advance. Several were computed. None was on the report the board read.

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# Capstone One — Meridian Care Records: the anatomy of a green programme that failed

**The programme.** A public-health authority rolls a shared clinical-records system to 40 clinics. Approved cost **USD 2,400,000**; eight-year appraisal life at a 7 % discount rate; cost of delay **USD 14,280 a week**; the honest business case assumes **70 %** clinician adoption worth **USD 685,440** a year against a theoretical maximum of **979,200**. Every figure below is a result printed in this volume and independently recomputed by the golden-answer suite.

## G.1.1 The completion report, and the report nobody wrote

Meridian's completion report said this, and all of it was true:

| DELIVERY MEASURE  | RESULT                                  |
|-------------------|-----------------------------------------|
| Clinics installed | <b>40 of 40</b>                         |
| Schedule          | on time                                 |
| Cost              | <b>3 % under</b> the approved 2,400,000 |
| Capital variance  | <b>+USD 72,000</b> , banked once        |

The programme verdict was taken separately, against benefit, and was also true. Measured adoption was **40 %**, not 70 %: **391,680** a year against **685,440**, a shortfall of  $979,200 \times (0.70 - 0.40) = \text{USD } 293,760$  a year.

| LINE                                  | USD         |
|---------------------------------------|-------------|
| Annual benefit shortfall              | 293,760     |
| Capital underspend                    | 72,000      |
| <b>Ratio, in the first year alone</b> | <b>4.08</b> |

Over Domain 2's eight-year appraisal life the shortfall is **2,350,080** undiscounted (**97.92 %** of what the entire programme cost to build), or **1,754,128.79** discounted at 7 % on an annuity factor of 5.971299, still **73.09 %** of the approved cost. Both statements went to the same board in the same month, and the one printed in bold was the 72,000.

Domain 1 is careful about why: the two verdicts are **not commensurable**, one being a stock banked once and the other a flow recurring as long as the service runs,

so the 4.08 is legitimate as a scale comparison and not as a net. That care is the point. The failure was not that anyone lied. It was that the commensurability problem was allowed to become a reporting convenience, a single figure in the delivery report and a flow that belonged to somebody else's business case.

### G.1.2 The same arithmetical error, twice, on the same dashboard

Meridian's board saw averages where products governed. It happened independently in two domains, and neither instance required any new information to detect.

**Domain 16, readiness.** Seven go-live conditions, averaging **90.71 %** on the programme dashboard. Readiness is a **conjunction** (every condition must hold), so the probability of a clean go-live is their product: **49.79 %**. The gap between the number reported and the number that mattered is **40.92 percentage points**. Domain 16 also settles the instinctive remedy: even all seven conditions at 0.98 leaves **13.19 %** of go-lives failing, which is why the answer to the residual is a rehearsed reversion plan rather than more assurance.

**Domain 15, commitments.** Region A's go-live had six independent predecessors, none worse than 0.85, giving **52.95 %**. Four such regions committed to one date stand at  $0.5295^4 = 7.86 \%$ , a commitment **92.14 %** likely to be missed **while all twenty-four components report green**.

These are the same mistake in two costumes, and the assembly is what makes that visible. A reader who meets the readiness conjunction in Domain 16 learns a technique; a reader who sees it beside Domain 15's commitment conjunction learns a **failure mode of reporting itself**: that a dashboard aggregating by mean will report health at exactly the moment a conjunction is failing, and will do so more confidently the more components it has. Both instances were available before go-live. Both are one multiplication.

**What the remediation ranking then buys.** Domain 16's ratio  $p'/p$  ranks the conditions by what fixing each is worth. Remediating **to the same target of 0.98** in each case, which is what makes the comparison fair, lifting the clinical champion from 0.80 is worth **11.20 points** of go-live probability, against **1.04** for lifting training from 0.96: a **10.7692×** difference on the same remedial target. Effort spent on the condition that already looks good is nearly worthless, and a dashboard sorted red-amber-green points a programme directly at it. The seven conditions, for the record, are training 0.96, data migration 0.90, interfaces 0.94, network and devices 0.98, workflow redesign 0.85, clinical champion 0.80 and rehearsed fallback 0.92 — summing to 6.35, hence the 90.71 % mean, and multiplying to 49.79 %.

### G.1.3 Every avoidable cost had a breakeven small enough to be embarrassing

Assemble Meridian's avoidable costs and the pattern is not that the interventions were expensive. It is that each was trivially cheap against what it prevented, and that the case for each rested on a **breakeven probability** rather than a ratio.

| DOMAIN | THE AVOIDABLE COST                                                                                                                                                                              | WHAT PREVENTION COST                              | BREAKEVEN                                                        |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|------------------------------------------------------------------|
| 11     | national reporting body's week-34 objection: <b>224,520</b> (9.0 weeks of critical path at 14,280, of which 4.0 weeks are governance latency, plus <b>96,000</b> of re-verification and rework) | a pre-consultation at <b>3,300</b>                | objection probability above <b>1.4698 %</b>                      |
| 6      | an approval modelled as a three-week label rather than against its governance calendar                                                                                                          | <b>1 week</b> of design work                      | worth <b>4 weeks</b> of schedule                                 |
| 13     | release sequencing: <b>386,920</b> of delay cost in the worst order against <b>231,880</b> ranked by delay-cost density                                                                         | nothing: it is a sequencing decision              | a <b>155,040</b> swing, <b>10.8571 weeks</b> of delay cost, free |
| 16     | going live unready                                                                                                                                                                              | holding <b>3 weeks</b> and spending <b>96,000</b> | dominated going                                                  |

Domain 11 is explicit about why the breakeven and not the ratio is what wins the argument: the prevention was 68.04 times cheaper than the consequence, and *a sponsor can dismiss a ratio and cannot assert one and a half per cent*. Read across the four rows, the finding generalises. Every one of these decisions was taken by someone who believed the risk was unlikely, and in every case the question they needed was not "how unlikely?" but "unlikelier than 1.47 %?" That is a question a professional can answer in a meeting, and it is the form the argument should always take.

Note the two governance figures in the first row. Of the nine weeks the objection cost, **four** were Domain 3's  $E[\text{wait}] = M/2 + L$  (half the meeting interval plus the whole paper lead time), which is to say that nearly half the price of a stakeholder failure was paid by the committee calendar, computable at design time, months before anyone objected to anything.

### G.1.4 The benefit was decided before delivery began

The temptation is to read G.1.1 as a delivery failure to be fixed by better change management. The ledger says otherwise, and Domain 2 says it first.

Meridian's parent authority scores an **alignment index** of  $\sum \min(\text{declared}, \text{funded})$  at **76.6667 %** on the spend that could be mapped to a declared objective and **59.0000 %** on all of it: the whole difference being the **40 %** of money that serves

no declared objective at all. Its **reallocation distance**, the money that would have to move to make the portfolio match its own strategy, is **USD 4,200,000: 1.75 Meridians**. The organisation that could not connect two fifths of its spending to an objective is the same organisation that later could not explain why clinicians were not using a records system.

Domain 5 supplies the mechanism, in a different public-sector programme so that the finding is not one project's anecdote. Case study B reports **612 requirements, 100 % verified, 34.0 %** of forecast benefit, because the **4,200** external applicants whose behaviour actually produced the benefit had no seat in a requirements process assembled from the organisational chart: only **47** of the 612, **7.6797 %**, addressed their journey. The correction cost **USD 285,000** and returned **USD 425,500** a year, a payback of **8.04 months**, for a conversation that had been available free eighteen months earlier.

Set the two together and Meridian's adoption shortfall stops looking like an operational surprise. **Verification asks whether we built what we specified; validation asks whether the specified thing produces the outcome, and the two fail independently.** A programme can pass every test it wrote for itself, which is precisely what 40 of 40 on time and 3 % under budget means, and fail the only test that was never written down.

### G.1.5 What the delivery organisation could and could not see

Two further numbers belong in the assembly because they bound what a competent leader could have known.

**The estate the decisions rested on.** Domain 14 finds Meridian's common data environment holding **14,100** records in six classes with **332** defects: a weighted mean defect rate of **2.3546 %** carrying **USD 172,820** of consequence-weighted exposure. The distribution is the finding: exposure looks nothing like the defect rate, and the lowest-rate class in the estate (0.8 %) is the second-highest by exposure per record at **USD 28.00**. Cleaning by defect rate would have started in the wrong place.

**The risk register's own completeness.** Domain 8 puts a floor under what was missing rather than assuming nothing was: two independent identification methods and their overlap imply **11.43** unidentified risks and **78.61 %** coverage. That is the number that finally gives management reserve a basis, and it is also an honest statement that roughly a fifth of the risk population was never written down, which is a fact about method, not about diligence.

**And the leader's capacity, which was never baselined.** Domain 12's figure is the one most likely to be recognised: recurring commitments of **37.0 hours** in a 45.0-hour planning week, **82.2 %**, leaving 8.0 hours of discretionary attention, of which 1.5 went to people work. Every intervention in G.1.3 had to come out of that 8.0 hours. A programme that plans a leader's calendar to 82.2 % occupancy has, in effect, decided not to do the cheapest work available to it.

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### G.1.6 The five questions this capstone equips a candidate to ask

1. **Is any number on this dashboard an average of things that must all be true?** If yes, compute the product before the meeting. Meridian's was 90.71 % reported against 49.79 % actual, and 24 green components against a 92.14 % probability of missing the date.
  2. **Which verdict is a stock and which is a flow, and are both on the same page?** A capital variance banked once (+72,000) and an annual benefit shortfall (293,760) are not comparable, and reporting only the first is the whole failure in miniature.
  3. **What is the breakeven probability, not the ratio?** 1.4698 % is an argument a sponsor cannot dismiss; "sixty-eight times cheaper" is one they can.
  4. **Do the people whose behaviour produces the benefit have a seat in the requirements process?** If the process was assembled from the organisational chart, they almost certainly do not, 7.6797 % of requirements in Domain 5's Case study B.
  5. **How much of the leader's week is uncommitted?** At 82.2 % occupancy, every preventive measure in this ledger competes with every other for the same eight hours.
-

# Capstone Two – Calderhall Metro

## Line 3: the date nobody in the programme owns

Meridian's failure was a **conjunction reported as a mean**: information that existed, aggregated by the wrong operation, on a dashboard nobody could read arithmetically. Calderhall's failure is not about information at all. Every number below was known, correctly aggregated and correctly reported. The programme still lost because of a **structural** fact: the date belonged to somebody outside the programme, the scope was not equally fixed, and the exchange rate between the two had been set years earlier by a decision that looked like a procurement choice and was in fact a schedule choice.

**The programme.** A metropolitan transport authority builds Line 3, a **19 km, 12-station** driverless extension, at a capital expenditure of **USD 2,880,000,000**. Notice to proceed is month 0. The line is committed to open at **month 63**: the date is set by a fixed-date international event the city is hosting, whose dates belong to the event's owner and cannot move, and which no one inside the programme can influence. All amounts are USD; schedule is in calendar months from notice to proceed; percentages are shown to four decimal places where a comparison depends on them; and money is computed at full precision and rounded for display, so re-multiplying two displayed figures may differ from the displayed product in its final digit. *(Calderhall, its authority, the line and the event are fictitious. Interface, possession and productivity figures are illustrative and do not describe any real railway. Whether a concession or grant condition permits an opening date to be renegotiated is a question of the specific instrument and of local law, and belongs to qualified counsel rather than to this arithmetic.)*

### G.2.1 The interface count, and the 0.2000 % decision

Domain 4 gives the count and the architecture choice: pairwise interfaces grow as  $n(n - 1)/2$  and a layered architecture holds them to  $n$ . Calderhall's baseline design has **16** controlled subsystems, track, traction power, signalling, rolling stock, platform screen doors, tunnel ventilation, fire and life safety, drainage, communications, passenger information, fare collection, lifts and escalators, the control centre, the depot, station civils, and the interface to the existing network.

|                                            | BASELINE (16<br>SUBSYSTEMS) | AFTER DESIGN GROWTH<br>(19) |
|--------------------------------------------|-----------------------------|-----------------------------|
| Possible pairwise interfaces<br>$n(n-1)/2$ | 120                         | 171                         |

|                                             | BASELINE (16<br>SUBSYSTEMS) | AFTER DESIGN GROWTH<br>(19) |
|---------------------------------------------|-----------------------------|-----------------------------|
| Interfaces the register actually requires   | 80                          | 126                         |
| Register as a share of the bound            | 66.6667 %                   | 73.6842 %                   |
| Interfaces under a layered architecture (n) | 16                          | 19                          |

Three subsystems arrive during detailed design: an interchange with an existing line, a third-party utility corridor, and a second depot connection. They add **51 possible** interfaces (**16, 17 and 18** as each in turn joins, which averages **17** and equals it for none of the three), and **46** required ones, because each of the three connects to nearly everything. This is Domain 4's caution honoured rather than ignored: the count that matters is the register, built from need, and the formula is the **bound** the register is tested against.

At a fully loaded **USD 145,000** to specify, design, build, test, witness and document one bilateral interface, and **USD 5,760,000** for a systems-integration authority with a controlled interface baseline and an integration test facility:

|                           | BILATERAL  | LAYERED   | SAVING         |
|---------------------------|------------|-----------|----------------|
| Baseline scope            | 11,600,000 | 8,080,000 | 3,520,000      |
| After growth              | 18,270,000 | 8,515,000 | 9,755,000      |
| Cost of the growth itself | 6,670,000  | 435,000   | ratio 15.3333× |

The layer remains worth building at baseline while its cost is below **USD 9,280,000**. But the third row is the one to read. **The same three subsystems cost 6,670,000 on a bilateral architecture and 435,000 on a layered one:** the architecture does not change what the growth is, only what it costs, and it changes that by a factor of fifteen. The 15.3333 is not an artefact of the unit cost, which cancels: it is  $46 \div 3$ , the ratio of the interface counts the two architectures make the growth into. One modelling choice is worth stating, because it cuts against the layer rather than for it: the 145,000 unit cost is held constant for a layer interface even though verifying one takes twice as many witnessed test-days (G.2.2), on the reasoning that a layer interface has one target and one protocol and is correspondingly cheaper to specify and build. Set the figures against the capital budget and the proportions become uncomfortable: the baseline interface work is **0.4028 %** of capital expenditure and the integration layer is **0.2000 %**, one part in five hundred. The decision that governs everything below was, on the programme's own cost report, a rounding error.

### G.2.2 The count becomes a duration, and the duration meets the date

The cost is not the mechanism. Interface verification sits on the critical path, and its duration is proportional to the count, because each interface must be witnessed on an energised, complete system. Calderhall can run **5** concurrent test streams (limited by one operational control centre, one energised test track and one authorised witnessing authority) for **20** working days a month: **100 witnessed test-days a month**. A bilateral interface takes **10** witnessed test-days, so the campaign verifies **10 interfaces a month**. A conformance test against a controlled interface baseline takes longer per interface (**20** test-days), because it is a full suite rather than a point check.

|                           | INTERFACES | TEST-DAYS | VERIFICATION CAMPAIGN |
|---------------------------|------------|-----------|-----------------------|
| Bilateral, baseline scope | 80         | 800       | <b>8.00 months</b>    |
| Bilateral, after growth   | 126        | 1,260     | <b>12.60 months</b>   |
| Layered, after growth     | 19         | 380       | <b>3.80 months</b>    |

The rest of the critical path is **53 months** (civils and station boxes 32, systemwide installation 14, trial running 4, independent safety assessment and authorisation 3), and none of it moves.

|                              | CRITICAL PATH       | AGAINST MONTH 63            |
|------------------------------|---------------------|-----------------------------|
| Baseline, bilateral          | <b>61.00 months</b> | <b>2.00 months of float</b> |
| After growth, bilateral      | <b>65.60 months</b> | <b>2.60 months late</b>     |
| After growth, <b>layered</b> | <b>56.80 months</b> | <b>6.20 months of float</b> |

Read the third row twice. A layered architecture carrying the **grown** scope finishes verification **4.20 months** earlier than a bilateral architecture carrying the **original** scope, and the same 4.20 months appears as the difference in float, 6.20 against 2.00, which is the check that the two readings agree. **The architecture decision was worth more than the entire scope growth.** The growth was survivable. What made it fatal was a choice about integration structure taken years before anyone knew the growth would happen, by people who believed they were choosing a procurement model.

### G.2.3 What the immovable date actually costs

Calderhall's cost of delay is **USD 3,150,000 a month**: deferred farebox and commercial revenue of **1,890,000** (45,000 boardings a day × 30 days × an average fare of **USD 1.40**) plus **1,260,000** of extended programme management, site establishment and financing carry. That is **USD 735,000 a week** on a 30-day month, **51.4706 times**



Meridian's 14,280 (G.1.3): the same discipline at fifty-one times the price of a week, which is why the metro's governance latency and its architecture decisions cannot be governed with a health programme's thresholds.

If the date could move, the 2.60-month overrun would cost **USD 8,190,000** and the conversation would end there. It cannot. So the overrun has to be repaid in **scope**, and the exchange rate is exact: at 100 test-days a month and 10 per interface, 2.60 months is **26 interfaces**, which is 2.60 months' worth of the 4.60 the growth added, the other 2.00 having been absorbed by the float that existed. Each deferred interface is therefore worth  $0.10 \text{ months} \times 3,150,000 = \text{USD } 315,000$  of avoided delay.

Deferred to what, though? After opening, the railway owns the night. Interface verification then happens in engineering possessions yielding **2.5** productive hours against a day shift's **10.0**: a productivity factor of **0.25**, so **4.0** possession nights replace one test-day.

| STEP                                                | RESULT                                        |
|-----------------------------------------------------|-----------------------------------------------|
| $26 \text{ interfaces} \times 10 \text{ test-days}$ | 260 test-day equivalents                      |
| $260 / 0.25$                                        | 1,040 possession nights                       |
| $1,040 / 200 \text{ nights available a year}$       | <b>5.20 years</b> of post-opening integration |
| $26 \times 145,000$ delivered in the programme      | 3,770,000                                     |
| $26 \times 145,000 / 0.25$ in possessions           | 15,080,000                                    |
| <b>Extra cost of deferring</b>                      | <b>11,310,000</b> (435,000 an interface)      |

Now put the two prices beside each other, per interface, because at that unit the count cancels:

| PRICE, PER INTERFACE                             | USD           |
|--------------------------------------------------|---------------|
| Cost of deferring one interface into possessions | 435,000       |
| Cost of the 0.10 month that same interface buys  | 315,000       |
| <b>Ratio</b>                                     | <b>1.3810</b> |

(The 435,000 here and the layered cost of the whole scope growth in G.2.1 are the same number for an arithmetical reason rather than a related one: each is three interface-units, one because the possession multiplier is 4.0, the other because three subsystems arrived. Unrelated quantities that happen to coincide.)

**Holding the date costs 1.3810 times what missing it would have cost, and the ratio does not depend on how many interfaces are deferred:** it is a property of the possession premium and the cost of delay, not of the programme's size. On the totals the same figure appears as  $11,310,000 \div 8,190,000$ . This is the finding a

reader could not have guessed from Capstone One, and it runs against the instinct that a committed date is defended first and paid for afterwards: **a date defended without an architecture is defended at a loss.**

Three qualifications keep the result honest, and none of them reverses it. The 4.0 multiplier assumes the cost is labour-dominated; possession working also carries premium rates, re-mobilisation and protection staff, so **11,310,000 is a floor**. Both figures are one-off totals caused by the same decision, which is the point of difference from G.1.1: there a stock was set beside a flow and only a scale comparison was available, whereas here the two are commensurable and may legitimately be subtracted. What they do not share is timing: the 8,190,000 falls in the 2.60 months before opening and the 11,310,000 across the 5.20 years of possessions after it, so at any positive discount rate the deferral premium is worth less than its undiscounted total, while the premium rates push the other way. And a fixed political date carries consequences the 3,150,000 does not price: a committed event, statutory undertakings, a public promise already made. Those consequences are exactly what the arithmetic hands the decision back to, because it now has a threshold: holding the date costs  $11,310,000 - 8,190,000 = \text{USD } 3,120,000$  more than moving it, so the date is worth defending if and only if the unpriced consequences of moving it are worth more than 3,120,000. That is a question a transport minister can answer and a spreadsheet cannot, and it is a question nobody was able to put until the two costs were computed.

#### G.2.4 The lever that is not available: adding people

Faced with 2.60 months, the programme proposes doubling the verification team from **60** engineers to **120**. This buys nothing, for two independent reasons, and the second is worse than the first.

**The campaign is stream-limited, not effort-limited.** Capacity is 100 witnessed test-days a month because each test consumes the control centre, the energised test track and the witnessing authority. A sixth stream needs a second energised section, which the traction-power staging cannot provide inside the remaining programme. Sixty additional engineers at **USD 1,450** a day over 20 days for 2.60 months cost **USD 4,524,000** and shorten the critical path by **zero months**.

**And the team gets smaller as it grows.** Domain 12's coordination arithmetic, applied unchanged:

| TEAM<br>SIZE | COMMUNICATION PATHS<br>$N(N-1)/2$ | COORDINATION SHARE<br>$C(N-1)/2H$ | NET CAPACITY HN –<br>$C \cdot N(N-1)/2$ |
|--------------|-----------------------------------|-----------------------------------|-----------------------------------------|
| 60           | 1,770                             | 36.8750 %                         | 1,515 hours a week                      |
| 120          | 7,140                             | 74.3750 %                         | 1,230 hours a week                      |

At Domain 12's calibrated link cost  $c = 0.5$  hours a week and a 40-hour week, doubling the team multiplies communication paths by **4.0339** and **reduces net capacity by 285 hours a week, 18.8119 %**. Net capacity peaks at **1,620 hours a week**, reached at 80 and 81 people alike (the continuous optimum is 80.5, so the two integers tie, exactly as Domain 12 records it), and at 120 the team forgoes **390 hours a week** against that peak. So the proposal spends 4,524,000 to remove 285 hours a week of productive capacity from the function on the critical path. Domain 12 states the rule; a programme under date pressure is where it is disbelieved.

### G.2.5 The last 0.80 months, and who owns them

Descoping 26 interfaces is a decision the programme is not free to take either. Of the 26, only **18** are demonstrably outside the safety-authorisation boundary: depot connections, commercial systems, the utility corridor's non-safety functions. The remaining **8** are inside it, and no arithmetic entitles a railway to open with an unverified fire, evacuation or signalling interface.

| STEP                        | RESULT                                                            |
|-----------------------------|-------------------------------------------------------------------|
| 18 interfaces × 0.10 months | 1.80 months bought by descoping                                   |
| Overrun                     | 2.60 months                                                       |
| <b>Residual</b>             | <b>0.80 months</b> , or <b>USD 2,520,000</b> , with nowhere to go |

G.2.3's exchange rate survives the narrowing, which is the test of whether it was a real finding or an artefact of the round 26: the 18 the programme may lawfully defer cost **7,830,000** in possessions to avoid **5,670,000** of delay, the same **1.3810**, and the last 0.80 months cannot be bought at any price because the only thing left to sell is a safety verification.

The only remaining levers are trial running (4 months) and authorisation (3 months), and **both belong to the independent safety assessor**, which has no accountability for the opening date and no incentive to shorten either. Count the quantities that determine whether Line 3 opens on the committed date (the date, the opening scope, the integration architecture, the trial-running duration, the authorisation duration), and the programme controls **two of the five**, and one of those two, the opening scope, only as far as the safety boundary allows: 18 of the 26 interfaces, **69.2308 %** of the lever it appears to hold. That is the structural statement of Calderhall's failure, and it is the door into Capstone Four.

### G.2.6 The five questions Calderhall adds to Capstone One

1. **What is the interface count, and what is it a count of?** The register, not the formula; the formula is the bound. Three subsystems added 51 possible

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interfaces and 46 required ones here, costing **15.3333 times** more on a bilateral architecture than on a layered one.

2. **Is any critical-path activity's duration proportional to a count that is still growing?** If so, the count is a schedule risk, not a cost risk. Verification here ran 8.00 months at the original scope, 12.60 at the grown scope and 3.80 at the grown scope on a layered architecture, and once the scope had grown, only the third made the date.
  3. **What is the exchange rate between schedule and scope?** With the date fixed, there is one, and it is arithmetic: 100 test-days a month, 10 an interface, so 2.60 months = **26 interfaces**. A programme that cannot state this rate is descoping without knowing the price.
  4. **Is holding the date cheaper than moving it?** Compute both. Here deferral cost **435,000** an interface against **315,000** of avoided delay (a ratio of **1.3810**, independent of the count), so the date was defended at a loss.
  5. **How many of the quantities that decide the outcome do I control?** Two of the five at Calderhall, and one of those two only in part. The answer bounds every plan built on the other three, and it is available on day one.
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# Capstone Three — Sable Plains Energy: the arithmetic that has no project in it

Capstones One and Two are single undertakings: one programme, one railway, one date. Sable Plains is the capstone that changes the unit of account. Every project in its portfolio is competently appraised and individually attractive, every business case is arithmetically correct, and the enterprise still forgoes more than two fifths of its build capacity to one portfolio decision and **39.3276 %** of every project's appraised value to another, because the decisions that govern it are **portfolio decisions that appear in no project's model**, and a reader who reasons project by project cannot see any of them. (The two losses are not additive and this capstone does not add them: they belong to different regimes, one steady-state and one episodic, and each is separately large.) This is the one that should break a habit.

**The developer.** Sable Plains Energy develops, builds and operates utility-scale solar with co-located storage. Its pipeline holds **48** named prospects, **2,400 MW** name-plate at **50 MW** a build. A build costs **USD 60,000,000** (**USD 1,200,000 a megawatt**), takes **one year** with the resource it needs, and earns **USD 7,200,000** a year of net contribution for a **25-year** contracted life — **USD 46.9667 a megawatt-hour** on **153,300 MWh** at a 35 % capacity factor. The discount rate is **8 %**. Measured throughput of the grid-connection and energisation function, the binding capability, is **4 energisations a year** (12 in the last three years). All amounts are USD; years run from each build's investment decision; capital is modelled as drawn in equal annual instalments ending at energisation, and that convention is load-bearing, see the caution in G.3.2. Every figure is computed at full precision and rounded for display, so re-multiplying two displayed figures may differ from the displayed product in its last few digits. *(Sable Plains, its pipeline and every figure below are fictitious and illustrative. Connection-queue rules, land-option law and the consequences of relinquishing a connection offer are jurisdiction-specific and are questions for qualified counsel.)*

## G.3.1 The pipeline is the asset, and it costs 1.9377 times what the projects report

Five development gates stand between origination and an investment decision. Each is passed or the prospect dies, so conversion is the **product**: G.1.2's finding in a new costume, and worth meeting twice because in a portfolio it decides the size of the whole enterprise rather than the honesty of one dashboard.

| GATE                                  | STAGE<br>COST | PASS<br>RATE | CUMULATIVE<br>SURVIVAL AFTER<br>THIS GATE | PROSPECTS ENTERING<br>THIS GATE, PER BUILT<br>PROJECT |
|---------------------------------------|---------------|--------------|-------------------------------------------|-------------------------------------------------------|
| Screening                             | 40,000        | 0.72         | 0.720000                                  | <b>5.7132</b>                                         |
| Land option                           | 120,000       | 0.65         | 0.468000                                  | <b>4.1135</b>                                         |
| Grid application<br>and offer         | 260,000       | 0.55         | 0.257400                                  | <b>2.6738</b>                                         |
| Consent                               | 640,000       | 0.80         | 0.205920                                  | <b>1.4706</b>                                         |
| Offtake and<br>investment<br>decision | 400,000       | 0.85         | <b>0.175032</b>                           | <b>1.1765</b>                                         |

Conversion is **17.5032 %** against an arithmetic mean pass rate of **71.4000 %**: a gap of **53.8968 percentage points**, wider than Meridian's 40.92 (G.1.2) on *fewer* factors, because the gap widens with both the count and the distance of each factor below one, and these five are weak where Meridian's seven were mostly strong. The geometric mean, **0.7057055**, is the number that actually behaves like a pass rate: raised to the fifth power it returns 17.5032 % on the digits printed here, which the arithmetic mean never does. To energise 4 builds a year the organisation must originate **22.8530 prospects a year**.

Now the finding, which is a cost and not a probability. The entrants column is the reciprocal of the downstream conjunction, 1.4706 prospects reach consent per build because  $1 \div (0.80 \times 0.85)$  of them must, and multiplying it by the stage costs gives the development cost the **portfolio** actually incurs for one built project:

| BASIS                                                                                                                                 | USD PER BUILT<br>PROJECT |
|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| The portfolio: $5.7132 \times 40,000 + 4.1135 \times 120,000 + 2.6738 \times 260,000 + 1.4706 \times 640,000 + 1.1765 \times 400,000$ | <b>2,829,105.53</b>      |
| The closing project's own path, on its charge code: $40,000 + 120,000 + 260,000 + 640,000 + 400,000$                                  | <b>1,460,000</b>         |
| <b>Understatement</b>                                                                                                                 | <b>1.9377×</b>           |

The entrant counts are displayed to four decimal places and the sum is computed from the exact reciprocals, so re-adding the displayed line will land a few dollars away. **A built project's development cost is 1.9377 times what the project reports**, because **USD 1,369,105.53** of it (**48.3936 %** of all portfolio development spend) is spent on prospects that never build. At four builds a year that is **USD 5,476,422.14** a year, or **USD 27,382.11 a megawatt**, carried by no business case in the company. PFL-AI's own capstone reaches the same conclusion from a cruder ratio, programme spend divided by closings, and the stage-resolved form here does

something the ratio cannot: it shows *where* the money goes, which is what a leader needs in order to move it.

### G.3.2 Twelve good projects and one bad decision, which was not about any of them

Twelve connection offers mature inside a two-year window. Each carries a contractual energisation milestone set by the network operator; miss it and the capacity is lost. Every one of the twelve has been appraised and approved on its own merits. The board starts all twelve.

Twelve builds sharing a function sized for four do not run at full speed. They run at roughly a third of it, which is Little's Law's content (Domain 13, KA 13.2.3, applied at portfolio level in KA 15.3.2): at a throughput of 4 a year, holding 12 in flight makes the average cycle time **3.00 years**. So compare the same twelve builds, the same twelve contracts and the same capacity under two sequencing policies.

|                      | ONE-YEAR BUILD       | THREE-YEAR BUILD               |
|----------------------|----------------------|--------------------------------|
| Capital drawn        | 60,000,000 at year 1 | 20,000,000 at years 1, 2 and 3 |
| PV of capital at 8 % | 55,555,555.56        | 51,541,939.74                  |
| PV of contribution   | 71,165,174.59        | 61,012,666.83                  |
| <b>NPV</b>           | <b>15,609,619.04</b> | <b>9,470,727.09</b>            |

Present value of contribution at energisation is  $7,200,000 \times AF(0.08, 25) = 7,200,000 \times 10.674776 = 76,858,388.56$ ; the three-year column discounts it three years and prices capital on  $AF(0.08, 3) = 2.577097$ . **Stretching the build from one year to three removes USD 6,138,891.95 — 39.3276 %: of every project's appraised value**, and it does so without changing one line of any project's contract, cost estimate, resource plan or offtake.

Then the portfolio comparison, which is the capstone's central finding:

| TWELVE BUILDS, TWELVE OFFERS, ONE CAPACITY       | PRESENT VALUE AT THE DECISION                           |
|--------------------------------------------------|---------------------------------------------------------|
| All twelve started together (each takes 3 years) | 12 × the three-year NPV = <b>113,648,725.02</b>         |
| Three cohorts of four, each at full speed        | 4 × the one-year NPV × 2.783265 = <b>173,782,809.45</b> |
| <b>Released by sequencing</b>                    | <b>+60,134,084.43, or +52.9122 %</b>                    |

The deferral factor **2.783265** is  $1 + 1.08^{-1} + 1.08^{-2}$ , the second and third cohorts starting a year and two years later. Two invariants keep this from being a conjuring trick, and both must be stated or the result is not believable. **Capacity**

**consumed is identical:** 4 a year for three years is twelve builds either way. (The simultaneous case has the scarce function spread thinly across all twelve rather than serving them in order, which is what produces both the three-year cycle time and twelve energisations arriving together. A first-in-first-out queue on the same capacity would finish four a year — the sequenced policy, reached by discipline instead of by decision.) **The last build energises in year three under both policies** — sequencing does not finish the programme earlier; it finishes eight of the twelve earlier and the last four at the same time. Per build, the sequenced plan is worth **1.5291 times** the simultaneous one (14,481,900.79 against 9,470,727.09), so the *gain* is 0.5291 of a simultaneous build's value, **USD 5,011,173.70** each, and it is bought entirely by declining to start work the organisation cannot progress.

The same fact in steady state, since Sable Plains is not a one-off: at four builds a year the policy forgoes **USD 24,555,567.80 a year** (10.2315 % of the **240,000,000** of capital it deploys annually), which capitalised at 8 % is **USD 306,944,597.50**. And the same fact on the balance sheet, on a deliberately different convention: with capital drawn uniformly, average capital in construction is **120,000,000** at four in flight and **360,000,000** at twelve, so the policy ties up an extra **240,000,000** permanently (the same figure as the annual deployment two sentences earlier, because eight extra builds half-drawn is arithmetically four builds' capital, and a different quantity entirely), carrying **19,200,000** a year at the cost of capital. Two conventions, two figures in the same range, neither derived from the other: **24,555,567.80** a year of value forgone on discounted cash flow against **19,200,000** a year of carry on capital held idle. They are not a reconciliation, they differ by **5,355,567.80** and nothing here bridges that, because they price different consequences of one policy, which is what PFL-AI's Domain 6 means by a basis bridge and this is not one. What two independent conventions landing in the same range do establish is narrower and worth stating exactly: the finding is not an artefact of either. One model supporting one number could not have shown that.

Three cautions. The equal-instalment capital convention is doing real work: a build that stretches also stretches its drawdown, which is why the three-year column's PV of capital is *lower* than the one-year's, and a front-loaded contract would produce a larger loss while a genuinely deferred one produces a smaller. **State the drawdown convention beside the result or the result is not interpretable.** Second, cycle time does not stretch exactly in proportion: dedicated construction contractors are not shared, so only the scarce functions queue, and 3.00 years is therefore an upper bound and 60,134,084.43 with it. Third, the sequenced plan requires eight connection offers to be re-timed or relinquished, and the value available to pay for that is **USD 7,516,760.55 an offer: 5.3691 times** a 1,400,000 deposit. That is the breakeven form G.1.3 argues for: not "sequencing is worth sixty million", which a board can dismiss, but "re-timing an offer is worth doing unless it costs more than seven and a half million", which it cannot.



### G.3.3 Two queues, and the one nobody manages

Why did twelve offers mature together in an organisation that can build four a year? Because the two halves of the enterprise are separate queues running at different rates, and only one of them has a manager.

Development is slow and its slowness is computable. Each stage takes elapsed time whether the prospect passes or fails, so expected development duration per prospect originated is the survival-weighted sum:

$$1.000000 \times 0.25 + 0.720000 \times 0.50 + 0.468000 \times 1.25 + 0.257400 \times 1.75 + 0.205920 \times 0.75 = \mathbf{1.79989 \text{ years in the system.}}$$

Little's Law again, now on the development pipeline: to originate 22.8530 prospects a year and hold each for 1.79989 years, the organisation must be carrying **41.1328 prospects in development at once**. It is resourced for **24**. So development completes  $24 \div 1.79989 = \mathbf{13.3341}$  prospects a year, which at 17.5032 % conversion feeds **2.3339 builds a year** against a build capacity of **4**.

**The binding constraint is development, and nobody is managing it.** The shortfall is **1.6661 builds a year: 83.3049 MW, USD 26,007,145.26 of NPV a year** that the construction organisation has the capacity to create and will never be given. Closing it needs **17.1328** more prospects in development, a **71.3868 %** increase, which at three prospects a manager is **5.7109** development managers at a loaded **USD 240,000** each — **USD 1,370,625.71** a year. Against a build worth 15,609,619.04, that spend breaks even at **0.0878 additional builds a year: one extra build every 11.3887 years**. The function that gates the entire enterprise pays for itself if it produces one more project every eleven years, and it is short by one and two thirds a year.

And the queue behind it ages. Twenty-four prospects wait for a development slot; at 13.3341 completions a year the wait is **1.7999 years**, so the average prospect spends **3.5998 years** in the pipeline all told, queue and development together. That average is not the figure to set against an instrument, because it mixes the survivors with the four fifths that leave, each of them sooner and more cheaply than a survivor even though 48.3936 % of the money is still theirs. The prospects holding land options are the ones that keep passing gates, and a prospect that passes all five spends the full **4.50 years** of stage durations in active development, having secured its option **0.75 years** in. Its option therefore has to hold for **3.75 years** against a life of **3**: it expires **0.75 years**, nine months, before the decision it exists to protect, and that is before the 1.7999 years of queue in front of it. **Every prospect that reaches an investment decision outlives its own land option**, which is why offers, options and consents expire in clusters and then a cohort of twelve arrives at once. (That the queue wait and the average time in development are both 1.7999 years follows from the queue being exactly as long as the active caseload, 24 either way; it is an artefact of these figures rather than a general identity.)

So the sequence runs: development is under-resourced, which starves the build function, which makes the build function look idle, which invites the board to start everything that is ready the moment offers mature. And and the concurrency then destroys 39.3276 % of each project's value. **The overload of one queue was caused by the starvation of the other.** No project's business case contains any part of that chain, and no project manager could have found it.

### G.3.4 The five questions Sable Plains adds to Capstones One and Two

1. **What does one delivered unit cost the portfolio, as against the project?**  
Here 2,829,105.53 against 1,460,000: **1.9377×**, because **48.3936 %** of development spend buys prospects that never build. A charge code that shows only the survivor always concludes development is cheap.
2. **What is the measured throughput, and how many things are in flight?** If work in progress exceeds throughput × the technical duration, cycle time is a management decision, not an attribute of the work: 12 against 4 here, costing **39.3276 %** of every project's NPV.
3. **Would sequencing the same work release value without delaying its completion?** Test it against the two invariants: same capacity consumed, same final completion date. At Sable Plains sequencing released **60,134,084.43, +52.9122 %**, and the last build still energised in year three.
4. **Which queue is actually binding, and is anyone managing it?** Development feeds **2.3339** builds a year into a capacity of 4. The constraint was in the function with no delivery accountability, and its breakeven was one extra build every **11.3887 years**.
5. **Does the pipeline age faster than its own instruments?** A survivor here needs its land option to hold **3.75 years** against a 3-year life (it expires nine months before the decision it protects), which guarantees expiry clusters, and an expiry cluster is what forces the concurrency in question 2.

# Capstone Four — the National Entitlement Service: twenty vetoes and one accountability

Calderhall controlled two of the five quantities that decided its date (G.2.5) and that was the structural finding hiding behind its arithmetic. The National Entitlement Service is that finding made the whole subject. Here the delivery organisation does not control the operating one, accountability is split across bodies that each hold a veto, and **the programme's dominant quantity is not any body's behaviour; it is the number of bodies**. That claim is easy to assert and it is usually asserted without arithmetic, which is why it never changes anything. This capstone computes it.

**The programme.** A national digital agency replaces a paper-based entitlement assessment with a digital service. Policy belongs to a ministry; money to a funding committee; data-sharing and privacy design to a national data authority; the operating model to a council representing **14** regional operating offices; service standards to a digital standards body; and statutory accessibility sign-off to an independent panel. The service handles **2,600,000** assessments a year at **USD 41.60** each manually and **USD 12.30** digitally. All amounts are USD; latency is in weeks; percentages and weeks are shown to four decimal places where a comparison depends on them, and money is computed from the exact latencies rather than the displayed ones, so re-multiplying a displayed week count by the cost of delay will differ from the printed money by a few tens of dollars. *(The service, the jurisdiction and all six bodies are fictitious. Statutory approval structures, and whether decision rights can lawfully be consolidated at all, differ entirely between jurisdictions and are questions for qualified counsel. The arithmetic below prices a governance design, it does not prescribe one.)*

## G.4.1 Six decision rights, priced

Domain 3's governance latency is  $E[\text{wait}] = M/2 + L$  for a body meeting every  $M$  weeks with papers closing  $L$  weeks ahead, and  $E[\text{wait}] = M/2 + L + (1/q - 1) \times M$  where the body returns an item for more information with probability  $1 - q$  (KA 3.1.2, KA 3.2.3). The first-pass rates below come from the programme's own decision log over 30 months.

| BODY              | BASE $M/2 + L$ |     |            | FIRST-PASS $Q$ | $E[\text{WAIT}]$ |
|-------------------|----------------|-----|------------|----------------|------------------|
|                   | $M$            | $L$ | $L$        |                |                  |
| Policy ministry   | 4              | 2   | <b>4.0</b> | 0.80           | <b>5.0000</b>    |
| Funding committee | 13             | 3   | <b>9.5</b> | 0.90           | <b>10.9444</b>   |

| BODY                             | BASE M/2 + |   | FIRST-PASS Q | E[WAIT]           |
|----------------------------------|------------|---|--------------|-------------------|
|                                  | M          | L |              |                   |
| National data authority          | 6          | 2 | 5.0          | 0.70              |
| Regional operations council      | 8          | 3 | 7.0          | 0.75              |
| Digital standards body           | 3          | 1 | 2.5          | 0.85              |
| Accessibility and equality panel | 10         | 2 | 7.0          | 0.95              |
| <b>Total</b>                     |            |   | 35.0         | product 30.5235 % |

One assumption is doing work in that total and should be visible before anything is built on it: the six sit in **series**, because each body's paper needs the last body's decision, money follows policy, the data authority's design follows the operating model, the accessibility panel signs off what the others settled. Where bodies are genuinely independent the right model is the **maximum** and not the sum, and the arithmetic then says to run them at the same time: the longest single wait is the funding committee's 10.9444 weeks, so a fully parallel path would save **32.7938 weeks**, more than either lever priced in G.4.2. That lever is available exactly to the extent the dependencies are imaginary, which is a question about this programme and not about arithmetic. What parallelism cannot touch is the other half of the finding: the probability of clearing six bodies first time is **30.5235 %** whether they sit in series or side by side, because a conjunction does not care about order.

Returns add **8.7383 weeks: 24.9665 %**, to a path that was already 35.0 weeks long before anyone disagreed with anything. And the conjunction is the same theorem as G.1.2 and G.3.1 in a third costume: the mean first-pass rate is **82.5000 %**, the probability of clearing all six first time is **30.5235 %**, and the gap is **51.9765 percentage points**. The geometric mean, **0.8205513**, returns the product at the sixth power on the digits printed here; the arithmetic mean reproduces nothing.

The programme's cost of delay is derived, not asserted:  $2,600,000 \times (41.60 - 12.30) = \text{USD } 76,180,000$  a year of operating saving, so **USD 1,465,000 a week**. (Citizen-service benefit is deliberately excluded, because it is contested and the finding does not need it.) The single most consequential decision the programme must take (the change to the assessment operating model) requires all six bodies, and therefore costs  $43.7383 \times 1,465,000 = \text{USD } 64,076,561.50$  of deferred benefit before a single body says no.

#### G.4.2 The finding: the count dominates the conduct

Every governance improvement programme in this situation attacks behaviour: better papers, better pre-briefing, better secretariat, a first-pass rate target. Price that

lever at its ceiling. Suppose every one of the six bodies reaches **0.95**, the best first-pass rate any of them has ever achieved:

| PATH                                                 | LATENCY                                  |
|------------------------------------------------------|------------------------------------------|
| At present                                           | 43.7383 weeks                            |
| With all six bodies at 0.95, $35.0 + 44/19$          | 37.3158 weeks                            |
| <b>Saved by every body at its best observed rate</b> | <b>6.4225 weeks, or USD 9,408,929.92</b> |

The  $44/19$  is exact: a 0.95 first-pass rate costs  $1/0.95 - 1 = 1/19$  of one meeting interval per body, and the six intervals sum to 44 weeks, so the whole return penalty falls to **2.3158 weeks**. That is the observed behavioural lever in full (an extraordinary improvement to ask of six independent institutions at once), and it also raises the clean-approval probability from 30.5235 % to **73.5092 %**, which is a real gain. Now price the other lever. Consolidate to three decision rights, keeping policy, money and data, and folding the standards assessment into the data authority's, the accessibility sign-off into a once-for-the-service determination, and the regional council's voice into the ministry's board:

| CONSOLIDATED TO THREE DECISION RIGHTS | VALUE                                      |
|---------------------------------------|--------------------------------------------|
| Latency on a three-body path          | 23.5159 weeks, or USD 34,450,753.97        |
| <b>Saved</b>                          | <b>20.2224 weeks, or USD 29,625,807.53</b> |
| First-pass probability rises to       | 50.4000 %                                  |

**Halving the number of decision rights is worth 3.1487 times the best observed improvement in every body's conduct.** And the result does not depend on where the behavioural ceiling is put: push conduct past anything observed, to a path on which no body ever returns anything ( $q = 1$  at all six, which no institution achieves), and the lever buys the entire return penalty, **8.7383 weeks**, against which the count lever is still **2.3142 times** larger.

Now the sharper form of the same finding, which is the one to carry away. The average decision right on this path costs  $43.7383 \div 6 = 7.2897$  **weeks**, or **USD 10,679,426.92**. Lifting all six bodies to the best rate any of them has achieved saves 6.4225 weeks — **88.1033 %** of what one single additional body costs, so **no attainable improvement in how these bodies behave pays for one more of them**. The unattainable version pays for one and very little more: abolishing returns altogether comes to **119.8712 %** of one decision right, which is the honest measure of the headroom; one extra approver costs about what every return in the system costs. Adding a seventh body resembling the six takes latency to **51.0280 weeks** and drops clean approval to **25.0461 %**; deleting one (taking the accessibility determination once for the service rather than for every change, which removes no scrutiny

anyone can name) saves **7.5263 weeks**, **USD 11,026,052.63**, and lifts clean approval to **32.1300 %**.

The reason is structural and generalises past this programme. Latency is **linear** in the number of decision rights, a sum of per-body waits, while clean-approval probability decays **geometrically** in it, as  $q^k$  on the geometric mean pass rate. Conduct moves the terms; topology moves how the terms are summed and leaves the 30.5235 % exactly where it was; count moves how many terms there are, and one of those is an exponent. **Only the count moves both.** That is why *the list of approvers is the governance design*, and why a review that examines each body's performance separately is examining the wrong variable however carefully it does so.

**G.4.3 The fourteen offices the delivery agency does not run**

Latency is only half of a split accountability. The other half is that the benefit is produced by the **14 regional operating offices**, none of which reports to the delivery agency and each of which can decline to deploy, decline to retrain, or keep the paper route open, a practical veto, in Domain 11's terms a consent risk that is a property of position rather than disposition (KA 11.1.2).

If each region independently adopts in year one with probability **0.85**, the expected number adopting is **11.90** and the probability that **all fourteen** do is **10.2770 %**. So a business case built on universal adoption overstates the annual benefit by exactly the complement of the adoption probability, on equal regional volumes, an assumption the realised case below shows to be itself optimistic: **15.0000 %**, or **USD 11,427,000 a year**, against an honest **64,753,000**. The 10.2770 % was computable before a line of code was written, and it is the same multiplication that G.1.2 found on Meridian's readiness dashboard and G.3.1 found in Sable Plains' funnel, three programmes, one operation. (The 15.0000 % is not that multiplication but its linear cousin: expected coverage is a *mean* of independent adoptions, while it is the *product* that decides whether all fourteen arrive. Both belong in the case, and they answer different questions.)

What actually happened: **9 of 14** regions switched in year one, handling **62 %** of national volume, slightly less than their share of the offices, because the larger offices were the more cautious, which is the ordinary direction of this error rather than bad luck. Realised saving **USD 47,231,600**; shortfall **USD 28,948,400 a year**, measured against full digital conversion rather than against the probabilistic case, so it is the whole gap and not the surprise in it. Set that against the entire national readiness programme (an operating-model workshop and two follow-ups in each region, **180 hours** at a blended **USD 95** an hour, **USD 17,100** a region, **USD 239,400** nationally):

| MEASURE                                  | VALUE            |
|------------------------------------------|------------------|
| Shortfall ÷ national readiness programme | <b>120.9206×</b> |

| MEASURE                                                  | VALUE                                                         |
|----------------------------------------------------------|---------------------------------------------------------------|
| Breakeven share of national volume: 239,400 / 76,180,000 | <b>0.3143 %</b>                                               |
| The same breakeven in volume                             | <b>8,170.6</b> assessments of 2,600,000                       |
| Per region, 17,100 / 29.30                               | <b>583.62</b> assessments of a regional average of 185,714.29 |

Domain 11's discipline is to lead with the breakeven and not the ratio, and here the two forms are so far apart that the choice decides the meeting. **121 times** invites a sponsor to argue about whether the regions would have adopted anyway. **\*\*0.3143 %** of volume (583.62 assessments in a region that handles 185,714.29) **\*\*** invites nothing; it is a number no operational director will claim a workshop cannot move. The national and per-region breakevens are the same figure computed two ways, which is the check that the allocation is coherent.

Now count the programme, which is the finding this capstone exists for:

|                                                                             | COUNT                                       |
|-----------------------------------------------------------------------------|---------------------------------------------|
| Bodies holding an approval veto                                             | <b>6</b> of the 20 parties that can stop it |
| Operating units holding a practical veto                                    | <b>14</b>                                   |
| <b>Parties able to stop or blunt the outcome</b>                            | <b>20</b>                                   |
| Parties with the 76,180,000 in their own objectives                         | <b>1</b> (the policy ministry)              |
| Parties holding a <i>delivery or operating</i> lever <b>and</b> the outcome | <b>0</b>                                    |

**Twenty vetoes and one accountability, and the one accountable party holds nothing but a veto.** The last row is the whole diagnosis, and it turns on a distinction worth naming: **a veto is a negative lever.** The ministry can refuse a policy change and cannot cause a region to adopt, cannot build the service and cannot retrain an assessor; the delivery agency holds the build levers and is measured on delivery rather than on benefit; the fourteen offices hold the adoption lever and are measured on casework. So the outcome has an owner with no instrument, and every instrument has an owner with no outcome. That is not a criticism of any of the twenty; each is discharging a duty it was given, and several of the vetoes exist for good reasons that would survive any review. It is a statement about a *design*, and the design's cost is now quantified twice and then confirmed after the event: **29,625,807.53** on one decision's latency from the count of approval rights; **11,427,000 a year** from adoption treated as certain rather than probable; and then, ex post, **28,948,400 a year** of realised shortfall against a **239,400** intervention that was never funded: the realised 38 % of volume standing where the probabilistic 15 % had been forecast.

Note the commensurability discipline G.1.1 insists on: **29,625,807.53 is a stock, banked once on one decision; 11,427,000 and 28,948,400 are flows, recurring as long as the service runs**; and those two flows are one flow, forecast and then realised, so the ledger has two entries and not three. They are not addable, the comparison is a scale comparison, and stating that is what stops this table becoming the reporting convenience Meridian's board was handed.

#### G.4.4 The four questions the National Entitlement Service adds

1. **How many parties can stop this, and how many are measured on the outcome?** Twenty and one here. The ratio is the governance design; every coordination mechanism on top of it is a workaround.
2. **What does one decision right cost?** 7.2897 weeks, USD 10,679,426.92 on this programme; and lifting every body to the best first-pass rate any of them has achieved recovers only **88.1033 %** of one, while abolishing their returns altogether recovers **119.8712 %**. Compute this before agreeing to add a body, because afterwards the argument is about the body's merits rather than its price.
3. **Is the improvement programme aimed at the count or the conduct?** Halving the count was worth **3.1487 times** the observed behavioural ceiling and **2.3142 times** an unattainable one. A review that assesses each body separately cannot discover this, because the quantity is a property of the list.
4. **Does the benefit case assume adoption or estimate it?** At 0.85 a region, universal adoption has a **10.2770 %** probability and the assumption overstates the benefit by **15.0000 %**. The correction is a stated coverage in the case, and a readiness programme whose breakeven (**0.3143 % of volume**) is small enough to end the argument.



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# PCI Standards

Mandatory professional rules established by PCI Global for work within a PCI certification scope. They are not legislation and do not displace applicable law, contract or authoritative standards, and each law is cited by its stable identifier — never by page. The foundational set binds every PCI credential; the certification set that follows it binds this one.

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# PCI Foundational Professional Standards

**Status:** The Foundational Standard set of the PCI Standard system — Level 1 of the hierarchy in the *PCI Standards Charter* §3. Fifteen standards, identifiers **PCI-FND-STD-01** to **PCI-FND-STD-15**, binding on every candidate for and holder of **PCL-AI** (PCI AI Project Controls Leader), **PFL-AI** (PCI AI Project Finance Leader) and **PML-AI** (PCI Project Management Leader – AI) within the scope of their PCI-certified work. Version 1.0, prepared 4 August 2026; **draft for approval** — see *Due-process record* below. Drafted under the *PCI Standards Charter* and the *PCI Standards Drafting Manual*, both of which govern. British English throughout.

## CAUTION

**PCI Standards are private professional certification requirements established by Project Controls Institute Global. They are not legislation, government regulation, legal advice or substitutes for applicable laws, contractual obligations, regulatory requirements or authoritative professional standards. Where an applicable legal, regulatory, contractual or authoritative requirement imposes a higher or different obligation, that requirement prevails.**

## How to read these standards

**Two documents govern, and this one conforms to both.** The *PCI Standards Charter* settles what a PCI Standard *is* — its status, the six-level hierarchy, the priority order when requirements conflict, due process, interpretation, amendment, exceptions and the consequences available to PCI. The *PCI Standards Drafting Manual* settles how a standard is *written* — the normative language system, the twenty-five-element structure, defined terms, external-reference classification, prohibited drafting patterns and the twenty-five audit questions. Where anything below is unclear, those documents prevail over it.

**The normative language, and the ISO mapping.** PCI uses modern must-drafting, exclusively. **must** states a mandatory PCI professional requirement; **must not** states a prohibited practice; **should** appears only in Recommended Practice; **may** grants permission; **can** describes capability and never permission. The ISO-style requirement verb is **deliberately not used anywhere in this file, in any field** — Drafting Manual §1 bars it because it has been read as both obligation and futurity, and a draft containing it fails gate. A reader who works to ISO/IEC drafting conventions therefore reads across as follows: **in a PCI Standard, must is the**

**requirement form**, and it carries the meaning an ISO document conveys with its own mandatory verb. **must** here marks a PCI requirement, never an external constraint.

**What is mandatory, and what is not.** Every numbered standard below is a **Mandatory PCI Standard** (Charter §3, Level 1). Every **Process Requirement** (PCI-FND-STD-NN-PR-NN) beneath a standard is a **Mandatory PCI Process Requirement** (Level 4) and is as binding as its parent. Nothing else in this file creates an obligation: purposes, notes, examples and the tables at the end are commentary, and a requirement appearing in one of them would be a drafting defect.

**One obligation per requirement.** Element 1 of each standard carries a single principal obligation. Where the subject matter needs more than one obligation, the others appear as numbered Process Requirements under that standard rather than as a bundled clause, so that each can be tested, breached and assessed separately.

**Terms that decide compliance are defined once**, in *Definitions* below, and used identically in every standard. No standard redefines a defined term. Element 4 of each standard lists the defined terms that bear on its compliance.

**External references are named and characterised, never reproduced**, and each is classified as exactly one of the ten categories in Drafting Manual §6. See *A note on external references* below.

**The suite principle** governs this set, in its one approved formulation:

#### CAUTION

**AI proposes; the professional verifies, decides and remains accountable.**

**No endorsement, affiliation or accreditation is claimed or implied.** Naming an external instrument means only that the instrument exists and is relevant to the subject under discussion. No standards body, professional institute, government, supervisory authority or financial institution has reviewed, approved, endorsed or accredited these standards, the credentials they support or PCI Global.

#### A note on external references

Each reference in element 17 of each standard records seven things: the issuing organisation, the title, the subject, the edition or effective date checked, the nature of the authority as one Drafting Manual §6 category, the verification date, and the applicability limitation. Five conventions apply, and they are not optional.

1. **Exactly one category per reference**, never combined. Categories are stated by number and name.

- 2. **Where the Manual's ten categories and the register's seven differ, the Manual governs here and both are shown.** The register (`../registries/EXTERNAL_AUTHORITIES.md`) uses a *voluntary framework* category that the Manual offers only in its environmental-and-social form (category 8). A voluntary framework that is neither environmental nor social — the NIST AI Risk Management Framework, the OECD AI Principles, SR 11-7 — is therefore recorded here as **category 10, illustrative practice**, which is accurate because this set relies on none of them for any requirement. The register's own category is given alongside so the two records reconcile.
- 3. **Nothing is invented.** No clause number, article, edition, effective date, judicial decision or requirement appears here that was not verified in the register on the date stated. Where a precise provision was not verified, the instrument is cited by name only.
- 4. **Only Regulation (EU) 2024/1689 (the EU AI Act) is described as legislation, and only within the European Union.** Every other instrument named is voluntary, contractual, professional or illustrative in its force. The General Data Protection Regulation is named in this set only as an illustrative reference within a jurisdictional caution; no requirement is sourced to it.
- 5. **The IFRS Conceptual Framework for Financial Reporting is not cited anywhere in this set.** The IASB states expressly that it is not a Standard and that nothing in it overrides a Standard. The superseded 14-standard set sourced no requirement to it either; this edition excludes it entirely so that the question cannot arise.

Every reference carries "verify current requirements" where the authority can change, and the official publication always governs.

Supersession

This edition **withdraws and replaces the fourteen-standard foundational set** identified `PCI-LAW-F-01` to `PCI-LAW-F-14`. Those identifiers are withdrawn and, per Charter §10, are never reused. The reason for withdrawal is structural, not substantive: the superseded set was written to an eighteen-field template, mixed two normative-language systems, and bundled several obligations into single clauses, none of which conforms to the Drafting Manual. Its substance on evidence, escalation and AI is carried forward. The mapping is:

| SUPERSEDED                                                                    | CARRIED FORWARD INTO                                        |
|-------------------------------------------------------------------------------|-------------------------------------------------------------|
| <code>PCI-LAW-F-01</code> Professional Accountability and the Suite Principle | <code>PCI-FND-STD-01</code>                                 |
| <code>PCI-LAW-F-02</code> Verification of AI Output Before Professional Use   | <code>PCI-FND-STD-03</code> and <code>PCI-FND-STD-14</code> |

| SUPERSEDED                                                                   | CARRIED FORWARD INTO                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PCI-LAW-F-03 Human Decision Authority                                        | PCI-FND-STD-04                                                                                                                                                                                                                           |
| PCI-LAW-F-04 Disclosure of Material AI Assistance                            | PCI-FND-STD-14-PR-01 to -PR-03                                                                                                                                                                                                           |
| PCI-LAW-F-05 Evidence and the Audit Trail                                    | PCI-FND-STD-02, PCI-FND-STD-07, PCI-FND-STD-12                                                                                                                                                                                           |
| PCI-LAW-F-06 Data Lineage and Integrity                                      | PCI-FND-STD-07                                                                                                                                                                                                                           |
| PCI-LAW-F-07 Honesty in Reporting and Forecasting                            | PCI-FND-STD-05, PCI-FND-STD-11                                                                                                                                                                                                           |
| PCI-LAW-F-08 Competence Boundaries and Referral                              | PCI-FND-STD-10                                                                                                                                                                                                                           |
| PCI-LAW-F-09 Confidentiality and Information Protection                      | PCI-FND-STD-09                                                                                                                                                                                                                           |
| PCI-LAW-F-10 Conflict-of-Interest Disclosure                                 | PCI-FND-STD-08                                                                                                                                                                                                                           |
| PCI-LAW-F-11 Duty to Escalate                                                | PCI-FND-STD-11                                                                                                                                                                                                                           |
| PCI-LAW-F-12 Record Retention                                                | PCI-FND-STD-12                                                                                                                                                                                                                           |
| PCI-LAW-F-13 Ethical Conduct Toward Candidates, Employers and the Public     | <b>Not carried forward at Level 1.</b> General conduct is a certification-conditions matter rather than a testable Foundational Standard; the specific duties it carried are in PCI-FND-STD-08, PCI-FND-STD-11-PR-05 and PCI-FND-STD-14. |
| PCI-LAW-F-14 No Misrepresentation of PCI Credentials or Accreditation Status | <b>Not carried forward at Level 1.</b> Credential-claim rules govern the relationship between PCI and the holder rather than the conduct of professional work, and belong in PCI's certification conditions.                             |

The two omissions are deliberate and are recorded here so the decision is visible rather than accidental. Neither obligation is abolished; each is relocated outside the Foundational Standard set. Until that relocation is published, the superseded text of PCI-LAW-F-13 and PCI-LAW-F-14 remains on the record and is not deleted.

### Due-process record for this edition

Charter §5 requires thirteen stages and requires a standard's file to record honestly which stages were performed and by whom, including where a stage was performed with AI assistance rather than by a named human. For this edition:

| STAGE                                           | STATUS FOR THIS EDITION                                                                                                                                                                            |
|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 — Problem definition                          | Partly performed. Inherited from the drafting brief and from the defects found in the superseded set; no independent evidence base was gathered.                                                   |
| 2 — Drafting instruction                        | Performed. The drafting brief set the identifiers, titles, intent and structure.                                                                                                                   |
| 3 — Initial draft                               | Performed, <b>with AI assistance and no named human drafter</b> .                                                                                                                                  |
| 4 — Technical review                            | <b>Not performed.</b> No subject specialist has confirmed that each requirement is right and achievable.                                                                                           |
| 5 — Standards and legal-characterisation review | Partly performed. Every external reference was checked against <code>../registries/EXTERNAL_AUTHORITIES.md</code> and its recorded verification date; no publisher was contacted for this edition. |
| 6 — Practitioner consultation                   | <b>Not performed.</b>                                                                                                                                                                              |
| 7 — Impact assessment                           | Partly performed. Burden, small-organisation impact and cross-jurisdiction usability were tested through audit questions 7, 22, 23 and 24 only.                                                    |
| 8 — Scenario testing                            | Partly performed, through audit questions 9, 23 and 24.                                                                                                                                            |
| 9 — Red-team challenge                          | Performed against all twenty-five audit questions; findings and resolutions are tabled at the end of this file.                                                                                    |
| 10 — Revision                                   | Performed. Every finding recorded in that table was resolved in this draft.                                                                                                                        |
| 11 — Approval                                   | <b>Not performed.</b> No approving body has approved this edition.                                                                                                                                 |
| 12 — Publication                                | <b>Not performed.</b> This is a draft for approval.                                                                                                                                                |
| 13 — Post-implementation review                 | Not applicable — the edition is not yet in force, so there is nothing to review.                                                                                                                   |

**This process must never be represented as equivalent in scale, independence or authority to that of a public standard-setter.** PCI is a small private certifier, and seven of the thirteen stages above have not been performed.

## Definitions

These terms decide whether a professional has complied. Each is defined once, here, and used identically in every standard below. None is defined by reference to itself. Where a standard's element 4 lists a term, the definition below is the one that applies.

**These definitions are the canonical ones for the whole corpus.** Drafting Manual §4 permits a term to be defined in the standard that uses it **or** in the *PCI*

*Standards Definitions Register*, and that register now exists at [PCI\\_STANDARDS\\_DEFINITIONS\\_REGISTER.md](#). Every term below that is also used in a certification volume carries the same wording there and in that volume; where a term legitimately carries different professional senses in different credentials, the register states each sense with the context it belongs to rather than forcing one. The register records what each file previously said, so every reconciliation is visible and reversible. It is a derived register and creates no obligation: **the definitions below govern.**

**D-01 · Adopting organisation.** The employer, client or engaging entity whose governance arrangement applies to the work in question. Where a professional works as a sole practitioner with no engaging entity's governance arrangement in force, the professional is their own adopting organisation and must record, once, the criteria and thresholds these standards leave to the adopting organisation to set.

**D-02 · AI system.** A machine-based system that infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, classifications or decisions. The term includes a general-purpose model accessed as a service, a model embedded inside another application, an automated agent that acts on its own outputs, and any such system whether licensed, self-hosted or free. PCI defines the term for its own purposes only. The OECD Recommendation on artificial intelligence and, within the European Union, Regulation (EU) 2024/1689 each define it for theirs; where either applies to a professional's work, its definition governs for that purpose.

**D-03 · Authorised system.** A system, service or tool that appears on the adopting organisation's published register of systems approved for information of the classification in question, for the purpose in question. A system is not authorised because it is widely used, because it is paid for, because a colleague uses it, or because no rule expressly forbids it.

**D-04 · Competent reviewer.** A named individual who, in relation to a particular item, satisfies all of the following: (a) their competence in the subject matter of that item is evidenced by a qualification in that subject matter, an assessed competence record held by the adopting organisation, or documented experience of comparable work recorded by the adopting organisation, and that competence is recorded for that class of work before the review begins; (b) they are able to state what would make the item wrong and which method would detect that error; and (c) they are able to perform the verification method the standard requires, and to reach a conclusion on the matter, without assistance from the preparer or reliance on the preparer's explanation of it. No limb alone is sufficient, and job title, seniority and availability establish none of them. The adopting organisation may quantify the experience required in (a); PCI does not. **Independence is not a limb of competence.** Where a standard requires the reviewer to be independent (D-12), that requirement is imposed by that standard's element 10 and is tested separately — a separation this set depends on, because [PCI-FND-STD-10](#) element 12's supervised-

acquisition exception needs a reviewer who is competent and is precisely *not* independent. For a review conclusion recorded on a PFL-AI engagement, the reviewer must additionally hold written authorisation from the engaging organisation to record a review conclusion on that subject.

**D-05 · Conflict.** A relationship, interest or duty of any of the following kinds, held by the professional or by a member of their household, which bears on a matter in which the professional acts: a financial interest in a party to the matter or in its outcome; employment, office, directorship or partnership with a party to the matter; a family or household relationship with an individual who is a party to, or decides, the matter; a gift, hospitality or benefit received from a party to the matter above the threshold the adopting organisation records; a past or prospective engagement with a party to the matter; a personal interest in a particular outcome, including an interest in the performance of work the professional themselves performed; a future role that depends on the outcome; and an interest in a tool, model, vendor or AI system that is being selected, priced, assessed or relied upon in the matter. A conflict is **actual** where it bears on a decision the professional is currently taking, **potential** where it would do so if a foreseeable event occurred, and **perceived** where a person outside the matter, on the facts recorded, would identify one of the listed relationships. **The appearance is part of the definition, not a softening of it**, and a conflict is a structural fact about a role rather than a lapse of character. The list is closed for the purposes of the disclosure duty: every conflict within it must be disclosed and there is no threshold. A judgement test stated in a standard — such as the reasonable-party test at [PCI-PFL-STD-01.02](#) — widens this list and never narrows it.

**D-06 · Contemporaneous.** Created at the time of the act it evidences, or as soon afterwards as establishing the facts permits, and never after the outcome the record bears on became known.

**D-07 · Control result.** The output of a check, validation, tolerance, limit, reconciliation, test, alert or approval gate that is designed to prevent, detect or flag an error, an exception or an unauthorised act — whether the check is automated, systemic or performed by a person.

**D-08 · Decision owner.** The single named individual who holds the authority, under the applicable governance arrangement or documented delegation schedule, to approve, reject, amend, defer or withhold the matter in question; who bears the consequence of that decision; and who answers for it afterwards. The accountability is held by one person, is not delegable, and is recorded before the output is relied upon. Where a body decides collectively, its named chair — or the named authority the delegation schedule assigns — is the decision owner for the purposes of these standards, and the record names that person. A decision owner is always a natural person: "the team", "management", "the business", "the sponsor", "the lenders", "the organisation", a role held by no named individual, a committee without a named chair, a system, a model and a vendor are never decision owners.



**D-09 · Engagement lead.** The professional accountable for accepting an engagement or role and for allocating work within it. Where the professional works alone, the professional is the engagement lead.

**D-10 · Escalation threshold.** The escalation threshold for a matter is reached at the earliest moment any of the following becomes true: the matter is material (D-15); it creates a risk to the safety of a person; it would change, or would have changed, a decision already taken or about to be taken; it affects an output already issued outside the professional's own organisation; it affects a contractual, regulatory, tax or financial-reporting position; or the escalation criteria published by the adopting organisation, stated in the finance documents, or recorded in the delegation schedule are met. **Any event a standard's element 13 states, any condition recorded before the work begins, and any documented value in a delegation schedule or in finance documents are additional to those six triggers and never in place of them** — a matter that reaches this threshold requires escalation under [PCI-FND-STD-11](#) whether or not it appears in any element 13. **The threshold names a destination and a time, and the absence of either does not remove the duty:** where no destination is documented the matter goes to the next authority above the decision owner (D-08) for it, and where no time is documented the time is the period defined in D-20. Reaching the threshold starts that period. The obligation does not depend on the professional's expectation of how the recipient will react, or on the matter being resolved afterwards.

**D-11 · Evidence.** A dated record that exists independently of the assertion it supports, that identifies its source, its version where it has one, and its author or issuing system, and that a person other than the author of the assertion is able to retrieve, examine and use to reach the same conclusion **without asking that author**. Primary source documents, executed contracts, system extracts identified by query and extraction time, measurement and inspection records, third-party confirmations, and recorded calculations showing their inputs and method are evidence. The following are **not** evidence for the purposes of these standards: an output of an AI system that does not identify the source of what it asserts; an AI-generated summary of a record, as evidence of the underlying fact — the underlying record is the evidence; a statement that a system, model or tool produced a figure, unaccompanied by the inputs and the method; a restatement of the assertion in a second document by the same author; a preparer's own statement offered in support of their own assertion; an unrecorded recollection; an unattributed file; an undated extract; an unversioned working copy; a screenshot with no source reference; and a dashboard or screen state that cannot be reproduced.

**D-12 · Independent.** A person is independent in relation to a specified matter if all of the following are true: (a) they did not perform the act, prepare the item or any part of it, or direct, specify or approve it; (b) they did not select, build or configure the tool, model or AI system that produced it; (c) they hold no conflict (D-05) in its outcome and no financial interest in the matter or in a party to it; (d) they receive no fee, bonus, continuing mandate, success payment or other benefit that varies with

the conclusion reached, and their remuneration, appraisal or continuation in the engagement is not determined by the outcome the item supports; (e) they are not accountable for the outcome the item reports on; and (f) they satisfy the reporting-line limb. **The reporting-line limb:** they do not report to the preparer in respect of the work under review, and are not in the reporting line of the person accountable for the outcome for the purpose of that matter. Reporting to the preparer's line manager on unrelated work does not by itself defeat independence; being appraised on the outcome does. **On a PFL-AI engagement a stricter limb operates** — the person is not in the preparer's reporting line at all, and does not report to a person whose performance is measured by the work's conclusion. Where two of these tests bear on the same act, the one producing the wider obligation governs. Independence is a fact about a relationship to a specified matter, never a state of mind, a job title or a permanent designation — which is why a permanently titled "independent assurance" function is not independent of an artefact it helped produce. Where the adopting organisation contains no person meeting this definition, independence may be satisfied by a named reviewer engaged from outside it, and where that too is impossible the exception process in the relevant standard applies.

**D-13 · Information owner.** The party entitled to decide how a given item of protected information may be used, disclosed, stored and processed — ordinarily the client, employer, counterparty or individual from whom or about whom it was obtained, or the party a contract names.

**D-14 · Issuing professional.** The professional who issues, signs, submits, recommends or approves a material output, or who authorises another person or a process to do so in their name.

**D-15 · Material.** An item is material if any of the following is true: (a) were it wrong, omitted or reversed, a decision within the scope of the work would or could have been taken differently; (b) it changes a reported figure by more than the quantified tolerance the adopting organisation has published for that figure; (c) it affects a contractual, regulatory, tax or financial-reporting position; (d) it affects the safety of a person; (e) it affects a party's reliance (D-24); or (f) it meets the adopting organisation's published materiality criteria. **A matter bearing on safety, legality, a licence or permission, a statutory duty, or the truth of a statement made to a decision-maker is material irrespective of size**, and no documented threshold reduces it. Where the adopting organisation has published no criteria, the decision owner records which of (a) to (e) applies, and why, before the output is issued. Materiality is determined on the position as known at the time of the act; a determination made after an error is discovered does not displace the determination required at the time. Materiality is judged twice: on the item alone, and on the accumulation of items of the same kind since the test was last applied. **Where limb (b) is measured** is an application rule and not a second test — for PCL-AI work, the recorded materiality rule stating the quantum, the basis and the person who set it, applied consistently between periods; for PFL-AI work, the engagement's materiality statement, stated before the work begins in the transaction's own metric, with the

finance documents overriding it where they state their own; for PML-AI work, the organisation's documented financial, schedule or exposure threshold, which reaches only matters purely of that character. **PCI sets no percentage.** The adjective *material* qualifies every noun it precedes in these standards — a material assumption, a material error, a material claim — by this same test.

**D-16 · Material output.** A deliverable, report, figure, estimate, forecast, model, schedule, certificate, valuation, opinion, recommendation or item of advice that is issued outside the professional's own working papers, or is relied upon for a material decision. A draft circulated for comment and marked as such is not a material output; a draft that a recipient is invited to rely upon is.

**D-17 · Override.** Proceeding contrary to a control result — by ignoring, bypassing, suppressing, dismissing, re-running, re-parameterising or disabling the control, or by accepting the item the control rejected.

**D-18 · PCI-certified work.** Work performed by a professional within the scope of a PCI certification they hold or for which they are a candidate — project controls, project management or project finance work, and the preparation, review, recommendation, approval or assurance of its outputs — whether performed as an employee, a contractor, a consultant, a secondee or a sole practitioner, and whether or not the professional's PCI credential was the reason they were engaged.

**D-19 · Professional.** A candidate for, or holder of, a PCL-AI, PFL-AI or PML-AI credential, when performing PCI-certified work. Where a standard states an obligation on "a professional", it binds that individual personally and is not discharged by their employer's compliance.

**D-20 · Promptly.** Without any delay beyond the time needed to establish the facts required to make the communication accurate, and in any event within the period the adopting organisation's published procedure states. Where no such period is published, "promptly" means **within one working day** where the matter creates a risk to the safety of a person or an ongoing financial loss, and **within five working days** in every other case. The period runs from the moment the professional first knows or suspects the matter, not from the moment they confirm it; where the facts cannot be established within the period, the professional communicates what is known within the period and completes the account afterwards.

**D-21 · Protected information.** Information obtained in professional work which is any of the following: classified by its owner as confidential, restricted or equivalent; subject to a confidentiality obligation in a contract, engagement letter or undertaking; personal data relating to an identifiable individual; price-sensitive or inside information; commercially sensitive information whose disclosure would harm a party's position; legally privileged; or subject to export, sanctions or security control. **Information whose classification the professional does not know is protected information until the information owner states otherwise.** Information does not cease to be protected because it has been paraphrased, partially

redacted, anonymised by removing names only, split across several submissions, or reduced to figures, where a person with access to the surrounding context is able to reconstruct the protected content from it.

**D-22 · Record.** Information in any medium, created or received in professional work, that evidences an act, a decision, an instruction, a communication or the basis of an output — including files, messages, model versions, system entries, prompts and responses exchanged with an AI system, and annotations on any of them.

**D-23 · Records owner.** The individual the adopting organisation designates as accountable for a system of record. Where none is designated, the issuing professional is the records owner for the records they control.

**D-24 · Reliance.** A person relies on an output when they take a decision, make a commitment, part with money, issue a further output, or represent a position to a third party on the basis of it. Reliance begins at the earliest of those acts, not at the moment the output is received.

**D-25 · Source.** A document, dataset, system, instrument, standard, contract, publication or person from which a professional takes a fact, a figure, a rule or an interpretation used in a material output.

**D-26 · Verified.** An item, figure, statement, extraction or machine output is verified when a named person who is a competent reviewer (D-04) for that item has applied to it, against evidence (D-11), at least one of the eight methods listed in [PCI-FND-STD-03-PR-01](#) — independent recomputation, source tracing, clause-to-summary comparison, sampling on a stated basis, reconciliation, boundary testing, sensitivity analysis, or named expert judgement recorded with its reasoning — and has recorded the method used, the source or population tested and, where a sample was used, its selection basis, the inputs used, the scope tested, the date, the result, and every difference found together with its resolution. Reading an output and finding it plausible is not verification. An item on which no such record exists is not verified, however carefully it was produced. **One deliberate credential-specific narrowing:** where the item is a figure produced by a financial model under PFL-AI, *named expert judgement* is not admissible — a modelled figure is reproducible, and an expert's opinion that it looks right is not a check on it — and *clause-to-summary comparison* is applied as *clause-to-output comparison*, to a model output rather than to a summary. Where a foundational obligation is being discharged, the eight-method list governs. PML-AI additionally uses *verification* in the engineering verification-and-validation sense; that is a different concept and is not this term.

**D-27 · Working day.** A day on which the adopting organisation ordinarily operates, in the location where the work is performed.

**D-28 · Working record.** The set of records that evidences how a material output was produced — its sources, data, assumptions, calculations, transformations, verifications, reviews, approvals and issue.

**D-29 · Approved.** A decision, document, figure or version is approved where the person holding authority for that decision under the applicable governance arrangement or recorded delegation of authority has given assent identifiably, recording the date, the version assented to, and the scope of the assent. Silence, absence of objection, unrecorded verbal assent, assent by a person outside their recorded authority, and assent recorded after the item was used are **not** approval.

**D-30 · Current.** A record, figure, extract, document or version is current where it is the latest version issued by the system or authority that owns it as at the deliverable's stated cut-off or, where the deliverable states none, as at its issue date — and where its version identifier and its extraction date and time are recorded on or with the deliverable. A record whose version cannot be identified is not current, whatever its age.

**A note on the two terms added at D-29 and D-30.** Drafting Manual §4 names *approved* and *current* among the terms that always require definition when they appear in an obligation. Both were used throughout this set and defined nowhere in it; only `PCL_AI_STANDARDS.md` carried a definition of either. The wording above is PCL-AI's, generalised so that it operates where a deliverable states no cut-off. Adding them states a test readers were already expected to apply; it imposes nothing new.

**A note on roles.** *Issuing professional, decision owner, competent reviewer, engagement lead, information owner* and *records owner* denote capacities, not necessarily different people. On a small engagement one individual may hold several of them, and these standards work in that case. What does not change is each standard's element 10: where a standard requires independence, holding two capacities does not create it, and the exception process in that standard's element 12 is the route.

#### PCI STANDARD

#### PCI-FND-STD-01

### Professional Accountability

**1. Normative requirement.** Before a material output is issued, the issuing professional must record their own acceptance of personal accountability for it, whether it was prepared by that professional, by another person, by software or by an AI system.

**2. Purpose.** This standard controls the risk that accountability for professional work dissolves — into a team, a tool, a vendor, a template or a process — at exactly the moment reliance is placed on it. Certification attaches to a person; if no person accepts the output, the credential guarantees nothing.

**3. Scope.** Governs every professional (D-19) performing PCI-certified work (D-18). Applies to every material output (D-16) and to the decisions to issue, recommend, approve or certify it. Applies on projects, programmes and transactions of any size, in any sector and under any delivery model. Applies to preparation, review, recommendation, approval and assurance alike: the

professional who reviews and approves accepts accountability for the approval, and the professional who prepares accepts accountability for the preparation.

**4. Defined terms.** D-14 issuing professional · D-15 material · D-16 material output · D-18 PCI-certified work · D-19 professional · D-02 AI system · D-24 reliance · D-28 working record. No term is redefined in this standard.

**5. Required actions.** Record, in the working record and dated on or before the issue date, an acceptance of accountability naming one individual. Ensure that individual is able to state the basis of the output and to identify the verification record required by [PCI-FND-STD-03](#). Identify, where parts of the output were prepared by others or by an AI system, which parts, and on whose verification the issuing professional relies.

**6. Prohibited actions.** Issuing a material output "as generated". Recording an acceptance of accountability on another individual's behalf. Attributing accountability for a material output to an AI system, a software supplier, a model, a tool configuration, a template, a team or an unnamed process. Issuing a material output the professional is not willing to be held accountable for.

**7. Required evidence.** An acceptance record for each material output, attributable to the accepting individual, dated on or before issue; where issue is delegated, the prior authorisation required by [PCI-FND-STD-01-PR-04](#); the working record showing what the accepting individual relied on others for.

**8. Responsible role.** The issuing professional (D-14). Where several professionals contribute, each is accountable for their own contribution and one named individual is accountable for the output as issued.

**9. Approval authority.** The issuing professional approves the issue of the output. Only the decision owner (D-08) for the matter may direct that an output be issued over the issuing professional's refusal, and that direction must be recorded and is itself a matter for escalation under [PCI-FND-STD-11](#).

**10. Independence requirement.** None for the acceptance itself – accountability is personal, and independence is not a component of it. Independence attaches to the verification the acceptance relies upon, and is required by [PCI-FND-STD-03](#) element 10.

**11. Materiality or threshold.** The obligation attaches to material outputs (D-16) as tested by D-15. Working papers, internal drafts marked as drafts, and exploratory analysis not offered for reliance are outside it. The adopting organisation (D-01) may set a lower threshold; it must not set a higher one.

**12. Exception and waiver. No exception is permitted.** Accountability for a material output cannot be waived, delegated away, disclaimed or transferred to a system or a supplier. A professional who is unwilling to accept accountability must not issue the output, and must escalate under [PCI-FND-STD-11](#).



**13. Escalation trigger.** The professional is unable to accept accountability honestly — because the output cannot be verified, exceeds their competence, or is being issued under instruction they do not accept; or a material output is discovered to have been issued with no individual accountable for it.

**14. AI application.** An AI system may draft, calculate, analyse, summarise, classify, structure the working record and assemble the evidence an accepting professional reviews before accepting.

**15. AI prohibition.** An AI system must not be named as the accountable party for any material output, must not record an acceptance of accountability, and must not be represented to any recipient as the author of a professional judgement.

**16. AI verification.** Before accepting accountability for an output to which an AI system materially contributed, the accepting professional must confirm by **source tracing** that every material factual claim in the output is supported by evidence (D-11) identified in the working record, and by **named approval** that the verification required by **PCI-FND-STD-03** was performed and recorded.

#### **17. External reference.**

- **ISO/IEC 17024** — *Conformity assessment — General requirements for bodies operating certification of persons*. Issuing organisation: ISO/IEC. Subject: requirements for bodies that certify individuals. Edition checked: ISO/IEC 17024:2026, superseding the 2012 edition. Nature of authority: **category 3 — international voluntary standard** (register category: international standard). Verification date: 2026-08-03 (register EXT-022). Applicability limitation: addresses certification bodies, not individual credential holders; voluntary unless a regulator or contract adopts it; no requirement in this standard is sourced to it. Verify current requirements.
- **Recommendation of the Council on Artificial Intelligence** (the "OECD AI Principles"), OECD/LEGAL/0449. Issuing organisation: OECD. Subject: intergovernmental principles for trustworthy AI, including accountability. Edition checked: adopted 2019, revised May 2024. Nature of authority: **category 10 — illustrative practice** (register category: voluntary framework). Verification date: 2026-08-03 (register EXT-081). Applicability limitation: a Council Recommendation, **not legislation and not binding even on adherents**; named here to illustrate an accountability principle, relied on for nothing.
- **Code of Ethics and Professional Conduct**. Issuing organisation: Project Management Institute. Subject: honesty, responsibility, respect and fairness for delivery practitioners. Edition checked: not independently verified — no edition asserted. Nature of authority: **category 6 —**

**ethical code** (register category: professional guidance). Verification date: not independently verified (register EXT-063). Applicability limitation: binding only on those the Project Management Institute's own membership or credential terms bind; it is not regulatory authority and creates no obligation for a PCI professional who is not subject to it.

**18. Jurisdictional caution.** Statutory regimes on professional liability, delegation, signature, certification and limitation of personal responsibility differ by jurisdiction and by profession, and some reserve certain sign-offs to licence holders. Obtain local legal advice before relying on any delegation, limitation or exclusion of personal responsibility, and before signing anything a local regime treats as a reserved act.

**19. Related PCI Standards.** [PCI-FND-STD-03](#) (the verification the acceptance relies on) · [PCI-FND-STD-04](#) (who decides) · [PCI-FND-STD-10](#) (competence to accept) · [PCI-FND-STD-11](#) (what to do when acceptance is impossible) · [PCI-FND-STD-14](#) (what an AI system must never be permitted to do).

**20. Related Body of Knowledge content.** PCL-AI Domain 13 — AI for Project Controls and PM: Concepts, Tools and Practice. PML-AI KA 1.2 — the project leader's accountability; KA 1.4 — ethics and the responsible use of AI; Domain 14 — digital delivery, data and responsible AI. PFL-AI KA 1.3 — ethics, fiduciary awareness and responsible AI; Domain 16 — data, automation and responsible AI in finance.

**21. Compliance test.** Compliance is demonstrated when, for a sample of material outputs **selected by the reviewer and not by the professional**, every one of the following holds for every sampled item: (a) an acceptance record exists, dated on or before the output's issue date; (b) it names exactly one individual, and that individual is a natural person, not a team, system, vendor or role held by nobody; (c) the record is attributable to that individual rather than entered by a third party on their behalf; (d) asked without prior notice, that individual identifies the verification record required by [PCI-FND-STD-03](#) for the output and states which parts of it were prepared by others or by an AI system; and (e) where issue was delegated, a prior authorisation exists that names the output or class of output, the scope and the period, and is dated before issue. A sampled item fails if any one of (a) to (e) is absent. The standard is complied with for the period under review only if no sampled item fails.

**22. Breach indicators.** Outputs issued with no named accountable individual. Acceptance records entered in bulk, after issue, or by an administrator on others' behalf. A named individual who cannot state the basis of the output.



Correspondence attributing an error to "the model", "the system", "the tool" or "the process". Delegated issue with no prior authorisation. Acceptance records whose timestamps cluster at period end rather than at issue.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure where breached in assessment; ethics review; certification investigation; suspension; withdrawal. Each is subject to due process and a right of appeal, and PCI claims no consequence beyond these.

**24. Examination application.** Tested by scenario judgement (where accountability sits when AI-assisted work proves wrong), by evidence selection (which record establishes who is accountable) and by ethical dilemma (a professional directed to issue work they cannot accept). Never by recalling standard numbers.

**25. Version and status.** Version 1.0 · approval date: not approved — draft submitted under Charter §5 Stage 11 · effective date: on approval · amendment note: supersedes [PCI-LAW-F-01](#), withdrawn. See *Due-process record*.

#### Process requirements under PCI-FND-STD-01

[PCI-FND-STD-01-PR-01](#). Each material output must have exactly one individual recorded as accountable for the output as issued, and that individual must be a natural person.

[PCI-FND-STD-01-PR-02](#). A professional must not attribute accountability for a material output to an AI system, a software supplier, a model, a tool configuration, a template, a team or an unnamed process, whether in the output, in correspondence, in a review, in an investigation or in evidence.

[PCI-FND-STD-01-PR-03](#). A professional who is unable to accept accountability for a material output must not issue it, and must escalate the reason under [PCI-FND-STD-11](#).

[PCI-FND-STD-01-PR-04](#). Where a material output is to be issued in a professional's name by another person or by an automated process, the professional must record their authorisation before the first such issue, naming the output or class of output, the scope, the period and the conditions on which the authorisation depends.

[PCI-FND-STD-01-PR-05](#). Where an output is issued jointly, each contributing professional must record accountability for their own contribution, identified by section, figure or workstream.

## Evidence Before Assertion

- 1. Normative requirement.** A professional must not present a material factual claim, benchmark, calculation, contractual interpretation or reference to an external instrument unless evidence (D-11) supporting it is identified in the working record and is retrievable by a competent reviewer at the time of presentation.
- 2. Purpose.** This standard controls the risk that assertions enter professional work on the strength of recollection, convention, plausibility or an AI system's fluency, and are then relied upon as though they had been established. An unevidenced claim is indistinguishable from an evidenced one to the reader; the distinction has to be made in the record.
- 3. Scope.** Governs every professional performing PCI-certified work. Applies to every material factual claim, benchmark, calculation, contractual interpretation and external-instrument reference in any material output, in any medium, including presentations, dashboards, models, correspondence relied upon, and oral advice recorded and acted on. Applies on projects, programmes and transactions of any size. Applies to preparation, review, recommendation, approval and assurance: a reviewer who approves an unevidenced claim breaches this standard as surely as the preparer who wrote it.
- 4. Defined terms.** D-11 evidence · D-04 competent reviewer · D-15 material · D-16 material output · D-25 source · D-28 working record. No term is redefined in this standard.
- 5. Required actions.** Link each material claim in the working record to the specific evidence supporting it, identified by document, system, extraction date and location within the source. Check quotations, clause references, figures and attributions against the instrument itself before presenting them. Where no evidence exists, either remove the claim or present it as an assumption under [PCI-FND-STD-05](#).
- 6. Prohibited actions.** Presenting an assumption as a fact. Citing a document, clause, figure or authority the professional has not read in the source. Attaching a document that does not support the claim as though it did. Presenting a benchmark without its source, population, date and basis of comparability. Presenting an AI system's output as the evidence for the claim it makes.
- 7. Required evidence.** The working record showing, for each material claim, the evidence relied on and where it sits; the dated check record required by [PCI-FND-STD-02-PR-02](#) for each external reference; retrievable copies of, or resolvable references to, the evidence itself.
- 8. Responsible role.** The issuing professional (D-14) for claims in outputs they issue. The competent reviewer (D-04) for claims they approve.

**9. Approval authority.** The decision owner (D-08) for the matter may approve presenting a claim as an assumption where evidence does not exist; nobody may approve presenting an unevidenced claim as a fact.

**10. Independence requirement.** None for the evidencing itself, which the preparer performs. Where the claim is a material calculation or an automated conclusion, the independent verification in [PCI-FND-STD-03](#) applies in addition.

**11. Materiality or threshold.** Attaches to material claims (D-15). A claim is material where its falsity would or could change a decision, a reported figure beyond the adopting organisation's published tolerance, a contractual or regulatory position, a party's reliance, or the safety of a person. The adopting organisation may publish a lower threshold. On a small engagement the working record may be a single referenced worksheet; the test in element 21 does not require any particular tool.

**12. Exception and waiver.** A time-limited exception is permitted only where a claim must be presented before its evidence can be retrieved and delay would itself create a risk to safety or an ongoing financial loss. The decision owner approves it; the justification, the evidence expected and the date by which it will be produced are recorded; it lasts no longer than five working days; the compensating control is that the claim is marked in the output as unevidenced at the date of issue; and the exception is reported to the engagement lead. An exception must not be used for a claim that will not later be evidenced.

**13. Escalation trigger.** Discovery that a material claim already relied upon has no supporting evidence; instruction to present an assumption as a fact; discovery that a cited clause, figure or authority does not exist or does not say what was attributed to it.

**14. AI application.** An AI system may locate candidate sources, extract citations for checking, draft the evidence index, and flag claims in a draft that carry no linked evidence.

**15. AI prohibition.** An AI system must not be the source of a factual claim, must not be cited as authority for one, and must not be represented as having confirmed that evidence exists.

**16. AI verification.** Every citation, quotation, figure, clause reference and attribution that originated with an AI system must be checked by **source tracing** — opening the cited source itself and confirming the cited content exists at the stated location and says what is attributed to it — and the check recorded with its date. Re-asking any AI system whether the citation is correct does not satisfy this element.

**17. External reference.**

- **Total Cost Management (TCM) Framework.** Issuing organisation: AACE International. Subject: an integrated process for cost and schedule management, including the documented basis of estimates and forecasts. Edition checked: not independently verified — no edition asserted. Nature of authority: **category 5 — professional framework** (register category: professional guidance). Verification date: not independently verified (register EXT-064). Applicability limitation: professional guidance from a membership body; it is not regulatory authority, binds nobody by its own force, and no requirement in this standard is sourced to it.
- **Financial Modelling Code.** Issuing organisation: ICAEW. Subject: principles for building, documenting and reviewing financial models. Edition checked: current at the register's verification date; no edition asserted. Nature of authority: **category 5 — professional framework** (register category: professional guidance). Verification date: 2026-08-03 (register EXT-125). Applicability limitation: principles-based guidance from a professional body; not a compliance standard, not certifiable, and binding only where a body, regulator or engagement adopts it.

**18. Jurisdictional caution.** What counts as evidence, what must be disclosed and what may be withheld in a dispute, an audit or a regulatory inquiry are questions of the governing law and of privilege, and they differ sharply between jurisdictions. Obtain local legal advice before withholding, redacting or restructuring the evidence behind a material claim, and before asserting privilege over it.

**19. Related PCI Standards.** [PCI-FND-STD-03](#) (verification of calculations) · [PCI-FND-STD-05](#) (what to do when there is no evidence — state it as an assumption) · [PCI-FND-STD-06](#) (whether the source is the right version) · [PCI-FND-STD-07](#) (tracing the figure back through its data) · [PCI-FND-STD-14](#) (AI-produced citations).

**20. Related Body of Knowledge content.** PCL-AI Domain 4 — performance management, variance analysis and reporting; Domain 13. PML-AI KA 9.4 — data quality and AI-output quality; KA 11.2 — executive communication and reporting. PFL-AI KA 6.4 — model checks, model audit and AI-assisted modelling controls; Domain 16.

**21. Compliance test.** The reviewer selects material claims from issued outputs, at their own choice and without notice to the professional. For each sampled claim: (a) the working record identifies specific evidence by document, system, extraction date and location; (b) the reviewer retrieves that

evidence without assistance from the professional; (c) **reading only that evidence and no explanation from the author, the reviewer reaches the same claim — the same magnitude, the same direction and the same qualification**; and (d) where the claim quotes or attributes anything to an external instrument, the quoted content is found in that instrument at the stated location. A sampled claim fails if the evidence cannot be retrieved, if it supports a different magnitude, direction or qualification, if it supports the claim only with an explanation the record does not contain, or if a quoted clause, figure or attribution cannot be located. The standard is complied with for the period under review only if no sampled claim fails.

**22. Breach indicators.** Claims traced to "industry standard", "typical", "as agreed" or "per the model" with no source. Benchmarks with no population or date. Citations to clauses that do not exist or whose numbering belongs to a different edition. Evidence folders whose contents do not match the claims that reference them. Round figures repeated across outputs with no origin. A working record assembled after the output was issued.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension; withdrawal. Each subject to due process and a right of appeal.

**24. Examination application.** Tested by evidence selection (which of four records actually supports the stated claim), by calculation review (a figure presented with a source that does not produce it) and by AI-verification case (a fluent paragraph containing a fabricated citation).

**25. Version and status.** Version 1.0 · approval date: not approved — draft submitted under Charter §5 Stage 11 · effective date: on approval · amendment note: new standard; carries forward part of the withdrawn [PCI-LAW-F-05](#).

### Process requirements under PCI-FND-STD-02

[PCI-FND-STD-02-PR-01](#). Each material claim must be linked in the working record to the specific evidence supporting it, identified by document or system, extraction date, and location within that source.

[PCI-FND-STD-02-PR-02](#). A quotation, clause reference, figure, edition, effective date or attribution to an external instrument must be checked against that instrument itself before presentation, and the check recorded with its date.

[PCI-FND-STD-02-PR-03](#). Where evidence for a material claim does not exist, the professional must either remove the claim from the output or present it expressly as an assumption under [PCI-FND-STD-05](#).

**PCI-FND-STD-02-PR-04.** A benchmark must be presented with its source, the population it was drawn from, its date and the basis on which the professional considers it comparable, or must not be presented.

**PCI-FND-STD-02-PR-05.** A contractual interpretation presented as a material claim must identify the executed contract and the provision relied on, and must state whether it is the professional's own reading or advice obtained under **PCI-FND-STD-10-PR-02**.

## PCI STANDARD

### PCI-FND-STD-03

## Independent Verification

**1. Normative requirement.** A material calculation, model output or automated conclusion must be verified (D-26) by a person independent (D-12) of its preparation before any person relies on it.

**2. Purpose.** This standard controls the risk that an error survives to the point of reliance because the only person who examined the work is the person who produced it — the case in which self-review blindness, a copied formula, an inherited assumption or an AI system's confident error is invisible from inside.

**3. Scope.** Governs every professional performing PCI-certified work. Applies to material calculations, model outputs and automated conclusions of every kind — estimates, forecasts, schedules, earned-value figures, cash-flow projections, risk quantifications, valuations, classifications and any figure produced by a script, a model or an AI system — on projects, programmes and transactions of any size. Applies to preparation (the preparer must obtain verification), review (the verifier must perform a named method) and approval (the decision owner must not rely before verification is recorded). Does not apply to exploratory analysis that nobody relies on.

**4. Defined terms.** D-26 verified · D-12 independent · D-04 competent reviewer · D-15 material · D-24 reliance · D-08 decision owner · D-01 adopting organisation. No term is redefined in this standard.

**5. Required actions.** Obtain verification before reliance. Record the verifier's name, the method used from the list in **PCI-FND-STD-03-PR-01**, the scope tested, the inputs used, the date and the result. Where verification fails, record the failure and withhold the item from reliance until a later verification succeeds.

**6. Prohibited actions.** Relying on an unverified material calculation, model output or automated conclusion. Recording as verification a re-run of the same tool, model or AI system on the same inputs. Recording as verification a review by the preparer, by the person who configured the producing tool, or

by a person with a conflict in the outcome. Recording a verification that was not performed. Discarding a failed verification and recording only a later successful one without the failure.

**7. Required evidence.** The verification record for each material item, dated on or before first reliance; the working papers of the verification itself, sufficient for the method to be re-performed; where an exception under element 12 was used, the waiver record and the disclosure made to recipients.

**8. Responsible role.** The preparer is responsible for obtaining verification before offering the item for reliance. The competent reviewer (D-04) named in the record is responsible for performing the method they record. The decision owner (D-08) is responsible for not relying before the record exists.

**9. Approval authority.** The decision owner for the matter may approve a waiver under element 12. No role may approve reliance on an item recorded as having failed verification.

**10. Independence requirement.** The verifier must be independent of preparation as defined in D-12: they must not have prepared the item, must not have selected, built or configured the tool, model or AI system that produced it, must hold no conflict in its outcome, and must not be appraised or remunerated on the outcome the item supports. Independence of line reporting is not required; independence of appraisal on the outcome is.

**11. Materiality or threshold.** Attaches to material items (D-15). The adopting organisation sets the quantified tolerance below which a calculation is not material for its own reporting, and PCI sets none — an invented percentage would be wrong on both a EUR 200,000 refurbishment and a EUR 20 billion programme. Where the adopting organisation publishes no tolerance, the decision owner records the D-15 limb relied on. Irrespective of any tolerance, an item is material where it supports an irreversible commitment, a payment, an external report or a decision bearing on the safety of a person.

**12. Exception and waiver.** A waiver is permitted **only** where the adopting organisation contains no person meeting D-12 and no external reviewer can be engaged in the time available. The decision owner approves it. The justification must record why no independent person is available and what was attempted. It lasts for one named output or for a period not exceeding three months, whichever ends first, and must be re-approved rather than rolled forward. The compensating controls are mandatory and cumulative: (a) the preparer applies a **second, different** method from the -PR-01 list and records both; (b) the limitation is stated in the issued output, in the same place as the conclusion; and (c) the waiver is reported to the engagement lead and recorded in a waiver log. **A waiver is not available where the item supports an irreversible commitment, a payment to a third party, an external regulatory or financial report, or a decision bearing on the safety of a person** — in each of those four cases the consequence of an error



is either unrecoverable or borne by someone outside the engagement, which is precisely when a second pair of eyes is worth most.

**One narrow allowance exists, and it is not a waiver.** Where an item cannot be verified in the time available and acting on it is unavoidable to protect the safety of a person, the professional may rely on it **to the extent necessary to protect that person**, and must then do all of the following: record the reliance, the necessity, and what was attempted to obtain verification, within one working day; state the limitation to every recipient of the item; obtain and record the verification as soon as it can be obtained; and communicate any resulting difference under [PCI-FND-STD-15](#). The decision owner must record their position on the reliance within five working days. The item must not be relied upon for any purpose beyond protecting that person until the verification is recorded. **This allowance does not extend to commercial urgency, a deadline, a reporting cycle or a funding date**, and the obligation to obtain the verification is not discharged by the emergency, only deferred by it.

**13. Escalation trigger.** A material item is discovered to have been relied upon without verification; a verification failure is overruled without a decision under [PCI-FND-STD-13](#); a verifier reports pressure to record a verification they did not perform; or waivers under element 12 recur for the same class of item, which indicates a governance defect rather than a one-off constraint.

**14. AI application.** An AI system may perform a cross-check, recompute independently in a different engine, reconcile two datasets, generate boundary and stress cases, flag anomalies and outliers, and draft the verification record for the human verifier to confirm.

**15. AI prohibition.** An AI system must not be recorded as the verifier, must not determine that verification is unnecessary, must not be the sole check on another AI system's output, and must not be represented to any recipient as having independently verified anything.

**16. AI verification.** Where an AI system produced the item, the human verifier must apply at least one of **independent recomputation** (reproducing the figure by a method that does not use the same system), **source tracing** (following each input to its origin), **reconciliation** (agreeing the output to an independent record) or **boundary testing** (running inputs at the limits of the stated range and confirming the behaviour is explicable), and must record which was used and what it showed. "Reviewed the AI output" is not a method and does not satisfy this element.

#### **17. External reference.**

- **Financial Modelling Code.** Issuing organisation: ICAEW. Subject: model construction, documentation and review discipline. Edition checked: current at the register's verification date; no edition asserted.



Nature of authority: **category 5 — professional framework** (register category: professional guidance). Verification date: 2026-08-03 (register EXT-125). Applicability limitation: not certifiable, not a compliance standard, binding only where adopted; no requirement here is sourced to it.

- **The FAST Standard.** Issuing organisation: FAST Standard Organisation. Subject: a structural convention for building transparent, reviewable financial models. Edition checked: current at the register's verification date; no edition asserted. Nature of authority: **category 7 — industry guidance** (register category: voluntary framework). Verification date: 2026-08-03 (register EXT-126). Applicability limitation: adopted voluntarily by modellers and firms; it imposes no obligation of its own and PCI imports none of it.
- **SR 11-7 / OCC 2011-12 — Supervisory Guidance on Model Risk Management.** Issuing organisation: US Federal Reserve Board and Office of the Comptroller of the Currency. Subject: model development, implementation, use and independent validation in supervised institutions. Edition checked: not independently verified — no edition asserted. Nature of authority: **category 10 — illustrative practice** (register category: illustrative reference; actual status: supervisory guidance). Verification date: not independently verified (register EXT-102). Applicability limitation: supervisory guidance with **no legal force of its own**, addressed to institutions supervised by those agencies in the United States; it does not apply to a project, a sponsor or an adviser by its own force, and is named here only to illustrate the independent-validation discipline.

**18. Jurisdictional caution.** Sector regulators in financial services, public procurement, safety- critical engineering and healthcare impose their own validation, independence and human-oversight obligations, and some reserve validation of certain models to licensed or approved persons. Obtain local regulatory advice where a verified output feeds a regulated activity, and treat any such obligation as prevailing over this standard under Charter §4.

**19. Related PCI Standards.** [PCI-FND-STD-01](#) (accountability the verification supports) · [PCI-FND-STD-02](#) (evidence for the inputs) · [PCI-FND-STD-04](#) (the decision that follows verification) · [PCI-FND-STD-07](#) (the lineage the verifier traces) · [PCI-FND-STD-13](#) (a verification failure must not be overridden silently) · [PCI-FND-STD-14](#) (the boundary of what an AI system may do). **Boundary with [PCI-FND-STD-14](#):** this standard governs

*whether an item has been checked; PCI-FND-STD-14 governs what an AI system must never be permitted or represented to do. They do not overlap.*

**20. Related Body of Knowledge content.** PCL-AI Domain 6 — earned value management and forecasting; Domain 13. PML-AI KA 9.4; Domain 14, especially KA 14.3 — AI use cases, prompting and verification. PFL-AI KA 6.4 — model checks, model audit and AI-assisted modelling controls; KA 13.2 — model audit; Domain 16, especially KA 16.3 — explainability, validation, bias and model risk.

**21. Compliance test.** The reviewer selects material calculations, model outputs and automated conclusions from issued work, at their own choice. For each sampled item, all of the following hold: (a) a verification record exists, dated on or before the date of first reliance (D-24); (b) it names the verifier, and the reviewer confirms from the working record that the verifier neither prepared the item nor configured the producing tool, and from the engagement's conflict declarations that the verifier held no conflict in the outcome; (c) it names one of the eight methods in -PR-01 and states the scope tested, the inputs used and the result; (d) **the reviewer re-performs the named method on the recorded inputs and obtains the recorded result within the tolerance the record states**; and (e) where the record shows a failed verification, no reliance is evidenced before a later successful verification. A sampled item fails if any one of (a) to (e) is absent, or if re-performance does not reproduce the recorded result and the record does not explain the difference. Where a waiver under element 12 was used, the sampled item is tested instead against the three cumulative compensating controls, all of which must be present.

**22. Breach indicators.** Verification records naming the preparer. Verification dated after the output was issued. Records naming a method but no scope, inputs or result. Identical verification wording repeated across unrelated items. Verifications recorded in a batch at period end. Model versions changed after verification with no re-verification. Waivers rolled forward for the same item across periods.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension; withdrawal. Each subject to due process and a right of appeal.

**24. Examination application.** Tested by calculation review (a model output with a verification record that cannot reproduce it), by evidence selection (which of several records constitutes verification) and by AI-verification case (an output "verified" by re-prompting the same system).

**25. Version and status.** Version 1.0 · approval date: not approved — draft submitted under Charter §5 Stage 11 · effective date: on approval · amendment note: carries forward the verification substance of the withdrawn PCI-LAW-

**F-02. Stage 9 amendment:** the second red-team pass found that the standard had no compliant route in the one case that will certainly arise — an unverified item that must be acted on to protect a person — and a standard with no route in that case is a standard that gets broken and teaches contempt for the rest; element 12 now carries a bounded allowance with recording, disclosure, deferred verification and the decision owner's after-the-fact position, and states why the four no-waiver cases are what they are.

### Process requirements under PCI-FND-STD-03

**PCI-FND-STD-03-PR-01.** The verification record must name the method used, and the method must be one of: **independent recomputation · source tracing · reconciliation · clause-to-summary comparison · sampling on a stated basis · boundary testing · sensitivity analysis · named expert judgement**. The record must state the scope tested, the inputs used, the date and the result.

**PCI-FND-STD-03-PR-02.** The verifier must not be the preparer of the item, and must not have selected, built or configured the tool, model or AI system that produced it.

**PCI-FND-STD-03-PR-03.** Re-running the same tool, model or AI system on the same inputs must not be recorded as verification, and asking any AI system to confirm its own or another AI system's output must not be recorded as verification.

**PCI-FND-STD-03-PR-04.** A verification that fails must be recorded together with its failure, and the item must not be relied upon until a later verification is recorded as successful.

**PCI-FND-STD-03-PR-05.** Where the item changes after verification — through a revised input, a changed assumption, a new model version or a corrected formula — the verification must be repeated in the scope affected before further reliance, and the repetition recorded.

## PCI STANDARD

### PCI-FND-STD-04

## Human Decision Authority

**1. Normative requirement.** A material professional decision must be recorded as taken by a named human decision owner (D-08) who holds authority for it under the applicable governance arrangement and is a competent reviewer (D-04) for its subject matter.

**2. Purpose.** This standard controls the risk that automation converts a recommendation pipeline into an unexamined decision pipeline — that outputs take

effect because nobody stopped them, rather than because an accountable person, competent in the subject, decided that they take effect.

**3. Scope.** Governs every professional performing PCI-certified work. Applies to material decisions, approvals, gate passages, releases, certifications, authorisations, payments and baseline changes, including those embedded in automated workflows, scripts, rules engines and agentic tools, on projects, programmes and transactions of any size. Applies to the design of decision workflows as well as to individual decisions, and to preparation, recommendation, approval and assurance of each.

**4. Defined terms.** D-08 decision owner · D-04 competent reviewer · D-15 material · D-02 AI system · D-24 reliance · D-01 adopting organisation. No term is redefined in this standard.

**5. Required actions.** Identify, for each material decision, the individual holding authority for it under the governance arrangement. Confirm that individual is a competent reviewer for its subject matter. Record the decision, its date, the decision owner, the material inputs considered, and any AI contribution. Insert an explicit human decision point ahead of any automated step that could otherwise give effect to a material recommendation.

**6. Prohibited actions.** Configuring a system to approve, certify, release, pay or commit on an automated or AI-generated output without a recorded human decision. Recording an adoption that did not involve an actual consideration of the matter. Ratifying an automated action after it has taken effect and presenting the ratification as the decision. Assigning a material decision to an individual who is not a competent reviewer for its subject matter. Removing or bypassing a human decision point to save time.

**7. Required evidence.** Decision records naming the human decision owner, dated, stating the decision, the material inputs considered and any AI contribution; workflow documentation identifying each human decision point in any automated chain; the record of the periodic workflow test required by [PCI-FND-STD-04-PR-05](#).

**8. Responsible role.** The decision owner (D-08) for the decision. The engagement lead (D-09) for ensuring that each material decision in the engagement has an identified decision owner. The professional who designs or configures an automated workflow is responsible for the decision points within it.

**9. Approval authority.** The decision owner approves, rejects, amends or defers. Where the decision exceeds their delegated authority, the next authority level under the governance arrangement decides, and the escalation is recorded.

**10. Independence requirement.** The decision owner need not be independent of preparation — deciding on one's own team's work is ordinary governance. The decision owner must hold no conflict (D-05) in the outcome;

where they do, [PCI-FND-STD-08-PR-01](#) applies and the decision passes to an independent decision owner.

**11. Materiality or threshold.** Attaches to material decisions (D-15). The adopting organisation's delegated-authority schedule sets the monetary and risk thresholds at which each level decides, and PCI sets none. Irrespective of that schedule, a decision is material where it commits funds irreversibly, alters a baseline, releases a payment, issues an external report, or bears on the safety of a person. On a small engagement the schedule may be a single recorded line stating who decides what; the standard does not require a governance framework, only an identified human decision owner.

**12. Exception and waiver. No exception is permitted** to the requirement that a material decision be taken by a named human decision owner. A time-limited exception to the *identity* of the decision owner is permitted where the named owner is unavailable and delay would create a risk to safety or an ongoing financial loss: a deputy named in advance in the governance arrangement decides, the substitution is recorded at the time, and the named owner reviews and records their position on the decision within five working days. There is no exception permitting an automated step to take effect on a material matter with no human decision at all.

**13. Escalation trigger.** Discovery that an automated or AI-generated output took effect on a material matter without a recorded human decision; instruction to remove a human decision point; a decision owner who declines to consider the matter but signs; discovery that decisions are being recorded by someone other than the decision owner.

**14. AI application.** An AI system may rank options, model consequences, draft the recommendation, pre-populate the approval package, flag exceptions, identify the correct decision owner and route the matter to them.

**15. AI prohibition.** An AI system must not decide, approve, certify, sign, release, authorise, waive or commit anything material, and must not close the path from recommendation to effect without an accountable human act in between.

**16. AI verification.** Before adopting an AI-generated recommendation on a material matter, the decision owner must record at least one **named-approval** entry identifying the verification performed under [PCI-FND-STD-03](#), and must record one factor they considered that is not stated in the recommendation itself — a **boundary test** of the recommendation against a constraint, a stakeholder position, a contractual term or a risk the recommendation does not address. This is the check that distinguishes a decision from a countersignature.

**17. External reference.**

- **Regulation (EU) 2024/1689** laying down harmonised rules on artificial intelligence (the "AI Act"). Issuing organisation: European Union. Subject: harmonised rules on AI systems placed on the EU market or used in the EU, including human-oversight duties for certain systems. Edition checked: in force since 1 August 2024; phased application — prohibitions from 2 February 2025, general-purpose AI obligations from 2 August 2025, general application from 2 August 2026, remaining rules by 2 August 2027. Nature of authority: **category 1 — applicable legislation or regulation, within the European Union only** (register category: illustrative reference, because the corpus relies on it for nothing). Verification date: 2026-08-03 (register EXT-100). Applicability limitation: **binding legislation only within the European Union and only on actors within its scope**; it imposes nothing outside the EU, PCI sources no requirement to it, and no clause, article or annex is asserted. Verify current requirements.
- **ISO/IEC 38507** — *Information technology — Governance of IT — Governance implications of the use of artificial intelligence by organizations*. Issuing organisation: ISO/IEC. Subject: what a governing body needs to consider when its organisation uses AI. Edition checked: ISO/IEC 38507:2022. Nature of authority: **category 3 — international voluntary standard**. Verification date: 2026-08-03 (register EXT-037). Applicability limitation: addressed to governing bodies rather than practitioners; guidance, not a certifiable requirements standard; voluntary unless a regulator or contract adopts it.
- **ISO/IEC 42001** — *Information technology — Artificial intelligence — Management system*. Issuing organisation: ISO/IEC. Subject: requirements for an AI management system. Edition checked: ISO/IEC 42001:2023, first edition. Nature of authority: **category 3 — international voluntary standard**. Verification date: 2026-08-03 (register EXT-021). Applicability limitation: a certifiable requirements standard, but voluntary unless adopted by regulation or contract; certification of an organisation's management system is not evidence that any individual decision complied with this standard.

**18. Jurisdictional caution.** Several jurisdictions and sectors regulate automated decision-making about individuals, including rights to human intervention, explanation or review, and some regulate automated trading, credit and safety decisions specifically. Obtain local legal advice before deploying any automated decision flow that affects people, and before relying on an automated decision in a regulated activity.

**19. Related PCI Standards.** *PCI-FND-STD-01* (accountability for the output the decision releases) · *PCI-FND-STD-03* (the verification the decision relies on) · *PCI-FND-STD-08* (a conflicted decision owner must not decide) · *PCI-FND-STD-10* (competence to decide) · *PCI-FND-STD-13* (a decision to override a control) · *PCI-FND-STD-14* (what an AI system must never be permitted to do). **Boundary with *PCI-FND-STD-14*:** this standard requires that a named, authorised, competent human owns each material decision; *PCI-FND-STD-14* prohibits an AI system being permitted or represented to hold a role reserved to a person. A workflow can satisfy one and breach the other.

**20. Related Body of Knowledge content.** PCL-AI Domain 13. PML-AI KA 3.2 — decision authorities and thresholds; KA 1.4; Domain 14, especially KA 14.4 — explainability, bias and human accountability. PFL-AI Domain 16, especially KA 16.4 — privacy, cybersecurity, human approval and AI governance.

**21. Compliance test.** Compliance is demonstrated when all four of the following hold. (a) **Record test:** for a sample of material decisions selected by the reviewer, each has a decision record naming one individual, dated, stating the decision and the material inputs considered, and that individual appears in the governance arrangement as holding authority at that threshold on that date. (b) **Competence test:** for each sampled decision, the named individual satisfies D-04 for its subject matter, evidenced by the record the adopting organisation holds. (c) **Walk-through test:** in a test instance of each automated workflow that carries material recommendations, the reviewer attempts to record a rejection and a deferral; both are accepted by the workflow and both appear in the decision log. (d) **Substance test:** for each sampled decision the record shows either a recorded rejection, amendment, deferral or condition, or a recorded factor considered that does not appear in the recommendation itself. A sampled decision fails if any of (a), (b) or (d) is absent; a workflow fails if (c) cannot be completed. The standard is complied with for the period under review only if no sampled decision and no workflow fails.

**22. Breach indicators.** Approval timestamps within seconds of recommendation generation, in volume. Decision logs containing no rejection, amendment, deferral or condition across a whole period. Approvals recorded under a generic or shared account. Automated releases with no preceding decision entry. A decision owner who cannot state the matter they approved. Workflow configuration in which the rejection path is absent or non-functional. Deputies deciding routinely rather than in the recorded absence of the named owner.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review;



certification investigation; suspension; withdrawal. Each subject to due process and a right of appeal.

**24. Examination application.** Tested by scenario judgement (where the human decision point must sit in an automated controls, delivery or finance workflow), by escalation decision (what to do on discovering an auto-approval) and by ethical dilemma (pressure to remove a decision point for throughput).

**25. Version and status.** Version 1.0 · approval date: not approved — draft submitted under Charter §5 Stage 11 · effective date: on approval · amendment note: supersedes [PCI-LAW-F-03](#), withdrawn.

#### Process requirements under PCI-FND-STD-04

[PCI-FND-STD-04-PR-01](#). An automated or AI-generated recommendation must not take effect on a material matter until the decision owner has recorded their adoption of it.

[PCI-FND-STD-04-PR-02](#). Before an automated workflow that could give effect to a material recommendation enters use, an explicit human decision point must be inserted ahead of the step that gives effect, and the workflow documentation must identify it.

[PCI-FND-STD-04-PR-03](#). The workflow must permit the decision owner to reject, amend and defer, and must record each of those outcomes; a workflow offering only approval must not be used for material matters.

[PCI-FND-STD-04-PR-04](#). The decision record must state the decision, its date, the decision owner, the material inputs considered and, where an AI system contributed, that fact and what it contributed.

[PCI-FND-STD-04-PR-05](#). Each automated workflow carrying material recommendations must be tested at an interval the adopting organisation states, and at least once before it enters use, to confirm that rejection and deferral are exercisable and recorded; the test and its result must be retained.

#### PCI STANDARD

##### PCI-FND-STD-05

### Transparent Assumptions

**1. Normative requirement.** A professional must not issue a forecast, model, appraisal or recommendation unless the document that states its conclusion also states, with equal prominence, the material assumptions on which that conclusion depends.



**2. Purpose.** This standard controls the risk that a conclusion travels further than the assumptions that produced it — that a range becomes a number, a number becomes a commitment, and the conditions on which it all rested stay behind in a working file nobody opens.

**3. Scope.** Governs every professional performing PCI-certified work. Applies to every forecast, model output, appraisal, valuation, estimate, business case and recommendation offered for reliance, on projects, programmes and transactions of any size, in internal and external use alike, including dashboards, slide material and AI-generated narrative. Applies to preparation (state them), review (test that they are stated and complete), recommendation and approval (do not approve a conclusion whose assumptions are not stated).

**4. Defined terms.** D-15 material · D-16 material output · D-11 evidence · D-24 reliance · D-08 decision owner · D-01 adopting organisation. No term is redefined in this standard.

**5. Required actions.** Identify which assumptions are material by the D-15 test. State them where the conclusion is stated. State each material limitation, each material dependency, and the sensitivity of the conclusion to each material assumption. Where a single figure is presented, state the range or confidence basis from which it was selected.

**6. Prohibited actions.** Presenting a target, an aspiration or a negotiating position as a forecast. Confining material assumptions to an appendix, a footnote, a linked file or a separate document. Presenting a list of assumptions so undifferentiated that the material ones cannot be identified from it. Presenting a single-point figure with no stated range or confidence basis. Omitting an assumption because stating it would weaken the case. Restating an assumption as a fact in a later document.

**7. Required evidence.** The issued document itself, showing the assumptions alongside the conclusion; the assumption log recording each material assumption, its basis, its owner and its review date; the sensitivity analysis supporting the statement required by [PCI-FND-STD-05-PR-02](#); the dependency list.

**8. Responsible role.** The issuing professional (D-14) for the document. The decision owner (D-08) for the matter is responsible for not relying on a conclusion whose assumptions are not stated.

**9. Approval authority.** The decision owner may approve a conclusion whose assumptions are stated and whose sensitivities are unfavourable; no role may approve the omission of a material assumption from the document that carries the conclusion.

**10. Independence requirement.** None for stating the assumptions, which the preparer does. Independence is required for the sensitivity analysis where the conclusion supports an irreversible commitment, a payment, an external

report or a decision bearing on the safety of a person, in which case **PCI-FND-STD-03** applies to the sensitivity computation as it does to any other material calculation.

**11. Materiality or threshold.** An assumption is material where a change in it, within the range the professional considers credible and records, would change the conclusion, its direction or its recommendation — or where it meets any other limb of D-15. This test scales: on a small refurbishment two or three assumptions will meet it; on a megaproject the same test may identify twenty, and the sensitivity statement in **-PR-02** bounds the work by requiring an effect statement only for those that meet it.

**12. Exception and waiver.** A time-limited exception is permitted where a conclusion must be issued before a sensitivity analysis can be completed and delay would create a risk to safety or an ongoing financial loss. The decision owner approves it; the justification and the date the analysis will be delivered are recorded; it lasts no longer than ten working days; the compensating control is that the document states, where the conclusion is stated, that sensitivities have not yet been tested and the conclusion must not be relied upon for an irreversible commitment until they are; and the exception is reported to the engagement lead. **No exception is permitted** to stating the material assumptions themselves.

**13. Escalation trigger.** Instruction to remove, soften or relocate a material assumption; discovery that a conclusion already relied upon rested on an assumption that was never stated; discovery that an assumption has failed and the conclusion has not been revisited under **PCI-FND-STD-15**.

**14. AI application.** An AI system may extract candidate assumptions from a model, generate scenarios and stress cases, compute sensitivities, draft the assumption register and flag conclusions whose stated assumptions do not reconcile with the model's inputs.

**15. AI prohibition.** An AI system must not decide which assumptions are material, must not decide what is disclosed and what is withheld, and must not approve a forecast, appraisal or recommendation for issue.

**16. AI verification.** Where an AI system generated the assumption list or the sensitivity statement, the professional must apply **sensitivity analysis** in person to at least each assumption the AI system did *not* identify as material, confirming by recomputation that changing it within the credible range does not change the conclusion — and must record the result. This tests the omission, which is where AI-generated assumption lists fail.

## **17. External reference.**

- **Total Cost Management (TCM) Framework.** Issuing organisation: AACE International. Subject: the documented basis of estimates and

forecasts. Edition checked: not independently verified — no edition asserted. Nature of authority: **category 5 — professional framework**. Verification date: not independently verified (register EXT-064). Applicability limitation: professional guidance from a membership body, binding by its own force on nobody; no requirement here is sourced to it.

- **Financial Modelling Code**. Issuing organisation: ICAEW. Subject: documenting model assumptions and their sensitivities. Edition checked: current at the register's verification date; no edition asserted. Nature of authority: **category 5 — professional framework**. Verification date: 2026-08-03 (register EXT-125). Applicability limitation: not certifiable, not a compliance standard, binding only where adopted.
- **ISO 31000 — Risk management — Guidelines**. Issuing organisation: ISO. Subject: principles and guidelines for managing risk, including uncertainty in decision-making. Edition checked: ISO 31000:2018, second edition; reviewed and confirmed 2023. Nature of authority: **category 3 — international voluntary standard**. Verification date: 2026-08-03 (register EXT-020). Applicability limitation: **guidance, not a certifiable requirements standard** — ISO states so expressly; voluntary unless a regulator or contract adopts it.

**18. Jurisdictional caution.** Securities, prospectus, public-procurement and financial-promotion regimes attach specific liability to forward-looking statements and to the adequacy of the warnings that accompany them, and some require prescribed cautionary wording. Obtain local legal advice before issuing a forecast into a regulated market, a public tender or an investor communication.

**19. Related PCI Standards.** [PCI-FND-STD-02](#) (an unevidenced claim becomes a stated assumption, not a fact) · [PCI-FND-STD-03](#) (verification of the sensitivity computation) · [PCI-FND-STD-06](#) (assumptions drawn from sources) · [PCI-FND-STD-11](#) (escalating a misleading presentation) · [PCI-FND-STD-15](#) (a failed assumption that invalidates issued work).

**20. Related Body of Knowledge content.** PCL-AI Domain 6 — earned value management and forecasting; Domain 4 — performance management, variance analysis and reporting; Domain 13. PML-AI KA 2.4 — kill criteria and honest gates; KA 11.2 — executive communication and reporting. PFL-AI KA 16.1 — forecast automation; Domain 16.

**21. Compliance test.** For a sample of issued forecasts, appraisals, models and recommendations selected by the reviewer: (a) the document stating the conclusion also states the assumptions, in the same document and not by reference to another; (b) **reading only the assumption statement, the**

**reviewer lists the assumptions whose failure would change the conclusion, and that list matches the set for which -PR-02 sensitivities are stated** — a mismatch in either direction is a failure, because it shows either an unstated material assumption or a sensitivity statement that does not describe the assumptions presented; (c) each stated sensitivity expresses the effect on the conclusion of a stated change in the assumption, in the units of the conclusion; (d) each material dependency names the external event, decision, approval or third-party act relied on; and (e) where the conclusion is a single figure, the range or confidence basis is stated. A sampled document fails if any of (a) to (e) is absent or if the test in (b) produces a mismatch. The standard is complied with for the period under review only if no sampled document fails.

**22. Breach indicators.** Assumption sections identical across unrelated documents. A hundred assumptions listed with no indication of which matter. Sensitivities expressed as percentages of an input rather than as an effect on the conclusion. Forecasts that equal the target to the pound. Single-point figures throughout with no range anywhere. Assumptions appearing only in an appendix or a linked model. Conclusions repeated in later documents with the assumptions dropped.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension; withdrawal. Each subject to due process and a right of appeal.

**24. Examination application.** Tested by scenario judgement (a forecast that matches the target exactly), by calculation review (a sensitivity statement that the model does not produce) and by ethical dilemma (instruction to move the assumptions to an appendix before the board pack goes out).

**25. Version and status.** Version 1.0 · approval date: not approved — draft submitted under Charter §5 Stage 11 · effective date: on approval · amendment note: carries forward part of the withdrawn [PCI-LAW-F-07](#).

### **Process requirements under PCI-FND-STD-05**

[PCI-FND-STD-05-PR-01](#). The document stating the conclusion must state each material limitation of the method, the data and the scope on which the conclusion rests.

[PCI-FND-STD-05-PR-02](#). The document must state, for each material assumption, the effect on the conclusion of a stated change in that assumption, expressed in the units of the conclusion.

[PCI-FND-STD-05-PR-03](#). The document must state each material dependency — the external event, decision, approval or third-party act that the conclusion assumes will occur — and by when it is assumed to occur.

**PCI-FND-STD-05-PR-04.** A single-point figure must not be presented as a forecast unless the document also states the range or the confidence basis on which that point was selected.

**PCI-FND-STD-05-PR-05.** Material assumptions must be identified as material within the assumption statement, and must not be presented in a manner that prevents a reader from distinguishing them from immaterial ones.

**PCI-FND-STD-05-PR-06.** Where a conclusion is repeated, summarised or carried into a later document, the material assumptions must be carried with it.

#### PCI STANDARD

#### PCI-FND-STD-06

### Source and Version Integrity

**1. Normative requirement.** A professional must not rely on a source (D-25) for a material purpose unless the working record contains a check, dated on or before the date of reliance, confirming that the source is the version in force for the matter to which it is applied.

**2. Purpose.** This standard controls the risk that work is performed correctly against the wrong instrument — a superseded standard, a previous contract revision, an unsigned draft, a stale rate table, a clause number that moved between editions, or a source that never existed at all.

**3. Scope.** Governs every professional performing PCI-certified work. Applies to every source relied on for a material purpose: external standards, contract documents and their amendments, rate and price tables, regulatory instruments, drawings and specifications, models and their versions, datasets, and internal procedures. Applies on projects, programmes and transactions of any size. Applies to preparation, review, recommendation, approval and assurance.

**4. Defined terms.** D-25 source · D-15 material · D-24 reliance · D-28 working record · D-11 evidence · D-01 adopting organisation. No term is redefined in this standard.

**5. Required actions.** Before reliance, record a check confirming which version of the source is in force for the matter, the date the check was made, and where the version was obtained. Confirm the source is authentic and that it applies to the entity, jurisdiction, period, contract or asset in question. Re-check before each subsequent issue where the instrument's status is known to be changing. Where the applicable version cannot be established, state that limitation wherever the conclusion is used.

**6. Prohibited actions.** Citing a clause, article, edition, effective date or requirement the professional has not read in the source itself. Presenting a

source as current without a dated check. Relying on a summary, a secondary description or an AI system's account of a source in place of the source. Applying an instrument to a period, entity or jurisdiction it does not cover. Carrying a citation forward from a previous deliverable without re-checking it.

**7. Required evidence.** A dated currency check for each source relied on for a material purpose; the retained copy of, or a resolvable reference to, the version relied on; the register of sources where the adopting organisation maintains one; the limitation statement where the version could not be established.

**8. Responsible role.** The professional who relies on the source. Where the adopting organisation maintains a controlled register of standards, contracts or rate tables, its records owner (D-23) is responsible for the register's currency; that does not discharge the relying professional's own check.

**9. Approval authority.** The decision owner (D-08) may approve reliance on a source whose applicable version cannot be established, on the conditions in element 12. No role may approve the citation of a provision nobody has read.

**10. Independence requirement.** Not applicable — a currency check is a factual comparison against the publisher's or issuer's own record, and its result does not depend on who performs it. Where the check result is disputed, [PCI-FND-STD-03](#) applies to the disputed conclusion.

**11. Materiality or threshold.** Attaches to sources relied on for a material purpose (D-15). It does not attach to background reading, orientation or context that no conclusion rests on. A source is always material where it fixes an entitlement, a price, a rate, a tolerance, a duty, a limit or a date used in a material output.

**12. Exception and waiver.** A time-limited exception is permitted where the applicable version cannot be established within the time available — for example where a contract amendment is executed but not yet circulated. The decision owner approves it; the justification records what was attempted and from whom the version was sought; it lasts until the version is obtained or thirty days, whichever is shorter; the compensating controls are that the output states the uncertainty where the conclusion is stated and identifies the version actually used, and that the conclusion is revisited under [PCI-FND-STD-15](#) once the correct version is obtained. **No exception is permitted to [PCI-FND-STD-06-PR-04](#):** an unread provision must never be cited.

**13. Escalation trigger.** Discovery that a material output relied on a superseded, unexecuted or inapplicable version; discovery that a cited clause, article, edition or effective date does not exist; discovery that a controlled register has issued a superseded version as current.

**14. AI application.** An AI system may search for the current version of an instrument, list the differences between two versions the professional supplies,

index a document set by version and date, and flag citations in a draft that do not resolve.

**15. AI prohibition.** An AI system must not be relied upon to establish which version of a source is in force, must not be cited as the authority for a version or an edition, and its output must not be recorded as the currency check required by element 1.

**16. AI verification.** Every version, edition, effective date, clause number and quotation that an AI system supplied must be confirmed by **source tracing** to the publisher's or issuer's own record — the publisher's catalogue, the executed contract, the issuing authority's register — and the confirmation recorded with its date and the place it was obtained. Where the publisher's record cannot be reached, the citation must be reduced to the instrument's name alone.

#### 17. External reference.

- **IFRS 18** *Presentation and Disclosure in Financial Statements* and **IAS 1** *Presentation of Financial Statements*. Issuing organisation: IFRS Foundation / IASB. Subject: presentation and disclosure in general purpose financial statements. Edition checked: IFRS 18 issued April 2024, **effective for annual reporting periods beginning on or after 1 January 2027**, earlier application permitted, and **replacing IAS 1** from that date; IAS 1 in force for periods beginning before it. Nature of authority: **category 2 — authoritative financial-reporting standard**. Verification date: 2026-08-03 (register EXT-003, EXT-004). Applicability limitation: mandatory only for entities applying IFRS Accounting Standards in a jurisdiction that has adopted them; named here as a live, verified example of a version cut-over with a stated effective date, and relied on for no requirement in this standard.
- **FIDIC suite of conditions of contract** — Red Book (*Construction*), Yellow Book (*Plant and Design-Build*), Silver Book (*EPC/Turnkey*). Issuing organisation: FIDIC (International Federation of Consulting Engineers). Subject: standard conditions of contract for works and plant. Edition checked: second editions 2017, reprinted 2022 with amendments; a further set of amendments published November 2022, effective 1 January 2023. Nature of authority: **category 4 — contract framework**. Verification date: 2026-08-03 (register EXT-050). Applicability limitation: **binds only the parties who adopt it, and only through the contract they sign**; it is not legislation. Named here because its clause numbering moved between editions, which is precisely the risk this standard addresses; **no clause number is cited**.
- `../registries/EXTERNAL_AUTHORITIES.md` — Suite External-Reference Register. Issuing organisation: PCI Global. Subject: the



registered identity, edition and category of every external authority named in the PCI corpus, with its verification date. Edition checked: compiled 2026-08-03, revised 2026-08-04. Nature of authority: **category 9 – PCI internal professional standard and instrument**. Verification date: 2026-08-04. Applicability limitation: a PCI internal record; **it is a snapshot, not a warranty**, and the official publication of any instrument always governs over it.

**18. Jurisdictional caution.** Which version of a standard, code or regulation applies to a project is frequently fixed by local law, by the building or permitting regime, or by the contract's governing-law clause, and transitional provisions differ. Obtain local legal advice where the applicable version affects an entitlement, a permit, a certification or a liability.

**19. Related PCI Standards.** [PCI-FND-STD-02](#) (evidence for the claim the source supports) · [PCI-FND-STD-07](#) (the data drawn from the source) · [PCI-FND-STD-12](#) (retaining the version relied on) · [PCI-FND-STD-14](#) (AI-produced citations) · [PCI-FND-STD-15](#) (what to do when the wrong version was used).

**20. Related Body of Knowledge content.** PCL-AI Domain 2 – financial reporting for project controls; Domain 7 – contract and commercial management; Domain 13. PML-AI Domain 14, especially KA 14.1 – data governance and common data environments. PFL-AI Domain 16.

**21. Compliance test.** For a sample of sources relied on in issued material outputs, selected by the reviewer: (a) a dated currency check exists in the working record, on or before the date of reliance, recording the version relied on and where it was obtained; (b) **the version recorded matches the version the reviewer independently obtains from the publisher or issuer on the review date, or the record explains the difference by reference to the period, entity or contract the work concerned**; (c) **every clause, article, edition, effective date and quotation appearing in the issued output is located by the reviewer in the version recorded, at the stated position**; and (d) where the record shows the applicable version could not be established, the limitation appears in the issued output where the conclusion is stated. A sampled source fails if the check is absent or post-dates reliance, if the version cannot be reconciled, if any cited clause, article, edition or date cannot be located in the recorded version, or if a required limitation statement is missing. The standard is complied with for the period under review only if no sampled source fails.

**22. Breach indicators.** Citations carrying clause numbers from a superseded edition. Editions or effective dates stated in prose with no check record. A



standard cited by number with no version anywhere in the file. Contract references to an unexecuted draft. Rate tables with no effective date. Currency checks dated after issue, or in a batch. Identical citation blocks copied between unrelated deliverables.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension; withdrawal. Each subject to due process and a right of appeal.

**24. Examination application.** Tested by evidence selection (which record establishes the applicable version), by scenario judgement (a contract amendment executed mid-assessment) and by AI-verification case (a citation that resolves to no such clause).

**25. Version and status.** Version 1.0 · approval date: not approved — draft submitted under Charter §5 Stage 11 · effective date: on approval · amendment note: new standard; no direct predecessor in the withdrawn set.

#### Process requirements under PCI-FND-STD-06

**PCI-FND-STD-06-PR-01.** The check must confirm that the source is authentic — obtained from the publisher or issuer, or from a channel the adopting organisation has recorded as authoritative — and must record where it was obtained.

**PCI-FND-STD-06-PR-02.** The check must confirm that the source applies to the entity, jurisdiction, period, contract or asset to which it is being applied.

**PCI-FND-STD-06-PR-03.** Where an instrument's status is known to be changing, the professional must record the date the currency was checked and must re-check before each subsequent issue that relies on it.

**PCI-FND-STD-06-PR-04.** A professional must not cite a clause, article, edition, effective date, judicial decision or requirement that they have not read in the source itself.

**PCI-FND-STD-06-PR-05.** Where the applicable version cannot be established, the professional must state that limitation wherever the conclusion is used, must identify the version actually applied, and must not present the source as current.

PCI STANDARD

PCI-FND-STD-07

## Data Lineage

- 1. Normative requirement.** A material output must be traceable in the working record from each source dataset, through every transformation and calculation, to the figure that was reviewed and approved, such that a competent reviewer is able to reproduce that figure from the sources without assistance from the preparer.
- 2. Purpose.** This standard controls the risk that a number cannot be explained – that between the system it came from and the report it appears in, a chain of extracts, adjustments, mappings and manual corrections has become impossible to retrace, so nobody can tell whether it is right, or what changed when it moves.
- 3. Scope.** Governs every professional performing PCI-certified work. Applies to the data underpinning every material output: cost actuals and commitments, progress and schedule data, estimates, financial statements and models, risk registers, and any dataset supplied to or produced by an AI system for a material purpose, on projects, programmes and transactions of any size. Applies to preparation, review (the reviewer performs the reproduction), recommendation, approval and assurance.
- 4. Defined terms.** D-04 competent reviewer · D-15 material · D-16 material output · D-28 working record · D-11 evidence · D-23 records owner · D-01 adopting organisation. No term is redefined in this standard.
- 5. Required actions.** Record, for each source dataset, its system or document of origin, its owner, the extraction date and time, and the extract's identifier or query. Record each transformation so it can be re-performed. Reconcile derived figures to their sources and retain the reconciliation. Label estimated, imputed, modelled and synthetic values wherever they appear. State the limitation where lineage cannot be established.
- 6. Prohibited actions.** Presenting a transformed figure as raw data. Presenting an estimated, imputed, modelled or synthetic value as observed. Adjusting data without recording the adjustment, its basis and its author. Excluding data points from a population without recording the exclusion and its basis. Rebuilding a lineage record after the fact and presenting it as contemporaneous. Supplying data to an AI system from a source outside the lineage the working record shows.
- 7. Required evidence.** The lineage record for each material output; retained extracts or reproducible queries; transformation records sufficient for re-performance; reconciliations of derived figures to their sources; the labelling of estimated, imputed, modelled and synthetic values in the output itself.

**8. Responsible role.** The professional who prepares the material output, for its lineage. The records owner (D-23) designated for a shared system of record, for that system's own extract and change history.

**9. Approval authority.** The decision owner (D-08) for the matter may approve issue where lineage for a part of the data cannot be established, on the conditions in element 12. No role may approve presenting unlabelled synthetic, imputed or modelled values as observed data.

**10. Independence requirement.** The reproduction test in element 21 must be performed by a competent reviewer independent (D-12) of the preparation of the output, because it is the reproduction, not the narrative, that establishes the lineage. Recording the lineage itself is the preparer's task and requires no independence.

**11. Materiality or threshold.** Attaches to data underpinning material outputs (D-15). It does not require field-level lineage for every value in a system; it requires that the **figures presented in the material output** be reproducible from their sources. **Depth is proportionate and technology-neutral:** on a small engagement a single worksheet naming the source file, its extraction date and the adjustments applied satisfies this standard; on a megaproject the same test may need a catalogued data pipeline. The test is reproducibility by a competent reviewer, never the possession of a particular tool.

**12. Exception and waiver.** A time-limited exception is permitted where lineage for part of the data genuinely cannot be established — for example legacy data inherited on transfer of an engagement, or a third-party feed supplied without provenance. The decision owner approves it; the justification records which data, why, and what was attempted. It lasts until the data is replaced or re-derived, and must be reviewed at each reporting period rather than left open. The compensating controls are that the affected figures are identified in the output as resting on data of unestablished lineage, that the limitation travels with the conclusion into every later use, and that the exception is recorded in a register the engagement lead reviews. **No exception is permitted** to labelling estimated, imputed, modelled or synthetic values.

**13. Escalation trigger.** Unexplained change in a system of record; discovery that a material conclusion rests on data of unknown origin; discovery that an AI system or automated pipeline consumed data from outside its authorised sources; a reconciliation that does not resolve and is issued anyway.

**14. AI application.** An AI system may capture lineage metadata, profile data quality, detect anomalies and duplicates, perform reconciliations for human review, and identify figures in a draft that no source supports.

**15. AI prohibition.** An AI system must not certify data as fit for purpose, must not be recorded as the source of data it generated, and its generated, imputed or inferred values must never be presented as observed data.

**16. AI verification.** Where an AI system produced, imputed or transformed data used in a material output, the professional must apply **source tracing** to a sample of the affected values, selected on a stated basis, following each back to the originating system or document, and **reconciliation** of the totals to an independent record; both must be recorded with their scope and result.

#### **17. External reference.**

- **ISO 8000** — *Data quality* (multi-part series). Issuing organisation: ISO. Subject: data-quality management, including provenance and data requirements. Edition checked: ISO 8000-1:2022 *Part 1: Overview*, with further parts issued separately. Nature of authority: **category 3 — international voluntary standard**. Verification date: 2026-08-03 (register EXT-026). Applicability limitation: a series rather than a single document, cited generically; voluntary unless a regulator or contract adopts it; no requirement here is sourced to it.
- **ISO/IEC 25012** — *Software engineering — Software product Quality Requirements and Evaluation (SQuaRE) — Data quality model*. Issuing organisation: ISO/IEC. Subject: a model of data-quality characteristics. Edition checked: ISO/IEC 25012:2008. Nature of authority: **category 3 — international voluntary standard**. Verification date: 2026-08-03 (register EXT-036). Applicability limitation: a long-standing edition within the SQuaRE series — check for supersession before relying on it; voluntary.
- **DAMA-DMBOK: Data Management Body of Knowledge**. Issuing organisation: DAMA International. Subject: a practitioner body of knowledge covering data governance, quality and lineage. Edition checked: second edition (2017); a revised printing exists. Nature of authority: **category 7 — industry guidance** (register category: industry practice). Verification date: 2026-08-03 (register EXT-092). Applicability limitation: a commercially published body of knowledge with no standard-setter's authority; **protected material — never reproduced or structurally mirrored**; named for orientation only.

**18. Jurisdictional caution.** Data-localisation, personal-data, banking-secrecy and financial record-keeping regimes constrain where data may be stored, processed and transferred, and cross-border transfer may require a specific legal basis. Obtain local legal advice before moving

project or personal data across a border or into a third-party service, including an AI service.

**19. Related PCI Standards.** *PCI-FND-STD-02* (evidence for the claim the figure supports) · *PCI-FND-STD-03* (verification of the calculation the lineage feeds) · *PCI-FND-STD-06* (whether the source is the right version) · *PCI-FND-STD-09* (what may be put into which system) · *PCI-FND-STD-12* (keeping the lineage record intact). **Boundary with *PCI-FND-STD-12*:** this standard governs whether a figure can be retraced; *PCI-FND-STD-12* governs whether the records survive, unaltered and attributable.

**20. Related Body of Knowledge content.** PCL-AI Domain 13. PML-AI KA 9.4 — data quality and AI-output quality; Domain 14, especially KA 14.1 — data governance and common data environments. PFL-AI KA 16.1 — financial-data architecture and anomaly detection; Domain 16.

**21. Compliance test.** The reviewer selects figures from issued material outputs and performs a **reproduction test**: using only the working record and without assistance from the preparer, the reviewer traces each sampled figure to its source datasets, re-applies the recorded transformations, and obtains the same figure to the tolerance the record states. For each sampled figure: (a) the reproduction succeeds; (b) each source dataset carries its origin, owner, extraction date and time, and query or identifier; (c) each transformation record states the rule, the parameters, the applying person or process and the date; (d) a retained reconciliation agrees the derived figure to its sources; and (e) any estimated, imputed, modelled or synthetic value in the figure is labelled as such in the output. A sampled figure fails if the reproduction does not succeed, if it succeeds only with an explanation the record does not contain, or if any of (b) to (e) is absent. The standard is complied with for the period under review only if no sampled figure fails.

**22. Breach indicators.** Extracts with no timestamp. Adjustment columns with no basis. Figures in a report that no query reproduces. Reconciliations carrying an unexplained balancing line. Data that changes between two issues of the same report with no change record. Synthetic or imputed values indistinguishable from observed ones. Lineage documentation created on the date of the audit.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension; withdrawal. Each subject to due process and a right of appeal.

**24. Examination application.** Tested by calculation review (a reported figure that the stated sources do not produce), by evidence selection (which

record establishes provenance) and by scenario judgement (a third-party feed arriving with no provenance mid-period).

**25. Version and status.** Version 1.0 · approval date: not approved — draft submitted under Charter §5 Stage 11 · effective date: on approval · amendment note: supersedes [PCI-LAW-F-06](#), withdrawn.

### Process requirements under PCI-FND-STD-07

[PCI-FND-STD-07-PR-01](#). The lineage record must state, for each source dataset, its system or document of origin, its owner, the extraction date and time, and the extract's identifier or query.

[PCI-FND-STD-07-PR-02](#). Each transformation must be recorded so that it can be re-performed — the rule applied, its parameters, the person or process that applied it, and the date.

[PCI-FND-STD-07-PR-03](#). Each derived figure presented in a material output must be reconciled to its sources, and the reconciliation retained with the output.

[PCI-FND-STD-07-PR-04](#). Estimated, imputed, modelled and synthetic values must be labelled as such wherever they appear, and must not be presented as observed data.

[PCI-FND-STD-07-PR-05](#). Where lineage cannot be established for data underpinning a material conclusion, the professional must state that limitation wherever the conclusion is used, and must carry it into every later document that repeats the conclusion.

[PCI-FND-STD-07-PR-06](#). The exclusion of data from a population, and any manual adjustment to extracted data, must be recorded with its basis and its author before the affected output is issued.

## PCI STANDARD

### PCI-FND-STD-08

## Conflict Disclosure

**1. Normative requirement.** A professional must disclose every conflict (D-05) capable of affecting their independent judgement, in writing, to the person or body to whom the professional duty is owed, before taking any act in the matter or, where the conflict arises later, on becoming aware of it.

**2. Purpose.** This standard controls the risk that a professional's judgement is shaped by an interest the people relying on that judgement do not know about — and the related risk that the conflicted person is also the person who decides that the conflict does not matter.

**3. Scope.** Governs every professional performing PCI-certified work. Applies to engagements, appointments, evaluations, procurements, awards, certifications, assessments, approvals and advice, on projects, programmes and transactions of any size, including interests held through a household member, and including an interest in a tool, model, vendor or AI system being selected, priced, assessed or relied upon. Applies to preparation, review, recommendation, approval and assurance: a reviewer's conflict is as disqualifying as a preparer's.

**4. Defined terms.** D-05 conflict · D-12 independent · D-08 decision owner · D-04 competent reviewer · D-09 engagement lead · D-01 adopting organisation. **The word "appropriately" is deliberately not used in this standard.** What counts as managing a conflict is fixed by [PCI-FND-STD-08-PR-01](#), which lists the four safeguards that satisfy it and states that no other arrangement does.

**5. Required actions.** Maintain a current written declaration of interests. Disclose a conflict in writing to the person or body to whom the duty is owed, before acting in the matter. Take no act in the matter until a safeguard under [PR-01](#) is recorded as operating. Refresh the declaration when interests or duties change, and before each decision in a matter where a safeguard is operating.

**6. Prohibited actions.** Concealing an interest. Deciding for oneself that a disclosed conflict needs no safeguard. Approving one's own work, one's own procurement or one's own performance assessment. Evaluating a tool, supplier, claim or AI system in which one holds an interest, without disclosure. Structuring an interest — through a household member, an intermediary or a timing arrangement — so that it falls outside a declaration. Acting in the matter between disclosure and the start of a safeguard.

**7. Required evidence.** The written declaration of interests and its date; the written disclosure for each matter; the record of which safeguard under [PR-01](#) was chosen, by whom, and when it began to operate; the conflict register where the adopting organisation maintains one; recusal and withdrawal records.

**8. Responsible role.** The conflicted professional owns the duty to disclose. The person or body to whom the duty is owed owns the choice of safeguard. The engagement lead (D-09) is responsible for confirming that declarations exist and are current before the engagement's material decisions are taken.

**9. Approval authority.** The person or body to whom the duty is owed decides which safeguard applies and whether the professional may continue in the matter. The conflicted professional has no authority over either question. Where the conflict is held by the most senior available decision owner, the next independent level of the governance arrangement decides.



**10. Independence requirement.** Where safeguard (b) or (d) of -PR-01 is used, the substitute decision owner or the reviewer must be independent (D-12) of the conflicted professional and of the interest. Where no such person exists inside the adopting organisation, a named external reviewer is engaged; where that too is impossible, safeguard (a) — withdrawal — is the only remaining option, and this standard admits no alternative.

**11. Materiality or threshold.** Every conflict within D-05 must be disclosed; there is no threshold below which an interest need not be declared. The adopting organisation sets the value at which a gift, benefit or hospitality becomes a declarable interest, and PCI sets no figure. The **choice of safeguard** is proportionate to the interest and to the significance of the matter; the **duty to disclose** is not.

**12. Exception and waiver.** **No exception is permitted** to the duty to disclose. A conflict is never cured by the professional's own view that it will not affect them, by long-standing practice, by the interest being small, or by the other party's likely indifference. The only permitted variation is in which of the four -PR-01 safeguards is applied, and that choice belongs to the person or body to whom the duty is owed. Where an engagement's terms forbid a safeguard the matter requires, the professional must withdraw from the matter.

**One narrow allowance exists, and it is not an exception to disclosure.** Where withdrawal under safeguard (a) would leave a matter bearing on the safety of a person with nobody able to act, the conflicted professional may act **to the extent necessary to protect that person**, and must then do all of the following: record the conflict, the necessity and each act taken, at the time; disclose in writing to the person or body to whom the duty is owed within one working day; and withdraw as soon as another person is able to act. That body must record its position on the acts taken within five working days, and must decide which -PR-01 safeguard operates from that point. **The allowance does not extend to commercial urgency, a deadline or the inconvenience of finding a substitute**, and it never permits the professional to choose their own safeguard or to keep the conflict off the record.

**13. Escalation trigger.** Discovery of an undisclosed conflict affecting a decision already taken; a conflicted professional resisting recusal or withdrawal; a conflict held by the most senior available decision owner; a party attempting to keep a disclosure off the record.

**14. AI application.** An AI system may screen an engagement's parties, suppliers and participants against a declared-interest register, flag name and entity matches for human assessment, and remind professionals when a declaration is due.

**15. AI prohibition.** An AI system must not determine whether a conflict exists, whether it is capable of affecting judgement, which safeguard applies,



or whether a professional may act. Those judgements belong to the person or body to whom the duty is owed.

**16. AI verification.** Where an AI system screened for conflicts, the professional must apply **sampling on a stated basis** to the matches the system cleared — not only to those it flagged — confirming by name-level comparison against the matter's participant, supplier and counterparty lists that no D-05 relationship was missed, and must record the sample basis and the result.

#### 17. External reference.

- **International Code of Ethics for Professional Accountants (including International Independence Standards).** Issuing organisation: IESBA / IFAC. Subject: fundamental ethical principles, threats to independence and safeguards. Edition checked: current at the register's verification date; no edition asserted. Nature of authority: **category 6 — ethical code.** Verification date: 2026-08-03 (register EXT-127). Applicability limitation: **binding only where a professional body, a regulator or an engagement has adopted it;** a PCI credential holder who is not subject to it is not made subject to it by this standard.
- **Code of Ethics and Professional Conduct.** Issuing organisation: Project Management Institute. Subject: fairness and conflict-of-interest disclosure for delivery practitioners. Edition checked: not independently verified — no edition asserted. Nature of authority: **category 6 — ethical code.** Verification date: not independently verified (register EXT-063). Applicability limitation: binding only on those the Project Management Institute's own terms bind; not regulatory authority.
- **ISO 37001 — Anti-bribery management systems — Requirements with guidance for use.** Issuing organisation: ISO. Subject: requirements for an anti-bribery management system, including conflict-of-interest and gift controls. Edition checked: ISO 37001:2025, superseding ISO 37001:2016. Nature of authority: **category 3 — international voluntary standard.** Verification date: 2026-08-04 (register EXT-133). Applicability limitation: certifiable, and voluntary unless a contract or regulator requires it; **certification is not a legal defence** — it is a third party's opinion about a management system at a point in time. No requirement here is sourced to it.

**18. Jurisdictional caution.** Public-procurement, anti-corruption, fiduciary, insider-dealing and public-official conduct regimes impose specific and sometimes criminal conflict, gift and disclosure duties, and their reach can be extraterritorial. Obtain local legal advice in public-sector,

regulated-finance and cross-border contexts before relying on any safeguard short of withdrawal.

**19. Related PCI Standards.** *PCI-FND-STD-03* (a conflicted person is not independent and cannot verify) · *PCI-FND-STD-04* (a conflicted decision owner must not decide) · *PCI-FND-STD-09* (information barriers) · *PCI-FND-STD-11* (escalating an undisclosed conflict) · *PCI-FND-STD-12* (the disclosure record).

**20. Related Body of Knowledge content.** PCL-AI Domain 13. PML-AI KA 12.4 — ethical leadership; KA 1.4. PFL-AI KA 1.3 — conflicts, independence, fiduciary awareness and responsible AI; Domain 16.

**21. Compliance test.** For a sample of matters selected by the reviewer: (a) a written declaration of interests exists for each professional acting in the matter, dated within the adopting organisation's declaration cycle or, where none is published, within the twelve months preceding their first act; (b) the declaration was current at the date of that first act; (c) where a conflict was declared, the record names which of the four *-PR-01* safeguards was applied, who chose it, and the date it began to operate; (d) the working record shows no act by the professional in the matter between the disclosure and that start date; and (e) **the reviewer compares each declaration against the matter's participant list, supplier and counterparty list, decision record and tool or vendor selection record, and finds no relationship of a kind listed in D-05 that the declaration omits.** A sampled matter fails if any of (a) to (e) is absent or if the comparison in (e) identifies an omitted D-05 relationship. The standard is complied with for the period under review only if no sampled matter fails.

**22. Breach indicators.** Declarations of interest signed years earlier and never refreshed. A safeguard recorded with no start date. The conflicted professional's own name against the choice of safeguard. Approvals of work the approver performed. Tool or vendor evaluations by a person with a holding in the vendor. Recusals recorded after the decision. Declarations that name no interests at all across a whole engagement team.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension; withdrawal. Where an undisclosed conflict affected a PCI assessment, the outcome of that assessment may be subject to examination failure and correction. Each is subject to due process and a right of appeal.

**24. Examination application.** Tested by ethical dilemma (an interest arising mid-engagement), by scenario judgement (which of the four safeguards the

facts permit) and by escalation decision (a conflict held by the only available decision owner).

**25. Version and status.** Version 1.0 · approval date: not approved — draft submitted under Charter §5 Stage 11 · effective date: on approval · amendment note: supersedes [PCI-LAW-F-10](#), withdrawn; the undefined judgement word "appropriately" used in that standard is replaced by the closed list of safeguards in [-PR-01](#). **Stage 9 amendment:** the second red-team pass found that where withdrawal is the only available safeguard and the matter bears on a person's safety, the standard required the only person able to act to stand aside; element 12 now carries a bounded allowance to act to the extent necessary to protect that person, with the disclosure duty untouched and the choice of safeguard still belonging to the person or body to whom the duty is owed.

### Process requirements under PCI-FND-STD-o8

[PCI-FND-STD-08-PR-01](#). A conflict must be managed by one of the following four safeguards, recorded in writing and operating, and by no other arrangement: **(a)** the professional withdraws from the matter; **(b)** the professional is recused from the specific decision and it is taken by a decision owner independent (D-12) of the interest; **(c)** an information barrier prevents the professional's access to the information that creates the conflict, and its operation is recorded; or **(d)** the professional's work in the matter is reviewed by a competent reviewer independent of the interest, and that review is recorded. **No other arrangement satisfies this standard.**

[PCI-FND-STD-08-PR-02](#). The choice of safeguard must be made by the person or body to whom the duty is owed, and never by the conflicted professional.

[PCI-FND-STD-08-PR-03](#). The professional must take no act in the matter between the moment the conflict arises and the moment a [-PR-01](#) safeguard is recorded as operating.

[PCI-FND-STD-08-PR-04](#). The declaration of interests must be refreshed when the professional's interests or duties change, and before each material decision in a matter where a safeguard is operating.

[PCI-FND-STD-08-PR-05](#). Where the conflict is held by the most senior available decision owner, the disclosure must be made to the next independent level of the governance arrangement, or to the party to whom the duty is owed.

## Confidentiality and Approved Technology

**1. Normative requirement.** A professional must not enter protected information (D-21) into a system that is not an authorised system (D-03) for information of that classification and for that purpose, including any AI system.

**2. Purpose.** This standard controls the risk that information entrusted to a professional leaves the boundary it was entrusted within — most commonly now by being pasted, uploaded or connected into a service whose retention, human-review and model-training terms nobody has read.

**3. Scope.** Governs every professional performing PCI-certified work. Applies to protected information in any medium, on any device including a personal one, during and after the engagement or employment in which it was obtained, on projects, programmes and transactions of any size. "Entering" includes typing, pasting, uploading, attaching, transmitting in a prompt, connecting through an integration, plug-in or agent, and permitting a system to read from a location where the information is held. Applies to preparation, review, recommendation, approval and assurance alike.

**4. Defined terms.** D-21 protected information · D-03 authorised system · D-13 information owner · D-02 AI system · D-01 adopting organisation · D-20 promptly. No term is redefined in this standard.

**5. Required actions.** Establish the classification of information before using it, and treat it as protected until the information owner states otherwise. Use only systems on the adopting organisation's register of authorised systems for that classification and purpose. Where no register exists, obtain and record the information owner's written authorisation before use. Record the check of an AI system's retention, human-review and model-training terms before protected information is entered into it.

**6. Prohibited actions.** Entering protected information into an unauthorised system. Assuming a system is authorised because it is widely used, paid for, used by a colleague or not expressly forbidden. Treating paraphrase, partial redaction, name removal, splitting across submissions or reduction to figures as curing the prohibition where the protected content can be reconstructed from context. Using protected information from one engagement for the benefit of another. Retaining protected information after the authorisation under which it was held has ended. Permitting protected information to be used to train, fine-tune or ground a model without the information owner's explicit authorisation.

**7. Required evidence.** The adopting organisation's register of authorised systems and its currency date, or the information owner's written authorisation where none exists; the recorded terms check for each AI system used with protected information; confidentiality undertakings; access and handling

records; incident records where protected information reached an unauthorised system.

**8. Responsible role.** The professional handling the information. The information owner (D-13) decides what may be done with it. Where the adopting organisation designates a role for maintaining the register of authorised systems, that role is responsible for its currency; that does not discharge the handling professional's own obligation.

**9. Approval authority.** The information owner authorises a use, a system or a transfer. No internal role may authorise a use the information owner has not permitted. Where a contract names an approver, that approver's authority is limited to what the contract grants.

**10. Independence requirement.** Not applicable to the individual act of handling, which is not a review. Where the adopting organisation adopts an AI system or external service for processing protected information at organisational scale, the assessment of that system must be performed by a competent reviewer independent (D-12) of the team procuring or sponsoring it, because a procurement sponsor is not independent of the procurement.

**11. Materiality or threshold.** No materiality threshold applies: this standard binds for any quantity of protected information, because the harm does not scale with volume — a single record can identify a person or move a price. The adopting organisation sets the classification scheme and the gift, retention and transfer thresholds within it; PCI sets none.

**12. Exception and waiver. PCI grants no waiver.** The only route by which protected information may enter a system that is not on the register is the information owner's written authorisation recorded in advance under [PCI-FND-STD-09-PR-01](#), which is not an exception to this standard but the mechanism the standard provides. Where an emergency makes advance authorisation impossible and delay would create a risk to the safety of a person, the professional may act to protect that person and must notify the information owner promptly (D-20) and record the act, the necessity and the information involved; that notification is not retrospective authorisation and does not cure a breach that was not necessary to protect a person.

**13. Escalation trigger.** Protected information is discovered in an unauthorised system or location; a request to use entrusted information for a purpose the owner has not authorised; discovery that an authorised system's terms have changed to permit training, human review or retention the owner did not accept; suspected or actual unauthorised disclosure.

**14. AI application.** An AI system that is an authorised system for the classification in question may be used for any professional purpose the information owner has authorised, subject to the rest of this standard set.

**15. AI prohibition.** An AI system must not be treated as confidential by default. Protected information must not be used to train, fine-tune, evaluate or ground any model without the information owner's explicit authorisation. An AI system must not determine whether information is protected, nor whether a system is authorised.

**16. AI verification.** Before protected information is entered into an AI system, the professional must perform a **clause-to-summary comparison** between the system's own published terms — retention period, human review, training use, sub-processing and geography — and the information owner's stated requirements, and must record the comparison, its date and its outcome. A vendor's marketing statement, a summary produced by the AI system itself, or a colleague's assurance does not satisfy this element.

#### 17. External reference.

- **ISO/IEC 27001** — *Information security, cybersecurity and privacy protection — Information security management systems — Requirements*. Issuing organisation: ISO/IEC. Subject: requirements for an information security management system. Edition checked: ISO/IEC 27001:2022, third edition, **plus Amd 1:2024**. Nature of authority: **category 3 — international voluntary standard**. Verification date: 2026-08-03 (register EXT-023). Applicability limitation: certifiable, but voluntary unless a regulator or contract requires it; an organisation's certification is not evidence that any individual act complied with this standard.
- **ISO/IEC 27701** — *Information security, cybersecurity and privacy protection — Privacy information management systems — Requirements and guidance*. Issuing organisation: ISO/IEC. Subject: privacy information management. Edition checked: ISO/IEC 27701:2025, **now a standalone requirements standard**; the withdrawn 2019 edition was an extension to ISO/IEC 27001 and 27002. Nature of authority: **category 3 — international voluntary standard**. Verification date: 2026-08-03 (register EXT-038). Applicability limitation: voluntary unless adopted; describing it as an extension is now incorrect.
- **General Data Protection Regulation** (Regulation (EU) 2016/679). Issuing organisation: European Union. Subject: processing of personal data. Edition checked: not independently verified — no edition, article or requirement asserted. Nature of authority: **category 10 — illustrative practice** (register category: illustrative reference). Verification date: not independently verified (register EXT-101). Applicability limitation: named here **only** as an example of a rights-based data-protection regime whose requirements a professional must confirm loc-

ally; **no requirement in this standard is sourced to it**, and this set describes only the EU AI Act as legislation.

**18. Jurisdictional caution.** Personal-data, banking-secrecy, state-secrecy, export-control, sanctions and sector confidentiality regimes impose specific and sometimes criminal obligations, and several restrict cross-border transfer and require a documented legal basis. Obtain local legal advice on personal-data processing, on cross-border transfer, and before entering any protected information into a service hosted outside the jurisdiction in which it was entrusted. Verify current requirements.

**19. Related PCI Standards.** *PCI-FND-STD-07* (data supplied to a system) · *PCI-FND-STD-08* (information barriers as a conflict safeguard) · *PCI-FND-STD-11* (escalating a disclosure) · *PCI-FND-STD-12* (retention and protection of the records) · *PCI-FND-STD-14* (AI use generally).

**20. Related Body of Knowledge content.** PCL-AI Domain 13. PML-AI Domain 14, especially KA 14.4 — cybersecurity and privacy; KA 1.4. PFL-AI Domain 16, especially KA 16.4 — privacy, cybersecurity, human approval and AI governance.

**21. Compliance test.** For a sample of engagements and of AI or external systems in use, selected by the reviewer: (a) a register of authorised systems exists with a currency date, or, where it does not, a written information-owner authorisation exists for each system used with protected information; (b) each system the professional used with protected information appears on that register for that classification and purpose, at the date of use; (c) for each AI system so used, a recorded terms comparison exists under element 16, dated before first use, covering retention, human review, training use, sub-processing and geography; (d) **the reviewer searches the professional's outbound traffic, prompt history, shared-drive and mailbox records for the period, using identifiers drawn from the engagement's protected information, and finds no instance of that information in a system absent from (b);** and (e) where an incident is recorded, the information owner was notified within the D-20 period. A sample fails if any of (a) to (e) is absent or if the search in (d) returns an instance. The standard is complied with for the period under review only if no sampled engagement or system fails.

**22. Breach indicators.** Prompt histories containing project, party or personal identifiers on systems not on the register. Personal accounts or devices used for engagement work. A register with no currency date, or one last updated before the tools in use were adopted. Terms checks recorded after first use. Documents "anonymised" by deleting names only. Protected information



retained after an engagement closed. Vendor assurances filed in place of the vendor's own terms.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension; withdrawal. Each subject to due process and a right of appeal.

**24. Examination application.** Tested by scenario judgement (an urgent request to summarise a confidential claim in a public AI service), by evidence selection (which record establishes that a system was authorised) and by ethical dilemma (a client's data would resolve another client's problem).

**25. Version and status.** Version 1.0 · approval date: not approved — draft submitted under Charter §5 Stage 11 · effective date: on approval · amendment note: supersedes [PCI-LAW-F-09](#), withdrawn.

### Process requirements under PCI-FND-STD-09

[PCI-FND-STD-09-PR-01](#). Where the adopting organisation maintains no register of authorised systems, the professional must obtain and record the information owner's written authorisation before entering protected information into any system that is not under the information owner's own control.

[PCI-FND-STD-09-PR-02](#). Before protected information is entered into an AI system, the professional must record a comparison of that system's published retention, human-review, training-use, sub-processing and geography terms against the information owner's requirements.

[PCI-FND-STD-09-PR-03](#). Information whose classification the professional does not know must be treated as protected information until the information owner states otherwise.

[PCI-FND-STD-09-PR-04](#). A professional must not use protected information obtained in one engagement for the benefit of another, and must not retain it after the authorisation under which it was held ends.

[PCI-FND-STD-09-PR-05](#). A professional who becomes aware that protected information has entered an unauthorised system must notify the information owner promptly (D-20) and must escalate under [PCI-FND-STD-11](#).

## PCI STANDARD

### PCI-FND-STD-10

## Competence and Limitation

**1. Normative requirement.** A professional must not accept, continue or issue work that requires competence they do not hold.



**2. Purpose.** This standard controls the risk of confident incompetence — and its modern form, in which fluent AI output supplies the vocabulary of a specialism the professional has not acquired, and the reader cannot tell the difference.

**3. Scope.** Governs every professional performing PCI-certified work. Applies to the acceptance, performance, delegation and issue of work, and to advice given in a professional capacity, on projects, programmes and transactions of any size and in any jurisdiction. Applies to preparation, review, recommendation, approval and assurance: reviewing work outside one's competence breaches this standard as surely as preparing it.

**4. Defined terms.** D-04 competent reviewer · D-09 engagement lead · D-19 professional · D-15 material · D-02 AI system · D-20 promptly · D-01 adopting organisation. No term is redefined in this standard.

**5. Required actions.** Before accepting a material engagement or role, record an assessment naming the competences the work requires and the evidence that each is held by a named individual on the engagement. Refer matters requiring specialist judgement outside that competence to a suitably qualified adviser, in writing, and record the referral and its answer. Disclose a competence gap discovered mid-engagement promptly. State the limits of one's competence in writing where a recipient could otherwise rely on them beyond it.

**6. Prohibited actions.** Holding oneself out as qualified in a specialism one does not hold. Issuing legal, tax, accounting, actuarial, insurance, regulatory or engineering conclusions on the strength of an AI system's output, a template or a reference work. Recording access to an AI system or a knowledge base as evidence of competence. Accepting work knowing it cannot be competently performed. Suppressing a competence limitation to win or keep an engagement. Continuing to issue affected outputs after a competence gap is discovered and before it is closed.

**7. Required evidence.** The competence assessment for each material engagement or role, naming competences and the evidence each is held; referral letters or instructions to qualified advisers and their responses; written statements of competence limits given to clients or employers; continuing professional development records maintained under PCI's certification requirements.

**8. Responsible role.** The professional accepting or performing the work. The engagement lead (D-09) for matching work to competence within the engagement, and for the assessment required by [PCI-FND-STD-10-PR-01](#).

**9. Approval authority.** The engagement lead approves acceptance of an engagement against the recorded competence assessment. No role may

approve the issue of an output requiring competence nobody on the engagement holds; the route is referral under -PR-02 or declining the work.

**10. Independence requirement.** The adviser to whom a specialist matter is referred must be independent (D-12) of the outcome — in particular, must not be remunerated on the conclusion reached. Where work proceeds near the edge of the professional's competence with no specialist involved, the review required by PCI-FND-STD-03 must be performed by a competent reviewer (D-04) in that subject matter, not by a colleague of equal unfamiliarity.

**11. Materiality or threshold.** The recorded assessment in -PR-01 attaches to material engagements and roles (D-15). The prohibition in element 1 has no threshold: work requiring absent competence must not be issued at any value. The adopting organisation may define competence frameworks and evidence standards; PCI requires only that the evidence named exists and relates to the competence claimed.

**12. Exception and waiver.** A time-limited exception is permitted where competence is being acquired under supervision. The engagement lead approves it; the justification names the supervising individual, who must be a competent reviewer (D-04) in the subject matter and must review every affected material output before issue; it lasts for the period recorded and must be re-approved rather than extended by silence; the compensating control is that the supervisor's review is recorded on each output; and the arrangement is reported to the client or employer where they would otherwise believe the work was performed by a competent professional unsupervised. **No exception is permitted** where a jurisdiction reserves the activity to a licensed person: in that case the work must be referred.

**13. Escalation trigger.** Discovery mid-engagement that the work requires competence not present; pressure to opine on a legal, tax, regulatory or safety question without qualified advice; discovery that an output already issued required competence nobody held; a client declining a necessary referral.

**14. AI application.** An AI system may support orientation in an unfamiliar area, summarise a field for context, suggest the questions to put to a qualified adviser, and help identify which specialism a matter requires.

**15. AI prohibition.** An AI system must not substitute for a qualified adviser where specialist judgement is required, must not be presented as the source of specialist advice, and access to it must not be recorded as evidence of competence.

**16. AI verification.** Where an AI system assisted work at or beyond the edge of the professional's competence, the verification required by PCI-FND-STD-03 must be performed by **named expert judgement** — a named competent reviewer in that subject matter — and the record must name that individual and the competence evidence relied on. Verification by the AI

system's user does not satisfy this element, because the user is by definition unable to detect the errors the specialism exists to detect.

#### 17. External reference.

- **ISO/IEC 17024** — *Conformity assessment — General requirements for bodies operating certification of persons*. Issuing organisation: ISO/IEC. Subject: competence-based certification of individuals, including the definition and evidencing of competence. Edition checked: ISO/IEC 17024:2026, superseding the 2012 edition; the new edition adds expectations on AI-based assessment tools, human oversight of AI-generated outcomes and competence in AI use. Nature of authority: **category 3 — international voluntary standard**. Verification date: 2026-08-03 (register EXT-022). Applicability limitation: addressed to certification bodies, not to individual credential holders; voluntary unless adopted by regulation or contract. Verify current requirements.
- **Code of Ethics and Professional Conduct**. Issuing organisation: Project Management Institute. Subject: honest representation of qualifications and competence. Edition checked: not independently verified — no edition asserted. Nature of authority: **category 6 — ethical code**. Verification date: not independently verified (register EXT-063). Applicability limitation: binding only on those its own terms bind; not regulatory authority.
- **International Code of Ethics for Professional Accountants (including International Independence Standards)**. Issuing organisation: IESBA / IFAC. Subject: the professional-competence and due-care principle. Edition checked: current at the register's verification date; no edition asserted. Nature of authority: **category 6 — ethical code**. Verification date: 2026-08-03 (register EXT-127). Applicability limitation: binding only where a body, regulator or engagement has adopted it.

**18. Jurisdictional caution.** Reserved-activity regimes — legal advice, statutory audit, tax practice, actuarial certification, engineering sign-off, insurance mediation — differ by jurisdiction, and performing a reserved activity without the licence can be an offence irrespective of actual competence. Obtain local advice on what only licensed persons may do before offering any adjacent service, and before signing anything a local regime treats as a reserved certificate.

**19. Related PCI Standards.** [PCI-FND-STD-01](#) (accountability for what is issued) · [PCI-FND-STD-03](#) (who is competent to verify) · [PCI-FND-STD-04](#)

(competence of the decision owner) · [PCI-FND-STD-11](#) (escalating a competence gap) · [PCI-FND-STD-14](#) (AI fluency is not competence).

**20. Related Body of Knowledge content.** PCL-AI Domain 13. PML-AI KA 1.2 — professional standards of care; KA 1.4. PFL-AI KA 1.3 — fiduciary and professional obligations; Domain 12's standing referral of jurisdiction-specific contract matters to qualified counsel; Domain 16.

**21. Compliance test.** For a sample of engagements and issued outputs selected by the reviewer: (a) a competence assessment exists, dated before acceptance, naming the competences the work requires; (b) for each named competence, the assessment identifies a named individual and the evidence that individual holds it — a qualification, an assessed competence record, or documented comparable experience — **and the reviewer confirms that the named evidence exists and relates to the competence claimed**; (c) no listed evidence consists of access to a tool, a knowledge base or an AI system; (d) for each matter in the engagement that required legal, tax, accounting, actuarial, insurance, regulatory or engineering judgement, a written referral and a response from a qualified adviser exist, or a record shows the matter was declined; and (e) where a supervised-acquisition exception was used, the supervisor's review is recorded on each affected output. A sampled engagement fails if any of (a) to (e) is absent or if the evidence in (b) cannot be located or does not relate to the competence claimed. The standard is complied with for the period under review only if no sampled engagement fails.

**22. Breach indicators.** Competence assessments listing tools rather than people. The same individual named as competent in unrelated specialisms across many engagements. Legal or tax conclusions in a controls or finance deliverable with no adviser named. Referral letters with no response on file. Supervision recorded at acceptance and never again. Scope creep into a specialism with no reassessment. Outputs whose vocabulary exceeds anything in the engagement's competence record.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension; withdrawal. Each subject to due process and a right of appeal.

**24. Examination application.** Tested by scenario judgement (a controls engagement that turns on a jurisdictional tax question), by escalation decision (a gap found after issue) and by ethical dilemma (a client who will not pay for a referral).

**25. Version and status.** Version 1.0 · approval date: not approved — draft submitted under Charter §5 Stage 11 · effective date: on approval · amendment note: supersedes [PCI-LAW-F-08](#), withdrawn.

## Process requirements under PCI-FND-STD-10

**PCI-FND-STD-10-PR-01.** Before accepting a material engagement or role, the professional must record an assessment naming the competences the work requires and, for each, the named individual who holds it and the evidence that they do.

**PCI-FND-STD-10-PR-02.** A matter requiring legal, tax, accounting, actuarial, engineering, insurance, regulatory or other specialist judgement outside the professional's competence must be referred in writing to a suitably qualified adviser, and the referral and its answer must be recorded.

**PCI-FND-STD-10-PR-03.** A professional must not present the output of an AI system, a template or a reference work as specialist advice, and access to such tools must not be recorded as evidence of competence under -PR-01.

**PCI-FND-STD-10-PR-04.** Where a competence gap is discovered after work has begun, the professional must disclose it promptly (D-20) to the engagement lead and to the client or employer, and must not issue the affected output until -PR-02 is satisfied or a competent individual takes the work over.

**PCI-FND-STD-10-PR-05.** The professional must state the limits of their competence in writing wherever a recipient could otherwise rely on them for a matter outside it.

## PCI STANDARD

### PCI-FND-STD-11

## Escalation of Material Misstatement

**1. Normative requirement.** A professional who knows or suspects that a material error, a material omission, an unsupported assumption or a misleading presentation exists in work that has been issued or is about to be issued must escalate it promptly (D-20) to the accountable governance level.

**2. Purpose.** This standard controls the risk that a known problem stays where it was found — that silence is treated as a neutral option, and the matter compounds while everyone who could have acted assumes someone else has.

**3. Scope.** Governs every professional performing PCI-certified work. Applies whether the professional caused the matter, discovered it, or was merely told of it; whether the work is their own or another's; and whether the matter concerns their own organisation, a client, a counterparty or a supplier. Applies on projects, programmes and transactions of any size. Applies to preparation, review, recommendation, approval and assurance, and **survives an instruction to stay silent**.

**4. Defined terms.** D-10 escalation threshold · D-20 promptly · D-15 material · D-08 decision owner · D-12 independent · D-27 working day · D-01 adopting organisation. No term is redefined in this standard.

**5. Required actions.** Know the escalation route for one's role before it is needed. On reaching the escalation threshold (D-10), raise the matter promptly to the level accountable for remedying it, with the evidence held at that moment. Record what was raised, to whom, when, with what evidence and with what response. Escalate further where the response does not remedy the matter within the applicable period.

**6. Prohibited actions.** Delaying escalation until the facts are complete. Reclassifying, downgrading, splitting or re-timing a matter so that it falls below a governance threshold. Escalating into a channel the professional knows to be captive to the matter. Escalating orally with no record. Participating in the suppression of another person's escalation. Retaliating against, or assisting retaliation against, a good-faith escalator.

**7. Required evidence.** The escalation record — what, to whom, when, with what evidence, and the response; the underlying evidence preserved under [PCI-FND-STD-12](#); the onward escalation record where the first response did not remedy the matter; the governance arrangement identifying the route.

**8. Responsible role.** The professional who knows or suspects. The decision owner (D-08) at the level receiving the escalation owns the remedy; receiving an escalation does not transfer the duty to escalate onward if it is not remedied.

**9. Approval authority.** No approval is required to escalate, and none may be required by any arrangement. The decision owner receiving the matter decides the remedy. Only an independent function may close an escalation the escalating professional still maintains.

**10. Independence requirement.** Where an escalation is closed without remedy and the escalating professional maintains the concern, the closure must be reviewed by a function independent (D-12) of the matter and of the person who closed it — internal audit, an audit or risk committee, the client, or PCI's conduct process where a PCI rule is engaged.

**11. Materiality or threshold.** The escalation threshold is D-10, which is reached at the earliest of six stated triggers, one of which is the adopting organisation's own published criteria. This design means the standard works where an organisation has a mature threshold framework and where it has none: on a small engagement the trigger will usually be "it would change a decision", which needs no framework at all. The adopting organisation may lower the threshold; it must not raise it above D-10.

**12. Exception and waiver. No exception is permitted.** The duty to escalate cannot be waived by an employer, a client, a contract, a confidentiality

undertaking or an instruction. Where a confidentiality obligation appears to conflict with the duty, the professional escalates within the boundary the obligation permits — to the party who owns the information, or to an independent internal function — and obtains legal advice before disclosing outside it. The duty to escalate internally is never suspended by that advice.

**13. Escalation trigger.** The escalation threshold (D-10). In addition, and independently: any attempt to penalise, deter or silence a good-faith escalation is itself a matter that must be escalated under this standard.

**14. AI application.** An AI system may detect conditions that meet the escalation threshold — anomalies, tolerance breaches, reconciliation failures, integrity flags — route them to the accountable level, and draft the escalation summary for the professional to check and send.

**15. AI prohibition.** An AI system must not decide that a matter need not be escalated, must not close an escalation, and must not be configured to filter, batch or suppress alerts in a way that prevents a matter meeting the escalation threshold from reaching a person.

**16. AI verification.** Where an AI system performs escalation-worthy detection, the professional must apply **sampling on a stated basis** to the items the system did *not* raise, at an interval the adopting organisation states, confirming by inspection that none met the D-10 threshold, and must record the sample basis, the period covered and the result. Testing only what the system raised tests the wrong population.

#### **17. External reference.**

- **ISO 31000 — Risk management — Guidelines.** Issuing organisation: ISO. Subject: risk communication, monitoring and review. Edition checked: ISO 31000:2018, second edition; reviewed and confirmed 2023. Nature of authority: **category 3 — international voluntary standard.** Verification date: 2026-08-03 (register EXT-020). Applicability limitation: **guidance, not a certifiable requirements standard;** voluntary unless adopted by regulation or contract.
- **FIDIC suite of conditions of contract.** Issuing organisation: FIDIC. Subject: contractual notice mechanisms, under which a party's failure to give timely notice can extinguish an entitlement. Edition checked: second editions 2017, reprinted 2022 with amendments; further amendments published November 2022, effective 1 January 2023. Nature of authority: **category 4 — contract framework.** Verification date: 2026-08-03 (register EXT-050). Applicability limitation: binds only the parties who adopt it, and only through the contract they sign; **no clause number is cited;** a contractual notice obligation is separate from, and additional to, this standard.



• **A Guide to the Project Management Body of Knowledge (PMBOK Guide).** Issuing organisation: Project Management Institute. Subject: governance, issue management and escalation practice. Edition checked: no edition asserted — the eighth edition is current, and citing without an edition remains correct across that change. Nature of authority: **category 5 — professional framework.** Verification date: 2026-08-03 (register EXT-060). Applicability limitation: professional guidance, not regulatory authority; no requirement here is sourced to it.

**18. Jurisdictional caution.** Whistleblower-protection regimes, mandatory-reporting duties and the consequences of external disclosure differ sharply between jurisdictions, and some matters must be reported to a regulator or authority within a fixed period. Obtain local legal advice before any disclosure outside the organisation, and before relying on any protection from detriment.

**19. Related PCI Standards.** [PCI-FND-STD-02](#) (an unevidenced claim) · [PCI-FND-STD-05](#) (a misleading presentation) · [PCI-FND-STD-12](#) (preserving the evidence) · [PCI-FND-STD-13](#) (an override that was not authorised) · [PCI-FND-STD-15](#) (correcting what was escalated). **Boundary with [PCI-FND-STD-15](#):** this standard governs raising the matter internally; [PCI-FND-STD-15](#) governs telling the people relying on the work and putting it right. Both can be engaged by one discovery, and satisfying one does not satisfy the other.

**20. Related Body of Knowledge content.** PCL-AI Domain 4 — performance management, variance analysis and reporting; Domain 13. PML-AI KA 3.3 — assurance, gates and escalation design; KA 1.4. PFL-AI Domain 16.

**21. Compliance test.** For a sample of matters that met the D-10 threshold in the period — identified by the reviewer from variance reports, reconciliation exceptions, audit findings, incident logs and correction records rather than from the escalation log itself: (a) an escalation record exists for each; (b) its date is within the D-20 period measured from the earliest evidence that the professional knew or suspected, which the reviewer establishes from correspondence and system records, not from the escalation record's own account; (c) the recipient named held authority under the governance arrangement to remedy the matter; (d) the record states what was raised and the evidence provided; (e) where the response did not remedy the matter within the applicable period, an onward escalation to the next level is recorded; and (f) where the escalation was closed and the professional maintained the concern, an independent review of the closure is recorded. A sampled matter fails if any of (a) to (f) is absent. **The test starts from the population of matters, not**



**from the population of escalations** — a clean escalation log with unescalated matters behind it is a failure, not a pass.

**22. Breach indicators.** A period's escalation log containing fewer entries than its exception, variance and correction records imply. Escalations dated after the matter was resolved by other means. Matters recorded just below a governance threshold in unusual numbers. Escalations addressed to the person the matter concerns. Verbal escalations referenced but never recorded. Departures of good-faith escalators following escalation. Alert configurations that batch or suppress the conditions the threshold describes.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension; withdrawal. Retaliation against a good-faith escalator is itself a matter for ethics review and certification investigation. Each is subject to due process and a right of appeal.

**24. Examination application.** Tested by escalation decision (when the threshold is reached and to whom the matter goes), by ethical dilemma (an instruction to hold the matter until after a board meeting) and by scenario judgement (an escalation route that includes the person responsible).

**25. Version and status.** Version 1.0 · approval date: not approved — draft submitted under Charter §5 Stage 11 · effective date: on approval · amendment note: supersedes [PCI-LAW-F-11](#), withdrawn, and carries forward part of [PCI-LAW-F-07](#).

### Process requirements under PCI-FND-STD-11

[PCI-FND-STD-11-PR-01](#). The escalation must be recorded, stating what was raised, to whom, when, with what evidence and with what response.

[PCI-FND-STD-11-PR-02](#). Where the response does not remedy the matter within the period the adopting organisation's procedure states, or within **ten working days** where it states none, the professional must escalate to the next independent level of the governance arrangement.

[PCI-FND-STD-11-PR-03](#). A professional must not participate in suppressing, delaying, downgrading, splitting, re-timing or reclassifying an escalation in order to keep a matter below a governance threshold.

[PCI-FND-STD-11-PR-04](#). Where the escalation route is itself implicated in the matter, the professional must escalate to a function independent of it — internal audit, an audit or risk committee, the client, or PCI's conduct process where a PCI rule is engaged.

[PCI-FND-STD-11-PR-05](#). A professional must not retaliate against, or assist retaliation against, a person who escalates in good faith.

**PCI-FND-STD-11-PR-06.** A professional must escalate on knowledge or suspicion, and must not wait until the matter is confirmed; where the facts are incomplete at the D-20 deadline, the professional must escalate what is known and complete the account afterwards.

## PCI STANDARD

### PCI-FND-STD-12

## Record Integrity

**1. Normative requirement.** A professional must keep the records (D-22) evidencing their material outputs and decisions in a manner that makes every record attributable to its author, fixes the time it was created, and makes any subsequent alteration or deletion detectable and attributable.

**2. Purpose.** This standard controls the risk that the account of what happened is written afterwards — that records are silently amended, backdated, replaced or lost, so that the work can no longer be reviewed, defended or learned from.

**3. Scope.** Governs every professional performing PCI-certified work. Applies to all records generated or received in that work, in any medium, including model versions, prompts and responses exchanged with an AI system for a material purpose, and records held in personal tools that form part of the professional record. Applies on projects, programmes and transactions of any size. Applies to preparation, review, recommendation, approval and assurance.

**4. Defined terms.** D-22 record · D-06 contemporaneous · D-23 records owner · D-16 material output · D-28 working record · D-01 adopting organisation · D-27 working day. No term is redefined in this standard.

**5. Required actions.** Create the record at the time of the act it evidences. Keep records where alteration is detectable and attributable and where prior versions survive. Retain records for the longest of the applicable periods. Suspend disposal immediately when a dispute, audit, investigation or PCI conduct process is under way or foreseeable. Record, where a record is created later than the act, both dates.

**6. Prohibited actions.** Presenting a record as contemporaneous when it is not. Altering or deleting a record outside an authorised process that retains the prior version and the identity of the person who changed it. Maintaining an official record that differs from the real basis of the decision. Conducting professional work on material matters through channels that create no retained record, for the purpose of avoiding one. Disposing of records under a hold. Disposing of the verification records for outputs still in reliance.

**7. Required evidence.** The records themselves, retrievable in a readable form; the version or change history showing author, time and alteration; the

applicable retention schedule and evidence of its application; disposal authorisations and logs; hold notices and evidence they were honoured; the periodic retrieval test required by [PCI-FND-STD-12-PR-07](#).

**8. Responsible role.** The professional for the records they control. The records owner (D-23) for a system of record. Where no records owner is designated, the issuing professional is the records owner for their own records.

**9. Approval authority.** The records owner authorises disposal under the applicable schedule. A hold may be imposed by any professional who becomes aware that a dispute, audit, investigation or conduct process is under way or foreseeable, and may be lifted only by the records owner on recorded advice that the matter is closed.

**10. Independence requirement.** The retrieval and integrity test in element 21 must be performed by a person independent (D-12) of the records' authors, because a person testing whether their own records can be retrieved and reconciled knows where to look. Creating and keeping the records requires no independence.

**11. Materiality or threshold.** Attaches to records evidencing material outputs and decisions (D-15). **The requirement is technology-neutral and proportionate:** it is satisfied by any arrangement in which alteration is detectable and attributable and prior versions survive — a versioned document store, a system with an audit trail, a dated document issued to a counterparty, or an email of record. **It does not require any particular product, platform or expenditure,** and a sole practitioner working in versioned files with dated issues complies. Retention periods are set by applicable law, the contract, the adopting organisation's schedule and PCI's certification rules, and the longest governs; PCI sets no free-standing period of its own.

**12. Exception and waiver.** A time-limited exception is permitted where a system in use cannot record alteration history and cannot be replaced within the engagement's timescale. The records owner approves it; the justification names the system and what was attempted; it lasts no longer than the engagement or twelve months, whichever is shorter, and must be re-approved rather than rolled forward. The compensating controls are that material outputs from that system are issued as fixed, dated documents to a recipient outside the professional's control, and that the working record retains the input data alongside each issue. **No exception is permitted to -PR-02:** a record must never be presented as contemporaneous when it is not, and no circumstance justifies it.

**13. Escalation trigger.** Discovery of unauthorised destruction, alteration or backdating; an instruction to delete or amend records connected to a

foreseeable dispute, audit or investigation; discovery that a system of record overwrites history silently; in-period records found to be irretrievable.

**14. AI application.** An AI system may classify records, apply a retention schedule, index and retrieve, maintain a disposal log, and identify records that appear to be missing from a set.

**15. AI prohibition.** An AI system must not dispose of, alter or destroy a record without a recorded human decision under [PCI-FND-STD-04](#), must not generate a record purporting to be contemporaneous, and must not be recorded as the author of a record a person authored.

**16. AI verification.** Where an AI system classifies records for retention or disposal, the professional must apply **sampling on a stated basis** to the records the system marked for disposal, before disposal occurs, confirming by inspection against the retention schedule and any hold that each is genuinely due, and must record the sample basis and the result. Disposal must not proceed on the classification alone.

#### **17. External reference.**

- **ISO 15489-1** — *Information and documentation — Records management — Part 1: Concepts and principles*. Issuing organisation: ISO. Subject: creating, capturing and managing records. Edition checked: ISO 15489-1:2016, revised from the 2001 edition. Nature of authority: **category 3 — international voluntary standard**. Verification date: 2026-08-03 (register EXT-025). Applicability limitation: voluntary unless a regulator or contract adopts it; no requirement here is sourced to it.
- **ISO/IEC 27001** — *Information security, cybersecurity and privacy protection — Information security management systems — Requirements*. Issuing organisation: ISO/IEC. Subject: protecting the integrity and availability of retained information. Edition checked: ISO/IEC 27001:2022, third edition, plus Amd 1:2024. Nature of authority: **category 3 — international voluntary standard**. Verification date: 2026-08-03 (register EXT-023). Applicability limitation: certifiable but voluntary; certification of a management system is not evidence that any particular record is intact.
- **ISO 19650 series** — *Organization and digitization of information about buildings and civil engineering works, including building information modelling (BIM) — Information management using building information modelling*. Issuing organisation: ISO. Subject: information management through a common data environment. Edition checked: multi-part — 19650-1:2018, -2:2018, -4:2022, -5:2020; cited as a series. Nature of authority: **category 3 — international voluntary standard**. Verification date: 2026-08-03 (register EXT-039). Applicability limitation:

voluntary unless imported by a contract or a client's information requirements; relevant to built-environment work and not to every engagement.

**18. Jurisdictional caution.** Statutory retention periods for tax, employment, financial, safety and personal-data records differ by jurisdiction and can conflict across borders, and litigation-hold and disclosure obligations attach at different moments in different legal systems. Obtain local legal advice before setting, shortening or applying a retention period, and before destroying, redacting or restructuring anything connected to a live or foreseeable dispute.

**19. Related PCI Standards.** [PCI-FND-STD-02](#) (the evidence that must survive) · [PCI-FND-STD-07](#) (the lineage record) · [PCI-FND-STD-09](#) (protecting the records' contents) · [PCI-FND-STD-11](#) (preserving what was escalated) · [PCI-FND-STD-13](#) (the override record) · [PCI-FND-STD-15](#) (retaining the superseded version).

**20. Related Body of Knowledge content.** PCL-AI Domain 13. PML-AI KA 3.3 — auditability and the decision record; KA 4.4 — the change board and the decision log; KA 16.4 — responsible archive and model or data retention. PFL-AI KA 13.2 — model audit; Domain 16.

**21. Compliance test.** The reviewer performs a **retrieval and integrity test** on a sample of records selected by reference to acts recorded elsewhere — decisions, issues, approvals and escalations — and not from a list the professional supplies. For each sampled record: (a) it is retrieved intact and in readable form; (b) it identifies its author and the time of creation; (c) where it has been altered, the prior version and the identity of the person who altered it are both retrievable; (d) its creation time is consistent with the act it evidences, and where it was created later, both dates are recorded; and (e) where the record falls under a hold, no disposal is evidenced. In addition, the reviewer tests **one record known to have been amended** and confirms the amendment history reconstructs the earlier state. A sampled record fails if any of (a) to (e) is absent, or if an amendment cannot be reconstructed. The standard is complied with for the period under review only if no sampled record fails.

**22. Breach indicators.** Documents whose stated date precedes their file metadata. Records whose author is a shared or generic account. Approval entries created in bulk after period end. Systems in which the current version is the only version. Material discussions conducted in ephemeral channels while formal records stay thin. Disposal logs that do not reconcile to the

retention schedule. Records that cannot be produced when an audit asks for them but were cited when the work was issued.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension; withdrawal. Where records were destroyed or altered to defeat review, PCI's quality and conduct processes may draw an adverse inference. Each is subject to due process and a right of appeal.

**24. Examination application.** Tested by evidence selection (which record set demonstrates an audit trail), by scenario judgement (an instruction to tidy the file before an audit) and by escalation decision (a system found to overwrite history).

**25. Version and status.** Version 1.0 · approval date: not approved — draft submitted under Charter §5 Stage 11 · effective date: on approval · amendment note: supersedes [PCI-LAW-F-12](#), withdrawn, and carries forward part of [PCI-LAW-F-05](#).

### Process requirements under PCI-FND-STD-12

[PCI-FND-STD-12-PR-01](#). A record must be created at the time of the act it evidences, or as soon afterwards as establishing the facts permits; a record created later must state both the date of creation and the date of the act.

[PCI-FND-STD-12-PR-02](#). A record must not be presented as contemporaneous (D-06) when it is not.

[PCI-FND-STD-12-PR-03](#). A record must remain retrievable in a readable form for the longest of the periods required by applicable law, the contract, the adopting organisation's retention schedule and PCI's certification rules.

[PCI-FND-STD-12-PR-04](#). A record must not be altered or deleted except through an authorised process that retains the prior version and the identity of the person who made the change.

[PCI-FND-STD-12-PR-05](#). Disposal must follow the authorised retention schedule, must be logged, and must be suspended immediately when a dispute, audit, investigation or PCI conduct process is under way or foreseeable.

[PCI-FND-STD-12-PR-06](#). Professional work on material matters must not be conducted through channels that create no retained record for the purpose of avoiding one.

[PCI-FND-STD-12-PR-07](#). The retrievability of in-period records must be tested at an interval the adopting organisation states, and at least once in each engagement, and the test and its result retained.

## PCI STANDARD

## PCI-FND-STD-13

## No Silent Override

**1. Normative requirement.** A control result (D-07) bearing on a material matter must not be overridden (D-17) unless a decision owner (D-08) authorised to do so has recorded the decision before the overridden output is relied upon.

**2. Purpose.** This standard controls the risk that the checks a system was built around stop functioning without anyone deciding that they stop — that a warning is dismissed, a tolerance widened or a validation switched off, and the organisation continues to believe it is protected by a control that no longer operates.

**3. Scope.** Governs every professional performing PCI-certified work. Applies to every control result on a material matter: validation failures, tolerance and limit breaches, reconciliation differences, failed checks, approval-gate rejections, model diagnostics, quality-assurance findings, safety interlocks and automated alerts, whether the control is systemic, automated or performed by a person. Applies on projects, programmes and transactions of any size. Applies to preparation, review, recommendation, approval, assurance and to the configuration of the controls themselves.

**4. Defined terms.** D-07 control result · D-17 override · D-08 decision owner · D-15 material · D-24 reliance · D-01 adopting organisation. No term is redefined in this standard.

**5. Required actions.** Record, before reliance, the control overridden, the reason, the evidence considered, the compensating action taken, the decision owner and the date. Time-limit every override and record its expiry. Review standing overrides at a stated interval. Review overrides in aggregate and escalate a control that is repeatedly overridden.

**6. Prohibited actions.** Proceeding past a control result on a material matter with no recorded decision. Disabling, re-parameterising, re-scoping or re-timing a control so that it does not produce the result that would otherwise require an override. Re-running a check until it passes and recording only the passing run. Clearing an exception automatically. Recording an override with no reason, no compensating action or no expiry. Approving one's own override.

**7. Required evidence.** The override record for each overridden control result; the control's own log showing the original result; the configuration change history for the control, showing who changed what and when; the standing-override review records; the aggregate override review and any escalation arising from it.



**8. Responsible role.** The decision owner (D-08) authorised for that control under the governance arrangement. The professional who configures a control is responsible for the change record required by [PCI-FND-STD-13-PR-02](#).

**9. Approval authority.** Only a decision owner authorised for that control may approve an override, and never the person whose work the control rejected. Where the authorised decision owner is the person whose work was rejected, the decision passes to the next authority level, and the passing is recorded.

**10. Independence requirement.** The approver of an override must not be the preparer of the item the control rejected. Full independence (D-12) is required where the overridden control bears on an irreversible commitment, a payment to a third party, an external regulatory or financial report, or the safety of a person; in every other case, independence of preparation is the minimum.

**11. Materiality or threshold.** Attaches to control results bearing on material matters (D-15). It does not attach to routine tooling messages, formatting warnings or informational notices that no control depends on. The adopting organisation classifies its controls and states which are material and who may override each; on a small engagement that classification may be a single recorded list. Where no classification exists, a control is material if the condition it detects would meet D-15.

**12. Exception and waiver. This standard is itself the exception mechanism for every control it covers,** so there is no waiver from it: an override is the authorised, recorded departure, and an unrecorded departure is a breach rather than an exception. One narrow allowance exists: where an override is unavoidable to protect the safety of a person and no authorised decision owner can be reached, the professional may act and must record the override, the necessity and the attempts to reach the decision owner within one working day, and the decision owner must record their position on it within five working days. That allowance does not apply to commercial urgency.

**13. Escalation trigger.** An override taken with no authorised decision; a control found to have been disabled, re-parameterised or re-scoped without a recorded decision; the same control overridden repeatedly across periods; an override that has passed its expiry and remains in effect; pressure to approve an override without the compensating action.

**14. AI application.** An AI system may present the control result and its context to the decision owner, draft the override record, track expiries, and analyse override patterns across a period for human review.

**15. AI prohibition.** An AI system must not authorise an override, must not clear, close or suppress a control exception, must not adjust a control's



parameters on a material matter without a recorded human decision, and must not be recorded as the decision owner for an override.

**16. AI verification.** Where a control is automated or AI-driven, the professional must apply **boundary testing** at an interval the adopting organisation states — submitting cases known to be outside the control's tolerance and confirming the control raises them — and must record the cases used and the result. A control that does not raise a known-bad case has failed silently, which is the condition this standard exists to detect.

#### 17. External reference.

- **Internal Control — Integrated Framework.** Issuing organisation: COSO (Committee of Sponsoring Organizations of the Treadway Commission). Subject: a framework of internal control components and principles, including the treatment of management override of controls. Edition checked: 2013, revising the 1992 original; seventeen principles across five components. Nature of authority: **category 5 — professional framework** (register category: voluntary framework). Verification date: 2026-08-03 (register EXT-084). Applicability limitation: voluntary in itself, though widely imported by regulators and internal audit functions; it is not legislation, and no requirement in this standard is sourced to it.
- **ISO 9001 — Quality management systems — Requirements.** Issuing organisation: ISO. Subject: requirements for a quality management system, including the control of nonconforming outputs. Edition checked: ISO 9001:2015, fifth edition; a revision is in progress at ISO/TC 176/SC 2. Nature of authority: **category 3 — international voluntary standard**. Verification date: 2026-08-03 (register EXT-033). Applicability limitation: certifiable, and voluntary unless a regulator or contract requires it; verify current requirements given the revision in progress.
- **SR 11-7 / OCC 2011-12 — Supervisory Guidance on Model Risk Management.** Issuing organisation: US Federal Reserve Board and Office of the Comptroller of the Currency. Subject: the governance of model overrides and of adjustments to model output in supervised institutions. Edition checked: not independently verified — no edition asserted. Nature of authority: **category 10 — illustrative practice**. Verification date: not independently verified (register EXT-102). Applicability limitation: supervisory guidance with no legal force of its own, addressed to institutions supervised by those agencies in the United States; named to illustrate override governance, relied on for nothing.

**18. Jurisdictional caution.** Safety, environmental, financial-reporting and market-conduct regimes attach specific and sometimes personal liability to the disabling or overriding of a required control, and some require an override to be notified to a regulator. Obtain local legal and regulatory advice before overriding any control that a permit, licence, safety case or regulatory regime requires.

**19. Related PCI Standards.** [PCI-FND-STD-03](#) (a failed verification is a control result) · [PCI-FND-STD-04](#) (the decision that an override requires) · [PCI-FND-STD-11](#) (escalating an unauthorised override) · [PCI-FND-STD-12](#) (the override record's integrity) · [PCI-FND-STD-14](#) (an AI system must not clear an exception).

**20. Related Body of Knowledge content.** PCL-AI Domain 4 – performance management, variance analysis and reporting; Domain 13. PML-AI KA 3.3 – assurance and gates; KA 4.4 – the change board and the decision log. PFL-AI KA 6.4 – model checks and controls; Domain 16.

**21. Compliance test.** The reviewer works from the **control logs, not the override log**. For a sample of control results in the period that did not lead to the item being rejected or corrected: (a) an override record exists for each, dated before the item was relied upon; (b) it names the control, the reason, the evidence considered, the compensating action, the decision owner and the date; (c) the named decision owner is authorised for that control under the governance arrangement and is not the preparer of the item the control rejected; (d) the override carries an expiry, and where the expiry has passed no continuing effect is evidenced; and (e) **the reviewer compares the control's configuration change history against the period's override records and finds no change that removed, widened, re-scoped or re-timed the control in place of an override**. A sampled control result fails if any of (a) to (d) is absent; the period fails if (e) identifies an unrecorded configuration change on a material control. The standard is complied with for the period under review only if no sampled result fails and (e) identifies nothing.

**22. Breach indicators.** Control tolerances widened shortly before a period end. Exception queues that empty with no corresponding decisions. Validation rules disabled "temporarily" and still disabled. The same override reason text repeated across many records. Overrides approved by the person whose work was rejected. Checks re-run repeatedly with only the final run retained. Standing overrides with no review date. A control whose exception count falls to zero with no change in the underlying process.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review;

certification investigation; suspension; withdrawal. Each subject to due process and a right of appeal.

**24. Examination application.** Tested by scenario judgement (a validation failure on the evening of a reporting deadline), by evidence selection (which record shows the override was authorised) and by ethical dilemma (an instruction to widen a tolerance rather than record an override).

**25. Version and status.** Version 1.0 · approval date: not approved — draft submitted under Charter §5 Stage 11 · effective date: on approval · amendment note: new standard; no direct predecessor in the withdrawn set.

### Process requirements under PCI-FND-STD-13

**PCI-FND-STD-13-PR-01.** The override record must state the control overridden, the reason, the evidence considered, the compensating action taken, the decision owner and the date.

**PCI-FND-STD-13-PR-02.** A control must not be disabled, re-parameterised, re-scoped or re-timed in order to prevent it producing the result that would otherwise require an override; every change to a material control's configuration must be recorded with its author, its reason and its date.

**PCI-FND-STD-13-PR-03.** An override must be time-limited and its expiry recorded; a standing override must be reviewed by the decision owner at an interval the adopting organisation states.

**PCI-FND-STD-13-PR-04.** Overrides must be reviewed in aggregate at an interval the adopting organisation states, and a control whose results are repeatedly overridden must be escalated under **PCI-FND-STD-11** as either a defective control or a defective process.

**PCI-FND-STD-13-PR-05.** An automated process must not clear, close or suppress a control exception on a material matter without a recorded human decision under **PCI-FND-STD-04**.

**PCI-FND-STD-13-PR-06.** A check must not be re-run until it passes with only the passing run retained; every run of a material control must remain in the control's log.

## PCI STANDARD

### PCI-FND-STD-14

## Responsible AI

**1. Normative requirement.** A professional must not permit an AI system (D-02) to decide, approve, certify, sign, waive or authorise a material matter,

and must not permit it to be represented to any person as having independently verified one.

**2. Purpose.** This standard controls the risk that the roles a profession reserves to accountable people are quietly occupied by a system — and the related risk that a recipient believes a figure was checked by a qualified person when it was produced, and endorsed, by the same machine.

**3. Scope.** Governs every professional performing PCI-certified work in which an AI system is used, at any stage, by anyone on the engagement. Applies to material outputs, decisions, approvals, certificates and representations, on projects, programmes and transactions of any size, whatever the tool, vendor, deployment model or apparent quality of the output. Applies to preparation, review, recommendation, approval, assurance and to the configuration of the tools themselves.

**4. Defined terms.** D-02 AI system · D-15 material · D-16 material output · D-26 verified · D-24 reliance · D-08 decision owner · D-11 evidence. No term is redefined in this standard.

**5. Required actions.** Keep the roles listed in element 1 with accountable people. Record an AI system's material contribution to a material output in the working record. Answer truthfully any question about whether and how an AI system contributed. Disclose the contribution to a recipient whose decision to rely could turn on how the output was produced. Check every AI-produced citation against its source.

**6. Prohibited actions.** Configuring or permitting a system to approve, certify, sign, release, pay, commit or waive on AI output. Representing an AI system's self-check, confidence score or second pass as independent verification. Denying an AI contribution that occurred. Presenting AI-generated material as wholly human-authored where the recipient's reliance could turn on authorship. Treating an AI system's fluency, confidence or formatting as evidence of correctness. Using one AI system as the sole check on another.

**7. Required evidence.** The AI-contribution record for each material output — the system, its version or model identifier where available, the date, and what it contributed; the disclosure made where [-PR-03](#) applies; the citation checks required by [-PR-04](#); the verification record required by [PCI-FND-STD-03](#).

**8. Responsible role.** The issuing professional (D-14) for the output. The decision owner (D-08) for the matter. The professional who configures an AI-assisted workflow is responsible for the boundaries within it.

**9. Approval authority.** The decision owner approves the matter; the AI system approves nothing. Adoption of an AI system for material work at organisational scale is approved by the decision owner for technology under

the governance arrangement, on the assessment required by [PCI-FND-STD-09](#) element 10 where protected information is involved.

**10. Independence requirement.** An AI system is never independent (D-12) for the purposes of these standards, because it does not hold the accountability that independence exists to protect. Where independence is required, a person satisfying D-12 must provide it, and an AI system's involvement neither creates nor substitutes for it.

**11. Materiality or threshold.** Attaches to material matters and material outputs (D-15, D-16). An AI system may be used freely on work that is not material and that nobody relies on — drafting internal notes, exploring options, learning a subject — and this standard does not burden that use. The adopting organisation may impose stricter limits; it must not permit anything element 1 prohibits.

**12. Exception and waiver. No exception is permitted.** No urgency, volume, cost pressure, staffing constraint, client instruction or vendor assurance permits an AI system to decide, approve, certify, sign, waive or authorise a material matter, or to be represented as having independently verified one. Where the work cannot be done without crossing that boundary, the work must not be done.

**13. Escalation trigger.** Discovery that an AI system decided, approved, certified, signed, waived or authorised a material matter; discovery that an AI system's output was represented as independently verified; an instruction to conceal or deny an AI contribution; discovery that an AI system in continued use produces systematically unreliable output.

**14. AI application.** An AI system may propose, analyse, classify, draft, summarise, translate, extract, reconcile, model, generate options and scenarios, check its own arithmetic for the professional to test, and assemble the material a professional reviews. **This is the whole of what the suite principle permits it to do, and it is a great deal.**

**15. AI prohibition.** As element 1, and in addition: an AI system must not determine that verification is unnecessary, must not be the sole verifier of another AI system's output, must not be named as the accountable party under [PCI-FND-STD-01](#), must not be recorded as a decision owner under [PCI-FND-STD-04](#), must not authorise an override under [PCI-FND-STD-13](#), and must not decide whether a matter is material, protected, conflicted or escalation-worthy.

**16. AI verification.** Every material AI contribution must be verified under [PCI-FND-STD-03](#) by a named person applying a named method, and in addition: every citation, quotation, figure, clause reference and attribution the AI system produced must be checked by **source tracing** against the cited source itself under [PCI-FND-STD-06-PR-04](#); and where the output classifies, ranks

or scores, the professional must apply **sampling on a stated basis** to the items the system did *not* flag, because a classifier's silence is the part no one inspects. Both checks must be recorded.

#### 17. External reference.

- **Regulation (EU) 2024/1689** laying down harmonised rules on artificial intelligence (the "AI Act"). Issuing organisation: European Union. Subject: harmonised rules on AI systems placed on the EU market or used in the EU, including transparency and human-oversight duties for certain systems. Edition checked: in force since 1 August 2024; phased application — prohibitions from 2 February 2025, general-purpose AI obligations from 2 August 2025, general application from 2 August 2026, remaining rules by 2 August 2027. Nature of authority: **category 1 — applicable legislation or regulation, within the European Union only**. Verification date: 2026-08-03 (register EXT-100). Applicability limitation: **binding legislation only within the European Union and only on actors within its scope**; it imposes nothing outside the EU, no clause, article or annex is asserted, and PCI sources no requirement to it. Verify current requirements.
- **ISO/IEC 42001** — *Information technology — Artificial intelligence — Management system*. Issuing organisation: ISO/IEC. Subject: requirements for an AI management system. Edition checked: ISO/IEC 42001:2023, first edition. Nature of authority: **category 3 — international voluntary standard**. Verification date: 2026-08-03 (register EXT-021). Applicability limitation: certifiable but voluntary unless adopted by regulation or contract; an organisation's certification is not evidence that any individual output complied with this standard.
- **ISO/IEC 23894** — *Information technology — Artificial intelligence — Guidance on risk management*. Issuing organisation: ISO/IEC. Subject: managing risk in AI-related activities. Edition checked: ISO/IEC 23894:2023. Nature of authority: **category 3 — international voluntary standard**. Verification date: 2026-08-03 (register EXT-024). Applicability limitation: **guidance, not requirements** — nothing can be certified against it; it sits alongside ISO/IEC 42001 rather than under it.
- **Artificial Intelligence Risk Management Framework (AI RMF 1.0)**, NIST AI 100-1. Issuing organisation: NIST, US Department of Commerce. Subject: governing, mapping, measuring and managing AI risk. Edition checked: AI RMF 1.0, January 2023. Nature of authority: **category 10 — illustrative practice** (register category: voluntary framework). Verification date: 2026-08-03 (register EXT-080). Applicability limitation: NIST states it is voluntary, rights-preserving and non-

sector-specific; it is **not a standard and not a regulation**, and PCI sources no requirement to it.

- **Recommendation of the Council on Artificial Intelligence** (the "OECD AI Principles"), OECD/LEGAL/0449. Issuing organisation: OECD. Subject: intergovernmental principles for trustworthy AI. Edition checked: adopted 2019, revised May 2024. Nature of authority: **category 10 – illustrative practice**. Verification date: 2026-08-03 (register EXT-o81). Applicability limitation: a Council Recommendation – **not legislation, and not binding even on adherents**.

**18. Jurisdictional caution.** AI-specific regimes are emerging at different speeds and with different scopes, and sectoral regulators in financial services, healthcare, employment, safety and public procurement impose their own duties on automated processing, explanation and human oversight. Obtain local legal advice before deploying an AI system into a regulated activity or into any process affecting individuals, and treat any such requirement as prevailing over this standard under Charter §4. Verify current requirements.

**19. Related PCI Standards.** [PCI-FND-STD-01](#) (who remains accountable) · [PCI-FND-STD-03](#) (how the contribution is verified) · [PCI-FND-STD-04](#) (who decides) · [PCI-FND-STD-06](#) (checking AI-produced citations) · [PCI-FND-STD-09](#) (what may be entered into an AI system) · [PCI-FND-STD-13](#) (an AI system must not clear an exception). **Boundaries:** [PCI-FND-STD-03](#) governs *whether an item was checked*; [PCI-FND-STD-04](#) governs *who owns the decision*; this standard governs *what an AI system must never be permitted to do or be represented as having done*. A workflow can satisfy any two and breach the third.

**20. Related Body of Knowledge content.** PCL-AI Domain 13 – AI for Project Controls and PM: Concepts, Tools and Practice. PML-AI KA 1.4 – ethics and the responsible use of AI; Domain 14, especially KA 14.3 – AI use cases, prompting and verification, and KA 14.4 – explainability, bias and human accountability; KA 11.4 – misinformation and AI-generated communication risks. PFL-AI Domain 16, especially KA 16.2 and KA 16.3 – explainability, validation, bias and model risk.

**21. Compliance test.** For a sample of material outputs and workflows selected by the reviewer: (a) **Configuration test** – in each AI-assisted workflow, the reviewer identifies every step that gives effect to an output and confirms from the configuration that a human decision under [PCI-FND-STD-04](#) precedes it; a workflow in which any effecting step can execute without one fails. (b) **Representation test** – the reviewer reads the issued



output and its covering communications and finds no statement, label, badge or implication that an AI system verified, approved, certified or signed anything. (c) **Record test** — where an AI system materially contributed, the working record names the system, its version or model identifier where available, the date and the contribution. (d) **Citation test** — every citation, quotation, figure and clause reference in the output is located by the reviewer in the source cited. (e) **Silence test** — where the output classifies, ranks or scores, a recorded sample of the items the system did **not** flag exists, with its stated basis and result. A sampled output fails if any of (b) to (e) is absent or is contradicted; a workflow fails on (a). The standard is complied with for the period under review only if nothing fails.

**22. Breach indicators.** Certificates, approvals or releases whose only signature is a system account. Output labelled "AI-verified", "automatically validated" or "checked by model". Verification records naming a model as verifier. Workflows in which the effecting step has no preceding human entry. Citations that resolve to nothing. A professional unable to say whether an AI system contributed to an output they issued. Sampling records that cover only flagged items. Denials of AI use contradicted by prompt or tool logs.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure where breached in assessment; ethics review; certification investigation; suspension; withdrawal. Each subject to due process and a right of appeal.

**24. Examination application.** Tested by AI-verification case (a fluent, plausible and wrong output with a fabricated citation), by scenario judgement (a workflow that auto-releases on a model score) and by ethical dilemma (a client who asks whether the report was written by a person).

**25. Version and status.** Version 1.0 · approval date: not approved — draft submitted under Charter §5 Stage 11 · effective date: on approval · amendment note: carries forward the boundary substance of the withdrawn [PCI-LAW-F-02](#) and the disclosure substance of the withdrawn [PCI-LAW-F-04](#).

### Process requirements under PCI-FND-STD-14

[PCI-FND-STD-14-PR-01](#). Where an AI system materially contributed to a material output, the working record must state the system used, its version or model identifier where available, the date, and what it contributed.

[PCI-FND-STD-14-PR-02](#). A professional must answer truthfully any question from a recipient about whether and how an AI system contributed to an output, and must not deny a contribution that occurred.

[PCI-FND-STD-14-PR-03](#). Where a recipient's decision to rely on an output could turn on how it was produced, the AI contribution must be disclosed to that recipient before reliance; where a contract, client rule, employer policy or



PCI examination condition imposes a stricter disclosure duty, that duty governs.

**PCI-FND-STD-14-PR-04.** A citation, quotation, figure, clause reference or attribution produced by an AI system must be checked against the cited source itself before presentation, under **PCI-FND-STD-06-PR-04**.

**PCI-FND-STD-14-PR-05.** An AI system must not be used as the sole check on another AI system's output, and must not determine that verification is unnecessary.

**PCI-FND-STD-14-PR-06.** A professional must not present or treat an AI system's fluency, confidence score, formatting or apparent authority as evidence that its output is correct.

#### PCI STANDARD

#### PCI-FND-STD-15

### Correction Duty

**1. Normative requirement.** A professional who discovers a material error in work they have issued must communicate it promptly (D-20) to every person known to be relying on that work.

**2. Purpose.** This standard controls the risk that an error, once discovered, is repaired quietly in the file while the people who acted on it continue to act on it — and the related risk that the correction arrives after the decision it would have changed.

**3. Scope.** Governs every professional performing PCI-certified work. Applies to every material output the professional has issued, recommended, approved or certified, and to errors of calculation, data, source version, assumption, omission and presentation alike, however the error is discovered and whoever caused it. Applies on projects, programmes and transactions of any size, and continues after the engagement or employment in which the work was issued has ended, for as long as the work remains in reliance.

**4. Defined terms.** D-20 promptly · D-15 material · D-16 material output · D-24 reliance · D-08 decision owner · D-22 record · D-27 working day. No term is redefined in this standard.

**5. Required actions.** Communicate the error promptly to every person known to be relying on the work. Record the error, its discovery date, its cause where known and its effect on each conclusion. Assess whether any decision already taken would have differed, and tell that decision's owner. Issue a corrected version identified as a correction. Retain the superseded version.

**6. Prohibited actions.** Correcting silently — replacing a figure, file or document with no record of the change. Waiting for a scheduled report to

carry the correction where the error meets the D-20 one-working-day condition. Communicating the correction only to those who ask. Withdrawing the superseded version from the record. Characterising a material error as a presentational refinement. Complying with an instruction not to tell the relying parties without escalating it.

**7. Required evidence.** The error record — what, when discovered, cause, effect; the reliance assessment and the communication to each affected decision owner; the corrected version, identified as such, stating what changed and which version it replaces; the retained superseded version; the notification under -PR-04 where the professional no longer controls the work; the escalation record where -PR-06 applies.

**8. Responsible role.** The issuing professional (D-14) who issued the work. Where that professional has left the engagement or the organisation, the duty attaches to them personally for the communication and to the engagement lead (D-09) for the correction of the record; neither discharges the other.

**9. Approval authority.** No approval is required to communicate an error to a person relying on the work, and none may be imposed. The decision owner (D-08) for each affected decision decides what to do about the decision. The engagement lead approves the form of the corrected issue.

**10. Independence requirement.** Where the error concerns a material calculation, model output or automated conclusion, the corrected version must be verified under PCI-FND-STD-03 by a competent reviewer independent (D-12) of the correction — the same discipline that applied to the original, applied again, because a correction made in haste is where the second error enters.

**11. Materiality or threshold.** Attaches to material errors (D-15), assessed on the position as it stood when the work was issued and as it stands on discovery, whichever produces the wider effect. The adopting organisation may set a lower threshold. **The urgency is set by D-20, not by the size of the error:** an error creating a risk to the safety of a person or an ongoing financial loss carries the one-working-day period, every other material error the five-working-day period.

**12. Exception and waiver. No exception is permitted** to communicating a material error to those relying on the work. Where a legal obligation, a regulator's direction or a court order restricts what may be communicated or to whom, the professional must obtain legal advice, must communicate to the fullest extent the restriction permits, must record the restriction and the advice, and must escalate under PCI-FND-STD-11. A confidentiality undertaking, a commercial concern, an employer's preference or the professional's own exposure is not such a restriction.

**13. Escalation trigger.** An instruction not to inform a relying party; a decision owner who declines to revisit a decision the corrected work would

have changed; discovery that a previous correction was not communicated; a relying party who cannot be identified or reached.

**14. AI application.** An AI system may identify which outputs, models and downstream documents used the erroneous figure, assess the arithmetic effect of the correction, draft the correction notice, and maintain the distribution list of relying parties.

**15. AI prohibition.** An AI system must not decide whether an error is material, whether it must be communicated, or to whom; must not approve a corrected version; and must not amend an issued output in place of an identified correction.

**16. AI verification.** Where an AI system identified the downstream uses of the erroneous figure, the professional must apply **source tracing** to at least the uses the system did *not* identify — searching the engagement's issued outputs for the figure and its derivatives — and must record the search basis and the result. Where an AI system computed the corrected figure, **PCI-FND-STD-03** applies to it in full.

#### **17. External reference.**

- **IAS 8.** Issuing organisation: IFRS Foundation / IASB. Subject: accounting for and disclosure of prior-period errors and changes in accounting estimates in financial statements. Edition checked: currently titled *Accounting Policies, Changes in Accounting Estimates and Errors*; **retitled *Basis of Preparation of Financial Statements from 1 January 2027*** as a consequential amendment to IFRS 18. Nature of authority: **category 2 — authoritative financial-reporting standard**. Verification date: 2026-08-03 (register EXT-005). Applicability limitation: mandatory only for entities applying IFRS Accounting Standards in a jurisdiction that has adopted them; it governs **financial reporting only** and imposes nothing on project records or professional deliverables. Cited by name; no paragraph or requirement is asserted.
- **ISO 9001 — *Quality management systems — Requirements*.** Issuing organisation: ISO. Subject: control of nonconforming outputs and corrective action. Edition checked: ISO 9001:2015, fifth edition; a revision is in progress. Nature of authority: **category 3 — international voluntary standard**. Verification date: 2026-08-03 (register EXT-033). Applicability limitation: certifiable but voluntary unless required by a regulator or contract; verify current requirements.
- **International Code of Ethics for Professional Accountants (including International Independence Standards).** Issuing organisation: IESBA / IFAC. Subject: the professional's response on becoming aware that information previously provided is misleading.

Edition checked: current at the register's verification date; no edition asserted. Nature of authority: **category 6 — ethical code**. Verification date: 2026-08-03 (register EXT-127). Applicability limitation: binding only where a professional body, a regulator or an engagement has adopted it; a PCI credential holder not subject to it is not made subject to it by this standard.

**18. Jurisdictional caution.** Correcting an issued statement can engage securities, prospectus, public-procurement, tax and financial-reporting obligations, restatement rules and notification duties, and in some regimes the timing and form of the correction are prescribed. Obtain local legal and accounting advice before correcting anything that has been published, filed, certified or relied upon by a lender, an investor or an authority.

**19. Related PCI Standards.** [PCI-FND-STD-01](#) (accountability for the work being corrected) · [PCI-FND-STD-03](#) (verifying the correction) · [PCI-FND-STD-05](#) (a failed assumption) · [PCI-FND-STD-11](#) (escalating the error internally) · [PCI-FND-STD-12](#) (retaining the superseded version). **Boundary with [PCI-FND-STD-11](#):** that standard is discharged by raising the matter internally; this one is discharged only by telling the people relying on the work and putting it right. One discovery ordinarily engages both.

**20. Related Body of Knowledge content.** PCL-AI Domain 4 — performance management, variance analysis and reporting; Domain 2 — financial reporting for project controls; Domain 13. PML-AI KA 11.2 — executive communication and reporting; KA 1.4. PFL-AI Domain 16.

**21. Compliance test.** The reviewer identifies material errors discovered in the period from correction records, restatements, re-issues, audit findings, change logs and version histories — **not from the correction register alone**. For each: (a) a communication to each person known to be relying on the work exists; (b) its date is within the D-20 period measured from the earliest evidence that the professional knew or suspected the error, established from correspondence and system records; (c) an error record exists stating the error, its discovery date, its cause where known and its effect on each conclusion; (d) a reliance assessment exists identifying each decision already taken that would have differed, and a communication to each of those decision owners; (e) a corrected version exists, identified as a correction, stating what changed and which version it replaces; and (f) the superseded version is retained and retrievable. A sampled error fails if any of (a) to (f) is absent, or if the reviewer finds the erroneous figure still in use in a later document with no correction. The standard is complied with for the period under review only if no sampled error fails.

**22. Breach indicators.** Re-issued documents with no correction notice. Figures that change between versions with no change record. Corrections carried in the next routine report rather than communicated. Distribution lists for corrections shorter than for the original issue. Superseded versions deleted. Error records whose discovery date matches the communication date exactly across many items. Downstream documents still carrying the erroneous figure after the correction.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension; withdrawal. Each subject to due process and a right of appeal.

**24. Examination application.** Tested by escalation decision (an error found the day after a funding decision), by ethical dilemma (an employer who prefers the correction to wait for the quarterly pack) and by calculation review (assessing whether a corrected figure would have changed the decision).

**25. Version and status.** Version 1.0 · approval date: not approved — draft submitted under Charter §5 Stage 11 · effective date: on approval · amendment note: new standard; carries forward the correction substance of the withdrawn PCI-LAW-F-07.

### Process requirements under PCI-FND-STD-15

PCI-FND-STD-15-PR-01. On discovery, the professional must record the error, its date of discovery, its cause where known, and its effect on each conclusion the work supports.

PCI-FND-STD-15-PR-02. The professional must assess and record whether any decision already taken in reliance on the work would have differed, and must communicate that assessment to the decision owner of each such decision.

PCI-FND-STD-15-PR-03. A corrected version must be issued, identified as a correction, stating what changed, why, and which earlier version it replaces; the superseded version must be retained in the record and must not be deleted.

PCI-FND-STD-15-PR-04. Where the professional no longer controls the issued work, they must notify the party who does, in writing, and must record the notification.

PCI-FND-STD-15-PR-05. A professional must not correct a material error silently — by replacing a figure, file or document without a record identifying the change.

PCI-FND-STD-15-PR-06. Where the professional is prevented from communicating the error by an instruction from an employer, a client or any other

person, they must record the instruction and escalate it under [PCI-FND-STD-11](#).

## Audit-question findings

Drafting Manual §9 requires all twenty-five questions to be answered before approval, and requires a standard failing one to be revised before approval, with the failure and its resolution recorded in the standard's file. This table is that record. Every question was worked against all fifteen standards; the rows below state what the question found and what was changed in response. A row reading "no change" means the question was answered without a defect being identified, not that it was skipped.

| # | QUESTION                                            | STANDARDS<br>AFFECTED | FINDING, AND WHAT WAS CHANGED                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---|-----------------------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | What exact failure does this standard prevent?      | All 15                | <b>Changed.</b> Element 2 of every standard was rewritten to name one professional failure in concrete terms rather than to restate the obligation as an aspiration. Two drafts failed: an early <a href="#">PCI-FND-STD-14</a> purpose said "to ensure responsible AI use", which names no failure, and an early <a href="#">PCI-FND-STD-12</a> purpose restated its own rule. Both were replaced with the specific failure mode.                                              |
| 2 | Is the requirement mandatory or only recommended?   | All 15                | <b>Changed.</b> <a href="#">should</a> was removed from every normative position; it appears nowhere in any element 1 or Process Requirement. The ISO requirement verb appears nowhere in the file, in any field, per Manual §1 — the superseded set used it in eleven Rule fields and this edition removes all of them. No Recommended Practice (Level 5) is created by this set.                                                                                              |
| 3 | Can a professional know whether it applies to them? | All 15                | <b>Changed.</b> Element 3 was standardised to answer four questions explicitly — who is governed, which decisions, which projects or transactions, and whether it bites at preparation, review, recommendation, approval or assurance. <a href="#">PCI-FND-STD-14</a> gained the clause "in which an AI system is used, at any stage, by anyone on the engagement", because a professional who did not personally use the tool was otherwise able to read themselves out of it. |

| # | QUESTION                                         | STANDARDS<br>AFFECTED                  | FINDING, AND WHAT WAS CHANGED                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---|--------------------------------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | Is the responsible person identifiable?          | 01, 02, 04, 07, 08, 10, 11, 12, 13, 15 | <b>Changed.</b> Element 8 never reads "the team", "management" or "the organisation". Six role terms were defined instead (D-08, D-09, D-13, D-14, D-23, plus D-04) and element 8 names one of them in every standard. <b>PCI-FND-STD-15</b> gained a split responsibility, because the issuing professional may have left the engagement while the duty to communicate remains personal.                                                                                   |
| 5 | Is the required action observable?               | 01, 04, 05, 08, 11, 14                 | <b>Changed.</b> Unobservable formulations were removed: "accept accountability" became <b>record</b> an acceptance (01); "manage the conflict" became one of four recorded safeguards (08); "consider the recommendation" became a recorded factor not stated in the recommendation itself (04); "disclose material AI assistance" became a record naming system, version, date and contribution (14).                                                                      |
| 6 | Is compliance provable?                          | All 15                                 | <b>Changed.</b> Element 7 in every standard names records that exist independently of the professional's own account. Where an element 1 obligation had no possible proof, it was reworded: <b>PCI-FND-STD-05's</b> original "must consider sensitivities" became "must state the effect on the conclusion of a stated change", which leaves a document behind.                                                                                                             |
| 7 | Is the required evidence proportionate?          | 02, 05, 07, 12                         | <b>Changed.</b> <b>PCI-FND-STD-07</b> element 11 now states that lineage is required for the <b>figures presented in the material output</b> , not field-level lineage for every value in a system. <b>PCI-FND-STD-12</b> element 11 was made technology-neutral. <b>PCI-FND-STD-05-PR-02</b> bounds the sensitivity work to material assumptions only. <b>PCI-FND-STD-02</b> element 11 permits a single referenced worksheet as the working record on a small engagement. |
| 8 | Can the standard be audited?                     | All 15                                 | <b>Changed.</b> Element 21 was rewritten in all fifteen standards to a sampling test with pass and fail conditions stated. See question 6 of this table and §5.1 of the Manual: five element-21 drafts were rejected as unauditably and replaced — see the Q21 row below for the pattern.                                                                                                                                                                                   |
| 9 | Can the standard be examined through a scenario? | All 15                                 | <b>No change required.</b> Element 24 in each standard names which of the Manual's six assessment modes applies and gives the scenario shape. No standard is assessable only by recalling its number.                                                                                                                                                                                                                                                                       |

| #  | QUESTION                                                           | STANDARDS<br>AFFECTED                          | FINDING, AND WHAT WAS CHANGED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----|--------------------------------------------------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10 | Can a professional technically comply while defeating its purpose? | 01, 03, 04, 05, 07, 08, 09, 11, 12, 13, 14, 15 | <b>Changed — the heaviest single source of revision in this edition.</b> Twelve gaming routes were found and closed: (i) naming an accountable person who never saw the work — closed by requiring the acceptance to be attributable to that individual and by the no-notice question in 01 element 21; (ii) "verifying" by re-running the same tool — closed by 03-PR-03; (iii) rubber-stamping — closed by the walk-through and substance limbs of 04 element 21; (iv) burying material assumptions among a hundred boilerplate ones — closed by the two-way matching test in 05 element 21(b) and by 05-PR-05; (v) narrative lineage that only the preparer can follow — closed by requiring reproduction <b>without assistance from the preparer</b> in 07 element 21; (vi) self-certifying that a conflict is harmless — closed by the closed list in 08-PR-01 and by 08-PR-02; (vii) redacting names and pasting the rest into an unauthorised tool — closed by the reconstruction clause in D-21; (viii) escalating into a captive channel or with no record — closed by 11-PR-01 and 11-PR-04, and by starting 11 element 21 from the population of <b>matters</b> rather than of escalations; (ix) working in ephemeral channels to avoid creating records — closed by 12-PR-06; (x) widening a tolerance instead of recording an override — closed by 13-PR-02 and by limb (e) of 13 element 21, which reads the configuration change history against the override log; (xi) re-running a check until it passes — closed by 13-PR-06; (xii) correcting silently and letting the next routine report carry it — closed by 15-PR-05 and by the prohibition in 15 element 6. |
| 11 | Does it conflict with another PCI standard?                        | 03, 04, 07, 11, 12, 14, 15                     | <b>Changed.</b> Four boundary statements were added to element 19 where two standards could be read as covering the same act: 03 against 14 (whether an item was checked, versus what an AI system may be permitted to do); 04 against 14 (who owns the decision, versus what an AI system may be represented as doing); 07 against 12 (retraceability versus survival of records); 11 against 15 (raising it internally versus telling those relying and putting it right). Each states that satisfying one does not satisfy the other.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |



| #  | QUESTION                                               | STANDARDS<br>AFFECTED          | FINDING, AND WHAT WAS CHANGED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----|--------------------------------------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12 | Does it duplicate an external standard unnecessarily?  | 02, 05, 07, 12, 13, 15         | <b>Changed.</b> Every element 17 entry now carries the sentence that PCI sources no requirement to the instrument, except where the instrument is genuinely the applicable authority. Three references present in early drafts were deleted as decorative: ISO 21500 and ISO 21502 under <b>PCI-FND-STD-01</b> , and ANSI/EIA-748 under <b>PCI-FND-STD-07</b> , none of which added an obligation or a limitation the standard needed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 13 | Does it misrepresent external authority?               | 01, 04, 06, 08, 09, 13, 14, 15 | <b>Changed, and this produced four corrections.</b> (i) The <b>IFRS Conceptual Framework is excluded from the whole set</b> — the IASB states it is not a Standard, and Manual §6 forbids sourcing a requirement to it. (ii) Only <b>Regulation (EU) 2024/1689</b> is described as legislation, and only within the EU; the <b>GDPR</b> is recorded as category 10 illustrative practice in 09 and relied on for nothing. (iii) <b>SR 11-7</b> is recorded as supervisory guidance with no legal force of its own, addressed to US-supervised institutions. (iv) <b>ISO/IEC 17024</b> is stated to address certification bodies, not individual credential holders — the superseded set cited it four times without that limitation. In addition, ISO 31000 and ISO/IEC 23894 are marked as guidance that nothing can be certified against, ISO/IEC 27701 is described as a standalone requirements standard rather than an extension, and ISO 37001 carries the note that certification is not a legal defence. |
| 14 | Does it require legal or jurisdiction-specific advice? | All 15                         | <b>No change required.</b> Element 18 in each standard names the specific regimes that displace or supplement it, and Charter §4 governs the priority. <b>PCI-FND-STD-10</b> element 12 additionally makes reserved activities a no-exception case.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 15 | Does it define the relevant materiality threshold?     | All 15                         | <b>Changed.</b> No percentage is invented anywhere in this set, per Manual §5.3. <i>Material</i> is defined once at D-15 by a six-limb test, with the adopting organisation's published tolerance as one limb and a recorded decision-owner determination as the fallback. Two standards state expressly that <b>no</b> threshold applies — <b>PCI-FND-STD-09</b> (a single record can identify a person) and <b>PCI-FND-STD-08</b> (every conflict within D-05 is declarable) — because a threshold there would have been a loophole.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 16 | Does it cover AI use?                                  | All 15                         | <b>No change required.</b> Elements 14, 15 and 16 are present in all fifteen standards, and element 16 names a specific method in every case rather than saying "review the AI output".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

| #  | QUESTION                               | STANDARDS<br>AFFECTED | FINDING, AND WHAT WAS CHANGED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----|----------------------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 17 | Does it preserve human accountability? | 01, 03, 04, 13, 14    | <b>Changed.</b> <i>PCI-FND-STD-14</i> element 10 was added to state that an AI system is <b>never</b> independent for the purposes of these standards, because it does not hold the accountability that independence exists to protect. Without it, a reader could have argued that a second model satisfied <i>PCI-FND-STD-03</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 18 | Does it contain an exception process?  | All 15                | <b>Changed.</b> Element 12 in every standard states either the exception process in full — approver, justification, duration, compensating controls, reporting — or that no exception is permitted. Six standards permit none: <i>01</i> , <i>08</i> (to disclosure), <i>11</i> , <i>14</i> , <i>15</i> (to communication), and <i>13</i> in substance, since an override is itself the recorded exception mechanism. <i>PCI-FND-STD-03</i> element 12 gained the carve-out that <b>no waiver is available</b> where the item supports an irreversible commitment, a payment, an external report or a safety decision.<br><b>Stage 9 addition:</b> the second red-team pass tested every element 12 against a genuine emergency and found two standards with no compliant route in the case that will certainly arise. <i>PCI-FND-STD-03</i> and <i>PCI-FND-STD-08</i> each gained a <b>narrow allowance, not a waiver</b> — act only to the extent necessary to protect the safety of a person, record it, disclose it within one working day, obtain the deferred verification or the substitute safeguard, and have the accountable person record their position within five working days. Neither allowance reaches commercial urgency. |
| 19 | Does it define escalation?             | All 15                | <b>Changed.</b> <i>Escalation threshold</i> is defined once at D-10 by six triggers, one of which is the adopting organisation's own criteria, so the standard functions with or without a threshold framework. Element 13 in every standard names the events that require escalation rather than saying "where appropriate".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| #  | QUESTION                                          | STANDARDS<br>AFFECTED      | FINDING, AND WHAT WAS CHANGED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----|---------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20 | Is every important term defined?                  | All 15                     | <b>Changed — the second-heaviest source of revision.</b> Twenty-eight terms are defined at D-01 to D-28 and used identically throughout. Undefined judgement words were removed from every obligation: " <b>appropriately</b> " is gone from <b>PCI-FND-STD-08</b> and replaced by the closed list at <b>08-PR-01</b> , as the drafting brief required; " <b>reasonable</b> " was removed from an early <b>08</b> element 21 and replaced with a comparison against the D-05 relationship list; " <b>adequate</b> ", " <b>relevant</b> ", " <b>timely</b> " and " <b>sufficient</b> " appear in no obligation. " <b>Promptly</b> " is defined at D-20 with a one-working-day and five-working-day default, and a stated start point at knowledge or suspicion rather than confirmation. No definition is circular: <i>material</i> , <i>independent</i> , <i>verified</i> and <i>evidence</i> are each defined by a test the reader applies, not by a synonym. |
| 21 | Is the language concrete and modern?              | All 15                     | <b>Changed.</b> Modern must-drafting only; the ISO requirement verb appears nowhere. Five element-21 compliance tests were rejected as too weak to produce agreement between two reviewers and rewritten: <b>01</b> (now a no-notice question to the named individual), <b>05</b> (now a two-way match between the assumption statement and the sensitivity statement), <b>07</b> (now a reproduction the reviewer performs unaided), <b>11</b> (now sampled from the population of matters, not of escalations) and <b>13</b> (now read from the control logs and the configuration change history, not from the override log).                                                                                                                                                                                                                                                                                                                               |
| 22 | Does it impose an impossible or excessive burden? | 03, 05, 07, 09, 10, 12, 14 | <b>Changed.</b> Six burdens were cut. (i) <b>PCI-FND-STD-03</b> originally required independent verification of every calculation; it now attaches to <b>material</b> items only and permits a bounded waiver where no independent person exists. (ii) <b>PCI-FND-STD-07</b> no longer implies field-level lineage. (iii) <b>PCI-FND-STD-12</b> no longer implies a records-management system; it requires only that alteration be detectable and attributable. (iv) <b>PCI-FND-STD-05</b> bounds sensitivity work to material assumptions. (v) <b>PCI-FND-STD-14</b> element 11 states expressly that non-material AI use — internal notes, exploration, learning — is not burdened by this standard at all. (vi) <b>PCI-FND-STD-10-PR-01</b> requires a competence assessment for <b>material</b> engagements and roles, not for every task.                                                                                                                 |

| #  | QUESTION                                                | STANDARDS AFFECTED         | FINDING, AND WHAT WAS CHANGED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----|---------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 23 | Can it operate on both small projects and megaprojects? | 01, 03, 04, 07, 09, 12, 13 | <b>Changed.</b> Each standard was tested against a two-person consultancy and a multi-billion programme, and seven failed on the small case before revision. (i) D-01 makes a sole practitioner their own adopting organisation and requires them to record, once, the criteria these standards leave to it. (ii) The note on roles states that one individual may hold several capacities. (iii) <b>PCI-FND-STD-03</b> element 12 provides the independence waiver with cumulative compensating controls, and forbids it on irreversible, payment, external-report and safety matters. (iv) <b>PCI-FND-STD-09-PR-01</b> supplies the information-owner authorisation route where no register of authorised systems exists. (v) <b>PCI-FND-STD-12</b> element 11 makes the requirement technology-neutral and states that versioned files with dated issues comply. (vi) <b>PCI-FND-STD-04</b> element 11 states that a single recorded line may be the whole delegated-authority schedule. (vii) <b>PCI-FND-STD-13</b> element 11 permits the control classification to be one recorded list. On the megaproject side, no standard needed relaxing; <b>PCI-FND-STD-05</b> element 11 notes that the same test simply identifies more assumptions. |
| 24 | Can it operate internationally?                         | All 15                     | <b>Changed.</b> No standard assumes a jurisdiction, a currency, a professional body, a regulator or a legal system. Element 18 in each standard identifies what needs local advice, and Charter §4 places applicable legislation and regulation above every PCI Standard. The single monetary illustration in the set ( <b>PCI-FND-STD-03</b> element 11) is expressly an illustration of why PCI sets no figure, not a threshold.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 25 | Is there a clear consequence within PCI's authority?    | All 15                     | <b>Changed.</b> Element 23 in every standard is drawn <b>only</b> from the Charter §9 list. The superseded set used the word "revocation" throughout; Charter §9 says <b>withdrawal</b> , and every occurrence was corrected. No standard claims a fine, a criminal consequence, a damages liability or any governmental enforcement power, and every consequence is stated as subject to due process and a right of appeal.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

### Two findings this audit could not resolve, recorded rather than hidden.

First, **stages 4, 6, 7, 8, 11, 12 and 13 of Charter §5 have not been performed** for this edition. The audit questions are a red-team exercise (stage 9); they are not a substitute for technical review by a subject specialist, for practitioner consultation, or for approval. Several element-11 thresholds and element-12 durations — the ten-

working-day onward-escalation period in [PCI-FND-STD-11-PR-02](#), the five-working-day default in D-20, the three-month waiver ceiling in [PCI-FND-STD-03](#) — are defensible drafting choices that have not been tested with practitioners and may prove wrong in use. They are configurable by the adopting organisation wherever that was possible, precisely because they are unvalidated.

Second, **the Drafting Manual's ten external-reference categories do not cleanly accommodate two kinds of instrument this set names**: a voluntary framework that is neither environmental nor social (the NIST AI RMF, the OECD AI Principles), and a national standard (ANSI/EIA-748, which this set consequently does not cite). Both are recorded here under category 10 or omitted, and the register's own category is shown alongside. The honest fix is a Manual amendment adding those two values, which is outside this file's authority to make.

A second red-team pass, run later across the whole four-file corpus, is recorded in [STANDARDS\\_RED\\_TEAM\\_REPORT.md](#). It produced two amendments to this file — the emergency allowances in [PCI-FND-STD-03](#) and [PCI-FND-STD-08](#) element 12, noted in their element 25 — and a body of findings against the three certification sets, including a set of definitional collisions between this file's D-01 to D-28 and the per-credential definitions.

**Those collisions are now closed.** The durable fix the red team recommended as its finding P-1 — a single **PCI Standards Definitions Register**, which Drafting Manual §4 already contemplates — has been built at [PCI\\_STANDARDS\\_DEFINITIONS\\_REGISTER.md](#), and the definitions above were reconciled to it. What changed in this file, and nothing else changed:

- **D-04 competent reviewer** — gains a recorded-before-the-review limb and a no-assistance-from-the-preparer limb, and states expressly that **independence is not part of competence**, so that [PCI-FND-STD-10](#) element 12's supervised-acquisition exception remains usable (red-team X-6 / R-5).
- **D-05 conflict** — gains the appearance limb and the future-role limb the PML-AI volume carried, and states that a judgement standard in a standard widens the closed list and never narrows it.
- **D-08 decision owner · D-11 evidence · D-12 independent · D-26 verified** — restated in the canonical wording, taking the union of the limbs and exclusions the four volumes carried between them. D-12 keeps its line-reporting carve-out and names the stricter PFL-AI limb as such; D-26 names the eight admissible methods in the definition itself and states PFL-AI's deliberate narrowing for a figure in a financial model.
- **D-10 escalation threshold** — states that a standard's element 13 events are additional to the six triggers, and supplies the destination-and-time fallback so the duty survives an organisation that documents neither.

- **D-15 material** — gains the *irrespective of size* sentence for safety, legality, a licence, a statutory duty and the truth of a statement, and records how limb (b) is measured in each credential.
- **D-29 approved** and **D-30 current** — added. Manual §4 names both as terms that always require definition, and neither was defined anywhere in this set.

No obligation was widened or narrowed by any of these edits: each states, in one place, the reading that the wider-obligation rule at the head of every certification volume already produced. The register records what each file previously said, so every change is visible and reversible. Four questions the reconciliation could not settle — **R-1**, **R-3**, **R-4** and **R-5** — remain with the Interpretation Panel.

## Index of Foundational Standards

| ID             | OFFICIAL TITLE              | PRINCIPAL OBLIGATION,<br>IN ONE LINE                                                                                                           | EXTERNAL-<br>REFERENCE<br>CATEGORIES USED                                            |
|----------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| PCI-FND-STD-01 | Professional Accountability | Record your own acceptance of personal accountability for a material output before it is issued, whoever or whatever prepared it.              | 3 (international voluntary standard) · 6 (ethical code) · 10 (illustrative practice) |
| PCI-FND-STD-02 | Evidence Before Assertion   | Do not present a material claim unless retrievable evidence for it is identified in the working record.                                        | 5 (professional framework)                                                           |
| PCI-FND-STD-03 | Independent Verification    | Have a material calculation, model output or automated conclusion verified by an independent competent reviewer before anyone relies on it.    | 5 (professional framework) · 7 (industry guidance) · 10 (illustrative practice)      |
| PCI-FND-STD-04 | Human Decision Authority    | Record every material decision as taken by a named, authorised, competent human decision owner.                                                | 1 (legislation — EU only) · 3 (international voluntary standard)                     |
| PCI-FND-STD-05 | Transparent Assumptions     | Do not issue a forecast, model, appraisal or recommendation unless the document carrying the conclusion also carries its material assumptions. | 3 (international voluntary standard) · 5 (professional framework)                    |

| ID             | OFFICIAL TITLE                          | PRINCIPAL OBLIGATION,<br>IN ONE LINE                                                                                                                          | EXTERNAL-<br>REFERENCE<br>CATEGORIES USED                                                      |
|----------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| PCI-FND-STD-06 | Source and Version Integrity            | Do not rely on a source without a dated check that it is the version in force for the matter.                                                                 | 2 (financial-reporting standard) · 4 (contract framework) · 9 (PCI internal instrument)        |
| PCI-FND-STD-07 | Data Lineage                            | Make every material output reproducible from its sources by a competent reviewer without the preparer's help.                                                 | 3 (international voluntary standard) · 7 (industry guidance)                                   |
| PCI-FND-STD-08 | Conflict Disclosure                     | Disclose every conflict capable of affecting your independent judgement, in writing, before acting in the matter.                                             | 3 (international voluntary standard) · 6 (ethical code)                                        |
| PCI-FND-STD-09 | Confidentiality and Approved Technology | Do not enter protected information into a system that is not authorised for it, including any AI system.                                                      | 3 (international voluntary standard) · 10 (illustrative practice)                              |
| PCI-FND-STD-10 | Competence and Limitation               | Do not accept, continue or issue work requiring competence you do not hold.                                                                                   | 3 (international voluntary standard) · 6 (ethical code)                                        |
| PCI-FND-STD-11 | Escalation of Material Misstatement     | Escalate a known or suspected material error, omission, unsupported assumption or misleading presentation promptly to the level accountable for remedying it. | 3 (international voluntary standard) · 4 (contract framework) · 5 (professional framework)     |
| PCI-FND-STD-12 | Record Integrity                        | Keep records so that authorship and creation time are fixed and any later alteration is detectable and attributable.                                          | 3 (international voluntary standard)                                                           |
| PCI-FND-STD-13 | No Silent Override                      | Do not override a control result on a material matter without an authorised decision owner's recorded decision, made before reliance.                         | 3 (international voluntary standard) · 5 (professional framework) · 10 (illustrative practice) |

| ID             | OFFICIAL TITLE  | PRINCIPAL OBLIGATION,<br>IN ONE LINE                                                                                                                          | EXTERNAL-<br>REFERENCE<br>CATEGORIES USED                                                     |
|----------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| PCI-FND-STD-14 | Responsible AI  | Never permit an AI system to decide, approve, certify, sign, waive or authorise a material matter, or to be represented as having independently verified one. | 1 (legislation — EU only) · 3 (international voluntary standard) · 10 (illustrative practice) |
| PCI-FND-STD-15 | Correction Duty | Communicate a material error in work you issued promptly to everyone known to be relying on it.                                                               | 2 (financial-reporting standard) · 3 (international voluntary standard) · 6 (ethical code)    |

**Set composition.** Fifteen standards · eighty-three Process Requirements · twenty-eight defined terms · twenty-five elements in every standard, in the Manual's order, none omitted.

CAUTION

AI proposes; the professional verifies, decides and remains accountable.

*Prepared 4 August 2026. Draft for approval under Charter §5 Stage 11 — not approved, not in force. External references checked against ../registries/EXTERNAL\_AUTHORITIES.md and its recorded verification dates; the official publication of every instrument named always governs, and where an authority can change, verify current requirements. British English throughout.*



# PML-AI Professional Standards — PCI Project Management Leader — AI

**Status:** Certification Standard set for the **PML-AI** credential (PCI Project Management Leader – AI). Version 2.0, drafted under the **PCI Standards Charter** (what a standard is) and the **PCI Standard Drafting Manual** (how a standard is written). A standard that does not conform to both does not pass gate. **Thirty-two standards and one hundred and forty-eight process requirements**, anchored across all sixteen domains of the PML-AI Body of Knowledge ([./pml-ai/](#)).

## CAUTION

**PCI Standards are private professional certification requirements established by Project Controls Institute Global. They are not legislation, government regulation, legal advice or substitutes for applicable laws, contractual obligations, regulatory requirements or authoritative professional standards. Where an applicable legal, regulatory, contractual or authoritative requirement imposes a higher or different obligation, that requirement prevails.**

## How to read these standards

### The two governing instruments

The **Charter** governs status, hierarchy, priority, due process, interpretation, amendment, exceptions and consequence. The **Drafting Manual** governs normative language, one-obligation clauses, identifiers, defined terms, the twenty-five-element structure, thresholds, external-reference classification, prohibited patterns and the twenty-five audit questions. Where any doubt arises, those two documents prevail over anything below.

### Identifiers

Each standard is cited as **PCI-PML-STD-DD.NN**, where **DD** is the two-digit Body-of-Knowledge domain of primary anchorage and **NN** a two-digit sequence within that domain. Each process requirement is cited as **PCI-PML-STD-DD.NN-PR-NN**. Identifiers are stable and are the only citation form; page numbers are never used, because pagination changes. A withdrawn identifier is never reused.

**This edition rennumbers the whole set.** The previous edition used [PML-LAW-DD-NN](#). Element 25 of each standard records the identifier it supersedes, so a citation made under the old scheme can be resolved. The old [PML-LAW-01-02](#) carried the title *Human Decision Authority in Delivery*, which collided with the foundational standard **Human Decision Authority**; it is republished here as [PCI-PML-STD-01.02](#) under the distinguishing title **Reserved Delivery Decisions and the Named Human Decider**, and its obligation is narrowed to what a delivery leader must do that the foundational standard does not already require.

**The normative language, and the ISO mapping**

PCI uses **modern must-drafting**, exclusively.

| WORD            | MEANING IN A PCI STANDARD                                       |
|-----------------|-----------------------------------------------------------------|
| <b>must</b>     | Mandatory PCI professional requirement                          |
| <b>must not</b> | Prohibited practice; a breach                                   |
| <b>should</b>   | Recommended practice; a justified alternative may be acceptable |
| <b>may</b>      | Permission                                                      |
| <b>can</b>      | Capability or possibility — never permission                    |

**The ISO mapping, stated for readers who work to ISO/IEC drafting conventions.** In an ISO or IEC deliverable, requirements are expressed with the auxiliary verb that ISO/IEC Directives Part 2 reserves for requirements, and a reader trained on those conventions may misread **must** as an external constraint rather than as the requirement itself. In a PCI Standard, **must** is the requirement form, and it occupies exactly the place that ISO/IEC drafting gives to its requirement verb. That verb is deliberately not printed anywhere in this file, in any field, including in quotations of PCI's own earlier drafts, because it has been read as both obligation and futurity and PCI's convention excludes it. A draft containing it fails gate. This file was mechanically checked for it before publication; the check is recorded in the audit table.

**Consequence for the reader of the previous edition:** every rule in the 2026 v1.0 set that was drafted with that verb has been re-expressed here with **must**. No obligation was weakened in the translation; several were strengthened, and each such change is noted in element 25 of the standard concerned.

**One obligation per requirement**

Element 1 of every standard states **one** principal obligation. Everything else the standard requires appears as a numbered **process requirement** under element 5. Process requirements are Level 4 instruments in the Charter's hierarchy: **they are mandatory**, they are breached independently of one another, and they are cited

independently. Nothing in a note, a rationale, a bracket or a commentary sentence in this file creates an obligation; if such a sentence reads as one, that is a drafting defect and the Interpretation Panel corrects it.

### The relationship to the Foundational Standards

Every standard below sits under the **PCI Foundational Standards**, which bind every credential holder in all three credentials. This file cites them in the Charter §3 identifier form `PCI-FND-STD-NN`. The foundational set is a **fifteen-standard** set, and its subjects are listed below. This is the same concordance the rebuilt PCL-AI Certification Standard set uses, so that `PCI-FND-STD-NN` denotes the same standard in every credential's standard file:

| ID                          | SUBJECT                      | ID                          | SUBJECT                                 |
|-----------------------------|------------------------------|-----------------------------|-----------------------------------------|
| <code>PCI-FND-STD-01</code> | Professional accountability  | <code>PCI-FND-STD-09</code> | Confidentiality and approved technology |
| <code>PCI-FND-STD-02</code> | Evidence before assertion    | <code>PCI-FND-STD-10</code> | Competence and limitation               |
| <code>PCI-FND-STD-03</code> | Independent verification     | <code>PCI-FND-STD-11</code> | Escalation of material misstatement     |
| <code>PCI-FND-STD-04</code> | Human decision authority     | <code>PCI-FND-STD-12</code> | Record integrity                        |
| <code>PCI-FND-STD-05</code> | Transparent assumptions      | <code>PCI-FND-STD-13</code> | No silent override                      |
| <code>PCI-FND-STD-06</code> | Source and version integrity | <code>PCI-FND-STD-14</code> | Responsible AI                          |
| <code>PCI-FND-STD-07</code> | Data lineage                 | <code>PCI-FND-STD-15</code> | Correction duty                         |
| <code>PCI-FND-STD-08</code> | Conflict disclosure          |                             |                                         |

**The published foundational file carries these fifteen standards under these identifiers.** Every `PCI-FND-STD-NN` citation below therefore resolves directly against `PCI_FOUNDATIONAL_STANDARDS.md`, and the subject it carries there is the subject intended here. The superseded `PCI-LAW-F-NN` identifiers are recorded, for historical traceability only, in `STANDARDS_CONCORDANCE.md`; no live citation uses them.

**A certification standard must add.** Where a standard below governs the same subject as a foundational standard, it states what leading projects, programmes and portfolios specifically requires that the foundational standard does not — a named artefact, a named authority, a named record, a named test. A certification standard that only restates its parent is a defect and is listed as such in the audit table.

## External references

Every external reference is classified as exactly **one** Drafting Manual §6 category, and the categories are never combined:

| #  | CATEGORY                                    | USED IN THIS FILE FOR                                                                                             |
|----|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 1  | Applicable legislation or regulation        | <i>(not used — applicability is unknown for a global readership)</i>                                              |
| 2  | Authoritative financial-reporting standard  | <i>(not used — this credential states no accounting treatment)</i>                                                |
| 3  | International voluntary standard            | ISO, IEC and ISO/IEC deliverables                                                                                 |
| 4  | Contract framework                          | FIDIC, NEC                                                                                                        |
| 5  | Professional framework                      | PMBOK Guide, AACE material, the Scrum Guide                                                                       |
| 6  | Ethical code                                | PMI Code of Ethics and Professional Conduct                                                                       |
| 7  | Industry guidance                           | <i>(not used)</i>                                                                                                 |
| 8  | Voluntary environmental or social framework | <i>(not used)</i>                                                                                                 |
| 9  | PCI internal professional standard          | the Foundational Standards and this set                                                                           |
| 10 | Illustrative practice                       | the EU AI Act, the GDPR, the NIST AI RMF, the OECD AI Principles, the G20/OECD Principles of Corporate Governance |

Four rules govern every entry, and each of them has been breached once somewhere in this corpus:

- **ISO 21500, ISO 21502, ISO 21503, ISO 21504, ISO 21505, ISO 31000, ISO 10006 and ISO 9000 are voluntary international standards, and each of them is guidance.** Nothing can be certified against any of them. **ISO 9001, ISO/IEC 42001, ISO/IEC 27001 and ISO 45001 are certifiable requirements standards**, and are described as such where they appear. **ISO 45003 is guidance, not a requirements standard**, and nothing can be certified against it — a point that matters in [PCI-PML-STD-12.02](#), where the temptation to dress guidance as a compliance obligation is greatest.
- **The Scrum Guide is a voluntary framework. Adoption is the whole of its force.** It binds a team that has adopted it and nobody else, and it is not a standard.
- **The PMBOK Guide and AACE material are professional frameworks, never regulatory authority.**
- **The EU AI Act and the GDPR are real legislation binding within their own jurisdictions.** They are named here to illustrate a regulatory shape and are

relied on for no requirement in this set; the applicable position for any given project is a question for qualified local counsel.

**No clause number, article, edition or effective date is asserted in this file.** Editions live in the suite External-Reference Register (`../registries/EXTERNAL_AUTHORITIES.md`), dated, with their verification status; each element 17 entry gives the register reference (`EXT-0NN`) and the date the currency was checked. Where the register records an entry as not independently verified, the standard says so. **A reader relying on any instrument named below must obtain the current edition from its publisher.** Nothing here is reproduced from any of them.

**The two taxonomies.** The suite register uses its own seven-category scheme; the Drafting Manual uses ten. Where the two differ — most visibly for the Scrum Guide, which the register calls a voluntary framework and the Manual's scheme reaches through category 5 — **Manual §6 governs this file**, and the register reference is given so a reader can see both.

### No endorsement

Naming an external instrument means only that the instrument exists and bears on the topic under discussion. No standards body, professional institute, government, supervisory authority or employer has reviewed, approved, endorsed or accredited these standards, the PML-AI credential or PCI Global.

### British English and the volume's forms

*Organisation, organisational, realisation, prioritisation, judgement, programme.* Spelling follows the suite editorial charter (`../registries/TERMINOLOGY.md`).

### The due-process record, stated honestly

Charter §5 requires thirteen stages and requires the file to record which were performed and by whom, **including where a stage was performed with AI assistance rather than by a named human.** For this edition:

| STAGE                                         | PERFORMED | BY                                                                                                                 |
|-----------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------|
| 1 Problem definition · 2 Drafting instruction | Yes       | Named human commissioner; the drafting brief is the reconstruction specification for this edition                  |
| 3 Initial draft                               | Yes       | AI-assisted drafting against the Manual's structure                                                                |
| 5 Standards and legal-characterisation review | Partial   | Checked against the suite External-Reference Register; <b>no external publisher was contacted for this edition</b> |

| STAGE                                                                                                                                 | PERFORMED | BY                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 8 Scenario testing                                                                                                                    | Partial   | Every standard tested against a six-person internal project and a multi-partner national programme (element 11); other scenario classes untested |
| 9 Red-team challenge                                                                                                                  | Partial   | The technical-compliance and unverifiable-requirement classes were worked; the full red-team is outstanding                                      |
| 10 Revision                                                                                                                           | Yes       | Findings recorded in the audit table at the end of this file                                                                                     |
| 4 Technical review · 6 Practitioner consultation · 7 Impact assessment · 11 Approval · 12 Publication · 13 Post-implementation review | No        | Outstanding                                                                                                                                      |

**This set is therefore a draft pending approval.** Element 25 of every standard says so. PCI is a small private certifier and this process must never be represented as equivalent in scale, independence or authority to that of a public standard-setter.

### The suite principle

#### CAUTION

**AI proposes; the professional verifies, decides and remains accountable.**

## Definitions

These definitions decide compliance. Every one is written as a test a reader can apply — what makes the thing what it is, measured against what, decided by whom — and none defines a term by itself. Where the suite terminology audit (`../registries/TERMINOLOGY_AUDIT.md`) records that a term carries genuinely different professional meanings across the three books, **both senses survive here and are flagged**; collapsing them would make the standards wrong.

**Where a term is also defined in the Foundational Standards.** *material, independent, evidence, competent reviewer, decision owner* and *escalation threshold* are also defined, at D-01 to D-30, in `PCI_FOUNDATIONAL_STANDARDS.md`. **They now carry the same wording in both places.** Each was reconciled to the canonical definition recorded in `PCI_STANDARDS_DEFINITIONS_REGISTER.md`, which also records what this volume previously said and why the change was made. The terms this set flags as carrying two genuine professional senses — *baseline, sponsor, verification,*

*governance* — are **not** collapsed by that reconciliation; the register carries both senses of each, with the context each belongs to, because collapsing a real collision would make this volume wrong.

Three reading rules remain, and they still matter for any term this reconciliation did not reach. First, **where a foundational standard states its own defined term by its D-NN number, that definition governs that foundational obligation**, and nothing here narrows it. Second, **where a definition here and a foundational definition both bear on the same act, the one producing the wider obligation governs** — Charter §4 states that a PCI Standard never lowers an obligation. Third, a term defined here and not there is a PML-AI term and carries only the sense given here.

**How element 21 samples are drawn.** Where a standard's element 21 tests "a stated sample", the sample is selected by the reviewer performing the test, not by the credential holder whose work is under review, and the reviewer records the basis of selection.

### Terms that decide compliance

**acceptance** — A named authority's recorded decision that a deliverable meets the acceptance criteria that were set and version-identified before the deliverable was produced. An acceptance record states what was tested, against which version of the criteria, what did not conform, and which nonconformities are carried forward with an owner and a date. **Silence, use of the deliverable, payment against it and the expiry of a review period are not acceptance.**

**baseline** — *Context flag: two senses, both used in this set, and both are correct professional usage.*

- **Control baseline** — the approved, version-controlled plan (scope, schedule or cost) against which performance is measured. This is the sense inherited binding from the suite style spine and is the sense meant wherever this set writes *baseline* unqualified in a planning, cost or change context.
- **Benefits baseline** — the measured pre-change position from which a benefit is measured, captured before the change and never reconstructed after it.

Where both senses could be meant in one sentence, this set writes the qualified form. A standard that says *baseline* in a benefits context and means the plan is a defect.

**benefit** — A measurable improvement resulting from an outcome, owned by a named individual outside the delivery team, measured against a benefits baseline. A benefit is what the organisation gets; *value* is the worth it attaches to that. **Neither is the Earned Value of a work package**, which is a budgeted money amount and a measure of work performed, not of worth delivered.

**competent reviewer** — (*Canonical* — D-04.) A named individual who, in relation to a particular item, satisfies all of: (a) their competence in the subject matter is

evidenced by a qualification, an assessed competence record or documented experience of comparable work, **recorded for that class of review before the review begins**; (b) they are able to state what would make the item wrong and which method would detect that error; and (c) they hold the technical knowledge and delivery experience to perform the verification method the standard requires and to reach a conclusion on the matter **without relying on the preparer's explanation of it**. Competence is recorded against the class of work, not against seniority, job title or availability. **Independence is not a limb of competence**: the requirement that the reviewer did not prepare, direct, specify or approve the thing being reviewed is *independence*, defined below, and it is imposed by each standard's element 10 rather than by this definition. The two were previously folded together here, which made [PCI-FND-STD-10](#) element 12's supervised-acquisition exception unusable, because that exception needs a reviewer who is competent and is precisely not independent — a supervisor. No standard loses an independence requirement by the change: every element 10 that required one still states it.

**decision owner** — (*Canonical — D-08.*) The single named individual holding authority, under the applicable governance arrangement or the organisation's documented delegation schedule, to approve, reject, amend, defer or withhold a specified decision; who bears its consequence; and who answers for it afterwards. The accountability is held by one person, is not delegable, and is recorded before the output is relied upon. **A committee is not a decision owner**. Where a body decides collectively, the decision owner is the named chair or the named authority the delegation schedule assigns, and the record names that person. A decision owner is always a natural person: "the team", "management", "the business", "the sponsor", "the lenders", "the organisation", a role held by no named individual, a system, a model and a vendor are never decision owners.

**detriment** — Any of the following acts occurring to a person after they raised a concern: removal from, or reduction of, a role, scope, responsibility, grade or pay; an adverse change to a performance rating, reference, bonus or renewal decision; exclusion from meetings, distributions or work they previously took part in; a reassignment they did not request; non-renewal or early termination of an assignment, secondment or contract; or a disciplinary step. **Detriment is defined by these observable acts, and not by how the person felt about them** — which is what makes an allegation of detriment capable of being investigated and either upheld or dismissed on evidence.

**evidence** — (*Canonical — D-11.*) A dated record that exists independently of the assertion it supports, attributable to an identified author or issuing system, version-identified where it has a version, and retained so that a person other than the author of the assertion — a person who was not present — can retrieve it, examine it and reach the same conclusion **without asking that author**, determining what was known, by whom, and when. The following are **not** evidence for the purposes of this set: an output of an AI system that does not identify the source of what it asserts; an AI-generated summary of a record, as evidence of the underlying fact — the



underlying record is; a statement that a system, model or tool produced a figure, unaccompanied by the inputs and the method; a restatement of the assertion in a second document by the same author; a preparer's own statement offered in support of their own assertion; a recollection; an undated summary or extract; an unattributed file; an unversioned extract or working copy; a screenshot with no source reference; and a dashboard state that cannot be reproduced.

**escalation threshold** — (*Canonical* — D-10.) The escalation threshold for a matter is reached at the earliest moment any of the following becomes true: the matter is *material*; it creates a risk to the safety of a person; it would change, or would have changed, a decision already taken or about to be taken; it affects an output already issued outside the credential holder's own organisation; it affects a contractual, regulatory, tax or financial-reporting position; or a documented value, condition or event stated in the organisation's delegation schedule is met. On reaching it the matter must be passed to a named higher authority within the time the threshold states. **Any documented value in the delegation schedule is additional to those six triggers and never in place of them**; a matter that reaches the threshold requires escalation under *PCI-FND-STD-11* whether or not the delegation schedule enumerated it. A threshold with no stated destination, or with no stated time, is an incomplete threshold and it is an aspiration until it is completed — but **the absence of a documented threshold does not remove an escalation duty**: where the delegation schedule names no destination, the matter goes to the next authority above the *decision owner* for it, and where it states no time, the foundational period at D-20 applies — one working day where the matter creates a risk to the safety of a person or an ongoing financial loss, five working days otherwise, running from the moment the credential holder first knows or suspects the matter. Reaching the threshold starts that period.

**gate** — A continuation decision taken by a named authority at a defined point in the delivery lifecycle, against criteria published before the evidence supporting the decision was assembled, and carrying the power to stop, hold, redirect or authorise the next commitment. A review that cannot stop the work is not a gate.

**independent** — (*Canonical* — D-12.) Of a person or function, **in relation to a specified matter**, where all of the following are true: (a) they did not perform the act, prepare the item or any part of it, or direct, specify or approve it; (b) they did not select, build or configure the tool, model or AI system that produced it; (c) they hold no conflict of interest in its outcome and no financial interest in the matter or in a party to it; (d) they receive no fee, bonus, continuing mandate, success payment or other benefit that varies with the conclusion reached, and their remuneration, appraisal or continuation is not determined by the outcome the item supports; (e) they are not accountable for the outcome it reports on; and (f) they are not in the reporting line of the person accountable for it *for the purpose of that matter*, and do not report to the preparer in respect of the work under review. Reporting to the preparer's line manager on unrelated work does not by itself defeat independence; being appraised on the outcome does. **Independence is a property of a relation-**

**ship to a specific matter, not a job title** — which is why a permanently titled "independent assurance" function loses independence over any artefact it helped produce.

**mandatory precondition** — A transition, gate or release condition recorded **met or not met**, with its approving authority named and dated, closed by the authority that owns it rather than by the project. A mandatory precondition carries no probability, no weighting, no score and no admission to economic trade against the cost of delay.

**material** — (*Canonical* — D-15.) A matter, item, error, omission, variance or difference is *material* if any of the following is true: (a) had the decision-maker known it at the time, it could have changed a decision within the scope of the work — its substance, its timing, its conditions, or the authority at which it had to be taken; (b) it changes a reported figure by more than the quantified tolerance published for that figure; (c) it affects a contractual, regulatory, tax or financial-reporting position; (d) it affects the safety of a person; (e) it affects a party's reliance; or (f) it meets the organisation's published materiality criteria. Materiality is assessed **against the decision**, not against a fixed sum, and it is judged twice: on the item alone, and on the accumulation of items of the same kind since the test was last applied. **In PML-AI work, limb (b) is measured by the documented threshold:** where the organisation's governance sets a documented financial, schedule or exposure threshold for a class of decision, that threshold applies to matters that are purely financial, schedule or exposure in character. **A matter bearing on safety, legality, a licence or permission, a statutory duty, or the truth of a statement made to a decision-maker is material irrespective of size**, and no documented threshold reduces it. **PCI sets no percentage.**

**sponsor** — *Context flag: the delivery and governance sense.* The single named individual accountable for the project's business outcome and mandate, who owns the business case, holds the authority reserved above the project leader, and answers for the benefits the project exists to enable. **The project-finance sense of "sponsor" — an equity investor promoting a project — belongs to the PFL-AI volume and is not used anywhere in this set.**

**approved** — (*Canonical* — D-29.) A decision, document, figure or version is *approved* where the person holding authority for that decision under the applicable governance arrangement or the documented delegation schedule has given assent identifiably, recording the date, the version assented to and the scope of the assent. Silence, absence of objection, unrecorded verbal assent, assent by a person outside their recorded authority, and assent recorded after the item was used are not approval. Distinguish from *acceptance* above, which is a named authority's recorded decision that a deliverable meets version-identified criteria.

**current** — (*Canonical* — D-30.) A record, figure, extract, document or version is *current* where it is the latest version issued by the system or authority that owns it as at the artefact's stated cut-off or, where it states none, as at its issue date — and where

its version identifier and its extraction date and time are recorded on or with the artefact. A record whose version cannot be identified is not current, whatever its age.

**material AI assistance** — (*Canonical — carried to the whole corpus by [PCI\\_STANDARDS\\_DEFINITIONS\\_REGISTER.md](#).)* AI assistance in producing an artefact is *material* where removing the AI-generated contribution would change a figure in the artefact by more than the applicable materiality measurement, or would change a recommendation, a classification that affects entitlement, coding, ranking or eligibility, or a stated conclusion. *Material AI contribution* means the same thing. Volume of use, licence cost and whether a human edited the artefact afterwards are irrelevant to the test. This set made compliance turn on the term at [PCI-PML-STD-14.02](#) element 21 without defining it; the definition is supplied here so the test can be applied.

### Supporting terms used in obligations

**abstention** — The interested party taking no part in the discussion or the decision on an item, minuted with the name of whoever decided in their place. An abstention that is not minuted is indistinguishable from participation when the file is read later.

**assurance** — Independent examination providing confidence that a control, process, plan or deliverable is what it is claimed to be. Distinguish from *verification* below.

**assurance capture** — An assurance function providing assurance on work it prepared, directed or specified, and therefore unable to challenge it. Capture is the least visible assurance failure, because the product still looks independent.

**concern** — A statement by any person working on or for a project that something in its delivery, conduct, safety, legality or reporting is wrong, is going wrong, or is being misrepresented. A concern is a concern whether or not it turns out to be correct; the correctness is what the response determines.

**concern route** — The named, published mechanism by which a concern reaches a named recipient. A route is a **bypass route** in relation to a person if that person is neither its recipient, nor able to see the concern before the recipient does, nor able to determine what happens to it.

**conflict of interest** — (*Canonical — [D-05](#), where the term is written conflict and means the same thing.*) A relationship, interest or duty of any of the kinds [D-05](#) lists, held by the person or by a member of their household or a connected person, which bears on a matter they are being asked to decide, advise on or assure — a financial stake or interest in a party or in the outcome; employment, office, directorship or partnership with a party; a personal, family or household connection; a gift, hospitality or benefit above the recorded threshold; a past or prospective engagement with a party; authorship of, or a personal interest in, the thing being assured; a future role that depends on the outcome; and an interest in a tool, model, vendor or AI system being selected, priced, assessed or relied upon. An **interest** is anything capable of affecting, or of appearing to affect, that person's judgement on

the matter. **The appearance is part of the definition, not a softening of it.** A conflict of interest is a structural fact about a role, not a lapse of character. The list is closed for the purposes of the disclosure duty: every conflict within it must be disclosed and there is no threshold.

**decision record** — The versioned, attributable log entry that converts a decision into an institutional fact. Its fields are the decision, the named decision owner, the date, the options considered, the reasoning, any recorded dissent, the declared interests or the nil return, and a **versioned reference to the information relied on**.

**delegation schedule** — The organisation's documented statement of which named role may take which class of decision up to which documented limit, what aggregates across related decisions, and where a decision above the limit goes.

**dependency** — A commitment by one party to give another party a defined thing by a defined date, recorded with the giver, the receiver, the thing, the date, the confidence and the consequence of a breach. **A dependency with no named owner on the giving side is not a dependency.**

**discretionary condition** — A transition or gate condition that genuinely admits degree at the decision date, is genuinely uncertain, and is genuinely tradeable against the cost of waiting. Its assessment carries a probability. It is the complement of a mandatory precondition, and the two never appear in the same block.

**gate block** — The set of mandatory preconditions sitting above a readiness assessment. While any item in it is recorded not met, the only decision available is hold.

**governance** — *Context flag.* **Project governance** (this set's default sense): the decision rights, accountabilities and information flows through which an organisation directs and controls a project, programme or portfolio. **Data governance**: ownership, definitions, access and traceability of data. This set writes *data governance* in full, always.

**nil return** — A positive record that a person had nothing to declare on a matter, so that silence in the file is evidence rather than an absence of evidence.

**product owner** — The single individual accountable for the order of a backlog within a value envelope set by the sponsor. Where the ordering right is exercised elsewhere, that other body is the product owner in fact and the record must say so.

**structural interest** — An interest touching the substance of a role rather than a single item. It is managed by changing the role or moving the work, never by a declaration followed by the decision being taken anyway.

**traceability** — A maintained, bidirectional link from each approved requirement to its source, to the design or backlog item that satisfies it, to the test or acceptance criterion that proves it, and to the deliverable that carries it — such that a reader can start at any one of those and reach the others without asking the author.

**verification** — *Context flag: two senses, both used in this set.*

- **V&V verification** — the check that what was built is what was specified. (Its partner, *validation*, asks whether what was built produces the outcome that was needed.)
- **AI verification** — a named human's check of machine output against source before reliance on it. This is the sense carried by the suite principle and by element 16 of every standard below. **The test is the canonical *verified at D-26*:** a named *competent reviewer* applies one of the eight admissible methods — independent recomputation, source tracing, clause-to-summary comparison, sampling on a stated basis, reconciliation, boundary testing, sensitivity analysis, or named expert judgement recorded with its reasoning — against *evidence*, and records the method, the source or population tested and its selection basis where a sample was used, the inputs, the scope, the date, the result, and every difference found with its resolution. Reading an output and finding it plausible is not verification.

Where a sentence could carry either sense, this set writes the qualified form. **The two senses are a declared collision, not a defect:** collapsing V&V verification into AI verification would make this set's delivery-assurance standards wrong. Both are recorded at `PCI_STANDARDS_DEFINITIONS_REGISTER.md` §4 with the context each belongs to, and at `../registries/TERMINOLOGY_AUDIT.md` Issue 9.

## Domain 1 — The Project Leadership Profession

### PCI STANDARD

#### PCI-PML-STD-01.01

### Leadership Accountability for Delivery Decisions

1. **Normative requirement.** A credential holder who leads a project, programme or portfolio must remain personally accountable for every delivery decision taken under their authority, including a decision informed, drafted, modelled or recommended by an AI system, an adviser, a supplier or a committee.
2. **Purpose.** Delivery leadership is an accountable role, not a coordinating one. Every other standard in this set assumes there is one identifiable person who answers for the decision. Where accountability is diffused across a governance diagram, a delivery partner and an analytics platform, nobody answers and the project drifts unowned until it fails visibly.
3. **Scope.** Every credential holder exercising project, programme or portfolio leadership, in every delivery model — predictive, adaptive or hybrid — whether the work is delivered in-house, through partners or through an

automated toolchain. It applies to preparation, recommendation, approval and assurance of delivery decisions, and to decisions taken in the leader's name by others.

**4. Defined terms.** *decision owner · decision record · delegation schedule · material · evidence · sponsor.* Additionally, **delivery decision** means a decision that changes what is built, when it is delivered, what it costs, what risk is accepted, who does the work, or what is represented to a decision-maker about any of those.

**5. Required actions — process requirements.** Each is mandatory and is breached independently.

- **PCI-PML-STD-01.01-PR-01 — Named acceptance.** The credential holder must accept the leadership accountability in a written instrument — charter, appointment letter or terms of reference — that names the individual, and must not begin exercising leadership authority before that instrument exists.
- **PCI-PML-STD-01.01-PR-02 — Known reservation boundary.** The credential holder must hold a current, dated statement of which decisions are personally reserved to them and which sit above them, drawn from the delegation schedule, and must re-read it at each gate.
- **PCI-PML-STD-01.01-PR-03 — Decision recording.** The credential holder must ensure that every material delivery decision taken under their authority produces a decision record carrying all of its defined fields, including the versioned reference to the information relied on.
- **PCI-PML-STD-01.01-PR-04 — Correction on discovery.** On discovering that a decision issued under their authority rested on a material error, the credential holder must record the error, notify every person who relied on the decision, and state what the corrected position is, within the notification time the delegation schedule sets for that decision class.

**6. Prohibited actions.** Presenting a committee, a model, a methodology, a framework or a supplier as the accountable party for a decision. Accepting a leadership title without the corresponding decision rights while allowing others to believe accountability is discharged. Signing a charter one does not intend to honour. Permitting a decision to be issued in one's name without one's knowledge and allowing the record to stand uncorrected once that is known.

**7. Required evidence.** The signed charter or appointment naming the accountable leader; the current delegation schedule; decision-record entries carrying the named owner, options, reasoning, recorded dissent and versioned information reference; correction and notification records for decisions later found defective.

**8. Responsible role.** The named credential holder appointed as project, programme or portfolio leader. Not "the team", not "the PMO", not "management".

**9. Approval authority.** The sponsor approves the leader's appointment and the reservation boundary. The governing body approves the delegation schedule the boundary is drawn from. Neither may be approved by the leader.

**10. Independence requirement.** Not applicable to the decision itself, because leadership accountability is by definition non-independent of delivery; independence attaches instead to the assurance of these records, which [PCI-PML-STD-01.03-PR-06](#) governs.

**11. Materiality or threshold.** This standard sets no number. The materiality of a delivery decision is decided by the definition of *material* above, applied against the organisation's documented delegation schedule. The credential holder must apply the documented thresholds; where the organisation has none for a decision class, the credential holder must record that gap and treat every decision in that class as reserved until a threshold is documented. *Six-person internal project:* the charter is one page, the delegation schedule is a five-row table, and the decision record is a shared log — the obligation is satisfied in minutes a week. *Multi-partner national programme:* the same obligations attach at each accountable tier, and the schedule states the aggregation rule across partners so a decision cannot be split below a limit.

**12. Exception and waiver.** No exception is permitted to the principal obligation: accountability cannot be waived, transferred or bounded by contract, disclaimer or tool configuration. An exception to [PR-01](#) — beginning to act before the written instrument exists — may be approved by the sponsor for a stated period not exceeding the next gate, on written justification, with the instrument's absence recorded as an open item and reported to the governing body.

**13. Escalation trigger.** Any material delivery decision with no identifiable accountable person; any instruction to act beyond or outside the recorded authority; discovery that a decision was issued in the leader's name without their knowledge or review.

**14. AI application.** AI may assemble decision packages, summarise options and trade-offs, maintain and age the decision record, flag entries missing an owner, a date, a reasoning field or a versioned information reference, and cluster the log to detect decisions being taken twice.

**15. AI prohibition.** An AI system must not hold, share or discharge accountability, be recorded as the decision maker or approver, be cited as the reason a decision was taken, or author a decision record entry wholesale. No configuration, disclaimer, procurement term or contract removes the credential holder's accountability.



**16. AI verification. Source tracing plus named approval.** Before a decision is taken on AI-assembled material, the credential holder must trace each figure and each stated constraint in the package to its cited source document and version, and must record their own name against the decision. Quarterly, the credential holder must apply **sampling with a stated basis** to the decision record — a stated number of entries drawn across the period — and confirm for each sampled entry that a named human owner, a reasoning field and a versioned information reference are present.

#### **17. External reference.**

- **ISO** · *ISO 21502, Project, programme and portfolio management — Guidance on project management* · relied on for: the existence of a distribution of accountabilities between a governing body, a sponsor and a project manager · edition not asserted here; recorded at **EXT-028** in the suite External-Reference Register · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: guidance only, voluntary unless imported by contract or regulation; nothing is certifiable against it; it is not the source of this standard's obligation.
- **ISO** · *ISO 21505, Project, programme and portfolio management — Guidance on governance* · relied on for: the location of delivery accountability inside an organisation's governance arrangements · **EXT-032**, recorded as **not independently verified — verify current requirements** · **Manual §6 category 3** · limitation: as above, and the register's verification status is open.
- **ISO/IEC** · *ISO/IEC 42001, Information technology — Artificial intelligence — Management system* · relied on for: the existence of an expectation that organisational roles and responsibilities for AI use are assigned · **EXT-021** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: a certifiable management-system standard, voluntary unless adopted; certification concerns a management system, not any individual decision.
- **OECD** · *Recommendation of the Council on Artificial Intelligence (the OECD AI Principles)* · relied on for: the proposition that human actors remain accountable for AI-influenced outcomes · **EXT-081** · **Manual §6 category 10 — illustrative practice** · currency checked 2026-08-03 · limitation: a Council Recommendation, not binding law even on adherents; never legislation.

**18. Jurisdictional caution.** Directors' duties, statutory delivery, safety, planning and licensing roles, employment law and the law of professional negligence determine **legal** responsibility independently of this profes-



sional requirement, and can place it on a person other than the credential holder. Obtain local legal advice on the statutory roles attaching to a specific project and entity.

**19. Related PCI Standards.** [PCI-FND-STD-01](#); [PCI-FND-STD-02](#); [PCI-FND-STD-04](#); [PCI-FND-STD-12](#); [PCI-FND-STD-15](#); [PCI-PML-STD-01.02](#); [PCI-PML-STD-03.02](#); [PCI-PML-STD-03.04](#).

**20. Related Body of Knowledge content.** PML-AI · Domain 1 · KA 1.2 The project leader's accountability · topics 1.2.1 accountability and responsibility, 1.2.2 the obligation set, 1.2.3 the professional standard of care; KA 1.4 topic 1.4.3 the leader's AI accountability. Also Domain 3 KA 3.2; Domain 14 KA 14.4.

**21. Compliance test.** Compliance is demonstrated when a reviewer can, for a stated sample of material delivery decisions taken in the period: (a) name the accountable individual from the charter or appointment instrument, without asking anyone; (b) find that individual's name in the decision record entry as owner; (c) open the versioned information reference in that entry and retrieve the document version cited; (d) match the decision's value or exposure against the delegation schedule and confirm it was taken at or below the authority the schedule assigns to that named individual; and (e) for any decision later corrected, find a correction record and a notification record dated within the schedule's notification time. A sample in which any entry fails (a), (b) or (d) fails the test.

**22. Breach indicators.** Decision entries whose owner field names a committee, a function or a tool. A charter with no named individual. Two people each stating the other decided. Corrections issued silently. A leader unable, when asked, to state the basis of a decision issued in their name within the previous quarter. A delegation schedule dated before the current organisational structure.

**23. Consequence within PCI authority.** Correction required; additional review; escalation; examination failure on the associated competency; ethics review; certification investigation; suspension or withdrawal of the PML-AI credential — each subject to due process and a right of appeal.

**24. Examination application.** Scenario judgement: a delivery decision goes wrong and responsibility is contested between a leader, a steering committee, a delivery partner and an analytics tool; the candidate locates accountability and states the next required act. Evidence selection: choosing, from five artefacts, the two that would establish who decided. Escalation decision: an instruction to act outside recorded authority.

**25. Version and status.** Version 2.0 · **not yet approved** (Charter §5 Stage 11 outstanding) · effective on approval · supersedes [PML-LAW-01-01 v1.0](#). Amendment note: renumbered; restructured to the twenty-five-element form;

legislative drafting removed; the former single rule split into one principal obligation and four process requirements; compliance test replaced with a performable five-part test.

#### PCI STANDARD

##### PCI-PML-STD-01.02

### Reserved Delivery Decisions and the Named Human Decider

**1. Normative requirement.** A credential holder must ensure that every delivery decision in a class the organisation has reserved to a human decider is taken by the named individual holding that authority, on evidence that individual has examined, and must not permit such a decision to be executed on the output of an automated or AI system without that examination.

**2. Purpose.** PCI-FND-STD-04 establishes that a human decides. This standard addresses what is specific to delivery: automation in delivery does not usually announce itself as a decision. A prioritisation engine reorders a backlog, a scheduling tool re-sequences a network, a resource optimiser reassigns people, a risk model closes a risk — and each of those is a decision that has already taken effect by the time a human sees it. The failure this standard prevents is **decision by default**: nobody decided, the system acted, and the record shows an approval that was never an examination.

**3. Scope.** Every credential holder in a delivery leadership or delivery governance role, for every decision class the organisation reserves to a human. It applies to decisions executed by scheduling, prioritisation, resource-allocation, risk-scoring, procurement-scoring and reporting automation, and to adaptive delivery where the ordering right sits with a product owner. It governs approval and execution, not analysis.

**4. Defined terms.** *decision owner · material · evidence · delegation schedule · AI verification.* Additionally, **reserved decision class** means a class of decision the organisation's delegation schedule states must be taken by a named human; **executed automatically** means taking effect in a system of record without a named human having recorded an examination first.

#### 5. Required actions — process requirements.

- **PCI-PML-STD-01.02-PR-01 — The reserved-class list exists.** The credential holder must maintain a dated list of the decision classes reserved to a human decider on their project, approved by the sponsor, covering at minimum: baseline change, scope acceptance, risk acceptance, contingency release, supplier selection and award, release or go-live, and any decision affecting safety, a licence, a permission or a statutory duty.

- **PCI-PML-STD-01.02-PR-02 — Automation inventory against the list.** The credential holder must maintain a dated inventory of the automated and AI systems in use on the project that are capable of changing anything in a reserved class, and must record for each whether its output takes effect automatically or on a named human's recorded action.
- **PCI-PML-STD-01.02-PR-03 — No automatic effect in a reserved class.** Where the inventory shows a system capable of taking effect automatically in a reserved class, the credential holder must have that capability disabled or gated, and must record the date it was disabled or gated and by whom.
- **PCI-PML-STD-01.02-PR-04 — Examination before decision.** For each reserved decision taken on AI-assisted material, the named decider must record what they examined, which version of it, and what they changed or confirmed — a record distinct from the approval itself.

**6. Prohibited actions.** Recording an approval that was a click on a system-generated recommendation with no examination record. Configuring or accepting a tool that changes a reserved-class item without a named human action. Describing an automated re-sequencing, re-prioritisation or reallocation as "the tool's output" when it has taken effect. Delegating a reserved decision to a supplier's system because the supplier holds the tool.

**7. Required evidence.** The approved reserved-class list with its date; the automation inventory with its per-system effect column; configuration records or screenshots with source references showing automatic effect disabled or gated, dated and attributed; examination records paired with approval records for a sample of reserved decisions.

**8. Responsible role.** The named credential holder leading the project, for maintaining [PR-01](#) to [PR-03](#). The named decider recorded in the delegation schedule for the class, for [PR-04](#).

**9. Approval authority.** The sponsor approves the reserved-class list. The governing body approves any addition to, or removal from, that list. A supplier or tool vendor must not approve either.

**10. Independence requirement.** The named decider must be independent of the configuration of the AI or automation system whose output they are examining, in the sense that they did not set its parameters, weights or thresholds for the decision in question. Where the same person unavoidably does both on a small project, [PCI-PML-STD-01.03-PR-05](#) applies and the arrangement is recorded and reported to the sponsor.

**11. Materiality or threshold.** This standard sets no number. The reserved-class list is the threshold instrument, and this standard requires that it exists, is approved, is dated, and is applied. The listed minimum classes above are the

floor; the organisation's governance sets everything else. *Six-person internal project*: the list is seven lines and the inventory is three rows — most such projects run one scheduling tool, one issue tracker and one spreadsheet with a model in it. *Multi-partner national programme*: each partner maintains an inventory for the systems it operates, the programme maintains the consolidated list, and the interface is a stated obligation in the partner agreement rather than an assumption.

**12. Exception and waiver.** An exception permitting automatic effect in a reserved class may be approved only by the governing body, only for a named system, only for a stated period not exceeding one gate interval, on a written justification stating the compensating control (at minimum: a daily exception report to the named decider naming every change the system made). Every such exception is reported to the sponsor and to assurance.

**13. Escalation trigger.** Discovery that a reserved-class item changed with no named human action. Any request to enable automatic effect in a reserved class. A supplier's refusal or inability to disclose whether its system takes effect automatically.

**14. AI application.** AI may generate options, rank them against stated criteria, model consequences, prepare the decision package, and detect and flag reserved-class changes that occurred with no paired examination record.

**15. AI prohibition.** An AI system must not take, approve, sign, certify, waive or authorise a reserved decision; must not be recorded as the decider; and must not be given effect in a reserved class by default configuration. An AI system's confidence score must not stand in for a human examination.

**16. AI verification. Boundary testing plus independent recomputation.** For each reserved decision class exposed to an AI or optimisation system, the named decider must test the system's behaviour at the boundary of the class — submit an input just inside and just outside the reserved threshold and confirm the system stops at the boundary — and must record the test date and result at least once per gate interval and after any configuration change. Where the system produces a number that carries the decision (a rank, a score, a date, a cost), the decider must recompute at least one such number by an independent method and record the comparison.

## **17. External reference.**

- **ISO/IEC · ISO/IEC 38507, Governance implications of the use of artificial intelligence by organizations** · relied on for: the existence of governing-body-level questions about where AI is permitted to act · **EXT-037** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: guidance aimed at governing bodies, not at practitioners; voluntary; not certifiable.

- **NIST (US Department of Commerce)** · *Artificial Intelligence Risk Management Framework (AI RMF 1.0)* · relied on for: the existence of a govern-map-measure-manage shape for AI risk · **EXT-080** · **Manual §6 category 10 — illustrative practice** · currency checked 2026-08-03 · limitation: NIST states it is voluntary, rights-preserving and non-sector-specific; **it is not a standard and not a regulation.**
- **European Union** · *Regulation (EU) 2024/1689 (the AI Act)* · relied on for: illustrating a risk-tiered regulatory shape that treats human oversight as a design question · **EXT-100** · **Manual §6 category 1 — applicable legislation or regulation** · currency checked 2026-08-03 · limitation: **binding legislation within the European Union only**, named here only to illustrate the shape; it is relied on for no requirement in this standard, and whether it applies to a given project is a question for qualified local counsel.

**18. Jurisdictional caution.** Whether a particular automated decision is lawful, whether it triggers a transparency, human-review or impact-assessment duty, and what disclosure is owed to an affected person are jurisdiction-specific questions. Obtain local legal advice before deploying automation that affects people's work allocation, pay, assessment or employment.

**19. Related PCI Standards.** [PCI-FND-STD-04](#); [PCI-FND-STD-03](#); [PCI-FND-STD-14](#); [PCI-FND-STD-13](#); [PCI-PML-STD-01.01](#); [PCI-PML-STD-03.02](#); [PCI-PML-STD-13.02](#); [PCI-PML-STD-14.02](#).

**20. Related Body of Knowledge content.** PML-AI · Domain 1 · KA 1.4 Ethics and the responsible use of AI · topics 1.4.2 the PCI responsible-AI principle, 1.4.3 the leader's AI accountability. Also Domain 14 KA 14.3 and KA 14.4; Domain 13 KA 13.1 topic 13.1.2 product ownership as a decision right.

**21. Compliance test.** Compliance is demonstrated when a reviewer can: (a) obtain the dated, sponsor-approved reserved-class list and confirm it contains the seven minimum classes; (b) obtain the automation inventory and confirm every system on the project capable of changing a reserved-class item appears on it; (c) for each such system, find either a dated record that automatic effect is disabled or gated, or a governing-body exception within its stated period; (d) draw a stated sample of reserved decisions from the period and find, for each, an examination record naming the version examined and dated **at or before** the approval record; and (e) find a boundary-test record for each exposed class dated within the current gate interval. Any reserved-class change in the systems of record with no paired named-human action fails the test outright.

**22. Breach indicators.** Approval timestamps that cluster within seconds of recommendation timestamps. An automation inventory that has not changed while three new tools were procured. A scheduling tool whose re-sequencing appears in the published plan with no change record. Suppliers reporting "the system decided". A reserved-class list that names no owner and no date.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure on the associated competency; ethics review; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** AI-verification case: a resource optimiser has reassigned four people across two workstreams overnight and the plan now shows a new critical path; the candidate states whether a decision has been taken, by whom, and what must happen next. Evidence selection: which artefact proves a human examined the recommendation rather than approved it.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-01-02](#) v1.0. Amendment note: **retitled** to remove the collision with the foundational standard *Human Decision Authority*; obligation narrowed to what delivery leadership adds — the reserved-class list, the automation inventory, the prohibition on automatic effect and the examination record — so the standard adds rather than restates; renumbered and restructured; legislative drafting removed.

## PCI STANDARD

### PCI-PML-STD-01.03

## Interests, Abstention and Assurance Independence

**1. Normative requirement.** A credential holder must not take, or participate in taking, a decision, recommendation, evaluation or assurance opinion on a matter in which they hold a conflict of interest.

**2. Purpose.** Until the current edition of the PML-AI manuscript set, the phrase *conflict of interest* appeared nowhere in it. The volume taught gates, steering committees, weighted scoring, supplier selection and three lines of assurance, and none of that machinery carried a rule about **who is barred from deciding**. That gap is now closed in the manuscript at Domain 1, KA 1.2.2a; this standard is its enforceable counterpart. The failure it prevents is the one the machinery cannot see: a decision that is procedurally perfect — criteria published, scores recorded, minutes taken — and taken by a person whose judgement was engaged elsewhere. Its hardest case is **assurance capture**, where the second line assures a plan it helped write and the product still looks independent.

**3. Scope.** Every credential holder, on every matter they decide, advise on, evaluate, score or assure: gate decisions, steering and change-board decisions, option appraisals and scoring models, tender evaluation and supplier selection and award, claims and disputes, assurance reviews and opinions, recruitment and assignment of people to roles they will later assess, and the setting or approval of thresholds. It applies to preparation, review, recommendation, approval and assurance alike.

**4. Defined terms.** *conflict of interest · structural interest · abstention · nil return · independent · assurance · assurance capture · decision record · competent reviewer · material.*

#### **5. Required actions — process requirements.**

- **PCI-PML-STD-01.03-PR-01 — Identify at appointment and at each convening.** The credential holder must ask themselves, in writing, at appointment to any role and again as each gate, steering meeting, evaluation panel, change board or assurance review is convened, whether they hold an interest in what is about to be decided. A single question at induction does not satisfy this requirement, because interests are acquired after induction.
- **PCI-PML-STD-01.03-PR-02 — Declare in writing before the item is taken.** A declaration must reach the chair in writing **before** the item is opened. A declaration made during or after the discussion does not satisfy this requirement; it is an explanation.
- **PCI-PML-STD-01.03-PR-03 — Record, including nil returns.** The decision record for every gate, steering, change-board, evaluation-panel and assurance decision must carry an interests field, and that field must carry either the declarations made or a positive **nil return** for each participant. A blank field does not satisfy this requirement.
- **PCI-PML-STD-01.03-PR-04 — Abstain, and name who decided instead.** The interested party must take no part in the discussion or the decision, and the minute must name the individual who decided in their place. An abstention that is not minuted does not satisfy this requirement.
- **PCI-PML-STD-01.03-PR-05 — Structural interests change the role.** Where an interest touches the substance of a role rather than a single item, the credential holder must move the person or move the work, and must record in the governance file that this was done and by whom. Managing a structural interest by declaration followed by the decision being taken anyway does not satisfy this requirement.
- **PCI-PML-STD-01.03-PR-06 — Assurance independence, tested by name.** A person or function must not provide an assurance opinion on a plan, control, estimate, schedule, business case or deliverable that the same person or function prepared, directed, specified or approved. The assurance record must name the assurer and must state, positively, that



the assurer's name does not appear in the authorship or approval record of the artefact assured.

- **PCI-PML-STD-01.03-PR-07 — Evaluation-panel exclusion.** A credential holder who holds an interest in any bidder, tenderer or candidate must not be a member of the evaluation panel for that procurement or appointment, must not see other evaluators' scores or comments, and must not moderate, chair or ratify the evaluation.

**6. Prohibited actions.** Deciding a matter in which one holds an interest. Declaring an interest and then participating "for continuity". Assuring one's own work, or one's own function's work. Chairing the moderation of scores one influenced. Recording an abstention with no named substitute decider. Leaving an interests field blank rather than recording a nil return. Treating an interest as extinguished because it was declared once, in a previous year, to a different body.

**7. Required evidence.** The interests register with entry dates; the written declarations with their timestamps relative to the meeting agenda; minutes carrying the interests field, the abstention and the substitute decider's name; role-change or work-transfer records for structural interests; assurance opinions carrying the assurer's name and the positive independence statement; evaluation-panel membership records with declarations against each member.

**8. Responsible role.** Each credential holder, for their own identification, declaration and abstention. The named chair of the body, for ensuring the interests field is completed and the abstention minuted. The named accountable owner of the assurance function, for [PR-06](#).

**9. Approval authority.** The chair of the body determines whether a declared interest requires abstention on an item. The sponsor determines a structural interest touching a project role. The governing body determines a structural interest touching the sponsor, the assurance function or the chair. **The interested party never determines their own case.**

**10. Independence requirement.** This standard is the independence requirement for the set. Its [PR-06](#) test is applied by matching names, not by relying on titles: a permanently titled independent function is not independent of an artefact whose authorship record carries its name.

**11. Materiality or threshold.** This standard sets no financial threshold, and one would defeat it — a personal connection has no value in currency. The test is the definition of *conflict of interest* above, applied by the chair, together with the daylight test the manuscript states at KA 1.4.1: would every party seeing the full picture still regard the decision as impartial? Where an organisation's governance sets a monetary de-minimis for **declarable financial** interests, that documented figure applies to financial interests only, and applies to



declaration, never to abstention. *Six-person internal project*: three of the six may hold interests in the same supplier because the population is small; the answer is a named substitute decider borrowed from elsewhere in the organisation, recorded in the minute, not a decision taken anyway. *Multi-partner national programme*: each partner maintains its own register, the programme holds the consolidated view for shared decision bodies, and a partner's refusal to disclose is itself the escalation trigger below.

**12. Exception and waiver.** No exception is permitted to the principal obligation or to PR-06. Where abstention would leave a decision with no competent decider — a genuine constraint on very small projects and in narrow specialisms — the matter must be escalated to the next authority in the delegation schedule and decided there; it must not be decided by the interested party with a disclaimer attached.

**13. Escalation trigger.** An interest that cannot be managed by abstention. A chair who declines to minute a declared interest or an abstention. An assurance opinion whose assurer appears in the authorship record. A party who refuses to state whether they hold an interest. Discovery that a completed decision, evaluation or assurance opinion was taken by an interested party.

**14. AI application.** AI may maintain the interests register, prompt the standing question at each convening, cross-match declared interests against a supplier or entity list, flag decision records with an empty interests field, and cross-match assurance-opinion signatories against document authorship and approval metadata to surface candidate capture cases.

**15. AI prohibition.** An AI system must not determine whether an interest exists, whether it is material, whether abstention is required, or whether an assurance arrangement is independent. It must not clear a conflict, and its failure to flag one is not evidence that none existed.

**16. AI verification. Reconciliation plus sampling with a stated basis.** Every AI-produced capture candidate must be confirmed against the artefact's own version history by a named human before it is recorded as a finding. Each quarter, a competent reviewer must reconcile the interests register against the membership lists of every decision body in the period and confirm that each member appears with either a declaration or a nil return, and must draw a stated sample of assurance opinions and trace each assurer's name against the assured artefact's authorship and approval record.

## 17. External reference.

- **Project Management Institute** · *Code of Ethics and Professional Conduct* · relied on for: the existence of a professional expectation of conflict disclosure and impartiality · **EXT-063**, recorded as **not independently verified** — **verify current requirements** · **Manual §6**

**category 6 — ethical code** · limitation: binding only where a body, regulator or engagement has adopted it; a PCI credential holder not subject to it is not made subject to it by this standard. No text is reproduced.

- **ISO/IEC** · *ISO/IEC 17024, Conformity assessment — General requirements for bodies operating certification of persons* · relied on for: the existence of impartiality requirements on bodies that certify people, which is why PCI holds itself to this standard as well · **EXT-022** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: it binds certification bodies, not credential holders; voluntary unless imported.
- **OECD (G20/OECD)** · *G20/OECD Principles of Corporate Governance* · relied on for: the existence of a governance expectation that conflicted parties do not decide · **EXT-128** · **Manual §6 category 10 — illustrative practice** · currency checked 2026-08-03 · limitation: a Council Recommendation; non-binding; **not legislation**.

**18. Jurisdictional caution.** Public-procurement law, anti-bribery law, company law, charity law and sector regulation impose their own conflict rules, registration duties and criminal consequences, and they differ by jurisdiction and by the nature of the buying entity. Compliance with this standard is not compliance with any of them. Obtain local legal advice before designing a declaration or evaluation process for a publicly funded procurement.

**19. Related PCI Standards.** [PCI-FND-STD-08](#) (the parent disclosure obligation); [PCI-FND-STD-01](#); [PCI-FND-STD-10](#); [PCI-PML-STD-03.03](#); [PCI-PML-STD-09.01](#); [PCI-PML-STD-10.01](#); [PCI-PML-STD-12.02](#). **What this standard adds to [PCI-FND-STD-08](#):** the foundational standard requires disclosure of a conflict. This standard requires **abstention with a named substitute decider**, requires a **nil return** so silence becomes evidence, requires a **role change** for structural interests, and applies a **name-matching test to assurance independence** — none of which the foundational standard states.

**20. Related Body of Knowledge content.** PML-AI · Domain 1 · KA 1.2 · topic 1.2.2a Interests, competence and confidentiality. Also Domain 3 KA 3.3 topics 3.3.2 assurance lines and 3.3.4 the decision record; Domain 2 KA 2.2 topic 2.2.3 selection and prioritisation models; Domain 10 KA 10.2 tendering and evaluation; Domain 9 KA 9.2 assurance and control.

**21. Compliance test.** Compliance is demonstrated when a reviewer, taking a stated sample of decision records drawn from **each** of gate decisions, steering decisions, evaluation panels and assurance opinions in the period, finds that: (a) every record carries a populated interests field, with a declaration or a nil

return against every participant named in the attendance list; (b) for every declared interest, the minute records the abstention and **names the individual who decided instead**; (c) for every assurance opinion, the assurer's name does **not** appear in the authorship or approval record of the artefact assured, and the opinion carries the positive statement to that effect; (d) for every structural interest recorded in the period, a dated role-change or work-transfer record exists; and (e) no evaluation-panel membership list contains a member with a declared interest in a bidder. Any sampled assurance opinion failing (c) fails the test outright, and the failure is reported as assurance capture rather than as a documentation defect.

**22. Breach indicators.** Interests fields blank rather than nil-returned. Declarations timestamped after the meeting. Abstentions minuted with no substitute decider. The same function appearing as author and assurer of a plan. A steering committee whose interests register has no entries across a year in a market where several members previously worked for the incumbent supplier. Evaluation moderation chaired by the sponsor of the winning option. A structural interest managed by an annual declaration.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure on the associated competency; ethics review; certification investigation; suspension or withdrawal of the PML-AI credential — each subject to due process and a right of appeal.

**24. Examination application.** Ethical dilemma: an assurance lead is asked to review the delivery plan their own team drafted for a project six weeks from a gate, and the alternative reviewer is unavailable for a month. Evidence selection: which two artefacts would show whether an award decision was taken by an interested party. Scenario judgement: a steering member's former employer is one of three bidders and the member offers to "stay in the room but not vote".

**25. Version and status.** Version 1.0 · **not yet approved** · effective on approval · **new standard** — there is no predecessor in the v1.0 set, and its absence was the completeness finding this standard answers. Amendment note: none.

## Domain 2 — Strategy, Selection and Business Alignment

PCI STANDARD

PCI-PML-STD-02.01

## Business-Case Integrity

**1. Normative requirement.** A credential holder must not present, endorse or rely on a business case whose stated benefits, costs, assumptions, dependencies or option comparison they know, or on the evidence available to them ought to know, to be materially misstated.

**2. Purpose.** The business case is the instrument that authorises everything downstream, and it is authored by the party that benefits from approval. Optimism is not the failure; **asymmetric optimism** is — a benefit case built on a best-case adoption rate against a cost case built on a mid-case contingency, compared to a do-nothing option deliberately drawn badly. The result survives approval, fails delivery, and cannot be corrected later because the approval it obtained has already consumed the funds.

**3. Scope.** Every credential holder preparing, reviewing, recommending, approving or assuring a business case, an option appraisal, a selection score or a re-approval, at initial approval and at every subsequent gate at which the case is restated. It applies to the case for continuing as much as to the case for starting.

**4. Defined terms.** *material · benefit · benefits baseline · evidence · sponsor · competent reviewer · decision owner*. Additionally, **option comparison** means the set of options appraised on a common basis, including the do-minimum option; **asymmetric treatment** means applying a different level of optimism, contingency, discount or evidence to one option or one side of the case than to another.

### 5. Required actions — process requirements.

- **PCI-PML-STD-02.01-PR-01 — Symmetric treatment.** The credential holder must apply the same estimating basis, the same contingency approach, the same optimism adjustment and the same evidence standard to every option and to both sides of the case, and must state in the case that this was done and where it was not.
- **PCI-PML-STD-02.01-PR-02 — Assumptions carry owner, source and date.** Every assumption on which a benefit or a cost materially depends must be recorded with a named owner, a source reference and the date it was last tested. An assumption with no owner is recorded as an open risk, not as an assumption.
- **PCI-PML-STD-02.01-PR-03 — The do-minimum option is drawn honestly.** The do-minimum or do-nothing option must be appraised on the same basis as the preferred option, and its stated consequences must be sourced, not asserted.
- **PCI-PML-STD-02.01-PR-04 — Restatement at each gate.** At each gate, the credential holder must restate the case against actual costs incurred, actual benefits evidence available and the current assumption

set, and must state whether the case still holds. A case carried forward unchanged from a previous gate, with no restatement, does not satisfy this requirement.

- **PCI-PML-STD-02.01-PR-05 — Kill criteria are stated before approval.** The case must state, before approval, the conditions under which the project would be stopped, and each such condition must be measurable against something the project will actually measure.

**6. Prohibited actions.** Presenting a benefit whose measurement method does not exist. Applying contingency to one option and not to its comparators. Restating a case by moving the benefit date rather than reducing the benefit. Presenting a sunk cost as a reason to continue. Removing a kill criterion once it is approaching. Suppressing a material adverse assumption change between gates because the next gate is close.

**7. Required evidence.** The versioned business case; the assumption register with owners, sources and test dates; the option appraisal showing a common basis; the gate-by-gate restatement record; the approved kill criteria and the measurement each is tested against; the record of every material change to an assumption between gates and its notification.

**8. Responsible role.** The **sponsor** owns the business case and answers for its integrity. The credential holder leading delivery answers for the accuracy of the delivery-side inputs — cost, schedule, risk, resource and dependency — and for stating any known material misstatement to the sponsor and, where the sponsor does not act, to the governing body.

**9. Approval authority.** The governing body approves the case at the authority level the delegation schedule sets for its value and exposure. The sponsor approves restatements within the tolerance the governing body sets. Neither the author of the case nor the delivery supplier may approve it.

**10. Independence requirement.** The gate restatement at [PR-04](#) must be reviewed by a competent reviewer independent of the case's preparation and independent of the delivery organisation, at the authority level the delegation schedule sets. On a project too small to supply such a reviewer internally, independence may be met by a named reviewer from another part of the organisation who did not contribute to the case, and the arrangement is recorded.

**11. Materiality or threshold.** This standard sets no percentage. A misstatement is material when it meets the definition of *material* above — it could reasonably have changed the approval, its conditions or the authority at which it was taken. The organisation's governance sets the documented approval thresholds and the documented tolerance for restatement, and this standard requires that those documented figures exist, are approved, and are applied. Where an organisation has no documented tolerance, the credential holder

must escalate every restatement. *Six-person internal project*: the case is four pages, the assumption register is eight rows and the kill criteria are two lines — and the restatement at each of two gates costs an afternoon. *Multi-partner national programme*: the same obligations apply per component and at the programme level, and the aggregation rule states which component restatements roll up to which programme decision, so that a portfolio of restated-but-still-green components cannot conceal an amber programme.

**12. Exception and waiver.** An exception to [PR-04](#) — carrying a case forward without restatement — may be approved only by the governing body, only where the interval since the last restatement is shorter than the organisation's documented minimum restatement interval, and only with a recorded statement that no material assumption has changed, signed by the sponsor. No exception is permitted to the principal obligation.

**13. Escalation trigger.** A benefit whose owner declines to accept it. An assumption that has become untrue and has not been reflected. A material adverse change with a gate more than the escalation time away. Instruction to remove or weaken a kill criterion. Discovery that the option comparison was not prepared on a common basis.

**14. AI application.** AI may test a case for internal arithmetic consistency, cross-check figures between the narrative and the model, extract the assumption set and flag assumptions with no owner or no source, compare the treatment of options for asymmetry, and track assumption changes between versions.

**15. AI prohibition.** An AI system must not approve a business case, decide whether a benefit is achievable, determine materiality, generate a benefit figure that is then presented as an estimate without a named human's derivation, or author the kill criteria.

**16. AI verification. Independent recomputation plus source tracing.** A competent reviewer must recompute the case's headline financial result by an independent method and reconcile it to the model without unexplained difference, and must trace every benefit figure and every cost figure that carries more than the documented materiality threshold to the source document and version it came from. Where AI produced the option comparison, the reviewer must additionally apply **sensitivity analysis**: vary each of the three assumptions with the largest effect and confirm the option ranking's stated robustness.

#### **17. External reference.**

- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of a business case as a governance instrument reviewed through the lifecycle · **EXT-028** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limita-

tion: guidance; voluntary; not certifiable; not the source of this obligation.

- **ISO** · *ISO 21504, Guidance on portfolio management* · relied on for: the existence of a portfolio-level test that a component still merits its funding · **EXT-031**, recorded as **not independently verified — verify current requirements** · **Manual §6 category 3** · limitation: as above, with an open verification status.
- **AACE International** · *Total Cost Management (TCM) Framework* · relied on for: the existence and purpose of a maturity-based progression in estimate reliability · **EXT-064**, recorded as **not independently verified — verify current requirements** · **Manual §6 category 5 — professional framework** · limitation: **professional framework, never regulatory authority**; no accuracy range, class table or recommended-practice text is reproduced.

**18. Jurisdictional caution.** Public-sector appraisal rules, subsidy control, state-aid regimes, listed-company disclosure duties and grant conditions impose their own appraisal, approval and disclosure requirements, and misstating a case to obtain public funds can engage criminal law. Obtain local legal advice where a case supports a regulated funding application or a market disclosure.

**19. Related PCI Standards.** *PCI-FND-STD-05*; *PCI-FND-STD-02*; *PCI-FND-STD-11*; *PCI-PML-STD-02.02*; *PCI-PML-STD-03.03*; *PCI-PML-STD-15.02*; *PCI-PML-STD-16.03*. **What this standard adds to *PCI-FND-STD-05*:** the foundational standard requires assumptions to be transparent. This standard requires **symmetric treatment across options**, an **owned and dated assumption register**, an **honestly drawn do-minimum, restatement at every gate**, and **kill criteria fixed before approval** — the specific mechanics by which a delivery business case goes wrong.

**20. Related Body of Knowledge content.** PML-AI · Domain 2 · KA 2.2 The business case and selection · topics 2.2.1 the business case as decision instrument, 2.2.2 options and appraisal, 2.2.3 selection and prioritisation models; KA 2.3 topic 2.3.4 assumption and dependency management; KA 2.4 Strategic termination · topics 2.4.2 sunk cost and escalation of commitment, 2.4.3 kill criteria and honest gates.

**21. Compliance test.** Compliance is demonstrated when the approved business case can be reconciled, without unexplained difference, to: the current cost estimate and its basis, the approved risk and contingency position, the benefits register with its named owners and its benefits baselines, the assumption register with owners and test dates, and the current schedule's stated benefit start dates — and when the gate file shows, for each gate held in



the period, a dated restatement, a named independent reviewer, and an explicit statement that the case does or does not still hold. A case that cannot be reconciled to the benefits register's owners fails the test.

**22. Breach indicators.** Benefits with no named owner. Assumptions with no date. A do-minimum option half a page long against a preferred option of twelve. Contingency present in one option only. A benefit whose start date has moved at every gate while its value has not. Kill criteria removed between versions. A restatement produced after the gate date.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Calculation review: a candidate is given an option comparison in which one option carries contingency and another does not, and states the effect on the ranking. Scenario judgement: an assumption underpinning 60 per cent of the benefit has become untrue four weeks before a gate. Ethical dilemma: a sponsor asks for the kill criteria to be "softened" because the project is close to one.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-02-01](#) v1.0. Amendment note: renumbered and restructured; legislative drafting removed; five process requirements separated out; symmetric treatment and honest do-minimum added as express obligations; compliance test replaced with a reconciliation test.

## PCI STANDARD

### PCI-PML-STD-02.02

## Benefits Ownership

**1. Normative requirement.** A credential holder must not allow a benefit to be claimed in a business case, a plan or a report unless a named individual outside the delivery team has accepted, in writing, accountability for realising it.

**2. Purpose.** A benefit the delivery team owns is a benefit nobody owns, because the delivery team disbands at handover and the benefit arrives afterwards. The failure this prevents is the standard one: a business case full of benefits, a project delivered on time and to budget, and no measurable improvement anywhere — with every party able to say truthfully that they did their part.

**3. Scope.** Every credential holder preparing, reviewing or approving a business case, a benefits register, a benefits map, a benefits profile or a benefits



realisation plan, on projects, programmes and portfolios, at approval and at every gate. It applies to enabling outputs as much as to end benefits.

**4. Defined terms.** *benefit · benefits baseline · sponsor · acceptance · evidence · material.* Additionally, **benefits owner** means the named individual, accountable in the receiving or operating organisation, who accepts that the improvement will be realised and measured in their area of accountability.

**5. Required actions — process requirements.**

- **PCI-PML-STD-02.02-PR-01 — Named written acceptance.** Every benefit in the register must carry the name and role of its benefits owner and a dated written acceptance by that individual. A name entered by the project on the owner's behalf does not satisfy this requirement.
- **PCI-PML-STD-02.02-PR-02 — A measurable definition before approval.** Every benefit must carry, before the case is approved, the measure that will evidence it, the source system or method that will produce the measure, the benefits baseline value, and the date the baseline was measured.
- **PCI-PML-STD-02.02-PR-03 — Baseline measured before the change.** The benefits baseline must be measured before the change is made. A baseline reconstructed after go-live must be labelled as reconstructed, with the method stated, and must not be presented as a measurement.
- **PCI-PML-STD-02.02-PR-04 — Refusal is recorded, not absorbed.** Where a proposed benefits owner declines to accept a benefit, the credential holder must record the refusal and its stated reason and escalate it, and must not transfer the benefit to the delivery team, to the sponsor by default, or to an unnamed function.

**6. Prohibited actions.** Listing a benefit with the delivery team, the project, the PMO or "the business" as its owner. Claiming a benefit whose measure does not yet exist. Reconstructing a baseline after the change and presenting it as measured. Double-counting the same improvement across two components of a programme. Presenting an enabling output — a system delivered, a process documented — as a benefit.

**7. Required evidence.** The benefits register with, per benefit: owner name and role, dated written acceptance, measure, source system, benefits baseline value and its measurement date, target value and target date. The refusal record and its escalation for any declined benefit. The double-count check for programmes.

**8. Responsible role.** The **sponsor** is accountable for securing a benefits owner for every claimed benefit. The credential holder leading delivery is

accountable for refusing to carry an unowned benefit in any document they issue, and for escalating any that remains unowned.

**9. Approval authority.** The governing body approves the benefits register at case approval. The sponsor approves changes to a benefit's target value or date within the documented tolerance; changes beyond it go to the governing body. The benefits owner alone accepts ownership; nobody may accept on their behalf.

**10. Independence requirement.** The benefits owner must be **independent of the delivery team** in the sense defined above: not accountable for delivering the output, and answering in the receiving or operating organisation. Measurement of realised benefits must be produced by, or verified by, a party independent of the delivery team — see [PCI-PML-STD-16.03](#).

**11. Materiality or threshold.** This standard sets no threshold, because ownership is binary: a benefit is owned or it is not. The organisation's governance sets which benefits are individually tracked and which are aggregated, and this standard requires that documented rule to exist and to be applied. A benefit below the documented individual-tracking threshold must still carry a named owner for its aggregate. *Six-person internal project*: one benefit, one owner, one measure already produced monthly by an existing report — the whole obligation is three cells in a table. *Multi-partner national programme*: the register is held at programme level, benefits owners sit in the operating organisations rather than in any partner, and the double-count check across components is a standing programme obligation because components have an incentive to claim the same improvement.

**12. Exception and waiver.** An exception permitting a benefit to be carried without an accepted owner may be approved only by the governing body, only until the next gate, and only where the case records that the benefit is excluded from the approval arithmetic while unowned. No exception permits an unowned benefit to count towards an approval.

**13. Escalation trigger.** A proposed benefits owner declining acceptance. A benefit whose measure cannot be produced by any existing or planned system. A benefit still unowned at the gate at which the case is approved. Discovery that the same improvement is claimed by two components.

**14. AI application.** AI may build a benefits register from case documents, flag benefits with no owner, no measure, no baseline or no date, detect candidate double-counts across a programme by clustering benefit descriptions, and check that each stated measure maps to a named source system.

**15. AI prohibition.** An AI system must not accept ownership of a benefit, assign an owner, decide whether a benefit is realisable, or generate a baseline value.

**16. AI verification. Source tracing plus named approval.** Every baseline value produced with AI assistance must be traced by a competent reviewer to the source system extract, with the extract date and the query or filter recorded, and the benefits owner must confirm the value in writing before it is recorded as the benefits baseline. Every AI-flagged double-count must be confirmed by a named human against both components' registers.

#### **17. External reference.**

- **ISO** · *ISO 21503, Guidance on programme management* · relied on for: the existence of a programme-level concern with benefits that no single component delivers · **EXT-030**, recorded as **not independently verified — verify current requirements** · **Manual §6 category 3 — international voluntary standard** · limitation: guidance; voluntary; not certifiable; open verification status.
- **ISO** · *ISO 21504, Guidance on portfolio management* · relied on for: the existence of a portfolio concern with the benefits a component is funded to produce · **EXT-031, not independently verified** · **Manual §6 category 3** · limitation: as above.
- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of benefit identification and realisation as lifecycle activities · **EXT-028** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.

**18. Jurisdictional caution.** Where benefits are claimed to a grant funder, a regulator, a market or a tax authority, the accuracy of the claim can engage funding-condition, disclosure or fraud law. Obtain local legal advice before a benefits statement is made outside the organisation.

**19. Related PCI Standards.** [PCI-FND-STD-02](#); [PCI-FND-STD-01](#); [PCI-FND-STD-12](#); [PCI-PML-STD-02.01](#); [PCI-PML-STD-15.02](#); [PCI-PML-STD-16.02](#); [PCI-PML-STD-16.03](#).

**20. Related Body of Knowledge content.** PML-AI · Domain 2 · KA 2.3 Benefits, value and sustainability · topics 2.3.1 benefits mapping, 2.3.2 measures and baselines. Also Domain 15 KA 15.2 benefits and portfolio balancing; Domain 16 KA 16.4 benefits measurement.

**21. Compliance test.** Compliance is demonstrated when a reviewer can open the benefits register and find, for **every** benefit carried in the approved case: a named individual and their role; a written acceptance dated on or before the approval date; a stated measure; a named source system that already produces or is planned to produce it; a benefits baseline value with the date it was measured; and a target with a date. The reviewer must then contact a stated

sample of named owners and confirm each is aware of and accepts the accountability recorded against them. **A register in which any sampled owner does not recognise their own entry fails the test**, regardless of the paperwork's completeness.

**22. Breach indicators.** Owner cells containing a team, a function or a job family rather than a person. Acceptance dates after the approval date. Baselines dated after go-live. The same improvement in two components' registers. A benefit whose measure is "to be defined". Owners who, when asked, say the project told them they were the owner.

**23. Consequence within PCI authority.** Correction required; additional review; escalation; examination failure; ethics review; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Evidence selection: which artefact establishes that a benefit is owned. Scenario judgement: the proposed benefits owner refuses, four days before the approval gate, and the sponsor proposes to record the PMO as owner. Calculation review: a programme's benefits total exceeds the sum of its components' distinct improvements.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-02-02](#) v1.0. Amendment note: renumbered and restructured; legislative drafting removed; the benefits-baseline sense of *baseline* separated from the control sense per the suite terminology audit; the compliance test strengthened to require confirmation with sampled owners, which is what makes the standard provable rather than documentary.

## Domain 3 — Governance, Organisation and Decision Rights

### PCI STANDARD

#### PCI-PML-STD-03.01

### Governance Authority Before Commitment

**1. Normative requirement.** A credential holder must not commit an organisation's funds, contractual obligations or people to delivery before the governance arrangements for that delivery — the named sponsor, the named decision bodies, the delegation schedule and the gate points — are documented and approved.

**2. Purpose.** Governance retrofitted after commitment is governance that cannot say no, because the money is spent, the contract is signed and the team

is hired. The failure this prevents is the project that reaches its first gate with nothing the gate can decide.

**3. Scope.** Every credential holder in a position to commit or to recommend commitment — funds, contracts, orders, hires or secondments — on projects, programmes and portfolios, in every delivery model. It applies to the initial commitment and to any commitment following a material change in the delivery structure, including the addition of a delivery partner.

**4. Defined terms.** *governance* (project sense) · *sponsor* · *delegation schedule* · *gate* · *decision owner* · *material*. Additionally, **commitment** means an act that creates a legal, financial or employment obligation the organisation cannot withdraw from without cost.

#### 5. Required actions — process requirements.

- **PCI-PML-STD-03.01-PR-01 — The four artefacts exist and are dated.** Before commitment, the credential holder must hold: a named sponsor with a written appointment; the decision bodies with their named members and their terms of reference; the delegation schedule; and the gate points with their criteria. Each artefact must carry a date and an approver.
- **PCI-PML-STD-03.01-PR-02 — The gate can stop the work.** The credential holder must confirm, in writing, that each defined gate carries the power to stop, hold or redirect, and must record any gate that does not, because a review that cannot stop the work is not a gate and must not be described as one.
- **PCI-PML-STD-03.01-PR-03 — Governance proportionate and recorded as such.** The credential holder must record the tailoring decision — which governance elements apply at what depth for this project's size and exposure — with the sponsor's approval, so that light governance is a decision rather than an omission.
- **PCI-PML-STD-03.01-PR-04 — Re-approval on structural change.** Where the delivery structure changes materially — a partner is added, a component is transferred, the sponsor changes, or the funding route changes — the credential holder must obtain re-approval of the four artefacts before further commitment.

**6. Prohibited actions.** Committing funds or contracts against a governance structure that is drafted but not approved. Describing a review with no stopping power as a gate. Operating with an unnamed sponsor or a sponsor who has not accepted the appointment. Allowing a delivery partner's internal governance to substitute for the client organisation's, without an approved instrument saying so.

**7. Required evidence.** The dated sponsor appointment; the terms of reference of each decision body with its membership; the approved delegation

schedule; the gate definitions with criteria and stopping powers; the tailoring decision and its approval; re-approval records at each structural change.

**8. Responsible role.** The named credential holder leading the project, for holding and applying the artefacts. The **sponsor**, for securing their approval. The governing body, for approving them.

**9. Approval authority.** The governing body approves the delegation schedule and the gate points. The sponsor approves the tailoring decision within the governing body's stated limits. The project leader approves nothing in this standard.

**10. Independence requirement.** The approval of the governance arrangements must be taken by a body independent of the delivery organisation in the sense defined above. Where a delivery partner sits on the approving body, that partner's members must abstain from the approval of the delegation schedule under [PCI-PML-STD-01.03-PR-04](#).

**11. Materiality or threshold.** This standard sets no financial trigger for "commitment", because organisations define commitment differently — order, contract signature, purchase requisition, hire. The credential holder must apply the organisation's documented definition of commitment and its documented approval thresholds; where none is documented, the credential holder must escalate before any commitment. *Six-person internal project:* the four artefacts are one page in total — a named sponsor, a fortnightly review with the power to stop, a three-row delegation table and two gates. The tailoring decision is one sentence. *Multi-partner national programme:* the artefacts exist at programme and component level, the delegation schedule states the aggregation rule across partners, and each partner's own governance is mapped to the programme's rather than substituted for it.

**12. Exception and waiver.** An exception permitting commitment before the artefacts are approved may be approved only by the governing body, only for a stated and capped amount, only for a stated period, on written justification, with the compensating control that every commitment made under the exception is individually reported to the governing body within its stated reporting time.

**13. Escalation trigger.** Pressure to commit before governance is approved. A gate defined without stopping power. A sponsor who has not accepted the appointment. A structural change made without re-approval. A delivery partner asserting that its own governance suffices.

**14. AI application.** AI may check a governance pack for missing artefacts, missing dates and missing approvers; test gate criteria for assessability and flag the unmeasurable ones; check a delegation schedule for decision classes with no accountable role or with more than one; and model the latency a proposed governance cadence produces.

**15. AI prohibition.** An AI system must not approve governance arrangements, appoint a sponsor, determine tailoring, or decide that a gate's criteria are assessable.

**16. AI verification. Clause-to-summary comparison plus named approval.** Where AI has summarised or assembled the governance pack, a competent reviewer must compare each summarised item against the source instrument it claims to summarise — the appointment letter, the terms of reference, the schedule — and confirm the summary states the same authority, the same limits and the same names. The sponsor must then approve the pack by name.

**17. External reference.**

- **ISO** · *ISO 21505, Guidance on governance of projects, programmes and portfolios* · relied on for: the existence of a governance framework distinct from management · **EXT-032, not independently verified — verify current requirements** · **Manual §6 category 3 — international voluntary standard** · limitation: guidance; voluntary; not certifiable; open verification status.
- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of pre-delivery governance establishment · **EXT-028** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.
- **OECD (G20/OECD)** · *G20/OECD Principles of Corporate Governance* · relied on for: the existence of a board-level expectation that authority is defined before it is exercised · **EXT-128** · **Manual §6 category 10 — illustrative practice** · currency checked 2026-08-03 · limitation: a Council Recommendation; non-binding; **not legislation**.

**18. Jurisdictional caution.** Company law, public-procurement law, financial regulation, delegated authority under statute and grant conditions determine who may lawfully bind an entity, and a professionally correct delegation schedule can still be legally ineffective. Obtain local legal advice on authority to contract for the specific entity and funding route.

**19. Related PCI Standards.** [PCI-FND-STD-01](#); [PCI-FND-STD-12](#); [PCI-PML-STD-03.02](#); [PCI-PML-STD-03.03](#); [PCI-PML-STD-03.04](#); [PCI-PML-STD-13.01](#).

**20. Related Body of Knowledge content.** PML-AI · Domain 3 · KA 3.1 Governance models · topics 3.1.1 what governance is for, 3.1.2 structures across organisational forms, 3.1.3 governance in agile and hybrid environments; KA



3.3 topic 3.3.1 stage gates. Also Domain 4 KA 4.1 charter and management plans.

**21. Compliance test.** Compliance is demonstrated when a reviewer, taking the date of the earliest commitment recorded in the project's financial or contractual system, finds that the sponsor appointment, the decision-body terms of reference, the delegation schedule and the gate definitions are each dated **on or before** that date and each carry a named approver; and when each gate definition states, in terms, the power to stop, hold or redirect. A commitment date earlier than any of the four artefact dates fails the test unless a governing-body exception covering that commitment exists.

**22. Breach indicators.** Governance packs dated after the first invoice. Gates described as "checkpoints" with no stated power. A sponsor named in a slide and nowhere else. A delegation schedule with decision classes carrying two accountable roles or none. Partner governance cited in place of the client's. Terms of reference with no membership list.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Scenario judgement: a leader is asked to place a long-lead order two weeks before the governance pack is due for approval. Evidence selection: which three artefacts must predate the first commitment. Escalation decision: a gate has been defined with no power to stop.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-03-01](#) v1.0. Amendment note: renumbered and restructured; legislative drafting removed; the "gate can stop the work" obligation and the re-approval-on-structural-change obligation added as express process requirements; compliance test replaced with a date-ordering test that a reviewer can perform from two systems.

## PCI STANDARD

### PCI-PML-STD-03.02

## Decision Rights and Delegated Authority

**1. Normative requirement.** A credential holder must take every decision at the authority level the organisation's documented delegation schedule assigns to it, and must not take, split, defer or aggregate a decision so as to avoid the authority the schedule requires.

**2. Purpose.** Delegation schedules fail in two directions, and both are common. Upward: everything goes to the steering committee, the committee



becomes a queue, and the escalation that matters arrives behind forty that do not. Downward: a decision above the limit is split into three below it, or taken under an urgency provision that becomes ordinary practice. This standard addresses the second, which is the one that produces exposure nobody approved.

**3. Scope.** Every credential holder taking, recommending, ratifying or recording a delivery decision, on projects, programmes and portfolios. It applies to financial, contractual, scope, schedule, risk, resource, quality and release decisions alike, and to urgent decisions taken outside the ordinary cadence.

**4. Defined terms.** *delegation schedule · decision owner · escalation threshold · material · decision record.* Additionally, **aggregation rule** means the schedule's provision that related decisions summing above a stated value within a stated period require the authority the schedule assigns to the aggregate; **relatedness class** means the set over which that rule sums.

#### 5. Required actions — process requirements.

- **PCI-PML-STD-03.02-PR-01 — Single accountable role per class.** The credential holder must confirm, before relying on the delegation schedule, that each decision class in it carries exactly one accountable role, and must escalate any class carrying two or none.
- **PCI-PML-STD-03.02-PR-02 — Universal registration.** Every decision in a class covered by the schedule must generate a decision-record entry **regardless of its value**, so that the aggregation rule has inputs. A rule that sums decisions it cannot see is not a control.
- **PCI-PML-STD-03.02-PR-03 — Aggregation applied by what the change touches.** The credential holder must apply the schedule's aggregation rule using its documented relatedness class, and where the organisation defines relatedness by what the changes touch — the same deliverable, the same assured control, the same interface — must not substitute a broader class such as requester or budget line.
- **PCI-PML-STD-03.02-PR-04 — Urgency provisions are time-boxed and reported.** A decision taken under an urgency or out-of-cycle provision must be recorded as such, with the reason, and must be reported to the authority that would ordinarily have taken it within the time the schedule states.
- **PCI-PML-STD-03.02-PR-05 — Escalation carries a destination and a time.** The credential holder must not rely on an escalation threshold that states no named destination and no time; where one is encountered, they must escalate the defect itself.

**6. Prohibited actions.** Splitting a decision to bring each part below a limit. Recording a decision at a lower value than it carries. Using an urgency provision as ordinary practice. Ratifying, after the event, a decision taken

without authority, without recording that it was taken without authority. Widening a relatedness class so that the aggregation rule trips constantly and is then disappplied.

**7. Required evidence.** The approved delegation schedule with its aggregation rule and relatedness class; the decision record showing universal registration; aggregation calculations for any class that approached or crossed the rule; urgency-decision records with reasons and their reporting; escalation records showing destination and elapsed time against the stated time.

**8. Responsible role.** The named credential holder for decisions within their authority; the named role the schedule assigns for each other class. The governing body owns the schedule itself.

**9. Approval authority.** The governing body approves the delegation schedule, its thresholds, its aggregation rule and its relatedness class. The sponsor approves nothing in this standard that would raise their own authority.

**10. Independence requirement.** Not applicable to the taking of a decision within delegated authority, because delegation exists precisely so that the accountable party decides; independence attaches to the periodic testing of the schedule's operation, which must be performed by a competent reviewer independent of the decisions tested.

**11. Materiality or threshold. This standard invents no figure and no percentage,** and doing so would be wrong: delegated authority levels are set by an organisation's own governance against its own balance sheet, risk appetite and decision rate. What this standard requires is that documented thresholds, a documented aggregation rule, a documented relatedness class and a documented period exist, and are applied. Where a threshold has been set without reference to the observed decision rate, so that it either never trips or trips on ordinary traffic, the credential holder must record that finding and escalate it — a rule that cannot fire and a rule that always fires are equally useless, and both look like controls. *Six-person internal project:* the schedule is three rows; the aggregation rule sums changes touching the same deliverable within a month; universal registration is the issue tracker they already use. *Multi-partner national programme:* the schedule exists at each tier, and the aggregation rule states explicitly whether it sums across partners — the case in which the exposure the rule exists to catch arrives as one coherent change split between four organisations.

**12. Exception and waiver.** An exception permitting a decision above the delegated level may be approved only by the authority the schedule assigns to that level or above, in writing, before the decision, and never retrospectively by the person who took it. A decision already taken without authority is a breach, is recorded as one, and is then ratified or reversed by the proper authority — ratification does not convert it into a compliant decision.

**13. Escalation trigger.** A decision class with two accountable roles or none. A proposal to split a decision. An urgency provision used more than the number of times the schedule permits. An aggregation that has crossed the rule. A threshold that has not fired in a period in which the observed decision rate says it should have.

**14. AI application.** AI may extract a decision register from minutes and flag entries missing an owner, a date or a versioned information reference; run integrity checks across a decision-rights matrix for classes with two accountable roles or none; compute aggregations across a relatedness class; identify decisions being re-decided by clustering decision text across a long log; and model the latency a governance cadence produces.

**15. AI prohibition.** An AI system must not take a delegated decision, set or change a threshold, determine a relatedness class, or ratify a decision taken without authority.

**16. AI verification. Independent recomputation plus sampling with a stated basis.** Every aggregation computed with AI assistance must be recomputed by hand by a competent reviewer before it is relied on to escalate or not to escalate — the arithmetic is a handful of operations and none of it should be taken on trust. Each quarter, a competent reviewer must draw a stated sample of decision entries and confirm against source minutes that the recorded value, class and authority match.

#### **17. External reference.**

- **ISO** · *ISO 21505, Guidance on governance of projects, programmes and portfolios* · relied on for: the existence of defined decision authority within governance · **EXT-032, not independently verified** · **Manual §6 category 3 — international voluntary standard** · limitation: guidance; voluntary; not certifiable; open verification status.
- **Project Management Institute** · *A Guide to the Project Management Body of Knowledge (PMBOK Guide)* · relied on for: the existence of change and decision authority structures in professional practice · **EXT-060** · **Manual §6 category 5 — professional framework** · currency checked 2026-08-03 · limitation: **a professional framework, never regulatory authority**; no edition is asserted and no text is reproduced.
- **COSO** · *Internal Control — Integrated Framework* · relied on for: the existence of an authorisation control concept · **EXT-084** · **Manual §6 category 10 — illustrative practice** · currency checked 2026-08-03 · limitation: voluntary in itself, though widely imported by regulators; it is not legislation and this standard does not rely on it for any requirement.

**18. Jurisdictional caution.** Whether a person has legal authority to bind an entity is determined by company law, delegated statutory authority, procurement regulation and the entity's own constitution, not by a delegation schedule. A decision within delegated professional authority can still be legally ineffective. Obtain local legal advice on binding authority.

**19. Related PCI Standards.** PCI-FND-STD-04; PCI-FND-STD-11; PCI-FND-STD-12; PCI-FND-STD-13; PCI-PML-STD-01.01; PCI-PML-STD-03.01; PCI-PML-STD-04.01; PCI-PML-STD-08.01.

**20. Related Body of Knowledge content.** PML-AI · Domain 3 · KA 3.2 Sponsorship and steering · topic 3.2.3 decision authorities and thresholds; KA 3.3 · topics 3.3.3 escalation design, 3.3.4 auditability and the decision record. Also Domain 4 KA 4.3 topic 4.3.3 baseline maintenance and KA 4.4 the change board.

**21. Compliance test.** Compliance is demonstrated when a reviewer can: (a) obtain the approved delegation schedule and confirm every decision class carries exactly one accountable role; (b) extract every decision in the period from the decision record and confirm that each was taken at or below the authority the schedule assigns to its class and value; (c) recompute the aggregation over the documented relatedness class and period and confirm that every aggregate crossing the rule was taken at the aggregate authority; (d) confirm that decisions below the individual threshold were nonetheless registered, by comparing the decision-record count against the change, order or issue counts in the operational systems; and (e) confirm every urgency decision was reported to the ordinary authority within the stated time. A material gap between (d)'s two counts fails the test, because an aggregation rule without universal registration cannot see its own inputs.

**22. Breach indicators.** Three change requests to the same deliverable in one month, each just below a limit. A decision register materially shorter than the change log. Urgency decisions clustering in the week before each steering meeting. An aggregation rule that has never fired. A relatedness class widened shortly after the rule first fired. Ratifications recorded as approvals.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Calculation review: seven related changes across a quarter, three of them below the individual limit, and the candidate determines the authority required. Scenario judgement: a supplier proposes to

raise two orders rather than one. Escalation decision: a decision class is found to carry two accountable roles a week before the gate.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes PML-LAW-03-02 v1.0. Amendment note: renumbered and restructured; legislative drafting removed; universal registration, the relatedness-class discipline and the time-boxing of urgency provisions added as express process requirements; element 11 rewritten to require documented thresholds rather than to state any.

#### PCI STANDARD

#### PCI-PML-STD-03.03

### Gate Evidence and the Gate Decision

**1. Normative requirement.** A credential holder must not recommend or take a gate decision except on evidence that is dated, attributable, version-identified and assessed against criteria published before the evidence was assembled.

**2. Purpose.** A gate whose criteria are written after the evidence is a gate that cannot fail, and a gate that cannot fail is a governance cost with no governance product. The failure this prevents is the retrospectively satisfiable gate: criteria adjusted to the evidence available, conditions attached with no owner and no date, and a "conditional pass" that is a pass.

**3. Scope.** Every credential holder preparing, reviewing, chairing, recommending, deciding or assuring a gate decision at any stage of a project, programme or portfolio lifecycle, including release gates in adaptive delivery and investment gates in a portfolio.

**4. Defined terms.** *gate · evidence · decision owner · material · competent reviewer · independent · acceptance.* Additionally, **conditional pass** means a continuation decision taken subject to stated conditions; **condition-closure** means the verified completion of such a condition.

#### 5. Required actions — process requirements.

- **PCI-PML-STD-03.03-PR-01 — Criteria published first.** The gate criteria must be published, dated and version-identified before the evidence supporting the gate is assembled, and the credential holder must not amend them after assembly without the approving authority's recorded decision and a statement of what changed and why.
- **PCI-PML-STD-03.03-PR-02 — Criteria are assessable.** Each criterion must be expressed so that two competent reviewers assessing the same evidence reach the same answer. The credential holder must identify any

criterion that cannot be assessed and must have it replaced or removed before the gate.

- **PCI-PML-STD-03.03-PR-03 — Evidence carries its provenance.** Every item in the gate pack must carry its date, its author or source system, and its version. An undated extract, an unversioned model output and a dashboard state that cannot be reproduced must not be admitted as evidence.
- **PCI-PML-STD-03.03-PR-04 — Conditions carry owner, date and consequence.** Every condition attached to a conditional pass must carry a named owner, a due date and the stated consequence of non-closure, and the credential holder must report condition-closure status at the following gate.
- **PCI-PML-STD-03.03-PR-05 — Dissent is recorded.** Where any member of the deciding body disagrees with the decision, the disagreement and its reason must be recorded in the decision record. A decision recorded as unanimous when it was not is a false record.
- **PCI-PML-STD-03.03-PR-06 — The criteria are set by the authority, not by the project.** The gate criteria must be approved, before publication, by the authority that holds the gate decision or by the authority the delegation schedule assigns to set them, and the approval must name that authority and its date. **The project being gated must not be the approver of its own gate criteria.** Where the project drafts them, the record must show that fact and the approving authority's assent to them as published.

**6. Prohibited actions.** Writing or altering criteria to fit the evidence. Admitting evidence with no date, no author and no version. Attaching a condition with no owner. Passing a gate on a promise that the evidence will follow. Recording a conditional pass and then never reporting closure. Presenting the absence of an assurance finding as an assurance opinion.

**7. Required evidence.** The published criteria with their version and publication date, and the approval of those criteria by the authority required by **PR-06**; the gate pack with per-item provenance; the decision record with the named decision owner, the decision, the conditions with owners and dates, and any recorded dissent; the condition-closure report at the following gate; the independent reviewer's opinion where one is required.

**8. Responsible role.** The named chair of the gate body decides and answers for the decision. The named credential holder leading the project answers for the completeness and provenance of the pack and for stating any material matter the pack does not cover.

**9. Approval authority.** The gate body's named chair, at the authority level the delegation schedule assigns. A condition may be waived only by the

authority that imposed it. The project must not close its own gate condition where the condition concerns an assurance, safety, licence or acceptance matter owned elsewhere.

**10. Independence requirement.** Where the delegation schedule requires assurance at the gate, the assurance opinion must be given by a competent reviewer independent of the pack's preparation, tested by the name-matching rule in [PCI-PML-STD-01.03-PR-06](#). **An assurance function must not opine on a plan it helped produce**, and the opinion must state positively that it did not.

**11. Materiality or threshold.** This standard sets no number. Which gates require independent assurance, what evidence depth each gate requires, and what tolerance triggers a hold are set by the organisation's governance and must be documented and applied. The credential holder must apply the documented depth; where a gate's evidence requirement is not documented, they must record the gap and escalate it. *Six-person internal project*: two gates, criteria of five lines each published at kick-off, a pack of six artefacts each carrying a date and a version, and assurance provided by a named colleague from another team. *Multi-partner national programme*: gates at component and programme level, criteria published against a common template so component evidence is comparable, and an assurance opinion at the programme gate independent of every partner rather than of one.

**12. Exception and waiver.** An exception permitting a gate to proceed on incomplete evidence may be approved only by the gate body's chair, only where the missing evidence is identified item by item, only with a named owner and a date for each missing item, and only where the decision record states what the decision would have been had the missing evidence been adverse. No exception permits criteria to be written after the evidence.

**13. Escalation trigger.** A request to alter criteria after evidence assembly. A criterion that cannot be assessed. Evidence offered without provenance. A condition from a previous gate still open at the next. An assurance opinion whose author appears in the pack's authorship record. Pressure to record a decision as unanimous when dissent was expressed.

**14. AI application.** AI may test criteria for assessability and flag the unmeasurable ones; check a gate pack for items missing a date, author or version; reconcile figures between pack documents; track condition closure and age open conditions; and detect criteria that changed between the published version and the version used.

**15. AI prohibition.** An AI system must not take a gate decision, give an assurance opinion, close a condition, or author a decision record entry wholesale — the record must show that a person applied judgement, and a wholly generated record cannot evidence that.



**16. AI verification. Reconciliation plus sampling with a stated basis.**

Every AI-produced gate reconciliation must be reproduced by hand for the figures that carry the decision. Every AI-flagged defect must be confirmed by a named human before it is reported as a finding. Where AI summarised a source document into the pack, a competent reviewer must perform a **clause-to-summary comparison** against the source and confirm the summary states the same position, including its qualifications.

**17. External reference.**

- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of stage-gate style continuation decisions in project governance · **EXT-028** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.
- **ISO** · *ISO 9001, Quality management systems — Requirements* · relied on for: the existence of a certifiable requirement for documented information and its control · **EXT-033** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: **this is the certifiable one in the ISO 9000 family**, but certification is voluntary and concerns a management system; a project is not certified against it and this standard does not import its requirements.
- **ISO** · *ISO 21505, Guidance on governance* · relied on for: the existence of assurance lines within governance · **EXT-032, not independently verified** · **Manual §6 category 3** · limitation: as above, with an open verification status.

**18. Jurisdictional caution.** Where a gate decision concerns a safety case, a licence, a permission, a consent or a statutory notification, the effective decision belongs to the body empowered to make it and a professional gate decision does not substitute for it. Obtain local legal advice on which approvals a specific project requires and who may grant them.

**19. Related PCI Standards.** [PCI-FND-STD-02](#); [PCI-FND-STD-06](#); [PCI-FND-STD-12](#); [PCI-PML-STD-01.03](#); [PCI-PML-STD-03.01](#); [PCI-PML-STD-09.01](#); [PCI-PML-STD-16.01](#).

**20. Related Body of Knowledge content.** PML-AI · Domain 3 · KA 3.3 Assurance, gates and escalation · topics 3.3.1 stage gates, 3.3.2 assurance lines, 3.3.4 auditability and the decision record. Also Domain 9 KA 9.2 assurance and control; Domain 16 KA 16.1 readiness.



**21. Compliance test.** Compliance is demonstrated when a reviewer can, for each gate held in the period: (a) obtain the criteria and confirm their publication date and version precede the earliest date on any evidence item in the pack; (b) confirm every pack item carries a date, an author or source system and a version; (c) take any two criteria at random, apply them to the pack, and reach the same answer the gate body recorded; (d) find every condition attached to the decision carrying a named owner, a date and a consequence; (e) find, in the following gate's pack, a closure status for each of those conditions; and (f) find, on the criteria themselves, the approval required by [PR-06](#), naming an approving authority who is neither the project being gated nor a person reporting to it for the purpose of that gate. A gate whose criteria post-date its evidence fails the test outright, and so does a gate whose criteria the project set for itself — **criteria published in time but written by the party they judge produce the same unfailable gate as criteria written late.**

**22. Breach indicators.** Criteria version dates later than the pack. Criteria phrased as "satisfactory", "adequate" or "on track" with no stated test. Conditions with no owner. Conditional passes never revisited. Gate packs assembled by the same person who signs the assurance opinion. Decisions recorded as unanimous in bodies where dissent is known to have been expressed. Evidence items that are screenshots with no source reference.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Evidence selection: from a pack of nine items, the candidate identifies the three that are not evidence and states why. Scenario judgement: the chair proposes to amend a criterion the pack does not meet. Escalation decision: a condition from two gates ago has never been closed and the project is at its final gate.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-03-03](#) v1.0. Amendment note: renumbered and restructured; legislative drafting removed; the criteria-before-evidence ordering, criterion assessability, per-item provenance, condition fields and recorded dissent separated into five process requirements; compliance test replaced with a date-ordering and re-application test two reviewers can independently perform. **Stage 9 amendment:** the standard fixed *when* the criteria are published but not *who* sets them, so a project could publish its own trivially satisfiable criteria before assembling evidence and comply in full while producing the unfailable gate element 2 describes; [PR-06](#) and step (f) of element 21 require the criteria to be approved by the authority that holds the gate decision.

## PCI STANDARD

## PCI-PML-STD-03.04

## Sponsor Accountability

**1. Normative requirement.** A credential holder must not lead delivery on a project, programme or portfolio that has no named individual sponsor who has accepted the sponsorship accountability in writing.

**2. Purpose.** The sponsor is the only role that can resolve the questions delivery cannot: whether the business case still holds, whether a benefit is still wanted, whether to stop. A vacant, nominal or rotating sponsorship transfers those questions to the delivery leader, who has neither the authority nor the standing to answer them, and the project continues by default long after the reason for it has gone.

**3. Scope.** Every credential holder in a delivery leadership role, from appointment to closure, including periods in which a sponsor is absent, has changed, or is acting. It governs the credential holder's obligation to secure, use and escalate about sponsorship; it does not purport to govern the sponsor, who might not be a credential holder.

**4. Defined terms.** *sponsor* (delivery sense) · *decision owner* · *benefit* · *material · escalation threshold* · *governance*. Additionally, **reserved sponsor decision** means a decision the delegation schedule assigns to the sponsor and to nobody below them.

**5. Required actions — process requirements.**

- **PCI-PML-STD-03.04-PR-01 — Written acceptance held.** The credential holder must hold the sponsor's dated written acceptance of the sponsorship accountability, naming the individual, before delivery commitment, and must escalate where it does not exist.
- **PCI-PML-STD-03.04-PR-02 — The reserved-decision list is agreed and used.** The credential holder must agree with the sponsor, in writing, the list of reserved sponsor decisions drawn from the delegation schedule, and must route each such decision to the sponsor rather than deciding it.
- **PCI-PML-STD-03.04-PR-03 — A stated decision service level.** The credential holder must record the time within which the sponsor undertakes to decide a reserved matter, and must escalate to the governing body where a reserved decision exceeds that time by the margin the escalation threshold states.
- **PCI-PML-STD-03.04-PR-04 — Sponsor change is a governance event.** On a change of sponsor, the credential holder must obtain the incoming sponsor's written acceptance, record a handover of the open reserved decisions and the current case position, and must not treat the

vacancy period as a period in which reserved decisions may be taken below the sponsor.

**6. Prohibited actions.** Taking a reserved sponsor decision because the sponsor is unavailable. Recording a committee as the sponsor. Continuing delivery through an unfilled sponsorship without escalating. Accepting a sponsor's verbal instruction on a reserved matter and recording it as a decision without written confirmation. Allowing a delivery supplier to supply the sponsor.

**7. Required evidence.** The sponsor's dated written acceptance; the agreed reserved-decision list; the decision-service-level record and the log of reserved decisions with elapsed times; escalations raised on breach; sponsor-change handover records.

**8. Responsible role.** The named credential holder leading delivery, for every obligation in this standard. The governing body appoints the sponsor and answers for the appointment.

**9. Approval authority.** The governing body appoints and replaces the sponsor and approves the reserved-decision list. The sponsor approves the decision service level. The credential holder approves nothing here.

**10. Independence requirement.** The sponsor must be independent of the delivery supplier: not employed by, contracted to, or remunerated by reference to the performance of the organisation delivering the work. Where the organisation is delivering to itself, the sponsor must sit outside the delivery reporting line for the purpose of the project.

**11. Materiality or threshold.** This standard sets no number. The reserved-decision list and the decision service level are set by the organisation's governance, and this standard requires that both are documented and applied. Where the organisation states no service level, the credential holder must propose one in writing and record the sponsor's response. *Six-person internal project:* the sponsor is a line director, the reserved list is four items, and the service level is "within one week" recorded in an email. *Multi-partner national programme:* there is one accountable senior responsible owner at programme level and a sponsor per component, the reserved-decision lists state which decisions rise from component to programme, and a partner-supplied sponsor is prohibited by element 10 rather than discouraged.

**12. Exception and waiver.** An exception permitting delivery to continue during a sponsorship vacancy may be approved only by the governing body, only for a stated period, only with an acting sponsor named in writing, and only where the reserved decisions that fall due in the period are listed and either taken by the acting sponsor or deferred with their consequences stated.

**13. Escalation trigger.** No written sponsor acceptance. A reserved decision exceeding the service level by the stated margin. A sponsor who declines to

decide a reserved matter. A sponsor change with no handover. A sponsor supplied by the delivery organisation. A sponsor whose stated position conflicts with the governing body's approved case.

**14. AI application.** AI may age the reserved-decision log against the service level and flag overdue items, maintain the reserved-decision list against the delegation schedule and flag divergence, and prepare the sponsor handover pack from the decision and case records.

**15. AI prohibition.** An AI system must not act as sponsor, take a reserved sponsor decision, accept sponsorship accountability, or be recorded as the approver of a reserved matter.

**16. AI verification. Reconciliation plus named approval.** The credential holder must reconcile the AI-maintained reserved-decision list against the approved delegation schedule at each gate and confirm that every reserved class appears; and the sponsor must confirm by name, at each gate, that the reserved list the project is operating is the list they accept.

#### **17. External reference.**

- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of a sponsor role distinct from the project manager · **EXT-028** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.
- **ISO** · *ISO 21505, Guidance on governance* · relied on for: the existence of an accountable owner within governance arrangements · **EXT-032, not independently verified** · **Manual §6 category 3** · limitation: as above, with an open verification status.
- **OECD (G20/OECD)** · *G20/OECD Principles of Corporate Governance* · relied on for: the existence of a board expectation that an accountable executive owner exists for major commitments · **EXT-128** · **Manual §6 category 10 — illustrative practice** · currency checked 2026-08-03 · limitation: a Council Recommendation; non-binding; **not legislation**.

**18. Jurisdictional caution.** In some jurisdictions and sectors the sponsor or senior responsible owner carries statutory or regulatory duties — for safety, for public funds, for data protection — that are distinct from this professional requirement and cannot be delegated. Obtain local legal advice on the statutory roles attaching to the specific project and entity.

**19. Related PCI Standards.** [PCI-FND-STD-01](#); [PCI-FND-STD-11](#); [PCI-PML-STD-01.01](#); [PCI-PML-STD-02.01](#); [PCI-PML-STD-02.02](#); [PCI-PML-STD-03.02](#).

**20. Related Body of Knowledge content.** PML-AI · Domain 3 · KA 3.2 Sponsorship and steering · topics 3.2.1 the sponsor role, 3.2.2 steering committees that work, 3.2.3 decision authorities and thresholds. Also Domain 1 KA 1.2 the leader's obligations; Domain 15 KA 15.4 transformation governance.

**21. Compliance test.** Compliance is demonstrated when a reviewer can (a) produce the sponsor's dated written acceptance naming an individual; (b) produce the agreed reserved-decision list and confirm that no decision in a reserved class appears in the decision record with a decider other than the sponsor or a governing-body-approved acting sponsor; (c) compute, from the reserved-decision log, the elapsed time of each reserved decision and confirm that every breach of the service level beyond the stated margin produced an escalation record; and (d) for any sponsor change in the period, produce the incoming sponsor's acceptance and the handover record. Any reserved-class decision taken below the sponsor fails the test.

**22. Breach indicators.** A sponsor named only in a slide pack. Reserved decisions decided by the project leader "in the interim". A reserved-decision log with no elapsed-time column. A sponsor employed by the delivery partner. Three sponsors in a year with no handover records. A steering committee minuted as "the sponsor".

**23. Consequence within PCI authority.** Correction required; additional review; escalation; examination failure; ethics review; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Scenario judgement: the sponsor has been unreachable for five weeks and a reserved contingency release is due. Evidence selection: which artefact establishes that sponsorship was accepted. Escalation decision: the incoming sponsor is a director of the delivery partner.

**25. Version and status.** Version 1.0 · **not yet approved** · effective on approval · **new standard** — sponsorship was addressed in the v1.0 set only through the *Related PCI standards* fields of other standards, which created no obligation. Amendment note: none.

## Domain 4 — Integration and Delivery Architecture

PCI STANDARD

PCI-PML-STD-04.01

## Change Authority and Integrated Change Control

**1. Normative requirement.** A credential holder must not permit a change to an approved control baseline to take effect before the change has been assessed for its integrated effect on scope, schedule, cost, risk, quality, benefits and dependencies, and approved by the authority the delegation schedule assigns to it.

**2. Purpose.** Baselines do not drift; they are moved, one uncontrolled change at a time, each of which looked small. The failure this prevents is the project whose plan is current, whose baseline is historical, and whose variance is therefore meaningless — after which nothing downstream, including every forecast and every gate, is measuring anything.

**3. Scope.** Every credential holder preparing, assessing, recommending, approving, implementing or assuring a change to an approved scope, schedule or cost baseline, in predictive, adaptive and hybrid delivery, including changes arising from a supplier, from a regulator and from a decision taken elsewhere in a programme.

**4. Defined terms.** *baseline* (control sense) · *delegation schedule* · *decision owner* · *material* · *evidence* · *dependency* · *benefit*. Additionally, **integrated effect** means the effect of a change across all seven dimensions listed in element 1, assessed together rather than separately; **take effect** means being reflected in a system of record that others plan or report from.

### 5. Required actions — process requirements.

- **PCI-PML-STD-04.01-PR-01 — One flow, no side doors.** Every change to an approved control baseline must enter through the defined change flow, whatever its origin, and the credential holder must not accept a change instructed directly into a delivery system, a supplier instruction or a meeting minute.
- **PCI-PML-STD-04.01-PR-02 — Seven-dimension assessment.** Each change must carry a written assessment of its effect on scope, schedule, cost, risk, quality, benefits and dependencies, with "none" stated positively where a dimension is unaffected. A blank dimension does not satisfy this requirement.
- **PCI-PML-STD-04.01-PR-03 — Baseline version integrity.** On approval, the credential holder must re-issue the affected baseline with a new version identifier and a change reference, and must retain the superseded version. Performance reported after the change must state which baseline version it is measured against.
- **PCI-PML-STD-04.01-PR-04 — Rejected and withdrawn changes retained.** Rejected, withdrawn and deferred changes must be retained in

the register with their reasons, because the pattern of what was refused is evidence at the next dispute and at the next gate.

**6. Prohibited actions.** Implementing a change before approval and regularising it afterwards. Assessing cost effect without schedule effect. Re-baselining to remove a variance rather than to reflect an approved change. Reporting performance against a baseline version that no longer exists. Accepting a supplier's implemented change because the work is already done. Deleting rejected changes from the register.

**7. Required evidence.** The change register with every entry's origin, seven-dimension assessment, recommendation, decision, decider, date and authority level; superseded and current baseline versions with change references; performance reports naming the baseline version; the rejected and withdrawn entries with reasons.

**8. Responsible role.** The named credential holder leading the project, for operating the flow and for the completeness of each assessment. The named change authority in the delegation schedule, for the decision.

**9. Approval authority.** The change authority the delegation schedule assigns, by value and by class. Changes crossing the aggregation rule go to the aggregate authority under [PCI-PML-STD-03.02-PR-03](#). A supplier may never approve a change to the client's baseline.

**10. Independence requirement.** The assessment of a change proposed by a supplier must be reviewed by a competent reviewer independent of that supplier before the decision. Where the change originates with the project leader, the assessment must be reviewed by a competent reviewer who did not prepare it.

**11. Materiality or threshold.** This standard states no figure. The change authority levels, the aggregation rule and the class of change requiring independent assessment are set by the organisation's governance; this standard requires that they are documented and applied, and that **every** change is registered regardless of value so the aggregation rule can see its inputs. *Six-person internal project*: one register, one weekly change slot, and a seven-column assessment that is usually six "none" entries and one line of substance. *Multi-partner national programme*: one flow per contract plus a programme-level flow for changes crossing an interface, with the interface owner named on both sides so a change cannot be approved by one partner into another partner's baseline.

**12. Exception and waiver.** An emergency change may be implemented before approval only where delay would cause injury, loss of life, breach of a statutory duty or irreversible loss, only on the credential holder's recorded decision naming the ground, and only if it is submitted through the flow within the time the delegation schedule states. The emergency provision must



not be used for commercial convenience, and its use is reported to the change authority each period.

**13. Escalation trigger.** A change implemented without approval. A supplier instruction that changes the baseline. A change whose assessment cannot be completed because another party will not supply the dependency effect. Re-baselining proposed with no approved change behind it. An aggregation crossing the delegation rule.

**14. AI application.** AI may draft the impact assessment across the seven dimensions for human completion, detect changes that appeared in a delivery system without a register entry by reconciling system logs against the register, cluster related changes to test the aggregation rule, and check that every issued report names a baseline version that exists.

**15. AI prohibition.** An AI system must not approve a change, determine its integrated effect as a final position, re-baseline a plan, or close a change record.

**16. AI verification. Reconciliation plus independent recomputation.** Each period, a competent reviewer must reconcile the change register against the schedule and cost systems' change logs and account for every difference. Any schedule or cost effect produced with AI assistance must be recomputed by an independent method before it is presented to the change authority, and the two results recorded.

#### **17. External reference.**

- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of integrated change control as a lifecycle practice · **EXT-028** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.
- **FIDIC** · *FIDIC suite of conditions of contract* · relied on for: the existence of contractual variation and claim mechanisms that run in parallel with, and do not replace, internal change control · **EXT-050** · **Manual §6 category 4 — contract framework** · currency checked 2026-08-03 · limitation: **binds only the parties who adopt it, through the contract they sign**; characterised generically, no clause numbers cited, no text reproduced.
- **NEC** · *NEC4 suite of contracts* · relied on for: the existence of a compensation-event mechanism with its own notification timescales · **EXT-051** · **Manual §6 category 4 — contract framework** · currency checked 2026-08-03 · limitation: as above; characterised generically.
- **Project Management Institute** · *PMBOK Guide* · relied on for: the existence of change control as professional practice · **EXT-060** ·



**Manual §6 category 5 — professional framework** · currency checked 2026-08-03 · limitation: **a professional framework, never regulatory authority**; no edition asserted, no text reproduced.

**18. Jurisdictional caution.** Contractual change and claim mechanisms carry notification periods, condition-precedent provisions and time bars whose effect is determined by the governing law of the contract, and a change correctly processed internally can still be barred contractually. Obtain legal advice on the applicable contract's notice requirements before relying on an internal change record.

**19. Related PCI Standards.** [PCI-FND-STD-06](#); [PCI-FND-STD-12](#); [PCI-FND-STD-13](#); [PCI-PML-STD-03.02](#); [PCI-PML-STD-05.01](#); [PCI-PML-STD-06.01](#); [PCI-PML-STD-07.01](#); [PCI-PML-STD-10.01](#).

**20. Related Body of Knowledge content.** PML-AI · Domain 4 · KA 4.3 Integrated baselines · topics 4.3.1 scope-schedule-cost integration, 4.3.2 configuration management, 4.3.3 baseline maintenance; KA 4.4 Integrated change control · topics 4.4.1 change flow, 4.4.2 impact assessment, 4.4.3 the change board and the decision log.

**21. Compliance test.** Compliance is demonstrated when a reviewer can reconcile, without unexplained difference: the current control baselines' version identifiers to the change register's approved entries; the change register's entry count to the change logs of the schedule and cost systems; each approved change's seven-dimension assessment to a completed field for every dimension; and each performance report issued in the period to a baseline version that exists in the version store. Any change present in a delivery system and absent from the register fails the test.

**22. Breach indicators.** Baseline version identifiers that do not increment. Reports citing a baseline "as at" a date rather than a version. Assessment fields blank in the schedule or dependency columns. Rejected changes absent from the register. Emergency provisions invoked more than a handful of times a year. Supplier progress that exceeds any approved scope.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Calculation review: a change approved on cost grounds alone is shown to consume all remaining float. Scenario judgement: a supplier has built to a verbal instruction. Evidence selection: which artefacts establish which baseline a variance was measured against.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-04-01](#) v1.0. Amendment note: renumbered and restructured; legislative drafting removed; the seven-dimension assessment, baseline version integrity and retention of rejected changes separated into process requirements; compliance test replaced with a four-way reconciliation.

## Domain 5 — Scope, Requirements and Value Definition

### PCI STANDARD

#### PCI-PML-STD-05.01

### Scope Integrity

**1. Normative requirement.** A credential holder must not allow work to be performed, or a deliverable to be produced, outside the approved scope baseline without an approved change.

**2. Purpose.** Scope creep is rarely a decision; it is an accumulation of accommodations, each helpful, each unpriced. The failure this prevents is the project that delivers more than it was funded for, later than it promised, and cannot show which of the two facts caused the other.

**3. Scope.** Every credential holder defining, approving, delivering, verifying or accepting scope, in predictive, adaptive and hybrid delivery. In adaptive delivery it applies to the scope envelope and the value envelope rather than to a fixed feature list — see [PCI-PML-STD-13.01](#).

**4. Defined terms.** *baseline* (control sense) · *acceptance* · *material* · *decision owner* · *evidence* · *traceability*. Additionally, **scope envelope** means the bounded set of outcomes within which an adaptive team may vary the specific work without a change; **out-of-scope work** means work performed that no approved scope item or scope envelope covers.

#### 5. Required actions — process requirements.

- [PCI-PML-STD-05.01-PR-01](#) — **Exclusions stated, not implied.** The approved scope baseline must state what is excluded as well as what is included, because a dispute about scope is almost always about something nobody wrote down either way.
- [PCI-PML-STD-05.01-PR-02](#) — **Acceptance criteria before production.** Every scope item must carry acceptance criteria, version-identified, agreed before the item is produced. An item whose criteria are written after it is built does not satisfy this requirement.

- **PCI-PML-STD-05.01-PR-03 — Out-of-scope work is refused or changed.** On identifying work outside the approved scope, the credential holder must either stop it or route it through the change flow, and must record which was done. Continuing it while "keeping an eye on it" does not satisfy this requirement.
- **PCI-PML-STD-05.01-PR-04 — Periodic scope reconciliation.** At an interval the governance sets and at every gate, the credential holder must reconcile work in progress and completed deliverables against the approved scope baseline and report any item with no approved scope reference.

**6. Prohibited actions.** Absorbing an addition because refusing it would be awkward. Delivering a deliverable with no acceptance criteria. Recording out-of-scope work as "clarification". Allowing a supplier to expand scope in exchange for time. Writing acceptance criteria to match what was built. Treating a stakeholder's expectation as scope without an approved change.

**7. Required evidence.** The approved scope baseline with inclusions and exclusions; acceptance criteria per item with their version and agreement date; the scope reconciliation at each interval and gate; the change records for every addition; records of work stopped as out of scope.

**8. Responsible role.** The named credential holder leading the project. The **sponsor** approves scope change beyond the credential holder's delegated authority.

**9. Approval authority.** The change authority the delegation schedule assigns for scope change. The named acceptance authority for acceptance criteria. A supplier must not approve either.

**10. Independence requirement.** Acceptance of a deliverable must be recorded by a person independent of its production, applying **PCI-PML-STD-09.01**. The scope reconciliation itself may be prepared by the project and must be reviewed at gates by a competent reviewer independent of the delivery organisation where the delegation schedule requires assurance at that gate.

**11. Materiality or threshold.** This standard sets no percentage of scope. The organisation's governance sets the change-authority thresholds and the reconciliation interval, and this standard requires that both are documented and applied. Every item of out-of-scope work is registered regardless of size, because the aggregation rule in **PCI-PML-STD-03.02** cannot operate on unregistered items. *Six-person internal project:* the baseline is a one-page inclusion and exclusion list, the criteria are two lines per deliverable, and the reconciliation is a fortnightly ten-minute check of the task board against that page. *Multi-partner national programme:* each contract carries its own scope baseline, the programme holds the interface scope explicitly so that a gap

between two partners' baselines is visible as a gap rather than assumed to be someone's, and the reconciliation runs at both levels.

**12. Exception and waiver.** An exception permitting work to proceed outside the approved baseline may be approved only by the change authority, only for a stated scope and period, and only where the cost and schedule effect is recorded at the time of approval rather than reconciled afterwards. No exception permits acceptance criteria to be written after production.

**13. Escalation trigger.** Work identified outside the baseline that the requesting party declines to route through change. A deliverable presented for acceptance with no criteria. A supplier's claim that a scope item was "always included" where the baseline is silent. Reconciliation showing a material quantity of unreferenced work.

**14. AI application.** AI may compare delivered artefacts and work-in-progress items against the scope baseline and flag those with no reference, draft acceptance criteria for human agreement, detect scope language drift between document versions, and maintain the exclusion list against change records.

**15. AI prohibition.** An AI system must not approve scope, accept a deliverable, decide that an item is within the envelope, or agree acceptance criteria.

**16. AI verification. Source tracing plus sampling with a stated basis.** Every AI-flagged unreferenced item must be traced by a named human to the scope baseline or to a change record before it is reported. Each reconciliation cycle, a competent reviewer must draw a stated sample of delivered items and confirm, item by item, that an approved scope reference and version-identified acceptance criteria exist and pre-date production.

#### **17. External reference.**

- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of scope definition and control as lifecycle practices · **EXT-028** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.
- **ISO** · *ISO 21500, Project, programme and portfolio management — Context and concepts* · relied on for: the vocabulary of project, programme and portfolio context · **EXT-027** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: since its current edition this document carries context and concepts, **not project-management guidance**, which moved to ISO 21502; it is cited here for concepts only.
- **Project Management Institute** · *PMBOK Guide* · relied on for: the existence of scope baselines and breakdown structures in professional practice · **EXT-060** · **Manual §6 category 5 — professional frame-**

**work** · currency checked 2026-08-03 · limitation: **never regulatory authority**; no edition asserted; no text reproduced.

**18. Jurisdictional caution.** Whether unpriced additional work is recoverable, and whether a variation was validly instructed, are contractual questions determined by the governing law and the contract's own mechanisms. Obtain legal advice before relying on an internal scope record in a commercial claim.

**19. Related PCI Standards.** [PCI-FND-STD-02](#); [PCI-FND-STD-12](#); [PCI-FND-STD-13](#); [PCI-PML-STD-04.01](#); [PCI-PML-STD-05.02](#); [PCI-PML-STD-09.01](#); [PCI-PML-STD-13.01](#).

**20. Related Body of Knowledge content.** PML-AI · Domain 5 · KA 5.1 Scope definition and the scope baseline; KA 5.4 Scope change, creep and verification/acceptance. Also Domain 4 KA 4.2 breakdown structures; Domain 9 KA 9.3 acceptance and nonconformance.

**21. Compliance test.** Compliance is demonstrated when a reviewer can take the list of deliverables produced and work packages in progress at a stated date and match **every one** to either an item in the approved scope baseline or an approved change record — with no residue — and can confirm, for a stated sample, that version-identified acceptance criteria exist with an agreement date **earlier than** the production start date recorded for that item. Any unmatched item, or any criteria dated after production began, fails the test.

**22. Breach indicators.** Deliverables the baseline does not mention. Acceptance criteria whose version date follows the delivery date. An exclusion list that has never been updated. Supplier progress reports describing work no change record covers. Reconciliations that report "no exceptions" every period on a project with an active change register.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Scenario judgement: a stakeholder's "small addition" has been in progress for three weeks. Evidence selection: which artefact establishes that a deliverable was in scope. Calculation review: reconciling a deliverable list against a baseline and identifying the residue.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-05-01 v1.0](#). Amendment note: renumbered and restructured; legislative drafting removed; exclusions, criteria-before-production, the stop-or-change obligation and periodic reconciliation

separated into process requirements; compliance test replaced with a no-residue matching test.

## PCI STANDARD

### PCI-PML-STD-05.02

## Requirements Traceability

**1. Normative requirement.** A credential holder must maintain traceability from every approved requirement to its source, to the work that satisfies it and to the test or acceptance criterion that proves it, for the life of the project.

**2. Purpose.** Without traceability nobody can answer three questions that decide whether a project succeeded: why is this requirement here, who asked for it, and is it still wanted; is anything being built that no requirement asked for; and is any requirement unproven at acceptance. Each unanswered question produces a distinct and expensive failure, and all three are cheap to prevent and impossible to reconstruct.

**3. Scope.** Every credential holder eliciting, approving, delivering, testing or accepting requirements, in predictive, adaptive and hybrid delivery, including requirements arising from regulation, from safety analysis and from interfaces with other projects.

**4. Defined terms.** *traceability · acceptance · evidence · material · baseline* (control sense) · *dependency*. Additionally, **orphan requirement** means an approved requirement with no work item satisfying it; **orphan deliverable** means delivered work satisfying no approved requirement.

### 5. Required actions — process requirements.

- **PCI-PML-STD-05.02-PR-01 — Every requirement carries a source.** Each approved requirement must record who raised it, the document or decision it derives from, and its date, so that its continued relevance can be tested rather than assumed.
- **PCI-PML-STD-05.02-PR-02 — Bidirectional links maintained.** The credential holder must maintain links in both directions — requirement to work item to test, and back — and must update them when a change is approved rather than at the end.
- **PCI-PML-STD-05.02-PR-03 — Orphans reported, not tidied.** At each reporting cycle and at each gate, the credential holder must report the count and identity of orphan requirements and orphan deliverables. Resolving an orphan by deleting the requirement, without a change record, does not satisfy this requirement.
- **PCI-PML-STD-05.02-PR-04 — Regulatory and safety requirements marked and never silently dropped.** Requirements deriving from a

statutory duty, a licence condition, a safety analysis or a consent must be marked as such in the register, and may be removed only by an approved change that names the authority permitting the removal.

**6. Prohibited actions.** Building work that satisfies no approved requirement and recording it as delivery. Closing a requirement as met with no test or acceptance evidence. Deleting an inconvenient requirement without a change record. Merging two requirements to hide that one is unmet. Marking a regulatory requirement as descoped on the project's own authority.

**7. Required evidence.** The requirements register with source, raiser, date and status per requirement; the traceability links to work items and to tests or acceptance criteria; the orphan report at each cycle and gate; change records for every requirement added, altered or removed; the marked subset of regulatory and safety requirements with their authority references.

**8. Responsible role.** The named credential holder leading the project, for maintaining traceability. The named requirement owner for each requirement's continued validity. The named acceptance authority for closure.

**9. Approval authority.** The change authority approves the addition, alteration or removal of an approved requirement. Only the authority that owns a regulatory, licence or safety requirement may permit its removal, and never the project.

**10. Independence requirement.** The confirmation that a requirement is proven must be given by a person independent of the work that satisfies it, under *PCI-PML-STD-09.01*. The traceability record itself is maintained by the project and reviewed at gates by a competent reviewer independent of the delivery organisation where the delegation schedule requires assurance.

**11. Materiality or threshold.** This standard sets no number of requirements and no coverage percentage, because a coverage percentage is exactly the figure that gets managed rather than met. What it requires is that **the orphan counts are reported**, that the trend is visible, and that the organisation's governance sets the tolerance at which an orphan count triggers escalation. *Six-person internal project:* the traceability matrix is a spreadsheet of forty rows with four columns, and the orphan report is a filter on it. *Multi-partner national programme:* requirements are held per partner with a programme-level register for cross-partner and interface requirements; the orphan report runs at both levels, and an interface requirement that is nobody's orphan is precisely the failure the programme-level report exists to catch.

**12. Exception and waiver.** An exception permitting a requirement to be accepted without a completed test may be approved only by the acceptance authority, only where the residual risk is recorded and owned, only for a stated period, and never for a requirement marked regulatory or safety-derived.



**13. Escalation trigger.** An orphan count exceeding the documented tolerance. A regulatory requirement proposed for removal. A requirement closed with no test evidence. A supplier declining to supply traceability contractually required. Discovery that traceability has not been maintained since a named date.

**14. AI application.** AI may build and maintain a traceability matrix from requirement, backlog and test artefacts; detect orphans in both directions; detect duplicate and contradictory requirements; propose candidate links for human confirmation; and flag requirements whose source reference does not resolve.

**15. AI prohibition.** An AI system must not approve a requirement, close one as met, decide that a test proves a requirement, or delete a link.

**16. AI verification. Sampling with a stated basis plus source tracing.** Every AI-proposed link must be confirmed by a named human before it is recorded as a trace. Each gate, a competent reviewer must draw a stated sample of requirements and trace each by hand from source document to work item to test evidence, and must additionally sample the **regulatory and safety** subset in full rather than by sample, because its population is small and its failure mode is severe.

#### **17. External reference.**

- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of requirements management within delivery practice · **EXT-028** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.
- **ISO** · *ISO 9001, Quality management systems — Requirements* · relied on for: the existence of a certifiable requirement that product requirements are determined, reviewed and verified · **EXT-033** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: voluntary unless imported; certification concerns a management system, not a project; this standard imports none of its requirements.
- **Project Management Institute** · *PMBOK Guide* · relied on for: the existence of a requirements traceability matrix as professional practice · **EXT-060** · **Manual §6 category 5 — professional framework** · currency checked 2026-08-03 · limitation: **never regulatory authority**; no edition asserted; no text reproduced.

**18. Jurisdictional caution.** Requirements deriving from statute, licence conditions, building or safety regulation, accessibility law or data-protection law are determined by those instruments and by the regulator,



not by the project's register. Obtain local legal and regulatory advice on which requirements are non-removable for the specific project.

**19. Related PCI Standards.** [PCI-FND-STD-06](#); [PCI-FND-STD-07](#); [PCI-FND-STD-12](#); [PCI-PML-STD-05.01](#); [PCI-PML-STD-09.01](#); [PCI-PML-STD-13.02](#); [PCI-PML-STD-16.02](#).

**20. Related Body of Knowledge content.** PML-AI · Domain 5 · KA 5.2 Requirements elicitation, analysis and traceability; KA 5.3 value definition and prioritisation. Also Domain 9 KA 9.3 acceptance and nonconformance; Domain 13 KA 13.2 backlogs.

**21. Compliance test.** Compliance is demonstrated when a reviewer can select any approved requirement at random and, without asking its author, reach its source document, the work item that satisfies it and the test or acceptance evidence that proves it — and can perform the same traverse in reverse from any delivered item — and when the orphan report for the current cycle can be reproduced from the register by an independent query returning the same counts. Any requirement marked regulatory or safety-derived that cannot be traversed in both directions fails the test outright.

**22. Breach indicators.** Traceability last updated at the previous gate. Orphan counts reported as zero in every period. Requirements closed on the same date they were raised. A test suite whose case count is unrelated to the requirement count. Requirements removed between register versions with no change reference. Source fields containing a person's name and no document.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Evidence selection: which artefacts prove a requirement was met. Scenario judgement: a safety-derived requirement is proposed for deferral to a later release. Calculation review: reconciling requirement, work-item and test counts to locate the orphans.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-05-02](#) v1.0. Amendment note: renumbered and restructured; legislative drafting removed; the regulatory and safety marking obligation added; orphan reporting made an express process requirement; compliance test replaced with a bidirectional traverse a reviewer performs unaided.

## Domain 6 – Planning, Scheduling and Delivery Flow

### PCI STANDARD

#### PCI-PML-STD-06.01

### Schedule Credibility

**1. Normative requirement.** A credential holder must not issue, endorse or rely on a schedule whose logic, durations, constraints or status they know, or ought to know, to misrepresent the achievable completion of the work.

**2. Purpose.** A schedule is a statement to other people about when they can rely on something. The failure this prevents is the schedule that meets the required date by construction — constraints imposed in place of logic, durations compressed without a resource or method change, progress claimed on activities that have not started — which is believed for exactly as long as it takes for the first dependent party to plan against it.

**3. Scope.** Every credential holder producing, reviewing, approving, statusing, forecasting or assuring a schedule at any level, in predictive, adaptive and hybrid delivery, including supplier schedules incorporated into a client plan and programme-level integrated schedules.

**4. Defined terms.** *baseline* (control sense) · *evidence* · *material* · *dependency* · *competent reviewer* · *escalation threshold*. Additionally, **hard constraint** means a date imposed on an activity that overrides its logic; **status date** means the date to which progress is stated; **negative float** means the condition in which the logic cannot achieve an imposed date.

#### 5. Required actions — process requirements.

- **PCI-PML-STD-06.01-PR-01 — Logic, not constraint, carries the dates.** Completion dates must be derived from activity logic and durations. Every hard constraint must be listed with the reason it exists and the authority that imposed it, and the credential holder must state the completion date the logic produces without it.
- **PCI-PML-STD-06.01-PR-02 — Compression states its mechanism.** Where a duration has been reduced, the credential holder must state the mechanism — added resource, changed method, changed scope, increased overlap — and its cost and risk consequence. A duration reduced with no stated mechanism is a target, not a duration, and must be labelled as one.
- **PCI-PML-STD-06.01-PR-03 — Progress is evidenced, not asserted.** Physical or output-based progress must be supported by evidence of the work performed. Progress recorded against an activity with no evidence of commencement does not satisfy this requirement.
- **PCI-PML-STD-06.01-PR-04 — Negative float is reported, never absorbed.** Negative float must be reported to the authority the escalation

threshold names, with its cause and the options, and must not be removed by re-imposing constraints, deleting logic, or extending the imposed date without an approved change.

- **PCI-PML-STD-06.01-PR-05 — Status date and baseline version stated on issue.** Every issued schedule must carry its status date and the control baseline version it is measured against.

**6. Prohibited actions.** Imposing a constraint to produce a required date and presenting the result as a forecast. Recording progress on activities not started. Deleting or reversing logic to remove negative float. Reporting a completion date the logic does not produce. Incorporating a supplier's schedule without reconciling its interfaces. Re-baselining to eliminate a variance with no approved change.

**7. Required evidence.** The schedule file with its logic, its constraint list and its status date; the constraint register with reasons and imposing authorities; the compression record with mechanisms and consequences; progress evidence per statused activity for a stated sample; the negative-float reports and their escalations; the baseline version reference on each issue.

**8. Responsible role.** The named credential holder leading the project, for the schedule issued in the project's name. The named planner or scheduler, for its technical preparation. Neither may be recorded as the other.

**9. Approval authority.** The change authority the delegation schedule assigns approves a change to the schedule baseline. The sponsor approves a change to a committed external date within their authority; above it, the governing body. A supplier must not approve a change to the client's schedule baseline.

**10. Independence requirement.** At each gate at which the delegation schedule requires assurance, the schedule must be reviewed by a competent reviewer independent of its preparation. Where the schedule supports a contractual claim or a committed external date, the review must also be independent of the party that benefits from the date.

**11. Materiality or threshold. This standard states no float threshold, no density figure and no percentage,** and inventing one would be indefensible: what counts as an acceptable constraint count or float profile depends on the contract, the sector and the planning method. The organisation's governance must document the schedule-quality criteria it applies, the tolerance at which a variance is escalated, and the sample size for progress evidence; this standard requires that they exist and are applied. Where the organisation applies a published schedule-assessment method, the credential holder must record which method and which criteria, so that a reported score can be interpreted. *Six-person internal project:* forty activities in a shared tool, three constraints each with a stated reason, and progress evidenced by the same artefacts the team produces anyway. *Multi-partner national programme:* an

integrated schedule assembled from partner schedules, with the interface milestones owned on both sides by name, and constraint and progress rules stated in the partner agreement so that a partner's internal convention cannot silently change the programme's dates.

**12. Exception and waiver.** An exception permitting a schedule to be issued with an unreconciled interface or an unsupported progress claim may be approved only by the sponsor, only for one reporting cycle, only where the affected activities are identified and the issued schedule states the limitation on its face. No exception permits a completion date to be presented as achievable when the logic does not produce it.

**13. Escalation trigger.** Negative float against a committed date. A required completion date that the logic cannot produce. An instruction to record progress not performed. A supplier schedule that cannot be reconciled at its interfaces. A constraint imposed with no stated authority. A compression with no mechanism.

**14. AI application.** AI may run schedule-quality checks, identify open ends, redundant logic, excessive lags and constraint concentrations, compare the current schedule against the baseline and summarise the differences, generate scenario schedules for human evaluation, and detect progress claims inconsistent with cost or delivery data.

**15. AI prohibition.** An AI system must not approve a schedule, decide a duration that is then issued without human derivation, remove a constraint, status an activity, or determine that a date is achievable.

**16. AI verification. Independent recomputation plus boundary testing.** A competent reviewer must recompute the critical path and the float on the issued schedule using a second method or tool and reconcile any difference before issue. Where AI generated or compressed durations, the reviewer must test the boundaries — the shortest and longest credible duration for the sampled activities — and record whether the issued value sits inside the range the mechanism supports. Progress statused with AI assistance must be confirmed by **sampling with a stated basis** against physical evidence.

#### **17. External reference.**

- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of schedule development and control as lifecycle practices · **EXT-028** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.
- **Project Management Institute** · *Practice Standard for Scheduling* · relied on for: the existence of recognised schedule-quality attributes in professional practice · **EXT-062** · **Manual §6 category 5 — profes-**

**sional framework** · currency checked 2026-08-03 · limitation: **never regulatory authority**; no metric, threshold or checklist from it is reproduced or relied on.

- **AACE International** · *Total Cost Management (TCM) Framework* · relied on for: the existence of a planning and scheduling process within a cost-management framework · **EXT-064, not independently verified — verify current requirements** · **Manual §6 category 5 — professional framework** · limitation: as above; no recommended-practice text is reproduced.

**18. Jurisdictional caution.** Where a schedule supports an extension-of-time claim, a delay analysis or a liquidated-damages position, the acceptable method and the evidential standard are determined by the contract and by the forum, and they differ between jurisdictions and between tribunals. Obtain legal advice before a schedule is relied on in a claim.

**19. Related PCI Standards.** [PCI-FND-STD-05](#); [PCI-FND-STD-02](#); [PCI-FND-STD-13](#); [PCI-PML-STD-04.01](#); [PCI-PML-STD-07.01](#); [PCI-PML-STD-08.01](#); [PCI-PML-STD-15.01](#).

**20. Related Body of Knowledge content.** PML-AI · Domain 6 · KA 6.1 Planning levels and logic networks; KA 6.2 The critical path and float; KA 6.3 Resources, constraints, milestones and rolling wave; KA 6.4 Delivery flow, recovery and forecasting. Also Domain 15 KA 15.1 dependency arithmetic.

**21. Compliance test.** Compliance is demonstrated when a reviewer can: (a) open the issued schedule and find its status date and baseline version on the face of it; (b) list every hard constraint and find, for each, a reason and an imposing authority in the constraint register; (c) remove the constraints in a copy of the schedule, recompute, and reconcile the resulting completion date to the date the credential holder stated under [PR-01](#); (d) draw a stated sample of statused activities and find, for each, evidence of the work performed; and (e) confirm that every period in which negative float existed produced an escalation record. A completion date that cannot be reproduced from the logic fails the test.

**22. Breach indicators.** Constraint counts rising as the committed date approaches. Float distributions with a large spike at zero. Activities at 90 per cent for several periods. Negative float that appears and disappears without a change record. Durations that changed with no mechanism recorded. Schedules issued with no status date. A supplier schedule whose interface dates differ from the client's by more than the reporting cycle.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Calculation review: a forward and backward pass on a small network with an imposed date, identifying the negative float and its cause. Scenario judgement: a sponsor asks for the completion date to be "brought back" without a scope or resource change. Evidence selection: which artefacts support a progress claim.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-06-01](#) v1.0. Amendment note: renumbered and restructured; legislative drafting removed; the constraint-transparency, compression-mechanism, progress-evidence, negative-float and status-date obligations separated into five process requirements; element 11 rewritten to require documented schedule-quality criteria rather than to assert any; compliance test replaced with a remove-the-constraints reproduction test.

## Domain 7 — Cost, Resources and Commercial Awareness

### PCI STANDARD

#### PCI-PML-STD-07.01

### Cost Stewardship

**1. Normative requirement.** A credential holder must report the project's cost position — committed, incurred, accrued and forecast — on a basis that is complete, current and reconcilable to the organisation's financial records.

**2. Purpose.** The two failures this prevents are opposite and equally common: the position that omits commitments and accruals, so the project looks affordable until the invoices arrive; and the forecast that is a target restated, so the overrun is discovered when the budget is exhausted rather than when it became inevitable. Both are failures of completeness, not of arithmetic.

**3. Scope.** Every credential holder preparing, reviewing, approving, reporting or assuring a project, programme or portfolio cost position or forecast, including contingency and reserve positions, in every delivery model.

**4. Defined terms.** *baseline* (control sense) · *material* · *evidence* · *competent reviewer* · *escalation threshold* · *decision owner*. Additionally, **committed** means value contractually committed but not yet incurred; **accrued** means value of work performed but not yet invoiced; **contingency** means provision

for identified risk held inside the control baseline; **management reserve** means provision for unidentified risk and scope change held outside it and released by the authority the governance plan names.

#### 5. Required actions — process requirements.

- **PCI-PML-STD-07.01-PR-01 — Four elements always present.** Every issued cost position must state committed, incurred, accrued and forecast values, and must not omit an element because it is incomplete; where an element is estimated, the basis is stated.
- **PCI-PML-STD-07.01-PR-02 — Reconciliation to the ledger.** At the interval the governance sets, the credential holder must reconcile the project cost position to the organisation's financial ledger and account for every difference, retaining the reconciliation.
- **PCI-PML-STD-07.01-PR-03 — Forecast method stated and consistent.** Each forecast must state the method used and the assumptions it rests on, and the method must not be changed between periods without recording the change, its reason and its effect on the reported figure.
- **PCI-PML-STD-07.01-PR-04 — Contingency and reserve reported separately.** Contingency and management reserve must be reported separately from the base cost and from each other, with drawdown recorded against the specific risk or change that justified it and the authority that released it.
- **PCI-PML-STD-07.01-PR-05 — Adverse movement reported in the period it is known.** A material adverse movement must be reported in the reporting cycle in which it becomes known, and must not be held for a later cycle, netted against an unrealised saving, or absorbed into contingency without a release decision.

**6. Prohibited actions.** Reporting incurred cost as the position while commitments are material and unstated. Drawing contingency to conceal a base-cost overrun. Presenting a forecast equal to the budget with no supporting change. Netting an adverse movement against an optimistic one. Changing forecast method to produce a better figure. Recording a management-reserve drawdown as a base-cost saving.

**7. Required evidence.** The cost report showing the four elements; the ledger reconciliation with its differences explained; the forecast with its stated method and assumptions and the record of any method change; the contingency and reserve registers with drawdowns, their justifying risk or change, and the releasing authority; the record of adverse movements with the date known and the date reported.

**8. Responsible role.** The named credential holder leading the project, for the position issued in the project's name. The named cost or finance role, for its preparation and for the ledger reconciliation.



**9. Approval authority.** The authority the governance plan names releases contingency; the authority it names — typically the sponsor or change authority, and this standard does not assume which — releases management reserve, through change control. Neither is released by the person who prepares the forecast.

**10. Independence requirement.** The ledger reconciliation must be prepared or verified by a person independent of the project's cost reporting. At each gate at which the delegation schedule requires assurance, the forecast must be reviewed by a competent reviewer independent of its preparation.

**11. Materiality or threshold.** This standard states no percentage. The organisation's governance sets the materiality threshold for adverse movement, the reconciliation interval, the contingency release authority and the forecast tolerance; this standard requires that each is documented and applied. Where the governance plan does not name the authority that releases management reserve, the credential holder must escalate that gap rather than assume the answer — the suite terminology audit records that different books name different roles, and assuming one is how a reserve is released by someone who could not release it. *Six-person internal project:* one cost line per person plus three purchase orders; the reconciliation is a monthly comparison against the finance extract and takes under an hour. *Multi-partner national programme:* the four elements are reported per partner and consolidated, the consolidation states which partner's accrual basis differs and by how much, and management reserve is held and released at programme level so that a component cannot fund its own overrun.

**12. Exception and waiver.** An exception permitting a cost position to be issued without a completed reconciliation may be approved by the sponsor for one cycle, only where the report states the limitation on its face and identifies the unreconciled amount. No exception permits a known material adverse movement to go unreported in the cycle in which it became known.

**13. Escalation trigger.** A forecast that exceeds the approved budget or the documented tolerance. A reconciliation difference that cannot be explained. Contingency drawdown against risks that were not in the register. Instruction to delay reporting an adverse movement. A supplier's commitment position that cannot be obtained.

**14. AI application.** AI may reconcile cost data across systems and surface unexplained differences, detect commitments absent from the position, produce several forecast methods for human selection, flag contingency drawdowns with no matching register entry, and detect forecast-method changes between periods.



**15. AI prohibition.** An AI system must not approve a cost position, select the forecast method that is issued, release contingency or reserve, or determine that a difference is immaterial.

**16. AI verification. Independent recomputation plus reconciliation.** The credential holder must recompute the headline forecast by an independent method and reconcile the two results before issue, recording both. Every AI-produced reconciliation must be checked by a competent reviewer against the ledger extract itself, with the extract date and filter recorded, and every AI-surfaced difference confirmed by a named human before it is reported as resolved.

#### 17. External reference.

- **AACE International** · *Total Cost Management (TCM) Framework* · relied on for: the existence and purpose of a cost-control cycle and of maturity-based estimate classification · **EXT-064, not independently verified — verify current requirements** · **Manual §6 category 5 — professional framework** · limitation: **a professional framework, never regulatory authority**; no class table, accuracy range or recommended-practice text is reproduced.
- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of cost planning and control as lifecycle practices · **EXT-028** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.
- **Project Management Institute** · *PMBOK Guide* · relied on for: the existence of cost baselines, reserves and forecasting in professional practice · **EXT-060** · **Manual §6 category 5** · currency checked 2026-08-03 · limitation: **never regulatory authority**; no edition asserted; no text reproduced.

**18. Jurisdictional caution.** How costs are recognised, when a liability is recorded, how contract assets and provisions are measured and what must be disclosed are determined by the financial-reporting framework the entity applies and by its auditors, not by a project cost report. **This standard states no accounting treatment.** Obtain qualified accounting advice for the specific entity and framework.

**19. Related PCI Standards.** PCI-FND-STD-02; PCI-FND-STD-05; PCI-FND-STD-07; PCI-PML-STD-04.01; PCI-PML-STD-06.01; PCI-PML-STD-07.02; PCI-PML-STD-08.01.

**20. Related Body of Knowledge content.** PML-AI · Domain 7 · KA 7.1 Estimating and budgeting; KA 7.2 The cost baseline, actuals and forecasting; KA 7.3 Earned value: measurement, variances and forecasting. Also Domain 8 KA 8.3 responses, reserves and governance.

**21. Compliance test.** Compliance is demonstrated when the issued cost position can be reconciled, without unexplained difference, to: the financial ledger extract at the stated date, the commitment register, the accrual schedule, the approved change register, the contingency and reserve registers with their release authorities, and the approved control baseline version named on the report — and when the forecast's stated method, applied to the same inputs by a second preparer, reproduces the reported figure within the tolerance the method itself implies. A position that omits commitments, or a forecast whose method cannot be reproduced, fails the test.

**22. Breach indicators.** Positions reporting incurred cost only. Forecasts equal to budget for consecutive periods on a project with an active risk register. Contingency drawdowns with no risk reference. Reconciliations with a persistent unexplained residual. A forecast method that changes in the period an overrun would otherwise appear. Adverse movements reported one cycle after the supplier notified them.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Calculation review: an earned-value suite in which the candidate selects and justifies a forecasting method and identifies why another is unsupportable. Scenario judgement: contingency is proposed to cover a base-cost overrun four weeks before a gate. Evidence selection: which artefacts establish the completeness of a reported position.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-07-01](#) v1.0, whose subject matter is split between this standard and [PCI-PML-STD-07.02](#). Amendment note: renumbered and restructured; legislative drafting removed; resource obligations moved out to their own standard so that one standard carries one principal obligation; the four-element completeness rule, ledger reconciliation, method consistency, separate reserve reporting and same-cycle adverse reporting separated into five process requirements.

PCI STANDARD

[PCI-PML-STD-07.02](#)

## Resource Decisions and the Commitment of People

**1. Normative requirement.** A credential holder must not commit a person, a team or a shared resource to a project plan without the agreement of the named individual who controls that resource's availability.

**2. Purpose.** The most common cause of a plan that cannot be delivered is a resource assumption nobody agreed to. The failure this prevents is the plan built on named people who are already committed elsewhere — a plan that is arithmetically sound, professionally presented, and false from the day it is issued, with the consequence landing on the individuals concerned rather than on the plan's author.

**3. Scope.** Every credential holder planning, committing, reallocating or reporting on human and shared resources — internal staff, seconded staff, supplier staff, specialist equipment and shared environments — on projects, programmes and portfolios, in every delivery model.

**4. Defined terms.** *decision owner · material · evidence · dependency · escalation threshold · baseline* (control sense). Additionally, **resource owner** means the named individual who controls whether a person or shared resource is available to a project; **committed availability** means the proportion of a resource's capacity that the resource owner has agreed to this project for a stated period.

### 5. Required actions — process requirements.

- **PCI-PML-STD-07.02-PR-01 — Named agreement before the plan is issued.** Each resource assumption material to the plan must carry the resource owner's name and a dated agreement to the committed availability, obtained before the plan is issued.
- **PCI-PML-STD-07.02-PR-02 — Aggregate demand tested against capacity.** The credential holder must test the plan's total demand on each shared resource against that resource's total committed availability across all claims on it, and must report any over-commitment rather than assuming resolution.
- **PCI-PML-STD-07.02-PR-03 — Withdrawal is a change, not an adjustment.** Where a committed resource is withdrawn or reduced, the credential holder must record it, assess the effect through **PCI-PML-STD-04.01**, and report the consequence, rather than re-planning silently around it.
- **PCI-PML-STD-07.02-PR-04 — Sustained overload is reported.** Where delivery depends on individuals working beyond the hours or duration the organisation's own policy permits, the credential holder must report that dependency to the sponsor as a delivery risk with its cause, and must not present the plan as achievable without stating it.

**6. Prohibited actions.** Naming people in a plan without their resource owner's agreement. Planning to an availability the resource owner has refused. Concealing over-commitment by planning at an aggregate level that hides it. Presenting sustained overtime as capacity. Reallocating a person committed to another project without that project's resource owner. Recording a supplier's named key personnel as committed when the contract does not secure them.

**7. Required evidence.** The resource plan with, per material resource, the owner's name and dated agreement; the aggregate demand-against-capacity analysis; withdrawal records and their change assessments; the reports of sustained overload with their causes; the contractual basis for any supplier key personnel recorded as committed.

**8. Responsible role.** The named credential holder leading the project, for obtaining and recording agreements and for reporting over-commitment. The named resource owner, for the availability they agree.

**9. Approval authority.** The resource owner agrees availability. The sponsor resolves competing claims within their authority; above it, the portfolio or programme authority the delegation schedule names. The project leader must not resolve a competing claim on another owner's resource.

**10. Independence requirement.** Not applicable to the agreement itself, because the resource owner's consent is by definition an interested party's consent to their own resource; independence attaches to the portfolio-level capacity test in [PCI-PML-STD-15.02](#), which must be performed by a function independent of any single competing project.

**11. Materiality or threshold.** This standard sets no percentage of a person's time. The organisation's governance sets the threshold at which a resource assumption is material enough to require a recorded agreement, the limits on working hours and duration, and the escalation route for competing claims; this standard requires that these exist and are applied. Where no threshold is documented, every named individual in the plan requires a recorded agreement. *Six-person internal project:* six agreements, one line each, in an email thread the leader retains — and the aggregate test is a single check that no one appears on two plans at once. *Multi-partner national programme:* resource owners sit in several organisations, the aggregate test runs at portfolio level under [PCI-PML-STD-15.02](#), and supplier key personnel are secured contractually rather than assumed, because a name in a tender is not committed availability.

**12. Exception and waiver.** An exception permitting a plan to be issued with an unagreed material resource assumption may be approved by the sponsor for a stated period not exceeding one reporting cycle, only where the plan states the unagreed assumption on its face and identifies the affected

activities. No exception permits an individual to be named in an issued plan against their resource owner's recorded refusal.

**13. Escalation trigger.** A resource owner's refusal or withdrawal of a committed resource. Aggregate demand exceeding committed availability on a shared resource. Delivery dependent on hours beyond policy. A supplier substituting named key personnel. A competing claim the sponsor cannot resolve.

**14. AI application.** AI may aggregate demand across plans and surface over-commitment, model the effect of alternative allocations, flag plan entries with no recorded owner agreement, and detect individuals whose aggregate allocation exceeds their committed availability.

**15. AI prohibition.** An AI system must not commit a person to a plan, agree availability, resolve a competing claim, or reallocate people between projects. Where an optimiser produces an allocation, it has produced a recommendation, and the decision is a reserved class under [PCI-PML-STD-01.02-PR-01](#).

**16. AI verification. Reconciliation plus named approval.** Every AI-produced allocation must be reconciled against the recorded owner agreements before it is issued, and each affected resource owner must confirm by name. Every AI-surfaced over-commitment must be confirmed by a named human against the competing plans before it is escalated.

#### 17. External reference.

- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of resource planning and of the negotiation of resources with those who control them · **EXT-028** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.
- **ISO** · *ISO 45001, Occupational health and safety management systems — Requirements with guidance for use* · relied on for: the existence of a certifiable management-system standard within which workload and fatigue are addressed as health and safety matters · **EXT-123** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: certifiable, but adoption is voluntary unless a contract or regulator requires it; this standard imports none of its requirements and states no fatigue rule of its own.
- **AACE International** · *Total Cost Management (TCM) Framework* · relied on for: the existence of resource economics within a cost-management framework · **EXT-064, not independently verified** · **Manual §6 category 5 — professional framework** · limitation: **never regulatory authority**; no text reproduced.

**18. Jurisdictional caution.** Working-time law, collective agreements, works-council consultation rights, secondment and agency-worker rules, and health-and-safety duties determine what may lawfully be required of people, and they differ sharply by jurisdiction. Obtain local legal and human-resources advice before planning on extended hours, cross-border deployment or supplier staff substitution.

**19. Related PCI Standards.** [PCI-FND-STD-01](#); [PCI-FND-STD-11](#); [PCI-PML-STD-06.01](#); [PCI-PML-STD-07.01](#); [PCI-PML-STD-12.01](#); [PCI-PML-STD-15.02](#).

**20. Related Body of Knowledge content.** PML-AI · Domain 7 · KA 7.4 Resource economics, procurement strategy and cash. Also Domain 6 KA 6.3 resources, constraints, milestones and rolling wave; Domain 12 KA 12.2 team formation and retention; Domain 15 KA 15.3 capacity and enterprise PMOs.

**21. Compliance test.** Compliance is demonstrated when a reviewer can take every named individual and shared resource in the issued plan whose contribution exceeds the documented materiality threshold and find, for each, a dated agreement from the named resource owner covering the committed availability the plan assumes; and can sum the plan's demand on each shared resource against all other current claims on it and reproduce the over-commitment figure the project reported. Any material resource assumption with no recorded owner agreement fails the test.

**22. Breach indicators.** Plans naming individuals whose owners have not been asked. Allocation percentages summing above 100 across concurrent plans. Delivery dates that depend on weekend working described nowhere. Supplier key personnel named in a plan and absent from the contract. Resource withdrawals resolved by re-planning with no change record. Resource owners who first learn of a commitment from the published plan.

**23. Consequence within PCI authority.** Correction required; additional review; escalation; examination failure; ethics review; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Calculation review: aggregate allocation across three concurrent plans showing a specialist committed at 160 per cent. Scenario judgement: a resource owner withdraws a key specialist a fortnight before a milestone. Ethical dilemma: a plan can only be met by sustained overtime the leader has not disclosed.

**25. Version and status.** Version 1.0 · **not yet approved** · effective on approval · **new standard**, carrying the resource obligations formerly bundled inside [PML-LAW-07-01](#) v1.0, which stated cost and resource obligations in one rule and could therefore be breached in two unrelated ways at once.

Amendment note: separated so that each standard carries one principal obligation.

## Domain 8 — Risk, Uncertainty and Resilience

### PCI STANDARD

#### PCI-PML-STD-08.01

### Risk Escalation

**1. Normative requirement.** A credential holder must escalate a risk whose exposure exceeds the escalation threshold the delegation schedule sets, to the named authority that threshold identifies, within the time it states.

**2. Purpose.** The value of an escalation is mostly in its timing: the same warning is worth a great deal early and almost nothing late, because what it buys is the option set. The failure this prevents is the escalation that arrives after the decision it was supposed to inform — technically compliant, entirely useless — and the one that never arrives because the person holding the risk hoped to resolve it first.

**3. Scope.** Every credential holder identifying, assessing, owning, responding to, reporting or assuring risk on a project, programme or portfolio, in every delivery model, including risks arising in a supplier's scope and risks transferred by contract.

**4. Defined terms.** *escalation threshold · material · decision owner · evidence · dependency · contingency* (as defined in [PCI-PML-STD-07.01](#)). Additionally, **exposure** means the assessed consequence of a risk combined with its assessed likelihood, on the basis the organisation's risk framework states; **decision action window** means the remaining duration less the escalation latency, expressed as a proportion of the remaining duration.

#### 5. Required actions — process requirements.

- [PCI-PML-STD-08.01-PR-01](#) — **Every risk has one named owner.** Each risk must carry exactly one named owner with the authority to act on it; a risk owned by a team, a function or a supplier organisation rather than a person does not satisfy this requirement.
- [PCI-PML-STD-08.01-PR-02](#) — **Escalate on threshold, not on comfort.** The credential holder must escalate when the documented threshold is met, whether or not they expect to resolve the risk, and must record the date the threshold was met alongside the date of escalation.
- [PCI-PML-STD-08.01-PR-03](#) — **Escalate with options.** Every escalation must carry the options open to the receiving authority, their



consequences, and the date after which each option closes. An escalation that reports a problem with no option set does not satisfy this requirement.

- **PCI-PML-STD-08.01-PR-04 — Test the action window.** Before escalating, the credential holder must state whether the receiving authority can still act in time, and where the window is negative must say so on the face of the escalation and route it out of cycle.
- **PCI-PML-STD-08.01-PR-05 — Contingency is drawn against a registered risk.** Contingency must be drawn only against a risk recorded in the register before the drawdown, with the releasing authority named.

**6. Prohibited actions.** Holding a risk above the threshold because a fix is expected. Reducing an assessed exposure without a recorded change to the underlying facts. Escalating a risk with no options. Recording a risk as closed because it stopped being discussed. Transferring a risk to a supplier by contract and removing it from the register while the consequence still lands on the client. Drawing contingency against a risk added to the register afterwards.

**7. Required evidence.** The risk register with one named owner per risk, assessed exposure and its basis, and status history; escalation records with the date the threshold was met, the date escalated, the options and the action window; out-of-cycle routing records; contingency drawdown records with the justifying register entry and the releasing authority.

**8. Responsible role.** The named risk owner for each risk. The named credential holder leading the project, for operating the escalation process and for escalating where the risk owner does not.

**9. Approval authority.** The authority the escalation threshold names receives and decides. The authority the governance plan names releases contingency. A risk may be closed only by its owner, with the reason recorded.

**10. Independence requirement.** At each gate at which the delegation schedule requires assurance, the risk register and the escalation record must be reviewed by a competent reviewer independent of the delivery organisation, whose review includes testing whether risks that met the threshold were escalated. That reviewer must not be the function that maintains the register.

**11. Materiality or threshold.** This standard sets no exposure figure, no matrix and no score, because risk appetite is an organisational decision and a number imported from elsewhere is a number nobody owns. It requires that the organisation documents its escalation thresholds, their named destinations and their times, and that the credential holder applies them. Where the organisation documents no threshold for a risk class, the credential holder must escalate every risk in that class until one exists. *Six-person internal pro-*



*ject*: one register of a dozen risks, a single threshold — anything that could delay the committed date or exceed the remaining contingency — and escalation by email to the sponsor within two working days. *Multi-partner national programme*: thresholds at component and programme level with a stated rule for which component risks rise, escalation destinations named as roles in the programme rather than organisations, and an explicit provision for risks that are below every component's threshold but correlated across components — which is the exposure a tiered threshold structure is worst at seeing.

**12. Exception and waiver.** An exception permitting an escalation to be deferred may be approved only by the authority that would have received it, in writing, before the escalation time expires, for a stated period, with the reason recorded. **A risk bearing on safety, a statutory duty, a licence condition or harm to a person must not be deferred**, and no exception is available for it.

**13. Escalation trigger.** Exposure meeting the documented threshold. A risk owner who declines to act or to escalate. A negative decision action window. A supplier risk whose consequence lands on the client. Correlated risks whose combined exposure exceeds a threshold that none of them individually meets. An instruction not to escalate.

**14. AI application.** AI may scan delivery, cost, schedule and issue data for early risk signals and propose candidate register entries, compute exposures on the organisation's stated basis, age unescalated risks against thresholds and flag them, compute the decision action window, and detect correlations between risks held in different registers.

**15. AI prohibition.** An AI system must not decide that a risk need not be escalated, close a risk, release contingency, own a risk, or assign a likelihood or consequence that is then reported without a named human's derivation.

**16. AI verification. Independent recomputation plus source tracing.** Every exposure computed with AI assistance must be recomputed by hand by the risk owner before it is used to decide escalation, and every AI-proposed risk must be traced by a named human to the underlying data before it enters the register. Where AI has aged risks against thresholds, a competent reviewer must confirm by **sampling with a stated basis** that no risk meeting a threshold was omitted from the flagged set.

#### **17. External reference.**

- **ISO** · *ISO 31000, Risk management — Guidelines* · relied on for: the existence of a risk-management process concept including escalation and review · **EXT-020** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: **guidance, and ISO states expressly that it is not a certifiable standard**; voluntary; this standard relies on it for no requirement.

- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of risk management within delivery practice · **EXT-028** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.
- **ISO/IEC** · *ISO/IEC 23894, Guidance on risk management for artificial intelligence* · relied on for: the existence of AI-specific risk guidance sitting alongside the AI management-system standard · **EXT-024** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: **guidance, not requirements**; voluntary; not certifiable.

**18. Jurisdictional caution.** Duties to report certain risks to a regulator, a safety authority, an auditor or a market are set by law and by licence conditions and carry their own timescales, which can be shorter than any internal threshold. Obtain local legal advice on mandatory reporting duties attaching to the specific project and sector.

**19. Related PCI Standards.** *PCI-FND-STD-11* (the parent duty to escalate); *PCI-FND-STD-05*; *PCI-FND-STD-02*; *PCI-PML-STD-03.02*; *PCI-PML-STD-07.01*; *PCI-PML-STD-08.02*; *PCI-PML-STD-12.02*; *PCI-PML-STD-15.01*. **What this standard adds to *PCI-FND-STD-11*:** the foundational standard creates the duty to escalate a material misstatement. This standard requires **one named risk owner**, escalation **on threshold rather than on expectation of resolution**, an **option set with closing dates**, an explicit **action-window test**, and **contingency drawn only against a pre-registered risk**.

**20. Related Body of Knowledge content.** PML-AI · Domain 8 · KA 8.1 Threats, opportunities and identification; KA 8.2 Analysis: from screening to quantification; KA 8.3 Responses, reserves and governance; KA 8.4 Resilience, bias and crisis leadership. Also Domain 3 KA 3.3 topic 3.3.3 escalation design.

**21. Compliance test.** Compliance is demonstrated when a reviewer can, for every risk in the period whose recorded exposure met the documented threshold: (a) find an escalation record naming the authority the threshold identifies; (b) compute the elapsed time between the date the threshold was met and the date escalated and confirm it is within the stated time; (c) find an option set with consequences and closing dates on the face of the escalation; and (d) find the receiving authority's recorded decision. The reviewer must additionally confirm that every contingency drawdown in the period references a risk whose register entry pre-dates the drawdown. A risk that met the threshold with no escalation record fails the test, and the absence of a later problem is not a defence.

**22. Breach indicators.** Exposures reassessed downward in the period before a gate with no change in the underlying facts. Escalation dates clustering

immediately before steering meetings. Escalations with no options. Risks closed with the reason "no longer relevant". Contingency drawdowns whose risk entry was created the same day. A shortening trend in escalation lead time. Risks whose owner is a supplier organisation.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Escalation decision: a risk meets the threshold six weeks before a committed date and the leader believes it can be resolved in four. Calculation review: computing a decision action window and showing the escalation is ceremonial. Ethical dilemma: a sponsor instructs that a risk be held below the threshold by splitting it.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-08-01](#) v1.0. Amendment note: renumbered and restructured; legislative drafting removed; single named ownership, escalation-on-threshold, the option set, the action-window test and the contingency rule separated into five process requirements; compliance test replaced with an elapsed-time test performed against two recorded dates.

## PCI STANDARD

### PCI-PML-STD-08.02

## Issue Management

**1. Normative requirement.** A credential holder must record every issue affecting delivery with a single named owner, a required resolution date and a stated consequence of non-resolution, and must not close an issue without recording how it was resolved.

**2. Purpose.** An issue is a risk that has already occurred: it is managed, not analysed. The failure this prevents is the issue log used as a repository rather than a control — hundreds of open items, no owners, no dates, and the two that matter indistinguishable from the rest. Where that happens the organisation discovers its issues through their consequences.

**3. Scope.** Every credential holder identifying, owning, resolving, reporting or assuring issues on a project, programme or portfolio, including issues raised by suppliers, by users and by assurance, and including issues escalated from a concern under [PCI-PML-STD-12.02](#).

**4. Defined terms.** *material · decision owner · escalation threshold · evidence · dependency · acceptance*. Additionally, **issue** means a risk that has occurred, or a present condition adversely affecting delivery; **resolution** means the recor-

ded action that removed the issue or, where it was accepted, the recorded acceptance of its consequence by the authority entitled to accept it.

##### 5. Required actions — process requirements.

- **PCI-PML-STD-08.02-PR-01 — Owner, date and consequence on entry.** Every issue must carry, at the point of recording, one named owner, a required resolution date and the stated consequence if it is not resolved by that date.
- **PCI-PML-STD-08.02-PR-02 — Ageing drives escalation.** Issues must be aged against their required resolution dates, and an issue past its date by the margin the escalation threshold states must be escalated to the named authority.
- **PCI-PML-STD-08.02-PR-03 — Closure states the resolution.** An issue may be closed only with a recorded statement of how it was resolved or, where the consequence was accepted, by whom and on what basis. Closure with no resolution statement does not satisfy this requirement.
- **PCI-PML-STD-08.02-PR-04 — Recurrence is analysed, not re-logged.** Where the same issue recurs within the period the governance sets, the credential holder must record it as a recurrence and route it to root-cause analysis under **PCI-PML-STD-09.02**, rather than opening an unconnected entry.

**6. Prohibited actions.** Closing an issue because it went quiet. Recording an issue with no owner or no date "pending assignment" beyond the period the governance allows. Re-logging a recurring issue as new. Accepting an issue's consequence on behalf of a party entitled to decide it. Downgrading an issue to a risk after it has occurred. Reporting an issue count without the ageing profile that gives it meaning.

**7. Required evidence.** The issue register with owner, raised date, required date, consequence, status and closure statement per entry; the ageing report per period; escalation records for overdue issues; the recurrence records and their routing to root-cause analysis; acceptance records for consequences accepted.

**8. Responsible role.** The named issue owner for each issue. The named credential holder leading the project, for operating the register, the ageing and the escalations.

**9. Approval authority.** The authority the escalation threshold names receives escalated issues. The consequence of an unresolved issue may be accepted only by the party that bears it, or by the authority the delegation schedule assigns; the project must not accept a consequence borne by an operating organisation or a user.

**10. Independence requirement.** Not applicable to issue ownership, which by design sits with the party able to act; independence attaches to the periodic review of closures, which must be sampled at each gate by a competent reviewer independent of the delivery organisation where the delegation schedule requires assurance.

**11. Materiality or threshold.** This standard states no issue count and no age limit. The organisation's governance sets the ageing margin that triggers escalation, the recurrence window and the closure review sample; this standard requires that they exist and are applied. Every issue is recorded regardless of size, because the ageing profile and the recurrence analysis both depend on a complete population. *Six-person internal project:* the register is the team's issue tracker with three extra fields, and the ageing report is a saved filter reviewed at the weekly meeting. *Multi-partner national programme:* each partner runs its own register, the programme holds the cross-partner issues explicitly with an owner named on the giving side, and the ageing profile is reported per partner so a single partner's backlog cannot be averaged away.

**12. Exception and waiver.** An exception permitting an issue to be closed without a resolution statement may be approved only by the authority the escalation threshold names, only where the issue is recorded as superseded by a specified change or decision, and only with that change or decision referenced. No exception permits closure by silence.

**13. Escalation trigger.** An issue past its required date by the documented margin. An issue whose owner declines ownership. A recurring issue at its second recurrence within the documented window. An issue whose consequence falls on a party that has not accepted it. An issue arising from a concern raised under [PCI-PML-STD-12.02](#) that has not been actioned.

**14. AI application.** AI may cluster issue text to detect recurrence, age the register and flag overdue items, propose owners from the routing history for human confirmation, detect issues with no consequence stated, and reconcile the register against defect, incident and change systems.

**15. AI prohibition.** An AI system must not own an issue, close one, accept a consequence, or determine that an issue is a recurrence for the purposes of [PR-04](#) — that determination is a judgement with a consequence, and a named human makes it.

**16. AI verification. Sampling with a stated basis plus source tracing.** Every AI-proposed recurrence cluster must be confirmed by a named human against both entries before it is treated as a recurrence. Each gate, a competent reviewer must draw a stated sample of closed issues and confirm the resolution statement against the evidence it cites.

**17. External reference.**

- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of issue management as a lifecycle practice distinct from risk management · **EXT-028** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.
- **ISO** · *ISO 31000, Risk management — Guidelines* · relied on for: the distinction between an uncertain event and one that has occurred · **EXT-020** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: **guidance, expressly not certifiable**; voluntary.
- **ISO** · *ISO 9001, Quality management systems — Requirements* · relied on for: the existence of a certifiable requirement to control nonconformity and to take corrective action · **EXT-033** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: voluntary unless imported; certification concerns a management system; this standard imports none of its requirements.

**18. Jurisdictional caution.** Some issues carry statutory reporting duties — safety incidents, data breaches, environmental events, regulatory breaches — with their own recipients and timescales that override any internal register. Obtain local legal advice on mandatory incident reporting for the sector and jurisdiction.

**19. Related PCI Standards.** [PCI-FND-STD-11](#); [PCI-FND-STD-12](#); [PCI-FND-STD-15](#); [PCI-PML-STD-08.01](#); [PCI-PML-STD-09.01](#); [PCI-PML-STD-09.02](#); [PCI-PML-STD-12.02](#).

**20. Related Body of Knowledge content.** PML-AI · Domain 8 · KA 8.1 topic on issues as occurred risks; KA 8.4 crisis leadership. Also Domain 9 KA 9.3 nonconformance and root-cause analysis; Domain 4 KA 4.4 the decision log.

**21. Compliance test.** Compliance is demonstrated when a reviewer can extract the issue register and confirm that: (a) every open entry carries one named owner, a required resolution date and a stated consequence; (b) the ageing profile reported to governance in each period can be reproduced from the register by an independent query returning the same counts; (c) every entry past its required date by the documented margin has a matching escalation record; (d) every closed entry carries a resolution statement whose cited evidence can be retrieved; and (e) for a stated sample of recurring issues, a root-cause record exists. An open entry with no owner, or a closed entry with no resolution statement, fails the test.

**22. Breach indicators.** A register whose open count never changes. Closures clustering at period end. Owners recorded as functions. Issues with required

dates in the past and no escalation. The same description appearing under four unconnected entries. An ageing profile reported as an average rather than as a distribution. Issues raised by users closed faster than issues raised by assurance.

**23. Consequence within PCI authority.** Correction required; additional review; escalation; examination failure; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Evidence selection: which fields make an issue register a control rather than a list. Scenario judgement: an issue has been open past its date for three cycles and the owner has left the organisation. Calculation review: reproducing an ageing profile and identifying the two entries that carry the exposure.

**25. Version and status.** Version 1.0 · **not yet approved** · effective on approval · **new standard** — the v1.0 set addressed risk and left issues to be inferred from it, which left the occurred-risk population ungoverned. Amendment note: none.

## Domain 9 — Quality, Assurance and Continuous Improvement

### PCI STANDARD

#### PCI-PML-STD-09.01

### Quality Acceptance

**1. Normative requirement.** A credential holder must not record a deliverable as accepted unless a named acceptance authority has decided, against the version-identified acceptance criteria set before the deliverable was produced, that it conforms.

**2. Purpose.** Acceptance is the point at which risk transfers — from the producer to the organisation, and often from the project to an operating team who were not in the room. The failure this prevents is acceptance by default: the deliverable used, invoiced, or simply not objected to within a review window, and thereafter treated as accepted by everyone except the people who have to live with it.

**3. Scope.** Every credential holder producing, reviewing, testing, accepting or assuring a deliverable on a project, programme or portfolio, in every delivery model, including deliverables produced by suppliers and deliverables accepted on behalf of an operating organisation.



**4. Defined terms.** *acceptance · evidence · competent reviewer · independent · material · decision owner.* Additionally, **nonconformity** means a failure to meet an acceptance criterion; **carried nonconformity** means a nonconformity accepted with the deliverable, recorded with an owner and a date.

**5. Required actions — process requirements.**

- **PCI-PML-STD-09.01-PR-01 — Criteria version-identified and pre-dated.** The acceptance criteria applied must carry a version identifier and an agreement date earlier than the deliverable's production start, and the acceptance record must cite that version.
- **PCI-PML-STD-09.01-PR-02 — Acceptance is a positive act.** Acceptance must be recorded as a dated decision by the named acceptance authority. Use of the deliverable, payment against it, and the expiry of a review period must not be recorded as acceptance.
- **PCI-PML-STD-09.01-PR-03 — Nonconformities carried, not lost.** Every nonconformity known at acceptance must be recorded with a named owner, a due date and the consequence of non-closure, and must be reported until closed.
- **PCI-PML-STD-09.01-PR-04 — The receiving party is in the decision.** Where a deliverable will be operated, maintained or used by an organisation other than the project, a named representative of that organisation must be recorded in the acceptance decision.
- **PCI-PML-STD-09.01-PR-05 — Acceptance evidence retrievable.** The test, inspection or review evidence cited in an acceptance record must be retained and retrievable for the retention period the organisation sets, and the acceptance record must reference it specifically enough to retrieve it.

**6. Prohibited actions.** Accepting against criteria written after production. Recording acceptance because a review window expired. Accepting on behalf of an operating organisation that was not consulted. Closing a nonconformity by removing the criterion. Accepting a deliverable whose test evidence cannot be produced. Splitting a deliverable so each part falls below the acceptance authority's threshold.

**7. Required evidence.** The version-identified acceptance criteria with their agreement date; the acceptance record naming the authority, the date, the criteria version and the outcome; the nonconformity register with owners, dates and consequences; the receiving organisation's named representative in the record; the retained test, inspection or review evidence.

**8. Responsible role.** The named acceptance authority decides. The named credential holder leading the project answers for the completeness of the acceptance pack and for refusing to record an acceptance that was not decided.



**9. Approval authority.** The acceptance authority the delegation schedule names, by deliverable class. A carried nonconformity may be accepted only by the authority that bears its consequence. A supplier may never accept its own deliverable.

**10. Independence requirement.** The acceptance authority must be **independent of the production** of the deliverable in the sense defined above. Where the project both produces and accepts — common on small internal projects — the acceptance authority must sit outside the producing team and the arrangement must be recorded; and where the deliverable will be operated by another organisation, [PR-04](#) supplies the independence that a small structure cannot.

**11. Materiality or threshold.** This standard states no defect count and no pass rate. The organisation's governance sets the acceptance authority by deliverable class, the classes requiring independent test, the retention period and the tolerance for carried nonconformities; this standard requires that these exist and are applied. A nonconformity bearing on safety, a licence condition or a statutory duty is not carried under any tolerance — it is a mandatory precondition under [PCI-PML-STD-16.01](#). *Six-person internal project:* the criteria are five lines per deliverable agreed at kick-off, acceptance is a dated line in the log signed by the receiving manager, and the evidence is the test output the team produced anyway. *Multi-partner national programme:* acceptance runs at component and integrated levels, the integrated acceptance names a representative of each receiving organisation, and the criteria are held against a common template so that a partner cannot accept against its own weaker set.

**12. Exception and waiver.** A deliverable may be accepted with an open nonconformity only where the nonconformity is recorded with an owner, a date and a consequence, and only by the authority bearing that consequence. No exception permits acceptance against criteria written after production, and no exception permits acceptance by expiry of a review window.

**13. Escalation trigger.** A deliverable presented with no pre-dated criteria. A receiving organisation that declines to be named in the acceptance. A nonconformity the project proposes to close by amending the criterion. Acceptance evidence that cannot be retrieved. Pressure to accept to release a payment milestone.

**14. AI application.** AI may check acceptance packs for missing criteria versions, missing evidence references and missing authorities; compare a deliverable against stated criteria and produce a candidate conformity assessment for human decision; age carried nonconformities; and detect acceptances recorded with no paired decision.

**15. AI prohibition.** An AI system must not accept a deliverable, decide conformity, close a nonconformity, or be recorded as the acceptance authority.

**16. AI verification. Independent recomputation or re-test plus sampling with a stated basis.** Where AI assessed conformity, a competent reviewer must independently re-perform the assessment on a stated sample of criteria and reconcile any difference before acceptance. Every AI-flagged nonconformity must be confirmed by a named human before it is recorded, and every AI-cleared criterion in the safety, licence or statutory subset must be re-checked in full rather than by sample.

**17. External reference.**

- **ISO · ISO 9001, Quality management systems — Requirements** · relied on for: the existence of a certifiable requirement to verify that outputs meet requirements before release · **EXT-033 · Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: **the certifiable member of its family**, but adoption is voluntary unless a contract or regulator requires it; a project is not certified against it; this standard imports none of its requirements and reproduces none of its text.
- **ISO · ISO 9000, Quality management systems — Fundamentals and vocabulary** · relied on for: the existence of an agreed vocabulary distinguishing conformity from nonconformity · **EXT-034, not independently verified — verify current requirements · Manual §6 category 3** · limitation: a vocabulary standard; **not certifiable**; voluntary.
- **ISO · ISO 10006, Quality management — Guidelines for quality management in projects** · relied on for: the existence of project-specific quality guidance · **EXT-035 · Manual §6 category 3** · currency checked 2026-08-03 · limitation: **guidelines, not requirements**; voluntary; not certifiable.

**18. Jurisdictional caution.** Whether acceptance has occurred, what it transfers, and what remedies survive it are contractual and statutory questions determined by the governing law — and in some sectors acceptance is regulated separately by a safety, health or licensing authority. Obtain legal advice before relying on an internal acceptance record in a commercial or regulatory position.

**19. Related PCI Standards.** PCI-FND-STD-02; PCI-FND-STD-12; PCI-PML-STD-01.03; PCI-PML-STD-05.01; PCI-PML-STD-05.02; PCI-PML-STD-16.01; PCI-PML-STD-16.02.

**20. Related Body of Knowledge content.** PML-AI · Domain 9 · KA 9.1 Quality planning; KA 9.2 Assurance and control; KA 9.3 Acceptance, nonconformance and root-cause analysis. Also Domain 5 KA 5.4 verification and acceptance; Domain 16 KA 16.1 handover and commissioning.

**21. Compliance test.** Compliance is demonstrated when a reviewer can take a stated sample of deliverables recorded as accepted and, for each: (a) retrieve the criteria version the acceptance record cites and confirm its agreement date precedes the production start date; (b) find a dated decision by the named acceptance authority, who does not appear in the deliverable's authorship or production record; (c) retrieve the test, inspection or review evidence from the reference in the record; (d) find a named representative of the receiving organisation where one operates the deliverable; and (e) find every nonconformity known at acceptance in the register with an owner, a date and a consequence. An acceptance recorded with no decision, or with criteria post-dating production, fails the test.

**22. Breach indicators.** Acceptance dates identical to payment dates. Criteria versions issued the week of acceptance. Acceptance authorities who also appear as deliverable authors. Nonconformity registers that empty at handover. Receiving organisations first seeing a deliverable after acceptance. Evidence references that name a folder rather than a document.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Evidence selection: which artefacts establish that acceptance occurred. Scenario judgement: a supplier claims acceptance because the client used the deliverable for six weeks. Ethical dilemma: a leader is asked to accept a deliverable with an open safety nonconformity to release a payment milestone.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-09-01](#) v1.0. Amendment note: renumbered and restructured; legislative drafting removed; the positive-act rule, the receiving-party rule and the evidence-retrievability rule separated into process requirements; compliance test replaced with a five-part retrieval test.

## PCI STANDARD

### PCI-PML-STD-09.02

## Lessons Learned and Organisational Retention

**1. Normative requirement.** A credential holder must convert each lesson accepted from a review into a change to a named standing artefact — a

process, a checklist, a template, an estimating basis, a criterion or a training item — with a named owner and a date.

**2. Purpose.** Lessons-learned processes are near-universal and largely ineffective, and the reason is structural rather than cultural: a lesson recorded in a repository nobody queries at the moment of decision has no route into behaviour. The failure this prevents is the well-run review whose output changes nothing, repeated on the next project by people who could not have found it.

**3. Scope.** Every credential holder conducting, contributing to, approving or assuring a retrospective, a post-implementation review, a post-project review or a root-cause analysis, on projects, programmes and portfolios, in every delivery model.

**4. Defined terms.** *evidence · decision owner · material · acceptance · competent reviewer.* Additionally, **standing artefact** means a document, model, template, checklist, criterion set or training item used on future work; **accepted lesson** means a lesson the review's approving authority has accepted as valid.

#### 5. Required actions — process requirements.

- **PCI-PML-STD-09.02-PR-01 — Each accepted lesson names its artefact.** Every accepted lesson must name the standing artefact it changes, the owner of that artefact and the date by which the change is made. A lesson recorded with no target artefact does not satisfy this requirement.
- **PCI-PML-STD-09.02-PR-02 — Review separated from assessment.** A post-project or post-implementation review must be conducted separately from any individual performance assessment, and its record must not be used as evidence in one. Where the two are combined, the review's output is not evidence of compliance with this standard.
- **PCI-PML-STD-09.02-PR-03 — Root cause where a defect recurs.** Where a defect, issue or nonconformity recurs within the window the governance sets, the credential holder must commission a root-cause analysis and must record its conclusion and the artefact change it produces.
- **PCI-PML-STD-09.02-PR-04 — Retrieval at the point of decision.** The credential holder must record where each changed artefact is held and how a future project reaches it at the moment it makes the corresponding decision. A lesson retrievable only by searching a repository does not satisfy this requirement.

**6. Prohibited actions.** Closing a review with a list of observations and no artefact changes. Using a review to attribute blame. Recording a lesson as "implemented" when only the repository entry changed. Deferring root-cause

analysis of a recurrence to the next project. Reporting a lesson count as evidence of learning.

**7. Required evidence.** The review record with attendance, method and accepted lessons; the artefact change record per accepted lesson with owner and date; the changed artefacts themselves, versioned; the root-cause analyses for recurrences with their conclusions; the retrieval route recorded for each changed artefact.

**8. Responsible role.** The named credential holder leading the project, for commissioning the review and for routing accepted lessons to artefacts. The named artefact owner, for making the change. The function that owns the standing artefact set, for its retrievability.

**9. Approval authority.** The review's approving authority accepts or rejects each lesson, with reasons. The artefact owner approves the change to their artefact. Neither may be the person whose work the lesson concerns, where the lesson attributes a cause to that work.

**10. Independence requirement.** A root-cause analysis under [PR-03](#) must be led by a competent reviewer independent of the work in which the defect arose. A retrospective within an adaptive team may be facilitated internally, and independence attaches instead to the periodic sampling of artefact changes at gates.

**11. Materiality or threshold.** This standard states no lesson count and no reuse rate. The organisation's governance sets the recurrence window that triggers root-cause analysis, the review points in the lifecycle and the retention period for review records; this standard requires that they exist and are applied. Where an organisation measures the reuse of lessons, the credential holder must record the basis of the measure so that it is interpretable rather than decorative. *Six-person internal project:* one retrospective per iteration and one at closure; accepted lessons change the team's own checklist, which is the standing artefact, and retrieval is that the checklist is opened at the start of the next piece of work. *Multi-partner national programme:* reviews run per component with a programme-level synthesis; artefact owners sit in the enterprise function rather than in a partner, because a lesson lodged only in a supplier's method leaves with the supplier.

**12. Exception and waiver.** An exception permitting an accepted lesson to be recorded with no artefact change may be approved by the review's approving authority only where the lesson is recorded as project-specific and incapable of generalisation, with that reason stated. No exception permits a recurrence to go without root-cause analysis where the governance's window has been met.

**13. Escalation trigger.** A recurrence meeting the documented window. An artefact owner who declines the change. A lesson concerning a safety, licence

or statutory matter. A review that cannot be held because the team has dispersed — which is itself a planning failure and is escalated as one.

**14. AI application.** AI may cluster a large lessons repository into recurring themes, propose candidate artefact targets for human decision, detect recurrences across issue and defect data, draft the change to a checklist or template for the artefact owner's approval, and test whether a proposed artefact change would have been retrievable at the decision point it targets.

**15. AI prohibition.** An AI system must not conduct the review, decide which lessons enter the standing process, accept or reject a lesson, or approve a change to a standing artefact. A review is partly an act of collective sense-making by the people who were there, and a generated summary is not that.

**16. AI verification. Source tracing plus named approval.** Every AI-produced theme must be traced by a named human to the underlying review entries before it is accepted as a lesson, and the artefact owner must approve the resulting change by name. Every AI-detected recurrence must be confirmed by a named human against both underlying records.

#### **17. External reference.**

- **ISO** · *ISO 9001, Quality management systems — Requirements* · relied on for: the existence of a certifiable requirement for corrective action addressing causes · **EXT-033** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: voluntary unless imported; certification concerns a management system; no requirement is imported here.
- **ISO** · *ISO 10006, Guidelines for quality management in projects* · relied on for: the existence of project-level improvement guidance · **EXT-035** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: **guidelines, not requirements**; voluntary; not certifiable.
- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of lessons capture as a lifecycle practice · **EXT-028** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.

**18. Jurisdictional caution.** Review records can be disclosable in litigation, arbitration, a regulatory investigation or a public inquiry, and privilege rules differ by jurisdiction. Obtain legal advice before conducting a review into an event with legal or regulatory exposure, and never let that advice become a reason to hold no review.

**19. Related PCI Standards.** PCI-FND-STD-12; PCI-FND-STD-02; PCI-PML-STD-08.02; PCI-PML-STD-09.01; PCI-PML-STD-12.02; PCI-PML-STD-16.02.

**20. Related Body of Knowledge content.** PML-AI · Domain 9 · KA 9.4 Lessons learned, continuous improvement, data quality and AI-output quality · topic 9.4.1 lessons that change behaviour. Also Domain 16 KA 16.3 knowledge transfer and the post-project review.

**21. Compliance test.** Compliance is demonstrated when a reviewer can take the accepted lessons from every review held in the period and, for each: (a) name the standing artefact it targets; (b) open that artefact and find the change, in a version dated on or before the recorded due date; (c) follow the recorded retrieval route and reach the changed artefact from the decision point it targets, without searching a repository; and (d) for every recurrence meeting the documented window, find a root-cause analysis led by someone independent of the work concerned. Accepted lessons with no artefact change fail the test, however many were recorded.

**22. Breach indicators.** Lesson counts reported with no artefact-change counts. Review outputs that are lists of observations. The same lesson recorded on consecutive projects. Root-cause analyses led by the team that produced the defect. Artefact changes dated long after their due dates. Review records referenced in appraisal documents.

**23. Consequence within PCI authority.** Correction required; additional review; escalation; examination failure; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Scenario judgement: a review produces eleven observations and the sponsor asks which of them are lessons. Evidence selection: which artefact proves a lesson changed behaviour. Calculation review: the reuse economics of a review, and where its value is actually lost.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes PML-LAW-09-02 v1.0 (*Organisational Learning*). Amendment note: renumbered, retitled and restructured; legislative drafting removed; the principal obligation changed from conducting reviews to **converting accepted lessons into named artefact changes**, which is the act that can be verified; retrieval at the point of decision added as an express process requirement.



## Domain 10 — Procurement, Contracts and Supply Networks

### PCI STANDARD

#### PCI-PML-STD-10.01

### Procurement Fairness

**1. Normative requirement.** A credential holder must evaluate every tender, bid or proposal against the evaluation criteria and weightings published to bidders before submissions were received, and must not apply any criterion, weighting or consideration that was not published.

**2. Purpose.** Procurement is where delivery meets money, incumbency and relationships, and it is the process most exposed to the interest that decides quietly. The failure this prevents is the evaluation that reaches the intended answer through a criterion introduced afterwards, a weighting adjusted at moderation, or a clarification granted to one bidder and not to others.

**3. Scope.** Every credential holder specifying, running, evaluating, moderating, recommending, awarding or assuring a procurement, and every credential holder administering a contract in a way that affects competition — including framework call-offs, single-source justifications, extensions and variations that displace competition.

**4. Defined terms.** *conflict of interest · independent · evidence · decision owner · material · acceptance.* Additionally, **published criteria** means the evaluation criteria and weightings issued to all bidders before the submission deadline; **moderation** means the reconciliation of evaluators' individual scores into a panel position.

#### 5. Required actions — process requirements.

- **PCI-PML-STD-10.01-PR-01 — Criteria and weightings published before submission.** The evaluation criteria and their weightings must be published to all bidders before the submission deadline, and must not be changed after it except by an addendum issued to all bidders with an extended deadline.
- **PCI-PML-STD-10.01-PR-02 — Individual scores recorded before moderation.** Each evaluator must record their own score and its written rationale against each published criterion before seeing any other evaluator's score, and those individual records must be retained.
- **PCI-PML-STD-10.01-PR-03 — Moderation changes are reasoned and retained.** Where moderation changes a score, the change and its reason must be recorded against the individual score it replaced. A



moderated score with no recorded reason does not satisfy this requirement.

- **PCI-PML-STD-10.01-PR-04 — Information symmetry.** Any clarification, correction or additional information given to one bidder that could affect a submission must be given to all bidders, and the record must show what was issued, to whom and when.
- **PCI-PML-STD-10.01-PR-05 — Panel interests cleared before evaluation.** Every panel member must have a declaration or a nil return recorded under **PCI-PML-STD-01.03-PR-03** before evaluation begins, and a member with an interest in a bidder must be excluded under **PCI-PML-STD-01.03-PR-07**.

**6. Prohibited actions.** Introducing an unpublished criterion, including "fit", "confidence" or "deliverability", unless it was published. Adjusting weightings after submissions. Moderating to a predetermined outcome. Giving one bidder information not given to others. Allowing an evaluator with an interest to score, moderate, chair or ratify. Accepting hospitality, employment discussion or any benefit from a bidder during a live procurement. Splitting a requirement to avoid a competition threshold.

**7. Required evidence.** The published criteria and weightings with their issue date; the individual evaluator score sheets with rationales and timestamps; the moderation record showing each change and its reason; the clarification log showing symmetric issue; the panel interests record; the award recommendation and the award decision with its authority; the debrief records issued to unsuccessful bidders where applicable.

**8. Responsible role.** The named credential holder running the procurement, for the process. Each named evaluator, for their own scores and declarations. The named award authority, for the award.

**9. Approval authority.** The award authority the delegation schedule assigns, applying the aggregation rule where a series of call-offs or extensions aggregates. A single-source or direct-award justification must be approved by the authority the schedule names for the aggregate value, never by the requesting project alone.

**10. Independence requirement.** Every evaluator must be independent of every bidder in the sense defined above. The moderation chair must be independent of the requesting project where the delegation schedule requires it. Assurance of the procurement must be provided by a competent reviewer independent of the panel and of the requesting project, and must satisfy the name-matching test in **PCI-PML-STD-01.03-PR-06**.

**11. Materiality or threshold.** This standard states no value threshold and no minimum bidder count. The organisation's governance — and, in the public sector, the applicable procurement regime — sets the competition thresholds,

the award authorities, the aggregation rule and the standstill or debrief obligations; this standard requires that the documented rules exist and are applied and that no requirement is split to fall below one. *Six-person internal project*: a single quotation exercise with three suppliers, criteria of four lines sent with the request, individual scores recorded in a shared sheet before discussion, and the leader's own declaration recorded before scoring begins. *Multi-partner national programme*: multiple concurrent competitions with evaluators drawn from several organisations; the interests register runs across all of them, the aggregation rule states whether call-offs across partners aggregate, and the assurance reviewer is independent of every partner rather than of one.

**12. Exception and waiver.** A departure from competition — single source, direct award, extension — may be approved only by the authority the delegation schedule assigns to the aggregate value, only on a written justification stating the ground, the alternatives considered and the period, and only where the justification is retained and reported. **No exception is permitted** to [PR-01](#), [PR-02](#) or [PR-05](#): criteria published after submissions, scores recorded after moderation, and undeclared panel interests are breaches in every case.

**13. Escalation trigger.** A proposal to change criteria or weightings after submissions. An evaluator's interest disclosed after scoring has begun. Information given to one bidder only. A moderation outcome that reverses the individual scores with no recorded reason. A bidder's approach to an evaluator outside the process. A series of call-offs aggregating above the award authority.

**14. AI application.** AI may check submissions for completeness against the published requirements, extract stated commitments for evaluator verification, compute weighted scores from evaluator inputs, detect divergence between individual and moderated scores, and check the clarification log for asymmetric issue.

**15. AI prohibition.** An AI system must not score a submission on a qualitative criterion in place of a named evaluator, decide an award, moderate scores, or determine that a bidder is non-compliant. Where AI produces a ranking, it has produced a recommendation, and the award is a reserved decision class under [PCI-PML-STD-01.02-PR-01](#).

**16. AI verification. Independent recomputation plus source tracing.** Every weighted score computed with AI assistance must be recomputed by hand before the recommendation is issued. Every extracted commitment must be traced by a named evaluator to the page and version of the submission it came from before it is scored. Where AI screened submissions for compliance, a competent reviewer must re-check in full every submission the system would exclude.

## 17. External reference.

- **FIDIC** · *FIDIC suite of conditions of contract* · relied on for: the existence of standard-form contractual structures a procurement may adopt · **EXT-050** · **Manual §6 category 4 — contract framework** · currency checked 2026-08-03 · limitation: **binds only the parties who adopt it, by signature**; characterised generically, no clause numbers cited, no text reproduced; it is not legislation.
- **NEC** · *NEC4 suite of contracts* · relied on for: the existence of an alternative standard-form structure with its own change and notification mechanics · **EXT-051** · **Manual §6 category 4** · currency checked 2026-08-03 · limitation: as above.
- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of procurement within delivery practice · **EXT-028** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.
- **Project Management Institute** · *Code of Ethics and Professional Conduct* · relied on for: the existence of a professional expectation of fairness and impartiality in supplier dealings · **EXT-063, not independently verified — verify current requirements** · **Manual §6 category 6 — ethical code** · limitation: binding only where a body, regulator or engagement has adopted it.

**18. Jurisdictional caution.** Public-procurement law, subsidy control, competition law, anti-bribery law and sector regulation impose binding, jurisdiction-specific rules on advertising, thresholds, standstill periods, debriefs, remedies and record-keeping, and breach can void an award or engage criminal liability. **This standard does not state any of those rules and compliance with it is not compliance with them.** Obtain local legal advice before running a regulated procurement.

**19. Related PCI Standards.** [PCI-FND-STD-08](#); [PCI-FND-STD-09](#); [PCI-FND-STD-01](#); [PCI-PML-STD-01.03](#); [PCI-PML-STD-03.02](#); [PCI-PML-STD-04.01](#); [PCI-PML-STD-09.01](#).

**20. Related Body of Knowledge content.** PML-AI · Domain 10 · KA 10.1 Make-or-buy and the procurement lifecycle; KA 10.2 Tendering and evaluation; KA 10.3 Contract strategy and supplier governance; KA 10.4 Claims, change and disputes; ethical sourcing; supply resilience.

**21. Compliance test.** Compliance is demonstrated when a reviewer can: (a) retrieve the published criteria and weightings and confirm their issue date precedes the submission deadline; (b) retrieve every evaluator's individual

score sheet and confirm each carries a rationale and a timestamp preceding the moderation meeting; (c) recompute the moderated result from the individual scores and the published weightings, and account for every difference by a recorded moderation reason; (d) confirm every clarification in the log was issued to all bidders; and (e) confirm every panel member has a declaration or nil return dated before evaluation began. A criterion in the evaluation that does not appear in the published set fails the test outright, whatever the merits of the winning bid.

**22. Breach indicators.** Moderated scores that reverse the individual ranking with brief or absent reasons. Criteria documents with version dates after the deadline. Clarifications issued to one bidder. Panel interests registers with no entries in a market with a long-standing incumbent. Award values clustering just below a competition threshold. Evaluators who scored and then joined the winning bidder. Debriefs that describe criteria the bidders were never given.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension or withdrawal of the PML-AI credential — each subject to due process and a right of appeal. **PCI cannot set aside an award, impose a fine or create any legal liability;** those consequences, where they exist, arise under the applicable procurement or criminal law and not under this standard.

**24. Examination application.** Ethical dilemma: at moderation the chair proposes to add a "deliverability" adjustment that was not published. Evidence selection: which artefacts would establish that an evaluation was fair. Scenario judgement: an evaluator discloses, after scoring, that they have been approached about a role with a bidder.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-10-01](#) v1.0. Amendment note: renumbered and restructured; legislative drafting removed; individual-scores-before-moderation, reasoned moderation changes, information symmetry and panel-interest clearance separated into process requirements; the link to [PCI-PML-STD-01.03](#) made explicit, which is the change that gives this standard an enforceable rule about who is barred from scoring; consequence field corrected to state expressly what PCI cannot do.

## Domain 11 — Stakeholders, Communication and Influence

## PCI-PML-STD-11.01

## Stakeholder Transparency

**1. Normative requirement.** A credential holder must not issue, endorse or allow to stand a report or communication about delivery status that omits a material adverse fact known to them at the time of issue.

**2. Purpose.** Reporting is where every other control either works or is neutralised, because a control that detects a problem and a report that omits it produce the same outcome as no control at all. The failure this prevents is the technically true report: every figure correct, every omission deliberate, and the reader left with a false picture they could not have detected.

**3. Scope.** Every credential holder preparing, approving, presenting or assuring a status report, dashboard, board paper, briefing or external communication about a project, programme or portfolio, and every credential holder who becomes aware that an issued report is materially misleading.

**4. Defined terms.** *material · evidence · decision owner · sponsor · escalation threshold · baseline* (control sense). Additionally, **materially misleading** means capable of causing a recipient reading it in the ordinary course of their role to form a view of status, risk or forecast that the known facts do not support; **status report** means any recurring or ad hoc statement of delivery position issued to a decision-maker.

**5. Required actions — process requirements.**

- **PCI-PML-STD-11.01-PR-01 — Adverse facts appear in the same document as the good ones.** A material adverse fact must appear in the report itself, not only in an appendix, a footnote, a verbal briefing or a separate document sent to a narrower list.
- **PCI-PML-STD-11.01-PR-02 — Status colours are defined and derived.** Where a report uses a rating, colour or index, its derivation rule must be documented and the reported value must follow it. A rating set by judgement against a documented rule must record the override, its reason and its author.
- **PCI-PML-STD-11.01-PR-03 — One version of status.** The credential holder must not issue different status positions to different audiences for the same period; where audiences require different depth, the underlying position must be identical and the difference must be one of detail only.
- **PCI-PML-STD-11.01-PR-04 — Correction of an issued report.** On discovering that an issued report was materially misleading, the credential holder must issue a correction to every recipient of the original, within the time the escalation threshold states.
- **PCI-PML-STD-11.01-PR-05 — Refusal is recorded.** Where the credential holder is instructed to issue a report they consider materially misleading, they must record the instruction, record their objection in

writing, and escalate under [PCI-FND-STD-11](#); they must not issue it in their own name.

**6. Prohibited actions.** Removing an adverse item because a decision on it is expected shortly. Reporting a rating that the documented rule does not produce, without recording the override. Briefing the sponsor on a position the board paper does not contain. Presenting a forecast as a range whose stated bounds exclude the outcome currently most likely. Attributing an omission to space. Allowing an issued report known to be misleading to stand uncorrected.

**7. Required evidence.** Issued reports with their dates, versions and distribution lists; the rating derivation rule and the override records; the correction notices with their distribution; the record of any instruction to report otherwise and the objection raised; the escalation record.

**8. Responsible role.** The named credential holder who issues or endorses the report. The **sponsor**, for the position communicated to the governing body. Neither may attribute the report to "the project".

**9. Approval authority.** The report's named approver, as the governance arrangements set. A rating override may be approved only by the named authority the derivation rule identifies. Nobody may approve the omission of a material adverse fact.

**10. Independence requirement.** Not applicable to routine status reporting, which is properly prepared by the delivery organisation; independence attaches at gates, where a competent reviewer independent of the delivery organisation must be able to reach a status position from the underlying data and compare it with the reported one.

**11. Materiality or threshold.** This standard states no variance percentage. Materiality is decided by the definition above, applied against the decisions the report informs, and the organisation's governance sets the reporting tolerances, the rating derivation rule and the correction time; this standard requires that these exist and are applied. **A fact bearing on safety, legality, a licence or a statutory duty is material at any size** and no tolerance excludes it. *Six-person internal project*: a weekly half-page with a stated rating rule of three lines and one distribution list, corrected by replying to the same thread. *Multi-partner national programme*: a consolidated report whose derivation rule states how component ratings combine — and states explicitly that a component's adverse item is not averaged away by others — with the same underlying position issued at every level and the depth varying, not the substance.

**12. Exception and waiver.** An adverse fact may be withheld from a wider distribution only where its disclosure would breach a legal duty of confidence, prejudice a legal position on written legal advice, or disclose personal data

unlawfully — and in every such case it must still be disclosed to the sponsor and to the governing body, with the ground for the restriction recorded. **Commercial embarrassment, an expected fix and the imminence of a decision are not grounds.**

**13. Escalation trigger.** An instruction to omit or soften a material adverse fact. A rating override that the derivation rule does not support. Discovery that an issued report is materially misleading. A divergence between the position briefed verbally and the position reported. A supplier's refusal to supply status data needed for a complete report.

**14. AI application.** AI may draft status narrative from underlying data, reconcile the narrative to the data and flag statements the data does not support, compute ratings from the documented derivation rule, compare successive reports for items that disappeared, and check distribution consistency across audiences.

**15. AI prohibition.** An AI system must not approve or issue a report, decide whether a fact is material, set a rating that overrides the documented rule, or generate a status narrative that is issued without a named human's verification against the underlying data.

**16. AI verification. Reconciliation plus clause-to-summary comparison.** Before issue, the credential holder must reconcile every figure in an AI-drafted report to its source system extract, with the extract date recorded, and must compare each summarised statement against the source it summarises to confirm it carries the same position **including its qualifications** — the failure mode of generated narrative being the quiet loss of a caveat. Ratings computed with AI assistance must be recomputed against the derivation rule by hand.



**17. External reference.**

- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of reporting and information management within delivery practice · **EXT-028** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.
- **ISO** · *ISO 21505, Guidance on governance* · relied on for: the existence of an information flow from delivery to the governing body · **EXT-032, not independently verified** · **Manual §6 category 3** · limitation: as above, with an open verification status.
- **Project Management Institute** · *PMBOK Guide* · relied on for: the existence of performance reporting in professional practice · **EXT-060** · **Manual §6 category 5 — professional framework** · currency checked 2026-08-03 · limitation: **never regulatory authority**; no edition asserted; no text reproduced.

**18. Jurisdictional caution.** Statements about a project can engage listed-company disclosure duties, prospectus and market-abuse rules, grant-condition reporting, public-inquiry duties and fraud law, and the threshold for a legally material statement is not the same as the professional one used here. Obtain legal advice before any statement about delivery status is made outside the organisation.

**19. Related PCI Standards.** **PCI-FND-STD-11** (the parent escalation obligation); **PCI-FND-STD-02**; **PCI-FND-STD-15**; **PCI-FND-STD-14**; **PCI-PML-STD-06.01**; **PCI-PML-STD-07.01**; **PCI-PML-STD-08.01**; **PCI-PML-STD-12.01**. **What this standard adds to PCI-FND-STD-11:** the foundational standard requires a material misstatement to be escalated. This standard adds the delivery-specific mechanics by which honest figures produce a dishonest picture — **same-document disclosure**, a **documented rating derivation with recorded overrides**, **one version of status across audiences**, a **correction duty to the original distribution**, and a **recorded refusal**.

**20. Related Body of Knowledge content.** PML-AI · Domain 11 · KA 11.1 Stakeholder systems and engagement strategies; KA 11.2 Executive communication and reporting; KA 11.4 Public and community stakeholders, cross-cultural communication and AI-generated communication risk. Also Domain 3 KA 3.3 the decision record.

**21. Compliance test.** Compliance is demonstrated when a reviewer can, for a stated sample of reporting periods: (a) rebuild the reported rating from the documented derivation rule and the period's underlying data, and account for



every difference by a recorded override with a named author; (b) list the material adverse facts recorded in the risk, issue, change and cost records in that period and find each one in the report itself rather than in a separate document; (c) compare every version of the period's report issued to different audiences and confirm that the underlying position is identical; and (d) for every correction issued, confirm its distribution matched the original's. A material adverse fact present in the underlying records and absent from the report fails the test.

**22. Breach indicators.** Ratings that stay green until the period before a gate. Adverse items appearing first in an appendix. Two decks for the same period with different messages. Corrections issued to a narrower list than the original. Overrides recorded with no author. Reports whose figures cannot be reproduced from the source systems. A pattern of items disappearing between draft and issue.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Ethical dilemma: the sponsor asks for an adverse item to be held for one cycle because a fix is expected. Evidence selection: which artefacts establish that a report was complete. Calculation review: rebuilding a status rating from its derivation rule and identifying the unrecorded override.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-11-01](#) v1.0. Amendment note: renumbered and restructured; legislative drafting removed; the same-document rule, the rating derivation rule, the one-version rule, the correction duty and the recorded refusal separated into five process requirements; compliance test replaced with a rebuild-and-compare test performed from source records.

## Domain 12 — Leadership, Teams and Organisational Behaviour

### PCI STANDARD

#### PCI-PML-STD-12.01

### Leadership Conduct

**1. Normative requirement.** A credential holder must not use their authority to cause another person to prepare, sign, present or withhold a professional

statement that the person has told them, or has recorded, is inaccurate or incomplete.

**2. Purpose.** Leadership conduct standards fail when they are written as character requirements, because character cannot be audited and an unverifiable requirement is not a requirement. This standard is therefore drafted around the single leadership act that is both observable and gravely damaging: **the use of positional authority to override another professional's judgement about their own work.** Every reporting, estimating, scheduling and assurance control in this set depends on the person nearest the facts being able to state them; this standard protects that, and leaves inspiration to the Body of Knowledge.

**3. Scope.** Every credential holder exercising authority over another person's professional output — line, project, matrix, contractual or client-side — including over supplier staff, seconded staff and assurance staff, and including instructions given verbally.

**4. Defined terms.** *material · evidence · decision owner · competent reviewer · detriment · independent.* Additionally, **professional statement** means an estimate, forecast, schedule, assessment, opinion, test result, assurance conclusion or status report issued by a person in their professional capacity; **override** means a leader's decision to issue a different position from the one the preparer holds.

#### **5. Required actions — process requirements.**

- **PCI-PML-STD-12.01-PR-01 — Overrides are recorded in writing.** Where a credential holder issues a professional statement different from the preparer's position, they must record the change, their reason and their own name, and must not present the result as the preparer's position.
- **PCI-PML-STD-12.01-PR-02 — The preparer's position travels with the output.** The preparer must be able to record their own position, and that record must accompany the output to the decision-maker. Removing, summarising away or declining to transmit a preparer's recorded position does not satisfy this requirement.
- **PCI-PML-STD-12.01-PR-03 — Attribution is accurate.** Analysis, drafting and professional conclusions must be attributed to the person who produced them, and a credential holder must not sign as preparer, author or reviewer of work they did not perform.
- **PCI-PML-STD-12.01-PR-04 — Assessment against criteria set in advance.** Where a credential holder contributes to a performance assessment of a team member, the assessment must be recorded against criteria set before the period assessed. An assessment altered after the person raised a concern is subject to **PCI-PML-STD-12.02-PR-04.**

- **PCI-PML-STD-12.01-PR-05 — First adverse report acknowledged in writing.** A material adverse fact reported to the credential holder by a team member must be acknowledged in writing within the time the governance sets, with a recorded statement of what will be done about it and by whom.

**6. Prohibited actions.** Instructing a person to change a professional statement they have recorded as accurate, without recording the override in one's own name. Removing a preparer's recorded position from a pack. Signing as author or reviewer of work one did not do. Attributing a team member's analysis to oneself. Assessing a person against criteria introduced after the period. Responding to a first report of bad news by questioning the reporter's competence in place of addressing the fact.

**7. Required evidence.** Override records with the changed position, the reason and the leader's name; preparers' recorded positions as transmitted to decision-makers; authorship and sign-off records; the performance-assessment criteria with their date of setting; the written acknowledgements of adverse reports with their response statements and dates.

**8. Responsible role.** The named credential holder exercising the authority. Where the instruction comes from above the credential holder, the credential holder's obligation is to record it and escalate under **PCI-FND-STD-11**, not to transmit it.

**9. Approval authority.** A credential holder may override a professional statement within their own authority and must record it. They must not approve the removal of the preparer's recorded position; only the receiving decision-maker may decide that they do not need it, and that decision is itself recorded.

**10. Independence requirement.** Where an override concerns an assurance conclusion, a test result or a safety assessment, the override must be reviewed by a competent reviewer independent of both the leader and the preparer before the statement is issued. For estimates, forecasts and schedules, independence attaches at gates under the standard governing that artefact.

**11. Materiality or threshold.** This standard sets no number. The organisation's governance sets the time for acknowledging an adverse report and the classes of statement whose override requires independent review; this standard requires that both exist and are applied. Every override of a professional statement is recorded regardless of size, because the pattern of overrides is the evidence, and a threshold would remove the population that reveals it. *Six-person internal project:* the override record is a line in the same document with the leader's name and a sentence of reason; the acknowledgement is a reply within one working day. *Multi-partner national programme:* the same obligation attaches at each tier and across organisational boundaries,

and the partner agreement states that a client-side leader's override of a supplier's professional statement is recorded in the client's record — the case where an override is otherwise invisible because it crossed a contract.

**12. Exception and waiver.** No exception is permitted to the principal obligation. An exception to [PR-02](#) — not transmitting the preparer's recorded position — may be granted only by the receiving decision-maker, in writing, and the fact that a position exists and was not transmitted must still be disclosed to them.

**13. Escalation trigger.** An instruction from any level to change a professional statement the preparer holds to be accurate. A preparer's position removed from a pack. A request to sign as author of work one did not do. An assessment altered after a concern was raised. An adverse report unacknowledged beyond the documented time.

**14. AI application.** AI may compare successive document versions and surface changes to professional conclusions, check that authorship metadata matches recorded sign-offs, age adverse reports against the acknowledgement time, and flag assessments whose criteria post-date the period assessed.

**15. AI prohibition.** An AI system must not override a professional statement, decide that an override is justified, author a professional statement attributed to a named person, or determine whether an adverse report requires action.

**16. AI verification. Clause-to-summary comparison plus named approval.** Where AI produced a summary of a professional statement for a decision pack, a named human must compare the summary against the statement and confirm it carries the same position and the same qualifications; and the preparer must confirm by name that the transmitted summary is theirs before it is issued as such.

#### **17. External reference.**

- **Project Management Institute** · *Code of Ethics and Professional Conduct* · relied on for: the existence of professional expectations of honesty, responsibility, respect and fairness · **EXT-063, not independently verified — verify current requirements** · **Manual §6 category 6 — ethical code** · limitation: binding only where a body, regulator or engagement has adopted it; a PCI credential holder not subject to it is not made subject to it by this standard; no text is reproduced.
- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of leadership and team-development activity within delivery practice · **EXT-028** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.

- **ISO** · *ISO 45001, Occupational health and safety management systems — Requirements* · relied on for: the existence of a certifiable management-system standard within which worker consultation and participation are addressed · **EXT-123** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: certifiable, but voluntary unless required by contract or regulator; this standard imports none of its requirements.

**18. Jurisdictional caution.** Employment law, collective agreements, professional-body rules and, in regulated professions, statutory duties govern what may be required of a professional and what protection they have when they refuse. Obtain local legal and human-resources advice before acting on an override dispute involving an employee, a contractor or a regulated professional.

**19. Related PCI Standards.** *PCI-FND-STD-13; PCI-FND-STD-01; PCI-FND-STD-10; PCI-FND-STD-11; PCI-PML-STD-11.01; PCI-PML-STD-12.02.*

**20. Related Body of Knowledge content.** PML-AI · Domain 12 · KA 12.1 Leadership theories in practice and emotional intelligence; KA 12.3 Delegation, coaching and difficult conversations; KA 12.4 Remote and multicultural teams, ethical leadership. Also Domain 1 KA 1.2 the professional standard of care; Domain 11 KA 11.3 negotiation and conflict.

**21. Compliance test.** Compliance is demonstrated when a reviewer can: (a) take a stated sample of professional statements issued in the period and, for each whose final position differs from the preparer's earlier version in the document history, find an override record carrying the leader's name and a reason; (b) confirm that the preparer's recorded position accompanied the output to the decision-maker, or that a written exception from the decision-maker exists disclosing that a position was withheld; (c) match authorship and sign-off records and find no statement signed by a person who did not perform the work; (d) confirm every performance-assessment criteria set in the period is dated before the period it assesses; and (e) compute, for a stated sample of adverse reports from team members, the elapsed time to written acknowledgement and confirm it is within the documented time. A changed professional conclusion with no override record fails the test.

**22. Breach indicators.** Document histories in which a preparer's figures change with no override record. Packs in which the preparer's caveats appear in early versions and not in the issued one. Sign-offs by people with no editing history on the document. Assessment criteria dated at the end of the period. Adverse reports with no acknowledgement. A pattern of estimates revised downward in the version immediately before issue.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension or withdrawal of the PML-AI credential — each subject to due process and a right of appeal.

**24. Examination application.** Ethical dilemma: a leader is instructed by a director to reissue a supplier's schedule with a completion date the supplier's planner has recorded as unachievable. Evidence selection: which artefacts would establish that an override occurred. Scenario judgement: a preparer asks for their dissent to be recorded and the leader considers it unhelpful to the decision.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-12-01](#) v1.0. Amendment note: renumbered, restructured and **substantially rewritten**. The v1.0 standard stated general conduct expectations that could not be audited — audit questions 5 and 6 both failed — and the principal obligation has been replaced with the single observable act of overriding another person's professional statement, supported by five process requirements each of which leaves a record.

## PCI STANDARD

### PCI-PML-STD-12.02

## Route to Raise a Concern and Freedom from Detriment

**1. Normative requirement.** A credential holder leading delivery must establish, publish and operate a route by which any person working on or for the project may raise a concern to a named recipient who is neither the subject of the concern nor in the subject's reporting line.

**2. Purpose.** Every control in this set consumes voluntary upward information. Risk identification, nonconformance reporting, interface issues and escalation lead time all fail silently in the same way when people learn that raising a problem is penalised, and the leader becomes the last person on the project to know. **This standard does not require a state of mind and does not require a leader to produce a feeling in another person.** A state of mind cannot be observed, cannot be audited and cannot be breached, so a standard written around one would be an aspiration wearing a standard's clothing — the exact defect the Drafting Manual prohibits. What this standard requires instead is three things that are entirely observable: **a route that goes around the person the concern is about, the absence of detriment to the person who used it, and a record showing the concern was answered.** Those are the conditions under which candour is possible; whether candour then occurs is a matter of leadership and is taught, not legislated.

**3. Scope.** Every credential holder leading a project, programme or portfolio team, including distributed, multicultural, seconded, contracted and supplier teams, and including concerns raised anonymously or through a third party. It applies to concerns about delivery, conduct, safety, legality and the accuracy of reporting. It does not displace an employer's statutory whistleblowing, grievance or safety-reporting arrangements, which run in parallel and take precedence where they apply.

**4. Defined terms.** *concern · concern route · detriment · evidence · material · independent · decision owner.* Additionally, **subject of a concern** means the person or organisation whose act, omission or decision the concern is about; **lookback window** means the period after a recorded concern during which an act meeting the definition of detriment against the person who raised it requires the review in [PR-04](#).

#### **5. Required actions — process requirements.**

- [PCI-PML-STD-12.02-PR-01](#) — **A published route with named recipients.** The credential holder must publish, to everyone working on or for the project, a route naming **at least two** recipients, the method of reaching each, what happens after a concern is raised, and the time within which a response is given. A route that names a role with no person, or that exists only in a policy nobody on the project has been given, does not satisfy this requirement.
- [PCI-PML-STD-12.02-PR-02](#) — **The bypass property.** For every person working on the project, at least one published recipient must be a person who is neither their line manager, nor in their reporting line, nor — where the concern is about the project leader — the project leader. The credential holder must be able to demonstrate this property person by person, and must add a recipient where it fails.
- [PCI-PML-STD-12.02-PR-03](#) — **A concern register with mandatory fields.** Every concern raised through the route must be recorded with: the date raised; the route used; whether the subject of the concern was a recipient of it (which must be recorded, and must be "no"); the substantive response given; the decision taken; the date closed; and the name of the person who responded. An entry missing any field is not a compliant record.
- [PCI-PML-STD-12.02-PR-04](#) — **No detriment, and any act resembling one is justified and reviewed.** The credential holder must not subject a person to detriment for raising a concern. Where an act meeting the definition of *detriment* falls on a person who raised a concern within the documented lookback window, the act must not take effect until: a written justification exists stating the grounds and the evidence; the decision is taken by a person who is **not** the subject of the concern; and a



competent reviewer independent of both parties has reviewed the justification and recorded their conclusion.

- **PCI-PML-STD-12.02-PR-05 — Anonymity is not defeated.** The credential holder must not attempt, and must not permit any person or system to attempt, to identify the author of an anonymous concern, and must not use monitoring, communications analysis or people analytics to identify or profile people who raise concerns.
- **PCI-PML-STD-12.02-PR-06 — The countable indicator is reported.** Each reporting period the credential holder must report the proportion of issues in the period that were **discovered by assurance, testing or audit** rather than **reported by the team**, and must treat a sustained rise in that proportion as an issue under **PCI-PML-STD-08.02** with a named owner. This is a count drawn from records the project already produces, and it is the one indicator of reporting health that does not depend on anybody's self-report.

**6. Prohibited actions.** Requiring concerns to be raised only through the person they are about, or only through a line manager. Responding to a concern by assessing the person who raised it. Subjecting a person to detriment for raising a concern. Attempting to identify an anonymous reporter. Closing a concern with no substantive response. Recording a concern only where it turned out to be correct. Treating dissent recorded in a review as disloyalty. Using a survey result as a substitute for operating the route.

**7. Required evidence.** The published route, with evidence of its distribution to everyone working on the project and the date; the bypass demonstration mapping each person to at least one qualifying recipient; the concern register with all seven fields per entry; the justification, decision and independent review record for any act meeting the definition of detriment inside the lookback window; the period-by-period indicator under **PR-06** and its trend.

**8. Responsible role.** The named credential holder leading the project, for establishing, publishing and operating the route and for the register. The named recipients, for responding. The employing organisation's designated function, for statutory whistleblowing and grievance matters, which this standard does not displace.

**9. Approval authority.** The sponsor approves the route and its recipients, and approves any change to them. The governing body approves the route where the project leader is a possible subject. **No act falling within PR-04 may be approved by the subject of the concern**, and no approval by the credential holder cures that.

**10. Independence requirement.** At least one published recipient must be independent of the project's delivery line. The reviewer of a justification under **PR-04** must be independent of both the person who raised the concern and its



subject, and must not be the person who decided the act. Where an allegation of detriment is made, the investigation must be conducted by a person independent of the project.

**11. Materiality or threshold.** This standard states no number of concerns, no target and no score, and any of those would be actively harmful: a concern target invites manufactured concerns and a survey score invites managed answers. **A survey result is not evidence of compliance with this standard, and a low score is not evidence of breach** — the evidence is the route, the register and the detriment review. The organisation's governance sets the response time, the lookback window and the reporting period; this standard requires that they exist and are applied. Where the organisation sets no lookback window, the credential holder must apply the period to the end of the project or twelve months, whichever is shorter, and record that they did so in the absence of a documented window. **That period is also the floor.** A documented lookback window shorter than it does not satisfy this standard, because a window an organisation can set to a fortnight is a protection the organisation can switch off; where the documented window is shorter, the credential holder applies the floor and records that they did so. *Six-person internal project:* two named recipients — the project leader and one named manager outside the team — a one-page route sent to all six on day one, and a register that may hold no entries for months. The bypass demonstration is six lines. **A register with no entries is not a breach;** a register with no entries alongside a rising PR-06 indicator is the thing to investigate. *Multi-partner national programme:* recipients named in each employing organisation plus one at programme level who is independent of every partner, because a supplier's engineer with a concern about the client's schedule has no usable route inside their own employer; the register is held at programme level with partner-level extracts; and the PR-06 indicator is reported per partner, because averaging conceals the one partner whose team has stopped reporting.

**12. Exception and waiver. No exception is permitted to PR-02, PR-04 or PR-05.** An exception to the response time in PR-01 may be granted by the sponsor where an investigation genuinely requires longer, provided the person who raised the concern is told, in writing, that it is being handled and when they will hear. Confidentiality of an investigation is not a ground for giving no response at all.

**13. Escalation trigger.** Any act meeting the definition of detriment against a person who raised a concern. Any attempt to identify an anonymous reporter. A concern about the project leader, the sponsor or a governing-body member — which is escalated outside the project on receipt. A concern that discloses a safety, legal or regulatory matter. A sustained rise in the PR-06 indicator. A concern unanswered beyond the documented time.

**14. AI application.** AI may age the concern register against the response time and flag overdue entries, check entries for missing mandatory fields, compute and trend the [PR-06](#) indicator from issue and assurance data, and summarise the themes of closed concerns for governance in aggregate form.

**15. AI prohibition.** An AI system must not identify or attempt to identify the author of an anonymous concern, score or profile individuals on attitude, engagement, sentiment or dissent, determine the outcome of a concern, decide whether an act constitutes detriment, or triage concerns so that some receive no human response. Sentiment or communications analysis must not be used on project communications for the purpose of locating dissent.

**16. AI verification. Sampling with a stated basis plus independent recomputation.** Each reporting period, a competent reviewer must draw a stated sample of concern entries and confirm against the underlying correspondence that the recorded response was given, was substantive, and came from the named responder; and must recompute the [PR-06](#) indicator by hand from the issue and assurance records for one period and reconcile it to the reported figure. Where any people-analytics tool is in use anywhere on the project, the credential holder must obtain and retain a written confirmation of its data sources and verify that concern-route data is not among them.

#### **17. External reference.**

- **ISO · ISO 45003, Occupational health and safety management — Psychological health and safety at work — Guidelines** · relied on for: the existence of psychosocial risk as a recognised subject within occupational health and safety management · **EXT-124 · Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: **guidance, not a requirements standard — nothing can be certified against it**, and no obligation in this standard derives from it. It is named because a reader will expect it to be addressed, and the correct treatment is to say what it is.
- **ISO · ISO 45001, Occupational health and safety management systems — Requirements with guidance for use** · relied on for: the existence of a certifiable management-system standard addressing worker consultation, participation and protection from reprisal · **EXT-123 · Manual §6 category 3** · currency checked 2026-08-03 · limitation: certifiable, but adoption is voluntary unless a contract or regulator requires it; this standard imports none of its requirements.
- **ISO · ISO 21502, Guidance on project management** · relied on for: the existence of team development and working environment within delivery practice · **EXT-028 · Manual §6 category 3** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.

**18. Jurisdictional caution.** Whistleblower protection, mandatory internal reporting channels, works- council consultation, employee monitoring, data-protection and anti-retaliation law differ sharply by jurisdiction and frequently impose obligations stricter and more specific than this standard — including on who may receive a report, how it must be handled, what records may be kept and for how long. **Obtain local legal advice before establishing a concern route, an anonymous channel, a register or any monitoring arrangement**, and where the applicable regime imposes a higher obligation, that regime governs.

**19. Related PCI Standards.** PCI-FND-STD-11 (the parent duty to escalate); PCI-FND-STD-01; PCI-FND-STD-09; PCI-PML-STD-08.02; PCI-PML-STD-09.02; PCI-PML-STD-12.01; PCI-PML-STD-14.01. **What this standard adds to PCI-FND-STD-11:** the foundational standard creates a duty to escalate, which binds the person who holds the information. This standard creates the **conditions under which they can discharge it** — a named recipient outside the line of the person concerned, protection from detriment tested by an independent review, and a record that the concern was answered.

**20. Related Body of Knowledge content.** PML-AI · Domain 12 · KA 12.2 Motivation, team formation and psychological safety · topic 12.2.3 psychological safety as a design variable, including its observable presence and absence signals and its stated evidential limits; KA 12.3 difficult conversations. Also Domain 3 KA 3.3 escalation design; Domain 8 KA 8.4 bias and crisis leadership; Domain 9 KA 9.4 lessons.

**21. Compliance test.** Compliance is demonstrated when a reviewer can perform all six of the following: (a) obtain the published route and the evidence of its distribution, and confirm it names at least two recipients with a stated method and response time; (b) take the project's current people list and confirm, person by person, that at least one named recipient is neither their line manager nor in their reporting line — and, for the project leader's own reports, that a recipient exists who is not the project leader; (c) open the concern register and confirm every entry carries all seven mandatory fields, and that the "subject was a recipient" field reads "no" in every case; (d) draw a stated sample of closed entries and confirm from the underlying correspondence that a substantive response was given by the named responder within the documented time; (e) cross-match the concern register against the project's assignment, role-change and assessment records for the lookback window and confirm that **every** act meeting the definition of *detriment* falling on a person who raised a concern carries a written justification, a decision taken by someone other than the subject of the concern, and an independent reviewer's recorded conclusion dated before the act took effect; and (f) recompute the PR-06 indicator for one period from the issue and assurance

records and reconcile it to the reported figure. **Test (e) is the operative one.** An act of detriment inside the lookback window with no independent review dated before it took effect fails the test, and no survey score, culture statement or leadership testimonial cures it.

**22. Breach indicators.** A route naming one recipient, who is the project leader. A route that exists in a corporate policy and has never been sent to the project. Register entries with the response field containing "noted". Concerns closed the day they were raised. A person removed from a workstream within weeks of raising a concern, with no justification record. Anonymous concerns followed by an inquiry into who wrote them. A **PR-06** indicator rising for three periods with no owner. A team whose register is empty while assurance is finding everything.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure on the associated competency; ethics review; certification investigation; suspension or withdrawal of the PML-AI credential — each subject to due process and a right of appeal. **PCI cannot compensate a person subjected to detriment, order reinstatement or impose any penalty on an employer;** those remedies, where they exist, arise under the applicable employment or whistleblowing law.

**24. Examination application.** Scenario judgement: a supplier's engineer raises a safety concern about the client's sequence and the only published recipient is their own project manager, who set the sequence. Evidence selection: which artefacts establish compliance, and why a staff-survey score is not among them. Ethical dilemma: a leader is asked to remove from a workstream a person who raised a concern three weeks earlier, for reasons the leader believes are genuine and unrelated. Escalation decision: an anonymous concern names the sponsor.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes **PML-LAW-12-02 v1.0** (*Psychological Safety*). Amendment note: renumbered, **retitled** and rewritten from the ground up. The v1.0 standard required a leader to "maintain conditions in which any team member can report bad news" — a state of affairs no reviewer could test, which failed audit questions 5 and 6. The principal obligation is now the **establishment and operation of a bypass route**, the protections are expressed as observable acts through a defined term for *detriment*, and the compliance test cross-matches two record sets that already exist. The concept of psychological safety remains the reason for the standard and is taught in Domain 12; it is not the requirement, because it is not observable. **Stage 9 amendment:** the lookback window was set wholly by the organisation, so the protection in **PR-04** could be switched off by documenting a fortnight; element 11 now makes the standard's own default the floor.

## Domain 13 — Agile, Adaptive and Hybrid Delivery

### PCI STANDARD

#### PCI-PML-STD-13.01

### Governance of Adaptive Delivery

**1. Normative requirement.** A credential holder using an adaptive or hybrid delivery approach must operate governance that produces the same decision rights, evidence and accountability as a predictive approach, expressed through the adaptive method's own artefacts.

**2. Purpose.** Adaptive delivery does not remove governance; it relocates it, from documents produced for gates to decisions taken continuously. The failure this prevents runs in both directions: the team that treats an iterative method as an exemption from evidence and authority, and the organisation that bolts a predictive gate structure onto an adaptive team and destroys the feedback the method exists to produce.

**3. Scope.** Every credential holder leading, governing, assuring or reporting on delivery using an adaptive or hybrid approach, at team, programme and portfolio level, including scaled arrangements and including contracts written for adaptive delivery.

**4. Defined terms.** *governance* (project sense) · *decision owner* · *delegation schedule* · *evidence* · *gate* · *acceptance* · *product owner*. Additionally, **value envelope** means the bounded authority within which a product owner may commit capacity without escalation; **release decision** means the decision to make an increment available to users.

#### 5. Required actions — process requirements.

- **PCI-PML-STD-13.01-PR-01 — Decision rights mapped to adaptive artefacts.** The credential holder must record which adaptive artefact or event carries each decision class in the delegation schedule, so that every class has a location. A class with no location is governed by the predictive route until one is recorded.
- **PCI-PML-STD-13.01-PR-02 — The value envelope is stated.** The product owner's value envelope must be documented and approved by the sponsor, and decisions exceeding it must escalate within the latency the delegation schedule states.
- **PCI-PML-STD-13.01-PR-03 — Release decisions carry acceptance and evidence.** A release decision must satisfy [PCI-PML-STD-09.01](#), and the increment's acceptance evidence must be retained. Iterative delivery does not reduce the evidence requirement; it changes its cadence.
- **PCI-PML-STD-13.01-PR-04 — Adaptive metrics are defined and not gamed.** Where flow, throughput, cycle-time or velocity measures are

reported to governance, their definition must be documented, applied consistently, and reported with the work-in-progress position, so that a rising throughput achieved by starting more work is visible as such.

- **PCI-PML-STD-13.01-PR-05 — Framework adoption is a recorded choice.** Where a published adaptive framework is adopted, the credential holder must record which framework, which parts of it are adopted, and which are not — because adoption is the whole of such a framework's force and an unrecorded partial adoption is unauditably.

**6. Prohibited actions.** Treating an adaptive method as removing the need for acceptance evidence, decision records or escalation. Reporting velocity as progress against a commitment. Increasing throughput by increasing work in progress and reporting the first without the second. Describing a team as following a named framework while omitting the parts that constrain it. Placing the ordering right with a committee while naming an individual as product owner.

**7. Required evidence.** The decision-rights map from classes to adaptive artefacts; the documented value envelope with its approval; release decisions with their acceptance records and retained evidence; the metric definitions and the reported series including work in progress; the framework-adoption record.

**8. Responsible role.** The named credential holder leading delivery, for the governance mapping and the reporting. The named **product owner**, for ordering decisions within the envelope. The named release authority, for release decisions.

**9. Approval authority.** The sponsor approves the value envelope. The delegation schedule's named authority approves decisions above it. The release authority approves releases. A committee must not be recorded as the product owner; where a committee holds the ordering right in fact, the record must say so and the product owner is named as a proxy.

**10. Independence requirement.** Acceptance of an increment must be recorded by a person independent of its production under **PCI-PML-STD-09.01**. Assurance of an adaptive delivery must be provided by a competent reviewer independent of the team, and must satisfy **PCI-PML-STD-01.03-PR-06** where that reviewer has contributed to the team's ways of working.

**11. Materiality or threshold.** This standard states no envelope value, no iteration length and no work-in-progress limit. The organisation's governance sets the value envelope, the escalation latency and the release authority; this standard requires that they are documented and applied. Where a team operates a work-in-progress limit, its basis must be recorded so that a change to it is visible as a decision. *Six-person internal project:* one team, one product owner with an envelope stated in a sentence, releases accepted by a named



user representative, and a metric definition of two lines. The decision-rights map is a five-row table. *Multi-partner national programme*: several teams across organisations with a stated rule for which decisions rise from team to programme, a programme-level release authority for anything crossing an interface, and metric definitions held at programme level so that partners' velocity figures are not added together as though they were comparable — which they are not.

**12. Exception and waiver.** An exception permitting a release without complete acceptance evidence may be approved only by the release authority, only where the missing evidence is identified item by item with owners and dates, and only where the release is reversible or the residual risk is accepted by the party bearing it. No exception permits a decision class to have no recorded location. **This exception does not reach a mandatory precondition.** Where the release is a release into operational use, *PCI-PML-STD-16.01* applies to it, and its element 12 permits no exception, waiver, dispensation or deferral in respect of a gate-block item — an iterative delivery model changes the cadence of the release, not the status of a safety case, a licence or a statutory notification.

**13. Escalation trigger.** A decision exceeding the value envelope. A decision class with no recorded location. A release proposed without acceptance evidence. A product owner who cannot exercise the ordering right. A throughput improvement accompanied by a rising work-in-progress position. A framework described as adopted whose constraining practices are not in use.

**14. AI application.** AI may reconcile the decision-rights map against the delegation schedule and flag unlocated classes, compute flow metrics with their work-in-progress position, detect backlog items that have never been reordered, and check release records for missing acceptance references.

**15. AI prohibition.** An AI system must not order a backlog as the operative decision, approve a release, set a value envelope, or act as product owner. A model that produces an order has produced a recommendation; the ordering right belongs to a named person.

**16. AI verification. Independent recomputation plus named approval.** Flow and throughput metrics produced with AI assistance must be recomputed by hand for at least one period per quarter and reconciled. Any AI-produced backlog order must be reviewed and adopted, altered or rejected by the named product owner, who records which they did.

#### **17. External reference.**

- **Ken Schwaber and Jeff Sutherland** · *The Scrum Guide* · relied on for: the existence of a defined set of accountabilities, events and artefacts in one widely used adaptive framework · **EXT-o86** · **Manual §6 cat-**

**egory 5 — professional framework** · currency checked 2026-08-03 · limitation: **a voluntary framework whose adoption is the whole of its force**. It binds a team that has adopted it and nobody else; it is **not a standard**, nothing can be certified against it as a requirements document, and no PCI obligation derives from it. Its concepts are described here in PCI's own words; no text is reproduced. The suite register classifies it as a voluntary framework; Manual §6 governs this file and places it at category 5.

- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of tailoring and of iterative life cycles within delivery practice · **EXT-028** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.
- **Project Management Institute** · *PMBOK Guide* · relied on for: the existence of adaptive and hybrid approaches in professional practice · **EXT-060** · **Manual §6 category 5 — professional framework** · currency checked 2026-08-03 · limitation: **never regulatory authority**; no edition asserted; no text reproduced.

**18. Jurisdictional caution.** Contracts written for adaptive delivery raise questions of what has been agreed, when payment is due and what has been accepted that are determined by the governing law of the contract. Obtain legal advice before adopting an outcome-based or capacity-based contract structure.

**19. Related PCI Standards.** [PCI-FND-STD-02](#); [PCI-FND-STD-12](#); [PCI-FND-STD-04](#); [PCI-PML-STD-03.01](#); [PCI-PML-STD-03.02](#); [PCI-PML-STD-05.01](#); [PCI-PML-STD-09.01](#); [PCI-PML-STD-13.02](#).

**20. Related Body of Knowledge content.** PML-AI · Domain 13 · KA 13.1 Agile principles and product ownership; KA 13.2 Backlogs, iteration planning, flow and Kanban; KA 13.3 Scaling considerations and hybrid governance; KA 13.4 Contracting for adaptive delivery, metrics and anti-patterns. Also Domain 3 KA 3.1 topic 3.1.3 governance in agile and hybrid environments.

**21. Compliance test.** Compliance is demonstrated when a reviewer can: (a) take every decision class in the delegation schedule and find, in the decision-rights map, the adaptive artefact or event that carries it; (b) retrieve the approved value envelope and confirm that every commitment above it in the period has an escalation record; (c) take a stated sample of releases and retrieve, for each, an acceptance record and the evidence it cites; (d) recompute one period's reported flow metric from the raw item data using the documented definition and reconcile it to the reported figure, and confirm the work-in-progress position was reported alongside it; and (e) compare the



framework-adoption record against observed practice for the constraining elements it claims to adopt. An unlocated decision class, or a release with no acceptance record, fails the test.

**22. Breach indicators.** Velocity reported as a commitment. Throughput rising while work in progress rises faster. Backlogs from which nothing is ever removed. A product owner whose decisions are routinely overturned. Releases with no acceptance record. A named framework in the delivery approach and none of its constraining events in the calendar. Assurance provided by the coach who designed the ways of working.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Calculation review: throughput and work-in-progress series showing an apparent improvement that is a queue. Scenario judgement: a steering committee requires a stage gate on an adaptive team six weeks into a quarterly cycle. Evidence selection: which artefacts show that an adaptive delivery is governed.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-13-01](#) v1.0. Amendment note: renumbered and restructured; legislative drafting removed; the decision-rights mapping, value envelope, release evidence, metric integrity and framework-adoption record separated into five process requirements; the Scrum Guide's characterisation corrected to state expressly that adoption is the whole of its force. **Stage 9 amendment:** element 12's release exception overlapped [PCI-PML-STD-16.01](#), whose element 12 admits no exception at all in respect of a mandatory precondition; element 12 now states that the exception does not reach one.

## PCI STANDARD

### PCI-PML-STD-13.02

## Product and Project Accountability

**1. Normative requirement.** A credential holder must ensure that, for every product or service being delivered, exactly one named individual holds the ordering right and exactly one named individual holds delivery accountability, and that the record names both.

**2. Purpose.** Where product and project accountability blur, two failures follow and they are opposite. Either nobody can order the work, so the backlog is arbitrated by whoever escalates hardest; or the product role is nominal, held by someone who must obtain approval for every decision, in

which case the authority sits with the approver and the record is false. The professional content of the role is the **decision right**, and this standard protects it.

**3. Scope.** Every credential holder in a delivery leadership, product ownership or governance role, in adaptive, hybrid and product-organised delivery, including arrangements where a product outlives the project that created it and arrangements where a supplier supplies the product role.

**4. Defined terms.** *product owner · decision owner · accountability · value envelope · evidence · material.* Additionally, **ordering right** means the authority to determine the sequence in which work is done and what is not done; **proxy** means a person recorded as holding a right that is in fact exercised elsewhere.

#### **5. Required actions — process requirements.**

- **PCI-PML-STD-13.02-PR-01 — Both roles named, and distinguished.** The record must name the individual holding the ordering right and the individual holding delivery accountability, and must state what each decides. Naming one person for both is permitted only where the record says so expressly and the sponsor has approved it.
- **PCI-PML-STD-13.02-PR-02 — Proxy arrangements are disclosed.** Where the ordering right is in fact exercised by a committee or by a person other than the named product owner, the credential holder must record that the named holder is a proxy and identify where the authority actually sits.
- **PCI-PML-STD-13.02-PR-03 — Refusals are recorded.** The product owner's decisions not to do work must be recorded, because a backlog from which nothing is ever removed is evidence that the ordering right is not being exercised.
- **PCI-PML-STD-13.02-PR-04 — Continuity across the project boundary.** Where a product continues beyond the project, the credential holder must record the transfer of the ordering right and of delivery accountability to named individuals in the receiving organisation before closure, under **PCI-PML-STD-16.02**.

**6. Prohibited actions.** Recording a committee as product owner. Naming a product owner who must obtain approval for every ordering decision without recording them as a proxy. Allowing a supplier to hold the ordering right for a client's product without an approved instrument. Closing a project leaving a live product with no named ordering right. Presenting a backlog with no refusals as evidence of prioritisation.

**7. Required evidence.** The record naming both individuals and their decision scopes; the sponsor's approval where one person holds both; the proxy

disclosure where applicable; the backlog history showing items removed or refused with dates; the transfer record at closure.

**8. Responsible role.** The named credential holder leading delivery, for maintaining the record and for escalating where a role is vacant or nominal. The **sponsor**, for appointing the ordering right and setting the value envelope.

**9. Approval authority.** The sponsor appoints the product owner and approves the combination of both roles in one person. The governing body approves a supplier holding the ordering right for a client product. The credential holder approves neither.

**10. Independence requirement.** Not applicable to the ordering right itself, which is a delivery decision right and is not intended to be independent; independence attaches to the acceptance of increments under [PCI-PML-STD-09.01](#) and to the assurance of the arrangement under [PCI-PML-STD-01.03-PR-06](#).

**11. Materiality or threshold.** This standard states no number. The value envelope and the escalation latency are set by the organisation's governance under [PCI-PML-STD-13.01-PR-02](#); this standard requires that the roles are named and that the record is true. The test of nominality is not a percentage but the observable one: whether decisions within the stated envelope are taken and stand. *Six-person internal project*: one person may hold both roles, and that is entirely proper — the obligation is that the record says so and the sponsor approved it, so that nobody later assumes a separation that never existed. *Multi-partner national programme*: the ordering right sits with the client organisation for each product, partners supply delivery accountability within their own scope, and the boundary is recorded per product rather than per contract, because a product delivered by three partners has one order and three delivery accountabilities.

**12. Exception and waiver.** An exception permitting the ordering right to be held collectively may be approved by the governing body for a stated period, only where the collective body is named, its chair is recorded as the decision owner under the definition above, and its decision latency is stated. No exception permits a product to run with no recorded ordering right.

**13. Escalation trigger.** A product with no named ordering right. A named product owner whose decisions are routinely overturned. A supplier exercising the ordering right without an approved instrument. Project closure approaching with a live product and no receiving owner. A backlog with no refusals across a period the governance considers material.

**14. AI application.** AI may detect backlog items that have never been reordered or refused, measure the decision wait between an item being raised and being ordered, flag products with no recorded owner, and compare recorded ordering decisions against the value envelope.

**15. AI prohibition.** An AI system must not hold the ordering right, order a backlog as the operative decision, or be recorded as a product owner or as delivery-accountable.

**16. AI verification. Named approval plus sampling with a stated basis.** Every AI-proposed ordering must be adopted, altered or rejected by the named product owner with a record of which. Each quarter, a competent reviewer must sample ordering decisions and confirm from the record that the named holder took them and that they stood.

#### **17. External reference.**

- **Ken Schwaber and Jeff Sutherland** · *The Scrum Guide* · relied on for: the existence of a single accountable ordering role in one widely used adaptive framework · **EXT-086** · **Manual §6 category 5 — professional framework** · currency checked 2026-08-03 · limitation: **a voluntary framework; adoption is the whole of its force**; it is not a standard and nothing can be certified against it as a requirements document; described in PCI's own words, no text reproduced. This standard's obligation is PCI's own and does not derive from it.
- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of defined roles and responsibilities within delivery · **EXT-028** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.
- **ISO** · *ISO 21505, Guidance on governance* · relied on for: the existence of accountability definition within governance arrangements · **EXT-032, not independently verified** · **Manual §6 category 3** · limitation: as above, with an open verification status.

**18. Jurisdictional caution.** Where a supplier holds decision rights over a client's product, the allocation of liability, intellectual property and data controllership is a contractual and regulatory question. Obtain legal advice before placing an ordering right outside the owning organisation.

**19. Related PCI Standards.** [PCI-FND-STD-01](#); [PCI-FND-STD-04](#); [PCI-FND-STD-12](#); [PCI-PML-STD-01.01](#); [PCI-PML-STD-01.02](#); [PCI-PML-STD-13.01](#); [PCI-PML-STD-16.02](#).

**20. Related Body of Knowledge content.** PML-AI · Domain 13 · KA 13.1 topic 13.1.2 product ownership as a decision right; KA 13.4 metrics and anti-patterns, including the proxy product owner. Also Domain 1 KA 1.2 accountability and responsibility; Domain 16 KA 16.2 operational transition.

**21. Compliance test.** Compliance is demonstrated when a reviewer can, for every product in delivery: (a) name the holder of the ordering right and the holder of delivery accountability from the record; (b) take a stated sample of ordering decisions in the period and confirm from the record that the named holder took them; (c) confirm that decisions within the stated value envelope were not subsequently overturned by another body without an escalation record; (d) find refusals or removals in the backlog history; and (e) where one person holds both roles, find the sponsor's recorded approval. A product whose ordering decisions are routinely retaken elsewhere, with the named holder unchanged and no proxy disclosure, fails the test.

**22. Breach indicators.** A product owner named in an organisation chart and absent from every ordering decision. A backlog that only grows. Median ordering decision waits of the same order as the escalation cycle. Products with two people each believing they hold the order. A project closing with a live product and an unnamed owner. Sponsors ordering the backlog directly while a product owner is named.

**23. Consequence within PCI authority.** Correction required; additional review; escalation; examination failure; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Scenario judgement: a named product owner must obtain steering-committee approval for every ordering decision, and the candidate states what the record must show. Evidence selection: which artefact establishes that the ordering right is exercised. Calculation review: a decision wait series showing a proxy arrangement.

**25. Version and status.** Version 1.0 · **not yet approved** · effective on approval · **new standard** — the v1.0 set governed adaptive delivery in one standard and left the product/project accountability boundary unstated, which is the boundary most often blurred in practice. Amendment note: none.

## Domain 14 — Digital Delivery, Data and Responsible AI

### PCI STANDARD

#### PCI-PML-STD-14.01

### Responsible Data Use in Delivery

**1. Normative requirement.** A credential holder must not collect, use, share, retain or expose data on a project beyond the purpose, the recipients and the retention period recorded for it.

**2. Purpose.** Delivery accumulates data faster than it governs it: common data environments, dashboards, supplier portals, personal data about workers and users, safety records, commercially confidential tenders. The failure this prevents is the quiet expansion of purpose — data collected for one reason, copied into a second system for a second reason, and left there — which is simultaneously a professional failure, a security exposure and, in many jurisdictions, unlawful.

**3. Scope.** Every credential holder specifying, commissioning, operating, sharing or closing a project data environment, dashboard, model, register or archive, including supplier-hosted environments and including data about people working on the project.

**4. Defined terms.** *evidence · material · decision owner · independent · detriment · data governance* (written in full, always, and distinct from project governance). Additionally, **purpose record** means the recorded statement of why a data set is held, who may see it and for how long; **common data environment** means the shared repository through which project information is exchanged.

#### 5. Required actions — process requirements.

- **PCI-PML-STD-14.01-PR-01 — A purpose record per data set.** Every project data set must carry a recorded purpose, a named owner, the classes of recipient permitted, and a retention period. A data set with no purpose record must not be created or, where inherited, must be recorded and escalated.
- **PCI-PML-STD-14.01-PR-02 — Access matches the record.** Access rights in each system must be reconciled against the purpose record at the interval the governance sets, and every access not supported by the record must be removed and the removal recorded.
- **PCI-PML-STD-14.01-PR-03 — Personal data is identified and minimised.** Data about identifiable people must be identified as such in the record, must be limited to what the recorded purpose requires, and must not be used to assess, rank or profile individuals except where the purpose record states that use and the applicable legal basis has been confirmed by the organisation's competent function.
- **PCI-PML-STD-14.01-PR-04 — Lineage for figures that carry decisions.** Any figure reported to a decision-maker must be traceable to its source system, its extract date and its transformation, and the credential holder must not report a figure whose lineage cannot be stated.
- **PCI-PML-STD-14.01-PR-05 — Closure disposes or transfers deliberately.** At project closure, every data set must be transferred to a named receiving owner, archived under the recorded retention period, or securely disposed of, with the decision recorded per data set.

**6. Prohibited actions.** Copying a data set into a second environment without extending the purpose record. Granting standing access "for convenience". Using worker or user data to profile individuals outside the recorded purpose. Reporting a figure whose source cannot be stated. Leaving a project environment live and unowned after closure. Sharing commercially confidential tender data with a party outside the evaluation. Exporting personal data to a supplier tool without the organisation's competent function confirming the basis.

**7. Required evidence.** The data register with purpose, owner, recipient classes and retention per set; the periodic access reconciliation with removals; the identification of personal data and the confirmed legal basis where used to assess individuals; lineage records for reported figures; the closure disposition record per data set.

**8. Responsible role.** The named credential holder leading the project, for the register, the reconciliation and the closure disposition. The named data-set owner, for each set. The organisation's competent data-protection or information-governance function, for the legal basis — **not** the project.

**9. Approval authority.** The organisation's information-governance authority approves a new purpose, a new recipient class and any extension of retention. The sponsor approves the closure disposition plan. The project must not approve its own extension of purpose.

**10. Independence requirement.** The access reconciliation must be performed or verified by a person independent of the team whose access is being reconciled. The confirmation of a legal basis must come from a function independent of the project.

**11. Materiality or threshold.** This standard states no data volume and no retention period, because retention is set by law, by contract and by the organisation's policy, and a number invented here would conflict with all three. It requires that the purpose record exists per data set, that the reconciliation interval is documented and met, and that the retention period recorded is the one the organisation's policy or the applicable law sets. *Six-person internal project:* six data sets, one register page, access reconciled quarterly by a named manager outside the team, and a closure decision of one line per set. *Multi-partner national programme:* a common data environment per partner plus a shared one, a register that records which partner is the owner and which are recipients for each set, and a closure disposition agreed contractually in advance — because the moment to agree what happens to a partner's data at exit is before the partner is exiting.

**12. Exception and waiver.** An exception permitting use beyond the recorded purpose may be approved only by the organisation's information-governance authority, in writing, before the use, for a stated period and recipient set, with the compensating control stated. **No exception may be approved by the**



**project alone**, and no exception permits use of personal data for an assessment purpose without a confirmed legal basis.

**13. Escalation trigger.** A data set found with no purpose record. Access that the record does not support. A proposal to use worker or user data to assess individuals. A figure reported whose lineage cannot be stated. A supplier tool found to be processing project personal data outside the record. A suspected or actual data breach — which is escalated immediately and outside the ordinary cycle.

**14. AI application.** AI may build the data register from system metadata, reconcile access lists against the register and flag divergence, detect probable personal data in unclassified stores, trace lineage across systems and flag figures with broken lineage, and produce the closure disposition list.

**15. AI prohibition.** An AI system must not determine a legal basis, approve a purpose extension, decide that data may be shared, profile individuals, or be the sole determinant that a data set contains no personal data.

**16. AI verification. Source tracing plus sampling with a stated basis.** Every AI-detected classification must be confirmed by a named human before it is acted on, and every AI-produced lineage must be traced by a competent reviewer to the source system and extract for a stated sample of reported figures. Where AI screened stores for personal data, a named human must review in full any store the system reports as clean but which the purpose record indicates is likely to contain personal data.

#### **17. External reference.**

- **ISO/IEC** · *ISO/IEC 27001, Information security management systems — Requirements* · relied on for: the existence of a certifiable requirement set for information-security management · **EXT-023** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: certifiable, but adoption is voluntary unless required by contract or regulator; this standard imports none of its controls.
- **ISO/IEC** · *ISO/IEC 27701, Privacy information management systems — Requirements and guidance* · relied on for: the existence of a privacy management-system standard · **EXT-038** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: **materially changed between editions** — the earlier edition was an extension to an information-security management system and the current one is a standalone standard certifiable in its own right. **Verify which edition any claim of conformity refers to.** Voluntary unless required.
- **ISO** · *ISO 8000 data-quality standards* (a multi-part series) · relied on for: the existence of a standardised treatment of data quality · **EXT-026** · **Manual §6 category 3** · currency checked 2026-08-03 ·



limitation: **a series, not one document**; voluntary; cited generically and relied on for no requirement.

- **ISO/IEC** · *ISO/IEC 25012, Data quality model* · relied on for: the existence of a defined set of data-quality characteristics · **EXT-036** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: voluntary; a model, not a requirements standard for a project; check for supersession within its series.
- **ISO** · *ISO 19650 series, Information management using building information modelling* · relied on for: the existence of a common data environment concept in the built environment · **EXT-039** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: **a series**; voluntary; sector-specific; cited generically.
- **European Union** · *General Data Protection Regulation* · relied on for: illustrating a rights-based data-protection approach · **EXT-101, not independently verified — verify current requirements** · **Manual §6 category 10 — illustrative practice** · limitation: **this is binding legislation within the European Union**, named here only to illustrate the shape of such a regime; it is relied on for no requirement in this standard, and whether it or any other data-protection regime applies is a question for qualified local counsel.

**18. Jurisdictional caution.** Data-protection law, employee-monitoring rules, works-council consultation, data-localisation requirements, sector confidentiality duties, breach-notification timescales and cross-border transfer restrictions are jurisdiction-specific and change frequently. **Compliance with this standard is not compliance with any of them.** Obtain local legal advice before collecting, sharing, exporting or retaining personal or confidential project data.

**19. Related PCI Standards.** [PCI-FND-STD-07](#); [PCI-FND-STD-09](#); [PCI-FND-STD-12](#); [PCI-PML-STD-11.01](#); [PCI-PML-STD-12.02](#); [PCI-PML-STD-14.02](#); [PCI-PML-STD-16.03](#).

**20. Related Body of Knowledge content.** PML-AI · Domain 14 · KA 14.1 Digital project environments, data governance and the common data environment; KA 14.4 Explainability, bias, human accountability, cybersecurity and privacy. Also Domain 9 KA 9.4 data quality; Domain 16 KA 16.4 responsible archive and model/data retention.

**21. Compliance test.** Compliance is demonstrated when a reviewer can: (a) list the project's data sets from the systems in use and match each to a purpose record with an owner, recipient classes and a retention period, with no residue; (b) extract the current access list from each system and reconcile it to the recipient classes in the record, with every difference explained by a dated

removal or a recorded approval; (c) select a stated sample of figures from reports issued in the period and trace each to its source system, extract date and transformation; (d) confirm that every use of data about identifiable people to assess or rank them carries a legal basis confirmed by the organisation's competent function; and (e) for a closed project, find a disposition decision for every data set. Any data set with no purpose record fails the test.

**22. Breach indicators.** Environments with more users than the project has people. Copies of a register in personal drives. Dashboards whose figures cannot be traced. Retention periods recorded as "indefinite". Personal data in a supplier's tool with no record. Access lists unchanged since project start. Closed projects with live environments.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal. **PCI cannot impose a data-protection penalty;** that consequence, where it exists, arises under the applicable law.

**24. Examination application.** Scenario judgement: a supplier proposes to copy the project's issue data into its own analytics platform to improve its service. Evidence selection: which artefacts establish that a reported figure is traceable. Ethical dilemma: a sponsor asks for individual productivity rankings from the delivery tool's activity data.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-14-01](#) v1.0. Amendment note: renumbered and restructured; legislative drafting removed; the purpose record, access reconciliation, personal-data minimisation, lineage and closure disposition separated into five process requirements; the GDPR's characterisation corrected to state expressly that it is binding legislation within its own jurisdiction and is relied on here for nothing.

## PCI STANDARD

### PCI-PML-STD-14.02

## Responsible AI in Delivery

**1. Normative requirement.** A credential holder must not rely on an AI system's output for a delivery decision, a report or a deliverable until a named human has verified it by a method recorded for that class of output.

**2. Purpose.** [PCI-FND-STD-03](#) requires verification before professional use. What delivery adds is that AI output arrives inside artefacts other people then rely on without knowing its origin: a risk register populated by a model, a schedule narrative generated from data, a supplier evaluation summary, a lessons theme. The failure this prevents is **unattributed reliance** — an output

nobody checked because everybody assumed somebody had, embedded in an artefact that carries the organisation's name.

**3. Scope.** Every credential holder using, commissioning, configuring, governing or assuring an AI system in delivery — including supplier-operated and embedded features in delivery tools — for any output that informs a decision, a report or a deliverable.

**4. Defined terms.** *AI verification* (the sense carried by the suite principle) · *evidence* · *material* · *competent reviewer* · *decision owner* · *independent*. Additionally, **AI use record** means the record of where AI is used on the project, for what, by whom, and with which verification method; **verification method** means one of the named methods in element 16, not a general instruction to review.

#### 5. Required actions — process requirements.

- **PCI-PML-STD-14.02-PR-01 — An AI use record exists.** The credential holder must maintain a dated record of every AI use on the project: the system, the output class, the person accountable for it, and the verification method applied. Embedded features in delivery tools must be included; an unlisted embedded feature is the most common gap.
- **PCI-PML-STD-14.02-PR-02 — A named method per output class, recorded before use.** Each output class must have a verification method recorded before the output is relied on, chosen from: independent recomputation, source tracing, clause-to-summary comparison, sampling with a stated basis, reconciliation, boundary testing, sensitivity analysis, expert judgement, or named approval. **"Review the AI output" is not a method** and does not satisfy this requirement.
- **PCI-PML-STD-14.02-PR-03 — Material AI assistance is disclosed.** Where AI assistance is material to a deliverable or a report, the credential holder must disclose it in the artefact, under **PCI-FND-STD-14**, in the form the organisation's governance sets.
- **PCI-PML-STD-14.02-PR-04 — Verification proportionate to reliance.** The depth of verification must be matched to the consequence of the output being wrong, and the credential holder must record the basis on which the depth was chosen for each output class.
- **PCI-PML-STD-14.02-PR-05 — Failures are recorded and fed back.** Where verification finds a material error in an AI output, the credential holder must record it, correct anything already issued in reliance on it under **PCI-PML-STD-11.01-PR-04**, and route the pattern to **PCI-PML-STD-09.02**.

**6. Prohibited actions.** Relying on an output because it is plausible and well-formed. Recording "AI reviewed" as a verification. Presenting AI-generated analysis as a named person's professional judgement. Using an AI system for

an output class with no recorded method. Treating a supplier's assurance about its own model as verification of the output. Suppressing a verification failure because the output was already issued.

**7. Required evidence.** The AI use record with output classes, accountable people and methods; the verification records per output class showing the method applied and by whom; disclosure statements in artefacts; the recorded basis for verification depth; the verification-failure log with corrections issued and the routing to lessons.

**8. Responsible role.** The named credential holder leading the project, for the AI use record and for the sufficiency of methods. The named person accountable for each output class, for performing the verification. Neither may be a supplier by default; where a supplier verifies, the credential holder records who in the supplier is accountable and retains the verification evidence.

**9. Approval authority.** The sponsor approves the AI use record and the verification methods. The governing body approves any use of AI in a reserved decision class under [PCI-PML-STD-01.02](#). A tool vendor approves nothing.

**10. Independence requirement.** The person verifying an AI output must be independent of the configuration of the system that produced it, in the sense that they did not set the parameters, prompts, weights or thresholds for that output. Where that is impossible on a small project, the arrangement must be recorded and a second named person must perform the verification for the highest-consequence output class. **Independence of configuration is the minimum this standard adds; it does not displace** [PCI-FND-STD-03](#), which requires a *material* calculation, model output or automated conclusion to be verified by a person independent of its preparation before any person relies on it. Where the verifier under this standard is also the preparer of the output, this standard is satisfied and the foundational verification is not.

**11. Materiality or threshold.** This standard states no accuracy figure and no confidence threshold. **A model's own confidence score is not a verification and must not be used as one.** The organisation's governance sets the output classes, the materiality at which AI assistance is disclosed, and the verification depth by consequence; this standard requires that these exist and are applied. *Six-person internal project:* three AI uses — drafting, summarising and a scheduling assistant — one line each in the record, with source tracing for figures, clause-to-summary comparison for summaries and named approval for anything issued. *Multi-partner national programme:* the record is consolidated across partners because a partner's embedded tool produces outputs that enter the programme's reports; verification methods are stated in the partner agreement; and the highest-consequence classes carry independent recomputation performed outside the partner that produced the output.

**12. Exception and waiver.** An exception permitting reliance on an unverified output may be approved only by the sponsor, only where the decision

cannot wait, only where the artefact states on its face that the output is unverified, and only with a named owner and a date for the verification to follow. **No exception is permitted** where the output bears on safety, a licence condition, a statutory duty or a person's rights.

**13. Escalation trigger.** An AI output relied on with no recorded method. A material verification failure. An embedded feature discovered producing outputs into project artefacts. A supplier declining to disclose whether AI produced a deliverable. Any proposal to use AI in a reserved decision class. An output that cannot be verified by any available method — which is a reason not to use it.

**14. AI application.** AI may assist across delivery — drafting, summarising, reconciling, classifying, detecting anomalies, modelling scenarios, checking documents for completeness — and every such use is permitted, recorded and verified rather than prohibited. AI may also maintain the AI use record itself and flag outputs relied on with no verification record.

**15. AI prohibition.** An AI system must not decide, approve, certify, sign, waive, authorise, accept, assure or be recorded as verifying anything. It must not verify another AI system's output for the purposes of this standard. It must not be represented as having independently verified anything, and no supplier statement converts its output into verified information.

**16. AI verification.** The method is named per output class, from this list, and recorded before use: independent recomputation; source tracing; clause-to-summary comparison; sampling with a stated basis; reconciliation; boundary testing; sensitivity analysis; expert judgement; named approval. For each class, the record states the method, who performs it, the sample size where sampling is used, and the tolerance at which a difference becomes a failure. Verification is performed **before** reliance, not after issue.

#### 17. External reference.

- **ISO/IEC** · *ISO/IEC 42001, Artificial intelligence management system* · relied on for: the existence of a management-system standard for AI governance, including assignment of roles · **EXT-021** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: certifiable as a management system, but adoption is voluntary; certification says nothing about any individual output; no requirement is imported here.
- **ISO/IEC** · *ISO/IEC 23894, Guidance on risk management for artificial intelligence* · relied on for: the existence of AI-specific risk guidance · **EXT-024** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: **guidance, not requirements**; voluntary; not certifiable.
- **ISO/IEC** · *ISO/IEC 38507, Governance implications of the use of artificial intelligence by organizations* · relied on for: the existence of governing-

body-level questions about AI use · **EXT-037** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: guidance for governing bodies; voluntary; not certifiable.

- **NIST (US Department of Commerce)** · *Artificial Intelligence Risk Management Framework (AI RMF 1.0)* · relied on for: the existence of a voluntary function-based framework for AI risk · **EXT-080** · **Manual §6 category 10 — illustrative practice** · currency checked 2026-08-03 · limitation: **voluntary, rights-preserving and non-sector-specific by NIST's own description; not a standard and not a regulation.**
- **OECD** · *Recommendation of the Council on Artificial Intelligence (the OECD AI Principles)* · relied on for: the existence of an international expectation of human accountability for AI-influenced outcomes · **EXT-081** · **Manual §6 category 10 — illustrative practice** · currency checked 2026-08-03 · limitation: a Council Recommendation; **never legislation**; non-binding even on adherents.
- **European Union** · *Regulation (EU) 2024/1689 (the AI Act)* · relied on for: illustrating a risk-tiered regulatory approach · **EXT-100** · **Manual §6 category 1 — applicable legislation or regulation** · currency checked 2026-08-03 · limitation: **binding legislation within the European Union only**, named here to illustrate the shape only; relied on for no requirement; applicability to any given project is a question for qualified local counsel.

**18. Jurisdictional caution.** Obligations on AI transparency, human oversight, impact assessment, automated decision-making, intellectual property in generated output and liability for AI-influenced harm are jurisdiction-specific and are changing quickly. Obtain local legal advice before deploying AI in a process affecting people's rights, employment, safety, or access to a service.

**19. Related PCI Standards.** **PCI-FND-STD-03** (the parent verification obligation); **PCI-FND-STD-14**; **PCI-FND-STD-04**; **PCI-FND-STD-01**; **PCI-PML-STD-01.02**; **PCI-PML-STD-11.01**; **PCI-PML-STD-14.01**. **What this standard adds to PCI-FND-STD-03:** the foundational standard requires independent verification before reliance. This standard requires an **AI use record covering embedded features, a named method per output class recorded before use, depth matched to consequence with the basis recorded, and a failure log routed into the organisation's standing artefacts.**

**20. Related Body of Knowledge content.** PML-AI · Domain 14 · KA 14.2 Dashboards, analytics, digital twins and automation; KA 14.3 AI use across the lifecycle, prompting and verification; KA 14.4 Explainability, bias, human



accountability, cybersecurity and privacy. Also Domain 1 KA 1.4 the leader's AI accountability; Domain 9 KA 9.4 AI-output quality.

**21. Compliance test.** Compliance is demonstrated when a reviewer can: (a) list the AI systems and embedded AI features actually in use on the project, by inspecting the tools rather than by asking, and match each to an entry in the AI use record; (b) for each output class, retrieve the recorded verification method and confirm it is one of the named methods with its parameters stated; (c) draw a stated sample of outputs relied on in the period and find, for each, a verification record naming the method, the person and the date, **dated before** the date of reliance; (d) confirm that every artefact with material AI assistance carries the disclosure; and (e) confirm every recorded verification failure produced a correction to anything already issued. An AI system in use and absent from the record fails the test, and an output relied on before its verification date fails it.

**22. Breach indicators.** An AI use record listing only the tools procured centrally. Verification fields containing "reviewed". Verification dates after issue dates. Identical verification records across dissimilar output classes. A supplier deliverable whose style changes abruptly with no disclosure. A model confidence score cited as the verification. Verification failures with no corrections.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure on the associated competency; ethics review; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** AI-verification case: a generated risk register of forty entries is offered for the gate pack, and the candidate states the method, the sample and what is done with the entries that fail. Ethical dilemma: an output cannot be verified by any available method and the decision is due. Evidence selection: which artefacts establish that verification preceded reliance.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-14-02](#) v1.0. Amendment note: renumbered and restructured; legislative drafting removed; the AI use record, the named-method rule, proportionate depth and the failure log separated into five process requirements; element 16 rewritten to name the permitted methods explicitly, because "review the AI output" was the defect the previous edition shared with most of the corpus. **Stage 9 amendment:** element 10 required independence only of the system's configuration, which allowed the preparer of a material output to be its verifier; element 10 now states that this is the minimum this standard adds and does not displace the independent-person verification [PCI-FND-STD-03](#) requires before reliance.

## Domain 15 — Programmes, Portfolios and Enterprise Delivery

### PCI STANDARD

#### PCI-PML-STD-15.01

### Programme Integration and Dependency Ownership

**1. Normative requirement.** A credential holder leading or governing a programme must ensure that every dependency between components is recorded with a named giver, a named receiver, the thing to be given, the date, and the consequence of a breach.

**2. Purpose.** Programmes fail at their joints. A dependency with no named owner on the giving side is not a dependency but a hope, and a milestone that requires six of them holds only as their joint probability, which is a great deal lower than any of them individually and lower than every dashboard suggests. The failure this prevents is the programme whose components are each green and whose integrated milestone is not achievable.

**3. Scope.** Every credential holder leading, integrating, governing, reporting on or assuring a programme or a multi-component delivery, including dependencies on parties outside the programme and dependencies between the programme and business-as-usual operations.

**4. Defined terms.** *dependency · decision owner · material · evidence · escalation threshold · baseline* (control sense). Additionally, **giver** means the named individual accountable for supplying the dependency; **integrated milestone** means a milestone whose achievement requires more than one component's contribution.

#### 5. Required actions — process requirements.

- **PCI-PML-STD-15.01-PR-01 — Named on both sides.** Every dependency must name an individual on the giving side and an individual on the receiving side, and the giver must have confirmed the commitment. A dependency naming organisations rather than people does not satisfy this requirement.
- **PCI-PML-STD-15.01-PR-02 — Integrated milestone confidence is computed as a conjunction.** Where an integrated milestone depends on several dependencies, its confidence must be reported as the joint probability of the dependencies holding, not as an average, a weighted average or a proportion of green items — and the correlation assumption used must be stated.
- **PCI-PML-STD-15.01-PR-03 — Decoupling is considered before improvement.** Where an integrated milestone's confidence is below the



tolerance the governance sets, the credential holder must record whether a dependency can be removed or deferred, because removing one dependency changes the arithmetic by more than improving any of them.

- **PCI-PML-STD-15.01-PR-04 — Breach of a dependency is an issue on the day it is known.** A dependency that will not be met must be raised as an issue under **PCI-PML-STD-08.02** on the day the giver knows, not on the date it was due.

**6. Prohibited actions.** Recording a dependency with an organisation as giver. Reporting an integrated milestone's confidence as an average of its dependencies. Accepting a receiver-side entry with no giver confirmation. Rolling up component RAG ratings into a programme rating with no stated rule. Reporting a dependency as met when a partial or conditional delivery was made. Holding a known dependency breach until the due date.

**7. Required evidence.** The dependency register with giver, receiver, thing, date, confidence and breach consequence per entry, and the giver's confirmation; the integrated milestone confidence computation with its stated correlation assumption; the decoupling analysis where tolerance was breached; the issue records for dependency breaches with the date known and the date raised.

**8. Responsible role.** The named credential holder leading the programme, for the register and the integrated computation. The named giver, for the commitment. The named receiver, for raising a breach.

**9. Approval authority.** The programme's governing body approves the dependency tolerance and any decoupling that changes a component's scope. A component must not release itself from a dependency.

**10. Independence requirement.** At each programme gate at which the delegation schedule requires assurance, the dependency register and the integrated confidence computation must be reviewed by a competent reviewer independent of every component, not merely of one — because a reviewer inside one component has an interest in where the dependency risk is recorded.

**11. Materiality or threshold.** This standard states no confidence figure. The organisation's governance sets the confidence tolerance for an integrated milestone, the escalation threshold for a dependency at risk and the reporting cadence; this standard requires that they are documented and applied, and that the **arithmetic is the conjunction** rather than an average — which is a method requirement, not a threshold. *Six-person internal project:* the standard applies only where the project is a component of a programme, and then its obligation is to name a giver and a receiver for each of its handful of external dependencies and to raise a breach the day it is known. *Multi-partner national programme:* the register is held at programme level with per-partner extracts,

givers are named individuals inside partner organisations with the commitment reflected in the partner agreement, and the integrated confidence is computed and published at programme level so that no partner computes its own.

**12. Exception and waiver.** An exception permitting an integrated milestone to be reported on a basis other than the conjunction may be approved only by the programme's governing body, only where the alternative basis and its assumption are stated on the face of the report, and only for a stated period. No exception permits a dependency with no named giver.

**13. Escalation trigger.** A dependency with no giver confirmation. An integrated milestone whose computed confidence is below tolerance. A dependency the giver states will not be met. A component proposing to change a date on which another component depends. A correlation between components' dependencies that the stated assumption does not cover.

**14. AI application.** AI may build and reconcile a dependency register from component schedules, detect dependencies present in one component's plan and absent from another's, compute integrated confidences and their sensitivities, identify which single dependency change most improves a milestone, and detect dependency entries that have not been confirmed by their giver.

**15. AI prohibition.** An AI system must not commit a dependency on behalf of a giver, decide that a dependency is met, set a confidence value that is reported without a named human's derivation, or approve a decoupling.

**16. AI verification. Independent recomputation plus named approval.** Every integrated confidence computed with AI assistance must be recomputed by hand before it is reported — it is a product of a handful of numbers and taking it on trust is indefensible — and every AI-proposed dependency must be confirmed by the named giver before it enters the register as a commitment.

#### **17. External reference.**

- **ISO** · *ISO 21503, Guidance on programme management* · relied on for: the existence of programme-level integration and dependency management · **EXT-030, not independently verified — verify current requirements** · **Manual §6 category 3 — international voluntary standard** · limitation: guidance; voluntary; not certifiable; open verification status.
- **ISO** · *ISO 21505, Guidance on governance* · relied on for: the existence of governance across multiple components · **EXT-032, not independently verified** · **Manual §6 category 3** · limitation: as above.
- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of interface and dependency management within delivery ·

**EXT-028 · Manual §6 category 3 · currency checked 2026-08-03 ·**  
 limitation: guidance; voluntary; not certifiable.

**18. Jurisdictional caution.** Where a dependency crosses a contractual boundary, whether it is enforceable, what notice is required and what remedy exists are contractual questions determined by the governing law. Obtain legal advice before relying on a dependency register in a commercial position.

**19. Related PCI Standards.** PCI-FND-STD-05; PCI-FND-STD-11; PCI-FND-STD-12; PCI-PML-STD-06.01; PCI-PML-STD-08.01; PCI-PML-STD-08.02; PCI-PML-STD-11.01; PCI-PML-STD-15.02.

**20. Related Body of Knowledge content.** PML-AI · Domain 15 · KA 15.1 Programme architecture and dependency management, including dependency arithmetic at programme scale and decoupling as the primary response. Also Domain 2 KA 2.3 topic 2.3.4 assumption and dependency management; Domain 6 KA 6.1 logic networks.

**21. Compliance test.** Compliance is demonstrated when a reviewer can: (a) take every entry in the dependency register and find a named individual on each side and a recorded confirmation from the giver; (b) take a stated integrated milestone, extract its dependencies' confidences and recompute the joint probability by hand, and reconcile it to the reported figure; (c) confirm the correlation assumption is stated on the report; (d) find, for every milestone below tolerance, a recorded decoupling analysis; and (e) for a stated sample of breached dependencies, compare the date the giver knew with the date the issue was raised. A reported integrated confidence that is an average rather than a product fails the test.

**22. Breach indicators.** Dependency registers whose giver column contains organisation names. Integrated milestones reported at the average of their dependencies. Programme ratings greener than every component computation supports. Dependency breaches raised on the due date. Registers where receivers created every entry and no giver confirmed one. The same dependency recorded differently in two components' plans.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Calculation review: six dependencies at stated confidences, and the candidate computes the milestone's joint probability and compares it with the dashboard's average. Scenario judgement: a component offers to improve one dependency from 0.85 to 0.92 and the candidate

evaluates it against removing a different dependency. Evidence selection: what makes a dependency register a control.

**25. Version and status.** Version 1.0 · **not yet approved** · effective on approval · **new standard** — the v1.0 set reached Domain 15 only through *Related book content* fields and therefore created no programme-level obligation. Amendment note: none.

## PCI STANDARD

### PCI-PML-STD-15.02

## Portfolio Prioritisation and Capacity Truth

**1. Normative requirement.** A credential holder advising or deciding a portfolio's composition must not allow the portfolio to hold more concurrent work than the organisation's assessed delivery capacity supports.

**2. Purpose.** A portfolio that is over-committed does not deliver less; it delivers everything later, and the delay is invisible because each component reports its own slippage as a local problem. The failure this prevents is the standing organisational condition in which every project is late for the same reason and no project can see it.

**3. Scope.** Every credential holder preparing, recommending, approving, reporting on or assuring a portfolio's composition, prioritisation, capacity plan or intake, including enterprise portfolio management offices and including intake outside the formal portfolio process.

**4. Defined terms.** *material · decision owner · benefit · evidence · independent · dependency · sponsor*. Additionally, **assessed capacity** means the organisation's recorded judgement of how much concurrent work each constrained resource pool can carry; **intake** means the admission of new work into the portfolio.

### 5. Required actions — process requirements.

- **PCI-PML-STD-15.02-PR-01 — Capacity is assessed and recorded per constrained pool.** The credential holder must hold a dated assessment of capacity for each constrained resource pool the portfolio depends on, with the basis of the assessment stated.
- **PCI-PML-STD-15.02-PR-02 — Intake is tested against capacity before approval.** No component may be approved into the portfolio without a recorded test of its demand against assessed capacity, and the result of that test must be presented to the approving authority.
- **PCI-PML-STD-15.02-PR-03 — Over-commitment is reported, not distributed.** Where demand exceeds assessed capacity, the credential holder must report the over-commitment and the components affected,

and must not resolve it by reducing every component's assumption proportionally.

- **PCI-PML-STD-15.02-PR-04 — Prioritisation criteria are published and applied.** The criteria by which components are ranked must be documented and published to the sponsors whose components are ranked, and the ranking must follow them; any departure must be recorded with its reason and its author.

**6. Prohibited actions.** Approving intake with no capacity test. Resolving over-commitment by assuming higher productivity. Ranking components against undisclosed criteria. Admitting work outside the portfolio process and reporting portfolio capacity as though it did not exist. Presenting a capacity plan whose protective capacity has been removed without stating that it was.

**7. Required evidence.** The dated capacity assessment per constrained pool with its basis; the intake test per approved component with its result; the over-commitment reports with affected components; the published prioritisation criteria and the ranking that applied them; the record of any departure from the criteria.

**8. Responsible role.** The named credential holder accountable for the portfolio process — typically the head of the portfolio or enterprise delivery function. The portfolio's governing body decides composition. Individual sponsors are not accountable for the portfolio's aggregate position.

**9. Approval authority.** The portfolio's governing body approves intake, composition and the capacity assessment. A component's sponsor must not approve their own component's admission. A departure from the published criteria may be approved only by the governing body, with the reason recorded.

**10. Independence requirement.** The capacity assessment and the intake test must be prepared by a function independent of any single competing component, and the credential holder performing them must not be the sponsor or leader of a component being ranked. Where the same person unavoidably holds both roles, they must abstain from the ranking of their own component under **PCI-PML-STD-01.03-PR-04**.

**11. Materiality or threshold.** This standard states no utilisation figure and no work-in-progress limit, because both are organisation-specific and a borrowed number is worse than none. The organisation's governance sets the capacity basis, the intake threshold and the concurrency limits it applies; this standard requires that these are documented, that the assessment is dated, and that intake is tested against them. Where protective capacity is held, its size and its basis must be recorded so that removing it is a visible decision. *Six-person internal project:* the standard applies to the organisation that funds it rather than to the project; for a small organisation the capacity assessment

may be a single table of six people and their committed availability, which is enough to make an intake test real. *Multi-partner national programme*: capacity is assessed per pool per organisation and consolidated, the intake test states which pools are constrained, and prioritisation criteria are published to every sponsor including those in partner organisations — because criteria disclosed only inside the client organisation produce exactly the disputes the criteria exist to prevent.

**12. Exception and waiver.** An exception permitting intake above assessed capacity may be approved only by the portfolio's governing body, only where the components that will be delayed are identified by name in the approval, and only where their sponsors are informed. **An over-commitment approved without naming who is delayed is not an exception; it is an undisclosed decision.**

**13. Escalation trigger.** Demand exceeding assessed capacity on any constrained pool. Intake approved with no capacity test. A ranking that departs from the published criteria with no recorded reason. Work admitted outside the portfolio process. A capacity assessment older than the period the governance sets.

**14. AI application.** AI may aggregate demand across components, model portfolio composition under a capacity constraint, compute the deferral cost of excess concurrency, test a ranking against the published criteria and flag departures, and detect work in delivery systems that has no portfolio record.

**15. AI prohibition.** An AI system must not decide portfolio composition, rank components as the operative decision, approve intake, or set the capacity assessment. An optimiser's allocation is a recommendation, and composition is a reserved decision class under [PCI-PML-STD-01.02-PR-01](#).

**16. AI verification. Independent recomputation plus sensitivity analysis.** Any AI-produced portfolio allocation must be recomputed against the recorded capacity assessment by a competent reviewer, and the reviewer must vary the two most influential capacity assumptions and record whether the recommended composition survives — because a composition that is optimal only at the assessment's point estimate is not a plan.

#### **17. External reference.**

- **ISO** · *ISO 21504, Guidance on portfolio management* · relied on for: the existence of portfolio balancing against capacity and strategy · **EXT-031, not independently verified — verify current requirements** · **Manual §6 category 3 — international voluntary standard** · limitation: guidance; voluntary; not certifiable; open verification status.

- **ISO · ISO 21503, *Guidance on programme management*** · relied on for: the existence of a programme layer between portfolio and project · **EXT-030, not independently verified · Manual §6 category 3** · limitation: as above.
- **ISO · ISO 21500, *Context and concepts*** · relied on for: the vocabulary distinguishing project, programme and portfolio · **EXT-027 · Manual §6 category 3** · currency checked 2026-08-03 · limitation: **context and concepts only** since its current edition; guidance moved to ISO 21502.

**18. Jurisdictional caution.** Portfolio decisions that change employment, close operations, or alter a regulated service can engage employment consultation, regulatory notification and public-law duties. Obtain local legal advice before a portfolio decision that stops work with those consequences.

**19. Related PCI Standards.** PCI-FND-STD-02; PCI-FND-STD-05; PCI-FND-STD-11; PCI-PML-STD-02.01; PCI-PML-STD-02.02; PCI-PML-STD-07.02; PCI-PML-STD-15.01; PCI-PML-STD-16.03.

**20. Related Body of Knowledge content.** PML-AI · Domain 15 · KA 15.2 Benefits and portfolio balancing; KA 15.3 Capacity and enterprise PMOs; KA 15.4 Transformation governance and strategic reporting. Also Domain 2 KA 2.2 topic 2.2.3 selection and prioritisation models; Domain 13 KA 13.2 flow.

**21. Compliance test.** Compliance is demonstrated when a reviewer can: (a) retrieve a dated capacity assessment for every constrained pool the portfolio depends on; (b) take every component admitted in the period and find a recorded intake test with its result presented to the approving authority; (c) sum the portfolio's current demand on each constrained pool and compare it with the assessed capacity, reproducing the over-commitment position the portfolio reported; (d) retrieve the published prioritisation criteria, re-rank a stated sample of components against them, and account for every difference from the published ranking by a recorded departure with a reason and an author; and (e) reconcile the portfolio record against the delivery systems and find no material work in progress with no portfolio entry. An intake approved with no capacity test fails the test.

**22. Breach indicators.** Capacity assessments older than the last reorganisation. Intake approvals with no test attached. Portfolio dashboards showing full capacity utilisation and no deferrals. Rankings that correlate with sponsor seniority rather than with the published criteria. Work in delivery systems absent from the portfolio. Protective capacity removed between versions with no decision record.



**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Calculation review: aggregate demand against a constrained pool, and the deferral the excess produces. Scenario judgement: a new component is directed into the portfolio by an executive with no capacity test. Ethical dilemma: a ranking is adjusted after publication to move one sponsor's component up.

**25. Version and status.** Version 1.0 · **not yet approved** · effective on approval · **new standard** — the v1.0 set contained no portfolio obligation. Amendment note: none.

## Domain 16 — Transition, Closeout and Benefits Realisation

### PCI STANDARD

#### PCI-PML-STD-16.01

### Transition Readiness and the Gate Block

**1. Normative requirement.** A credential holder must not permit a transition, go-live or release into operational use to proceed while any mandatory precondition for it is recorded as not met.

**2. Purpose.** Transition is the decision that is irreversible for the people who did not take it. Some of its conditions are not assessments of likelihood at all — a safety case closed by the authority that owns it, a licence or regulatory approval granted, a privacy or data-protection assessment signed, statutory notifications made, takeover certificates issued. The failure this prevents is the one a single readiness template invites: **every condition entered as a probability**, at which point a forbidden thing has been silently converted into a chance that it is permissible, and the resulting number can be raised by improving something else entirely.

**3. Scope.** Every credential holder preparing, assessing, recommending, deciding or assuring a transition, go-live, commissioning, handover or operational release, in every delivery model and at every scale, including phased and pilot releases.

**4. Defined terms.** *mandatory precondition · gate block · discretionary condition · acceptance · evidence · decision owner · independent · material.* Additionally, **readiness assessment** means the record of whether the conditions for



transition are satisfied; **go/hold economics** means the comparison of the cost of proceeding against the cost of waiting.

#### 5. Required actions — process requirements.

- **PCI-PML-STD-16.01-PR-01 — Two blocks, separated before assessment.** The readiness assessment must separate a gate block of mandatory preconditions from a block of discretionary conditions, and the separation must be made and approved **before** the assessment is populated. A single undifferentiated condition list does not satisfy this requirement.
- **PCI-PML-STD-16.01-PR-02 — Gate-block items are binary, authority-named and dated.** Each mandatory precondition must be recorded met or not met, with the approving authority named and the date given, and must be closed by the authority that owns it rather than by the project.
- **PCI-PML-STD-16.01-PR-03 — No probability, no weight, no score on a gate-block item.** A mandatory precondition must not be expressed as a probability, a percentage, a confidence, a weighting, a red-amber-green rating or any other quantity admitting degree, and must not appear among the discretionary conditions' probabilities or in any aggregate computed from them.
- **PCI-PML-STD-16.01-PR-04 — No gate-block item in the go/hold economics.** A mandatory precondition must not be entered into the go/hold economics, priced against the cost of delay, or traded against any other condition. **Expressing it as a cost concedes that some cost of delay would be large enough to buy it, and that is precisely what the item exists to forbid.**
- **PCI-PML-STD-16.01-PR-05 — Discretionary readiness is a conjunction, and its assumption is stated.** The probability of a clean transition across the discretionary conditions must be reported as their joint probability, not as an average, a weighted average or the proportion of conditions rated green, and the correlation assumption used must be stated on the face of the assessment.

**6. Prohibited actions.** Entering a safety case, a licence approval, a privacy assessment, a statutory notification or a takeover certificate as a probability, a percentage or a rating. Including a mandatory precondition in a weighted readiness score. Pricing a mandatory precondition against the cost of delay. Producing an option set while the gate block is open. The project recording a gate-block item as met on its own authority. Reporting a readiness average and describing it as the probability of a clean transition.

**7. Required evidence.** The readiness assessment in two blocks, with the separation approved and dated before population; the gate-block record with each item's met/not-met status, approving authority and date, and the

underlying certificate, approval, signature or notification; the discretionary block with its probabilities, its joint computation and its stated correlation assumption; the go/hold economics showing only discretionary conditions; the transition decision with its named decision owner.

**8. Responsible role.** The named credential holder leading the transition, for producing the assessment in the required form and for refusing to proceed while the gate block is open. The named transition decision owner, for the decision. The **authority that owns each mandatory precondition**, for closing it.

**9. Approval authority.** The transition decision owner the delegation schedule names decides, and may decide only *hold* while any gate-block item is not met. Each mandatory precondition is closed by its owning authority — clinical or safety governance, the regulator or licensing body, the accountable privacy authority, the notifying authority, the certifying party — and **by nobody else**.

**10. Independence requirement.** Every mandatory precondition must be closed by an authority independent of the project in the sense defined above: not the preparer of the thing being approved, not accountable for the delivery date, and holding no interest in the transition proceeding. The readiness assessment itself must be reviewed at the transition gate by a competent reviewer independent of the delivery organisation, who must satisfy [PCI-PML-STD-01.03-PR-06](#).

**11. Materiality or threshold. This standard states no readiness percentage, and any such figure would be the defect it exists to prevent.** The gate block is binary and admits no threshold at all. For the discretionary block, the organisation's governance sets the confidence tolerance at which transition may be recommended and the escalation route below it; this standard requires that the tolerance is documented, that the computation is the conjunction, and that the assumption is stated. Which items belong in the gate block is set by law, licence, contract and the organisation's governance — not by this standard and not by the project — and the credential holder's obligation is to obtain that list from the owning authorities and record it. *Six-person internal project*: the gate block may contain a single item — an information-security sign-off, say — recorded met with the name of the person who signed it and the date; the discretionary block may hold four conditions whose product is computed in one line. The obligation costs almost nothing and removes the failure entirely. *Multi-partner national programme*: the gate block is assembled from every jurisdiction, regulator and operating organisation the transition touches, each item names its own owning authority, and no partner may close another's item. The discretionary conjunction is computed at the level at which the transition actually happens — per site, per region — rather than nationally, because a national average conceals the site that is not ready.

**12. Exception and waiver. No exception, waiver, dispensation or deferral is permitted** in respect of a mandatory precondition. There is no authority within PCI's system that may approve a transition while a gate-block item is not met, and a credential holder must not seek, recommend or record one. Where the organisation believes an item has been wrongly classified as mandatory, the remedy is to have the **owning authority** reclassify it, in writing, before the assessment — not to trade it at the gate.

**The emergency case, stated so that the standard survives it.** Where continuing without the transition is itself the greater danger — a failed system that cannot be restored, an asset that must be taken out of service — the answer is still not a trade at the gate. It is that **the authority that owns the item exercises its own emergency instrument**: a regulator's derogation, a safety authority's temporary approval, a licensing body's dispensation, an interim certificate. An item closed or varied through the owning authority's own emergency process is a **met** item, recorded under **PR-02** with that authority's instrument, its scope and its expiry, and reported at the next governance meeting. What a credential holder must not do is seek such an instrument from anyone other than the owning authority, treat its absence as a probability, or permit the transition while the item stands not met. Where the owning authority declines or cannot be reached, the decision remains **hold**, and the consequences of holding are escalated under **PCI-FND-STD-11** rather than resolved at the gate.

**13. Escalation trigger.** Any gate-block item not met as the transition date approaches. Any proposal to express a mandatory precondition as a probability, a rating or a cost. Any request for a dispensation against a gate-block item. Discovery that a gate-block item was closed by the project rather than by its owning authority. A discretionary conjunction below the documented tolerance. A readiness dashboard reporting an average.

**14. AI application.** AI may assemble the readiness assessment from source records, check that every gate-block item carries a named authority and a date, flag any gate-block item that has acquired a numeric value, compute the discretionary conjunction and its sensitivities, and identify which discretionary condition most improves the joint probability.

**15. AI prohibition.** An AI system must not decide readiness, close a mandatory precondition, assign a probability to one, recommend transition, or produce a readiness figure that is reported without a named human's derivation. **A model asked to score readiness across an undifferentiated condition list will produce exactly the defect this standard prohibits**, and the credential holder must not use one in that form.

**16. AI verification. Independent recomputation plus boundary testing.** The discretionary conjunction must be recomputed by hand before the transition decision and reconciled to the reported figure. The credential holder must

additionally test the assessment's structure by boundary case: set one gate-block item to not met and confirm that the assessment's output is *hold* and that no option set or economic comparison is produced. An assessment that returns a number in that test is structurally non-compliant and must be corrected before use.

#### 17. External reference.

- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of transition and handover as lifecycle activities · **EXT-028** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.
- **ISO** · *ISO 9001, Quality management systems — Requirements* · relied on for: the existence of a certifiable requirement that outputs are verified before release · **EXT-033** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: voluntary unless imported; certification concerns a management system; no requirement is imported here.
- **ISO** · *ISO 10006, Guidelines for quality management in projects* · relied on for: the existence of project-specific quality guidance covering closure · **EXT-035** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: **guidelines, not requirements**; voluntary; not certifiable.
- **ISO** · *ISO 45001, Occupational health and safety management systems — Requirements* · relied on for: the existence of a certifiable management-system standard within which operational readiness for safety is addressed · **EXT-123** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: certifiable, but voluntary unless required; **it does not determine which items are mandatory preconditions for any given transition** — law, licence and contract do.

**18. Jurisdictional caution.** Which approvals, certificates, notifications and assessments are legally required before an asset, system or service enters use — and which body may grant each — are determined by statute, licence conditions, sector regulation and contract, and they differ by jurisdiction and by sector. **This standard does not state which items are mandatory for any project, and a professional must not infer the list from the examples given.** Obtain local legal and regulatory advice to establish the gate block for the specific transition.

**19. Related PCI Standards.** [PCI-FND-STD-02](#); [PCI-FND-STD-11](#); [PCI-FND-STD-13](#); [PCI-PML-STD-03.03](#); [PCI-PML-STD-09.01](#); [PCI-PML-STD-15.01](#); [PCI-PML-STD-16.02](#).

**20. Related Body of Knowledge content.** PML-AI · Domain 16 · KA 16.1 Handover, commissioning and readiness — the two-block readiness model, the gate block of mandatory preconditions, the conjunction across discretionary conditions and its contrast with the averaged dashboard. Also Domain 14 KA 14.4, which states the same boundary for security and privacy controls; Domain 9, which states it for quality, where the economic optimum is taken among compliant options rather than across them.

**21. Compliance test.** Compliance is demonstrated when a reviewer can: (a) obtain the readiness assessment and confirm it is in two blocks, with the separation approved and dated before the assessment was populated; (b) confirm that **every** gate-block item carries a met/not-met value, a named approving authority and a date, and carries **no** numeric value of any kind; (c) confirm that no gate-block item appears in the discretionary probability list, in any weighted score, or in the go/hold economics; (d) recompute the discretionary joint probability by hand from the stated conditions and reconcile it to the reported figure, and find the correlation assumption stated; and (e) for any period in which a gate-block item was not met, confirm that the recorded decision was **hold** and that no option set or economic comparison was produced. **A probability, percentage, weighting or rating attached to any mandatory precondition fails this test outright**, whatever the transition's outcome, and so does an option set produced while the gate block was open.

**22. Breach indicators.** A single readiness template with one probability column. A safety case shown at "95 per cent". A dashboard reporting readiness as a weighted average. A cost-of-delay analysis that includes a licence approval among its variables. A gate-block item signed by the project's own transition manager. A readiness figure that improved without any condition changing. Requests for a "dispensation" against a safety or licence item.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure on the associated competency; ethics review; certification investigation; suspension or withdrawal of the PML-AI credential — each subject to due process and a right of appeal. **PCI cannot authorise a transition, grant an approval, or override any regulator or safety authority**, and no PCI process substitutes for the approvals this standard requires to be obtained from their owning authorities.

**24. Examination application.** Calculation review: seven discretionary conditions whose product is computed and compared with the dashboard's average, with the candidate stating why the two differ and in which direction. Scenario judgement: a safety case is not yet closed, the cost of delay is large, and the candidate states the option set — which is *hold*, and there is no option set below the gate block. Ethical dilemma: an executive asks for the safety case to be shown as a probability so that the readiness figure "reflects reality".

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-16-01](#) v1.0. Amendment note: renumbered, restructured and **substantially strengthened** to match the current Domain 16 readiness model. The v1.0 standard required readiness to be assessed and evidenced but did not distinguish mandatory preconditions from discretionary conditions, and therefore did not prohibit the central failure — a forbidden item entered as a probability. The two-block separation, the binary authority-named recording, the prohibitions on probability and on economic trade, and the conjunction rule are new, and element 12 now states expressly that no exception exists. **Stage 9 amendment:** the red-team's emergency case — the transition that must happen because continuing is the greater danger — had no compliant route, and a standard with no route in the case that will arise is a standard that gets broken; element 12 now states the route without conceding the duty, by directing the credential holder to the owning authority's own derogation or temporary-approval instrument, requiring it to be recorded under [PR-02](#) with its scope and expiry, and leaving *hold* as the decision where that authority declines or cannot be reached.

#### PCI STANDARD

#### PCI-PML-STD-16.02

### Operational Acceptance and Handover

**1. Normative requirement.** A credential holder must not close a project until the receiving organisation has recorded, by a named individual, that it accepts operational responsibility for what has been transferred.

**2. Purpose.** Projects end on their own schedule; operations begin on theirs. The failure this prevents is the closure that transfers an asset, a system or a service to an organisation that has not agreed to take it, has not been given what it needs to run it, and discovers both facts after the project team has dispersed — at which point reconstructing what the team knew costs several times what recording it would have cost.

**3. Scope.** Every credential holder closing a project, phase, contract or component that transfers something to be operated, maintained, supported or further developed by another organisation or team, including transfers to a supplier's operations, to business-as-usual, and to a successor project.

**4. Defined terms.** *acceptance · evidence · decision owner · material · independent · mandatory precondition.* Additionally, **receiving organisation** means the team or organisation that will operate, maintain or support what is transferred; **operational responsibility** means accountability for running it, including for its faults.

## 5. Required actions — process requirements.

- **PCI-PML-STD-16.02-PR-01 — A named acceptance from the receiver.** Operational acceptance must be recorded by a named individual in the receiving organisation, with the date, and must state what is accepted. Acceptance by the project on the receiver's behalf does not satisfy this requirement.
- **PCI-PML-STD-16.02-PR-02 — The handover set is defined and delivered.** The credential holder must agree with the receiving organisation, before closure, what must be handed over — as-built documentation, configuration baselines, runbooks, licences, credentials, contracts, open defects, known workarounds, contacts — and must record delivery of each item.
- **PCI-PML-STD-16.02-PR-03 — Open items transfer with owners and dates.** Every open defect, nonconformity, warranty item, contractual obligation and carried risk must transfer with a named owner in the receiving organisation and a date, and must not be closed at project closure for the reason that the project is closing.
- **PCI-PML-STD-16.02-PR-04 — Capability is transferred, not just documents.** The credential holder must record how the receiving organisation was made able to operate what it receives — training delivered, procedures rehearsed, a period of supported operation — and the receiving organisation's confirmation that it was.

**6. Prohibited actions.** Closing a project with an unaccepted transfer. Recording acceptance signed by the project. Closing open defects at closure to clear the register. Handing over documentation in place of capability. Transferring a live product with no named ordering right under **PCI-PML-STD-13.02-PR-04**. Retaining credentials, licences or contracts the receiver needs. Treating the disbanding of the team as the closure event.

**7. Required evidence.** The operational acceptance record with the named receiver individual and the date; the agreed handover set with delivery confirmation per item; the transferred open-item register with receiving owners and dates; the capability-transfer record with the receiver's confirmation; the closure decision with its named decision owner.

**8. Responsible role.** The named credential holder leading the project, for securing acceptance and delivering the handover set. The named individual in the receiving organisation, for the acceptance. The sponsor, for resolving a refusal.

**9. Approval authority.** The receiving organisation's named acceptance authority accepts. The sponsor approves closure. The governing body approves closure where operational acceptance has not been obtained — and that approval must state what is being transferred unaccepted and to whom.



**10. Independence requirement.** The acceptance authority must be independent of the project in the sense defined above: accountable within the receiving organisation, not the project's own representative, and not remunerated by reference to the project's closure. Where the project and the receiver sit in the same small organisation, the acceptance authority must at least sit outside the project team and the arrangement must be recorded.

**11. Materiality or threshold.** This standard states no number. The organisation's governance sets the closure criteria, the handover set for each class of transfer, the supported-operation period and the tolerance for open items transferred; this standard requires that they exist and are applied. An open item that is a mandatory precondition under *PCI-PML-STD-16.01* does not transfer — it is closed before transition or the transition does not happen. *Six-person internal project*: the handover set is six items, acceptance is a dated line signed by the receiving manager, and capability transfer is a two-hour walkthrough recorded as delivered. *Multi-partner national programme*: acceptance is recorded per receiving organisation rather than once, because a national transfer has many receivers, and the open-item register is split by receiver so that no organisation inherits items it never saw.

**12. Exception and waiver.** Closure without operational acceptance may be approved only by the governing body, only where the unaccepted transfer is described item by item, only where the receiving organisation is notified in writing, and only where a named owner outside the project holds the residual responsibility until acceptance. No exception permits an open safety, licence or statutory item to transfer unaccepted.

**13. Escalation trigger.** A receiving organisation that declines acceptance. A handover item the project cannot supply. An open safety or regulatory item approaching closure. A receiver with no named individual able to accept. Closure proposed while the team that holds the knowledge has already dispersed.

**14. AI application.** AI may assemble the handover set from project records and flag missing items, extract open items from the defect, risk and issue registers for transfer, draft runbooks from configuration and test records for human verification, and check that every transferred item has a receiving owner.

**15. AI prohibition.** An AI system must not accept operational responsibility, decide that a handover is complete, close an open item, or be recorded as the receiving party.

**16. AI verification. Source tracing plus named approval.** Every AI-drafted runbook or as-built record must be traced by a competent reviewer to the configuration baseline and test evidence it claims to reflect, and the receiving



organisation must confirm by name that each handover item is what it needs and that it received it.

#### 17. External reference.

- **ISO** · *ISO 21502, Guidance on project management* · relied on for: the existence of transition and closure activities within delivery practice · **EXT-028** · **Manual §6 category 3 — international voluntary standard** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.
- **ISO** · *ISO 9001, Quality management systems — Requirements* · relied on for: the existence of a certifiable requirement to control documented information and to release outputs · **EXT-033** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: voluntary unless imported; no requirement is imported here.
- **ISO** · *ISO 15489-1, Records management — Concepts and principles* · relied on for: the existence of records-management principles governing the retention and transfer of records · **EXT-025** · **Manual §6 category 3** · currency checked 2026-08-03 · limitation: voluntary; guidance on principles; not a certifiable requirement for a project.

**18. Jurisdictional caution.** Whether operational responsibility, statutory duties, licences, safety duties, data controllership and warranties transfer — and when — is determined by law, licence and contract, not by a handover record. Obtain legal advice on what transfers and what remains, particularly for regulated assets and for personal data.

**19. Related PCI Standards.** [PCI-FND-STD-02](#); [PCI-FND-STD-12](#); [PCI-PML-STD-09.01](#); [PCI-PML-STD-13.02](#); [PCI-PML-STD-14.01](#); [PCI-PML-STD-16.01](#); [PCI-PML-STD-16.03](#).

**20. Related Body of Knowledge content.** PML-AI · Domain 16 · KA 16.2 Operational transition and contract closeout; KA 16.3 Knowledge transfer and post-project review, including the cost of reconstructing as-built knowledge after the team disperses. Also Domain 9 KA 9.3 acceptance; Domain 13 KA 13.1 product ownership.

**21. Compliance test.** Compliance is demonstrated when a reviewer can, for a closed project: (a) produce an operational acceptance record naming an individual in the receiving organisation, dated on or before the closure date; (b) take the agreed handover set and confirm a delivery record for every item; (c) take the project's open defect, risk, nonconformity and warranty registers at closure and find every open item in the receiving organisation's registers with a named owner and a date, with no item closed on the closure date for

that reason; (d) find the capability-transfer record and the receiver's confirmation; and (e) confirm no mandatory precondition under **PCI-PML-STD-16.01** transferred open. Closure with no receiver acceptance and no governing-body exception fails the test.

**22. Breach indicators.** Acceptance records signed by project staff. Registers that empty on the closure date. Handover sets consisting only of documents. Receiving teams raising defects the project had recorded and closed. Credentials still held by former project staff months after closure. Closure dates that precede the receiver's first involvement.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Scenario judgement: the receiving operations manager refuses acceptance a week before the project's funding ends. Evidence selection: which artefacts establish that capability, not just documentation, was transferred. Calculation review: the cost of reconstructing as-built knowledge against the cost of recording it.

**25. Version and status.** Version 1.0 · **not yet approved** · effective on approval · **new standard** — the v1.0 set covered transition readiness and benefits measurement and left the acceptance of operational responsibility between them unstated. Amendment note: none.

## PCI STANDARD

### PCI-PML-STD-16.03

## Benefits Measurement

**1. Normative requirement.** A credential holder must not state that a benefit has been realised except on a measurement taken by the method, from the source and against the benefits baseline recorded for that benefit before the change was made.

**2. Purpose.** Benefits reporting is the least policed statement an organisation makes about itself, and it is made after the people who could contradict it have moved on. The failure this prevents is the realised benefit that was measured differently from the way it was defined, against a baseline reconstructed afterwards, by the party whose case depended on it.

**3. Scope.** Every credential holder measuring, reporting, reviewing or assuring benefits realisation on a project, programme or portfolio, at closure and through the post-closure measurement period, including benefits reported to funders, boards and the public.

**4. Defined terms.** *benefit · benefits baseline · evidence · material · independent · decision owner · sponsor.* Additionally, **realisation statement** means an assertion that a benefit has been achieved in whole or in part; **measurement period** means the period after transition over which the benefit is measured.

**5. Required actions — process requirements.**

- **PCI-PML-STD-16.03-PR-01 — Measure by the recorded method and source.** Each realisation statement must use the measure, the source system and the calculation recorded for that benefit under **PCI-PML-STD-02.02-PR-02**, and any change to any of them must be recorded with its reason and its effect on the reported figure.
- **PCI-PML-STD-16.03-PR-02 — Compare against the pre-change baseline.** The comparison must be against the benefits baseline measured before the change. Where a baseline was reconstructed, the statement must say so and state the reconstruction method.
- **PCI-PML-STD-16.03-PR-03 — Attribution is stated, not assumed.** The statement must record what else changed in the measurement period that could have produced the movement, and must not attribute the whole movement to the project where other causes are known.
- **PCI-PML-STD-16.03-PR-04 — Shortfalls are reported with the same prominence as achievements.** A benefit measured below its target must be reported to the same audiences, in the same document, and in the same period as the benefits that met theirs.

**6. Prohibited actions.** Changing a measure to one that shows a better result. Comparing against a reconstructed baseline without saying so. Attributing an improvement with several causes wholly to the project. Reporting only the benefits that were achieved. Extending a measurement period until a target is met, without recording the extension. Reporting a proxy measure as the defined benefit.

**7. Required evidence.** The benefits register entries with method, source, baseline and target; the source system extracts with dates and filters; the realisation statements with their comparisons; the record of any method, source or period change with reasons; the attribution statement; the reporting record showing shortfalls issued to the same audiences.

**8. Responsible role.** The named **benefits owner** for each benefit, for the measurement and the statement. The **sponsor**, for the aggregate position reported to the governing body. The credential holder who prepared the measurement, for its integrity.

**9. Approval authority.** The governing body approves the closure of a benefit as realised or unrealised. The sponsor approves a change to a measurement

period or method within their documented authority; above it, the governing body. The delivery organisation approves nothing here.

**10. Independence requirement.** The measurement must be produced by, or verified by, a party independent of the delivery organisation and independent of the benefits owner where the owner's own performance is assessed on the benefit. Where an organisation is too small to supply that, the measurement must be taken directly from an unmodified source system extract and the extract retained.

**11. Materiality or threshold.** This standard states no tolerance for a shortfall. The organisation's governance sets the measurement period, the tolerance at which a shortfall is escalated, and the point at which a benefit is closed as unrealised; this standard requires that these exist and are applied and that a shortfall inside tolerance is still reported. *Six-person internal project:* one benefit, measured from a report the operating team already produces, compared with a baseline captured before go-live, reported once at three months and once at twelve. *Multi-partner national programme:* benefits are measured by the operating organisations rather than by any partner, the double-count check from [PCI-PML-STD-02.02](#) runs again at measurement, and the attribution statement is required per benefit because a national programme's measurement period contains many other causes.

**12. Exception and waiver.** An exception permitting a benefit to be reported on a method or source other than the recorded one may be approved only by the governing body, only where the change and its effect on the figure are stated in the report, and only where the original method's result is reported alongside it where it can still be produced. No exception permits a shortfall to go unreported.

**13. Escalation trigger.** A benefit measured materially below target. A source system that no longer produces the recorded measure. A proposal to change a method or extend a measurement period. A benefits owner who declines to report a shortfall. Discovery that a baseline was reconstructed after the change.

**14. AI application.** AI may extract measurements from source systems on the recorded basis, detect changes of method or source between periods, compute the comparison against baseline, identify concurrent changes in the measurement period as candidate alternative causes for human assessment, and check that every benefit in the register has a statement for the period.

**15. AI prohibition.** An AI system must not state that a benefit has been realised, decide attribution, select the measure or source, or determine that a shortfall is immaterial.

**16. AI verification. Independent recomputation plus source tracing.** Every reported realisation figure produced with AI assistance must be

recomputed from the source extract by a competent reviewer, with the extract date and filter recorded, and traced to the register entry that defines the method. Where AI proposed alternative causes for attribution, a named human must assess each and record the conclusion.

### 17. External reference.

- **ISO · ISO 21504, *Guidance on portfolio management*** · relied on for: the existence of benefits tracking at portfolio level · **EXT-031, not independently verified — verify current requirements · Manual §6 category 3 — international voluntary standard** · limitation: guidance; voluntary; not certifiable; open verification status.
- **ISO · ISO 21503, *Guidance on programme management*** · relied on for: the existence of benefits realisation as a programme concern · **EXT-030, not independently verified · Manual §6 category 3** · limitation: as above.
- **ISO · ISO 21502, *Guidance on project management*** · relied on for: the existence of benefits identification and realisation within delivery practice · **EXT-028 · Manual §6 category 3** · currency checked 2026-08-03 · limitation: guidance; voluntary; not certifiable.
- **ISO · ISO 15489-1, *Records management — Concepts and principles*** · relied on for: the existence of records-management principles governing retention of the evidence a measurement rests on · **EXT-025 · Manual §6 category 3** · currency checked 2026-08-03 · limitation: voluntary; principles guidance.

**18. Jurisdictional caution.** Benefits statements made to funders, regulators, markets or the public can engage grant-condition, disclosure, procurement and fraud law, and the legal standard for such a statement is not the professional one used here. Obtain legal advice before a benefits claim is made outside the organisation.

**19. Related PCI Standards.** PCI-FND-STD-02; PCI-FND-STD-06; PCI-FND-STD-12; PCI-PML-STD-02.01; PCI-PML-STD-02.02; PCI-PML-STD-11.01; PCI-PML-STD-15.02; PCI-PML-STD-16.02.

**20. Related Body of Knowledge content.** PML-AI · Domain 16 · KA 16.4 Benefits measurement, responsible archive and model/data retention. Also Domain 2 KA 2.3 topics 2.3.1 benefits mapping and 2.3.2 measures and baselines; Domain 15 KA 15.2 benefits and portfolio balancing.

**21. Compliance test.** Compliance is demonstrated when a reviewer can, for every benefit reported as realised in the period: (a) retrieve the register entry defining its measure, source, baseline and target, and confirm the reported

figure used that measure and that source; (b) reproduce the reported figure from the source system extract, using the recorded calculation, and reconcile any difference; (c) confirm the comparison is against a baseline whose measurement date precedes the change date, or that the report states the baseline was reconstructed and how; (d) find the attribution statement listing concurrent changes; and (e) confirm that every benefit measured below target in the period appears in the same report, to the same distribution, as those that met theirs. A realisation figure that cannot be reproduced from the recorded source fails the test.

**22. Breach indicators.** Measures that changed between the case and the measurement. Baselines dated after go-live. Reports containing only achieved benefits. Measurement periods extended without a decision record. Attribution statements absent from programmes with large concurrent change. Benefits reported as realised whose owners cannot produce the extract.

**23. Consequence within PCI authority.** Correction required; output withheld; additional review; escalation; examination failure; ethics review; certification investigation; suspension or withdrawal — each subject to due process and a right of appeal.

**24. Examination application.** Calculation review: a reported improvement recomputed from source, showing a method change between the case and the measurement. Ethical dilemma: a sponsor asks for the measurement period to be extended by a quarter because the trend is improving. Evidence selection: which artefacts establish that a benefit was realised.

**25. Version and status.** Version 2.0 · **not yet approved** · effective on approval · supersedes [PML-LAW-16-02](#) v1.0. Amendment note: renumbered and restructured; legislative drafting removed; the method-and-source rule, the pre-change baseline rule, the attribution statement and the equal-prominence rule for shortfalls separated into four process requirements; the benefits-baseline sense of *baseline* separated from the control sense per the suite terminology audit.

## Audit findings — the twenty-five questions worked over the set

Drafting Manual §9 requires that every question below is answered before approval, and that a standard failing one is revised, with the failure and its resolution recorded. This table records the working for the set as a whole, names the standards that failed a question in draft, and states what was changed. **Every failure listed was resolved before this edition was assembled**; the questions with an open finding are marked and carried into the outstanding due-process stages.

The front-matter stage record above notes that the Stage 9 red-team was only partly worked for this edition. It has since been completed across the whole four-file corpus, and its findings and their disposition are recorded in [STANDARDS\\_RED\\_TEAM\\_REPORT.md](#). The amendments that pass produced in this volume are noted in the element 25 of each standard changed, and the definitional reading rules it added sit at the head of the Definitions above.

| # | QUESTION                                            | FINDING ACROSS THE SET                                                                                                                                                                                                                                 | ACTION TAKEN                                                                                                                                                                                                           |
|---|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | What exact failure does this standard prevent?      | Answered in element 2 of all 32 standards. Three drafts stated a purpose that was really a benefit of compliance rather than a failure — <a href="#">12.01</a> , <a href="#">12.02</a> , <a href="#">13.02</a> .                                       | Element 2 rewritten in those three to name the failure: authority used to override professional judgement; information that stops flowing upward; the ordering right exercised by someone other than its named holder. |
| 2 | Is the requirement mandatory or only recommended?   | All 32 principal obligations and all 148 process requirements are expressed with <b>must</b> or <b>must not</b> . No <a href="#">should</a> carries an obligation anywhere in the set; no Recommended Practice is published in this edition.           | None needed. Checked mechanically.                                                                                                                                                                                     |
| 3 | Can a professional know whether it applies to them? | Element 3 of each standard names the roles, the decisions and the delivery models. <a href="#">15.01</a> and <a href="#">15.02</a> initially read as though they bound every credential holder, when they bind those in programme and portfolio roles. | Element 3 of both narrowed, and element 11 of each states expressly what the obligation amounts to for a small internal project.                                                                                       |
| 4 | Is the responsible person identifiable?             | Element 8 of all 32 names an individual role. No standard uses "the team", "management", "relevant people" or "the organisation" as a responsible role. Two drafts used "the PMO" — <a href="#">03.03</a> and <a href="#">09.02</a> .                  | Replaced with the named credential holder and the named artefact owner respectively.                                                                                                                                   |

| # | QUESTION                                         | FINDING ACROSS THE SET                                                                                                                                                                                                                                                                                               | ACTION TAKEN                                                                                                                                                                                                                                                                                                                                                                                            |
|---|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 | Is the required action observable?               | The set's hardest question, and the one the previous edition failed. 12.01 required conduct ("act with integrity, model the behaviour") and 12.02 required a state of affairs ("maintain conditions in which any team member can report bad news"). Neither can be observed and therefore neither could be required. | 12.01 rebuilt around one <b>observable act</b> — overriding another person's professional statement — with five process requirements that each leave a record. 12.02 rebuilt around <b>three observable things</b> : a published route with named recipients, the bypass property demonstrated person by person, and a concern register with seven mandatory fields. Both now pass.                     |
| 6 | Is compliance provable?                          | Element 21 of every standard is now a test a reviewer performs against records that exist. The behavioural standards were the exposure: a leadership standard provable only by testimony proves nothing.                                                                                                             | 12.02 element 21 test (e) <b>cross-matches two record sets that already exist</b> — the concern register against assignment, role-change and assessment records — so that detriment is found by matching dates and names rather than by asking anybody how they felt. 12.01 element 21 test (a) uses <b>document version history</b> , which is produced automatically by every document system in use. |
| 7 | Is the required evidence proportionate?          | Element 7 was tested against a six-person project throughout. Two standards required evidence a small project cannot economically produce — an earlier 05.02 required full traceability tooling and an earlier 14.02 required a verification log per output instance.                                                | 05.02 now requires a maintained matrix in any form and full traversal only for the regulatory and safety subset; 14.02 requires a method per <b>output class</b> , not per output.                                                                                                                                                                                                                      |
| 8 | Can the standard be audited?                     | Yes for all 32. Each element 21 names the artefacts to obtain and the comparison to make.                                                                                                                                                                                                                            | None needed.                                                                                                                                                                                                                                                                                                                                                                                            |
| 9 | Can the standard be examined through a scenario? | Element 24 of every standard gives at least two examination modes drawn from the Manual's list. No standard is examinable only by recalling its number.                                                                                                                                                              | None needed.                                                                                                                                                                                                                                                                                                                                                                                            |



| #  | QUESTION                                                           | FINDING ACROSS THE SET                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ACTION TAKEN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10 | Can a professional technically comply while defeating its purpose? | Six standards were vulnerable and were changed. 03.02 — register only decisions above the limit, so the aggregation rule sees nothing. 05.02 — resolve orphans by deleting requirements. 08.01 — reassess exposure downward instead of escalating. 13.01 — raise throughput by starting more work. 16.01 — enter a mandatory precondition as a high probability so the readiness figure stays true-looking. 12.02 — publish a route whose only recipient is the person concerns are about. | 03.02-PR-02 universal registration; 05.02-PR-03 orphans reported, not tidied; 08.01-PR-02 escalate on threshold rather than on expectation of resolution, with <b>the date the threshold was met</b> recorded beside the date escalated; 13.01-PR-04 metrics reported with the work-in-progress position; 16.01-PR-03 and -PR-04 prohibit probability and prohibit economic trade, and element 16 adds a <b>boundary test of the assessment's structure</b> ; 12.02-PR-02 the bypass property, demonstrated person by person. |
| 11 | Does it conflict with another PCI standard?                        | No conflict found. Two overlaps were resolved by narrowing: 01.02 against PCI-FND-STD-04, and 05.01 against 04.01 where a scope change is also a baseline change.                                                                                                                                                                                                                                                                                                                          | 01.02 retitled and narrowed to the reserved-class list, the automation inventory and the examination record. 05.01-PR-03 routes scope additions <b>into</b> 04.01 rather than duplicating its assessment.                                                                                                                                                                                                                                                                                                                     |
| 12 | Does it duplicate an external standard unnecessarily?              | No standard imports a requirement from an external instrument. Every element 17 entry states what the instrument is relied on <b>for</b> — usually the existence of a concept — and states that the obligation is PCI's own.                                                                                                                                                                                                                                                               | The phrase "it is not the source of this standard's obligation" or its equivalent appears in the limitation column throughout.                                                                                                                                                                                                                                                                                                                                                                                                |
| 13 | Does it misrepresent external authority?                           | Four risks were identified and closed. ISO 21500 is <b>context and concepts</b> since its current edition, not project-management guidance. ISO 45003 is <b>guidance and nothing can be certified against it</b> . The Scrum Guide is a <b>voluntary framework whose adoption is the whole of its force</b> , not a standard. The EU AI Act and the GDPR are <b>binding legislation in their own jurisdictions</b> , named here to illustrate a shape and relied on for nothing.           | All four stated expressly at the point of use. No clause number, article, edition or effective date is asserted anywhere; editions are held in the suite register with their verification status, and rows recorded there as <b>not independently verified</b> are marked as such in the standard that cites them.                                                                                                                                                                                                            |

| #  | QUESTION                                               | FINDING ACROSS THE SET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ACTION TAKEN                                                                                                                                                                                            |
|----|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14 | Does it require legal or jurisdiction-specific advice? | Element 18 of all 32 states what needs local advice. The heaviest are 10.01 (procurement regimes), 12.02 (whistleblowing, monitoring, works councils), 14.01 (data protection) and 16.01 (which approvals are legally required, and who may grant them).                                                                                                                                                                                                                                                                               | 16.01 element 18 states expressly that <b>the professional must not infer the gate-block list from the examples given</b> — the examples are the manuscript's, and the list is the owning authorities'. |
| 15 | Does it define the relevant materiality threshold?     | Yes, and <b>no standard in this set invents a percentage</b> . Element 11 of every standard requires that the organisation's own documented threshold exists and is applied, and states the professional-judgement criteria and the applying role where no number is defensible. Three standards state expressly that a number would be harmful: 12.02 (a concern target invites manufactured concerns), 15.02 (a borrowed utilisation figure is worse than none) and 16.01 (the gate block is binary and admits no threshold at all). | Element 11 of every standard carries a <i>six-person internal project</i> line and a <i>multi-partner national programme</i> line, which is where the thresholds were tested.                           |
| 16 | Does it cover AI use?                                  | Elements 14, 15 and 16 appear in all 32 standards.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | None needed.                                                                                                                                                                                            |
| 17 | Does it preserve human accountability?                 | Element 15 of every standard prohibits an AI system from deciding, approving, certifying, signing, waiving, authorising or being recorded as verifying. 01.01, 01.02 and 14.02 carry the load.                                                                                                                                                                                                                                                                                                                                         | 14.02 element 15 adds that <b>an AI system must not verify another AI system's output</b> for the purposes of this set.                                                                                 |
| 18 | Does it contain an exception process?                  | Element 12 of all 32. <b>Four standards state that no exception is permitted</b> , and each gives the reason: 01.03 (accountability for a conflicted decision cannot be waived; escalate instead), 12.01 (the principal obligation), 12.02 (PR-02, PR-04, PR-05) and 16.01 (no exception exists in respect of a mandatory precondition, and none may be sought).                                                                                                                                                                       | 16.01 element 12 also states the correct remedy where a classification is disputed: have <b>the owning authority</b> reclassify it in writing before the assessment.                                    |

| #  | QUESTION                             | FINDING ACROSS THE SET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ACTION TAKEN                                                                                                                                                                                                                                                                                                                                                                   |
|----|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19 | Does it define escalation?           | Element 13 of all 32. The defined term <i>escalation threshold</i> requires a named destination <b>and</b> a time; 03.02-PR-05 makes encountering one without them an escalable defect in itself.                                                                                                                                                                                                                                                                                                                                                                                                                                  | None needed.                                                                                                                                                                                                                                                                                                                                                                   |
| 20 | Is every important term defined?     | Thirteen compliance-deciding terms are defined at the head of the file, plus nineteen supporting terms. Terms the suite terminology audit records as carrying legitimately different meanings across the three books are <b>kept apart and flagged</b> , not collapsed: <i>sponsor</i> (delivery sense used; the project-finance sense is PFL-AI's), <i>baseline</i> (control sense and benefits sense, both defined), <i>verification</i> (V&V sense and AI-assurance sense, both defined), <i>governance</i> (project sense and data governance, written in full), and <i>benefit</i> against <i>value</i> against Earned Value. | The definition of <i>material</i> was rewritten twice: it now states that safety, legality, licence and truthfulness matters are material <b>irrespective of size</b> and that no documented threshold reduces them.                                                                                                                                                           |
| 21 | Is the language concrete and modern? | <b>Zero occurrences of the legislative auxiliary verb</b> in the file, in any field, checked mechanically. No <i>may not</i> is used for a prohibition. No undefined judgement word carries an obligation: <i>appropriate</i> , <i>adequate</i> , <i>reasonable</i> , <i>timely</i> and <i>sufficient</i> were removed from every obligation or replaced with a stated test.                                                                                                                                                                                                                                                       | The Manual §1 requirement to state the ISO mapping was met <b>without printing the prohibited word</b> : the mapping is stated by reference to the auxiliary that ISO/IEC Directives Part 2 reserves for requirements. This is recorded here because it is a deliberate resolution of a tension between two Manual provisions, and the Interpretation Panel should confirm it. |

| #  | QUESTION                                                 | FINDING ACROSS THE SET                                                                                                                                                                                                                                                                                                                                                                                       | ACTION TAKEN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 22 | <b>Does it impose an impossible or excessive burden?</b> | Tested standard by standard against a six-person internal project. Three drafts were excessive: an earlier 03 . 01 required four separate approved artefacts on any project of any size; an earlier 10 . 01 required a three-person panel; an earlier 16 . 03 required independent measurement in all cases.                                                                                                 | 03 . 01-PR-03 makes proportionate tailoring an express, recorded decision so light governance is a choice rather than an omission; 10 . 01 element 11 scales to a three-supplier quotation exercise; 16 . 03 element 10 permits, for an organisation too small to supply independence, measurement taken directly from an unmodified source extract that is retained. <b>The residual burden is honestly stated:</b> 01 . 03, 12 . 02 and 16 . 01 each add real recording work, and each was retained because the failure it prevents is severe. |
| 23 | Can it operate on both small projects and megaprojects?  | Yes — element 11 of every standard states both cases explicitly, and this is where the set was most often rewritten.                                                                                                                                                                                                                                                                                         | Recorded in element 11 throughout.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 24 | Can it operate internationally?                          | No standard depends on a single jurisdiction's legal concepts. Element 18 of every standard states what is jurisdiction-specific, and no standard states a legal position.                                                                                                                                                                                                                                   | 12 . 02 element 18 states expressly that where a national whistleblowing or monitoring regime imposes a higher or more specific obligation, <b>that regime governs.</b>                                                                                                                                                                                                                                                                                                                                                                          |
| 25 | Is there a clear consequence within PCI's authority?     | Element 23 of all 32 draws only from Charter §9. Three standards state expressly what PCI <b>cannot</b> do, because those are the three where a reader might otherwise assume a wider power: 10 . 01 (PCI cannot set aside an award or impose a fine), 12 . 02 (PCI cannot compensate, order reinstatement or penalise an employer) and 16 . 01 (PCI cannot authorise a transition or override a regulator). | Added to those three; the remainder rely on the Charter §9 list, which is stated in full in each.                                                                                                                                                                                                                                                                                                                                                                                                                                                |

### Definitions reconciliation

The red team's structural finding **P-1** — no PCI Standards Definitions Register, so each volume built its own and seven compliance-deciding terms diverged — has since been closed. The register is published at [PCI\\_STANDARDS\\_DEFINITIONS\\_REGISTER.md](#) and *Terms that decide compliance* above were

reconciled to it: *material*, *independent*, *evidence*, *competent reviewer*, *decision owner*, *escalation threshold* and *conflict of interest* now carry the canonical wording, and *approved*, *current* and *material AI assistance* were added. **No obligation changed.** Three notes:

- **The four collisions this set flags are preserved, not collapsed** — *baseline* (control versus benefits), *sponsor* (delivery versus project-finance), *verification* (V&V versus AI) and *governance* (project versus data). Each is carried in the register with both senses and the context each belongs to. Forcing one definition on any of them would make this volume wrong, which is worse than the divergence.
- **Competent reviewer no longer folds independence into competence.** The limb requiring that the reviewer did not prepare, direct, specify or approve the thing reviewed is *independence*, imposed by each standard's element 10 and tested separately. That separation is what makes [PCI-FND-STD-10](#) element 12's supervised-acquisition exception usable, and no element 10 loses an independence requirement.
- **Material AI assistance was undefined here** while [PCI-PML-STD-14.02](#) element 21(d) made compliance turn on it. The canonical definition is supplied so the test can be applied; the Interpretation Panel should confirm it, because it decides which artefacts must carry the disclosure.

Open findings carried forward

| FINDING                                                                                                                                                                                                                                                        | WHY IT IS OPEN                                                                     | WHERE IT GOES                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Six external-reference rows are recorded as "not independently verified" in the suite register</b><br>— ISO 21503, ISO 21504, ISO 21505, ISO 9000, the PMBOK Guide's companion <i>Code of Ethics and Professional Conduct</i> , and the AACE TCM Framework. | Charter §5 Stage 5 was performed against the register, not against the publishers. | Verification before publication; each citing standard states the open status in element 17. |

| FINDING                                                                     | WHY IT IS OPEN                                                                                                                                                                                                                                                       | WHERE IT GOES                                                                                                                      |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| ~~The published foundational file lags the concordance.~~<br>Closed.        | PCI_FUNDATIONAL_STANDARDS .md now carries fifteen standards under the Charter §3 form PCI-FND-STD-01 to PCI-FND-STD-15, with the subjects listed in <i>How to read these standards</i> above. Every citation in this set resolves against the published file itself. | Closed on the foundational rebuild. The superseded PCI-LAW-F-NN identifiers survive only as history, in STANDARDS_CONCORDANCE .md. |
| Charter §5 Stages 4, 6, 7, 11, 12 and 13 have not been performed.           | Technical review, practitioner consultation, impact assessment, approval, publication and post-implementation review are outstanding.                                                                                                                                | Element 25 of every standard records the set as <b>not yet approved</b> , with the effective date on approval.                     |
| The ISO-mapping statement resolves a tension between two Manual provisions. | Manual §1 both prohibits the legislative auxiliary everywhere and requires the ISO mapping to be stated. This file states the mapping by describing the auxiliary rather than printing it.                                                                           | Interpretation Panel confirmation under Charter §6.                                                                                |

## Index of PML-AI Professional Standards

**Thirty-two standards · one hundred and forty-eight process requirements · sixteen anchor domains.** External-reference categories are Drafting Manual §6 categories: **3** international voluntary standard · **4** contract framework · **5** professional framework · **6** ethical code · **10** illustrative practice. Every standard also relates to category **9**, PCI internal professional standard, through its element 19, and that category is not repeated in the table.

| ID                  | OFFICIAL TITLE                                   | ANCHOR DOMAIN                          | PRINCIPAL OBLIGATION                                                                                              | EXTERNAL REFERENCE CATEGORIES |
|---------------------|--------------------------------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------|
| PCI-PML-STD-01 . 01 | Leadership Accountability for Delivery Decisions | D1 — The project leadership profession | Remain personally accountable for every delivery decision taken under one's authority, including AI-informed ones | 3, 10                         |

| ID                | OFFICIAL<br>TITLE                                                      | ANCHOR<br>DOMAIN                                              | PRINCIPAL<br>OBLIGATION                                                                                                               | EXTERNAL<br>REFERENCE<br>CATEGORIES |
|-------------------|------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| PCI-PML-STD-01.02 | Reserved<br>Delivery<br>Decisions and<br>the Named<br>Human<br>Decider | D1                                                            | Ensure a<br>decision in a<br>reserved class<br>is taken by<br>the named<br>human on<br>evidence they<br>examined                      | 3, 10                               |
| PCI-PML-STD-01.03 | Interests,<br>Abstention<br>and Assurance<br>Independence              | D1                                                            | Do not decide,<br>advise on,<br>evaluate or<br>assure a<br>matter in<br>which one<br>holds a<br>conflict of<br>interest               | 3, 6, 10                            |
| PCI-PML-STD-02.01 | Business-Case<br>Integrity                                             | D2 — Strategy,<br>selection and<br>business<br>alignment      | Do not<br>present,<br>endorse or<br>rely on a<br>materially<br>misstated<br>business case                                             | 3, 5                                |
| PCI-PML-STD-02.02 | Benefits<br>Ownership                                                  | D2                                                            | Do not allow<br>a benefit to be<br>claimed<br>unless a<br>named<br>individual<br>outside<br>delivery has<br>accepted it in<br>writing | 3                                   |
| PCI-PML-STD-03.01 | Governance<br>Authority<br>Before<br>Commitment                        | D3 —<br>Governance,<br>organisation<br>and decision<br>rights | Do not<br>commit funds,<br>contracts or<br>people before<br>governance is<br>documented<br>and approved                               | 3, 10                               |

| ID                | OFFICIAL<br>TITLE                              | ANCHOR<br>DOMAIN                              | PRINCIPAL<br>OBLIGATION                                                                                            | EXTERNAL<br>REFERENCE<br>CATEGORIES |
|-------------------|------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| PCI-PML-STD-03.02 | Decision Rights and Delegated Authority        | D3                                            | Take every decision at the authority the delegation schedule assigns; do not split, defer or aggregate to avoid it | 3, 5, 10                            |
| PCI-PML-STD-03.03 | Gate Evidence and the Gate Decision            | D3                                            | Do not take a gate decision except on dated, attributable, versioned evidence against criteria published first     | 3                                   |
| PCI-PML-STD-03.04 | Sponsor Accountability                         | D3                                            | Do not lead delivery with no named individual sponsor who has accepted the accountability in writing               | 3, 10                               |
| PCI-PML-STD-04.01 | Change Authority and Integrated Change Control | D4 — Integration and delivery architecture    | Do not let a baseline change take effect before integrated assessment and approval at the assigned authority       | 3, 4, 5                             |
| PCI-PML-STD-05.01 | Scope Integrity                                | D5 — Scope, requirements and value definition | Do not allow work outside the approved scope baseline without an approved change                                   | 3, 5                                |



| ID                | OFFICIAL<br>TITLE                                        | ANCHOR<br>DOMAIN                              | PRINCIPAL<br>OBLIGATION                                                                                    | EXTERNAL<br>REFERENCE<br>CATEGORIES |
|-------------------|----------------------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------|-------------------------------------|
| PCI-PML-STD-05.02 | Requirements<br>Traceability                             | D5                                            | Maintain traceability from every approved requirement to source, to satisfying work, and to proving test   | 3, 5                                |
| PCI-PML-STD-06.01 | Schedule<br>Credibility                                  | D6 — Planning, scheduling and delivery flow   | Do not issue or rely on a schedule that misrepresents achievable completion                                | 3, 5                                |
| PCI-PML-STD-07.01 | Cost<br>Stewardship                                      | D7 — Cost, resources and commercial awareness | Report the cost position — committed, incurred, accrued, forecast — completely, currently and reconcilably | 3, 5                                |
| PCI-PML-STD-07.02 | Resource<br>Decisions and the<br>Commitment<br>of People | D7                                            | Do not commit a person or shared resource to a plan without the named resource owner's agreement           | 3, 5                                |
| PCI-PML-STD-08.01 | Risk<br>Escalation                                       | D8 — Risk, uncertainty and resilience         | Escalate a risk meeting the documented threshold, to the named authority, within the stated time           | 3                                   |

| ID                | OFFICIAL<br>TITLE                                     | ANCHOR<br>DOMAIN                                             | PRINCIPAL<br>OBLIGATION                                                                                                                    | EXTERNAL<br>REFERENCE<br>CATEGORIES |
|-------------------|-------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| PCI-PML-STD-08.02 | Issue<br>Management                                   | D8                                                           | Record every<br>issue with one<br>named owner,<br>a required<br>date and a<br>stated<br>consequence;<br>close only<br>with a<br>resolution | 3                                   |
| PCI-PML-STD-09.01 | Quality<br>Acceptance                                 | D9 — Quality,<br>assurance and<br>continuous<br>improvement  | Do not record<br>acceptance<br>unless a<br>named<br>authority<br>decided<br>conformity<br>against pre-<br>dated criteria                   | 3                                   |
| PCI-PML-STD-09.02 | Lessons<br>Learned and<br>Organisational<br>Retention | D9                                                           | Convert each<br>accepted<br>lesson into a<br>change to a<br>named<br>standing<br>artefact, with<br>an owner and<br>a date                  | 3                                   |
| PCI-PML-STD-10.01 | Procurement<br>Fairness                               | D10 —<br>Procurement,<br>contracts and<br>supply<br>networks | Evaluate only<br>against<br>criteria and<br>weightings<br>published to<br>bidders before<br>submission                                     | 3, 4, 6                             |
| PCI-PML-STD-11.01 | Stakeholder<br>Transparency                           | D11 —<br>Stakeholders,<br>communication<br>and influence     | Do not issue<br>or let stand a<br>delivery<br>report that<br>omits a<br>material<br>adverse fact<br>known at<br>issue                      | 3, 5                                |

| ID                | OFFICIAL<br>TITLE                                            | ANCHOR<br>DOMAIN                                                 | PRINCIPAL<br>OBLIGATION                                                                                                                  | EXTERNAL<br>REFERENCE<br>CATEGORIES |
|-------------------|--------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| PCI-PML-STD-12.01 | Leadership<br>Conduct                                        | D12 —<br>Leadership,<br>teams and<br>organisational<br>behaviour | Do not use<br>authority to<br>cause another<br>person to<br>state, or<br>withhold,<br>what they<br>have recorded<br>as inaccurate        | 3, 6                                |
| PCI-PML-STD-12.02 | Route to Raise<br>a Concern and<br>Freedom from<br>Detriment | D12                                                              | Establish,<br>publish and<br>operate a<br>concern route<br>to a named<br>recipient<br>outside the<br>subject's line                      | 3                                   |
| PCI-PML-STD-13.01 | Governance of<br>Adaptive<br>Delivery                        | D13 — Agile,<br>adaptive and<br>hybrid delivery                  | Operate<br>governance<br>producing the<br>same decision<br>rights,<br>evidence and<br>accountability<br>through<br>adaptive<br>artefacts | 3, 5                                |
| PCI-PML-STD-13.02 | Product and<br>Project<br>Accountability                     | D13                                                              | Ensure exactly<br>one named<br>holder of the<br>ordering right<br>and one of<br>delivery<br>accountability,<br>both recorded             | 3, 5                                |
| PCI-PML-STD-14.01 | Responsible<br>Data Use in<br>Delivery                       | D14 — Digital<br>delivery, data<br>and<br>responsible AI         | Do not collect,<br>use, share,<br>retain or<br>expose data<br>beyond its<br>recorded<br>purpose,<br>recipients and<br>retention          | 3, 10                               |

| ID                | OFFICIAL<br>TITLE                                          | ANCHOR<br>DOMAIN                                                 | PRINCIPAL<br>OBLIGATION                                                                                                                       | EXTERNAL<br>REFERENCE<br>CATEGORIES |
|-------------------|------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| PCI-PML-STD-14.02 | Responsible AI<br>in Delivery                              | D14                                                              | Do not rely on<br>an AI output<br>until a named<br>human has<br>verified it by a<br>method<br>recorded for<br>that output<br>class            | 3, 10                               |
| PCI-PML-STD-15.01 | Programme<br>Integration<br>and<br>Dependency<br>Ownership | D15 —<br>Programmes,<br>portfolios and<br>enterprise<br>delivery | Record every<br>inter-<br>component<br>dependency<br>with a named<br>giver, receiver,<br>thing, date<br>and breach<br>consequence             | 3                                   |
| PCI-PML-STD-15.02 | Portfolio<br>Prioritisation<br>and Capacity<br>Truth       | D15                                                              | Do not let the<br>portfolio hold<br>more<br>concurrent<br>work than<br>assessed<br>delivery<br>capacity<br>supports                           | 3                                   |
| PCI-PML-STD-16.01 | Transition<br>Readiness and<br>the Gate<br>Block           | D16 —<br>Transition,<br>closeout and<br>benefits<br>realisation  | Do not permit<br>transition<br>while any<br>mandatory<br>precondition<br>is recorded<br>not met                                               | 3                                   |
| PCI-PML-STD-16.02 | Operational<br>Acceptance<br>and Handover                  | D16                                                              | Do not close a<br>project until a<br>named<br>individual in<br>the receiving<br>organisation<br>has accepted<br>operational<br>responsibility | 3                                   |

| ID                | OFFICIAL<br>TITLE       | ANCHOR<br>DOMAIN | PRINCIPAL<br>OBLIGATION                                                                                   | EXTERNAL<br>REFERENCE<br>CATEGORIES |
|-------------------|-------------------------|------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------|
| PCI-PML-STD-16.03 | Benefits<br>Measurement | D16              | Do not state a benefit is realised except on the recorded method, source and pre-change benefits baseline | 3                                   |

## Distribution

Every one of the sixteen PML-AI domains carries at least one standard. Domain 3 carries four, because governance is where decision rights, gate evidence and sponsorship are taught; Domain 16 carries three, because transition, operational acceptance and benefits measurement are three separate decisions that fail in three separate ways; Domains 1, 2, 5, 7, 8, 9, 12, 13, 14 and 15 carry two or three; Domains 4, 6, 10 and 11 carry the single standard their material makes mandatory.

**Eight standards are new in this edition** — 01.03, 03.04, 07.02, 08.02, 13.02, 15.01, 15.02 and 16.02 — and two were rebuilt because they could not be verified: 12.01 and 12.02. 01.02 was retitled to end a collision with the foundational standard *Human Decision Authority*, and 09.02 was retitled and given a new principal obligation. The remainder were renumbered, restructured to the twenty-five-element form, re-drafted in must-form, split into one principal obligation plus process requirements, and given a compliance test that can actually be performed.

Every standard in this set operates under the Foundational Standards [PCI-FND-STD-01](#) to [PCI-FND-STD-15](#), under the **PCI Standards Charter** and the **PCI Standards Drafting Manual**, and under one principle:

### CAUTION

**AI proposes; the professional verifies, decides and remains accountable.**

# Glossary

Consolidated from every Knowledge Area's key-terms table. The Knowledge Area reference after each entry is the authority — the gloss points to the treatment and does not replace it.

## A

**Abdication** — Transferred authority without the information or criteria to exercise it. (KA 12.3)

**Abnormally low tender** — A bid materially below its benchmark; the benchmark must be stated, since the bid spread and the buyer's estimate can disagree. (KA 10.2)

**Abstention** — The interested party taking no part in the discussion or the decision, minuted with the name of whoever decided in their place. (KA 1.2)

**AC** — Actual Cost of the work performed, including period accruals. (KA 7.2)

**Acceptance** — A formal decision by a named authority that a deliverable meets its criteria and may pass on. (KA 9.3)

**Acceptance by silence** — Deemed acceptance through the expiry of a review period; not a decision. (KA 9.3)

**Acceptance sampling** — Accepting or rejecting a population on the evidence of a sample. (KA 9.3)

**Acceptance-criteria expansion** — Creep entering through a growing criterion while the requirement text is unchanged. (KA 5.4)

**Accountability** — The obligation to answer; single-holder, non-delegable. (KA 1.2)

**Accountability laundering** — Presenting a model's output as a neutral input to avoid owning the judgement it embodies. (KA 14.4)

**Accrual** — Cost of work received but not yet invoiced, recognised in the period. (KA 7.2)

**Accuracy class** — The definition-linked range accompanying an estimate; never optional. (KA 7.1)

**Adaptive delivery** — A delivery mode fixing capacity and cadence and varying scope, correcting direction from working output at short intervals. (KA 13.1)

**Adequacy ratio** — Remaining reserve ÷ the confidence-level requirement of the register still open; below 1.00 is a governance trigger. (KA 8.3)

**Adoption** — The outcome measure linking an output to any benefit at all. (KA 1.3)

**Aggregate-capacity illusion** — Judging feasibility on total capacity when the constraint binds period by period. (KA 15.2)

**Aggregation illusion** — Treating a framework call-off as competitively priced because the framework was competed; the rate may have drifted from the market. (KA 10.2)

**Aggregation trap** — Averaging component ratios instead of summing their numerators and denominators; frequently reverses the sign. (KA 15.4)

**AI use register** — The row-per-use record of use, decision informed, accountable person, consequence, measurements, tier, versions, boundary, failure route and decisions to reopen on failure. (KA 14.4)

**Alignment decay** — The erosion of strategic fit after approval, as strategy and understanding move. (KA 2.1)

**Alignment half-life** —  $\ln(0.5) \div \ln(1 - h)$ : the delivery duration beyond which alignment is more likely lost than kept. (KA 2.1)

**Alignment index** —  $\sum \min(\text{declared weight}, \text{funded share})$ , the overlap between stated priorities and funded spend. (KA 2.1)

**Already-committed benefit** — Benefit claimed against a baseline that already assumes a saving committed elsewhere. (KA 15.2)

**Amber compression** — The collapse of traffic-light status into a single uninformative colour, caused by author discretion over undefined thresholds. (KA 11.2)

**Analogous / parametric / bottom-up** — Scale an analogue  $\cdot$  rate  $\times$  parameter  $\cdot$  sum the packages. (KA 7.1)

**Analytics ladder** — Descriptive  $\rightarrow$  diagnostic  $\rightarrow$  predictive  $\rightarrow$  prescriptive; each rung depends on the one below being right. (KA 14.2)

**Approval percentile** — The position of the approved number inside its own declared range; Auriga's baseline is a P33. (KA 7.1)

**Assessed total impact** — Direct + schedule + rework + interface re-verification + regression + documentation + risk + benefit. (KA 4.4)

**Assumption** — Something taken as true without being controlled, carrying an owner and a resolution date. (KA 5.1)

**Assumption exposure ratio** —  $\sum \text{EMV} \div \text{NPV}$ ; as it approaches one, the case is a bet rather than a value proposition. (KA 2.3)

**Assumption register** — Statement, basis, impact-if-false, owner, test date, falsifying trigger. (KA 2.3)

**Assurance capture** — An assurance function assuring work it helped produce, and therefore unable to challenge it. (KA 3.3)

**Assurance map** — Risks and controls against assurance lines, marking coverage, gaps and duplication. (KA 3.3)

**Asynchronous artefact** — A written product complete enough to remove an exchange rather than accelerate it. (KA 12.4)

**Attitude** — Supportive, neutral, sceptical or opposed, assessed on evidence; predicts what an engagement contact will achieve. (KA 11.1)

**Attributable act** — An output that confers authority or discharges an obligation, and therefore cannot be produced by a tool at any verification depth. (KA 14.3)

**Attributable improvement** — Raw improvement less the counterfactual improvement; the difference of the two differences. (KA 2.3)

**Attribution** — The portion of a measured improvement plausibly caused by this project. (KA 2.3)

**Authority bounds** — The spend, scope, resource and commitment limits within which the leader decides alone. (KA 4.1)

**Avoid / reduce / transfer / accept** — The threat-response families; transfer is a price, not a disappearance. (KA 8.3)

## B

**BAC** — Budget at Completion: the time-phased cost baseline's total. (KA 7.1)

**Back-translation** — Independent re-translation to the source language as a check on anything carrying a commitment, instruction or safety implication. (KA 11.4)

**Backward traceability** — Design or test  $\rightarrow$  approved requirement; detects unrequested scope, and costs one query. (KA 5.2)

**Base rate** — The prevalence of the condition a monitor looks for; with the saving-to-investigation ratio it decides how a monitor should be tuned. (KA 8.4)

**Base-rate difference** — Unequal prevalence between groups, which makes simultaneous equalisation of all error measures impossible. (KA 14.4)

**Baseline** — The measured pre-change position; must be measured before, not reconstructed after. (KA 2.3)

**Baseline drift** — Cumulative movement through individually authorised small changes, with no decision on the total. (KA 4.3)

**Baseline maintenance** — Applying changes by accumulation, with traceability to authority, preserving the original. (KA 4.3)

**Basis of an attitude entry** — The observed act and its source, recorded instead of a characterisation of the person; the form an assessment takes in a record that names identified individuals. (KA 11.1)

**Benefit per unit of adoption-hours (U)** — clinics  $\times$  valuation rate  $\times$  operating weeks, the constant in  $\text{benefit} = U \times a \times h$ . (KA 16.4)

**Benefits bridge** — The reconciliation from gross claimed benefit through eliminations and adoption to net realistic benefit. (KA 15.2)

**Benefits map** — Output  $\rightarrow$  enabling change  $\rightarrow$  outcome  $\rightarrow$  benefit  $\rightarrow$  objective, with owners and measures. (KA 2.3)

**Benefits measurement plan** — Measure definition, baseline and date, target profile, method, frequency, owner, **fund type**, reporting route and end date, per benefit. (KA 16.4)

**Benefits owner** — The accountable person outside the project who will realise the benefit. (KA 2.1)

**Benefits profile** — The time-phased benefit stream; ramps in reality, flat in most business cases. (KA 2.2)

**Best alternative** — What happens absent agreement, priced fully: invoice cost plus delay at the cost of delay plus internal effort plus risk as an **EMV**. (KA 11.3)

**Binding constraint** — The resource that actually limits the portfolio, usually a scarce capability, rarely money. (KA 15.2)

**Binding step** — The step whose yield improvement raises **RTY** most; identifiable from the yields alone. (KA 9.4)

**Blended rate** — Weighted average cost per hour of a mixed resource pool. (KA 7.4)

**Blocked work in progress** — Blocked arrival rate  $\times E[\text{wait}]$ : the items a team holds but cannot work on, invisible on a board. (KA 13.3)

**Boundary exposure** — The priced consequence of an unstated extent: additional units  $\times$  unit cost. (KA 5.1)

**Bounded envelope** — Stated limits within which a delivery team decides freely, reviewed rather than approved decision by decision. (KA 3.1)

**Bounded funding envelope** — Authority to spend a stated number of iterations before continuation requires an explicit decision. (KA 13.4)

**Breakeven adoption** — The adoption level at which NPV is zero, a more useful board sentence than NPV. (KA 2.2)

**Breakeven breach cost** — Value forgone by reserving capacity  $\div$  the improvement in failure probability it buys. (KA 15.3)

**Breakeven defective fraction (p)** —  $c/u$ : verification cost  $\div$  escaped-defect cost; above it, full verification pays. Independent of **n**. (KA 9.3)

**Breakeven dispute probability** — The dispute rate at which writing testable criteria exactly pays for itself; **2.67 %** at Meridian against an observed 25 %. (KA 5.4)

**Breakeven elicitation spend** — The investment at which the saving from earlier detection is exactly recovered; expressed either as a sum or as a tolerable delay. (KA 5.2)

**Breakeven elimination rate** — The elimination rate at which a portfolio just satisfies its investment decision rule. (KA 15.2)

**Breakeven error probability (P)** —  $C_v \div [(1 - q) \times L]$ : the error rate above which a verification pays for itself. (KA 1.4)

**Breakeven fidelity** — Twin cost  $\div$  gross benefit at perfect fidelity: the fidelity below which the twin destroys value. (KA 14.2)

**Breakeven retrieval rate** — review cost  $\div$  (lessons  $\times$  avoided cost per applied lesson): the threshold above which reviewing pays. (KA 16.3)



**Breakeven rework probability** –  $p^* = \text{good outcome} + (\text{good} - \text{bad})$ ; the probability at which an overlap stops paying. (KA 6.4)

**Breakeven settlement premium** –  $\text{carrying cost} \times \text{months saved} + (\text{EMV of determination} - \text{own assessed value})$ . (KA 16.2)

**Breakeven value of a released hour** –  $\text{Incremental direct cost} \div \text{released hours, USD}$  12.65 on Auriga, 6.836 % of the leader's rate. (KA 12.3)

**Breakeven volume (Q)** – The volume at which two options cost the same:  $(F_{\text{make}} - F_{\text{buy}}) / (v_{\text{buy}} - v_{\text{make}})$ . (KA 10.1)

## C

**Capability / conduct / circumstance** – The three causes of under-performance, requiring training, consequence and support respectively. (KA 12.3)

**Capability-specific capacity** – Capacity modelled per scarce capability, since the portfolio consumes them in component-specific mixes. (KA 15.3)

**Capacity-based contract** – A funded team at a rate for a cadence; the buyer bears quantity risk, change is free at the boundary. (KA 13.4)

**Capture-recapture estimate** – Register completeness inferred from the overlap of two independent identification methods:  $\hat{N} = n_1 n_2 / m$ . (KA 8.1)

**Carrying cost of an open item** – Advisory + internal management + opportunity cost of locked cash, per period; invisible in every ledger. (KA 16.2)

**Cascade (enabler) double count** – An enabling component claiming benefit already claimed by the components it enables. (KA 15.2)

**Cash-releasing / non-cash-releasing** – Benefits that reduce spend versus those that release capacity. (KA 2.3)

**Cashability confirmation** – The written confirmation, from the finance owner of the affected budget, that a claimed saving is budget-removable and from which period. Until it exists the claim is carried as capacity-released. (KA 15.2)

**Causation** – The demonstrated link from the contractual basis to the effect claimed; where most claims are decided. (KA 10.4)

**Cause concentration** – The share of non-conformances attributable to the single largest root cause. (KA 9.3)

**Cause → event → consequence** – The three-part risk statement; each part maps to a response type. (KA 8.1)

**Ceiling price** – The cap on the buyer's total exposure under an incentive contract. (KA 7.4)

**Change** – An assessed, authorised, recorded movement of the scope baseline (Domain 4, KA 4.4). (KA 5.4)

**Change board** – The body authorised to approve baseline changes within limits, including a technical authority. (KA 4.4)

**Change vs clarification vs defect** – Baseline moves · baseline already covered it · work fails to meet the baseline. (KA 4.4)

**Channel load** –  $\text{Channels} \times \text{hours per channel per period}$ ; compared with engagement capacity to give the sustainable party count. (KA 11.2)

**Charter** – The document authorising the project and conferring bounded authority on its leader. (KA 4.1)

**Claim boundary** – What a sustainability or social-value measure counts and excludes, over what period and against which baseline; without it the number is not checkable. (KA 2.3)

**Closing account** – Reconciliation of realised value against the promise, decomposed into its causes and reported to the approving body. (KA 16.4)

**Coaching time per report** –  $(R - e(n - b)) / n$ ; 65.00 minutes at 6 reports on Auriga, 17.50 at 12. (KA 12.3)

**Commensurability** — Whether two figures may be compared directly; a capital variance and an annual benefit may not, without a horizon and a rate. (KA 1.1)

**Commissioning** — Progressive proving of a deliverable in its operating environment, from component test to performance demonstration. (KA 16.1)

**Commitment** — A contractual obligation (e.g. a PO); a funding fact, not yet a cost. (KA 7.2)

**Commitment coverage** —  $(AC + \text{open commitments}) \div EAC$ : the share of the forecast whose price is already fixed; 74.72 % on Auriga. (KA 7.2)

**Commitment exposure** — Range width  $\times$  cost of delay, what a date quoted from a ranged package risks. (KA 6.3)

**Commitment lead-time rule** — The rolling-wave horizon must be at least as long as the longest commitment lead time inside it. (KA 4.1)

**Committee capacity** — Meetings per year  $\times$  substantive items per meeting; compared against decision demand, average, peak and off-peak. (KA 3.2)

**Common data environment (CDE)** — A single agreed information store with explicit item states and authorised, recorded transitions. (KA 14.1)

**Communication architecture** — The designed set of channels between parties (mesh, routed or hybrid) with its load computed against capacity. (KA 11.2)

**Communication path** — One pairwise relationship in a team; counted by  $n(n - 1) / 2$  (Domain 4, KA 4.2.3). (KA 12.2)

**Community stakeholder** — An unbounded, heterogeneous group with a legitimate interest in a project's effects, often with statutory engagement rights that vary by jurisdiction. (KA 11.4)

**Comparison cohort** — Comparable units measured at baseline and review without receiving the change, used to net out other causes. (KA 16.4)

· Also read at KA 2.3: A group that did not receive the change, used to measure the counterfactual improvement.

**Competence limit** — The boundary beyond which a leader may not sign an opinion, estimate or assessment; the professional act is to obtain the competence or to decline in writing. (KA 1.2)

**Component** — A project or enabling activity within a programme, with a named owner and a stated consequence of cancellation. (KA 15.1)

**Composite fitness** — The product of dimension conformance rates; the share of records fit for use. (KA 9.4)

**Concentration line** — The share of portfolio cost sitting in components below the alert threshold; the companion to any aggregate index. (KA 15.4)

**Concession (use as is)** — Acceptance of a nonconformance as it stands, authorised by the party bearing the consequence. (KA 9.3)

**Condition-based handover** — Handover triggered by a satisfied condition set rather than a calendar date. (KA 16.1)

**Condition-closure rate (r)** — The share of a gate's conditions verified closed by the following gate. (KA 3.3)

**Conditional acceptance** — Acceptance with open items; priced by moving each item up the correction ladder by **severity class**. (KA 5.4)

**Conditional pass** — Proceeding subject to conditions: a real instrument only if the conditions have owners, dates and consequences. (KA 3.3)

**Confidence bound (p\_upper)** —  $1 - \alpha^{(1/n)}$ , the largest defective fraction consistent with a clean sample of  $n$ . (KA 9.3)

**Confidentiality duty** — The standing obligation to hold entrusted information for the purpose it was given for and no other; it survives the project and the tool, and predates both. (KA 1.2)

**Configuration audit** — Verification that the actual state matches the recorded state. (KA 4.3)

**Configuration management** — Knowing which version of every controlled item is approved, and how it came to be. (KA 4.3)

**Confirmation rule** — The provision that stops a tolerance trigger firing on one period's measurement noise. (KA 3.3)

**Conflict of interest** — An interest that bears on a matter the person is being asked to decide, advise on or assure. Not a lapse of character but a structural fact about a role: the professional act is to declare and abstain, not to claim to have none. Where abstention cannot manage it, the role changes rather than the record staying silent. (KA 1.2)

**Conflict source** — Structural, resource, information, interest or interpersonal, each with a different remedy. (KA 11.3)

**Consent risk** — Whether the party holds a veto, approval, licence, signature or practical ability to withhold cooperation; a property of position, not disposition. (KA 11.1)

**Consent risk pricing** —  $EMV$  of a consent failure, presented with its breakeven cost and required probability reduction, never as a point estimate alone. (KA 11.4)

**Consent-risk floor** — A minimum engagement allocation to a party that can stop the work, justified by breakeven probability rather than by benefit rate. (KA 11.1)

**Consequence per defect ( $u_i$ )** — The internally assessed cost of one defect of a class reaching use; assessed, owned and stated. (KA 14.1)

**Consequence threshold** — A point where the consequence function steps rather than scales; where resilience is bought. (KA 8.4)

**Consequence-weighted exposure** —  $\sum n_i d_i u_i$ : the expected cost of a data estate's defects. (KA 14.1)

**Consistency check** — A specific question testing whether two subsidiary plans agree. (KA 4.1)

**Consolidation time (C)** — Elapsed time from data cut-off to pack issue. (KA 11.2)

**Constraint** — An externally imposed date; modelled as logic, not as a pin. (KA 6.3)

**Containment layer** — A review, test or inspection with a stated coverage, detection rate and unit correction cost. (KA 9.2)

**Contestability** — The affected party's ability to challenge a decision and have it examined by a human with authority to change it. (KA 14.4)

**Contingency draw index** —  $(\text{Contingency drawn} \div \text{contingency}) \div \text{percent complete}$ ; above 1.00 the reserve is depleting faster than the work is being done. (KA 7.1)

**Contingency reserve** — Inside the baseline; funds identified risks; PM-controlled. (KA 7.1)

**Continuation decision** — The periodic judgement that the next increment's cost is worth its expected value; the capacity model's brake. (KA 13.4)

**Contract growth** — Final account against original sum, variations *and* settlements, not variations alone. (KA 16.2)

**Control account** — The level at which performance is measured and reported (Domain 7's earned value attaches here). (KA 4.2)  
· *Also read at KA 7.1*: The WBS  $\times$  organisation point where scope, budget and actuals integrate.

**Control schedule** — The L3 CPM network under change control; home of float and progress. (KA 6.1)

**Convergence point** — A node where paths merge; slippage compounds there. (KA 6.2)

**Coordination share** —  $c(n - 1) / (2h)$ , coordination overhead as a fraction of team capacity;  $(n - 1) / 160$  at  $c = 0.5$ ,  $h = 40$ . (KA 12.2)

**Correction asymmetry** — The observed tendency of corrections to propagate less well than the original claim; real, and not reduced here to a multiplier. (KA 11.4)

**Correction ladder** — The cost of correcting one requirement defect by the stage found; Meridian's observed ladder rises  $\times 4$  per stage. (KA 5.2)

**Correction-cost ladder** — The rising cost of correcting a defect by the layer at which it is found. (KA 9.2)

**Correlation** — Shared drivers that raise aggregate variance; independence is usually optimistic. (KA 8.2)

**Cost of conformance** — Prevention + appraisal: money spent so defects do not occur or do not escape. (KA 9.1)

**Cost of delay** — The benefit forgone per period of lateness. (KA 1.3)

**Cost of nonconformance** — Internal + external failure: money lost because they did. (KA 9.1)

**Cost of quality (CoQ)** — Prevention + appraisal + internal failure + external failure. (KA 9.1)

**Cost per crew-week of capacity** — The priced rungs of Domain 6's capacity ladder, from 15,200 at straight time to 45,000 for accepting the delay (7.4.1c). (KA 7.4)

**Cost per productive hour** — Burdened pay  $\div (1 - \text{non-productive share}) \times (1 + \text{overhead})$ ; USD 100.79 against a USD 95.00 charge-out rate. (KA 7.4)

**Cost slope** — An activity's crash cost per week, with a technical limit on weeks available. (KA 6.4)

**Count view / size view** — Progress by item count against progress by size delivered; the first overstates when sequencing is optimal. (KA 13.3)

**Counterfactual cost** — Cost incurred *only* in the do-nothing case, and therefore value created by avoiding it. (KA 2.2)

**Counterfactual test** — What happens to the benefit if this bundle is not delivered? A zero answer means zero attributed value. (KA 5.3)

**Counterparty risk (priced)** — The secondary EMV of a transfer:  $p \times P(\text{cannot pay}) \times \text{amount recovered}$ , usually correlated with the event transferred. (KA 8.3)

**Coverage** — Distinct risks identified as a share of the estimated population; an optimistic figure, since shared blind spots deflate the population and so inflate it. (KA 8.1)

**CPI SPI** —  $EV/AC \cdot EV/PV$ , as ratios. (KA 7.3)

**Crashing / fast-tracking** — Buying duration with money / with overlap risk. (KA 6.4)

**Crisis declaration** — The named role's act that changes stated authorities (a stop-work authority, a stated fund against a named reserve, a standing cadence), and which someone other than the crisis lead stands down. (KA 8.4)

**Criterion influence** —  $(\text{score range}) \times \text{weight}$ : the most a criterion can move a total, and therefore what the model cannot decide. (KA 2.2)

**Critical chain** — The binding sequence once logic *and* capacity are honoured; need not be the critical path. (KA 6.3)

**Critical path** — The zero-float longest path; the project's shortest possible duration. (KA 6.2)

**Crossover horizon** —  $R \times (1 + d/g)$ : the remaining work below which reinforcement finishes later, not earlier. (KA 1.3)

**Crossover weighting (w)** — The price weight at which two bidders' weighted scores are equal. (KA 10.2)

**Cumulative test** — The delegation-schedule provision that related decisions aggregating above X within period P require the authority appropriate to the aggregate. Setting X and P is arithmetic, not judgement, and is derived in Domain 4, KA 4.3.3, a round number chosen without reference to the observed decision rate provides the appearance of a control and none of the function. (KA 3.3) · *Also read at KA 4.3:* A threshold on aggregated related changes over a stated period, set from the observed change rate.

**Cumulative-effect test** — The rule that related concessions aggregating above a threshold require the authority appropriate to the aggregate. Operable only once five things are fixed in the quality plan: who sets the threshold (sponsor or design authority, at baseline), how it is derived, the aggregation period, the relatedness test, and the named owner of the running total. (KA 9.3)

**Custody handover** — The recorded transfer of each record class to a continuing custodian at closure, so records outlive the temporary organisation. (KA 3.3)

**CV SV** —  $EV - AC \cdot EV - PV$ , in currency. (KA 7.3)

**Cycle time (C)** — Elapsed time from an item starting to finishing; a consequence of  $W$  at a given  $T$ . (KA 13.2)

**Cycle-time decomposition** — Splitting average cycle time into its blocked and unblocked populations to locate the delay. (KA 13.3)

## D

**Dangle** — An activity missing a predecessor or successor; a network defect. (KA 6.1)

**Data class schedule** — One row per data class: owner, definition, source of record, quality standard, access and retention. (KA 14.1)

**Data minimisation** — Holding only what the project needs, which reduces the security and privacy problem rather than defending it. (KA 14.4)

**Data quality dimensions** — Completeness, accuracy, validity, consistency, timeliness, uniqueness. (KA 9.4)

**Date constraint** — A pinned date overriding logic; each one weakens the model. (KA 6.1)

**Daylight test** — Would every party, seeing the full picture, still regard this as impartial? (KA 1.4)

**Decidability** — The property that every decision has exactly one accountable decision-maker with sufficient authority. (KA 3.1)

**Decidable scope** — Enumerated deliverables, quantified extent, named interfaces, explicit exclusions. (KA 5.1)

**Decision action window** —  $(\text{remaining duration} - \text{escalation latency}) \div \text{remaining duration}$ ; negative means the escalation is ceremonial. (KA 3.3)

**Decision breakeven adoption** — The adoption at which one priced decision stops paying, 36.76 % for Meridian's acceleration. Distinct from Domain 2's **breakeven adoption**, which is the level at which the whole investment's NPV is zero (41.05 %). (KA 1.3)

**Decision convention** — Whether a meeting decides or ratifies, and whether authority is individual or collective. (KA 11.4)

**Decision record** — The versioned, attributable log that converts a decision from a memory into an institutional fact. (KA 3.3)

**Decision right** — The authority to make a specified class of decision, bounded by value, scope and reversibility. (KA 3.1)

**Decision tree** — Explicit rollback of decisions and chance events to a present choice. (KA 8.2)

**Declared interest** — Anything capable of affecting, or of appearing to affect, judgement on a matter before a person (financial, employment, personal, authorship of what is being assured, or a future role dependent on the outcome) disclosed in writing to the chair before the item is taken. *Context flag: the stakeholder sense of "interest" (attention paid) is Domain 11, KA 11.1.* (KA 1.2)

**Declining the signature** — The refusal to sign an acceptance, readiness certificate, forecast or account the leader does not believe, stated in writing with what would have to change. (KA 12.4)

**Decoupling** — Removing a predecessor from a milestone by designing an operable interim arrangement, multiplying the probability by  $1/p_i$ . (KA 15.1)

**Defect rate by class ( $d_i$ )** — The share of records in a class failing that class's defined pass criteria. (KA 14.1)

**Defects-liability period** — The interval after acceptance in which the contractor must rectify defects that appear. (KA 16.2)

**Definition of done** — The pre-agreed, binary, testable standard that gives the word "done" one meaning across every report. (KA 13.1)

**Delay-cost density** — Cost of delay per week ÷ effort in team-weeks; sequencing in decreasing order minimises total delay cost. (KA 13.1)

**Delegation** — Transfer of authority to decide and act, with the information and criteria to do so; accountability is retained. (KA 12.3)

**Delegation schedule** — The statement of authority by decision class, on value, reversibility and externality. (KA 3.2)

**Deletion point** — The stated end of the register: what is deleted at closure, what is retained, for how long and on whose authority, with departed role holders' assessments deleted at each refresh. (KA 11.1)

**Delivery verdict** — The project-level judgment: was the output delivered to time, cost and quality? A one-off variance. (KA 1.1)

**Depth and kind** — Two separate verification choices: how hard to look, and what sort of looking finds this class of error. (KA 1.4)

**Derived audit tolerance** — The item-level tolerance obtained by inverting  $p^k$  from an interface-level target:  $p = q^{(1/k)}$ . (KA 4.3)

**Descope exchange rate** —  $1/CPI$ , the forecast cash saved per unit of budgeted scope removed; USD 1.10 per dollar on Auriga. (KA 7.3)

**Detection rate ( $d_i$ )** — The share of defects reaching a layer that the layer finds. (KA 9.2)

**Detection rate ( $q$ )** — The share of material errors present that a given review depth finds; a measured quantity, never asserted. (KA 14.3)

**Deterministic bias** —  $t_e - m = (o - 2m + p)/6$ , one sixth of the estimate's skew. (KA 6.4)

**Deterrence cap vs compensation cap** — The service-credit cap needed to exceed the supplier's cost of compliance, against the cap needed to cover the buyer's loss; they differ. (KA 10.3)

**Differential error** — A systematic difference in error rates between identifiable groups; measurable only by disaggregated measurement. (KA 14.4)

**Digital delivery environment** — The set of systems holding a project's authoritative information and mediating its work. (KA 14.1)

**Digital twin** — A model kept synchronised with a real asset through a data connection; without synchronisation it is a drawing. (KA 14.2)

**Disclosed claim** — A benefit reported outside the organisation; needs a stated boundary and method, a named owner, retained evidence and the signing function's approval, and is never approved by the project. (KA 2.3)

**Discovery arrivals** — New items found necessary during build; subtracted from throughput to give the net drain. (KA 13.2)

**Discretionary attention** — The leader's working week less recurring commitments; the resource all leadership acts are paid from. (KA 12.1)

**Discretionary condition** — A readiness condition that admits degree, is genuinely uncertain at the decision date, and is therefore assessed as a probability. (KA 16.1)

**Displaced judgment** — Ceasing to think because a tool is watching. (KA 8.4)

**Disruption / measured mile** — Lost productivity on unchanged work, established by comparing an unimpacted period's productivity with the impacted period's. (KA 10.4)

**Dissent record** — The dated record of the fact stated, when and to whom, the decision taken, and the risk the leader believes remains — filed in the decision record, with a copy to a party other than the decision-maker. (KA 12.4)



**Distributed team** — A team whose members' working hours do not fully coincide; time distance, not spatial distance, is the governing variable. (KA 12.4)

**Do-minimum option** — The cheapest compliant alternative where inaction is not permissible; the honest comparator, and an option in its own right. (KA 2.2)

**Do-nothing baseline** — The comparison against which all benefits are measured; without it value is unmeasurable. (KA 2.2)

**Double-counting** — The same benefit claimed by more than one case; one claimant per benefit. (KA 2.3)

**Draw protocol** — The published rules for releasing contingency. (KA 8.3)

**Driver** — The reason the work exists; determines what success means and how it is measured. (KA 2.1)

**Duration multiplier** —  $1/(1 - r)$ , the hyperbolic effect of rework on elapsed time. (KA 9.2)

**Duty-engaged flag** — A mandatory register field marking a row whose realisation would engage a duty arising under law or regulation, an operating licence or a safety obligation. Such rows leave the EMV comparison before it is run: the duty is not transferable and acceptance is not available. Whether a duty is engaged is a question for counsel and the regulator, and the flag records who answered it. (KA 8.3)

## E

**EAC ETC** — Estimate at / to complete;  $EAC = AC + ETC$ . (KA 7.3)

**Early-warning indicator** — A leading measure tied to a risk, firing before the event. (KA 8.3)

**Earning rule** — The convention by which physical progress becomes EV; fixed before the period it measures. On Auriga's ten-package group the choice moved EV by 15.00 % of budget. (KA 7.3)

**Effective detection ( $q_{eff}$ )** — A gate's detection rate multiplied by its condition-closure rate:  $q_{eff} = q \times r$ . (KA 3.3)

**Elicitation** — Deliberate discovery of need, as distinct from collection of requests. (KA 5.2)

**Elimination rate** — Eliminations ÷ gross claimed benefit; invariant to the adoption assumption. (KA 15.2)

**Emergency change route** — A named authority, a short deadline and mandatory retrospective ratification. (KA 4.4)

**Emergency-route share** — Emergency changes ÷ total requests, a latency diagnostic, so set a review trigger on it rather than a target. (KA 4.4)

**Emotional intelligence** — Four practised operations (self-awareness, self-regulation, social awareness, relationship management) skills, not a trait score. (KA 12.1)

**Employment-records provisions** — Custodian, access list, copy to the individual, retention period from the organisation's policy, and disposal — settled with human resources, legal and the data-protection accountability holder, whose requirements differ by territory and are not stated in this book. (KA 12.4)

**EMV** — probability × impact; the average of outcomes that will not individually occur. (KA 8.2)

**Enabling change** — The non-project work (training, workflow, adoption support) without which no benefit occurs. (KA 2.3)

**Engagement objective** — A stated future state of a relationship with a date and a test, never an activity count. (KA 11.1)

**Enterprise decision architecture** — The set of governance tiers, the decision classes routed through them, and the resulting latency distribution. (KA 15.4)

**Enterprise delivery capacity** — The measured rate at which an organisation completes change, in the unit of its binding capability. (KA 15.3)

**Enterprise PMO** — The function that maintains portfolio data, runs allocation, measures capacity and gives second-line assurance; no decision rights. (KA 15.3)

**Entitlement** — The contractual basis for a claim; without it, merits are irrelevant. (KA 10.4)

**Enumeration** — Evaluating every feasible subset under the constraint; the optimum where the candidate set is small enough. (KA 5.3)

**Envelope coverage** — The share of a stream's above-authority decisions the delegated envelope contains. (KA 3.1)

**Error escape rate (e)** — The share of units on which a route lets a material error through; differs between manual and automated routes and must be measured. (KA 14.2)

**ES EF LS LF** — Earliest/latest start and finish per activity. (KA 6.2)

**Escalation class** — A defined trigger, destination, decision, latency and out-of-cycle route. (KA 3.3)

**Escalation duty** — The obligation to raise, with options, a conflict beyond one's authority. (KA 1.2)

**Escalation lead time** — How far before impact an escalation arrives; a shortening trend signals a decaying mechanism. (KA 3.3)

**Escalation of commitment** — Continuing because of sunk cost; countered by forward-looking framing. (KA 8.4)

· Also read at KA 2.4: Continuing because of what is invested; countered structurally.

**Escalation-grade message** — Exception, cause, impact priced, options, recommendation by name, decision required by whom and by when, and the date the leader knew. (KA 11.2)

**Escape** — A defect reaching the next stage, or the customer, undetected. (KA 9.1)

**Escape fraction** —  $\Pi(1 - d_i)$  across the chain, the share of introduced defects reaching the customer. (KA 9.2)

**Escape-cost asymmetry** —  $(1 - q_p)u_p \div (1 - q_o)u_o$ , why a plausible wrong number out-damages an obviously wrong one. (KA 14.3)

**Evaluation model** — Weights, normalisation, scoring scale and moderation rules; a piece of arithmetic that materially selects the winner. (KA 10.2)

**Excess work in progress cost** — Annual benefit run rate  $\times$  excess work in progress  $\div$  throughput, forgone once and never recovered. (KA 15.3)

**Exclusion** — A written statement of what is outside the boundary; the highest-return line in the document. (KA 5.1)

**Expected annual loss (EAL)** — Incident probability  $\times$  assessed impact; the residual after a control is the basis of comparison. (KA 14.4)

**Expected containment cost per introduced defect** — (Internal + external failure)  $\div$  defects introduced; the breakeven price of prevention. (KA 9.2)

**Expected cost of disruption** —  $\sum P(\text{state}) \times \text{consequence}(\text{state})$ ; compared against the certain cost of the structure that reduces it. (KA 10.4)

**Explainability (decision sense)** — The property that a named human can state why an output says what it says, in the vocabulary of the obligation. (KA 14.4)

**Exposed loss** —  $(1 - q) \times L$ : the loss a verification is actually protecting against, after any downstream control. (KA 1.4)

**Exposure per record** —  $d_{iu1}$ , the quantity that ranks remediation effort across classes. (KA 14.1)

**External acceptance** — Where an approval regime applies, a body outside the project may need to be notified of, or to accept, a repair, concession or regrade before use. It sits outside the project's authority and differs by sector and jurisdiction; the record names whether it is engaged and who established that. (KA 9.3)



**External-failure unit cost** — The average cost of one escaped defect in the operating context; set by context, not the project. (KA 9.1)

## F

**Factual-claims register** — Each claim in an external communication with its source document, retained with the issued item. (KA 11.4)

**Failure route** — The pre-written sequence for an output found materially wrong after reliance: stop and suspend, enumerate the decisions that cited it, notify in writing that day, reopen above the stated materiality threshold with the original authority, record whether anyone outside must be told and on whose determination, and re-measure under a named authority before resumption. (KA 14.4)

**False stop / false continue** — Stopping a project that would have created value; continuing one that will not. The two errors a criterion trades. (KA 2.4)

**Feasible frontier** — The top few feasible sets and what distinguishes them, the honest output of an enumeration. (KA 15.2)

**Feedback and delay** — Systems properties that make interventions look wrong before they look right. (KA 1.3)

**FFP / FPIF / CPFF / CPIF / T&M** — Contract models ordered by who carries cost risk. (KA 7.4)

**Final account** — The single agreed statement of the original sum, variations, settlements, deductions and final payment, signed as full and final. (KA 16.2)

**First-instance rule** — Hold the conversation at the first clear instance, when it is about an event rather than a person. (KA 12.3)

**First-time-right yield ( $y_i$ )** — The share of items passing a step without rework. (KA 9.4)

**Fitness for purpose** — Whether conforming, correctly graded output serves the use it was built for. (KA 9.1)

**Flat-equivalent basis** — Breakeven stated as the level a profile flat from year one would need (Meridian 41.05 %). (KA 2.2)

**Flip point** — The smallest weight shift that reverses a ranking; report it with the ranking or the ranking is unqualified. (KA 2.2)

**Flow efficiency** — Active work time as a share of cycle time; typically low, and where the improvement lies. (KA 13.2)

**Flow metric set** — Throughput, cycle-time distribution, work in progress with its blocked and aged shares, flow efficiency. (KA 13.4)

**Forward / backward pass** — Left-to-right ES/EF computation; right-to-left LS/LF computation. (KA 6.2)

**Forward breakeven** — The adoption or benefit level at which the *remaining* cost is just repaid; the only defensible level for a gate criterion. (KA 2.4)

**Forward exit cost** — The cost of stopping: termination, demobilisation, disposal; a forward cost, never sunk. (KA 2.4)

**Forward traceability** — Requirement → design → test → accepted deliverable; detects unmet scope. (KA 5.2)

**Forward-looking NPV** — Remaining benefit PV less remaining cost; the only basis for continuation. (KA 2.4)

**Free float FF** — Slip available without delaying any successor;  $FF \leq TF$ . (KA 6.2)

**FS / SS / FF / SF** — The four dependency types. (KA 6.1)

**Fund type** — Cashable (a named budget line is removed), capacity-released (time or space freed, no budget moves), cost-avoidance (a future outlay not incurred) or non-financial. Mandatory, because a total that mixes them cannot be tested against a cash rule. (KA 15.2)

**Fund type (carried from approval) –**

Cashable with the budget line named and the finance owner's written confirmation, capacity-released, cost-avoidance or non-financial (Domain 15, KA 15.2.1). It decides what the realised number may be compared against, and an unconfirmed cashable claim is measured as capacity-released. (KA 16.4)

**Funding absorbed** – Cash paid out – cash received; equals client exposure – payables + loss to date. Best stated in days of spend (44.80 on Auriga). (KA 7.4)

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## G

**Gate block** – The set of mandatory preconditions sitting above the readiness arithmetic; while any item is not met the only available decision is hold. (KA 16.1)

**Governance** – The decision rights, accountabilities and information flows through which an organisation directs and controls a project. (KA 3.1)

**Governance artefact** – A report, portal or template, evidence of governance, never a substitute for decision rights. (KA 3.1)

**Governance calendar** – The meeting and paper-deadline dates an approval event must be scheduled against, rather than approximated by a lag. (KA 6.1)

**Governance latency** – The expected wait for a committee decision:  $E[\text{wait}] = M/2 + L$ . (KA 3.2)

**Grade** – The level of capability or specification the requirements ask for; independent of quality. (KA 9.1)

**Greedy ranking** – Taking candidates in ratio order while they fit; a heuristic that strands capacity when candidates are lumpy. (KA 5.3)

**Gross latency-weeks** –  $\Sigma$  over decision classes of count  $\times$  cumulative tier latency; the portfolio's total waiting. (KA 15.4)

**Gross residual** – The sum of the reconciliation's error classes before netting. The figure that carries the finding. (KA 4.3)

**Grounding** – Supplying and identifying the material an output must be derived from, with nothing outside it permitted. (KA 14.3)

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## H

**Hallucination** – Fluent, confidently-stated content that is false. (KA 1.4)

**Handover** – Transfer of custody and operational accountability for a deliverable to a named receiving organisation, on stated conditions. (KA 16.1)

**Handover artefact** – What was done, what is in flight, what is blocked, what the next party must decide. (KA 12.4)

**Hard / soft constraint** – Physics, law, contract vs preference and convention; only soft ones are tradeable. (KA 2.1)

**Honest attribution (AI)** – Stating what was AI-generated and what was verified, and how. (KA 12.4)

**Honest gate** – Real stop option, pre-set evidence standard, independent deciders, recorded reasons. (KA 2.4)

**Honesty asymmetry** – Bad news costs the messenger and helps the project; leaders must make it safe. (KA 1.2)

**Hundred-per-cent rule** – The children of any element sum to exactly that element: an arithmetic invariant, and checkable. (KA 4.2)

**Hybrid boundary** – The named line where feedback stops being cheap and fast, and the control regime changes. (KA 13.3)

**Hybrid delivery** – A deliberate combination of predictive and adaptive methods under one governance frame, with a named boundary between them. (KA 13.1)

**Hypercare** – A bounded period of elevated support after go-live, with measured exit criteria and a receiving-organisation owner. (KA 16.1)

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## I

**Implied ETC efficiency** – The efficiency an EAC assumes for the remaining work; equal to the TCPI computed to that EAC. (KA 7.3)

**Increment** — A slice of product complete against the definition of done and therefore judgeable, releasable or rejectable alone. (KA 13.1)

**Influence** — Capacity to affect the project's decisions, resources or permissions; predicts the damage a disagreement does. (KA 11.1)

**Influence test** — A risk should be allocated to the party that can influence it; a risk allocated elsewhere is priced, not managed. (KA 10.3)

**Information age** —  $R/2 + G$ , the expected staleness of a fact at publication, for reporting period  $R$  and production lag  $G$ ; the registered  $E[\text{wait}]$  identity applied to information. (KA 14.2)

**Integrated change control** — Assessing and authorising a change across scope, schedule and cost together. (KA 4.4)

**Integrating point (hub)** — The function through which routed communication passes; carries its own overhead and its own distortion risk. (KA 11.2)

**Integration layer** — An architecture or interface contract that replaces  $n(n-1)/2$  pairwise agreements with  $n$ . (KA 13.3)

**Integration milestone** — The boundary event where an adaptive stream's ranged forecast enters a predictive network. (KA 13.3)  
· Also read at KA 6.4: The CPM node where an agile stream's delivery enters the network.

**Interaction term** — The product of both factor movements; the amount that shifts between decomposition lines when the order changes. (KA 16.4)

**Interest** — Attention paid, not stake held; predicts how quickly a party notices. (KA 11.1)

**Interest duty** — The four steps of 1.2.2a: identify at appointment and at each gate · declare in writing before the item is taken · record in the decision record, nil returns included · abstain, with the minute naming who decided instead. (KA 1.2)

**Interface** — A defined relationship across a boundary at which something must be agreed and verified. (KA 4.2)

**Interface agreement** — The parties, content, format, direction, timing, exceptions, verifier and version of one interface. (KA 4.2)

**Interruption load** — Observed unplanned escalation and interruption time; the largest and most designable line in the attention audit. (KA 12.1)

**Irrecoverable cost** — The share of a formal dispute's cost that cannot be recovered whatever the outcome; the figure that usually decides whether fighting can pay. (KA 10.4)

**Irreversibility** — The rising cost of correction as commitments harden, projects' defining failure mode. (KA 1.1)

**Issue** — A risk that has occurred; managed, not analysed. (KA 8.1)

## J

**Joint gain** —  $\Sigma(\text{value to one party}) - \Sigma(\text{cost to the other})$  over the issues traded; negative for issues that destroy value. (KA 11.3)

## K

**Key-person concentration** — The share of procedures executable by exactly one named individual; an availability risk no project register records. (KA 16.3)

**Kill criteria** — Pre-agreed stop conditions; powerful only because agreed in advance, and only useful if calibrated. (KA 2.4)

**Knowledge transfer** — Conversion of project knowledge into documentation, trained people and rehearsed procedures usable without the project team. (KA 16.3)

## L

**Lag / lead** — Imposed delay / overlap on a dependency. (KA 6.1)

**Last responsible moment** — The latest point a decision can be taken without foreclosing a needed option. (KA 1.3)

**Latency-to-cycle ratio** — Governance latency ÷ the delivery cadence it governs; above 1.0 the answer arrives after the increment that asked. (KA 3.1)

**Leadership (delivery context)** — The production of willing, aligned and informed effort under conditions the plan did not anticipate. (KA 12.1)

**Least-cost duration** — The duration maximising net saving; found where the marginal week's cost first exceeds the value of a week. (KA 6.4)

**Legal liability** — Who must compensate whom, set by contract and jurisdiction; distinct from professional accountability and capable of sitting elsewhere. (KA 1.2)

**Legitimately product-free work** — Integration, testing, migration, training delivery, commissioning; real work with no product, and the class a product-derived WBS omits by construction. (KA 4.2)

**Level of effort** — Earning by calendar ( $EV \equiv PV$ ); can never show schedule variance. (KA 7.3)

**Level variance** — The value effect of a wrong steady-state adoption or benefit-per-unit assumption. (KA 16.4)

**Level-of-effort distortion** —  $s \times (1 - SPI\_d)$ : the amount a level-of-effort share  $s$  adds to a reported  $SPI$ ; 0.28 at a 70 % share with discrete  $SPI$  0.60. (KA 7.3)

**Lineage** — The reconstructable path from source record to a displayed value. (KA 14.1)

**Link cost  $c$**  — Hours of total team time consumed per active pairwise relationship per week; a locally calibrated planning figure. (KA 12.2)

**Little's Law** —  $W = T \times C$  for a system in steady state, independent of arrival and service distributions. (KA 13.2)

**Local optimisation** — Function-level efficiency that degrades whole-system performance. (KA 1.3)

**Logic density** — Links ÷ activities; the surplus over  $n - 1$  counts the network's convergences. (KA 6.1)

**Logrolling (trading whole issues)** — Conceding entirely on issues valued asymmetrically, rather than partially on all of them. (KA 11.3)

**Long-lead item** — A component whose procurement chain exceeds the time available after baseline approval, and must therefore be committed earlier. (KA 10.1)

## M

**Management (delivery context)** — The allocation and control of resources against a plan. (KA 12.1)

**Management reserve** — Outside the baseline; unknown-unknowns; sponsor-controlled. (KA 7.1)

**Mandatory precondition** — A go-live condition recorded met or not met, with its approving authority named; it carries no probability and admits no economic trade. (KA 16.1)

**Marginal capacity of the  $n$ th member** —  $h - c(n - 1)$ ; zero at  $n = 81$  under the same parameters. (KA 12.2)

**Marginal value of the constraint** —  $V(n) - V(n-1)$  from enumeration; lumpy, non-monotone, and a property of the candidate set. (KA 2.2)

**Marginal-dollar allocation identity** — For any payment mechanism, buyer share + supplier share of the next dollar of cost = 1; a contract allocates cost, it does not remove it. (KA 10.3)

**Material error rate ( $p$ )** — The share of items of a class containing a material error before verification; measured per class and per configuration. (KA 14.3)

**Maximum blindness** —  $P + C + L$ , how long a problem arising just after a cut-off stays invisible to the decision body. (KA 11.2)

**Maximum sustainable span** —  $n \leq (R + eb) / (T + e)$ ; 7.6 on Auriga, so 7 reports. (KA 12.3)

**Meeting interval ( $M$ )** — The period between meetings of a decision body. (KA 3.2)

**Member vs attendee** — Members decide; attendees inform. Conflating them enlarges the body and reduces its authority. (KA 3.2)

**Mesh vs layered** —  $n(n-1)/2$  possible interfaces against  $n$ , the architecture choice priced in WE 4.2.3. (KA 4.2)

**Milestone** — Zero-duration event with an owner and a done-definition; its date is computed. (KA 6.3)

**Mixed-stream test** — An investment rule applied to a benefit stream of more than one fund type. Permitted as a committee's own convention, but only where the composition of the denominator is disclosed with the result. (KA 15.2)

**Model retention record** — Model identifier, version and date alongside input, output, verifier and decision; the minimum that makes an AI-informed decision explainable. (KA 16.4)

**Monte Carlo simulation** — Repeated sampling to produce an outcome distribution rather than a point. (KA 8.2)

**Multiplication rule** — The probability a milestone requiring all of  $k$  independent predecessors is met is the product of their probabilities. (KA 15.1)

**Must-have inflation** — Originator-assigned categories collapsing into near-universal "must have", destroying the scheme's information content. (KA 5.3)

## N

**Named owner** — The human who verified an AI-assisted artefact and answers for it. (KA 1.4)

**Near-critical band** — The float threshold below which a path joins the monitored set; a decision, not a convention. (KA 6.2)

**Need-by time** — The point beyond which a decision no longer changes what the team does; the test of whether a body can serve it. (KA 3.1)

**Net drain** — Gross throughput minus arrival rate, the rate at which a backlog actually shrinks. (KA 13.2)

**Net team capacity** —  $hn - c \cdot n(n-1)/2$ ; maximised at 80–81 people under the illustrative parameters. (KA 12.2)

**Non-financial ceiling** — The working-time, rest, fatigue, safety-case, permit or agreement limit bounding a compression lever. Supplied by the safety function, HR or counsel, jurisdiction- and sector-specific, and where it binds the lever is unavailable at any price. (KA 6.4)

**Nonconformance** — Output that does not meet its specified requirement. (KA 9.3)

**Notice provision** — The required form and period for notifying a claim; in some contracts and jurisdictions a condition precedent. (KA 10.4)

**Notice-bar exposure** — The value of the heads of claim that depend on valid notice, grossed up for overhead and profit. (KA 10.4)

**Notification map** — The one-page record, written before any incident, of which external notifications could fall due, who determines that they do, to whom, within what period and who may make them. Whether any obligation applies is for the relevant authority and qualified counsel, never for the project alone. (KA 8.4)

## O

**Obligation item** — A backlog item carrying a safety, regulatory, statutory, contractual or security-remediation obligation. A constraint on the order, not a candidate within it: scheduled to its required date by the authority accountable for the obligation and excluded from the delay-cost-density ranking. (KA 13.1)

**Omission credit** — The rate at which removed scope is credited back, typically below the bid rate, and the asymmetry that prices a net-zero change. (KA 13.4)

**Operational transition** — Establishment of a funded, staffed, permanent capability to run and improve the delivered service. (KA 16.2)

**Ordering right, and its two bounds** — The product owner orders *discretionary* value, within a value envelope above which decisions escalate and outside the obligation items the ordering right does not reach. Deprioritising an obligation item exceeds the right rather than exercising it. (KA 13.1)

**Original-to-current reconciliation** — Original baseline +  $\Sigma$  approved changes = current baseline, exactly, every reporting period. (KA 4.3)

**Orphan** — A design element or test case tracing to no approved requirement. (KA 5.2)

**Orphan work package** — Budgeted work producing no product anyone asked for: waste that still reports 100 % complete. (KA 4.2)

**Out-of-cycle mechanism** — A written-resolution or delegated-authority route used when the ordinary cadence is too slow. (KA 3.3)

**Out-of-cycle route** — Domain 3's mechanism (KA 3.3.3) applied at enterprise scale, where it is worth most: the saving is largest at the slowest tier, because  $E[\text{wait}] = M/2 + L$  grows with  $M$ . (KA 15.4)  
 · Also read at KA 10.3: Domain 3's mechanism (KA 3.3.3) applied across a contract boundary: named authorities on **both** sides, since a route that exists for only one party is not a route.  
 · Also read at KA 13.3: Domain 3's mechanism (KA 3.3.3) applied where governance latency exceeds the iteration length, the specific mismatch KA 13.3 quantifies.

**Outcome / deliverable / activity registers** — The three levels routinely conflated; scope is the deliverable register, bounded by outcomes and decomposed into activities. (KA 5.1)

**Output / outcome / benefit / value** — Delivered thing · change in how things work · measurable improvement · benefit against cost and risk. (KA 1.3)

**Output vs input specification** — Specifying what must be achieved against how it must be built; determines who is answerable for fitness for purpose. (KA 10.1)

**Outturn ratio** — Final cost ÷ approved cost for a comparable completed project; the unit of a reference class. (KA 8.4)

**Over-claim share** — *counterfactual* ÷ *raw*, invariant to the valuation rate, the operating year and the number of adopters. (KA 2.3)

**Over-trust through fluency** — Polished output suppressing the scrutiny rough output would attract. (KA 1.4)

**Overlap window** — A deliberately created shared period, bought with unsocial hours that must be explicitly allocated. (KA 12.4)

## P

**P50 / P80** — Confidence levels, amounts covering the aggregate outcome with stated probability. (KA 8.2)

**Paid-to-productive multiplier** —  $1 / (1 - n)$ , 1.17647 at a 15 % non-productive share; the commonest silent under-funding in a labour estimate. (KA 7.4)

**Pairwise configuration integrity** —  $p^k$ , the conformance of a relationship spanning  $k$  controlled items, always worse than the item rate  $p$ . (KA 4.3)

**Paper lead time (L)** — How far before a meeting submissions close. (KA 3.2)

**Partial horizon** — One element planned to work-package level ahead of the wave because its lead time demands it, labelled as such. (KA 4.1)

**Party readiness (p)** — The probability that a party arrives at a cycle able to approve; countable as the share of items approved at first presentation. (KA 3.1)

**Path float** —  $PD$  – path length; the slip a whole chain can absorb, shared by every activity on it. (KA 6.2)

**Path migration** — The critical path moving as durations change; why passes are re-run. (KA 6.4)

**People-work capacity** — Discretionary attention less the personal planning reserve: 6.5 h/week on Auriga. (KA 12.1)



**Performance measurement baseline —**

The authorised scope, arranged in time, with cost attached, one statement in three documents. (KA 4.3)

**Plan of plans —** The integrated set of subsidiary plans and the baselines they produce.

(KA 4.1)

**Plan survival probability —** The product over periods of the probability that each period's loss does not exceed its slack. (KA 15.3)

**Plan-set consistency audit —** Reading the subsidiary plans against one another and pricing the findings, before baseline approval (WE 4.1.2). (KA 4.1)

**Planning package —** A far-horizon element planned above work-package level, with its uncertainty stated. (KA 4.1)

· Also read at KA 6.3: A far-wave summary activity with ranged duration awaiting elaboration.

**Pod (structured team) —** A sub-team with end-to-end ownership and one named path outward: the human form of Domain 4's layered architecture. (KA 12.2)

**Point of total assumption (PTA) —** The cost above which the supplier bears 100 % of further overrun (Domain 7, KA 7.4.3); read here as the point at which behaviour turns from efficiency to entitlement. (KA 10.3)

· Also read at KA 7.4: The cost above which the seller bears 100 % of further overrun.

**Portfolio —** The set of programmes and projects an organisation chooses to run within a finite capacity and a strategy. (KA 15.1)

· Also read at KA 1.1: The funded, balanced set of projects and programmes.

**Portfolio benefits register —** The authoritative statement of portfolio benefits, organised by benefit rather than by component, each carrying a measure, a baseline, a receiving-organisation owner, a fund type and its contributing components. (KA 15.2)

**Portfolio one-page view —** The report supporting exactly three decisions: continue/change/stop, reallocate, escalate. (KA 15.4)

**Portfolio process —** Comparable description, explicit criteria, funding to capacity, periodic re-decision. (KA 2.1)

**Portfolio work in progress —** The number of initiatives in flight; by Little's Law it sets cycle time, not throughput. (KA 15.3)

**Post-project review —** Structured examination of what happened and why, to improve future decisions, separated from performance assessment. (KA 16.3)

**Pre-decision consultation —** Consultation capable of changing the design, engagement; as distinct from post-decision notification. (KA 11.4)

**Pre-mortem —** Imagining failure before committing, to license dissent and surface missed risk. (KA 8.4)

**Precedence rule —** The pre-agreed rule for which authority prevails in a matrix conflict. (KA 3.1)

**Precision —** True alerts ÷ all alerts; worth investigating while precision exceeds investigation cost ÷ saving. (KA 8.4)

**Preserve rather than tidy —** From declaration, working papers, logs, messages and superseded drafts are retained as they are; the mess is the evidence that the account was reasonable on what was known. (KA 8.4)

**Price normalisation —** The formula converting prices into scores; the ratio and linear conventions give different scores and different crossovers. (KA 10.2)

**Price of confidence —** The cost per percentage point of moving a reserve between confidence levels. (KA 8.2)

**Priority rule —** The heuristic (commonly minimum total float first) used to break resource conflicts; beatable, and known to be. (KA 6.3)

**Procurement —** Converting a need the project cannot meet internally into an enforceable obligation held by another organisation, and managing it to closure. (KA 10.1)

**Procurement lead time** — Process legs + governance latency + physical legs; a schedule predecessor, not an administrative overhead. (KA 10.1)

**Product breakdown structure** — Decomposition of the thing delivered; drives configuration management and acceptance. (KA 4.2)

**Product owner** — The single individual accountable for the order of the backlog within a value envelope set by the sponsor. (KA 13.1)

**Product-to-work cross-match** — The traceability test that every product has work and every work package a product: a candidate list, not a finding. (KA 4.2)

**Production lag (G)** — Elapsed time from data cut-off to publication; the more powerful of the two staleness levers, day for day. (KA 14.2)

**Programme** — A temporary organisation delivering a coherent outcome through components that cannot deliver it individually. (KA 15.1)

· *Also read at KA 1.1:* Related projects managed together for outcomes no one project delivers.

**Programme dependency register** — The record of cross-boundary dependencies, whose critical column is the owner on the giving side. (KA 15.1)

**Programme verdict** — The programme-level judgment: did the outcome and its benefit materialise? An annual flow. (KA 1.1)

**Progressive elaboration** — Adding detail as it becomes knowable, at a stated horizon and cadence. (KA 4.1)

**Project** — Temporary endeavour producing a defined result. (KA 1.1)

**Prolongation** — Time-related costs of extended duration; requires a demonstrated critical-path effect. (KA 10.4)

**Propagation rate** — The weekly cost of a materialised risk continuing to contaminate downstream work while a decision is pending; the term that makes early notification arithmetically obvious and is usually omitted. (KA 8.4)

**Proportional shortfall** — The ratio  $p'/p$  by which lifting a condition multiplies the conjunction: the correct remediation ranking. (KA 16.1)

**Protective capacity** — Capacity deliberately withheld from allocation so that a period's plan can absorb variance. (KA 15.3)

**Provenance citation** — Each material assertion in an output referenced to the supplied source location it came from. (KA 14.3)

**Psychological safety** — A shared belief that the team is safe for interpersonal risk-taking, candour without penalty. (KA 12.2)

**PTA headroom** — The distance from the seller's current cost forecast to the PTA; 5.59 % on Auriga, and the figure to trend monthly. (KA 7.4)

**PV EV AC** — Budgeted cost of work scheduled · performed · the cost actually incurred. (KA 7.3)

## Q

**Qualified alternate** — A second supplier approved and kept exercisable, receiving no volume; the structure that actually buys resilience. (KA 10.4)

**Quality** — The degree to which delivered work conforms to its stated requirements. (KA 9.1)

**Quality assurance** — Process-directed, preventive activity: is the way we work capable of producing conforming output? (KA 9.2)

**Quality control** — Product-directed, detective activity: does this output conform? (KA 9.2)

**Quality premium test** — Comparing a higher-quality bid's price premium with the expected cost it demonstrably avoids. (KA 10.2)



**Quality-adjusted breakeven volume** —  $F / [(m + e_{mu}) - (a + e_{au})]$ : the volume at which an automation pays once escaped errors are priced. (KA 14.2)

**Quantification threshold** — The impact above which a screened risk is quantified individually; a screen without one has replaced quantification. (KA 8.2)

**Quantum** — The claim's money, built up by head of cost and evidenced contemporaneously. (KA 10.4)

**Quoted direct cost** — The figure on the form: typically a minority of the true cost, and the wrong basis for a threshold. (KA 4.4)

## R

**Ramp-up loss** — Weeks of a replacement's tenure minus the sum of their weekly effectiveness, valued at a person-week. (KA 12.2)

**Ramped basis** — Breakeven stated as the *steady state* a proportionally ramped profile must reach (Meridian 45.01 %). (KA 2.2)

**Range forecast** — Backlog ÷ the observed high and low sustained net drains, with its assumptions stated. (KA 13.2)

**Range of optimality** — The band of an uncertain input over which the recommended option stays recommended. (KA 9.1)

**Re-baselining** — Domain 4's instrument (KA 4.3.3) seen from earned value: it resets the PV curve every index is measured against, which is exactly why it cannot be used to retire an adverse variance. (KA 7.2)

· *Also read at KA 4.3:* Replacing a baseline that no longer conveys information, legitimate, and the most abused instrument in project control.

**Re-decision tax** — The committee slots and delay cost consumed by questions that had already been decided. (KA 3.3)

**Readiness conjunction** —  $P(\text{clean transition}) = \prod p_i$  over the  $k$  necessary discretionary conditions; always at or below the averaged dashboard figure. (KA 16.1)

**Reallocation distance** — The money that must move to close the gaps:  $(1 - \text{index}) \times \text{denominator}$ ; deficits equal surpluses. (KA 2.1)

**Recall (by group)** — Flagged at-risk cases ÷ actual at-risk cases within a group. (KA 14.4)

**Record class, custodian, retention period** — The three fields that make "the evidence is retained" real: what class of record this is, the named role accountable for it existing, and how long it is held with the source of that period named. Periods come from the applicable regime and the contract, not from this book. (KA 9.3)

**Record custodian** — The named role accountable for a record class existing, being complete and being retrievable: one role per class, so that no class is custodied by "the project". (KA 3.3)

**Recovery plan** — Target date, ranked moves, costs, risks, non-financial ceilings, and **both** decision authorities: the one spending the money and the safety or approval-holding authority where one is engaged. (KA 6.4)

**Recurrence rate** — Repeat nonconformances as a share of all; the test of whether causes are being removed. (KA 9.3)  
· *Also read at KA 9.4:* Repeat nonconformances as a share of all; the honest measure of an improvement loop.

**Recurring versus one-off value** — The distinction that decides whether a support function survives its second year. (KA 15.3)

**Reference-class forecasting** — Estimating from comparable completed projects to counter optimism bias. (KA 8.4)

**Refusal permission** — An explicit instruction that "the source does not say" is an acceptable and expected answer. (KA 14.3)

**Register access list** — The stated set of people who may read the stakeholder register; it is not a general-circulation document. (KA 11.1)

**Registered measure** — The single definition of a benefit measure fixed before measurement; variants are identified, not substituted. (KA 16.4)

**Regrade** — Use of nonconforming output for a lesser purpose for which it conforms, with traceability. (KA 9.3)

**Reinforcement trough** — The transient period after adding people in which progress is worse than it would have been; exists when supervision load per newcomer exceeds their ramp productivity. (KA 1.3)

**Rejection entry** — The log record of a change not approved: the entry most often missing and the reason requests recur. (KA 4.4)

**Relatedness class** — The set over which a cumulative test sums; widening it multiplies false trips and re-centralises the delegation. (KA 3.3)

**Relatedness class (concessions)** — The set over which the cumulative total is summed, defined by what the concessions touch, never by who requested them; widening it multiplies false trips. (KA 9.3)

**Releasability** — The property that an increment can go live alone; it is what converts sequencing into value. (KA 13.1)

**Released hours** — Capacity freed, not cash; a benefit only once redeployed to a stated use. (KA 1.3)

**Released leader hours** — Status-quo leader hours less retained oversight; the actual product a delegation buys. (KA 12.3)

**Reperformance** — Recomputing a result from its inputs by an independent route; the only effective control against plausible error. (KA 14.3)

**Reporting period (P)** — The interval of activity a report covers. (KA 11.2)

**Required draw efficiency** — Remaining contingency ÷ (demonstrated draw rate × remaining progress) — the TCPI of the reserve. (KA 7.1)

**Required per-dependency reliability** — The uniform probability each of  $k$  dependencies must carry to reach a target milestone confidence: the  $k$ -th root of the target. (KA 15.1)

**Required rate** — Backlog ÷ target weeks, plus arrivals, the throughput a committed date presupposes. (KA 13.2)

**Required throughput** — Backlog ÷ (successor's LS – stream start): what the network asks a cadence team to sustain. (KA 6.4)

**Requirement bundle** — The smallest set of requirements that together delivers a measurable benefit, the unit of value attribution. (KA 5.3)

**Requirement classes** — Functional · non-functional · constraint · data · transition · regulatory: reported separately because they fail differently. (KA 5.2)

**Requirement inflation** — Elaborating need into a count that impresses and disperses meaning, defeating traceability and prioritisation. (KA 5.2)

**Requirement-count reconciliation** — Baseline count + approved additions = traced count; the only cheap control that detects creep. (KA 5.4)

**Reservation value** — The worst terms a party would rationally accept, derived from its priced alternative; a maximum for a buyer, a minimum for a seller. (KA 11.3)

**Reserved matter** — A decision class that requires the highest (often unanimous) authority; kept deliberately short. (KA 3.1)

**Residual exposure** —  $\sum p_i \prod (1 - q_{ij}) u_i$  across an assurance map; the only quantity that ranks its rows. (KA 3.3)

**Resilience** — Capability to absorb unidentified risk; buffers, optionality, modularity, redundancy, fast detection. (KA 8.4)

**Resolution route** — The chosen mode, its cost, and the escalation path with its latency if the mode fails. (KA 11.3)

**Resource histogram** — Demand per period from the loaded schedule. (KA 6.3)

**Resource-feasible schedule** — A schedule in which no period's demand exceeds any resource cap. (KA 6.3)

**Responsibility** — The obligation to do; shareable and delegable. (KA 1.2)

**Responsible-AI principle** — AI proposes; the professional verifies, decides and remains accountable. (KA 1.1)

**Restated criterion** — A criterion that repeats the requirement instead of stating a measurable condition: worse than an absent one, and commoner. (KA 5.4)

**Retention** — A withheld percentage of payment, released on completion criteria; Auriga's 5 % is 55.00 % of the bid margin at completion. (KA 7.4)  
· Also read at KA 16.2: A proportion of the contract sum withheld as security, released against defined events.

**Retention period** — How long a record class is held, set in advance at the longest of contractual limitation, applicable retention requirement, benefits-realisation horizon and records policy; a schedule by class, never one rule, and never a legal position stated by this book. (KA 3.3)

**Retention schedule** — Class-by-class statement of what is kept, for how long, on what authority, and what disposal or de-identification follows. (KA 16.4)

**Retrieval rate** — The proportion of captured lessons retrieved before a comparable decision; the term in which almost all lesson value is lost. (KA 16.3)

**Reversal-cost ratio ( $\rho$ )** — Cost to undo a decision  $\div$  its value; authority reads on `max(value, cost to undo)`. (KA 3.2)

**Reversion plan** — The rehearsed route back, with a trigger, a decision-maker, a time limit and a latest-possible-reversion point. (KA 16.1)

**Rework allowance** — The capacity a plan reserves for rework; a forecast, and a decision. (KA 9.2)

**Rework share of capacity ( $r$ )** — The fraction of a team's available effort consumed correcting output. (KA 9.2)

**Risk / opportunity** — Uncertain event affecting an objective, adversely / favourably. (KA 8.1)

**Risk appetite / threshold** — Accepted exposure, expressed operably enough to apply consistently. (KA 8.1)

**Risk loading** — The premium a risk-averse supplier adds above the risk-neutral price; the arguable, visible price of certainty. (KA 10.3)

**Risk retirement** — Closing a risk whose window has passed; frees contingency for return, often far more than the risk's EMV. (KA 8.3)

**Risk-neutral fixed price** — Expected cost plus the required margin; at this price, transferring risk costs the buyer nothing in expectation and buys variance reduction only. (KA 10.3)

**Risk-tiered diligence** — Concentrating supply-chain assurance effort where breach probability is highest, rather than screening uniformly. (KA 10.4)

**Rolled throughput yield (RTY)** —  $\prod y_i$ , the share of items passing every sequential step first time. (KA 9.4)

**Rolling wave** — Near-term detail, far-term packages, elaborated under change control. (KA 6.3)

**Route of last resort** — For a risk of harm to people: next governance tier, then independent assurance, then the board or its audit or risk committee — stated in advance, each step in writing. (KA 12.4)

**Route to market** — The competitive structure chosen for a procurement; each route buys a different amount of price discovery at a different cost in time. (KA 10.2)

**Runbook adequacy test** — Whether someone who was not on the project can execute it under supervision: the only proof documentation works. (KA 16.3)

## S

**Safety-standards pairing** — Safety governs whether people can speak; standards govern what they are held to. Both are required. (KA 12.2)

**Salience** — Influence  $\times$  interest, a useful first cut and an inadequate allocation rule. (KA 11.1)

**Same-pool over-claim** — Aggregate claims on a resource pool exceeding the pool; capped at the pool and allocated. (KA 15.2)

**Scenario analysis** — Complete network re-runs under coherent stated assumptions. (KA 6.4)

**Schedule hierarchy (L1–L4)** — Nested plans for different audiences, derived from one logic. (KA 6.1)

**Scope baseline** — The approved scope statement + WBS + WBS dictionary, one of Domain 4's three baseline dimensions. (KA 5.1)

**Scope creep** — Movement of delivered content with no baseline movement and no record; invisible to every change-log instrument. (KA 5.4)

**Scope statement** — The authorised description of what will and will not be delivered, specific enough to be tested against. (KA 5.1)

**Scope-based contract** — A firm price for a defined body of work; the supplier bears effort risk, the buyer pays for change. (KA 13.4)

**Screening ruling** — The recorded classification: the baseline clause relied on, the outcome, and a named technical authority; the only artefact that makes a classification contestable later. (KA 4.4)

**Secondary risk** — Risk created by a response; a register without them is incomplete. (KA 8.3)

**Self-consistent sampling plan** — One whose confidence bound lies below its own breakeven defective fraction. (KA 9.3)

**Service acceptance** — The receiving organisation's formal agreement that the service meets specification. (KA 16.1)

**Share ratio** — The agreed buyer/seller split of over- and under-run. (KA 7.4)

**Shared benefit** — One benefit claimed by two components that both genuinely contribute to it; counted once. (KA 15.2)

**Silent measure** — An outcome measure reported by two parties, owned by neither, and never accompanied by a proposed action. (KA 1.2)

**Simple payback** — Cost  $\div$  annual benefit, in periods; scales as the reciprocal of adoption. (KA 1.3)

**Simultaneous operations** — Two disciplines working the same plant at once; a safety decision requiring its own review and approval before it is an arithmetic one. (KA 6.4)

**Single-A test** — The check that each decision class has exactly one Accountable role. (KA 3.3)

**Situational diagnosis** — Deciding what direction and support this person needs on this task before choosing a behaviour. (KA 12.1)

**Smoothing / levelling** — Flattening demand within float / accepting a later date under a hard cap. (KA 6.3)

**Solution capture** — Recording a stakeholder's proposed solution as the requirement, locking the design before the problem is understood. (KA 5.2)

**Source of record** — The one system whose value for a fact prevails; all other holdings are copies. (KA 14.1)

**Source-dated figure** — A reported value carrying the date of its own source rather than the date of the pack. (KA 15.4)

**Sponsor** — The individual accountable for the project's business outcome and mandate. (KA 3.2)

**Sponsor load** — The annual diary hours the sponsor obligations imply; a portfolio allocation constraint, not a personal one. (KA 3.2)

**Sponsor turnaround** — The measured interval from a decision reaching the sponsor to its return; read as a distribution, not a mean. (KA 3.2)

**Stage gate** — A continuation decision by a named authority against pre-set criteria, with power to stop, hold or redirect. (KA 3.3)

**Stakeholder** — A party whose decisions, consent, resources, behaviour or objection can materially affect the project's outputs, outcomes or legitimacy, or who is materially affected by them. (KA 11.1)

**Stakeholder system** — The stakeholder set together with the relationships between its members, who influences whom, and which pairs align under pressure. (KA 11.1)

**Standard of care** — The care a competent practitioner would exercise in the circumstances. (KA 1.2)

**Status accounting** — The record of the current version of every controlled item. (KA 4.3)

**Steering committee** — The body that makes decisions exceeding the project leader's authority, in time for them to matter. (KA 3.2)

**Stranded capacity** — Unused constrained resource that no remaining candidate is small enough to use, the signature of greedy failure. (KA 5.3)

**Straw options** — An options set built so the preferred option cannot lose. (KA 2.2)

**Structural interest** — An interest touching the substance of a role rather than a single item; managed by changing the role or the work, never by a declaration followed by the decision being taken anyway. (KA 1.2)

**Sub-additivity of controls** — Combined avoided loss is less than the sum of individual avoided losses, because reductions multiply. (KA 14.4)

**Sub-tier concentration** — A shared dependency below the first tier that correlates suppliers' failures and destroys a dual-sourcing case. (KA 10.4)

**Sufficiency convention T** — The minimum weekly coaching time an organisation states as meaningful; 45 minutes on Auriga, a convention not a measurement. (KA 12.3)

**Sum test** — Attributed benefit across bundles must not exceed the business case, and any shortfall is explained, not distributed. (KA 5.3)

**Sunk cost** — Unrecoverable spend carrying no information beyond the forecast; irrelevant to a continuation decision. (KA 2.4)

**Supersession hazard** — The annual probability that a stated strategic priority is replaced, merged or de-prioritised. (KA 2.1)

**Surplus** — Own reservation value less the price agreed (buyer), or price less own reservation value (seller). (KA 11.3)

**Suspension on breach** — The consequence that gives the cumulative test teeth: further concessions in that class stop pending a decision by the authority appropriate to the aggregate, taken through the escalation machinery. (KA 9.3)

**Synthetic content risk** — Convincing reproduction of a programme's identity, spokespeople, voice or documents at negligible cost. (KA 11.4)

**Systematic error** — An error repeating consistently across similar items, the characteristic shape of AI error, requiring stratified sampling. (KA 9.4)

## T

**T&M/fixed-price breakeven** —  $\text{fixed price} \div \text{hourly rate}$ ; the premium as a percentage of the estimate equals the hours overrun at which the two options cross. (KA 7.4)

**Tailoring** — A recorded, owned decision to adapt method, never a silent omission. (KA 4.1)

**Tailoring (of a message)** — Varying depth and vocabulary by audience while holding the facts identical. **Context flag:** distinct from *tailoring of method* (Domain 4, KA 4.1.3), which is a recorded decision to adapt process; the shared word carries two different concepts and neither reading substitutes for the other. (KA 11.2)

**Target fee vs expected fee** — The fee at the target cost, against the probability-weighted fee actually expected; any real overrun risk makes the second lower than the first. (KA 10.3)

**Target operating state** — The stated description of the organisation once the change has landed; the test of component coherence. (KA 15.1)

**TCPI** — Efficiency the remaining work must achieve to hit a stated target. (KA 7.3)

**Temporary organisation** — A team with no accumulated trust or shared norms; both must be designed. (KA 1.1)

**Test-order rule** — Rank assumptions by *EMV* ÷ *cost to test*, so the free tests are cleared before the paper is written. (KA 2.3)

**Three lines of assurance** — Management controls; independent oversight typically inside management; independent audit typically reporting outside it. A model of roles published by a professional body, revised by its owner, voluntary rather than required, a lens for finding gaps and duplication, not an obligatory structure. (KA 3.3)

**Three-point / PERT estimate** —  $t_e = (o + 4m + p) / 6$ ,  $\sigma = (p - o) / 6$ . (KA 6.4)

**Threshold mismatch** — Client and supplier change-approval thresholds set independently, so changes between them arrive committed on one side and unapproved on the other. (KA 10.1)

**Throughput (T)** — Completed items per unit of time, a measurement, never an aspiration. (KA 13.2)

**Tier traversal** — One decision passing one tier; the unit on which a paper-lead-time saving accrues. (KA 15.4)

**Time-phased cost baseline** — The *PV* curve; it must move whenever the schedule does. (KA 4.3)

**Timing value of an escalation** — Weeks of earliness × the run rate the remedy recovers: what the *timing* alone is worth, separate from the remedy. (KA 1.2)

**Timing variance** — The value effect of the benefit profile arriving later than the case assumed. (KA 16.4)

**Tolerance** — The amount by which an objective may be exceeded before appetite is breached; the room inside which contingency sits. (KA 8.3)  
· *Also read at KA 9.1:* The range within which output is conforming; a specification without one is untestable.

**Total cost of ownership** — Whole-life cost of the service, including the operating cost that must appear in the receiving organisation's budget before closeout. (KA 16.2)

**Total cost of ownership (TCO)** — Transition-in + operating cost over the period of need + transition-out; the only valid basis for make-or-buy. (KA 10.1)

**Total float TF** — *LS* – *ES*; slip available without delaying the project. (KA 6.2)

**Total funding requirement** — *BAC* + management reserve: the cash the sponsor must have available; Auriga's is USD 4,240,000. (KA 7.1)

**Tranche** — A group of components delivered together to reach a usable intermediate state at which benefit begins. (KA 15.1)

**Transition-in / transition-out** — The cost of entering and of leaving an arrangement; the exit column is the one most often scored at zero. (KA 10.1)

**Transition-out** — Dissolution of the project organisation: authority withdrawn, accounts closed, access revoked, records placed, people released well. (KA 16.3)

**Turnover cost** — Recruitment + ramp-up + lost productivity, excluding the saved pay of a vacant post. (KA 12.2)



**Twin fidelity** — The share of test questions on which the twin's answer matches the asset within a stated tolerance. (KA 14.2)

**Two-envelope evaluation** — Scoring quality before price is revealed, to remove price anchoring from quality judgement. (KA 10.2)

**Two-person rule** — A named drafter and a named verifier for every external communication. (KA 11.4)

## U

**Uncertainty** — Conditions that cannot be meaningfully enumerated with probabilities. (KA 8.1)

**Uncommitted forecast balance** —  $EAC - AC$  — open commitments; its excess over the baseline's uncommitted balance equals \ (KA 7.2)

**Uniform yield requirement** —  $RTY^{(1/k)}$ : the per-step yield an end-to-end target implies over  $k$  steps. (KA 9.4)

**Uniform-target defect** — Setting one rate target across classes of unequal consequence, which can cut defect counts while raising expected cost. (KA 14.1)

**Unsocial-hours burden** — The hours per group per week spent outside normal local working time; equalised by rotation. (KA 12.4)

## V

**VAC** —  $BAC - EAC$ , forecast variance at completion. (KA 7.3)

**Validation** — Does what we built produce the outcome we needed? A project can pass the first completely and fail the second completely. (KA 5.4)

**Value envelope** — The bounded authority within which a product owner may commit capacity without escalation. (KA 13.1)

**Value of information** — The reduction in expected cost that knowing enables; zero if it changes no action. (KA 8.2)

**Value of perfect information** — The most any signal could be worth: expected cost without information less expected cost knowing the truth free. (KA 8.2)

**Value per unit of constrained effort** — Attributed benefit  $\div$  effort in the scarce resource: the ranking that engages the constraint. (KA 5.3)

**Value per unit of constraint** — Ranking basis when a scarce resource binds; a heuristic, not an optimum. (KA 2.2)

**Value-led core** — Engagement effort sized from the benefit arithmetic rather than from the salience score. (KA 11.1)

**Variance contribution** —  $p(1-p)I^2$  for one risk; quadratic in impact, so it ranks a register differently from *EMV*. (KA 8.2)

**Velocity** — Throughput weighted by team-relative size units, a planning input inside a team, never a performance measure. (KA 13.4)

**Verification** — Did we build what we specified? (KA 5.4)

**Verification proportionality** — Depth of checking matched to the stakes of reliance. (KA 1.4)

**Verification standard proportional to consequence** — A rule setting review depth from the cost of an escaped error, not from convenience. (KA 9.4)

**Verification tier** — A defined review depth with a cost per item  $v$  and a measured detection rate  $q$ , performed by a named human. (KA 14.3)

**Verification tier threshold** —  $u^*_k = \Delta v_k / (p \cdot \Delta q_k)$ : the escaped-error consequence at which the next tier begins to pay. (KA 14.3)

**Voluntary framework vs disclosure regime** — A framework is adopted by choice; a regime applies on its own terms. Adopting the first says nothing about the second, and both are questions for the reporting function and qualified counsel. (KA 2.3)

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## W

**WBS** — Deliverable-oriented hierarchical decomposition of the total scope of work.

(KA 4.2)

**WBS dictionary** — Per work package: work, deliverable, acceptance basis, owner, estimate, dependencies, where scope actually lives.

(KA 5.1)

**Weighted scoring** — Ranking by  $\Sigma(\text{weight} \times \text{ordinal score})$ ; forces criteria into the open, steerable by whoever sets weights. (KA 2.2)

**Within-cell span** — The *EMV* range a single matrix cell contains: the product of its two band ratios. (KA 8.2)

**Within-group variation** — The reason national-culture averages cannot predict an individual's behaviour. (KA 12.4)

**Work in progress (W)** — Items started and not finished; a managed quantity with an explicit limit. (KA 13.2)

**Work package** — The lowest WBS level: one owner, a verifiable outcome, an endorsed estimate, an assessable duration. (KA 4.2)

**Work-in-progress limit** — An explicit cap on *W*, changed only deliberately; the control the flow arithmetic justifies. (KA 13.2)

**Working-hour overlap** —  $\max(0, \min(\text{end}) - \max(\text{start}))$  with all windows converted to a common reference. (KA 12.4)

**Works test / service test** — Whether the thing was built as specified / whether the organisation can actually run it. (KA 16.1)

---

## Z

**Zero-fee cost** —  $\text{target cost} + \text{target fee} \div \text{seller share}$ , where the incentive fee is exhausted; on Auriga USD 2,500,000, above the *PTA*, which is why the ceiling binds first. (KA 7.4)

**Zero-overlap pair** — Two groups with no common working hours; each exchange costs at least one working day. (KA 12.4)

**Zone of possible agreement (ZOPA)** — The interval between the two reservation values; empty means no deal at any level of skill.

(KA 11.3)



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